

21-00-01 Environmental Control System Miscellaneous Cards (ECSMC)
21-00-01-01 Passenger

Interval	Installed	Required	Procedure
C	2	1	(M) (O)

One may be inoperative provided:

- a. Associated ECSMC is deactivated.
- b. Both ARINC Signal Gateway cards in the opposite cardfile operate normally.
- c. Opposite equipment cooling controller operates normally.
- d. Opposite equipment cooling supply fan operates normally.
- e. Opposite lavatory/galley fan operates normally.
- f. One pack operates normally.
- g. Appropriate performance adjustments are applied.

[For FCAC installed:](#)

- h. For right ECSMC inoperative, FCAC remains OFF.

MAINTENANCE (M)

NOTE: For a pack to be operating normally, certain items associated with that system must be operative. See DDG item 21-51-01 for a listing of those items.

Deactivate the inoperative ECSMC and for Forward Cargo Air Conditioning installed with right ECSMC inoperative, position FWD CARGO A/C control OFF (AMM 21-00-00/901).

1. Deactivate the inoperative ECSMC (AMM 21-00-00/901).

NOTE 1: For left ECSMC deactivated: The left lower and forward upper recirculation fans are inoperative and also considered deactivated. The RECIRC FAN status message will be displayed. The aft cargo heat is inoperative and the CARGO HEAT AFT advisory message will be displayed.

NOTE 2: For right ECSMC deactivated: The right lower and aft upper recirculation fans are inoperative and also considered deactivated. The RECIRC FAN status message will be displayed. The bulk cargo heat is inoperative and the CARGO HEAT BULK advisory message will be displayed.

- A. Gain access to the right systems cardfile (P84) or left systems cardfile (P85) in the main equipment center.
- B. Set the power switch for the inoperative ECSMC to the OFF position.

For Forward Cargo Air Conditioning (FCAC) installed:

2. For right ECSMC inoperative, position Overhead Panel (P5) FWD CARGO A/C control OFF.

OPERATIONS (O)

1. For left ECSMC deactivated:
 - A. Aft cargo heat is inoperative and the CARGO HEAT AFT advisory message will be displayed.
 - B. Use DDG item 21-44-01 (O) Note for predicting aft cargo compartment temperatures. Temperature sensitive cargo should not be carried in the aft cargo compartment.

NOTE: Actual aft cargo compartment temperature will not be available on the Air Synoptic.
 - C. Left lower and forward upper recirculation fans are inoperative. Increase flight planning fuel by 0.7%.
2. For right ECSMC deactivated:
 - A. Bulk cargo heat is inoperative and the CARGO HEAT BULK advisory message will be displayed.
 - B. Use DDG item 21-44-03 (O) Note for predicting bulk cargo compartment temperatures. Temperature sensitive cargo should not be carried in the bulk cargo compartment.

NOTE: Actual bulk cargo compartment temperature will not be available on the Air Synoptic.
 - C. Right lower and aft upper recirculation fans are inoperative. Increase flight planning fuel by 0.7%.

For Forward Cargo Air Conditioning (FCAC) installed:

- D. FCAC is inoperative and must remain OFF. Temperature sensitive cargo should not be carried in the forward cargo compartment.

21-00-01 Environmental Control System Miscellaneous Cards (ECSMC)

21-00-01-02 777F

Interval	Installed	Required	Procedure
C	2	1	(M) (O)

One may be inoperative provided:

- a. Associated ECSMC is deactivated.
- b. Both ARINC Signal Gateway cards in the opposite cardfile operate normally.
- c. Opposite equipment cooling controller operates normally.
- d. Opposite equipment cooling supply fan operates normally.
- e. Opposite lavatory/galley fan operates normally.
- f. One pack operates normally.

- g. Both engine bleed systems operate normally.
 - h. Appropriate performance adjustments are applied.
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MAINTENANCE (M)

NOTE 1: For a pack to be operating normally, certain items associated with that system must be operative. See DDG item 21-51-01 for a listing of those items.

NOTE 2: For an engine bleed system to be operating normally, certain items associated with that system must be operative. See DDG item 36-00-01 for a listing of those items.

Deactivate the inoperative ECSMC (AMM 21-00-00/901).

NOTE 1: For left ECSMC deactivated: The left lower recirculation fan is inoperative and also considered deactivated. The RECIRC FAN status message will be displayed. The aft cargo heat is inoperative and the CARGO HEAT AFT advisory message will be displayed.

NOTE 2: For right ECSMC deactivated: The right lower recirculation fan is inoperative and also considered deactivated. The RECIRC FAN status message will be displayed. The bulk cargo heat is inoperative. For airplanes without AIMS V17B or later installed, the CARGO HEAT BULK advisory message will be displayed.

1. Gain access to the associated right systems cardfile (P84), or left systems cardfile (P85), in the main equipment center.
2. Set the power switch for the inoperative ECSMC to the OFF position.

OPERATIONS (O)

Use DDG item 21-25-01 (O) procedure.

For Aft Cargo Air Conditioning (ACAC) not installed:

1. For left ECSMC deactivated:
 - A. Lower aft cargo heat is inoperative and the CARGO HEAT AFT advisory message will be displayed.
 - B. Use DDG item 21-44-01 (O) Note for predicting lower aft cargo compartment temperatures. Temperature sensitive cargo should not be carried in the lower aft cargo compartment.

NOTE: Actual lower aft cargo compartment temperature will not be available on the Air Synoptic.

2. For right ECSMC deactivated:
 - A. For airplanes without AIMS V17B or later installed, the CARGO HEAT BULK advisory message will be displayed. Disregard this nuisance message.

**21-24-01 Gasper Fan (Passenger)**

Interval	Installed	Required	Procedure
D	1	0	(M)

Gasper fan may be inoperative provided it is deactivated.

MAINTENANCE (M)

Deactivate the gasper fan.

1. Open and collar the P110 panel GSPR FAN CTRL circuit breaker.
2. Open and collar the P110 panel GSPR FAN circuit breaker.

21-24-01 Gasper Fan (Passenger)

21-24-01-01 GASPER Switch ON Light

Interval	Installed	Required	Procedure
C	1	0	

**21-25-01 Recirculation Fans****21-25-01-01 Passenger**

Interval	Installed	Required	Procedure
C	4	0	(M) (O)

May be inoperative provided associated recirculation fan is deactivated.

MAINTENANCE (M)

Deactivate the inoperative recirculation fan (AMM 21-00-00/901).

NOTE: The RECIRC FAN status message indicates one or more fans are inoperative.

1. Determine which recirculation fan is inoperative:
 - A. Position both Air Conditioning Panel (P5) PACK switches to OFF.
 - B. Position both Air Conditioning Panel (P5) RECIRC FANS switches to ON.
 - C. On the Air Conditioning Maintenance page, use the recirculation fan status to determine the inoperative fan.
 - D. Position both Air Conditioning Panel (P5) PACK switches to AUTO.
2. Deactivate the inoperative recirculation fan.
 - A. For forward upper recirculation fan inoperative, open and collar the P110 panel FWD UPR RECIRC FAN circuit breaker.
 - B. For aft upper recirculation fan inoperative, open and collar the P210 panel AFT UPR RECIRC FAN circuit breaker.
 - C. For left lower recirculation fan inoperative, open and collar the P110 panel L LWR RECIRC FAN circuit breaker.

For -200/-200ER/-200LR airplanes without Forward Cargo Air Conditioning (FCAC) installed:

- D. For right lower recirculation fan inoperative, open and collar the P210 panel R LWR RECIRC FAN circuit breaker.

For -200/-200ER/-200LR airplanes with Forward Cargo Air Conditioning (FCAC) installed, and for -300/-300ER airplanes:

- E. For right lower recirculation fan inoperative:
 - 1) Open the P210 panel AFT UPR/R LWR RECIRC FAN CTRL circuit breaker.
 - 2) Gain access to the lower recirculation fans by removing the aft endwall in the forward cargo compartment.
 - 3) Disconnect, cap and stow the electrical connector (DB21252) from the right lower recirculation fan.
 - 4) Re-install the endwall.



- 5) Close the P210 panel AFT UPR/R LWR RECIRC FAN CTRL circuit breaker.

OPERATIONS (O)

Fuel penalties are negligible with one recirculation fan inoperative. For two recirculation fans inoperative, increase flight planning fuel by 0.7%, plus 0.3% for each additional inoperative fan.

For Forward Cargo Air Conditioning (FCAC) installed:

NOTE: For one or both lower recirculation fans inoperative, temperature control in the forward cargo compartment may be degraded.

21-25-01 Recirculation Fans

21-25-01-01 Passenger

21-25-01-01-01 RECIRC FANS Switch ON Lights

Interval	Installed	Required	Procedure
C	2	0	

21-25-01 Recirculation Fans

21-25-01-02 777F

Interval	Installed	Required	Procedure
C	2	0	(M) (O)

May be inoperative provided:

- Associated recirculation fan is deactivated.
- Both engine bleed systems operate normally.

MAINTENANCE (M)

NOTE: For an engine bleed system to be operating normally, certain items associated with that system must be operative. See DDG item 36-00-01 for a listing of those items.

Deactivate the inoperative recirculation fan (AMM 21-00-00/901).

- For left recirculation fan inoperative, open and collar the P110 panel L LWR RECIRC FAN CTRL circuit breaker.
- For right recirculation fan inoperative, open and collar the P210 panel R LWR RECIRC FAN CTRL circuit breaker.

**OPERATIONS (O)****For ACAC not installed:**

NOTE 1: For M/D FLOW set to HIGH flow mode, FCAC temperature control capability is degraded. Temperature sensitive cargo should not be carried in the lower forward cargo compartment."

For ACAC installed:

NOTE 2: For M/D FLOW set to HIGH flow mode, FCAC and ACAC temperature control capability is degraded. Temperature sensitive cargo should not be carried in the lower cargo compartments."

For both recirculation fans inoperative:

1. For overhead panel (P5) M/D FLOW selector set to NORM, increase flight planning fuel by 0.5%.
2. For overhead panel (P5) M/D FLOW selector set to HIGH, increase flight planning fuel by 0.8%.

21-25-01 Recirculation Fans**21-25-01-02 777F****21-25-01-02-01 RECIRC FANS Switch ON Light**

Interval	Installed	Required	Procedure
C	1	0	

**21-26-01 Lavatory/Galley Ventilation Fans**

Interval	Installed	Required	Procedure
C	2	1	(M)

One may be inoperative provided:

- a. Inoperative ventilation fan is deactivated.
- b. Opposite ECSMC operates normally.

MAINTENANCE (M)

Deactivate the inoperative lavatory/galley ventilation fan (AMM 21-00-00/901).

1. For the left lavatory/galley ventilation fan inoperative:

NOTE: The right ECSMC must operate normally.

- A. Open and collar the P110 panel L LAV/GLY FAN CTRL circuit breaker.
- B. For P110 panel L LAV/GLY FAN circuit breaker installed, open and collar the circuit breaker.
- C. For P110 panel L LAV/GLY FAN circuit breaker not installed:
 - 1) Gain access to the left lavatory/galley ventilation fan by removing the left endwall in the bulk cargo compartment.
 - 2) Disconnect, cap and stow the electrical connector (DB21262) from the left lavatory/galley ventilation fan.
 - 3) Re-install the endwall.

2. For the right lavatory/galley ventilation fan inoperative:

NOTE: The left ECSMC must operate normally.

- A. Open and collar the P210 panel R LAV/GLY FAN CTRL circuit breaker.
- B. For P210 panel R LAV/GLY FAN circuit breaker installed, open and collar the circuit breaker.
- C. For P210 panel R LAV/GLY FAN circuit breaker not installed:
 - 1) Gain access to the right lavatory/galley ventilation fan by removing the left endwall in the bulk cargo compartment.
 - 2) Disconnect, cap and stow the electrical connector (DB21261) from the right lavatory/galley ventilation fan.
 - 3) Re-install the endwall.

**21-26-02 Chiller Boost Fan (Passenger)**

Interval	Installed	Required	Procedure
C	1	0	(M)

Chiller boost fan may be inoperative provided it is deactivated.

MAINTENANCE (M)

Deactivate the inoperative Chiller Boost Fan (AMM 21-00-00/901).

1. Open and collar the P210 panel CHLR BOOST FAN CTRL circuit breaker.
2. Open and collar the P210 panel CHLR BOOST FAN circuit breaker.



21-26-03 Bulk Cargo Ventilation Fan
21-26-03-01 With Chiller Boost Fan Installed

Interval	Installed	Required	Procedure
C	1	0	(M)

May be inoperative provided:

- Bulk cargo ventilation fan is deactivated.
 - Chiller boost fan is deactivated.
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MAINTENANCE (M)

Deactivate the Bulk Cargo Ventilation Fan and the Chiller Boost Fan (AMM 21-00-00/901).

- Deactivate the Bulk Cargo Ventilation Fan.
 - Open and collar the P210 panel BULK CGO VENT FAN CTRL circuit breaker.
 - Open and collar the P210 panel BULK CGO VENT FAN circuit breaker.
- Deactivate the Chiller Boost Fan.
 - Open and collar the P210 panel CHLR BOOST FAN CTRL circuit breaker.
 - Open and collar the P210 panel CHLR BOOST FAN circuit breaker.

OPERATIONS NOTE

Bulk cargo ventilation is not available. Live animals should not be carried in the bulk cargo compartment.

21-26-03 Bulk Cargo Ventilation Fan
21-26-03-02 Without Chiller Boost Fan Installed

Interval	Installed	Required	Procedure
C	1	0	(M)

May be inoperative provided bulk cargo ventilation fan is deactivated.

MAINTENANCE (M)

Deactivate the Bulk Cargo Ventilation Fan (AMM 21-00-00/901).

- Open and collar the P210 panel BULK CGO VENT FAN CTRL circuit breaker.
- Open and collar the P210 panel BULK CGO VENT FAN circuit breaker.

**OPERATIONS NOTE**

Bulk cargo ventilation is not available. Live animals should not be carried in the bulk cargo compartment.

**21-26-04 Cabin Smoking Area Ventilation Exhaust Valves**

Interval	Installed	Required	Procedure
D	-	0	(M)

May be inoperative provided the inoperative exhaust valve(s) is deactivated closed.

MAINTENANCE (M)

Deactivate the inoperative cabin smoking area ventilation exhaust valve closed (AMM 21-00-00/901).

1. For the aft ventilation exhaust valve inoperative:
 - A. Open and collar P110 panel circuit breaker AFT CABIN EXHAUST VALVE.
 - B. Gain access to the aft exhaust valve located in the crown above the ceiling panels near Door 4R at STA 1954, WL 326, RBL 27 by removing the ceiling panels and moisture shroud.
 - C. Manually position the aft exhaust valve closed.
 - D. Re-install ceiling panels and moisture shroud.
2. For the forward ventilation exhaust valve inoperative:
 - A. Open and collar P110 panel circuit breaker FWD CABIN EXHAUST VALVE.
 - B. Gain access to the forward exhaust valve located in the crown above the ceiling panels near Door 2R at STA 739, WL 318, RBL 51 by removing the ceiling panels and moisture shroud.
 - C. Manually position the forward exhaust valve closed.
 - D. Re-install ceiling panels and moisture shroud.

21-26-05 Forward Cargo Air Conditioning (FCAC) Exhaust Fan

21-26-05A FCAC Used

Interval	Installed	Required	Procedure
C	1	0	(M)

FCAC exhaust fan may be inoperative provided it is deactivated.

MAINTENANCE (M)

Deactivate the FCAC exhaust fan (AMM 21-00-00/901).

1. Open and collar P110 panel FWD CGO EXHAUST FAN CTRL circuit breaker.
2. Open and collar P110 panel FWD CGO EXHAUST FAN circuit breaker.

OPERATIONS NOTE

Odors from the lower forward cargo compartment may migrate to the other areas of the airplane when FCAC is used.

21-26-05 Forward Cargo Air Conditioning (FCAC) Exhaust Fan

21-26-05B FCAC Not Used

Interval	Installed	Required	Procedure
D	1	0	(M)

May be inoperative provided:

- a. FCAC exhaust fan is deactivated.
- b. FCAC remains off.

MAINTENANCE (M)

Deactivate the FCAC exhaust fan (AMM 21-00-00/901).

1. Open and collar P110 panel FWD CGO EXHAUST FAN CTRL circuit breaker.
2. Open and collar P110 panel FWD CGO EXHAUST FAN circuit breaker.

For Passenger, and 777F without Aft Cargo Air Conditioning (ACAC) installed:

3. Position Overhead Panel (P5) FWD CARGO A/C control OFF.

For 777F with Aft Cargo Air Conditioning (ACAC) installed:

4. Position Overhead Panel (P5) LWR CARGO TEMP FWD A/C control OFF.

**OPERATIONS NOTE**

For Passenger, and 777F without Aft Cargo Air Conditioning (ACAC) installed:

1. Position FWD CARGO A/C control OFF.

NOTE: Lower forward cargo air conditioning is not available.
Temperature sensitive cargo should not be carried in the lower forward cargo compartment.

For 777F with Aft Cargo Air Conditioning (ACAC) installed:

2. Position LWR CARGO TEMP FWD A/C control OFF.

NOTE: Lower forward cargo air conditioning is not available.
Temperature sensitive cargo should not be carried in the lower forward cargo compartment.

**21-26-06 Lavatory/Galley Ventilation Shutoff Valve (777F)**

Interval	Installed	Required	Procedure
C	1	0	(M)

May be inoperative provided:

- Valve is deactivated open.
- Flight is conducted pressurized.

MAINTENANCE (M)

NOTE 1: At least one pack and two engine bleed systems must operate normally, or both packs and at least one engine bleed system must operate normally.

NOTE 2: For a pack to be operating normally, certain items associated with that system must be operative. See DDG item 21-51-01 for a listing of those items.

NOTE 3: For an engine bleed system to be operating normally, certain items associated with that system must be operative. See DDG item 36-00-01 for a listing of those items.

NOTE 4: Dispatch is not allowed using the unpressurized dispatch option for MMEL items: 21-31-01, 21-31-02, 21-31-03, 21-31-05, 21-31-06, 21-31-07, 21-32-01, 21-32-02, and 21-61-01.

Deactivate the inoperative lavatory/galley ventilation shutoff valve open (AMM 21-00-00/901).

- Open and collar the P110 panel CB LGVS SOV circuit breaker.
- Gain access to the lavatory/galley ventilation valve.
- Position the lavatory/galley ventilation valve open.

**21-26-07 Alternate Ventilation System Fan (777F)**

Interval	Installed	Required	Procedure
C	1	0	(M)

May be inoperative provided:

- Fan is deactivated.
- Flight is conducted pressurized.

MAINTENANCE (M)

NOTE 1: At least one pack and two engine bleed systems must operate normally, or both packs and at least one engine bleed system must operate normally.

NOTE 2: For a pack to be operating normally, certain items associated with that system must be operative. See DDG item 21-51-01 for a listing of those items.

NOTE 3: For an engine bleed system to be operating normally, certain items associated with that system must be operative. See DDG item 36-00-01 for a listing of those items.

NOTE 4: Dispatch is not allowed using the unpressurized dispatch option for MMEL items: 21-31-01, 21-31-02, 21-31-03, 21-31-05, 21-31-06, 21-31-07, 21-32-01, 21-32-02, and 21-61-01.

Deactivate the alternate ventilation system fan (AMM 21-00-00/901).

- Open and collar the P210 panel CB AVS FAN CTRL circuit breaker.
- Open and collar the P210 panel CB AVS FAN circuit breaker.

**21-26-08 Alternate Ventilation System Shutoff Valves (777F)**

Interval	Installed	Required	Procedure
C	2	0	(M)

May be inoperative provided:

- Associated valve is deactivated closed.
- Flight is conducted pressurized.

MAINTENANCE (M)

NOTE 1: At least one pack and two engine bleed systems must operate normally, or both packs and at least one engine bleed system must operate normally.

NOTE 2: For a pack to be operating normally, certain items associated with that system must be operative. See DDG item 21-51-01 for a listing of those items.

NOTE 3: For an engine bleed system to be operating normally, certain items associated with that system must be operative. See DDG item 36-00-01 for a listing of those items.

NOTE 4: Dispatch is not allowed using the unpressurized dispatch option for MMEL items: 21-31-01, 21-31-02, 21-31-03, 21-31-05, 21-31-06, 21-31-07, 21-32-01, 21-32-02, and 21-61-01.

Deactivate the inoperative alternate ventilation system shutoff valve closed (AMM 21-00-00/901).

- For an inoperative aft alternate ventilation system shutoff valve, open and collar the P110 panel CB AFT AVS SOV circuit breaker.
- For an inoperative forward alternate ventilation system shutoff valve, open and collar the P110 panel CB FWD AVS SOV circuit breaker.
- Gain access to the inoperative alternate ventilation system shutoff valve.
- Position the inoperative alternate ventilation system shutoff valve closed.

**21-26-09 Alternate Ventilation System ALTN VENT Switch (777F)**

Interval	Installed	Required	Procedure
C	1	0	

May be inoperative provided flight is conducted pressurized.

MAINTENANCE NOTE

NOTE 1: At least one pack and two engine bleed systems must operate normally, or both packs and at least one engine bleed system must operate normally.

NOTE 2: For a pack to be operating normally, certain items associated with that system must be operative. See DDG item 21-51-01 for a listing of those items.

NOTE 3: For an engine bleed system to be operating normally, certain items associated with that system must be operative. See DDG item 36-00-01 for a listing of those items.

NOTE 4: Dispatch is not allowed using the unpressurized dispatch option for MMEL items: 21-31-01, 21-31-02, 21-31-03, 21-31-05, 21-31-06, 21-31-07, 21-32-01, 21-32-02, and 21-61-01.

21-26-09 Alternate Ventilation System ALTN VENT Switch (777F)**21-26-09-01 FAULT Light**

Interval	Installed	Required	Procedure
C	1	0	

21-26-09 Alternate Ventilation System ALTN VENT Switch (777F)**21-26-09-02 ON Light**

Interval	Installed	Required	Procedure
C	1	0	

**21-27-01 Equipment Cooling Air Filter**

Interval	Installed	Required	Procedure
C	1	0	(M)

May be operated with filter removed.

MAINTENANCE (M)

Remove the equipment cooling air filter (AMM 21-00-00/901).

1. Gain access to the equipment cooling air filter by removing the left sidewall liner between STA 571 and STA 592 in lower forward cargo compartment.
2. Remove the equipment cooling air filter (AMM 21-27-02/401).
3. Re-install the left sidewall liner.

**21-27-02 Equipment Cooling Supply Fans**

Interval	Installed	Required	Procedure
C	2	1	(M)

One may be inoperative provided:

- Inoperative supply fan is deactivated.
- Opposite ECSMC operates normally.
- Both override valve motors operate normally.
- Both equipment cooling controllers operate normally.
- Flight is conducted pressurized.

MAINTENANCE (M)

NOTE 1: At least one pack and two engine bleed systems must operate normally, or both packs and at least one engine bleed system must operate normally.

NOTE 2: For a pack to be operating normally, certain items associated with that system must be operative. See DDG item 21-51-01 for a listing of those items.

NOTE 3: For an engine bleed system to be operating normally, certain items associated with that system must be operative. See DDG item 36-00-01 for a listing of those items.

NOTE 4: Dispatch is not allowed using the unpressurized dispatch option for MMEL items: 21-31-01, 21-31-02, 21-31-03, 21-31-05, 21-31-06, 21-31-07, 21-32-01, 21-32-02, and 21-61-01.

Deactivate the inoperative equipment cooling supply fan (AMM 21-00-00/901).

- For the left equipment cooling supply fan inoperative:
 - Open and collar the P110 panel EQPT CLG SPLY FAN 1 CTRL circuit breaker.
 - Gain access to the left equipment cooling supply fan located in the lower forward cargo compartment behind the left sidewall panel near the forward ball panel.
 - Disconnect, cap and stow the electrical connector (DB21581) from the left equipment cooling supply fan.
 - Re-install the left sidewall panel.
- For the right equipment cooling supply fan inoperative:
 - Open and collar the P210 panel EQPT CLG SPLY FAN 2 CTRL circuit breaker.
 - Gain access to the right equipment cooling supply fan located in the lower forward cargo compartment behind the left sidewall panel near the forward ball panel.



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- C. Disconnect, cap and stow the electrical connector (DB21582) from the right equipment cooling supply fan.
 - D. Re-install the left sidewall panel.

**21-27-03 Equipment Cooling Override Valve Motors**

Interval	Installed	Required	Procedure
C	2	1	(M)

One may be inoperative provided:

- Inoperative override valve motor is deactivated.
 - Both equipment cooling supply fans operate normally.
 - Both equipment cooling controllers operate normally.
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MAINTENANCE (M)

Deactivate the inoperative override valve motor (AMM 21-00-00/901).

- For left override valve motor inoperative, open and collar the P310 panel EQPT CLG SMOKE OVRD VLV MOT 1 circuit breaker.
- For right override valve motor inoperative, open and collar the P310 EQPT CLG SMOKE OVRD VLV MOT 2 circuit breaker.

21-27-04 Equipment Cooling Low Flow Sensors
21-27-04-01 Passenger

Interval	Installed	Required	Procedure
C	2	1	(M)

One flow sensor may be inoperative provided it is deactivated.

MAINTENANCE (M)

Deactivate the inoperative equipment cooling low flow sensor (AMM 21-00-00/901).

1. For main equipment bay low flow sensor inoperative, open and collar the P310 panel EQPT BAY LOW FLOW DET circuit breaker.
2. For flight deck low flow sensor inoperative, open and collar the P210 panel F/D LOW FLOW DET circuit breaker.

21-27-04 Equipment Cooling Low Flow Sensors
21-27-04-02 777F
21-27-04-02-01 Main Equipment Center Low Flow Sensor
21-27-04-02-01A Three-Way Valve Operation Verified

Interval	Installed	Required	Procedure
C	1	0	(M)

May be inoperative provided:

- a. Sensor is deactivated.
- b. Flight deck low flow sensor operates normally.
- c. Equipment cooling three-way valve is verified to operate normally before each flight.

MAINTENANCE (M)

Deactivate the inoperative main equipment center low flow sensor (AMM 21-00-00/901).

1. Open and collar the P310 panel EQPT BAY LOW FLOW DET circuit breaker.

Verify the equipment cooling three-way valve operates normally before each flight (AMM 21-00-00/901).

1. Provide electrical power on the airplane (AMM 24-22-00/201).
2. Gain access to the equipment cooling three-way valve through the Right Main Equipment Electronic Center.
3. Establish communications with the flight crew.



4. Set MAIN DECK CARGO FIRE ARM switch on the P5 overhead panel to ARMED.
5. Verify the three-way valve position indicator is in the C-B position (A/C-Mode).
6. Set MAIN DECK CARGO FIRE ARM switch on the P5 overhead panel off.
7. Verify the three-way valve position indicator is in the A-B position (EE-Mode).

OPERATIONS NOTE

The EQUIP FLOW DET and EQUIP 3-WAY VALVE status messages will be displayed.

21-27-04 Equipment Cooling Low Flow Sensors**21-27-04-02 777F****21-27-04-02-01 Main Equipment Center Low Flow Sensor****21-27-04-02-01B Three-Way Valve Considered Inoperative**

Interval	Installed	Required	Procedure
C	1	0	(M)

May be inoperative provided:

- a. Sensor is deactivated.
- b. Flight deck low flow sensor operates normally.
- c. Equipment cooling three-way valve is considered inoperative.

NOTE: The airplane must also be dispatched using MMEL item 21-27-15.

MAINTENANCE (M)

Deactivate the inoperative main equipment center low flow sensor (AMM 21-00-00/901).

1. Open and collar the P310 panel EQPT BAY LOW FLOW DET circuit breaker.

21-27-04 Equipment Cooling Low Flow Sensors**21-27-04-02 777F****21-27-04-02-02 Flight Deck Low Flow Sensor**

Interval	Installed	Required	Procedure
C	1	0	(M)

May be inoperative provided:

- a. Sensor is deactivated.
- b. Main equipment center low flow sensor operates normally.



- c. Equipment cooling three-way valve is considered inoperative.
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NOTE: The airplane must also be dispatched using MMEL item 21-27-15.

MAINTENANCE (M)

NOTE: The flight deck low flow sensor is interchangeable with the main equipment center low flow sensor.

Deactivate the inoperative flight deck low flow sensor (AMM 21-00-00/901).

1. Open and collar the P210 panel F/D LOW FLOW DET circuit breaker.

**21-27-05 Equipment Cooling Vent Fan**

Interval	Installed	Required	Procedure
C	1	0	(M)

May be inoperative provided:

- Fan is deactivated.
 - For ground operations with OAT 30 degrees C or higher, both packs are selected on or airplane is supplied with conditioned air.
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MAINTENANCE (M)

Deactivate the equipment cooling vent fan (AMM 21-00-00/901).

- Open and collar the P110 panel EQPT CLG VENT FAN CTRL circuit breaker.
- Disconnect, cap and stow the electrical connector (DB21583) from the equipment cooling vent fan.
- For ground operations with OAT at or above 30 degrees C, select both packs on or supply conditioned air to the airplane.

OPERATIONS NOTE

For Forward Cargo Air Conditioning (FCAC) installed:

- FCAC will operate normally.

For Forward Cargo Air Conditioning (FCAC) not installed:

- Forward cargo heat is not available. Temperature sensitive cargo should not be carried in the forward cargo compartment. Use DDG item 21-27-07 (O) Note for predicting forward cargo compartment temperatures.

21-27-06 Equipment Cooling Vent Valve
21-27-06-01 Passenger

Interval	Installed	Required	Procedure
C	1	0	(M)

May be inoperative provided:

- a. Valve is deactivated closed.
- b. Equipment cooling vent fan is deactivated.
- c. For ground operations with OAT 30 degrees C or higher, both packs are selected on or airplane is supplied with conditioned air.

For FCAC installed:

- d. FCAC exhaust fan is deactivated.

MAINTENANCE (M)

Deactivate the equipment cooling vent valve closed, deactivate the equipment cooling vent fan and if installed, deactivate the FCAC exhaust fan (AMM 21-00-00/901).

1. Deactivate the equipment cooling vent valve closed.
 - A. Open and collar the P110 panel EQPT CLG VENT VLV CTRL circuit breaker.
 - B. Open and collar the P110 panel EQPT CLG VENT VLV circuit breaker.
 - C. Gain access to the equipment cooling vent valve (V21584) located in the area below the forward cargo compartment floor on the left side by removing the floor panels and moisture shroud (AMM 25-52-10/401).
 - D. Manually position the equipment cooling vent valve closed.
 - E. Re-install the moisture shroud and floor panels.
2. Deactivate the equipment cooling vent fan.
 - A. Open and collar the P110 panel EQPT CLG VENT FAN CTRL circuit breaker.
 - B. Disconnect, cap and stow the electrical connector (DB21583) from the equipment cooling vent fan.
3. For ground operations with OAT at or above 30 degrees C, select both packs on or supply conditioned air to the airplane.

For Forward Cargo Air Conditioning (FCAC) installed:

4. Deactivate the FCAC exhaust fan.
 - A. Open and collar P110 panel FWD CGO EXHAUST FAN CTRL circuit breaker.

- B. Open and collar P110 panel FWD CGO EXHAUST FAN circuit breaker.

OPERATIONS NOTE

For Forward Cargo Air Conditioning (FCAC) not installed:

Forward cargo heat is not available. Temperature sensitive cargo should not be carried in the forward cargo compartment. Use DDG Item 21-27-07 (O) Note for predicting forward cargo compartment temperatures.

For Forward Cargo Air Conditioning (FCAC) installed:

1. For dispatch with FCAC used, forward cargo odors may migrate to the passenger cabin and flight deck.
2. For dispatch with FCAC not used, forward cargo heat and air conditioning are not available. Temperature sensitive cargo should not be carried in the forward cargo compartment. Use DDG Item 21-27-07 (O) Note for predicting forward cargo compartment temperatures.

21-27-06 Equipment Cooling Vent Valve

21-27-06-02 777F

Interval	Installed	Required	Procedure
C	1	0	(M)

May be inoperative provided:

- a. Valve is deactivated closed.
- b. FCAC exhaust fan is deactivated.
- c. For ground operations with OAT 30 degrees C or higher, both packs are selected on or airplane is supplied with conditioned air.

MAINTENANCE (M)

Deactivate the equipment cooling vent valve closed and deactivate the FCAC exhaust fan (AMM 21-00-00/901).

1. Deactivate the equipment cooling vent valve closed.
 - A. Open and collar the P110 panel EQPT CLG VENT VLV CTRL circuit breaker.
 - B. Open and collar the P110 panel EQPT CLG VENT VLV circuit breaker.
 - C. Gain access to the equipment cooling vent valve (V21584) located in the area below the lower forward cargo compartment floor on the left side by removing the floor panels and moisture shroud (AMM 25-52-10/401).
 - D. Manually position the equipment cooling vent valve closed.
 - E. Re-install the moisture shroud and floor panels.



2. Deactivate the FCAC exhaust fan.
 - A. Open and collar P110 panel FWD CGO EXHAUST FAN CTRL circuit breaker.
 - B. Open and collar P110 panel FWD CGO EXHAUST FAN circuit breaker.
3. For ground operations with OAT at or above 30 degrees C, select both packs on or supply conditioned air to the airplane.

OPERATIONS NOTE

1. For dispatch with FCAC used, lower forward cargo odors may migrate to the supernumerary area and flight deck.
2. For dispatch with FCAC not used, lower forward cargo heat and air conditioning are not available. Temperature sensitive cargo should not be carried in the lower forward cargo compartment. Use DDG Item 21-27-07 (O) Note for predicting lower forward cargo compartment temperatures.

**21-27-07 Forward Cargo Heat Valve**

Interval	Installed	Required	Procedure
C	1	0	(M)

Forward cargo heat valve may be inoperative provided it is deactivated closed.

MAINTENANCE (M)

Deactivate the forward cargo heat valve closed (AMM 21-00-00/901).

For Forward Cargo Air Conditioning (FCAC) not installed:

1. Open and collar the P110 panel EQPT CLG CGO HT VLV CTRL circuit breaker.
2. Open and collar the P110 panel EQPT CLG CGO HT VLV circuit breaker.
3. Gain access to the forward cargo heat valve located in the area below the forward cargo compartment floor on the left side by removing the floor panels and the moisture shroud (AMM 25-52-10/401).
4. Manually position the forward cargo heat valve closed.
5. Re-install the moisture shroud and cargo floor panels.

For Forward Cargo Air Conditioning (FCAC) installed:

1. Open and collar the P110 panel EQPT CLG CGO HT VLV CTRL circuit breaker.
2. Gain access to the forward cargo heat valve located in the area below the forward cargo compartment floor on the left side by removing the floor panels the moisture shroud (AMM 25-52-10/401).
3. Manually position the forward cargo heat valve closed.
4. Re-install the moisture shroud and cargo floor panels.

OPERATIONS NOTE**For Forward Cargo Air Conditioning (FCAC) installed:**

1. FCAC will operate normally.

For Forward Cargo Air Conditioning (FCAC) installed and not used:

2. Forward cargo heat is not used. Temperature sensitive cargo should not be carried in the forward cargo compartment. The following tables may be used for predicting forward cargo compartment temperature. Temperature variations may occur.

Airport OAT Degrees C	Forward Cargo Compartment Temperature Degrees F / Degrees C
-40	30 / -1
16	75 / 24
39	117 / 47



NOTE: Ground operation - Steady state temperatures with air conditioning packs on.

TAT Degrees C	Forward Cargo Compartment Temperature Degrees F / Degrees C
-44	35 / 2
-28	48 / 9
-3	68 / 20

NOTE: In-flight - Steady state temperatures after approximately 2.5 hours cruise at FL350.

For Forward Cargo Air Conditioning (FCAC) not installed:

3. Forward cargo heat is not available. Temperature sensitive cargo should not be carried in the forward cargo compartment. The following tables may be used for predicting forward cargo compartment temperature. Temperature variations may occur.

Airport OAT Degrees C	Forward Cargo Compartment Temperature Degrees F / Degrees C
-40	30 / -1
16	75 / 24
39	117 / 47

NOTE: Ground operation - Steady state temperatures with air conditioning packs on.

TAT Degrees C	Forward Cargo Compartment Temperature Degrees F / Degrees C
-44	35 / 2
-28	48 / 9
-3	68 / 20

NOTE: In-flight - Steady state temperatures after approximately 2.5 hours cruise at FL350.

**21-27-08 Equipment Cooling Controllers**

Interval	Installed	Required	Procedure
C	2	1	

Left controller may be inoperative provided:

- Right equipment cooling supply fan operates normally.
 - Right ECSMC operates normally.
 - Both override valve motors operate normally.
 - Flight deck low flow detector operates normally.
-

MAINTENANCE NOTE

The left and right equipment cooling controllers are interchangeable (AMM 21-27-01/401).

**21-27-09 SATCOM Backup Cooling Fans (Passenger)**

21-27-09A Both Lavatory/Galley Vent Fans Operational

Interval	Installed	Required	Procedure
C	-	0	(M)

May be inoperative provided:

- a. Inoperative cooling fan(s) is deactivated.
- b. Both lavatory/galley ventilation fans operate normally.

MAINTENANCE (M)

Deactivate the inoperative SATCOM backup cooling fan (AMM 21-00-00/901).

1. For the left backup cooling fan inoperative:
 - A. Open and collar the P110 panel L SATCOM BACKUP CLG FAN CTRL circuit breaker.
 - B. Open and collar the P110 panel L SATCOM BACKUP CLG FAN circuit breaker.
2. For right backup cooling fan inoperative:
 - A. Open and collar the P210 panel R SATCOM BACKUP CLG FAN CTRL circuit breaker.
 - B. Open and collar the P210 panel R SATCOM BACKUP CLG FAN circuit breaker.

21-27-09 SATCOM Backup Cooling Fans (Passenger)

21-27-09B Lavatory/Galley Fan(s) Inoperative

Interval	Installed	Required	Procedure
C	-	0	(M) (O)

May be inoperative provided:

- a. Inoperative SATCOM backup cooling fan(s) is deactivated.
- b. SATCOM remains off.
- c. Alternate procedures are established and used.
- d. Flight remains within 180 minutes of landing at a suitable airport.

**MAINTENANCE (M)**

Deactivate the inoperative SATCOM backup cooling fan and SATCOM system (AMM 21-00-00/901).

1. For the left backup cooling fan inoperative:
 - A. Open and collar the P110 panel L SATCOM BACKUP CLG FAN CTRL circuit breaker.
 - B. Open and collar the P110 panel L SATCOM BACKUP CLG FAN circuit breaker.
 - C. Open and collar the P110 panel SATL HIGH L circuit breaker.
 - D. Open and collar the P110 panel SATL SDU L circuit breaker.
 - E. For a low gain antenna system installed, open and collar P110 panel SATL LOW circuit breaker.
2. For right backup cooling fan inoperative:
 - A. Open and collar the P210 panel R SATCOM BACKUP CLG FAN CTRL circuit breaker.
 - B. Open and collar the P210 panel R SATCOM BACKUP CLG FAN circuit breaker.
 - C. Open and collar the P210 panel SATL HIGH R circuit breaker.
 - D. Open and collar the P210 panel SATL SDU R circuit breaker.

OPERATIONS (O)

1. Flight must remain within 180 minutes of landing at a suitable airport.
2. Alternate procedures must be established and used.

21-27-09 SATCOM Backup Cooling Fans (Passenger)

21-27-09C Procedures Do Not Require Use of SATCOM

Interval	Installed	Required	Procedure
D	-	0	(M)

May be inoperative provided:

- a. Inoperative SATCOM backup cooling fan(s) is deactivated.
- b. Procedures do not require the use of SATCOM.

MAINTENANCE (M)

Deactivate the inoperative SATCOM backup cooling fan and SATCOM system (AMM 21-00-00/901).

1. For the left backup cooling fan inoperative:
 - A. Open and collar the P110 panel L SATCOM BACKUP CLG FAN CTRL circuit breaker.
 - B. Open and collar the P110 panel L SATCOM BACKUP CLG FAN circuit breaker.



- C. Open and collar the P110 panel SATL HIGH L circuit breaker.
 - D. Open and collar the P110 panel SATL SDU L circuit breaker.
 - E. For a low gain antenna system installed, open and collar P110 panel SATL LOW circuit breaker.
2. For right backup cooling fan inoperative:
- A. Open and collar the P210 panel R SATCOM BACKUP CLG FAN CTRL circuit breaker.
 - B. Open and collar the P210 panel R SATCOM BACKUP CLG FAN circuit breaker.
 - C. Open and collar the P210 panel SATL HIGH R circuit breaker.
 - D. Open and collar the P210 panel SATL SDU R circuit breaker.

21-27-10 In-Flight Entertainment System (IFES) Equipment Cooling Fan

21-27-10-01 All

Interval	Installed	Required	Procedure
D	-	0	(M)

May be inoperative provided associated IFES equipment is deactivated.

NOTE: Operators should determine and deactivate the equipment cooled by the IFES cooling fan.

NOTE: The following information is applicable to the Boeing system. One cooling fan is installed.

MAINTENANCE (M)

Deactivate the In-Flight Entertainment System (IFES) and cooling fan (AMM 21-00-00/901).

1. Open and collar P110 panel IFE CLG FAN circuit breaker.
2. Open and collar P110 panel IFE CLG FAN CTRL circuit breaker (If installed).
3. Operators should determine and deactivate the equipment cooled by the IFES cooling fan.

NOTE: The following circuit breakers may or may not be installed.

- A. Open and collar P210 panel VIDEO PGMS & CTRL circuit breaker.
- B. Open and collar P210 panel VIDEO PGMS & CTRL B circuit breaker.
- C. Open and collar P210 panel CABIN PRINTER circuit breaker.
- D. Open and collar P110 panel VOD circuit breaker.

21-27-10 In-Flight Entertainment System (IFES) Equipment Cooling Fan

21-27-10-02 STC ST02657NY-D

Interval	Installed	Required	Procedure
C	2	1	(M)

May be inoperative provided:

- a. Inoperative fan is deactivated.
- b. Remaining fan is verified to operate normally once each flight day.



NOTE: The IFES cooling fan installation defined by STC ST02657NY-D is not a Boeing system. Information applicable to this installation is not provided.

**21-27-11 In-Flight Entertainment System (IFES) Equipment Cooling Smoke Detector**

21-27-11A One Operative

Interval	Installed	Required	Procedure
D	2	1	

21-27-11 In-Flight Entertainment System (IFES) Equipment Cooling Smoke Detector

21-27-11B All Inoperative

Interval	Installed	Required	Procedure
D	-	0	(M)

May be inoperative provided associated IFES equipment is deactivated.

NOTE: Operators should determine and deactivate the equipment cooled by the IFES cooling fan.

MAINTENANCE (M)

Deactivate the In-Flight Entertainment System (IFES), cooling fan and smoke detection (AMM 21-00-00/901).

1. Open and collar P110 panel IFE CLG SMK DET circuit breaker.
2. Open and collar P110 panel IFE CLG FAN circuit breaker.
3. Open and collar P110 panel IFE CLG FAN CTRL circuit breaker (If installed).
4. Operators should determine and deactivate the equipment cooled by the IFES cooling fan.

NOTE: Some of the following circuit breakers may or may not be installed.

- A. Open and collar P210 panel VIDEO PGMS & CTRL circuit breaker.
- B. Open and collar P210 panel VIDEO PGMS & CTRL B circuit breaker.
- C. Open and collar P210 panel CABIN PRINTER circuit breaker.
- D. Open and collar P110 panel VOD circuit breaker.

**21-27-12 Equipment Cooling Duct Pressure Sensors**

Interval	Installed	Required	Procedure
C	4	0	

May be inoperative provided both equipment cooling low flow sensors operate normally.

MAINTENANCE NOTE

For inoperative equipment cooling duct pressure sensor causing ground call horn to activate on ground, deactivate the inoperative sensor (AMM 21-00-00/901).

1. For the E4 or the F/D pressure sensor, open P110 panel E4/FLTDK PRESS XDCR circuit breaker.
2. For the E5 or the VSCF pressure sensor, open P210 panel E5/VSCF PRESS XDCR circuit breaker.
3. Gain access to the inoperative pressure sensor(s) (AMM 21-27-08/401).
4. Disconnect, cap and stow the electrical connector from the associated duct pressure sensor(s).
5. Cap the electrical receptacle on the pressure sensor(s).
6. Close the associated circuit breaker(s).

**21-27-13 Equipment Cooling Divert Valve**

Interval	Installed	Required	Procedure
C	1	0	(M)

May be inoperative provided:

- Valve is deactivated closed.
 - Equipment cooling vent fan is deactivated.
 - For ground operations with OAT 30 degrees C or higher, both packs are selected on or airplane is supplied with conditioned air.
-

MAINTENANCE (M)

Deactivate the equipment cooling divert valve closed and deactivate the equipment cooling vent fan (AMM 21-00-00/901).

- Deactivate the equipment cooling divert valve closed.
 - Open and collar P210 panel EQPT CLG DIVERT VLV CTRL circuit breaker.
 - Open and collar P210 panel EQT CLG DIVERT VLV circuit breaker.
 - Gain access to the equipment cooling divert valve.
 - Manually position the equipment cooling divert valve closed.
 - Re-install the moisture shroud and the floor panels.
- Deactivate the equipment cooling vent fan.
 - Open and collar the P110 panel EQPT CLG VENT FAN CTRL circuit breaker.
 - Disconnect, cap and stow the electrical connector (DB21583) from the equipment cooling vent fan.
- For ground operations with OAT at or above 30 degrees C, select both packs on or supply conditioned air to the airplane.

OPERATIONS NOTE

Forward cargo air conditioning will operate normally.

21-27-14 Equipment Cooling Inboard Valve

21-27-14-01 Passenger

Interval	Installed	Required	Procedure
C	1	0	(M)

May be inoperative provided:

- a. Valve is deactivated closed.
- b. Equipment cooling vent fan is deactivated.
- c. For ground operations with OAT 30 degrees C or higher, both packs are selected on or airplane is supplied with conditioned air.

MAINTENANCE (M)

Deactivate the equipment cooling inboard valve closed and deactivate the equipment cooling vent fan (AMM 21-00-00/901).

1. Deactivate the equipment cooling inboard valve closed.
 - A. Open and collar P210 panel circuit breaker EQT CLG INBD VLV.
 - B. Gain access to the equipment cooling inboard valve.
 - C. Manually position the equipment cooling inboard valve closed.
 - D. Re-install the moisture shroud and the floor panels.
2. Deactivate the equipment cooling vent fan.
 - A. Open and collar the P110 panel EQPT CLG VENT FAN CTRL circuit breaker.
 - B. Disconnect, cap and stow the electrical connector (DB21583) from the equipment cooling vent fan.
3. For ground operations with OAT at or above 30 degrees C, select both packs on or supply conditioned air to the airplane.

OPERATIONS NOTE

Forward cargo air conditioning will operate normally.

21-27-14 Equipment Cooling Inboard Valve

21-27-14-02 777F

Interval	Installed	Required	Procedure
C	1	0	(M)

May be inoperative provided:

- a. Valve is deactivated closed.



- b. For ground operations with OAT 30 degrees C or higher, both packs are selected on or airplane is supplied with conditioned air.
-

MAINTENANCE (M)

Deactivate the equipment cooling inboard valve closed (AMM 21-00-00/901).

1. Deactivate the equipment cooling inboard valve closed.
 - A. Open and collar P210 panel circuit breaker EQT CLG INBD VLV.
 - B. Gain access to the equipment cooling inboard valve.
 - C. Manually position the equipment cooling inboard valve closed.
 - D. Re-install the moisture shroud and the floor panels.
2. For ground operations with OAT at or above 30 degrees C, select both packs on or supply conditioned air to the airplane.

OPERATIONS NOTE

Lower forward cargo air conditioning will operate normally.



21-27-15 Equipment Cooling Three-Way Valve (777F)
21-27-15-01 Without PRR 62273 or Production Equivalent
Incorporated

Interval	Installed	Required	Procedure
C	1	0	(M) (O)

May be inoperative provided:

- Valve is deactivated closed.
- Procedures are established and used to verify main deck cargo compartment remains empty or contains only ballast, empty cargo handling equipment (ballast may be loaded in ULDs), or fly away kits.

NOTE: Operator MELs must define which items are approved for inclusion in the fly away kits and which materials can be used as ballast.

NOTE: PRR 62273 is applicable to 777F airplane line number 892 and-on.
Service Bulletin 777-21-0148 provides the production equivalent of PRR 62273.

MAINTENANCE (M)

Deactivate the equipment cooling three-way valve closed (AMM 21-00-00/901).

- Open and collar P110 panel 3-WAY VALVE circuit breaker.
- Gain access to the inoperative equipment cooling three-way valve.
- Manually position the equipment cooling three-way valve closed in EE-Mode (A-B valve position).

OPERATIONS (O)

Main deck cargo compartments must remain empty, except for ballast, empty cargo containers (ballast may be loaded in ULDs), fly away kits, pallets, and cargo restraint components.

21-27-15 Equipment Cooling Three-Way Valve (777F)
21-27-15-02 With PRR 62273 or Production Equivalent
Incorporated

Interval	Installed	Required	Procedure
C	1	0	(M) (O)

May be inoperative provided:

- Valve is deactivated open.
 - Both packs are operated continuously while valve is deactivated open.
-



NOTE: PRR 62273 is applicable to 777F airplane line number 892 and-on. Service Bulletin 777-21-0148 provides the production equivalent of PRR 62273.

MAINTENANCE (M)

Deactivate the equipment cooling three-way valve open (AMM 21-00-00/901).

1. Open and collar P110 panel 3-WAY VALVE circuit breaker.
2. Gain access to the inoperative equipment cooling three-way valve.
3. With both packs operating, manually position the equipment cooling three-way valve open in A/C-Mode (C-B valve position).

OPERATIONS (O)

Operate both packs continuously for flight instrument cooling.

| NOTE: Operations with packs selected OFF may not be conducted.

**21-27-16 Flight Deck Equipment Cooling System (777F)**

Interval	Installed	Required	Procedure
C	1	0	(M) (O)

May be inoperative provided:

- Equipment cooling three-way valve is deactivated closed.
- Procedures are established and used to verify main deck cargo compartment remains empty or contains only ballast, empty cargo handling equipment (ballast may be loaded in ULDs), or fly away kits.

NOTE: Operator MELs must define which items are approved for inclusion in the fly away kits and which materials can be used as ballast.

MAINTENANCE (M)

Deactivate the equipment cooling three-way valve closed (AMM 21-00-00/901).

- Open and collar P110 panel 3-WAY VALVE circuit breaker.
- Gain access to the equipment cooling three-way valve.
- Manually position the equipment cooling three-way valve closed in EE-Mode (A-B valve position).

OPERATIONS (O)

Main deck cargo compartments must remain empty, except for ballast, empty cargo containers (ballast may be loaded in ULDs), fly away kits, pallets, and cargo restraint components.



21-27-17 EQUIP COOLING Switch Lights**21-27-17-01 OVRD Light**

Interval	Installed	Required	Procedure
C	1	0	

21-27-17 EQUIP COOLING Switch Lights**21-27-17-02 AUTO Light**

Interval	Installed	Required	Procedure
C	1	0	

**21-28-01 Forward Cargo Air Conditioning (FCAC) Shutoff Valve (Passenger)**

Interval	Installed	Required	Procedure
D	1	0	(M)

May be inoperative provided:

- Valve is deactivated closed.
- FCAC remains OFF.

MAINTENANCE (M)

Deactivate the FCAC shutoff valve closed (AMM 21-00-00/901).

- Position Overhead Panel (P5) FWD CARGO A/C control OFF.
- Open and collar P210 panel FWD CGO A/C SOV CTRL circuit breaker.
- Open and collar P210 panel FWD CGO A/C SOV circuit breaker.
- Gain access to the FCAC shutoff valve located aft of the forward cargo compartment in the right side of the mix bay area below the recirculation filters.
- Manually position the FCAC shutoff valve closed.

OPERATIONS NOTE

- Position FWD CARGO A/C control OFF.
- Forward cargo air conditioning is not available. Temperature sensitive cargo should not be carried in the Forward cargo compartment.

**21-28-02 FCAC Flow Regulating Valve (777F)**

Interval	Installed	Required	Procedure
C	1	0	(M)

May be inoperative provided:

- Valve is deactivated closed.
 - FCAC remains OFF.
-

MAINTENANCE (M)

Deactivate the lower forward cargo flow regulating valve closed (AMM 21-00-00/901).

- Open P210 panel R CTC CHAN 1 (AC) circuit breaker.
- Open P210 panel R CTC CHAN 1 (DC) circuit breaker.
- Open P210 panel R CTC CHAN 2 (AC) circuit breaker.
- Open P210 panel R CTC CHAN 2 (DC) circuit breaker.
- Gain access to the lower forward cargo flow regulating valve.
- Disconnect, cap, and stow the electrical connector from valve.
- Manually position the lower forward cargo flow regulating valve closed.
- Close P210 panel R CTC CHAN 1 (AC) circuit breaker.
- Close P210 panel R CTC CHAN 1 (DC) circuit breaker.
- Close P210 panel R CTC CHAN 2 (AC) circuit breaker.
- Close P210 panel R CTC CHAN 2 (DC) circuit breaker.

For Aft Cargo Air Conditioning (ACAC) not installed:

- Position Overhead Panel (P5) FWD CARGO A/C control OFF.

For Aft Cargo Air Conditioning (ACAC) installed:

- Position Overhead Panel (P5) LWR CARGO TEMP FWD A/C control OFF.

OPERATIONS NOTE

For Aft Cargo Air Conditioning (ACAC) not installed:

- Position FWD CARGO A/C control OFF.
- Lower forward cargo air conditioning is not available. Temperature sensitive cargo should not be carried in the lower forward cargo compartment.

For Aft Cargo Air Conditioning (ACAC) installed:

- Position LWR CARGO TEMP FWD A/C control OFF.
- Lower forward cargo air conditioning is not available. Temperature sensitive cargo should not be carried in the lower forward cargo compartment.

**21-28-03 ACAC Flow Regulating Valve (777F)**

Interval	Installed	Required	Procedure
C	1	0	(M)

May be inoperative provided:

- Valve is deactivated closed.
- ACAC is not operated in AUTO.

MAINTENANCE (M)

Deactivate the lower aft cargo flow regulating valve closed (AMM 21-00-00/901).

- Position Overhead Panel (P5) LWR CARGO TEMP AFT A/C control OFF.
- Open P210 panel R CTC CHAN 1 (AC) circuit breaker.
- Open P210 panel R CTC CHAN 1 (DC) circuit breaker.
- Open P210 panel R CTC CHAN 2 (AC) circuit breaker.
- Open P210 panel R CTC CHAN 2 (DC) circuit breaker.
- Gain access to the lower aft cargo flow regulating valve.
- Disconnect, cap, and stow the electrical connector from valve.
- Manually position the lower aft cargo flow regulating valve closed.
- Close P210 panel R CTC CHAN 1 (AC) circuit breaker.
- Close P210 panel R CTC CHAN 1 (DC) circuit breaker.
- Close P210 panel R CTC CHAN 2 (AC) circuit breaker.
- Close P210 panel R CTC CHAN 2 (DC) circuit breaker.

OPERATIONS NOTE

- Position LWR CARGO TEMP AFT A/C control LOW, HIGH or OFF.
- Lower aft cargo heat is available, however lower aft cargo air conditioning is not available. Temperature sensitive cargo should not be carried in the lower aft cargo compartment.

**21-28-04 ACAC Exhaust Shutoff Valve (777F)**

Interval	Installed	Required	Procedure
C	1	0	(M)

May be inoperative provided:

- Valve is deactivated closed.
 - ACAC is not operated in AUTO.
-

MAINTENANCE (M)

Deactivate the lower aft cargo exhaust shutoff valve closed (AMM 21-00-00/901).

- Position Overhead Panel (P5) LWR CARGO TEMP AFT A/C control OFF.
- Open and collar P110 panel ACAC EXHAUST SOV circuit breaker.
- Gain access to the lower aft cargo exhaust shutoff valve.
- Manually position the lower aft cargo exhaust shutoff valve closed.

OPERATIONS NOTE

- Position LWR CARGO TEMP AFT A/C control LOW, HIGH or OFF.
- Lower aft cargo heat is available, however lower aft cargo air conditioning is not available. Temperature sensitive cargo should not be carried in the lower aft cargo compartment.

**21-29-01 Lower Lobe Attendant Rest (LLAR) Shutoff Valve**

Interval	Installed	Required	Procedure
C	1	0	(M)

May be inoperative provided:

- Valve is deactivated closed.
- LLAR is considered inoperative.

NOTE: The airplane must also be dispatched using MMEL item 25-29-04.

MAINTENANCE (M)

Deactivate LLAR shutoff valve and LLAR closed (AMM 21-00-00/901).

- Open and collar P110 panel LLAR SOV/SMK ALARM CTRL circuit breaker.
- Open and collar P320 panel LLAR SOV SEC PWR circuit breaker.
- Gain access to the LLAR shutoff valve by removing the maintenance access panel on the left forward wall of the LLAR and the cargo lining behind the access panel.
- Manually position LLAR shutoff valve (V21001) closed.
- Re-install the cargo lining and access panel.

**21-29-02 Main Deck Crew Rest (MDCR)/Main Deck Flight Crew Rest (MDFCR) Shutoff Valve**

Interval	Installed	Required	Procedure
C	1	0	(M)

May be inoperative provided:

- Valve is deactivated closed.
- MDCR/MDFCR is considered inoperative.

NOTE: The airplane must also be dispatched using MMEL item 25-29-04.

MAINTENANCE (M)

Deactivate the MDCR/MDFCR shutoff valve and associated MDCR/MDFCR closed (AMM 21-00-00/901).

- Open and collar P110 panel MDCR SOV or MDCR SUPPLY SOV circuit breaker.
- Gain access to the MDCR/MDFCR shutoff valve (AMM 21-29-10-000-801).
- Manually position MDCR/MDFCR shutoff valve (V21006) closed.

**21-29-03 Main Deck Crew Rest (MDCR) Exhaust Valve**

Interval	Installed	Required	Procedure
C	1	0	(M)

May be inoperative provided:

- Valve is deactivated closed.
- MDCR is considered inoperative.

NOTE: The airplane must also be dispatched using MMEL item 25-29-04.

MAINTENANCE (M)

Deactivate the MDCR exhaust valve and MDCR closed (AMM 21-00-00/901).

- Open and collar P110 panel MDCR EXH VLV circuit breaker.
- Gain access to the MDCR exhaust valve located above the MDCR ceiling hatch.
- Manually position MDCR exhaust valve (V21007) closed.

**21-29-04 Overhead Flight Crew/Attendant Rest Supply Shutoff Valves (Includes FSI Installation)**

21-29-04-01 Overhead Flight Crew Rest (OFCR)

Interval	Installed	Required	Procedure
C	1	0	(M)

May be inoperative provided:

- OFCR supply shutoff valve is deactivated closed.
- OFCR is considered inoperative.

NOTE 1: The overhead crew rest installation manufactured by Flight Structures, Inc. (FSI) is not a Boeing system. Information applicable to the FSI installation is not provided.

NOTE 2: The airplane must also be dispatched using MMEL item 25-29-04.

MAINTENANCE (M)

Deactivate the OFCR shutoff valve (AMM 21-00-00/901).

- Open the P210 panel UPR FWD REST SOV/EXH VALVE circuit breaker.
- Gain access to the OFCR shutoff valve (AMM 21-29-30/401).
- Disconnect, cap and stow the shutoff valve electrical connector.
- Manually position the shutoff valve closed.
- Close the P210 panel UPR FWD REST SOV/EXH VALVE circuit breaker.

21-29-04 Overhead Flight Crew/Attendant Rest Supply Shutoff Valves (Includes FSI Installation)

21-29-04-02 Overhead Flight Attendant Rest (OFAR)

Interval	Installed	Required	Procedure
C	1	0	(M)

May be inoperative provided:

- OFAR supply shutoff valve is deactivated closed.
- OFAR is considered inoperative.

NOTE 1: The overhead attendant rest installation manufactured by Flight Structures, Inc. (FSI) is not a Boeing system. Information applicable to the FSI installation is not provided.

NOTE 2: The airplane must also be dispatched using MMEL item 25-29-04.

MAINTENANCE (M)

Deactivate the OFAR shutoff valve (AMM 21-00-00/901).

1. Open the P210 panel UPR AFT REST SOV/EXH VALVE circuit breaker.
2. Gain access to the OFAR shutoff valve (AMM 21-29-20/401).
3. Disconnect, cap and stow the shutoff valve electrical connector.
4. Manually position the shutoff valve closed.
5. Close the P210 panel UPR AFT REST SOV/EXH VALVE circuit breaker.

**21-29-05 Overhead Flight Crew/Attendant Rest Exhaust Valves (Includes FSI Installation)**

21-29-05-01 Overhead Flight Crew Rest (OFCR)

Interval	Installed	Required	Procedure
C	1	0	(M)

May be inoperative provided:

- OFCR exhaust valve is deactivated closed.
- OFCR is considered inoperative.

NOTE 1: The overhead crew rest installation manufactured by Flight Structures, Inc. (FSI) is not a Boeing system. Information applicable to the FSI installation is not provided.

NOTE 2: The airplane must also be dispatched using MMEL item 25-29-04.

MAINTENANCE (M)

Deactivate the OFCR exhaust valve (AMM 21-00-00/901).

- Open the P210 panel UPR FWD REST SOV/EXH VALVE circuit breaker.
- Gain access to the OFCR exhaust valve (AMM 21-29-31/401).
- Disconnect, cap and stow the exhaust valve electrical connector.
- Manually position the exhaust valve closed.
- Close the P210 panel UPR FWD REST SOV/EXH VALVE circuit breaker.

21-29-05 Overhead Flight Crew/Attendant Rest Exhaust Valves (Includes FSI Installation)

21-29-05-02 Overhead Flight Attendant Rest (OFAR)

Interval	Installed	Required	Procedure
C	-	0	(M)

May be inoperative provided:

- OFAR exhaust valve(s) is deactivated closed.
- OFAR is considered inoperative.

NOTE 1: The overhead attendant rest installation manufactured by Flight Structures, Inc. (FSI) is not a Boeing system. Information applicable to the FSI installation is not provided.

NOTE 2: The airplane must also be dispatched using MMEL item 25-29-04.

**MAINTENANCE (M)**

Deactivate the OFAR exhaust valve (AMM 21-00-00/901).

1. Open the P210 panel UPR AFT REST SOV/EXH VALVE circuit breaker.
2. Gain access to the OFAR exhaust valve (AMM 21-29-21/401).
3. Disconnect, cap and stow the exhaust valve electrical connector.
4. Manually position the exhaust valve closed.
5. Close the P210 panel UPR AFT REST SOV/EXH VALVE circuit breaker.

**21-29-06 Overhead Flightcrew Rest Supply Shutoff Valves
(Taxi, Takeoff & Landing Installation)**

21-29-06-01 Primary Supply Shutoff Valve

Interval	Installed	Required	Procedure
C	1	0	(M)

May be inoperative provided:

- Primary supply shutoff valve is deactivated closed.
- Secondary supply shutoff valve operates normally.
- Left air conditioning pack operates normally.

MAINTENANCE (M)

NOTE: For a pack to be operating normally, certain items associated with that system must be operative. See DDG item 21-51-01 for a listing of those items.

Deactivate the primary OFCR supply shutoff valve closed (AMM 21-00-00/901).

- Open P210 panel UPR FWD REST SOV/EXH VLV circuit breaker.
- Gain access to the primary shutoff valve.
- Disconnect, cap and stow the primary shutoff valve electrical connector.
- Manually position the primary shutoff valve closed.
- Close P210 panel UPR FWD REST SOV/EXH VLV circuit breaker.

OPERATIONS NOTE

With the primary shutoff valve disabled, the OFCR may become warm. To mitigate a warm OFCR, select a cooler flight deck area temperature. To mitigate the cooler flight deck temperature, the shoulder and foot heaters in the flight deck may be selected on.

**21-29-06 Overhead Flightcrew Rest Supply Shutoff Valves
(Taxi, Takeoff & Landing Installation)**

21-29-06-02 Secondary Supply Shutoff Valve

Interval	Installed	Required	Procedure
C	1	0	(M) (O)

May be inoperative provided:

- Secondary supply shutoff valve is deactivated closed.
- Primary supply shutoff valve operates normally.
- Left air conditioning pack operates normally.

- d. OFCR is not occupied at FL 250 or below.

MAINTENANCE (M)

NOTE: For a pack to be operating normally, certain items associated with that system must be operative. See DDG item 21-51-01 for a listing of those items.

Deactivate the secondary OFCR supply shutoff valve closed (AMM 21-00-00/901).

1. Open P210 panel UPR FWD REST SEC SOV circuit breaker.
2. Gain access to the secondary shutoff valve.
3. Disconnect, cap and stow the secondary shutoff valve electrical connector.
4. Manually position the secondary shutoff valve closed.
5. Close P210 panel UPR FWD REST SEC SOV circuit breaker.

OPERATIONS (O)

The OFCR is not available for in-flight use during taxi, takeoff and landing. Do not enter OFCR until altitude is above FL 250.

21-29-06 Overhead Flightcrew Rest Supply Shutoff Valves (Taxi, Takeoff & Landing Installation)

21-29-06-03 Primary and Secondary Supply Shutoff Valves

Interval	Installed	Required	Procedure
C	2	0	(M)

May be inoperative provided:

- a. Primary and secondary supply shutoff valves are deactivated closed.
- b. OFCR is considered inoperative.

NOTE: The airplane must also be dispatched using MMEL item 25-29-02.

MAINTENANCE (M)

NOTE: The Taxi, Takeoff & Landing (TT&L) configuration incorporates a Primary and a Secondary supply shutoff valve.

Deactivate both OFCR shutoff valves closed (AMM 21-00-00/901).

1. Open P210 panel UPR FWD REST SOV/EXH VLV circuit breaker.
2. Open P210 panel UPR FWD REST SEC SOV circuit breaker.
3. Gain access to the OFCR shutoff valves.
4. Disconnect, cap and stow both of the shutoff valve electrical connectors.
5. Manually position both of the shutoff valves closed.
6. Close the associated circuit breakers.



**21-29-07 Overhead Flight Crew Rest (OFCR) Pressure Sensor
(Taxi, Takeoff & Landing Installation)**

Interval	Installed	Required	Procedure
C	1	0	

21-31-01 Auto Cabin Pressure Controls (L and R)
21-31-01-01 Passenger With P/N 3676-GRS-014-00 or Later ASCPC Software Installed
21-31-01-01A Manual Control Operates on Both Outflow Valves

Interval	Installed	Required	Procedure
C	2	1	(M)

One may be inoperative provided:

- a. Manual cabin pressure control is verified to operate normally on both outflow valves before each departure.
- b. Left or right AFDC operates normally.

MAINTENANCE (M)

NOTE: ASCPC software P/N 3676-GRS-014-00 is installed by Boeing Service Bulletin 777-36-0013. All software P/Ns installed in ASCPC-2 are considered later versions than software P/Ns installed in ASCPC-1.

Before each departure, verify manual cabin pressure control operates normally on both outflow valves (AMM 21-00-00/901).

1. Position Pressurization Panel (P5) OUTFLOW VALVE switches to MAN.

NOTE: Outflow valve position indications on EICAS may blank when MAN is selected.

2. Hold each Pressurization Panel (P5) OUTFLOW VALVE MANUAL switch CLOSE until the EICAS outflow valve position indicator is CL, or if EICAS indication is not available hold the switch CLOSE for 30 seconds.
3. Confirm the outflow valve doors are closed.
4. Hold each Pressurization Panel (P5) OUTFLOW VALVE MANUAL switch OPEN until the EICAS outflow valve position indicator is OP, or if EICAS indication is not available hold the switch OPEN for 30 seconds.
5. Confirm the outflow valve doors are full open.
6. Position Pressurization Panel (P5) OUTFLOW VALVE switches to AUTO.

OPERATIONS NOTE

Leave OUTFLOW VALVE switches in AUTO unless required to accomplish a Non-Normal Checklist.

- 21-31-01 Auto Cabin Pressure Controls (L and R)**
- 21-31-01-01 Passenger With P/N 3676-GRS-014-00 or Later ASCPC Software Installed**
- 21-31-01-01B One Outflow Valve Deactivated / Manual Control Operates on Other Valve**

Interval	Installed	Required	Procedure
C	2	1	(M) (O)

One may be inoperative provided:

- One outflow valve is deactivated 7% open.
- Manual cabin pressure control is verified to operate normally on the remaining outflow valve.
- Left or right AFDC operates normally.
- One pack operates normally, and the other pack operates normally or in standby cooling mode.
- Both CTCs operate normally.
- Both engine bleed systems operate normally.

For OFCR installed:

- For aft outflow valve deactivated 7% open, the OFCR is considered inoperative.

For OFAR installed:

- For aft outflow valve deactivated 7% open, the OFAR is considered inoperative.

For OFCR or OFAR installed:

NOTE: For aft outflow valve deactivated, the airplane must also be dispatched using MMEL item 25-29-04.

MAINTENANCE (M)

Use DDG item 21-31-03-02 (M) procedure.

OPERATIONS (O)

- Use DDG item 21-31-03-02 (O) procedure.
- Leave OUTFLOW VALVE switches in AUTO unless required to accomplish a Non-Normal Checklist.

- 21-31-01 Auto Cabin Pressure Controls (L and R)**
- 21-31-01-02 777F**

Interval	Installed	Required	Procedure
C	2	1	(M)

One may be inoperative provided:

- a. Manual cabin pressure control is verified to operate normally on both outflow valves before each departure.
- b. Left or right AFDC operates normally.

MAINTENANCE (M)

Before each departure, verify manual cabin pressure control operates normally on both outflow valves (AMM 21-00-00/901).

1. Position Pressurization Panel (P5) OUTFLOW VALVE switches to MAN.

NOTE: Outflow valve position indications on EICAS may blank when MAN is selected.

2. Hold each Pressurization Panel (P5) OUTFLOW VALVE MANUAL switch CLOSE until the EICAS outflow valve position indicator is CL, or if EICAS indication is not available hold the switch CLOSE for 30 seconds.
3. Confirm the outflow valve doors are closed.
4. Hold each Pressurization Panel (P5) OUTFLOW VALVE MANUAL switch OPEN until the EICAS outflow valve position indicator is OP, or if EICAS indication is not available hold the switch OPEN for 30 seconds.
5. Confirm the outflow valve doors are full open.
6. Position Pressurization Panel (P5) OUTFLOW VALVE switches to AUTO.

OPERATIONS NOTE

Leave OUTFLOW VALVE switches in AUTO unless required to accomplish a Non-Normal Checklist.

21-31-01 Auto Cabin Pressure Controls (L and R)

21-31-01-03 All

Interval	Installed	Required	Procedure
C	2	0	(M) (O)

May be inoperative provided:

- a. Flight is conducted unpressurized.
- b. Procedures are established and used to verify cargo compartments remain empty or contain only ballast, empty cargo handling equipment (ballast may be loaded in ULDs), or fly away kits.
- c. For extended overwater flight, manual cabin pressure control is verified to operate normally on both outflow valves.

For LLAR, OFCR, OFAR, MDCR or MDPCR installed:

- d. The rest area(s) is considered inoperative.



NOTE: Operator MELs must define which items are approved for inclusion in the fly away kits and which materials can be used as ballast.

For LLAR, OFCR, OFAR, MDCR or MDFCR installed:

NOTE: The airplane must also be dispatched using MMEL item 25-29-04.

MAINTENANCE (M)

For extended overwater flight, verify manual cabin pressure control operates normally on both outflow valves (AMM 21-00-00/901).

1. Position Pressurization Panel (P5) OUTFLOW VALVE switches to MAN.
NOTE: Outflow valve position indications on EICAS may blank when MAN is selected.
2. Hold each Pressurization Panel (P5) OUTFLOW VALVE MANUAL switch CLOSE until the EICAS outflow valve position indicator is CL, or if EICAS indication is not available hold the switch CLOSE for 30 seconds.
3. Confirm the outflow valve doors are closed.
4. Hold each Pressurization Panel (P5) OUTFLOW VALVE MANUAL switch OPEN until the EICAS outflow valve position indicator is OP, or if EICAS indication is not available hold the switch OPEN for 30 seconds.
5. Confirm the outflow valve doors are full open.
6. Leave Pressurization Panel (P5) OUTFLOW VALVE switches in MAN.

OPERATIONS (O)

1. Cargo compartments must remain empty, except for ballast, empty cargo containers (ballast may be loaded in ULDs), fly away kits, pallets, and cargo restraint components.
2. Position both outflow valves full open.
 - A. Position both OUTFLOW VALVE switches to MAN.
NOTE 1: The CABIN ALTITUDE AUTO caution message will be displayed.
NOTE 2: Outflow valve position indications on EICAS may blank when MAN is selected.
 - B. Hold each OUTFLOW VALVE MANUAL switch OPEN for 30 seconds.
3. Maintain the outflow valves full open unless required to accomplish a Non-Normal Checklist.
4. For flight with passengers:
 - A. Limit enroute altitude to 10,000 feet.
 - B. To avoid passenger discomfort, limit climb (after takeoff obstacle clearance has been assured) and descent rates to 500 fpm.

21-31-02 Manual Cabin Pressure Outflow Valve Controls (FWD and AFT)
21-31-02-01 Passenger

Interval	Installed	Required	Procedure
C	2	1	(M) (O)

One may be inoperative provided:

- Associated outflow valve is deactivated 7% open.
- Manual cabin pressure control is verified to operate normally on the remaining outflow valve.
- One pack operates normally, and the other pack operates normally or in standby cooling mode.
- Both CTCs operate normally.
- Both engine bleed systems operate normally.

For OFCR installed:

- For aft outflow valve deactivated 7% open, the OFCR is considered inoperative.

For OFAR installed:

- For aft outflow valve deactivated 7% open, the OFAR is considered inoperative.

For OFCR or OFAR installed:

NOTE: For aft outflow valve deactivated, the airplane must also be dispatched using MMEL item 25-29-04.

MAINTENANCE (M)

NOTE 1: For a pack to be operating normally, certain items associated with that system must be operative. See DDG item 21-51-01 for a listing of those items.

NOTE 2: A pack will operate in standby cooling mode with an inoperative ACM, or an inoperative ECV, or an inoperative Condenser Inlet Temperature Control System.

NOTE 3: For an engine bleed system to be operating normally, certain items associated with that system must be operative. See DDG item 36-00-01 for a listing of those items.

Deactivate the associated valve 7% open and verify manual cabin pressure control operates normally on the remaining outflow valve using applicable DDG item 21-31-03-01 or 21-31-03-02 (M) procedure (AMM 21-00-00/901).

OPERATIONS (O)

Use applicable DDG item 21-31-03-01 or 21-31-03-02 (O) procedure.

**21-31-02 Manual Cabin Pressure Outflow Valve Controls (FWD and AFT)****21-31-02-02 777F****21-31-02-02-01 With P/N 3677-GRS-105-00 ASCPC Software Installed**

Interval	Installed	Required	Procedure
C	2	1	(M) (O)

One may be inoperative provided:

- Associated outflow valve is deactivated 7% open.
 - Manual cabin pressure control is verified to operate normally on the remaining outflow valve.
 - One pack operates normally, and the other pack operates normally or in standby cooling mode.
 - Both CTCs operate normally.
 - Both engine bleed systems operate normally.
 - Rigid cargo barrier doors remain closed inflight.
-

MAINTENANCE (M)

NOTE 1: For a pack to be operating normally, certain items associated with that system must be operative. See DDG item 21-51-01 for a listing of those items.

NOTE 2: A pack will operate in standby cooling mode with an inoperative ACM, or an inoperative ECV, or an inoperative Condenser Inlet Temperature Control System.

NOTE 3: For an engine bleed system to be operating normally, certain items associated with that system must be operative. See DDG item 36-00-01 for a listing of those items.

Deactivate the associated valve 7% open and verify manual cabin pressure control operates normally on the remaining outflow valve using DDG item 21-31-03-03-01 (M) procedure (AMM 21-00-00/901).

OPERATIONS (O)

Use DDG item 21-31-03-03-01 (O) procedure.

21-31-02 Manual Cabin Pressure Outflow Valve Controls (FWD and AFT)
21-31-02-02 777F
21-31-02-02-02 With P/N 3674-GRS-106-00 or Later ASCPC Software Installed

Interval	Installed	Required	Procedure
C	2	1	(M) (O)

One may be inoperative provided:

- Associated outflow valve is deactivated 7% open.
- Manual cabin pressure control is verified to operate normally on the remaining outflow valve.
- One pack operates normally, and the other pack operates normally or in standby cooling mode.
- Both CTCs operate normally.
- Both engine bleed systems operate normally.

MAINTENANCE (M)

NOTE 1: PRR 62273, Part C, installs ASCPC-2 software P/N 3674-GRS-106-00 on L/N 998 and-on. Service Bulletin 777-36-0032 provides the production equivalent of PRR 62273 by installing ASCPC-2 software P/N 3674-GRS-106-00.

NOTE 2: For a pack to be operating normally, certain items associated with that system must be operative. See DDG item 21-51-01 for a listing of those items.

NOTE 3: A pack will operate in standby cooling mode with an inoperative ACM, or an inoperative ECV, or an inoperative Condenser Inlet Temperature Control System.

NOTE 4: For an engine bleed system to be operating normally, certain items associated with that system must be operative. See DDG item 36-00-01 for a listing of those items.

Deactivate the associated valve 7% open and verify manual cabin pressure control operates normally on the remaining outflow valve using DDG item 21-31-03-03-02 (M) procedure (AMM 21-00-00/901).

OPERATIONS (O)

Use DDG item 21-31-03-03-02 (O) procedure.

21-31-02 Manual Cabin Pressure Outflow Valve Controls (FWD and AFT)
21-31-02-03 All

Interval	Installed	Required	Procedure
C	2	0	(M) (O)

May be inoperative provided:

- Flight is conducted unpressurized.
- Procedures are established and used to verify cargo compartments remain empty or contain only ballast, empty cargo handling equipment (ballast may be loaded in ULDs), or fly away kits.
- Extended overwater flight is prohibited.

For LLAR, OFCR, OFAR, MDCR or MDPCR installed:

- The rest area(s) is considered inoperative.

For first class suites installed:

- First class suites are considered inoperative.

NOTE: Operator MELs must define which items are approved for inclusion in the fly away kits and which materials can be used as ballast.

NOTE 1: First class suites refers to the High Wall First Class Passenger Suite installation (option number 2525E407E69).

For LLAR, OFCR, OFAR, MDCR or MDPCR installed:

NOTE 2: The airplane must also be dispatched using MMEL item 25-29-04.

For first class suites installed:

NOTE 3: The airplane must also be dispatched using MMEL item 25-25-05.

MAINTENANCE (M)

Deactivate both outflow valves full open (AMM 21-31-00/201).

- Determine position of outflow valves.
- Open and collar the following circuit breakers:
 - P210 panel R FWD OFL VLV circuit breaker.
 - P310 panel L FWD OFL VLV circuit breaker.
 - P210 panel R AFT OFL VLV circuit breaker.
 - P310 panel L AFT OFL VLV circuit breaker.
- For outflow valve(s) not in the full open position, also do the following:
 - Gain access to the associated outflow valve assembly.
 - Remove the primary and secondary pushrods.
 - Position the valve door(s) in the full open position.
 - Install the hold-open rod, part number J21006.



4. Position the associated Pressurization Panel (P5) OUTFLOW VALVE switches to AUTO.

OPERATIONS (O)

NOTE: The CABIN ALTITUDE AUTO caution message will be displayed.

1. The flight is conducted unpressurized.
2. Extended overwater flight is prohibited.
3. Cargo compartments must remain empty, except for ballast, empty cargo containers (ballast may be loaded in ULDs), fly away kits, pallets, and cargo restraint components.
4. For flight with passengers:
 - A. Limit enroute altitude to 10,000 feet.
 - B. To avoid passenger discomfort, limit climb (after takeoff obstacle clearance has been assured) and descent rates to 500 fpm.

21-31-03 Outflow Valves (FWD and AFT)**21-31-03-01 Passenger Without P/N 3676-GRS-014-00 or Later ASCPC Software Installed**

Interval	Installed	Required	Procedure
C	2	1	(M) (O)

One may be inoperative provided:

- Valve is deactivated 7% open.
- Manual cabin pressure control is verified to operate normally on the remaining outflow valve.
- Both auto cabin pressure controls operate normally.
- One pack operates normally, and the other pack operates normally or in standby cooling mode.
- Both CTCs operate normally.
- Both engine bleed systems operate normally.

For OFCR installed:

- For aft outflow valve deactivated 7% open, the OFCR is considered inoperative.

For OFAR installed:

- For aft outflow valve deactivated 7% open, the OFAR is considered inoperative.

For OFCR or OFAR installed:

NOTE: For aft outflow valve deactivated, the airplane must also be dispatched using MMEL item 25-29-04.

MAINTENANCE (M)

NOTE 1: ASCPC software P/N 3676-GRS-014-00 is installed by Boeing Service Bulletin 777-36-0013. All software P/Ns installed in ASCPC-2 are considered later versions than software P/Ns installed in ASCPC-1.

NOTE 2: For a pack to be operating normally, certain items associated with that system must be operative. See DDG item 21-51-01 for a listing of those items.

NOTE 3: A pack will operate in standby cooling mode with an inoperative ACM, or an inoperative ECV, or an inoperative Condenser Inlet Temperature Control System.

NOTE 4: For an engine bleed system to be operating normally, certain items associated with that system must be operative. See DDG item 36-00-01 for a listing of those items.



Deactivate inoperative outflow valve 7% open and verify manual cabin pressure control operates normally on the remaining operative outflow valve (AMM 21-00-00/901).

NOTE: For ground operations with OAT at or above 24 degrees C and forward outflow valve deactivated 7% open, supply conditioned air to the airplane.

1. Supply electrical power on the airplane (AMM 24-22-00/201).
2. Deactivate inoperative outflow valve 7% open.
 - A. For manual control operative and valve position displayed on the Air Supply Maintenance page:
 - 1) Position the associated Pressurization Panel (P5) OUTFLOW VALVE switch to MAN.
 - 2) Using the associated Pressurization Panel (P5) OUTFLOW VALVE MANUAL switch, position the inoperative outflow valve to indicate 0.07 on the Air Supply Maintenance page.
 - 3) For the forward outflow valve inoperative:
 - a. Open and collar the P210 panel R FWD OFL VLV circuit breaker.
 - b. Open and collar the P310 panel L FWD OFL VLV circuit breaker.
 - 4) For the aft outflow valve inoperative:
 - a. Open and collar the P210 panel R AFT OFL VLV circuit breaker.
 - b. Open and collar the P310 panel L AFT OFL VLV circuit breaker.
 - 5) Position the associated Pressurization Panel (P5) OUTFLOW VALVE switch to AUTO.
 - 6) Confirm the gap between the two outflow doors is 0.375 ± 0.031 inches.
 - B. For OFV (P/N 2214420-X) installed with manual control inoperative or valve position on the Air Supply Maintenance page not available:
 - 1) For the forward outflow valve inoperative:
 - a. Open and collar the P210 panel R FWD OFL VLV circuit breaker.
 - b. Open and collar the P310 panel L FWD OFL VLV circuit breaker.
 - 2) For the aft outflow valve inoperative:
 - a. Open and collar the P210 panel R AFT OFL VLV circuit breaker.
 - b. Open and collar the P310 panel L AFT OFL VLV circuit breaker.
 - 3) Gain access to the associated outflow valve assembly.

- 4) Disconnect the electrical connector from the back of the valve control unit.
 - 5) Remove one DC motor from the outflow valve assembly.
 - 6) Move the outflow valve doors until the gap between the two doors is 0.375 ± 0.031 inches.
 - 7) Re-install the DC motor.
 - 8) Reconnect the electrical connector on the back of the valve control unit.
 - 9) Position the associated Pressurization Panel (P5) OUTFLOW VALVE switch to AUTO.
3. Verify manual cabin pressure control operates normally on the remaining outflow valve.
- A. Position the associated Pressurization Panel (P5) OUTFLOW VALVE switch to MAN.

NOTE: Outflow valve position indications on EICAS may blank when MAN is selected.
 - B. Hold the associated Pressurization Panel (P5) OUTFLOW VALVE MANUAL switch CLOSE until the EICAS outflow valve position indicator is CL, or if EICAS indication is not available hold the switch CLOSE for 30 seconds.
 - C. Confirm the associated outflow valve door is closed.
 - D. Hold the associated Pressurization Panel (P5) OUTFLOW VALVE MANUAL switch OPEN until the EICAS outflow valve position indicator is OP, or if EICAS indication is not available hold the switch OPEN for 30 seconds.
 - E. Confirm the associated outflow valve door is full open.
 - F. Position the associated Pressurization Panel (P5) OUTFLOW VALVE switch to AUTO.

OPERATIONS (O)

NOTE 1: Aft cabin temperature control may be degraded when dispatching with the aft outflow valve deactivated 7% open.

NOTE 2: The associated OUTFLOW VALVE AFT or FWD advisory message will be displayed.

NOTE 3: The outflow valve position indication pointer will be missing for the deactivated outflow valve on the EICAS pressurization indications.

1. During passenger loading, if both packs are operating, select right pack OFF prior to closing the main entry door(s). This avoids uncomfortable changes in cabin pressure.
2. After all main entry doors are closed at the parking stand or departure gate, if both packs are operating, select right pack OFF before reopening a main entry door. After leaving the parking stand or gate, select right pack AUTO.



3. If the operative outflow valve fails during flight and cannot be controlled manually it may be necessary to select packs OFF to depressurize the airplane for landing.
 - A. During descent when below 15,000 ft. MSL, select one pack OFF.
 - B. During descent when below 10,000 ft. MSL, select remaining pack OFF.

For Forward Cargo Air Conditioning (FCAC) installed:

4. For aft outflow valve deactivated and FIRE CARGO AFT warning message, accomplish the associated non-normal checklist and also set FWD CARGO A/C control OFF.
5. Prior to stand or gate arrival, if both packs are operating, select right pack OFF. After main entry doors are open at the stand or gate, select packs as desired.

21-31-03 Outflow Valves (FWD and AFT)

21-31-03-02 Passenger With P/N 3676-GRS-014-00 or Later ASCPC Software Installed

Interval	Installed	Required	Procedure
C	2	1	(M) (O)

One may be inoperative provided:

- a. Valve is deactivated 7% open.
- b. Manual cabin pressure control is verified to operate normally on the remaining outflow valve.
- c. Left or right auto cabin pressure control operates normally.
- d. One pack operates normally, and the other pack operates normally or in standby cooling mode.
- e. Both CTCs operate normally.
- f. Both engine bleed systems operate normally.

For OFCR installed:

- g. For aft outflow valve deactivated 7% open, the OFCR is considered inoperative.

For OFAR installed:

- h. For aft outflow valve deactivated 7% open, the OFAR is considered inoperative.

For OFCR or OFAR installed:

NOTE: For aft outflow valve deactivated, the airplane must also be dispatched using MMEL item 25-29-04.

MAINTENANCE (M)

NOTE 1: ASCPC software P/N 3676-GRS-014-00 is installed by Boeing Service Bulletin 777-36-0013. All software P/Ns installed in ASCPC-2 are considered later versions than software P/Ns installed in ASCPC-1.

NOTE 2: For a pack to be operating normally, certain items associated with that system must be operative. See DDG item 21-51-01 for a listing of those items.

NOTE 3: A pack will operate in standby cooling mode with an inoperative ACM, or an inoperative ECV, or an inoperative Condenser Inlet Temperature Control System.

NOTE 4: For an engine bleed system to be operating normally, certain items associated with that system must be operative. See DDG item 36-00-01 for a listing of those items.

Deactivate inoperative outflow valve 7% open and verify manual cabin pressure control operates normally on the remaining outflow valve (AMM 21-00-00/901).

1. Supply electrical power on the airplane (AMM 24-22-00/201).
2. Deactivate inoperative outflow valve 7% open.

NOTE: For ground operations with OAT at or above 24 degrees C and forward outflow valve deactivated 7% open, supply conditioned air to the airplane.

- A. For manual control operative and valve position displayed on the Air Supply Maintenance page:
 - 1) Position the associated Pressurization Panel (P5) OUTFLOW VALVE switch to MAN.
 - 2) Using the associated Pressurization Panel (P5) OUTFLOW VALVE MANUAL switch, position the inoperative outflow valve to indicate 0.07 on the Air Supply Maintenance page.
 - 3) For the forward outflow valve inoperative:
 - a. Open and collar the P210 panel R FWD OFL VLV circuit breaker.
 - b. Open and collar the P310 panel L FWD OFL VLV circuit breaker.
 - 4) For the aft outflow valve inoperative:
 - a. Open and collar the P210 panel R AFT OFL VLV circuit breaker.
 - b. Open and collar the P310 panel L AFT OFL VLV circuit breaker.
 - 5) Position the associated Pressurization Panel (P5) OUTFLOW VALVE switch to AUTO.
 - 6) Confirm the gap between the two outflow doors is 0.375 ± 0.031 inches.



- B. For OFV (P/N 2214420-X) installed with manual control inoperative or valve position on the Air Supply Maintenance page not available:
- 1) For the forward outflow valve inoperative:
 - a. Open and collar the P210 panel R FWD OFL VLV circuit breaker.
 - b. Open and collar the P310 panel L FWD OFL VLV circuit breaker.
 - 2) For the aft outflow valve inoperative:
 - a. Open and collar the P210 panel R AFT OFL VLV circuit breaker.
 - b. Open and collar the P310 panel L AFT OFL VLV circuit breaker.
 - 3) Gain access to the associated outflow valve assembly.
 - 4) Disconnect the electrical connector from the back of the valve control unit.
 - 5) Remove one DC motor from the outflow valve assembly.
 - 6) Move the outflow valve doors until the gap between the two doors is 0.375 ± 0.031 inches.
 - 7) Re-install the DC motor.
 - 8) Reconnect the electrical connector on the back of the valve control unit.
 - 9) Position the associated Pressurization Panel (P5) OUTFLOW VALVE switch to AUTO.
3. Verify manual cabin pressure control operates normally on the remaining outflow valve.
- A. Position the associated Pressurization Panel (P5) OUTFLOW VALVE switch to MAN.
- NOTE: Outflow valve position indications on EICAS may blank when MAN is selected.
- B. Hold the associated Pressurization Panel (P5) OUTFLOW VALVE MANUAL switch CLOSE until the EICAS outflow valve position indicator is CL, or if EICAS indication is not available hold the switch CLOSE for 30 seconds.
- C. Confirm the associated outflow valve door is closed.
- D. Hold the associated Pressurization Panel (P5) OUTFLOW VALVE MANUAL switch OPEN until the EICAS outflow valve position indicator is OP, or if EICAS indication is not available hold the switch OPEN for 30 seconds.
- E. Confirm the associated outflow valve door is full open.
- F. Position the associated Pressurization Panel (P5) OUTFLOW VALVE switch to AUTO.

OPERATIONS (O)

NOTE 1: Aft cabin temperature control may be degraded when dispatching with the aft outflow valve deactivated 7% open.

NOTE 2: The associated OUTFLOW VALVE AFT or FWD advisory messages will be displayed.

NOTE 3: The outflow valve position indication pointer will be missing for the deactivated outflow valve on the EICAS pressurization indications.

1. During passenger loading, if both packs are operating, select right pack OFF prior to closing the main entry door(s). This avoids uncomfortable changes in cabin pressure.
2. After all main entry doors are closed at the parking stand or departure gate, if both packs are operating, select right pack OFF before reopening a main entry door. After leaving the parking stand or gate, select right pack AUTO.
3. If the operative outflow valve fails during flight and cannot be controlled manually it may be necessary to select packs OFF to depressurize the airplane for landing.
 - A. During descent when below 15,000 ft. MSL, select one pack OFF.
 - B. During descent when below 10,000 ft. MSL, select remaining pack OFF.

For Forward Cargo Air Conditioning (FCAC) installed:

4. For aft outflow valve deactivated and FIRE CARGO AFT warning message, accomplish the associated non-normal checklist and also set FWD CARGO A/C control OFF.
5. Prior to stand or gate arrival, if both packs are operating, select right pack OFF. After main entry doors are open at the stand or gate, select packs as desired.

21-31-03 Outflow Valves (FWD and AFT)

21-31-03-03 777F

21-31-03-03-01 With P/N 3677-GRS-105-00 ASCPC Software Installed

Interval	Installed	Required	Procedure
C	2	1	(M) (O)

One may be inoperative provided:

- a. Valve is deactivated 7% open.
- b. Manual cabin pressure control is verified to operate normally on the remaining outflow valve.
- c. Left or right auto cabin pressure control operates normally.
- d. One pack operates normally, and the other pack operates normally or in standby cooling mode.
- e. Both CTCs operate normally.

- f. Both engine bleed systems operate normally.
 - g. Rigid cargo barrier doors remain closed inflight.
-

MAINTENANCE (M)

NOTE 1: For a pack to be operating normally, certain items associated with that system must be operative. See DDG item 21-51-01 for a listing of those items.

NOTE 2: A pack will operate in standby cooling mode with an inoperative ACM, or an inoperative ECV, or an inoperative Condenser Inlet Temperature Control System.

NOTE 3: For an engine bleed system to be operating normally, certain items associated with that system must be operative. See DDG item 36-00-01 for a listing of those items.

Deactivate inoperative outflow valve 7% open and verify manual cabin pressure control operates normally on the remaining operative outflow valve (AMM 21-00-00/901).

NOTE: For ground operations with OAT at or above 24 degrees C and forward outflow valve deactivated 7% open, supply conditioned air to the airplane.

1. Supply electrical power on the airplane (AMM 24-22-00/201).
2. Deactivate inoperative outflow valve 7% open.
 - A. For manual control operative and valve position displayed on the Air Supply Maintenance page:
 - 1) Position the associated Pressurization Panel (P5) OUTFLOW VALVE switch to MAN.
 - 2) Using the associated Pressurization Panel (P5) OUTFLOW VALVE MANUAL switch, position the inoperative outflow valve to indicate 0.07 on the Air Supply Maintenance page.
 - 3) For the forward outflow valve inoperative:
 - a. Open and collar the P210 panel R FWD OFL VLV circuit breaker.
 - b. Open and collar the P310 panel L FWD OFL VLV circuit breaker.
 - 4) For the aft outflow valve inoperative:
 - a. Open and collar the P210 panel R AFT OFL VLV circuit breaker.
 - b. Open and collar the P310 panel L AFT OFL VLV circuit breaker.
 - 5) Position the associated Pressurization Panel (P5) OUTFLOW VALVE switch to AUTO.

- 6) Confirm the gap between the two outflow doors is 0.375 ± 0.031 inches.
- B. For OFV (P/N 2214420-X) installed with manual control inoperative, or valve position on the Air Supply Maintenance page not available:
 - 1) For the forward outflow valve inoperative:
 - a. Open and collar the P210 panel R FWD OFL VLV circuit breaker.
 - b. Open and collar the P310 panel L FWD OFL VLV circuit breaker.
 - 2) For the aft outflow valve inoperative:
 - a. Open and collar the P210 panel R AFT OFL VLV circuit breaker.
 - b. Open and collar the P310 panel L AFT OFL VLV circuit breaker.
 - 3) Gain access to the associated outflow valve assembly.
 - 4) Disconnect the electrical connector from the back of the valve control unit.
 - 5) Remove one DC motor from the outflow valve assembly.
 - 6) Move the outflow valve doors until the gap between the two doors is 0.375 ± 0.031 inches.
 - 7) Re-install the DC motor.
 - 8) Reconnect the electrical connector on the back of the valve control unit.
 - 9) Position the associated Pressurization Panel (P5) OUTFLOW VALVE switch to AUTO.
3. Verify manual cabin pressure control operates normally on the remaining outflow valve.
 - A. Position the associated Pressurization Panel (P5) OUTFLOW VALVE switch to MAN.

NOTE: Outflow valve position indications on EICAS may blank when MAN is selected.
 - B. Hold the associated Pressurization Panel (P5) OUTFLOW VALVE MANUAL switch CLOSE until the EICAS outflow valve position indicator is CL, or if EICAS indication is not available hold the switch CLOSE for 30 seconds.
 - C. Confirm the associated outflow valve door is closed.
 - D. Hold the associated Pressurization Panel (P5) OUTFLOW VALVE MANUAL switch OPEN until the EICAS outflow valve position indicator is OP, or if EICAS indication is not available hold the switch OPEN for 30 seconds.
 - E. Confirm the associated outflow valve door is full open.

- F. Position the associated Pressurization Panel (P5) OUTFLOW VALVE switch to AUTO.

OPERATIONS (O)

NOTE 1: The associated OUTFLOW VALVE AFT or FWD advisory message will be displayed.

NOTE 2: The outflow valve position indication pointer will be missing for the deactivated outflow valve on the EICAS pressurization indications.

1. During passenger loading, if both packs are operating, select right pack OFF prior to closing the main entry door(s). This avoids uncomfortable changes in cabin pressure.
2. After all main entry doors are closed at the parking stand or departure gate, if both packs are operating, select right pack OFF before reopening a main entry door. After leaving the parking stand or gate, select right pack AUTO.
3. Rigid cargo barrier doors remain closed inflight.
4. If the operative outflow valve fails during flight and cannot be controlled manually it may be necessary to select packs OFF to depressurize the airplane for landing.
 - A. During descent when below 15,000 ft. MSL, select one pack OFF.
 - B. During descent when below 10,000 ft. MSL, select remaining pack OFF.
5. For forward outflow valve deactivated:
 - A. For FIRE CARGO MAIN DECK warning message, the FWD OUTFLOW VALVE is already considered closed.
6. For aft outflow valve deactivated:

For Aft Cargo Air Conditioning (ACAC) installed:

- B. For FIRE CARGO FWD warning message, accomplish the associated non-normal checklist and also set LWR CARGO TEMP AFT A/C control OFF.

For Aft Cargo Air Conditioning (ACAC) not installed:

- A. For FIRE CARGO AFT warning message, accomplish the associated non-normal checklist and also set LWR CARGO TEMP FWD A/C control OFF.
 - B. For FIRE CARGO AFT warning message, accomplish the associated non-normal checklist and also set FWD CARGO A/C control OFF.
7. Prior to stand or gate arrival, if both packs are operating, select right pack OFF. After main entry doors are open at the stand or gate, select packs as desired.

21-31-03 Outflow Valves (FWD and AFT)**21-31-03-03 777F****21-31-03-03-02 With P/N 3674-GRS-106-00 or Later ASCPC Software Installed**

Interval	Installed	Required	Procedure
C	2	1	(M) (O)

One may be inoperative provided:

- Valve is deactivated 7% open.
- Manual cabin pressure control is verified to operate normally on the remaining outflow valve.
- Left or right auto cabin pressure control operates normally.
- One pack operates normally, and the other pack operates normally or in standby cooling mode.
- Both CTCs operate normally.
- Both engine bleed systems operate normally.

MAINTENANCE (M)

NOTE 1: PRR 62273, Part C, installs ASCPC-2 software P/N 3674-GRS-106-00 on L/N 998 and-on. Service Bulletin 777-36-0032 provides the production equivalent of PRR 62273 by installing ASCPC-2 software P/N 3674-GRS-106-00.

NOTE 2: For a pack to be operating normally, certain items associated with that system must be operative. See DDG item 21-51-01 for a listing of those items.

NOTE 3: A pack will operate in standby cooling mode with an inoperative ACM, or an inoperative ECV, or an inoperative Condenser Inlet Temperature Control System.

NOTE 4: For an engine bleed system to be operating normally, certain items associated with that system must be operative. See DDG item 36-00-01 for a listing of those items.

Deactivate inoperative outflow valve 7% open and verify manual cabin pressure control operates normally on the remaining operative outflow valve (AMM 21-00-00/901).

NOTE: For ground operations with OAT at or above 24 degrees C and forward outflow valve deactivated 7% open, supply conditioned air to the airplane.

- Supply electrical power on the airplane (AMM 24-22-00/201).

2. Deactivate inoperative outflow valve 7% open.
 - A. For manual control operative and valve position displayed on the Air Supply Maintenance page:
 - 1) Position the associated Pressurization Panel (P5) OUTFLOW VALVE switch to MAN.
 - 2) Using the associated Pressurization Panel (P5) OUTFLOW VALVE MANUAL switch, position the inoperative outflow valve to indicate 0.07 on the Air Supply Maintenance page.
 - 3) For the forward outflow valve inoperative:
 - a. Open and collar the P210 panel R FWD OFL VLV circuit breaker.
 - b. Open and collar the P310 panel L FWD OFL VLV circuit breaker.
 - 4) For the aft outflow valve inoperative:
 - a. Open and collar the P210 panel R AFT OFL VLV circuit breaker.
 - b. Open and collar the P310 panel L AFT OFL VLV circuit breaker.
 - 5) Position the associated Pressurization Panel (P5) OUTFLOW VALVE switch to AUTO.
 - 6) Confirm the gap between the two outflow doors is 0.375 ± 0.031 inches.
 - B. For OFV (P/N 2214420-X) installed with manual control inoperative, or valve position on the Air Supply Maintenance page not available:
 - 1) For the forward outflow valve inoperative:
 - a. Open and collar the P210 panel R FWD OFL VLV circuit breaker.
 - b. Open and collar the P310 panel L FWD OFL VLV circuit breaker.
 - 2) For the aft outflow valve inoperative:
 - a. Open and collar the P210 panel R AFT OFL VLV circuit breaker.
 - b. Open and collar the P310 panel L AFT OFL VLV circuit breaker.
 - 3) Gain access to the associated outflow valve assembly.
 - 4) Disconnect the electrical connector from the back of the valve control unit.
 - 5) Remove one DC motor from the outflow valve assembly.
 - 6) Move the outflow valve doors until the gap between the two doors is 0.375 ± 0.031 inches.
 - 7) Re-install the DC motor.

- 8) Reconnect the electrical connector on the back of the valve control unit.
 - 9) Position the associated Pressurization Panel (P5) OUTFLOW VALVE switch to AUTO.
3. Verify manual cabin pressure control operates normally on the remaining outflow valve.
- A. Position the associated Pressurization Panel (P5) OUTFLOW VALVE switch to MAN.
- NOTE: Outflow valve position indications on EICAS may blank when MAN is selected.
- B. Hold the associated Pressurization Panel (P5) OUTFLOW VALVE MANUAL switch CLOSE until the EICAS outflow valve position indicator is CL, or if EICAS indication is not available hold the switch CLOSE for 30 seconds.
 - C. Confirm the associated outflow valve door is closed.
 - D. Hold the associated Pressurization Panel (P5) OUTFLOW VALVE MANUAL switch OPEN until the EICAS outflow valve position indicator is OP, or if EICAS indication is not available hold the switch OPEN for 30 seconds.
 - E. Confirm the associated outflow valve door is full open.
 - F. Position the associated Pressurization Panel (P5) OUTFLOW VALVE switch to AUTO.

OPERATIONS (O)

NOTE 1: The associated OUTFLOW VALVE AFT or FWD advisory message will be displayed.

NOTE 2: The outflow valve position indication pointer will be missing for the deactivated outflow valve on the EICAS pressurization indications.

1. During passenger loading, if both packs are operating, select right pack OFF prior to closing the main entry door(s). This avoids uncomfortable changes in cabin pressure.
2. After all main entry doors are closed at the parking stand or departure gate, if both packs are operating, select right pack OFF before reopening a main entry door. After leaving the parking stand or gate, select right pack AUTO.
3. If the operative outflow valve fails during flight and cannot be controlled manually it may be necessary to select packs OFF to depressurize the airplane for landing.
 - A. During descent when below 15,000 ft. MSL, select one pack OFF.
 - B. During descent when below 10,000 ft. MSL, select remaining pack OFF.
4. For forward outflow valve deactivated:

For Aft Cargo Air Conditioning (ACAC) installed:

- A. For FIRE CARGO FWD warning message, accomplish the associated non-normal checklist and also set LWR CARGO TEMP AFT A/C control OFF.
5. For aft outflow valve deactivated:

For Aft Cargo Air Conditioning (ACAC) installed:

- A. For FIRE CARGO AFT warning message, accomplish the associated non-normal checklist and also set LWR CARGO TEMP FWD A/C control OFF.

For Aft Cargo Air Conditioning (ACAC) not installed:

- B. For FIRE CARGO AFT warning message, accomplish the associated non-normal checklist and also set FWD CARGO A/C control OFF.
6. Prior to stand or gate arrival, if both packs are operating, select right pack OFF. After main entry doors are open at the stand or gate, select packs as desired.

21-31-03 Outflow Valves (FWD and AFT)
21-31-03-04 All

Interval	Installed	Required	Procedure
C	2	0	(M) (O)

May be inoperative provided:

- a. Flight is conducted unpressurized.
- b. Procedures are established and used to verify cargo compartments remain empty or contain only ballast, empty cargo handling equipment (ballast may be loaded in ULDs), or fly away kits.
- c. Extended overwater flight is prohibited.

For LLAR, OFCR, OFAR, MDCR or MDFCR installed:

- d. The rest area(s) is considered inoperative.

For first class suites installed:

- e. First class suites are considered inoperative.

NOTE: Operator MELs must define which items are approved for inclusion in the fly away kits and which materials can be used as ballast.

NOTE 1: First class suites refers to the High Wall First Class Passenger Suite installation (option number 2525E407E69).

For LLAR, OFCR, OFAR, MDCR or MDFCR installed:

NOTE 2: The airplane must also be dispatched using MMEL item 25-29-04.

For first class suites installed:

NOTE 3: The airplane must also be dispatched using MMEL item 25-25-05.

MAINTENANCE (M)

NOTE: For both outflow valve status messages displayed due to faults in a single channel in both valves, it may be possible to achieve operation of one outflow valve on both channels by exchanging components. Consult airplane CMC data, and appropriate fault isolation manual (FIM) and AMM.

Deactivate both outflow valves full open (AMM 21-31-00/201).

1. Determine position of the outflow valves.
2. For outflow valve(s) in the full open position:
 - A. Open and collar the P210 panel R FWD OFL VLV circuit breaker.
 - B. Open and collar the P310 panel L FWD OFL VLV circuit breaker.
 - C. Open and collar the P210 panel R AFT OFL VLV circuit breaker.
 - D. Open and collar the P310 panel L AFT OFL VLV circuit breaker.
 - E. Position the associated Pressurization Panel (P5) OUTFLOW VALVE switch(es) to AUTO.
3. For outflow valve(s) not in the full open position and manual control is available:
 - A. Position the associated Pressurization Panel (P5) OUTFLOW VALVE switch(es) to MAN.
 - B. Hold the associated Pressurization Panel (P5) OUTFLOW VALVE MANUAL switch(es) OPEN until the EICAS outflow valve position indicator is OP, or if EICAS indication is not available hold the switch OPEN for 30 seconds.
 - C. For the EICAS outflow valve position indication(s) not available, confirm the associated outflow door(s) is full open.
 - D. Open and collar the P210 panel R FWD OFL VLV circuit breaker.
 - E. Open and collar the P310 panel L FWD OFL VLV circuit breaker.
 - F. Open and collar the P210 panel R AFT OFL VLV circuit breaker.
 - G. Open and collar the P310 panel L AFT OFL VLV circuit breaker.
 - H. Position the associated Pressurization Panel (P5) OUTFLOW VALVE switch(es) to AUTO.
4. For outflow valve(s) not in the full open position and manual control is not available:
 - A. Open and collar the P210 panel R FWD OFL VLV circuit breaker.
 - B. Open and collar the P310 panel L FWD OFL VLV circuit breaker.
 - C. Open and collar the P210 panel R AFT OFL VLV circuit breaker.
 - D. Open and collar the P310 panel L AFT OFL VLV circuit breaker.
 - E. Gain access to the associated outflow valve assembly.
 - F. Remove the primary and secondary pushrods.
 - G. Position the valve door(s) in the full open position.
 - H. Install the hold-open rod, part number J21006.



- I. Position the associated Pressurization Panel (P5) OUTFLOW VALVE switch(es) to AUTO.

OPERATIONS (O)

NOTE: The CABIN ALTITUDE AUTO caution message will be displayed.

1. The flight is conducted unpressurized.
2. Extended overwater flight is prohibited.
3. Cargo compartments must remain empty, except for ballast, empty cargo containers (ballast may be loaded in ULDs), fly away kits, pallets, and cargo restraint components.
4. For flight with passengers:
 - A. Limit enroute altitude to 10,000 feet.
 - B. To avoid passenger discomfort, limit climb (after takeoff obstacle clearance has been assured) and descent rates to 500 fpm.

21-31-03 Outflow Valves (FWD and AFT)**21-31-03-05 OUTFLOW VALVE Switch Lights****21-31-03-05-01 MAN Lights**

Interval	Installed	Required	Procedure
C	2	0	

21-31-03 Outflow Valves (FWD and AFT)**21-31-03-05 OUTFLOW VALVE Switch Lights****21-31-03-05-02 AUTO Lights**

Interval	Installed	Required	Procedure
C	2	0	

**21-31-04 Remote Cabin Pressure Sensor**

Interval	Installed	Required	Procedure
C	1	0	

May be inoperative provided both Auto Cabin Pressure Controls operate normally.

21-31-05 Cabin Rate Of Climb Indication
21-31-05A Both Auto Cabin Pressure Controls Operative

Interval	Installed	Required	Procedure
C	1	0	

May be inoperative provided both Auto Cabin Pressure Controls operate normally.

21-31-05 Cabin Rate Of Climb Indication
21-31-05B Unpressurized Flight

Interval	Installed	Required	Procedure
C	1	0	(M) (O)

May be inoperative provided:

- Flight is conducted unpressurized.
- Procedures are established and used to verify cargo compartments remain empty or contain only ballast, empty cargo handling equipment (ballast may be loaded in ULDs), or fly away kits.
- For extended overwater flight, manual cabin pressure control is verified to operate normally on both outflow valves.

[For LLAR, OFCR, OFAR, MDCR or MDFCR installed:](#)

- The rest area(s) is considered inoperative.

[For first class suites installed:](#)

- First class suites are considered inoperative.

NOTE: Operator MELs must define which items are approved for inclusion in the fly away kits and which materials can be used as ballast.

NOTE 1: First class suites refers to the High Wall First Class Passenger Suite installation (option number 2525E407E69).

[For LLAR, OFCR, OFAR, MDCR or MDFCR installed:](#)

NOTE 2: The airplane must also be dispatched using MMEL item 25-29-04.

[For first class suites installed:](#)

NOTE 3: The airplane must also be dispatched using MMEL item 25-25-05.

MAINTENANCE (M)

Use DDG item 21-31-01-03 (M) procedure.

OPERATIONS (O)

Use DDG item 21-31-01-03 (O) procedure.

21-31-06 Cabin Differential Pressure Indication
21-31-06A Cabin Altitude Indication Operative

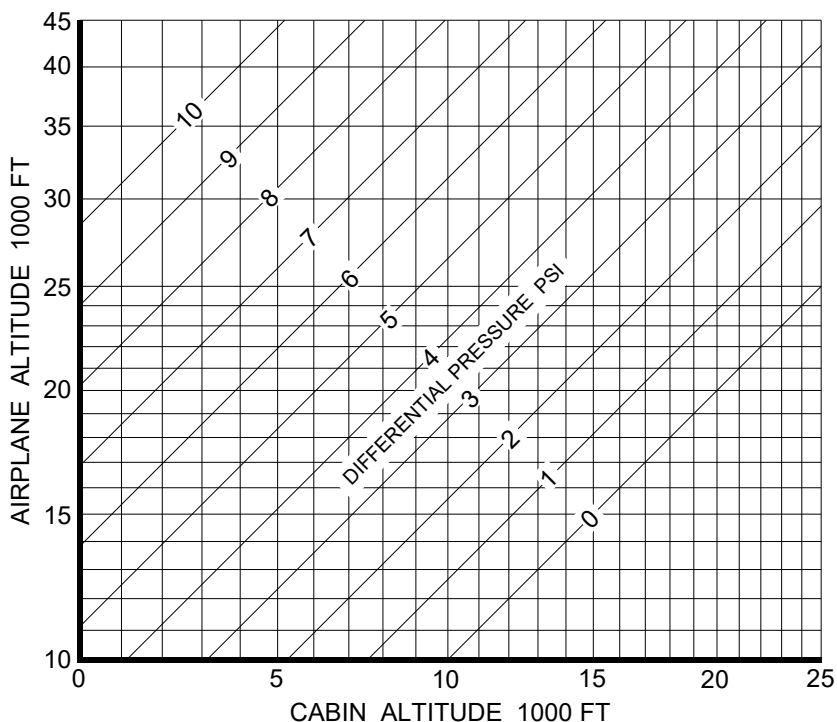
Interval	Installed	Required	Procedure
C	1	0	(O)

May be inoperative provided:

- Cabin altitude indication operates normally.
- A chart is provided to convert cabin altitude to cabin differential pressure.

OPERATIONS (O)

Cabin differential pressure may be derived from the following chart using indicated values of airplane altitude and cabin altitude.



**21-31-06 Cabin Differential Pressure Indication****21-31-06B Unpressurized Flight**

Interval	Installed	Required	Procedure
C	1	0	(M) (O)

May be inoperative provided:

- Flight is conducted unpressurized.
- Procedures are established and used to verify cargo compartments remain empty or contain only ballast, empty cargo handling equipment (ballast may be loaded in ULDs), or fly away kits.
- For extended overwater flight, manual cabin pressure control is verified to operate normally on both outflow valves.

[For LLAR, OFCR, OFAR, MDCR or MDPCR installed:](#)

- The rest area(s) is considered inoperative.

[For first class suites installed:](#)

- First class suites are considered inoperative.

NOTE: Operator MELs must define which items are approved for inclusion in the fly away kits and which materials can be used as ballast.

NOTE 1: First class suites refers to the High Wall First Class Passenger Suite installation (option number 2525E407E69).

[For LLAR, OFCR, OFAR, MDCR or MDPCR installed:](#)

NOTE 2: The airplane must also be dispatched using MMEL item 25-29-04.

[For first class suites installed:](#)

NOTE 3: The airplane must also be dispatched using MMEL item 25-25-05.

MAINTENANCE (M)

Use DDG item 21-31-01-03 (M) procedure.

OPERATIONS (O)

Use DDG item 21-31-01-03 (O) procedure.

21-31-07 Cabin Altitude Indication
21-31-07A Cabin Differential Pressure Indication Operative

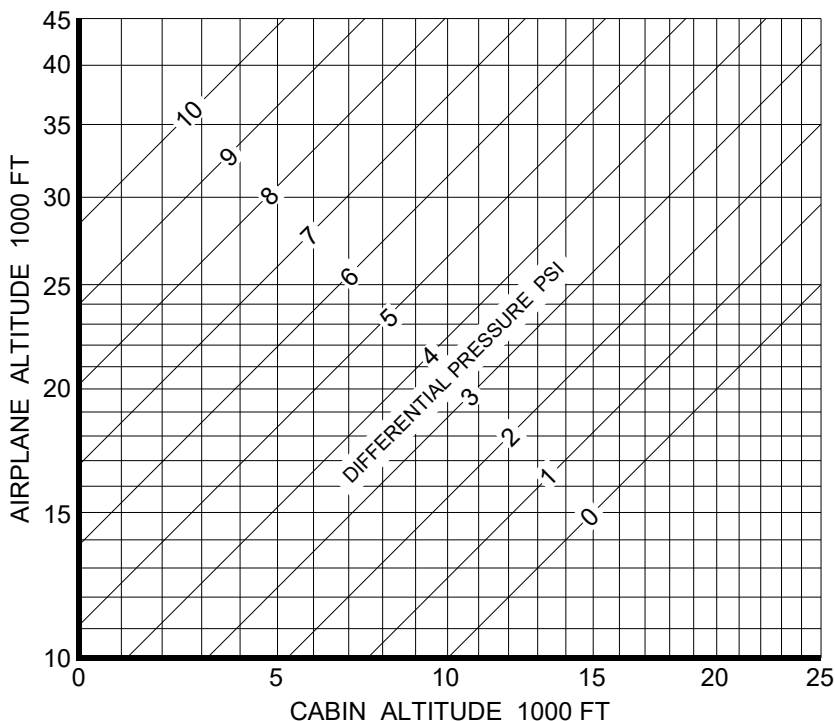
Interval	Installed	Required	Procedure
C	1	0	(O)

May be inoperative provided:

- Cabin differential pressure indication operates normally.
- A chart is provided to convert cabin differential pressure to cabin altitude.

OPERATIONS (O)

Cabin altitude may be derived from the following chart using indicated values of cabin differential pressure and airplane altitude.



**21-31-07 Cabin Altitude Indication****21-31-07B Unpressurized Flight**

Interval	Installed	Required	Procedure
C	1	0	(M) (O)

May be inoperative provided:

- Flight is conducted unpressurized.
- Procedures are established and used to verify cargo compartments remain empty or contain only ballast, empty cargo handling equipment (ballast may be loaded in ULDs), or fly away kits.
- For extended overwater flight, manual cabin pressure control is verified to operate normally on both outflow valves.

[For LLAR, OFCR, OFAR, MDCR or MDPCR installed:](#)

- The rest area(s) is considered inoperative.

[For first class suites installed:](#)

- First class suites are considered inoperative.

NOTE: Operator MELs must define which items are approved for inclusion in the fly away kits and which materials can be used as ballast.

NOTE 1: First class suites refers to the High Wall First Class Passenger Suite installation (option number 2525E407E69).

[For LLAR, OFCR, OFAR, MDCR or MDPCR installed:](#)

NOTE 2: The airplane must also be dispatched using MMEL item 25-29-04.

[For first class suites installed:](#)

NOTE 3: The airplane must also be dispatched using MMEL item 25-25-05.

MAINTENANCE (M)

Use DDG item 21-31-01-03 (M) procedure.

OPERATIONS (O)

Use DDG item 21-31-01-03 (O) procedure.

**21-31-08 Outflow Valve Position Indications**

Interval	Installed	Required	Procedure
C	2	0	

OPERATIONS NOTE

To manually position the outflow valve(s) full open or full closed, position the OUTFLOW VALVE Switch to MAN and press and hold the OUTFLOW VALVE MANUAL Switch OPEN or CLOSE for 30 seconds.

21-32-01 Positive Pressure Relief Valves
21-32-01A One Valve Inoperative

Interval	Installed	Required	Procedure
C	2	1	(M)

One positive pressure relief valve may be inoperative provided it is deactivated closed.

MAINTENANCE (M)

Deactivate the associated positive pressure relief valve in the closed position (AMM 21-00-00/901).

1. Gain access to the valve by removing the applicable sidewall liner on the left side of the lower forward cargo compartment.
2. If flapper doors are open, from outside the airplane press the flapper doors closed.
3. Disconnect and cap the remote ambient sense hose and the primary metering section cap.
4. Disconnect and cap the integral ambient sense tube and the secondary metering section cap.
5. Re-install the sidewall liner.

21-32-01 Positive Pressure Relief Valves
21-32-01B Both Valves Inoperative

Interval	Installed	Required	Procedure
C	2	0	(M) (O)

May be inoperative provided:

- a. Flight is conducted unpressurized.
- b. Extended overwater flight is prohibited.
- c. Both outflow valves are deactivated open.
- d. Procedures are established and used to verify cargo compartments remain empty or contain only ballast, empty cargo handling equipment (ballast may be loaded in ULDs), or fly away kits.

[For LLAR, OFCR, OFAR, MDCR or MDFCR installed:](#)

- e. The crew/attendant rest area(s) is considered inoperative.

[For first class suites installed:](#)

- f. First class suites are considered inoperative.



Use DDG item 21-31-03-04 (O) procedure.

21-32-02 Negative Pressure Relief Vents

Interval	Installed	Required	Procedure
C	4	0	(M) (O)

May be inoperative provided:

- a. Flight is conducted unpressurized.
- b. Extended overwater flight is prohibited.
- c. Both outflow valves are deactivated open.
- d. Procedures are established and used to verify cargo compartments remain empty or contain only ballast, empty cargo handling equipment (ballast may be loaded in ULDs), or fly away kits.

For LLAR, OFCR, OFAR, MDCR or MDFCR installed:

- e. The rest area(s) is considered inoperative.

For first class suites installed:

- f. First class suites are considered inoperative.

NOTE 1: Operator MELs must define which items are approved for inclusion in the fly away kits and which materials can be used as ballast.

NOTE 2: These provisions are not intended to prohibit crew/attendant rest inspections by crewmembers.

NOTE 1: First class suites refers to the High Wall First Class Passenger Suite installation (option number 2525E407E69).

For LLAR, OFCR, OFAR, MDCR or MDFCR installed:

NOTE 2: The airplane must also be dispatched using MMEL item 25-29-04.

For first class suites installed:

NOTE 3: The airplane must also be dispatched using MMEL item 25-25-05.

MAINTENANCE (M)

Use DDG item 21-31-03-04 (M) procedure.

OPERATIONS (O)

Use DDG item 21-31-03-04 (O) procedure.

**21-40-01 Main Deck Cargo Shutoff Valves (777F)****21-40-01-01 Forward Shutoff Valves 1 and 2**

Interval	Installed	Required	Procedure
C	2	0	(M) (O)

May be inoperative provided:

- Inoperative valve is deactivated closed.
 - Forward main deck cargo zone trim air modulation valve is deactivated closed.
 - Aft main deck cargo shutoff valves 1 and 2 operate normally.
 - Forward and aft main deck cargo shutoff valve 3 operate normally.
 - Both packs operate normally.
 - Both engine bleed systems operate normally.
-

MAINTENANCE (M)

NOTE: The Forward Main Deck Cargo Shutoff Valves 1 and 2 are interchangeable with the Forward or Aft Main Deck Cargo Shutoff Valves 3.

Deactivate the inoperative forward main deck cargo shutoff valve closed and deactivate the forward main deck cargo zone trim air modulation valve closed (AMM 21-00-00/901).

- Deactivate the inoperative forward main deck cargo shutoff valve closed.
 - Open P110 panel FWD M/D SOV #1/2 - TRMD circuit breaker.
 - Access the inoperative forward main deck cargo shutoff valve.
 - Disconnect, cap, and stow the electrical connector from the valve.
 - Turn the manual open/close knob until the position indicator is in the CLOSED position.
 - Close P110 panel FWD M/D SOV #1/2 - TRMD circuit breaker.
- Deactivate the forward main deck cargo zone trim modulation valve closed.
 - Open P110 panel L CTC CHAN 2 (AC) circuit breaker.
 - Open P110 panel L CTC CHAN 2 (DC) circuit breaker.
 - Access the forward main deck cargo zone trim air modulating valve.
 - Disconnect, cap, and stow the electrical connector from the valve.
 - Turn the manual open/close knob until the position indicator is in the CLOSED position.
 - Close P110 panel L CTC CHAN 2 (AC) circuit breaker.
 - Close P110 panel L CTC CHAN 2 (DC) circuit breaker.

OPERATIONS (O)

NOTE: Main deck cargo air conditioning is degraded forward of STA 1278 in the main cargo compartment. Temperature sensitive cargo should not be carried in the forward area of the main deck cargo compartment.

Position M/D FLOW switch to HIGH.

21-40-01 Main Deck Cargo Shutoff Valves (777F)
21-40-01-02 Aft Shutoff Valves 1 and 2

Interval	Installed	Required	Procedure
C	2	0	(M) (O)

May be inoperative provided:

- a. Inoperative valve is deactivated closed.
- b. Aft main deck cargo zone trim air modulation valve is deactivated closed.
- c. Forward main deck cargo shutoff valves 1 and 2 operate normally.
- d. Forward and aft main deck cargo shutoff valve 3 operate normally.
- e. Both packs operate normally.
- f. Both engine bleed systems operate normally.

MAINTENANCE (M)

NOTE: The Aft Main Deck Cargo Shutoff Valves 1 and 2 are interchangeable with the Forward or Aft Main Deck Cargo Shutoff Valves 3.

Deactivate the inoperative aft main deck cargo shutoff valve closed and deactivate the aft main deck cargo zone trim air modulation valve closed (AMM 21-00-00/901).

1. Deactivate the inoperative aft main deck cargo shutoff valve closed.
 - A. Open P210 panel AFT M/D SOV #1/2 - TRMD circuit breaker.
 - B. Access the inoperative aft main deck cargo shutoff valve.
 - C. Disconnect, cap, and stow the electrical connector from the valve.
 - D. Turn the manual open/close knob until the position indicator is in the CLOSED position.
 - E. Close P210 panel AFT M/D SOV #1/2 - TRMD circuit breaker.
2. Deactivate the aft main deck cargo zone trim modulation valve closed.
 - A. Open P210 panel R CTC CHAN 2 (AC) circuit breaker.
 - B. Open P210 panel R CTC CHAN 2 (DC) circuit breaker.
 - C. Access the aft main deck cargo zone trim air modulating valve.
 - D. Disconnect, cap, and stow the electrical connector from the valve.
 - E. Turn the manual open/close knob until the position indicator is in the CLOSED position.
 - F. Close P210 panel R CTC CHAN 2 (AC) circuit breaker.



G. Close P210 panel R CTC CHAN 2 (DC) circuit breaker.

OPERATIONS (O)

NOTE: Main deck cargo air conditioning is degraded aft of STA 1278 in the main cargo compartment. Temperature sensitive cargo should not be carried in the aft area of the main deck cargo compartment.

Position M/D FLOW switch to HIGH.

21-40-01 Main Deck Cargo Shutoff Valves (777F)

21-40-01-03 Forward and Aft Shutoff Valve 3

Interval	Installed	Required	Procedure
C	2	0	(M)

May be inoperative provided:

- Inoperative valve is deactivated closed.
- Forward main deck cargo shutoff valves 1 and 2 operate normally.
- Aft main deck cargo shutoff valves 1 and 2 operate normally.

MAINTENANCE (M)

NOTE: The Forward or Aft Main Deck Cargo Shutoff Valves 3 are interchangeable with the Forward or Aft Main Deck Cargo Shutoff Valves 1 and 2.

Deactivate the inoperative main deck cargo shutoff valve 3 closed (AMM 21-00-00/901).

- Open P110 panel FWD/AFT M/D SOV #3 - UNTRMD circuit breaker.
- Access the inoperative main deck cargo shutoff valve 3.
- Disconnect, cap, and stow the electrical connector from the valve.
- Turn the manual open/close knob until the position indicator is in the CLOSED position.
- Close P110 panel FWD/AFT M/D SOV #3 - UNTRMD circuit breaker.

OPERATIONS NOTE

The M/D FLOW HIGH setting will not be available.

**21-41-01 Lower Lobe Attendant Rest (LLAR) Electric Heaters**

Interval	Installed	Required	Procedure
C	2	0	

OPERATIONS NOTE

1. For one heater inoperative, the temperature in the LLAR may be colder than normal.
2. For both heaters inoperative, temperature control will be degraded and the LLAR may not be usable.

**21-41-02 Flight Crew Rest (FCR) Electric Heater Systems**

Interval	Installed	Required	Procedure
C	-	0	

OPERATIONS NOTE

1. The temperature in the associated FCR may be colder than normal.
2. For augmented flight crew operations, flight crew duty periods should be adjusted appropriately (14 CFR 117).

**21-41-03 Main Deck Crew Rest (MDCR)/Main Deck Flight Crew Rest (MDFCR) Electric Heater System**

Interval	Installed	Required	Procedure
C	1	0	(M)

Electric heater system may be inoperative provided it is deactivated.

MAINTENANCE (M)

Deactivate the MDCR/MDFCR heaters (AMM 21-00-00/901).

1. Open and collar P110 panel MDCR HTR/TEMP CTRL circuit breaker.

OPERATIONS NOTE

1. Temperature control will be degraded and the MDCR/MDFCR may not be usable.
2. For augmented flight crew operations, flight crew duty periods should be adjusted appropriately (14 CFR 117).

21-41-03 Main Deck Crew Rest (MDCR)/Main Deck Flight Crew Rest (MDFCR) Electric Heater System

21-41-03-01 Temperature Sensors

Interval	Installed	Required	Procedure
C	2	0	

OPERATIONS NOTE

The MDCR/MDFCR temperature control will be degraded.

**21-41-04 Overhead Flight Crew/Attendant Rest Electric Heater Systems (Includes FSI Installation)**

21-41-04-01 Overhead Flight Crew Rest (OFCR) Heater Systems

Interval	Installed	Required	Procedure
C	-	0	(M)

May be inoperative provided:

- Associated heater system is deactivated.
- OFCR is considered inoperative.

NOTE 1: The overhead crew rest installation manufactured by Flight Structures, Inc. (FSI) is not a Boeing system. Information applicable to the FSI installation is not provided.

NOTE 2: The airplane must also be dispatched using MMEL item 25-29-04.

MAINTENANCE (M)

Deactivate the associated OFCR electric heaters (AMM 21-00-00/901).

- Open and collar P19 panel CB - HEATER 2 - OFCR BUNK 1 circuit breaker.
- Open and collar P19 panel CB - HEATER 3 - OFCR BUNK 2 circuit breaker.
- Open and collar P19 panel CB - HEATER 1 - OFCR COMM circuit breaker.
- For second common area heater installed, open and collar P19 panel CB - HEATER 2 COM - OFCR circuit breaker.

21-41-04 Overhead Flight Crew/Attendant Rest Electric Heater Systems (Includes FSI Installation)

21-41-04-01 Overhead Flight Crew Rest (OFCR) Heater Systems**21-41-04-01-01 Bunk Heater Systems**

Interval	Installed	Required	Procedure
C	-	0	(M)

May be inoperative provided:

- Associated bunk heater system is deactivated.
- One common area heater system operates normally.

NOTE: The overhead crew rest installation manufactured by Flight Structures, Inc. (FSI) is not a Boeing system. Information applicable to the FSI installation is not provided.

MAINTENANCE (M)

Deactivate the inoperative OFCR bunk electric heater (AMM 21-00-00/901).

1. For the single bunk installation, open and collar P19 panel CB - HEATER 1 COMM - OFCR circuit breaker.
2. For the two bunk installation:
 - A. For the bunk 1 electric heater, open and collar P19 panel CB - HEATER 2 - OFCR BUNK 1 circuit breaker.
 - B. For the bunk 2 electric heater, open and collar P19 panel CB - HEATER 3 - OFCR BUNK 2 circuit breaker.

21-41-04 Overhead Flight Crew/Attendant Rest Electric Heater Systems (Includes FSI Installation)

21-41-04-01 Overhead Flight Crew Rest (OFCR) Heater Systems

21-41-04-01-02 Common Area Heater System(s)

Interval	Installed	Required	Procedure
C	-	0	(M)

May be inoperative provided:

- a. Associated common area heater system(s) is deactivated.
- b. Bunk heater system(s) operate normally.

NOTE: The overhead crew rest installation manufactured by Flight Structures, Inc. (FSI) is not a Boeing system. Information applicable to the FSI installation is not provided.

MAINTENANCE (M)

Deactivate the OFCR common area heater (AMM 21-00-00/901).

1. Open and collar P19 panel CB - HEATER 1 - OFCR COMM circuit breaker.
2. For second common area heater installed, open and collar P19 panel CB - HEATER 2 COM - OFCR circuit breaker.

21-41-04 Overhead Flight Crew/Attendant Rest Electric Heater Systems (Includes FSI Installation)

21-41-04-02 Overhead Flight Attendant Rest (OFAR) Heater Systems

21-41-04-02A One Inoperative

Interval	Installed	Required	Procedure
C	2	1	(M)

One may be inoperative provided associated heater system is deactivated.

NOTE: The overhead attendant rest installation manufactured by Flight Structures, Inc. (FSI) is not a Boeing system. Information applicable to the FSI installation is not provided.

MAINTENANCE (M)

Deactivate the inoperative OFAR heater (AMM 21-00-00/901).

1. For electric heater 1, open and collar P20 panel CB - HEATER 1 - OFAR circuit breaker.
2. For electric heater 2, open and collar P20 panel CB - HEATER 2 - OFAR circuit breaker.

21-41-04 Overhead Flight Crew/Attendant Rest Electric Heater Systems (Includes FSI Installation)

21-41-04-02 Overhead Flight Attendant Rest (OFAR) Heater Systems

21-41-04-02B Both Inoperative

Interval	Installed	Required	Procedure
C	2	0	(M)

May be inoperative provided:

- a. Heater systems are deactivated.
- b. OFAR is considered inoperative.

NOTE 1: The overhead attendant rest installation manufactured by Flight Structures, Inc. (FSI) is not a Boeing system. Information applicable to the FSI installation is not provided.

NOTE 2: The airplane must also be dispatched using MMEL item 25-29-04.

**MAINTENANCE (M)**

Deactivate the inoperative OFAR heaters (AMM 21-00-00/901).

1. Open and collar P20 panel CB - HEATER 1 - OFAR circuit breaker.
2. Open and collar P20 panel CB - HEATER 2 - OFAR circuit breaker.

21-44-01 Aft Cargo Heat Valves

Interval	Installed	Required	Procedure
C	2	0	(M)

May be inoperative provided system is deactivated closed.

MAINTENANCE (M)

Deactivate the lower aft cargo heat system closed (AMM 21-00-00/901).

1. Position Maintenance Panel (P61) AFT CARGO TEMP selector OFF.
2. Open and collar the P110 panel AFT CGO HT TCV circuit breaker.
3. Open and collar the P110 panel AFT CGO HT SOV circuit breaker.
4. Gain access to the lower aft cargo heat valves (AMM 25-52-06/401).
5. Confirm the lower aft cargo compartment Temperature Control Valve (TCV) or the lower aft cargo compartment Shutoff Valve (SOV) remains in the closed position when the pneumatic duct is pressurized.
 - A. Pressurize the airplane pneumatic system (AMM 36-00-00/201).
 - B. Confirm at least one lower aft cargo heat valve remains closed when the APU duct is pressurized.
 - C. Remove pressure from the pneumatic system.
6. If both valves remain open, one of the valves must be replaced (AMM 21-44-01/401).

NOTE 1: If only one of the inoperative valves is replaced, do not accomplish the valve installation test required by the installation procedure.

NOTE 2: All four of the lower aft and bulk cargo TCVs and SOVs are interchangeable.

7. Re-install the sidewall liner.

OPERATIONS NOTE

For all Passenger airplanes, and 777F without ACAC:

1. Lower aft cargo compartment heat is not available. Temperature sensitive cargo should not be carried in the lower aft cargo compartment.
2. The following tables may be used for predicting lower aft cargo compartment temperature. Temperature variations may occur.



Airport OAT Degrees C	Lower Aft Cargo Compartment Temperature Degrees F / Degrees C
-40	26 / -3
16	76 / 24
39	115 / 46

NOTE: Ground operation - Steady state temperatures with air conditioning packs on.

TAT Degrees C	Lower Aft Cargo Compartment Temperature Degrees F / Degrees C
-44	29 / -2
-28	41 / 5
-3	63 / 17

NOTE: In-flight - Steady state temperatures after approximately 2.5 hours cruise at FL350.

**21-44-02 Aft Cargo Compartment Temperature Sensor**

Interval	Installed	Required	Procedure
C	1	0	

May be inoperative provided system remains off.

OPERATIONS NOTE

For all Passenger airplanes, and 777F without ACAC:

1. Position Maintenance Panel (P61) AFT CARGO TEMP selector OFF.
2. Lower aft cargo compartment heat is not available. Temperature sensitive cargo should not be carried in the lower aft cargo compartment.
3. Use DDG item 21-44-01 (O) Note for predicting lower aft cargo compartment temperatures.
4. Lower aft cargo compartment temperature will not be available on the Air Synoptic.

**21-44-03 Bulk Cargo Heat Valves (Passenger)**

Interval	Installed	Required	Procedure
C	2	0	(M)

May be inoperative provided system is deactivated closed.

MAINTENANCE (M)

Deactivate the bulk cargo heat system closed (AMM 21-00-00/901).

1. Position Maintenance Panel (P61) BULK CARGO TEMP selector OFF.
2. Open and collar the P210 panel BULK CGO HT TCV circuit breaker.
3. Open and collar the P210 panel BULK CGO HT SOV circuit breaker.
4. Gain access to the bulk cargo heat valves (AMM 25-52-06/401).
5. Confirm the bulk cargo compartment Temperature Control Valve (TCV) or the bulk cargo compartment Shutoff Valve (SOV) remains in the closed position when the pneumatic duct is pressurized.
 - A. Pressurize the airplane pneumatic system (AMM 36-00-00/201).
 - B. Confirm at least one bulk cargo heat valve remains closed when the APU duct is pressurized.
 - C. Depressurize the pneumatic system (AMM 36-00-00/201).
6. If both valves remain open, one of the valves must be replaced (AMM 21-44-03/401).

NOTE 1: If only one of the inoperative valves is replaced, do not accomplish the valve installation test required by the installation procedure.

NOTE 2: All four of the aft and bulk cargo TCVs and SOVs are interchangeable.

7. Re-install the sidewall liner.

OPERATIONS NOTE

1. Bulk cargo compartment heat is not available. Temperature sensitive cargo should not be carried in the bulk cargo compartment.
2. The following tables may be used for predicting cargo compartment temperature. Temperature variations may occur.

Airport OAT Degrees C	Bulk Cargo Compartment Temperature Degrees F / Degrees C
-40	38 / 3
16	78 / 25
39	117 / 47



NOTE: Ground operation - Steady state temperatures with air conditioning packs on.

TAT Degrees C	Bulk Cargo Compartment Temperature Degrees F / Degrees C
-44	32 / 0
-28	44 / 7
-3	63 / 17

NOTE: In-flight - Steady state temperatures after approximately 2.5 hours cruise at FL350.

**21-44-04 Bulk Cargo Compartment Temperature Sensor
(Passenger)**

Interval	Installed	Required	Procedure
C	1	0	

May be inoperative provided system remains off.

OPERATIONS NOTE

1. Position Overhead Maintenance Panel (P61) BULK CARGO TEMP selector OFF.
2. Bulk cargo compartment heat is not available. Temperature sensitive cargo should not be carried in the bulk cargo compartment.
3. Use DDG item 21-44-03 (O) Note for predicting bulk cargo compartment temperatures.
4. Bulk cargo compartment temperature will not be available on the Air Synoptic.

**21-45-01 Flight Deck Foot Heaters**

Interval	Installed	Required	Procedure
C	2	0	(M)

May be inoperative OFF.

MAINTENANCE (M)

Deactivate the inoperative foot heater (AMM 21-00-00/901).

1. Position the associated Sidewall Panel (P13-1/P14-1) foot heater selector OFF.
2. For a Captain's foot heater failed on:

NOTE: Captain's shoulder heater will be deactivated. The airplane must also be dispatched using MMEL item 21-45-02.

- A. Open and collar P110 panel CAPT FT & SHLDR HTR CTRL circuit breaker.
 - B. Open and collar P110 panel CAPT FT & SHLDR HTR circuit breaker.
3. For a First Officer's foot heater failed on:

NOTE: First Officer's shoulder heater will be deactivated. The airplane must also be dispatched using MMEL item 21-45-02.

- A. Open and collar P210 panel F/O FT & SHLDR HTR CTRL circuit breaker.
- B. Open and collar P210 panel F/O FT & SHLDR HTR circuit breaker.

**21-45-02 Flight Deck Shoulder Heaters**

Interval	Installed	Required	Procedure
C	2	0	(M)

May be inoperative OFF.

MAINTENANCE (M)

Deactivate the inoperative shoulder heater (AMM 21-00-00/901).

1. Position the associated Sidewall Panel (P13-1/P14-1) shoulder heater control OFF.
2. For a Captain's shoulder heater failed on:

NOTE: Captain's foot heater will be deactivated. The airplane must also be dispatched using MMEL item 21-45-01.
 - A. Open and collar P110 panel CAPT FT & SHLDR HTR CTRL circuit breaker.
 - B. Open and collar P110 panel CAPT FT & SHLDR HTR circuit breaker.
3. For a First Officer's shoulder heater failed on:

NOTE: First Officer's footer heat will be deactivated. The airplane must also be dispatched using MMEL item 21-45-01.
 - A. Open and collar P210 panel F/O FT & SHLDR HTR CTRL circuit breaker.
 - B. Open and collar P210 panel F/O FT & SHLDR HTR circuit breaker.



21-51-01 Air Conditioning Packs
21-51-01-01 -200/-200ER (Without PRR 61980 or Production Equivalent Incorporated)

Interval	Installed	Required	Procedure
C	2	1	(M) (O)

One may be inoperative provided:

- Remaining pack operates normally.
- Both engine bleed systems operate normally.
- Both outflow valves operate normally.
- Airplane remains at FL 350 or below.
- Flight remains within 60 minutes of landing at a suitable airport.
- Appropriate performance adjustments are applied.
- For FSI OHAR installed, the FSI OHAR is deactivated closed.
- JAMCO flight deck security door part number XXX 9721-1 is not installed.

NOTE: These provisions are not intended to prohibit attendant rest facility inspections by crew members.

NOTE 1: The Overhead Attendant Rest (OHAR) installation manufactured by Flight Structures, Inc. (FSI) is not a Boeing system. Information applicable to the FSI installation is not provided.

NOTE 2: PRR 61980 installs enhancements for dispatch with a single pack, or dispatch with a single engine bleed system, on airplane L/N's 423, 429, 454, 455, 457 thru 461, and 463 and-on. Installation of the following Service Bulletins provides the production equivalent of PRR 61980.

Boeing Service Bulletin	Item
777-21-0075	ECSMC Software P/N 210E-BCE-01A-09 or later
777-21-0086	Return Air Grill Baffles
777-25-0227 or 777-25-0327	Fwd Cargo Floor Panels and Roller Trays
777-29-0027	HYDIM Software P/N 297A-BCD-000-27 or later
777-31-0086	WEU Software P/N 3105-BCE-01C-10 or later
777-36-0016	ASCPC-1 Software P/N 3677-GRS-015-00 or later

MAINTENANCE (M)

NOTE 1: For a pack to be operating normally, the following associated items must be operative (AMM 21-00-00/901):



M MEL Item	Associated Status Message
21-51-01	PACK L or R
21-51-02	PACK FCV L or R
21-52-01	PACK ACM L or R
21-52-02	ECON COOL VALVE L or R
21-52-03	CMPSRSR SENSOR L or R
21-52-04	PACK OUT TEMP SNSR L or R
21-52-05	RAM AIR INLET DOOR L or R
21-52-06	RAM AIR EXIT DOOR L or R
21-52-07	COND TEMP SENSOR L or R

NOTE 2: For an engine bleed system to be operating normally, certain items associated with that system must be operative. See DDG item 36-00-01 for a listing of those items.

1. For APU pneumatic duct pressure seal P/N 93C10353-9 or earlier installed, inspect the APU pneumatic duct pressure seal (boot) per 777-SL-36-005.

NOTE: Airplanes L/N 212 and on were delivered with improved seal P/N 93C10353-13 installed and do not require this inspection.

OPERATIONS (O)

NOTE 1: Aft and bulk cargo heating may be affected. Depending on cargo type and ambient conditions, compartment temperatures may be as cold as 55 degrees F (13 degrees C) with HIGH selected.

NOTE 2: If the air cycle machine fails on the remaining pack, the pack will operate continuously in standby cooling mode, regardless of altitude or outside air temperature, to maintain cabin pressurization. A single pack in standby cooling mode is capable of maintaining cabin pressurization at cruise altitude.

For Lower Lobe Attendant Rest (LLAR), Overhead Flight Crew Rest (OFCR), Overhead Flight Attendant Rest (OFAR), Main Deck Crew Rest (MDCR) or Main Deck Flight Crew Rest (MDFCR) installed:

NOTE 3: For left pack inoperative or off, temperature control in the crew/attendant rest(s) may be degraded.

For Forward Cargo Air Conditioning (FCAC) installed:

NOTE 4: FCAC temperature control is degraded. Temperature sensitive cargo should not be carried in the forward cargo compartment.



NOTE 5: For APU-to-Pack takeoff function installed, if left pack is inoperative the APU-to-Pack Takeoff Supplementary Procedure is not allowed.

1. Position associated pack switch OFF.
2. Reduce performance limited weights by the appropriate adjustments.

NOTE: The takeoff performance limited weight reductions may be avoided by performing a Packs Off Takeoff or for a right pack inoperative, performing an APU-to-Pack Takeoff (if installed) Supplementary Procedure.

Performance Limited Weights	Takeoff & Landing	Enroute Climb
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For -200 or -200ER/GE90-76B thru -90B:

thru 632,500 lb (286,902 kg)	14,300 lb (6,486 kg)	7,700 lb (3,493 kg)
over 632,500 lb (286,902 kg)	20,700 lb (9,389 kg)	7,900 lb (3,583 kg)

For -200 or -200ER/GE90-94B:

thru 632,500 lb (286,902 kg)	14,300 lb (6,486 kg)	9,400 lb (4,264 kg)
over 632,500 lb (286,902 kg)	22,400 lb (10,160 kg)	9,400 lb (4,264 kg)

For -200 or -200ER/PW4074 thru PW4084:

All	21,800 lb (9,888 kg)	8,600 lb (3,901 kg)
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For -200 or -200ER/PW4090:

thru 632,500 lb (286,902 kg)	21,500 lb (9,752 kg)	7,200 lb (3,266 kg)
over 632,500 lb (286,902 kg)	26,500 lb (12,020 kg)	7,200 lb (3,266 kg)

For -200 or -200ER/RR875 thru RR892:

thru 632,500 lb (286,902 kg)	6,400 lb (2,903 kg)	No penalty
over 632,500 lb (286,902 kg)	8,200 lb (3,720 kg)	No penalty

For -200 or -200ER/RR895:

All	6,800 lb (3,084 kg)	No penalty
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NOTE: The enroute climb weight penalties are based on single engine operating speeds that approximate maximum lift-to-drag ratio speed. To account for the difference in level off altitude when operating at other speeds, multiply the enroute climb weight penalty listed above by the appropriate factor listed on page 3.00-01-00.2 in the ENROUTE DIVERSION SPEED EFFECTS paragraph.

3. Flight must remain within 60 minutes of landing at a suitable airport.
4. Flight must remain at or below FL 350.
5. Operate the wing anti-ice manually.
 - A. Position WING ANTI-ICE selector OFF.
 - B. For flaps up and wing anti-ice required (do not use during takeoff and landing):
 - 1) Position the ENG Bleed switch associated with the inoperative pack OFF.

NOTE: The ENG IDLE DISAGREE advisory message will be displayed when idle thrust is selected inflight.
 - 2) Position WING ANTI-ICE selector ON, then OFF as required.
 - 3) Prior to landing, position the ENG Bleed switch associated with the inoperative pack ON.
6. For a FIRE CARGO AFT or FIRE CARGO FWD warning message, accomplish the associated Non-Normal Checklist and the following:
 - A. If above FL 330, descend to FL 330.
 - B. Position both ENGINE ANTI-ICE selectors ON at top-of-descent.

For Overhead Flight Attendant Rest (OFAR) installed:

7. For smoke detected in OFAR:
 - A. Position AFT and BULK CARGO TEMP selectors OFF.

For Forward Cargo Air Conditioning (FCAC) installed:

- B. Position FWD CARGO A/C control OFF.

21-51-01 Air Conditioning Packs

21-51-01-02 -300 (Without PRR 61980 or Production Equivalent Incorporated)

Interval	Installed	Required	Procedure
C	2	1	(O)

One may be inoperative provided:

- a. Remaining pack operates normally.
- b. Both engine bleed systems operate normally.
- c. Both outflow valves operate normally.



- d. Procedures are established and used to verify forward cargo compartment remains empty or contains only ballast, empty cargo handling equipment, (ballast may be loaded in ULDs), or fly away kits.
- e. Flight remains within 60 minutes of landing at a suitable airport.
- f. Appropriate performance adjustments are applied.

NOTE: Operator MELs must define which items are approved for inclusion in the fly away kits and which materials can be used as ballast.

NOTE: PRR 61980 installs enhancements for dispatch with a single pack, or dispatch with a single engine bleed system, on airplane L/N's 423, 429, 454, 455, 457 thru 461, and 463 and-on. Installation of the following Service Bulletins provides the production equivalent of PRR 61980.

Boeing Service Bulletin	Item
777-21-0075	ECSCMC Software P/N 210E-BCE-01A-09 or later
777-21-0086	Return Air Grill Baffles
777-25-0227 or 777-25-0327	Fwd Cargo Floor Panels and Roller Trays
777-29-0027	HYDIM Software P/N 297A-BCD-000-27 or later
777-31-0086	WEU Software P/N 3105-BCE-01C-10 or later
777-36-0016	ASCPC-1 Software P/N 3677-GRS-015-00 or later

MAINTENANCE NOTE

1. For a pack to be operating normally, the following associated items must be operative:

MMEL Item	Associated Status Message
21-51-01	PACK L or R
21-51-02	PACK FCV L or R
21-52-01	PACK ACM L or R
21-52-02	ECON COOL VALVE L or R
21-52-03	CMPRSR SENSOR L or R
21-52-04	PACK OUT TEMP SNSR L or R
21-52-05	RAM AIR INLET DOOR L or R
21-52-06	RAM AIR EXIT DOOR L or R
21-52-07	COND TEMP SENSOR L or R

2. For an engine bleed system to be operating normally, certain items associated with that system must be operative. See DDG item 36-00-01 for a listing of those items.

OPERATIONS (O)

NOTE 1: Aft and bulk cargo heating may be affected. Depending on cargo type and ambient conditions, compartment temperatures may be as cold as 55 degrees F (13 degrees C) with HIGH selected.

NOTE 2: If the air cycle machine fails on the remaining pack, the pack will operate continuously in standby cooling mode, regardless of altitude or outside air temperature, to maintain cabin pressurization. A single pack in standby cooling mode is capable of maintaining cabin pressurization at cruise altitude.

For Lower Lobe Attendant Rest (LLAR), Overhead Flight Crew Rest (OFCR), Overhead Flight Attendant Rest (OFAR), Main Deck Crew Rest (MDCR) or Main Deck Flight Crew Rest (MDFCR) installed:

NOTE 3: For left pack inoperative or off, temperature control in the associated crew rest may be degraded.

NOTE 4: For APU-to-Pack takeoff function installed, if left pack is inoperative the APU-to-Pack Takeoff Supplementary Procedure is not allowed.

1. Position associated pack switch OFF.
2. Reduce takeoff performance limited weights by the appropriate adjustments.

NOTE: The takeoff performance limited weight reductions may be avoided by performing a Packs Off Takeoff or for a right pack inoperative, performing an APU-to-Pack Takeoff (if installed) Supplementary Procedure.

Performance Limited Weights	Takeoff & Landing	Enroute Climb
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For -300/PW4090:

All	21,800 lb (9,888 kg)	7,200 lb (3,266 kg)
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For -300/RR892:

All	7,000 lb (3,175 kg)	No penalty
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NOTE: The enroute climb weight penalties are based on single engine operating speeds that approximate maximum lift-to-drag ratio speed. To account for the difference in level off altitude when operating at other speeds, multiply the enroute climb weight penalty listed above by the appropriate factor listed on page 3.00-01-00.2 in the ENROUTE DIVERSION SPEED EFFECTS paragraph.

3. Flight must remain within 60 minutes of landing at a suitable airport.
4. The forward cargo compartment must remain empty, except for ballast, empty cargo containers (ballast may be loaded in ULDs), fly away kits, pallets, and cargo restraint components.
5. Operate the wing anti-ice manually.
 - A. Position WING ANTI-ICE selector OFF.
 - B. For flaps up and wing anti-ice required (do not use during takeoff and landing):
 - 1) Position the ENG Bleed switch associated with the inoperative pack OFF.

NOTE: The ENG IDLE DISAGREE advisory message will be displayed when idle thrust is selected inflight.
 - 2) Position WING ANTI-ICE selector ON, then OFF as required.
 - 3) Prior to landing, position the ENG Bleed switch associated with the inoperative pack ON.
6. For a FIRE CARGO AFT or FIRE CARGO FWD warning message, accomplish the associated Non-Normal Checklist and the following:
 - A. If above FL 330, descend to FL 330.
 - B. Position both ENGINE ANTI-ICE selectors ON at top-of-descent.

For Overhead Flight Attendant Rest (OFAR) installed:

7. For smoke detected in OFAR:
 - A. Position AFT and BULK CARGO TEMP selectors OFF.

For Forward Cargo Air Conditioning (FCAC) installed:

- B. Position FWD CARGO A/C control OFF.

21-51-01 Air Conditioning Packs

21-51-01-03 -200/-200ER/-300 (With PRR 61980 or Production Equivalent Incorporated)

Interval	Installed	Required	Procedure
C	2	1	(M) (O)

One may be inoperative provided:

- a. Remaining pack operates normally.
- b. Both engine bleed systems operate normally.



- c. Both outflow valves operate normally.
- d. Flight remains within 60 minutes of landing at a suitable airport.
- e. Appropriate performance adjustments are applied.
- f. For FSI OHAR installed on -200, the FSI OHAR is deactivated closed.

NOTE: These provisions are not intended to prohibit attendant rest facility inspections by crewmembers.

NOTE 1: The Overhead Attendant Rest (OHAR) installation manufactured by Flight Structures, Inc. (FSI) is not a Boeing system. Information applicable to the FSI installation is not provided.

NOTE 2: PRR 61980 installs enhancements for dispatch with a single pack, or dispatch with a single engine bleed system, on airplane L/N's 423, 429, 454, 455, 457 thru 461, and 463 and-on. Installation of the following Service Bulletins provides the production equivalent of PRR 61980.

Boeing Service Bulletin	Item
777-21-0075	ECSMC Software P/N 210E-BCE-01A-09 or later
777-21-0086	Return Air Grill Baffles
777-25-0227 or 777-25-0327	Fwd Cargo Floor Panels and Roller Trays
777-29-0027	HYDIM Software P/N 297A-BCD-000-27 or later
777-31-0086	WEU Software P/N 3105-BCE-01C-10 or later
777-36-0016	ASCP-1 Software P/N 3677-GRS-015-00 or later

MAINTENANCE (M)

NOTE 1: For a pack to be operating normally, the following associated items must be operative (AMM 21-00-00/901):

MMEL Item	Associated Status Message
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21-51-01	PACK L or R
21-51-02	PACK FCV L or R
21-52-01	PACK ACM L or R
21-52-02	ECON COOL VALVE L or R
21-52-03	CMPRSR SENSOR L or R
21-52-04	PACK OUT TEMP SNSR L or R
21-52-05	RAM AIR INLET DOOR L or R
21-52-06	RAM AIR EXIT DOOR L or R
21-52-07	COND TEMP SENSOR L or R

NOTE 2: For an engine bleed system to be operating normally, certain items associated with that system must be operative. See DDG item 36-00-01 for a listing of those items.

For -200/-200ER:

1. For APU pneumatic duct pressure seal P/N 93C10353-9 or earlier installed, inspect the APU pneumatic duct pressure seal (boot) per 777-SL-36-005.

NOTE: Airplanes L/N 212 and on were delivered with improved seal P/N 93C10353-13 installed and do not require this inspection.

OPERATIONS (O)

NOTE 1: Aft and bulk cargo heating may be affected. Depending on cargo type and ambient conditions, compartment temperatures may be as cold as 55 degrees F (13 degrees C) with HIGH selected.

NOTE 2: If the air cycle machine fails on the remaining pack, the pack will operate continuously in standby cooling mode, regardless of altitude or outside air temperature, to maintain cabin pressurization. A single pack in standby cooling mode is capable of maintaining cabin pressurization at cruise altitude.

For Lower Lobe Attendant Rest (LLAR), Overhead Flight Crew Rest (OFCR), Overhead Flight Attendant Rest (OFAR), Main Deck Crew Rest (MDCR) or Main Deck Flight Crew Rest (MDFCR) installed:

NOTE 3: For left pack inoperative or off, temperature control in the associated crew rest may be degraded.

For Forward Cargo Air Conditioning (FCAC) installed:

NOTE 4: FCAC temperature control is degraded. Temperature sensitive cargo should not be carried in the forward cargo compartment.



NOTE 5: For APU-to-Pack takeoff function installed, if left pack is inoperative the APU-to-Pack Takeoff Supplementary Procedure is not allowed.

1. Position associated pack switch OFF.
2. Reduce performance limited weights by the appropriate adjustments.

NOTE: The takeoff performance limited weight reductions may be avoided by performing a Packs Off Takeoff or for a right pack inoperative, performing an APU-to-Pack Takeoff (if installed) Supplementary Procedure.

Performance Limited Weights	Takeoff & Landing	Enroute Climb
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For -200 or -200ER/GE90-76B thru -90B:

thru 632,500 lb (286,902 kg)	14,300 lb (6,486 kg)	7,700 lb (3,493 kg)
over 632,500 lb (286,902 kg)	20,700 lb (9,389 kg)	7,900 lb (3,583 kg)

For -200 or -200ER/GE90-94B:

thru 632,500 lb (286,902 kg)	14,300 lb (6,486 kg)	9,400 lb (4,264 kg)
over 632,500 lb (286,902 kg)	22,400 lb (10,160 kg)	9,400 lb (4,264 kg)

For -200 or -200ER/PW4074 thru PW4084:

All	21,800 lb (9,888 kg)	8,600 lb (3,901 kg)
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For -200 or -200ER/PW4090:

thru 632,500 lb (286,902 kg)	21,500 lb (9,752 kg)	7,200 lb (3,266 kg)
over 632,500 lb (286,902 kg)	26,500 lb (12,020 kg)	7,200 lb (3,266 kg)

For -200 or -200ER/RR875 thru RR892:

thru 632,500 lb (286,902 kg)	6,400 lb (2,903 kg)	No penalty
over 632,500 lb (286,902 kg)	8,200 lb (3,720 kg)	No penalty

For -200 or -200ER/RR895:

All	6,800 lb (3,084 kg)	No penalty
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For -300/PW4090:

All	21,800 lb (9,888 kg)	7,200 lb (3,266 kg)
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For -300/RR892:

All	7,000 lb (3,175 kg)	No penalty
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NOTE: The enroute climb weight penalties are based on single engine operating speeds that approximate maximum lift-to-drag ratio speed. To account for the difference in level off altitude when operating at other speeds, multiply the enroute climb weight penalty listed above by the appropriate factor listed on page 3.00-01-00.2 in the ENROUTE DIVERSION SPEED EFFECTS paragraph.

3. Flight must remain within 60 minutes of landing at a suitable airport.
4. Operate the wing anti-ice manually.
 - A. Position WING ANTI-ICE selector OFF.
 - B. For flaps up and wing anti-ice required (do not use during takeoff and landing):
 - 1) Position the ENG Bleed switch associated with the inoperative pack OFF.

NOTE: The ENG IDLE DISAGREE advisory message will be displayed when idle thrust is selected inflight.
 - 2) Position WING ANTI-ICE selector ON, then OFF as required.
 - 3) Prior to landing, position the ENG Bleed switch associated with the inoperative pack ON.
5. For a FIRE CARGO AFT or FIRE CARGO FWD warning message, accomplish the associated Non-Normal Checklist and the following:
 - A. If above FL 330, descend to FL 330.
 - B. Position both ENGINE ANTI-ICE selectors ON at top-of-descent.

For Overhead Flight Attendant Rest (OFAR) installed:

6. For smoke detected in OFAR:
 - A. Position AFT and BULK CARGO TEMP selectors OFF.

For Forward Cargo Air Conditioning (FCAC) installed:

- B. Position FWD CARGO A/C control OFF.

21-51-01 Air Conditioning Packs

21-51-01-04 -200LR/-300ER

Interval	Installed	Required	Procedure
C	2	1	(O)

One may be inoperative provided:

- a. Remaining pack operates normally.

- b. Both engine bleed systems operate normally.
- c. Both outflow valves operate normally.
- d. Flight remains within 60 minutes of landing at a suitable airport.
- e. Appropriate performance adjustments are applied.

MAINTENANCE NOTE

1. For a pack to be operating normally, the following associated items must be operative:

MMEL Item	Associated Status Message
21-51-01	PACK L or R
21-51-02	PACK FCV L or R
21-52-01	PACK ACM L or R
21-52-02	ECON COOL VALVE L or R
21-52-03	CMPRSR SENSOR L or R
21-52-04	PACK OUT TEMP SNSR L or R
21-52-05	RAM AIR INLET DOOR L or R
21-52-06	RAM AIR EXIT DOOR L or R
21-52-07	COND TEMP SENSOR L or R

2. For an engine bleed system to be operating normally, certain items associated with that system must be operative. See DDG item 36-00-01 for a listing of those items.

OPERATIONS (O)

NOTE 1: Aft and bulk cargo heating may be affected. Depending on cargo type and ambient conditions, compartment temperatures may be as cold as 55 degrees F (13 degrees C) with HIGH selected.

NOTE 2: If the air cycle machine fails on the remaining pack, the pack will operate continuously in standby cooling mode, regardless of altitude or outside air temperature, to maintain cabin pressurization. A single pack in standby cooling mode is capable of maintaining cabin pressurization at cruise altitude.

For Lower Lobe Attendant Rest (LLAR), Overhead Flight Crew Rest (OFCR), Overhead Flight Attendant Rest (OFAR), Main Deck Crew Rest (MDCR) or Main Deck Flight Crew Rest (MDFCR) installed:

NOTE 3: For left pack inoperative or off, temperature control in the associated crew rest may be degraded.

**For Forward Cargo Air Conditioning (FCAC) installed:**

NOTE 4: FCAC temperature control is degraded. Temperature sensitive cargo should not be carried in the forward cargo compartment.

NOTE 5: For APU-to-Pack takeoff function installed, if left pack is inoperative the APU-to-Pack Takeoff Supplementary Procedure is not allowed.

1. Position associated pack switch OFF.
2. Reduce performance limited weights by the appropriate adjustments.

NOTE: The takeoff performance limited weight reductions may be avoided by performing a Packs Off Takeoff or for a right pack inoperative, performing an APU-to-Pack Takeoff (if installed) Supplementary Procedure.

Performance Limited Weights	Takeoff & Landing	Enroute Climb
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For -200LR/GE90-100 Series:

All	14,800 lb (6,713 kg)	No penalty
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For -300ER/GE90-115B:

All	11,600 lb (5,262 kg)	No penalty
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3. Flight must remain within 60 minutes of landing at a suitable airport.
4. Operate the wing anti-ice manually.
 - A. Position WING ANTI-ICE selector OFF.
 - B. For flaps up and wing anti-ice required (do not use during takeoff and landing):
 - 1) Position the ENG Bleed switch associated with the inoperative pack OFF.

NOTE: The ENG IDLE DISAGREE advisory message will be displayed when idle thrust is selected inflight.
 - 2) Position WING ANTI-ICE selector ON, then OFF as required.
 - 3) Prior to landing, position the ENG Bleed switch associated with the inoperative pack ON.
5. For a FIRE CARGO AFT or FIRE CARGO FWD warning message, accomplish the associated Non-Normal Checklist and the following:
 - A. If above FL 330, descend to FL 330.
 - B. Position both ENGINE ANTI-ICE selectors ON at top-of-descent.

For Overhead Flight Attendant Rest (OFAR) installed:

6. For smoke detected in OFAR:
 - A. Position AFT and BULK CARGO TEMP selectors OFF.



For Forward Cargo Air Conditioning (FCAC) installed:

B. Position FWD CARGO A/C control OFF.

21-51-01 Air Conditioning Packs**21-51-01-05 777F****21-51-01-05A AVS Operative**

Interval	Installed	Required	Procedure
C	2	1	(O)

One may be inoperative provided:

- Remaining pack operates normally.
- Both engine bleed systems operate normally.
- Both outflow valves operate normally.
- Alternate ventilation system operates normally.
- For left pack inoperative, right CTC operates normally.
- Appropriate performance adjustments are applied.

MAINTENANCE NOTE

NOTE 1: For a pack to be operating normally, the following associated items must be operative:

MMEL Item	Associated Status Message
21-51-01	PACK L or R
21-51-02	PACK FCV L or R
21-52-01	PACK ACM L or R
21-52-02	ECON COOL VALVE L or R
21-52-03	CMPRSR SENSOR L or R
21-52-04	PACK OUT TEMP SNSR L or R
21-52-05	RAM AIR INLET DOOR L or R
21-52-06	RAM AIR EXIT DOOR L or R
21-52-07	COND TEMP SENSOR L or R

NOTE 2: For an engine bleed system to be operating normally, certain items associated with that system must be operative. See DDG item 36-00-01 for a listing of those items.

OPERATIONS (O)

NOTE 1: If the air cycle machine fails on the remaining pack, the pack will operate continuously in standby cooling mode, regardless of altitude or outside air temperature, to maintain cabin pressurization. A single pack in standby cooling mode is capable of maintaining cabin pressurization at cruise altitude.

NOTE 2: For APU-to-Pack takeoff function installed, if left pack is inoperative the APU-to-Pack Takeoff Supplementary Procedure is not allowed.

For Forward Cargo Air Conditioning (FCAC) installed:

NOTE 3: FCAC temperature control is degraded. Temperature sensitive cargo should not be carried in the forward cargo compartment.

For Aft Cargo Air Conditioning (ACAC) installed:

NOTE 4: ACAC temperature control is degraded. Temperature sensitive cargo should not be carried in the aft cargo compartment.

1. Position associated pack switch OFF.
2. Reduce performance limited weights by the appropriate adjustments.

NOTE: The takeoff performance limited weight reductions may be avoided by performing a Packs Off Takeoff or for a right pack inoperative, performing an APU-to-Pack Takeoff (if installed) Supplementary Procedure.

Performance Limited Weights	Takeoff & Landing	Enroute Climb
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For 777F/GE90-100 Series:

All	14,800 lb (6,713 kg)	No penalty
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3. Operate the wing anti-ice manually.
 - A. Position WING ANTI-ICE selector OFF.
 - B. For flaps up and wing anti-ice required (do not use during takeoff and landing):
 - 1) Position the ENG Bleed switch associated with the inoperative pack OFF.

NOTE: The ENG IDLE DISAGREE advisory message will be displayed when idle thrust is selected inflight.
 - 2) Position WING ANTI-ICE selector ON, then OFF as required.
 - 3) Prior to landing, position the ENG Bleed switch associated with the inoperative pack ON.
4. For a FIRE CARGO AFT or FIRE CARGO FWD warning message, accomplish the associated Non-Normal Checklist and the following:
 - A. If above FL 330, descend to FL 330.



B. Position both ENGINE ANTI-ICE selectors ON at top-of-descent.

21-51-01 Air Conditioning Packs**21-51-01-05 777F****21-51-01-05B AVS Inoperative**

Interval	Installed	Required	Procedure
C	2	1	(O)

One may be inoperative provided:

- Remaining pack operates normally.
- Both engine bleed systems operate normally.
- Both outflow valves operate normally.
- Flight remains within 60 minutes of landing at a suitable airport.
- For left pack inoperative, right CTC operates normally.
- Appropriate performance adjustments are applied.

MAINTENANCE NOTE

NOTE 1: For a pack to be operating normally, the following associated items must be operative:

M MEL Item	Associated Status Message
21-51-01	PACK L or R
21-51-02	PACK FCV L or R
21-52-01	PACK ACM L or R
21-52-02	ECON COOL VALVE L or R
21-52-03	CMPRSR SENSOR L or R
21-52-04	PACK OUT TEMP SNSR L or R
21-52-05	RAM AIR INLET DOOR L or R
21-52-06	RAM AIR EXIT DOOR L or R
21-52-07	COND TEMP SENSOR L or R

NOTE 2: For an engine bleed system to be operating normally, certain items associated with that system must be operative. See DDG item 36-00-01 for a listing of those items.

**OPERATIONS (O)**

NOTE 1: If the air cycle machine fails on the remaining pack, the pack will operate continuously in standby cooling mode, regardless of altitude or outside air temperature, to maintain cabin pressurization. A single pack in standby cooling mode is capable of maintaining cabin pressurization at cruise altitude.

NOTE 2: For APU-to-Pack takeoff function installed, if left pack is inoperative the APU-to-Pack Takeoff Supplementary Procedure is not allowed.

For Forward Cargo Air Conditioning (FCAC) installed:

NOTE 3: FCAC temperature control is degraded. Temperature sensitive cargo should not be carried in the forward cargo compartment.

For Aft Cargo Air Conditioning (ACAC) installed:

NOTE 4: ACAC temperature control is degraded. Temperature sensitive cargo should not be carried in the aft cargo compartment.

1. Position associated pack switch OFF.
2. Reduce performance limited weights by the appropriate adjustments.

NOTE: The takeoff performance limited weight reductions may be avoided by performing a Packs Off Takeoff or for a right pack inoperative, performing an APU-to-Pack Takeoff (if installed) Supplementary Procedure.

Performance Limited Weights	Takeoff & Landing	Enroute Climb
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For 777F/GE90-100 Series:

All	14,800 lb (6,713 kg)	No penalty
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3. Flight must remain within 60 minutes of landing at a suitable airport.
4. Operate the wing anti-ice manually.
 - A. Position WING ANTI-ICE selector OFF.
 - B. For flaps up and wing anti-ice required (do not use during takeoff and landing):
 - 1) Position the ENG Bleed switch associated with the inoperative pack OFF.

NOTE: The ENG IDLE DISAGREE advisory message will be displayed when idle thrust is selected inflight.

- 2) Position WING ANTI-ICE selector ON, then OFF as required.
- 3) Prior to landing, position the ENG Bleed switch associated with the inoperative pack ON.



5. For a FIRE CARGO AFT or FIRE CARGO FWD warning message, accomplish the associated Non-Normal Checklist and the following:
- A. If above FL 330, descend to FL 330.
 - B. Position both ENGINE ANTI-ICE selectors ON at top-of-descent.

21-51-01 Air Conditioning Packs**21-51-01-06 DELETED (All)**

Interval	Installed	Required	Procedure

Deleted relief in revision 23

21-51-01 Air Conditioning Packs**21-51-01-07 PACK Switch Lights****21-51-01-07-01 OFF Lights**

Interval	Installed	Required	Procedure
C	2	0	

21-51-01 Air Conditioning Packs**21-51-01-07 PACK Switch Lights****21-51-01-07-02 AUTO Lights**

Interval	Installed	Required	Procedure
C	2	0	

**21-51-02 Pack Flow Control/Shutoff Valves (FCVs)****21-51-02-01 Upper FCVs**

Interval	Installed	Required	Procedure
C	2	0	(M)

May be inoperative provided:

- a. Inoperative upper FCV(s) is locked closed.
- b. Associated lower FCV operates normally.

MAINTENANCE (M)

Lock the inoperative upper FCV closed (AMM 21-00-00/901).

1. Lock the inoperative upper FCV closed.
 - A. Depressurize the pneumatic system (AMM 36-00-00/201).
 - B. For left pack inoperative FCV:
 - 1) Open the P110 panel L CTC CHAN 1 (AC) circuit breaker.
 - 2) Open the P110 panel L CTC CHAN 1 (DC) circuit breaker.
 - 3) Open the P110 panel L CTC CHAN 2 (AC) circuit breaker.
 - 4) Open the P110 panel L CTC CHAN 2 (DC) circuit breaker.
 - 5) Open the P110 panel L ECV/FCV POS SNSR circuit breaker.
 - 6) Gain access to the FCV by opening ECS access door 195EL.
 - C. For right pack inoperative FCV:
 - 1) Open the P210 panel R CTC CHAN 1 (AC) circuit breaker.
 - 2) Open the P210 panel R CTC CHAN 1 (DC) circuit breaker.
 - 3) Open the P210 panel R CTC CHAN 2 (AC) circuit breaker.
 - 4) Open the P210 panel R CTC CHAN 2 (DC) circuit breaker.
 - 5) Open the P210 panel R ECV/FCV POS SNSR circuit breaker.
 - 6) Gain access to the FCV by opening ECS access door 196ER.
 - D. Lock the inoperative upper FCV closed.
 - 1) Remove the locking plug from the FCV actuator boss.
 - 2) Turn the manual wrench until the FCV lock indicates CL.
 - 3) Install the locking plug in the manual lock and tighten according to the instructions and caution statement noted in AMM 21-00-00-040-849.
 - E. Close the ECS access door.
 - F. Close the associated circuit breakers.
2. Confirm the associated lower FCV operates normally.
 - A. Pressurize the pneumatic system (AMM 36-00-00/201).
 - B. Position Air Conditioning Panel (P5) associated PACK switch to AUTO.



- C. For airplanes without AIMS V14 or later installed, confirm the associated PACK L or R advisory and status messages are not displayed.
- NOTE: If the PACK L or R advisory and status messages are displayed, the lower FCV was inoperative. Unlock the upper FCV and dispatch using MMEL item 21-51-02-02.
- D. For airplanes with AIMS V14 or later installed, confirm the associated PACK L or R advisory and status and PACK FCV LWR L or R status message are not displayed.
- NOTE: If the PACK L or R advisory and status messages and the PACK FCV LWR L or R status message are displayed, the lower FCV was inoperative. Unlock the upper FCV and dispatch using MMEL item 21-51-02-02.
- E. Depressurize the pneumatic system (AMM 36-00-00/201).

OPERATIONS NOTE

NOTE: For APU-to-Pack takeoff function installed, if left upper FCV is inoperative the APU-to-Pack takeoff Supplementary Procedure is not allowed.

21-51-02 Pack Flow Control/Shutoff Valves (FCVs)**21-51-02-02 Lower FCVs**

Interval	Installed	Required	Procedure
C	2	1	(M)

One may be inoperative provided:

- Inoperative lower FCV is locked closed.
- Opposite pack operates normally.
- Associated upper FCV operates normally.

NOTE: The airplane must also be dispatched using MMEL Item 21-73-01.

MAINTENANCE (M)

NOTE 1: For a pack to be operating normally, certain items associated with that system must be operative. See DDG item 21-51-01 for a listing of those items.

NOTE 2: For a lower FCV locked closed, the associated ozone converter is inoperative.

NOTE 3: The lower FCV is interchangeable with the upper FCV.



Lock the inoperative lower FCV closed (AMM 21-00-00/901).

1. Lock the inoperative lower FCV closed.
 - A. Depressurize the pneumatic system (AMM 36-00-00/201).
 - B. For left pack inoperative FCV:
 - 1) Open the P110 panel L CTC CHAN 1 (AC) circuit breaker.
 - 2) Open the P110 panel L CTC CHAN 1 (DC) circuit breaker.
 - 3) Open the P110 panel L CTC CHAN 2 (AC) circuit breaker.
 - 4) Open the P110 panel L CTC CHAN 2 (DC) circuit breaker.
 - 5) Open the P110 panel L ECV/FCV POS SNSR circuit breaker.
 - 6) Gain access to the FCV by opening ECS access door 195EL.
 - C. For right pack inoperative FCV:
 - 1) Open the P210 panel R CTC CHAN 1 (AC) circuit breaker.
 - 2) Open the P210 panel R CTC CHAN 1 (DC) circuit breaker.
 - 3) Open the P210 panel R CTC CHAN 2 (AC) circuit breaker.
 - 4) Open the P210 panel R CTC CHAN 2 (DC) circuit breaker.
 - 5) Open the P210 panel R ECV/FCV POS SNSR circuit breaker.
 - 6) Gain access to the FCV by opening ECS access door 196ER.
 - D. Lock the inoperative lower FCV closed.
 - 1) Remove the locking plug from the FCV actuator boss.
 - 2) Turn the manual wrench until the FCV lock indicates CL.
 - 3) Install the locking plug in the manual lock and tighten according to the instructions and caution statement noted in AMM 21-00-00-040-849.
 - E. Close the ECS access door.
 - F. Close the associated circuit breakers.
2. Confirm the associated upper FCV operates normally.
 - A. Pressurize the pneumatic system (AMM 36-00-00/201).
 - B. Position Air Conditioning Panel (P5) associated PACK switch to AUTO.
 - C. For airplanes without AIMS V14 or later installed, confirm the associated PACK L or R advisory and status messages are not displayed.

NOTE: If the PACK L or R advisory and status messages are displayed, the upper FCV was inoperative. Unlock the lower FCV and dispatch using MMEL item 21-51-02-01.
 - D. For airplanes with AIMS V14 or later installed, confirm the associated PACK L or R advisory and status and PACK FCV UPR L or R status message are not displayed.



NOTE: If the PACK L or R advisory and status messages and the PACK FCV UPR L or R status message are displayed, the upper FCV was inoperative. Unlock the lower FCV and dispatch using MMEL item 21-51-02-01.

- E. Depressurize the pneumatic system (AMM 36-00-00/201).

OPERATIONS NOTE

For APU-to-Pack takeoff function installed, if left lower FCV is inoperative the APU-to-Pack takeoff Supplementary Procedure is not allowed.

**21-51-03 Flight Deck Flow Regulating Valve (777F)**

Interval	Installed	Required	Procedure
C	1	0	(M)

May be inoperative provided valve is deactivated open.

MAINTENANCE (M)

Deactivate the flight deck flow regulating valve open (AMM 21-00-00/901).

1. Open P110 panel L CTC CHAN 1 (AC) circuit breaker.
2. Open P110 panel L CTC CHAN 1 (DC) circuit breaker.
3. Open P110 panel L CTC CHAN 2 (AC) circuit breaker.
4. Open P110 panel L CTC CHAN 2 (DC) circuit breaker.
5. Gain access to the flight deck flow regulating valve.
6. Disconnect, cap and stow the electrical connector from valve.
7. Manually position the flight deck flow regulating valve open.
8. Close P110 panel L CTC CHAN 1 (AC) circuit breaker.
9. Close P110 panel L CTC CHAN 1 (DC) circuit breaker.
10. Close P110 panel L CTC CHAN 2 (AC) circuit breaker.
11. Close P110 panel L CTC CHAN 2 (DC) circuit breaker.

21-52-01 Air Cycle Machines (ACM)

Interval	Installed	Required	Procedure
C	2	1	(M) (O)

One may be inoperative provided:

- Associated second stage turbine bypass valve is locked open.
- Ram air inlet and exit doors operate normally or are secured open.
- Opposite pack operates normally.
- Both engine bleed systems operate normally.
- Appropriate performance adjustments are applied.

MAINTENANCE (M)

NOTE 1: For a pack to be operating normally, certain items associated with that system must be operative. See DDG item 21-51-01 for a listing of those items.

NOTE 2: For an engine bleed system to be operating normally, certain items associated with that system must be operative. See DDG item 36-00-01 for a listing of those items.

Lock the associated second stage turbine bypass valve open (AMM 21-00-00/901).

- Depressurize the pneumatic system (AMM 36-00-00/201).
- Gain access to the inoperative second stage turbine bypass valve, located inboard of the pack low limit valve, using the ECS access door 195EL or 196ER.
- Disconnect, cap, and stow the electrical connector from the inoperative valve.
- Turn the manual open/close knob until the valve position indicator shows the valve is open (O).
- Close the ECS access door.

OPERATIONS (O)

NOTE 1: The pack with the inoperative ACM may remain off on the ground and at low altitudes with the PACK MODE advisory message displayed and PACK OFF light illuminated. The pack will operate in standby cooling mode when appropriate altitude and OAT conditions exist with the PACK OFF light extinguished.

NOTE 2: If the opposite pack fails inflight, the pack operating in standby cooling mode will continue to operate regardless of altitude or OAT to maintain cabin pressurization. A pack operating in standby cooling mode is capable of maintaining cabin pressurization at cruise altitude.



For Lower Lobe Attendant Rest (LLAR), Overhead Flight Crew Rest (OFCR), Overhead Flight Attendant Rest (OFAR), Main Deck Crew Rest (MDCR) or Main Deck Flight Crew Rest (MDFCR) installed:

NOTE 3: For left pack operating in standby cooling mode, ventilation and temperature control in the associated crew rest may be degraded.

For Forward Cargo Air Conditioning (FCAC) installed:

NOTE 4: FCAC performance will be degraded. Temperature sensitive cargo should not be carried in the forward cargo compartment.

For Aft Cargo Air Conditioning (ACAC) installed:

NOTE 5: ACAC performance will be degraded. Temperature sensitive cargo should not be carried in the aft cargo compartment.

NOTE 6: For APU-to-Pack takeoff function installed, if Left ACM is inoperative the APU-to-Pack takeoff Supplementary Procedure is not allowed.

1. Reduce performance limited weights by the appropriate adjustments:

NOTE: The takeoff performance limited weight reductions may be avoided by performing a Packs Off Takeoff or for a right ACM inoperative, performing an APU-to-Pack Takeoff (if installed) Supplementary Procedure.

Performance Limited Weights	Takeoff & Landing	Enroute Climb
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For -200 or -200ER/GE90-76B thru -90B:

thru 632,500 lb (286,902 kg)	14,300 lb (6,486 kg)	No penalty
over 632,500 lb (286,902 kg)	20,700 lb (9,389 kg)	No penalty

For -200 or -200ER/GE90-94B:

thru 632,500 lb (286,902 kg)	14,300 lb (6,486 kg)	No penalty
over 632,500 lb (286,902 kg)	22,400 lb (10,160 kg)	No penalty

For -200 or -200ER/PW4074 thru PW4084:

All	21,800 lb (9,888 kg)	No penalty
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For -200 or -200ER/PW4090:

thru 632,500 lb (286,902 kg)	21,500 lb (9,752 kg)	No penalty
over 632,500 lb (286,902 kg)	26,500 lb (12,020 kg)	No penalty

**For -200 or -200ER/RR875 thru RR892:**

thru 632,500 lb (286,902 kg)	6,400 lb (2,903 kg)	No penalty
over 632,500 lb (286,902 kg)	8,200 lb (3,720 kg)	No penalty

For -200 or -200ER/RR895:

All	6,800 lb (3,084 kg)	No penalty
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For -200LR or 777F/GE90-100 Series:

All	14,800 lb (6,713 kg)	No penalty
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For -300/PW4090:

All	21,800 lb (9,888 kg)	No penalty
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For -300/RR892:

All	7,000 lb (3,175 kg)	No penalty
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For -300ER/GE90-115B:

All	11,600 lb (5,262 kg)	No penalty
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2. Operate the wing anti-ice manually.
 - A. Position WING ANTI-ICE selector OFF.
 - B. For flaps up and wing anti-ice required (do not use during takeoff and landing):
 - 1) Position the ENG Bleed switch associated with the inoperative ACM OFF.

NOTE: The ENG IDLE DISAGREE advisory message will be displayed when idle thrust is selected inflight.
 - 2) Position WING ANTI-ICE selector ON, then OFF as required.
 - 3) Prior to landing, position the ENG Bleed switch associated with the inoperative ACM ON.

**21-52-02 Economy Cooling Valves (ECV)****21-52-02A One Valve Inoperative**

Interval	Installed	Required	Procedure
C	2	1	(O)

One may be inoperative provided:

- Opposite pack operates normally.
- Both engine bleed systems operate normally.
- Appropriate performance adjustments are applied.

MAINTENANCE NOTE

NOTE 1: For a pack to be operating normally, certain items associated with that system must be operative. See DDG item 21-51-01 for a listing of those items.

NOTE 2: For an engine bleed system to be operating normally, certain items associated with that system must be operative. See DDG item 36-00-01 for a listing of those items.

OPERATIONS (O)

NOTE 1: The pack with the inoperative ECV may remain off on the ground and at low altitudes with the PACK MODE advisory message displayed and PACK OFF light illuminated. The pack will operate in standby cooling mode when appropriate altitude and OAT conditions exist with the PACK OFF light extinguished.

NOTE 2: If the opposite pack fails inflight, the pack operating in standby cooling mode will continue to operate, regardless of altitude or OAT, to maintain cabin pressurization. A pack in standby cooling mode is capable of maintaining cabin pressurization at cruise altitude.

For Lower Lobe Attendant Rest (LLAR), Overhead Flight Crew Rest (OFCR), Overhead Flight Attendant Rest (OFAR), Main Deck Crew Rest (MDCR) or Main Deck Flight Crew Rest (MDFCR) installed:

NOTE 3: For left pack operating in standby cooling mode, ventilation and temperature control in the associated crew rest may be degraded.

For Forward Cargo Air Conditioning (FCAC) installed:

NOTE 4: FCAC performance will be degraded. Temperature sensitive cargo should not be carried in the forward cargo compartment.

For Aft Cargo Air Conditioning (ACAC) installed:

NOTE 5: ACAC performance will be degraded. Temperature sensitive cargo should not be carried in the aft cargo compartment.



NOTE 6: For APU-to-Pack takeoff function installed, if Left ECV is inoperative the APU-to-Pack takeoff Supplementary Procedure is not allowed.

1. Reduce performance limited weights by the appropriate adjustments:

NOTE: The takeoff performance limited weight reductions may be avoided by performing a Packs Off Takeoff or for a right ECV inoperative, performing an APU-to-Pack Takeoff (if installed) Supplementary Procedure.

Performance Limited Weights	Takeoff & Landing	Enroute Climb
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For -200 or -200ER/GE90-76B thru -90B:

thru 632,500 lb (286,902 kg)	14,300 lb (6,486 kg)	No penalty
over 632,500 lb (286,902 kg)	20,700 lb (9,389 kg)	No penalty

For -200 or -200ER/GE90-94B:

thru 632,500 lb (286,902 kg)	14,300 lb (6,486 kg)	No penalty
over 632,500 lb (286,902 kg)	22,400 lb (10,160 kg)	No penalty

For -200 or -200ER/PW4074 thru PW4084:

All	21,800 lb (9,888 kg)	No penalty
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For -200 or -200ER/PW4090:

thru 632,500 lb (286,902 kg)	21,500 lb (9,752 kg)	No penalty
over 632,500 lb (286,902 kg)	26,500 lb (12,020 kg)	No penalty

For -200 or -200ER/RR875 thru RR892:

thru 632,500 lb (286,902 kg)	6,400 lb (2,903 kg)	No penalty
over 632,500 lb (286,902 kg)	8,200 lb (3,720 kg)	No penalty

For -200 or -200ER/RR895:

All	6,800 lb (3,084 kg)	No penalty
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For -200LR or 777F/GE90-100 Series:

All	14,800 lb (6,713 kg)	No penalty
-----	----------------------	------------

For -300/PW4090:

All	21,800 lb (9,888 kg)	No penalty
-----	----------------------	------------

For -300/RR892:

All	7,000 lb (3,175 kg)	No penalty
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For -300ER/GE90-115B:

All	11,600 lb (5,262 kg)	No penalty
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2. Operate the wing anti-ice manually.
 - A. Position WING ANTI-ICE selector OFF.
 - B. For flaps up and wing anti-ice required (do not use during takeoff and landing):
 - 1) Position the ENG Bleed switch associated with the inoperative ECV OFF.

NOTE: The ENG IDLE DISAGREE advisory message will be displayed when idle thrust is selected inflight.
 - 2) Position WING ANTI-ICE selector ON, then OFF as required.
 - 3) Prior to landing, position the ENG Bleed switch associated with the inoperative ECV ON.

21-52-02 Economy Cooling Valves (ECV)**21-52-02B One or Both Valves Inoperative Deactivated Closed**

Interval	Installed	Required	Procedure
C	2	0	(M)

May be inoperative provided the inoperative valve(s) is locked closed.

MAINTENANCE (M)

Lock the inoperative ECV closed (AMM 21-00-00/901).

1. Depressurize the pneumatic system (AMM 36-00-00/201).
2. Open P110 panel L ECV/FCV POS SNSR circuit breaker.
3. Open P210 panel R ECV/FCV POS SNSR circuit breaker.
4. Gain access to the inoperative ECV using ECS access door 195EL or 196ER.
5. Lock the ECV closed (AMM 21-00-00-040-851).
6. Close the ECS access door.
7. Close P110 panel L ECV/FCV POS SNSR circuit breaker
8. Close P210 panel R ECV/FCV POS SNSR circuit breaker.

**21-52-03 Compressor Discharge Temperature Sensors**

Interval	Installed	Required	Procedure
C	6	2	(M)

May be inoperative provided one sensor per pack is verified to operate normally.

MAINTENANCE (M)

Verify one sensor for each pack operates normally (AMM 21-00-00/901).

1. On the Air Conditioning Maintenance page, confirm a CPRSR OUT TEMP indication is displayed for both packs.

**21-52-04 Pack Discharge Temperature Sensors**

Interval	Installed	Required	Procedure
C	6	2	(M)

May be inoperative provided one sensor per pack is verified to operate normally.

MAINTENANCE (M)

Verify one sensor for each pack operates normally (AMM 21-00-00/901).

1. On the Air Conditioning Maintenance page, confirm a PACK OUT TEMP indication is displayed for both packs.

21-52-05 Ram Air Inlet Door Systems
21-52-05A Associated Pack Used

Interval	Installed	Required	Procedure
C	2	1	(M) (O)

One may be inoperative provided:

- a. Associated door is secured full open.
- b. Appropriate performance adjustments are applied.

NOTE: To avoid cold flight deck temperatures on passenger airplanes, the left Trim Air Control System and left Trim Air PRSOV should operate normally when the left Ram Air Inlet Door is secured full open.

MAINTENANCE (M)

Secure the inoperative ram air inlet door full open (AMM 21-00-00/901).

1. For left ram air inlet door inoperative:
 - A. Open P110 panel L CTC CHAN 1 (AC) circuit breaker.
 - B. Open P110 panel L CTC CHAN 1 (DC) circuit breaker.
 - C. Open P110 panel L CTC CHAN 2 (AC) circuit breaker.
 - D. Open P110 panel L CTC CHAN 2 (DC) circuit breaker.
 - E. Open the ram air inlet actuator door 191LL.
2. For right ram air inlet door inoperative:
 - A. Open P210 panel R CTC CHAN 1 (AC) circuit breaker.
 - B. Open P210 panel R CTC CHAN 1 (DC) circuit breaker.
 - C. Open P210 panel R CTC CHAN 2 (AC) circuit breaker.
 - D. Open P210 panel R CTC CHAN 2 (DC) circuit breaker.
 - E. Open the ram air inlet actuator door 192LR.
3. Disconnect, cap, and stow the electrical connector from the actuator.
4. Disconnect the actuator rod end from the bellcrank.
5. Loosen the screws on the fixed link, and rotate the fixed link 180 degrees.
6. Position the inlet door to the fully open position.
7. Connect the fixed link to the bellcrank.
8. Tighten the screws on the fixed link to 18-28 pound-inches.
9. Close the ram air inlet actuator access door.
10. Close the associated circuit breakers.

OPERATIONS (O)

1. Reduce performance limited weights by the appropriate adjustments.

Performance Limited Weights	Takeoff & Landing	Enroute Climb
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**For -200 or -200ER/GE90-76B thru -90B:**

thru 632,500 lb (286,902 kg)	400 lb (181 kg)	2,100 lb (953 kg)
over 632,500 lb (286,902 kg)	500 lb (227 kg)	2,200 lb (998 kg)

For -200 or -200ER/GE90-94B:

thru 632,500 lb (286,902 kg)	400 lb (181 kg)	2,600 lb (1,179 kg)
over 632,500 lb (286,902 kg)	700 lb (317 kg)	2,600 lb (1,179 kg)

For -200 or -200ER/PW4074 thru PW4084:

All	400 lb (181 kg)	2,300 lb (1,043 kg)
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For -200 or -200ER/PW4090:

thru 632,500 lb (286,902 kg)	400 lb (181 kg)	2,300 lb (1,043 kg)
over 632,500 lb (286,902 kg)	500 lb (227 kg)	2,300 lb (1,043 kg)

For -200 or -200ER/RR875 thru RR895:

thru 632,500 lb (286,902 kg)	400 lb (181 kg)	2,200 lb (998 kg)
over 632,500 lb (286,902 kg)	600 lb (272 kg)	2,200 lb (998 kg)

For -200LR or 777F/GE90-100 Series:

All	500 lb (227 kg)	2,300 lb (1,043 kg)
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For -300/PW4090:

All	400 lb (181 kg)	2,300 lb (1,043 kg)
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For -300/RR892:

All	400 lb (181 kg)	2,100 lb (953 kg)
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For -300ER/GE90-115B:

All	400 lb (181 kg)	2,300 lb (1,043 kg)
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NOTE: The enroute climb weight penalties are based on single engine operating speeds that approximate maximum lift-to-drag ratio speed. To account for the difference in level off altitude when operating at other speeds, multiply the enroute climb weight penalty listed above by the appropriate factor listed on page 3.00-01-00.2 in the ENROUTE DIVERSION SPEED EFFECTS paragraph.

2. Increase flight planning fuel by 0.7%

21-52-05 Ram Air Inlet Door Systems
21-52-05B Associated Pack Not Used

Interval	Installed	Required	Procedure
C	2	1	(M) (O)

One may be inoperative provided:

- a. Associated inlet door is verified closed.
- b. Associated pack is not used.

MAINTENANCE (M)

Verify the ram air inlet door is closed for dispatch.

1. On the AIR CONDITIONING maintenance page, verify that the associated RAM AIR INLET indicates 0.00.

NOTE: If the ram air inlet door can not be closed for dispatch, install a ram air inlet door cover plate (AMM 21-00-00/901)

OPERATIONS (O)

Position associated pack switch OFF.

NOTE: The airplane must also be dispatched using MMEL item 21-51-01 Operations procedure.

21-52-06 Ram Air Exhaust Door Systems

21-52-06-01 -200/-200ER/-300/-300ER
21-52-06-01A Associated Pack(s) Used

Interval	Installed	Required	Procedure
C	2	0	(M)

May be inoperative provided associated door is secured full open.

MAINTENANCE (M)

Secure the inoperative ram air exhaust door full open (AMM 21-00-00/901).

1. For left ram air exhaust door inoperative:
 - A. Open P110 panel L CTC CHAN 1 (AC) circuit breaker.
 - B. Open P110 panel L CTC CHAN 1 (DC) circuit breaker.
 - C. Open P110 panel L CTC CHAN 2 (AC) circuit breaker.
 - D. Open P110 panel L CTC CHAN 2 (DC) circuit breaker.
 - E. Open the ram air exhaust blowout door 195QL.
2. For right ram air exhaust door inoperative:
 - A. Open P210 panel R CTC CHAN 1 (AC) circuit breaker.
 - B. Open P210 panel R CTC CHAN 1 (DC) circuit breaker.
 - C. Open P210 panel R CTC CHAN 2 (AC) circuit breaker.
 - D. Open P210 panel R CTC CHAN 2 (DC) circuit breaker.
 - E. Open the ram air exhaust blowout door 196QR.
3. Disconnect, cap, and stow the electrical connector from the actuator.
4. Disconnect the actuator rod end from the actuator arm assembly.
5. Loosen the screws on the fixed link, and rotate the fixed link 180 degrees.
6. Position the ram air exhaust door to the fully open position.
7. Connect the fixed link to the actuator arm assembly.
8. Tighten the screws on the fixed link to 18-28 pound-inches.
9. Close the ram air exhaust blowout door.
10. Close the associated circuit breakers.

21-52-06 Ram Air Exhaust Door Systems

21-52-06-01 -200/-200ER/-300/-300ER
21-52-06-01B Associated Pack Not Used

Interval	Installed	Required	Procedure
C	2	1	(O)

One may be inoperative provided associated pack is not used.

OPERATIONS (O)

Position associated pack switch OFF.

NOTE: The airplane must also be dispatched using MMEL item 21-51-01 Operations procedure.

21-52-06 Ram Air Exhaust Door Systems

21-52-06-02 -200LR and 777F
21-52-06-02A Associated Pack(s) Used

Interval	Installed	Required	Procedure
C	2	0	(M) (O)

May be inoperative provided:

- a. Associated door is secured full open.
- b. Appropriate performance adjustments are applied.

MAINTENANCE (M)

Secure the inoperative ram air exhaust door full open (AMM 21-00-00/901).

1. For left ram air exhaust door inoperative:
 - A. Open P110 panel L CTC CHAN 1 (AC) circuit breaker.
 - B. Open P110 panel L CTC CHAN 1 (DC) circuit breaker.
 - C. Open P110 panel L CTC CHAN 2 (AC) circuit breaker.
 - D. Open P110 panel L CTC CHAN 2 (DC) circuit breaker.
 - E. Open the ram air exhaust blowout door 195QL.
2. For right ram air exhaust door inoperative:
 - A. Open P210 panel R CTC CHAN 1 (AC) circuit breaker.
 - B. Open P210 panel R CTC CHAN 1 (DC) circuit breaker.
 - C. Open P210 panel R CTC CHAN 2 (AC) circuit breaker.
 - D. Open P210 panel R CTC CHAN 2 (DC) circuit breaker.



E. Open the ram air exhaust blowout door 196QR.

3. Disconnect, cap, and stow the electrical connector from the actuator.
4. Disconnect the actuator rod end from the actuator arm assembly.
5. Loosen the screws on the fixed link, and rotate the fixed link 180 degrees.
6. Position the ram air exhaust door to the fully open position.
7. Connect the fixed link to the actuator arm assembly.
8. Tighten the screws on the fixed link to 18-28 pound-inches.
9. Close the ram air exhaust blowout door.
10. Close the associated circuit breakers.

OPERATIONS (O)

Reduce performance limited weights by the appropriate adjustments for each inoperative open door.

Performance Limited Weights	Takeoff & Landing	Enroute Climb
All	200 lb (91 kg)	450 lb (205 kg)

NOTE: The enroute climb weight penalties listed above are based on single engine operating speeds that approximate maximum lift-to-drag ratio speed. To account for the difference in level off altitude when operating at other speeds, multiply the enroute climb weight penalty listed above by the appropriate factor listed on page 3.00-01-00.2 in the ENROUTE DIVERSION SPEED EFFECTS paragraph.

21-52-06 Ram Air Exhaust Door Systems

21-52-06-02 -200LR and 777F

21-52-06-02B Associated Pack Not Used

Interval	Installed	Required	Procedure
C	2	1	(O)

One may be inoperative provided associated pack is not used.

OPERATIONS (O)

Position associated pack switch OFF.

NOTE: The airplane must also be dispatched using MMEL item 21-51-01 Operations procedure.

21-52-07 Condenser Inlet Temperature Control Systems

Interval	Installed	Required	Procedure
C	2	1	(M) (O)

One may be inoperative provided:

- Associated condenser low limit valve is locked open.
- Opposite pack operates normally.
- Both engine bleed systems operate normally.
- Appropriate performance adjustments are applied.

MAINTENANCE (M)

NOTE 1: For a pack to be operating normally, certain items associated with that system must be operative. See DDG item 21-51-01 for a listing of those items.

NOTE 2: For an engine bleed system to be operating normally, certain items associated with that system must be operative. See DDG item 36-00-01 for a listing of those items.

Lock the associated condenser low limit valve open (AMM 21-00-00/901).

NOTE: Locking the low limit valve open may result in display of the associated ECON COOL VALVE L or R status message. No additional maintenance action is required for the ECON COOL VALVE L or R status message.

- Depressurize the pneumatic system (AMM 36-00-00/201).
- Gain access to the associated condenser low limit valve, located outboard of the pack second stage turbine bypass valve, using ECS access door 195EL or 196ER.
- Disconnect, cap, and stow the electrical connectors from the condenser low limit valve.
- Turn the manual open/close knob until the valve position indicator shows the valve is open (O).
- Close the ECS access door.

OPERATIONS (O)

NOTE 1: The pack with the inoperative condenser inlet temperature control system may remain off on the ground and at low altitudes with the PACK MODE advisory message displayed and PACK OFF light illuminated. The pack will operate in standby cooling mode when appropriate altitude and OAT conditions exist with the PACK MODE advisory message and PACK OFF light extinguished.

NOTE 2: If the opposite pack fails inflight, the pack operating in standby cooling mode will continue to operate regardless of altitude or OAT to maintain cabin pressurization. A pack operating in standby cooling mode is capable of maintaining cabin pressurization at cruise altitude.

For Lower Lobe Attendant Rest (LLAR), Overhead Flight Crew Rest (OFCR), Overhead Flight Attendant Rest (OFAR), Main Deck Crew Rest (MDCR) or Main Deck Flight Crew Rest (MDFCR) installed:

NOTE 3: For left pack operating in standby cooling mode, ventilation and temperature control in the associated crew rest may be degraded.

For Forward Cargo Air Conditioning (FCAC) installed:

NOTE 4: FCAC performance will be degraded. Temperature sensitive cargo should not be carried in the forward cargo compartment.

For Aft Cargo Air Conditioning (ACAC) installed:

NOTE 5: ACAC performance will be degraded. Temperature sensitive cargo should not be carried in the aft cargo compartment.

NOTE 6: For APU-to-Pack takeoff function installed, if left condenser inlet temperature control is inoperative the APU-to-Pack takeoff Supplementary Procedure is not allowed.

1. Reduce performance limited weights by the appropriate adjustments:

NOTE: The takeoff performance limited weight reductions may be avoided by performing a Packs Off Takeoff or for a right condenser inlet temperature control inoperative, performing an APU-to-Pack Takeoff (if installed) Supplementary Procedure.

Performance Limited Weights	Takeoff & Landing	Enroute Climb
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For -200 or -200ER/GE90-76B thru -90B:

thru 632,500 lb (286,902 kg)	14,300 lb (6,486 kg)	No penalty
over 632,500 lb (286,902 kg)	20,700 lb (9,389 kg)	No penalty

For -200 or -200ER/GE90-94B:

thru 632,500 lb (286,902 kg)	14,300 lb (6,486 kg)	No penalty
over 632,500 lb (286,902 kg)	22,400 lb (10,160 kg)	No penalty

**For -200 or -200ER/PW4074 thru PW4084:**

All	21,800 lb (9,888 kg)	No penalty
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For -200 or -200ER/PW4090:

thru 632,500 lb (286,902 kg)	21,500 lb (9,752 kg)	No penalty
over 632,500 lb (286,902 kg)	26,500 lb (12,020 kg)	No penalty

For -200 or -200ER/RR875 thru RR892:

thru 632,500 lb (286,902 kg)	6,400 lb (2,903 kg)	No penalty
over 632,500 lb (286,902 kg)	8,200 lb (3,720 kg)	No penalty

For -200 or -200ER/RR895:

All	6,800 lb (3,084 kg)	No penalty
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For -200LR or 777F/GE90-100 Series:

All	14,800 lb (6,713 kg)	No penalty
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For -300/PW4090:

All	21,800 lb (9,888 kg)	No penalty
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For -300/RR892:

All	7,000 lb (3,175 kg)	No penalty
-----	---------------------	------------

For -300ER/GE90-115B:

All	11,600 lb (5,262 kg)	No penalty
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2. Operate the wing anti-ice manually.
 - A. Position WING ANTI-ICE selector OFF.
 - B. For flaps up and wing anti-ice required (do not use during takeoff and landing):
 - 1) Position the ENG Bleed switch associated with the inoperative condenser inlet temperature control system OFF.



NOTE: The ENG IDLE DISAGREE advisory message will be displayed when idle thrust is selected inflight.

- 2) Position WING ANTI-ICE selector ON, then OFF as required.
- 3) Prior to landing, position the ENG Bleed switch associated with the inoperative condenser inlet temperature control system ON.

**21-52-08 Lower Lobe Attendant Rest (LLAR) Duct Temperature Sensors**

21-52-08A One Sensor Inoperative

Interval	Installed	Required	Procedure
D	2	1	

21-52-08 Lower Lobe Attendant Rest (LLAR) Duct Temperature Sensors

21-52-08B Both Sensors Inoperative

Interval	Installed	Required	Procedure
C	2	0	

OPERATIONS NOTE

Temperature control will be degraded.



21-61-01 Cabin Temperature Controllers (CTC)
21-61-01-01 -200/-200ER (Without PRR 61980 or Production Equivalent Incorporated)

Interval	Installed	Required	Procedure
C	2	1	(M) (O)

One may be inoperative provided:

- Associated pack is not used.
- Opposite pack operates normally.
- Both engine bleed systems operate normally.
- Both outflow valves operate normally.
- Airplane remains at FL 350 or below.
- Flight remains within 60 minutes of landing at a suitable airport.
- Appropriate performance adjustments are applied.
- For the right CTC inoperative, electrical power on the right AC Transfer Bus is verified before each departure.

For FCAC installed:

- For right CTC inoperative, FCAC remains OFF.

NOTE: PRR 61980 installs enhancements for dispatch with a single pack, or dispatch with a single engine bleed system, on airplane L/N's 423, 429, 454, 455, 457 thru 461, and 463 and-on. For production equivalent information, see DDG item 21-51-01.

MAINTENANCE (M)

NOTE 1: For a pack to be operating normally, certain items associated with that system must be operative. See DDG item 21-51-01 for a listing of those items.

NOTE 2: For an engine bleed system to be operating normally, certain items associated with that system must be operative. See DDG item 36-00-01 for a listing of those items.

- For right CTC inoperative, before each departure, use the Electrical Maintenance page to verify the Right AC Transfer Bus is powered by confirming the R AC XFR bus status is ON (AMM 21-00-00/901).
- For APU pneumatic duct pressure seal P/N 93C10353-9 or earlier installed, inspect the APU pneumatic duct pressure seal (boot) per 777-SL-36-005.

NOTE: Airplanes L/N 212 and on were delivered with improved seal P/N 93C10353-13 installed and do not require this inspection.

For Forward Cargo Air Conditioning (FCAC) installed:

3. For right CTC inoperative, position Air Conditioning Panel (P5) FWD CARGO A/C control OFF.

OPERATIONS (O)

1. Position associated pack switch OFF. Use DDG Item 21-51-01-01 (O) procedure.
2. For APU-to-Pack takeoff function installed, the APU-to-Pack Takeoff Supplementary Procedure is not allowed.

For Forward Cargo Air Conditioning (FCAC) installed:

3. For right CTC inoperative, position FWD CARGO A/C control OFF.

NOTE: Forward cargo air conditioning is not available. Temperature sensitive cargo should not be carried in the forward cargo compartment.

21-61-01 Cabin Temperature Controllers (CTC)

21-61-01-02 -300 (Without PRR 61980 or Production Equivalent Incorporated)

Interval	Installed	Required	Procedure
C	2	1	(M) (O)

One may be inoperative provided:

- a. Associated pack is not used.
- b. Opposite pack operates normally.
- c. Both engine bleed systems operate normally.
- d. Both outflow valves operate normally.
- e. Procedures are established and used to verify forward cargo compartment remains empty or contains only ballast, empty cargo handling equipment (ballast may be loaded in ULDs), or fly away kits.
- f. Flight remains within 60 minutes of landing at a suitable airport.
- g. Appropriate performance adjustments are applied.
- h. For the right CTC inoperative, electrical power on the right AC Transfer Bus is verified before each departure.

For FCAC installed:

- i. For right CTC inoperative, FCAC remains OFF.

NOTE: Operator MELs must define which items are approved for inclusion in the fly away kits and which materials can be used as ballast.

NOTE: PRR 61980 installs enhancements for dispatch with a single pack, or dispatch with a single engine bleed system, on airplane L/N's 423, 429, 454, 455, 457 thru 461, and 463 and-on. For production equivalent information, see DDG item 21-51-01.

MAINTENANCE (M)

NOTE 1: For a pack to be operating normally, certain items associated with that system must be operative. See DDG item 21-51-01 for a listing of those items.

NOTE 2: For an engine bleed system to be operating normally, certain items associated with that system must be operative. See DDG item 36-00-01 for a listing of those items.

1. For right CTC inoperative, before each departure, use the Electrical Maintenance page to verify the Right AC Transfer Bus is powered by confirming the R AC XFR bus status is ON (AMM 21-00-00/901).
2. For APU pneumatic duct pressure seal P/N 93C10353-9 or earlier installed, inspect the APU pneumatic duct pressure seal (boot) per 777-SL-36-005.

NOTE: Airplanes L/N 212 and on were delivered with improved seal P/N 93C10353-13 installed and do not require this inspection.

For Forward Cargo Air Conditioning (FCAC) installed:

3. For right CTC inoperative, position Air Conditioning Panel (P5) FWD CARGO A/C control OFF.

OPERATIONS (O)

1. Position associated pack switch OFF. Use DDG Item 21-51-01-02 (O) procedure.
2. For APU-to-Pack takeoff function installed, the APU-to-Pack Takeoff Supplementary Procedure is not allowed.

For Forward Cargo Air Conditioning (FCAC) installed:

3. For right CTC inoperative, position FWD CARGO A/C control OFF.

NOTE: Forward cargo air conditioning is not available. Temperature sensitive cargo should not be carried in the forward cargo compartment.

21-61-01 Cabin Temperature Controllers (CTC)

21-61-01-03 -200/-200ER/-300 (With PRR 61980 or Production Equivalent Incorporated)

Interval	Installed	Required	Procedure
C	2	1	(M) (O)

One may be inoperative provided:

- a. Associated pack is not used.
- b. Opposite pack operates normally.
- c. Both engine bleed systems operate normally.
- d. Both outflow valves operate normally.

- e. Flight remains within 60 minutes of landing at a suitable airport.
- f. Appropriate performance adjustments are applied.
- g. For the right CTC inoperative, electrical power on the right AC Transfer Bus is verified before each departure.

For FCAC installed:

- h. For right CTC inoperative, FCAC remains OFF.
-

NOTE: PRR 61980 installs enhancements for dispatch with a single pack, or dispatch with a single engine bleed system, on airplane L/N's 423, 429, 454, 455, 457 thru 461, and 463 and-on. For production equivalent information, see DDG item 21-51-01.

MAINTENANCE (M)

NOTE 1: For a pack to be operating normally, certain items associated with that system must be operative. See DDG item 21-51-01 for a listing of those items.

NOTE 2: For an engine bleed system to be operating normally, certain items associated with that system must be operative. See DDG item 36-00-01 for a listing of those items.

- 1. For right CTC inoperative, before each departure, use the Electrical Maintenance page to verify the Right AC Transfer Bus is powered by confirming the R AC XFR bus status is ON (AMM 21-00-00/901).
- 2. For APU pneumatic duct pressure seal P/N 93C10353-9 or earlier installed, inspect the APU pneumatic duct pressure seal (boot) per 777-SL-36-005.

NOTE: Airplanes L/N 212 and on were delivered with improved seal P/N 93C10353-13 installed and do not require this inspection.

For Forward Cargo Air Conditioning (FCAC) installed:

- 3. For right CTC inoperative, position Air Conditioning Panel (P5) FWD CARGO A/C control OFF.

OPERATIONS (O)

- 1. Position associated pack switch OFF. Use DDG Item 21-51-01-03 (O) procedure.
- 2. For APU-to-Pack takeoff function installed, the APU-to-Pack Takeoff Supplementary Procedure is not allowed.

For Forward Cargo Air Conditioning (FCAC) installed:

- 3. For right CTC inoperative, position FWD CARGO A/C control OFF.

NOTE: Forward cargo air conditioning is not available. Temperature sensitive cargo should not be carried in the forward cargo compartment.

**21-61-01 Cabin Temperature Controllers (CTC)****21-61-01-04 -200LR/-300ER**

Interval	Installed	Required	Procedure
C	2	1	(M) (O)

One may be inoperative provided:

- Associated pack is not used.
- Opposite pack operates normally.
- Both engine bleed systems operate normally.
- Both outflow valves operate normally.
- Flight remains within 60 minutes of landing at a suitable airport.
- Appropriate performance adjustments are applied.
- For the right CTC inoperative, electrical power on the right AC Transfer Bus is verified before each departure.

For FCAC installed:

- For right CTC inoperative, FCAC remains OFF.

MAINTENANCE (M)

NOTE 1: For a pack to be operating normally, certain items associated with that system must be operative. See DDG item 21-51-01 for a listing of those items.

NOTE 2: For an engine bleed system to be operating normally, certain items associated with that system must be operative. See DDG item 36-00-01 for a listing of those items.

- For right CTC inoperative, before each departure, use the Electrical Maintenance page to verify the Right AC Transfer Bus is powered by confirming the R AC XFR bus status is ON (AMM 21-00-00/901).

For Forward Cargo Air Conditioning (FCAC) installed:

- For right CTC inoperative, position Air Conditioning Panel (P5) FWD CARGO A/C control OFF.

OPERATIONS (O)

- Position associated pack switch OFF. Use DDG Item 21-51-01-04 (O) procedure.
- For APU-to-Pack takeoff function installed, the APU-to-Pack Takeoff Supplementary Procedure is not allowed.

For Forward Cargo Air Conditioning (FCAC) installed:

- For right CTC inoperative, position FWD CARGO A/C control OFF.

NOTE: Forward cargo air conditioning is not available. Temperature sensitive cargo should not be carried in the forward cargo compartment.

21-61-01 Cabin Temperature Controllers (CTC)**21-61-01-05 777F****21-61-01-05-01 Without PRR 62273 or Production Equivalent Incorporated****21-61-01-05-01A AVS Operative**

Interval	Installed	Required	Procedure
C	2	1	(M) (O)

Right CTC may be inoperative provided:

- Right pack is not used.
- Left pack operates normally.
- Both engine bleed systems operate normally.
- Both outflow valves operate normally.
- Procedures are established and used to verify main deck cargo compartment remains empty or contains only ballast, empty cargo handling equipment (ballast may be loaded in ULDs), or fly away kits.
- Alternate ventilation system operates normally.
- Equipment cooling three-way valve is deactivated closed.
- FCAC remains OFF.
- Lower forward cargo flow regulating valve is deactivated closed.
- Electrical power on the right AC Transfer Bus is verified before each departure.
- Appropriate performance adjustments are applied.

For ACAC installed:

- Lower aft cargo flow regulating valve is deactivated closed.

NOTE: Operator MELs must define which items are approved for inclusion in the fly away kits and which materials can be used as ballast.

NOTE: PRR 62273 is applicable to 777F airplane line number 892 and-on. Service Bulletin 777-21-0148 provides the production equivalent of PRR 62273.

MAINTENANCE (M)

NOTE 1: For a pack to be operating normally, certain items associated with that system must be operative. See DDG item 21-51-01 for a listing of those items.

NOTE 2: For an engine bleed system to be operating normally, certain items associated with that system must be operative. See DDG item 36-00-01 for a listing of those items.

Deactivate the equipment cooling three-way valve closed, deactivate the lower forward cargo flow regulating valve closed, deactivate the lower aft cargo flow regulating valve closed (if installed), and before each departure verify the Right AC Transfer Bus is powered (AMM 21-00-00/901).

1. Deactivate the equipment cooling three-way valve closed.
 - A. Open and collar P110 panel 3-WAY VALVE circuit breaker.
 - B. Gain access to the equipment cooling three-way valve.
 - C. Manually position the equipment cooling three-way valve closed in EE-Mode (A-B valve position).
2. Deactivate the lower forward cargo flow regulating valve closed.

NOTE: The CGO A/C FRV LWR FWD status message will be displayed for a deactivated valve.

- A. Position Overhead Panel (P5) LWR CARGO TEMP FWD A/C control OFF.
 - B. Open P210 panel R CTC CHAN 1 (AC) circuit breaker.
 - C. Open P210 panel R CTC CHAN 1 (DC) circuit breaker.
 - D. Open P210 panel R CTC CHAN 2 (AC) circuit breaker.
 - E. Open P210 panel R CTC CHAN 2 (DC) circuit breaker.
 - F. Gain access to the lower forward cargo flow regulating valve.
 - G. Disconnect, cap, and stow the electrical connector from valve.
 - H. Manually position the lower forward cargo flow regulating valve closed.
 - I. Close P210 panel R CTC CHAN 1 (AC) circuit breaker.
 - J. Close P210 panel R CTC CHAN 1 (DC) circuit breaker.
 - K. Close P210 panel R CTC CHAN 2 (AC) circuit breaker.
 - L. Close P210 panel R CTC CHAN 2 (DC) circuit breaker.
3. Before each departure, use the Electrical Maintenance page to verify the Right AC Transfer Bus is powered by confirming the R AC XFR bus status is ON.

For Aft Cargo Air Conditioning (ACAC) installed:

4. Deactivate the lower aft cargo flow regulating valve closed.

NOTE: The CGO A/C FRV LWR AFT status message will be displayed for a deactivated valve.

- A. Position Overhead Panel (P5) LWR CARGO TEMP AFT A/C control OFF.
 - B. Open P210 panel R CTC CHAN 1 (AC) circuit breaker.
 - C. Open P210 panel R CTC CHAN 1 (DC) circuit breaker.
 - D. Open P210 panel R CTC CHAN 2 (AC) circuit breaker.

- E. Open P210 panel R CTC CHAN 2 (DC) circuit breaker.
- F. Gain access to the lower aft cargo flow regulating valve.
- G. Disconnect, cap, and stow the electrical connector from valve.
- H. Manually position the lower aft cargo flow regulating valve closed.
- I. Close P210 panel R CTC CHAN 1 (AC) circuit breaker.
- J. Close P210 panel R CTC CHAN 1 (DC) circuit breaker.
- K. Close P210 panel R CTC CHAN 2 (AC) circuit breaker.
- L. Close P210 panel R CTC CHAN 2 (DC) circuit breaker.

OPERATIONS (O)

1. Position right pack switch OFF. Use DDG Item 21-51-01-05A (O) procedure.
2. Main deck cargo compartment must remain empty, except for ballast, empty cargo containers (ballast may be loaded in ULDs), fly away kits, pallets, and cargo restraint components.
3. Position LWR CARGO TEMP FWD A/C control OFF.

NOTE: Lower forward cargo air conditioning is not available.
Temperature sensitive cargo should not be carried in the lower forward cargo compartment.

4. For APU-to-Pack takeoff function installed, the APU-to-Pack Takeoff Supplementary Procedure is not allowed.

For Aft Cargo Air Conditioning (ACAC) installed:

5. Position LWR CARGO TEMP AFT A/C control LOW, HIGH or OFF.

NOTE: Lower aft cargo heat is available, however lower aft cargo air conditioning is not available. Temperature sensitive cargo should not be carried in the lower aft cargo compartment.

21-61-01 Cabin Temperature Controllers (CTC)

21-61-01-05 777F

21-61-01-05-01 Without PRR 62273 or Production Equivalent Incorporated

21-61-01-05-01B AVS Inoperative

Interval	Installed	Required	Procedure
C	2	1	(M) (O)

Right CTC may be inoperative provided:

- a. Right pack is not used.
- b. Left pack operates normally.
- c. Both engine bleed systems operate normally.
- d. Both outflow valves operate normally.

- e. Procedures are established and used to verify main deck cargo compartment remains empty or contains only ballast, empty cargo handling equipment (ballast may be loaded in ULDs), or fly away kits.
- f. Flight remains within 60 minutes of landing at a suitable airport.
- g. Equipment cooling three-way valve is deactivated closed.
- h. FCAC remains OFF.
- i. Lower forward cargo flow regulating valve is deactivated closed.
- j. Electrical power on the right AC Transfer Bus is verified before each departure.
- k. Appropriate performance adjustments are applied.

For ACAC installed:

- l. Lower aft cargo flow regulating valve is deactivated closed.

NOTE: Operator MELs must define which items are approved for inclusion in the fly away kits and which materials can be used as ballast.

NOTE: PRR 62273 is applicable to 777F airplane line number 892 and-on. Service Bulletin 777-21-0148 provides the production equivalent of PRR 62273.

MAINTENANCE (M)

NOTE 1: For a pack to be operating normally, certain items associated with that system must be operative. See DDG item 21-51-01 for a listing of those items.

NOTE 2: For an engine bleed system to be operating normally, certain items associated with that system must be operative. See DDG item 36-00-01 for a listing of those items.

Deactivate the equipment cooling three-way valve closed, deactivate the lower forward cargo flow regulating valve closed, deactivate the lower aft cargo flow regulating valve closed (if installed), and before each departure verify the Right AC Transfer Bus is powered (AMM 21-00-00/901).

1. Deactivate the equipment cooling three-way valve closed.
 - A. Open and collar P110 panel 3-WAY VALVE circuit breaker.
 - B. Gain access to the equipment cooling three-way valve.
 - C. Manually position the equipment cooling three-way valve closed in EE-Mode (A-B valve position).
2. Deactivate the lower forward cargo flow regulating valve closed.

NOTE: The CGO A/C FRV LWR FWD status message will be displayed for a deactivated valve.

- A. Position Overhead Panel (P5) LWR CARGO TEMP FWD A/C control OFF.
- B. Open P210 panel R CTC CHAN 1 (AC) circuit breaker.

- C. Open P210 panel R CTC CHAN 1 (DC) circuit breaker.
 - D. Open P210 panel R CTC CHAN 2 (AC) circuit breaker.
 - E. Open P210 panel R CTC CHAN 2 (DC) circuit breaker.
 - F. Gain access to the lower forward cargo flow regulating valve.
 - G. Disconnect, cap, and stow the electrical connector from valve.
 - H. Manually position the lower forward cargo flow regulating valve closed.
 - I. Close P210 panel R CTC CHAN 1 (AC) circuit breaker.
 - J. Close P210 panel R CTC CHAN 1 (DC) circuit breaker.
 - K. Close P210 panel R CTC CHAN 2 (AC) circuit breaker.
 - L. Close P210 panel R CTC CHAN 2 (DC) circuit breaker.
3. Before each departure, use the Electrical Maintenance page to verify the Right AC Transfer Bus is powered by confirming the R AC XFR bus status is ON.

For Aft Cargo Air Conditioning (ACAC) installed:

4. Deactivate the lower aft cargo flow regulating valve closed
- NOTE: The CGO A/C FRV LWR AFT status message will be displayed for a deactivated valve.
- A. Position Overhead Panel (P5) LWR CARGO TEMP AFT A/C control OFF.
 - B. Open P210 panel R CTC CHAN 1 (AC) circuit breaker.
 - C. Open P210 panel R CTC CHAN 1 (DC) circuit breaker.
 - D. Open P210 panel R CTC CHAN 2 (AC) circuit breaker.
 - E. Open P210 panel R CTC CHAN 2 (DC) circuit breaker.
 - F. Gain access to the lower aft cargo flow regulating valve.
 - G. Disconnect, cap, and stow the electrical connector from valve.
 - H. Manually position the lower aft cargo flow regulating valve closed.
 - I. Close P210 panel R CTC CHAN 1 (AC) circuit breaker.
 - J. Close P210 panel R CTC CHAN 1 (DC) circuit breaker.
 - K. Close P210 panel R CTC CHAN 2 (AC) circuit breaker.
 - L. Close P210 panel R CTC CHAN 2 (DC) circuit breaker.

OPERATIONS (O)

- 1. Position right pack switch OFF. Use DDG Item 21-51-01-05B (O) procedure.
- 2. Main deck cargo compartment must remain empty, except for ballast, empty cargo containers (ballast may be loaded in ULDs), fly away kits, pallets, and cargo restraint components.
- 3. Position LWR CARGO TEMP FWD A/C control OFF.



NOTE: Lower forward cargo air conditioning is not available. Temperature sensitive cargo should not be carried in the lower forward cargo compartment.

4. For APU-to-Pack takeoff function installed, the APU-to-Pack Takeoff Supplementary Procedure is not allowed.

For Aft Cargo Air Conditioning (ACAC) installed:

5. Position LWR CARGO TEMP AFT A/C control LOW, HIGH or OFF.

NOTE: Lower aft cargo heat is available, however lower aft cargo air conditioning is not available. Temperature sensitive cargo should not be carried in the lower aft cargo compartment.

21-61-01 Cabin Temperature Controllers (CTC)

21-61-01-05 777F

21-61-01-05-02 With PRR 62273 or Production Equivalent Incorporated

21-61-01-05-02A AVS Operative

Interval	Installed	Required	Procedure
C	2	1	(M) (O)

Right CTC may be inoperative provided:

- Right pack is not used.
- Left pack operates normally.
- Both engine bleed systems operate normally.
- Both outflow valves operate normally.
- Alternate ventilation system operates normally.
- FCAC remains OFF.
- Lower forward cargo flow regulating valve is deactivated closed.
- Electrical power on the right AC Transfer Bus is verified before each departure.
- Appropriate performance adjustments are applied.

For ACAC installed:

- Lower aft cargo flow regulating valve is deactivated closed.

NOTE: PRR 62273 is applicable to 777F airplane line number 892 and-on. Service Bulletin 777-21-0148 provides the production equivalent of PRR 62273.

MAINTENANCE (M)

NOTE 1: For a pack to be operating normally, certain items associated with that system must be operative. See DDG item 21-51-01 for a listing of those items.

NOTE 2: For an engine bleed system to be operating normally, certain items associated with that system must be operative. See DDG item 36-00-01 for a listing of those items.

Deactivate the lower forward cargo flow regulating valve closed, deactivate the lower aft cargo flow regulating valve closed (if installed), and before each departure verify the Right AC Transfer Bus is powered (AMM 21-00-00/901).

1. Deactivate the lower forward cargo flow regulating valve closed.

NOTE: The CGO A/C FRV LWR FWD status message will be displayed for a deactivated valve.

- A. Position Overhead Panel (P5) LWR CARGO TEMP FWD A/C control OFF.
 - B. Open P210 panel R CTC CHAN 1 (AC) circuit breaker.
 - C. Open P210 panel R CTC CHAN 1 (DC) circuit breaker.
 - D. Open P210 panel R CTC CHAN 2 (AC) circuit breaker.
 - E. Open P210 panel R CTC CHAN 2 (DC) circuit breaker.
 - F. Gain access to the lower forward cargo flow regulating valve.
 - G. Disconnect, cap, and stow the electrical connector from valve.
 - H. Manually position the lower forward cargo flow regulating valve closed.
 - I. Close P210 panel R CTC CHAN 1 (AC) circuit breaker.
 - J. Close P210 panel R CTC CHAN 1 (DC) circuit breaker.
 - K. Close P210 panel R CTC CHAN 2 (AC) circuit breaker.
 - L. Close P210 panel R CTC CHAN 2 (DC) circuit breaker.
2. Before each departure, use the Electrical Maintenance page to verify the Right AC Transfer Bus is powered by confirming the R AC XFR bus status is ON.

For Aft Cargo Air Conditioning (ACAC) installed:

3. Deactivate the lower aft cargo flow regulating valve closed.

NOTE: The CGO A/C FRV LWR AFT status message will be displayed for a deactivated valve.

- A. Position Overhead Panel (P5) LWR CARGO TEMP AFT A/C control OFF.
- B. Open P210 panel R CTC CHAN 1 (AC) circuit breaker.
- C. Open P210 panel R CTC CHAN 1 (DC) circuit breaker.
- D. Open P210 panel R CTC CHAN 2 (AC) circuit breaker.
- E. Open P210 panel R CTC CHAN 2 (DC) circuit breaker.
- F. Gain access to the lower aft cargo flow regulating valve.
- G. Disconnect, cap, and stow the electrical connector from valve.
- H. Manually position the lower aft cargo flow regulating valve closed.
- I. Close P210 panel R CTC CHAN 1 (AC) circuit breaker.

- J. Close P210 panel R CTC CHAN 1 (DC) circuit breaker.
- K. Close P210 panel R CTC CHAN 2 (AC) circuit breaker.
- L. Close P210 panel R CTC CHAN 2 (DC) circuit breaker.

OPERATIONS (O)

1. Position right pack switch OFF. Use DDG Item 21-51-01-05A (O) procedure.
2. Position LWR CARGO TEMP FWD A/C control OFF.

NOTE: Lower forward cargo air conditioning is not available.
Temperature sensitive cargo should not be carried in the lower forward cargo compartment.
3. For APU-to-Pack takeoff function installed, the APU-to-Pack Takeoff Supplementary Procedure is not allowed.

For Aft Cargo Air Conditioning (ACAC) installed:

4. Position LWR CARGO TEMP AFT A/C control LOW, HIGH or OFF.

NOTE: Lower aft cargo heat is available, however lower aft cargo air conditioning is not available. Temperature sensitive cargo should not be carried in the lower aft cargo compartment.

21-61-01 Cabin Temperature Controllers (CTC)**21-61-01-05 777F****21-61-01-05-02 With PRR 62273 or Production Equivalent Incorporated****21-61-01-05-02B AVS Inoperative**

Interval	Installed	Required	Procedure
C	2	1	(M) (O)

Right CTC may be inoperative provided:

- a. Right pack is not used.
- b. Left pack operates normally.
- c. Both engine bleed systems operate normally.
- d. Both outflow valves operate normally.
- e. Flight remains within 60 minutes of landing at a suitable airport.
- f. FCAC remains OFF.
- g. Lower forward cargo flow regulating valve is deactivated closed.
- h. Electrical power on the right AC Transfer Bus is verified before each departure.
- i. Appropriate performance adjustments are applied.

For ACAC installed:

- j. Lower aft cargo flow regulating valve is deactivated closed.
-

NOTE: PRR 62273 is applicable to 777F airplane line number 892 and-on. Service Bulletin 777-21-0148 provides the production equivalent of PRR 62273.

MAINTENANCE (M)

NOTE 1: For a pack to be operating normally, certain items associated with that system must be operative. See DDG item 21-51-01 for a listing of those items.

NOTE 2: For an engine bleed system to be operating normally, certain items associated with that system must be operative. See DDG item 36-00-01 for a listing of those items.

Deactivate the lower forward cargo flow regulating valve closed, deactivate the lower aft cargo flow regulating valve closed (if installed), and before each departure verify the Right AC Transfer Bus is powered (AMM 21-00-00/901).

1. Deactivate the lower forward cargo flow regulating valve closed.

NOTE: The CGO A/C FRV LWR FWD status message will be displayed for a deactivated valve.

- A. Position Overhead Panel (P5) LWR CARGO TEMP FWD A/C control OFF.
 - B. Open P210 panel R CTC CHAN 1 (AC) circuit breaker.
 - C. Open P210 panel R CTC CHAN 1 (DC) circuit breaker.
 - D. Open P210 panel R CTC CHAN 2 (AC) circuit breaker.
 - E. Open P210 panel R CTC CHAN 2 (DC) circuit breaker.
 - F. Gain access to the lower forward cargo flow regulating valve.
 - G. Disconnect, cap, and stow the electrical connector from valve.
 - H. Manually position the lower forward cargo flow regulating valve closed.
 - I. Close P210 panel R CTC CHAN 1 (AC) circuit breaker.
 - J. Close P210 panel R CTC CHAN 1 (DC) circuit breaker.
 - K. Close P210 panel R CTC CHAN 2 (AC) circuit breaker.
 - L. Close P210 panel R CTC CHAN 2 (DC) circuit breaker.
2. Before each departure, use the Electrical Maintenance page to verify the Right AC Transfer Bus is powered by confirming the R AC XFR bus status is ON.

For Aft Cargo Air Conditioning (ACAC) installed:

3. Deactivate the lower aft cargo flow regulating valve closed

NOTE: The CGO A/C FRV LWR AFT status message will be displayed for a deactivated valve.

- A. Position Overhead Panel (P5) LWR CARGO TEMP AFT A/C control OFF.
- B. Open P210 panel R CTC CHAN 1 (AC) circuit breaker.
- C. Open P210 panel R CTC CHAN 1 (DC) circuit breaker.
- D. Open P210 panel R CTC CHAN 2 (AC) circuit breaker.
- E. Open P210 panel R CTC CHAN 2 (DC) circuit breaker.
- F. Gain access to the lower aft cargo flow regulating valve.
- G. Disconnect, cap, and stow the electrical connector from valve.
- H. Manually position the lower aft cargo flow regulating valve closed.
- I. Close P210 panel R CTC CHAN 1 (AC) circuit breaker.
- J. Close P210 panel R CTC CHAN 1 (DC) circuit breaker.
- K. Close P210 panel R CTC CHAN 2 (AC) circuit breaker.
- L. Close P210 panel R CTC CHAN 2 (DC) circuit breaker.

OPERATIONS (O)

1. Position right pack switch OFF. Use DDG Item 21-51-01-05B (O) procedure.
2. Position LWR CARGO TEMP FWD A/C control OFF.

NOTE: Lower forward cargo air conditioning is not available.
Temperature sensitive cargo should not be carried in the lower forward cargo compartment.

3. For APU-to-Pack takeoff function installed, the APU-to-Pack Takeoff Supplementary Procedure is not allowed.

For Aft Cargo Air Conditioning (ACAC) installed:

4. Position LWR CARGO TEMP AFT A/C control LOW, HIGH or OFF.

NOTE: Lower aft cargo heat is available, however lower aft cargo air conditioning is not available. Temperature sensitive cargo should not be carried in the lower aft cargo compartment.

21-61-01 Cabin Temperature Controllers (CTC)

21-61-01-05 777F

21-61-01-05-03 CTC Channels

21-61-01-05-03-01 Left CTC channel-1

Interval	Installed	Required	Procedure
C	1	0	

May be inoperative provided:

- a. Right CTC channel-1 operates normally.
- b. Left CTC channel-2 operates normally.



- c. Supernumerary zone trim air modulation valve is considered inoperative.
-

NOTE: The airplane must also be dispatched using MMEL item 21-61-20.

21-61-01 Cabin Temperature Controllers (CTC)

21-61-01-05 777F

21-61-01-05-03 CTC Channels

21-61-01-05-03-02 Right CTC channel-1

Interval	Installed	Required	Procedure
C	1	0	

May be inoperative provided:

- Left CTC channel-1 operates normally.
 - Right CTC channel-2 operates normally.
 - Equipment cooling trim air modulation valve is considered inoperative.
-

NOTE: The airplane must also be dispatched using MMEL item 21-61-18.

21-61-01 Cabin Temperature Controllers (CTC)

21-61-01-05 777F

21-61-01-05-03 CTC Channels

21-61-01-05-03-03 CTC channel-1

Interval	Installed	Required	Procedure
C	2	0	

May be inoperative provided:

- CTC channel-2 operates normally.
 - Supernumerary zone trim air modulation valve is considered inoperative.
 - Equipment cooling trim air modulation valve is considered inoperative.
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NOTE 1: The airplane must also be dispatched using MMEL item 21-61-20.

NOTE 2: The airplane must also be dispatched using MMEL item 21-61-18.

**21-61-01 Cabin Temperature Controllers (CTC)****21-61-01-05 777F****21-61-01-05-03 CTC Channels****21-61-01-05-03-04 CTC channel-2**

Interval	Installed	Required	Procedure
C	2	0	

May be inoperative provided left and right CTC channel-1 operate normally.

OPERATIONS NOTE

1. A MAIN DECK A/C advisory message displayed without a CABIN TEMP CTRL status message may indicate an inoperative channel-2 in the left CTC, or an inoperative channel-2 in the right CTC.
2. For channel-2 inoperative in the left CTC, main deck cargo air conditioning is inoperative forward of STA 1278. Temperature sensitive cargo should not be carried in the main deck cargo compartment forward of STA 1278.
3. For channel-2 inoperative in the right CTC, main deck cargo air conditioning is inoperative aft of STA 1278. Temperature sensitive cargo should not be carried in the main deck cargo compartment aft of STA 1278.

21-61-01 Cabin Temperature Controllers (CTC)**21-61-01-06 DELETED (All)**

Interval	Installed	Required	Procedure

Deleted relief in revision 23.

21-61-02 Flight Deck Zone Temperature Control System
21-61-02-01 Passenger

Interval	Installed	Required	Procedure
C	1	0	

May be inoperative provided left trim air switch remains off.

OPERATIONS NOTE

1. The L TRIM AIR switch must remain off.
 - A. Flight deck temperatures may be colder than normal or slow to respond to control position. To improve flight deck temperatures, position FLIGHT DECK TEMP control to AUTO Warm.
 - B. To minimize adjusting the cabin temperature selection, request flight attendants allow additional time for cabin temperature stabilization.
2. For APU-to-Pack takeoff function installed, the APU-to-Pack takeoff Supplementary Procedure is not allowed.

21-61-02 Flight Deck Zone Temperature Control System
21-61-02-01 Passenger
21-61-02-01-01 Auto Mode

Interval	Installed	Required	Procedure
C	1	0	(M)

May be inoperative provided MAN control mode is verified to operate normally.

MAINTENANCE (M)

Verify manual temperature control operates normally (AMM 21-00-00/901).

1. Position Air Conditioning Panel (P5) FLIGHT DECK TEMP control to the MAN C position.
2. On the ECS Maintenance page confirm F/D TRIM VLV indicates approximately 0.0 or on the Air Synoptic confirm the F/D TRIM indicator is full cold (C).
3. Position Air Conditioning Panel (P5) FLIGHT DECK TEMP control to the MAN W position.
4. On the ECS Maintenance page confirm F/D TRIM VLV indicates approximately 1.0 or on the Air Synoptic confirm the F/D TRIM indicator is full warm (W).
5. Position Air Conditioning Panel (P5) FLIGHT DECK TEMP control to the desired setting.

**21-61-02 Flight Deck Zone Temperature Control System****21-61-02-02 777F****21-61-02-02-01 AUTO Mode**

Interval	Installed	Required	Procedure
C	1	0	(M)

May be inoperative provided MAN control mode is verified to operate normally.

MAINTENANCE (M)

Verify manual temperature control operates normally (AMM 21-00-00/901).

1. Position Air Conditioning Panel (P5) FLIGHT DECK TEMP control to the MAN C position.
2. On the ECS Maintenance page confirm F/D TRIM VLV indicates approximately 0.0 or on the Air Synoptic confirm the F/D TRIM indicator is full cold (C).
3. Position Air Conditioning Panel (P5) FLIGHT DECK TEMP control to the MAN W position.
4. On the ECS Maintenance page confirm F/D TRIM VLV indicates approximately 1.0 or on the Air Synoptic confirm the F/D TRIM indicator is full warm (W).
5. Position Air Conditioning Panel (P5) FLIGHT DECK TEMP control to the desired setting.

21-61-03 Cabin Zone Temperature Control Systems (Passenger)
21-61-03A Valve Deactivation Required

Interval	Installed	Required	Procedure
C	6	0	(M)

May be inoperative provided associated zone trim air modulation valve is deactivated closed.

MAINTENANCE (M)

Deactivate the associated cabin zone trim air modulation valve closed (AMM 21-00-00/901).

1. For cabin zones A, C or E, deactivate the associated trim air modulation valve closed.
 - A. Open P110 panel L CTC CHAN 1 (AC) circuit breaker.
 - B. Open P110 panel L CTC CHAN 1 (DC) circuit breaker.
 - C. Open P110 panel L CTC CHAN 2 (AC) circuit breaker.
 - D. Open P110 panel L CTC CHAN 2 (DC) circuit breaker.
 - E. Open the ECS access door 195EL.
 - F. Disconnect, cap, and stow the electrical connector from the applicable zone trim air modulation valve.
 - G. Turn the associated manual open/close knob until the valve position indicator is in the closed (CL) position.
 - H. Close the ECS access door 195EL.
 - I. Close P110 panel L CTC CHAN 1 (AC) circuit breaker.
 - J. Close P110 panel L CTC CHAN 1 (DC) circuit breaker.
 - K. Close P110 panel L CTC CHAN 2 (AC) circuit breaker.
 - L. Close P110 panel L CTC CHAN 2 (DC) circuit breaker.
2. For cabin zones B, D or F, deactivate the associated trim air modulation valve closed.
 - A. Open P210 panel R CTC CHAN 1 (AC) circuit breaker.
 - B. Open P210 panel R CTC CHAN 1 (DC) circuit breaker.
 - C. Open P210 panel R CTC CHAN 2 (AC) circuit breaker.
 - D. Open P210 panel R CTC CHAN 2 (DC) circuit breaker.
 - E. Open the ECS access door 196ER.
 - F. Disconnect, cap, and stow the electrical connector from the applicable zone trim air modulation valve.
 - G. Turn the associated manual open/close knob until the valve position indicator is in the closed (CL) position.
 - H. Close the ECS access door 196ER.



- I. Close P210 panel R CTC CHAN 1 (AC) circuit breaker.
- J. Close P210 panel R CTC CHAN 1 (DC) circuit breaker.
- K. Close P210 panel R CTC CHAN 2 (AC) circuit breaker.
- L. Close P210 panel R CTC CHAN 2 (DC) circuit breaker.

21-61-03 Cabin Zone Temperature Control Systems (Passenger)

21-61-03B Trim Air Switch(es) Remains Off

Interval	Installed	Required	Procedure
C	6	0	(O)

May be inoperative with the associated zone trim air modulation valve in any position provided:

- a. Associated trim air switch remains off.

[For FCAC installed:](#)

- b. For right trim air switch selected off, FCAC remains OFF.

OPERATIONS (O)

Position associated TRIM AIR switch off.

1. For left trim air switch off, flight deck temperatures may be colder than normal, or slow to respond to control position. To improve flight deck temperatures, position FLIGHT DECK TEMP Control to AUTO Warm.
2. To minimize adjusting the cabin temperature selection, request flight attendants allow additional time for cabin temperature stabilization.

[For Forward Cargo Air Conditioning \(FCAC\) installed:](#)

3. For right trim air switch off, position FWD CARGO A/C control OFF.

NOTE: Forward cargo air conditioning is not available. Temperature sensitive cargo should not be carried in the forward cargo compartment.

21-61-04 Cabin Temperature Control (Passenger)

Interval	Installed	Required	Procedure
C	1	0	

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**21-61-05 FCAC Duct Temperature Sensing System**

21-61-05-01 Passenger

Interval	Installed	Required	Procedure
D	1	0	(M)

May be inoperative provided:

- Forward cargo zone trim air modulation valve is deactivated closed.
- FCAC remains OFF.

MAINTENANCE (M)

Deactivate the forward cargo zone trim air modulation valve closed (AMM 21-00-00/901).

NOTE: The CARGO TRIM VALVE FWD status message will be displayed for a deactivated valve.

- Position Overhead Panel (P5) FWD CARGO A/C control OFF.
- Deactivate the forward cargo zone trim air modulation valve closed.
 - Open P210 panel R CTC CHAN 1 (AC) circuit breaker.
 - Open P210 panel R CTC CHAN 1 (DC) circuit breaker.
 - Open P210 panel R CTC CHAN 2 (AC) circuit breaker.
 - Open P210 panel R CTC CHAN 2 (DC) circuit breaker.
 - Access the forward cargo zone trim air modulation valve.
 - Disconnect, cap, and stow the electrical connector from the forward cargo zone trim air modulation valve.
 - Turn the manual open/close knob until the position indicator is in the CLOSED position.
 - Close P210 panel R CTC CHAN 1 (AC) circuit breaker.
 - Close P210 panel R CTC CHAN 1 (DC) circuit breaker.
 - Close P210 panel R CTC CHAN 2 (AC) circuit breaker.
 - Close P210 panel R CTC CHAN 2 (DC) circuit breaker.

OPERATIONS NOTE

- Position FWD CARGO A/C control OFF.
- Forward cargo air conditioning is not available. Temperature sensitive cargo should not be carried in the forward cargo compartment.

21-61-05 FCAC Duct Temperature Sensing System

21-61-05-02 777F

Interval	Installed	Required	Procedure
C	1	0	(M)

May be inoperative provided:

- a. Lower forward cargo zone trim air modulation valve is deactivated closed.
- b. FCAC remains OFF.

MAINTENANCE (M)

Deactivate the lower forward cargo zone trim air modulating valve closed (AMM 21-00-00/901).

NOTE: The CGO TRM VLV LWR FWD status message will be displayed for a deactivated valve.

1. Deactivate the lower forward cargo zone trim air modulating valve closed.
 - A. Open P210 panel R CTC CHAN 1 (AC) circuit breaker.
 - B. Open P210 panel R CTC CHAN 1 (DC) circuit breaker.
 - C. Open P210 panel R CTC CHAN 2 (AC) circuit breaker.
 - D. Open P210 panel R CTC CHAN 2 (DC) circuit breaker.
 - E. Access the lower forward cargo zone trim air modulating valve.
 - F. Disconnect, cap, and stow the electrical connector from the lower forward cargo zone trim air modulating valve.
 - G. Turn the manual open/close knob until the position indicator is in the CLOSED position.
 - H. Close P210 panel R CTC CHAN 1 (AC) circuit breaker.
 - I. Close P210 panel R CTC CHAN 1 (DC) circuit breaker.
 - J. Close P210 panel R CTC CHAN 2 (AC) circuit breaker.
 - K. Close P210 panel R CTC CHAN 2 (DC) circuit breaker.

For Aft Cargo Air Conditioning (ACAC) not installed:

2. Position Overhead Panel (P5) FWD CARGO A/C control OFF.

For Aft Cargo Air Conditioning (ACAC) installed:

3. Position Overhead Panel (P5) LWR CARGO TEMP FWD A/C control OFF.

OPERATIONS NOTE

For Aft Cargo Air Conditioning (ACAC) not installed:

1. Position FWD CARGO A/C control OFF.
2. Lower forward cargo air conditioning is not available. Temperature sensitive cargo should not be carried in the lower forward cargo compartment.



For Aft Cargo Air Conditioning (ACAC) installed:

1. Position LWR CARGO TEMP FWD A/C control OFF.
2. Lower forward cargo air conditioning is not available. Temperature sensitive cargo should not be carried in the lower forward cargo compartment.

**21-61-06 FCAC Cargo Compartment Temperature Sensing System**

21-61-06-01 Passenger

Interval	Installed	Required	Procedure
D	1	0	

May be inoperative provided FCAC remains OFF.

OPERATIONS NOTE

1. Position FWD CARGO A/C control OFF.
2. Forward cargo air conditioning is not available. Temperature sensitive cargo should not be carried in the forward cargo compartment.

21-61-06 FCAC Cargo Compartment Temperature Sensing System

21-61-06-02 777F

Interval	Installed	Required	Procedure
C	1	0	

May be inoperative provided FCAC remains OFF.

OPERATIONS NOTE

For Aft Cargo Air Conditioning (ACAC) not installed:

1. Position FWD CARGO A/C control OFF.
2. Lower forward cargo air conditioning is not available. Temperature sensitive cargo should not be carried in the lower forward cargo compartment.

For Aft Cargo Air Conditioning (ACAC) installed:

1. Position LWR CARGO TEMP FWD A/C control OFF.
2. Lower forward cargo air conditioning is not available. Temperature sensitive cargo should not be carried in the lower forward cargo compartment.

**21-61-07 FCAC Cargo Zone Trim Air Modulation Valve**

21-61-07-01 Passenger

Interval	Installed	Required	Procedure
D	1	0	(M)

May be inoperative provided:

- Valve is deactivated closed.
- FCAC remains OFF.

MAINTENANCE (M)

Deactivate the forward cargo zone trim air modulation valve closed (AMM 21-00-00/901).

- Position Overhead Panel (P5) FWD CARGO A/C control OFF.
- Deactivate the forward cargo zone trim air modulation valve closed.
 - Open P210 panel R CTC CHAN 1 (AC) circuit breaker.
 - Open P210 panel R CTC CHAN 1 (DC) circuit breaker.
 - Open P210 panel R CTC CHAN 2 (AC) circuit breaker.
 - Open P210 panel R CTC CHAN 2 (DC) circuit breaker.
 - Access the forward cargo zone trim air modulation valve.
 - Disconnect, cap, and stow the electrical connector from the forward cargo zone trim air modulation valve.
 - Turn the manual open/close knob until the position indicator is in the CLOSED position.
 - Close P210 panel R CTC CHAN 1 (AC) circuit breaker.
 - Close P210 panel R CTC CHAN 1 (DC) circuit breaker.
 - Close P210 panel R CTC CHAN 2 (AC) circuit breaker.
 - Close P210 panel R CTC CHAN 2 (DC) circuit breaker.

OPERATIONS NOTE

- Position FWD CARGO A/C control OFF.
- Forward cargo air conditioning is not available. Temperature sensitive cargo should not be carried in the forward cargo compartment.

21-61-07 FCAC Cargo Zone Trim Air Modulation Valve

21-61-07-02 777F

Interval	Installed	Required	Procedure
C	1	0	(M)

May be inoperative provided:

- a. Valve is deactivated closed.
 - b. FCAC remains OFF.
-

MAINTENANCE (M)

Deactivate the lower forward cargo zone trim air modulation valve closed (AMM 21-00-00/901).

1. Deactivate the lower forward cargo zone trim air modulation valve closed.
 - A. Open P210 panel R CTC CHAN 1 (AC) circuit breaker.
 - B. Open P210 panel R CTC CHAN 1 (DC) circuit breaker.
 - C. Open P210 panel R CTC CHAN 2 (AC) circuit breaker.
 - D. Open P210 panel R CTC CHAN 2 (DC) circuit breaker.
 - E. Access the lower forward cargo zone trim air modulation valve.
 - F. Disconnect, cap, and stow the electrical connector from the valve.
 - G. Turn the manual open/close knob until the position indicator is in the CLOSED position.
 - H. Close P210 panel R CTC CHAN 1 (AC) circuit breaker.
 - I. Close P210 panel R CTC CHAN 1 (DC) circuit breaker.
 - J. Close P210 panel R CTC CHAN 2 (AC) circuit breaker.
 - K. Close P210 panel R CTC CHAN 2 (DC) circuit breaker.

For Aft Cargo Air Conditioning (ACAC) not installed:

2. Position Overhead Panel (P5) FWD CARGO A/C control OFF.

For Aft Cargo Air Conditioning (ACAC) installed:

3. Position Overhead Panel (P5) LWR CARGO TEMP FWD A/C control OFF.

OPERATIONS NOTE

For Aft Cargo Air Conditioning (ACAC) not installed:

1. Position FWD CARGO A/C control OFF.
2. Lower forward cargo air conditioning is not available. Temperature sensitive cargo should not be carried in the lower forward cargo compartment.

For Aft Cargo Air Conditioning (ACAC) installed:

1. Position LWR CARGO TEMP FWD A/C control OFF.
2. Lower forward cargo air conditioning is not available. Temperature sensitive cargo should not be carried in the lower forward cargo compartment.

**21-61-08 FCAC Cargo Temperature Control**

21-61-08-01 Passenger

Interval	Installed	Required	Procedure
D	1	0	(M)

May be inoperative provided:

- FCAC shutoff valve is deactivated closed.
- FCAC remains OFF.

MAINTENANCE (M)

Deactivate the FCAC shutoff valve closed (AMM 21-00-00/901).

NOTE: The CARGO A/C SOV FWD status message will be displayed for a deactivated valve.

- Position Overhead Panel (P5) FWD CARGO A/C control OFF.
- Open and collar P210 panel FWD CGO A/C SOV CTRL circuit breaker.
- Open and collar P210 panel FWD CGO A/C SOV circuit breaker.
- Gain access to the FCAC shutoff valve located aft of the forward cargo compartment in the right side of the mix bay area below the recirculation filters.
- Manually position the FCAC shutoff valve closed.

OPERATIONS NOTE

- Position FWD CARGO A/C control OFF.
- Forward cargo air conditioning is not available. Temperature sensitive cargo should not be carried in the forward cargo compartment.

21-61-08 FCAC Cargo Temperature Control

21-61-08-02 777F

Interval	Installed	Required	Procedure
C	1	0	(M)

May be inoperative provided:

- Lower forward cargo flow regulating valve is deactivated closed.
- FCAC remains OFF.

MAINTENANCE (M)

Deactivate the lower forward cargo flow regulating valve closed (AMM 21-00-00/901).

NOTE: The CGO A/C FRV LWR FWD status message will be displayed for a deactivated valve.

1. Open P210 panel R CTC CHAN 1 (AC) circuit breaker.
2. Open P210 panel R CTC CHAN 1 (DC) circuit breaker.
3. Open P210 panel R CTC CHAN 2 (AC) circuit breaker.
4. Open P210 panel R CTC CHAN 2 (DC) circuit breaker.
5. Gain access to the lower forward cargo flow regulating valve.
6. Disconnect, cap, and stow the electrical connector from valve.
7. Manually position the lower forward cargo flow regulating valve closed.
8. Close P210 panel R CTC CHAN 1 (AC) circuit breaker.
9. Close P210 panel R CTC CHAN 1 (DC) circuit breaker.
10. Close P210 panel R CTC CHAN 2 (AC) circuit breaker.
11. Close P210 panel R CTC CHAN 2 (DC) circuit breaker.

For Aft Cargo Air Conditioning (ACAC) not installed:

12. Position Overhead Panel (P5) FWD CARGO A/C control OFF.

For Aft Cargo Air Conditioning (ACAC) installed:

13. Position Overhead Panel (P5) LWR CARGO TEMP FWD A/C control OFF.

OPERATIONS NOTE

For Aft Cargo Air Conditioning (ACAC) not installed:

1. Position FWD CARGO A/C control OFF.
2. Lower forward cargo air conditioning is not available. Temperature sensitive cargo should not be carried in the lower forward cargo compartment.

For Aft Cargo Air Conditioning (ACAC) installed:

1. Position LWR CARGO TEMP FWD A/C control OFF.
2. Lower forward cargo air conditioning is not available. Temperature sensitive cargo should not be carried in the lower forward cargo compartment.

**21-61-09 ACAC Duct Temperature Sensing System (777F)**

Interval	Installed	Required	Procedure
D	1	0	(M)

May be inoperative provided:

- Lower aft cargo zone trim air modulation valve is deactivated closed.
- ACAC remains OFF.

MAINTENANCE (M)

Deactivate the lower aft cargo zone trim air modulating valve closed (AMM 21-00-00/901).

NOTE: The CGO TRM VLV LWR AFT status message will be displayed for a deactivated valve.

- Position Overhead Panel (P5) LWR CARGO TEMP AFT A/C control OFF.
- Deactivate the lower aft cargo zone trim air modulating valve closed.
 - Open P110 panel L CTC CHAN 1 (AC) circuit breaker.
 - Open P110 panel L CTC CHAN 1 (DC) circuit breaker.
 - Open P110 panel L CTC CHAN 2 (AC) circuit breaker.
 - Open P110 panel L CTC CHAN 2 (DC) circuit breaker.
 - Access the lower aft cargo zone trim air modulating valve.
 - Disconnect, cap, and stow the electrical connector from the valve.
 - Turn the manual open/close knob until the position indicator is in the CLOSED position.
 - Close P110 panel L CTC CHAN 1 (AC) circuit breaker.
 - Close P110 panel L CTC CHAN 1 (DC) circuit breaker.
 - Close P110 panel L CTC CHAN 2 (AC) circuit breaker.
 - Close P110 panel L CTC CHAN 2 (DC) circuit breaker.

OPERATIONS NOTE

- Position LWR CARGO TEMP AFT A/C control OFF.
- Lower aft cargo air conditioning is not available. Temperature sensitive cargo should not be carried in the lower aft cargo compartment.

**21-61-10 ACAC Cargo Compartment Temperature Sensing System (777F)**

Interval	Installed	Required	Procedure
D	1	0	

May be inoperative provided ACAC remains OFF.

OPERATIONS NOTE

1. Position LWR CARGO TEMP AFT A/C control OFF.
2. Lower aft cargo air conditioning is not available. Temperature sensitive cargo should not be carried in the lower aft cargo compartment.

**21-61-11 ACAC Cargo Zone Trim Air Modulation Valve (777F)**

Interval	Installed	Required	Procedure
D	1	0	(M)

May be inoperative provided:

- Valve is deactivated closed.
- ACAC remains OFF.

MAINTENANCE (M)

Deactivate the lower aft cargo zone trim air modulation valve closed (AMM 21-00-00/901).

- Position Overhead Panel (P5) LWR CARGO TEMP AFT A/C control OFF.
- Deactivate the lower aft cargo zone trim air modulation valve closed.
 - Open P110 panel L CTC CHAN 1 (AC) circuit breaker.
 - Open P110 panel L CTC CHAN 1 (DC) circuit breaker.
 - Open P110 panel L CTC CHAN 2 (AC) circuit breaker.
 - Open P110 panel L CTC CHAN 2 (DC) circuit breaker.
 - Access the lower aft cargo zone trim air modulation valve.
 - Disconnect, cap, and stow the electrical connector from the valve.
 - Turn the manual open/close knob until the position indicator is in the CLOSED position.
 - Close P110 panel L CTC CHAN 1 (AC) circuit breaker.
 - Close P110 panel L CTC CHAN 1 (DC) circuit breaker.
 - Close P110 panel L CTC CHAN 2 (AC) circuit breaker.
 - Close P110 panel L CTC CHAN 2 (DC) circuit breaker.

OPERATIONS NOTE

- Position LWR CARGO TEMP AFT A/C control OFF.
- Lower aft cargo air conditioning is not available. Temperature sensitive cargo should not be carried in the lower aft cargo compartment.

**21-61-12 ACAC Cargo Temperature Control (777F)**

Interval	Installed	Required	Procedure
D	1	0	(M)

May be inoperative provided:

- Lower aft cargo flow regulating valve is deactivated closed.
- ACAC remains OFF.

MAINTENANCE (M)

Deactivate the lower aft cargo flow regulating valve closed (AMM 21-00-00/901).

NOTE: The CGO A/C FRV LWR AFT status message will be displayed for a deactivated valve.

- Position Overhead Panel (P5) LWR CARGO TEMP AFT A/C control OFF.
- Open P210 panel R CTC CHAN 1 (AC) circuit breaker.
- Open P210 panel R CTC CHAN 1 (DC) circuit breaker.
- Open P210 panel R CTC CHAN 2 (AC) circuit breaker.
- Open P210 panel R CTC CHAN 2 (DC) circuit breaker.
- Gain access to the lower aft cargo flow regulating valve.
- Disconnect, cap, and stow the electrical connector from valve.
- Manually position the lower aft cargo flow regulating valve closed.
- Close P210 panel R CTC CHAN 1 (AC) circuit breaker.
- Close P210 panel R CTC CHAN 1 (DC) circuit breaker.
- Close P210 panel R CTC CHAN 2 (AC) circuit breaker.
- Close P210 panel R CTC CHAN 2 (DC) circuit breaker.

OPERATIONS NOTE

- Position LWR CARGO TEMP AFT A/C control OFF.
- Lower aft cargo air conditioning is not available. Temperature sensitive cargo should not be carried in the lower aft cargo compartment.

**21-61-13 MDAC Duct Temperature Sensing System (777F)**

Interval	Installed	Required	Procedure
C	2	0	(M)

May be inoperative provided associated main deck cargo zone trim air modulation valve is deactivated closed.

MAINTENANCE (M)

Deactivate the associated main deck cargo zone trim air modulation valve closed (AMM 21-00-00/901).

1. For an inoperative forward main deck cargo duct temperature sensing system:

NOTE: The CGO TRM VLV MAIN FWD status message will be displayed for a deactivated forward valve.

- A. Open P110 panel L CTC CHAN 2 (AC) circuit breaker.
- B. Open P110 panel L CTC CHAN 2 (DC) circuit breaker.
- C. Access the forward main deck cargo zone trim air modulation valve.
- D. Disconnect, cap, and stow the electrical connector from the valve.
- E. Turn the manual open/close knob until the position indicator is in the CLOSED position.
- F. Close P110 panel L CTC CHAN 2 (AC) circuit breaker.
- G. Close P110 panel L CTC CHAN 2 (DC) circuit breaker.

2. For an inoperative aft main deck cargo duct temperature sensing system:

NOTE: The CGO TRM VLV MAIN AFT status message will be displayed for a deactivated aft valve.

- A. Open P210 panel R CTC CHAN 2 (AC) circuit breaker.
- B. Open P210 panel R CTC CHAN 2 (DC) circuit breaker.
- C. Access the aft main deck cargo zone trim air modulation valve.
- D. Disconnect, cap, and stow the electrical connector from the valve.
- E. Turn the manual open/close knob until the position indicator is in the CLOSED position.
- F. Close P210 panel R CTC CHAN 2 (AC) circuit breaker.
- G. Close P210 panel R CTC CHAN 2 (DC) circuit breaker.

OPERATIONS NOTE

1. For an inoperative forward main deck cargo duct temperature sensing system, main deck cargo air conditioning is not available forward of STA 1278.
2. For an inoperative aft main deck cargo duct temperature sensing system, main deck cargo air conditioning is not available aft of STA 1278.



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3. Temperature sensitive cargo should not be carried in the associated main deck cargo compartment area.

**21-61-14 MDAC Cargo Compartment Temperature Sensing System (777F)**

Interval	Installed	Required	Procedure
C	2	0	

OPERATIONS NOTE

1. For an inoperative forward main deck cargo compartment temperature sensing system, main deck cargo air conditioning is not available forward of STA 1278.
2. For an inoperative aft main deck cargo compartment temperature sensing system, main deck cargo air conditioning is not available aft of STA 1278.
3. Temperature sensitive cargo should not be carried in the associated main deck cargo compartment area.

**21-61-15 MDAC Cargo Zone Trim Air Modulation Valves (777F)**

Interval	Installed	Required	Procedure
C	2	0	(M)

May be inoperative provided valve is deactivated closed.

MAINTENANCE (M)

Deactivate the associated main deck cargo zone trim air modulation valve closed (AMM 21-00-00/901).

1. For an inoperative forward main deck cargo zone trim modulation valve:
 - A. Open P110 panel L CTC CHAN 2 (AC) circuit breaker.
 - B. Open P110 panel L CTC CHAN 2 (DC) circuit breaker.
 - C. Access the forward main deck cargo zone trim air modulation valve.
 - D. Disconnect, cap, and stow the electrical connector from the valve.
 - E. Turn the manual open/close knob until the position indicator is in the CLOSED position.
 - F. Close P110 panel L CTC CHAN 2 (AC) circuit breaker.
 - G. Close P110 panel L CTC CHAN 2 (DC) circuit breaker.
2. For an inoperative aft main deck cargo zone trim modulation valve:
 - A. Open P210 panel R CTC CHAN 2 (AC) circuit breaker.
 - B. Open P210 panel R CTC CHAN 2 (DC) circuit breaker.
 - C. Access the aft main deck cargo zone trim air modulation valve.
 - D. Disconnect, cap, and stow the electrical connector from the valve.
 - E. Turn the manual open/close knob until the position indicator is in the CLOSED position.
 - F. Close P210 panel R CTC CHAN 2 (AC) circuit breaker.
 - G. Close P210 panel R CTC CHAN 2 (DC) circuit breaker.

OPERATIONS NOTE

1. For an inoperative forward main deck cargo zone trim air modulation valve, main deck cargo air conditioning is not available forward of STA 1278.
2. For an inoperative aft main deck cargo zone trim air modulation valve, main deck cargo air conditioning is not available aft of STA 1278.
3. Temperature sensitive cargo should not be carried in the associated main deck cargo compartment area.

**21-61-16 MDAC Cargo Temperature Controls (777F)**

Interval	Installed	Required	Procedure
C	2	0	

OPERATIONS NOTE

For a single inoperative temperature control, the main deck cargo temperature will default to the operative main deck cargo temperature control setting. For both temperature controls inoperative, the main deck cargo area temperature will default to 40 deg F (4 deg C).

21-61-17 Equipment Cooling Duct Temperature Sensing System (777F)
21-61-17-01 Without PRR 62273 or Production Equivalent Incorporated

Interval	Installed	Required	Procedure
C	1	0	(M) (O)

May be inoperative provided:

- a. Equipment cooling trim air modulation valve is deactivated closed.
- b. Equipment cooling three-way valve is deactivated closed.
- c. Procedures are established and used to verify main deck cargo compartment remains empty or contains only ballast, empty cargo handling equipment (ballast may be loaded in ULDs), or fly away kits.

NOTE: Operator MELs must define which items are approved for inclusion in the fly away kits and which materials can be used as ballast.

NOTE: PRR 62273 is applicable to 777F airplane line number 892 and-on. Service Bulletin 777-21-0148 provides the production equivalent of PRR 62273.

MAINTENANCE (M)

NOTE: The Equipment Cooling Duct Temperature Sensors are interchangeable with the FCAC Duct Temperature Sensors, the ACAC Duct Temperature Sensors, the MDAC Duct Temperature Sensors and the Supernumerary Duct Temperature Sensors.

Deactivate the equipment cooling trim air modulation valve closed and deactivate the equipment cooling three-way valve closed (AMM 21-00-00/901).

1. Deactivate the equipment cooling trim air modulation valve closed

NOTE: The TRIM VLV EQUIP COOL status message will be displayed for a deactivated valve.

- A. Open P210 panel R CTC CHAN 1 (AC) circuit breaker.
- B. Open P210 panel R CTC CHAN 1 (DC) circuit breaker.
- C. Gain access to the equipment cooling trim air modulation valve.
- D. Disconnect, cap, and stow the electrical connector from the valve.
- E. Manually position the equipment cooling trim air modulation valve closed.
- F. Close P210 panel R CTC CHAN 1 (AC) circuit breaker.
- G. Close P210 panel R CTC CHAN 1 (DC) circuit breaker.
2. Deactivate the equipment cooling three-way valve closed.
 - A. Open and collar P110 panel 3-WAY VALVE circuit breaker.



- B. Gain access to the equipment cooling three-way valve.
- C. Manually position the equipment cooling three-way valve closed in EE-Mode (A-B valve position).

OPERATIONS (O)

The main deck cargo compartment must remain empty, except for ballast, empty cargo containers (ballast may be loaded in ULDs), fly away kits, pallets, and cargo restraint components.

21-61-17 Equipment Cooling Duct Temperature Sensing System (777F)

21-61-17-02 With PRR 62273 or Production Equivalent Incorporated

Interval	Installed	Required	Procedure
C	1	0	(M) (O)

May be inoperative provided:

- a. Equipment cooling trim air modulation valve is deactivated closed.
- b. Equipment cooling three-way valve is deactivated open.
- c. Both packs are operated continuously while three-way valve is deactivated open.

NOTE: PRR 62273 is applicable to 777F airplane line number 892 and-on. Service Bulletin 777-21-0148 provides the production equivalent of PRR 62273.

MAINTENANCE (M)

Deactivate the equipment cooling trim air modulation valve closed and deactivate the equipment cooling three-way valve open (AMM 21-00-00/901).

1. Deactivate the equipment cooling trim air modulation valve closed.

NOTE: The TRIM VLV EQUIP COOL status message will be displayed for a deactivated valve.

- A. Open P210 panel R CTC CHAN 1 (AC) circuit breaker.
 - B. Open P210 panel R CTC CHAN 1 (DC) circuit breaker.
 - C. Gain access to the equipment cooling trim air modulation valve.
 - D. Disconnect, cap, and stow the electrical connector from the valve.
 - E. Manually position the equipment cooling trim air modulation valve closed.
 - F. Close P210 panel R CTC CHAN 1 (AC) circuit breaker.
 - G. Close P210 panel R CTC CHAN 1 (DC) circuit breaker.
2. Deactivate the equipment cooling three-way valve open.
 - A. Open and collar P110 panel 3-WAY VALVE circuit breaker.



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- B. Gain access to the equipment cooling three-way valve.
 - C. With both packs operating, manually position the equipment cooling three-way valve open in A/C-Mode (C-B valve position).

OPERATIONS (O)

Operate both packs continuously for flight instrument cooling.

NOTE: Operations with packs selected OFF may not be conducted.



21-61-18 Equipment Cooling Trim Air Modulation Valve (777F)
21-61-18-01 Without PRR 62273 or Production Equivalent
Incorporated

Interval	Installed	Required	Procedure
C	1	0	(M) (O)

May be inoperative provided:

- Valve is deactivated closed.
- Equipment cooling three-way valve is deactivated closed.
- Procedures are established and used to verify main deck cargo compartment remains empty or contains only ballast, empty cargo handling equipment (ballast may be loaded in ULDs), or fly away kits.

NOTE: Operator MELs must define which items are approved for inclusion in the fly away kits and which materials can be used as ballast.

NOTE: PRR 62273 is applicable to 777F airplane line number 892 and-on.
Service Bulletin 777-21-0148 provides the production equivalent of PRR 62273.

MAINTENANCE (M)

NOTE: The Equipment Cooling Trim Air Modulation Valve is interchangeable with the FCAC Cargo Zone Trim Air Modulation Valve, the ACAC Cargo Zone Trim Air Modulation Valve, the MDAC Cargo Zone Trim Air Modulation Valves and the Supernumerary Zone Trim Air Modulation Valve.

Deactivate the equipment cooling trim air modulation valve closed and deactivate the equipment cooling three-way valve closed (AMM 21-00-00/901).

- Deactivate the equipment cooling trim air modulation valve closed
 - Open P210 panel R CTC CHAN 1 (AC) circuit breaker.
 - Open P210 panel R CTC CHAN 1 (DC) circuit breaker.
 - Gain access to the equipment cooling trim air modulation valve.
 - Disconnect, cap, and stow the electrical connector from the valve.
 - Manually position the equipment cooling trim air modulation valve closed.
 - Close P210 panel R CTC CHAN 1 (AC) circuit breaker.
 - Close P210 panel R CTC CHAN 1 (DC) circuit breaker.
- Deactivate the equipment cooling three-way valve closed.
 - Open and collar P110 panel 3-WAY VALVE circuit breaker.
 - Gain access to the equipment cooling three-way valve.



- C. Manually position the equipment cooling three-way valve closed in EE-Mode (A-B valve position).

OPERATIONS (O)

The main deck cargo compartment must remain empty, except for ballast, empty cargo containers (ballast may be loaded in ULDs), fly away kits, pallets, and cargo restraint components.

21-61-18 Equipment Cooling Trim Air Modulation Valve (777F)

21-61-18-02 With PRR 62273 or Production Equivalent Incorporated

Interval	Installed	Required	Procedure
C	1	0	(M) (O)

May be inoperative provided:

- Valve is deactivated closed.
- Equipment cooling three-way valve is deactivated open.
- Both packs are operated continuously while three-way valve is deactivated open.

NOTE: PRR 62273 is applicable to 777F airplane line number 892 and-on.
Service Bulletin 777-21-0148 provides the production equivalent of PRR 62273.

MAINTENANCE (M)

Deactivate the equipment cooling trim air modulation valve closed and deactivate the equipment cooling three-way valve open (AMM 21-00-00/901).

- Deactivate the equipment cooling trim air modulation valve closed.
 - Open P210 panel R CTC CHAN 1 (AC) circuit breaker.
 - Open P210 panel R CTC CHAN 1 (DC) circuit breaker.
 - Gain access to the equipment cooling trim air modulation valve.
 - Disconnect, cap, and stow the electrical connector from the valve.
 - Manually position the equipment cooling trim air modulation valve closed.
 - Close P210 panel R CTC CHAN 1 (AC) circuit breaker.
 - Close P210 panel R CTC CHAN 1 (DC) circuit breaker.
- Deactivate the equipment cooling three-way valve open.
 - Open and collar P110 panel 3-WAY VALVE circuit breaker.
 - Gain access to the equipment cooling three-way valve.
 - With both packs operating, manually position the equipment cooling three-way valve open in A/C-Mode (C-B valve position).

**OPERATIONS (O)**

Operate both packs continuously for flight instrument cooling.

| NOTE: Operations with pack(s) selected OFF may not be conducted.

**21-61-19 Supernumerary Duct Temperature Sensing System
(777F)**

Interval	Installed	Required	Procedure
C	1	0	(M)

May be inoperative provided:

- Supernumerary zone trim air modulation valve is deactivated closed.
- Supernumerary area is not occupied.

[For FCR installed:](#)

- Rest area is considered inoperative.

NOTE: The airplane must also be dispatched using MMEL item 25-29-04.

MAINTENANCE (M)

Deactivate the supernumerary zone trim air modulation valve closed (AMM 21-00-00/901).

NOTE: The SUPRNMRY TRIM VALVE status message will be displayed for a deactivated valve.

- Open P110 panel L CTC CHAN 1 (AC) circuit breaker.
- Open P110 panel L CTC CHAN 1 (DC) circuit breaker.
- Gain access to the supernumerary zone trim air modulation valve.
- Disconnect, cap, and stow the electrical connector from the valve.
- Manually position the supernumerary zone trim air modulation valve closed.
- Close P110 panel L CTC CHAN 1 (AC) circuit breaker.
- Close P110 panel L CTC CHAN 1 (DC) circuit breaker.

**21-61-20 Supernumerary Zone Trim Air Modulation Valve
(777F)**

Interval	Installed	Required	Procedure
C	1	0	(M)

May be inoperative provided:

- Valve is deactivated closed.
- Supernumerary area is not occupied.

For FCR installed:

- Rest area is considered inoperative.

For FCR installed:

NOTE: The airplane must also be dispatched using MMEL item 25-29-04.

MAINTENANCE (M)

Deactivate the inoperative supernumerary zone trim air modulating valve closed (AMM 21-00-00/901).

- Open P110 panel L CTC CHAN 1 (AC) circuit breaker.
- Open P110 panel L CTC CHAN 1 (DC) circuit breaker.
- Gain access to the supernumerary zone trim air modulation valve.
- Disconnect, cap, and stow the electrical connector from the valve.
- Manually position the supernumerary zone trim air modulation valve closed.
- Close P110 panel L CTC CHAN 1 (AC) circuit breaker.
- Close P110 panel L CTC CHAN 1 (DC) circuit breaker.



21-61-21 Supernumerary Temperature Control (777F)

Interval	Installed	Required	Procedure
C	1	0	

**21-62-01 Trim Air Control Systems****21-62-01-01 Passenger****21-62-01-01A One Inoperative**

Interval	Installed	Required	Procedure
C	2	1	(O)

One may be inoperative provided:

- Associated trim air switch remains off.

[For FCAC installed:](#)

- For right trim air switch selected off, FCAC remains OFF.

OPERATIONS (O)

- Position associated TRIM AIR switch off.

NOTE 1: For left trim air switch off, flight deck temperature may be colder than normal, or slow to respond to control position. To improve flight deck temperature, position FLIGHT DECK TEMP control to AUTO Warm. To minimize adjusting the cabin temperature selection, request flight attendants allow additional time for cabin temperature stabilization.

[For Forward Cargo Air Conditioning \(FCAC\) installed:](#)

NOTE 2: For left trim air switch off, FCAC temperature control is degraded. Temperature sensitive cargo should not be carried in the lower forward cargo compartment.

[For Forward Cargo Air Conditioning \(FCAC\) installed:](#)

- For right trim air switch off, position FWD CARGO A/C control OFF.

NOTE: Lower forward cargo air conditioning is not available. Temperature sensitive cargo should not be carried in the lower forward cargo compartment.

21-62-01 Trim Air Control Systems**21-62-01-01 Passenger****21-62-01-01B Both Inoperative**

Interval	Installed	Required	Procedure
C	2	0	(O)

May be inoperative provided:

- Both trim air switches remain off.
- Both packs operate normally.

For FCAC installed:

- c. FCAC remains OFF.

MAINTENANCE NOTE

For a pack to be operating normally, certain items associated with that system must be operative. See DDG item 21-51-01 for a listing of those items.

OPERATIONS (O)

Position TRIM AIR switches off.

1. Flight deck temperature may be colder than normal, or slow to respond to control position. To improve flight deck temperature, position FLIGHT DECK TEMP control to AUTO Warm.
2. To minimize adjusting the cabin temperature selection, request flight attendants allow additional time for cabin temperature stabilization.

For Forward Cargo Air Conditioning (FCAC) installed:

3. For right trim air switch off, position FWD CARGO A/C control OFF.

NOTE 1: Lower forward cargo air conditioning is not available.
Temperature sensitive cargo should not be carried in the lower forward cargo compartment.

NOTE 2: For left trim air switch off, FCAC temperature control is degraded. Temperature sensitive cargo should not be carried in the lower forward cargo compartment.

21-62-01 Trim Air Control Systems
21-62-01-02 777F
21-62-01-02-01 Without PRR 62273 or Production Equivalent Incorporated

Interval	Installed	Required	Procedure
C	2	1	(M) (O)

Right trim air control system may be inoperative provided:

- a. Right trim air switch remains off.
- b. Equipment cooling three-way valve is deactivated closed.
- c. FCAC remains OFF.
- d. Procedures are established and used to verify main deck cargo compartment remains empty or contains only ballast, empty cargo handling equipment (ballast may be loaded in ULDs), or fly away kits.

NOTE: Operator MELs must define which items are approved for inclusion in the fly away kits and which materials can be used as ballast.



NOTE: PRR 62273 is applicable to 777F airplane line number 892 and-on.
Service Bulletin 777-21-0148 provides the production equivalent of PRR 62273.

MAINTENANCE (M)

Deactivate the equipment cooling three-way valve closed and position FCAC control OFF (AMM 21-00-00/901).

1. Open and collar P110 panel 3-WAY VALVE circuit breaker.
2. Gain access to the inoperative equipment cooling three-way valve.
3. Manually position the equipment cooling three-way valve closed in EE-Mode (A-B valve position).

For Aft Cargo Air Conditioning (ACAC) not installed:

4. Position Overhead Panel (P5) FWD CARGO A/C control OFF.

For Aft Cargo Air Conditioning (ACAC) installed:

5. Position Overhead Panel (P5) LWR CARGO TEMP FWD A/C control OFF.

OPERATIONS (O)

1. Position right TRIM AIR switch off.
2. The main deck cargo compartment must remain empty, except for ballast, empty cargo containers (ballast may be loaded in ULDs), fly away kits, pallets, and cargo restraint components.

For Aft Cargo Air Conditioning (ACAC) not installed:

3. Position FWD CARGO A/C control OFF.

NOTE: Lower forward cargo air conditioning is not available.
Temperature sensitive cargo should not be carried in the lower forward cargo compartment.

For Aft Cargo Air Conditioning (ACAC) installed:

4. Position LWR CARGO TEMP FWD A/C control OFF.

NOTE: Lower forward cargo air conditioning is not available.
Temperature sensitive cargo should not be carried in the lower forward cargo compartment.

21-62-01 Trim Air Control Systems

21-62-01-02 777F

21-62-01-02-02 With PRR 62273 or Production Equivalent Incorporated

Interval	Installed	Required	Procedure
C	2	1	(M) (O)

Right trim air control system may be inoperative provided:

- a. Right trim air switch remains off.
- b. Equipment cooling three-way valve is deactivated open.

- c. Both packs are operated continuously while three-way valve is deactivated open.
 - d. FCAC remains OFF.
-

NOTE: PRR 62273 is applicable to 777F airplane line number 892 and-on.
Service Bulletin 777-21-0148 provides the production equivalent of PRR 62273.

MAINTENANCE (M)

Deactivate the equipment cooling three-way valve open and position FCAC control OFF (AMM 21-00-00/901).

1. Open and collar P110 panel 3-WAY VALVE circuit breaker.
2. Gain access to the equipment cooling three-way valve.
3. With both packs operating, manually position the equipment cooling three-way valve open in A/C-Mode (C-B valve position).

For Aft Cargo Air Conditioning (ACAC) not installed:

4. Position Overhead Panel (P5) FWD CARGO A/C control OFF.

For Aft Cargo Air Conditioning (ACAC) installed:

5. Position Overhead Panel (P5) LWR CARGO TEMP FWD A/C control OFF.

OPERATIONS (O)

1. Operate both packs continuously for flight instrument cooling.
NOTE: Operations with pack(s) selected OFF may not be conducted.
2. Position right TRIM AIR switch off.

For Aft Cargo Air Conditioning (ACAC) not installed:

3. Position FWD CARGO A/C control OFF.

NOTE: Lower forward cargo air conditioning is not available.
Temperature sensitive cargo should not be carried in the lower forward cargo compartment.

For Aft Cargo Air Conditioning (ACAC) installed:

4. Position LWR CARGO TEMP FWD A/C control OFF.

NOTE: Lower forward cargo air conditioning is not available.
Temperature sensitive cargo should not be carried in the lower forward cargo compartment.

**21-62-02 Trim Air Pressure Regulating/Shutoff Valves (PRSOVs)****21-62-02-01 Passenger****21-62-02-01A One Inoperative**

Interval	Installed	Required	Procedure
C	2	1	(M) (O)

One may be inoperative provided:

- a. Associated valve is locked closed.

For FCAC installed:

- b. For right trim air PRSOV locked closed, FCAC remains OFF.

MAINTENANCE (M)

Deactivate the inoperative trim air PRSOV closed and for FCAC installed and right trim air PRSOV closed, position FCAC control OFF (AMM 21-00-00/901).

1. Depressurize the pneumatic system (AMM 36-00-00/201).
2. For left trim air PRSOV:
 - A. Open P110 panel L CTC CHAN 1 (AC) circuit breaker.
 - B. Open P110 panel L CTC CHAN 1 (DC) circuit breaker.
 - C. Open P110 panel L CTC CHAN 2 (AC) circuit breaker.
 - D. Open P110 panel L CTC CHAN 2 (DC) circuit breaker.
 - E. Access the left trim air PRSOV.
3. For right trim air PRSOV:
 - A. Open P210 panel R CTC CHAN 1 (AC) circuit breaker.
 - B. Open P210 panel R CTC CHAN 1 (DC) circuit breaker.
 - C. Open P210 panel R CTC CHAN 2 (AC) circuit breaker.
 - D. Open P210 panel R CTC CHAN 2 (DC) circuit breaker.
 - E. Access the right trim air PRSOV.
4. Lock the inoperative trim air PRSOV closed.
 - A. Remove the locking plug from the trim air PRSOV actuator boss.
 - B. Turn the manual wrench until the manual lock indicator is in the closed (CL) position.
 - C. Install the locking plug in the hole and tighten according to the instructions and caution statement noted in AMM 21-00-00-040-863.
5. Close the opened circuit breakers.

For Forward Cargo Air Conditioning (FCAC) installed:

6. For right trim air PRSOV locked closed, position Overhead Panel (P5) FWD CARGO A/C control OFF.

OPERATIONS (O)

NOTE: The AIR synoptic may not agree with the position of the inoperative Trim Air PRSOV.

1. Position associated TRIM AIR switch off.

NOTE 1: For left trim air switch off, flight deck temperature may be colder than normal, or slow to respond to control position. To improve flight deck temperature, position FLIGHT DECK TEMP control to AUTO Warm. To minimize adjusting the cabin temperature selection, request flight attendants allow additional time for cabin temperature stabilization.

For Forward Cargo Air Conditioning (FCAC) installed:

NOTE 2: For left trim air switch off, FCAC temperature control is degraded. Temperature sensitive cargo should not be carried in the lower forward cargo compartment.

For Forward Cargo Air Conditioning (FCAC) installed:

2. For right trim air switch off, position FWD CARGO A/C control OFF.

NOTE: Lower forward cargo air conditioning is not available. Temperature sensitive cargo should not be carried in the lower forward cargo compartment.

21-62-02 Trim Air Pressure Regulating/Shutoff Valves (PRSOVs)

21-62-02-01 Passenger

21-62-02-01B Both Inoperative

Interval	Installed	Required	Procedure
C	2	0	(M) (O)

May be inoperative provided:

- a. Both valves are locked closed.
- b. Both packs operate normally.

For FCAC installed:

- c. FCAC remains OFF.

MAINTENANCE (M)

NOTE: For a pack to be operating normally, certain items associated with that system must be operative. See DDG item 21-51-01 for a listing of those items.

Deactivate the trim air PRSOVs closed and for FCAC installed, position FCAC control OFF (AMM 21-00-00/901).

1. Depressurize the pneumatic system (AMM 36-00-00/201).

2. Open the following circuit breakers:
 - A. P110 panel L CTC CHAN 1 (AC) circuit breaker.
 - B. P110 panel L CTC CHAN 1 (DC) circuit breaker.
 - C. P110 panel L CTC CHAN 2 (AC) circuit breaker.
 - D. P110 panel L CTC CHAN 2 (DC) circuit breaker.
 - E. P210 panel R CTC CHAN 1 (AC) circuit breaker.
 - F. P210 panel R CTC CHAN 1 (DC) circuit breaker.
 - G. P210 panel R CTC CHAN 2 (AC) circuit breaker.
 - H. P210 panel R CTC CHAN 2 (DC) circuit breaker.
3. Access the trim air PRSOVs.
4. Lock the trim air PRSOVs closed.
 - A. Remove the locking plug from the trim air PRSOV actuator boss.
 - B. Turn the manual wrench until the manual lock indicator is in the closed (CL) position.
 - C. Install the locking plug in the hole and tighten according to the instructions and caution statement noted in AMM 21-00-00-040-863.
5. Close the opened circuit breakers.

For Forward Cargo Air Conditioning (FCAC) installed:

6. Position Overhead Panel (P5) FWD CARGO A/C control OFF.

OPERATIONS (O)

NOTE: The AIR synoptic may not agree with the position of the Trim Air PRSOVs.

Position TRIM AIR switches off.

1. Flight deck temperature may be colder than normal, or slow to respond to control position. To improve flight deck temperature, position FLIGHT DECK TEMP control to AUTO Warm.
2. To minimize adjusting the cabin temperature selection, request flight attendants allow additional time for cabin temperature stabilization.

For Forward Cargo Air Conditioning (FCAC) installed:

3. For right trim air switch off, position FWD CARGO A/C control OFF.

NOTE: Lower forward cargo air conditioning is not available.
Temperature sensitive cargo should not be carried in the lower forward cargo compartment.

21-62-02 Trim Air Pressure Regulating/Shutoff Valves (PRSOVs)
21-62-02-02 777F
21-62-02-02-01 Without PRR 62273 or Production Equivalent Incorporated

Interval	Installed	Required	Procedure
C	2	1	(M) (O)

Right trim air PRSOV may be inoperative provided:

- a. Right trim air PRSOV is locked closed.
- b. Equipment cooling three-way valve is deactivated closed.
- c. FCAC remains OFF.
- d. Procedures are established and used to verify main deck cargo compartment remains empty or contains only ballast, empty cargo handling equipment (ballast may be loaded in ULDs), or fly away kits.

NOTE: Operator MELs must define which items are approved for inclusion in the fly away kits and which materials can be used as ballast.

NOTE: PRR 62273 is applicable to 777F airplane line number 892 and-on. Service Bulletin 777-21-0148 provides the production equivalent of PRR 62273.

MAINTENANCE (M)

Deactivate the right trim air PRSOV closed, deactivate the equipment cooling three-way valve closed and position FCAC control OFF (AMM 21-00-00/901).

1. Deactivate the right trim air PRSOV closed
 - A. Depressurize the pneumatic system (AMM 36-00-00/201).
 - B. Open P210 panel R CTC CHAN 1 (AC) circuit breaker.
 - C. Open P210 panel R CTC CHAN 1 (DC) circuit breaker.
 - D. Open P210 panel R CTC CHAN 2 (AC) circuit breaker.
 - E. Open P210 panel R CTC CHAN 2 (DC) circuit breaker.
 - F. Access the right trim air PRSOV.
 - G. Lock the inoperative trim air PRSOV closed.
 - 1) Remove the locking plug from the trim air PRSOV actuator boss.
 - 2) Turn the manual wrench until the manual lock indicator is in the closed (CL) position.
 - 3) Install the locking plug in the hole and tighten according to the instructions and caution statement noted in AMM 21-00-00-040-863.
 - H. Close the opened circuit breakers.



2. Deactivate the equipment cooling three-way valve closed.
 - A. Open and collar P110 panel 3-WAY VALVE circuit breaker.
 - B. Gain access to the equipment cooling three-way valve.
 - C. Manually position the equipment cooling three-way valve closed in EE-Mode (A-B valve position).

For Aft Cargo Air Conditioning (ACAC) not installed:

3. Position Overhead Panel (P5) FWD CARGO A/C control OFF.

For Aft Cargo Air Conditioning (ACAC) installed:

4. Position Overhead Panel (P5) LWR CARGO TEMP FWD A/C control OFF.

OPERATIONS (O)

1. Position right TRIM AIR switch off.
2. The main deck cargo compartment must remain empty, except for ballast, empty cargo containers (ballast may be loaded in ULDs), fly away kits, pallets, and cargo restraint components.

For Aft Cargo Air Conditioning (ACAC) not installed:

3. Position FWD CARGO A/C control OFF.

NOTE: Lower forward cargo air conditioning is not available.
Temperature sensitive cargo should not be carried in the lower forward cargo compartment.

For Aft Cargo Air Conditioning (ACAC) installed:

4. Position LWR CARGO TEMP FWD A/C control OFF.

NOTE: Lower forward cargo air conditioning is not available.
Temperature sensitive cargo should not be carried in the lower forward cargo compartment.

21-62-02 Trim Air Pressure Regulating/Shutoff Valves (PRSOVs)

21-62-02-02 777F

21-62-02-02-02 With PRR 62273 or Production Equivalent Incorporated

Interval	Installed	Required	Procedure
C	2	1	(M) (O)

Right trim air PRSOV may be inoperative provided:

- a. Right trim air PRSOV is locked closed.
 - b. Equipment cooling three-way valve is deactivated open.
 - c. Both packs are operated continuously while three-way valve is deactivated open.
 - d. FCAC remains OFF.
-

NOTE: PRR 62273 is applicable to 777F airplane line number 892 and on.
Service Bulletin 777-21-0148 provides the production equivalent of PRR 62273.

MAINTENANCE (M)

Deactivate the right trim air PRSOV closed, deactivate the equipment cooling three-way valve open and position FCAC control OFF (AMM 21-00-00/901).

1. Deactivate the right trim air PRSOV closed.
 - A. Depressurize the pneumatic system (AMM 36-00-00/201).
 - B. Open P210 panel R CTC CHAN 1 (AC) circuit breaker.
 - C. Open P210 panel R CTC CHAN 1 (DC) circuit breaker.
 - D. Open P210 panel R CTC CHAN 2 (AC) circuit breaker.
 - E. Open P210 panel R CTC CHAN 2 (DC) circuit breaker.
 - F. Access the right trim air PRSOV.
 - G. Lock the inoperative trim air PRSOV closed.
 - 1) Remove the locking plug from the trim air PRSOV actuator boss.
 - 2) Turn the manual wrench until the manual lock indicator is in the closed (CL) position.
 - 3) Install the locking plug in the hole and tighten according to the instructions and caution statement noted in AMM 21-00-00-040-863.
 - H. Close the opened circuit breakers.
2. Deactivate the equipment cooling three-way valve open.
 - A. Open and collar P110 panel 3-WAY VALVE circuit breaker.
 - B. Gain access to the inoperative equipment cooling three-way valve.
 - C. With both packs operating, manually position the equipment cooling three-way valve open in A/C-Mode (C-B valve position).

For Aft Cargo Air Conditioning (ACAC) not installed:

3. Position Overhead Panel (P5) FWD CARGO A/C control OFF.

For Aft Cargo Air Conditioning (ACAC) installed:

4. Position Overhead Panel (P5) LWR CARGO TEMP FWD A/C control OFF.

OPERATIONS (O)

1. Operate both packs continuously for flight instrument cooling.

NOTE: Operations with pack(s) selected OFF may not be conducted.
2. Position right TRIM AIR switch off.

For Aft Cargo Air Conditioning (ACAC) not installed:

3. Position FWD CARGO A/C control OFF.

NOTE: Lower forward cargo air conditioning is not available.
Temperature sensitive cargo should not be carried in the lower forward cargo compartment.



[For Aft Cargo Air Conditioning \(ACAC\) installed:](#)

4. Position LWR CARGO TEMP FWD A/C control OFF.

NOTE: Lower forward cargo air conditioning is not available.
Temperature sensitive cargo should not be carried in the lower
forward cargo compartment.

**21-62-03 TRIM AIR Switch Lights****21-62-03-01 FAULT Lights**

Interval	Installed	Required	Procedure
C	2	0	

21-62-03 TRIM AIR Switch Lights**21-62-03-02 ON Lights**

Interval	Installed	Required	Procedure
C	2	0	

**21-71-01 Flight Deck Humidification System**

Interval	Installed	Required	Procedure
D	1	0	(M)

May be inoperative provided:

- Humidification system is off.
- Associated water supply is shut off.

MAINTENANCE (M)

Shutoff the flight deck humidification system water supply (AMM 21-00-00/901).

- Position Overhead Panel (P5) HUMID switch off.
- Gain access to the humidifier water manual shutoff valve.
- Close the humidifier water manual shutoff valve.

21-71-01 Flight Deck Humidification System

21-71-01-01 HUMID Switch ON Light

Interval	Installed	Required	Procedure
D	1	0	

**21-73-01 Ozone Converters****21-73-01-01 Passenger**

Interval	Installed	Required	Procedure
C	2	0	

As required by 14 CFR.

OPERATIONS NOTE

See 14 CFR 121.578 or other appropriate regulations.

21-73-01 Ozone Converters**21-73-01-02 777F**

Interval	Installed	Required	Procedure
D	2	0	



Intentionally
Blank

**22-11-01 Autopilot Flight Director Computers (AFDC)**22-11-01A *One Inoperative*22-11-01B *Two Inoperative*22-11-01C *All Inoperative***22-11-02 Autopilot Backdrive Actuator Systems**22-11-02A *One Inoperative*22-11-02B *Both Inoperative***22-11-03 Autopilot Mode Control Panel Lanes****22-11-04 Mode Control Panel Windows**

22-11-04-01 Airspeed (IAS-MACH)

22-11-04-02 Heading (HDG-TRK)

22-11-04-03 Vertical Speed (V/S-FPA)

22-11-04-04 Altitude (ALTITUDE)

22-11-04-05 Window Lighting

22-11-05 Mode Control Panel Selectors

22-11-05-01 V/S - FPA Selector (DOWN & UP)

22-11-05-02 BANK LIMIT Selector (AUTO, 5, 10, 15, 20, 25)

22-11-05-03 Altitude Increment Selector (AUTO, 1000)

22-11-05-04 Selector Push Functions

22-11-05-04-01 IAS - MACH

22-11-05-04-02 HDG - TRK SEL

22-11-05-04-03 ALTITUDE

22-11-06 Mode Control Panel Switches

22-11-06-01 A/P Engage Switches

22-11-06-01A *One Inoperative*22-11-06-01B *Both Inoperative*

22-11-06-02 Autothrottle Arm Switches (A/T ARM L, R)

22-11-06-02-01 PW and RR

22-11-06-02-01A *One Inoperative*22-11-06-02-01B *Both Inoperative*

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22-11-06-02-02	GE
22-11-06-02-02A	<i>One Inoperative</i>
22-11-06-02-02B	<i>Both Inoperative</i>
22-11-06-03	A/T Engage Switch
22-11-06-03-01	PW and RR
22-11-06-03-02	GE
22-11-06-04	F/D Switches
22-11-06-05	IAS - MACH Reference Switch
22-11-06-06	HDG - TRK Reference Switch
22-11-06-07	V/S - FPA Reference Switch
22-11-06-08	APP Switch
22-11-06-09	CLB CON, LNAV, VNAV, FLCH, VS/FPA, Heading/Track HOLD, Altitude HOLD, and LOC Switches
22-11-06-10	Arm/Engage Lights
22-11-07	Automatic Landing System (Autoland)
22-11-07-01	Triple Channel Autoland (LAND 3)
22-11-08	Control Wheel Autopilot Disconnect Switches
22-11-08A	<i>One Inoperative</i>
22-11-08B	<i>Both Inoperative</i>
22-11-09	Takeoff/Go-Around (TO/GA) Switches
22-11-09A	<i>One Inoperative</i>
22-11-09B	<i>Both Inoperative</i>
22-11-10	Flight Director Systems
22-31-01	Autothrottle System
22-31-01-01	PW and RR
22-31-01-02	GE
22-31-02	Autothrottle Servo Motors
22-31-02-01	PW and RR
22-31-02-01A	<i>One Inoperative</i>
22-31-02-01B	<i>Both Inoperative</i>

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22-31-02-02 GE

22-31-02-02A *One Inoperative*22-31-02-02B *Both Inoperative***22-31-03 Autothrottle Disconnect Switches**

22-31-03-01 PW and RR

22-31-03-01A *One Inoperative*22-31-03-01B *Both Inoperative*

22-31-03-02 GE

22-31-03-02A *One Inoperative*22-31-03-02B *Both Inoperative*



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22-11-01 Autopilot Flight Director Computers (AFDC)
22-11-01A One Inoperative

Interval	Installed	Required	Procedure
C	3	2	(M) (O)

One may be inoperative provided:

- a. Associated AFDC is deactivated.
- b. Associated AFDC backdrive actuator is deactivated.
- c. Approach minimums do not require use of the associated autopilot.

MAINTENANCE (M)

NOTE: The AFDCs are interchangeable. The failed AFDC may be moved to the center position to maintain left and right backdrive functions operational.

Deactivate the inoperative AFDC and associated backdrive actuator (AMM 22-00-00/901).

1. For the center AFDC inoperative, open and collar the P11 panel AFDC C circuit breaker.
2. For the left AFDC inoperative:
 - A. Open and collar the P11 panel AFDC L circuit breaker.
 - B. Open and collar the P11 panel AFDC BACK DRIVE L circuit breaker.
3. For the right AFDC inoperative:
 - A. Open and collar the P11 panel AFDC R circuit breaker.
 - B. Open and collar the P11 panel AFDC BACK DRIVE R circuit breaker.

OPERATIONS (O)

1. Align the ADIRU.
2. For the left or right AFDC inoperative, confirm the autopilot system is operational by engaging and disengaging the autopilot on the ground with flaps UP before each departure.
3. The autoland system is not fail-operational and the NO LAND 3 advisory message will be displayed.

22-11-01 Autopilot Flight Director Computers (AFDC)
22-11-01B Two Inoperative

Interval	Installed	Required	Procedure
C	3	1	(M) (O)

Center and one other AFDC may be inoperative provided:

- a. Associated AFDC is deactivated.

- b. For left or right AFDC inoperative, the associated AFDC backdrive actuator is deactivated.
- c. Radio altimeter associated with the operative AFDC operates normally.
- d. Both flap/slat control lanes operate normally.
- e. All warning electronics system channels operate normally.
- f. Approach minimums do not require use of the autopilots.

MAINTENANCE (M)

NOTE: The AFDCs are interchangeable.

Deactivate the inoperative AFDCs and associated backdrive actuator (AMM 22-00-00/901).

1. Open and collar the P11 panel AFDC C circuit breaker.
2. For the left AFDC inoperative:
 - A. Open and collar the P11 panel AFDC L circuit breaker.
 - B. Open and collar the P11 panel AFDC BACK DRIVE L circuit breaker.
3. For the right AFDC inoperative:
 - A. Open and collar the P11 panel AFDC R circuit breaker.
 - B. Open and collar the P11 panel AFDC BACK DRIVE R circuit breaker.

OPERATIONS (O)

1. Align the ADIRU.
2. Confirm the autopilot system is operational by engaging and disengaging the autopilot on the ground with flaps UP before each departure.
3. The autoland system is inoperative and the NO AUTOLAND advisory message will be displayed. Autopilot may be engaged, however only one flight director is available.
4. For PBN operations:
 - A. RNP AR APCH operations are not allowed.

NOTE: Refer to DDG Section 4 for additional information.

22-11-01 Autopilot Flight Director Computers (AFDC)

22-11-01C All Inoperative

Interval	Installed	Required	Procedure
B	3	0	(M) (O)

May be inoperative provided:

- a. All AFDCs are deactivated.
- b. Both AFDC backdrive actuators are deactivated.
- c. Approach minimums do not require use of the autopilots.
- d. Number of flight segments and segment duration is acceptable to flight crew.



- e. Enroute operations do not require use of the autopilots.
-

MAINTENANCE (M)

Deactivate the inoperative AFDCs and associated backdrive actuators (AMM 22-00-00/901).

1. Open and collar the P11 panel AFDC C circuit breaker.
2. Open and collar the P11 panel AFDC L circuit breaker.
3. Open and collar the P11 panel AFDC BACK DRIVE L circuit breaker.
4. Open and collar the P11 panel AFDC R circuit breaker.
5. Open and collar the P11 panel AFDC BACK DRIVE R circuit breaker.
6. Simultaneously open and close the P11 panel MCP L and MCP R circuit breakers to set the MCP IAS-MACH, HDG-TRK and V/S-FPA windows to their default states (IAS, HDG and V/S).

OPERATIONS (O)

1. Autopilot cannot be engaged, flight director is not available, and the autoland system is inoperative. No advisory messages will be displayed.
2. The selected speed and the speed bug will not be available on the PFD.
3. The autothrottle is available. However, the autothrottle engaged light will not illuminate when the autothrottle is engaged.
4. For PBN operations:
 - A. RNP AR APCH operations are not allowed.

NOTE: Refer to DDG Section 4 for additional information.

**22-11-02 Autopilot Backdrive Actuator Systems****22-11-02A One Inoperative**

Interval	Installed	Required	Procedure
C	2	1	(M) (O)

One may be inoperative provided:

- Associated AFDC backdrive actuator is deactivated.
 - Opposite AFDC operates normally.
 - Approach minimums do not require use of the associated autopilot.
-

MAINTENANCE (M)

Deactivate the associated AFDC backdrive actuator (AMM 22-00-00/901).

- For the left backdrive actuator system inoperative, open and collar P11 panel AFDC BACK DRIVE L circuit breaker.
- For the right backdrive actuator system inoperative, open and collar P11 panel AFDC BACK DRIVE R circuit breaker.

OPERATIONS (O)

- Align the ADIRU.
- Confirm the autopilot system is operational by engaging and disengaging the autopilot on the ground with flaps UP before each departure.
- Autoland system is not fail-operational and the NO LAND 3 advisory message will be displayed.

22-11-02 Autopilot Backdrive Actuator Systems**22-11-02B Both Inoperative**

Interval	Installed	Required	Procedure
B	2	0	(M) (O)

May be inoperative provided:

- Both AFDC backdrive actuators are deactivated.
 - Approach minimums do not require use of the autopilots.
 - Number of flight segments and segment duration is acceptable to flight crew.
 - Enroute operations do not require use of the autopilots.
-

MAINTENANCE (M)

Deactivate the AFDC backdrive actuators (AMM 22-00-00/901).

- Open and collar P11 panel AFDC BACK DRIVE L circuit breaker.
- Open and collar P11 panel AFDC BACK DRIVE R circuit breaker.

**OPERATIONS (O)**

1. Autopilot cannot be engaged, but flight director is available.
2. Autoland system is inoperative and the NO AUTOLAND advisory message will be displayed.

**22-11-03 Autopilot Mode Control Panel Lanes**

Interval	Installed	Required	Procedure
C	2	1	

MAINTENANCE NOTE

1. For AUTOTHROTTLE SERVO L or R status message and AUTOTHROTTLE L or R advisory message displayed, the airplane must also be dispatched using MMEL item 22-31-02.
2. For SEL SPD displayed on any PFD, the airplane must also be dispatched using MMEL item 22-11-05-04-01.
3. For Flight Director Bars not displayed on any PFD, the airplane must also be dispatched using MMEL item 22-11-10.

**22-11-04 Mode Control Panel Windows****22-11-04-01 Airspeed (IAS-MACH)**

Interval	Installed	Required	Procedure
C	1	0	

May be inoperative provided selected airspeed indications on both PFDs operate normally.

22-11-04 Mode Control Panel Windows**22-11-04-02 Heading (HDG-TRK)**

Interval	Installed	Required	Procedure
C	1	0	

May be inoperative provided selected heading indications on both PFDs operate normally.

22-11-04 Mode Control Panel Windows**22-11-04-03 Vertical Speed (V/S-FPA)**

Interval	Installed	Required	Procedure
C	1	0	

May be inoperative provided selected vertical speed indications on both PFDs operate normally.

22-11-04 Mode Control Panel Windows**22-11-04-04 Altitude (ALTITUDE)**

Interval	Installed	Required	Procedure
C	1	0	

May be inoperative provided selected altitude indications on both PFDs operate normally.

**22-11-04 Mode Control Panel Windows****22-11-04-05 Window Lighting**

Interval	Installed	Required	Procedure
B	1	0	

May be inoperative provided:

- Selected airspeed indications on both PFDs operate normally.
 - Selected heading indications on both PFDs operate normally.
 - Selected vertical speed indications on both PFDs operate normally.
 - Selected altitude indications on both PFDs operate normally.
-

**22-11-05 Mode Control Panel Selectors****22-11-05-01 V/S - FPA Selector (DOWN & UP)**

Interval	Installed	Required	Procedure
C	1	0	

22-11-05 Mode Control Panel Selectors**22-11-05-02 BANK LIMIT Selector (AUTO, 5, 10, 15, 20, 25)**

Interval	Installed	Required	Procedure
C	1	0	

22-11-05 Mode Control Panel Selectors**22-11-05-03 Altitude Increment Selector (AUTO, 1000)**

Interval	Installed	Required	Procedure
C	1	0	(O)

May be inoperative provided selector is verified to be in AUTO position.

OPERATIONS (O)

Verify the Altitude Increment selector operates in the AUTO position.

1. Turn the Altitude selector in either direction.
2. Observe the altitude displayed in the altitude window. If altitude changes in increments of 100 feet, the AUTO position is operating normally.

22-11-05 Mode Control Panel Selectors**22-11-05-04 Selector Push Functions****22-11-05-04-01 IAS - MACH**

Interval	Installed	Required	Procedure
C	1	0	



22-11-05 Mode Control Panel Selectors**22-11-05-04 Selector Push Functions****22-11-05-04-02 HDG - TRK SEL**

Interval	Installed	Required	Procedure
C	1	0	

22-11-05 Mode Control Panel Selectors**22-11-05-04 Selector Push Functions****22-11-05-04-03 ALTITUDE**

Interval	Installed	Required	Procedure
C	1	0	

**22-11-06 Mode Control Panel Switches****22-11-06-01 A/P Engage Switches****22-11-06-01A One Inoperative**

Interval	Installed	Required	Procedure
C	2	1	

22-11-06 Mode Control Panel Switches**22-11-06-01 A/P Engage Switches****22-11-06-01B Both Inoperative**

Interval	Installed	Required	Procedure
B	2	0	(O)

May be inoperative provided:

- Approach minimums do not require use of the autopilots.
- Number of flight segments and segment duration is acceptable to flight crew.
- Enroute operations do not require use of the autopilots.

OPERATIONS (O)

- Autopilot cannot be engaged. Flight director may be used if available.
- For PBN operations:
PBN operations may be conducted using flight director provided it is suitable for the selected PBN operation

NOTE: Refer to DDG Section 4 for additional information.

22-11-06 Mode Control Panel Switches**22-11-06-02 Autothrottle Arm Switches (A/T ARM L, R)****22-11-06-02-01 PW and RR****22-11-06-02-01A One Inoperative**

Interval	Installed	Required	Procedure
C	2	1	(M) (O)

One may be inoperative provided:

- Associated servo motor is deactivated.
- Both thrust levers are manually set for takeoff and go-around thrust.
- Flight remains within 180 minutes of landing at a suitable airport.

MAINTENANCE (M)

Deactivate the associated autothrottle servo motor (AMM 22-00-00/901).

1. For the left A/T ARM switch inoperative, open and collar P11 panel A/T SERVO L circuit breaker.
2. For the right A/T ARM switch inoperative, open and collar P11 panel A/T SERVO R circuit breaker.

OPERATIONS (O)

1. The AUTOTHROTTLE L or R advisory message will be displayed.
2. The autothrottle cannot be engaged on the ground.
3. Set both thrust levers manually for takeoff and go-around.
4. The flight must remain within 180 minutes of landing at a suitable airport.
5. For an inoperative switch failed to the armed state, the EPR trim function will operate normally, provided the operative autothrottle arm switch is positioned to A/T ARM.

NOTE: The autothrottle disconnect caution will be annunciated if the flight crew selects FLCH, VNAV, or TO/GA. Pressing the autothrottle disconnect switch will clear the caution.

6. For an inoperative switch failed to the off state, autothrottle functionality is available in flight for the operative autothrottle.

22-11-06 Mode Control Panel Switches**22-11-06-02 Autothrottle Arm Switches (A/T ARM L, R)****22-11-06-02-01 PW and RR****22-11-06-02-01B Both Inoperative**

Interval	Installed	Required	Procedure
C	2	0	

May be inoperative provided:

- a. Approach minimums do not require autothrottle use.
 - b. Flight remains within 180 minutes of landing at a suitable airport.
-

OPERATIONS NOTE

1. Autothrottle is inoperative.
2. The flight must remain within 180 minutes of landing at a suitable airport.

22-11-06 Mode Control Panel Switches
22-11-06-02 Autothrottle Arm Switches (A/T ARM L, R)
22-11-06-02-02 GE
22-11-06-02-02A One Inoperative

Interval	Installed	Required	Procedure
C	2	1	(M) (O)

One may be inoperative provided:

- Associated servo motor is deactivated.
- EEC normal mode operates normally.
- Both thrust levers are manually set for takeoff and go-around thrust.
- Flight remains within 180 minutes of landing at a suitable airport.

MAINTENANCE (M)

Deactivate the associated autothrottle servo motor (AMM 22-00-00/901).

- For the left A/T ARM switch inoperative, open and collar P11 panel A/T SERVO L circuit breaker.
- For the right A/T ARM switch inoperative, open and collar P11 panel A/T SERVO R circuit breaker.

OPERATIONS (O)

- The AUTOTHROTTLE L or R advisory message will be displayed.
- The autothrottle cannot be engaged on the ground.
- Set both thrust levers manually for takeoff and go-around.
- The flight must remain within 180 minutes of landing at a suitable airport.
- For an inoperative switch failed to the armed state, the N1 trim function will operate normally, provided the operative autothrottle arm switch is positioned to A/T ARM.

NOTE: The autothrottle disconnect caution will be annunciated if the flight crew selects FLCH, VNAV, or TO/GA. Pressing the autothrottle disconnect switch will clear the caution.

- For an inoperative switch failed to the off state, autothrottle functionality is available in flight for the operative autothrottle.

**22-11-06 Mode Control Panel Switches****22-11-06-02 Autothrottle Arm Switches (A/T ARM L, R)****22-11-06-02-02 GE****22-11-06-02-02B Both Inoperative**

Interval	Installed	Required	Procedure
C	2	0	

May be inoperative provided:

- EEC normal mode operates normally.
 - Approach minimums do not require autothrottle use.
 - Flight remains within 180 minutes of landing at a suitable airport.
-

OPERATIONS NOTE

- Autothrottle is inoperative.
- The flight must remain within 180 minutes of landing at a suitable airport.

22-11-06 Mode Control Panel Switches**22-11-06-03 A/T Engage Switch****22-11-06-03-01 PW and RR**

Interval	Installed	Required	Procedure
C	1	0	

May be inoperative provided:

- Approach minimums do not require autothrottle use.
 - Flight remains within 180 minutes of landing at a suitable airport.
-

OPERATIONS NOTE

- Any autothrottle mode which operates normally may be used.
- The flight must remain within 180 minutes of landing at a suitable airport.

22-11-06 Mode Control Panel Switches**22-11-06-03 A/T Engage Switch****22-11-06-03-02 GE**

Interval	Installed	Required	Procedure
C	1	0	

May be inoperative provided:

- EEC normal mode operates normally.



- b. Approach minimums do not require autothrottle use.
 - c. Flight remains within 180 minutes of landing at a suitable airport.
-

OPERATIONS NOTE

- 1. Any autothrottle mode which operates normally may be used.
- 2. The flight must remain within 180 minutes of landing at a suitable airport.

22-11-06 Mode Control Panel Switches**22-11-06-04 F/D Switches**

Interval	Installed	Required	Procedure
C	2	0	

May be inoperative provided approach minimums do not require flight director use.

OPERATIONS NOTE

For PBN operations:

- 1. One F/D switch inoperative:
 - A. RNP AR APCH operations are not allowed.
 - B. RNP APCH operations may be allowed if flight director for pilot flying (PF) is available.
- 2. Both F/D switches inoperative:
 - A. RNP AR APCH and RNP APCH operations are not allowed.
 - B. Other PBN operations may be allowed if autopilot is available.

NOTE: Refer to DDG Section 4 for additional information.

22-11-06 Mode Control Panel Switches**22-11-06-05 IAS - MACH Reference Switch**

Interval	Installed	Required	Procedure
C	1	0	

May be inoperative provided IAS is displayed in associated window.

MAINTENANCE NOTE

For MACH displayed in the associated MCP window, simultaneously open and close P11 panel MCP L and MCP R circuit breakers to cause IAS to be displayed.

**22-11-06 Mode Control Panel Switches****22-11-06-06 HDG - TRK Reference Switch**

Interval	Installed	Required	Procedure
C	1	0	

May be inoperative provided HDG is displayed in associated window.

MAINTENANCE NOTE

For TRK displayed in the associated MCP window, simultaneously open and close P11 panel MCP L and MCP R circuit breakers to cause HDG to be displayed.

22-11-06 Mode Control Panel Switches**22-11-06-07 V/S - FPA Reference Switch**

Interval	Installed	Required	Procedure
C	1	0	

MAINTENANCE NOTE

For FPA displayed in the associated MCP window, simultaneously open and close P11 panel MCP L and MCP R circuit breakers to cause V/S to be displayed.

22-11-06 Mode Control Panel Switches**22-11-06-08 APP Switch**

Interval	Installed	Required	Procedure
C	1	0	

May be inoperative provided approach minimums do not require use of the autopilot flight director system approach mode.

OPERATIONS NOTE

For PBN operations:

RNP AR APCH and RNP APCH operations are not allowed.

NOTE: Refer to DDG Section 4 for additional information.

22-11-06 Mode Control Panel Switches
22-11-06-09 CLB CON, LNAV, VNAV, FLCH, VS/FPA, Heading/Track HOLD, Altitude HOLD, and LOC Switches

Interval	Installed	Required	Procedure
C	8	0	

May be inoperative provided enroute operations do not require their use.

OPERATIONS NOTE

- For enroute operations such as RVSM, any one of the following autopilot pitch modes will provide acceptable altitude hold capability: Altitude Hold (ALT), Vertical Speed (V/S) or Flight Path Angle (FPA) with 000 selected, Vertical Navigation (VNAV) with FMC flight plan, or VNAV with FMC flight plan combined with altitude intervention.
- For lateral navigation (LNAV) switch inoperative:
PBN operations are not allowed.
- For vertical navigation switch (VNAV) inoperative:
 - RNP AR APCH is not allowed.
 - For RNP APCH, other vertical modes may be used, provided appropriate approach minimums are used.

22-11-06 Mode Control Panel Switches
22-11-06-10 Arm/Engage Lights

Interval	Installed	Required	Procedure
C	11	0	

May be inoperative provided the associated mode indications on both PFDs operate normally.



22-11-07 Automatic Landing System (Autoland)

Interval	Installed	Required	Procedure
C	1	0	

May be inoperative provided approach minimums do not require its use.

22-11-07 Automatic Landing System (Autoland)**22-11-07-01 Triple Channel Autoland (LAND 3)**

Interval	Installed	Required	Procedure
C	1	0	

May be inoperative provided approach minimums do not require its use.

**22-11-08 Control Wheel Autopilot Disconnect Switches****22-11-08A One Inoperative**

Interval	Installed	Required	Procedure
C	2	1	

One may be inoperative provided:

- Autopilot is not used at 1,500 feet AGL or below.
- Approach minimums do not require the use of the autopilot.

NOTE: Capability to perform Performance Based Navigation operations may be affected.

OPERATIONS NOTE

- Do not use autopilot below 1,500 feet AGL.
- For PBN operations:
PBN operations may be conducted provided flight director is used at 1,500 feet AGL or below.

22-11-08 Control Wheel Autopilot Disconnect Switches**22-11-08B Both Inoperative**

Interval	Installed	Required	Procedure
B	2	0	

May be inoperative provided:

- Autopilot is not used.
- Approach minimums do not require use of the autopilot.
- Number of flight segments and segment duration is acceptable to flight crew.
- Enroute operations do not require use of the autopilot.

**22-11-09 Takeoff/Go-Around (TO/GA) Switches****22-11-09A One Inoperative**

Interval	Installed	Required	Procedure
C	2	1	

One may be inoperative provided approach minimums do not require its use.

OPERATIONS NOTE

1. The autoland system is not fail-operational. The NO LAND 3 advisory message may be displayed.
2. For PBN operations:
RNP AR APCH operations are not allowed.

NOTE: Refer to DDG Section 4 for additional information.

22-11-09 Takeoff/Go-Around (TO/GA) Switches**22-11-09B Both Inoperative**

Interval	Installed	Required	Procedure
C	2	0	

May be inoperative provided:

- a. Both thrust levers are operated manually for takeoff and go-around.
- b. Autopilot and flight director are not used on approach below 500 feet AGL or MDA, whichever is higher.

NOTE: Flight director go-around and windshear guidance are not available with both TO/GA switches inoperative.

OPERATIONS NOTE

1. The autoland system is not fail-operational. The NO LAND 3 or NO AUTOLAND advisory message may be displayed.
2. Flight director go-around and windshear guidance are not available with both TO/GA switches inoperative.
3. For PBN operations:
PBN operations that require either autopilot or flight director use below 500 feet AGL or MDA, whichever is higher, may not be used. RNP AR APCH, RNP APCH, RNP 1, P-RNAV, RNAV 1 and Advanced RNP operations may not be allowed.

NOTE: Refer to DDG Section 4 for additional information.

**22-11-10 Flight Director Systems**

Interval	Installed	Required	Procedure
C	2	0	

May be inoperative provided approach minimums do not require their use.

OPERATIONS NOTE

For PBN operations:

1. One Flight Director inoperative:
 - A. RNP AR APCH operations are not allowed.
 - B. RNP APCH operations may be allowed if flight director for pilot flying (PF) is available.
2. Both Flight Directors inoperative:
 - A. RNP AR APCH and RNP APCH operations are not allowed.
 - B. Other PBN operations may be allowed if autopilot is available.

NOTE: Refer to DDG Section 4 for additional information.

**22-31-01 Autothrottle System****22-31-01-01 PW and RR**

Interval	Installed	Required	Procedure
C	1	0	

May be inoperative provided:

- Approach minimums do not require its use.
- Flight remains within 180 minutes of landing at a suitable airport.

NOTE: Any autothrottle mode which operates normally may be used.

OPERATIONS NOTE

Some failure modes of the autothrottle system may cause the EPR reference/target indications to be inoperative.

22-31-01 Autothrottle System**22-31-01-02 GE**

Interval	Installed	Required	Procedure
C	1	0	

May be inoperative provided:

- EEC normal mode operates normally.
- Approach minimums do not require its use.
- Flight remains within 180 minutes of landing at a suitable airport.

NOTE: Any autothrottle mode which operates normally may be used.

OPERATIONS NOTE

Some failure modes of the autothrottle system may cause the N1 reference/target indications to be inoperative.

22-31-02 Autothrottle Servo Motors
22-31-02-01 PW and RR
22-31-02-01A One Inoperative

Interval	Installed	Required	Procedure
C	2	1	(M) (O)

One may be inoperative provided:

- a. Associated servo motor is deactivated.
- b. Both thrust levers are manually set for takeoff and go-around thrust.
- c. Flight remains within 180 minutes of landing at a suitable airport.

MAINTENANCE (M)

Deactivate the inoperative autothrottle servo motor (AMM 22-00-00/901).

1. For the left servo motor inoperative:
 - A. Open and collar the P11 panel A/T SERVO L circuit breaker.
 - B. Disconnect, cap and stow the electrical connector from the left autothrottle servo motor.
2. For the right servo motor inoperative:
 - A. Open and collar the P11 panel A/T SERVO R circuit breaker.
 - B. Disconnect, cap and stow the electrical connector from the right autothrottle servo motor.

OPERATIONS (O)

NOTE 1: The AUTOTHROTTLE L or R advisory message will be displayed.

NOTE 2: The autothrottle cannot be engaged on the ground.

NOTE 3: The EPR trim function of the EEC will operate normally, provided both autothrottle arm switches are positioned to A/T ARM.

NOTE 4: For both autothrottle arm switches positioned to A/T ARM, the autothrottle disconnect caution will be annunciated if the flight crew selects FLCH, VNAV, or TO/GA. Pressing the autothrottle disconnect switch will clear the caution.

1. The flight must remain within 180 minutes of landing at a suitable airport.
2. For takeoff:
 - A. Set both autothrottle switches to A/T ARM.
 - B. Set both thrust levers manually.
3. After takeoff, if desired set the deactivated autothrottle servo motor A/T ARM switch OFF to restore functionality of the operative autothrottle servo motor.
4. For go-around, set both thrust levers manually.

**22-31-02 Autothrottle Servo Motors****22-31-02-01 PW and RR****22-31-02-01B Both Inoperative**

Interval	Installed	Required	Procedure
C	2	0	(M) (O)

May be inoperative provided:

- Both servo motors are deactivated.
 - Approach minimums do not require use of the autothrottles.
 - Flight remains within 180 minutes of landing at a suitable airport.
-

MAINTENANCE (M)

Deactivate the inoperative autothrottle servo motors (AMM 22-00-00/901).

- Open and collar the P11 panel A/T SERVO L circuit breaker.
- Open and collar the P11 panel A/T SERVO R circuit breaker.

OPERATIONS (O)

NOTE 1: The AUTOTHROTTLE L and R advisory messages will be displayed.

NOTE 2: The EPR trim function of the EEC will operate normally, provided both autothrottle arm switches are positioned to A/T ARM.

NOTE 3: For both autothrottle arm switches positioned to A/T ARM, the autothrottle disconnect caution will be annunciated if the flight crew selects FLCH, VNAV, or TO/GA. Pressing the autothrottle disconnect switch will clear the caution.

- The flight must remain within 180 minutes of landing at a suitable airport.
 - For takeoff, set both autothrottle switches to A/T ARM.
-

22-31-02 Autothrottle Servo Motors**22-31-02-02 GE****22-31-02-02A One Inoperative**

Interval	Installed	Required	Procedure
C	2	1	(M) (O)

One may be inoperative provided:

- EEC normal mode operates normally.
 - Associated servo motor is deactivated.
 - Both thrust levers are manually set for takeoff and go-around thrust.
 - Flight remains within 180 minutes of landing at a suitable airport.
-

MAINTENANCE (M)

Deactivate the inoperative autothrottle servo motor (AMM 22-00-00/901).

1. For the left servo motor inoperative:
 - A. Open and collar the P11 panel A/T SERVO L circuit breaker.
 - B. Disconnect, cap and stow the electrical connector from the left autothrottle servo motor.
2. For the right servo motor inoperative:
 - A. Open and collar the P11 panel A/T SERVO R circuit breaker.
 - B. Disconnect, cap and stow the electrical connector from the right autothrottle servo motor.

OPERATIONS (O)

NOTE 1: The AUTOTHROTTLE L or R advisory message will be displayed.

NOTE 2: The autothrottle cannot be engaged on the ground.

NOTE 3: The N1 trim function of the EEC will operate normally, provided both autothrottle arm switches are positioned to A/T ARM.

NOTE 4: For both autothrottle arm switches positioned to A/T ARM, the autothrottle disconnect caution will be annunciated if the flight crew selects FLCH, VNAV, or TO/GA. Pressing the autothrottle disconnect switch will clear the caution.

1. The flight must remain within 180 minutes of landing at a suitable airport.
2. For takeoff:
 - A. Set both autothrottle switches to A/T ARM.
 - B. Set both thrust levers manually.
3. After takeoff, if desired set the deactivated autothrottle servo motor A/T ARM switch OFF to restore functionality of the operative autothrottle servo motor.
4. For go-around, set both thrust levers manually.

22-31-02 Autothrottle Servo Motors

22-31-02-02 GE

22-31-02-02B Both Inoperative

Interval	Installed	Required	Procedure
C	2	0	(M) (O)

May be inoperative provided:

- a. EEC normal mode operates normally.
- b. Both servo motors are deactivated.
- c. Approach minimums do not require use of the autothrottles.



- d. Flight remains within 180 minutes of landing at a suitable airport.
-

MAINTENANCE (M)

Deactivate the inoperative autothrottle servo motors (AMM 22-00-00/901).

1. Open and collar the P11 panel A/T SERVO L circuit breaker.
2. Open and collar the P11 panel A/T SERVO R circuit breaker.

OPERATIONS (O)

NOTE 1: The AUTOTHROTTLE L and R advisory messages will be displayed.

NOTE 2: The N1 trim function of the EEC will operate normally, provided both autothrottle arm switches are positioned to A/T ARM.

NOTE 3: For both autothrottle arm switches positioned to A/T ARM, the autothrottle disconnect caution will be annunciated if the flight crew selects FLCH, VNAV, or TO/GA. Pressing the autothrottle disconnect switch will clear the caution.

1. The flight must remain within 180 minutes of landing at a suitable airport.
2. For takeoff, set both autothrottle switches to A/T ARM.

**22-31-03 Autothrottle Disconnect Switches****22-31-03-01 PW and RR****22-31-03-01A One Inoperative**

Interval	Installed	Required	Procedure
C	2	1	

One may be inoperative provided both A/T ARM switches operate normally.

22-31-03 Autothrottle Disconnect Switches**22-31-03-01 PW and RR****22-31-03-01B Both Inoperative**

Interval	Installed	Required	Procedure
C	2	0	

May be inoperative provided:

- Autothrottles are not armed.
 - Approach minimums do not require use of autothrottles.
 - Flight remains within 180 minutes of landing at a suitable airport.
-

22-31-03 Autothrottle Disconnect Switches**22-31-03-02 GE****22-31-03-02A One Inoperative**

Interval	Installed	Required	Procedure
C	2	1	

One may be inoperative provided both A/T ARM switches operate normally.

22-31-03 Autothrottle Disconnect Switches**22-31-03-02 GE****22-31-03-02B Both Inoperative**

Interval	Installed	Required	Procedure
C	2	0	

May be inoperative provided:

- EEC normal mode operates normally.



-
- b. Autothrottles are not armed.
 - c. Approach minimums do not require use of autothrottles.
 - d. Flight remains within 180 minutes of landing at a suitable airport.
-



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23-11-01A Not Required by 14 CFR

Interval	Installed	Required	Procedure
D	-	-	

Any in excess of those required by 14 CFR may be inoperative.

MAINTENANCE NOTE

If desired, the inoperative HF radio can be deactivated by opening and collaring the associated P11 panel L HF COMM or R HF COMM circuit breaker.

OPERATIONS NOTE

See 14 CFR 121.345 through 121.351 or other appropriate regulations.

23-11-01 High Frequency (HF) Communications System

23-11-01B SATVOICE Operative

Interval	Installed	Required	Procedure
C	-	1	(O)

May be inoperative while conducting operations that require two LRCS provided:

- Aircraft SATVOICE system operates normally.
- SATVOICE services are available as a LRCS over the intended route of flight.
- The ICAO Flight Plan is updated (as required) to notify ATC of the communications equipment status of the aircraft.
- Alternate procedures are established and used.

MAINTENANCE NOTE

If desired, the inoperative HF radio can be deactivated by opening and collaring the associated P11 panel L HF COMM or R HF COMM circuit breaker.

OPERATIONS (O)

- Ensure SATVOICE services are available as a LRCS over the intended route of flight.
- Alternate procedures must be established and used.

**23-11-01 High Frequency (HF) Communications System**

23-11-01-01 HF Data Link**23-11-01-01A Alternate Procedures Required**

Interval	Installed	Required	Procedure
C	1	0	(O)

May be inoperative provided alternate procedures are established and used.

OPERATIONS (O)

Alternate procedures must be established and used.

23-11-01 High Frequency (HF) Communications System

23-11-01-01 HF Data Link**23-11-01-01B Procedures Do Not Require Use**

Interval	Installed	Required	Procedure
D	1	0	

May be inoperative provided procedures do not require its use.



23-12-01 VHF Communications Systems

Interval	Installed	Required	Procedure
D	3	-	

Any in excess of those required by 14 CFR may be inoperative provided left VHF radio operates normally.

OPERATIONS NOTE

See 14 CFR 121.345 through 121.351 or other appropriate regulations.

23-12-01 VHF Communications Systems**23-12-01-01 VHF Data Link****23-12-01-01A Alternate Procedures Required**

Interval	Installed	Required	Procedure
C	1	0	(O)

May be inoperative provided alternate procedures are established and used.

OPERATIONS (O)

Alternate procedures must be established and used.

23-12-01 VHF Communications Systems**23-12-01-01 VHF Data Link****23-12-01-01B Procedures Do Not Require Use**

Interval	Installed	Required	Procedure
D	1	0	

May be inoperative provided procedures do not require its use.



23-15-01 Satellite Communication (SATCOM) Systems*********23-15-01A One Operative**

Interval	Installed	Required	Procedure
D	2	1	

23-15-01 Satellite Communication (SATCOM) Systems*********23-15-01B All Inoperative**

Interval	Installed	Required	Procedure
C	-	0	(O)

May be inoperative provided:

- Alternate procedures are established and used.
 - Flight remains within 180 minutes of landing at a suitable airport.
-

OPERATIONS (O)

NOTE: The SATCOM advisory message will be displayed when the status message SATCOM SYSTEM is displayed.

- Alternate procedures must be established and used.
 - The flight must remain within 180 minutes of landing at a suitable airport.
-

23-15-01 Satellite Communication (SATCOM) Systems*********23-15-01C Procedures Do Not Require Use**

Interval	Installed	Required	Procedure
D	-	0	

May be inoperative provided procedures do not require its use.

**23-15-01 Satellite Communication (SATCOM) Systems**
*********23-15-01-01 SATCOM Voice Systems****23-15-01-01A Alternate Procedures Required**

Interval	Installed	Required	Procedure
C	-	0	(O)

May be inoperative provided:

- Alternate procedures are established and used.
 - Flight remains within 180 minutes of landing at a suitable airport.
-

OPERATIONS (O)

NOTE: The SATCOM VOICE advisory message will be displayed when the status message SATCOM HI GAIN is displayed.

- Alternate procedures must be established and used.
 - The flight must remain within 180 minutes of landing at a suitable airport.
-

23-15-01 Satellite Communication (SATCOM) Systems
*********23-15-01-01 SATCOM Voice Systems****23-15-01-01B Procedures Do Not Require Use**

Interval	Installed	Required	Procedure
D	-	0	

May be inoperative provided procedures do not require its use.

23-15-01 Satellite Communication (SATCOM) Systems
*********23-15-01-02 SATCOM Data Link****23-15-01-02A Alternate Procedures Required**

Interval	Installed	Required	Procedure
C	1	0	(O)

May be inoperative provided alternate procedures are established and used.

OPERATIONS (O)

Alternate procedures must be established and used.

**23-15-01 Satellite Communication (SATCOM) Systems**

23-15-01-02 SATCOM Data Link**23-15-01-02B Procedures Do Not Require Use**

Interval	Installed	Required	Procedure
D	1	0	

May be inoperative provided procedures do not require its use.



23-15-02 Automated Flight Information Reporting System (AFIRS) (STC ST03079NY)

23-15-02A Alternate Procedures Required

Interval	Installed	Required	Procedure
C	1	0	(O)

May be inoperative provided alternate procedures are established and used.

NOTE: Any portion of system that operates normally may be used.

NOTE: The automated flight information reporting system (AFIRS) based on STC ST03079NY is not a Boeing system. Information applicable to the STC ST03079NY installation is not provided.

23-15-02 Automated Flight Information Reporting System (AFIRS) (STC ST03079NY)

23-15-02B Procedures Do Not Require Use

Interval	Installed	Required	Procedure
D	1	0	

May be inoperative provided procedures do not require its use.

NOTE: Any portion of system that operates normally may be used.

23-15-02 Automated Flight Information Reporting System (AFIRS) (STC ST03079NY)**23-15-02-01 Global Voice SATCOM**

23-15-02-01A Alternate Procedures Required

Interval	Installed	Required	Procedure
C	1	0	(O)

May be inoperative provided alternate procedures are established and used.

NOTE: The automated flight information reporting system (AFIRS) based on STC ST03079NY is not a Boeing system. Information applicable to the STC ST03079NY installation is not provided.



23-15-02 Automated Flight Information Reporting System (AFIRS) (STC ST03079NY)**23-15-02-01 Global Voice SATCOM***********23-15-02-01B Procedures Do Not Require Use**

Interval	Installed	Required	Procedure
D	1	0	

May be inoperative provided procedures do not require its use.
- - - - -

23-15-02 Automated Flight Information Reporting System (AFIRS) (STC ST03079NY)**23-15-02-01 Global Voice SATCOM****23-15-02-01-01 Cockpit Dialer Pad***********23-15-02-01-01A Alternate Procedures Required**

Interval	Installed	Required	Procedure
C	1	0	(O)

May be inoperative provided alternate procedures are established and used.
- - - - -

NOTE: The automated flight information reporting system (AFIRS) based on STC ST03079NY is not a Boeing system. Information applicable to the STC ST03079NY installation is not provided.

23-15-02 Automated Flight Information Reporting System (AFIRS) (STC ST03079NY)**23-15-02-01 Global Voice SATCOM****23-15-02-01-01 Cockpit Dialer Pad***********23-15-02-01-01B Procedures Do Not Require Use**

Interval	Installed	Required	Procedure
D	1	0	

May be inoperative provided procedures do not require its use.
- - - - -

**23-15-02 Automated Flight Information Reporting System
(AFIRS) (STC ST03079NY)**

23-15-02-01 Global Voice SATCOM

23-15-02-01-02 Handset

23-15-02-01-02A Alternate Procedures Required

Interval	Installed	Required	Procedure
C	1	0	(O)

May be inoperative provided alternate procedures are established and used.

NOTE: The automated flight information reporting system (AFIRS) based on STC ST03079NY is not a Boeing system. Information applicable to the STC ST03079NY installation is not provided.

23-15-02 Automated Flight Information Reporting System (AFIRS) (STC ST03079NY)

23-15-02-01 Global Voice SATCOM

23-15-02-01-02 Handset

23-15-02-01-02B Procedures Do Not Require Use

Interval	Installed	Required	Procedure
D	1	0	

May be inoperative provided procedures do not require its use.

23-15-02 Automated Flight Information Reporting System (AFIRS) (STC ST03079NY)

23-15-02-02 Global Messaging

23-15-02-02A Alternate Procedures Required

Interval	Installed	Required	Procedure
C	1	0	(O)

May be inoperative provided alternate procedures are established and used.

NOTE: Any portion of system that operates normally may be used.



NOTE: The automated flight information reporting system (AFIRS) based on STC ST03079NY is not a Boeing system. Information applicable to the STC ST03079NY installation is not provided.

23-15-02 Automated Flight Information Reporting System (AFIRS) (STC ST03079NY)

23-15-02-02 Global Messaging

23-15-02-02B Procedures Do Not Require Use

Interval	Installed	Required	Procedure
D	1	0	

May be inoperative provided procedures do not require its use.

NOTE: Any portion of system that operates normally may be used.

**23-21-01 Selective Call System (SELCAL)**

Interval	Installed	Required	Procedure
C	1	0	(O)

May be inoperative provided alternate procedures are established and used.

OPERATIONS (O)

Alternate procedures must be established and used.

**23-24-01 Radio Tuning Panels**

Interval	Installed	Required	Procedure
C	3	2	

One may be inoperative provided left radio tuning panel operates normally.

NOTE: Any function that operates normally may be used.

MAINTENANCE NOTE

The Radio Tuning Panels are interchangeable.

**23-24-02 Emergency Locator Transmitter (ELT) (Fixed)**

23-24-02A Required By 14 CFR - Inoperative

Interval	Installed	Required	Procedure
A	1	0	(M)

May be inoperative provided:

- System is deactivated.
- Repairs are made within 90 consecutive calendar-days.

NOTE: Dispatch relief for the Portable (survival) ELT's is in MMEL item 25-63-03.

MAINTENANCE (M)

Deactivate the inoperative ELT (AMM 23-00-00/901).

- Gain access to the inoperative ELT-DT in the section 47 crown.
- Disconnect, cap and stow the associated control/indication electrical connector.
- Position the ELT front panel switch to ARM.

OPERATIONS NOTE

See 14 CFR 91.207, 121.353, 121.339 or other appropriate regulations.

23-24-02 Emergency Locator Transmitter (ELT) (Fixed)

23-24-02B Required By 14 CFR - Missing

Interval	Installed	Required	Procedure
A	1	0	(M)

May be missing provided:

- Placard stating "ELT not installed" is placed in view of the pilot.
- Repairs are made within 90 consecutive calendar-days.

NOTE: Dispatch relief for the Portable (survival) ELT's is in MMEL item 25-63-03.

MAINTENANCE (M)

Placard "ELT not installed" and place it in view of the pilot (AMM DDG 23-00-00/901).

OPERATIONS NOTE

See 14 CFR 91.207, 121.353, 121.339 or other appropriate regulations.

**23-24-02 Emergency Locator Transmitter (ELT) (Fixed)**

23-24-02C Not Required By 14 CFR - Inoperative

Interval	Installed	Required	Procedure
D	1	0	(M)

Any in excess of those required by 14 CFR may be inoperative provided system is deactivated.

NOTE: Dispatch relief for the Portable (survival) ELT's is in MMEL item 25-63-03.

MAINTENANCE (M)

Deactivate the inoperative ELT (AMM 23-00-00/901).

1. Gain access to the inoperative ELT-DT in the section 47 crown.
2. Disconnect, cap and stow the associated control/indication electrical connector.
3. Position the ELT front panel switch to ARM.

OPERATIONS NOTE

See 14 CFR 91.207, 121.353, 121.339 or other appropriate regulations.

23-24-02 Emergency Locator Transmitter (ELT) (Fixed)

23-24-02D Not Required By 14 CFR - Missing

Interval	Installed	Required	Procedure
D	1	0	

Any in excess of those required by 14 CFR may be missing.

NOTE: Dispatch relief for the Portable (survival) ELT's is in MMEL item 25-63-03.

OPERATIONS NOTE

See 14 CFR 91.207, 121.353, 121.339 or other appropriate regulations.

23-24-02 Emergency Locator Transmitter (ELT) (Fixed)

23-24-02-01 Remote ELT Switch

Interval	Installed	Required	Procedure
D	1	0	(M)



May be inoperative provided:

- a. Remote ELT Switch is deactivated.
 - b. ELT Switch is placed in the ARMED mode.
-

MAINTENANCE (M)

Deactivate the remote ELT switch and set ELT front panel switch to ARM (AMM 23-00-00/901).

1. Position the remote ELT switch to ARMED.
2. Gain access to the inoperative ELT in the section 47 crown.
3. Position the ELT front panel switch to ARM.



23-24-03

Low Frequency Underwater Locating Device (ULD)

Interval	Installed	Required	Procedure

Dispatch relief for this equipment moved to item 23-25-01 in revision 23.

**23-24-04 Aircraft Autonomous Distress Tracking (ADT) System**
*********23-24-04-01 Emergency Locator Transmitter - Distress Tracking (ELT-DT)****23-24-04-01A Required By 14 CFR - Inoperative**

Interval	Installed	Required	Procedure
A	1	0	(M)

May be inoperative provided:

- System is deactivated.
 - Repairs are made within 90 consecutive calendar-days.
-

MAINTENANCE (M)

Deactivate the inoperative ELT-DT (AMM 23-00-00/901).

- Gain access to the ELT-DT in the section 47 crown.
 - For ACR model ELT-DT, position the ELT-DT front panel switch to OFF.
 - For Dukane-Seacom model ELT-DT, disconnect the ELT-DT (LRU) J1 connector (D3007) and secure for flight.
-

23-24-04 Aircraft Autonomous Distress Tracking (ADT) System
*********23-24-04-01 Emergency Locator Transmitter - Distress Tracking (ELT-DT)****23-24-04-01B Required By 14 CFR - Missing**

Interval	Installed	Required	Procedure
A	1	0	(M)

May be missing provided:

- Placard stating "ELT-DT not installed" is placed in view of the pilot.
 - Repairs are made within 90 consecutive calendar-days.
-

MAINTENANCE (M)

Placard "ELT-DT not installed" and place it in view of the pilot (AMM DDG 23-00-00/901).

**23-24-04 Aircraft Autonomous Distress Tracking (ADT) System**

23-24-04-01 Emergency Locator Transmitter - Distress Tracking (ELT-DT)**23-24-04-01C Not Required By 14 CFR - Inoperative**

Interval	Installed	Required	Procedure
D	1	0	(M)

Any in excess of those required by 14 CFR may be inoperative provided system is deactivated.

MAINTENANCE (M)

Deactivate the inoperative ELT-DT (AMM 23-00-00/901).

1. Gain access to the inoperative ELT-DT in the section 47 crown.
2. For ACR model ELT-DT, position the ELT-DT front panel switch to OFF.
3. For Dukane-Seacom model ELT-DT, disconnect the ELT-DT (LRU) J1 connector (D3007) and secure for flight.

23-24-04 Aircraft Autonomous Distress Tracking (ADT) System

23-24-04-01 Emergency Locator Transmitter - Distress Tracking (ELT-DT)**23-24-04-01D Not Required By 14 CFR - Missing**

Interval	Installed	Required	Procedure
D	1	0	

Any in excess of those required by 14 CFR may be missing.

23-24-04 Aircraft Autonomous Distress Tracking (ADT) System

23-24-04-01 Emergency Locator Transmitter - Distress Tracking (ELT-DT)**23-24-04-01-01 Remote ELT Switch**

Interval	Installed	Required	Procedure
D	1	0	(M)

May be inoperative provided:

- a. Remote ELT Switch is deactivated.
- b. ELT-DT Switch is placed in the ARMED mode.

NOTE: The remote ELT switch is meant to refer to the panel in the flight deck and is not to be confused with the remote function panel (RFP) which is a maintenance device provided by the ELT-DT supplier.

MAINTENANCE (M)

NOTE: If remote ELT switch is failed so that the ELT-DT system is activated, dispatch using MMEL 23-24-04-01A or 23-24-04-01C.

Deactivate the remote ELT switch and set ELT-DT front panel switch to ARM (AMM 23-00-00/901).

1. Position the remote ELT switch to ARMED.
2. Gain access to the inoperative ELT-DT in the section 47 crown.
3. Position the ELT-DT front panel switch to ARM.

23-24-04 Aircraft Autonomous Distress Tracking (ADT) System ***

23-24-04-02 ADT Trigger Unit (ATU)

23-24-04-02A Required By 14 CFR - Inoperative

Interval	Installed	Required	Procedure
A	1	0	(M)

May be inoperative provided:

- a. System is deactivated.
- b. ELT-DT is set to fixed ELT mode.
- c. Repairs are made within 90 consecutive calendar-days.

MAINTENANCE (M)

Deactivate the ATU and set the ELT-DT to fixed ELT mode (AMM 23-00-00/901).

1. Open and collar the P110 panel GADSS ATU circuit breaker.
2. Gain access to the ATU Lockout toggle switch in the aft cargo compartment (E6-1 shelf).
3. Position the ATU Lockout toggle switch to ATU LOCKOUT to set the ELT-DT to fixed ELT mode.

**23-24-04 Aircraft Autonomous Distress Tracking (ADT) System**

23-24-04-02 ADT Trigger Unit (ATU)**23-24-04-02B Required By 14 CFR - Missing**

Interval	Installed	Required	Procedure
A	1	0	(M)

May be missing provided:

- ELT-DT is set to fixed ELT mode.
 - Placard stating "ELT-DT not installed" is placed in view of the pilot.
 - Repairs are made within 90 consecutive calendar-days.
-

MAINTENANCE (M)

Set the ELT-DT to fixed ELT mode (AMM 23-00-00/901).

- Open and collar the P110 panel GADSS ATU circuit breaker.
- Gain access to the ATU Lockout toggle switch in the aft cargo compartment (E6-1 shelf).
- Position the ATU Lockout toggle switch to ATU LOCKOUT to set the ELT-DT to fixed ELT mode.

23-24-04 Aircraft Autonomous Distress Tracking (ADT) System

23-24-04-02 ADT Trigger Unit (ATU)**23-24-04-02C Not Required By 14 CFR - Inoperative**

Interval	Installed	Required	Procedure
D	1	0	(M)

Any in excess of those required by 14 CFR may be inoperative provided:

- System is deactivated.
 - ELT-DT is set to fixed ELT mode.
-

MAINTENANCE (M)

Deactivate the ATU and set the ELT-DT to fixed ELT mode (AMM 23-00-00/901).

- Open and collar the P110 panel GADSS ATU circuit breaker.
- Gain access to the ATU Lockout toggle switch in the aft cargo compartment (E6-1 shelf).
- Position the ATU Lockout toggle switch to ATU LOCKOUT to set the ELT-DT to fixed ELT mode.

**23-24-04 Aircraft Autonomous Distress Tracking (ADT) System**
*********23-24-04-02 ADT Trigger Unit (ATU)****23-24-04-02D Not Required By 14 CFR - Missing**

Interval	Installed	Required	Procedure
D	1	0	(M)

Any in excess of those required by 14 CFR may be missing provided ELT-DT is set to fixed ELT mode.

MAINTENANCE (M)

Set the ELT-DT to fixed ELT mode (AMM 23-00-00/901).

1. Open and collar the P110 panel GADSS ATU circuit breaker.
2. Gain access to the ATU Lockout toggle switch in the aft cargo compartment (E6-1 shelf).
3. Position the ATU Lockout toggle switch to ATU LOCKOUT to set the ELT-DT to fixed ELT mode.

**23-25-01 Low Frequency Underwater Locating Device (ULD)**

23-25-01A Not Required - Inoperative or Missing

Interval	Installed	Required	Procedure
D	-	-	

May be inoperative or missing provided operations do not require its use.

23-25-01 Low Frequency Underwater Locating Device (ULD)

23-25-01B Required - Inoperative or Missing

Interval	Installed	Required	Procedure
C	-	0	

May be inoperative or missing.

**23-27-01 Data Communication Management System****23-27-01A SATCOM Voice Inoperative**

Interval	Installed	Required	Procedure
C	1	0	(O)

May be inoperative provided:

- Alternate procedures are established and used.
- Flight remains within 180 minutes of landing at a suitable airport.

MAINTENANCE NOTE

For DCMS inoperative, the Flight Deck Printer, Maintenance Access Terminal (MAT), Portable Maintenance Access Terminal, the Side Display access to the Central Maintenance Computer and ACARS may be inoperative.

OPERATIONS (O)

NOTE 1: For all AIMS-1, and AIMS-2 prior to BP V14, the 8.33 kHz VHF radio tuning (if installed) is not available.

NOTE 2: The DATALINK SYS advisory message will be displayed.

- Alternate procedures must be established and used.
- The flight must remain within 180 minutes of landing at a suitable airport.

23-27-01 Data Communication Management System**23-27-01B SATCOM Voice Operative**

Interval	Installed	Required	Procedure
C	1	0	(O)

May be inoperative provided:

- Alternate procedures are established and used.
- SATCOM voice operates normally.

MAINTENANCE NOTE

For DCMS inoperative, the Flight Deck Printer, Maintenance Access Terminal (MAT), Portable Maintenance Access Terminal, the Side Display access to the Central Maintenance Computer and ACARS may be inoperative.

OPERATIONS (O)

NOTE 1: For all AIMS-1, and AIMS-2 prior to BP V14, the 8.33 kHz VHF radio tuning (if installed) is not available.

NOTE 2: The DATALINK SYS advisory message will be displayed.



Alternate procedures must be established and used.

23-27-01 Data Communication Management System**23-27-01-01 Automatic Dependent Surveillance - Contract (ADS - C) Function**

Interval	Installed	Required	Procedure
C	1	0	(O)

May be inoperative provided alternate procedures are established and used.

MAINTENANCE NOTE

The Automatic Dependent Surveillance - Contract (ADS-C) Function is hosted in the FMCF with its flight deck interface provided by the DCMS.

OPERATIONS (O)

Alternate procedures must be established and used.

23-27-01 Data Communication Management System**23-27-01-02 ACPT/CANC/RJCT Switch Lights**

Interval	Installed	Required	Procedure
C	6	0	

**23-27-02 Flight Deck Communications System (ACARS Data Link)****23-27-02A SATCOM Voice Inoperative**

Interval	Installed	Required	Procedure
C	1	0	(O)

May be inoperative provided:

- Alternate procedures are established and used.
- Flight remains within 180 minutes of landing at a suitable airport.

OPERATIONS (O)

NOTE 1: For AIMS-2 with BP V14 and on, 8.33 kHz VHF radio tuning (if installed) is not available.

NOTE 2: Interface to ACARS Communications Manager on MFD will be inoperative.

- Alternate procedures must be established and used.
- The flight must remain within 180 minutes of landing at a suitable airport.

23-27-02 Flight Deck Communications System (ACARS Data Link)**23-27-02B SATCOM Voice Operative**

Interval	Installed	Required	Procedure
C	1	0	(O)

May be inoperative provided:

- Alternate procedures are established and used.
- SATCOM voice operates normally.

OPERATIONS (O)

NOTE 1: For AIMS-2 with BP V14 and on, 8.33 kHz VHF radio tuning (if installed) is not available.

NOTE 2: Interface to ACARS Communications Manager on MFD will be inoperative.

Alternate procedures must be established and used.

**23-27-02 Flight Deck Communications System (ACARS Data Link)****23-27-02C Procedures Do Not Require Use**

Interval	Installed	Required	Procedure
D	1	0	

May be inoperative provided procedures do not require its use.

23-27-02 Flight Deck Communications System (ACARS Data Link)**23-27-02-01 ATC Databases****23-27-02-01A Alternate Procedures Required**

Interval	Installed	Required	Procedure
C	2	0	(O)

May be inoperative provided alternate procedures are established and used.

OPERATIONS (O)

NOTE: Aeronautical Telecommunication Network (ATN) addresses are not available for Controller/Pilot Data Link Communications (CPDLC) function.

Use voice communication procedures.

23-27-02 Flight Deck Communications System (ACARS Data Link)**23-27-02-01 ATC Databases****23-27-02-01B Procedures Do Not Require Use**

Interval	Installed	Required	Procedure
D	2	0	

May be inoperative provided procedures do not require its use.

23-31-01 Passenger Address System (Passenger)

Interval	Installed	Required	Procedure
B	1	0	(O)

May be inoperative provided:

- Alternate, normal and emergency procedures, and/or operating restrictions are established and used.
- Flight attendant chime and call lights operate normally.

NOTE: Any passenger address function that operates normally may be used.

OPERATIONS (O)

Alternate procedures may be based on the use of the Cabin/Service Interphone System. Prior to flight, the pilot in command must brief cabin crew members on the emergency procedures which will be used.

23-31-01 Passenger Address System (Passenger)
23-31-01-01 Passenger Address Controller Modes

Interval	Installed	Required	Procedure
C	2	1	(M)

One may be inoperative provided operating controller mode is selected.

MAINTENANCE (M)

- Position the PA NORM/ALT switch on the Attendant Switch Panel (ASP) to ALT.
- Confirm the PASSENGER ADDRESS status message is not displayed.

23-31-01 Passenger Address System (Passenger)
23-31-01-02 Lavatory Speakers

Interval	Installed	Required	Procedure
C	-	-	(O)

May be inoperative provided alternate procedures are established and used.

OPERATIONS (O)

Alternate procedures must be established and used.

**23-31-01 Passenger Address System (Passenger)****23-31-01-03 Cabin Speakers****23-31-01-03A Inoperative Speakers Not Adjacent**

Interval	Installed	Required	Procedure
C	-	-	(M)

May be inoperative provided inoperative speakers are not adjacent to each other.

MAINTENANCE (M)

NOTE: Certain cabin speaker drive module faults may result in PASSENGER ADDRESS status message being displayed when the associated speakers operate normally.

Confirm inoperative cabin speakers are not adjacent (forward to aft).

23-31-01 Passenger Address System (Passenger)**23-31-01-03 Cabin Speakers****23-31-01-03B Inoperative Speakers Adjacent**

Interval	Installed	Required	Procedure
C	-	-	(M)

No passenger seat, cabin attendant seat or crew rest area bunk may be occupied from which passenger address system is not audible and intelligible; seat must be blocked and placarded DO NOT OCCUPY.

MAINTENANCE (M)

NOTE: Certain cabin speaker drive module faults may result in PASSENGER ADDRESS status message being displayed when the associated speakers operate normally.

Affected seats must be blocked and placarded DO NOT OCCUPY.

23-31-01 Passenger Address System (Passenger)**23-31-01-04 Ambient Noise Sensor (ANS) System**

Interval	Installed	Required	Procedure
C	-	-	(O)

May be inoperative provided ANS is selected off.

**OPERATIONS (O)**

Select the ANS off using the CACP or CSCP.

23-31-01 Passenger Address System (Passenger)**23-31-01-05 Prerecorded Passenger Announcement System***********23-31-01-05A Alternate Procedures Used**

Interval	Installed	Required	Procedure
C	1	0	(O)

May be inoperative provided alternate procedures are established and used.

OPERATIONS (O)

Alternate procedures must be established and used. Prior to flight, the pilot in command must brief cabin crew members on the emergency procedures which will be used.

23-31-01 Passenger Address System (Passenger)**23-31-01-05 Prerecorded Passenger Announcement System***********23-31-01-05B Procedures Do Not Require Use**

Interval	Installed	Required	Procedure
D	1	0	

May be inoperative provided procedures do not require its use.

**23-31-02 Personnel Address System (777F)****23-31-02A Alternate Procedures Required**

Interval	Installed	Required	Procedure
C	1	0	(O)

May be inoperative provided

- Alternate, normal and emergency procedures, and/or operating restrictions are established and used.
- Supernumerary call lights/chimes systems operate normally.

NOTE: Any personnel address system function that operates normally may be used.

OPERATIONS (O)

Alternate procedures must be established and used.

23-31-02 Personnel Address System (777F)**23-31-02B Procedures Do Not Require Use**

Interval	Installed	Required	Procedure
D	1	0	

May be inoperative provided procedures do not require its use.



23-39-01 Cabin Area Control Panels (CACP)/Cabin System Control Panels (CSCP) (Passenger)

Interval	Installed	Required	Procedure
C	-	1	

NOTE: Any CACP/CSCP function that operates normally may be used.

**23-41-01 Service Interphone System****23-41-01-01 Nose Gear Jack****23-41-01-01A Flight Interphone Jack Operative**

Interval	Installed	Required	Procedure
C	1	0	(O)

Service interphone flight deck to ground/ground to flight deck function may be inoperative provided:

- Nose gear flight interphone jack operates normally.
- Alternate procedures are established and used.

OPERATIONS (O)

Use flight interphone for communication with ground crew.

23-41-01 Service Interphone System**23-41-01-01 Nose Gear Jack****23-41-01-01B Flight Interphone Jack Inoperative**

Interval	Installed	Required	Procedure
B	1	0	(O)

May be inoperative provided alternate procedures are established and used.

OPERATIONS (O)

Alternate procedures must be established and used. These procedures may include hand signals.

23-41-01 Service Interphone System**23-41-01-02 Other Than Nose Gear Jacks**

Interval	Installed	Required	Procedure
D	-	0	

May be inoperative provided procedures do not require its use.

23-42-01 Cabin Interphone Systems (Passenger)
23-42-01-01 Flight Deck to Cabin, Cabin to Flight Deck, Functions
23-42-01-01A Required by 14 CFR

Interval	Installed	Required	Procedure
B	-	-	(O)

May be inoperative provided:

- Flight deck to cabin and cabin to flight deck interphone functions operate normally on at least fifty percent of the cabin handsets.
- Flight deck to cabin and cabin to flight deck interphone functions operate normally on at least one handset per exit door pair.
- Alternate communications procedures between affected flight attendants station(s) and flight deck are established and used.

NOTE: Any cabin interphone function that operates normally may be used.

MAINTENANCE NOTE

Dispatch relief for cabin interphone equipment associated with the observer seats is in MMEL item 25-11-02.

OPERATIONS (O)

Alternate procedures may be based on the use of the Passenger Address System. Prior to flight, the pilot in command must brief cabin crew members on the alternate emergency procedures that will be used.

23-42-01 Cabin Interphone Systems (Passenger)
23-42-01-01 Flight Deck to Cabin, Cabin to Flight Deck, Functions
23-42-01-01B Not Required by 14 CFR

Interval	Installed	Required	Procedure
C	1	-	(O)

May be inoperative provided:

- Crewmember interphone system not required by 14 CFR.
- Alternate, normal and emergency procedures, and/or operating restrictions are established and used.

NOTE: Any cabin interphone function that operates normally may be used.

MAINTENANCE NOTE

Dispatch relief for cabin interphone equipment associated with the observer seats is in MMEL item 25-11-02.

**OPERATIONS (O)**

Alternate procedures may be based on the use of the Passenger Address System. Prior to flight, the pilot in command must brief cabin crew members on the alternate emergency procedures that will be used.

23-42-01 Cabin Interphone Systems (Passenger)**23-42-01-02 Cabin to Cabin Functions**

Interval	Installed	Required	Procedure
B	-	-	(O)

May be inoperative provided:

- Cabin to cabin interphone functions operate normally on at least fifty percent of the cabin handsets.
- Cabin to cabin interphone functions operate normally on at least one handset per exit door pair.
- Alternate communications procedures between affected flight attendants station(s) are established and used.

NOTE: Any cabin interphone function that operates normally may be used.

OPERATIONS (O)

Alternate procedures may be based on the use of the Passenger Address System. Prior to flight, the pilot in command must brief cabin crew members on the alternate emergency procedures that will be used.

23-42-01 Cabin Interphone Systems (Passenger)**23-42-01-03 Flight Deck/Cabin to Crew Rest, Crew Rest to Flight Deck/Cabin, Functions**

Interval	Installed	Required	Procedure
B	-	0	(O)

May be inoperative provided alternate procedures are established and used.

NOTE: Any cabin interphone function that operates normally may be used.

OPERATIONS (O)

Alternate procedures must be established and used.

**23-42-01 Cabin Interphone Systems (Passenger)****23-42-01-04 Cabin Interphone Controller Modes**

Interval	Installed	Required	Procedure
C	2	1	(M)

One may be inoperative provided operating controller mode is selected.

MAINTENANCE (M)

Select operating controller (AMM 23-00-00/901).

1. Position the Attendant Switch Panel (ASP) CI NORM/ALT switch to ALT.
2. Confirm the CABIN INTERPHONE status message is not displayed.

**23-42-02 Cabin Interphone Alerting Systems (Passenger)****23-42-02-01 Flight Deck Call System****23-42-02-01-01 Flight Deck Call Lights and EICAS Messages**

Interval	Installed	Required	Procedure
B	-	0	(O)

May be inoperative provided:

- Flight deck chime operates normally.
- Alternate procedures are established and used to differentiate between normal and emergency calls.

OPERATIONS (O)

Alternate procedures must be established and used.

23-42-02 Cabin Interphone Alerting Systems (Passenger)**23-42-02-01 Flight Deck Call System****23-42-02-01-02 Flight Deck Chime**

Interval	Installed	Required	Procedure
B	1	0	

May be inoperative provided flight deck call lights and EICAS messages operate normally.

23-42-02 Cabin Interphone Alerting Systems (Passenger)**23-42-02-02 Flight Attendant Call Lights****23-42-02-02A Passenger Address (PA) System Operative**

Interval	Installed	Required	Procedure
B	-	0	(O)

May be inoperative provided:

- PA system operates normally.
- Alternate procedures for contacting flight attendants are established and used.

NOTE 1: Passenger to attendant call system is considered a Non-Essential Equipment and Furnishings (NEF) item.

NOTE 2: Any visual alerting system function that operates normally may be used.

**OPERATIONS (O)**

Alternate procedures must be established and used.

23-42-02 Cabin Interphone Alerting Systems (Passenger)**23-42-02-02 Flight Attendant Call Lights****23-42-02-02B Flight Attendant Chime Operative**

Interval	Installed	Required	Procedure
B	-	0	(O)

May be inoperative provided:

- Flight attendant chime operates normally.
- Alternate procedures for contacting flight attendants are established and used.

NOTE 1: Passenger to attendant call system is considered a Non-essential Equipment and Furnishings (NEF).

NOTE 2: Any visual alerting system function that operates normally may be used.

OPERATIONS (O)

Alternate procedures must be established and used.

23-42-02 Cabin Interphone Alerting Systems (Passenger)**23-42-02-03 Flight Attendant Chime****23-42-02-03A Passenger Address (PA) System Operative**

Interval	Installed	Required	Procedure
B	-	0	(O)

May be inoperative provided:

- PA system operates normally.
- Alternate procedures for contacting flight attendants are established and used.

NOTE 1: Passenger to attendant call system is considered a Non-Essential Equipment and Furnishings (NEF) item.

NOTE 2: Any audio alerting system function that operates normally may be used.

OPERATIONS (O)

Alternate procedures must be established and used.

**23-42-02 Cabin Interphone Alerting Systems (Passenger)****23-42-02-03 Flight Attendant Chime****23-42-02-03B Flight Attendant Call Lights Operative**

Interval	Installed	Required	Procedure
B	-	0	(O)

May be inoperative provided:

- Flight attendant call lights operate normally.
- Alternate procedures for contacting flight attendants are established and used.

NOTE 1: Passenger to attendant call system is considered a Non-essential Equipment and Furnishings (NEF).

NOTE 2: Any audio alerting system function that operates normally may be used.

OPERATIONS (O)

Alternate procedures must be established and used.

23-42-02 Cabin Interphone Alerting Systems (Passenger)**23-42-02-04 Crew Rest Call Lights/Chimes Systems**

Interval	Installed	Required	Procedure
B	-	0	(O)

May be inoperative provided:

- Associated crew rest cabin interphone handset system operates normally.
- Alternate procedures for contacting crew rest occupants are established and used.

NOTE: Any alerting system function that operates normally may be used.

OPERATIONS (O)

Alternate procedures must be established and used.

**23-42-03 Cabin Interphone Handset Systems (Passenger)****23-42-03-01 Flight Deck****23-42-03-01A Alternate Procedures Required**

Interval	Installed	Required	Procedure
C	1	0	(O)

May be inoperative provided:

- Flight deck to cabin communication operates normally.
 - Alternate procedures are established and used.
-

OPERATIONS (O)

Alternate procedures must be established and used.

23-42-03 Cabin Interphone Handset Systems (Passenger)**23-42-03-01 Flight Deck****23-42-03-01B Procedures Do Not Require Use**

Interval	Installed	Required	Procedure
D	1	0	

May be inoperative provided procedures do not require its use.

23-42-03 Cabin Interphone Handset Systems (Passenger)**23-42-03-02 Cabin**

Interval	Installed	Required	Procedure
B	-	-	(O)

May be inoperative provided:

- Fifty percent of cabin handsets operate normally.
- One handset must operate normally at each pair of exit doors.
- Alternate communications procedures between the affected flight attendant station(s) are established and used.

NOTE 1: An operative handset at an inoperative flight attendant seat shall not be counted to satisfy the fifty percent requirement.

NOTE 2: Any handset function that operates normally may be used.

**OPERATIONS (O)**

Alternate procedures must be established and used.

23-42-03 Cabin Interphone Handset Systems (Passenger)**23-42-03-03 Crew Rests**

Interval	Installed	Required	Procedure
C	-	0	(O)

May be inoperative provided alternate procedures are established and used.

OPERATIONS (O)

Alternate procedures must be established and used.



23-43-01 Ground Crew Call System

Interval	Installed	Required	Procedure
C	1	0	(O)

May be inoperative provided alternate procedures are established and used.

OPERATIONS (O)

NOTE: An inoperative ground crew call horn will not annunciate loss of
 equipment cooling or when the inertial reference system is operating on
 battery power.

Alternate procedures must be established and used.

**23-45-01 Crew Communication System (777F)****23-45-01-01 Flight Deck to Supernumerary, Supernumerary to Flight Deck, Functions****23-45-01-01A Alternate Procedures Required**

Interval	Installed	Required	Procedure
C	1	0	(O)

May be inoperative provided alternate, normal and emergency procedures, and/or operating restrictions are established and used.

OPERATIONS (O)

Alternate procedures must be established and used.

23-45-01 Crew Communication System (777F)**23-45-01-01 Flight Deck to Supernumerary, Supernumerary to Flight Deck, Functions****23-45-01-01B Supernumerary/Crew Rest Not Occupied**

Interval	Installed	Required	Procedure
D	1	0	

May be inoperative provided procedures do not require its use.

23-45-01 Crew Communication System (777F)**23-45-01-02 Flight Deck to Crew Rest, Crew Rest to Flight Deck, Functions****23-45-01-02A Alternate Procedures Required**

Interval	Installed	Required	Procedure
C	1	0	(O)

May be inoperative provided alternate, normal and emergency procedures, and/or operating restrictions are established and used.

OPERATIONS (O)

Alternate procedures must be established and used.

**23-45-01 Crew Communication System (777F)****23-45-01-02 Flight Deck to Crew Rest, Crew Rest to Flight Deck, Functions****23-45-01-02B Supernumerary/Crew Rest Not Occupied**

Interval	Installed	Required	Procedure
D	1	0	

May be inoperative provided procedures do not require its use.

23-45-01 Crew Communication System (777F)**23-45-01-03 Call Switch Lights**

Interval	Installed	Required	Procedure
C	6	0	

**23-45-02 Crew Communication Alerting Systems (777F)****23-45-02-01 Flight Deck Call System (Lights and EICAS Messages)****23-45-02-01A Supernumerary Area/Crew Rest Occupied**

Interval	Installed	Required	Procedure
B	1	0	

NOTE: The flight deck chime must be operative.

23-45-02 Crew Communication Alerting Systems (777F)**23-45-02-01 Flight Deck Call System (Lights and EICAS Messages)****23-45-02-01B Supernumerary Area/Crew Rest Not Occupied**

Interval	Installed	Required	Procedure
D	1	0	

May be inoperative provided:

- Supernumerary area remains unoccupied.
 - Flightcrew rest facility is considered inoperative.
-

NOTE: The airplane must also be dispatched using MMEL item 25-29-04.

23-45-02 Crew Communication Alerting Systems (777F)**23-45-02-01 Flight Deck Call System (Lights and EICAS Messages)****23-45-02-01C Alternate Procedures Required**

Interval	Installed	Required	Procedure
C	1	0	(O)

May be inoperative provided alternate procedures are established and used.

OPERATIONS (O)

Alternate procedures must be established and used.

**23-45-02 Crew Communication Alerting Systems (777F)****23-45-02-02 Supernumerary Call Lights/Chimes Systems****23-45-02-02A Personnel Address Operative**

Interval	Installed	Required	Procedure
B	1	0	(O)

May be inoperative provided:

- Personnel address system operates normally.
- Alternate procedures are established and used.

NOTE: Any alerting system function that operates normally may be used.

OPERATIONS (O)

Alternate procedures must be established and used.

23-45-02 Crew Communication Alerting Systems (777F)**23-45-02-02 Supernumerary Call Lights/Chimes Systems****23-45-02-02B Supernumerary Area Not Occupied**

Interval	Installed	Required	Procedure
D	1	0	

May be inoperative provided supernumerary area remains unoccupied.

23-45-02 Crew Communication Alerting Systems (777F)**23-45-02-03 Crew Rest Call Lights/Chimes Systems****23-45-02-03A Personnel Address Operative**

Interval	Installed	Required	Procedure
B	1	0	(O)

May be inoperative provided:

- Personnel address system operates normally.
- Alternate procedures are established and used.

NOTE: Any alerting system function that operates normally may be used.

OPERATIONS (O)

Alternate procedures must be established and used.

**23-45-02 Crew Communication Alerting Systems (777F)****23-45-02-03 Crew Rest Call Lights/Chimes Systems****23-45-02-03B Crew Rest Not Occupied**

Interval	Installed	Required	Procedure
D	1	0	

May be inoperative provided crew rest facility is considered inoperative.

NOTE: The airplane must also be dispatched using MMEL item 25-29-04.

**23-45-03 Crew Communication Handset Systems (777F)****23-45-03-01 Supernumerary Handset****23-45-03-01A Alternate Procedures Required**

Interval	Installed	Required	Procedure
C	-	0	(O)

May be inoperative provided alternate procedures for communication with supernumerary occupants are established and used.

OPERATIONS (O)

Alternate procedures must be established and used.

23-45-03 Crew Communication Handset Systems (777F)**23-45-03-01 Supernumerary Handset****23-45-03-01B Supernumerary Area Not Occupied**

Interval	Installed	Required	Procedure
D	-	0	

May be inoperative provided procedures do not require its use.

23-45-03 Crew Communication Handset Systems (777F)**23-45-03-02 Crew Rest Handset****23-45-03-02A Alternate Procedures Required**

Interval	Installed	Required	Procedure
C	1	0	(O)

May be inoperative provided alternate procedures for communication with crew rest occupants are established and used.

OPERATIONS (O)

Alternate procedures must be established and used.

**23-45-03 Crew Communication Handset Systems (777F)****23-45-03-02 Crew Rest Handset****23-45-03-02B Crew Rest Not Occupied**

Interval	Installed	Required	Procedure
D	1	0	

May be inoperative provided procedures do not require its use.



23-46-01 Cargo Interphone/Intercom System (777F)

23-46-01A Alternate Procedures Required

Interval	Installed	Required	Procedure
C	1	0	(O)

May be inoperative provided alternate procedures are established and used.

NOTE: Any cargo interphone/intercom function that operates normally may be used.

OPERATIONS (O)

Alternate procedures must be established and used.

23-46-01 Cargo Interphone/Intercom System (777F)

23-46-01B Procedures Do Not Require Use

Interval	Installed	Required	Procedure
D	1	0	

May be inoperative provided procedures do not require its use.

**23-51-01 Flight Interphone System****23-51-01-01 Nose Gear Jack****23-51-01-01A Service Interphone Jack Operative**

Interval	Installed	Required	Procedure
C	1	0	(O)

Flight interphone flight deck to ground/ground to flight deck function may be inoperative provided:

- Nose gear service interphone jack operates normally.
- Alternate procedures are established and used.

OPERATIONS (O)

Use service interphone for communication with ground crew during ground handling operations.

23-51-01 Flight Interphone System**23-51-01-01 Nose Gear Jack****23-51-01-01B Service Interphone Jack Inoperative**

Interval	Installed	Required	Procedure
B	1	0	(O)

May be inoperative provided alternate procedures are established and used.

OPERATIONS (O)

Alternate procedures must be established and used. These procedures may include hand signals.

23-51-01 Flight Interphone System**23-51-01-02 Main Equipment Center (MEC) Jacks**

Interval	Installed	Required	Procedure
D	-	0	

May be inoperative provided procedures do not require its use.

**23-51-02 Flight Deck Speakers**

Interval	Installed	Required	Procedure
C	2	0	

May be inoperative provided:

- Procedures do not require their use.
- Associated headset earphones or headphones are installed and operate normally.

**23-51-03 Flight Deck Hand Microphones**

23-51-03A Associated Boom Microphone Operates Normally

Interval	Installed	Required	Procedure
C	-	0	

May be inoperative or missing provided associated boom microphone operates normally.

23-51-03 Flight Deck Hand Microphones

23-51-03B Not Required By Regulation

Interval	Installed	Required	Procedure
D	-	0	

Any in excess of those required by regulation may be inoperative.



23-51-04 Flight Deck Headsets/Headphones

Interval	Installed	Required	Procedure
D	-	-	

Any in excess of those required by regulation may be inoperative.

OPERATIONS NOTE

See 14 CFR Sections 121.359(g), 125.227(e) or other appropriate regulations.

23-51-04 Flight Deck Headsets/Headphones**23-51-04-01 Headset Boom Microphones****23-51-04-01A Required by Regulation**

Interval	Installed	Required	Procedure
A	-	0	

May be inoperative provided:

- Associated hand microphone is installed and operates normally.
 - Repairs are made within 3 flight days.
-

23-51-04 Flight Deck Headsets/Headphones**23-51-04-01 Headset Boom Microphones****23-51-04-01B Any In Excess Of Those Required By Regulation**

Interval	Installed	Required	Procedure
D	-	-	

Any in excess of those required by regulation may be inoperative.

23-51-04 Flight Deck Headsets/Headphones**23-51-04-02 Headset Earphones/Headphones**

Interval	Installed	Required	Procedure
C	-	1	

Either Captain's or First Officer's earphone/headphones may be inoperative provided associated flight deck speaker operates normally.



23-51-04-02-01 Active Noise Canceling/Reduction Function

Interval	Installed	Required	Procedure
D	-	0	

May be inoperative provided normal audio function operates normally.

23-51-05 Audio Control Panels
23-51-05-01 Captain's Audio Control Panel

Interval	Installed	Required	Procedure
C	1	0	(O)

May be inoperative provided first observer's audio control panel operates normally.

NOTE: Any function that operates normally may be used.

MAINTENANCE NOTE

The Audio Control Panels (ACP) are interchangeable.

OPERATIONS (O)

The captain must use the first observer ACP. Position aft aisle stand Observer Audio (OBS AUDIO) selector CAPT.

23-51-05 Audio Control Panels
23-51-05-02 First Observer's Audio Control Panel

Interval	Installed	Required	Procedure
A	1	0	

May be inoperative provided:

- a. Captain's audio control panel operates normally.
- b. Repairs are made within 2 flight days.

NOTE: Any function that operates normally may be used.

MAINTENANCE NOTE

The Audio Control Panels (ACP) are interchangeable.

23-51-05 Audio Control Panels
23-51-05-03 Second Observer's Audio Control Panel

Interval	Installed	Required	Procedure
D	-	0	

MAINTENANCE NOTE

The Audio Control Panels are interchangeable.

**23-51-05 Audio Control Panels****23-51-05-04 MIC, CALL, Receiver Lights**

Interval	Installed	Required	Procedure
C	-	0	

Audio control panel lights may be inoperative provided procedures do not require its use.

MAINTENANCE NOTE

The Audio Control Panels are interchangeable.

23-51-06 Microphone (MIC)/Interphone Switches
23-51-06-01 Control Wheel MIC/Interphone Switches

Interval	Installed	Required	Procedure
C	2	1	(M)

One may be inoperative provided:

- a. Associated audio control panel MIC/interphone switch operates normally.
- b. Affected switch is deactivated open.

MAINTENANCE (M)

Deactivate the inoperative Control Wheel MIC/interphone switch, confirm it is deactivated open and verify the associated pilot's Audio Control Panel (ACP) MIC/interphone switch operates normally (AMM 23-00-00/901).

1. Deactivate the inoperative control wheel MIC/interphone switch (AMM 23-51-06/401).
 - A. For Captain's switch inoperative, open P11 panel INTERPHONE-CAPTAIN circuit breaker.
 - B. For First Officer's switch inoperative, open P11 panel INTERPHONE-F/O circuit breaker.
 - C. Remove the two screws which mount the inoperative MIC/interphone switch to the control wheel.
 - D. Pull the inoperative MIC/interphone switch out of the control wheel.
 - E. Disconnect, cap and stow the wires from the inoperative MIC/interphone switch.
 - F. Install the associated MIC/interphone switch in the control wheel and secure with the two screws.
 - G. For Captain's switch inoperative, close P11 panel INTERPHONE-CAPTAIN circuit breaker.
 - H. For First Officer's switch inoperative, close P11 panel INTERPHONE-F/O circuit breaker.
2. Confirm the inoperative Control Wheel MIC/interphone switch is deactivated open and verify the associated pilot's ACP MIC/interphone switch operates normally.
 - A. Provide electrical power on the airplane (AMM 24-22-00/201).
 - B. Push the FLT Transmitter Select Switch on the Captain's and F/O's ACPs and ensure the FLT MIC light is illuminated on both panels.
 - C. Ensure the FLT Receiver Light is illuminated and Receiver Volume Control is properly adjusted on both ACPs.
 - D. Speak into the associated pilot's microphone while simultaneously pushing the deactivated Control Wheel MIC/interphone switch. No transmission should be heard on the other pilot's headset.



- E. Speak into the associated pilot's microphone while simultaneously pushing the associated pilot's ACP MIC switch. The transmission should be clearly heard on the other pilot's headset.

23-51-06 Microphone (MIC)/Interphone Switches**23-51-06-02 Flight Crew Audio Control Panel MIC/Interphone Switches**

Interval	Installed	Required	Procedure
C	2	1	(M)

One may be inoperative provided:

- Associated control wheel MIC/interphone switch operates normally.
- Affected switch is verified inoperative open.

MAINTENANCE (M)

Confirm the inoperative Audio Control Panel (ACP) MIC/interphone switch is inoperative open and verify the associated pilot's Control Wheel MIC/interphone switch operates normally (AMM 23-00-00/901).

- Provide electrical power on the airplane (AMM 24-22-00/201).
- Push the PA Transmitter Select Switch on the associated ACP and ensure the PA MIC light is illuminated.
- Without pressing MIC on the Control Wheel, ACP or Glareshield, speak into the associated microphone and determine if voice transmission can be heard over the passenger/personnel address (PA) system. If no voice transmission is heard, the inoperative ACP MIC/interphone switch is in the open position.
- Speak into the associated pilot's microphone while simultaneously pushing the inoperative ACP MIC/interphone switch. No transmission should be heard on the other pilot's headset.
- Speak into the associated pilot's microphone while simultaneously pushing the associated pilot's Control Wheel MIC/interphone switch. The transmission should be clearly heard on the other pilot's headset.

23-51-06 Microphone (MIC)/Interphone Switches**23-51-06-03 Glareshield MIC/Interphone Switches**

Interval	Installed	Required	Procedure
C	2	0	(M)

May be inoperative provided the affected switch is deactivated open.

MAINTENANCE (M)

Deactivate the inoperative Glareshield MIC/Interphone switch and confirm it is deactivated open (AMM 23-00-00/901).

1. Deactivate the inoperative Glareshield MIC/Interphone switch.
 - A. For Captain's switch inoperative, open P11 panel INTERPHONE-CAPTAIN circuit breaker.
 - B. For First Officer's switch inoperative, open P11 panel INTERPHONE-F/O circuit breaker.
 - C. Gain access to the inoperative Glareshield MIC/Interphone switch in the CLOCK/MAP/MIC panel on the glareshield.
 - D. Remove the inoperative Glareshield MIC/Interphone switch from the back of the glareshield panel.
 - E. Disconnect, cap and stow the wires from the inoperative MIC/Interphone switch.
 - F. Install the associated switch in the glareshield panel and secure with the two screws.
 - G. For Captain's switch inoperative, close P11 panel INTERPHONE-CAPTAIN circuit breaker.
 - H. For First Officer's switch inoperative, close P11 panel INTERPHONE-F/O circuit breaker.
2. Confirm the inoperative Glareshield MIC/Interphone switch is deactivated open.
 - A. Provide electrical power on the airplane (AMM 24-22-00/201).
 - B. Push the FLT Transmitter Select Switch on the Captain's and F/O's ACPs and ensure the FLT MIC light is illuminated on both panels.
 - C. Ensure the FLT Receiver Light is illuminated and Receiver Volume Control is properly adjusted on both ACPs.
 - D. Speak into the associated pilot's microphone while simultaneously pushing the deactivated Glareshield MIC/Interphone switch. No transmission should be heard on the other pilot's headset.

**23-70-01 Flight Deck Door Visual Surveillance Systems (Passenger)**

23-70-01A Viewing Port Not Used

Interval	Installed	Required	Procedure
A	1	0	(O)

May be inoperative provided:

- Alternate procedures are established and used.
- Repairs are made within 3 flight days.

OPERATIONS (O)

Alternate procedures must be established and used.

23-70-01 Flight Deck Door Visual Surveillance Systems (Passenger)

23-70-01B Viewing Port Used

Interval	Installed	Required	Procedure
C	1	0	(O)

May be inoperative provided:

- A flight deck door viewing port is installed and operates normally.
- Alternate procedures are established and used.

NOTE: Any visual surveillance system function that operates normally may be used.

OPERATIONS (O)

Alternate procedures must be established and used.

23-70-01 Flight Deck Door Visual Surveillance Systems (Passenger)

23-70-01C Procedures Do Not Require Use

Interval	Installed	Required	Procedure
D	1	0	

May be inoperative provided procedures do not require its use.



23-70-02 Direct View Camera System

Interval	Installed	Required	Procedure
C	1	0	(O)

May be inoperative provided alternate procedures are established and used.

OPERATIONS (O)

Alternate procedures must be established and used.

**23-70-03 Cabin Video Monitoring System (CVMS)**

Interval	Installed	Required	Procedure
D	1	0	

NOTE: Any portion of the system that operates normally may be used.

NOTE 1: Dispatch relief for the Flight Deck Door Visual Surveillance Systems is available in 23-70-01.

NOTE 2: Dispatch relief for the Direct View Camera System is available in 23-70-02.

NOTE 3: Dispatch relief for the Ground Maneuver Camera System is available in 31-61-10.



23-71-01 Cockpit Voice Recorder (CVR) System

Interval	Installed	Required	Procedure
A	1	0	

May be inoperative provided:

- Flight Data Recorder (FDR) operates normally.
 - Repairs are made within 3 flight days.
-

23-71-01 Cockpit Voice Recorder (CVR) System**23-71-01-01 Backup Battery Power**

Interval	Installed	Required	Procedure
C	1	0	

**23-93-01 Overhead Panel Bus Controllers (OPBC)****23-93-01-01 Passenger****23-93-01-01-01 Left OPBC**

Interval	Installed	Required	Procedure
B	1	0	(M) (O)

May be inoperative provided:

- The right Overhead Panel ARINC 629 System (OPAS) is verified to operate normally before each departure.
 - One air conditioning pack operates normally.
 - Both auto cabin pressure controls operate normally.
 - One center system hydraulic demand pump is selected ON for takeoff and landing.
 - Left and right pitot and static air data modules operate normally.
 - Passenger Information Signs are considered inoperative.
-

NOTE: The airplane must also be dispatched using MMEL item 33-24-01.

MAINTENANCE (M)

NOTE 1: The Generator Drive lights will be inoperative.

NOTE 2: For a pack to be operating normally, certain items associated with that system must be operative. See DDG item 21-51-01 for a listing of those items.

Verify the right Overhead Panel ARINC 629 System (OPAS) operates normally (AMM 23-00-00/901).

NOTE: The left and right OPBCs are interchangeable (AMM 23-93-01/401), the left and right PDCUs are interchangeable (AMM 23-93-03/401), and the OPICs are interchangeable (AMM 23-93-02/201).

- Confirm the OPBC R, PDCU R, OPIC RA2 and OPIC LA2 status messages are not displayed.
- Open the P11 panel OPAS 2 circuit breaker and wait at least 5 seconds.
- Close the P11 panel OPAS 2 circuit breaker and wait at least 10 seconds.
- Confirm the OPBC R, PDCU R, OPIC RA2 and OPIC LA2 status messages are not displayed.

OPERATIONS (O)

NOTE: The Generator Drive lights will be inoperative.

- Routes requiring a true course reference, or flight near the North or South Poles or in latitudes greater than 82 degrees N and S (polar keyholes) should be avoided.

2. Position one center hydraulic system Demand pump ON for takeoff and landing.

NOTE: Do not position center system Demand pump selector ON before engine start.
3. For loss of the remaining OPBC, the following effects should be expected:

NOTE: Most of the flight deck switch positions will not be available to the using systems and most of the annunciator lights will be inoperative.
 - A. The left and right hydraulic system Demand pumps and the center hydraulic Primary pumps default to off. The center hydraulic system Demand in AUTO will default to off.
 - B. HYD PRESS SYS C, HYD PRESS DEM L, HYD PRESS DEM R, SPOILERS and NO AUTOLAND alerting messages will be displayed.
 - C. Position one center hydraulic system Demand pump ON.
 - D. HYD PRESS DEM L, HYD PRESS DEM R, HYD PRESS PRI C1, HYD PRESS PRI C2 and HYD PRESS DEM C2 (if C1 selected ON) alerting messages will be displayed.
 - E. The Display Select Panel will be inoperative. Use the CDU to select displays as required.
 - F. True heading/track cannot be flown outside of the polar keyholes. Autopilot roll modes other than LNAV cannot be flown in the polar keyholes.
 - G. The Cursor Control Device (CCD) Cursor Location lights will be inoperative.
 - H. Electronic Checklist closed loop items with input from OPAS will require override.
 - I. Master Brightness (MASTER BRIGHT) is not available for the primary displays and CDUs.
 - J. The air conditioning system target temperature will default to 75 degrees F (23 degrees C) and all Recirculation Fans will be off.
 - K. All synoptic displays may have missing or incomplete information.
 - L. The fuel jettison system is inoperative.
 - M. Predictive windshear system warning and caution alerts will be inoperative.
 - N. The Landing Altitude (LDG ALT) selector will be inoperative.
 - O. The Landing Light switches will be inoperative.

23-93-01 Overhead Panel Bus Controllers (OPBC)
23-93-01-01 Passenger
23-93-01-01-02 Right OPBC

Interval	Installed	Required	Procedure
B	1	0	(M) (O)

May be inoperative provided:

- The left Overhead Panel ARINC 629 System (OPAS) is verified to operate normally before each departure.
- One air conditioning pack operates normally.
- Both auto cabin pressure controls operate normally.
- One center system hydraulic demand pump is selected ON for takeoff and landing.

MAINTENANCE (M)

NOTE 1: The Generator Drive lights will be inoperative.

NOTE 2: For a pack to be operating normally, certain items associated with that system must be operative. See DDG item 21-51-01 for a listing of those items.

Verify the left Overhead Panel ARINC 629 System (OPAS) operates normally (AMM 23-00-00/901).

NOTE: The left and right OPBCs are interchangeable (AMM 23-93-01/401), the left and right PDCUs are interchangeable (AMM 23-93-03/401), and the OPICs are interchangeable (AMM 23-93-02/201).

- Confirm the OPBC L, PDCU L, OPIC LA1 and OPIC RA1 status messages are not displayed.
- Open the P11 panel OPAS 1 circuit breaker and wait at least 5 seconds.
- Close the P11 panel OPAS 1 circuit breaker and wait at least 10 seconds.
- Confirm the OPBC L, PDCU L, OPIC LA1 and OPIC RA1 status messages are not displayed.

OPERATIONS (O)

NOTE: The Generator Drive lights will be inoperative.

- Routes requiring a true course reference, or flight near the North or South Poles or in latitudes greater than 82 degrees N and S (polar keyholes) should be avoided.
- Position one center hydraulic system Demand pump ON for takeoff and landing.

NOTE: Do not position center system Demand pump selector ON before engine start.

3. For loss of the remaining OPBC, the following effects should be expected:

NOTE: Most of the flight deck switch positions will not be available to the using systems and most of the annunciator lights will be inoperative.

- A. The left and right hydraulic system Demand pumps and the center hydraulic Primary pumps default to off. The center hydraulic system Demand in AUTO will default to off.
- B. HYD PRESS SYS C, HYD PRESS DEM L, HYD PRESS DEM R, SPOILERS and NO AUTOLAND alerting messages will be displayed.
- C. Position one center hydraulic system Demand pump ON.
- D. HYD PRESS DEM L, HYD PRESS DEM R, HYD PRESS PRI C1, HYD PRESS PRI C2 and HYD PRESS DEM C2 (if C1 selected ON) alerting messages will be displayed.
- E. The Display Select Panel will be inoperative. Use the CDU to select displays as required.
- F. True heading/track cannot be flown outside of the polar keyholes. Autopilot roll modes other than LNAV cannot be flown in the polar keyholes.
- G. The Cursor Control Device (CCD) Cursor Location lights will be inoperative.
- H. Electronic Checklist closed loop items with input from OPAS will require override.
- I. Master Brightness (MASTER BRIGHT) is not available for the primary displays and CDUs.
- J. The air conditioning system target temperature will default to 75 degrees F (23 degrees C) and all Recirculation Fans will be off.
- K. All synoptic displays may have missing or incomplete information.
- L. The fuel jettison system is inoperative.
- M. Predictive windshear system warning and caution alerts will be inoperative.
- N. The Landing Altitude (LDG ALT) selector will be inoperative.
- O. The Landing Light switches will be inoperative.

23-93-01 Overhead Panel Bus Controllers (OPBC)

23-93-01-02 777F

23-93-01-02-01 Left OPBC

Interval	Installed	Required	Procedure
B	1	0	(M) (O)

May be inoperative provided:

- a. The right Overhead Panel ARINC 629 System (OPAS) is verified to operate normally before each departure.



- b. One air conditioning pack operates normally.
 - c. Both auto cabin pressure controls operate normally.
 - d. One center system hydraulic demand pump is selected ON for takeoff and landing.
 - e. Left and right pitot and static air data modules operate normally.
 - f. Supernumerary/Courier Area Lighted Information Signs are considered inoperative.
-

NOTE: The airplane must also be dispatched using MMEL item 33-24-02.

MAINTENANCE (M)

NOTE 1: The Generator Drive lights will be inoperative.

NOTE 2: For a pack to be operating normally, certain items associated with that system must be operative. See DDG item 21-51-01 for a listing of those items.

Verify the right Overhead Panel ARINC 629 System (OPAS) operates normally (AMM 23-00-00/901).

NOTE: The left and right OPBCs are interchangeable (AMM 23-93-01/401), the left and right PDCUs are interchangeable (AMM 23-93-03/401), and the OPICs are interchangeable (AMM 23-93-02/201).

1. Confirm the OPBC R, PDCU R, OPIC RA2 and OPIC LA2 status messages are not displayed.
2. Open the P11 panel OPAS 2 circuit breaker and wait at least 5 seconds.
3. Close the P11 panel OPAS 2 circuit breaker and wait at least 10 seconds.
4. Confirm the OPBC R, PDCU R, OPIC RA2 and OPIC LA2 status messages are not displayed.

OPERATIONS (O)

NOTE: The Generator Drive lights will be inoperative.

1. Routes requiring a true course reference, or flight near the North or South Poles or in latitudes greater than 82 degrees N and S (polar keyholes) should be avoided.
2. Position one center hydraulic system Demand pump ON for takeoff and landing.

NOTE: Do not position center system Demand pump selector ON before engine start.

3. For loss of the remaining OPBC, the following effects should be expected:

NOTE: Most of the flight deck switch positions will not be available to the using systems and most of the annunciator lights will be inoperative.

- A. The left and right hydraulic system Demand pumps and the center hydraulic Primary pumps default to off. The center hydraulic system Demand in AUTO will default to off.
- B. HYD PRESS SYS C, HYD PRESS DEM L, HYD PRESS DEM R, SPOILERS and NO AUTOLAND alerting messages will be displayed.
- C. Position one center hydraulic system Demand pump ON.
- D. HYD PRESS DEM L, HYD PRESS DEM R, HYD PRESS PRI C1, HYD PRESS PRI C2 and HYD PRESS DEM C2 (if C1 selected ON) alerting messages will be displayed.
- E. The Display Select Panel will be inoperative. Use the CDU to select displays as required.
- F. True heading/track cannot be flown outside of the polar keyholes. Autopilot roll modes other than LNAV cannot be flown in the polar keyholes.
- G. The Cursor Control Device (CCD) Cursor Location lights will be inoperative.
- H. Electronic Checklist closed loop items with input from OPAS will require override.
- I. Master Brightness (MASTER BRIGHT) is not available for the primary displays and CDUs.
- J. The air conditioning system target temperature will default to 75 degrees F (23 degrees C) and all Recirculation Fans will be off.
- K. All synoptic displays may have missing or incomplete information.
- L. The fuel jettison system is inoperative.
- M. Predictive windshear system warning and caution alerts will be inoperative.
- N. The Landing Altitude (LDG ALT) selector will be inoperative.
- O. The Landing Light switches will be inoperative.

23-93-01 Overhead Panel Bus Controllers (OPBC)
23-93-01-02 777F
23-93-01-02-02 Right OPBC

Interval	Installed	Required	Procedure
B	1	0	(M) (O)

May be inoperative provided:

- a. The left Overhead Panel ARINC 629 System (OPAS) is verified to operate normally before each departure.
- b. One air conditioning pack operates normally.

- c. Both auto cabin pressure controls operate normally.
 - d. One center system hydraulic demand pump is selected ON for takeoff and landing.
-

MAINTENANCE (M)

NOTE 1: The Generator Drive lights will be inoperative.

NOTE 2: For a pack to be operating normally, certain items associated with that system must be operative. See DDG item 21-51-01 for a listing of those items.

Verify the left Overhead Panel ARINC 629 System (OPAS) operates normally (AMM 23-00-00/901).

NOTE: The left and right OPBCs are interchangeable (AMM 23-93-01/401), the left and right PDCUs are interchangeable (AMM 23-93-03/401), and the OPICs are interchangeable (AMM 23-93-02/201).

1. Confirm the OPBC L, PDCU L, OPIC LA1 and OPIC RA1 status messages are not displayed.
2. Open the P11 panel OPAS 1 circuit breaker and wait at least 5 seconds.
3. Close the P11 panel OPAS 1 circuit breaker and wait at least 10 seconds.
4. Confirm the OPBC L, PDCU L, OPIC LA1 and OPIC RA1 status messages are not displayed.

OPERATIONS (O)

NOTE: The Generator Drive lights will be inoperative.

1. Routes requiring a true course reference, or flight near the North or South Poles or in latitudes greater than 82 degrees N and S (polar keyholes) should be avoided.
2. Position one center hydraulic system Demand pump ON for takeoff and landing.

NOTE: Do not position center system Demand pump selector ON before engine start.

3. For loss of the remaining OPBC, the following effects should be expected:

NOTE: Most of the flight deck switch positions will not be available to the using systems and most of the annunciator lights will be inoperative.

- A. The left and right hydraulic system Demand pumps and the center hydraulic Primary pumps default to off. The center hydraulic system Demand in AUTO will default to off.
- B. HYD PRESS SYS C, HYD PRESS DEM L, HYD PRESS DEM R, SPOILERS and NO AUTOLAND alerting messages will be displayed.
- C. Position one center hydraulic system Demand pump ON.



- D. HYD PRESS DEM L, HYD PRESS DEM R, HYD PRESS PRI C1, HYD PRESS PRI C2 and HYD PRESS DEM C2 (if C1 selected ON) alerting messages will be displayed.
- E. The Display Select Panel will be inoperative. Use the CDU to select displays as required.
- F. True heading/track cannot be flown outside of the polar keyholes. Autopilot roll modes other than LNAV cannot be flown in the polar keyholes.
- G. The Cursor Control Device (CCD) Cursor Location lights will be inoperative.
- H. Electronic Checklist closed loop items with input from OPAS will require override.
- I. Master Brightness (MASTER BRIGHT) is not available for the primary displays and CDUs.
- J. The air conditioning system target temperature will default to 75 degrees F (23 degrees C) and all Recirculation Fans will be off.
- K. All synoptic displays may have missing or incomplete information.
- L. The fuel jettison system is inoperative.
- M. Predictive windshear system warning and caution alerts will be inoperative.
- N. The Landing Altitude (LDG ALT) selector will be inoperative.
- O. The Landing Light switches will be inoperative.

**23-93-02 Overhead Panel Interface Cards (OPIC)****23-93-02A ADPs Operative**

Interval	Installed	Required	Procedure
C	4	3	

One may be inoperative provided:

- One air conditioning pack operates normally.
 - Both center system hydraulic demand pumps operate normally.
-

MAINTENANCE NOTE

NOTE: For a pack to be operating normally, certain items associated with that system must be operative. See DDG item 21-51-01 for a listing of those items.

- For OPIC LA1 status message displayed, the left hydraulic system Primary pump FAULT light, the center hydraulic system Primary C1 pump FAULT light and the passenger or supernumerary oxygen ON light will be inoperative.
- For OPIC LA2 status message displayed, the center hydraulic system Demand C1 pump FAULT light and the right hydraulic system Demand pump FAULT light will be inoperative.
- For OPIC RA1 status message displayed, the left hydraulic system Demand pump FAULT light, the center hydraulic system Primary C2 pump FAULT light and ram air turbine UNLKD light will be inoperative.
- For OPIC RA2 status message displayed, the right hydraulic system Primary pump FAULT light and center hydraulic system Demand C2 FAULT light will be inoperative.

23-93-02 Overhead Panel Interface Cards (OPIC)**23-93-02B ADP Inoperative**

Interval	Installed	Required	Procedure
C	4	3	

One may be inoperative provided;

- One air conditioning pack operates normally.
 - One center system hydraulic demand pump is selected ON for takeoff and landing.
-

MAINTENANCE NOTE

NOTE: For a pack to be operating normally, certain items associated with that system must be operative. See DDG item 21-51-01 for a listing of those items.

1. For OPIC LA1 status message displayed, the left hydraulic system Primary pump FAULT light, the center hydraulic system Primary C1 pump FAULT light and the passenger or supernumerary oxygen ON light will be inoperative
2. For OPIC LA2 status message displayed, the center hydraulic system Demand C1 pump FAULT light and the right hydraulic system Demand pump FAULT light will be inoperative.
3. For OPIC RA1 status message displayed, the left hydraulic system Demand pump FAULT light, the center hydraulic system Primary C2 pump FAULT light and ram air turbine UNLKD light will be inoperative.
4. For OPIC RA2 status message displayed, the right hydraulic system Primary pump FAULT light and center hydraulic system Demand C2 FAULT light will be inoperative.

OPERATIONS NOTE

Position one center hydraulic system Demand pump selector ON for takeoff and landing.

NOTE 1: Do not position center system Demand pump selector ON before engine start.

NOTE 2: To reduce noise, center system Demand pump may be positioned OFF for the other phases of flight.

**23-93-03 Panel Data Concentrator Units (PDCU)**

Interval	Installed	Required	Procedure
C	2	1	



24-00-01 Electrical Synoptic Display

24-00-02 IFE/PASS SEATS Power Switch Lights (Passenger)

24-00-02-01 OFF Light

24-00-02-02 ON Light

24-00-03 CABIN/UTILITY Power Switch Lights

24-00-03-01 OFF Light

24-00-03-02 ON Light

24-09-01 Electrical Load Management System (ELMS) Power Management Channels

24-09-01-01 P110 Channels

24-09-01-02 P210 Channels

24-09-01-03 P310 Channels

24-11-01 Engine Driven Generator Systems (IDG, GCB)

24-11-01-01 GEN CTRL Switch Lights

24-11-01-01-01 OFF Lights

24-11-01-01-02 ON Lights

24-11-01-02 DRIVE DISC Switch DRIVE Lights

24-11-02 Integrated Drive Generator (IDG) Oil Pressure Indication Systems

24-21-01 APU Driven Generator System (Generator, AGCU, APB)

24-21-01-01 APU GEN Switch Lights

24-21-01-01-01 OFF Light

24-21-01-01-02 ON Light

24-22-01 Relays - AC

24-22-01-01 Ground Handling

24-22-01-02 Ground Service Select

24-22-01-03 Ground Service Transfer

24-25-01 Backup Electrical Power System (Backup Generators, Converter and CCBs)

24-25-01A *Generator Installed*



Table of Contents**24-25-01B *Generator Removed*****24-25-01-01 BACKUP GEN Switch Lights**

24-25-01-01-01 OFF Lights

24-25-01-01-02 ON Lights

24-28-01 BUS TIE Switch Lights

24-28-01-01 ISLN Lights

24-28-01-02 AUTO Lights

24-31-01 APU Battery24-31-01A *APU Required*24-31-01B *APU Not Required*24-31-01C *APU Not Required, Battery Removed***24-31-02 APU Battery Charger**24-31-02A *APU Required*24-31-02B *APU Not Required***24-31-03 BATTERY Switch Lights**

24-31-03-01 OFF Light

24-31-03-02 ON Light

24-41-01 External Power Systems

24-41-01-01 AVAIL Lights (Flight Deck)

24-41-01-02 ON Lights (Flight Deck)

24-41-01-03 External Power Connected Lights (External Power Panel)

24-41-01-04 External Power Not In Use Lights (External Power Panel)

24-41-02 Bus Power Control Unit**24-61-01 Ground Handling Transformer Rectifier Unit**

**24-00-01 Electrical Synoptic Display**

Interval	Installed	Required	Procedure
C	1	0	

OPERATIONS NOTE

For missing data on the Synoptic Display, selecting an alternate location for the display (L INBD, LWR CTR, or R INBD) may restore the missing data. Synoptic Displays containing missing data may continue to be used to the extent remaining data is useful. Airplane system faults will be annunciated by alerting and status messages.



24-00-02 IFE/PASS SEATS Power Switch Lights (Passenger)
24-00-02-01 OFF Light

Interval	Installed	Required	Procedure
C	1	0	

24-00-02 IFE/PASS SEATS Power Switch Lights (Passenger)
24-00-02-02 ON Light

Interval	Installed	Required	Procedure
C	1	0	

**24-00-03 CABIN/UTILITY Power Switch Lights****24-00-03-01 OFF Light**

Interval	Installed	Required	Procedure
C	1	0	

24-00-03 CABIN/UTILITY Power Switch Lights**24-00-03-02 ON Light**

Interval	Installed	Required	Procedure
C	1	0	

**24-09-01 Electrical Load Management System (ELMS) Power Management Channels****24-09-01-01 P110 Channels**

Interval	Installed	Required	Procedure
C	2	1	(M) (O)

One channel may be inoperative provided:

- Remaining channel is verified to operate normally before the first flight of each day.
- Forward center-of-gravity remains within limits with the weight of center tank fuel added to airplane zero fuel weight.

MAINTENANCE (M)

Verify the remaining channel in the P110 panel is operating normally before the first flight of each day (AMM DDG 24-00-00/901).

- Supply electrical power on airplane (24-22-00/201).
- Manually initiate self-test to verify associated ELMS panel operates normally (AMM 24-09-00/501).
- Verify ELMS P110 PANEL status message is not displayed.

OPERATIONS (O)

- Ensure the airplane center-of-gravity is within the forward limit with the weight of the center tank fuel added to the airplane zero fuel weight.

24-09-01 Electrical Load Management System (ELMS) Power Management Channels**24-09-01-02 P210 Channels**

Interval	Installed	Required	Procedure
C	2	1	(M) (O)

One channel may be inoperative provided:

- Remaining channel is verified to operate normally before the first flight of each day.
- Forward center-of-gravity remains within limits with the weight of center tank fuel added to airplane zero fuel weight.

MAINTENANCE (M)

Verify the remaining channel in the P210 panel is operating normally before the first flight of each day (AMM DDG 24-00-00/901).

- Supply electrical power on airplane (24-22-00/201).



2. Manually initiate self-test to verify associated ELMS panel operates normally (AMM 24-09-00/501).
3. Verify ELMS P210 PANEL status message is not displayed.

OPERATIONS (O)

1. Ensure the airplane center-of-gravity is within the forward limit with the weight of the center tank fuel added to the airplane zero fuel weight.

24-09-01 Electrical Load Management System (ELMS) Power Management Channels**24-09-01-03 P310 Channels**

Interval	Installed	Required	Procedure
C	2	1	(M)

One channel may be inoperative provided the remaining channel is verified to operate normally before the first flight of each day.

MAINTENANCE (M)

NOTE: One half of the refueling valves may be inoperative and the time to refuel may be increased.

Verify the remaining channel in the P310 panel is operating normally before the first flight of each day (AMM DDG 24-00-00/901).

1. Supply electrical power on airplane (24-22-00/201).
2. Manually initiate self-test to verify associated ELMS panel operates normally (AMM 24-09-00/501).
3. Verify ELMS P310 PANEL status message is not displayed.

OPERATIONS NOTE

For some ELMS P310 faults which result in the ELMS P310 CHANNEL status message, cabin lighting may remain in the last selected setting. For these faults, changing the cabin lighting intensity at the cabin system control panel or at a cabin area control panel is not possible. The regular and supplemental night lights are automatically selected on for these faults.

24-11-01 Engine Driven Generator Systems (IDG, GCB)

Interval	Installed	Required	Procedure
B	2	1	(M) (O)

One may be inoperative provided:

- Associated IDG is disconnected.
- APU driven generator operates normally and is used to supply buses of the inoperative channel throughout the flight.
- All generator control units, including the APU, are verified to operate normally.
- Backup AC power system is verified to operate normally before each departure.
- Flight remains within 180 minutes of landing at a suitable airport.

MAINTENANCE (M)

Verify left, right and APU Generator Control Units (GCU) operate normally and disconnect the inoperative IDG (AMM 24-00-00/901).

- Verify left, right and APU Generator Control Units (GCU) operate normally.
 - Start APU (AMM 49-11-00/201). Wait 60 seconds.
 - Position Electrical panel (P5) APU GEN switch to ON and confirm OFF light extinguishes.
 - Position Electrical panel (P5) L and R BUS TIE switches to AUTO and confirm both amber ISLN lights extinguish.
 - Position Electrical panel (P5) L and R GEN CTRL switches to OFF and confirm both amber OFF lights illuminate.
 - Select Electrical panel (P5) PRI EXT PWR off and confirm ON light extinguishes.
 - Confirm the APU generator is providing power to the left and right main AC busses.
 - On Electrical Maintenance page 1/2, confirm the only AC electrical source with a Load readout greater than 0.0 is the APU generator.
 - On Electrical Maintenance page 2/2, confirm the AC power for both L and R Main Busses is ON.
 - Confirm the ELEC AC BUS L and ELEC AC BUS R caution messages are not displayed.
 - Position Electrical Panel (P5) L BUS TIE switch to ISLN and confirm amber ISLN light illuminates and ELEC AC BUS L caution message is displayed.



- H. Position Electrical Panel (P5) L BUS TIE switch to AUTO and confirm amber ISLN light extinguishes and ELEC AC BUS L caution message is not displayed.
- I. Position Electrical Panel (P5) R BUS TIE switch to ISLN and confirm amber ISLN light illuminates and ELEC AC BUS R caution message is displayed.
- J. Position Electrical Panel (P5) R BUS TIE switch to AUTO and confirm amber ISLN light extinguishes and ELEC AC BUS R caution message is not displayed.
- K. Position operative Electrical Panel (P5) GEN CTRL switch to ON.

For GE:

- 2. Disconnect the inoperative IDG.

NOTE 1: During the authorized repair interval, if the associated engine is operated for more than 50 hours with the IDG disconnected, then the IDG should be removed for shop inspection prior to returning the airplane to service.

NOTE 2: To prevent oil loss, cap the IDG/AGB seal pad drain if oil leakage beyond AMM allowable limits is observed from the engine drain mast, IDG/AGB drain port (AMM 71-71-00/601).

- A. On Electrical Maintenance page 1/2, confirm the Oil Level for the associated IDG is NORMAL.

NOTE: The oil level can also be checked by viewing the IDG oil level sightglass.

- B. On Electrical Maintenance page 1/2, confirm the oil filter status for the backup generators is NORMAL.
- C. Start associated engine (AMM 71-00-00/210).
- D. When engine is at idle or higher, push and release Electrical Panel (P5) associated generator DRIVE DISC switch and confirm Electrical Panel (P5) associated generator DRIVE light is illuminated.
- E. Shut down associated engine (AMM 71-00-00/201).

For PW or RR:

- 3. Disconnect the inoperative IDG.

NOTE: During the authorized repair interval, if the associated engine is operated for more than 50 hours with the IDG disconnected, then the IDG should be removed for shop inspection prior to returning the airplane to service.

- A. On Electrical Maintenance page 1/2, confirm the Oil Level for the associated IDG is NORMAL.

NOTE: The oil level can also be checked by viewing the IDG oil level sightglass.



- B. On Electrical Maintenance page 1/2, confirm the oil filter status for the backup generators is NORMAL.
- C. Start associated engine (AMM 71-00-00/210).
- D. When engine is at idle or higher, push and release Electrical Panel (P5) associated generator DRIVE DISC switch and confirm Electrical Panel (P5) associated generator DRIVE light is illuminated.
- E. Shut down associated engine (AMM 71-00-00/201).

OPERATIONS (O)

1. The flight must remain within 180 minutes of landing at a suitable airport.
2. Adjust flight planning fuel as follows:
 - A. Increase flight planning fuel by 2.5% to account for the open APU air inlet door.
 - B. Increase flight planning fuel by 200 lb (90 kg) per hour of APU usage in flight and 535 lb (240 kg) per hour of APU usage on the ground.
3. Verify the Backup AC electrical power system operates normally. Confirm the ELEC BACKUP SYS or ELEC BACKUP GEN L or R status and advisory messages are not displayed 30 seconds after both engines have been started.
4. Start the APU before departure and operate the APU continuously supplying the main AC bus associated with the inoperative IDG for the entire flight.

24-11-01 Engine Driven Generator Systems (IDG, GCB)**24-11-01-01 GEN CTRL Switch Lights****24-11-01-01-01 OFF Lights**

Interval	Installed	Required	Procedure
C	2	0	

24-11-01 Engine Driven Generator Systems (IDG, GCB)**24-11-01-01 GEN CTRL Switch Lights****24-11-01-01-02 ON Lights**

Interval	Installed	Required	Procedure
C	2	0	



24-11-01 Engine Driven Generator Systems (IDG, GCB)

24-11-01-02 DRIVE DISC Switch DRIVE Lights

Interval	Installed	Required	Procedure
C	2	0	



**24-11-02 Integrated Drive Generator (IDG) Oil Pressure
Indication Systems**

Interval	Installed	Required	Procedure
C	2	0	

MAINTENANCE NOTE

To determine if the associated ELEC GEN DRIVE L or R advisory message is available, review the Electrical Synoptic page. For amber DRIVE indication displayed, the associated ELEC GEN DRIVE L or R advisory message will be operative. For amber DRIVE indication not displayed, the associated ELEC GEN DRIVE L or R advisory message will be inoperative.

24-21-01 APU Driven Generator System (Generator, AGCU, APB)

Interval	Installed	Required	Procedure
C	1	0	(M) (O)

May be inoperative provided:

- a. Procedures do not require use of the APU for electrical power.
- b. Auxiliary Power Breaker (APB) remains open.
- c. Both engine driven generator systems operate normally.
- d. Backup AC power system is verified to operate normally before each departure.
- e. Flight remains within 180 minutes of landing at a suitable airport.

NOTE: APU may be used as a pneumatic source.

MAINTENANCE (M)

Position Electrical Panel (P5) APU GEN switch OFF (AMM 24-00-00/901).

OPERATIONS (O)

1. Dispatch is not allowed if the APU is required by other procedures. MMEL dispatch procedures for the following inoperative equipment may require APU electrical availability.

MMEL Item	Associated Status Message
24-11-01	ELEC GEN SYS L or R IDG VALVE CLOSED L or R
24-25-01	ELEC BACKUP GEN L or R ELEC BACKUP SYS

2. Set APU GEN switch OFF.
3. The flight must remain within 180 minutes of landing at a suitable airport.
4. Verify the Backup AC electrical power system operates normally. Confirm the ELEC BACKUP SYS or ELEC BACKUP GEN L or R status and advisory messages are not displayed 30 seconds after both engines have been started.



24-21-01 APU Driven Generator System (Generator, AGCU, APB)**24-21-01-01 APU GEN Switch Lights****24-21-01-01-01 OFF Light**

Interval	Installed	Required	Procedure
C	1	0	

24-21-01 APU Driven Generator System (Generator, AGCU, APB)**24-21-01-01 APU GEN Switch Lights****24-21-01-01-02 ON Light**

Interval	Installed	Required	Procedure
C	1	0	

24-22-01 Relays - AC
24-22-01-01 Ground Handling

Interval	Installed	Required	Procedure
C	1	0	(M)

May be inoperative provided the equipment on the Ground Handling Bus is deactivated before each departure.

MAINTENANCE (M)

Deactivate the equipment on the Ground Handling Bus before each departure (AMM 24-00-00/901).

NOTE: Any ground handling bus function that operates normally may be used on the ground. If the ground handling bus can be powered by a power source, it is acceptable to close the circuit breakers on the ground.

1. Open and collar the P300 panel CARGO HDLG AFT COMPT L circuit breaker.
2. Open and collar the P300 panel CARGO HDLG AFT COMPT R circuit breaker.
3. Open and collar the P300 panel CARGO HDLG FWD COMPT L circuit breaker.
4. Open and collar the P300 panel CARGO HDLG FWD COMPT R circuit breaker.
5. Open and collar the P300 panel GND HDLG AC BUS DISTR P320 circuit breaker.

24-22-01 Relays - AC
24-22-01-02 Ground Service Select

Interval	Installed	Required	Procedure
C	1	0	(M)

Control of the relay may be inoperative provided the relay remains in the engine running position.

MAINTENANCE (M)

Verify the Ground Service Select Relay is in the engine running position (AMM 24-00-00/901).

1. Position the Attendant Switch Panel (ASP) GROUND SERVICE switch to off.

2. Verify the inoperative relay is in the engine running position using secondary external power, or by operating an engine.
 - A. For using secondary external power:
 - 1) Supply secondary external electrical power (AMM 24-22-00/201).
 - a. Position Electrical Panel (P5) BATTERY switch ON.
 - b. Apply secondary external power.
 - 2) For APU running, position the Electrical Panel (P5) APU GEN Control switch OFF.
 - 3) Disconnect primary external power.

NOTE: The primary external power cord must be removed from the receptacle. Selecting the Electrical Panel (P5) PRIMARY EXT PWR switch off is not sufficient.
 - 4) Confirm ELEC AC BUS R caution message is not displayed.
 - 5) On Electrical Maintenance page 2/2, confirm all GND SVC are ON.
 - B. For using an engine:
 - 1) Start an engine (AMM 71-00-00/201).
 - 2) Position associated Electrical Panel (P5) GEN CTRL switch ON.
 - 3) For left engine started, position Electrical Panel (P5) L and R BUS TIE switches AUTO.
 - 4) For APU running, position Electrical Panel (P5) APU GEN Control switch OFF.
 - 5) Disconnect primary external power.

NOTE: The primary external power cord must be removed from the receptacle. Selecting the Electrical panel (P5) PRIMARY EXT PWR switch off is not sufficient.
 - 6) Confirm ELEC AC BUS R caution message is not displayed.
 - 7) On Electrical Maintenance page 2/2, confirm all GND SVC are ON.
 - 8) Shut down the engine (AMM 71-00-00/201).

24-22-01 Relays - AC
24-22-01-03 Ground Service Transfer

Interval	Installed	Required	Procedure
C	1	0	(M)

Control of the relay may be inoperative provided the relay remains in the engine running position.

MAINTENANCE (M)

Verify the Ground Service Transfer Relay is in the engine running position (AMM 24-00-00/901).

1. Position the Attendant Switch Panel (ASP) GROUND SERVICE switch to off.
2. Verify the inoperative relay is in the engine running position using secondary external power, or by operating an engine.

A. For using secondary external power:

- 1) Supply secondary external electrical power (AMM 24-22-00/201).
 - a. Position Electrical Panel (P5) BATTERY switch ON.
 - b. Apply secondary external power.
- 2) For APU running, position the Electrical Panel (P5) APU GEN Control switch OFF.
- 3) Disconnect primary external power.

NOTE: The primary external power cord must be removed from the receptacle. Selecting the Electrical Panel (P5) PRIMARY EXT PWR switch off is not sufficient.

- 4) Confirm ELEC AC BUS R caution message is not displayed.
- 5) On Electrical Maintenance page 2/2, confirm all GND SVC are ON.

B. For using an engine:

- 1) Start an engine (AMM 71-00-00/201).
- 2) Position associated Electrical Panel (P5) GEN CTRL switch ON.
- 3) For left engine started, position Electrical Panel (P5) L and R BUS TIE switches AUTO.
- 4) For APU running, position Electrical Panel (P5) APU GEN Control switch OFF.
- 5) Disconnect primary external power.

NOTE: The primary external power cord must be removed from the receptacle. Selecting the Electrical panel (P5) PRIMARY EXT PWR switch off is not sufficient.

- 6) Confirm ELEC AC BUS R caution message is not displayed.
- 7) On Electrical Maintenance page 2/2, confirm all GND SVC are ON.
- 8) Shut down the engine (AMM 71-00-00/201).

**24-25-01 Backup Electrical Power System (Backup Generators, Converter and CCBs)****24-25-01A Generator Installed**

Interval	Installed	Required	Procedure
C	1	0	

May be inoperative provided:

- Both engine driven generators operate normally.
- APU generator operates normally.

MAINTENANCE NOTE

- A sheared generator shaft is indicated by associated maintenance message 24-46000 or 24-46010, and the PFCS INTERFACE status message.
- For the PFCS INTERFACE status message displayed, the airplane must also be dispatched using MMEL item 27-02-04.
- To prevent damage to the inoperative backup generator, confirm the generator's oil system status is normal when its associated shaft is not sheared.
 - Confirm oil level status and oil filter status indicate NORMAL using Electrical Maintenance page 1/2 or using the sightglass.
 - Confirm associated 24-46080 or 24-46090 maintenance message for low oil pressure is not displayed or check the differential pressure indicator at the backup generator.

NOTE: Backup generator P/N 757066G (S281W401-12) and P/N 762620D (S281W401-15) and earlier require compliance with Airworthiness Directive (AD) 2000-13-04. Information for various backup generator configurations, service bulletins, PRRs and AD compliance is contained in Boeing Service Letter 777-SL-24-030-A.

OPERATIONS NOTE

- The Electrical Synoptic Display may have incomplete or missing information.
- For ELEC BACKUP SYS advisory message displayed after engine start, confirm FLIGHT CONTROL SYS status message is not displayed.

24-25-01 Backup Electrical Power System (Backup Generators, Converter and CCBs)

24-25-01B Generator Removed

Interval	Installed	Required	Procedure
C	1	0	(M)

May be inoperative provided:

- One backup generator is removed and an appropriate dispatch kit is installed.
- Both engine driven generators operate normally.
- APU generator operates normally.

NOTE 1: The airplane must also be dispatched using MMEL item 27-02-04.

NOTE 2: The airplane must have PRR 61703 installed, or the service equivalent SB 777-27-0023 installed, to remove a backup generator.

MAINTENANCE (M)

Remove inoperative backup generator and install associated dispatch kit (AMM 24-00-00/901).

- Remove the backup generator (AMM 24-25-01/401).

NOTE: The PFCS INTERFACE status message will be displayed.

- Remove the union (if installed) and check valve from the oil-in port on the removed backup generator.
- Install dispatch kit (AMM 24-25-16/401).

For PW:

- Kit J24012.

For GE:

- Kit J24013.

For RR:

- Kit J24011.

OPERATIONS NOTE

- The following weight and balance information is provided:

- Backup generator:
 - Weight: 43.5 lb (19.8 kg)
 - Balance Arm: 979.8 in.
- Dispatch kit:

For PW and RR:

- Weight: 9.1 lb (4.2 kg)
- Balance Arm: 983.6 in.

**For GE:**

- 1) Weight: 14 lb (6.3 kg)
- 2) Balance Arm: 983.6 in.
2. The Electrical Synoptic Display may have incomplete or missing information.
3. For ELEC BACKUP SYS advisory message displayed after engine start, confirm FLIGHT CONTROL SYS status message is not displayed.

24-25-01 Backup Electrical Power System (Backup Generators, Converter and CCBs)**24-25-01-01 BACKUP GEN Switch Lights****24-25-01-01-01 OFF Lights**

Interval	Installed	Required	Procedure
C	2	0	

24-25-01 Backup Electrical Power System (Backup Generators, Converter and CCBs)**24-25-01-01 BACKUP GEN Switch Lights****24-25-01-01-02 ON Lights**

Interval	Installed	Required	Procedure
C	2	0	

**24-28-01 BUS TIE Switch Lights****24-28-01-01 ISLN Lights**

Interval	Installed	Required	Procedure
C	2	0	

24-28-01 BUS TIE Switch Lights**24-28-01-02 AUTO Lights**

Interval	Installed	Required	Procedure
C	2	0	

**24-31-01 APU Battery****24-31-01A APU Required**

Interval	Installed	Required	Procedure
C	1	0	(O)

May be inoperative provided APU is started before departure and is operated for the entire flight.

OPERATIONS (O)

1. Increase flight planning fuel by 2.5% to account for the open APU air inlet door.
2. Increase flight planning fuel by 200 lb (90 kg) per hour of APU usage in flight and 535 lb (240 kg) per hour of APU usage on the ground.
3. Start the APU before departure and operate the APU continuously throughout the flight.

24-31-01 APU Battery**24-31-01B APU Not Required**

Interval	Installed	Required	Procedure
C	1	0	(O)

May be inoperative provided:

- a. Other procedures do not require use of the APU.
- b. Backup AC power system is verified to operate normally before each departure.
- c. Flight remains within 180 minutes of landing at a suitable airport.

OPERATIONS (O)

1. Dispatch is not allowed if the APU is required by other procedures. MMEL dispatch procedures for the following inoperative equipment may require APU availability.



M MEL Item	Associated Status Message
24-11-01	ELEC GEN SYS L or R IDG VALVE CLOSED L or R
24-25-01	ELEC BACKUP GEN L or R ELEC BACKUP SYS
29-11-03	HYD PUMP DEM C1 or C2
36-11-01	BLEED PRSOV L or R
36-11-02	BLEED HP SOV L or R
36-11-03	None
36-12-01	BLEED ISLN VALVE L or R
36-12-02	BLEED ISLN VALVE C
36-22-01	BLEED TEMP SENSOR L or R

- The flight must remain within 180 minutes of landing at a suitable airport.
- Verify the Backup AC electrical power system operates normally. Confirm the ELEC BACKUP SYS or ELEC BACKUP GEN L or R status and advisory messages are not displayed 30 seconds after both engines have been started.

24-31-01 APU Battery**24-31-01C APU Not Required, Battery Removed**

Interval	Installed	Required	Procedure
C	1	0	(M) (O)

May be inoperative removed provided:

- APU battery charger is deactivated.
- Other procedures do not require use of the APU.
- Backup AC power system is verified to operate normally before each departure.
- Flight remains within 180 minutes of landing at a suitable airport.

MAINTENANCE (M)

Deactivate APU Battery Charger (AMM 24-00-00/901).

- Open and collar P320 panel APU BATTERY CHARGER POWER circuit breaker.

**OPERATIONS (O)**

1. Dispatch is not allowed if the APU is required by other procedures. MMEL dispatch procedures for the following inoperative equipment may require APU availability.

MMEL Item	Associated Status Message
24-11-01	ELEC GEN SYS L or R IDG VALVE CLOSED L or R
24-25-01	ELEC BACKUP GEN L or R ELEC BACKUP SYS
29-11-03	HYD PUMP DEM C1 or C2
36-11-01	BLEED PRSOV L or R
36-11-02	BLEED HPSOV L or R
36-11-03	None
36-12-01	BLEED ISLN VALVE L or R
36-12-02	BLEED ISLN VALVE C
36-22-01	BLEED TEMP SENSOR L or R

2. The flight must remain within 180 minutes of landing at a suitable airport.
3. Verify the Backup AC electrical power system operates normally. Confirm the ELEC BACKUP SYS or ELEC BACKUP GEN L or R status and advisory messages are not displayed 30 seconds after both engines have been started.
4. The following weight and balance information is provided:

For -200/-200ER/-200LR/777F:

- A. Weight: 102.6 lb (46.5kg)
- B. Balance Arm: 1950 in.

For -300/-300ER:

- A. Weight: 102.6 lb (46.5kg)
- B. Balance Arm: 2139 in.

24-31-02 APU Battery Charger
24-31-02A APU Required

Interval	Installed	Required	Procedure
C	1	0	(M) (O)

May be inoperative provided:

- APU battery charger is deactivated.
- APU is started before departure and is operated for the entire flight.

MAINTENANCE (M)

Deactivate APU Battery Charger (AMM 24-00-00/901).

- Open and collar P320 panel APU BATTERY CHARGER POWER circuit breaker.

OPERATIONS (O)

- Increase flight planning fuel by 2.5% to account for the open APU air inlet door.
- Increase flight planning fuel by 200 lb (90 kg) per hour of APU usage in flight and 535 lb (240 kg) per hour of APU usage on the ground.
- Start the APU before departure and operate the APU continuously for the entire flight.

24-31-02 APU Battery Charger
24-31-02B APU Not Required

Interval	Installed	Required	Procedure
C	1	0	(M) (O)

May be inoperative provided:

- Other procedures do not require use of the APU.
- Backup AC power system is verified to operate normally before each departure.
- APU battery charger is deactivated.
- Flight remains within 180 minutes of landing at a suitable airport.

MAINTENANCE (M)

Deactivate APU Battery Charger (AMM 24-00-00/901).

- Open and collar P320 panel APU BATTERY CHARGER POWER circuit breaker.

**OPERATIONS (O)**

1. Dispatch is not allowed if the APU is required by other procedures. MMEL dispatch procedures for the following inoperative equipment may require APU availability.

MMEL Item	Associated Status Message
24-11-01	ELEC GEN SYS L or R IDG VALVE CLOSED L or R
24-25-01	ELEC BACKUP GEN L or R ELEC BACKUP SYS
29-11-03	HYD PUMP DEM C1 or C2
36-11-01	BLEED PRSOV L or R
36-11-02	BLEED HPSOV L or R
36-11-03	None
36-12-01	BLEED ISLN VALVE L or R
36-12-02	BLEED ISLN VALVE C
36-22-01	BLEED TEMP SENSOR L or R

2. The flight must remain within 180 minutes of landing at a suitable airport.
3. Verify the Backup AC electrical power system operates normally. Confirm the ELEC BACKUP SYS or ELEC BACKUP GEN L or R status and advisory messages are not displayed 30 seconds after both engines have been started.



24-31-03 BATTERY Switch Lights

24-31-03-01 OFF Light

Interval	Installed	Required	Procedure
C	1	0	

24-31-03 BATTERY Switch Lights

24-31-03-02 ON Light

Interval	Installed	Required	Procedure
C	1	0	

**24-41-01 External Power Systems**

Interval	Installed	Required	Procedure
C	2	0	

MAINTENANCE NOTE

Electrical power may not be available for any of the galley complexes (AMM Sections 24-09-00 and 24-20-00).

24-41-01 External Power Systems**24-41-01-01 AVAIL Lights (Flight Deck)**

Interval	Installed	Required	Procedure
C	2	0	(O)

May be inoperative provided alternate procedures are established and used.

OPERATIONS (O)

Use alternate procedures to communicate status of the power system connection.

24-41-01 External Power Systems**24-41-01-02 ON Lights (Flight Deck)**

Interval	Installed	Required	Procedure
C	2	0	(O)

May be inoperative provided alternate procedures are established and used.

OPERATIONS (O)

Use alternate procedures to communicate status of the power system connection.

24-41-01 External Power Systems**24-41-01-03 External Power Connected Lights (External Power Panel)**

Interval	Installed	Required	Procedure
C	2	0	(O)

May be inoperative provided alternate procedures are established and used.

OPERATIONS (O)

Use alternate procedures to communicate status of the power system connection.

24-41-01 External Power Systems

24-41-01-04 External Power Not In Use Lights (External Power Panel)

Interval	Installed	Required	Procedure
C	2	0	(O)

May be inoperative provided alternate procedures are established and used.

OPERATIONS (O)

Use alternate procedures to communicate status of the power system connection.

**24-41-02 Bus Power Control Unit**

Interval	Installed	Required	Procedure
C	1	0	

MAINTENANCE NOTE

Electrical power for the Ground Handling and Ground Service Busses will not be available.



24-61-01 Ground Handling Transformer Rectifier Unit

Interval	Installed	Required	Procedure
C	1	0	

MAINTENANCE NOTE

Equipment powered by the DC ground handling bus will be inoperative.



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25-11-01 Flight Crew Seats

25-11-01-01 Powered Adjustment Systems

25-11-01-01-01 Powered Lumbar

25-11-01-02 Manual Adjustment Systems

25-11-01-02-01 Recline Systems

25-11-01-02-02 Vertical Adjustments

25-11-01-02-02A *Seat Acceptable to Crewmember*

25-11-01-02-02B *Powered Adjustment Operates Normally*

25-11-01-02-03 Armrests

25-11-01-02-04 Lumbar/Thigh Supports

25-11-01-02-05 Headrests

25-11-01-02-06 Seat Pan Tilt

25-11-01-02-07 Armrest Extensions

25-11-01-02-08 Thigh Support Extensions

25-11-02 Observer Seat(s)

25-11-02-01 First Observer Seat (Including Associated Equipment)

25-11-02-01A *Passenger Seat Available*

25-11-02-01B *Second Observer Seat Available*

25-11-02-01C *Associated Safety Equipment Acceptable*

25-11-02-02 Second Observer Seat(s) (Including Associated Equipment)

25-18-01 Flotation Equipment

25-20-01 Non-Essential Equipment and Furnishings (NEF)

25-25-01 Flight Attendant Seat Assembly (Single or Dual Position)

25-25-01-01 Required Flight Attendant Seats (Passenger)

25-25-01-01A *Passengers Carried*

25-25-01-01B *No Passengers Carried*

25-25-01-02 Excess Flight Attendant Seats (Passenger)

25-25-01-03 777F

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- 25-25-02-01 Passenger Seats (includes all Configurations and Locations)
- 25-25-02-02 Positioning Controls for Taxi, Takeoff, and Landing (TTL) (Mechanical and/or Electrical)
 - 25-25-02-02A *Seat Secured*
 - 25-25-02-02B *Seat Immovable*
- 25-25-02-03 Underseat Baggage Restraining System
- 25-25-02-04 Armrests
 - 25-25-02-04-01 With Seat Positioning Controls for Taxi, Takeoff, and Landing (TTL) and/or Other Controls
 - 25-25-02-04-02 Without Seat Positioning Controls for Taxi, Takeoff, and Landing (TTL) and/or Other Controls
- 25-25-02-05 Seat Belt/Air Bag Restraint Systems
 - 25-25-02-05-01 Seat Belt/Air Bags Required by 14 CFR
 - 25-25-02-05-02 Seat Belt/Air Bags Not Required by 14 CFR
- 25-25-02-06 Delethalization Pads

25-25-03 Supernumerary Seats (777F)

- 25-25-03-01 Recline Mechanism
 - 25-25-03-01A *Seat Secured Up-Right*
 - 25-25-03-01B *Immovable Seat Back*
- 25-25-03-02 Armrests
- 25-25-03-03 Leg Rests

25-25-04 Crew/Attendant Rest Seats

- 25-25-04-01 Recline Mechanism
 - 25-25-04-01A *Seat Secured Up-Right*
 - 25-25-04-01B *Immovable Seat Back*
- 25-25-04-02 Armrests
- 25-25-04-03 Leg Rests

25-25-05 First Class Suites (option 2525E407E69)

- 25-25-05-01 Entry Door
 - 25-25-05-01A *Seat Occupied*



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	25-25-05-01B	<i>Seat Not Occupied</i>
25-25-06	Passenger Mini-Suite	
25-25-06-01	Entry Door(s)	
	25-25-06-01A	<i>Seat Occupied</i>
	25-25-06-01B	<i>Seat Not Occupied</i>
25-28-01	Storage Bin(s)/Cabin, Galley and Lavatory Storage Compartments/Closets	
	25-28-01A	<i>Secured Closed</i>
	25-28-01B	<i>Door Removed or Retractable Door Secured Open</i>
	25-28-01C	<i>Closed</i>
25-28-01-01	Multi Latch/Quarter Turn Lug Installations	
25-28-01-02	Storage Compartment Key Locks	
25-28-02	Overhead Cross Aisle Stowage (OCAS) System	
25-28-02-01	Electric Extend/Retract Function	
25-29-01	Flight Crew Rest (FCR) Door	
	25-29-01A	<i>FCR Deactivated Closed</i>
	25-29-01B	<i>Door Removed</i>
25-29-01-01	Door Lock	
25-29-02	Main Deck Crew Rest (MDCR)/Main Deck Flight Crew Rest (MDFCR) Door	
	25-29-02A	<i>MDCR/MDFCR Deactivated Closed</i>
	25-29-02B	<i>Door Removed</i>
25-29-02-01	Door Lock	
25-29-03	Overhead Flight Crew/Attendant Rest Door (Includes FSI Installation)	
25-29-03-01	Door Lock	
25-29-04	Crew/Attendant Rest Facilities (OFCR, OFAR, FCR, MDCR, MDFCR, LLAR)	
	25-29-04A	<i>Deactivated Closed</i>
	25-29-04B	<i>Procedures Do Not Require Its Use</i>
25-29-04-01	Bunks	



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25-29-05-01 Door Lock

25-30-01 Galley/Cabin Waste Container Access Door/Covers**25-41-01 Exterior Lavatory Door Ashtrays**

25-41-01-01 Passenger

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25-41-02 Lavatory Waste Container Flapper/Access Doors**25-52-01 Lower Cargo Compartment Lining (Forward and Aft)****25-53-01 Lower Cargo Handling Systems****25-53-02 Lower Cargo Restraint Systems**25-53-02A *Cargo Compartment Used*25-53-02B *Cargo Compartment(s) Remains Empty*25-53-02C *Individual Cargo Areas***25-57-01 Main Deck Cargo Handling Systems (777F)****25-58-01 Main Deck Cargo Restraint Systems (777F)**25-58-01A *Cargo Compartment Used*25-58-01B *Cargo Compartment(s) Remains Empty*25-58-01C *Individual Cargo Areas***25-59-01 Main Deck Cargo Compartment Lining (777F)****25-63-01 Emergency Evacuation Signal System****25-63-02 FASTEN SEAT BELT WHILE SEATED Placards****25-63-03 Portable Emergency Locator Transmitter (ELT)****25-64-01 Megaphones (Passenger)**25-64-01A *Passengers Carried*25-64-01B *No Passengers Carried*

25-64-01-01 Tamper Seals or Tags

25-64-02 Portable Flashlights/Flashlight Holders25-64-02A *Required by 14 CFR*

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25-64-02B *Any In Excess Of Those Required By 14 CFR*

25-64-02C *No Passenger Carried*

25-64-02-01 Tamper Seals or Tags

25-64-03 Emergency Medical Equipment

25-64-03-01 First Aid Kit (FAK) and/or Associated Equipment

25-64-03-01A *Required By 14 CFR*

25-64-03-01B *Any In Excess Of Those Required By 14 CFR*

25-64-03-01-01 Tamper Seals or Tags

25-64-03-02 Emergency Medical Kit (EMK) and/or Associated Equipment

25-64-03-02A *Required By 14 CFR*

25-64-03-02B *Any In Excess Of Those Required By 14 CFR*

25-64-03-02-01 Tamper Seals or Tags

25-64-03-03 Automated External Defibrillators (AED) and/or Associated Equipment

25-64-03-03A *Required By 14 CFR*

25-64-03-03B *Any In Excess Of Those Required By 14 CFR*

25-64-03-03-01 Tamper Seals or Tags

25-64-04 Smoke Vision System (SVS)



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**25-11-01 Flight Crew Seats****25-11-01-01 Powered Adjustment Systems**

Interval	Installed	Required	Procedure
D	2	0	(M)

Powered adjustment system(s) may be inoperative provided it is deactivated.

MAINTENANCE (M)

Deactivate the inoperative flight crew seat powered adjustment system (AMM 25-00-00/901).

1. For Captain's seat, open and collar P110 panel CAPT SEAT circuit breaker.
2. For First Officer's seat, open and collar P210 panel F/O SEAT circuit breaker.

25-11-01 Flight Crew Seats**25-11-01-01 Powered Adjustment Systems****25-11-01-01-01 Powered Lumbar**

Interval	Installed	Required	Procedure
D	2	0	(M)

May be inoperative provided:

- a. Inoperative powered lumbar adjustment system(s) is deactivated.
- b. Seat is acceptable to the affected crewmember.

MAINTENANCE (M)

Deactivate the associated powered lumbar adjustment system. (AMM 25-00-00/901).

1. For Captain's seat:
 - A. Open P110 panel CAPT SEAT circuit breaker.
 - B. Disconnect, cap and stow the main seat power harness (2A474-0603) connector (J7) from the lumbar control switch box (1A482-0479) under the Captain's seat.
 - C. Close P110 panel CAPT SEAT circuit breaker.
2. For First Officer's seat:
 - A. Open P210 panel F/O SEAT circuit breaker.



- B. Disconnect, cap and stow the main seat power harness (2A474-0603) connector (J7) from the lumbar control switch box (1A482-0480) under the First Officer's seat.
- C. Close P210 panel F/O SEAT circuit breaker.

25-11-01 Flight Crew Seats**25-11-01-02 Manual Adjustment Systems****25-11-01-02-01 Recline Systems**

Interval	Installed	Required	Procedure
A	2	0	(M)

May be inoperative provided:

- a. Seat is secured in an upright position acceptable to the affected crewmember.
- b. Repairs are made within 2 flight days.

MAINTENANCE (M)

Secure the associated seat backrest in an upright position that is acceptable to the affected flight crew member (AMM 25-00-00/901).

25-11-01 Flight Crew Seats**25-11-01-02 Manual Adjustment Systems****25-11-01-02-02 Vertical Adjustments****25-11-01-02-02A Seat Acceptable to Crewmember**

Interval	Installed	Required	Procedure
A	2	0	

May be inoperative provided:

- a. Seat is acceptable to the affected crewmember.
- b. Repairs are made within 2 flight days.

25-11-01 Flight Crew Seats**25-11-01-02 Manual Adjustment Systems****25-11-01-02-02 Vertical Adjustments****25-11-01-02-02B Powered Adjustment Operates Normally**

Interval	Installed	Required	Procedure
C	2	0	



May be inoperative provided associated powered vertical adjustment system operates normally.

25-11-01 Flight Crew Seats**25-11-01-02 Manual Adjustment Systems****25-11-01-02-03 Armrests**

Interval	Installed	Required	Procedure
B	4	0	(M)

May be inoperative provided:

- Affected armrest is stowed in the retracted position or removed.
 - Seat is acceptable to affected crewmember.
-

MAINTENANCE (M)

Remove or stow the armrest in the retracted position (AMM 25-00-00/901).

25-11-01 Flight Crew Seats**25-11-01-02 Manual Adjustment Systems****25-11-01-02-04 Lumbar/Thigh Supports**

Interval	Installed	Required	Procedure
C	4	0	

May be inoperative provided seat is acceptable to the affected crewmember.

25-11-01 Flight Crew Seats**25-11-01-02 Manual Adjustment Systems****25-11-01-02-05 Headrests**

Interval	Installed	Required	Procedure
C	2	0	

May be inoperative provided seat is acceptable to the affected crewmember.

MAINTENANCE NOTE

[For enhanced comfort flight deck seats installed:](#)

- If desired, the headrest up/down adjustment may be secured in the fully down or mid position. (AMM 25-00-00/901).



2. If desired, the headrest fore/aft adjustment may be secured in the fully aft position. (AMM 25-00-00/901).

25-11-01 Flight Crew Seats**25-11-01-02 Manual Adjustment Systems****25-11-01-02-06 Seat Pan Tilt**

Interval	Installed	Required	Procedure
C	2	0	(M)

May be inoperative provided:

- Affected seat pan is secured in the horizontal (untilted) position.
- Seat is acceptable to affected crewmember.

MAINTENANCE (M)

Secure seat in the horizontal (untilted) position using an appropriate lockout pin (AMM 25-00-00/901).

25-11-01 Flight Crew Seats**25-11-01-02 Manual Adjustment Systems****25-11-01-02-07 Armrest Extensions**

Interval	Installed	Required	Procedure
D	4	0	

May be inoperative provided seat is acceptable to affected crewmember.

25-11-01 Flight Crew Seats**25-11-01-02 Manual Adjustment Systems****25-11-01-02-08 Thigh Support Extensions**

Interval	Installed	Required	Procedure
D	4	0	

May be inoperative provided seat is acceptable to affected crewmember.

**MAINTENANCE NOTE**

1. If desired, the thigh support extension fore/aft adjustment may be secured in the fully aft position. (AMM 25-00-00/901).
2. If desired, the thigh support extension pitch adjustment may be secured in the fully down position. (AMM 25-00-00/901).

**25-11-02 Observer Seat(s)****25-11-02-01 First Observer Seat (Including Associated Equipment)****25-11-02-01A Passenger Seat Available**

Interval	Installed	Required	Procedure
A	1	0	

May be inoperative provided:

- A passenger seat in the passenger cabin is made available to an FAA inspector for the performance of official duties.
 - Repairs are made within 2 flight days.
-

25-11-02 Observer Seat(s)**25-11-02-01 First Observer Seat (Including Associated Equipment)****25-11-02-01B Second Observer Seat Available**

Interval	Installed	Required	Procedure
A	1	0	

May be inoperative provided:

- Second observer seat is available, and acceptable to an FAA inspector for the performance of official duties.
 - Repairs are made within 2 flight days.
-

25-11-02 Observer Seat(s)**25-11-02-01 First Observer Seat (Including Associated Equipment)****25-11-02-01C Associated Safety Equipment Acceptable**

Interval	Installed	Required	Procedure
A	1	0	

May be inoperative provided:

- First observer seat is acceptable to the FAA inspector for the performance of official duties.
- Required minimum safety equipment (safety belt and oxygen) is available.
- Repairs are made within 2 flight days.



NOTE 1: These provisos are intended to provide for occupancy of the above seats by an FAA inspector when the minimum safety equipment (safety belt and oxygen) is functional and the inspector determines the conditions to be acceptable.

NOTE 2: The Pilot-in-Command will determine if the minimum safety equipment is functional for the other persons authorized to occupy an observer seat(s).

25-11-02 Observer Seat(s)
25-11-02-02 Second Observer Seat(s) (Including Associated Equipment)

Interval	Installed	Required	Procedure
D	-	0	

NOTE: The Pilot-in-Command will determine if the minimum safety equipment is functional for the other persons authorized to occupy an observer seat(s).

**25-18-01 Flotation Equipment**

Interval	Installed	Required	Procedure
D	-	-	

Any in excess of those required by 14 CFR may be inoperative or missing provided required distribution is maintained.

OPERATIONS NOTE

See 14 CFR 121.339, 121.340 or other appropriate regulations.



25-20-01 Non-Essential Equipment and Furnishings (NEF)

Interval	Installed	Required	Procedure
	-	0	

May be inoperative, damaged or missing provided that the item(s) is deferred in accordance with the operator's NEF deferral program. The NEF program, procedures and processes are outlined in the operator's (insert name) Manual. (M) and (O) procedures, if required, must be available to the flight crew and included in the operator's appropriate document.

NOTE: Exterior lavatory door ash trays are not considered NEF items.

**25-25-01 Flight Attendant Seat Assembly (Single or Dual Position)****25-25-01-01 Required Flight Attendant Seats (Passenger)****25-25-01-01A Passengers Carried**

Interval	Installed	Required	Procedure
B	-	-	(M) (O)

One seat position or assembly (dual position) may be inoperative provided:

- Affected seat position or seat assembly is not occupied.
- Flight attendant(s) displaced by inoperative seat(s) occupies either an adjacent flight attendant seat or the passenger seat which is most accessible to the inoperative seat(s), so as to most effectively perform assigned duties.
- Alternate procedures are established and used as published in crewmember manuals.
- Folding type seat stows automatically or is secured in the retracted position.
- Passenger seat assigned to flight attendant is placarded FOR FLIGHT ATTENDANT USE ONLY.

NOTE 1: An automatic folding seat that will not stow automatically is considered inoperative.

NOTE 2: A seat position with an inoperative or missing restraint system is considered inoperative.

NOTE 3: Individual operators, when operating with inoperative seats, will consider the locations and combinations of seats to ensure that proximity to exits and distribution requirements of the applicable 14 CFRs are met.

NOTE 4: If one side of a dual seat assembly is inoperative and a flight attendant is displaced to the adjacent seat, the adjacent seat must operate normally.

MAINTENANCE (M)

For automatic stowing inoperative, secure seat in stowed position (AMM 25-00-00/901).

OPERATIONS (O)

The flight attendant assigned to the affected seat must occupy a passenger seat as close to, or closer than, the nearest seated passenger to the associated exit. The flight attendant must be able to reach the assigned emergency exit in essentially the same time as from the normally assigned seat, i.e., a 2 or 3 second time difference is considered "essentially the same time".

**25-25-01 Flight Attendant Seat Assembly (Single or Dual Position)****25-25-01-01 Required Flight Attendant Seats (Passenger)****25-25-01-01B No Passengers Carried**

Interval	Installed	Required	Procedure
C	-	0	(M) (O)

May be inoperative provided:

- No passengers are carried.
- A maximum of 19 persons authorized by 14 CFR for non-passenger carrying operations are carried.
- Folding type seat stows automatically or is secured in the retracted position.
- Affected seat position or seat assembly is not occupied.
- Alternate procedures are established and used.

NOTE 1: An automatic folding seat that will not stow automatically is considered inoperative.

NOTE 2: A seat position with an inoperative or missing restraint system is considered inoperative.

MAINTENANCE (M)

For automatic stowing inoperative, secure seat in stowed position (AMM 25-00-00/901).

OPERATIONS (O)

- No passengers may be carried.
- A maximum of 19 persons authorized by 14 CFR 121.583 may be carried.
- Establish alternate procedures for use when flight attendant seats are not available.

25-25-01 Flight Attendant Seat Assembly (Single or Dual Position)**25-25-01-02 Excess Flight Attendant Seats (Passenger)**

Interval	Installed	Required	Procedure
C	-	-	(M)

May be inoperative provided:

- Affected seat position or seat assembly is not occupied.
- Folding type seat stows automatically or is secured in the retracted position.

NOTE 1: An automatic folding seat that will not stow automatically is considered inoperative.



NOTE 2: A seat position with an inoperative or missing restraint system is considered inoperative.

MAINTENANCE (M)

For automatic stowing inoperative, secure seat in stowed position (AMM 25-00-00/901).

25-25-01 Flight Attendant Seat Assembly (Single or Dual Position)

25-25-01-03 777F

Interval	Installed	Required	Procedure
D	-	-	

May be inoperative provided affected seat position or seat assembly is not occupied.

25-25-02 Passenger Seats (Passenger)
25-25-02-01 Passenger Seats (includes all Configurations and Locations)

Interval	Installed	Required	Procedure
D	-	-	(M)

May be inoperative provided:

- Seat does not restrict access to any emergency exit, egress route, or main aisle.
- The affected seat(s) is blocked and placarded "DO NOT OCCUPY".

NOTE 1: A seat with an inoperative seat belt or shoulder harness is considered inoperative.

NOTE 2: Affected seat(s) may include the seat(s) behind and/or adjacent outboard seats.

NOTE 3: Inoperative seats do not affect the required number of flight attendants.

MAINTENANCE (M)

Block the affected seat (AMM 25-00-00/901).

25-25-02 Passenger Seats (Passenger)
25-25-02-02 Positioning Controls for Taxi, Takeoff, and Landing (TTL) (Mechanical and/or Electrical)
25-25-02-02A Seat Secured

Interval	Installed	Required	Procedure
D	-	-	(M)

May be inoperative and seat occupied provided seat is secured in the taxi, takeoff, and landing (TTL) position.

MAINTENANCE (M)

Secure the associated seat in the taxi, takeoff, and landing (TTL) position (AMM 25-00-00/901).

**25-25-02 Passenger Seats (Passenger)****25-25-02-02 Positioning Controls for Taxi, Takeoff, and Landing (TTL) (Mechanical and/or Electrical)****25-25-02-02B Seat Immovable**

Interval	Installed	Required	Procedure
D	-	-	

May be inoperative and seat occupied provided seat is immovable in the taxi, takeoff, and landing (TTL) position.

25-25-02 Passenger Seats (Passenger)**25-25-02-03 Underseat Baggage Restraining System**

Interval	Installed	Required	Procedure
C	-	-	(O)

May be inoperative provided:

- Baggage is not stowed under seat with inoperative restraining system.
 - Associated seat is placarded "DO NOT STOW BAGGAGE UNDER THIS SEAT".
 - Procedures are established to alert cabin crew of inoperative restraining system.
-

OPERATIONS (O)

Establish procedures to identify to the cabin crew the seats with inoperative underseat baggage restraining systems.

25-25-02 Passenger Seats (Passenger)**25-25-02-04 Armrests****25-25-02-04-01 With Seat Positioning Controls for Taxi, Takeoff, and Landing (TTL) and/or Other Controls**

Interval	Installed	Required	Procedure
D	-	-	(M)

May be inoperative or missing and seat occupied provided:

- Armrest does not restrict access to any emergency exit, egress route, or main aisle.

- b. If armrest with seat control is missing or removed, seat is secured in taxi, takeoff, and landing (TTL) position.

MAINTENANCE (M)

If required, secure the associated seat in the taxi, takeoff, and landing (TTL) position (AMM 25-00-00/901).

25-25-02 Passenger Seats (Passenger)
25-25-02-04 Armrests
25-25-02-04-02 Without Seat Positioning Controls for Taxi, Takeoff, and Landing (TTL) and/or Other Controls

Interval	Installed	Required	Procedure
D	-	-	

May be inoperative or missing and seat occupied provided it does not restrict access to any emergency exit, egress route, or main aisle.

25-25-02 Passenger Seats (Passenger)
25-25-02-05 Seat Belt/Air Bag Restraint Systems

25-25-02-05-01 Seat Belt/Air Bags Required by 14 CFR

Interval	Installed	Required	Procedure
D	-	-	(M)

May be inoperative provided affected seat is blocked and placarded "DO NOT OCCUPY".

MAINTENANCE (M)

Block the affected seat (AMM 25-00-00/901).

25-25-02 Passenger Seats (Passenger)
25-25-02-05 Seat Belt/Air Bag Restraint Systems

25-25-02-05-02 Seat Belt/Air Bags Not Required by 14 CFR

Interval	Installed	Required	Procedure
D	-	-	(M)



May be inoperative or disconnected provided seat belt operates normally.

MAINTENANCE (M)

Seat belt/air bags which are not required may be disconnected. (AMM 25-00-00/901).

25-25-02 Passenger Seats (Passenger)

25-25-02-06 Delethalization Pads

Interval	Installed	Required	Procedure
D	-	-	(M)

May be inoperative or missing provided affected seat is blocked and placarded "DO NOT OCCUPY".

MAINTENANCE (M)

Block the affected seat (AMM 25-00-00/901).



25-25-03 Supernumerary Seats (777F)

Interval	Installed	Required	Procedure
D	-	0	(M)

May be inoperative provided:

- Seat is not occupied.
- Affected seat(s) is blocked and placarded DO NOT OCCUPY.

NOTE: A seat with an inoperative seat belt is considered inoperative.

MAINTENANCE (M)

Block the affected seat (AMM 25-00-00/901).

25-25-03 Supernumerary Seats (777F)**25-25-03-01 Recline Mechanism****25-25-03-01A Seat Secured Up-Right**

Interval	Installed	Required	Procedure
D	-	-	(M)

May be inoperative and seat occupied provided seat is secured in the full up-right position.

MAINTENANCE (M)

Secure the associated seat in the upright position (AMM 25-00-00/901).

25-25-03 Supernumerary Seats (777F)**25-25-03-01 Recline Mechanism****25-25-03-01B Immovable Seat Back**

Interval	Installed	Required	Procedure
D	-	-	

May be inoperative and seat occupied provided seat back is immovable in the full up-right position.

**25-25-03 Supernumerary Seats (777F)****25-25-03-02 Armrests**

Interval	Installed	Required	Procedure
D	-	0	(M)

May be inoperative provided armrest is stowed in retracted position or removed.

MAINTENANCE (M)

Stow an inoperative armrest in the retracted position or remove the armrest.
(AMM 25-00-00/901).

25-25-03 Supernumerary Seats (777F)**25-25-03-03 Leg Rests**

Interval	Installed	Required	Procedure
D	-	0	(M)

May be inoperative and seat occupied provided leg rest is stowed in retracted position or removed.

MAINTENANCE (M)

Stow an inoperative leg rest in the retracted position or remove the leg rest. (AMM 25-00-00/901).

**25-25-04 Crew/Attendant Rest Seats**

Interval	Installed	Required	Procedure
D	-	0	(M)

May be inoperative provided:

- Seat is not occupied.
- Affected seat(s) is blocked and placarded DO NOT OCCUPY.

NOTE: A seat with an inoperative seat belt is considered inoperative.

MAINTENANCE (M)

Block the affected seat (AMM 25-00-00-040-810).

OPERATIONS NOTE

For augmented flight crew operations, flight crew duty periods should be adjusted appropriately (14 CFR 117).

25-25-04 Crew/Attendant Rest Seats

25-25-04-01 Recline Mechanism**25-25-04-01A Seat Secured Up-Right**

Interval	Installed	Required	Procedure
D	-	-	(M)

May be inoperative and seat occupied provided seat is secured in the full up-right position.

MAINTENANCE (M)

Secure the associated seat in the upright position (AMM 25-00-00-040-810).

OPERATIONS NOTE

For augmented flight crew operations, flight crew duty periods should be adjusted appropriately (14 CFR 117).

**25-25-04 Crew/Attendant Rest Seats**

25-25-04-01 Recline Mechanism**25-25-04-01B Immovable Seat Back**

Interval	Installed	Required	Procedure
D	-	-	

May be inoperative and seat occupied provided seat back is immovable in the full up-right position.

OPERATIONS NOTE

For augmented flight crew operations, flight crew duty periods should be adjusted appropriately (14 CFR 117).

25-25-04 Crew/Attendant Rest Seats

25-25-04-02 Armrests

Interval	Installed	Required	Procedure
D	-	0	(M)

May be inoperative provided armrest is stowed in retracted position or removed.

MAINTENANCE (M)

Stow an inoperative armrest in the retracted position or remove the armrest (AMM 25-00-00-040-810).

25-25-04 Crew/Attendant Rest Seats

25-25-04-03 Leg Rests

Interval	Installed	Required	Procedure
D	-	0	(M)

May be inoperative and seat occupied provided:

- Leg rest is stowed in the retracted position.
- Seat is acceptable to the affected crewmember.

MAINTENANCE (M)

Stow an inoperative leg rest in the retracted position (AMM 25-00-00/901).

**25-25-05 First Class Suites (option 2525E407E69)**

Interval	Installed	Required	Procedure
D	-	0	

May be inoperative provided:

- Affected seat is not occupied.
- Affected seat is considered inoperative

NOTE: The airplane must also be dispatched using MMEL item 25-25-02-01.

25-25-05 First Class Suites (option 2525E407E69)

25-25-05-01 Entry Door**25-25-05-01A Seat Occupied**

Interval	Installed	Required	Procedure
D	-	0	(M)

One or more may be inoperative and seat occupied provided the affected door(s) is secured in the fully open position or removed.

MAINTENANCE (M)

Secure the associated suite entry door in the fully open position (AMM DDG 25-00-00/901).

25-25-05 First Class Suites (option 2525E407E69)

25-25-05-01 Entry Door**25-25-05-01B Seat Not Occupied**

Interval	Installed	Required	Procedure
D	-	0	

One or more may be inoperative provided:

- Affected seat is not occupied.
- Affected seat is considered inoperative

NOTE: The airplane must also be dispatched using MMEL item 25-25-02-01.

**25-25-06 Passenger Mini-Suite**

25-25-06-01 Entry Door(s)**25-25-06-01A Seat Occupied**

Interval	Installed	Required	Procedure
D	-	0	(M)

One or more may be inoperative and seat occupied provided that the affected door(s) is secured in the fully open position or removed.

MAINTENANCE (M)

Secure the associated suite entry door(s) in the fully open position (AMM DDG 25-00-00 /901).

25-25-06 Passenger Mini-Suite

25-25-06-01 Entry Door(s)**25-25-06-01B Seat Not Occupied**

Interval	Installed	Required	Procedure
D	-	0	

One or more may be inoperative provided:

- Affected seat is not occupied.
- Affected seat is considered inoperative

25-28-01 Storage Bin(s)/Cabin, Galley and Lavatory Storage Compartments/Closets
25-28-01A Secured Closed

Interval	Installed	Required	Procedure
C	-	-	(M)

May be inoperative provided:

- Procedures are established to secure the affected bin, compartment or closet in the closed position,
- Associated bin, compartment or closet is prominently placarded DO NOT USE,
- Any emergency equipment located in affected bin, compartment or closet is considered inoperative.
- Affected bin, compartment or closet is not used for storage of any items except for those permanently affixed.

NOTE 1: For overhead bins, if no partitions are installed, the entire overhead bin is considered inoperative.

NOTE 2: Proviso is not intended to preclude crewmember inspections.

MAINTENANCE (M)

Establish procedures to secure and placard inoperative bins, compartments or closets closed (AMM 25-00-00-040-811).

OPERATIONS NOTE

For emergency equipment located in affected compartment, the airplane must also be dispatched using the applicable emergency equipment MMEL item.

25-28-01 Storage Bin(s)/Cabin, Galley and Lavatory Storage Compartments/Closets
25-28-01B Door Removed or Retractable Door Secured Open

Interval	Installed	Required	Procedure
C	-	-	(M) (O)

May be inoperative provided:

- For non-retractable doors, affected door is removed.
- For retractable doors, affected door is removed or secured in the retracted (fully open) position.
- Affected bin, compartment or closet is not used for storage of any items, except those permanently affixed.
- Affected bin, compartment or closet is prominently placarded DO NOT USE.



- e. Procedures are established and used to alert crew members and passengers of inoperative bins, compartments or closets.
- f. Passengers are briefed that affected bin, compartment or closet is not to be used.

NOTE 1: For overhead bins, if no partitions are installed, the entire overhead bin is considered inoperative.

NOTE 2: Any emergency equipment located in the affected bin, compartment or closet (permanently affixed) is available for use.

MAINTENANCE (M)

Remove affected door and placard associated bin, compartment or closet (AMM 25-00-00-040-811).

OPERATIONS (O)

Ensure crew members and passengers are briefed that associated bin, compartment or closet is not used.

25-28-01 Storage Bin(s)/Cabin, Galley and Lavatory Storage Compartments/Closets

25-28-01C Closed

Interval	Installed	Required	Procedure
C	-	-	

May be inoperative provided:

- a. Affected bin, compartment, or closet is closed.
- b. Affected bin, compartment, or closet is prominently placarded "DO NOT USE".
- c. Any emergency equipment located in affected bin, compartment, or closet is considered inoperative.
- d. Location placarding for any emergency equipment stored in affected bin, compartment, or closet is removed or obscured.

NOTE: Use of this proviso may be dependent upon an operator's aircraft security program, as appropriate.



25-28-01 Storage Bin(s)/Cabin, Galley and Lavatory Storage Compartments/Closets

25-28-01-01 Multi Latch/Quarter Turn Lug Installations

Interval	Installed	Required	Procedure
C	-	-	

- One latch/lug per compartment may be inoperative provided:
- a. Remaining latch(es)/lug(s) on affected compartments operates normally.
 - b. If affected compartment is used for a galley cart, the cart remains empty.
-

25-28-01 Storage Bin(s)/Cabin, Galley and Lavatory Storage Compartments/Closets

25-28-01-02 Storage Compartment Key Locks

Interval	Installed	Required	Procedure
D	-	0	(M)

- May be inoperative provided:
- a. Storage compartment is unlocked.
 - b. Doors can be secured by other means.
-

MAINTENANCE (M)

Establish procedures to secure inoperative doors (AMM 25-00-00-040-811).

**25-28-02 Overhead Cross Aisle Stowage (OCAS) System**

Interval	Installed	Required	Procedure
D	-	0	(M)

May be inoperative provided affected OCAS is deactivated closed.

MAINTENANCE (M)

Deactivate the associated OCAS closed (AMM 25-00-00/901).

1. Remove all items, except those permanently affixed, from the OCAS.
2. Stow the OCAS closed.

25-28-02 Overhead Cross Aisle Stowage (OCAS) System

25-28-02-01 Electric Extend/Retract Function

Interval	Installed	Required	Procedure
D	-	0	(M)

May be inoperative provided associated electric motor is deactivated.

MAINTENANCE (M)

NOTE: Manual override storage mechanism may be used.

Deactivate the associated OCAS electric motor by opening the associated OCAS circuit breaker(s) on the associated Galley Electrical Control Panel (AMM 25-00-00/901).

25-29-01 Flight Crew Rest (FCR) Door

25-29-01A FCR Deactivated Closed

Interval	Installed	Required	Procedure
C	-	0	

May be inoperative provided associated FCR is considered inoperative.

NOTE: The airplane must also be dispatched using MMEL item 25-29-04.

25-29-01 Flight Crew Rest (FCR) Door

25-29-01B Door Removed

Interval	Installed	Required	Procedure
C	-	0	(M)

May be inoperative provided associated FCR door is removed.

MAINTENANCE (M)

Remove the inoperative FCR door.

OPERATIONS NOTE

For augmented flight crew operations, flight crew duty periods should be adjusted appropriately (14 CFR 117).

25-29-01 Flight Crew Rest (FCR) Door

25-29-01-01 Door Lock

Interval	Installed	Required	Procedure
C	-	0	(M)

May be inoperative provided:

- Associated FCR door lock is deactivated in the unlocked position.
- Associated FCR door opens and closes normally.

MAINTENANCE (M)

Deactivate the FCR door lock in the unlocked position (AMM 25-00-00/901).

NOTE: If installed, the Lock Assembly Override Feature may be used to deactivate the door lock.

**25-29-02 Main Deck Crew Rest (MDCR)/Main Deck Flight Crew Rest (MDFCR) Door**

25-29-02A MDCR/MDFCR Deactivated Closed

Interval	Installed	Required	Procedure
C	-	0	

May be inoperative provided MDCR/MDFCR is considered inoperative.

NOTE: The airplane must also be dispatched using MMEL item 25-29-04.

25-29-02 Main Deck Crew Rest (MDCR)/Main Deck Flight Crew Rest (MDFCR) Door

25-29-02B Door Removed

Interval	Installed	Required	Procedure
C	-	0	(M)

May be inoperative provided MDCR/MDFCR door is removed.

MAINTENANCE (M)

Remove the inoperative MDCR/MDFCR door.

OPERATIONS NOTE

For augmented flight crew operations, flight crew duty periods should be adjusted appropriately (see 14 CFR 117).

25-29-02 Main Deck Crew Rest (MDCR)/Main Deck Flight Crew Rest (MDFCR) Door

25-29-02-01 Door Lock

Interval	Installed	Required	Procedure
C	-	0	(M)

May be inoperative provided:

- Associated MDCR/MDFCR door lock is deactivated in the unlocked position.
 - Associated MDCR/MDFCR door opens and closes normally.
-

**MAINTENANCE (M)**

Deactivate the MDCR/MDFCR door lock in the unlocked position (AMM 25-00-00/901).

NOTE: If installed, the Lock Assembly Override Feature may be used to deactivate the door lock.

**25-29-03 Overhead Flight Crew/Attendant Rest Door (Includes FSI Installation)**

Interval	Installed	Required	Procedure
C	-	0	

May be inoperative provided associated flightcrew/attendant rest facility is considered inoperative.

NOTE 1: The overhead crew rest installation manufactured by Flight Structures, Inc. (FSI) is not a Boeing system. Information applicable to the FSI installation is not provided.

NOTE 2: The airplane must also be dispatched using MMEL item 25-29-04.

25-29-03 Overhead Flight Crew/Attendant Rest Door (Includes FSI Installation)

25-29-03-01 Door Lock

Interval	Installed	Required	Procedure
C	-	0	(M)

May be inoperative provided:

- Associated OFCR/OFAR door lock is deactivated in the unlocked position.
- Associated OFCR/OFAR door opens and closes normally.

NOTE: The overhead crew rest installation manufactured by Flight Structures, Inc. (FSI) is not a Boeing system. Information applicable to the FSI installation is not provided.

MAINTENANCE (M)

Deactivate the OFCR/OFAR door lock in the unlocked position (AMM 25-00-00/901).

- For keyed lock installed, the lock assembly override may be used.
- For keyless lock installed, the lock assembly override may be used, or the lock assembly may be removed.

25-29-04 Crew/Attendant Rest Facilities (OFCR, OFAR, FCR, MDCR, MDFCR, LLAR)

25-29-04A Deactivated Closed

Interval	Installed	Required	Procedure
C	-	0	(M) (O)

May be inoperative provided:

- Associated flightcrew/attendant rest facility is deactivated closed.

For Flight Crew Rest (OFCR, FCR, MDFCR) inoperative:

- Appropriate adjustments to flightcrew FDP times are applied.

NOTE: This provision is not intended to prohibit flightcrew/attendant rest facility inspections by crewmembers.

MAINTENANCE (M)

Deactivate the associated crew/attendant rest closed AMM 25-00-00/901).

- Remove all baggage and personal items from the associated crew/attendant rest.

NOTE: Blankets, pillows and other items normally used in the crew rest may remain.

- Close the associated crew/attendant rest entrance door.

OPERATIONS (O)

- The crew/attendant rest is not available for inflight use. Remove personal items and do not enter except for inspections by crewmembers.
- For augmented flight crew operations, flight crew duty periods should be adjusted appropriately (14 CFR 117).

25-29-04 Crew/Attendant Rest Facilities (OFCR, OFAR, FCR, MDCR, MDFCR, LLAR)

25-29-04B Procedures Do Not Require Its Use

Interval	Installed	Required	Procedure
C	-	0	

May be inoperative provided procedures do not require their use.

**25-29-04 Crew/Attendant Rest Facilities (OFCR, OFAR, FCR, MDCR, MDFCR, LLAR)**

25-29-04-01 Bunks

Interval	Installed	Required	Procedure
C	-	0	

May be inoperative provided bunk is not occupied.

NOTE: A bunk with an inoperative or missing restraint system is considered inoperative.

OPERATIONS NOTE

1. For augmented flight crew operations, flight crew duty periods should be adjusted appropriately (14 CFR 117).

**25-29-05 Lower Lobe Attendant Rest (LLAR) Entrance Enclosure Door**

Interval	Installed	Required	Procedure
C	1	0	

May be inoperative provided LLAR is considered inoperative.

NOTE: The airplane must also be dispatched using MMEL item 25-29-04.

25-29-05 Lower Lobe Attendant Rest (LLAR) Entrance Enclosure Door

25-29-05-01 Door Lock

Interval	Installed	Required	Procedure
C	1	0	(M)

May be inoperative provided:

- LLAR entrance enclosure door lock is deactivated in the unlocked position.
 - LLAR entrance enclosure door opens and closes normally.
-

MAINTENANCE (M)

Deactivate the LLAR entrance enclosure door lock in the unlocked position (AMM 25-00-00/901).

NOTE: If installed, the Lock Assembly Override Feature may be used to deactivate the door lock.

**25-30-01 Galley/Cabin Waste Container Access Door/Covers**

Interval	Installed	Required	Procedure
C	-	-	(M) (O)

May be inoperative provided:

- Associated waste container is empty and access is secured to prevent waste introduction into the waste container.
 - Procedures are established to ensure that sufficient galley/cabin waste containers are available to accommodate all waste that may be generated on a flight.
-

MAINTENANCE (M)

Empty the receptacle and secure the access door/cover in a manner that will prevent waste from being deposited in the waste compartment/receptacle (AMM 25-00-00/901).

OPERATIONS (O)

Establish procedures to ensure sufficient galley waste containers are available to accommodate all the waste that may be generated on a flight.

**25-41-01 Exterior Lavatory Door Ashtrays****25-41-01-01 Passenger**

Interval	Installed	Required	Procedure
A	-	-	

Fifty percent may be inoperative or missing provided repairs are made within 10 calendar days.

25-41-01 Exterior Lavatory Door Ashtrays**25-41-01-02 777F**

Interval	Installed	Required	Procedure
A	1	0	

May be inoperative or missing provided repair is made within 10 calendar days.

**25-41-02 Lavatory Waste Container Flapper/Access Doors**

Interval	Installed	Required	Procedure
C	-	-	(M)

May be inoperative provided:

- Associated waste container is empty and access is secured to prevent waste introduction into the waste container.
- Lavatory is used only by crewmembers.
- Associated lavatory entrance door is locked closed and placarded, INOPERATIVE - DO NOT ENTER.

NOTE: These provisions are not intended to prohibit lavatory use or inspections by crewmembers.

MAINTENANCE (M)

Empty waste container, secure container access and close and lock lavatory door (AMM 25-00-00/901).

- Empty the associated lavatory waste receptacle.
- Secure the flapper/access door in a manner that will prevent waste from being deposited into the receptacle.
- Close and lock the associated lavatory door.

OPERATIONS NOTE

Lavatory door should remain locked closed, except for use or inspection by crewmembers.

25-52-01 Lower Cargo Compartment Lining (Forward and Aft)

Interval	Installed	Required	Procedure
C	-	-	(O)

May be damaged or missing provided procedures are established and used to verify the associated cargo compartment remains empty or contains only ballast, empty cargo handling equipment (ballast may be loaded in ULDs), or fly away kits.

NOTE: Operator MELs must define which items are approved for inclusion in the fly away kits and which materials can be used as ballast.

NOTE: Lower Cargo Compartment Linings include: sloping sidewall liners, all endwalls, decompression panels, ceiling liners and mounted equipment, all cargo door liners, vertical sidewall liners including cheek panels and mounted equipment, waste tank enclosure panels, access panels, ceiling light assemblies, the main equipment center bulkhead liner, firestops, bilge curtains, bulkhead liners, endwall/enclosure liners and panels. In addition, some of the cargo floor panels and walkways are included in this item because they are containment barriers for smoke and fire suppression. The forward cargo compartment floor panels and walkways that are containment barriers are located at the aft end of the forward compartment. The aft compartment floor panels and walkways that are containment barriers are located at the aft end of the aft compartment, just forward of the bulk cargo compartment. All of the bulk cargo compartment floor panels are containment barriers.

OPERATIONS (O)

The associated cargo compartment must remain empty, except for ballast, empty cargo containers (ballast may be loaded in ULDs), fly away kits, pallets, and cargo restraint components.

NOTE: Both the aft and bulk cargo compartments must remain empty when there are damaged or missing linings in either the aft or the bulk cargo compartment.

**25-53-01 Lower Cargo Handling Systems**

Interval	Installed	Required	Procedure
D	-	0	

NOTE: Any portion of the system which operates normally may be used.

25-53-02 Lower Cargo Restraint Systems**25-53-02A Cargo Compartment Used**

Interval	Installed	Required	Procedure
A	-	-	(M)

May be inoperative or missing provided:

- Approved cargo loading limits are observed. The only source documents are the Type Certificate (TC), Supplemental Type Certificate (STC), Airplane Flight Manual (AFM), Airplane Flight Manual Supplement (AFMS) or the TC/STC Weight and Balance Manual (WBM).
- Repairs are made within 120 consecutive calendar-days.

MAINTENANCE (M)

Use acceptable cargo loading limits from an approved source.

- Refer to the Weight and Balance Manual for cargo loading limits.
- If the number of inoperative or missing restraint components exceeds the allowance in the Weight and Balance Manual, the affected cargo compartment cannot be used. Refer to MMEL Item 25-53-02B.
- Deactivation procedures may or may not be required. Develop deactivation procedures depending on system configuration.

25-53-02 Lower Cargo Restraint Systems**25-53-02B Cargo Compartment(s) Remains Empty**

Interval	Installed	Required	Procedure
A	-	-	

May be inoperative or missing provided:

- Cargo compartment remains empty.
- Repairs are made within 120 consecutive calendar-days.

25-53-02 Lower Cargo Restraint Systems**25-53-02C Individual Cargo Areas**

Interval	Installed	Required	Procedure
A	-	-	

Individual cargo areas may be inoperative provided:

- Aircraft is operated in accordance with Original Equipment Manufacturer (OEM) W&B source document.



b. Repairs are made within 120 consecutive calendar-days.



25-57-01 Main Deck Cargo Handling Systems (777F)

Interval	Installed	Required	Procedure
D	-	0	

NOTE: Any portion of the system which operates normally may be used.

**25-58-01 Main Deck Cargo Restraint Systems (777F)****25-58-01A Cargo Compartment Used**

Interval	Installed	Required	Procedure
A	-	-	(M)

May be inoperative or missing provided:

- Approved cargo loading limits are observed. The only source documents are the Type Certificate (TC), Supplemental Type Certificate (STC), Airplane Flight Manual (AFM), Airplane Flight Manual Supplement (AFMS) or the TC/STC Weight and Balance Manual (WBM).
 - Repairs are made within 120 consecutive calendar days.
-

MAINTENANCE (M)

Use acceptable cargo loading limits from an approved source (AMM 25-00-00/901).

- Refer to the Weight and Balance Manual for cargo loading limits.
- If the number of inoperative or missing restraint components exceeds the allowance in the Weight and Balance Manual, the affected cargo compartment cannot be used. Refer to MMEL Item 25-58-01B.
- Deactivation procedures may or may not be required. Develop deactivation procedures depending on system configuration.

25-58-01 Main Deck Cargo Restraint Systems (777F)**25-58-01B Cargo Compartment(s) Remains Empty**

Interval	Installed	Required	Procedure
A	-	-	

May be inoperative or missing provided:

- Cargo compartment remains empty.
 - Repairs are made within 120 consecutive calendar days.
-

25-58-01 Main Deck Cargo Restraint Systems (777F)**25-58-01C Individual Cargo Areas**

Interval	Installed	Required	Procedure
A	-	-	

Individual cargo areas may be inoperative provided:

- Aircraft is operated in accordance with Original Equipment Manufacturer (OEM) W&B source document.



b. Repairs are made within 120 consecutive calendar-days.

**25-59-01 Main Deck Cargo Compartment Lining (777F)**

Interval	Installed	Required	Procedure
C	-	-	(O)

May be damaged or missing provided procedures are established and used to verify the main deck cargo compartment remains empty or contains only ballast, empty cargo handling equipment (ballast may be loaded in ULDs), or fly away kits.

NOTE: Operator MELs must define which items are approved for inclusion in the fly away kits and which materials can be used as ballast.

NOTE: Main Deck Cargo Compartment Linings include: Ceilings, sidewalls, cover strips, endwalls, dado panels, window reveals, insulation, decompression panel, cargo handling control panels, electrical panels, cargo door edge trim, cargo door lining, smoke detector assemblies, markers, liner support rails and brackets, fabric parts, ECS duct and ceiling light assemblies.

OPERATIONS (O)

The main deck cargo compartment must remain empty, except for ballast, empty cargo containers (ballast may be loaded in ULDs), fly away kits, pallets, and cargo restraint components.



25-63-01 Emergency Evacuation Signal System

Interval	Installed	Required	Procedure
C	1	0	(O)

May be inoperative provided alternate procedures are used as primary means of initiating an emergency evacuation.

OPERATIONS (O)

Establish alternate procedures to initiate an emergency evacuation. Prior to flight, the pilot in command should brief cabin crew members on the alternate procedures.



25-63-02 FASTEN SEAT BELT WHILE SEATED Placards

Interval	Installed	Required	Procedure
C	-	-	

One or more signs or placards may be illegible or missing provided a legible sign or placard is visible from each occupied passenger seat.

**25-63-03 Portable Emergency Locator Transmitter (ELT)**

Interval	Installed	Required	Procedure
D	-	-	

Any in excess of those required by 14 CFR may be inoperative or missing.

NOTE: Dispatch relief for the Fixed type ELT's is in MMEL item 23-24-02.

OPERATIONS NOTE

See 14-CFR 91.207, 121.353, 121.339 or other appropriate regulations.

**25-64-01 Megaphones (Passenger)****25-64-01A Passengers Carried**

Interval	Installed	Required	Procedure
D	-	-	

Any in excess of those required by 14 CFR may be inoperative or removed provided:

- Inoperative megaphone remains in a certified location until removed from the aircraft at the next suitable maintenance facility.
- Location placarding is removed or obscured.
- Required distribution is maintained.

OPERATIONS NOTE

See 14 CFR 121.309 or other appropriate regulations.

25-64-01 Megaphones (Passenger)**25-64-01B No Passengers Carried**

Interval	Installed	Required	Procedure
C	-	0	(O)

May be inoperative or missing provided:

- No passengers are carried.
- A maximum of 19 persons authorized by 14 CFR for non-passenger carrying operations are carried.
- Alternate procedures are established and used.

OPERATIONS (O)

- No passengers may be carried.
- A maximum of 19 persons authorized by 14 CFR 121.583 may be carried.
- Establish alternate procedures for use when megaphones are not available.

25-64-01 Megaphones (Passenger)**25-64-01-01 Tamper Seals or Tags**

Interval	Installed	Required	Procedure
C	-	-	(O)



May be inoperative, damaged, or missing provided proper installation and operation is verified at each preflight.

OPERATIONS (O)

Before each departure, verify affected item(s) is properly installed and operate(s) normally.

**25-64-02 Portable Flashlights/Flashlight Holders****25-64-02A Required by 14 CFR**

Interval	Installed	Required	Procedure
C	-	-	

May be inoperative or removed provided:

- Crewmember assigned to the affected position has an equivalent operative flashlight readily available.
 - Inoperative flashlight remains in a certified location or is removed from the aircraft.
 - Location placarding is removed or obscured.
-

25-64-02 Portable Flashlights/Flashlight Holders**25-64-02B Any In Excess Of Those Required By 14 CFR**

Interval	Installed	Required	Procedure
D	-	-	

Any in excess of those required by 14 CFR may be inoperative or removed provided:

- Inoperative flashlight remains in a certified location until removed from the aircraft at the next suitable maintenance facility.
 - Location placarding is removed or obscured.
-

25-64-02 Portable Flashlights/Flashlight Holders**25-64-02C No Passenger Carried**

Interval	Installed	Required	Procedure
C	-	0	(O)

May be inoperative or missing provided:

- No passengers are carried.
 - A maximum of 19 persons authorized by 14 CFR for non-passenger carrying operations are carried.
 - Alternate procedures are established and used.
-

OPERATIONS (O)

- No passengers may be carried.



2. A maximum of 19 persons authorized by 14 CFR 121.583 may be carried.
3. Establish alternate procedures for use when flashlights are not available.

25-64-02 Portable Flashlights/Flashlight Holders**25-64-02-01 Tamper Seals or Tags**

Interval	Installed	Required	Procedure
C	-	-	(O)

May be inoperative, damaged, or missing provided proper installation and operation is verified at each preflight.

OPERATIONS (O)

Before each departure, verify affected item(s) is properly installed and operate(s) normally.



25-64-03 Emergency Medical Equipment
25-64-03-01 First Aid Kit (FAK) and/or Associated Equipment
25-64-03-01A Required By 14 CFR

Interval	Installed	Required	Procedure
A	-	-	(O)

If more than one is required by 14 CFR, only one of the required first aid kits may be incomplete, missing or inoperative provided:

- FAK is resealed in a manner that will identify it as a unit that cannot be mistaken for a fully serviceable unit.
- Location placarding is removed or obscured.
- Repairs or replacements are made within one flight.

NOTE: Medical equipment installed in the aircraft as part of an EMS operation is not considered part of the normal complement of equipment. No MMEL relief applies to that equipment and 14 CFR maintenance and inspection requirements do not apply.

OPERATIONS (O)

NOTE: See 14 CFR 121.803 or other appropriate regulations.

Each operator must develop their own procedures based on the emergency medical equipment installed on their airplanes.

25-64-03 Emergency Medical Equipment
25-64-03-01 First Aid Kit (FAK) and/or Associated Equipment
25-64-03-01B Any In Excess Of Those Required By 14 CFR

Interval	Installed	Required	Procedure
D	-	-	

Any in excess of those required by 14 CFR may be incomplete, inoperative or removed.

OPERATIONS NOTE

See 14 CFR 121.803 or other appropriate regulations.

25-64-03 Emergency Medical Equipment
25-64-03-01 First Aid Kit (FAK) and/or Associated Equipment
25-64-03-01-01 Tamper Seals or Tags

Interval	Installed	Required	Procedure
C	-	-	(O)

May be inoperative, damaged, or missing provided proper FAK servicing is verified at each preflight.

OPERATIONS (O)

Before each departure, verify FAK has been properly serviced.

25-64-03 Emergency Medical Equipment
25-64-03-02 Emergency Medical Kit (EMK) and/or Associated Equipment

25-64-03-02A Required By 14 CFR

Interval	Installed	Required	Procedure
A	-	0	(O)

May be incomplete, missing or inoperative provided:

- a. EMK is resealed in a manner that will identify it as a unit that cannot be mistaken for a fully serviceable unit.
- b. Location placarding is removed or obscured.
- c. Repairs or replacements are made within one flight.

NOTE: Medical equipment installed in the aircraft as part of an EMS operation is not considered part of the normal complement of equipment. No MMEL relief applies to that equipment and 14 CFR maintenance and inspection requirements do not apply.

OPERATIONS (O)

NOTE: See 14 CFR 121.803 or other appropriate regulations.

Each operator must develop their own procedures based on the emergency medical equipment installed on their airplanes.

**25-64-03 Emergency Medical Equipment****25-64-03-02 Emergency Medical Kit (EMK) and/or Associated Equipment**

25-64-03-02B Any In Excess Of Those Required By 14 CFR

Interval	Installed	Required	Procedure
D	-	-	

Any in excess of those required by 14 CFR may be incomplete, inoperative or removed.

OPERATIONS NOTE

See 14 CFR 121.803 or other appropriate regulations.

25-64-03 Emergency Medical Equipment**25-64-03-02 Emergency Medical Kit (EMK) and/or Associated Equipment****25-64-03-02-01 Tamper Seals or Tags**

Interval	Installed	Required	Procedure
C	-	-	(O)

May be inoperative, damaged, or missing provided proper EMK servicing is verified at each preflight.

OPERATIONS (O)

Before each departure, verify EMK has been properly serviced.

25-64-03 Emergency Medical Equipment**25-64-03-03 Automated External Defibrillators (AED) and/or Associated Equipment**

25-64-03-03A Required By 14 CFR

Interval	Installed	Required	Procedure
A	-	0	(O)

May be incomplete, missing or inoperative provided:

- AED is resealed in a manner that will identify it as a unit that cannot be mistaken for a fully serviceable unit.
- Location placarding is removed or obscured.



- c. Repairs or replacements are made within one flight.

NOTE: Medical equipment installed in the aircraft as part of an Emergency Medical Service (EMS) operation is not considered part of the normal complement of equipment. No MMEL relief applies to that equipment and 14 CFR maintenance and inspection requirements do not apply.

OPERATIONS (O)

NOTE: See 14 CFR 121.803 or other appropriate regulations.

Each operator must develop their own procedures based on the emergency medical equipment installed on their airplanes.

25-64-03 Emergency Medical Equipment

25-64-03-03 Automated External Defibrillators (AED) and/or Associated Equipment

25-64-03-03B Any In Excess Of Those Required By 14 CFR

Interval	Installed	Required	Procedure
D	-	-	

Any in excess of those required by 14 CFR may be incomplete, inoperative or removed.

OPERATIONS NOTE

See 14 CFR 121.803 or other appropriate regulations.

25-64-03 Emergency Medical Equipment

25-64-03-03 Automated External Defibrillators (AED) and/or Associated Equipment

25-64-03-03-01 Tamper Seals or Tags

Interval	Installed	Required	Procedure
C	-	-	(O)

May be inoperative, damaged, or missing provided proper servicing is verified at each preflight.

OPERATIONS (O)

Before each departure, verify affected item(s) has been properly serviced.

**25-64-04 Smoke Vision System (SVS)**

Interval	Installed	Required	Procedure
D	2	0	

May be inoperative or missing.



26-11-01 Engine Fire Detector Loops

26-11-01-01 PW & GE

26-11-01-02 RR

26-11-01-02-01 SB 777-26-0011 (PRR 61202-38) and SB
777-26-0014 (PRR 61202-39) Incorporated

26-11-01-02-02 All Others

26-11-02 Engine Overheat Circuits

26-13-01 Lavatory Smoke Detection Systems

26-13-01-01 Passenger

26-13-01-01A *Smoke Detection System(s) Inoperative*

26-13-01-01B *Any In Excess Of Those Required By 14
CFR*

26-13-01-01-01 Lavatory Call Light Smoke Detected Function

26-13-01-02 777F

26-13-02 Overhead Cross Aisle Stowage (OCAS) Smoke Detection System (Detector, Horn and Light)

26-14-01 Lower Lobe Attendant Rest (LLAR) Smoke Detection System

26-14-01-01 Remote Smoke Detector Channels

26-14-02 Flight Crew Rest (FCR) Smoke Detection Systems

26-14-02-01 Two Bunk FCR - Bunk Smoke Detectors

26-14-02-02 Single Bunk FCR - Smoke Detectors

26-14-02-03 Seat Only FCR - Smoke Detectors

26-14-03 Main Deck Crew Rest (MDCR)/Main Deck Flight Crew Rest (MDFCR) Smoke Detection System

26-14-03-01 Bunk Smoke Detectors (MDCR)

26-14-04 Overhead Flight Crew/Attendant Rest Smoke Detection Systems

26-14-04-01 Bunk Smoke Detectors

26-14-04-02 Common Area Smoke Detectors

26-14-04-02-01 Overhead Flight Crew Rest (OFCR)

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26-14-04-02-02	Overhead Flight Attendant Rest (OFAR)
26-14-04-03	Aisle Smoke Detectors (OFAR)
26-14-04-04	Closet Smoke Detector(s)
26-14-04-05	Vanity Module (Includes Sink With Drink Storage Area) Smoke Detector (OFCR)
26-14-04-06	Stairwell Smoke Detector
26-14-04-06-01	OFCR Without Lavatory
26-14-04-06-02	OFCR With Lavatory
26-14-04-06-03	OFAR
26-14-04-07	Changing Area Smoke Detector (OFAR)
26-14-04-08	Lavatory Smoke Detector (OFCR)
26-14-05	Overhead Flight Crew/Attendant Rest Smoke Detection Systems (FSI Installation)
26-14-05-01	Bunk Smoke Detectors
26-14-05-01-01	Overhead Flight Crew Rest (OFCR)
26-14-05-01-02	Overhead Attendant Rest (OHAR) Forward and Aft Bunks
26-14-05-02	Common Area Smoke Detector
26-14-05-02-01	Overhead Flight Crew Rest (OFCR)
26-14-05-02-02	Overhead Attendant Rest (OHAR)
26-14-05-03	Vestibule Smoke Detector
26-14-05-03-01	Overhead Flight Crew Rest (OFCR)
26-14-05-03-02	Overhead Attendant Rest (OHAR)
26-15-01	APU Fire Detection System
26-15-01A	<i>APU Not Used</i>
26-15-01B	<i>APU Used On Ground</i>
26-15-01-01	APU Fire Detector Loops
26-16-01	Lower Cargo Compartment Smoke Detection Systems (Forward and Aft) (Passenger)
26-16-01-01	Lower Cargo Compartment Smoke Detector Channels
26-16-01-02	Lower Cargo Compartment Smoke Detector Fans
26-16-01-03	Lower Cargo Compartment Smoke Detector Zones

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26-16-01-04	Lower Cargo Compartment Smoke Detector Heaters
26-16-01-04A	<i>Humid Cargo Not Carried</i>
26-16-01-04B	<i>Humid Cargo Carried</i>
26-16-02	Lower Cargo Compartment Smoke Detection Systems (777F)
26-16-02-01	Forward
26-16-02-01-01	Detectors
26-16-02-01-02	Detector Zones
26-16-02-02	Aft
26-16-02-02-01	Detectors
26-16-02-02-02	Detector Zones
26-16-03	Main Deck Cargo Compartment Smoke Detection Systems (777F)
26-16-03-01	Detectors
26-16-03-02	Detector Zones
26-16-04	Cargo Smoke Detection (CSD) System Controller (777F)
26-16-04-01	CSD Controller Power Bus
26-16-04-02	CSD Controller Channels
26-16-04-03	CSD Controller Area Network Bus
26-17-01	Wheel Well Fire Detection System
26-17-01A	<i>BTMS Operative</i>
26-17-01B	<i>BTMS Inoperative</i>
26-18-01	Wing Duct Leak Detector Loops
26-18-02	Body Duct Leak Detector Loops
26-18-03	Strut Overheat Detector Loops
26-18-04	Engine Fan Case Overheat Detection Systems
26-18-04-01	PW & GE
26-18-04-02	RR
26-19-01	E/E Cooling Smoke Detector Channels

Table of Contents**26-19-02 Equipment Cooling Smoke Detectors (777F)**

26-19-02-01 Supply Smoke Detector

26-19-02-02 Exhaust Smoke Detector

26-21-01 Fire BTL DISCH Lights (Engine, APU, Cargo)**26-21-02 Fire Extinguishing Squib Test System**

26-21-02-01 Cargo Squib Test System

26-21-02-02 Lower Lobe Attendant Rest (LLAR) Squib Test System

26-21-02-02A *Squib Continuity Verified*26-21-02-02B *Squib Continuity Not Verified***26-22-01 APU Fire Extinguisher System****26-22-02 APU Auto Discharge****26-23-01 Lower Cargo Compartment Fire Extinguisher Bottles**

26-23-01-01 Without LLAR Installed

26-23-01-02 With LLAR Installed

26-23-01-03 Metered Bottles

26-23-01-04 Bottle Pressure Switches

26-23-02 Lower Cargo Compartment Fire Extinguisher Flow Valves (Forward or Aft)26-23-02A *One Inoperative*26-23-02B *Both Inoperative***26-24-01 Lavatory Fire Extinguisher Systems**

26-24-01-01 Passenger

26-24-01-01A *Smoke Detection System(s) Operative*26-24-01-01B *Smoke Detection System(s) Inoperative*26-24-01-01C *Any In Excess Of Those Required By 14 CFR*

26-24-01-02 777F

26-25-01 Lower Lobe Attendant Rest (LLAR) Fire Extinguisher Bottle**26-26-01 Portable Fire Extinguishers**

26-26-01-01 Tamper Seals or Tags

Table of Contents**26-27-01 Main Deck Cargo Fire Suppression System (FSS) (STC ST02375LA)**

26-27-01-01 Fire Control Hub

26-27-01-02 Fire Control Panel

26-27-01-02-01 Surfactant Tank Quantity Indication

26-27-01-02-01A

26-27-01-02-01B

26-27-01-02-01C

26-27-01-02-02 Inert Gas Cylinder (Argon) Pressure Indication -
Electrical Function

26-27-01-02-02A

26-27-01-02-02B

26-27-01-03 Fire Control Unit

26-27-01-04 Fire Detection Unit

26-27-01-05 Storage and Release System

26-27-01-05-01 Surfactant Tank

26-27-01-05-02 Inert Gas Cylinder (Argon) Pressure Indication -
Mechanical Function

26-27-01-05-02A

26-27-01-05-02B

26-27-01-05-03 Penetrator Modules

26-27-01-06 Lower Forward Instrument Panel FSS DISCH
Annunciator



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Intentionally
Blank



26-11-01 Engine Fire Detector Loops**26-11-01-01 PW & GE**

Interval	Installed	Required	Procedure
C	4	2	

One per engine may be inoperative.

26-11-01 Engine Fire Detector Loops**26-11-01-02 RR****26-11-01-02-01 SB 777-26-0011 (PRR 61202-38) and SB 777-26-0014 (PRR 61202-39) Incorporated**

Interval	Installed	Required	Procedure
C	4	2	

One per engine may be inoperative.

26-11-01 Engine Fire Detector Loops**26-11-01-02 RR****26-11-01-02-02 All Others**

Interval	Installed	Required	Procedure
B	4	2	

One per engine may be inoperative.



26-11-02 Engine Overheat Circuits

Interval	Installed	Required	Procedure
C	4	0	

26-13-01 Lavatory Smoke Detection Systems
26-13-01-01 Passenger
26-13-01-01A Smoke Detection System(s) Inoperative

Interval	Installed	Required	Procedure
C	-	-	(M) (O)

For each lavatory, may be inoperative provided associated:

- Lavatory waste container is empty.
- Lavatory door is locked closed and placarded, INOPERATIVE - DO NOT ENTER.
- Lavatory is used only by crewmembers.

NOTE: These provisions are not intended to prohibit lavatory use or inspections by crewmembers.

MAINTENANCE (M)

Deactivate the associated lavatory (AMM 26-00-00/901).

- Check that the lavatory waste receptacle in the waste container is empty.
- When desired, deactivate the inoperative smoke detector.

- Open the P110 panel LAV SMOKE DETECT-L circuit breaker.
- Open the P210 panel LAV SMOKE DETECTOR RIGHT circuit breaker.
- Loosen the screws on the face plate of the inoperative smoke detector and remove the face plate.

NOTE: The smoke detector is installed in ceiling of each lavatory.

- Remove the mounting screws that hold the smoke detector to the ceiling bracket.
- Disconnect, cap and stow the electrical connector.
- Re-attach the smoke detector to the ceiling bracket with the mounting screws and tighten the mounting screws.
- Re-install the smoke detector face plate and tighten the face plate mounting screws.
- Close the P110 panel LAV SMOKE DETECT-L circuit breaker.
- Close the P210 panel LAV SMOKE DETECTOR RIGHT circuit breaker.

- Close and lock the lavatory door.

OPERATIONS (O)

Lavatory door should remain locked closed, except for use or inspection by crewmembers.

**26-13-01 Lavatory Smoke Detection Systems****26-13-01-01 Passenger****26-13-01-01B Any In Excess Of Those Required By 14 CFR**

Interval	Installed	Required	Procedure
D	-	0	

Any in excess of that required by 14 CFR may be inoperative.

26-13-01 Lavatory Smoke Detection Systems**26-13-01-01 Passenger****26-13-01-01-01 Lavatory Call Light Smoke Detected Function**

Interval	Installed	Required	Procedure
C	-	0	

NOTE: Attendant call and occupancy indications are considered NEF.

26-13-01 Lavatory Smoke Detection Systems**26-13-01-02 777F**

Interval	Installed	Required	Procedure
D	1	0	



26-13-02 Overhead Cross Aisle Stowage (OCAS) Smoke
Detection System (Detector, Horn and Light)

Interval	Installed	Required	Procedure
D	-	0	

MAINTENANCE NOTE

The associated OCAS may be used.

**26-14-01 Lower Lobe Attendant Rest (LLAR) Smoke Detection System**

Interval	Installed	Required	Procedure
C	1	0	(M)

May be inoperative provided:

- LLAR shutoff valve is deactivated closed.
- LLAR is considered inoperative.

NOTE: The airplane must also be dispatched using MMEL item 25-29-04.

MAINTENANCE (M)

NOTE: The LLAR Smoke Detection System is inoperative when both DET CHAN 1 C/R LWR and DET CHAN 2 C/R LWR status messages are displayed.

Deactivate the LLAR shutoff valve using DDG item 21-29-01 (M) procedure.

26-14-01 Lower Lobe Attendant Rest (LLAR) Smoke Detection System

26-14-01-01 Remote Smoke Detector Channels

Interval	Installed	Required	Procedure
C	2	1	

One channel may be inoperative provided both aft cargo compartment smoke detector channels operate normally.

26-14-02 Flight Crew Rest (FCR) Smoke Detection Systems

Interval	Installed	Required	Procedure
C	-	0	(M)

May be inoperative provided:

- a. Associated FCR smoke detection system is deactivated.
 - b. Associated FCR is considered inoperative.
-

NOTE: The airplane must also be dispatched using MMEL item 25-29-04.

MAINTENANCE (M)

Deactivate the associated smoke detection system and the FCR closed (AMM 26-00-00/901).

1. Deactivate the associated FCR smoke detection system.
 - A. For an FCR installed near Door 1L, open and collar P110 panel DR 1 CREW REST CTRL/SMK DET PWR circuit breaker.
 - B. For an FCR installed near Door 3, open and collar P110 panel MN DK CR SMK DET PWR circuit breaker.

26-14-02 Flight Crew Rest (FCR) Smoke Detection Systems

26-14-02-01 Two Bunk FCR - Bunk Smoke Detectors

Interval	Installed	Required	Procedure
C	2	1	(M) (O)

One may be inoperative provided:

- a. Associated bunk is not used and personal items are removed.
 - b. A conspicuous barrier strap or rope is placed across the associated bunk with a placard attached stating that the bunk is not to be used.
-

MAINTENANCE (M)

Deactivate associated bunk (AMM 26-00-00/901).

1. Remove all personal items from the associated bunk.
2. Place a barrier strap or rope across the associated bunk with a placard attached that states, DO NOT USE.

OPERATIONS (O)

1. The bunk associated with the inoperative smoke detector is not available for inflight use. Remove personal items from the bunk.



2. For augmented flight crew operations, flight crew duty periods should be adjusted appropriately (14 CFR 117).

26-14-02 Flight Crew Rest (FCR) Smoke Detection Systems
*********26-14-02-02 Single Bunk FCR - Smoke Detectors**

Interval	Installed	Required	Procedure
C	-	1	

May be inoperative and the bunk used provided one smoke detector operates normally in the associated FCR.

26-14-02 Flight Crew Rest (FCR) Smoke Detection Systems
*********26-14-02-03 Seat Only FCR - Smoke Detectors**

Interval	Installed	Required	Procedure
C	2	1	

26-14-03 Main Deck Crew Rest (MDCR)/Main Deck Flight Crew Rest (MDFCR) Smoke Detection System

Interval	Installed	Required	Procedure
C	1	0	(M)

May be inoperative provided:

- Smoke detection system is deactivated.
- MDCR/MDFCR is considered inoperative.

NOTE: The airplane must also be dispatched using MMEL item 25-29-04.

MAINTENANCE (M)

Deactivate the smoke detection system and the MDCR/MDFCR closed (AMM 26-00-00/901).

- Open and collar P110 panel MN DK CR SMK DET PWR circuit breaker.

26-14-03 Main Deck Crew Rest (MDCR)/Main Deck Flight Crew Rest (MDFCR) Smoke Detection System

26-14-03-01 Bunk Smoke Detectors (MDCR)

Interval	Installed	Required	Procedure
C	4	2	(M) (O)

May be inoperative provided:

- Associated bunk is not used and personal items are removed.
- A conspicuous barrier strap or rope is placed across the associated bunk with a placard attached stating that the bunk is not to be used.

MAINTENANCE (M)

Deactivate the associated bunk (AMM 26-00-00/901).

- Remove all personal items from the associated bunk.
- Place a barrier strap or rope across the associated bunk with a placard attached that states, DO NOT USE.

OPERATIONS (O)

- The bunk associated with the inoperative smoke detector is not available for inflight use. Remove personal items from the bunk.
- For augmented flight crew operations, flight crew duty periods should be adjusted appropriately (14 CFR 117).

**26-14-04 Overhead Flight Crew/Attendant Rest Smoke Detection Systems**

Interval	Installed	Required	Procedure
C	2	0	(M)

May be inoperative provided:

- a. Inoperative smoke detection system is deactivated.
- b. Associated crew rest facility is considered inoperative.

NOTE: The airplane must also be dispatched using MMEL item 25-29-04.

MAINTENANCE (M)

Deactivate the inoperative smoke detection system (AMM 26-00-00/901).

1. Deactivate the inoperative smoke detection system.
 - A. For the forward overhead flight crew rest, open and collar P210 panel UPR FWD MDL SMK DET circuit breaker.
 - B. For the aft overhead flight attendant rest, open and collar P210 panel UPR AFT MDL SMK DET circuit breaker.

NOTE: If the associated crew rest SMOKE XXXX caution message remains displayed, temporarily disconnect the electrical connectors from all smoke detectors in the associated crew rest, then re-install each connector after the message has cleared.

26-14-04 Overhead Flight Crew/Attendant Rest Smoke Detection Systems

26-14-04-01 Bunk Smoke Detectors

Interval	Installed	Required	Procedure
C	-	0	

May be inoperative provided:

- a. Adjacent aisle/common area smoke detector(s) operates normally.
- b. For bunk curtain installed, associated curtain is secured open or removed.

26-14-04 Overhead Flight Crew/Attendant Rest Smoke Detection Systems

26-14-04-02 Common Area Smoke Detectors
26-14-04-02-01 Overhead Flight Crew Rest (OFCR)

Interval	Installed	Required	Procedure
C	-	0	

May be inoperative provided:

- Bunk smoke detector(s) operate normally.
- For common area or bunk curtain(s) installed, curtain(s) is secured open or removed.

26-14-04 Overhead Flight Crew/Attendant Rest Smoke Detection Systems

26-14-04-02 Common Area Smoke Detectors
26-14-04-02-02 Overhead Flight Attendant Rest (OFAR)

Interval	Installed	Required	Procedure
C	-	1	

May be inoperative provided no more than one bunk smoke detector in adjacent bunk area(s) is inoperative.

26-14-04 Overhead Flight Crew/Attendant Rest Smoke Detection Systems

26-14-04-03 Aisle Smoke Detectors (OFAR)

Interval	Installed	Required	Procedure
C	-	-	

May be inoperative provided adjacent aisle/common area smoke detector(s) operates normally.

**26-14-04 Overhead Flight Crew/Attendant Rest Smoke Detection Systems**

26-14-04-04 Closet Smoke Detector(s)

Interval	Installed	Required	Procedure
C	-	0	(M) (O)

May be inoperative provided:

- Closet is not used and personal items are removed.
 - A conspicuous barrier strap or rope is placed across the closet door with a placard attached stating the closet is not to be used.
-

MAINTENANCE (M)

Deactivate associated closet (AMM 26-00-00/901).

- Remove all personal items from the associated closet.
- Place a barrier strap or rope across the associated closet door with a placard attached that states, DO NOT USE.

OPERATIONS (O)

The closet associated with the inoperative smoke detector is not available for inflight use. Remove personal items from the closet.

26-14-04 Overhead Flight Crew/Attendant Rest Smoke Detection Systems

26-14-04-05 Vanity Module (Includes Sink With Drink Storage Area) Smoke Detector (OFCR)

Interval	Installed	Required	Procedure
C	1	0	

May be inoperative provided:

- All common area smoke detectors operate normally.
 - Stairwell smoke detector operates normally.
 - For common area curtain installed, common area curtain is secured open or removed.
-

**26-14-04 Overhead Flight Crew/Attendant Rest Smoke Detection Systems**

26-14-04-06 Stairwell Smoke Detector**26-14-04-06-01 OFCR Without Lavatory**

Interval	Installed	Required	Procedure
C	1	0	

May be inoperative provided:

- All common area smoke detectors operate normally.
 - For common area curtain installed, common area curtain is secured open or removed.
-

26-14-04 Overhead Flight Crew/Attendant Rest Smoke Detection Systems

26-14-04-06 Stairwell Smoke Detector**26-14-04-06-02 OFCR With Lavatory**

Interval	Installed	Required	Procedure
C	1	0	(M)

May be inoperative provided:

- All common area smoke detectors operate normally.
 - For common area curtain installed, common area curtain is secured open or removed.
 - Waste container is empty and access is secured to prevent waste introduction into the waste container.
-

MAINTENANCE (M)

Empty the waste receptacle and secure its access door in a manner that will prevent waste from being deposited into the receptacle (AMM 26-00-00/901).

**26-14-04 Overhead Flight Crew/Attendant Rest Smoke Detection Systems**

26-14-04-06 Stairwell Smoke Detector**26-14-04-06-03 OFAR**

Interval	Installed	Required	Procedure
C	1	0	

May be inoperative provided:

- All common area smoke detectors operate normally.
 - For common area curtain installed, common area curtain is secured open or removed.
-

26-14-04 Overhead Flight Crew/Attendant Rest Smoke Detection Systems

26-14-04-07 Changing Area Smoke Detector (OFAR)

Interval	Installed	Required	Procedure
C	1	0	

May be inoperative provided:

- All common area smoke detectors operate normally.
 - Stairwell smoke detector operates normally.
 - For changing area curtain installed, changing area curtain is secured open or removed.
-

26-14-04 Overhead Flight Crew/Attendant Rest Smoke Detection Systems

26-14-04-08 Lavatory Smoke Detector (OFCR)

Interval	Installed	Required	Procedure
C	1	0	

May be inoperative provided:

- All common area smoke detectors operate normally.
- Stairwell smoke detector operates normally.



- c. For common area curtain installed, common area curtain is secured open or removed.



26-14-05 Overhead Flight Crew/Attendant Rest Smoke Detection Systems (FSI Installation)

Interval	Installed	Required	Procedure
C	2	0	(M) (O)

May be inoperative provided:

- Inoperative smoke detection system is deactivated.
- Associated crew rest facility is deactivated closed.

NOTE: These provisions are not intended to prohibit crew rest facility inspections by crewmembers.

NOTE: The overhead crew rest installation manufactured by Flight Structures, Inc. (FSI) is not a Boeing system. Information applicable to the FSI installation is not provided.

26-14-05 Overhead Flight Crew/Attendant Rest Smoke Detection Systems (FSI Installation)

26-14-05-01 Bunk Smoke Detectors**26-14-05-01-01 Overhead Flight Crew Rest (OFCR)**

Interval	Installed	Required	Procedure
C	2	0	(M) (O)

May be inoperative provided:

- Common area smoke detection system operates normally.
 - Bunk curtains remain open.
-

NOTE: The overhead crew rest installation manufactured by Flight Structures, Inc. (FSI) is not a Boeing system. Information applicable to the FSI installation is not provided.

26-14-05 Overhead Flight Crew/Attendant Rest Smoke Detection Systems (FSI Installation)

26-14-05-01 Bunk Smoke Detectors
26-14-05-01-02 Overhead Attendant Rest (OHAR) Forward and Aft Bunks

Interval	Installed	Required	Procedure
C	2	0	(M) (O)

May be inoperative provided:

- a. Common area smoke detection system operates normally.
- b. Forward and aft bunks (1, 2, 7 and 8) are not used and personal items are removed.
- c. A conspicuous barrier strap or rope is placed across the associated bunk with a placard attached stating that the bunk is not to be used.

NOTE: The overhead crew rest installation manufactured by Flight Structures, Inc. (FSI) is not a Boeing system. Information applicable to the FSI installation is not provided.

26-14-05 Overhead Flight Crew/Attendant Rest Smoke Detection Systems (FSI Installation)

26-14-05-02 Common Area Smoke Detector
26-14-05-02-01 Overhead Flight Crew Rest (OFCR)

Interval	Installed	Required	Procedure
C	1	0	

May be inoperative provided OFCR is considered inoperative.

NOTE 1: The overhead crew rest installation manufactured by Flight Structures, Inc. (FSI) is not a Boeing system. Information applicable to the FSI installation is not provided.

NOTE 2: The airplane must also be dispatched using MMEL item 25-29-04.

**26-14-05 Overhead Flight Crew/Attendant Rest Smoke Detection Systems (FSI Installation)**

26-14-05-02 Common Area Smoke Detector**26-14-05-02-02 Overhead Attendant Rest (OHAR)**

Interval	Installed	Required	Procedure
C	1	0	(M)

May be inoperative provided OHAR is deactivated closed.

NOTE: This provision is not intended to prohibit OHAR inspections by crewmembers.

NOTE: The overhead crew rest installation manufactured by Flight Structures, Inc. (FSI) is not a Boeing system. Information applicable to the FSI installation is not provided.

26-14-05 Overhead Flight Crew/Attendant Rest Smoke Detection Systems (FSI Installation)

26-14-05-03 Vestibule Smoke Detector**26-14-05-03-01 Overhead Flight Crew Rest (OFCR)**

Interval	Installed	Required	Procedure
C	1	0	

May be inoperative provided common and bunk area smoke detectors operate normally.

NOTE: The overhead crew rest installation manufactured by Flight Structures, Inc. (FSI) is not a Boeing system. Information applicable to the FSI installation is not provided.

26-14-05 Overhead Flight Crew/Attendant Rest Smoke Detection Systems (FSI Installation)

26-14-05-03 Vestibule Smoke Detector**26-14-05-03-02 Overhead Attendant Rest (OHAR)**

Interval	Installed	Required	Procedure
C	1	0	



May be inoperative provided common and bunk area smoke detectors operate normally.

NOTE: The overhead crew rest installation manufactured by Flight Structures, Inc. (FSI) is not a Boeing system. Information applicable to the FSI installation is not provided.

**26-15-01 APU Fire Detection System****26-15-01A APU Not Used**

Interval	Installed	Required	Procedure
C	1	0	(O)

May be inoperative provided:

- Other procedures do not require use of APU.
 - Backup AC power system is verified to operate normally before each departure.
 - APU selector switch remains in the OFF position.
 - Flight remains within 180 minutes of landing at a suitable airport.
-

MAINTENANCE NOTE

The APU fire detection system is inoperative when both FIRE LOOP 1 APU and FIRE LOOP 2 APU status messages are displayed.

OPERATIONS (O)

- Dispatch is not allowed if the APU is required by other procedures. MMEL dispatch procedures for the following inoperative equipment may require APU availability:

MMEL Item	Associated Status Message
24-11-01	ELEC GEN SYS L or R IDG VALVE CLOSED L or R
24-25-01	ELEC BACKUP GEN L or R ELEC BACKUP SYS
29-11-03	HYD PUMP DEM C1 or C2
36-11-01	BLEED PRSOV L or R
36-11-02	BLEED HPSOV L or R
36-11-03	None
36-12-01	BLEED ISLN VALVE L or R
36-12-02	BLEED ISLN VALVE C
36-22-01	BLEED TEMP SENSOR L or R

- The following restrictions imposed for an inoperative APU generator must be observed:
 - The flight must remain within 180 minutes of landing at a suitable airport.

- B. Verify that the Backup AC electrical power system operates normally. Confirm that ELEC BACKUP SYS or ELEC BACKUP GEN L or R status and advisory messages are not displayed 30 seconds after both engines have been started.
3. Position the APU selector OFF.

26-15-01 APU Fire Detection System
26-15-01B APU Used On Ground

Interval	Installed	Required	Procedure
C	1	0	(M) (O)

May be inoperative provided:

- a. Other procedures do not require use of APU.
- b. Backup AC power system is verified to operate normally before each departure.
- c. APU is used for ground operations only.
- d. APU is continuously monitored by ground personnel when operating.
- e. APU external control system operates normally.
- f. APU is not used during taxi.
- g. Flight remains within 180 minutes of landing at a suitable airport.

MAINTENANCE (M)

NOTE: The APU fire detection system is inoperative when both FIRE LOOP 1 APU and FIRE LOOP 2 APU status messages are displayed.

The APU must be continually visually monitored when the APU is in use on the ground. If an APU fire is observed, the APU fire extinguisher must be manually discharged using either the APU fire shutdown switch on the P40 service and APU shutdown panel (nose gear) or the APU and Cargo Fire Panel (P5) APU fire switch (AMM 26-00-00/901).

OPERATIONS (O)

NOTE: APU fire bottle discharge may occur during on ground APU operation if intermittent fire detector loop shorts have not been resolved.

1. Dispatch is not allowed if the APU is required by other procedures. MMEL dispatch procedures for the following inoperative equipment may require APU availability:



M MEL Item	Associated Status Message
24-11-01	ELEC GEN SYS L or R IDG VALVE CLOSED L or R
24-25-01	ELEC BACKUP GEN L or R ELEC BACKUP SYS
29-11-03	HYD PUMP DEM C1 or C2
36-11-01	BLEED PRSOV L or R
36-11-02	BLEED HPSOV L or R
36-11-03	None
36-12-01	BLEED ISLN VALVE L or R
36-12-02	BLEED ISLN VALVE C
36-22-01	BLEED TEMP SENSOR L or R

2. The following restrictions imposed for an inoperative APU generator must be observed:
 - A. The flight must remain within 180 minutes of landing at a suitable airport.
 - B. Verify that the Backup AC electrical power system operates normally. Confirm that ELEC BACKUP SYS or ELEC BACKUP GEN L or R status and advisory messages are not displayed 30 seconds after both engines have been started.
3. For APU use during ground operations, position a fire guard near the tail of the airplane to monitor the APU.
4. Position the APU selector OFF and do not use the APU inflight or during taxi operations.

26-15-01 APU Fire Detection System

26-15-01-01 APU Fire Detector Loops

Interval	Installed	Required	Procedure
C	2	1	

OPERATIONS NOTE

APU fire bottle discharge may occur during on ground APU operation if intermittent fire detector loop shorts have not been resolved.

26-16-01 Lower Cargo Compartment Smoke Detection Systems (Forward and Aft) (Passenger)

Interval	Installed	Required	Procedure
C	2	0	(M) (O)

May be inoperative provided:

- a. Procedures are established and used to verify the associated compartment remains empty or contains only ballast, empty cargo handling equipment (ballast may be loaded in ULDs), or fly away kits.

For LLAR installed:

- b. For aft compartment system inoperative, LLAR is considered inoperative.

For FCAC installed:

- c. For forward compartment system inoperative, FCAC remains OFF.

NOTE: Operator MELs must define which items are approved for inclusion in the fly away kits and which materials can be used as ballast.

For LLAR installed and aft compartment system inoperative:

NOTE: The airplane must also be dispatched using MMEL item 25-29-04.

MAINTENANCE (M)

NOTE: The smoke detection system for a compartment is inoperative when both channels or both fans are inoperative.

Configure the airplane (AMM 26-00-00/901).

1. Associated cargo compartment(s) must remain empty, except for ballast, empty cargo containers (ballast may be loaded in ULDs), fly away kits, pallets, and cargo restraint components.
2. If desired, deactivate the aft smoke detection system:
 - A. Open and collar the P110 panel AFT CGO SMOKE CTRL CHAN 1 circuit breaker.
 - B. Open and collar the P210 panel AFT CGO SMOKE CTRL CHAN 2 circuit breaker.
 - C. If installed, open and collar the P210 panel SMOKE DETECTOR HTR-AFT circuit breaker.

NOTE: Do not deactivate the forward smoke detection system.

For Lower Lobe Attendant Rest (LLAR) installed:

3. For aft smoke detection system inoperative, deactivate the LLAR shutoff valve closed using DDG item 21-29-01 (M) procedure.

For Forward Cargo Air Conditioning (FCAC) installed:

4. For forward smoke detection system inoperative, position Overhead Panel (P5) FWD CARGO A/C control OFF.

**OPERATIONS (O)**

1. Associated cargo compartment(s) must remain empty, except for ballast, empty cargo containers (ballast may be loaded in ULDs), fly away kits, pallets, and cargo restraint components.

For Forward Cargo Air Conditioning (FCAC) installed:

2. For forward smoke detection system inoperative, Position FWD CARGO A/C control OFF.

26-16-01 Lower Cargo Compartment Smoke Detection Systems (Forward and Aft) (Passenger)**26-16-01-01 Lower Cargo Compartment Smoke Detector Channels**

Interval	Installed	Required	Procedure
C	4	2	

One channel per compartment may be inoperative.

26-16-01 Lower Cargo Compartment Smoke Detection Systems (Forward and Aft) (Passenger)**26-16-01-02 Lower Cargo Compartment Smoke Detector Fans**

Interval	Installed	Required	Procedure
C	4	2	

One fan per compartment may be inoperative.

26-16-01 Lower Cargo Compartment Smoke Detection Systems (Forward and Aft) (Passenger)**26-16-01-03 Lower Cargo Compartment Smoke Detector Zones**

Interval	Installed	Required	Procedure
C	-	0	(M) (O)

Smoke detection in any zone may be inoperative provided:

- a. Procedures are established and used to verify the associated compartment zone and adjacent compartment zone(s) remains empty or contains only ballast, empty cargo handling equipment (ballast may be loaded in ULDs), or fly away kits.

For LLAR installed:

- b. For aft compartment Zone 2 or Zone 3 detector inoperative, LLAR is considered inoperative.

For FCAC installed:

- c. For forward compartment system zone detector inoperative, FCAC remains OFF.

NOTE: Operator MELs must define which items are approved for inclusion in the fly away kits and which materials can be used as ballast.

For LLAR installed and aft compartment Zone 2 or Zone 3 detector inoperative:

NOTE: The airplane must also be dispatched using MMEL item 25-29-04.

MAINTENANCE (M)

Configure the airplane (AMM 26-00-00/901).

1. Associated cargo compartment zone and adjacent zone(s) must remain empty, except for ballast, empty cargo containers (ballast may be loaded in ULDs), fly away kits, pallets, and cargo restraint components.

For -200/-200ER/-200LR:**A. Aft and bulk cargo compartments:**

- 1) Zone 1 is from bulk compartment aft wall to bulk compartment cargo divider net.
- 2) Zone 2 is from bulk compartment cargo divider net to STA 1664 floor beam.
- 3) Zone 3 is from STA 1664 floor beam to forward wall.

B. Forward compartment:

- 1) Zone 1 is from aft wall to STA 772 floor beam.
- 2) Zone 2 is from STA 772 floor beam to STA 592 floor beam.
- 3) Zone 3 is from STA 592 floor beam to forward wall.

For -300/-300ER:**A. Aft and bulk cargo compartments:**

- 1) Zone 1 is from bulk compartment aft wall to bulk compartment cargo divider net.
- 2) Zone 2 is from bulk compartment cargo divider net to STA 1664 floor beam.
- 3) Zone 3 is from STA 1664 floor beam to STA 1434 + 189 floor beam.
- 4) Zone 4 is from STA 1434 + 189 floor beam to forward wall.

B. Forward compartment:

- 1) Zone 1 is from aft wall to STA 825 + 147 floor beam.
- 2) Zone 2 is from STA 825 + 147 floor beam to STA 772 floor beam.
- 3) Zone 3 is from STA 772 floor beam to STA 592 floor beam.
- 4) Zone 4 is from STA 592 floor beam to forward wall.

**For Lower Lobe Attendant Rest (LLAR) installed:**

2. For aft compartment zone 2 or 3 detector inoperative, deactivate the LLAR shutoff valve closed using DDG item 21-29-01 (M) procedure.

For Forward Cargo Air Conditioning (FCAC) installed:

3. For forward compartment zone detector inoperative, position Overhead Panel (P5) FWD CARGO A/C control OFF.

OPERATIONS (O)

1. Associated cargo compartment zone and adjacent zone(s) must remain empty, except for ballast, empty cargo containers (ballast may be loaded in ULDs), fly away kits, pallets, and cargo restraint components.

For Forward Cargo Air Conditioning (FCAC) installed:

2. For forward compartment zone detector inoperative, position FWD CARGO A/C control OFF.

26-16-01 Lower Cargo Compartment Smoke Detection Systems (Forward and Aft) (Passenger)**26-16-01-04 Lower Cargo Compartment Smoke Detector Heaters****26-16-01-04A Humid Cargo Not Carried**

Interval	Installed	Required	Procedure
C	-	0	

May be inoperative provided humid cargo is not carried in the associated compartment.

26-16-01 Lower Cargo Compartment Smoke Detection Systems (Forward and Aft) (Passenger)**26-16-01-04 Lower Cargo Compartment Smoke Detector Heaters****26-16-01-04B Humid Cargo Carried**

Interval	Installed	Required	Procedure
C	-	0	(O)

May be inoperative provided procedures are established and used when humid cargo is carried in the associated compartment.

OPERATIONS (O)

NOTE: Cargo such as fruit, flowers or live animals may cause false fire warnings due to condensation of moisture inside the smoke detectors.

Appropriate procedures should be established to reduce the possibility of certain cargo causing condensation of moisture inside the smoke detectors.

26-16-02 Lower Cargo Compartment Smoke Detection Systems (777F)
26-16-02-01 Forward

Interval	Installed	Required	Procedure
C	1	0	(M) (O)

May be inoperative provided:

- a. Procedures are established and used to verify the forward compartment remains empty or contains only ballast, empty cargo handling equipment (ballast may be loaded in ULDs), or fly away kits.
- b. FCAC remains OFF.

NOTE: Operator MELs must define which items are approved for inclusion in the fly away kits and which materials can be used as ballast.

MAINTENANCE (M)

NOTE: For DET FIRE CARGO FWD advisory message displayed, all smoke detection zones in the lower forward cargo compartment are inoperative and compartment must remain empty.

Configure the airplane (AMM 26-00-00/901).

1. The lower forward cargo compartment must remain empty, except for ballast, empty cargo containers (ballast may be loaded in ULDs), fly away kits, pallets, and cargo restraint components.

For Aft Cargo Air Conditioning (ACAC) not installed:

2. Position Overhead Panel (P5) FWD CARGO A/C control OFF.

For Aft Cargo Air Conditioning (ACAC) installed:

3. Position Overhead Panel (P5) LWR CARGO TEMP FWD A/C control OFF.

OPERATIONS (O)

1. The lower forward cargo compartment must remain empty, except for ballast, empty cargo containers (ballast may be loaded in ULDs), fly away kits, pallets, and cargo restraint components.

For Aft Cargo Air Conditioning (ACAC) not installed:

2. Position FWD CARGO A/C control OFF.

For Aft Cargo Air Conditioning (ACAC) installed:

3. Position LWR CARGO TEMP FWD A/C control OFF.

**26-16-02 Lower Cargo Compartment Smoke Detection Systems (777F)****26-16-02-01 Forward****26-16-02-01-01 Detectors**

Interval	Installed	Required	Procedure
C	15	12	

One detector in each zone may be inoperative provided adjacent smoke detectors operate normally.

MAINTENANCE NOTE

1. Adjacent smoke detectors are detectors immediately next to the inoperative detector either fore or aft, to the side, or diagonally.
2. Smoke detectors are interchangeable.

26-16-02 Lower Cargo Compartment Smoke Detection Systems (777F)**26-16-02-01 Forward****26-16-02-01-02 Detector Zones**

Interval	Installed	Required	Procedure
C	3	-	(M) (O)

May be inoperative provided:

- a. All detectors in a loaded zone operate normally.
- b. Procedures are established and used to verify the associated compartment zone remains empty or contains only ballast, empty cargo handling equipment (ballast may be loaded in ULDs), or fly away kits.

NOTE: Operator MELs must define which items are approved for inclusion in the fly away kits and which materials can be used as ballast.

MAINTENANCE (M)

NOTE: For DET FIRE CARGO FWD advisory message displayed, all smoke detection zones in the lower forward cargo compartment are inoperative and compartment must remain empty.

The associated lower forward cargo compartment zone must remain empty, except for ballast, empty cargo containers (ballast may be loaded in ULDs), fly away kits, pallets, and cargo restraint components (AMM 26-00-00/901).

1. Zone 1 is from the aft wall to STA 772 floor beam.
2. Zone 2 is from STA 772 floor beam to STA 592 floor beam.

3. Zone 3 is from STA 592 floor beam to the forward wall.

OPERATIONS (O)

The associated lower forward cargo compartment zone must remain empty, except for ballast, empty cargo containers (ballast may be loaded in ULDs), fly away kits, pallets, and cargo restraint components.

26-16-02 Lower Cargo Compartment Smoke Detection Systems (777F)

26-16-02-02 Aft

Interval	Installed	Required	Procedure
C	1	0	(M) (O)

May be inoperative provided:

- a. Procedures are established and used to verify the aft compartment remains empty or contains only ballast, empty cargo handling equipment (ballast may be loaded in ULDs), or fly away kits.

For ACAC installed:

- b. ACAC remains OFF.

NOTE: Operator MELs must define which items are approved for inclusion in the fly away kits and which materials can be used as ballast.

MAINTENANCE (M)

NOTE: For DET FIRE CARGO AFT advisory message displayed, all smoke detection zones in the lower aft cargo compartment are inoperative and compartment must remain empty.

Configure the airplane (AMM 26-00-00/901).

1. The lower aft cargo compartment must remain empty, except for ballast, empty cargo containers (ballast may be loaded in ULDs), fly away kits, pallets, and cargo restraint components.

For Aft Cargo Air Conditioning (ACAC) installed:

2. Position Overhead Panel (P5) LWR CARGO TEMP AFT A/C control OFF.

OPERATIONS (O)

1. The lower aft cargo compartment must remain empty, except for ballast, empty cargo containers (ballast may be loaded in ULDs), fly away kits, pallets, and cargo restraint components.

For Aft Cargo Air Conditioning (ACAC) installed:

2. Position LWR CARGO TEMP AFT A/C control OFF.

**26-16-02 Lower Cargo Compartment Smoke Detection Systems (777F)****26-16-02-02 Aft****26-16-02-02-01 Detectors**

Interval	Installed	Required	Procedure
C	15	12	

One detector in each zone may be inoperative provided adjacent smoke detectors operate normally.

MAINTENANCE NOTE

1. Adjacent smoke detectors are detectors immediately next to the inoperative detector either fore or aft, to the side, or diagonally.
2. Smoke detectors are interchangeable.

26-16-02 Lower Cargo Compartment Smoke Detection Systems (777F)**26-16-02-02 Aft****26-16-02-02-02 Detector Zones**

Interval	Installed	Required	Procedure
C	3	-	(M) (O)

May be inoperative provided:

- a. All detectors in a loaded zone operate normally.
- b. Procedures are established and used to verify the associated compartment zone remains empty or contains only ballast, empty cargo handling equipment (ballast may be loaded in ULDs), or fly away kits.

NOTE: Operator MELs must define which items are approved for inclusion in the fly away kits and which materials can be used as ballast.

MAINTENANCE (M)

NOTE: For DET FIRE CARGO AFT advisory message displayed, all smoke detection zones in the lower aft cargo compartment are inoperative and compartment must remain empty.

The associated lower aft cargo compartment zone must remain empty, except for ballast, empty cargo containers (ballast may be loaded in ULDs), fly away kits, pallets, and cargo restraint components (AMM 26-00-00/901).

1. Zone 1 is from the aft wall to STA 1895 floor beam.
2. Zone 2 is from STA 1895 floor beam to STA 1643 floor beam.



-
3. Zone 3 is from STA 1643 floor beam to the forward wall.

OPERATIONS (O)

The associated lower aft cargo compartment zone must remain empty, except for ballast, empty cargo containers (ballast may be loaded in ULDs), fly away kits, pallets, and cargo restraint components.

**26-16-03 Main Deck Cargo Compartment Smoke Detection Systems (777F)**

Interval	Installed	Required	Procedure
C	1	0	(M) (O)

May be inoperative provided procedures are established and used to verify the main deck compartment remains empty or contains only ballast, empty cargo handling equipment (ballast may be loaded in ULDs), or fly away kits.

NOTE: Operator MELs must define which items are approved for inclusion in the fly away kits and which materials can be used as ballast.

MAINTENANCE (M)

NOTE: For DET FIRE CARGO MAIN advisory message displayed, the main deck cargo compartment smoke detection system is inoperative and compartment must remain empty.

The main deck cargo compartment must remain empty, except for ballast, empty cargo containers (ballast may be loaded in ULDs), fly away kits, pallets, and cargo restraint components (AMM 26-00-00/901).

OPERATIONS (O)

The main deck cargo compartment must remain empty, except for ballast, empty cargo containers (ballast may be loaded in ULDs), fly away kits, pallets, and cargo restraint components.

26-16-03 Main Deck Cargo Compartment Smoke Detection Systems (777F)**26-16-03-01 Detectors**

Interval	Installed	Required	Procedure
C	46	38	

One detector in each zone may be inoperative provided adjacent smoke detectors operate normally.

MAINTENANCE NOTE

1. Adjacent smoke detectors are detectors immediately next to the inoperative detector either fore or aft, to the side, or diagonally.
2. Smoke detectors are interchangeable.

26-16-03 Main Deck Cargo Compartment Smoke Detection Systems (777F)
26-16-03-02 Detector Zones

Interval	Installed	Required	Procedure
C	8	-	(M) (O)

May be inoperative provided:

- All detectors in a loaded zone operate normally.
- Procedures are established and used to verify the associated compartment zone remains empty or contains only ballast, empty cargo handling equipment (ballast may be loaded in ULDs), or fly away kits.

NOTE: Operator MELs must define which items are approved for inclusion in the fly away kits and which materials can be used as ballast.

MAINTENANCE (M)

NOTE: For DET FIRE CARGO MAIN advisory message displayed, the main deck cargo compartment smoke detection system is inoperative and compartment must remain empty.

The associated main deck cargo compartment zone must remain empty, except for ballast, empty cargo containers (ballast may be loaded in ULDs), fly away kits, pallets, and cargo restraint components (AMM 26-00-00/901).

- Zone 1 is from the forward wall to STA 522.5 floor beam.
- Zone 2 is from STA 522.5 floor beam to STA 774.5 floor beam.
- Zone 3 is from STA 774.5 to STA 1026.5 floor beam.
- Zone 4 is from STA 1026.5 to STA 1278.5 floor beam.
- Zone 5 is from STA 1278.5 to STA 1530.5 floor beam.
- Zone 6 is from STA 1530.5 to STA 1794.5 floor beam.
- Zone 7 is from STA 1794.5 to STA 2046.5 floor beam.
- Zone 8 is from STA 2046.5 to the aft wall.

OPERATIONS (O)

The associated main deck cargo compartment zone must remain empty, except for ballast, empty cargo containers (ballast may be loaded in ULDs), fly away kits, pallets, and cargo restraint components.

**26-16-04 Cargo Smoke Detection (CSD) System Controller (777F)****26-16-04-01 CSD Controller Power Bus**

Interval	Installed	Required	Procedure
C	2	1	

26-16-04 Cargo Smoke Detection (CSD) System Controller (777F)**26-16-04-02 CSD Controller Channels**

Interval	Installed	Required	Procedure
C	2	1	

One may be inoperative provided warning electronics system channels operate normally.

26-16-04 Cargo Smoke Detection (CSD) System Controller (777F)**26-16-04-03 CSD Controller Area Network Bus**

Interval	Installed	Required	Procedure
C	8	4	

MAINTENANCE NOTE

1. For Controller Area Network (CAN) busses Fwd-CH A and Fwd-CH B inoperative, the DET FIRE CARGO FWD advisory message will be displayed. The airplane must be dispatched using MMEL item 26-16-02-01.
2. For Controller Area Network (CAN) busses Aft-CH A and Aft-CH B inoperative, the DET FIRE CARGO AFT advisory message will be displayed. The airplane must be dispatched using MMEL item 26-16-02-02.
3. For Controller Area Network (CAN) Main-1 CH A and Main-2 CH B inoperative, or for CAN busses Main-3 CH A and Main-4 CH B inoperative, all eight detector zone status messages will be displayed. The airplane must be dispatched using MMEL item 26-16-03.

26-17-01 Wheel Well Fire Detection System
26-17-01A BTMS Operative

Interval	Installed	Required	Procedure
C	1	0	(M) (O)

May be inoperative provided:

- Both wheel well fire detection loops are deactivated.
- Brake temperature monitoring system (BTMS) operates normally.

NOTE: Avoid the possibility of retracting an overheated wheel by monitoring brake temperature indications.

MAINTENANCE (M)

Deactivate wheel well fire detection loops 1 and 2 (AMM DDG 26-00-00/901).

NOTE: DET FIRE WHEEL WELL status message will be displayed when the wheel well fire detection system is deactivated.

- Access the wheel well fire detectors in the upper area of the left and right bays of the main wheel well.
- Disconnect, cap and stow the electrical connectors (D7589P and D7615P).

OPERATIONS (O)

Avoid the possibility of retracting an overheated wheel by monitoring brake temperature indications.

26-17-01 Wheel Well Fire Detection System
26-17-01B BTMS Inoperative

Interval	Installed	Required	Procedure
C	1	0	(M) (O)

May be inoperative provided:

- Both wheel well fire detection loops are deactivated.
- Landing gear remains extended for ten minutes after takeoff.
- Appropriate performance adjustments are applied.

NOTE: In case of engine failure after V1, landing gear should be retracted until takeoff obstacles are cleared.

MAINTENANCE (M)

Deactivate wheel well fire detection loops 1 and 2 (AMM DDG 26-00-00/901).



NOTE: DET FIRE WHEEL WELL status message will be displayed when the wheel well fire detection system is deactivated.

1. Access the wheel well fire detectors in the upper area of the left and right bays of the main wheel well.
2. Disconnect, cap and stow the electrical connectors (D7589P and D7615P).

OPERATIONS (O)

1. It is not necessary to apply landing gear extended takeoff performance adjustments when dispatching under this item.
2. Increase flight planning fuel by 4,500 lbs (2,041 kg) to account for the landing gear extended for ten minutes after takeoff.
3. Leave landing gear extended for 10 minutes after takeoff. For engine failure after V₁, retract landing gear until obstacles are cleared.



26-18-01 Wing Duct Leak Detector Loops

Interval	Installed	Required	Procedure
C	4	2	

One per side may be inoperative.

**26-18-02 Body Duct Leak Detector Loops**

Interval	Installed	Required	Procedure
C	2	1	

One may be inoperative provided both ASG cards in the opposite cardfile operate normally.

MAINTENANCE NOTE

1. For BLEED LOOP 1 BODY message, both ASG cards in the Right Cardfile must operate normally. Both ASG cards in the Right Cardfile are operating normally if the RSCF ASG CARD L or R messages are not displayed.
2. For BLEED LOOP 2 BODY message, both ASG cards in the Left Cardfile must operate normally. Both ASG cards in the Left Cardfile are operating normally if the LSCF ASG CARD L or R messages are not displayed.



26-18-03 Strut Overheat Detector Loops

Interval	Installed	Required	Procedure
C	4	2	

One per side may be inoperative.

**26-18-04 Engine Fan Case Overheat Detection Systems****26-18-04-01 PW & GE**

Interval	Installed	Required	Procedure
C	2	1	

One may be inoperative provided:

- Associated engine anti-ice selector remains OFF.
 - Airplane is not operated in known or forecast icing conditions.
 - Flight remains within 120 minutes of landing at a suitable airport.
-

26-18-04 Engine Fan Case Overheat Detection Systems**26-18-04-02 RR**

Interval	Installed	Required	Procedure
C	2	1	

One may be inoperative provided one engine overheat circuit operates normally on the associated engine.



26-19-01 E/E Cooling Smoke Detector Channels

Interval	Installed	Required	Procedure
C	2	1	

**26-19-02 Equipment Cooling Smoke Detectors (777F)****26-19-02-01 Supply Smoke Detector**

Interval	Installed	Required	Procedure
C	1	0	

May be inoperative provided:

- Equipment cooling vent fan operates normally.
 - Equipment cooling exhaust smoke detector operates normally.
-

26-19-02 Equipment Cooling Smoke Detectors (777F)**26-19-02-02 Exhaust Smoke Detector**

Interval	Installed	Required	Procedure
C	1	0	(M)

May be inoperative provided:

- Equipment cooling vent fan is deactivated.
 - Equipment cooling supply smoke detector operates normally.
 - For ground operations with OAT 30 degrees C or higher, both packs are selected on or airplane is supplied with conditioned air.
-

MAINTENANCE (M)

Deactivate the equipment cooling vent fan (AMM 21-00-00/901).

- Open and collar the P110 panel EQPT CLG VENT FAN CTRL circuit breaker.
- Disconnect, cap and stow the electrical connector (DB21583) from the equipment cooling vent fan.
- For ground operations with OAT at or above 30 degrees C, select both packs on or supply conditioned air to the airplane.

26-21-02 Fire Extinguishing Squib Test System

Interval	Installed	Required	Procedure
C	1	0	(M)

May be inoperative provided squib electrical continuity is verified once each flight day.

MAINTENANCE (M)

NOTE: The Squib Test Control relay may be inoperative when all of the applicable squib test status messages are displayed. The airplane may be dispatched using this item by replacing the squib test control relay with an operative relay from the ELMS cabinet, successfully accomplishing the fire extinguishing squib test, and re-installing the operative relay back in the ELMS cabinet.

For the Squib Test Control relay inoperative, verify fire extinguishing bottle squib electrical continuity once each flight day (AMM 26-00-00/901).

1. Open the P310 panel FIRE EXT APU circuit breaker.
2. Open the P310 panel FIRE EXT CGO BTL 1A & VLV SQB circuit breaker.
3. Remove the Squib Test Control relay K26010 from the P310 Panel.

For -200/-200ER/-200LR:

4. Use relay K21452, R DR 1 & 2 HEAT as a replacement relay.
 - A. Open the P210 panel R DR 1 HTR circuit breaker
 - B. Open the P210 panel R DR 2 HTR circuit breaker.
 - C. Open the P210 panel R DR 1-4 HTR CTRL circuit breaker.
 - D. Remove relay K21452 and install it in the Squib Test Control relay location.

For -300/-300ER:

5. Use relay K21452, R DR 1 & 2 HEAT as a replacement relay.
 - A. Open the P210 panel R DR 1 HTR circuit breaker
 - B. Open the P210 panel R DR 2 HTR circuit breaker.
 - C. Open the P210 panel R DR 1-5 HTR CTRL circuit breaker.
 - D. Remove relay K21452 and install it in the Squib Test Control relay location.

For 777F:

6. Use relay K21452, R DOOR HTR 1 as a replacement relay.
 - A. Open the P210 panel R DR 1 HTR circuit breaker
 - B. Open the P210 panel R DR 1 HTR CONT circuit breaker.
 - C. Remove relay K21452 and install it in the Squib Test Control relay location.

7. Close the P310 panel FIRE EXT APU circuit breaker.
8. Close the P310 panel FIRE EXT CGO BTL 1A & VLV SQB circuit breaker.
9. Use the MAT to initiate a fire extinguisher squib test.
10. Confirm that all of the applicable squib test status messages are not displayed.
11. Open the P310 panel FIRE EXT APU circuit breaker.
12. Open the P310 panel FIRE EXT CGO BTL 1A & VLV SQB circuit breaker.
13. Remove the replacement relay from the Squib Test Control relay location and re-install it in its original location.
14. Close the P210 panel circuit breaker(s) opened for the replacement relay..
15. Do the applicable operational test for the re-installed relay K21452.
16. Re-install the inoperative Squib Test Control relay K26010 in the P310 panel.

NOTE: The applicable squib test status messages will be displayed after an automatic fire bottle test occurs.

17. Close the P310 panel FIRE EXT APU circuit breaker.
18. Close the P310 panel FIRE EXT CGO BTL 1A & VLV SQB circuit breaker.

26-21-02 Fire Extinguishing Squib Test System

26-21-02-01 Cargo Squib Test System

Interval	Installed	Required	Procedure
C	1	0	(M) (O)

May be inoperative provided:

- a. Procedures are established and used to verify lower cargo compartments remain empty or contain only ballast, empty cargo handling equipment (ballast may be loaded in ULDs), or fly away kits.

For Lower Lobe Attendant Rest (LLAR) installed:

- b. LLAR is considered inoperative.

NOTE: Operator MELs must define which items are approved for inclusion in the fly away kits and which materials can be used as ballast.

For LLAR installed:

NOTE: The airplane must also be dispatched using MMEL item 25-29-04.

MAINTENANCE (M)

1. Lower cargo compartments must remain empty, except for ballast, empty cargo containers (ballast may be loaded in ULDs), fly away kits, pallets, and cargo restraint components.

For Lower Lobe Attendant Rest (LLAR) installed:

- Deactivate the LLAR shutoff valve closed using DDG item 21-29-01 (M) procedure.

OPERATIONS (O)

- Lower cargo compartments must remain empty, except for ballast, empty cargo containers (ballast may be loaded in ULDs), fly away kits, pallets, and cargo restraint components.

26-21-02 Fire Extinguishing Squib Test System

26-21-02-02 Lower Lobe Attendant Rest (LLAR) Squib Test System

26-21-02-02A Squib Continuity Verified

Interval	Installed	Required	Procedure
C	1	0	(M)

May be inoperative provided squib electrical continuity is verified once each flight day using LLAR Smoke Test switch.

MAINTENANCE (M)

Verify the LLAR squib electrical continuity once each flight day by pressing the LLAR Smoke Test switch in the LLAR Entrance Enclosure for 5 seconds and observing that the Bottle Test light in the LLAR Entrance Enclosure illuminates (AMM 26-00-00/901)

26-21-02 Fire Extinguishing Squib Test System

26-21-02-02 Lower Lobe Attendant Rest (LLAR) Squib Test System

26-21-02-02B Squib Continuity Not Verified

Interval	Installed	Required	Procedure
C	1	0	(M)

May be inoperative provided LLAR is considered inoperative.

NOTE: The airplane must also be dispatched using MMEL item 25-29-04.

MAINTENANCE (M)

Deactivate the LLAR shutoff valve closed using DDG item 21-29-01 (M) procedure.

**26-22-01 APU Fire Extinguisher System**

Interval	Installed	Required	Procedure
C	1	0	(O)

May be inoperative provided:

- Other procedures do not require use of APU.
- Backup AC power system is verified to operate normally before each departure.
- APU selector switch remains in the OFF position.
- Flight remains within 180 minutes of landing at a suitable airport.

OPERATIONS (O)

- Dispatch is not allowed if the APU is required by other procedures. MMEL dispatch procedures for the following inoperative equipment may require APU availability:

MMEL Item	Associated Status Message
24-11-01	ELEC GEN SYS L or R IDG VALVE CLOSED L or R
24-25-01	ELEC BACKUP GEN L or R ELEC BACKUP SYS
29-11-03	HYD PUMP DEM C1 or C2
36-11-01	BLEED PRSOV L or R
36-11-02	BLEED HP SOV L or R
36-11-03	None
36-12-01	BLEED ISLN VALVE L or R
36-12-02	BLEED ISLN VALVE C
36-22-01	BLEED TEMP SENSOR L or R

- The following restrictions imposed for an inoperative APU generator must be observed:
 - The flight must remain within 180 minutes of landing at a suitable airport.
 - Verify the Backup AC electrical power system operates normally. Confirm the ELEC BACKUP SYS or ELEC BACKUP GEN L or R status and advisory messages are not displayed 30 seconds after both engines have been started.
- Position the APU selector OFF.

**26-22-02 APU Auto Discharge**

Interval	Installed	Required	Procedure
C	1	0	(M)

May be inoperative provided APU is monitored during APU ground operations for fire warning.

MAINTENANCE (M)

During APU ground operations, monitor the APU for fire warnings. For a fire warning, the APU fire extinguisher must be manually discharged using either the APU fire shutdown switch on the P40 service and APU shutdown panel (nose gear) or the APU and Cargo Fire Panel (P5) APU fire switch (AMM 26-00-00/901).

26-23-01 Lower Cargo Compartment Fire Extinguisher Bottles
26-23-01-01 Without LLAR Installed

Interval	Installed	Required	Procedure
C	-	0	(O)

May be inoperative provided procedures are established and used to verify cargo compartments remain empty or contain only ballast, empty cargo handling equipment (ballast may be loaded in ULDs), or fly away kits.

NOTE: Operator MELs must define which items are approved for inclusion in the fly away kits and which materials can be used as ballast.

OPERATIONS (O)

The lower cargo compartments must remain empty, except for ballast, empty cargo containers (ballast may be loaded in ULDs), fly away kits, pallets, and cargo restraint components.

26-23-01 Lower Cargo Compartment Fire Extinguisher Bottles
26-23-01-02 With LLAR Installed

Interval	Installed	Required	Procedure
C	-	0	(M) (O)

May be inoperative provided:

- a. LLAR shutoff valve is deactivated closed.
- b. Procedures are established and used to verify cargo compartments remain empty or contain only ballast, empty cargo handling equipment (ballast may be loaded in ULDs), or fly away kits.
- c. LLAR is considered inoperative.

NOTE: Operator MELs must define which items are approved for inclusion in the fly away kits and which materials can be used as ballast.

NOTE: The airplane must also be dispatched using MMEL item 25-29-04.

MAINTENANCE (M)

Deactivate the LLAR shutoff valve closed using DDG item 21-29-01 (M) procedure.

OPERATIONS (O)

1. The lower cargo compartments must remain empty, except for ballast, empty cargo containers (ballast may be loaded in ULDs), fly away kits, pallets, and cargo restraint components.

26-23-01 Lower Cargo Compartment Fire Extinguisher Bottles
26-23-01-03 Metered Bottles

Interval	Installed	Required	Procedure
C	-	1	(M) (O)

May be inoperative provided:

- Bottles 1A and 1B operate normally.
- Metered bottle 2A operates normally.
- Associated bottle is deactivated.
- Airplane is operated pressurized.
- Appropriate flight planning adjustments are applied to account for remaining fire suppression capability.

MAINTENANCE (M)

Deactivate the inoperative fire extinguishing bottle (AMM 26-00-00/901).

- Gain access to the right side of the forward cargo compartment and remove the sidewall lining to expose the inoperative bottle (AMM 25-52-06/401).
- Install the access platform near the inoperative bottle.
- For bottle 2B or 2C, open the P310 Panel FIRE EXT CGO BTL 2A, 2B & 2C circuit breaker.
- For bottle 2D, open the P310 Panel FIRE EXT CGO BTL 2D, 2E & 2F circuit breaker.
- Disconnect the flex hose assembly from the inoperative bottle and plumbing. Plug (MS21913V10P, AS5231V10P or equivalent) the ends and stow the hose near the bottle.
- Cap the discharge head of the bottle with a pressure cap (AS5233V10P), or equivalent).
- Cap the plumbing connection with a pressure cap (AS5233V10P or equivalent).
- Disconnect, cap and stow the electrical connector from the squib.
 - For bottle 2B (B26016), electrical connector DB26016A.
 - For bottle 2C (B26017), electrical connector DB26017A.
 - For bottle 2D (B26026), electrical connector DB26026A.
- Cap the squib (protective cover is attached to the bottle). Do not use a shunt plug. A shunt plug may damage the squib pins.
- Close the P310 Panel FIRE EXT CGO BTL 2A, 2B & 2C and FIRE EXT CGO BTL 2D, 2E & 2F circuit breaker.
- Remove the access platform.
- Re-install the sidewall lining (AMM 25-52-06/401).



13. Initiate a fire extinguisher squib test using one of the following methods:
 - A. Use a Maintenance Access Terminal (MAT) to do the operational test of the cargo squibs (AMM 26-23-00/501). The TEST CONDITION on the MAT will show FAILED due to the deactivated bottle.
 - B. Cycle power to the ELMS P310 panel (airplane power applied) by opening P310 panel circuit breakers PSU B PRI, PSU A PRI, PSU B ALTN, and PSU A ALTN and then closing them. Wait two minutes.
14. Verify fire bottle status message is only displayed for the deactivated bottle.

OPERATIONS (O)

For ETOPS flight planning, use the following total fire suppression times for planning time-limited system alternates minus 15 minutes (14 CFR 121.633).

1. For three metered bottles installed (2A, 2B, and 2C):

NOTE: Bottles 1A, 1B and 2A must operate normally.

Filter/Regulator	Total Fire Suppression Time (minutes)	
	One Bottle Inoperative (2B/2C)	Two Bottles Inoperative (2B/2C)

For -200/-200ER/-200LR:

473995-1/-2/-3 (S218W601-31) 72-lb Bottles	148	84
473995-4/-5 (S218W601-31) 72-lb Bottles	188	104
473995-4/-5 (S218W601-31) 80-lb Bottles	208	114

For -200/-200ER/-200LR/777F:

473494-1/-2/-3 (S218W601-3) 72-lb Bottles	123	71
473494-4/-5 (S218W601-3) 72-lb Bottles	137	78

**For -200LR with Gaseous Oxygen System:**

473995-1/-2/-3 (S218W601-31) 72-lb Bottles	138	74
473995-4/-5 (S218W601-31) 72-lb Bottles	178	94
474688 (S218W601-32) 72-lb Bottles	160	85

For -200LR:

474688 (S218W601-32) 72-lb Bottles	170	95
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For -200LR with Aux Fuel Tank System:

474688 (S218W601-32) 72-lb Bottles	155	80
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For 777F:

474688 (S218W601-32) 72-lb Bottles	171	96
475083 (S218W601-33) 80-lb Bottles	160	90
474688 (S218W601-32) 80-lb Bottles	189	105

For -300/-300ER:

473857-4/-5 (S218W601-30) 80-lb Bottles	141	80
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2. For four metered bottles installed (2A, 2B, 2C and 2D):

NOTE: Bottles 1A, 1B and 2A must operate normally.



Filter/Regulator	Total Fire Suppression Time (minutes)		
	One Bottle Inoperative (2B/2C/2D)	Two Bottles Inoperative (2B/2C/2D)	Three Bottles Inoperative (2B/2C/2D)

For -300/-300ER:

473857-1/-2/-3 (S218W601-30) 80-lb Bottles	185	130	75
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For -300ER:

473857-4/-5 (S218W601-30) 80-lb Bottles	202	141	80
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26-23-01 Lower Cargo Compartment Fire Extinguisher Bottles

26-23-01-04 Bottle Pressure Switches

Interval	Installed	Required	Procedure
C	-	0	(M)

May be inoperative provided:

- Associated bottle is verified full.
- Associated bottle squib firing circuit is verified operating normally before each departure.

MAINTENANCE (M)

NOTE: For a bottle pressure switch failed closed when the bottle pressure is normal or the bottle pressure is low, or the bottle squib fails a squib test, the associated bottle status message will be displayed.

Verify the associated bottle is full and verify the associated squib operates normally before each departure (AMM26-00-00/901).

- Confirm the associated bottle pressure switch is failed closed.
 - Display the Fire Protection maintenance page.
 - Confirm the associated squib condition under the Cargo Fire Extinguishing System is listed as NORM.
 - Confirm the associated bottle pressure condition under the Cargo Fire Extinguishing System is listed as LOW.

2. Remove, weigh and reinstall the associated bottle.
 - A. Remove the associated bottle from the airplane (AMM 26-23-01/401).
 - B. Weigh the bottle. The bottle must weigh the same as the weight on the data plate +/- 0.3 lb.

NOTE: The measured weight of the bottle includes the charged bottle, the inspection tag on the bottle, and the swivel assemblies. If the squib cartridges are not installed on the fire extinguisher bottle, do not install them. Weigh the squib cartridges as loose parts and include the weight of the squib cartridges in the measured weight. If the squib cartridges are installed, weigh the bottle with the cartridges installed. Remove all protective covers when the parts are weighed.

- C. Re-install the bottle in the airplane (AMM 26-23-01/401). It is not necessary to perform the bottle pressure switch operational test associated with this procedure. Confirm the bottle squib operational test condition shows PASSED.
3. Before each departure, verify the associated bottle squib is operating normally.
 - A. If an automatic squib test has not occurred recently (less than 12 hours), manually initiate a fire extinguisher squib test using one of the following methods.

NOTE: A squib test is automatically initiated upon airplane power-up or after landing.

- 1) Use a Maintenance Access Terminal (MAT) to do the operational test of the cargo squibs (AMM 26-23-00/501). The TEST CONDITION on the MAT will show PASSED.
 - 2) Cycle power to the ELMS P310 panel (airplane power applied) by opening P310 panel circuit breakers PSU B PRI, PSU A PRI, PSU B ALTN, and PSU A ALTN and then closing them. Wait two minutes.
 - B. Display the Fire Protection maintenance page and verify the associated squib condition under the Cargo Fire Extinguishing System is listed as NORM.

26-23-02 Lower Cargo Compartment Fire Extinguisher Flow Valves (Forward or Aft)
26-23-02A One Inoperative

Interval	Installed	Required	Procedure
C	2	1	(M) (O)

One may be inoperative provided:

- Associated flow valve is capped.
- Associated flow valve squib electrical connectors are capped and stowed.
- Procedures are established and used to verify the associated cargo compartment remains empty or contains only ballast, empty cargo handling equipment (ballast may be loaded in ULDs), or fly away kits.

For Lower Lobe Attendant Rest (LLAR) installed:

- For aft compartment flow valve inoperative, LLAR is considered inoperative.

NOTE: Operator MELs must define which items are approved for inclusion in the fly away kits and which materials can be used as ballast.

For LLAR installed and aft compartment flow valve inoperative:

NOTE: The airplane must also be dispatched using MMEL item 25-29-04.

MAINTENANCE (M)

Deactivate the inoperative flow valve (AMM 26-00-00/901).

- Gain access to the right side of the lower forward cargo compartment and open the appropriate access panel to expose the inoperative fire extinguishing flow valve (AMM 25-52-06/401).
- Install the access platform near the inoperative flow valve.
- Open the P310 Panel FIRE EXT CGO BTL 1A & VLV SQB circuit breaker.
- Open the P310 Panel FIRE EXT CGO BTL 1B & VLV SQB circuit breaker.
- Disconnect the inlet tube from the inoperative flow valve and the plumbing connection. Plug (MS21913-12, AS5231V12P or equivalent) both ends and stow the inlet tube near the flow valve.
- Cap the inlet port of the flow valve with a pressure cap (BACC14AD12, or equivalent).
- Cap the plumbing connection with a pressure cap (BACC14AD12, or equivalent).
- Disconnect, cap and stow the electrical connectors from the flow valve squibs (two squibs on each flow valve).
 - For forward flow valve (V26201), electrical connectors DV26201A and DV26201B.
 - For aft flow valve (V26202), electrical connectors DV26202A and DV26202B.



9. Cap the squibs (protective covers are attached to the valve). Do not use a shunt plug. A shunt plug may damage the squib pins.
10. Close the P310 Panel FIRE EXT CGO BTL 1A & VLV SQB circuit breaker.
11. Close the P310 Panel FIRE EXT CGO BTL 1B & VLV SQB circuit breaker.
12. Remove the access platform.
13. Close the access panel (AMM 25-52-06/401).
14. Initiate a fire extinguisher squib test by opening P310 panel PSU B PRI, PSU A PRI, PSU B ALTN, and PSU A ALTN circuit breakers and then closing them (with airplane powered). Wait two minutes.
15. Verify the status message for the remaining flow valve is not displayed (only for inoperative flow valve).
16. The associated lower cargo compartment must remain empty, except for ballast, empty cargo containers (ballast may be loaded in ULDs), fly away kits, pallets, and cargo restraint components.

For Lower Lobe Attendant Rest (LLAR) installed:

17. Deactivate the LLAR shutoff valve closed using DDG item 21-29-01 (M) procedure.

OPERATIONS (O)

The associated lower cargo compartment must remain empty, except for ballast, empty cargo containers (ballast may be loaded in ULDs), fly away kits, pallets, and cargo restraint components.

26-23-02 Lower Cargo Compartment Fire Extinguisher Flow Valves (Forward or Aft)

26-23-02B Both Inoperative

Interval	Installed	Required	Procedure
C	2	0	(M) (O)

May be inoperative provided:

- a. Procedures are established and used to verify cargo compartments remain empty or contain only ballast, empty cargo handling equipment (ballast may be loaded in ULDs), or fly away kits.

For Lower Lobe Attendant Rest (LLAR) installed:

- b. LLAR is considered inoperative.

NOTE: Operator MELs must define which items are approved for inclusion in the fly away kits and which materials can be used as ballast.

For LLAR installed:

NOTE: The airplane must also be dispatched using MMEL item 25-29-04.

MAINTENANCE (M)

1. Lower cargo compartments must remain empty, except for ballast, empty cargo containers (ballast may be loaded in ULDs), fly away kits, pallets, and cargo restraint components.

For Lower Lobe Attendant Rest (LLAR) installed:

2. Deactivate the LLAR shutoff valve closed using DDG item 21-29-01 (M) procedure.

OPERATIONS (O)

The lower cargo compartments must remain empty, except for ballast, empty cargo containers (ballast may be loaded in ULDs), fly away kits, pallets, and cargo restraint components.



26-24-01 Lavatory Fire Extinguisher Systems
26-24-01-01 Passenger
26-24-01-01A Smoke Detection System(s) Operative

Interval	Installed	Required	Procedure
C	-	-	

May be inoperative for each lavatory provided associated lavatory smoke detection system operates normally.

26-24-01 Lavatory Fire Extinguisher Systems
26-24-01-01 Passenger
26-24-01-01B Smoke Detection System(s) Inoperative

Interval	Installed	Required	Procedure
C	-	-	(M) (O)

May be inoperative for each lavatory provided associated:

- Lavatory waste container is empty.
- Lavatory door is locked closed and placarded, INOPERATIVE - DO NOT ENTER.
- Lavatory is used only by crewmembers.

NOTE: These provisions are not intended to prohibit lavatory use or inspections by crewmembers.

MAINTENANCE (M)

Deactivate the associated lavatory AMM 26-00-00/901).

- Check that the lavatory waste receptacle in the waste container is empty.
- Close and lock the lavatory door.

OPERATIONS (O)

Lavatory door should remain locked closed, except for use or inspection by crewmembers.

26-24-01 Lavatory Fire Extinguisher Systems
26-24-01-01 Passenger
26-24-01-01C Any In Excess Of Those Required By 14 CFR

Interval	Installed	Required	Procedure
D	-	0	



Any in excess of that required by 14 CFR may be inoperative.

26-24-01 Lavatory Fire Extinguisher Systems
26-24-01-02 777F

Interval	Installed	Required	Procedure
D	1	0	

**26-25-01 Lower Lobe Attendant Rest (LLAR) Fire Extinguisher Bottle**

Interval	Installed	Required	Procedure
C	1	0	(M)

May be inoperative provided:

- LLAR shutoff valve is deactivated closed.
- LLAR is considered inoperative.

NOTE: The airplane must also be dispatched using MMEL item 25-29-04.

MAINTENANCE (M)

Deactivate the LLAR shutoff valve closed using DDG item 21-29-01 (M) procedure.

**26-26-01 Portable Fire Extinguishers**

Interval	Installed	Required	Procedure
D	-	-	

Any in excess of those required by 14 CFR may be inoperative or removed provided:

- Inoperative fire extinguisher remains in a certified location until removed from the aircraft at the next suitable maintenance facility.
- Location placarding is removed or obscured.
- Required distribution is maintained.

NOTE: Inoperative fire extinguishers removed from a certified location or removed from the aircraft, are subject to 49 CFR dangerous goods regulations.

MAINTENANCE NOTE

- Stow the out-of-service extinguisher in a place (possibly in a bag) where it will not be used by mistake, until it can be removed from the airplane at the next available maintenance base.
- Ensure required distribution of serviceable extinguishers is maintained throughout the airplane.

OPERATIONS NOTE

See 14 CFR 121.309 or other appropriate regulations.

26-26-01 Portable Fire Extinguishers

26-26-01-01 Tamper Seals or Tags

Interval	Installed	Required	Procedure
C	-	-	(O)

May be inoperative, damaged or missing provided proper installation and servicing is verified at each preflight.

OPERATIONS (O)

Verify proper tamper seal(s) or tag(s) installation and servicing at each preflight.



**26-27-01 Main Deck Cargo Fire Suppression System (FSS)
(STC ST02375LA)**

Interval	Installed	Required	Procedure
C	1	0	(M)

May be inoperative.

NOTE: The main deck cargo fire suppression system installation defined by STC ST02375LA is not a Boeing system. Information applicable to this installation is not provided.

**26-27-01 Main Deck Cargo Fire Suppression System (FSS)
(STC ST02375LA)**

26-27-01-01 Fire Control Hub

Interval	Installed	Required	Procedure
C	1	0	(M)

May be inoperative provided FSS is considered inoperative.

NOTE: The main deck cargo fire suppression system installation defined by STC ST02375LA is not a Boeing system. Information applicable to this installation is not provided.

**26-27-01 Main Deck Cargo Fire Suppression System (FSS)
(STC ST02375LA)**

26-27-01-02 Fire Control Panel

Interval	Installed	Required	Procedure
C	1	0	(M)

May be inoperative provided surfactant tank and argon tank capacities are verified by an alternative method.

NOTE: The main deck cargo fire suppression system installation defined by STC ST02375LA is not a Boeing system. Information applicable to this installation is not provided.

**26-27-01 Main Deck Cargo Fire Suppression System (FSS)
(STC ST02375LA)**

26-27-01-02 Fire Control Panel
26-27-01-02-01 Surfactant Tank Quantity Indication
26-27-01-02-01A

Interval	Installed	Required	Procedure
C	1	0	(M)

May be inoperative provided FSS is considered to have reduced capability.

NOTE: The main deck cargo fire suppression system installation defined by STC ST02375LA is not a Boeing system. Information applicable to this installation is not provided.

**26-27-01 Main Deck Cargo Fire Suppression System (FSS)
(STC ST02375LA)**

26-27-01-02 Fire Control Panel
26-27-01-02-01 Surfactant Tank Quantity Indication
26-27-01-02-01B

Interval	Installed	Required	Procedure
C	1	0	(M)

May be inoperative provided FCP status light is operative.

NOTE: The main deck cargo fire suppression system installation defined by STC ST02375LA is not a Boeing system. Information applicable to this installation is not provided.

**26-27-01 Main Deck Cargo Fire Suppression System (FSS)
(STC ST02375LA)**

26-27-01-02 Fire Control Panel
26-27-01-02-01 Surfactant Tank Quantity Indication
26-27-01-02-01C

Interval	Installed	Required	Procedure
C	1	0	(M)



May be inoperative provided surfactant level is verified once each flight day.

NOTE: The main deck cargo fire suppression system installation defined by STC ST02375LA is not a Boeing system. Information applicable to this installation is not provided.

**26-27-01 Main Deck Cargo Fire Suppression System (FSS)
(STC ST02375LA)**

26-27-01-02 Fire Control Panel

**26-27-01-02-02 Inert Gas Cylinder (Argon) Pressure Indication -
Electrical Function**

26-27-01-02-02A

Interval	Installed	Required	Procedure
C	1	0	(M)

Electrical function may be inoperative provided FSS is considered inoperative.

NOTE: The main deck cargo fire suppression system installation defined by STC ST02375LA is not a Boeing system. Information applicable to this installation is not provided.

**26-27-01 Main Deck Cargo Fire Suppression System (FSS)
(STC ST02375LA)**

26-27-01-02 Fire Control Panel

**26-27-01-02-02 Inert Gas Cylinder (Argon) Pressure Indication -
Electrical Function**

26-27-01-02-02B

Interval	Installed	Required	Procedure
C	1	0	(M)

Electrical function may be inoperative provided:

- Mechanical indicator is operative.
 - Pressure is verified once each flight day.
-

NOTE: The main deck cargo fire suppression system installation defined by STC ST02375LA is not a Boeing system. Information applicable to this installation is not provided.



**26-27-01 Main Deck Cargo Fire Suppression System (FSS)
(STC ST02375LA)**

26-27-01-03 Fire Control Unit

Interval	Installed	Required	Procedure
C	13	0	

May be inoperative provided FSS is considered inoperative for the affected position.

NOTE: The main deck cargo fire suppression system installation defined by STC ST02375LA is not a Boeing system. Information applicable to this installation is not provided.

**26-27-01 Main Deck Cargo Fire Suppression System (FSS)
(STC ST02375LA)**

26-27-01-04 Fire Detection Unit

Interval	Installed	Required	Procedure
C	47	0	

May be inoperative provided FSS is considered inoperative for the affected position.

NOTE: The main deck cargo fire suppression system installation defined by STC ST02375LA is not a Boeing system. Information applicable to this installation is not provided.

**26-27-01 Main Deck Cargo Fire Suppression System (FSS)
(STC ST02375LA)**

26-27-01-05 Storage and Release System**26-27-01-05-01 Surfactant Tank**

Interval	Installed	Required	Procedure
C	1	0	(M)

May be inoperative provided FSS is considered to have reduced capability.

NOTE: The main deck cargo fire suppression system installation defined by STC ST02375LA is not a Boeing system. Information applicable to this installation is not provided.

**26-27-01 Main Deck Cargo Fire Suppression System (FSS)
(STC ST02375LA)**

26-27-01-05 Storage and Release System**26-27-01-05-02 Inert Gas Cylinder (Argon) Pressure Indication -
Mechanical Function****26-27-01-05-02A**

Interval	Installed	Required	Procedure
C	1	0	(M)

May be inoperative provided FSS is considered inoperative.

NOTE: The main deck cargo fire suppression system installation defined by STC ST02375LA is not a Boeing system. Information applicable to this installation is not provided.

**26-27-01 Main Deck Cargo Fire Suppression System (FSS)
(STC ST02375LA)**

26-27-01-05 Storage and Release System**26-27-01-05-02 Inert Gas Cylinder (Argon) Pressure Indication -
Mechanical Function****26-27-01-05-02B**

Interval	Installed	Required	Procedure
C	1	0	(M)

May be inoperative provided FSS is considered inoperative for affected position.

NOTE: The main deck cargo fire suppression system installation defined by STC ST02375LA is not a Boeing system. Information applicable to this installation is not provided.



**26-27-01 Main Deck Cargo Fire Suppression System (FSS)
(STC ST02375LA)**

26-27-01-05 Storage and Release System**26-27-01-05-03 Penetrator Modules**

Interval	Installed	Required	Procedure
C	13	0	

May be inoperative provided FSS is considered inoperative for affected position.

NOTE: The main deck cargo fire suppression system installation defined by STC ST02375LA is not a Boeing system. Information applicable to this installation is not provided.

**26-27-01 Main Deck Cargo Fire Suppression System (FSS)
(STC ST02375LA)**

**26-27-01-06 Lower Forward Instrument Panel FSS DISCH
Annunciator**

Interval	Installed	Required	Procedure
C	2	0	

May be inoperative.

NOTE: The main deck cargo fire suppression system installation defined by STC ST02375LA is not a Boeing system. Information applicable to this installation is not provided.



Intentionally
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- 27-00-01 Flight Controls Synoptic Display**
- 27-02-01 Bank Angle Protection**
 - 27-02-01-01 Without Standby Attitude Indicator P/N S231W120-4 or Equivalent Installed
 - 27-02-01-02 With Standby Attitude Indicator P/N S231W120-4 or Equivalent Installed
- 27-02-02 Primary Flight Computer (PFC) Lanes**
- 27-02-03 Primary Flight Computer Channels**
- 27-02-04 PFCS Interface**
- 27-02-05 Actuator Pressure Sensors**
- 27-02-06 Flight Control Hydraulic Shutoff Valve Systems**
- 27-02-07 Primary Flight Computer DISC Light**
- 27-02-08 Thrust Asymmetry Compensation (TAC) Function (Except -200LR/GE90-115B and 777F/GE90-115B)**
 - 27-02-08-01 THRUST ASYM COMP Switch Lights
 - 27-02-08-01-01 OFF Light
 - 27-02-08-01-02 AUTO Light
- 27-03-01 Flap/Slat Control Lanes**
- 27-03-02 Flap Secondary Control**
- 27-03-03 Slat Secondary Control**
- 27-11-01 Aileron Trim System**
- 27-11-02 Control Wheel Position Transducers**
- 27-11-03 Aileron Power Control Units (PCUs)**
 - 27-11-03-01 Passenger (With PRRs 61384-15 and 62200 or Equivalent Installed)
 - 27-11-03-02 777F
- 27-11-04 Flaperon Power Control Units (PCUs)**
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27-61-03-01 Passenger (With PRRs 61384-15 and 62200, or Equivalent Installed)

27-61-03-02 777F

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**27-00-01 Flight Controls Synoptic Display**

Interval	Installed	Required	Procedure
C	1	0	

OPERATIONS NOTE

For missing data on the Synoptic Display, selecting an alternate location for the display (L INBD, LWR CTR, or R INBD) may restore the missing data. Synoptic Displays containing missing data may continue to be used to the extent remaining data is useful. Airplane system faults will be annunciated by alerting and status messages.



27-02-01 Bank Angle Protection
27-02-01-01 Without Standby Attitude Indicator P/N S231W120-4 or Equivalent Installed

Interval	Installed	Required	Procedure
C	1	0	

May be inoperative provided flight is conducted in day VMC only.

MAINTENANCE NOTE

Service Bulletin 777-34-0025 (PRR 61760 equivalent) installs improved Standby Attitude Indicator (SAI) part number S231W120-4. The optional Integrated Standby Flight Display (ISFD) is considered equivalent to the S231W120-4 SAI.

27-02-01 Bank Angle Protection
27-02-01-02 With Standby Attitude Indicator P/N S231W120-4 or Equivalent Installed

Interval	Installed	Required	Procedure
C	1	0	

MAINTENANCE NOTE

Service Bulletin 777-34-0025 (PRR 61760 equivalent) installs improved Standby Attitude Indicator (SAI) part number S231W120-4. The optional Integrated Standby Flight Display (ISFD) is considered equivalent to the S231W120-4 SAI.

27-02-02 Primary Flight Computer (PFC) Lanes

Interval	Installed	Required	Procedure
C	9	7	

.....

**27-02-03 Primary Flight Computer Channels**

Interval	Installed	Required	Procedure
A	3	2	(M)

One may be inoperative provided:

- Inoperative PFC channel is deactivated.
- Operations are limited to three flights before repairs are made.
- Approach minimums do not require its use.

MAINTENANCE (M)

NOTE: FLIGHT CONTROL SYS may display with hydraulics depressurized. Pressurize hydraulics and assure FLIGHT CONTROL SYS does not display for dispatch.

Determine which PFC channel is inoperative, then deactivate the inoperative channel (AMM 27-00-00/901).

- Use the Maintenance Access Terminal (MAT) to review CMC data to determine which PFC is inoperative.
 - Using the MAT, select EXISTING FLIGHT DECK EFFECTS.
 - Locate the displayed PFC CHANNEL status message and the correlated maintenance messages.
 - For 23-81001, 27-14018, 27-15000, 27-15001, 27-15002, 27-15100, 27-19101, 27-19201 or 27-19301 maintenance message displayed, the associated L PFC channel is inoperative.
 - For 23-81002, 27-14019, 27-15003, 27-15004, 27-15005, 27-15101, 27-19102, 27-19202 or 27-19302 maintenance message displayed, the associated C PFC channel is inoperative.
 - For 23-81003, 27-14020, 27-15006, 27-15007, 27-15008, 27-15102, 27-19103, 27-19203 or 27-19303 maintenance message displayed, the associated R PFC channel is inoperative.
- Deactivate the inoperative PFC channel.
 - For L PFC channel inoperative, open and collar the PFC LANE 1, PFC LANE 2, and PFC LANE 3 circuit breakers located on PSA-L in the E1-6 rack.
 - For C PFC channel inoperative, open and collar the PFC LANE 1, PFC LANE 2, and PFC LANE 3 circuit breakers located on PSA-C in the E2-6 rack.



-
- C. For R PFC channel inoperative, open and collar the PFC LANE 1, PFC LANE 2, and PFC LANE 3 circuit breakers located on PSA-R in the E5-1 rack.
 3. When electrical power is shutdown, open the BATT INTLK circuit breaker on the front panel of the associated PSA (prevents discharge of the PSA battery).
 4. When electrical power is restored, close the BATT INTLK circuit breaker on the front panel of the associated PSA.

OPERATIONS NOTE

The NO LAND 3 advisory message will be displayed.

**27-02-04 PFCS Interface**

Interval	Installed	Required	Procedure
C	1	0	(M)

May be inoperative provided landing gear truck tilt pressure is verified to be normal before each departure.

MAINTENANCE (M)

Verify truck tilt pressure is normal before each departure (AMM 27-00-00/901).

1. Ensure landing gear lever is in the DOWN position.
2. Pressurize the center hydraulic system (AMM 29-11-00/201).
3. After 2 minutes, verify pressure readings for the L TRUCK TILT and R TRUCK TILT hydraulic lines are 2700 to 3200 psi on the LANDING GEAR ACTN/INDN Maintenance Page.

OPERATIONS NOTE

[For -200LR/777F/-300/-300ER:](#)

The NO LAND 3 advisory message may be displayed.



27-02-05 Actuator Pressure Sensors

Interval	Installed	Required	Procedure
B	7	6	

One flaperon or rudder sensor may be inoperative.

27-02-06 Flight Control Hydraulic Shutoff Valve Systems

Interval	Installed	Required	Procedure
C	6	3	(M)

May be inoperative provided the associated valve is deactivated open.

MAINTENANCE (M)

NOTE: The flight control hydraulic shutoff valve system includes the SOV and associated VALVE CLOSED light and FLT CONTROL VALVE advisory message.

Deactivate the associated hydraulic shutoff valve (SOV) in the open position (AMM 27-00-00/901).

1. Position associated overhead Maintenance Panel (P61) FLT CONTROL HYD POWER valve switch in the NORM position.
2. Open and collar the circuit breaker for the associated flight control hydraulic shutoff valve.
 - A. For the L TAIL valve, open and collar P110 panel L FC SOV TAIL circuit breaker.
 - B. For the L WING valve, open and collar P110 panel L FC SOV WG circuit breaker.
 - C. For the C TAIL valve, open and collar P110 panel C FC SOV TAIL circuit breaker.
 - D. For the C WING valve, open and collar P110 panel C FC SOV WG circuit breaker.
 - E. For the R TAIL valve, open and collar P210 panel R FC SOV TAIL circuit breaker.
 - F. For the R WING valve, open and collar P210 panel R FC SOV WG circuit breaker.
3. Gain access to the associated shutoff valve.

NOTE 1: The L TAIL shutoff valve is forward of the inboard end of the flaperon on the left wing.

NOTE 2: The L WING shutoff valve is forward of the center of the flaperon on the left wing.

NOTE 3: The C TAIL and C WING shutoff valves are located on the aft bulkhead of the left main landing gear wheel well.

NOTE 4: The R TAIL shutoff valve is located in the right main landing gear wheel well, on the top panel near the aft outboard corner.

NOTE 5: The R WING shutoff valve is forward of the center of the flaperon on the right wing.



4. Manually position the lever on the associated shutoff valve to the open position.
5. Install lockwire through the holes provided on the lever and valve body to hold the shutoff valve lever in the open position.
6. If the FLT CONTROL VALVE advisory message remains displayed or the associated shutoff valve VALVE CLOSED light on the overhead Maintenance Panel (P61) remains illuminated after the valve is deactivated open, electrically disconnect the deactivated valve.
 - A. For the Left Tail shutoff valve, disconnect, cap and stow the electrical cable and connector DV27101.
 - B. For the Left Wing shutoff valve, disconnect, cap and stow the electrical cable and connector DV27100.
 - C. For the Right Tail shutoff valve, disconnect, cap and stow the electrical cable and connector DV27102.
 - D. For the Right Wing shutoff valve, disconnect, cap and stow the electrical cable and connector DV27098.
 - E. For the Center Tail shutoff valve, disconnect, cap and stow the electrical cable and connector DV27103.
 - F. For the Center Wing shutoff valve, disconnect, cap and stow the electrical cable and connector DV27099.
7. Re-install the panels removed for valve access.

**27-02-07 Primary Flight Computer DISC Light**

Interval	Installed	Required	Procedure
C	1	0	



**27-02-08 Thrust Asymmetry Compensation (TAC) Function
(Except -200LR/GE90-115B and 777F/GE90-115B)**

Interval	Installed	Required	Procedure
C	1	0	

**27-02-08 Thrust Asymmetry Compensation (TAC) Function
(Except -200LR/GE90-115B and 777F/GE90-115B)****27-02-08-01 THRUST ASYM COMP Switch Lights****27-02-08-01-01 OFF Light**

Interval	Installed	Required	Procedure
C	1	0	

**27-02-08 Thrust Asymmetry Compensation (TAC) Function
(Except -200LR/GE90-115B and 777F/GE90-115B)****27-02-08-01 THRUST ASYM COMP Switch Lights****27-02-08-01-02 AUTO Light**

Interval	Installed	Required	Procedure
C	1	0	

**27-03-01 Flap/Slat Control Lanes**

Interval	Installed	Required	Procedure
C	2	1	(M)

One may be inoperative provided:

- a. Flap lever control position sensor is verified to operate normally.
 - b. Alternate control is verified to operate normally before each departure.
-

MAINTENANCE (M)

Verify the flap lever control position sensor operates normally, and before each departure, verify the alternate control for the flaps and slats operates normally. (AMM 27-00-00/901).

1. Verify the flap lever control position sensor operates normally.
 - A. Supply electrical power.
 - B. Position flap control lever UP.
 - C. On the FLAP/SLAT Maintenance page, confirm all four FLAP LEVER position indications are greater than 9.44.
2. Verify the alternate control for the flaps and slats operates normally before each departure.
 - A. Use the alternate flap control to extend the flaps and slats (AMM 27-51-00/201).
 - 1) Push the control stand (P10) ALTN FLAPS ARM switch to ALTN.
 - 2) Position the control stand (P10) ALTN FLAPS selector to EXT.
 - B. Verify the flaps extend to position 20 and the slats are in the midrange position on the EICAS display.
 - C. Use the alternate flap control to retract the flaps and slats (AMM 27-51-00/201).
 - 1) Push the control stand (P10) ALTN FLAPS ARM switch to ALTN.
 - 2) Position the control stand (P10) ALTN FLAPS selector to RET.
 - D. Verify the flaps and the slats are retracted on the EICAS display.
 - E. Push the control stand ALTN FLAPS ARM switch to off.



27-03-02 Flap Secondary Control

Interval	Installed	Required	Procedure
C	1	0	(M)

May be inoperative provided Slat Secondary Control is verified to operate normally.

MAINTENANCE (M)

Verify Slat Secondary Control operates normally (AMM 27-00-00/901).

- 1. Supply electrical power to the airplane.
- 2. Make sure that the flaps and slats are in the retracted position.
- 3. Make sure that the flap control lever is in the UP position.
- 4. Make sure that the thrust reverser cowl are in the proper position.

WARNING: **Operation of the flaps and slats with the thrust reverser cowl open too far can cause serious injury to persons and damage to equipment.**

- 5. Make sure that the Air Driven Pump (ADP) access door is fully open or fully closed.
- 6. Use a MAT to do the system test of the high lift system secondary controls.
- 7. Repeat Step 6 with the other FSEU.

**27-03-03 Slat Secondary Control**

Interval	Installed	Required	Procedure
C	1	0	(M)

May be inoperative provided Flap Secondary Control is verified to operate normally.

MAINTENANCE (M)

Verify Flap Secondary Control operates normally (AMM 27-00-00/901).

1. Supply electrical power to the airplane.
2. Make sure that the flaps and slats are in the retracted position.
3. Make sure that the flap control lever is in the UP position.
4. Make sure that the thrust reverser cowl is in the proper position.

WARNING: **Operation of the flaps and slats with the thrust reverser cowl open too far can cause serious injury to persons and damage to equipment.**

5. Make sure that the Air Driven Pump (ADP) access door is fully open or fully closed.
6. Use a MAT to do the system test of the high lift system secondary controls.
7. Repeat Step 6 with the other FSEU.

27-11-01 Aileron Trim System

Interval	Installed	Required	Procedure
C	1	0	(M) (O)

May be inoperative provided:

- Aileron trim system is deactivated.
- Aileron trim system is verified to be centered before departure.
- Autopilot roll modes operate normally.

MAINTENANCE (M)

NOTE: For autopilot roll modes to operate normally, the following items must operate normally:

MMEL Item	Associated Status Message
22-11-01C	AFDC L, C, R
22-11-02B	AUTOPILOT BACKDRV L, R
22-11-05-04-02	None
22-11-06-01B	None
22-11-06-08	None
22-11-06-09	None

Deactivate the aileron trim system (AMM 27-00-00/901).

- For aileron trim centered (both control wheels at 0 ± 0.5 units of trim), open and collar the P110 panel AIL TRIM ACTR circuit breaker.
- For aileron trim not centered (both control wheels not at 0 ± 0.5 units of trim), replace the trim actuator with a tool that holds the control wheels in the zero trim position.
 - Remove the aileron trim actuator, saving the bolts, washers, nuts and bushings (AMM 27-11-09/401).
 - Install the aileron trim actuator dummy link A27088-18.
 - Adjust the dummy link to a length of 6 15/16 to 7 1/16 inches (6.937 to 7.063 inches) from center to center between the bolt holes.
 - Install the dummy link in the position of the inoperative actuator, using the same bolts, washers, nuts, and bushings.
 - Secure the trim actuator electrical connector to the dummy link with tape or wire tie.
- Verify that both control wheels are centered (0 ± 0.5 units of trim).

**OPERATIONS (O)**

Before departure, verify that both control wheels are centered (0 ± 0.5 units of trim).

27-11-02 Control Wheel Position Transducers

Interval	Installed	Required	Procedure
C	6	5	

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**27-11-03 Aileron Power Control Units (PCUs)****27-11-03-01 Passenger (With PRRs 61384-15 and 62200 or Equivalent Installed)**

Interval	Installed	Required	Procedure
A	4	3	(M)

One may be inoperative provided:

- Inoperative PCU is deactivated in the bypass mode.
 - Inoperative PCU blocking mode is verified to operate normally.
 - Ailerons are verified to operate normally.
 - Repairs are made within 3 flight days.
-

NOTE: PRRs 61384-15 and 62200 Part-B are applicable to airplane production L/Ns 718, 772, 774 and 776 and-on. This item is also applicable to airplanes with the AIMS-1 or AIMS-2 platforms with BP V14 or later AIMS OPS installed together with the following PFC software installed: PFC OPS 2762-PFC-741-00, or PFC OPS 276D-PFC-841-00, or PFC OPS 2768-PFC-910-00 or later installed.

MAINTENANCE (M)

Deactivate the inoperative aileron PCU in the bypass mode, verify blocking mode operates normally and verify ailerons operate normally (AMM 27-00-00/901).

- For an inoperative left aileron PCU:
 - Open the Power Supply Assembly Center (PSA-C) in E2-6 rack AIL PCU circuit breaker.
 - Open P110 panel AIL PCU LIB(BLK)/LOB(BYP) circuit breaker.
 - Gain access to the left aileron PCUs.
 - Verify blocking mode operates normally on the inoperative PCU.
 - Deactivate the inoperative PCU in bypass mode.
 - For an inoperative left inboard aileron PCU, collar the Power Supply Assembly Center (PSA-C) in E2-6 rack AIL PCU circuit breaker.
 - For an inoperative left outboard aileron PCU, collar the P110 panel AIL PCU LIB(BLK)/LOB(BYP) circuit breaker.
 - Close the operative aileron PCU circuit breaker.
- For an inoperative right aileron PCU:
 - Open the Power Supply Assembly Right (PSA-R) in E5-1 rack AIL PCU circuit breaker.
 - Open the Power Supply Assembly Left (PSA-L) in E1-6 rack AIL PCU circuit breaker.
 - Gain access to the right aileron PCUs.



- D. Verify blocking mode operates normally on the inoperative PCU.
- E. Deactivate the inoperative PCU in bypass mode.
 - 1) For an inoperative right inboard aileron PCU, collar the Power Supply Assembly Right (PSA-R) in E5-1 rack AIL PCU circuit breaker.
 - 2) For an inoperative right outboard aileron PCU, collar the Power Supply Assembly Left (PSA-L) in E1-6 rack AIL PCU circuit breaker.
 - 3) Close the operative aileron PCU circuit breaker.
- 3. Check the operation of the ailerons.

NOTE: Do not do an Actuator Confidence Test, or an ACE Monitor Test, or L or R Aileron Actuator Confidence Tests.
- 4. Confirm the FLIGHT CONTROL SYS status message is not displayed.

OPERATIONS NOTE

When airspeed exceeds approximately 250 KIAS, air loads may cause the associated aileron to move slightly to a trailing-edge-up position while the opposite aileron will normally be commanded to a slight trailing-edge-down position. Some trim input may be necessary to correct for the resulting roll.

27-11-03 Aileron Power Control Units (PCUs)

27-11-03-02 777F

Interval	Installed	Required	Procedure
A	4	3	(M) (O)

One may be inoperative provided:

- a. Inoperative PCU is deactivated in the bypass mode.
- b. Inoperative PCU blocking mode is verified to operate normally.
- c. Ailerons are verified to operate normally.
- d. Appropriate performance adjustments are applied.
- e. Repairs are made within 3 flight days.

MAINTENANCE (M)

Deactivate the inoperative aileron PCU in the bypass mode, verify blocking mode operates normally and verify ailerons operate normally (AMM 27-00-00/901).

- 1. For an inoperative left aileron PCU:
 - A. Open the Power Supply Assembly Center (PSA-C) in E2-6 rack AIL PCU circuit breaker.
 - B. Open P110 panel AIL PCU LIB(BLK)/LOB(BYP) circuit breaker.
 - C. Gain access to the left aileron PCUs.
 - D. Verify blocking mode operates normally on the inoperative PCU.

-
- E. Deactivate the inoperative PCU in bypass mode.
- 1) For an inoperative left inboard aileron PCU, collar the Power Supply Assembly Center (PSA-C) in E2-6 rack AIL PCU circuit breaker.
 - 2) For an inoperative left outboard aileron PCU, collar the P110 panel AIL PCU LIB(BLK)/LOB(BYP) circuit breaker.
 - 3) Close the operative aileron PCU circuit breaker.
2. For an inoperative right aileron PCU:
- A. Open the Power Supply Assembly Right (PSA-R) in E5-1 rack AIL PCU circuit breaker.
 - B. Open the Power Supply Assembly Left (PSA-L) in E1-6 rack AIL PCU circuit breaker.
 - C. Gain access to the right aileron PCUs.
 - D. Verify blocking mode operates normally on the inoperative PCU.
 - E. Deactivate the inoperative PCU in bypass mode.
 - 1) For an inoperative right inboard aileron PCU, collar the Power Supply Assembly Right (PSA-R) in E5-1 rack AIL PCU circuit breaker.
 - 2) For an inoperative right outboard aileron PCU, collar the Power Supply Assembly Left (PSA-L) in E1-6 rack AIL PCU circuit breaker.
 - 3) Close the operative aileron PCU circuit breaker.
3. Check the operation of the ailerons.
- NOTE: Do not do an Actuator Confidence Test, or an ACE Monitor Test, or L or R Aileron Actuator Confidence Tests.
4. Confirm the FLIGHT CONTROL SYS status message is not displayed.

OPERATIONS (O)

NOTE: When airspeed exceeds approximately 250 KIAS, air loads may cause the associated aileron to move slightly to a trailing-edge-up position while the opposite aileron will normally be commanded to a slight trailing-edge-down position. Some trim input may be necessary to correct for the resulting roll.

1. Observe Maximum Gross Weight limits.
 - A. Taxi Weight limit is 722,300 lb (327,629 kg).
 - B. Takeoff Weight limit is 720,300 lb (326,722 kg).
 - C. Landing Weight limit is 575,000 lb (260,815 kg).
 - D. Zero Fuel Weight limit is 547,000 lb (248,115 kg).
2. Observe Center of Gravity (CG) limits.
 - A. Forward Takeoff and Landing CG limits:
 - 1) 300,000 lb (136,077 kg) at 20.0 % MAC.
 - 2) 722,300 lb (327,629 kg) at 20.0 % MAC.



- B. Forward Zero Fuel CG limits:
 - 1) 300,000 lb (136,077 kg) at 20.0 % MAC.
 - 2) 460,000 lb (208,562 kg) at 20.0 % MAC.
 - 3) 547,000 lb (248,115 kg) at 28.0 % MAC.
- C. Aft Takeoff, Landing, and Zero Fuel CG limits:
 - 1) 300,000 lb (136,077 kg) at 29.2 % MAC.
 - 2) 321,900 lb (146,011 kg) at 33.3 % MAC.
 - 3) 450,000 lb (204,115 kg) at 39.5 % MAC.
 - 4) 609,000 lb (276,237 kg) at 44.0 % MAC.
 - 5) 705,300 lb (319,918 kg) at 41.2 % MAC.
 - 6) 722,300 lb (327,629 kg) at 39.9 % MAC.

27-11-04 Flaperon Power Control Units (PCUs)
27-11-04-01 Passenger (With PRRs 61384-15 and 62200, or Equivalent Installed)

Interval	Installed	Required	Procedure
A	4	3	(M)

One may be inoperative provided:

- Inoperative PCU is deactivated in the bypass mode.
- Flaperons are verified to operate normally.
- Repairs are made within 3 flight days.

NOTE: PRRs 61384-15 and 62200 Part-B are applicable to airplane production L/Ns 718, 772, 774 and 776 and-on. This item is also applicable to airplanes with the AIMS-1 or AIMS-2 platforms with BP V14 or later AIMS OPS installed together with the following PFC software installed: PFC OPS 2762-PFC-741-00, or PFC OPS 276D-PFC-841-00, or PFC OPS 2768-PFC-910-00 or later installed.

MAINTENANCE (M)

Deactivate the inoperative flaperon PCU in the bypass mode and verify flaperons operate normally (AMM 27-00-00/901).

- For an inoperative left flaperon outboard PCU, open and collar the Power Supply Assembly Left (PSA-L) in E1-6 rack FLPRN PCU circuit breaker.
- For an inoperative left flaperon inboard PCU, open and collar the Power Supply Assembly Right (PSA-R) in E5-1 rack FLPRN PCU circuit breaker.
- For an inoperative right flaperon outboard PCU, open and collar the Power Supply Assembly Center (PSA-C) in E2-6 rack FLPRN PCU circuit breaker.
- For an inoperative right flaperon inboard PCU:
 - Open P110 panel FLPRN PCU RIB(BYP)/SPLR 4 & 11 circuit breaker.
 - Open P110 panel ACE-L2 PWR circuit breaker.
 - Set Maintenance Panel (P61) left, center and right WING FLT CONTROL HYD POWER switches to SHUT OFF.
 - Disconnect, cap and stow the PCU electrical connector D4422P (WD 27-11-18). The connector is located in the Main Equipment Center on the left side of the E1-1 shelf near the forward cargo access door (station 408, connector position 71).

NOTE: The ACTUATOR PRESS SNSR status message will be displayed. The airplane must also be dispatched using MMEL item 27-02-05.

- E. Close the open circuit breakers.
 - F. Set Maintenance Panel (P61) left, center and right WING FLT CONTROL HYD POWER switches to NORM.
5. Check the operation of the flaperons.
- NOTE: Do not do an Actuator Confidence Test, or an ACE Monitor Test, or L or R Flaperon Actuator Confidence Tests.
6. Confirm the FLIGHT CONTROL SYS status message is not displayed.

27-11-04 Flaperon Power Control Units (PCUs)
27-11-04-02 777F

Interval	Installed	Required	Procedure
A	4	3	(M)

One may be inoperative provided:

- a. Inoperative PCU is deactivated in the bypass mode.
- b. Flaperons are verified to operate normally.
- c. Repairs are made within 3 flight days.

MAINTENANCE (M)

Deactivate the inoperative flaperon PCU in the bypass mode and verify flaperons operate normally (AMM 27-00-00/901).

1. For an inoperative left flaperon outboard PCU, open and collar the Power Supply Assembly Left (PSA-L) in E1-6 rack FLPRN PCU circuit breaker.
2. For an inoperative left flaperon inboard PCU, open and collar the Power Supply Assembly Right (PSA-R) in E5-1 rack FLPRN PCU circuit breaker.
3. For an inoperative right flaperon outboard PCU, open and collar the Power Supply Assembly Center (PSA-C) in E2-6 rack FLPRN PCU circuit breaker.
4. For an inoperative right flaperon inboard PCU:
 - A. Open P110 panel FLPRN PCU RIB(BYP)/SPLR 4 & 11 circuit breaker.
 - B. Open P110 panel ACE-L2 PWR circuit breaker.
 - C. Set Maintenance Panel (P61) left, center and right WING FLT CONTROL HYD POWER switches to SHUT OFF.
 - D. Disconnect, cap and stow the PCU electrical connector D4422P (WD 27-11-18). The connector is located in the Main Equipment Center on the left side of the E1-1 shelf near the forward cargo access door (station 408, connector position 71).

NOTE: The ACTUATOR PRESS SNSR status message will be displayed. The airplane must also be dispatched using MMEL item 27-02-05.



-
- E. Close the open circuit breakers.
 - F. Set Maintenance Panel (P61) left, center and right WING FLT CONTROL HYD POWER switches to NORM.

- 5. Check the operation of the flaperons.

NOTE: Do not do an Actuator Confidence Test, or an ACE Monitor Test, or L or R Flaperon Actuator Confidence Tests.

- 6. Confirm the FLIGHT CONTROL SYS status message is not displayed.



27-21-01 Rudder Trim System**27-21-01-01 Rudder Trim Control High Rate Function**

Interval	Installed	Required	Procedure
C	1	0	

27-21-01 Rudder Trim System**27-21-01-02 Rudder MANUAL TRIM CANCEL Switch**

Interval	Installed	Required	Procedure
C	1	0	

**27-21-02 Rudder Trim Indicator**

Interval	Installed	Required	Procedure
C	1	0	(O)

May be inoperative provided rudder trim is verified centered before each departure.

OPERATIONS (O)

Before each departure, verify rudder trim is centered.

1. Push the rudder trim Manual Trim Cancel switch or rotate Rudder Trim selector as required to center the rudder pedals.
2. Verify the rudder is centered by confirming the left and right rudder pedals are evenly aligned or the Flight Control Synoptic display RUDDER TRIM indication is zero (within 0.2 units left or right).



27-21-03 Rudder Pedal Adjust Crank Handle Spring (With PRR 62539-1R or Equivalent Incorporated)

Interval	Installed	Required	Procedure
C	2	0	(O)

May be inoperative provided rudder pedal adjust crank handle is manually stowed.

OPERATIONS (O)

NOTE: PRR 62539-1R installs the rudder pedal adjust pull-and-turn crank handle, which automatically stows after pedal adjustment. Service Bulletin 777-27-0122 provides the production equivalent of PRR 62539-1R.

When adjusting rudder pedal position, manually stow the affected rudder pedal adjust crank handle.

**27-31-01 Control Column Position Transducers**

Interval	Installed	Required	Procedure
C	6	5	



27-32-01 Stall Warning Systems

Interval	Installed	Required	Procedure
C	2	1	

27-32-01 Stall Warning Systems

27-32-01-01 Stick Shakers

Interval	Installed	Required	Procedure
C	2	1	

**27-41-01 Control Wheel Pitch Trim Switches**

Interval	Installed	Required	Procedure
C	2	1	



27-41-02 Alternate Pitch Trim Position Switches

Interval	Installed	Required	Procedure
C	6	5	

One may be inoperative provided both control wheel pitch trim switches operate normally.

**27-41-03 Stabilizer Trim Rate Solenoid Valves**

Interval	Installed	Required	Procedure
C	2	1	



27-48-01 STAB Position Indicators

Interval	Installed	Required	Procedure
C	2	1	(O)

One may be inoperative provided it is blanked or covered.

OPERATIONS (O)

1. For a blank inoperative indicator, no action is required.
2. For an illuminated inoperative indicator, cover with tape or other material.



27-48-02 Stabilizer Position Transducer Modules

Interval	Installed	Required	Procedure
C	3	2	

**27-48-03 Nose Gear Pressure Switch**

Interval	Installed	Required	Procedure
C	1	0	(O)

May be inoperative provided stabilizer trim position is verified to be properly set before each departure.

MAINTENANCE NOTE

For a Nose Gear Pressure switch leaking pneumatically, isolate the switch (AMM 27-00-00/901).

1. Install Landing Gear downlock pins (AMM 32-00-30/201).
2. Install Landing Gear Door safety pins (AMM 32-00-15/201).
3. Set the nose gear oleo service and pressure switch line shutoff valve to the fully closed position.
4. Acquire a rig pin, J20003-6, and remove the attached flag.
5. Install the rig pin in the rig pin hole to hold the shutoff valve in the closed position, and lockwire (MS20995C32) the rig pin to the valve body using the rig pin flag attachment hole.
6. Remove Landing Gear Door safety pins (AMM 32-00-15/201).
7. Remove Landing Gear downlock pins (AMM 32-00-30/201).

OPERATIONS (O)

Verify the stabilizer trim position is correct for the takeoff thrust, takeoff flaps, gross weight and center of gravity before each departure.

NOTE: The STAB GREENBAND advisory message may be displayed.

1. Confirm the FMC takeoff data has been correctly initialized (takeoff thrust, takeoff flaps, gross weight, center of gravity).
2. Confirm the stabilizer trim is properly set to the position computed for the takeoff thrust, takeoff flaps, gross weight and center of gravity.

27-48-04 Nose Gear Pressure Transducer

Interval	Installed	Required	Procedure
C	1	0	(O)

May be inoperative provided stabilizer trim position is verified to be properly set before each departure.

MAINTENANCE NOTE

For Nose Gear Pressure Transducer leaking pneumatically, isolate the transducer (AMM 27-00-00/901).

1. Install Landing Gear downlock pins (AMM 32-00-30/201).
2. Install Landing Gear Door safety pins (AMM 32-00-15/201).
3. Set the nose gear oleo service and pressure switch line shutoff valve to the fully closed position.
4. Acquire a rig pin, J20003-6, and remove the attached flag.
5. Install the rig pin in the rig pin hole to hold the shutoff valve in the closed position, and lockwire (MS20995C32) the rig pin to the valve body using the rig pin flag attachment hole.
6. Remove Landing Gear Door safety pins (AMM 32-00-15/201).
7. Remove Landing Gear downlock pins (AMM 32-00-30/201).

OPERATIONS (O)

Verify the stabilizer trim position is correct for the takeoff thrust, takeoff flaps, gross weight and center of gravity before each departure.

NOTE: The STAB GREENBAND advisory message may be displayed.

1. Confirm the FMC takeoff data has been correctly initialized (takeoff thrust, takeoff flaps, gross weight, center of gravity).
2. Confirm the stabilizer trim is properly set to the position computed for the takeoff thrust, takeoff flaps, gross weight and center of gravity.

**27-51-01 Inboard Flap, Outboard Support Mechanism Springs
(No. 3 & No. 6 Supports)**

Interval	Installed	Required	Procedure
A	4	2	(M)

One may be broken or missing on each support provided:

- a. Broken springs are removed.
- b. Remaining spring on each support is verified in place and not broken before each departure.
- c. Operations are limited to fifteen departures before repairs are made.

MAINTENANCE (M)

Remove the broken springs and verify remaining spring is in place and not broken before each departure (AMM 27-00-00/901).

1. Supply electrical power (AMM 24-22-00/201).
2. Ensure the thrust reverser cowl is not open farther than the hold-open rod position.
3. Ensure the air-driven pump (ADP) door is fully open or fully closed.
4. Pressurize the center hydraulic system (AMM 29-11-00/201).
5. Extend the trailing edge flaps to the 30-unit position (AMM 27-51-00/201).
6. Remove power from the center hydraulic system (AMM 29-11-00/201).
7. Deactivate the trailing edge flaps (AMM 27-51-11/201).
8. Remove any broken No. 3 or No. 6 flap support springs (AMM 27-51-14/401).

NOTE: For airplanes incorporating SB 777-57-0025 (PRR 61777-53 factory installed equivalent), a damper is installed on the spring assembly. For dispatch with only one spring in place, remove the damper and retain for subsequent repair action.

9. Verify the remaining spring on the associated No. 3 or No. 6 flap support mechanism is in place and not broken before each departure.
10. Return the airplane to its usual condition.
11. Reactivate the trailing edge flaps (AMM 27-51-00/201).
12. Ensure the air-driven pump (ADP) door is fully open or fully closed.
13. Pressurize the center hydraulic system (AMM 29-11-00/201).
14. Retract the trailing edge flaps (AMM 27-51-00/201).
15. Remove center hydraulic system power (AMM 29-11-00/201).



27-51-02

ALTN FLAPS ARM Switch ALTN Light

Interval	Installed	Required	Procedure
C	1	0	

27-59-01 Flap/Slat Skew Sensors

Interval	Installed	Required	Procedure
C	28	14	

**27-61-01 Speedbrake Lever Position Transducers**

Interval	Installed	Required	Procedure
C	4	3	

**27-61-02 Spoiler Auxiliary Retract Device (SARD) (-300/-300ER)**

Interval	Installed	Required	Procedure
A	2	1	

One may be inoperative provided:

- Associated passenger entry door three is considered inoperative.
- Repairs are made within 1 flight day.

OPERATIONS NOTE

For spoiler 7 SARD inoperative, overwing door 3L must be considered inoperative. For spoiler 8 SARD inoperative, overwing door 3R must be considered inoperative. The airplane must also be dispatched using MMEL item 52-11-01.

27-61-03 Spoiler Power Control Units (PCUs)
27-61-03-01 Passenger (With PRRs 61384-15 and 62200, or Equivalent Installed)

Interval	Installed	Required	Procedure
A	14	12	(M) (O)

One symmetrical pair (except 4/11) may be inoperative provided:

- Symmetrical pair is deactivated in the retracted position.
- Remaining spoilers are verified to operate normally.
- Appropriate performance adjustments are applied.
- Repairs are made within 3 flight days.

NOTE 1: Dispatch relief does not apply for a fault condition that is reported to cause asymmetric roll input, or for a spoiler panel detached from the actuator, or for a spoiler not in the retracted position.

NOTE 2: PRRs 61384-15 and 62200 Part-B are applicable to airplane production L/Ns 718, 772, 774 and 776 and-on. This item is also applicable to airplanes with the AIMS-1 or AIMS-2 platforms with BP V14 or later AIMS OPS installed together with the following PFC software installed: PFC OPS 2762-PFC-741-00, or PFC OPS 276D-PFC-841-00, or PFC OPS 2768-PFC-910-00 or later installed.

MAINTENANCE (M)

Deactivate the symmetrical spoiler pair in the retracted position and verify remaining spoilers operate normally (AMM 27-00-00/901).

- Confirm speedbrake lever is in DOWN position.
- For spoilers 1/14 or 7/8, open the Power Supply Assembly Center (PSA-C) in E2-6 rack ACE PWR circuit breaker.
- For spoilers 2/13, open the Power Supply Assembly Left (PSA-L) in E1-6 rack ACE PWR circuit breaker.
- For spoilers 3/12 or 6/9, open the Power Supply Assembly Right (PSA-R) in E5-1 rack ACE PWR circuit breaker.
- For spoilers 5/10, open P110 panel ACE-L2 PWR circuit breaker.
- Depressurize all three hydraulic systems.
- Set Maintenance Panel (P61) left, center and right WING FLT CONTROL HYD POWER switches to SHUT OFF.
- Gain access to the spoiler pair PCUs.
- Disconnect, cap and stow the inoperative PCU electrical connectors.
- Close any opened access panels.
- Close the open circuit breakers.



12. Set Maintenance Panel (P61) left, center and right WING FLT CONTROL HYD POWER switches to NORM.

13. Check the operation of the spoilers.

NOTE: Do not do the ACE Monitor Test.

14. Confirm the FLIGHT CONTROL SYS status message is not displayed.

OPERATIONS (O)

NOTE: The SPOILERS advisory message will be displayed.

Apply the appropriate performance adjustments.

Takeoff Runway/Obstacle	Minimum Takeoff Field Length	Landing Field Length	Minimum Landing Field Length
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For -200 or -200ER/GE90-76B thru -90B:

11,500 lb (5,216 kg)	6,900 ft (2,103 m)	25,000 lb (11,340 kg)	6,600 ft (2,012 m)
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For -200 or -200ER/GE90-94B:

11,000 lb (4,990 kg)	7,200 ft (2,195 m)	25,000 lb (11,340 kg)	7,100 ft (2,164 m)
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For -200 or -200ER/PW4074 thru PW4084:

11,000 lb (4,990 kg)	6,100 ft (1,859 m)	25,000 lb (11,340 kg)	No penalty (No penalty)
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For -200 or -200ER/PW4090:

11,000 lb (4,990 kg)	6,300 ft (1,920 m)	25,000 lb (11,340 kg)	6,700 ft (2,042 m)
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For -200 or -200ER/RR875 thru RR892:

11,000 lb (4,990 kg)	6,200 ft (1,890 m)	24,000 lb (10,886 kg)	6,500 ft (1,981 m)
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For -200 or -200ER/RR895:

11,000 lb (4,990 kg)	6,600 ft (2,012 m)	25,000 lb (11,340 kg)	6,800 ft (2,073 m)
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**For -200LR/GE90-110B:**

11,000 lb (4,990 kg)	7,600 ft (2,316 m)	31,000 lb (14,061 kg)	6,900 ft (2,103 m)
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For -200LR/GE90-115B:

12,000 lb (5,443 kg)	7,800 ft (2,377 m)	27,000 lb (12,247 kg)	7,100 ft (2,164 m)
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For -300/PW4090:

12,000 lb (5,443 kg)	5,000 ft (1,524 m)	26,000 lb (11,794 kg)	No penalty (No penalty)
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For -300/RR892:

11,000 lb (4,990 kg)	4,800 ft (1,463 m)	25,000 lb (11,340 kg)	No penalty (No penalty)
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For -300ER/GE90-115B:

12,000 lb (5,443 kg)	6,100 ft (1,859 m)	31,000 lb (14,062 kg)	No penalty (No penalty)
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27-61-03 Spoiler Power Control Units (PCUs)**27-61-03-02 777F**

Interval	Installed	Required	Procedure
A	14	12	(M) (O)

One symmetrical pair (except 4/11) may be inoperative provided:

- Symmetrical pair is deactivated in the retracted position.
- Remaining spoilers are verified to operate normally.
- Appropriate performance adjustments are applied.
- Repairs are made within 3 flight days.

NOTE: Dispatch relief does not apply for a fault condition that is reported to cause asymmetric roll input, or for a spoiler panel detached from the actuator, or for a spoiler not in the retracted position.

MAINTENANCE (M)

Deactivate the symmetrical spoiler pair in the retracted position and verify remaining spoilers operate normally (AMM 27-00-00/901).

1. Confirm speedbrake lever is in DOWN position.
2. For spoilers 1/14 or 7/8, open the Power Supply Assembly Center (PSA-C) in E2-6 rack ACE PWR circuit breaker.
3. For spoilers 2/13, open the Power Supply Assembly Left (PSA-L) in E1-6 rack ACE PWR circuit breaker.
4. For spoilers 3/12 or 6/9, open the Power Supply Assembly Right (PSA-R) in E5-1 rack ACE PWR circuit breaker.
5. For spoilers 5/10, open P110 panel ACE-L2 PWR circuit breaker.
6. Depressurize all three hydraulic systems.
7. Set Maintenance Panel (P61) left, center and right WING FLT CONTROL HYD POWER switches to SHUT OFF.
8. Gain access to the spoiler pair PCUs.
9. Disconnect, cap and stow the inoperative PCU electrical connectors.
10. Close any opened access panels.
11. Close the open circuit breakers.
12. Set Maintenance Panel (P61) left, center and right WING FLT CONTROL HYD POWER switches to NORM.
13. Check the operation of the spoilers.

NOTE: Do not do the ACE Monitor Test.

14. Confirm the FLIGHT CONTROL SYS status message is not displayed.

OPERATIONS (O)

NOTE: The SPOILERS advisory message will be displayed.

1. Observe Maximum Gross Weight limits.
 - A. Taxi Weight limit is 722,300 lb (327,629 kg).
 - B. Takeoff Weight limit is 720,300 lb (326,722 kg).
 - C. Landing Weight limit is 575,000 lb (260,815 kg).
 - D. Zero Fuel Weight limit is 547,000 lb (248,115 kg).
2. Observe Center of Gravity (CG) limits.
 - A. Forward Takeoff and Landing CG limits:
 - 1) 300,000 lb (136,077 kg) at 20.0% MAC.
 - 2) 722,300 lb (327,629 kg) at 20.0% MAC.
 - B. Forward Zero Fuel CG limits:
 - 1) 300,000 lb (136,077 kg) at 20.0% MAC.
 - 2) 460,000 lb (208,562 kg) at 20.0% MAC.
 - 3) 547,000 lb (248,115 kg) at 28.0% MAC.
 - C. Aft Takeoff, Landing, and Zero Fuel CG limits:
 - 1) 300,000 lb (136,077 kg) at 29.2% MAC.



- 2) 321,900 lb (146,011 kg) at 33.3% MAC.
- 3) 450,000 lb (204,115 kg) at 39.5% MAC.
- 4) 609,000 lb (276,237 kg) at 44.0% MAC.
- 5) 705,300 lb (319,918 kg) at 41.2% MAC.
- 6) 722,300 lb (327,629 kg) at 39.9% MAC.

3. Apply appropriate performance adjustments.

Takeoff Runway/Obstacle	Minimum Takeoff Field Length	Landing Field Length	Minimum Landing Field Length
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For 777F/GE90-110B:

11,000 lb (4,990 kg)	7,600 ft (2,316 m)	31,000 lb (14,061 kg)	6,900 ft (2,103 m)
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For 777F/GE90-115B:

12,000 lb (5,443 kg)	7,800 ft (2,377 m)	27,000 lb (12,247 kg)	7,100 ft (2,164 m)
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27-62-01 Automatic Speedbrake Function

Interval	Installed	Required	Procedure
C	1	0	(M) (O)

May be inoperative provided:

- Automatic speedbrake function is deactivated.
- Landing gear truck tilt pressure is verified to be normal before each flight.
- Appropriate performance adjustments are applied.

MAINTENANCE (M)

Deactivate the automatic speedbrake and verify truck tilt pressure is normal before each flight (AMM 27-00-00/901).

- Open and collar left power supply assembly (PSA-L) AUTO SPDBRK (L, R) circuit breaker.

NOTE: The PSA-L is located in the E1-6 rack in the Main Equipment Center.

- Open and collar right power supply assembly (PSA-R) AUTO SPDBRK (L, R) circuit breaker.

NOTE: The PSA-R is located in the E5-1 rack aft of the forward cargo door.

- Ensure landing gear lever is in the DOWN position.
- Pressurize the center hydraulic system.
- After 2 minutes, verify pressure readings for the L TRUCK TILT and R TRUCK TILT hydraulic lines are 2700 to 3200 psi on the LANDING GEAR ACTN/INDN EICAS Maintenance Page.

OPERATIONS (O)

- Base landing performance on manual speedbrakes.
- Extend speedbrakes manually for rejected takeoff and landing.
 - For rejected takeoff:
 - Simultaneously close the thrust levers, disengage the autothrottles and apply maximum manual wheel brakes or verify operation of RTO autobrakes.
 - Manually raise SPEED BRAKE lever.
 - Apply the maximum amount of reverse thrust consistent with conditions.
 - For landing, use the AUTO SPEEDBRAKE Non-Normal Checklist.



Intentionally
Blank

**28-11-01 Sump Drain Valves**

28-11-01-01 Surge Tanks

28-11-01-02 Main and Center Tanks

28-11-01-03 Auxiliary Tank(s)

28-11-01-03A *Auxiliary Tank(s) Usable*28-11-01-03B *Auxiliary Tank(s) Empty***28-14-01 Auxiliary Tank Fuel Vent Valves**28-14-01A *Auxiliary Tank(s) Usable*28-14-01B *Auxiliary Tank(s) Empty***28-14-02 Auxiliary Tank Refuel Isolation Valves**28-14-02A *One inoperative*28-14-02B *Auxiliary Tank(s) Empty***28-14-03 Auxiliary Tank Refuel Transfer Valve(s)**28-14-03A *Auxiliary Tank(s) Usable*28-14-03B *Auxiliary Tank(s) Empty***28-14-04 Auxiliary Tank Fuel Pump(s)**28-14-04A *Auxiliary Tank(s) Usable*28-14-04B *Auxiliary Tank(s) Empty***28-14-05 Auxiliary Tank Fuel Pump Isolation Valve(s)**28-14-05A *Auxiliary Tank(s) Usable*28-14-05B *Auxiliary Tank(s) Empty***28-14-06 Auxiliary Tank Fuel Transfer Valves**28-14-06A *One Inoperative*28-14-06B *Auxiliary Tank(s) Empty***28-14-07 Auxiliary Tank Fuel Backup Transfer System**28-14-07A *Auxiliary Tank(s) Usable*28-14-07B *Auxiliary Tank(s) Empty***28-14-08 Auxiliary Tank Fuel Pressure Hold Mode**28-14-08A *Auxiliary Tank(s) Usable*

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28-21-01-02 Integrated Refuel Panel

28-21-01-02-01 Fuel Quantity Indicators

28-21-01-02-02 Load Select System

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28-21-01-02-08 Refuel Valve Switches

28-21-02 Auxiliary Tank Fuel Quantity Indication (Auxiliary Refuel Panel)

28-21-02A *Auxiliary Tank(s) Usable*

28-21-02B *Auxiliary Tank(s) Empty*

28-21-03 Refuel Adapters**28-22-01 Main Tank Fuel Pumps**

28-22-01-01 PUMPS Switch PRESS Lights

28-22-01-02 PUMPS Switch ON Lights

28-22-02 Center Tank Fuel Override/Jettison Pumps

28-22-02A *One Pump Inoperative*

28-22-02B *Both Pumps Inoperative*

28-22-02-01 PUMPS Switch PRESS Lights

28-22-02-02 PUMPS Switch ON Lights

28-22-03 Fuel Crossfeed Valves

28-22-03-01 CROSSFEED Switch VALVE Lights

28-22-04 Fuel Scavenge Systems

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28-31-02-02B	<i>Main Tank Jettison Operative</i>
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28-31-02-03A	<i>Fuel Jettison System Deactivated</i>
28-31-02-03B	<i>Main Tank Jettison Operative</i>
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28-31-03-01	Without PRR 62777-79S or Equivalent Installed

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28-31-04-01B	<i>Both Inoperative</i>
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28-31-04-01-02	Switch ON Lights
28-31-04-02	With PRR 62777-79S or Equivalent Installed
28-31-04-02A	<i>One Inoperative</i>
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28-45-01-01	Auxiliary Tank 1
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28-45-01-03	Auxiliary Tank 3

28-11-01 Sump Drain Valves
28-11-01-01 Surge Tanks

Interval	Installed	Required	Procedure
C	2	0	(M)

May be inoperative provided there is no evidence of leakage.

MAINTENANCE (M)

Visually inspect exterior surface location of the inoperative sump drain valve to verify that there is no evidence of fuel leakage (AMM 28-00-00/901).

28-11-01 Sump Drain Valves
28-11-01-02 Main and Center Tanks

Interval	Installed	Required	Procedure
C	4	3	(M)

One may be inoperative provided:

- There is no evidence of leakage.
- Alternate procedures are used to prevent water accumulation in associated tank.

MAINTENANCE (M)

Verify that there is no evidence of leakage and use alternate procedures to prevent water accumulation in associated tank (AMM 28-26-00/201).

- Visually inspect exterior surface location of the inoperative sump drain valve to verify that there is no evidence of fuel leakage.
- Use alternate procedures to prevent water accumulation in the associated tank.

28-11-01 Sump Drain Valves
28-11-01-03 Auxiliary Tank(s)

28-11-01-03A Auxiliary Tank(s) Usable

Interval	Installed	Required	Procedure
C	-	0	(M)

May be inoperative provided:

- There is no evidence of leakage.

- b. Alternate procedures are used to prevent water accumulation in associated tank(s).
-

MAINTENANCE (M)

Verify that there is no evidence of leakage and use alternate procedures to prevent water accumulation in associated tank (AMM 28-26-00/201).

1. Visually inspect exterior surface location of the inoperative sump drain valve to verify that there is no evidence of fuel leakage.
2. Use alternate procedures to prevent water accumulation in the auxiliary fuel tank(s).

28-11-01 Sump Drain Valves**28-11-01-03 Auxiliary Tank(s)***********28-11-01-03B Auxiliary Tank(s) Empty**

Interval	Installed	Required	Procedure
D	-	0	

May be inoperative provided all auxiliary tanks remain empty.

MAINTENANCE NOTE

For auxiliary tank(s) not empty, defuel the auxiliary tank(s) (AMM 28-26-00/201).

28-14-01 Auxiliary Tank Fuel Vent Valves

28-14-01A Auxiliary Tank(s) Usable

Interval	Installed	Required	Procedure
C	2	0	(M) (O)

May be inoperative provided:

- a. Inoperative valve is deactivated open.
- b. Auxiliary tank fuel pump transfer system operates normally.
- c. Auxiliary tank fuel quantity indication operates normally.
- d. Maximum zero fuel weight limit is reduced by the weight of auxiliary tank fuel.

MAINTENANCE (M)

NOTE: For an auxiliary tank fuel pump transfer system to be operating normally, the following items associated with that system must be operative:

MMEL Item	Associated Status Message
28-14-04	AUX FUEL PUMP 1, 2, 3
28-14-05	AUX PUMP ISLN VLV 1, 2, 3
28-14-06	AUX XFR VLV 1, 2
28-22-07	None
28-42-01	AUX PRESS SNSR PUMP 1, 2, 3

Deactivate the inoperative vent valve open (AMM 28-00-00/901).

1. For inoperative auxiliary vent valve 1:
 - A. Open and collar P110 panel AUX FUEL VENT VLV 1 PWR circuit breaker.
 - B. Open and collar P310 panel AUX FUEL VENT VLV PWR HOT BAT circuit breaker.
 - C. Open and collar P320 panel AUX FUEL VLVS GND PWR circuit breaker.
2. For inoperative auxiliary vent valve 2:
 - A. Open and collar P310 panel AUX FUEL VENT VLV 2 PWR circuit breaker.
 - B. Open and collar P310 panel AUX FUEL VENT VLV PWR HOT BAT circuit breaker.
 - C. Open and collar P320 panel AUX FUEL VLVS GND PWR circuit breaker.
3. Gain access to the inoperative auxiliary tank fuel vent valve.



4. Manually open the inoperative auxiliary tank fuel vent valve.

OPERATIONS (O)

Include auxiliary tank fuel as part of Zero Fuel Weight.

28-14-01 Auxiliary Tank Fuel Vent Valves*********28-14-01B Auxiliary Tank(s) Empty**

Interval	Installed	Required	Procedure
D	2	0	

May be inoperative provided all auxiliary tanks remain empty.

MAINTENANCE NOTE

For auxiliary tank(s) not empty, defuel the auxiliary tank(s) (AMM 28-26-00/201).

**28-14-02 Auxiliary Tank Refuel Isolation Valves**

28-14-02A One inoperative

Interval	Installed	Required	Procedure
C	2	1	

28-14-02 Auxiliary Tank Refuel Isolation Valves

28-14-02B Auxiliary Tank(s) Empty

Interval	Installed	Required	Procedure
D	2	0	

May be inoperative provided all auxiliary tanks remain empty.

MAINTENANCE NOTE

For auxiliary tank(s) not empty, defuel the auxiliary tank(s) (AMM 28-26-00/201).

**28-14-03 Auxiliary Tank Refuel Transfer Valve(s)**

28-14-03A Auxiliary Tank(s) Usable

Interval	Installed	Required	Procedure
C	-	0	(M) (O)

May be inoperative provided:

- Inoperative valve is deactivated closed.
- The auxiliary tank fuel pump transfer system operates normally.
- Auxiliary tank fuel quantity indication operates normally.
- Maximum zero fuel weight limit is reduced by the weight of the associated auxiliary tank fuel.

MAINTENANCE (M)

NOTE: For the auxiliary tank fuel pump transfer system to be operating normally, certain items associated with that system must be operative. See DDG item 28-14-01 for a listing of those items.

I Deactivate the inoperative refuel transfer valve closed (AMM 28-26-00/201).

- For auxiliary tank 1 refuel transfer valve inoperative, open and collar P310 panel AUX RFUEL XFER VLV 1 circuit breaker.
- For auxiliary tank 2 installed with refuel transfer valve inoperative, open and collar P310 panel AUX RFUEL XFER VLV 2 circuit breaker.
- For auxiliary tank 3 installed with refuel transfer valve inoperative, open and collar P310 panel AUX RFUEL XFER VLV 3 circuit breaker.
- Gain access to the inoperative auxiliary tank refuel transfer valve.
- Manually close the inoperative auxiliary tank refuel transfer valve.

NOTE: Valve may be manually opened for refuel.

OPERATIONS (O)

Reduce the maximum fuel weight by the weight of the auxiliary tank fuel.

28-14-03 Auxiliary Tank Refuel Transfer Valve(s)

28-14-03B Auxiliary Tank(s) Empty

Interval	Installed	Required	Procedure
D	-	0	

May be inoperative provided all auxiliary tanks remain empty.

**MAINTENANCE NOTE**

For auxiliary tank(s) not empty, defuel the auxiliary tank(s) (AMM 28-26-00/201). |

**28-14-04 Auxiliary Tank Fuel Pump(s)**

28-14-04A Auxiliary Tank(s) Usable

Interval	Installed	Required	Procedure
C	-	0	(M) (O)

May be inoperative provided:

- Inoperative auxiliary tank fuel pump is deactivated.
- Auxiliary tank fuel backup transfer system operates normally.
- Auxiliary tank fuel quantity indication operates normally.
- Maximum zero fuel weight limit is reduced by the weight of the associated auxiliary tank fuel.

MAINTENANCE (M)

NOTE: For an auxiliary tank fuel backup transfer system to be operating normally, the following items associated with that system must be operative:

MMEI Item	Associated Status Message
28-14-01	AUX VENT VLV 1, 2
28-14-03	AUX RFUEL XFR VLV 1, 2, 3
28-14-06	AUX XFR VLV 1, 2
28-14-07	AUX FUEL BACKUP SYS
28-22-07	None
28-45-01-01	AUX PRESS SNSR CELL 1

Deactivate the inoperative auxiliary tank fuel pump (AMM 28-00-00/901).

- For auxiliary tank 1 pump inoperative, open and collar P210 panel AUX TANK PUMP 1 PWR circuit breaker.
- For auxiliary tank 2 installed with pump inoperative, open and collar P210 panel AUX TANK PUMP 2 PWR circuit breaker.
- For auxiliary tank 3 installed with pump inoperative, open and collar P210 panel AUX TANK PUMP 3 PWR circuit breaker.

OPERATIONS (O)

Reduce the maximum fuel weight by the weight of the auxiliary tank fuel.

28-14-04 Auxiliary Tank Fuel Pump(s)

28-14-04B Auxiliary Tank(s) Empty

Interval	Installed	Required	Procedure
D	-	0	(M)

May be inoperative provided:

- a. Inoperative auxiliary tank fuel pump is deactivated.
- b. All auxiliary tanks remain empty.

MAINTENANCE (M)

Deactivate the inoperative auxiliary tank fuel pump and defuel the auxiliary tank(s) (AMM 28-00-00/901).

1. For auxiliary tank 1 pump inoperative, open and collar P210 panel AUX TANK PUMP 1 PWR circuit breaker.
2. For auxiliary tank 2 installed with pump inoperative, open and collar P210 panel AUX TANK PUMP 2 PWR circuit breaker.
3. For auxiliary tank 3 installed with pump inoperative, open and collar P210 panel AUX TANK PUMP 3 PWR circuit breaker.
4. For auxiliary tank(s) not empty, defuel the auxiliary tank(s).

**28-14-05 Auxiliary Tank Fuel Pump Isolation Valve(s)**

28-14-05A Auxiliary Tank(s) Usable

Interval	Installed	Required	Procedure
C	-	0	(M) (O)

May be inoperative provided:

- Inoperative valve is deactivated closed.
- Associated auxiliary tank fuel pump is deactivated.
- Auxiliary tank fuel backup transfer system operates normally.
- Auxiliary tank fuel quantity indication operates normally.
- Maximum zero fuel weight limit is reduced by the weight of the associated auxiliary tank fuel.

MAINTENANCE (M)

NOTE: For the auxiliary tank fuel backup transfer system to be operating normally, certain items associated with that system must be operative. See DDG item 28-14-04 for a listing of those items.

Deactivate the inoperative auxiliary tank fuel pump isolation valve closed, and deactivate the associated auxiliary fuel tank pump (AMM 28-00-00/901).

- For auxiliary tank 1 pump isolation valve inoperative:
 - Open and collar P210 panel AUX FUEL PUMP ISLN VLV 1 PWR circuit breaker.
 - Open and collar P210 panel AUX TANK PUMP 1 PWR circuit breaker.
 - Gain access to the inoperative isolation valve.
 - Manually close the inoperative isolation valve.
- For auxiliary tank 2 installed with pump isolation valve inoperative:
 - Open and collar P210 panel AUX FUEL PUMP ISLN VLV 2 PWR circuit breaker.
 - Open and collar P210 panel AUX TANK PUMP 2 PWR circuit breaker.
 - Gain access to the inoperative isolation valve.
 - Manually close the inoperative isolation valve.
- For auxiliary tank 3 installed with pump isolation valve inoperative:
 - Open and collar P210 panel AUX FUEL PUMP ISLN VLV 3 PWR circuit breaker.
 - Open and collar P210 panel AUX TANK PUMP 3 PWR circuit breaker.
 - Gain access to the inoperative isolation valve.

- D. Manually close the inoperative isolation valve.

OPERATIONS (O)

Include auxiliary tank fuel from associated tank(s) as part of Zero Fuel Weight.

28-14-05 Auxiliary Tank Fuel Pump Isolation Valve(s)

28-14-05B Auxiliary Tank(s) Empty

Interval	Installed	Required	Procedure
D	-	0	(M)

May be inoperative provided:

- a. Inoperative valve is deactivated closed.
- b. Associated auxiliary tank fuel pump is deactivated.
- c. All auxiliary tanks remain empty

MAINTENANCE (M)

Deactivate the inoperative auxiliary tank fuel pump isolation valve closed, deactivate the associated auxiliary fuel tank pump and defuel the auxiliary tank(s) (AMM 28-00-00/901).

1. For auxiliary tank 1 pump isolation valve inoperative:
 - A. Open and collar P210 panel AUX FUEL PUMP ISLN VLV 1 PWR circuit breaker.
 - B. Open and collar P210 panel AUX TANK PUMP 1 PWR circuit breaker.
 - C. Gain access to the inoperative isolation valve.
 - D. Manually close the inoperative isolation valve.
2. For auxiliary tank 2 installed with pump isolation valve inoperative:
 - A. Open and collar P210 panel AUX FUEL PUMP ISLN VLV 2 PWR circuit breaker.
 - B. Open and collar P210 panel AUX TANK PUMP 2 PWR circuit breaker.
 - C. Gain access to the inoperative isolation valve.
 - D. Manually close the inoperative isolation valve.
3. For auxiliary tank 3 installed with pump isolation valve inoperative:
 - A. Open and collar P210 panel AUX FUEL PUMP ISLN VLV 3 PWR circuit breaker.
 - B. Open and collar P210 panel AUX TANK PUMP 3 PWR circuit breaker.
 - C. Gain access to the inoperative isolation valve.
 - D. Manually close the inoperative isolation valve.
4. For auxiliary tank(s) not empty, defuel the auxiliary tank(s).

**28-14-06 Auxiliary Tank Fuel Transfer Valves**

28-14-06A One Inoperative

Interval	Installed	Required	Procedure
C	2	1	(M) (O)

One may be inoperative provided:

- Inoperative valve is deactivated closed.
 - Maximum zero fuel weight limit is reduced by the weight of auxiliary tank fuel.
-

MAINTENANCE (M)

Deactivate the inoperative transfer valve closed (AMM 28-00-00/901).

- For auxiliary tank fuel transfer valve 1 inoperative, open and collar P110 panel AUX FUEL XFR VLV 1 PWR circuit breaker.
- For auxiliary tank fuel transfer valve 2 inoperative, open and collar P310 panel AUX FUEL XFR VLV 2 PWR circuit breaker.
- Gain access to the inoperative auxiliary tank fuel transfer valve.
- Manually close the inoperative auxiliary tank fuel transfer valve.

OPERATIONS (O)

Reduce the maximum fuel weight by the weight of the auxiliary tank fuel.

28-14-06 Auxiliary Tank Fuel Transfer Valves

28-14-06B Auxiliary Tank(s) Empty

Interval	Installed	Required	Procedure
D	2	0	(M)

May be inoperative provided:

- Inoperative valve is deactivated closed.
 - All auxiliary tanks remain empty.
-

MAINTENANCE (M)

Deactivate the inoperative transfer valve closed and defuel the auxiliary tank(s) (AMM 28-00-00/901).

- For auxiliary tank fuel transfer valve 1 inoperative, open and collar P110 panel AUX FUEL XFR VLV 1 PWR circuit breaker.
- For auxiliary tank fuel transfer valve 2 inoperative, open and collar P310 panel AUX FUEL XFR VLV 2 PWR circuit breaker.
- Gain access to the inoperative auxiliary tank fuel transfer valve.



4. Manually close the inoperative auxiliary tank fuel transfer valve.
5. For auxiliary tank(s) not empty, defuel the auxiliary tank(s).

**28-14-07 Auxiliary Tank Fuel Backup Transfer System**

28-14-07A Auxiliary Tank(s) Usable

Interval	Installed	Required	Procedure
C	1	0	(M) (O)

May be inoperative provided:

- Associated auxiliary tank fuel backup transfer PRSOV is deactivated.
- Auxiliary tank fuel vent valves operate normally.
- Auxiliary tank fuel pump transfer system operates normally.
- Maximum zero fuel weight limit is reduced by the weight of auxiliary tank fuel.

MAINTENANCE (M)

NOTE: For the auxiliary tank fuel pump transfer system to be operating normally, certain items associated with that system must be operative. See DDG item 28-14-01 for a listing of those items.

Deactivate the associated auxiliary tank fuel backup transfer PRSOV (AMM

28-26-00/201).

- Open and collar P110 panel AIR SUPPLY-AUX FUEL TANK PRESS CTRL circuit breaker.

OPERATIONS (O)

Reduce the maximum fuel weight by the weight of the auxiliary tank fuel.

28-14-07 Auxiliary Tank Fuel Backup Transfer System

28-14-07B Auxiliary Tank(s) Empty

Interval	Installed	Required	Procedure
D	1	0	

May be inoperative provided:

- Auxiliary tank fuel vent valves operate normally.
- All auxiliary tanks remain empty.

MAINTENANCE NOTE

For auxiliary tank(s) not empty, defuel the auxiliary tank(s) (AMM 28-26-00/201).

**28-14-08 Auxiliary Tank Fuel Pressure Hold Mode**

28-14-08A Auxiliary Tank(s) Usable

Interval	Installed	Required	Procedure
C	1	0	(O)

May be inoperative provided:

- There is no evidence of fuel leaking from the drain mast.
 - Auxiliary tank fuel pump transfer system operates normally.
 - Maximum zero fuel weight limit is reduced by the weight of auxiliary tank fuel.
-

OPERATIONS (O)

Include auxiliary tank fuel as part of Zero Fuel Weight.

28-14-08 Auxiliary Tank Fuel Pressure Hold Mode

28-14-08B Auxiliary Tank(s) Empty

Interval	Installed	Required	Procedure
D	1	0	

May be inoperative provided all auxiliary tanks remain empty.

MAINTENANCE NOTE

For auxiliary tank(s) not empty, defuel the auxiliary tank(s) (AMM 28-26-00/201). |

28-21-01 Pressure Fueling System**28-21-01-01 Refuel Valves****28-21-01-01A Refuel Valve Closed**

Interval	Installed	Required	Procedure
C	6	0	(M)

May be inoperative provided:

- Inoperative refuel valve(s) is closed.
- Alternate procedures are used for refueling.

MAINTENANCE (M)

- For one refuel valve inoperative closed per tank, use normal refueling procedures (refueling time for the associated tank will be increased).
- For more than one refuel valve inoperative closed per tank (AMM 12-11-01/301):
 - Verify that fuel load for the associated tank does not exceed the tank capacity.
 - Manually open one or both of associated tank's refuel valves.

NOTE: OVERFILL PROTECTION IS INOPERATIVE. THE FUEL SYSTEM WILL NOT STOP THE REFUEL OPERATION AT A SET FUEL QUANTITY. STOP THE FUELING SOURCE AT THE DESIRED QUANTITY FOR THE ASSOCIATED TANK.

- Refuel the associated tank, stopping the fueling source at the desired fuel quantity.
- Manually close the refuel valve(s).

28-21-01 Pressure Fueling System**28-21-01-01 Refuel Valves****28-21-01-01B Refuel Valve Open**

Interval	Installed	Required	Procedure
C	6	5	(M)

One main tank refuel valve may be inoperative provided:

- Inoperative refuel valve is open.
- Fuel jettison system is considered inoperative.
- Alternate procedures are used for refueling.

NOTE 1: There are 6 refuel valves installed on the 777 airplane, 2 for the left main tank, 2 for the center tank, and 2 for the right main tank. However, item MEL/DDG item 28-21-01-01B for an inoperative open refuel valve offers dispatch relief only for either the left or right main tank refuel valves. Dispatch is not allowed with an inoperative open center tank refuel valve. Dispatch relief for an inoperative center tank refuel valve is only available per item 28-21-01-01A with the center tank refuel valve inoperative closed.

NOTE 2: The airplane must also be dispatched using MMEL item 28-31-01.

MAINTENANCE (M)

Use alternate refueling procedures (AMM 12-11-01/301).

1. Verify that fuel load for the associated tank does not exceed the tank capacity.

NOTE: OVERFILL PROTECTION IS INOPERATIVE. THE FUEL SYSTEM WILL NOT STOP THE REFUEL OPERATION AT A SET FUEL QUANTITY. STOP THE FUELING SOURCE AT THE DESIRED QUANTITY FOR THE ASSOCIATED TANK.

2. Refuel the associated tank, stopping the fueling source at the desired fuel quantity.

28-21-01 Pressure Fueling System

28-21-01-02 Integrated Refuel Panel

28-21-01-02-01 Fuel Quantity Indicators

Interval	Installed	Required	Procedure
C	4	0	(M)

May be inoperative provided alternate procedures are used for refueling.

MAINTENANCE (M)

NOTE: Global Alternative Method of Compliance (GAMOC) to AD 2020-11-11 (approved by FAA letter 784-20-9463) provides maintenance procedures to follow when refueling the Center Wing Tank with the IRP Center Fuel Quantity indicator or Total Fuel Quantity indicator inoperative.

Refuel airplane using one of the following alternate procedures (AMM 28-00-00/901).

1. Refuel using the Integrated Refuel Panel (P28) TOTAL/BACKUP fuel quantity indicator (if operating).
2. Refuel using the flight deck EICAS or Fuel Synoptic Display fuel quantity indications (if operating).

3. Determine the associated tank quantity using the FMC calculated Fuel on CDU PROGRESS page from the previous flight and the operating fuel quantity indications. Then refuel the associated tank with a known quantity of fuel. A fuel tanker's fuel indication may be used if verified by the associated tank's fuel measuring sticks.
4. Defuel the associated tank (AMM 28-26-00/201) and service with a known quantity of fuel (AMM 12-11-01/301).
5. Determine the fuel quantity using the associated tank's measuring sticks (Fuel Measuring Stick Manual).

NOTE: The fuel quantity measuring sticks may not give accurate results until 5 to 15 minutes after refueling operations are complete.

28-21-01 Pressure Fueling System**28-21-01-02 Integrated Refuel Panel****28-21-01-02-02 Load Select System**

Interval	Installed	Required	Procedure
C	1	0	(M)

May be inoperative provided alternate procedures are used for refueling.

NOTE: Any function which operates normally may be used.

MAINTENANCE (M)

NOTE: The Load Select System includes the four Load Select Indicators, the Load Select Quantity Selector, the four Load Select Set Switches and the Total/Backup Display Tank Select Switch.

Use alternate procedures for refueling (AMM 12-11-01/301).

1. Utilize individual operative components of pressure refueling system to refuel.

28-21-01 Pressure Fueling System**28-21-01-02 Integrated Refuel Panel****28-21-01-02-03 Refuel Valve Lights**

Interval	Installed	Required	Procedure
C	6	0	(M)

May be inoperative provided:

- a. Overfill function operates normally.
 - b. Associated refuel valve is verified closed after each refueling.
-

MAINTENANCE (M)

Verify the associated valve is closed after each refueling (AMM 28-00-00/901).

1. Monitor and ensure associated refuel valve is closed after refueling.

28-21-01 Pressure Fueling System
28-21-01-02 Integrated Refuel Panel
28-21-01-02-04 Overfill Function (Normal/Reset Switch, Light and Test)

Interval	Installed	Required	Procedure
C	1	0	(M)

May be inoperative provided:

- a. Refuel valve lights operate normally.
- b. Verify refuel valves close when appropriate during refueling.

MAINTENANCE (M)

Verify refuel valves close when appropriate during refueling (AMM 28-00-00/901).

1. Monitor and ensure the refuel valves close appropriately during refueling.

28-21-01 Pressure Fueling System
28-21-01-02 Integrated Refuel Panel
28-21-01-02-05 Defuel Valve Function (Switch and Open Light)

Interval	Installed	Required	Procedure
C	1	0	

May be inoperative provided defuel valve is considered inoperative.

MAINTENANCE NOTE

The airplane must also be dispatched using MMEL item 28-26-01.

28-21-01 Pressure Fueling System
28-21-01-02 Integrated Refuel Panel
28-21-01-02-06 Power Switch - BATT Position

Interval	Installed	Required	Procedure
C	1	0	

**MAINTENANCE NOTE**

Refueling using battery mode will not be available.

28-21-01 Pressure Fueling System**28-21-01-02 Integrated Refuel Panel****28-21-01-02-07 Test Functions (Indication and System)**

Interval	Installed	Required	Procedure
C	2	0	

MAINTENANCE NOTE

The integrated refuel panel also accomplishes an indication test during initial power up.

28-21-01 Pressure Fueling System**28-21-01-02 Integrated Refuel Panel****28-21-01-02-08 Refuel Valve Switches**

Interval	Installed	Required	Procedure
C	3	0	

May be inoperative provided:

- Load select system operates normally.
- Overfill function operates normally.

MAINTENANCE NOTE

If one or more refuel valve switch(es) are inoperative closed, refer to DDG item 28-21-01-01A (M) procedure to refuel the associated tank(s).

**28-21-02 Auxiliary Tank Fuel Quantity Indication (Auxiliary Refuel Panel)**

28-21-02A Auxiliary Tank(s) Usable

Interval	Installed	Required	Procedure
C	1	0	(M)

May be inoperative provided alternate procedures are used for refueling.

MAINTENANCE (M)

Verification of auxiliary tank fuel quantity may be accomplished using one of the following methods (AMM 28-26-00/201):

1. Determine the fuel quantity using the associated tank's Electronic Fuel Level Indication (EFLI).
2. Determine the fuel quantity using the flight deck fuel quantity indication system.
3. Defuel the associated tank and service with a known quantity of fuel.

28-21-02 Auxiliary Tank Fuel Quantity Indication (Auxiliary Refuel Panel)

28-21-02B Auxiliary Tank(s) Empty

Interval	Installed	Required	Procedure
D	1	0	

May be inoperative provided all auxiliary tanks remain empty.

MAINTENANCE NOTE

For auxiliary tank(s) not empty, defuel the auxiliary tank(s) (AMM 28-26-00/201).

**28-21-03 Refuel Adapters**

Interval	Installed	Required	Procedure
C	-	1	(M)

May be inoperative provided there are no fuel leaks.

MAINTENANCE (M)

Immediately after refuel operations, inspect the area around the associated refuel adapter and verify fuel leaks are not present.

NOTE 1: Dispatch is not allowed for an inoperative poppet valve. A poppet valve is not operating normally if it leaks immediately after refueling.

NOTE 2: This item is applicable to failures of the refuel/defuel handle, break-away flange, a break-away flange that is beyond wear limits, or a refuel adapter that has failed the bonding resistance check.

NOTE 3: An inoperative adapter should not be used in refuel or defuel operations.

28-22-01 Main Tank Fuel Pumps

Interval	Installed	Required	Procedure
C	4	3	(M)

One may be inoperative provided:

- a. Inoperative main tank fuel pump is deactivated.
- b. Both main tank quantity indications operate normally.
- c. Flight remains within 120 minutes of landing at a suitable airport.

MAINTENANCE (M)

Deactivate the inoperative fuel pump (AMM 28-00-00/901).

1. For left main tank forward pump inoperative:
 - A. Open and collar P110 panel L FWD FUEL BOOST PUMP circuit breaker.
 - B. Open and collar P310 panel L FWD FUEL BOOST PUMP CTRL circuit breaker.
 - C. Open and collar P320 panel L FWD FUEL BOOST PUMP GND SVCE circuit breaker.
2. For left main tank aft pump inoperative:
 - A. Open and collar P110 panel L AFT FUEL BOOST PUMP circuit breaker.
 - B. Open and collar P110 panel L AFT FUEL BOOST PUMP CTRL circuit breaker.
3. For right main tank forward pump inoperative:
 - A. Open and collar P110 panel R FWD FUEL BOOST PUMP L AC SECT 2 circuit breaker.
 - B. Open and collar P110 panel R FWD FUEL BOOST PUMP CTRL circuit breaker.
 - C. Open and collar P210 panel R FWD FUEL BOOST PUMP R XFR circuit breaker.
4. For right main tank aft pump inoperative:
 - A. Open and collar P210 panel R AFT FUEL BOOST PUMP circuit breaker.
 - B. Open and collar P210 panel R AFT FUEL BOOST PUMP CTRL circuit breaker.

**28-22-01 Main Tank Fuel Pumps****28-22-01-01 PUMPS Switch PRESS Lights**

Interval	Installed	Required	Procedure
C	4	0	

28-22-01 Main Tank Fuel Pumps**28-22-01-02 PUMPS Switch ON Lights**

Interval	Installed	Required	Procedure
C	4	0	

**28-22-02 Center Tank Fuel Override/Jettison Pumps****28-22-02A One Pump Inoperative**

Interval	Installed	Required	Procedure
C	2	1	(M) (O)

One may be inoperative provided:

- Inoperative pump is deactivated.
- Center tank quantity indication operates normally.
- With center tank fueled, fuel quantity remaining in main wing tanks is adequate to reach a suitable airport if remaining center pump fails at any time.
- Center tank fuel is accounted for in the airplane weight and balance in the event center tank fuel cannot be used.

MAINTENANCE (M)

Deactivate the inoperative pump (AMM 28-00-00/901).

- For left center tank pump inoperative, open and collar P110 panel CTR L OVRD FUEL PUMP CTRL circuit breaker.
- For right center tank pump inoperative, open and collar P210 panel CTR R OVRD FUEL PUMP CTRL circuit breaker.

OPERATIONS (O)

- For center tank fuel:
 - For center tank fuel required, ensure suitable airports can be reached with fuel in the main wing tanks in the event remaining center tank pump fails.
 - Reduce the maximum zero fuel weight limit by the amount of center tank fuel.
 - Adjust the zero fuel weight center of gravity forward limit based on the amount of center tank fuel.
 - When operative center tank pump switch is ON, set one crossfeed switch open.
 - When operative center tank pump switch is OFF, set crossfeed switches closed.

28-22-02 Center Tank Fuel Override/Jettison Pumps**28-22-02B Both Pumps Inoperative**

Interval	Installed	Required	Procedure
C	2	0	(M) (O)



May be inoperative provided:

- Pumps are deactivated.
- Center tank quantity indication operates normally.
- Center tank fuel is considered unusable and is accounted for in the airplane weight and balance.

NOTE: AFM fuel loading and usage limitations are for usable fuel.

MAINTENANCE (M)

Deactivate the inoperative pumps (AMM 28-00-00/901).

- For left center tank pump inoperative, open and collar P110 panel CTR L OVRD FUEL PUMP CTRL circuit breaker.
- For right center pump inoperative, open and collar P210 panel CTR R OVRD FUEL PUMP CTRL circuit breaker.

OPERATIONS (O)

- For center tank fuel:

NOTE 1: Center tank fuel is considered unusable. FMC fuel prediction for INSUFFICIENT FUEL message on CDU will not be accurate.

NOTE 2: FUEL IN CENTER advisory message may be displayed.

- Reduce the maximum zero fuel weight limit by the amount of center tank fuel.
- Adjust the zero fuel weight center of gravity forward limit based on the amount of center tank fuel.

28-22-02 Center Tank Fuel Override/Jettison Pumps

28-22-02-01 PUMPS Switch PRESS Lights

Interval	Installed	Required	Procedure
C	2	0	

28-22-02 Center Tank Fuel Override/Jettison Pumps

28-22-02-02 PUMPS Switch ON Lights

Interval	Installed	Required	Procedure
C	2	0	

28-22-03 Fuel Crossfeed Valves

Interval	Installed	Required	Procedure
C	2	1	(M) (O)

One may be inoperative provided:

- a. Inoperative fuel crossfeed valve is deactivated closed.
- b. Remaining valve is verified to operate normally before each departure.

MAINTENANCE (M)

NOTE: The fuel imbalance pointer may appear if the affected crossfeed valve is in the open position when deactivated.

Deactivate the affected crossfeed valve in the closed position (AMM 28-00-00/901).

1. Open the landing gear doors and install safety pins (AMM 32-00-15/201).
2. For forward crossfeed valve inoperative:
 - A. Open and collar P110 panel FUEL XFEED VLV FWD circuit breaker.
 - B. Gain access to the crossfeed valve (left main wheel well - rear spar of the left center tank).
 - C. Disconnect, cap and stow the electrical connector (DV28163).
 - D. Move the manual override handle to the CLOSED position and lockwire.
3. For aft crossfeed valve inoperative:
 - A. Open and collar P310 panel FUEL XFEED VLV AFT circuit breaker.
 - B. Gain access to the crossfeed valve (left main wheel well - rear spar of the left center tank).
 - C. Disconnect, cap and stow the electrical connector (DV28263).
 - D. Move the manual override handle to the CLOSED position and lockwire.
4. Remove the landing gear door safety pins and close the landing gear doors (AMM 32-00-15/201).

OPERATIONS (O)

NOTE 1: The fuel imbalance pointer may appear.

NOTE 2: The associated FUEL CROSSFEED AFT or FWD advisory message may be displayed for the crossfeed valve deactivated in the closed position. After the fuel pumps are ON, the fuel synoptic may incorrectly display a green flow line for the associated crossfeed valve. The EICAS display may have expanded fuel indications.



NOTE 3: Before engine start, if the APU is used, the FUEL PRESS ENG L caution message and the FUEL PUMP R AFT and FUEL PUMP R FWD advisory messages may be displayed.

Before each departure, perform the following fuel crossfeed valve check:

(Required by AD 2019-19-10)

1. Position operative crossfeed valve on and verify associated FUEL CROSSFEED AFT or FWD advisory message does not display.
2. Position operative crossfeed valve off and verify associated FUEL CROSSFEED AFT or FWD advisory message does not display.

28-22-03 Fuel Crossfeed Valves

28-22-03-01 CROSSFEED Switch VALVE Lights

Interval	Installed	Required	Procedure
C	2	0	

28-22-04 Fuel Scavenge Systems

Interval	Installed	Required	Procedure
C	2	0	(O)

May be inoperative provided:

- a. Center tank quantity indication operates normally.
- b. Appropriate amount of center tank fuel is considered unusable and is accounted for in the airplane weight and balance.

NOTE: AFM fuel loading and usage limitations are for usable fuel.

NOTE 1: For the -200 without Service Bulletins 777-28-0078 (ELMS2) or 777-28-0082 (ELMS1) installed, and for the -200ER/-200LR/-300/-300ER/777F airplanes without PRR 62777-79S or without Service Bulletins 777-28-0078 (ELMS2) or 777-28-0082 (ELMS1) installed, the fuel scavenge system includes the jet pump and two shutoff valves in each system. If the jet pump is inoperative or either shutoff valve is inoperative closed, the associated system is inoperative. For more than one system shutoff valve inoperative open, use MMEL item 28-22-06B.

NOTE 2: For -200ER/-200LR/-300/-300ER/777F with PRR 62777-79S, the main tank fuel jettison system's pumps provide the motive power to the fuel scavenge system's jet pumps, and the shutoff valves are not installed. If a jet pump is inoperative, or if the associated main tank jettison pump is inoperative, or if the fuel jettison system is deactivated, the associated fuel scavenge system is inoperative. The PRR 62777-79S is applicable to airplane line numbers 1099 and-on. Service Bulletin 777-28-0078 (ELMS2), and 777-28-0082 (ELMS1) provide the production equivalent of PRR 62777-79S for the -200/-200ER/-200LR/-300/-300ER/777F airplanes.

NOTE 3: For additional information on PRR 61998-2, refer to 777-FTD-28-06004.

OPERATIONS (O)

NOTE: For one scavenge system inoperative, fuel imbalance may occur when center tank fuel is scavenged.

For -200:

1. For center tank fuel required:
 - A. Increase flight planning fuel by 800 lb (362 kg).
 - B. Reduce the maximum zero fuel weight limit by 800 lb (362 kg).
 - C. Adjust the zero fuel weight center of gravity forward limit 0.1% MAC aft.

2. For center tank fuel not required:
 - A. Center tank fuel at or above 800 lb (362 kg):
 - 1) Reduce the maximum zero fuel weight limit by 800 lb (362 kg).
 - 2) Adjust the zero fuel weight center of gravity forward limit 0.1% MAC aft.
 - B. Center tank fuel less than 800 lb (362 kg):
 - 1) Reduce the maximum zero fuel weight limit by the amount of center tank fuel.
 - 2) Adjust the zero fuel weight center of gravity forward limit based on the amount of center tank fuel.

For -200ER without PRR 61998-2 or equivalent installed:

1. For center tank fuel required:
 - A. Increase flight planning fuel by 1,400 lb (636 kg).
 - B. Reduce the maximum zero fuel weight limit by 1,400 lb (636 kg).
 - C. Adjust the zero fuel weight center of gravity forward limit 0.1% MAC aft.
2. For center tank fuel not required:
 - A. Center tank fuel at or above 1,400 lb (636 kg):
 - 1) Reduce the maximum zero fuel weight limit by 1,400 lb (636 kg).
 - 2) Adjust the zero fuel weight center of gravity forward limit 0.1% MAC aft.
 - B. Center tank fuel less than 1,400 lb (636 kg):
 - 1) Reduce the maximum zero fuel weight limit by the amount of center tank fuel.
 - 2) Adjust the zero fuel weight center of gravity forward limit based on the amount of center tank fuel.

For -200ER with PRR 61998-2 or equivalent installed:

1. For center tank fuel required:
 - A. Increase flight planning fuel by 2,400 lb (1,090 kg).
 - B. Reduce the maximum zero fuel weight limit by 2,400 lb (1,090 kg).
 - C. Adjust the zero fuel weight center of gravity forward limit 0.2% MAC aft.
2. For center tank fuel not required:
 - A. Center tank fuel at or above 2,400 lb (1,090 kg):
 - 1) Reduce the maximum zero fuel weight limit by 2,400 lb (1,090 kg).
 - 2) Adjust the zero fuel weight center of gravity forward limit 0.2% MAC aft.

- B. Center tank fuel less than 2,400 lb (1,090 kg):
 - 1) Reduce the maximum zero fuel weight limit by the amount of center tank fuel.
 - 2) Adjust the zero fuel weight center of gravity forward limit based on the amount of center tank fuel.

For -200LR/777F:

- 1. For center tank fuel required:
 - A. Increase flight planning fuel by 2,700 lb (1,225 kg).
 - B. Reduce the maximum zero fuel weight limit by 2,700 lb (1,225 kg).
 - C. Adjust the zero fuel weight center of gravity forward limit 0.2% MAC aft.
- 2. For center tank fuel not required:
 - A. Center tank fuel at or above 2,700 lb (1,225 kg):
 - 1) Reduce the maximum zero fuel weight limit by 2,700 lb (1,225 kg).
 - 2) Adjust the zero fuel weight center of gravity forward limit 0.2% MAC aft.
 - B. Center tank fuel less than 2,700 lb (1,225 kg):
 - 1) Reduce the maximum zero fuel weight limit by the amount of center tank fuel.
 - 2) Adjust the zero fuel weight center of gravity forward limit based on the amount of center tank fuel.

For -300 without PRR 61998-2 or equivalent installed:

- 1. For center tank fuel required:
 - A. Increase flight planning fuel by 1,400 lb (636 kg).
 - B. Reduce the maximum zero fuel weight limit by 1,400 lb (636 kg).
 - C. Adjust the zero fuel weight center of gravity forward limit 0.1% MAC aft.
- 2. For center tank fuel not required:
 - A. Center tank fuel at or above 1,400 lb (636 kg):
 - 1) Reduce the maximum zero fuel weight limit by 1,400 lb (636 kg).
 - 2) Adjust the zero fuel weight center of gravity forward limit 0.1% MAC aft.
 - B. Center tank fuel less than 1,400 lb (636 kg):
 - 1) Reduce the maximum zero fuel weight limit by the amount of center tank fuel.
 - 2) Adjust the zero fuel weight center of gravity forward limit based on the amount of center tank fuel.

**For -300 with PRR 61998-2 or equivalent installed:**

1. For center tank fuel required:
 - A. Increase flight planning fuel by 2,400 lb (1,090 kg).
 - B. Reduce the maximum zero fuel weight limit by 2,400 lb (1,090 kg).
 - C. Adjust the zero fuel weight center of gravity forward limit 0.1% MAC aft.
2. For center tank fuel not required:
 - A. Center tank fuel at or above 2,400 lb (1,090 kg):
 - 1) Reduce the maximum zero fuel weight limit by 2,400 lb (1,090 kg).
 - 2) Adjust the zero fuel weight center of gravity forward limit 0.1% MAC aft.
 - B. Center tank fuel less than 2,400 lb (1,090 kg):
 - 1) Reduce the maximum zero fuel weight limit by the amount of center tank fuel.
 - 2) Adjust the zero fuel weight center of gravity forward limit based on the amount of center tank fuel.

For -300ER without PRR 61998-2 or equivalent installed:

1. For center tank fuel required:
 - A. Increase flight planning fuel by 2,400 lb (1,090 kg).
 - B. Reduce the maximum zero fuel weight limit by 2,400 lb (1,090 kg).
 - C. Adjust the zero fuel weight center of gravity forward limit 0.1% MAC aft.
2. For center tank fuel not required:
 - A. Center tank fuel at or above 2,400 lb (1,090 kg):
 - 1) Reduce the maximum zero fuel weight limit by 2,400 lb (1,090 kg).
 - 2) Adjust the zero fuel weight center of gravity forward limit 0.1% MAC aft.
 - B. Center tank fuel less than 2,400 lb (1,090 kg):
 - 1) Reduce the maximum zero fuel weight limit by the amount of center tank fuel.
 - 2) Adjust the zero fuel weight center of gravity forward limit based on the amount of center tank fuel.

For -300ER with PRR 61998-2 or equivalent installed:

1. For center tank fuel required:
 - A. Increase flight planning fuel by 2,700 lb (1,225 kg).
 - B. Reduce the maximum zero fuel weight limit by 2,700 lb (1,225 kg).
 - C. Adjust the zero fuel weight center of gravity forward limit 0.1% MAC aft.



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2. For center tank fuel not required:
 - A. Center tank fuel at or above 2,700 lb (1,225 kg):
 - 1) Reduce the maximum zero fuel weight limit by 2,700 lb (1,225 kg).
 - 2) Adjust the zero fuel weight center of gravity forward limit 0.1% MAC aft.
 - B. Center tank fuel less than 2,700 lb (1,225 kg):
 - 1) Reduce the maximum zero fuel weight limit by the amount of center tank fuel.
 - 2) Adjust the zero fuel weight center of gravity forward limit based on the amount of center tank fuel.

OPERATIONS NOTE

NOTE: Boeing Alternative Method of Compliance (AMOC) to AD 2016-11-03 (approved by FAA letter 140S-17-48) provides the option to carry extra fuel to account for failures of the existing center tank fuel scavenge system instead of installing SB 777-28-0078. Operators using the AMOC and dispatching with fuel scavenge inoperative may reduce the DDG fuel weight penalty by 700 lbs (320 kg), the additional reserve fuel carried in compliance with the AMOC.



28-22-05 Fuel Shutoff Valve Battery

Interval	Installed	Required	Procedure
C	1	0	

**28-22-06 Fuel Scavenge Shutoff Valves (Without PRR 62777-79S or Equivalent Installed)****28-22-06A One Inoperative**

Interval	Installed	Required	Procedure
C	4	3	(M)

One valve may be inoperative provided:

- Inoperative valve is open.
- Fuel quantity system operates normally.
- Remaining valve on that side is verified to operate normally.

NOTE: The PRR 62777-79S is applicable to airplane line numbers 1099 and-on. Service Bulletins 777-28-0078 (ELMS2) and 777-28-0082 (ELMS1) each provide the production equivalent of PRR 62777-79S.

MAINTENANCE (M)

Verify the fuel scavenge shutoff valve on the associated side operates normally (AMM 28-00-00/901).

- For an outlet fuel scavenge shutoff valve inoperative open, verify proper operation of the inlet shutoff valve.

NOTE: For an outlet shutoff valve inoperative closed, use MMEL item 28-22-04.

- A. Defuel (AMM 28-26-00/201) the main fuel tank on the same side as the inoperative shutoff valve if necessary to achieve the following fuel quantity:

For -200:

- Less than 21,000 lb (9,500 kg).

For -200ER/-300:

- Less than 30,200 lb (13,700 kg).

For -200LR/777F/-300ER:

- Less than 53,900 lb (24,400 kg).

- B. Refuel (AMM 12-11-01/301) the center fuel tank if necessary to achieve the following fuel quantity:

For -200:

- More than 13,200 lb (6,000 kg).

For -200ER/-300:

- More than 49,300 lb (22,400 kg).

For -200LR/777F/-300ER:

- More than 48,100 lb (21,800 kg).

- C. For left side shutoff valve inoperative, position Fuel Panel (P5) L PUMPS FWD and AFT switches ON.
 - 1) For APU operating, position Fuel Panel (P5) CENTER PUMPS L switch ON.
 - D. For right side shutoff valve inoperative, position Fuel Panel (P5) R PUMPS FWD and AFT switches ON.
 - E. Allow the pumps to operate for 30 minutes.
 - 1) Verify the main tank fuel quantity on the same side as the failed valve does not increase.
 - 2) Verify the center tank fuel quantity does not increase.
 - F. Position Fuel Panel (P5) Fuel Pump switches off.
2. For an inlet fuel scavenge shutoff valve inoperative open, verify proper operation of the outlet shutoff valve.

NOTE: For an inlet shutoff valve inoperative closed, use MMEL item 28-22-04.

- A. Refuel (AMM 12-11-01/301) the center fuel tank if necessary to achieve a fuel quantity greater than 1,000 lb (455 kg).
- B. Refuel (AMM 12-11-01/301) the main fuel tank on the same side as the inoperative shutoff valve if necessary to achieve the following fuel quantity:

For -200:

- 1) More than 23,700 lb (10,800 kg).

For -200ER/-300:

- 2) More than 33,200 lb (15,100 kg).

For -200LR/777F/-300ER:

- 3) More than 56,350 lb (25,500 kg).

- C. For left side shutoff valve inoperative, position Fuel Panel (P5) L PUMPS FWD and AFT switches ON.
 - 1) For APU operating, position Fuel Panel (P5) CENTER PUMPS L switch ON.
- D. For right side shutoff valve inoperative, position Fuel Panel (P5) R PUMPS FWD and AFT switches ON.
- E. Allow the pumps to operate for 30 minutes.
 - 1) Verify the main tank fuel quantity on the same side as the failed valve does not increase.
 - 2) Verify the center tank fuel quantity does not increase.
- F. Position Fuel Panel (P5) Fuel Pump switches off.

**28-22-06 Fuel Scavenge Shutoff Valves (Without PRR 62777-79S or Equivalent Installed)****28-22-06B More Than One Inoperative**

Interval	Installed	Required	Procedure
C	4	0	(M)

May be inoperative provided:

- Inoperative valve(s) is open.
- Center tank remains empty.
- Center tank quantity indication operates normally.

NOTE: The PRR 62777-79S is applicable to airplane line numbers 1099 and-on. Service Bulletins 777-28-0078 (ELMS2) and 777-28-0082 (ELMS1) each provide the production equivalent of PRR 62777-79S.

MAINTENANCE (M)

For center tank fuel, defuel the center tank (AMM 28-00-00/901).

- Defuel the center tank to a quantity less than 400 lb (180 kg) by the suction defueling procedure (AMM 28-26-00/201).
- If required, drain fuel to achieve required fuel amount.

**28-22-07 Auxiliary (AUX) Fuel Switch**

Interval	Installed	Required	Procedure
D	1	0	

May be inoperative provided:

- AUX fuel switch is in the OFF position.
- All auxiliary tanks remain empty.

MAINTENANCE NOTE

For auxiliary tank(s) not empty, defuel the auxiliary tank(s) (AMM 28-26-00/201).

28-22-07 Auxiliary (AUX) Fuel Switch**28-22-07-01 PRESS Light**

Interval	Installed	Required	Procedure
D	1	0	

28-22-07 Auxiliary (AUX) Fuel Switch**28-22-07-02 ON Light**

Interval	Installed	Required	Procedure
D	1	0	



28-25-01 APU DC Fuel Pump

Interval	Installed	Required	Procedure
C	1	0	(M)

May be inoperative provided fuel pump is deactivated.

MAINTENANCE (M)

Deactivate the APU DC fuel pump (AMM 28-00-00/901).

- 1. Open and collar P310 panel DC FUEL PUMP circuit breaker.

**28-25-02 APU Fuel Shutoff Valve**

Interval	Installed	Required	Procedure
C	1	0	(M) (O)

May be inoperative provided:

- a. Valve is deactivated closed.
 - b. Other procedures do not require the use of the APU.
 - c. Backup AC power system is verified to operate normally before each departure.
 - d. Flight remains within 180 minutes of landing at a suitable airport.
-

MAINTENANCE (M)

Deactivate the APU fuel shutoff valve in the closed position (AMM 28-00-00/901).

1. Open the landing gear doors and install safety pins (AMM 32-00-15/201).
2. Open and collar P11 panel APU FUEL S/O VALVE circuit breaker.
3. Gain access to the APU shutoff valve actuator located in the left main wheel well (rear spar of left center tank).
4. Disconnect, cap and stow the electrical connector (DV28023).
5. Move the manual override handle to the CLOSED position and lockwire.
6. Remove the landing gear door safety pins and close the landing gear doors (AMM 32-00-15/201).

OPERATIONS (O)

1. Dispatch is not allowed if the APU is required by other procedures. MMEL dispatch procedures for the following inoperative equipment may require APU availability:



M MEL Item	Associated Status Message
24-11-01	ELEC GEN SYS L or R IDG VALVE CLOSED L or R
24-25-01	ELEC BACKUP GEN L or R ELEC BACKUP SYS
29-11-03	HYD PUMP DEM C1 or C2
36-11-01	BLEED PRSOV L or R
36-11-02	BLEED HPSOV L or R
36-11-03	None
36-12-01	BLEED ISLN VALVE L or R
36-12-02	BLEED ISLN VALVE C
36-22-01	BLEED TEMP SENSOR L or R

2. The flight must remain within 180 minutes of landing at a suitable airport.
3. Verify the backup AC electrical power system operates normally. Confirm the ELEC BACKUP SYS or ELEC BACKUP GEN L or R status and advisory messages are not displayed 30 seconds after both engines have been started.

**28-25-03 APU Fuel Isolation Valve**

Interval	Installed	Required	Procedure
C	1	0	(M)

May be inoperative provided isolation valve is deactivated closed.

MAINTENANCE (M)

Deactivate the APU fuel isolation valve in the closed position (AMM 28-00-00/901).

1. Open the landing gear doors and install safety pins (AMM 32-00-15/201).
2. Open and collar P310 panel APU FUEL ISLN VLV circuit breaker.
3. Gain access to the APU fuel isolation valve actuator located in the left main wheel well (rear spar of the left center tank).
4. Disconnect, cap and stow the electrical connector (DV28031).
5. Move the manual override handle to the CLOSED position and lockwire.
6. Remove the landing gear door safety pins and close the landing gear doors (AMM 32-00-15/201).

**28-26-01 Defuel Valve**

Interval	Installed	Required	Procedure
C	1	0	(M)

Defuel valve may be inoperative provided it is secured closed.

MAINTENANCE (M)

Deactivate the defuel valve in the closed position (AMM 28-00-00/901).

1. Open the landing gear doors and install safety pins (AMM 32-00-15/201).
2. Open and collar P310 panel DFUEL VLVS circuit breaker.
3. Gain access to the defuel valve actuator located in the right main wheel well (rear spar of the right center tank).
4. Disconnect, cap and stow the electrical connector (DV28310).
5. Move the manual override handle to the CLOSED position and lockwire.
6. Remove the landing gear door safety pins and close the landing gear doors (AMM 32-00-15/201).

**28-31-01 Fuel Jettison System****28-31-01-01 Without PRR 62777-79S or Equivalent Installed**

Interval	Installed	Required	Procedure
C	1	0	(M) (O)

May be inoperative provided:

- Jettison system is deactivated.
 - Jettison nozzle valves are secured closed.
 - Appropriate performance adjustments are applied.
-

NOTE: The PRR 62777-79S is applicable to airplane line numbers 1099 and-on. Service Bulletins 777-28-0078 (ELMS2) and 777-28-0082 (ELMS1) each provide the production equivalent of PRR 62777-79S.

MAINTENANCE (M)

Deactivate the jettison system and secure the jettison nozzle valves in the closed position (AMM 28-00-00/901).

- Open and collar P110 panel L FUEL JETT PUMP CTRL, L FUEL JETT ISLN VLV and L FUEL JETT NOZ VLV circuit breakers.
- Open and collar P210 panel R FUEL JETT PUMP CTRL, R FUEL JETT ISLN VLV and R FUEL JETT NOZ VLV circuit breakers.
- For left jettison nozzle valve:
 - Gain access to the left jettison nozzle valve actuator through associated access door.
 - Disconnect, cap and stow the electrical connector (DV28166).
 - Confirm the manual override handle is in the CLOSED position and lockwire.
 - Close access door.
- For right jettison nozzle valve:
 - Gain access to the right jettison nozzle valve actuator through associated access door.
 - Disconnect, cap and stow the electrical connector (DV28266).
 - Confirm the manual override handle is in the CLOSED position and lockwire.
 - Close access door.

OPERATIONS (O)

NOTE: The Fuel Synoptic will incorrectly indicate jettison flow from the center tank when the center tank override pumps are on with fuel in the center tank. The FUEL JETT NOZZLE L and R advisory messages will be displayed and overhead panel L and R fuel jettison nozzle VALVE lights will be illuminated.

For -200ER/-300:

1. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 4,000 lb (1,800 kg) (for fuel that would be burned during a return-to-land procedure).

For -200LR/777F/-300ER:

2. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 5,400 lb (2,450 kg) (for fuel that would be burned during a return-to-land procedure).

28-31-01 Fuel Jettison System
28-31-01-01 Without PRR 62777-79S or Equivalent Installed
28-31-01-01-01 ARM Switch FAULT Light

Interval	Installed	Required	Procedure
C	1	0	

28-31-01 Fuel Jettison System
28-31-01-01 Without PRR 62777-79S or Equivalent Installed
28-31-01-01-02 ARM Switch ARMED Light

Interval	Installed	Required	Procedure
C	1	0	

28-31-01 Fuel Jettison System
28-31-01-02 With PRR 62777-79S or Equivalent Installed

Interval	Installed	Required	Procedure
C	1	0	(M) (O)

May be inoperative provided:

- a. Jettison system is deactivated.
- b. Jettison nozzle valves are secured closed.
- c. Fuel scavenge system is considered inoperative.
- d. Appropriate performance adjustments are applied.

NOTE 1: The PRR 62777-79S is applicable to airplane line numbers 1099 and-on.
Service Bulletins 777-28-0078 (ELMS2) and 777-28-0082 (ELMS1) each provide the production equivalent of PRR 62777-79S.

NOTE 2: The airplane must also be dispatched using MMEL item 28-22-04.

MAINTENANCE (M)

Deactivate the jettison system and secure the jettison nozzle valves in the closed position (AMM 28-00-00/901).

1. Open and collar P110 panel L FUEL JETT PUMP CTRL, L FUEL JETT ISLN VLV and L FUEL JETT NOZ VLV circuit breakers.
2. Open and collar P210 panel R FUEL JETT PUMP CTRL, R FUEL JETT ISLN VLV and R FUEL JETT NOZ VLV circuit breakers.
3. For left jettison nozzle valve:
 - A. Gain access to the left jettison nozzle valve actuator through associated access door.
 - B. Disconnect, cap and stow the electrical connector (DV28166).
 - C. Confirm the manual override handle is in the CLOSED position and lockwire.
 - D. Close access door.
4. For right jettison nozzle valve:
 - A. Gain access to the right jettison nozzle valve actuator through associated access door.
 - B. Disconnect, cap and stow the electrical connector (DV28266).
 - C. Confirm the manual override handle is in the CLOSED position and lockwire.
 - D. Close access door.

OPERATIONS (O)

NOTE: The Fuel Synoptic will incorrectly indicate jettison flow from the center tank when the center tank override pumps are on with fuel in the center tank. The FUEL JETT NOZZLE L and R advisory messages will be displayed and overhead panel L and R fuel jettison nozzle VALVE lights will be illuminated.

For -200ER/-300:

1. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 4,000 lb (1,800 kg) (for fuel that would be burned during a return-to-land procedure).

For -200LR/777F/-300ER:

2. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 5,400 lb (2,450 kg) (for fuel that would be burned during a return-to-land procedure).

**28-31-01 Fuel Jettison System****28-31-01-02 With PRR 62777-79S or Equivalent Installed****28-31-01-02-01 ARM Switch FAULT Light**

Interval	Installed	Required	Procedure
C	1	0	

28-31-01 Fuel Jettison System**28-31-01-02 With PRR 62777-79S or Equivalent Installed****28-31-01-02-02 ARM Switch ARMED Light**

Interval	Installed	Required	Procedure
C	1	0	

**28-31-02 Center Tank Jettison Isolation Valves****28-31-02-01 -200**

Interval	Installed	Required	Procedure
C	2	0	(M)

May be inoperative provided inoperative valve(s) is secured closed.

MAINTENANCE (M)

Deactivate and secure the inoperative jettison isolation valve in the closed position (AMM 28-00-00/901).

1. Open the landing gear doors and install safety pins (AMM 32-00-15/201).
2. For left jettison isolation valve inoperative:
 - A. Open and collar P110 panel L FUEL JETT ISLN VLV circuit breaker.
 - B. Gain access to the jettison isolation valve actuator located in the left main wheel well (rear spar of the left center tank).
 - C. Disconnect, cap and stow the electrical connector (DV28165).
 - D. Move the manual override handle to the CLOSED position and lockwire.
3. For right jettison isolation valve inoperative:
 - A. Open and collar P210 panel R FUEL JETT ISLN VLV circuit breaker.
 - B. Gain access to the jettison isolation valve actuator located in the right main wheel well (rear spar of the right center tank).
 - C. Disconnect, cap and stow the electrical connector (DV28265).
 - D. Move the manual override handle to the CLOSED position and lockwire.
4. Remove the landing gear door safety pins and close the landing gear doors (AMM 32-00-15/201).

28-31-02 Center Tank Jettison Isolation Valves**28-31-02-02 -200ER/-200LR/-300/-300ER/777F (Without PRR 62777-79S or Equivalent Installed)****28-31-02-02A Fuel Jettison System Deactivated**

Interval	Installed	Required	Procedure
C	2	0	(M)

May be inoperative provided:

- a. Inoperative valve(s) is secured closed.
- b. Fuel jettison system is considered inoperative.



NOTE 1: The PRR 62777-79S is applicable to airplane line numbers 1099 and-on. Service Bulletins 777-28-0078 (ELMS2) and 777-28-0082 (ELMS1) each provide the production equivalent of PRR 62777-79S.

NOTE 2: The airplane must also be dispatched using MMEL item 28-31-01.

MAINTENANCE (M)

Secure the inoperative jettison isolation valve in the closed position (AMM 28-00-00/901).

1. Open the landing gear doors and install safety pins (AMM 32-00-15/201).
2. For left jettison isolation valve:
 - A. Open and collar P110 panel L FUEL JETT ISLN VLV circuit breaker.
 - B. Gain access to the jettison isolation valve actuator located in the left main wheel well (rear spar of the left center tank).
 - C. Disconnect, cap and stow the electrical connector (DV28165).
 - D. Move the manual override handle to the CLOSED position and lockwire.
3. For right jettison isolation valve:
 - A. Open and collar P210 panel R FUEL JETT ISLN VLV circuit breaker.
 - B. Gain access to the jettison isolation valve actuator located in the right main wheel well (rear spar of the right center tank).
 - C. Disconnect, cap and stow the electrical connector (DV28265).
 - D. Move the manual override handle to the CLOSED position and lockwire.
4. Remove the landing gear door safety pins and close the landing gear doors (AMM 32-00-15/201).

28-31-02 Center Tank Jettison Isolation Valves

28-31-02-02 -200ER/-200LR/-300/-300ER/777F (Without PRR 62777-79S or Equivalent Installed)

28-31-02-02B Main Tank Jettison Operative

Interval	Installed	Required	Procedure
C	2	0	(M) (O)

May be inoperative provided:

- a. Inoperative valve(s) is secured closed.
- b. Appropriate gross weight and center-of-gravity adjustments are applied.

NOTE: The PRR 62777-79S is applicable to airplane line numbers 1099 and-on. Service Bulletins 777-28-0078 (ELMS2) and 777-28-0082 (ELMS1) each provide the production equivalent of PRR 62777-79S.

**MAINTENANCE (M)**

Deactivate and secure the inoperative jettison isolation valve in the closed position (AMM 28-00-00/901).

1. Open the landing gear doors and install safety pins (AMM 32-00-15/201).
2. For left jettison isolation valve inoperative:
 - A. Open and collar P110 panel L FUEL JETT ISLN VLV circuit breaker.
 - B. Gain access to the jettison isolation valve actuator located in the left main wheel well (rear spar of the left center tank).
 - C. Disconnect, cap and stow the electrical connector (DV28165).
 - D. Move the manual override handle to the CLOSED position and lockwire.
3. For right jettison isolation valve inoperative:
 - A. Open and collar P210 panel R FUEL JETT ISLN VLV circuit breaker.
 - B. Gain access to the jettison isolation valve actuator located in the right main wheel well (rear spar of the right center tank).
 - C. Disconnect, cap and stow the electrical connector (DV28265).
 - D. Move the manual override handle to the CLOSED position and lockwire.
4. Remove the landing gear door safety pins and close the landing gear doors (AMM 32-00-15/201).

OPERATIONS (O)**For -200ER:**

1. For dispatch with center tank fuel greater than 22,000 lbs (9,980 kg) and Gross Weight exceeding 460,000 lb (208,650 kg), Center-of-Gravity (CG) must be aft of 22% MAC for takeoff.
2. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 4,000 lb (1,800 kg) (for fuel that would be burned during a return-to-land procedure).

For -200LR/777F:

1. For dispatch with center tank fuel and Gross Weight exceeding 523,000 lb (237,230 kg), Center-of-Gravity (CG) must be aft of 25% MAC for takeoff.
2. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 5,400 lb (2,450 kg) (for fuel that would be burned during a return-to-land procedure).

For -300:

1. For dispatch with center tank fuel and Gross Weight exceeding 524,000 lb (238,000 kg), Center-of-Gravity (CG) must be aft of 17% MAC for takeoff.
2. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 4,000 lb (1,800 kg) (for fuel that would be burned during a return-to-land procedure).

For -300ER:

1. For dispatch with center tank fuel and Gross Weight exceeding 556,000 lb (252,200 kg), Center-of-Gravity (CG) must be aft of 21% MAC for takeoff.
2. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 5,400 lb (2,450 kg) (for fuel that would be burned during a return-to-land procedure).

28-31-02 Center Tank Jettison Isolation Valves
28-31-02-03 -200ER/-200LR/-300/-300ER/777F (With PRR 62777-79S or Equivalent Installed)
28-31-02-03A Fuel Jettison System Deactivated

Interval	Installed	Required	Procedure
C	2	0	(M)

May be inoperative provided:

- a. Inoperative valve(s) is secured closed.
- b. Fuel jettison system is considered inoperative.
- c. Fuel scavenge system is considered inoperative.

NOTE 1: The PRR 62777-79S is applicable to airplane line numbers 1099 and-on. Service Bulletins 777-28-0078 (ELMS2) and 777-28-0082 (ELMS1) each provide the production equivalent of PRR 62777-79S.

NOTE 2: The airplane must also be dispatched using MMEL item 28-31-01.

MAINTENANCE (M)

Secure the inoperative jettison isolation valve in the closed position (AMM 28-00-00/901).

1. Open the landing gear doors and install safety pins (AMM 32-00-15/201).
2. For left jettison isolation valve:
 - A. Open and collar P110 panel L FUEL JETT ISLN VLV circuit breaker.
 - B. Gain access to the jettison isolation valve actuator located in the left main wheel well (rear spar of the left center tank).
 - C. Disconnect, cap and stow the electrical connector (DV28165).
 - D. Move the manual override handle to the CLOSED position and lockwire.
3. For right jettison isolation valve:
 - A. Open and collar P210 panel R FUEL JETT ISLN VLV circuit breaker.
 - B. Gain access to the jettison isolation valve actuator located in the right main wheel well (rear spar of the right center tank).
 - C. Disconnect, cap and stow the electrical connector (DV28265).



- D. Move the manual override handle to the CLOSED position and lockwire.
4. Remove the landing gear door safety pins and close the landing gear doors (AMM 32-00-15/201).

28-31-02 Center Tank Jettison Isolation Valves**28-31-02-03 -200ER/-200LR/-300/-300ER/777F (With PRR 62777-79S or Equivalent Installed)****28-31-02-03B Main Tank Jettison Operative**

Interval	Installed	Required	Procedure
C	2	0	(M) (O)

May be inoperative provided:

- a. Inoperative valve(s) is secured closed.
- b. Appropriate gross weight and center-of-gravity adjustments are applied.

NOTE: The PRR 62777-79S is applicable to airplane line numbers 1099 and-on. Service Bulletins 777-28-0078 (ELMS2) and 777-28-0082 (ELMS1) each provide the production equivalent of PRR 62777-79S.

MAINTENANCE (M)

Deactivate and secure the inoperative jettison isolation valve in the closed position (AMM 28-00-00/901).

- Open the landing gear doors and install safety pins (AMM 32-00-15/201).
- For left jettison isolation valve inoperative:
 - Open and collar P110 panel L FUEL JETT ISLN VLV circuit breaker.
 - Gain access to the jettison isolation valve actuator located in the left main wheel well (rear spar of the left center tank).
 - Disconnect, cap and stow the electrical connector (DV28165).
 - Move the manual override handle to the CLOSED position and lockwire.
- For right jettison isolation valve inoperative:
 - Open and collar P210 panel R FUEL JETT ISLN VLV circuit breaker.
 - Gain access to the jettison isolation valve actuator located in the right main wheel well (rear spar of the right center tank).
 - Disconnect, cap and stow the electrical connector (DV28265).
 - Move the manual override handle to the CLOSED position and lockwire.
- Remove the landing gear door safety pins and close the landing gear doors (AMM 32-00-15/201).

**OPERATIONS (O)****For -200ER:**

1. For dispatch with center tank fuel greater than 22,000 lbs (9,980 kg) and Gross Weight exceeding 460,000 lb (208,650 kg), Center-of-Gravity (CG) must be aft of 22% MAC for takeoff.
2. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 4,000 lb (1,800 kg) (for fuel that would be burned during a return-to-land procedure).

For -200LR/777F:

1. For dispatch with center tank fuel and Gross Weight exceeding 523,000 lb (237,230 kg), Center-of-Gravity (CG) must be aft of 25% MAC for takeoff.
2. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 5,400 lb (2,450 kg) (for fuel that would be burned during a return-to-land procedure).

For -300:

1. For dispatch with center tank fuel and Gross Weight exceeding 524,000 lb (238,000 kg), Center-of-Gravity (CG) must be aft of 17% MAC for takeoff.
2. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 4,000 lb (1,800 kg) (for fuel that would be burned during a return-to-land procedure).

For -300ER:

1. For dispatch with center tank fuel and Gross Weight exceeding 556,000 lb (252,200 kg), Center-of-Gravity (CG) must be aft of 21% MAC for takeoff.
2. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 5,400 lb (2,450 kg) (for fuel that would be burned during a return-to-land procedure).

**28-31-03 Main Tanks Jettison Pumps****28-31-03-01 Without PRR 62777-79S or Equivalent Installed**

Interval	Installed	Required	Procedure
C	2	0	(M) (O)

May be inoperative provided:

- Inoperative pump is deactivated.
- Appropriate performance adjustments are applied.

NOTE: The PRR 62777-79S is applicable to airplane line numbers 1099 and-on. Service Bulletins 777-28-0078 (ELMS2) and 777-28-0082 (ELMS1) each provide the production equivalent of PRR 62777-79S.

MAINTENANCE (M)

Deactivate the inoperative pump(s) (AMM 28-00-00/901).

- For left main tank jettison pump inoperative, open and collar P110 panel L FUEL JETT PUMP CTRL circuit breaker.
- For right main tank jettison pump inoperative, open and collar P210 panel R FUEL JETT PUMP CTRL circuit breaker.

OPERATIONS (O)

NOTE: For one main tank jettison pump inoperative, fuel jettison of the opposite jettison pump is inhibited to prevent fuel imbalance.

For -200ER/-300:

Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus center tank fuel (which would be jettisoned during a return-to-land procedure), plus 4,000 lb (1,800 kg) (for fuel that would be burned during a return-to-land procedure).

For -200LR/777F/-300ER:

Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus center tank fuel (which would be jettisoned during a return-to-land procedure), plus 5,400 lb (2,450 kg) (for fuel that would be burned during a return-to-land procedure).

28-31-03 Main Tanks Jettison Pumps**28-31-03-02 With PRR 62777-79S or Equivalent Installed**

Interval	Installed	Required	Procedure
C	2	0	(M) (O)

May be inoperative provided:

- Inoperative pump is deactivated.

- b. Fuel scavenge system is considered inoperative.
 - c. Appropriate performance adjustments are applied.
-

NOTE 1: The PRR 62777-79S is applicable to airplane line numbers 1099 and-on. Service Bulletins 777-28-0078 (ELMS2) and 777-28-0082 (ELMS1) each provide the production equivalent of PRR 62777-79S.

NOTE 2: The airplane must also be dispatched using MMEL item 28-22-04.

MAINTENANCE (M)

Deactivate the inoperative pump(s) (AMM 28-00-00/901).

1. For left main tank jettison pump inoperative, open and collar P110 panel L FUEL JETT PUMP CTRL circuit breaker.
2. For right main tank jettison pump inoperative, open and collar P210 panel R FUEL JETT PUMP CTRL circuit breaker.

OPERATIONS (O)

NOTE: For one main tank jettison pump inoperative:

1. Fuel jettison of the opposite jettison pump is inhibited to prevent fuel imbalance.
2. Fuel scavenge of the opposite jettison pump operates normally.
3. Fuel scavenge rate is approximately half the normal fuel scavenge rate.

For -200ER/-300:

Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus center tank fuel (which would be jettisoned during a return-to-land procedure), plus 4,000 lb (1,800 kg) (for fuel that would be burned during a return-to-land procedure).

For -200LR/777F/-300ER:

Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus center tank fuel (which would be jettisoned during a return-to-land procedure), plus 5,400 lb (2,450 kg) (for fuel that would be burned during a return-to-land procedure).

**28-31-04 Fuel Jettison Nozzle Valves****28-31-04-01 Without PRR 62777-79S or Equivalent Installed****28-31-04-01A One Inoperative**

Interval	Installed	Required	Procedure
C	2	1	(M)

One nozzle valve may be inoperative provided it is secured closed.

NOTE: The PRR 62777-79S is applicable to airplane line numbers 1099 and-on. Service Bulletins 777-28-0078 (ELMS2) and 777-28-0082 (ELMS1) each provide the production equivalent of PRR 62777-79S.

MAINTENANCE (M)

Secure the inoperative fuel jettison nozzle valve in the closed position (AMM 28-00-00/901).

1. For left jettison nozzle valve inoperative:
 - A. Open and collar P110 panel L FUEL JETT NOZ VLV circuit breaker.
 - B. Gain access to the left jettison nozzle valve actuator.
 - C. Disconnect, cap and stow the electrical connector (DV28166).
 - D. Confirm the manual override handle is in the CLOSED position and lockwire.
2. For right jettison nozzle valve inoperative:
 - A. Open and collar P210 panel R FUEL JETT NOZ VLV circuit breaker.
 - B. Gain access to the right jettison nozzle valve actuator.
 - C. Disconnect, cap and stow the electrical connector (DV28266).
 - D. Confirm the manual override handle is in the CLOSED position and lockwire.

28-31-04 Fuel Jettison Nozzle Valves**28-31-04-01 Without PRR 62777-79S or Equivalent Installed****28-31-04-01B Both Inoperative**

Interval	Installed	Required	Procedure
C	2	0	(M) (O)

May be inoperative provided:

- a. Both jettison nozzle valves are secured closed.
- b. Jettison system is deactivated.
- c. Appropriate performance penalties are applied.



NOTE: The PRR 62777-79S is applicable to airplane line numbers 1099 and-on. Service Bulletins 777-28-0078 (ELMS2) and 777-28-0082 (ELMS1) each provide the production equivalent of PRR 62777-79S.

MAINTENANCE (M)

Secure the fuel jettison nozzle valves in the closed position and deactivate the jettison system (AMM 28-26-00/201).

1. For left jettison nozzle valve:
 - A. Open and collar P110 panel L FUEL JETT NOZ VLV circuit breaker.
 - B. Gain access to the left jettison nozzle valve actuator.
 - C. Disconnect, cap and stow the electrical connector (DV28166).
 - D. Confirm the manual override handle is in the CLOSED position and lockwire.
2. For right jettison nozzle valve:
 - A. Open and collar P210 panel R FUEL JETT NOZ VLV circuit breaker.
 - B. Gain access to the right jettison nozzle valve actuator.
 - C. Disconnect, cap and stow the electrical connector (DV28266).
 - D. Confirm the manual override handle is in the CLOSED position and lockwire.
3. Deactivate the fuel jettison system.
 - A. Open and collar P110 panel L FUEL JETT PUMP CTRL and L FUEL JETT ISLN VLV circuit breakers.
 - B. Open and collar P210 panel R FUEL JETT PUMP CTRL and R FUEL JETT ISLN VLV circuit breakers.

OPERATIONS (O)

NOTE: The Fuel Synoptic will incorrectly indicate jettison flow from the center tank when the center tank override pumps are on with fuel in the center tank. The FUEL JETT NOZZLE L and R advisory messages will be displayed and overhead panel L and R fuel jettison nozzle VALVE lights will be illuminated.

For -200ER/-300:

Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 4,000 lb (1,800 kg) (for fuel that would be burned during a return-to-land procedure).

For -200LR/777F/-300ER:

Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 5,400 lb (2,450 kg) (for fuel that would be burned during a return-to-land procedure).

**28-31-04 Fuel Jettison Nozzle Valves****28-31-04-01 Without PRR 62777-79S or Equivalent Installed****28-31-04-01-01 Switch VALVE Lights**

Interval	Installed	Required	Procedure
C	2	0	

28-31-04 Fuel Jettison Nozzle Valves**28-31-04-01 Without PRR 62777-79S or Equivalent Installed****28-31-04-01-02 Switch ON Lights**

Interval	Installed	Required	Procedure
C	2	0	

28-31-04 Fuel Jettison Nozzle Valves**28-31-04-02 With PRR 62777-79S or Equivalent Installed****28-31-04-02A One Inoperative**

Interval	Installed	Required	Procedure
C	2	1	(M)

One nozzle valve may be inoperative provided it is secured closed.

NOTE: The PRR 62777-79S is applicable to airplane line numbers 1099 and-on. Service Bulletins 777-28-0078 (ELMS2) and 777-28-0082 (ELMS1) each provide the production equivalent of PRR 62777-79S.

MAINTENANCE (M)

Secure the inoperative fuel jettison nozzle valve in the closed position (AMM 28-00-00/901).

1. For left jettison nozzle valve inoperative:
 - A. Open and collar P110 panel L FUEL JETT NOZ VLV circuit breaker.
 - B. Gain access to the left jettison nozzle valve actuator.
 - C. Disconnect, cap and stow the electrical connector (DV28166).
 - D. Confirm the manual override handle is in the CLOSED position and lockwire.
2. For right jettison nozzle valve inoperative:
 - A. Open and collar P210 panel R FUEL JETT NOZ VLV circuit breaker.



- B. Gain access to the right jettison nozzle valve actuator.
- C. Disconnect, cap and stow the electrical connector (DV28266).
- D. Confirm the manual override handle is in the CLOSED position and lockwire.

28-31-04 Fuel Jettison Nozzle Valves**28-31-04-02 With PRR 62777-79S or Equivalent Installed****28-31-04-02B Both Inoperative**

Interval	Installed	Required	Procedure
C	2	0	(M) (O)

May be inoperative provided:

- a. Both jettison nozzle valves are secured closed.
- b. Jettison system is deactivated.
- c. Fuel scavenge system is considered inoperative.
- d. Appropriate performance penalties are applied.

NOTE 1: The PRR 62777-79S is applicable to airplane line numbers 1099 and-on.
Service Bulletins 777-28-0078 (ELMS2) and 777-28-0082 (ELMS1) each provide the production equivalent of PRR 62777-79S.

NOTE 2: The airplane must also be dispatched using MMEL item 28-22-04.

MAINTENANCE (M)

Secure the fuel jettison nozzle valves in the closed position and deactivate the jettison system (AMM 28-26-00/201).

1. For left jettison nozzle valve:
 - A. Open and collar P110 panel L FUEL JETT NOZ VLV circuit breaker.
 - B. Gain access to the left jettison nozzle valve actuator.
 - C. Disconnect, cap and stow the electrical connector (DV28166).
 - D. Confirm the manual override handle is in the CLOSED position and lockwire.
2. For right jettison nozzle valve:
 - A. Open and collar P210 panel R FUEL JETT NOZ VLV circuit breaker.
 - B. Gain access to the right jettison nozzle valve actuator.
 - C. Disconnect, cap and stow the electrical connector (DV28266).
 - D. Confirm the manual override handle is in the CLOSED position and lockwire.
3. Deactivate the fuel jettison system.
 - A. Open and collar P110 panel L FUEL JETT PUMP CTRL and L FUEL JETT ISLN VLV circuit breakers.



- B. Open and collar P210 panel R FUEL JETT PUMP CTRL and R FUEL JETT ISLN VLV circuit breakers.

OPERATIONS (O)

NOTE: The Fuel Synoptic will incorrectly indicate jettison flow from the center tank when the center tank override pumps are on with fuel in the center tank. The FUEL JETT NOZZLE L and R advisory messages will be displayed and overhead panel L and R fuel jettison nozzle VALVE lights will be illuminated.

For -200ER/-300:

Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 4,000 lb (1,800 kg) (for fuel that would be burned during a return-to-land procedure).

For -200LR/777F/-300ER:

Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 5,400 lb (2,450 kg) (for fuel that would be burned during a return-to-land procedure).

28-31-04 Fuel Jettison Nozzle Valves

28-31-04-02 With PRR 62777-79S or Equivalent Installed

28-31-04-02-01 Switch VALVE Lights

Interval	Installed	Required	Procedure
C	2	0	

28-31-04 Fuel Jettison Nozzle Valves

28-31-04-02 With PRR 62777-79S or Equivalent Installed

28-31-04-02-02 Switch ON Lights

Interval	Installed	Required	Procedure
C	2	0	

**28-40-01 Fuel Synoptic Display**

Interval	Installed	Required	Procedure
C	1	0	

May be inoperative provided individual fuel quantity indications required for dispatch operate normally.

OPERATIONS NOTE

If the Synoptic Display is missing data, selecting an alternate location for the display (L INBD, LWR CTR, or R INBD) may restore the missing data. Synoptic Displays containing missing data may continue to be used to the extent remaining data is useful. Airplane system faults will be annunciated by alerting and status messages.

28-41-01 Fuel Quantity Indications (Flight Deck)
28-41-01-01 Center Tank

Interval	Installed	Required	Procedure
B	1	0	(M) (O)

May be inoperative provided:

- Main tank fuel quantity indications operate normally.
- Fuel quantity in center tank is verified by an alternate procedure.
- All pumps for the center tank operate normally.
- Fuel flow Indications operate normally.
- FMC FUEL is initialized with the known total fuel quantity.
- Flight remains within 180 minutes of landing at a suitable airport.

For 200ER, -200LR, 777F, -300 and -300ER:

- Appropriate gross weight and center-of-gravity adjustments are applied.

NOTE: The airplane must also be dispatched using MMEL item 28-41-03.

MAINTENANCE (M)

Verify the fuel quantity in the center tank using one of the following methods (AMM 28-00-00/901):

- Use the center tank's Integrated Refuel Panel (IRP) (P28) quantity indication (if operating).
- Refuel using the FMC Calculated Fuel on CDU Progress page 2 from previous flight and operating fuel quantity indications to determine the center tank quantity. Then refuel the center tank with a known quantity of fuel. A fuel tanker's fuel indication may be used if verified by the center tank's fuel measuring sticks.
- Defuel the center tank (AMM 28-26-00/201) and service with a known quantity of fuel (AMM 12-11-01/301).
- Determine the fuel quantity using the center tank's measuring sticks (Fuel Measuring Stick Manual).

NOTE: The fuel quantity measuring sticks may not give accurate results until 5 to 15 minutes after refueling operations are complete.

OPERATIONS (O)

NOTE: FUEL IN CENTER and FUEL LOW CENTER advisory messages will not be displayed when appropriate.

- Total fuel quantity must be calculated and manually initialized on the FMC PERF INIT page.

NOTE 1: The total fuel quantity indication may be inoperative or inaccurate.

NOTE 2: The FMC CALCULATED fuel quantity will be blank prior to manual initialization.

- A. Manually enter the total fuel quantity on the PERF INIT page. MANUAL will be displayed next to the entered fuel quantity.

NOTE: The manually entered total fuel quantity and the takeoff speeds may blank after engine start.

- B. Before and after engine start, verify the fuel quantity entered is displayed under the CALCULATED fuel quantity display on the PROGRESS page 2.
2. Flight crew should monitor FMC Calculated fuel and the operative fuel tank quantity indications to ensure usages are normal and to derive the amount of fuel in the tank with the inoperative quantity indication when necessary.
3. Apply appropriate gross weight and center-of-gravity restrictions.

For -200ER:

- A. For gross weight exceeding 495,000 lb (225,000 kg), Center-of-Gravity must be aft of 20% MAC for takeoff.

For -200LR/777F:

- B. For gross weight exceeding 563,000 lb (255,000 kg), Center-of-Gravity must be aft of 22.2% MAC for takeoff.

For -300:

- C. For gross weight exceeding 535,000 lb (243,000 kg), Center-of-Gravity must be aft of 16% MAC for takeoff.

For -300ER:

- D. For gross weight exceeding 586,000 lb (266,000 kg), Center-of-Gravity must be aft of 19.4% MAC for takeoff.
4. The flight must remain with 180 minutes of landing at a suitable airport.
5. The FUEL LOW CENTER advisory message will not operate. When FUEL PUMP CENTER L or R advisory messages are displayed, accomplish the FUEL PUMP CENTER L or R Non-Normal Checklist.
6. If it is necessary to jettison fuel enroute:

NOTE: Selection of fuel jettison with a fuel quantity indication inoperative causes all FMC performance and VNAV data to become invalid. If VNAV is the current active pitch mode of the Autopilot Flight Director System (AFDS), the AFDS pitch mode will fail with a resulting Master Caution light, AUTOPILOT caution message and amber bar through the pitch annunciation on the Primary Flight Display (PFD).

- A. Fuel jettison system maximum landing weight and fuel to remain automatic shutoff functions will be inoperative.
- B. FUEL AUTO JETTISON caution message will be displayed. Accomplish the Manual FUEL AUTO JETTISON Non-Normal Checklist.

**28-41-01 Fuel Quantity Indications (Flight Deck)****28-41-01-02 Main Tanks****28-41-01-02-01 Without PRR 62777-79S or Equivalent Installed**

Interval	Installed	Required	Procedure
B	2	1	(M) (O)

One may be inoperative provided:

- Center tank fuel quantity indication operates normally.
- Fuel quantity in associated tank is verified by an alternate procedure.
- All pumps for the associated tank operate normally.
- Fuel flow Indications operate normally.
- FMC FUEL is initialized with the known total fuel quantity.
- Flight remains within 180 minutes of landing at a suitable airport.

For 200ER, -200LR, 777F, -300 and -300ER:

- Appropriate gross weight and center-of-gravity adjustments are applied.

NOTE 1: The airplane must also be dispatched using MMEL item 28-41-03.

NOTE 2: The PRR 62777-79S is applicable to airplane line numbers 1099 and on.
Service Bulletins 777-28-0078 (ELMS2) and 777-28-0082 (ELMS1) each provide the production equivalent of PRR 62777-79S.

MAINTENANCE (M)

Verify the fuel quantity in the tank associated with the inoperative quantity indication using one of the following methods (AMM 28-00-00/901):

- Use the associated tank's Integrated Refuel Panel (IRP) (P28) quantity indication (if operating).
- Refuel using the FMC Calculated Fuel on CDU Progress page 2 from previous flight and operating fuel quantity indications to determine the associated tank quantity. Then refuel the associated tank with a known quantity of fuel. A fuel tanker's fuel indication may be used if verified by the associated tank's fuel measuring sticks.
- Defuel the associated tank (AMM 28-26-00/201) and service with a known quantity of fuel (AMM 12-11-01/301).
- Determine the fuel quantity using the associated tank's measuring sticks (Fuel Measuring Stick Manual).

NOTE: The fuel quantity measuring sticks may not give accurate results until 5 to 15 minutes after refueling operations are complete.

OPERATIONS (O)

NOTE 1: FUEL IMBALANCE advisory message may be displayed erroneously or may not display with an actual fuel imbalance.

NOTE 2: If the FUEL IMBALANCE advisory message displays, to prevent unnecessary fuel balancing, the crew should not accomplish the FUEL IMBALANCE checklist unless an actual fuel imbalance can be confirmed.

1. Total fuel quantity must be calculated and manually initialized on the FMC PERF INIT page.

NOTE 1: The total fuel quantity indication may be inoperative or inaccurate.

NOTE 2: The FMC CALCULATED fuel quantity will be blank prior to manual initialization.

- A. Manually enter the total fuel quantity on the PERF INIT page. MANUAL will be displayed next to the entered fuel quantity.

NOTE: The manually entered total fuel quantity and the takeoff speeds may blank after engine start.

- B. Before and after engine start, verify the fuel quantity entered is displayed under the CALCULATED fuel quantity display on the PROGRESS page 2.
2. Flight crew should monitor FMC Calculated fuel and the operative fuel tank quantity indications to ensure usages are normal and to derive the amount of fuel in the tank with the inoperative quantity indication when necessary.
3. Apply appropriate gross weight and center-of-gravity restrictions.

For -200ER:

- A. For gross weight exceeding 495,000 lb (225,000 kg), Center-of-Gravity must be aft of 20% MAC for takeoff.

For -200LR/777F:

- B. For gross weight exceeding 563,000 lb (255,000 kg), Center-of-Gravity must be aft of 22.2% MAC for takeoff.

For -300:

- C. For gross weight exceeding 535,000 lb (243,000 kg), Center-of-Gravity must be aft of 16% MAC for takeoff.

For -300ER:

- D. For gross weight exceeding 586,000 lb (266,000 kg), Center-of-Gravity must be aft of 19.4% MAC for takeoff.
4. The flight must remain with 180 minutes of landing at a suitable airport.
5. If it is necessary to jettison fuel enroute:

NOTE: Selection of fuel jettison with a fuel quantity indication inoperative causes all FMC performance and VNAV data to become invalid. If VNAV is the current active pitch mode of the Autopilot Flight Director System (AFDS), the AFDS pitch mode will fail with a resulting Master Caution light, AUTOPILOT caution message and amber bar through the pitch annunciation on the Primary Flight Display (PFD).

- A. Fuel jettison system maximum landing weight and fuel to remain automatic shutoff functions will be inoperative.
- B. FUEL AUTO JETTISON caution message will be displayed. Accomplish the Manual FUEL AUTO JETTISON Non-Normal Checklist.

28-41-01 Fuel Quantity Indications (Flight Deck)
28-41-01-02 Main Tanks
28-41-01-02-02 With PRR 62777-79S or Equivalent Installed

Interval	Installed	Required	Procedure
B	2	1	(M) (O)

One may be inoperative provided:

- a. Center tank fuel quantity indication operates normally.
- b. Fuel quantity in associated tank is verified by an alternate procedure.
- c. All pumps for the associated tank operate normally.
- d. Fuel flow Indications operate normally.
- e. FMC FUEL is initialized with the known total fuel quantity.
- f. Flight remains within 180 minutes of landing at a suitable airport.
- g. Fuel scavenge system is considered inoperative.

For 200ER, -200LR, 777F, -300 and -300ER:

- h. Appropriate gross weight and center-of-gravity adjustments are applied.

NOTE 1: The airplane must also be dispatched using MMEL item 28-41-03.

NOTE 2: The airplane must also be dispatched using MMEL item 28-22-04.

NOTE 3: The PRR 62777-79S is applicable to airplane line numbers 1099 and-on. Service Bulletins 777-28-0078 (ELMS2) and 777-28-0082 (ELMS1) each provide the production equivalent of PRR 62777-79S.

MAINTENANCE (M)

Verify the fuel quantity in the tank associated with the inoperative quantity indication using one of the following methods (AMM 28-00-00/901):

- 1. Use the associated tank's Integrated Refuel Panel (IRP) (P28) quantity indication (if operating).

2. Refuel using the FMC Calculated Fuel on CDU Progress page 2 from previous flight and operating fuel quantity indications to determine the associated tank quantity. Then refuel the associated tank with a known quantity of fuel. A fuel tanker's fuel indication may be used if verified by the associated tank's fuel measuring sticks.
3. Defuel the associated tank (AMM 28-26-00/201) and service with a known quantity of fuel (AMM 12-11-01/301).
4. Determine the fuel quantity using the associated tank's measuring sticks (Fuel Measuring Stick Manual).

NOTE: The fuel quantity measuring sticks may not give accurate results until 5 to 15 minutes after refueling operations are complete.

OPERATIONS (O)

NOTE 1: FUEL IMBALANCE advisory message may be displayed erroneously or may not display with an actual fuel imbalance.

NOTE 2: If the FUEL IMBALANCE advisory message displays, to prevent unnecessary fuel balancing, the crew should not accomplish the FUEL IMBALANCE checklist unless an actual fuel imbalance can be confirmed.

1. Total fuel quantity must be calculated and manually initialized on the FMC PERF INIT page.

NOTE 1: The total fuel quantity indication may be inoperative or inaccurate.

NOTE 2: The FMC CALCULATED fuel quantity will be blank prior to manual initialization.

- A. Manually enter the total fuel quantity on the PERF INIT page. MANUAL will be displayed next to the entered fuel quantity.

NOTE: The manually entered total fuel quantity and the takeoff speeds may blank after engine start.

- B. Before and after engine start, verify the fuel quantity entered is displayed under the CALCULATED fuel quantity display on the PROGRESS page 2.
2. Flight crew should monitor FMC Calculated fuel and the operative fuel tank quantity indications to ensure usages are normal and to derive the amount of fuel in the tank with the inoperative quantity indication when necessary.
3. Apply appropriate gross weight and center-of-gravity restrictions.

For -200ER:

- A. For gross weight exceeding 495,000 lb (225,000 kg), Center-of-Gravity must be aft of 20% MAC for takeoff.

For -200LR/777F:

- B. For gross weight exceeding 563,000 lb (255,000 kg), Center-of-Gravity must be aft of 22.2% MAC for takeoff.

**For -300:**

- C. For gross weight exceeding 535,000 lb (243,000 kg), Center-of-Gravity must be aft of 16% MAC for takeoff.

For -300ER:

- D. For gross weight exceeding 586,000 lb (266,000 kg), Center-of-Gravity must be aft of 19.4% MAC for takeoff.
4. The flight must remain with 180 minutes of landing at a suitable airport.
5. If it is necessary to jettison fuel enroute:

NOTE: Selection of fuel jettison with a fuel quantity indication inoperative causes all FMC performance and VNAV data to become invalid. If VNAV is the current active pitch mode of the Autopilot Flight Director System (AFDS), the AFDS pitch mode will fail with a resulting Master Caution light, AUTOPILOT caution message and amber bar through the pitch annunciation on the Primary Flight Display (PFD).

- A. Fuel jettison system maximum landing weight and fuel to remain automatic shutoff functions will be inoperative.
- B. FUEL AUTO JETTISON caution message will be displayed. Accomplish the Manual FUEL AUTO JETTISON Non-Normal Checklist.

28-41-01 Fuel Quantity Indications (Flight Deck)**28-41-01-03 Auxiliary Tank(s)*********

Interval	Installed	Required	Procedure
B	1	0	(M) (O)

May be inoperative provided:

- Center and main tank fuel quantity indications operate normally.
 - Fuel quantity in auxiliary tank(s) is verified by an alternate procedure.
 - Auxiliary tank fuel pump transfer system operates normally.
 - Auxiliary tank fuel backup transfer system operates normally.
 - Fuel flow indications operate normally.
 - FMC FUEL is initialized with the known total fuel quantity.
 - Flight remains within 180 minutes of landing at a suitable airport.
 - Appropriate gross weight and center-of-gravity adjustments are applied.
-

NOTE: The airplane must also be dispatched using MMEL item 28-41-03.

MAINTENANCE (M)

NOTE 1: For the auxiliary tank fuel pump transfer system to be operating normally, certain items associated with that system must be operative. See DDG item 28-14-01 for a listing of those items.

NOTE 2: For the auxiliary tank fuel backup transfer system to be operating normally, certain items associated with that system must be operative. See DDG item 28-14-04 for a listing of those items.

Verify the fuel quantity in auxiliary tank(s) and prior to each flight confirm auxiliary fuel cell configuration fields are present on the FQIS Maintenance Page (AMM 28-00-00/901).

1. Verify the fuel quantity in auxiliary tank(s) using one of the following methods:
 - A. Use the associated tank's auxiliary refuel panel quantity indication (if operating).
 - B. Determine the fuel quantity using the associated tank's Electronic Fuel Level Indication (EFLI).
 - C. Defuel the associated tank(s) and service with a known quantity of fuel.
2. Prior to each flight, confirm the appropriate auxiliary fuel cell configuration fields are present on the Fuel Quantity Maintenance Page 2/3.

OPERATIONS (O)

NOTE: FUEL IN AUX and FUEL LOW AUX advisory messages and FUEL AUX XFR Caution message will not be displayed when appropriate.

1. Total fuel quantity must be calculated and manually initialized on the FMC PERF INIT page.

NOTE 1: The total fuel quantity indication may be inoperative or inaccurate. |

NOTE 2: The FMC CALCULATED fuel quantity will be blank prior to manual initialization.

- A. Manually enter the total fuel quantity on the PERF INIT page. MANUAL will be displayed next to the entered fuel quantity.

NOTE: The manually entered total fuel quantity and the takeoff speeds may blank after engine start.

- B. Before and after engine start, verify the fuel quantity entered is displayed under the CALCULATED fuel quantity display on the PROGRESS page 2.
2. Flight Crew must confirm that auxiliary tank fuel transfer has actually occurred by monitoring the center tank fuel quantity.

NOTE 1: Fuel transfer from the auxiliary tank(s) to the center tank begins 55 minutes after takeoff.



NOTE 2: The fuel system synoptic may also be used to monitoring fuel transfer to center tank.

3. Apply appropriate gross weight and center-of-gravity restrictions.

For -200LR:

- A. For gross weight exceeding 563,000 lb (255,000 kg), Center-of-Gravity must be aft of 22.2% MAC for takeoff.

4. If it is necessary to jettison fuel enroute:

NOTE: Selection of fuel jettison with a fuel quantity indication inoperative causes all FMC performance and VNAV data to become invalid. If VNAV is the current active pitch mode of the Autopilot Flight Director System (AFDS), the AFDS pitch mode will fail with a resulting Master Caution light, AUTOPILOT caution message and amber bar through the pitch annunciation on the Primary Flight Display (PFD).

- A. Fuel jettison system maximum landing weight and fuel to remain automatic shutoff functions will be inoperative.
- B. FUEL AUTO JETTISON caution message will be displayed. Accomplish the Manual FUEL AUTO JETTISON Non-Normal Checklist.

**28-41-02 Fuel Quantity Processor Input/Output (I/O) Channels**

Interval	Installed	Required	Procedure
B	2	1	(O)

One channel may be inoperative provided:

- All flight deck fuel indications operate normally.
- Fuel flow indications operate normally.
- FMC TOTALIZER FUEL quantity agrees with fuel quantity indications.

OPERATIONS (O)

Verify that the fuel quantities loaded agree with the quantity indications on Fuel Synoptic, FMC and EICAS display.

28-41-03 Total Fuel Quantity Indication

Interval	Installed	Required	Procedure
C	1	0	(O)

May be inoperative provided:

- Fuel flow indications operate normally.
- FMC FUEL is initialized with the known total fuel quantity.
- Flight remains within 180 minutes of landing at a suitable airport.

OPERATIONS (O)

- Total fuel quantity must be calculated and manually initialized on the FMC PERF INIT page.

NOTE 1: The total fuel quantity indication will be inoperative.

NOTE 2: The FMC CALCULATED fuel quantity will be blank prior to manual initialization.

- Manually enter the total fuel quantity on the PERF INIT page. MANUAL will be displayed next to the entered fuel quantity.

NOTE: The manually entered total fuel quantity and the takeoff speeds may blank after engine start.

- Before and after engine start, verify the fuel quantity entered is displayed under the CALCULATED fuel quantity display on the PROGRESS page 2.
- Flight crew should monitor FMC Calculated fuel and the operative fuel tank quantity indications to ensure usages are normal.
 - The flight must remain within 180 minutes of landing at a suitable airport.
 - If it is necessary to jettison fuel enroute:

NOTE: Selection of fuel jettison with Total Fuel Quantity Indication inoperative causes all FMC Performance and VNAV data to become invalid. If VNAV is the current active pitch mode of the Autopilot Flight Director System (AFDS), the AFDS pitch mode will fail with a resulting Master Caution light, AUTOPILOT caution message and amber bar through the pitch annunciation on the Primary Flight Display (PFD).

- Fuel jettison system maximum landing weight and fuel to remain automatic shutoff functions will be inoperative.
- FUEL AUTO JETTISON caution message will be displayed. Accomplish the FUEL AUTO JETTISON Non-Normal Checklist.

**28-42-01 Auxiliary Tank Fuel Pump Pressure Transducer(s)**
*****28-42-01A Auxiliary Tank(s) Usable**

Interval	Installed	Required	Procedure
C	-	0	(M) (O)

May be inoperative provided:

- Associated auxiliary tank fuel pump is deactivated.
- Auxiliary tank fuel backup transfer system operates normally.
- Auxiliary tank fuel quantity indication operates normally.
- Maximum zero fuel weight limit is reduced by the weight of the associated auxiliary tank fuel.

MAINTENANCE (M)

NOTE: For the auxiliary tank fuel backup transfer system to be operating normally, certain items associated with that system must be operative. See DDG item 28-14-04 for a listing of those items.

Deactivate the associated auxiliary tank fuel pump (AMM 28-00-00/901).

- For auxiliary tank 1 pump pressure sensor inoperative, open and collar P210 panel AUX TANK PUMP 1 PWR circuit breaker.
- For auxiliary tank 2 installed with pump pressure sensor inoperative, open and collar P210 panel AUX TANK PUMP 2 PWR circuit breaker.
- For auxiliary tank 3 installed with pump pressure sensor inoperative, open and collar P210 panel AUX TANK PUMP 3 PWR circuit breaker.

OPERATIONS (O)

Include auxiliary tank fuel from associated tank(s) as part of Zero Fuel Weight

28-42-01 Auxiliary Tank Fuel Pump Pressure Transducer(s)
*****28-42-01B Auxiliary Tank(s) Empty**

Interval	Installed	Required	Procedure
D	-	0	(M)

May be inoperative provided:

- Associated auxiliary tank fuel pump is deactivated.
- All auxiliary tanks remain empty.

**MAINTENANCE (M)**

Deactivate the associated auxiliary tank fuel pump and defuel the auxiliary tank(s) (AMM 28-00-00/901).

1. For auxiliary tank 1 pump pressure sensor inoperative, open and collar P210 panel AUX TANK PUMP 1 PWR circuit breaker.
2. For auxiliary tank 2 installed with pump pressure sensor inoperative, open and collar P210 panel AUX TANK PUMP 2 PWR circuit breaker.
3. For auxiliary tank 3 installed with pump pressure sensor inoperative, open and collar P210 panel AUX TANK PUMP 3 PWR circuit breaker.
4. For auxiliary tank(s) not empty, defuel the auxiliary tank(s).



28-43-01 Fuel Temperature Indicating System

Interval	Installed	Required	Procedure
C	1	0	(O)

May be inoperative provided Total Air Temperature (TAT) is used as an indication of fuel temperature.

OPERATIONS (O)

Monitor fuel temperature using TAT.

**28-44-01 Measuring Sticks**

Interval	Installed	Required	Procedure
C	-	0	

May be inoperative provided fuel quantity is determined by other acceptable means.

**28-45-01 Auxiliary Tank Fuel Cell Pressure Transducers**

Interval	Installed	Required	Procedure
D	-	0	

May be inoperative provided all auxiliary tanks remain empty.

MAINTENANCE NOTE

For auxiliary tank(s) not empty, defuel the auxiliary tank(s) (AMM 28-26-00/201).

28-45-01 Auxiliary Tank Fuel Cell Pressure Transducers

28-45-01-01 Auxiliary Tank 1

Interval	Installed	Required	Procedure
C	1	0	(O)

May be inoperative provided:

- Auxiliary tank fuel pump transfer system operates normally.
- Auxiliary tank fuel quantity indication operates normally.
- Maximum zero fuel weight limit is reduced by the weight of the associated auxiliary tank fuel.

MAINTENANCE NOTE

For the auxiliary tank fuel pump transfer system to be operating normally, certain items associated with that system must be operative. See DDG item 28-14-01 for a listing of those items.

OPERATIONS (O)

Reduce the maximum fuel weight by the weight of the auxiliary tank fuel.

28-45-01 Auxiliary Tank Fuel Cell Pressure Transducers

28-45-01-02 Auxiliary Tank 2

Interval	Installed	Required	Procedure
C	1	0	

**28-45-01 Auxiliary Tank Fuel Cell Pressure Transducers**

28-45-01-03 Auxiliary Tank 3

Interval	Installed	Required	Procedure
C	1	0	



- 29-11-01 Engine Driven Pumps (EDP) Depressurization Function**
- 29-11-02 Center System Primary Alternating Current Motor Pumps (ACMP)**
- 29-11-03 Center System Demand Air Driven Pumps (ADP)**
 - 29-11-03A ADP Verified Not Running*
 - 29-11-03B ADP Deactivated*
 - 29-11-03C Other ADP Auto/Heater Function Inoperative*
 - 29-11-03-01 ADP Auto/Heater Function (-200/-200ER With PRR 61570 or SB 777-29-0004 Incorporated, and -200LR/-300/-300ER/777F)
 - 29-11-03-01A Other ADP Operates Normally*
 - 29-11-03-01B Other ADP Inoperative*
 - 29-11-03-01C Both Auto/Heater Functions Inoperative*
- 29-11-04 Demand Pump Selectors**
 - 29-11-04-01 Center System AUTO Position
 - 29-11-04-02 Left & Right System AUTO Position
 - 29-11-04-03 ON Position
- 29-11-05 Hydraulic Interface Module (HYDIM) Cards**
- 29-11-06 Center Hydraulic Isolation System**
- 29-18-01 Reservoir Quantity Gage (Remote)**
- 29-18-02 Hydraulic System Accumulators**
 - 29-18-02-01 Charging Valves
 - 29-18-02-02 Charging Gages
- 29-21-01 RAT Heater System**
 - 29-21-01-01 -200/-200ER With PRR 61570 or SB 777-29-0004 Incorporated, and -200LR/-300/-300ER/777F
 - 29-21-01-02 -200/-200ER Without PRR 61570 or SB 777-29-0004 Incorporated
- 29-21-02 RAM AIR TURBINE Switch Lights**
 - 29-21-02-01 UNLKD Light



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**29-11-02 Center System Primary Alternating Current Motor Pumps (ACMP)**

Interval	Installed	Required	Procedure
C	2	1	(M)

One may be inoperative provided:

- Inoperative center ACMP is deactivated.
- Both center ADPs operate normally.

MAINTENANCE (M)

Deactivate the inoperative center system primary pump (AMM 29-00-00/901).

- For center system PRIMARY C1 ELEC pump inoperative:
 - Position Hydraulic Panel (P5) PRIMARY C1 ELEC switch off.
 - Open P110 panel C1 HYD PUMP CTRL circuit breaker.

NOTE: Opening the C1 HYD PUMP CTRL circuit breaker results in the display of the HYD ISLN VALVE status message. This status message clears when the circuit breaker is closed.
 - Disconnect, cap and stow electrical connector D10008P on P100 panel and cap the P100 panel jack (J8).
 - Close P110 panel C1 HYD PUMP CTRL circuit breaker.
- For center system PRIMARY C2 ELEC pump inoperative:
 - Position Hydraulic Panel (P5) PRIMARY C2 ELEC switch off.
 - Open and collar P210 panel C2 HYD PUMP CTRL circuit breaker.

OPERATIONS NOTE

Turning the nose wheel more than 13 degrees by tiller input or towing, or moving a flight control surface, may cause a low center system hydraulic pressure condition resulting in the display of EICAS messages while on the ground. Verify that these messages correlate to the low pressure condition and that normal hydraulic pressure returns.

- For center system PRIMARY C1 ELEC pump inoperative, position PRIMARY C1 ELEC switch off.
- For center system PRIMARY C2 ELEC pump inoperative, position PRIMARY C2 ELEC switch off.

29-11-03 Center System Demand Air Driven Pumps (ADP)
29-11-03A ADP Verified Not Running

Interval	Installed	Required	Procedure
C	2	1	(M) (O)

One may be inoperative provided:

- a. Inoperative ADP is selected off.
- b. Both center system ACMPs operate normally.
- c. Remaining ADP heater operates normally.
- d. Remaining ADP AUTO function operates normally.
- e. Associated pump is verified not running when selected OFF.
- f. APU is used as ADP air source during takeoff.
- g. Appropriate performance adjustments are applied.

MAINTENANCE (M)

Verify the inoperative center system demand ADP is not running when selected OFF and the remaining center system demand ADP heater operates normally.

1. Verify the inoperative center system demand ADP is not running when selected OFF.
 - A. Position Hydraulic Panel (P5) DEMAND AIR C1 and C2 selectors OFF.
 - B. Position Hydraulic Panel (P5) PRIMARY ELEC C1 and C2 switches off.
 - C. On the Air Synoptic page, confirm the left and right DUCT PRESS are greater than 21.
 - D. On the Hydraulic Synoptic page, confirm the center system hydraulic system pressure is less than 200, or on the Hydraulic Maintenance page, confirm the associated Demand Pump PRESS is less than 200 and RUN is NO.

 NOTE: For ADP running when selected OFF, the pump must be deactivated. Airplane may be dispatched using MMEL item 29-11-03B.
 - E. Position the operative Hydraulic Panel (P5) DEMAND AIR C1 or C2 selector AUTO.
 - F. Position Hydraulic Panel (P5) PRIMARY C1 ELEC and PRIMARY C2 ELEC switches ON.
2. Verify the remaining center system demand ADP heater is operating normally.



For airplanes with SB 777-29-0004 (PRR 61570) incorporated:

- A. Confirm the associated HYD PUMP DEM C1 or C2 status message is not displayed.

For airplanes with SB 777-29-0004 (PRR 61570) not incorporated:

- A. Press and release E6-2 shelf RAT GEN HEATER TEST switch.
- B. Within 60 seconds of pressing RAT GEN HEATER TEST switch, confirm the operative ADP heater current is 1.5 to 2.2 amps.

NOTE 1: C1 ADP wire number is W5601-1004-16 and ground block is GB46015.

NOTE 2: C2 ADP wire number is W5135-1004-16 and ground block is GB46012.

OPERATIONS (O)

NOTE: For APU-to-Pack takeoff function installed, the APU-to-Pack Takeoff Supplementary Procedure is not allowed.

1. Reduce the performance limited weights by the appropriate adjustments.

Performance Limited Weights	Takeoff Climb	Takeoff Runway/Obstacle
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For -200 or -200ER/GE or PW or RR:

All	1,400 lb (636 kg)	9,600 lb (4,355 kg)
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For -200LR or 777F/GE90-100 Series or -300ER/GE90-115B:

All	1,700 lb (771 kg)	9,600 lb (4,355 kg)
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For -300/PW4090 or RR:

All	400 lb (181 kg)	3,700 lb (1,678 kg)
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2. Position the inoperative center system DEMAND pump selector OFF.
3. For takeoff, configure the pneumatic system with the APU supplying air for the remaining operative center system demand pump.
 - A. Position APU Bleed air switch AUTO.
 - B. Start the APU if it is not operating.
 - C. Position Bleed Isolation L ISLN and R ISLN switches CLOSED.

NOTE: The BLEED ISLN CLOSED L and BLEED ISLN CLOSED R advisory messages will be displayed.
 - D. After takeoff is completed, position Bleed Isolation L ISLN and R ISLN switches AUTO and configure the APU as required.

29-11-03 Center System Demand Air Driven Pumps (ADP)
29-11-03B ADP Deactivated

Interval	Installed	Required	Procedure
C	2	1	(M) (O)

One may be inoperative provided:

- a. Inoperative ADP is deactivated.
- b. Both center system ACMPs operate normally.
- c. Remaining ADP heater operates normally.
- d. Remaining ADP AUTO function operates normally.
- e. APU is used as ADP air source during takeoff.
- f. Appropriate performance adjustments are applied.

MAINTENANCE (M)

Deactivate the inoperative ADP and verify the remaining center system demand ADP heater operates normally (AMM 29-00-00/901).

1. Deactivate the inoperative ADP.
 - A. Depressurize the center hydraulic system (AMM 29-11-00/201).
 NOTE: It is not necessary to depressurize the reservoir.
 - B. Pressurize the pneumatic system (AMM 36-00-00/201).
 - C. Disconnect the Logic Speed Control Unit (LSCU).
 - 1) For C1 ADP inoperative:
 - a. Open the P310 panel C HYD AIR PUMP 1 circuit breaker.
 - b. Gain access to the right systems card file (P84) in the main equipment center and position the power switch for the HYDIM CR to OFF.
 - c. Gain access to the C1 ADP by opening access door 197JL.
 - d. Disconnect, cap and stow electrical connector DM29306A from the C1 ADP LSCU. Cap the LSCU receptacle.
 - e. Close access door 197JL.
 - f. Position the power switch for the HYDIM CR to ON.
 - g. Close the P310 panel C HYD AIR PUMP 1 circuit breaker.
 - 2) For C2 ADP inoperative:
 - a. Open the P310 panel C HYD AIR PUMP 2 circuit breaker.
 - b. Gain access to the left systems card file (P85) and position the power switch for the HYDIM CL to OFF.
 - c. Gain access to the C2 ADP by opening access door 197JL.



- d. Disconnect, cap and stow electrical connector DM29321A from the C2 ADP LSCU. Cap the LSCU receptacle.
 - e. Close access door 197JL.
 - f. Position the power switch for the HYDIM CL to ON.
 - g. Close the P310 panel C HYD AIR PUMP 2 circuit breaker.
 - 3) On the Hydraulic Maintenance page, confirm the associated Demand Pump PRESS is less than 200 and RUN is NO, or on the Hydraulic Synoptic page, confirm the center system hydraulic system pressure is less than 200.

D. Depressurize the pneumatic pressure (AMM 36-00-00/201).

2. Verify the remaining center system demand ADP heater is operating normally.

For airplanes with SB 777-29-0004 (PRR 61570) incorporated:

- A. Confirm the associated HYD PUMP DEM C1 or C2 status message is not displayed.

For airplanes with SB 777-29-0004 (PRR 61570) not incorporated:

- A. Press and release E6-2 shelf RAT GEN HEATER TEST switch.
- B. Within 60 seconds of pressing RAT GEN HEATER TEST switch, confirm the operative ADP heater current is 1.5 to 2.2 amps.

NOTE 1: C1 ADP wire number is W5601-1004-16 and ground block is GB46015.

NOTE 2: C2 ADP wire number is W5135-1004-16 and ground block is GB46012.

OPERATIONS (O)

NOTE: For APU-to-Pack takeoff function installed, the APU-to-Pack Takeoff Supplementary Procedure is not allowed.

1. Reduce the performance limited weights by the appropriate adjustments.

Performance Limited Weights	Takeoff Climb	Takeoff Runway/Obstacle
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For -200 or -200ER/GE or PW or RR:

All	1,400 lb (636 kg)	9,600 lb (4,355 kg)
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For -200LR or 777F/GE90-100 Series or -300ER/GE90-115B:

All	1,700 lb (771 kg)	9,600 lb (4,355 kg)
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For -300/PW4090 or RR:

All	400 lb (181 kg)	3,700 lb (1,678 kg)
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2. Position the inoperative center system DEMAND pump selector OFF.
3. For takeoff, configure the pneumatic system with the APU supplying air for the remaining operative center system demand pump.
 - A. Position APU Bleed air switch AUTO.
 - B. Start the APU if it is not operating.
 - C. Position Bleed Isolation L ISLN and R ISLN switches CLOSED.

NOTE: The BLEED ISLN CLOSED L and BLEED ISLN CLOSED R advisory messages will be displayed.
 - D. After takeoff is completed, position Bleed Isolation L ISLN and R ISLN switches AUTO and configure the APU as required.

29-11-03 Center System Demand Air Driven Pumps (ADP)**29-11-03C Other ADP Auto/Heater Function Inoperative**

Interval	Installed	Required	Procedure
C	2	1	(M) (O)

One may be inoperative provided:

- a. Inoperative ADP is selected off.
- b. Both center system ACMPs operate normally.
- c. Associated pump is verified not running when selected OFF.
- d. Remaining ADP is selected ON for the entire flight.
- e. APU is used as ADP air source during takeoff.
- f. Appropriate performance adjustments are applied.

MAINTENANCE (M)

Verify the inoperative center system demand ADP is not running when selected OFF. If required, disconnect the ADP so it does not run (AMM 29-00-00/901).

1. Depressurize the center hydraulic system (AMM 29-11-00/201).

NOTE: It is not necessary to depressurize the reservoir.

2. Pressurize the pneumatic system (AMM 36-00-00/201).
3. On the Hydraulic Maintenance page, confirm the associated Demand Pump PRESS is less than 200 and RUN is NO, or on the Hydraulic Synoptic page, confirm the center system hydraulic system pressure is less than 200
4. For the ADP running when selected OFF, disconnect the Logic Speed Control Unit (LSCU).
 - A. For C1 ADP inoperative:
 - 1) Open the P310 panel C HYD AIR PUMP 1 circuit breaker.



- 2) Gain access to the right systems card file (P84) in the main equipment center and position the power switch for the HYDIM CR to OFF.
 - 3) Gain access to the C1 ADP by opening access door 197JL.
 - 4) Disconnect, cap and stow electrical connector DM29306A from the C1 ADP LSCU. Cap the LSCU receptacle.
 - 5) Close access door 197JL.
 - 6) Position the power switch for the HYDIM CR to ON.
 - 7) Close the P310 panel C HYD AIR PUMP 1 circuit breaker.
- B. For C2 ADP inoperative:
- 1) Open the P310 panel C HYD AIR PUMP 2 circuit breaker.
 - 2) Gain access to the left systems card file (P85) and position the power switch for the HYDIM CL to OFF.
 - 3) Gain access to the C2 ADP by opening access door 197JL.
 - 4) Disconnect, cap and stow electrical connector DM29321A from the C2 ADP LSCU. Cap the LSCU receptacle.
 - 5) Close access door 197JL.
 - 6) Position the power switch for the HYDIM CL to ON.
 - 7) Close the P310 panel C HYD AIR PUMP 2 circuit breaker.
- C. On the Hydraulic Maintenance page, confirm the associated Demand Pump PRESS is less than 200 and RUN is NO, or on the Hydraulic Synoptic page, confirm the center system hydraulic system pressure is less than 200.
5. Depressurize the pneumatic pressure (AMM 36-00-00/201).

OPERATIONS (O)

NOTE: For APU-to-Pack takeoff function installed, the APU-to-Pack Takeoff Supplementary Procedure is not allowed.

1. Reduce the performance limited weights by the appropriate adjustments.

Performance Limited Weights	Takeoff Climb	Takeoff Runway/Obstacle
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For -200 or -200ER/GE or PW or RR:

All	1,400 lb (636 kg)	9,600 lb (4,355 kg)
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For -200LR or 777F/GE90-100 Series or -300ER/GE90-115B:

All	1,700 lb (771 kg)	9,600 lb (4,355 kg)
-----	-------------------	---------------------

For -300/PW4090 or RR:

All	400 lb (181 kg)	3,700 lb (1,678 kg)
-----	-----------------	---------------------

2. Position the inoperative center system DEMAND pump selector OFF.
3. Position remaining operative center system DEMAND pump selector ON prior to takeoff and leave ON for the entire flight.

NOTE: Do not position center system DEMAND pump selector ON before engine start.
4. For takeoff, configure the pneumatic system with the APU supplying air for the remaining operative center system demand pump.
 - A. Position APU Bleed air switch AUTO.
 - B. Start the APU if it is not operating.
 - C. Position Bleed Isolation L ISLN and R ISLN switches CLOSED.

NOTE: The BLEED ISLN CLOSED L and BLEED ISLN CLOSED R advisory messages will be displayed.
 - D. After takeoff is completed, position Bleed Isolation L ISLN and R ISLN switches AUTO and configure the APU as required.

29-11-03 Center System Demand Air Driven Pumps (ADP)
29-11-03-01 ADP Auto/Heater Function (-200/-200ER With PRR 61570 or SB 777-29-0004 Incorporated, and -200LR/-300/-300ER/777F)
29-11-03-01A Other ADP Operates Normally

Interval	Installed	Required	Procedure
C	2	1	(M) (O)

One may be inoperative provided:

- a. Remaining ADP operates normally.
- b. Associated ADP ON and OFF selector positions are verified to operate normally.
- c. Associated ADP is selected ON for the entire flight.

MAINTENANCE (M)

Verify the associated center system demand ADP selector ON and OFF positions operate normally (AMM 29-00-00/901).

1. Supply electrical power on the airplane (AMM 24-22-00/201).
2. Pressurize the pneumatic system (AMM 36-00-00/201).
3. For the associated HYD PUMP DEM C1 or C2 status message displayed:
 - A. Position the associated Hydraulic Panel (P5) DEMAND AIR C1 or C2 selector ON.
 - B. Confirm the associated HYD PUMP DEM C1 or C2 status message clears.



- C. Position the associated Hydraulic Panel (P5) DEMAND AIR C1 or C2 selector OFF.
 - D. Confirm the associated HYD PUMP DEM C1 or C2 status message is displayed.
 - E. Depressurize the pneumatic system (AMM 36-00-00/201).
4. For the associated HYD PUMP DEM C1 or C2 status message not displayed:
- A. Position the associated Hydraulic Panel (P5) DEMAND AIR C1 or C2 selector ON.
 - B. On the Hydraulic Maintenance page, confirm the associated Demand Pump PRESS is greater than 2800, or on the Hydraulic Synoptic page, confirm the center system hydraulic system pressure is greater than 2800.
 - C. Position the associated Hydraulic Panel (P5) DEMAND AIR C1 or C2 selector OFF.
 - D. On the Hydraulic Maintenance page, confirm the associated Demand Pump PRESS is less than 200, or on the Hydraulic Synoptic page, confirm the center system hydraulic system pressure is less than 200.
 - E. Depressurize the pneumatic system (AMM 36-00-00/201).

OPERATIONS (O)

1. Position associated center system DEMAND pump selector ON prior to takeoff and leave ON for the entire flight.

NOTE: Do not position center system DEMAND pump selector ON before engine start.
2. Position remaining center system DEMAND pump selector AUTO.

29-11-03 Center System Demand Air Driven Pumps (ADP)**29-11-03-01 ADP Auto/Heater Function (-200/-200ER With PRR 61570 or SB 777-29-0004 Incorporated, and -200LR/-300/-300ER/777F)****29-11-03-01B Other ADP Inoperative**

Interval	Installed	Required	Procedure
C	2	1	(M) (O)

One may be inoperative provided:

- a. Both center system ACMPs operate normally.
- b. Associated ADP ON and OFF selector positions are verified to operate normally.
- c. Associated ADP is selected ON for the entire flight.
- d. APU is used as ADP air source during takeoff.

- e. Appropriate performance adjustments are applied.

MAINTENANCE (M)

Verify the associated center system demand ADP selector ON and OFF positions operate normally (AMM 29-00-00/901).

1. Supply electrical power on the airplane (AMM 24-22-00/201).
2. Pressurize the pneumatic system (AMM 36-00-00/201).
3. For the associated HYD PUMP DEM C1 or C2 status message displayed:
 - A. Position the associated Hydraulic Panel (P5) DEMAND AIR C1 or C2 selector ON.
 - B. Confirm the associated HYD PUMP DEM C1 or C2 status message clears.
 - C. Position the associated Hydraulic Panel (P5) DEMAND AIR C1 or C2 selector OFF.
 - D. Confirm the associated HYD PUMP DEM C1 or C2 status message is displayed.
 - E. Depressurize the pneumatic system (AMM 36-00-00/201).
4. For the associated HYD PUMP DEM C1 or C2 status message not displayed:
 - A. Position the associated Hydraulic Panel (P5) DEMAND AIR C1 or C2 selector ON.
 - B. On the Hydraulic Maintenance page, confirm the associated Demand Pump PRESS is greater than 2800, or on the Hydraulic Synoptic page, confirm the center system hydraulic system pressure is greater than 2800.
 - C. Position the associated Hydraulic Panel (P5) DEMAND AIR C1 or C2 selector OFF.
 - D. On the Hydraulic Maintenance page, confirm the associated Demand Pump PRESS is less than 200, or on the Hydraulic Synoptic page, confirm the center system hydraulic system pressure is less than 200.
 - E. Depressurize the pneumatic system (AMM 36-00-00/201).

OPERATIONS (O)

NOTE: For APU-to-Pack takeoff function installed, the APU-to-Pack Takeoff Supplementary Procedure is not allowed.

1. Reduce the performance limited weights by the appropriate adjustments.

NOTE: For airplane also dispatched using MMEL item 29-11-03B, it is not required to add the following penalties.

Performance Limited Weights	Takeoff Climb	Takeoff Runway/Obstacle
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For -200 or -200ER/GE or PW or RR:

All	1,400 lb (636 kg)	9,600 lb (4,355 kg)
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For -200LR or 777F/GE90-100 Series or -300ER/GE90-115B:

All	1,700 lb (771 kg)	9,600 lb (4,355 kg)
-----	-------------------	---------------------

For -300/PW4090 or RR:

All	400 lb (181 kg)	3,700 lb (1,678 kg)
-----	-----------------	---------------------

2. Position inoperative center system DEMAND pump selector OFF.
3. Position associated center system DEMAND pump selector ON prior to takeoff and leave ON for the entire flight.

NOTE: Do not position center system DEMAND pump selector ON before engine start.

4. For takeoff, configure the pneumatic system with the APU supplying air for the operating center system demand pump.

- A. Position APU Bleed air switch AUTO.
- B. Start the APU if it is not operating.
- C. Position Bleed Isolation L ISLN and R ISLN switches CLOSED.

NOTE: The BLEED ISLN CLOSED L and BLEED ISLN CLOSED R advisory messages will be displayed.

- D. After takeoff is completed, position Bleed Isolation L ISLN and R ISLN switches AUTO and configure the APU as required.

29-11-03 Center System Demand Air Driven Pumps (ADP)

29-11-03-01 ADP Auto/Heater Function (-200/-200ER With PRR 61570 or SB 777-29-0004 Incorporated, and -200LR/-300/-300ER/777F)

29-11-03-01C Both Auto/Heater Functions Inoperative

Interval	Installed	Required	Procedure
C	2	0	(M) (O)

May be inoperative provided:

- a. Both center system ACMPs operate normally.
- b. ADP ON and OFF selector positions are verified to operate normally for both pumps.
- c. One ADP is selected ON for the entire flight.
- d. APU is used as ADP air source during takeoff.

- e. Appropriate performance adjustments are applied.

MAINTENANCE (M)

Verify both center system demand ADP selector's ON and OFF positions operate normally (AMM 29-00-00/901).

1. Supply electrical power on the airplane (AMM 24-22-00/201).
2. Pressurize the pneumatic system (AMM 36-00-00/201).
3. For the associated HYD PUMP DEM C1 and/or C2 status message displayed:
 - A. Position the associated Hydraulic Panel (P5) DEMAND AIR C1 or C2 selector ON.
 - B. Confirm the associated HYD PUMP DEM C1 or C2 status message clears.
 - C. Position the associated Hydraulic Panel (P5) DEMAND AIR C1 or C2 selector OFF.
 - D. Confirm the associated HYD PUMP DEM C1 or C2 status message is displayed.
 - E. Depressurize the pneumatic system (AMM 36-00-00/201).
4. For the associated HYD PUMP DEM C1 and/or C2 status message not displayed:
 - A. Position the associated Hydraulic Panel (P5) DEMAND AIR C1 or C2 selector ON.
 - B. On the Hydraulic Maintenance page, confirm the associated Demand Pump PRESS is greater than 2800, or on the Hydraulic Synoptic page, confirm the center system hydraulic system pressure is greater than 2800.
 - C. Position the associated Hydraulic Panel (P5) DEMAND AIR C1 or C2 selector OFF.
 - D. On the Hydraulic Maintenance page, confirm the associated Demand Pump PRESS is less than 200, or on the Hydraulic Synoptic page, confirm the center system hydraulic system pressure is less than 200.
 - E. Depressurize the pneumatic system (AMM 36-00-00/201).

OPERATIONS (O)

NOTE: For APU-to-Pack takeoff function installed, the APU-to-Pack Takeoff Supplementary Procedure is not allowed.

1. Reduce the performance limited weights by the appropriate adjustments.

Performance Limited Weights	Takeoff Climb	Takeoff Runway/Obstacle
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**For -200 or -200ER/GE or PW or RR:**

All	1,400 lb (636 kg)	9,600 lb (4,355 kg)
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For -200LR or 777F/GE90-100 Series or -300ER/GE90-115B:

All	1,700 lb (771 kg)	9,600 lb (4,355 kg)
-----	-------------------	---------------------

For -300/PW4090 or RR:

All	400 lb (181 kg)	3,700 lb (1,678 kg)
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2. Position one center system DEMAND pump selector ON prior to takeoff and leave ON for the entire flight.
NOTE: Do not position center system DEMAND pump selector ON before engine start.
3. Position remaining center system DEMAND pump selector AUTO.
4. For takeoff, configure the pneumatic system with the APU supplying air for the operating center system demand pump.
 - A. Position APU Bleed air switch AUTO.
 - B. Start the APU if it is not operating.
 - C. Position Bleed Isolation L ISLN and R ISLN switches CLOSED.
NOTE: The BLEED ISLN CLOSED L and BLEED ISLN CLOSED R advisory messages will be displayed.
 - D. After takeoff is completed, position Bleed Isolation L ISLN and R ISLN switches AUTO and configure the APU as required.

**29-11-04 Demand Pump Selectors****29-11-04-01 Center System AUTO Position**

Interval	Installed	Required	Procedure
C	2	0	

May be inoperative provided associated demand ADP auto function is considered inoperative.

NOTE 1: For -200/-200ER with PRR 61570 or SB 777-29-0004 incorporated, and for -200LR/-300/-300ER/777F, the airplane must be dispatched using MMEL item 29-11-03-01.

NOTE 2: For -200/-200ER without PRR 61570 or SB 777-29-0004 incorporated, the airplane must be dispatched using MMEL item 29-21-01-02.

29-11-04 Demand Pump Selectors**29-11-04-02 Left & Right System AUTO Position**

Interval	Installed	Required	Procedure
C	2	0	(O)

May be inoperative provided:

- ON and OFF positions operate normally.
- Associated pump selector is ON for takeoff and landing.

OPERATIONS (O)

Position the associated DEMAND pump selector ON for takeoff and landing.

NOTE: Selector may be placed in ON position for the entire flight if desired.

29-11-04 Demand Pump Selectors**29-11-04-03 ON Position**

Interval	Installed	Required	Procedure
C	4	0	

May be inoperative provided the AUTO and OFF positions for associated pump operate normally.

29-11-05 Hydraulic Interface Module (HYDIM) Cards

Interval	Installed	Required	Procedure
C	4	3	(M)

One may be inoperative provided:

- Inoperative card is in the HYDIM center right (HCR) card position.
- Individual center system pump pressure indications operate normally.
- Center system demand ADP C1 auto function is considered inoperative.

NOTE 1: For -200/-200ER with PRR 61570 or SB 777-29-0004 incorporated, and for -200LR/-300/-300ER/777F, the airplane must be dispatched using MMEL item 29-11-03-01 for one inoperative.

NOTE 2: For -200/-200ER without PRR 61570 or SB 777-29-0004 incorporated, the airplane must be dispatched using MMEL item 29-21-01-02.

MAINTENANCE (M)

Inoperative HYDIM card must be in the HYDIM Center Right (CR) card position. Configure HYDIM cards as required (AMM 29-00-00/901).

- Gain access to HYDIM cards in the right systems cardfile (P84) and left systems cardfile (P85) in the main equipment center (AMM 29-11-50/401).
- For HYDIM Center Left (CL) card inoperative:
 - Set the P85 power switch for HYDIM CL card to the OFF position.
 - Set the P84 power switch for HYDIM CR card to the OFF position.
 - Open P11 panel RAT AUTO CRTL circuit breaker.
 - Open P110 panel C1 HYD PUMP CTRL circuit breaker.
 - Open P310 panel C HYD AIR PUMP 1, C HYD AIR PUMP 2, RAT MAN/AUTO CTRL and NOSE GEAR ISLN VALVE circuit breakers.
 - Exchange HYDIM CL card (P85 position A4) with the serviceable HYDIM CR card (P84 position A4).
 - Set the P85 power switch for HYDIM CL card to the ON position.

NOTE: HYDIM CR card power switch remains off.

 - Close P11, P110 and P310 panel circuit breakers.
- For HYDIM Center Right (CR) card inoperative:
 - Set the P84 power switch for HYDIM CR card to the OFF position.
- For HYDIM Left (L) card inoperative:
 - Set the P85 power switch for HYDIM L card to the OFF position.
 - Set the P84 power switch for HYDIM CR card to the OFF position.
 - Open P11 panel L ENG EDP SUPPLY VALVE, LDG GEAR RETR AUTO OFF and RAT AUTO CRTL circuit breakers.
 - Open P110 panel L D-PRESS VLV SPLY circuit breaker.



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- E. Open P310 C HYD AIR PUMP 1 circuit breaker.
 - F. Exchange HYDIM L card (P85 position A17) with the serviceable HYDIM CR card (P84 position A4).
 - G. Set the P85 power switch for HYDIM L card to the ON position.
NOTE: HYDIM CR card power switch remains off.
 - H. Close P11, P110 and P310 panel circuit breakers.
5. For HYDIM Right (R) card inoperative:
- A. Set the P84 power switches for HYDIM R card and HYDIM CR card to the OFF positions.
 - B. Open P11 panel LDG GEAR RETR AUTO OFF, R ENG EDP SUPPLY VALVE, and RAT AUTO CTRL circuit breakers
 - C. Open P210 panel R D-PRESS VLV SPLY circuit breaker.
 - D. Open P310 panel C HYD AIR PUMP 1 circuit breaker.
 - E. Exchange HYDIM R card (P84 position A17) with the serviceable HYDIM CR card (P84 position A4).
 - F. Set the P84 power switch for HYDIM R card to the ON position.
NOTE: HYDIM CR card power switch remains off.
 - G. Close P11, P210 and P310 panel circuit breakers.
6. Use the maintenance access terminal (MAT) to perform an LRU replacement test of the exchanged serviceable HYDIM card. Following this test, only the HYDIM CARD HCR status message should be displayed.

OPERATIONS NOTE

The Hydraulic Synoptic Display may have incomplete or missing information.

29-11-06 Center Hydraulic Isolation System

Interval	Installed	Required	Procedure
C	1	0	(M)

May be inoperative provided center hydraulic isolation system is deactivated open..

MAINTENANCE (M)

Deactivate center hydraulic isolation system with the reserve and nose gear isolation valves in the open position (AMM 29-00-00/901).

1. Open the landing gear doors and install safety pins (AMM 32-00-15/201).
2. Open P110 panel C1 HYD PUMP CTRL circuit breaker.
3. Open and collar P310 panel NOSE GEAR ISLN VALVE circuit breaker.
4. Gain access to the reserve and nose gear isolation valves (right wheel well).
5. Deactivate the reserve isolation valve in the open position.
 - A. Disconnect, cap and stow electrical connector (DV29302).
 - B. Move the manual override handle to the OPEN position and lockwire.
6. Deactivate the nose gear isolation valve in the open position.
 - A. Disconnect, cap and stow electrical connector (DV29303).
 - B. Move the manual override handle to the OPEN position and lockwire.
7. Close P110 panel C1 HYD PUMP CTRL circuit breaker.
8. Remove the landing gear door safety pins and close the landing gear doors (AMM 32-00-15/201).

OPERATIONS NOTE

The RESERVE BRAKES/STRG advisory message will be displayed. Do not accomplish the RESERVE BRAKES/STRG Non-Normal Checklist.

For failures resulting in HYD PRESS SYS C caution message being displayed, accomplish the HYD PRESS SYS C and the RESERVE BRAKES/STRG Non-Normal Checklists.



29-18-01 Reservoir Quantity Gage (Remote)

Interval	Installed	Required	Procedure
C	1	0	



29-18-02 Hydraulic System Accumulators

Interval	Installed	Required	Procedure
C	-	0	

29-18-02 Hydraulic System Accumulators**29-18-02-01 Charging Valves**

Interval	Installed	Required	Procedure
C	3	0	

29-18-02 Hydraulic System Accumulators**29-18-02-02 Charging Gages**

Interval	Installed	Required	Procedure
C	3	0	

**29-21-01 RAT Heater System****29-21-01-01 -200/-200ER With PRR 61570 or SB 777-29-0004 Incorporated, and -200LR/-300/-300ER/777F**

Interval	Installed	Required	Procedure
A	1	0	

May be inoperative provided repairs are made within 3 flight days.

MAINTENANCE NOTE

The RAT heater system is considered to be operating normally when the heater switch (E6 bay) is moved from its existing position to the other position (HEATER 1 or HEATER 2) and the RAT GEN HEAT status message clears when the RAT heater is tested (AMM 29-21-00/501).

29-21-01 RAT Heater System**29-21-01-02 -200/-200ER Without PRR 61570 or SB 777-29-0004 Incorporated**

Interval	Installed	Required	Procedure
A	1	0	(O)

May be inoperative provided:

- Both center system ACMPs operate normally.
- One ADP is selected ON for the entire flight.
- APU is used as ADP air source during takeoff.
- Appropriate performance adjustments are applied.
- Repairs are made within 3 flight days.

NOTE: For airplanes that do not have PRR 61570 or SB 777-29-0004 installed, the RAT GEN HEAT status message may also indicate demand ADP heaters and auto functions are not operating normally.

OPERATIONS (O)

NOTE: For APU-to-Pack takeoff function installed, the APU-to-Pack Takeoff Supplementary Procedure is not allowed.

- Reduce the performance limited weights by the appropriate adjustments.

Performance Limited Weights	Takeoff Climb	Takeoff Runway/Obstacle
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For -200 or -200ER:

All	1,400 lb (636 kg)	9,600 lb (4,355 kg)
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2. Position one center system DEMAND pump selector ON prior to takeoff and leave ON for the entire flight.

NOTE: Do not position center system DEMAND pump selector ON before engine start.
3. Position remaining center system DEMAND pump selector AUTO.
4. For takeoff, configure the pneumatic system with the APU supplying air for the remaining operative center system demand pump.
 - A. Position APU Bleed air switch AUTO.
 - B. Start the APU if it is not operating.
 - C. Position Bleed Isolation L ISLN and R ISLN switches CLOSED.

NOTE: The BLEED ISLN CLOSED L and BLEED ISLN CLOSED R advisory messages will be displayed.
 - D. After takeoff is completed, position Bleed Isolation L ISLN and R ISLN switches AUTO and configure the APU as required.



29-21-02 RAM AIR TURBINE Switch Lights
29-21-02-01 UNLKD Light

Interval	Installed	Required	Procedure
C	1	0	

29-21-02 RAM AIR TURBINE Switch Lights
29-21-02-02 PRESS Light

Interval	Installed	Required	Procedure
C	1	0	

**29-30-01 Hydraulic Pump Lights****29-30-01-01 FAULT Lights**

Interval	Installed	Required	Procedure
C	8	5	

One in each system may be inoperative.

29-30-01 Hydraulic Pump Lights**29-30-01-02 ON Lights**

Interval	Installed	Required	Procedure
C	4	0	

**29-30-02 Hydraulic Synoptic Display**

Interval	Installed	Required	Procedure
C	1	0	

OPERATIONS NOTE

For missing data on the Synoptic Display, selecting an alternate location for the display (L INBD, LWR CTR, or R INBD) may restore the missing data. Synoptic Displays containing missing data may continue to be used to the extent remaining data is useful. Airplane system faults will be annunciated by alerting and status messages.

**29-31-01 Hydraulic System Pressure Transducers**

Interval	Installed	Required	Procedure
C	3	0	

May be inoperative provided all associated system pump pressure indications operate normally.

29-31-02 Pump Pressure Indication Systems
29-31-02-01 Left & Right Systems

Interval	Installed	Required	Procedure
C	4	2	(M)

One in each system may be inoperative provided:

- a. Associated system pressure transducer operates normally.
- b. Associated pump operation is verified normal before each departure.

NOTE: The pump transducer may fail such that the associated pump status message is displayed rather than the pump pressure indication status message.

MAINTENANCE (M)

Verify the associated pump is operating normally before each departure. For pump status message displayed, verify the associated pump pressure indication transducer is the malfunction and disconnect that transducer (AMM 29-00-00/901).

1. Verify the associated pump is operating normally before each departure.
 - A. Supply electrical power on the airplane (AMM 24-22-00/201).
 - B. Pressurize the associated hydraulic system with the associated pump.
 - 1) For an Alternating Current Motor Pump (ACMP), position the associated Hydraulic Panel (P5) DEMAND pump selector ON.
 - 2) For an Engine Driven Pump (EDP):
 - a. Dry motor the associated engine, or start the associated engine (AMM 71-00-00/201).
 - b. Position the associated Hydraulic Panel (P5) PRIMARY pump switch ON.
 - C. On Hydraulic Maintenance page, confirm the associated system pressure is normal after the pump has been operated for at least 60 seconds.
 - 1) For ACMP, 2700 to 3200 psi.
 - 2) For EDP, 2900 to 3200 psi.
 - D. Position the associated Hydraulic Panel (P5) pump selector or switch off.
 - E. If an engine was started, shutdown the engine (AMM 71-00-00/201).
2. For pump status message displayed instead of pump pressure indication status message, verify the associated pump pressure indication transducer is the malfunction and disconnect that transducer.
 - A. Supply electrical power on the airplane (AMM 24-22-00/201).

- B. Pressurize the associated hydraulic system with the associated pump.
- 1) For an Alternating Current Motor Pump (ACMP), position the associated Hydraulic Panel (P5) DEMAND pump selector ON.
 - 2) For an Engine Driven Pump (EDP):
 - a. Dry motor the associated engine, or start the associated engine (AMM 71-00-00/201).
 - b. Position the associated Hydraulic Panel (P5) PRIMARY pump switch ON.

- C. On Hydraulic Maintenance page, confirm the following pump pressure values are displayed:

NOTE: Intermittent readings or a blank pump pressure indication also indicate a malfunctioning transducer.

- 1) For ACMP, 0 to 2600 psi, or 3250 to 4000 psi.
 - 2) For EDP, 0 to 2800 psi, or 3250 to 4000 psi.
- D. Position the associated Hydraulic Panel (P5) pump selector or switch off.
- E. If an engine was started, shutdown the engine (AMM 71-00-00/201).
- F. If the pneumatic system was pressurized, depressurize the pneumatic system (AMM 36-00-00/201).
- G. Disconnect, cap and stow the applicable pump pressure transducer connector and associated wiring, and install a cap on the transducer electrical receptacle.

NOTE: The hydraulic pump status message will continue to be displayed, the pump pressure on the Hydraulic maintenance page will be blank and the pump pressure indication status message will be displayed. The hydraulic synoptic flow bars may not indicate correctly.

- 1) For an ACMP, refer to AMM 29-31-02/401 pressure transducer removal procedure.
- 2) For an EDP, refer to AMM 29-00-00/901 pressure transducer deactivation procedure.

OPERATIONS NOTE

For a pump pressure transducer disconnected, the associated pump pressure advisory message will be displayed and the hydraulic synoptic flow bars may not indicate correctly.

**29-31-02 Pump Pressure Indication Systems****29-31-02-02 Center System****29-31-02-02A Pump Operative**

Interval	Installed	Required	Procedure
C	4	3	(M)

One may be inoperative provided:

- Associated system pressure transducer operates normally.
- Associated pump operation is verified normal before each departure.

MAINTENANCE (M)

Verify the associated pump is operating normally before each departure (AMM 29-00-00/901).

- Supply electrical power on the airplane (AMM 24-22-00/201).
- Pressurize the associated hydraulic system with the associated pump.
 - For an Alternating Current Motor Pump (ACMP), position the associated Hydraulic Panel (P5) PRIMARY pump switch ON.
 - For an Air Driven Pump (ADP):
 - Pressurize the pneumatic system (AMM 36-00-00/201)
 - Position the associated Hydraulic Panel (P5) DEMAND pump selector ON.
- On Hydraulic Maintenance page, confirm the associated system pressure is normal after the pump has been operated for at least 60 seconds.
 - For ACMP, 2700 to 3200 psi.
 - For ADP, 2900 to 3200 psi.
- Position the associated Hydraulic Panel (P5) pump switch or selector off.
- If the pneumatic system was pressurized, depressurize the pneumatic system (AMM 36-00-00/201).

29-31-02 Pump Pressure Indication Systems**29-31-02-02 Center System****29-31-02-02B Pump Inoperative**

Interval	Installed	Required	Procedure
C	4	3	

One may be inoperative for an associated inoperative pump.

**29-32-01 Pump Temperature Indications****29-32-01-01 Left & Right Systems**

Interval	Installed	Required	Procedure
C	4	2	(M)

One in each system may be inoperative provided:

- Associated pump pressure indications operate normally.
- Associated pump operation is verified normal before each departure.

MAINTENANCE (M)

Verify the associated pump is operating normally by verifying the pump pressure is normal before each departure (AMM 29-00-00/901).

- Supply electrical power on the airplane (AMM 24-22-00/201).
- For an Alternating Current Motor Pump (ACMP) temperature indication inoperative, position the associated Hydraulic Panel (P5) DEMAND pump selector ON.
- For an Engine Driven Pump (EDP) temperature indication inoperative:
 - Dry motor the associated engine, or start the associated engine (AMM 71-00-00/201)
 - Position the associated Hydraulic Panel (P5) PRIMARY pump switch ON
- On Hydraulic Maintenance page, confirm the associated system pressure is normal after the pump has been operated for at least 60 seconds.
 - For ACMP: 2700 to 3200 psi.
 - For EDP: 2900 to 3200 psi.
- Position the associated pump Hydraulic Panel (P5) selector or switch off.
- If an engine was started, shutdown the engine (AMM 71-00-00/201).
- If the pneumatic system was pressurized, depressurize the pneumatic system (AMM 36-00-00/201).

29-32-01 Pump Temperature Indications**29-32-01-02 Center System****29-32-01-02A Pump Operative**

Interval	Installed	Required	Procedure
C	4	3	(M)

One may be inoperative provided:

- Associated pump pressure indications operate normally.



- b. Associated pump operation is verified normal before each departure.
-

MAINTENANCE (M)

Verify the associated pump is operating normally by verifying the pump pressure is normal before each departure (AMM 29-00-00/901).

1. Supply electrical power on the airplane (AMM 24-22-00/201).
2. For an Alternating Current Motor Pump (ACMP) temperature indication inoperative, position the associated Hydraulic Panel (P5) PRIMARY pump switch ON.
3. For an Air Driven Pump (ADP) temperature indication inoperative:
 - A. Pressurize the pneumatic system (AMM 36-00-00/201)
 - B. Position the associated Hydraulic Panel (P5) DEMAND pump selector ON
4. On Hydraulic Maintenance page, confirm the associated system pressure is normal after the pump has been operated for at least 60 seconds.
 - A. For ACMP: 2700 to 3200 psi.
 - B. For ADP: 2900 to 3200 psi.
5. Position the associated pump Hydraulic Panel (P5) switch or selector off.
6. If the pneumatic system was pressurized, depressurize the pneumatic system (AMM 36-00-00/201).

29-32-01 Pump Temperature Indications

29-32-01-02 Center System

29-32-01-02B Pump Inoperative

Interval	Installed	Required	Procedure
C	4	3	

One may be inoperative for an associated inoperative pump.

29-33-01 Hydraulic Fluid Quantity Indications (Flight Deck)

Interval	Installed	Required	Procedure
C	3	1	(M)

Two may be inoperative provided:

- a. Reservoir level is verified normal before each departure.
- b. Associated system pressure indication operates normally.
- c. For the center system indication inoperative, center hydraulic isolation system is considered inoperative.

NOTE: For the center system indication inoperative, the airplane must also be dispatched using MMEL item 29-11-06.

MAINTENANCE (M)

Verify the associated system reservoir quantity level is normal before each departure, and for center quantity indication inoperative, deactivate Center Hydraulic Isolation System (AMM 29-00-00/901).

1. Verify associated reservoir quantity level is normal using one of the following:
 - A. Reservoir quantity gage (if operating):
 - 1) Supply electrical power on the airplane (AMM 24-22-00/201).
 - 2) Open the Hydraulic Ground Service Bay Access Door (198DR).
 - 3) Position the handle of the reservoir fill selector valve to the associated system.
 - 4) Verify the reservoir quantity gage indicates in the green band.
 - 5) Close the Hydraulic Ground Service Bay Access Door.
 - B. Hydraulic Reservoir Sight Glass:
 - 1) For left or right hydraulic quantity indications inoperative, deactivate the associated engine thrust reverser for ground maintenance (AMM 78-31-00/201).
 - 2) For center hydraulic quantity indication inoperative, open the landing gear doors and install safety pins (AMM 32-00-15/201).
 - 3) Gain access to the associated reservoir:
 - a. For left system, open the aft strut Fairing Doors (434CL and 434CR).
 - b. For right system, open the aft strut Fairing Doors (444CL and 444CR).
 - c. For center system, right wheel well.
 - 4) Verify the lower sight glass (refill level) is dark.

NOTE: Red indicates fluid is below the sight glass level.



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- 5) For left system, close the aft strut Fairing Doors (434CL and 434CR).
 - 6) For right system, close the aft strut Fairing Doors (444CL and 444CR).
 - 7) For center hydraulic quantity indication inoperative, remove the landing gear door safety pins and close the landing gear doors (AMM 32-00-15/201).
 - 8) For left or right hydraulic quantity indications inoperative, activate the associated engine thrust reverser (AMM 78-31-00/201).



Intentionally
Blank

**30-11-01 Wing Anti-Ice Systems****30-11-02 Wing Anti-Ice Control (Card and Switch)****30-11-03 Wing Anti-Ice Indications (WAI)****30-21-01 Engine Anti-Ice Systems**

30-21-01-01 PW and RR

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30-21-01-02 GE

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Deactivation Kit Installed**30-21-02 Engine Anti-Ice Indications (EAI)****30-31-01 Pitot Probe Heater Systems**

30-31-01-01 Right Pitot Heater

30-31-01-02 Left Pitot Heater

30-31-01-03 Center Pitot Heater

30-33-01 Total Air Temperature (TAT) Probe Heater System

30-33-01-01 Single TAT System

30-33-01-02 Dual TAT System

30-33-01-02A One Inoperative

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**30-11-01 Wing Anti-Ice Systems**

Interval	Installed	Required	Procedure
C	2	0	(M)

May be inoperative provided:

- a. Associated valve is locked closed.
- b. Wing anti-ice control switch remains OFF.
- c. Airplane is not operated in known or forecast icing conditions.

MAINTENANCE (M)

Lock the associated Wing Anti-Ice (WAI) valve(s) in the closed position (AMM 30-00-00/901).

1. Open P210 panel WAI CTRL circuit breaker.
2. Deactivate the leading edge slats (AMM 27-81-00/201).
3. Gain access to the appropriate WAI valve:
 - A. For left WAI valve, open Access Panel 521AB.
 - B. For right WAI valve, open Access Panel 621AB.
4. Lock the WAI valve in the closed position:
 - A. Remove the locking screw from the normal position.
 - B. Set the crank to the locked (CL) position.
 - C. Install the locking screw through the crank into the locked position.

NOTE: The usual running torque for the locking screw is 10-15 pound-inches (1.1-1.7 newton-meters).

5. Close the opened Access Panel(s).
6. Activate the leading edge slats (AMM 27-81-00/201).
7. Position the ANTI-ICE Panel (P5) Wing Anti-Ice selector OFF.
8. Close P210 panel WAI CTRL circuit breaker.

**30-11-02 Wing Anti-Ice Control (Card and Switch)**

Interval	Installed	Required	Procedure
C	1	0	(M)

May be inoperative provided:

- a. Both wing anti-ice valves are locked closed.
 - b. Wing anti-ice control switch remains OFF.
 - c. Airplane is not operated in known or forecast icing conditions.
-

MAINTENANCE (M)

Lock both Wing Anti-Ice (WAI) valves in the closed position (AMM 30-00-00/901).

1. Open P210 panel WAI CTRL circuit breaker.
2. Deactivate the leading edge slats (AMM 27-81-00/201).
3. Gain access to the both WAI valves:
 - A. For left WAI valve, open Access Panel 521AB.
 - B. For right WAI valve, open Access Panel 621AB.
4. Lock the WAI valves in the closed position:
 - A. Remove the locking screw from the normal position.
 - B. Set the crank to the locked (CL) position.
 - C. Install the locking screw through the crank into the locked position.

NOTE: The usual running torque for the locking screw is 10-15 pound-inches (1.1-1.7 newton-meters).

5. Close Access Panels 521AB and 621AB.
6. Activate the leading edge slats (AMM 27-81-00/201).
7. Position the ANTI-ICE Panel (P5) Wing Anti-Ice selector OFF.
8. Close P210 panel WAI CTRL circuit breaker.



Interval	Installed	Required	Procedure
C	2	0	

**30-21-01 Engine Anti-Ice Systems**

Interval	Installed	Required	Procedure
C	2	1	(M)

One may be inoperative provided:

- Valve is locked closed.
 - Associated engine anti-ice control switch remains OFF.
 - Airplane is not operated in known or forecast icing conditions.
 - Flight remains within 120 minutes of landing at a suitable airport.
-

MAINTENANCE (M)

Lock associated EAI valve in the closed position (AMM 30-00-00/901).

- Deactivate the associated EAI valve controller:
 - For the right EAI valve, open P210 panel EAI CTRL R circuit breaker.
 - For the left EAI valve, open P310 panel EAI CTRL L circuit breaker.

For GE:

- Deactivate the leading edge slats in the retracted position (AMM 27-81-00/201).
- Deactivate the associated thrust reverser for ground maintenance (AMM 78-31-00/201).
- Lock the EAI valve in the closed position.
 - Remove the locking screw from the normal position.
 - Set the crank to the locked (closed) position.
 - Install the locking screw through the crank into the locked position.
 - Position associated ANTI-ICE Panel (P5) Engine Anti-Ice selector OFF.
- Close the associated circuit breaker(s).
- Activate the thrust reverser (AMM 78-31-00/201).

For GE:

- Activate the leading edge slats (AMM 27-81-00/201).
- Activate the associated EAI valve controller by closing the associated circuit breaker.
- Position the associated ANTI-ICE Panel (P5) Engine Anti-Ice selector OFF.

30-21-01 Engine Anti-Ice Systems
30-21-01-01 PW and RR
30-21-01-01A OAT 38 Degrees C or Below, With Deactivation Kit Installed

Interval	Installed	Required	Procedure
C	2	1	(M) (O)

For operations at ambient temperature 38 degrees C or below, one may be inoperative provided:

- Valve is deactivated open using deactivation kit.
- Engine fan case overheat detection system operates normally.
- Appropriate performance adjustments are applied.

MAINTENANCE (M)

Deactivate the associated EAI valve by installing the EAI valve lock-open kit, P/N K28703 (AMM 30-00-00/901).

- Remove inoperative EAI valve (AMM 30-21-02/401).
- Modify valve with lock-open kit (AMM 30-00-00/901).
- Re-install modified valve (AMM 30-21-02/401).

For PW:

- Visually inspect the area near the Cowl Thermal Anti-Ice duct for signs of leaks and the fan cowl in general for flammable fluid leaks. Dispatch is not allowed if any of the duct clamps or supports are not secure or any air leaks from the EAI duct are present, or if significant flammable fluid leakage is present.

OPERATIONS (O)

- Reduce performance limited weights by the appropriate adjustments.

Performance Limited Weights	Takeoff & Landing	Enroute Climb
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For -200 or -200ER/PW4074 thru PW4084:

All	19,600 lb (8,891 kg)	19,400 lb (8,800 kg)
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For -200 or -200ER/PW4090:

thru 632,500 lb (286,902 kg)	12,200 lb (5,534 kg)	13,800 lb (6,260 kg)
over 632,500 lb (286,902 kg)	17,300 lb (7,847 kg)	13,800 lb (6,260 kg)

**For -200 or -200ER/RR875 thru RR892:**

thru 632,500 lb (286,902 kg)	20,100 lb (9,117 kg)	21,600 lb (9,798 kg)
over 632,500 lb (286,902 kg)	26,500 lb (12,020 kg)	21,600 lb (9,798 kg)

For -200 or -200ER/RR895:

All	21,700 lb (9,843 kg)	23,000 lb (10,433 kg)
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For -300/PW4090:

All	12,400 lb (5,625 kg)	13,000 lb (5,897 kg)
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For -300/RR892:

All	22,000 lb (9,979 kg)	23,500 lb (10,659 kg)
-----	----------------------	-----------------------

NOTE: The enroute climb weight penalties listed above are based on single engine operating speeds that approximate maximum lift-to-drag ratio speed. To account for the difference in level off altitude when operating at other speeds, multiply the enroute climb weight penalty listed above by the appropriate factor listed on page 3.00-01-00.2 in the ENROUTE DIVERSION SPEED EFFECTS paragraph.

For PW:

2. Increase flight planning fuel by 1.0% for an inoperative EAI Valve or EAI Pressure Sensor, or by 1.3% for an inoperative EAI Card.

For RR:

3. Increase flight planning fuel by 0.5% for an inoperative EAI Valve or EAI Pressure Sensor, or by 0.8% for an inoperative EAI Card.
4. For ETOPS flight planning:

For PW:

- A. Increase the ETOPS engine inoperative Critical Fuel Reserves by 1.6%.

For RR:

- B. Increase the ETOPS engine inoperative Critical Fuel Reserves by 0.9%.
5. Position ENGINE ANTI-ICE selector(s) as follows:
 - A. For EAI Valve or EAI Pressure Sensor inoperative, position associated EAI selector ON.

NOTE 1: When an EAI selector is positioned to ON and the temperature is greater than 10 degrees C, the ANTI-ICE ON advisory message will be displayed. Do not accomplish the ANTI-ICE ON Non-Normal Checklist.

NOTE 2: Inflight, the ENG IDLE DISAGREE advisory message may be displayed when engine thrust is reduced to idle.

- B. For EAI Card inoperative, position both EAI selectors ON.

NOTE 1: When an EAI selector is positioned to ON and the temperature is greater than 10 degrees C, the ANTI-ICE ON advisory message will be displayed. Do not accomplish the ANTI-ICE ON Non-Normal Checklist.

NOTE 2: The associated ANTI-ICE ENG L or R advisory message will be displayed, the EAI indication will not be displayed on EICAS, and the associated EAI valve will be displayed as invalid (low intensity white) and the green flow bar will not be displayed on the Air synoptic.

6. For the inoperative EAI valve, if the ANTI-ICE LEAK ENG L or R caution message is subsequently displayed:

For PW:

- A. Do not accomplish the two minute delay associated with the ANTI-ICE LEAK ENG L or R Non-Normal Checklist. Complete the remainder of the checklist as soon as practical.

For RR:

- B. Do not accomplish the initial one minute delay associated with the ANTI-ICE LEAK ENG L or R Non-Normal Checklist. Complete the remainder of the checklist as soon as practical.

30-21-01 Engine Anti-Ice Systems

30-21-01-01 PW and RR

30-21-01-01B OAT 38 Degrees C or Below, Without Deactivation Kit Installed

Interval	Installed	Required	Procedure
C	2	1	(M) (O)

For operations at ambient temperature 38 degrees C or below, one may be inoperative provided:

- Valve is electrically disconnected and verified open.
- Engine fan case overheat detection system operates normally.
- Associated EAI pressure sensor system operates normally.
- Associated EAI card operates normally.

- e. Appropriate performance adjustments are applied.

MAINTENANCE (M)

For EAI VALVE L or R status message displayed, deactivate the EAI valve by disconnecting the electrical connector and verify EAI valve is open when the associated engine is running (AMM 30-00-00/901).

1. Deactivate the associated EAI valve controller.
 - A. For the right EAI valve, open P210 panel EAI CTRL R circuit breaker.
 - B. For the left EAI valve, open P310 panel EAI CTRL L circuit breaker.
2. Deactivate the associated thrust reverser for ground maintenance (AMM 78-31-00/201).
3. Deactivate the EAI valve in the open position.
 - A. On the associated engine, disconnect, cap and stow the torque motor electrical connector DM30211 or DM30211A from the EAI valve controller, M30211.
 - B. Install a protective cover over the electrical connector on the valve controller.

For PW:

- C. Visually inspect the area near the Cowl Thermal Anti-Ice duct for signs of leaks and the fan cowl in general for flammable fluid leaks. Dispatch is not allowed if any of the duct clamps or supports are not secure or any air leaks from the EAI duct are present, or if significant flammable fluid leakage is present.
4. Activate the thrust reverser (AMM 78-31-00/201).
5. Activate the associated EAI valve controller by closing the associated circuit breaker.
6. Verify the EAI valve is open when the associated engine is running.
 - A. Position associated ANTI-ICE Panel (P5) Engine Anti-Ice selector OFF.
 - B. Start the associated engine (71-00-00/201).
 - C. Confirm the EAI indication (green EAI) is displayed above the N1 indication on the EICAS display for the associated engine (indicates the EAI valve is open).

NOTE: If the EAI indication is not displayed, the valve may be stuck closed and a thrust setting slightly above minimum idle may open the valve.

- D. Shutdown the associated engine (71-00-00/201).

OPERATIONS (O)

1. Reduce performance limited weights by the appropriate adjustments.

Performance Limited Weights	Takeoff & Landing	Enroute Climb
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**For -200 or -200ER/PW4074 thru PW4084:**

All	19,600 lb (8,891 kg)	19,400 lb (8,800 kg)
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For -200 or -200ER/PW4090:

thru 632,500 lb (286,902 kg)	12,200 lb (5,534 kg)	13,800 lb (6,260 kg)
over 632,500 lb (286,902 kg)	17,300 lb (7,847 kg)	13,800 lb (6,260 kg)

For -200 or -200ER/RR875 thru RR892:

thru 632,500 lb (286,902 kg)	20,100 lb (9,117 kg)	21,600 lb (9,798 kg)
over 632,500 lb (286,902 kg)	26,500 lb (12,020 kg)	21,600 lb (9,798 kg)

For -200 or -200ER/RR895:

All	21,700 lb (9,843 kg)	23,000 lb (10,433 kg)
-----	----------------------	-----------------------

For -300/PW4090:

All	12,400 lb (5,625 kg)	13,000 lb (5,897 kg)
-----	----------------------	----------------------

For -300/RR892:

All	22,000 lb (9,979 kg)	23,500 lb (10,659 kg)
-----	----------------------	-----------------------

NOTE: The enroute climb weight penalties listed above are based on single engine operating speeds that approximate maximum lift-to-drag ratio speed. To account for the difference in level off altitude when operating at other speeds, multiply the enroute climb weight penalty listed above by the appropriate factor listed on page 3.00-01-00.2 in the ENROUTE DIVERSION SPEED EFFECTS paragraph.

For PW:

2. Increase flight planning fuel by 1.0%.

For RR:

3. Increase flight planning fuel by 0.5%.
4. For ETOPS flight planning:

For PW:

- A. Increase the ETOPS engine inoperative Critical Fuel Reserves by 1.6%.

For RR:

- B. Increase the ETOPS engine inoperative Critical Fuel Reserves by 0.9%.

- 5. Verify the EAI valve is open when the associated engine is running before each departure.

- A. Position associated ANTI-ICE Panel (P5) Engine Anti-Ice selector OFF.
- B. After the associated engine has been started, confirm the associated EAI indication (green EAI) is displayed above the N1 indication.

NOTE 1: If the EAI indication is not displayed, the valve may be stuck closed and a thrust setting slightly above minimum idle may open the valve.

NOTE 2: On the Air synoptic, the EAI valve will indicate failed open because the EAI selector is OFF and the green flow bar will be displayed.

- 6. Position associated ENGINE ANTI-ICE selector ON.

NOTE 1: When an EAI selector is positioned to ON and the temperature is greater than 10 degrees C, the ANTI-ICE ON advisory message will be displayed. Do not accomplish the ANTI-ICE ON Non-Normal Checklist.

NOTE 2: Inflight, the ENG IDLE DISAGREE advisory message may be displayed when engine thrust is reduced to idle.

NOTE 3: The ANTI-ICE ENG advisory message may be displayed during ground operations prior to takeoff. Flight crew should accomplish the associated non-normal checklist. If ANTI-ICE ENG clears, takeoff is allowed.

- 7. For the inoperative EAI valve, if the ANTI-ICE LEAK ENG L or R caution message is subsequently displayed:

For PW:

- A. Do not accomplish the two minute delay associated with the ANTI-ICE LEAK ENG L or R Non-Normal Checklist. Complete the remainder of the checklist as soon as practical.

For RR:

- B. Do not accomplish the initial one minute delay associated with the ANTI-ICE LEAK ENG L or R Non-Normal Checklist. Complete the remainder of the checklist as soon as practical.

30-21-01 Engine Anti-Ice Systems
30-21-01-01 PW and RR
30-21-01-01C OAT Above 38 Degrees C, With Deactivation Kit Installed

Interval	Installed	Required	Procedure
A	2	1	(M) (O)

For operations at ambient temperature above 38 degrees C, one may be inoperative provided:

- Valve is deactivated open using deactivation kit.
- Engine fan case overheat detection system operates normally.
- Appropriate performance adjustments are applied.
- Repairs are made within 3 flight days.

MAINTENANCE (M)

Deactivate the associated EAI valve by installing the EAI valve lock-open kit, P/N K28703 (AMM 30-00-00/901).

- Remove inoperative EAI valve (AMM 30-21-02/401).
- Modify valve with lock-open kit (AMM 30-00-00/901).
- Re-install modified valve (AMM 30-21-02/401).

For PW:

- Visually inspect the area near the Cowl Thermal Anti-Ice duct for signs of leaks and the fan cowl in general for flammable fluid leaks. Dispatch is not allowed if any of the duct clamps or supports are not secure or any air leaks from the EAI duct are present, or if significant flammable fluid leakage is present.

OPERATIONS (O)

- Reduce performance limited weights by the appropriate adjustments.

Performance Limited Weights	Takeoff & Landing	Enroute Climb
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For -200 or -200ER/PW4074 thru PW4084:

All	19,600 lb (8,891 kg)	19,400 lb (8,800 kg)
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**For -200 or -200ER/PW4090:**

thru 632,500 lb (286,902 kg)	12,200 lb (5,534 kg)	13,800 lb (6,260 kg)
over 632,500 lb (286,902 kg)	17,300 lb (7,847 kg)	13,800 lb (6,260 kg)

For -200 or -200ER/RR875 thru RR892:

thru 632,500 lb (286,902 kg)	20,100 lb (9,117 kg)	21,600 lb (9,798 kg)
over 632,500 lb (286,902 kg)	26,500 lb (12,020 kg)	21,600 lb (9,798 kg)

For -200 or -200ER/RR895:

All	21,700 lb (9,843 kg)	23,000 lb (10,433 kg)
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For -300/PW4090:

All	12,400 lb (5,625 kg)	13,000 lb (5,897 kg)
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For -300/RR892:

All	22,000 lb (9,979 kg)	23,500 lb (10,659 kg)
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NOTE: The enroute climb weight penalties listed above are based on single engine operating speeds that approximate maximum lift-to-drag ratio speed. To account for the difference in level off altitude when operating at other speeds, multiply the enroute climb weight penalty listed above by the appropriate factor listed on page 3.00-01-00.2 in the ENROUTE DIVERSION SPEED EFFECTS paragraph.

For PW:

2. Increase flight planning fuel by 1.0% for an inoperative EAI Valve or EAI Pressure Sensor, or by 1.3% for an inoperative EAI Card.

For RR:

3. Increase flight planning fuel by 0.5% for an inoperative EAI Valve or EAI Pressure Sensor, or by 0.8% for an inoperative EAI Card.
4. For ETOPS flight planning:

For PW:

- A. Increase the ETOPS engine inoperative Critical Fuel Reserves by 1.6%.

For RR:

- B. Increase the ETOPS engine inoperative Critical Fuel Reserves by 0.9%.
5. Position ENGINE ANTI-ICE selector(s) as follows:
- A. For EAI Valve or EAI Pressure Sensor inoperative, position associated EAI selector ON.
- NOTE 1: When an EAI selector is positioned to ON and the temperature is greater than 10 degrees C, the ANTI-ICE ON advisory message will be displayed. Do not accomplish the ANTI-ICE ON Non-Normal Checklist.
- NOTE 2: Inflight, the ENG IDLE DISAGREE advisory message may be displayed when engine thrust is reduced to idle.
- B. For EAI Card inoperative, position both EAI selectors ON.
- NOTE 1: When an EAI selector is positioned to ON and the temperature is greater than 10 degrees C, the ANTI-ICE ON advisory message will be displayed. Do not accomplish the ANTI-ICE ON Non-Normal Checklist.
- NOTE 2: The associated ANTI-ICE ENG L or R advisory message will be displayed, the EAI indication will not be displayed on EICAS, and the associated EAI valve will be displayed as invalid (low intensity white) and the green flow bar will not be displayed on the Air synoptic.
6. For the inoperative EAI valve, if the ANTI-ICE LEAK ENG L or R caution message is subsequently displayed:

For PW:

- A. Do not accomplish the two minute delay associated with the ANTI-ICE LEAK ENG L or R Non-Normal Checklist. Complete the remainder of the checklist as soon as practical.

For RR:

- B. Do not accomplish the initial one minute delay associated with the ANTI-ICE LEAK ENG L or R Non-Normal Checklist. Complete the remainder of the checklist as soon as practical.

30-21-01 Engine Anti-Ice Systems
30-21-01-01 PW and RR
30-21-01-01D OAT Above 38 Degrees C, Without Deactivation Kit Installed

Interval	Installed	Required	Procedure
A	2	1	(M) (O)

For operations at ambient temperature above 38 degrees C, one may be inoperative provided:

- a. Valve is electrically disconnected and verified open.
 - b. Engine fan case overheat detection system operates normally.
 - c. Associated EAI pressure sensor system operates normally.
 - d. Associated EAI card operates normally.
 - e. Appropriate performance adjustments are applied.
 - f. Repairs are made within 3 flight days.
-

MAINTENANCE (M)

For EAI VALVE L or R status message displayed, deactivate the EAI valve by disconnecting the electrical connector and verify EAI valve is open when the associated engine is running (AMM 30-00-00/901).

1. Deactivate the associated EAI valve controller.
 - A. For the right EAI valve, open P210 panel EAI CTRL R circuit breaker.
 - B. For the left EAI valve, open P310 panel EAI CTRL L circuit breaker.
2. Deactivate the associated thrust reverser for ground maintenance (AMM 78-31-00/201).
3. Deactivate the EAI valve in the open position.
 - A. On the associated engine, disconnect, cap and stow the torque motor electrical connector DM30211 or DM30211A from the EAI valve controller, M30211.
 - B. Install a protective cover over the electrical connector on the valve controller.

For PW:

- C. Visually inspect the area near the Cowl Thermal Anti-Ice duct for signs of leaks and the fan cowl in general for flammable fluid leaks. Dispatch is not allowed if any of the duct clamps or supports are not secure or any air leaks from the EAI duct are present, or if significant flammable fluid leakage is present.
4. Activate the thrust reverser (AMM 78-31-00/201).
5. Activate the associated EAI valve controller by closing the associated circuit breaker.
6. Verify the EAI valve is open when the associated engine is running.
 - A. Position associated ANTI-ICE Panel (P5) Engine Anti-Ice selector OFF.
 - B. Start the associated engine (71-00-00/201).
 - C. Confirm the EAI indication (green EAI) is displayed above the N1 indication on the EICAS display for the associated engine (indicates the EAI valve is open).



NOTE: If the EAI indication is not displayed, the valve may be stuck closed and a thrust setting slightly above minimum idle may open the valve.

D. Shutdown the associated engine (71-00-00/201).

OPERATIONS (O)

1. Reduce performance limited weights by the appropriate adjustments.

Performance Limited Weights	Takeoff & Landing	Enroute Climb
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For -200 or -200ER/PW4074 thru PW4084:

All	19,600 lb (8,891 kg)	19,400 lb (8,800 kg)
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For -200 or -200ER/PW4090:

thru 632,500 lb (286,902 kg)	12,200 lb (5,534 kg)	13,800 lb (6,260 kg)
over 632,500 lb (286,902 kg)	17,300 lb (7,847 kg)	13,800 lb (6,260 kg)

For -200 or -200ER/RR875 thru RR892:

thru 632,500 lb (286,902 kg)	20,100 lb (9,117 kg)	21,600 lb (9,798 kg)
over 632,500 lb (286,902 kg)	26,500 lb (12,020 kg)	21,600 lb (9,798 kg)

For -200 or -200ER/RR895:

All	21,700 lb (9,843 kg)	23,000 lb (10,433 kg)
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For -300/PW4090:

All	12,400 lb (5,625 kg)	13,000 lb (5,897 kg)
-----	----------------------	----------------------

For -300/RR892:

All	22,000 lb (9,979 kg)	23,500 lb (10,659 kg)
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NOTE: The enroute climb weight penalties listed above are based on single engine operating speeds that approximate maximum lift-to-drag ratio speed. To account for the difference in level off altitude when operating at other speeds, multiply the enroute climb weight penalty listed above by the appropriate factor listed on page 3.00-01-00.2 in the ENROUTE DIVERSION SPEED EFFECTS paragraph.

For PW:

2. Increase flight planning fuel by 1.0%.

For RR:

3. Increase flight planning fuel by 0.5%.
4. For ETOPS flight planning:

For PW:

- A. Increase the ETOPS engine inoperative Critical Fuel Reserves by 1.6%.

For RR:

- B. Increase the ETOPS engine inoperative Critical Fuel Reserves by 0.9%.
5. Verify the EAI valve is open when the associated engine is running before each departure.
 - A. Position associated ANTI-ICE Panel (P5) Engine Anti-Ice selector OFF.
 - B. After the associated engine has been started, confirm the associated EAI indication (green EAI) is displayed above the N1 indication.

NOTE 1: If the EAI indication is not displayed, the valve may be stuck closed and a thrust setting slightly above minimum idle may open the valve.

NOTE 2: On the Air synoptic, the EAI valve will indicate failed open because the EAI selector is OFF and the green flow bar will be displayed.

6. Position associated ENGINE ANTI-ICE selector ON.

NOTE 1: When an EAI selector is positioned to ON and the temperature is greater than 10 degrees C, the ANTI-ICE ON advisory message will be displayed. Do not accomplish the ANTI-ICE ON Non-Normal Checklist.

NOTE 2: Inflight, the ENG IDLE DISAGREE advisory message may be displayed when engine thrust is reduced to idle.

NOTE 3: The ANTI-ICE ENG advisory message may be displayed during ground operations prior to takeoff. Flight crew should accomplish the associated non-normal checklist. If ANTI-ICE ENG clears, takeoff is allowed.

7. For the inoperative EAI valve, if the ANTI-ICE LEAK ENG L or R caution message is subsequently displayed:

For PW:

- A. Do not accomplish the two minute delay associated with the ANTI-ICE LEAK ENG L or R Non-Normal Checklist. Complete the remainder of the checklist as soon as practical.

For RR:

- B. Do not accomplish the initial one minute delay associated with the ANTI-ICE LEAK ENG L or R Non-Normal Checklist. Complete the remainder of the checklist as soon as practical.

30-21-01 Engine Anti-Ice Systems
30-21-01-02 GE
30-21-01-02A OAT 38 Degrees C or Below, With Deactivation Kit Installed

Interval	Installed	Required	Procedure
C	2	1	(M) (O)

For operations at ambient temperature 38 degrees C or below, one may be inoperative provided:

- a. Valve is deactivated open using deactivation kit.
- b. Engine fan case overheat detection system operates normally.
- c. Turbine case cooling air flow systems operate normally.
- d. Appropriate performance adjustments are applied.

Except for GE90-100 Series:

- e. EECs must be operated in the normal mode.

MAINTENANCE (M)

Deactivate the associated EAI valve by installing the EAI valve lock-open kit, P/N K28703 (AMM 30-00-00/901).

1. Remove inoperative EAI valve (AMM 30-21-02/401).
2. Modify valve with lock-open kit (AMM 30-00-00/901).
3. Re-install modified valve (AMM 30-21-02/401).
4. Visually inspect the area near the Cowl Thermal Anti-Ice duct for signs of leaks and the fan cowl in general for flammable fluid leaks. Dispatch is not allowed if any of the duct clamps or supports are not secure or any air leaks from the EAI duct are present, or if significant flammable fluid leakage is present.

OPERATIONS (O)

1. Reduce performance limited weights by the appropriate adjustments.



Performance Limited Weights	Takeoff & Landing	Enroute Climb
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For -200 or -200ER/GE90-76B thru -90B:

thru 632,500 lb (286,902 kg)	9,100 lb (4,128 kg)	11,800 lb (5,352 kg)
over 632,500 lb (286,902 kg)	13,200 lb (5,987 kg)	11,800 lb (5,352 kg)

For -200 or -200ER/GE90-94B:

thru 632,500 lb (286,902 kg)	9,100 lb (4,128 kg)	13,900 lb (6,305 kg)
over 632,500 lb (286,902 kg)	14,300 lb (6,486 kg)	13,900 lb (6,305 kg)

For -200LR or 777F/GE90-100 Series:

All	13,300 lb (6,033 kg)	12,100 lb (5,489 kg)
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For -300ER/GE90-115B:

All	10,400 lb (4,717 kg)	11,800 lb (5,353 kg)
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NOTE: The enroute climb weight penalties listed above are based on single engine operating speeds that approximate maximum lift-to-drag ratio speed. To account for the difference in level off altitude when operating at other speeds, multiply the enroute climb weight penalty listed above by the appropriate factor listed on page 3.00-01-00.2 in the ENROUTE DIVERSION SPEED EFFECTS paragraph.

2. Increase flight planning fuel by 1.0% for an inoperative EAI Valve or EAI Pressure Sensor, or by 1.3% for an inoperative EAI Card.
3. For ETOPS flight planning, increase engine inoperative Critical Fuel Reserves by 1.8%.
4. Position ENGINE ANTI-ICE selector(s) as follows:
 - A. For EAI Valve or EAI Pressure Sensor inoperative, position associated EAI selector ON.

NOTE 1: When an EAI selector is positioned to ON and the temperature is greater than 10 degrees C, the ANTI-ICE ON advisory message will be displayed. Do not accomplish the ANTI-ICE ON Non-Normal Checklist.

NOTE 2: Inflight, the ENG IDLE DISAGREE advisory message may be displayed when engine thrust is reduced to idle.

- B. For EAI Card inoperative, position both EAI selectors ON.

NOTE 1: When an EAI selector is positioned to ON and the temperature is greater than 10 degrees C, the ANTI-ICE ON advisory message will be displayed. Do not accomplish the ANTI-ICE ON Non-Normal Checklist.

NOTE 2: The associated ANTI-ICE ENG L or R advisory message will be displayed, the EAI indication will not be displayed on EICAS, and the associated EAI valve will be displayed as invalid (low intensity white) and the green flow bar will not be displayed on the Air synoptic.

5. For the inoperative EAI valve, if the ANTI-ICE LEAK ENG L or R caution message is subsequently displayed, do not accomplish the two minute delay associated with the ANTI-ICE LEAK ENG L or R Non-Normal Checklist. Complete the remainder of the checklist as soon as practical.

30-21-01 Engine Anti-Ice Systems

30-21-01-02 GE

30-21-01-02B OAT 38 Degrees C or Below, Without Deactivation Kit Installed

Interval	Installed	Required	Procedure
C	2	1	(M) (O)

For operations at ambient temperature 38 degrees C or below, one may be inoperative provided:

- a. Valve is electrically disconnected and verified open.
- b. Engine fan case overheat detection system operates normally.
- c. Associated EAI pressure sensor system operates normally.
- d. Associated EAI card operates normally.
- e. Turbine case cooling air flow systems operate normally.
- f. Appropriate performance adjustments are applied.

[Except for GE90-100 Series:](#)

- g. EECs must be operated in the normal mode.

MAINTENANCE (M)

For EAI VALVE L or R status message displayed, deactivate the EAI valve by disconnecting the electrical connector and verify EAI valve is open when the associated engine is running (AMM 30-00-00/901).

1. Deactivate the associated EAI valve controller.
 - A. For the right EAI valve, open P210 panel EAI CTRL R circuit breaker.



- B. For the left EAI valve, open P310 panel EAI CTRL L circuit breaker.
2. Deactivate the leading edge slats in the retracted position (AMM 27-81-00/201).
 3. Deactivate the associated thrust reverser for ground maintenance (AMM 78-31-00/201).
 4. Deactivate the EAI valve in the open position.
 - A. On the associated engine, disconnect, cap and stow the torque motor electrical connector DM30211 or DM30211A from the EAI valve controller, M30211.
 - B. Install a protective cover over the electrical connector on the valve controller.
 - C. Visually inspect the area near the Cowl Thermal Anti-Ice duct for signs of leaks and the fan cowl in general for flammable fluid leaks. Dispatch is not allowed if any of the duct clamps or supports are not secure or any air leaks from the EAI duct are present, or if significant flammable fluid leakage is present.
 5. Activate the thrust reverser (AMM 78-31-00/201).
 6. Activate the leading edge slats (AMM 27-81-00/201).
 7. Activate the associated EAI valve controller by closing the associated circuit breaker.
 8. Verify the EAI valve is open when the associated engine is running.
 - A. Position associated ANTI-ICE Panel (P5) Engine Anti-Ice selector OFF.
 - B. Start the associated engine (71-00-00/201).
 - C. Confirm the EAI indication (green EAI) is displayed above the N1 indication on the EICAS display for the associated engine (indicates the EAI valve is open).
- NOTE: If the EAI indication is not displayed, the valve may be stuck closed and a thrust setting slightly above minimum idle may open the valve.
- D. Shutdown the associated engine (71-00-00/201).

OPERATIONS (O)

1. Reduce performance limited weights by the appropriate adjustments.

Performance Limited Weights	Takeoff & Landing	Enroute Climb
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For -200 or -200ER/GE90-76B thru -90B:

thru 632,500 lb (286,902 kg)	9,100 lb (4,128 kg)	11,800 lb (5,352 kg)
over 632,500 lb (286,902 kg)	13,200 lb (5,987 kg)	11,800 lb (5,352 kg)

For -200 or -200ER/GE90-94B:

thru 632,500 lb (286,902 kg)	9,100 lb (4,128 kg)	13,900 lb (6,305 kg)
over 632,500 lb (286,902 kg)	14,300 lb (6,486 kg)	13,900 lb (6,305 kg)

For -200LR or 777F/GE90-100 Series:

All	13,300 lb (6,033 kg)	12,100 lb (5,489 kg)
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For -300ER/GE90-115B:

All	10,400 lb (4,717 kg)	11,800 lb (5,353 kg)
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NOTE: The enroute climb weight penalties listed above are based on single engine operating speeds that approximate maximum lift-to-drag ratio speed. To account for the difference in level off altitude when operating at other speeds, multiply the enroute climb weight penalty listed above by the appropriate factor listed on page 3.00-01-00.2 in the ENROUTE DIVERSION SPEED EFFECTS paragraph.

2. Increase flight planning fuel by 1.0%.
3. For ETOPS flight planning, increase engine inoperative Critical Fuel Reserves by 1.8%.
4. Verify the EAI valve is open when the associated engine is running before each departure.
 - A. Position associated ANTI-ICE Panel (P5) Engine Anti-Ice selector OFF.
 - B. After the associated engine has been started, confirm the associated EAI indication (green EAI) is displayed above the N1 indication.

NOTE 1: If the EAI indication is not displayed, the valve may be stuck closed and a thrust setting slightly above minimum idle may open the valve.

NOTE 2: On the Air synoptic, the EAI valve will indicate failed open because the EAI selector is OFF and the green flow bar will be displayed.

5. Position associated ENGINE ANTI-ICE selector ON.

NOTE 1: When an EAI selector is positioned to ON and the temperature is greater than 10 degrees C, the ANTI-ICE ON advisory message will be displayed. Do not accomplish the ANTI-ICE ON Non-Normal Checklist.

NOTE 2: Inflight, the ENG IDLE DISAGREE advisory message may be displayed when engine thrust is reduced to idle.

NOTE 3: The ANTI-ICE ENG advisory message may be displayed during ground operations prior to takeoff. Flight crew should accomplish the associated non-normal checklist. If ANTI-ICE ENG clears, takeoff is allowed.

6. For the inoperative EAI valve, if the ANTI-ICE LEAK ENG L or R caution message is subsequently displayed, do not accomplish the two minute delay associated with the ANTI-ICE LEAK ENG L or R Non-Normal Checklist. Complete the remainder of the checklist as soon as practical.

30-21-01 Engine Anti-Ice Systems

30-21-01-02 GE

30-21-01-02C OAT Above 38 Degrees C, With Deactivation Kit Installed

Interval	Installed	Required	Procedure
A	2	1	(M) (O)

For operations at ambient temperature above 38 degrees C, one may be inoperative provided:

- a. Valve is deactivated open using deactivation kit.
- b. Engine fan case overheat detection system operates normally.
- c. Turbine case cooling air flow systems operate normally.
- d. Appropriate performance adjustments are applied.
- e. Repairs are made within 3 flight days.

Except for GE90-100 Series:

- f. EECs must be operated in the normal mode.

MAINTENANCE (M)

Deactivate the associated EAI valve by installing the EAI valve lock-open kit, P/N K28703 (AMM 30-00-00/901).

1. Remove inoperative EAI valve (AMM 30-21-02/401).
2. Modify valve with lock-open kit (AMM 30-00-00/901).
3. Re-install modified valve (AMM 30-21-02/401).
4. Visually inspect the area near the Cowl Thermal Anti-Ice duct for signs of leaks and the fan cowl in general for flammable fluid leaks. Dispatch is not allowed if any of the duct clamps or supports are not secure or any air leaks from the EAI duct are present, or if significant flammable fluid leakage is present.

OPERATIONS (O)

1. Reduce performance limited weights by the appropriate adjustments.



Performance Limited Weights	Takeoff & Landing	Enroute Climb
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For -200 or -200ER/GE90-76B thru -90B:

thru 632,500 lb (286,902 kg)	9,100 lb (4,128 kg)	11,800 lb (5,352 kg)
over 632,500 lb (286,902 kg)	13,200 lb (5,987 kg)	11,800 lb (5,352 kg)

For -200 or -200ER/GE90-94B:

thru 632,500 lb (286,902 kg)	9,100 lb (4,128 kg)	13,900 lb (6,305 kg)
over 632,500 lb (286,902 kg)	14,300 lb (6,486 kg)	13,900 lb (6,305 kg)

For -200LR or 777F/GE90-100 Series:

All	13,300 lb (6,033 kg)	12,100 lb (5,489 kg)
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For -300ER/GE90-115B:

All	10,400 lb (4,717 kg)	11,800 lb (5,353 kg)
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NOTE: The enroute climb weight penalties listed above are based on single engine operating speeds that approximate maximum lift-to-drag ratio speed. To account for the difference in level off altitude when operating at other speeds, multiply the enroute climb weight penalty listed above by the appropriate factor listed on page 3.00-01-00.2 in the ENROUTE DIVERSION SPEED EFFECTS paragraph.

2. Increase flight planning fuel by 1.0% for an inoperative EAI Valve or EAI Pressure Sensor, or by 1.3% for an inoperative EAI Card.
3. For ETOPS flight planning, increase engine inoperative Critical Fuel Reserves by 1.8%.
4. Position ENGINE ANTI-ICE selector(s) as follows:
 - A. For EAI Valve or EAI Pressure Sensor inoperative, position associated EAI selector ON.

NOTE 1: When an EAI selector is positioned to ON and the temperature is greater than 10 degrees C, the ANTI-ICE ON advisory message will be displayed. Do not accomplish the ANTI-ICE ON Non-Normal Checklist.

NOTE 2: Inflight, the ENG IDLE DISAGREE advisory message may be displayed when engine thrust is reduced to idle.

- B. For EAI Card inoperative, position both EAI selectors ON.

NOTE 1: When an EAI selector is positioned to ON and the temperature is greater than 10 degrees C, the ANTI-ICE ON advisory message will be displayed. Do not accomplish the ANTI-ICE ON Non-Normal Checklist.

NOTE 2: The associated ANTI-ICE ENG L or R advisory message will be displayed, the EAI indication will not be displayed on EICAS, and the associated EAI valve will be displayed as invalid (low intensity white) and the green flow bar will not be displayed on the Air synoptic.

5. For the inoperative EAI valve, if the ANTI-ICE LEAK ENG L or R caution message is subsequently displayed, do not accomplish the two minute delay associated with the ANTI-ICE LEAK ENG L or R Non-Normal Checklist. Complete the remainder of the checklist as soon as practical.

30-21-01 Engine Anti-Ice Systems

30-21-01-02 GE

30-21-01-02D OAT Above 38 Degrees C, Without Deactivation Kit Installed

Interval	Installed	Required	Procedure
A	2	1	(M) (O)

For operations at ambient temperature above 38 degrees C, one may be inoperative provided:

- Valve is electrically disconnected and verified open.
- Engine fan case overheat detection system operates normally.
- Associated EAI pressure sensor system operates normally.
- Associated EAI card operates normally.
- Turbine case cooling air flow systems operate normally.
- Appropriate performance adjustments are applied.
- Repairs are made within 3 flight days.

Except for GE90-100 Series:

- EECs must be operated in the normal mode.

MAINTENANCE (M)

For EAI VALVE L or R status message displayed, deactivate the EAI valve by disconnecting the electrical connector and verify EAI valve is open when the associated engine is running (AMM 30-00-00/901).

1. Deactivate the associated EAI valve controller.
 - A. For the right EAI valve, open P210 panel EAI CTRL R circuit breaker.
 - B. For the left EAI valve, open P310 panel EAI CTRL L circuit breaker.
2. Deactivate the leading edge slats in the retracted position (AMM 27-81-00/201).
3. Deactivate the associated thrust reverser for ground maintenance (AMM 78-31-00/201).
4. Deactivate the EAI valve in the open position.
 - A. On the associated engine, disconnect, cap and stow the torque motor electrical connector DM30211 or DM30211A from the EAI valve controller, M30211.
 - B. Install a protective cover over the electrical connector on the valve controller.
 - C. Visually inspect the area near the Cowl Thermal Anti-Ice duct for signs of leaks and the fan cowl in general for flammable fluid leaks. Dispatch is not allowed if any of the duct clamps or supports are not secure or any air leaks from the EAI duct are present, or if significant flammable fluid leakage is present.
5. Activate the thrust reverser (AMM 78-31-00/201).
6. Activate the leading edge slats (AMM 27-81-00/201).
7. Activate the associated EAI valve controller by closing the associated circuit breaker.
8. Verify the EAI valve is open when the associated engine is running.
 - A. Position associated ANTI-ICE Panel (P5) Engine Anti-Ice selector OFF.
 - B. Start the associated engine (71-00-00/201).
 - C. Confirm the EAI indication (green EAI) is displayed above the N1 indication on the EICAS display for the associated engine (indicates the EAI valve is open).

NOTE: If the EAI indication is not displayed, the valve may be stuck closed and a thrust setting slightly above minimum idle may open the valve.

- D. Shutdown the associated engine (71-00-00/201).

OPERATIONS (O)

1. Reduce performance limited weights by the appropriate adjustments.

Performance Limited Weights	Takeoff & Landing	Enroute Climb
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**For -200 or -200ER/GE90-76B thru -90B:**

thru 632,500 lb (286,902 kg)	9,100 lb (4,128 kg)	11,800 lb (5,352 kg)
over 632,500 lb (286,902 kg)	13,200 lb (5,987 kg)	11,800 lb (5,352 kg)

For -200 or -200ER/GE90-94B:

thru 632,500 lb (286,902 kg)	9,100 lb (4,128 kg)	13,900 lb (6,305 kg)
over 632,500 lb (286,902 kg)	14,300 lb (6,486 kg)	13,900 lb (6,305 kg)

For -200LR or 777F/GE90-100 Series:

All	13,300 lb (6,033 kg)	12,100 lb (5,489 kg)
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For -300ER/GE90-115B:

All	10,400 lb (4,717 kg)	11,800 lb (5,353 kg)
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NOTE: The enroute climb weight penalties listed above are based on single engine operating speeds that approximate maximum lift-to-drag ratio speed. To account for the difference in level off altitude when operating at other speeds, multiply the enroute climb weight penalty listed above by the appropriate factor listed on page 3.00-01-00.2 in the ENROUTE DIVERSION SPEED EFFECTS paragraph.

2. Increase flight planning fuel by 1.0%.
3. For ETOPS flight planning, increase engine inoperative Critical Fuel Reserves by 1.8%.
4. Verify the EAI valve is open when the associated engine is running before each departure.
 - A. Position associated ANTI-ICE Panel (P5) Engine Anti-Ice selector OFF.
 - B. After the associated engine has been started, confirm the associated EAI indication (green EAI) is displayed above the N1 indication.

NOTE 1: If the EAI indication is not displayed, the valve may be stuck closed and a thrust setting slightly above minimum idle may open the valve.

NOTE 2: On the Air synoptic, the EAI valve will indicate failed open because the EAI selector is OFF and the green flow bar will be displayed.

-
5. Position associated ENGINE ANTI-ICE selector ON.

NOTE 1: When an EAI selector is positioned to ON and the temperature is greater than 10 degrees C, the ANTI-ICE ON advisory message will be displayed. Do not accomplish the ANTI-ICE ON Non-Normal Checklist.

NOTE 2: Inflight, the ENG IDLE DISAGREE advisory message may be displayed when engine thrust is reduced to idle.

NOTE 3: The ANTI-ICE ENG advisory message may be displayed during ground operations prior to takeoff. Flight crew should accomplish the associated non-normal checklist. If ANTI-ICE ENG clears, takeoff is allowed.

6. For the inoperative EAI valve, if the ANTI-ICE LEAK ENG L or R caution message is subsequently displayed, do not accomplish the two minute delay associated with the ANTI-ICE LEAK ENG L or R Non-Normal Checklist. Complete the remainder of the checklist as soon as practical.

**30-21-02 Engine Anti-Ice Indications (EAI)**

Interval	Installed	Required	Procedure
C	2	0	

30-31-01 Pitot Probe Heater Systems
30-31-01-01 Right Pitot Heater

Interval	Installed	Required	Procedure
B	1	0	(M)

May be inoperative provided:

- a. Right pitot air data module is deactivated.
- b. Left and center pitot air data modules operate normally.
- c. Left and center pitot probe heater systems operate normally.
- d. Left static air data module operates normally.
- e. Standby airspeed indication operates normally.
- f. Right AIR DATA/ATT instrument source switch operates normally.
- g. Approach minimums do not require its use.
- h. SAARU data is verified to be available to the right PFD before each departure.
- i. Left and center pitot probes are inspected before each departure.

NOTE: The airplane must also be dispatched using MMEL items 34-21-04-01 and 34-21-06.

MAINTENANCE (M)

NOTE: The left, center and right Pitot and Static ADMs are interchangeable.

Deactivate the right Pitot Air Data Module (ADM), inspect the left and center Pitot Probes before each departure, and verify SAARU data is available to the right Primary Flight Display (PFD) before each departure (AMM 30-00-00/901).

1. Deactivate the right Pitot ADM.

NOTE 1: The PITOT ADM R status message will be displayed.

NOTE 2: The AOA VANE R status message will be displayed.

[For Collins weather radar installed:](#)

NOTE 3: AUTOTILT FAIL will be annunciated on the ND.

- A. Gain access to the Right Flight Controls Power Supply Assembly (PSA) on Equipment Rack E5-1.
- B. Open and collar the Right Flight Controls Power Supply Assembly (PSA) front panel PITOT ADM circuit breaker.
2. Prior to each departure, visually examine the left and center pitot probe external installations for damage and unwanted material in the pitot probe's opening. If damage or unwanted material is noted, perform Pitot Probe - Inspection /Check (AMM 34-11-01/601).



3. Verify SAARU data is available to the right Primary Flight Display (PFD) before each departure.
 - A. With the airplane powered, verify SAARU alignment.

For ISFD not installed:

NOTE: The SAARU is also the attitude source for the Standby Attitude indicator and it should display sky/ground (blue over brown) shading.
 - B. Position right AIR DATA/ATT switch to ALTN (non-normal position - switch in).
 - C. Verify the Attitude Indication on the right PFD displays the sky/ground (blue over brown) shading and pitch scale.
 - D. Position right AIR DATA/ATT switch off (normal position - switch out).

OPERATIONS NOTE

1. The autoland system is not fail-operational and the NO LAND 3 advisory message will be displayed.
2. If the Captain's AIR DATA/ATT instrument source switch is positioned to ALTN, airspeed will be flagged on the left PFD. This effect is normal for this dispatch condition.
3. An additional failure, such as a plugged or damaged pitot probe, may result in the two remaining pitot sources miscomparing and voted airspeed being declared invalid (if the center Pitot Heat is inoperative, data from the center Pitot ADM is not used in the voted airspeed determination). Lack of valid voted airspeed will result in display of the NAV AIR DATA SYS advisory or AIR DATA SYS Caution message and single channel airspeed and altitude will be displayed on the PFDs.

NOTE: For a right PFD airspeed flagged, selection of the First Officer's AIR DATA/ATT switch to ALTN will display the left PFD airspeed on the right PFD. This single airspeed condition will be indicated by the SGL SOURCE AIR DATA advisory message.

The following display effects with regard to airspeed (A/S) are possible:

Pitot Heat Inop/Pitot ADM Deact. At Dispatch	Pitot Probe Plugged, Damaged, etc.	L PFD A/S Display	Standby A/S Display	R PFD A/S Display
Right	Left	In Error (plugged)	Correct	Flag (ADM deact.)
Right	Center	Correct	In Error (plugged)	Flag (ADM deact.)

30-31-01 Pitot Probe Heater Systems
30-31-01-02 Left Pitot Heater

Interval	Installed	Required	Procedure
B	1	0	(M)

May be inoperative provided:

- Left pitot air data module is deactivated.
- Right and center pitot air data modules operate normally.
- Right and center pitot probe heater systems operate normally.
- Right static air data module operates normally.
- Standby airspeed indication operates normally.
- Left AIR DATA/ATT instrument source switch operates normally.
- Approach minimums do not require its use.
- SAARU data is verified to be available to the left PFD before each departure.
- Right and center pitot probes are inspected before each departure.

NOTE: The airplane must also be dispatched using MMEL items 34-21-04-02, 34-21-06 and 34-46-01-02.

MAINTENANCE (M)

NOTE: The left, center and right Pitot and Static ADMs are interchangeable.

Deactivate the left Pitot Air Data Module (ADM), inspect the right and center Pitot Probes before each departure, and verify SAARU data is available to the left Primary Flight Display (PFD) before each departure.

- Deactivate the left Pitot ADM (AMM 30-00-00/901).

NOTE 1: The PITOT ADM L status message will be displayed.

NOTE 2: The AOA VANE L status message will be displayed.

NOTE 3: The WINDSHEAR REAC status message will be displayed.

NOTE 4: The SATCOM SYSTEM and SATCOM HI GAIN status messages may be displayed. For a SATCOM SYSTEM or SATCOM HI GAIN status message displayed, the airplane must also be dispatched using MMEL item 23-15-01.

For Collins weather radar installed:

NOTE 5: AUTOTILT FAIL will be annunciated on the ND.

- Gain access to the Left Flight Controls Power Supply Assembly (PSA) on Equipment Rack E1-6.
- Open and collar the Left Flight Controls Power Supply Assembly (PSA) front panel PITOT ADM circuit breaker.

2. Prior to each departure, visually examine the right and center pitot probe external installations for damage and unwanted material in the pitot probe's opening. If damage or unwanted material is noted, perform Pitot Probe - Inspection /Check (AMM 34-11-01/601).
3. Verify SAARU data is available to the left Primary Flight Display (PFD) before each departure.

A. With the airplane powered, verify SAARU alignment.

For ISFD not installed:

NOTE: The SAARU is also the attitude source for the Standby Attitude indicator and it should display sky/ground (blue over brown) shading.

- B. Position left AIR DATA/ATT switch to ALTN (non-normal position - switch in).
- C. Verify the Attitude Indication on the left PFD displays the sky/ground (blue over brown) shading and pitch scale.
- D. Position left AIR DATA/ATT switch off (normal position - switch out).

OPERATIONS NOTE

1. The autoland system is not fail-operational and the NO LAND 3 advisory message will be displayed.
2. If the First Officer's AIR DATA/ATT instrument source switch is positioned to ALTN, airspeed will be flagged on the right PFD. This effect is normal for this dispatch condition.
3. The WINDSHEAR SYS advisory message will be displayed.
4. An additional failure, such as a plugged or damaged pitot probe, may result in the two remaining pitot sources miscomparing and voted airspeed being declared invalid (if the center Pitot Heat is inoperative, data from the center Pitot ADM is not used in the voted airspeed determination). Lack of valid voted airspeed will result in display of the NAV AIR DATA SYS advisory or AIR DATA SYS Caution message and single channel airspeed and altitude will be displayed on the PFDs.

NOTE: For a left PFD airspeed flagged, selection of the Captain's AIR DATA/ATT switch to ALTN will display the right PFD airspeed on the left PFD. This single airspeed condition will be indicated by the SGL SOURCE AIR DATAV advisory message.

The following display effects with regard to airspeed (A/S) are possible:

Pitot Heat Inop/Pitot ADM Deact. At Dispatch	Pitot Probe Plugged, Damaged, etc.	L PFD A/S Display	Standby A/S Display	R PFD A/S Display
Left	Center	Flag (ADM deact.)	In Error (plugged)	Correct
Left	Right	Flag (ADM deact.)	Correct	In Error (plugged)

30-31-01 Pitot Probe Heater Systems
30-31-01-03 Center Pitot Heater

Interval	Installed	Required	Procedure
B	1	0	(M)

May be inoperative provided:

- Left and right pitot air data modules operate normally.
- Left and right pitot probe heater systems operate normally.
- Standby airspeed indication operates normally.
- Approach minimums do not require its use.
- Airplane is not operated in known or forecast icing conditions or visible moisture.
- Left and right pitot probes are inspected before each departure.
- Flight remains within 120 minutes of landing at a suitable airport.

MAINTENANCE (M)

NOTE: The left, center and right Pitot and Static ADMs are interchangeable.

Prior to each departure, visually examine the left and right pitot probe external installations for damage and unwanted material in the pitot probe's opening. If damage or unwanted material is noted, perform Pitot Probe - Inspection /Check (AMM 34-11-01/601).

OPERATIONS NOTE

- The autoland system is not fail-operational and the NO LAND 3 advisory message will be displayed.



2. Data from the center Pitot ADM will not be used in the voted airspeed determination. An additional failure, such as a plugged or damaged pitot probe, may result in the two remaining pitot sources miscomparing and voted airspeed being declared invalid. Lack of valid voted airspeed will result in display of the NAV AIR DATA SYS advisory or AIR DATA SYS Caution message and single channel airspeed and altitude will be displayed on the PFDs.

The following display effects with regard to airspeed (A/S) are possible:

Pitot Heat Inoperative	Pitot Probe Plugged, Damaged, etc.	L PFD A/S Display	Standby A/S Display	R PFD A/S Display
Center*	Left	In Error (plugged)	Unheated *	Correct
Center*	Right	Correct	Unheated *	In Error (plugged)

* Restricted to non-icing conditions



30-33-01 Total Air Temperature (TAT) Probe Heater System
30-33-01-01 Single TAT System

Interval	Installed	Required	Procedure
C	1	0	

May be inoperative provided:

- Airplane is not operated in known or forecast icing conditions.
 - Flight remains within 120 minutes of landing at a suitable airport.
-

30-33-01 Total Air Temperature (TAT) Probe Heater System
30-33-01-02 Dual TAT System

30-33-01-02A One Inoperative

Interval	Installed	Required	Procedure
C	2	1	

MAINTENANCE NOTE

Position the GPWS RA/TAT Panel (P61) TAT selector to the position with an operative heater. This action clears the status message.

30-33-01 Total Air Temperature (TAT) Probe Heater System
30-33-01-02 Dual TAT System

30-33-01-02B Both Inoperative

Interval	Installed	Required	Procedure
C	2	0	

May be inoperative provided:

- Airplane is not operated in known or forecast icing conditions.
 - Flight remains within 120 minutes of landing at a suitable airport.
-



30-34-01 Engine PT2/TT2 Probe Heater Systems (PW & RR)
30-34-01-01 PW

Interval	Installed	Required	Procedure
C	2	0	(O)

May be inoperative provided:

- Both engines must be operated in the alternate (ALTN) mode.
 - Appropriate performance adjustments are applied.
-

MAINTENANCE NOTE

For engines without PW SB 4G-112-73-29 or production equivalent installed (EEC software SCN8C or later) and EEC C1 faults present, use DDG item 73-21-02-01 Maintenance Note procedure.

OPERATIONS (O)

Use DDG item 73-21-02-01 Operations procedure.

30-34-01 Engine PT2/TT2 Probe Heater Systems (PW & RR)
30-34-01-02 RR

Interval	Installed	Required	Procedure
C	2	0	(O)

May be inoperative provided:

- N2 indication on associated engine operates normally.
 - Both engines must be operated in the alternate (ALTN) mode.
 - Appropriate performance adjustments are applied.
-

OPERATIONS (O)

Use DDG item 73-21-02-03 Operations procedure.



30-41-01 Flight Deck Forward Window Primary Heater Systems

Interval	Installed	Required	Procedure
C	2	1	

One may be inoperative provided:

- a. Both flight deck side (No. 2) window heater systems operate normally.
- b. Airplane is not operated in known or forecast icing conditions.
- c. Associated switch remains OFF.
- d. Flight remains within 120 minutes of landing at a suitable airport.

OPERATIONS NOTE

If cracks are observed in the associated window, notify maintenance to inspect the window per AMM 56-11-00/601.

**30-41-02 Flight Deck Side (No. 2) Window Heater Systems**

Interval	Installed	Required	Procedure
C	2	1	

One may be inoperative provided:

- a. Both flight deck forward window primary heater systems operate normally.
- b. Associated switch remains OFF.

Flight Deck Aft (No. 3) Window Heater Systems

Interval	Installed	Required	Procedure
D	2	0	



30-41-04 WINDOW HEAT Switch Lights**30-41-04-01 INOP Lights**

Interval	Installed	Required	Procedure
C	4	0	

30-41-04 WINDOW HEAT Switch Lights**30-41-04-02 ON Lights**

Interval	Installed	Required	Procedure
C	4	0	

**30-42-01 Windshield Wipers**

Interval	Installed	Required	Procedure
C	2	0	

May be inoperative provided:

- Airplane is not operated in precipitation within 5 statute miles of the airport of departure or intended landing.
 - Approach minimums do not require its use.
-

30-42-01 Windshield Wipers**30-42-01-01 Low Speed Functions**

Interval	Installed	Required	Procedure
C	2	0	

May be inoperative provided associated high speed function operates normally.

30-42-01 Windshield Wipers**30-42-01-02 High Speed Functions****30-42-01-02A One Inoperative**

Interval	Installed	Required	Procedure
C	2	1	

One may be inoperative provided associated low speed function operates normally.

30-42-01 Windshield Wipers**30-42-01-02 High Speed Functions****30-42-01-02B Both Inoperative**

Interval	Installed	Required	Procedure
C	2	0	

May be inoperative provided:

- Both low speed functions operate normally.



- b. Airplane is not operated in known or forecast precipitation of moderate or greater intensity within 5 statute miles of the airport of departure or intended landing.
-

30-42-01 Windshield Wipers**30-42-01-03 Intermittent Functions**

Interval	Installed	Required	Procedure
C	2	0	

**30-71-01 Drain Mast Heaters**

Interval	Installed	Required	Procedure
C	2	0	(M)

May be inoperative provided:

- Associated galley service and lavatory sinks are not used.
- Associated galley and lavatory sink water supply shutoff valves are closed.

MAINTENANCE (M)

Shut off the water supplies associated with the inoperative drain mast heater (AMM 30-00-00/901).

For -200/-200ER/-200LR:

- For the forward drain mast heater inoperative, close the water supply shut off valves to the galley sinks, lavatory sinks and drinking fountains (if installed) forward of, but not including, the door 3 service area.
- For the aft drain mast heater inoperative, close the water supply shut off valves to the galley sinks, lavatory sinks and drinking fountains (if installed) aft of, and including, the door 3 service area.

For -300/-300ER:

- For the forward drain mast heater inoperative, close the water supply shut off valves to the galley sinks, lavatory sinks and drinking fountains (if installed) forward of and including the door 3 service area, except outboard lavatories installed on the aft side of the door 3 service area.
- For the aft drain mast heater inoperative, close the water supply shut off valves to the galley sinks, lavatory sinks and drinking fountains (if installed) aft of the door 3 service area including outboard lavatories installed on the aft side of the door 3 service area.

For 777F:

- For the forward drain mast heater inoperative, close the water supply shut off valves to the galley and lavatory sinks.

**30-81-01 Ice Detectors**

Interval	Installed	Required	Procedure
C	2	0	(O)

May be inoperative provided engine and wing anti-ice systems are operated manually.

OPERATIONS (O)

If one or both ice detectors are inoperative, operate the Wing and Engine Anti-Ice systems manually.

**31-09-01 Systems Cardfile ARINC Signal Gateway (ASG) Cards**

31-09-01-01 Left Systems Cardfile ASG Cards

31-09-01-02 Right Systems Cardfile ASG Cards

31-09-02 Systems Cardfile Power Supplies

31-09-02-01 Left Systems Cardfile (LSCF) Power Supplies

31-09-02-02 Right Systems Cardfile (RSCF) Power Supplies

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31-61-12 Head-Up Display (HUD)/Enhanced Flight Vision System (EFVS) (STC ST03981AT)

31-61-12-01 Head-Up Display (HUD) System

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31-09-01 Systems Cardfile ARINC Signal Gateway (ASG) Cards

Interval	Installed	Required	Procedure
C	4	2	(O)

Two ASG cards and/or the associated ARINC 629 couplers may be inoperative provided:

- One inoperative ASG card and/or associated ARINC 629 coupler is in the left systems cardfile.
- One inoperative ASG card and/or associated ARINC 629 coupler is in the right systems cardfile.
- Left and right hydraulic demand pump ON and OFF selectors operate normally.
- Left and right hydraulic demand pump selectors are ON for takeoff and landing.
- Center system hydraulic demand pump C1 or C2 is ON for takeoff and landing.

OPERATIONS (O)

NOTE 1: The ENG IDLE DISAGREE advisory message may be displayed for an inoperative ASG card.

NOTE 2: For LSCF ASG CARD L and RSCF ASG CARD R status messages displayed, both engines will operate at approach idle in the air whenever idle thrust is selected.

- Position Hydraulic Panel (P5) DEMAND L and R ELEC selectors ON for takeoff and landing.
- Position Hydraulic Panel (P5) DEMAND AIR C1 or C2 selector ON for takeoff and landing.

NOTE: Do not position center system demand pump selector ON before engine start.

31-09-01 Systems Cardfile ARINC Signal Gateway (ASG) Cards
31-09-01-01 Left Systems Cardfile ASG Cards

Interval	Installed	Required	Procedure
C	2	1	(O)

One ASG card and/or the associated ARINC 629 coupler may be inoperative in the left systems cardfile provided:

- Left hydraulic demand pump ON and OFF selector operates normally.
- Left hydraulic demand pump selector is ON for takeoff and landing.



- c. Both right systems cardfile ASG cards operate normally.
-

OPERATIONS (O)

Position Hydraulic Panel (P5) DEMAND L ELEC selector ON for takeoff and landing.

NOTE: For LSCF ASG CARD L or LSCF ASG CARD R status message displayed, the ENG IDLE DISAGREE advisory message may be displayed.

31-09-01 Systems Cardfile ARINC Signal Gateway (ASG) Cards

31-09-01-02 Right Systems Cardfile ASG Cards

Interval	Installed	Required	Procedure
C	2	1	(O)

One ASG card and/or the associated ARINC 629 coupler may be inoperative in the right systems cardfile provided:

- Right hydraulic demand pump ON and OFF selector operates normally.
 - Right hydraulic demand pump selector is ON for takeoff and landing.
 - Both left systems cardfile ASG cards operate normally.
-

OPERATIONS (O)

Position Hydraulic Panel (P5) DEMAND R ELEC selector ON for takeoff and landing.

NOTE: For RSCF ASG CARD R status message displayed, the ENG IDLE DISAGREE advisory message may be displayed.

**31-09-02 Systems Cardfile Power Supplies****31-09-02-01 Left Systems Cardfile (LSCF) Power Supplies**

Interval	Installed	Required	Procedure
C	2	1	(M) (O)

One power supply may be inoperative provided it is deactivated.

MAINTENANCE (M)

Deactivate the inoperative power supply (AMM 31-00-00/901).

NOTE: The power supply preregulator cards are interchangeable and the linear monitor cards are interchangeable.

1. For an inoperative left systems cardfile power supply 1, open and collar P310 panel LSCF PS 1 circuit breaker.
2. For an inoperative left systems cardfile power supply 2, open and collar P310 panel LSCF PS 2 circuit breaker.

OPERATIONS (O)

For the LSCF power supply 2 inoperative, do not position the Battery switch OFF until the APU has spooled down. After initiating normal APU shutdown wait two minutes before positioning the Battery switch OFF.

31-09-02 Systems Cardfile Power Supplies**31-09-02-02 Right Systems Cardfile (RSCF) Power Supplies****31-09-02-02A Power Supply 2 Inoperative**

Interval	Installed	Required	Procedure
C	2	1	(M)

Power supply 2 may be inoperative provided it is deactivated.

MAINTENANCE (M)

Deactivate the right systems cardfile power supply 2 (AMM 31-00-00/901).

NOTE: The power supply preregulator cards are interchangeable and the linear monitor cards are interchangeable.

1. Open and collar P210 panel RSCF PS 2 circuit breaker.

**31-09-02 Systems Cardfile Power Supplies****31-09-02-02 Right Systems Cardfile (RSCF) Power Supplies****31-09-02-02B Power Supply 1 Inoperative**

Interval	Installed	Required	Procedure
C	2	1	(M)

Power supply 1 may be inoperative provided:

- Power supply 1 is deactivated.
- Left air/ground system operates normally.

MAINTENANCE (M)

Deactivate the right systems cardfile power supply 1 (AMM 31-00-00/901).

NOTE: The power supply preregulator cards are interchangeable and the linear monitor cards are interchangeable.

- Open and collar P310 panel RSCF PS 1 circuit breaker.



31-20-01 Countdown Timer

Interval	Installed	Required	Procedure
D	1	0	

**31-25-01****Clocks*********

Interval	Installed	Required	Procedure
C	2	1	



Interval	Installed	Required	Procedure
C	2	1	

**31-31-01 Flight Data Recorder (FDR) System****31-31-01A One Operative**

Interval	Installed	Required	Procedure
C	-	1	

Any in excess of those required by 14 CFR may be inoperative.

OPERATIONS NOTE

See 14 CFR 121.344 or other appropriate regulations.

31-31-01 Flight Data Recorder (FDR) System**31-31-01B All Inoperative**

Interval	Installed	Required	Procedure
A	-	0	

May be inoperative provided:

- Cockpit Voice Recorder (CVR) operates normally.
- Airplane is not dispatched from a designated airport as listed in the operator's MEL unless:
 - The FDR failure occurs after pushback but prior to takeoff, or
 - The FDR repair was attempted but was not successful.
- In those cases where repair is attempted but not successful, the airplane may be dispatched on a flight or a series of flights until the next designated airport where repair must be accomplished prior to dispatch.
- Repairs are made within 3 flight days.

31-31-01 Flight Data Recorder (FDR) System**31-31-01-01 FDR Recording Parameters Required by 14 CFR**

Interval	Installed	Required	Procedure
A	-	-	

May be inoperative provided:

- Cockpit Voice Recorder (CVR) operates normally.
- Repairs are made within 20 calendar days.

OPERATIONS NOTE

See 14 CFR 121.344 or other appropriate regulations.

**31-31-01 Flight Data Recorder (FDR) System****31-31-01-02 FDR Recording Parameters Not Required by 14 CFR**

Interval	Installed	Required	Procedure
A	-	-	

May be inoperative provided repairs are made prior to the completion of the next heavy maintenance visit.

OPERATIONS NOTE

See 14 CFR 121.344 or other appropriate regulations.

**31-33-01 Airborne Printer (Flight Deck)****31-33-01A Alternate Procedures Used**

Interval	Installed	Required	Procedure
C	1	0	(O)

May be inoperative provided alternate procedures are established and used.

MAINTENANCE NOTE

If desired, the Airborne Printer can be deactivated by opening and collaring the P210 panel FLT DECK PRINTER circuit breaker.

OPERATIONS (O)

Alternate procedures must be established and used.

31-33-01 Airborne Printer (Flight Deck)**31-33-01B Procedures Do Not Require Use**

Interval	Installed	Required	Procedure
D	1	0	

May be inoperative provided procedures do not require its use.

MAINTENANCE NOTE

If desired, the Airborne Printer can be deactivated by opening and collaring the P210 panel FLT DECK PRINTER circuit breaker.

31-33-01 Airborne Printer (Flight Deck)**31-33-01-01 Miscellaneous Features**

Interval	Installed	Required	Procedure
D	-	-	

MAINTENANCE NOTE

Miscellaneous Features refers to those features, which when inoperative, do not affect the printer's primary functions. The paper cutter would be an example of a Miscellaneous Feature.

**31-35-01 Quick Access Recorder**

31-35-01A Alternate Procedures Used

Interval	Installed	Required	Procedure
C	1	0	(O)

May be inoperative provided alternate procedures are established and used.

OPERATIONS (O)

Alternate procedures must be established and used.

31-35-01 Quick Access Recorder

31-35-01B Procedures Do Not Require Use

Interval	Installed	Required	Procedure
D	1	0	

May be inoperative provided procedures do not require its use.



31-35-02

Airplane Condition Monitoring Function (ACMF)

Interval	Installed	Required	Procedure
D	1	0	



31-41-01 Airplane Information Management System (AIMS)

31-41-01-01 AIMS-2 Cabinet Bus

Interval	Installed	Required	Procedure
C	1	0	

May be dispatched with AIMS CABINET BUS faults.

**31-51-01 Master Warning Lights (Glareshield)**

Interval	Installed	Required	Procedure
C	2	1	

One may be inoperative provided the aural for the master warning system operates normally.



31-51-02 Master Caution Lights (Glareshield)

Interval	Installed	Required	Procedure
C	2	1	

One may be inoperative provided the aural for the master caution system operates normally.

**31-51-03 Aural Warning Speaker Systems**

Interval	Installed	Required	Procedure
C	2	1	

One may be inoperative provided master warning and master caution lights operate normally.

31-51-04 Warning Electronics System (WES) Channels
31-51-04-01 Passenger
31-51-04-01A Channel Not Deactivated

Interval	Installed	Required	Procedure
C	4	3	

One may be inoperative provided:

- a. Both stabilizer position indicators operate normally.
- b. SPEED LIM flag is not displayed adjacent to the airspeed indication on either PFD.

MAINTENANCE NOTE

The stabilizer position indicator operates normally if the stabilizer position and green band are displayed on the indicator (indicators not blanked).

31-51-04 Warning Electronics System (WES) Channels
31-51-04-01 Passenger
31-51-04-01B Channel Deactivated

Interval	Installed	Required	Procedure
C	4	3	(M)

One WES channel may be inoperative provided it is deactivated.

MAINTENANCE (M)

NOTE: The warning electronic system (WES) consists of four warning electronic unit (WEU) channels.

Deactivate the inoperative WEU channel (AMM 31-00-00/901).

1. For WEU CHANNEL L1, open and collar ELMS P310 Panel WARNING ELECTRONIC - L1 circuit breaker.

NOTE: STALL WARNING SYS L and WARNING SPEAKER L status messages will be displayed. The airplane must also be dispatched using MMEL items 27-32-01 and 31-51-03.

2. For WEU CHANNEL L2, open and collar ELMS P310 Panel WARNING ELECTRONIC - L2 circuit breaker.

NOTE: STALL WARNING SYS L status message will be displayed. The airplane must also be dispatched using MMEL item 27-32-01.

3. For WEU CHANNEL R1, open and collar ELMS P210 Panel WARNING ELECTRONIC - R1 circuit breaker.



NOTE: STALL WARNING SYS R and WARNING SPEAKER R status messages will be displayed. The airplane must also be dispatched using MMEL items 27-32-01 and 31-51-03.

4. For WEU CHANNEL R2, open and collar ELMS P210 Panel WARNING ELECTRONIC - R2 circuit breaker.

NOTE: STALL WARNING SYS R status message will be displayed. The airplane must also be dispatched using MMEL item 27-32-01.

31-51-04 Warning Electronics System (WES) Channels**31-51-04-02 777F****31-51-04-02A Channel Not Deactivated**

Interval	Installed	Required	Procedure
C	4	3	

One may be inoperative provided:

- CSD controller channels operate normally.
 - All ASG cards operate normally.
 - Both stabilizer position indicators operate normally.
 - SPEED LIM flag is not displayed adjacent to the airspeed indication on either PFD.
-

MAINTENANCE NOTE

The stabilizer position indicator operates normally if the stabilizer position and green band are displayed on the indicator (indicators not blanked).

31-51-04 Warning Electronics System (WES) Channels**31-51-04-02 777F****31-51-04-02B Channel Deactivated**

Interval	Installed	Required	Procedure
C	4	3	(M)

One may be inoperative provided:

- Inoperative channel is deactivated.
 - CSD controller channels operate normally.
 - All ASG cards operate normally.
-

MAINTENANCE (M)

NOTE: The warning electronic system (WES) consists of four warning electronic unit (WEU) channels.



Deactivate the inoperative WEU channel (AMM 31-00-00/901).

1. For WEU CHANNEL L1, open and collar ELMS P310 Panel WARNING ELECTRONIC - L1 circuit breaker.

NOTE: STALL WARNING SYS L and WARNING SPEAKER L status messages will be displayed. The airplane must also be dispatched using MMEL items 27-32-01 and 31-51-03.

2. For WEU CHANNEL L2, open and collar ELMS P310 Panel WARNING ELECTRONIC - L2 circuit breaker.

NOTE: STALL WARNING SYS L status message will be displayed. The airplane must also be dispatched using MMEL item 27-32-01.

3. For WEU CHANNEL R1, open and collar ELMS P210 Panel WARNING ELECTRONIC - R1 circuit breaker.

NOTE: STALL WARNING SYS R and WARNING SPEAKER R status messages will be displayed. The airplane must also be dispatched using MMEL items 27-32-01 and 31-51-03.

4. For WEU CHANNEL R2, open and collar ELMS P210 Panel WARNING ELECTRONIC - R2 circuit breaker.

NOTE: STALL WARNING SYS R status message will be displayed. The airplane must also be dispatched using MMEL item 27-32-01.

31-51-05 **Warning Electronic System (WES) Autopilot Disconnect Warning Function**

Interval	Installed	Required	Procedure
B	1	0	(O)

May be inoperative provided:

- a. Autopilots are not used.
- b. Approach minimums do not require use of the autopilots.
- c. Number of flight segments and segment duration is acceptable to flight crew.
- d. Enroute operations do not require use of the autopilots.
- e. Flight remains within 180 minutes of landing at a suitable airport.

OPERATIONS (O)

Do not use the autopilots and observe the following restrictions:

1. Approach minimums must not require autopilot use.
2. Enroute operations must not require autopilot use.
3. The flight must remain within 180 minutes of landing at a suitable airport.
4. For PBN operations:
PBN operations may be conducted provided autopilot is not used. Flight director may be used if suitable for the intended PBN operation.

NOTE: Refer to DDG Section 4 for additional information.

31-51-05 **Warning Electronic System (WES) Autopilot Disconnect Warning Function**

31-51-05-01 **WES Autopilot Interfaces**

Interval	Installed	Required	Procedure
C	3	2	(O)

One may be inoperative provided the WES autopilot disconnect warning function is verified to operate normally.

OPERATIONS (O)

Verify the Warning Electronic System (WES) autopilot disconnect warning function operates normally.

1. Engage, then manually disengage the autopilots.
2. Confirm illumination of the master warning lights, activation of an aural indication and display of the AUTOPILOT DISC warning message.



31-51-06 TAKEOFF Configuration CHECK Switch*********31-51-06A Procedures Require Use**

Interval	Installed	Required	Procedure
C	1	0	

31-51-06 TAKEOFF Configuration CHECK Switch*********31-51-06B Procedures Do Not Require Use**

Interval	Installed	Required	Procedure
D	1	0	

May be inoperative provided procedures do not require its use.

**31-61-01 Display Units (DU)**

Interval	Installed	Required	Procedure
C	6	5	

One may be inoperative in the lower center DU position.

MAINTENANCE NOTE

1. The six display units are interchangeable. If the inoperative display unit is in a location other than the lower center position, it can be replaced with the operative unit from the lower center position.

For -300/-300ER:

2. For a non-video capable display unit installed in the left or right inboard or in the lower center display locations, ground maneuver camera video will not be available on that display.

For Flight Deck Door Visual Surveillance System installed:

3. For a non-video capable display unit installed in the left or right inboard or in the lower center display locations, Flight Deck Door Visual Surveillance System video will not be available on that display.
4. Refer to AMM 31-61-01/401 for DU removal and installation procedures.

**31-61-02 Cursor Control Devices (CCD)****31-61-02A One Inoperative**

Interval	Installed	Required	Procedure
C	2	1	

31-61-02 Cursor Control Devices (CCD)**31-61-02B Both Inoperative**

Interval	Installed	Required	Procedure
C	2	0	(O)

May be inoperative provided alternate procedures are established and used.

OPERATIONS (O)

Paper checklist procedures must be used. Data communications functions such as ACARS are not accessible on an MFD and require alternate procedures.

31-61-02 Cursor Control Devices (CCD)**31-61-02-01 Cursor Location Lights**

Interval	Installed	Required	Procedure
C	6	0	

31-61-02 Cursor Control Devices (CCD)**31-61-02-02 Cursor Location Switch Lights**

Interval	Installed	Required	Procedure
C	6	0	

**31-61-03 Remote Light Sensor (RLS) System**

Interval	Installed	Required	Procedure
B	1	0	

May be inoperative provided all manual display brightness controls operate normally.

**31-61-04 Display Select Panel (DSP)**

Interval	Installed	Required	Procedure
C	1	0	(M) (O)

May be inoperative provided:

- The left and right CDU DSP backup features are verified to operate normally.

[For Ground Maneuver Camera System installed:](#)

- The ground maneuver camera system is considered inoperative.

[For Flight Deck Door Visual Surveillance System installed:](#)

- The flight deck door visual surveillance system is considered inoperative.

[For Ground Maneuver Camera System installed:](#)

NOTE 1: The airplane must also be dispatched using MMEL item 31-61-10.

[For Flight Deck Door Visual Surveillance System installed:](#)

NOTE 2: The airplane must also be dispatched using MMEL item 23-70-01.

MAINTENANCE (M)

Verify the left and right CDU DSP backup features operate normally (AMM 31-00-00/901).

- Press DSP CTL (Display Select Panel Control select key) on the main menu of the CDU to ON.
- Confirm that the DSP prompt is displayed on the main menu of the CDU beside the Display Select Panel page select key. This indicates display select control is available through the CDU.

OPERATIONS (O)

Use the CDU Display Select Panel (DSP) backup feature.

31-61-04 Display Select Panel (DSP)**31-61-04-01 Multifunction Display Lights**

Interval	Installed	Required	Procedure
C	3	0	

**31-61-04 Display Select Panel (DSP)****31-61-04-02 Illuminated Switch Lights****31-61-04-02-01 -200/-200ER/-200LR/777F**

Interval	Installed	Required	Procedure
C	15	0	

31-61-04 Display Select Panel (DSP)**31-61-04-02 Illuminated Switch Lights****31-61-04-02-02 -300/-300ER**

Interval	Installed	Required	Procedure
C	16	0	

**31-61-05 EICAS Status Messages****31-61-05A System(s) Used**

Interval	Installed	Required	Procedure
C	-	0	(M) (O)

May be inoperative provided a procedure is used before each flight to verify that associated equipment operates normally.

MAINTENANCE (M)

Verify the associated equipment operates normally before each flight (AMM 31-00-00/901).

NOTE: There is no acceptable procedure available for verifying the Flight Control System operates normally when the FLIGHT CONTROL SYS status message is displayed.

OPERATIONS (O)

Verify the associated equipment operates normally before each flight.

31-61-05 EICAS Status Messages**31-61-05B System(s) Considered Inoperative**

Interval	Installed	Required	Procedure
C	-	0	

May be inoperative provided associated equipment is considered inoperative.

31-61-06 Inboard Display Selector Positions

Interval	Installed	Required	Procedure
B	8	7	(M) (O)

One may be inoperative provided:

- a. Lower center display unit operates normally.
- b. Remaining positions are verified to operate normally on associated selector.
- c. Alternate procedures are used when required.

MAINTENANCE (M)

Verify that the inboard display (INBOARD DSPL) selector with the inoperative position operates normally in the remaining positions (AMM 31-00-00/901).

1. Supply electrical power (AMM 24-22-00/201).
2. For verifying that the PFD position operates normally:
 - A. Position the INBOARD DSPL selector with the inoperative position to PFD.
 - B. Verify the associated side's inboard displays the PFD and the outboard display blanks.
 - C. On the glareshield panel (P55), use the Display Select Panel (DSP) to select the inboard display unit associated with the inoperative INBOARD DSPL selector (L INBD or R INBD).
 - D. On the DSP, verify that no other DSP selections will overwrite the PFD displayed on the inboard display.
3. For verifying that the MFD position operates normally:
 - A. Position INBOARD DSPL selector with the inoperative position to MFD.
 - B. On the glareshield panel (P55), use the Display Select Panel (DSP) to select the inboard display unit associated with the inoperative INBOARD DSPL selector (L INBD or R INBD).
 - C. On the DSP, select the NAV position.
 - 1) Ensure the navigation display (ND) format appears on the associated inboard display unit.
 - 2) Use the EFIS Control Panel to adjust the range and modes on the ND to confirm the associated inboard display unit is properly responding to the EFIS CONTROL PANEL.
 - D. On the DSP, one at a time, select the COMM and CHKL, and at least two synoptic formats. As each DSP selection is made, ensure the selected format appears on the associated inboard display unit.
4. For verifying that the NAV position operates normally:
 - A. Position INBOARD DSPL selector with the inoperative position to NAV.



- B. Ensure the ND format appears on the associated inboard display unit.
 - C. Use the EFIS Control Panel to adjust the range and modes on the ND to confirm the associated inboard display unit is properly responding to the EFIS CONTROL PANEL.
 - D. On the DSP, select any format other than NAV and ensure that the associated inboard display unit remains in the ND format.
5. For verifying that the EICAS position operates normally:
- A. Position the INBOARD DSPL selector with the inoperative position to EICAS.
 - B. Verify the associated side's inboard displays EICAS and the upper center display blanks.
 - C. On the glareshield panel (P55), use the Display Select Panel (DSP) to select the inboard display unit associated with the inoperative INBOARD DSPL selector (L INBD or R INBD).
 - D. Verify that display selections made from the DSP do not overwrite the EICAS, except ENG, FUEL and AIR displays.

OPERATIONS (O)

Use alternate procedures if remaining display unit capability differs from established operator procedures.

NOTE: AIMS display redundancy management will automatically provide PFD or EICAS formats to the proper display units when an associated outboard display unit failure or an upper center display unit failure is detected. Since the PFD and EICAS positions are not normally used, no alternate procedures are required when either or both positions are inoperative.

1. For an inoperative INBOARD DSPL selector in the NAV position, select the MFD position and select the desired MFD format on the associated inboard display unit using the DSP.
2. For an inoperative INBOARD DSPL selector in the MFD position, select the NAV position and use lower center display and opposite-side inboard display units for displaying required MFD formats.

**31-61-07 Electronic Checklist (ECL) System**

Interval	Installed	Required	Procedure
C	1	0	(M) (O)

May be inoperative provided electronic checklist is deactivated and established suitable checklist procedures are used.

MAINTENANCE (M)

Deactivate the electronic checklist (AMM 31-00-00/901).

1. Display the Maintenance Task Maintenance page.
2. Toggle the Checklist Function ENABLE/DISABLE button to DISABLE.

OPERATIONS (O)

Use established paper checklists.

NOTE: The rectangular icon symbol that precedes some alert messages will not be displayed.

31-61-07 Electronic Checklist (ECL) System**31-61-07-01 ECL Closed Loop Switch Indications**

Interval	Installed	Required	Procedure
C	-	0	(O)

May be inoperative provided ECL line item override procedures are used when required to complete checklists.

OPERATIONS (O)

Use the Line Item Override (ITEM OVRD) Key as required.



31-61-08 EFIS Control Panels

Interval	Installed	Required	Procedure
C	2	1	(M) (O)

One may be inoperative provided the associated CDU EFIS control panel backup feature is verified to operate normally.

MAINTENANCE (M)

Verify the associated (same side) CDU EFIS control backup feature operates normally (AMM 31-00-00/901).

1. Press the associated CDU Menu key.
2. Press EFIS Control Select key to ON.
3. Press the EFIS Page Select key to ON.
4. Verify the EFIS Control page is displayed.

OPERATIONS (O)

Use associated (same side) CDU to control the functions of the inoperative EFIS control panel.

31-61-08 EFIS Control Panels**31-61-08-01 Map Switch Lights**

Interval	Installed	Required	Procedure
C	14	0	

**31-61-09 Instrument Source Switches****31-61-09-01 NAV and DSPL CTRL Switches**

Interval	Installed	Required	Procedure
C	5	0	

May be inoperative provided associated switches are not moved in flight.

31-61-09 Instrument Source Switches**31-61-09-02 AIR DATA/ATT Switches****31-61-09-02A One Inoperative**

Interval	Installed	Required	Procedure
C	2	1	

One may be inoperative provided the associated switch is not moved in flight.

MAINTENANCE NOTE

For AIMS Input/Output Module (IOM) #4 in the left AIMS cabinet inoperative, selecting the left AIR DATA/ATT switch to the ALTN position will display failure flags on the left PFD for attitude, airspeed and altitude. For IOM #2 in the right AIMS cabinet inoperative, selecting the right AIR DATA/ATT switch to the ALTN position will display an attitude fail flag on the right PFD.

31-61-09 Instrument Source Switches**31-61-09-02 AIR DATA/ATT Switches****31-61-09-02B Both Inoperative**

Interval	Installed	Required	Procedure
C	2	0	(M)

May be inoperative provided:

- One AIR DATA/ATT switch is in the normal position.
 - Associated display units are verified to operate from selected sources.
 - Associated switches are not moved in flight.
-

MAINTENANCE (M)

NOTE: For AIMS Input/Output Module (IOM) #4 in the left AIMS cabinet inoperative, selecting the left AIR DATA/ATT switch to the ALTN position will display failure flags on the left PFD for attitude, airspeed and altitude. For IOM #2 in the right AIMS cabinet inoperative, selecting the right AIR DATA/ATT switch to the ALTN position will display an attitude fail flag on the right PFD.

Verify display units operate from selected sources (AMM 31-00-00/901).

1. Position both AIR DATA/ATT switches off (normal position - switch out).
2. Verify the ALTN ATTITUDE advisory message is not displayed.

**31-61-10 Ground Maneuver Camera System (-300/-300ER)**

Interval	Installed	Required	Procedure
D	1	0	

OPERATIONS NOTE

Any operative portion of the camera system may be used.

31-61-11 Heading Reference (HDG REF) Switch
31-61-11-01 TRUE Function

Interval	Installed	Required	Procedure
C	1	0	(O)

May be inoperative provided enroute or contingency procedures including diversion airports do not require its use.

OPERATIONS (O)

Operation is not allowed in routes where True North referencing is required.

31-61-11 Heading Reference (HDG REF) Switch
31-61-11-02 Switch Lights
31-61-11-02-01 TRUE Light

Interval	Installed	Required	Procedure
C	1	0	

31-61-11 Heading Reference (HDG REF) Switch
31-61-11-02 Switch Lights
31-61-11-02-02 NORM Light

Interval	Installed	Required	Procedure
C	1	0	



- 31-61-12 Head-Up Display (HUD)/Enhanced Flight Vision System (EFVS) (STC ST03981AT)**
- 31-61-12-01 Head-Up Display (HUD) System**
- 31-61-12-01A Takeoff and Approach Minimums Do Not Require Use**

Interval	Installed	Required	Procedure
C	1	0	(O)

May be inoperative provided takeoff and approach minimums do not require its use.

NOTE: Any mode which operates normally may be used.

NOTE: The HUD/EFVS installation defined by STC ST03981AT is not a Boeing system. Information applicable to this installation is not provided.

- 31-61-12 Head-Up Display (HUD)/Enhanced Flight Vision System (EFVS) (STC ST03981AT)**
- 31-61-12-01 Head-Up Display (HUD) System**
- 31-61-12-01B Procedures Do Not Require Use**

Interval	Installed	Required	Procedure
D	1	0	

May be inoperative provided procedures do not require its use.

NOTE: The HUD/EFVS installation defined by STC ST03981AT is not a Boeing system. Information applicable to this installation is not provided.

- 31-61-12 Head-Up Display (HUD)/Enhanced Flight Vision System (EFVS) (STC ST03981AT)**
- 31-61-12-01 Head-Up Display (HUD) System**
- 31-61-12-01-01 HUD Normal/Declutter Functions (Control Wheel Switch)**

Interval	Installed	Required	Procedure
C	1	0	

NOTE: The HUD/EFVS installation defined by STC ST03981AT is not a Boeing system. Information applicable to this installation is not provided.

**31-61-12 Head-Up Display (HUD)/Enhanced Flight Vision System (EFVS) (STC ST03981AT)****31-61-12-01 Head-Up Display (HUD) System****31-61-12-01-02 HUD Guidance Annunciator****31-61-12-01-02A Takeoff Minima Does Not Require Use**

Interval	Installed	Required	Procedure
C	1	0	(O)

May be inoperative provided takeoff minimums do not require its use.

NOTE: The HUD/EFVS installation defined by STC ST03981AT is not a Boeing system. Information applicable to this installation is not provided.

31-61-12 Head-Up Display (HUD)/Enhanced Flight Vision System (EFVS) (STC ST03981AT)**31-61-12-01 Head-Up Display (HUD) System****31-61-12-01-02 HUD Guidance Annunciator****31-61-12-01-02B Procedures Do Not Require Use**

Interval	Installed	Required	Procedure
D	1	0	

May be inoperative provided procedures do not require its use.

NOTE: The HUD/EFVS installation defined by STC ST03981AT is not a Boeing system. Information applicable to this installation is not provided.

31-61-12 Head-Up Display (HUD)/Enhanced Flight Vision System (EFVS) (STC ST03981AT)**31-61-12-02 Enhanced Flight Vision System (EFVS)****31-61-12-02A Takeoff and Approach Minimums Do Not Require Use**

Interval	Installed	Required	Procedure
C	1	0	(O)

May be inoperative provided takeoff and approach minimums do not require its use.

NOTE: Any mode which operates normally may be used.



NOTE: The HUD/EFVS installation defined by STC ST03981AT is not a Boeing system. Information applicable to this installation is not provided.

31-61-12 Head-Up Display (HUD)/Enhanced Flight Vision System (EFVS) (STC ST03981AT)

31-61-12-02 Enhanced Flight Vision System (EFVS)

31-61-12-02B Procedures Do Not Require Use

Interval	Installed	Required	Procedure
D	1	0	

May be inoperative provided procedures do not require its use.

NOTE: The HUD/EFVS installation defined by STC ST03981AT is not a Boeing system. Information applicable to this installation is not provided.

31-61-12 Head-Up Display (HUD)/Enhanced Flight Vision System (EFVS) (STC ST03981AT)

31-61-12-02 Enhanced Flight Vision System (EFVS)

31-61-12-02-01 EFVS Hide/Show Functions (Control Wheel Switch)

Interval	Installed	Required	Procedure
C	1	0	

May be inoperative provided EFVS is considered inoperative.

NOTE: The HUD/EFVS installation defined by STC ST03981AT is not a Boeing system. Information applicable to this installation is not provided.

31-61-12 Head-Up Display (HUD)/Enhanced Flight Vision System (EFVS) (STC ST03981AT)

31-61-12-03 HUD Combiner Control Panel Functions

31-61-12-03-01 HUD Brightness Functions (BRT)

Interval	Installed	Required	Procedure
C	1	0	

May be inoperative provided brightness is set at a usable level that is acceptable to the affected crew member.

NOTE: The HUD/EFVS installation defined by STC ST03981AT is not a Boeing system. Information applicable to this installation is not provided.

31-61-12 Head-Up Display (HUD)/Enhanced Flight Vision System (EFVS) (STC ST03981AT)
31-61-12-03 HUD Combiner Control Panel Functions
31-61-12-03-02 HUD Automatic/Manual Brightness Functions (AUTO/MAN)
31-61-12-03-02A MAN Function Operates Normally

Interval	Installed	Required	Procedure
C	1	0	

AUTO function may be inoperative provided MAN function operates normally.

NOTE: The HUD/EFVS installation defined by STC ST03981AT is not a Boeing system. Information applicable to this installation is not provided.

31-61-12 Head-Up Display (HUD)/Enhanced Flight Vision System (EFVS) (STC ST03981AT)
31-61-12-03 HUD Combiner Control Panel Functions
31-61-12-03-02 HUD Automatic/Manual Brightness Functions (AUTO/MAN)
31-61-12-03-02B AUTO Function Operates Normally

Interval	Installed	Required	Procedure
C	1	0	

MAN function may be inoperative provided AUTO function operates normally.

NOTE: The HUD/EFVS installation defined by STC ST03981AT is not a Boeing system. Information applicable to this installation is not provided.

31-61-12 Head-Up Display (HUD)/Enhanced Flight Vision System (EFVS) (STC ST03981AT)
31-61-12-03 HUD Combiner Control Panel Functions
31-61-12-03-03 EFVS Brightness/Contrast Functions (BRT/CONT)

Interval	Installed	Required	Procedure
C	1	0	

May be inoperative provided brightness/contrast is set at a usable level that is acceptable to the affected crew member.



NOTE: The HUD/EFVS installation defined by STC ST03981AT is not a Boeing system. Information applicable to this installation is not provided.

31-61-12 Head-Up Display (HUD)/Enhanced Flight Vision System (EFVS) (STC ST03981AT)

31-61-12-04 EFVS Window Heater

Interval	Installed	Required	Procedure
C	1	0	(O)

May be inoperative provided takeoff and approach minimums do not require use of the EFVS in known or forecast icing conditions.

NOTE: The HUD/EFVS installation defined by STC ST03981AT is not a Boeing system. Information applicable to this installation is not provided.



32-00-01	Landing Gear Synoptic Display
32-08-01	Proximity Sensor Electronic Unit (PSEU) Channels
32-08-01A	<i>One Inoperative - PSEU 1</i>
32-08-01B	<i>One Inoperative - PSEU 2</i>
32-08-01C	<i>Two Inoperative - PSEU 1 On Left System Bus, PSEU 2 On Right System Bus</i>
32-08-01D	<i>Two Inoperative - PSEU 1 On Right System Bus, PSEU 2 On Left System Bus</i>
32-08-01E	<i>Two Inoperative - Gear Down Dispatch</i>
32-09-01	Air/Ground System - Weight on Wheels
32-09-01-01	Left System
32-09-01-02	Right System
32-11-01	Nose Landing Gear Strut Upper Air Chamber Pressure Indicator (-300ER)
32-30-01	Landing Gear Actuation System
32-30-02	Semi-Lever Gear (SLG) System (-300ER)
32-31-01	Landing Gear Lever Lock Solenoid
32-31-02	Landing Gear Selector Valve Electrical Control Circuits
32-32-01	Main Gear Door Uplock Spring Assemblies
32-32-01-01	Spring Retention Cords
32-32-02	Main Gear Uplock Springs
32-32-03	Main Gear Side Brace Springs (PRR 61350-34 Not Incorporated)
32-35-01	Landing Gear Alternate Extend System
32-35-01-01	Alternate Extend Hydraulic Pressure Switch
32-35-02	Ground Door Release Control System
32-35-02-01	Door Open Control Switches
32-41-01	Brake Accumulator Pressure Indicator (Wheel Well)
32-41-02	Brake Accumulator Pressure Indicator (Flight Deck)

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**32-00-01 Landing Gear Synoptic Display**

Interval	Installed	Required	Procedure
C	1	0	

OPERATIONS NOTE

If the Synoptic Display is missing data, selecting an alternate location for the display (L INBD, LWR CTR, or R INBD) may restore the missing data. Synoptic Displays containing missing data may continue to be used to the extent remaining data is useful. Airplane system faults will be annunciated by alerting and status messages.

**32-08-01 Proximity Sensor Electronic Unit (PSEU) Channels****32-08-01A One Inoperative - PSEU 1**

Interval	Installed	Required	Procedure
C	4	3	

One may be inoperative in PSEU 1 provided the flight lock system is considered inoperative.

32-08-01 Proximity Sensor Electronic Unit (PSEU) Channels**32-08-01B One Inoperative - PSEU 2**

Interval	Installed	Required	Procedure
C	4	3	

One may be inoperative in PSEU 2.

32-08-01 Proximity Sensor Electronic Unit (PSEU) Channels**32-08-01C Two Inoperative - PSEU 1 On Left System Bus, PSEU 2 On Right System Bus**

Interval	Installed	Required	Procedure
C	4	2	(M) (O)

One may be inoperative in each PSEU provided:

- PSEU 1 transmission is verified on the left ARINC 629 systems bus.
- PSEU 2 transmission is verified on the right ARINC 629 systems bus.
- Flight lock system is considered inoperative.

NOTE: The airplane must also be dispatched using MMEL item 52-11-05.

MAINTENANCE (M)

Verify PSEU 1 transmission is on the left ARINC 629 systems bus and PSEU 2 transmission is on the right ARINC 629 systems bus (AMM 32-00-00/901).

- Verify PSEU 1 and PSEU 2 data is displayed on the Landing Gear ACTN/INDN maintenance page.
- Open P110 panel PSEU 1 PRI PWR circuit breaker.
- Verify PSEU 1 data is removed from the Landing Gear ACTN/INDN display. PSEU 1 is transmitting on the left ARINC 629 systems bus.
- Open P310 panel PSEU 2 ALTN PWR circuit breaker.

5. Verify PSEU 2 data is removed from the Landing Gear ACTN/INDN display. PSEU 2 is transmitting on the right ARINC 629 systems bus.
6. Close the PSEU 1 PRI PWR and the PSEU 2 ALTN PWR circuit breakers.

OPERATIONS (O)

The autobrake selector may not automatically move from RTO to OFF after takeoff. If autobrake selector remains in RTO, manually move to OFF.

32-08-01 Proximity Sensor Electronic Unit (PSEU) Channels

32-08-01D Two Inoperative - PSEU 1 On Right System Bus, PSEU 2 On Left System Bus

Interval	Installed	Required	Procedure
C	4	2	(M) (O)

One may be inoperative in each PSEU provided:

- a. PSEU 1 transmission is verified on the right ARINC 629 systems bus.
- b. PSEU 2 transmission is verified on the left ARINC 629 systems bus.
- c. After takeoff, gear remains down for two minutes before retraction.
- d. Takeoff performance is based on landing gear extended.
- e. Flight lock system is considered inoperative

NOTE: The airplane must also be dispatched using MMEL item 52-11-05.

MAINTENANCE (M)

Verify that PSEU 1 transmission is on the right ARINC 629 systems bus and PSEU 2 transmission is on the left ARINC 629 systems bus (AMM 32-00-00/901).

1. Verify PSEU 1 and PSEU 2 data is displayed on the Landing Gear ACTN/INDN maintenance page.
2. Open P310 panel PSEU 1 ALTN PWR circuit breaker.
3. Verify PSEU 1 data is removed from the Landing Gear ACTN/INDN display. PSEU 1 is transmitting on the right ARINC 629 bus.
4. Open P210 panel PSEU 2 PRI PWR circuit breaker.
5. Verify PSEU 2 data is removed from the Landing Gear ACTN/INDN display. PSEU 2 is transmitting on the left ARINC 629 systems bus.
6. Close the PSEU 1 ALTN PWR and PSEU 2 PRI PWR circuit breakers.

OPERATIONS (O)

1. The autobrake selector may not automatically move from RTO to OFF after takeoff. If autobrake selector remains in RTO, manually move to OFF.
2. After takeoff, leave the landing gear extended for a minimum of two minutes to allow the wheels to spin down prior to gear retraction.



3. Base takeoff performance on landing gear extended. Takeoff performance for the landing gear extended is available using AFM-DPI or Boeing Takeoff Module (BTM).
4. Increase flight planning fuel by 1,900 lbs (861 kg).

32-08-01 Proximity Sensor Electronic Unit (PSEU) Channels**32-08-01E Two Inoperative - Gear Down Dispatch**

Interval	Installed	Required	Procedure
C	4	2	(M) (O)

One may be inoperative in each PSEU provided:

- a. Landing gear are secured in the down position.
- b. Airplane is dispatched in accordance with the appropriate AFM Gear Extended Appendix.
- c. Flight lock system is considered inoperative

NOTE: The airplane must also be dispatched using MMEL item 52-11-05.

MAINTENANCE (M)

Secure the landing gear in the down position (AMM 32-00-00/901).

1. Install landing gear downlock pins, including installation procedure steps for airplane flight with landing gear locked in the extended position (AMM 32-00-30/201).
2. Place Alternate Gear Down Dispatch switch (Main Equipment Center above lower access door) to VMO position.

NOTE: The Warning Electronics System (WES) will set VMO/MMO to 270 KIAS/.73 Mach and VMO GEAR DOWN memo message will be displayed.

3. Deactivate the lever lock solenoid for the landing gear control lever by opening and collaring the P210 panel LG LEVER LOCK circuit breaker.

OPERATIONS (O)

1. The autobrake selector may not automatically move from RTO to OFF after takeoff. If autobrake selector remains in RTO, manually move to OFF.
2. Observe AFM Landing Gear Extended certificate limitations and operational limits.

NOTE: Dispatch requires authorization by the AFM Landing Gear Extended Appendix.

3. Base flight plan on landing gear extended.
4. The Warning Electronics System (WES) will set VMO/MMO to 270 KIAS/.73 Mach and VMO GEAR DOWN memo message will be displayed.

**32-09-01 Air/Ground System - Weight on Wheels****32-09-01-01 Left System**

Interval	Installed	Required	Procedure
C	1	0	(M)

May be inoperative provided:

- Associated system is deactivated.
- Right air/ground system operates normally.
- Right system cardfile power supply 1 operates normally.

MAINTENANCE (M)

Deactivate the left Air/Ground System (AMM 32-00-00/901).

- Open and collar the P11 panel AIR/GND 1 circuit breaker.

NOTE 1: The WINDSHEAR SYS advisory message and WINDSHEAR PRED status message may be displayed. The airplane must also be dispatched using MMEL item 34-43-01-01.

For EFB installed:

NOTE 2: Some on-ground functionality of the left EFB will be inoperative. The airplane must also be dispatched using MMEL item 46-11-02-01.

- Gain access to the P85 Left System Cardfile in the Main Equipment Center.
- Set the power switch for the L Weight on Wheels card (P85 position A16) to the OFF position.

OPERATIONS NOTE

If available, select the right ATC transponder.

32-09-01 Air/Ground System - Weight on Wheels**32-09-01-02 Right System**

Interval	Installed	Required	Procedure
C	1	0	(M)

May be inoperative provided:

- Associated system is deactivated.
- Left air/ground system operates normally.
- All galley/lavatory water supply shutoff valves are closed.

MAINTENANCE (M)

NOTE: The forward and aft drain mast heaters and the gray water drain line heaters default to low heat. Low heat inflight is likely to result in ice plugging drain masts or lines which can cause overflowing sinks. The left and right WOW cards are interchangeable (AMM 20-10-01/401).

Deactivate the right Air/Ground System and shutoff the water supply to the galley sinks, lavatory sinks and if installed drinking fountains (AMM 32-00-00/901).

1. Open and collar the P11 panel AIR/GND 2 and AIR/GND 3 circuit breakers.

NOTE 1: The WINDSHEAR SYS advisory message and WINDSHEAR PRED status message may be displayed. The airplane must also be dispatched using MMEL item 34-43-01-01.

For EFB installed:

NOTE 2: Some on-ground functionality of the right EFB will be inoperative. The airplane must also be dispatched using MMEL item 46-11-02-01.

2. Gain access to the P84 Right System Cardfile in the Main Equipment Center.
3. Set the power switch for the R Weight-On-Wheels card (P84 position A16) to the OFF position.
4. Close the water supply shut off valves to the galley sinks, lavatory sinks and drinking fountains (if installed).

OPERATIONS NOTE

1. If available, select the left ATC transponder.
2. Water for the galley sinks, lavatory sinks and drinking fountains (if installed) will not be available.



32-11-01 **Nose Landing Gear Strut Upper Air Chamber
Pressure Indicator (-300ER)**

Interval	Installed	Required	Procedure
D	1	0	

**32-30-01 Landing Gear Actuation System**

Interval	Installed	Required	Procedure
B	1	0	(M) (O)

May be inoperative provided:

- Inoperative components are properly secured by an accepted procedure.
 - Landing gear is secured in the down position.
 - Airplane is dispatched in accordance with the appropriate AFM Gear Extended Appendix.
-

MAINTENANCE (M)

Secure the landing gear in the down position (AMM 32-00-00/901).

- Install landing gear downlock pins, including installation procedure steps for airplane flight with landing gear locked in the extended position (AMM 32-00-30/201).
- Place Alternate Gear Down Dispatch switch (Main Equipment Center above the lower access door) to VMO position.

NOTE: The Warning Electronics System (WES) will set VMO/MMO to 270 KIAS/.73 Mach and VMO GEAR DOWN memo message will be displayed.

- Deactivate the lever lock solenoid for the landing gear control lever by opening and collaring the P210 panel LG LEVER LOCK circuit breaker.

OPERATIONS (O)

- Observe AFM Landing Gear Extended certificate limitations and operational limits.

NOTE: Dispatch requires authorization by the AFM Landing Gear Extended Appendix.

- Base flight plan on landing gear extended.
- The Warning Electronics System (WES) will set VMO/MMO to 270 KIAS/.73 Mach and VMO GEAR DOWN memo message will be displayed.

32-30-02 Semi-Lever Gear (SLG) System (-300ER)

Interval	Installed	Required	Procedure
C	1	0	(M) (O)

May be inoperative provided:

- a. SLG system is deactivated.
- b. Appropriate performance adjustments are applied.

MAINTENANCE (M)

Deactivate the semi-lever gear system (AMM 32-00-00/901).

1. For the SEMI LEVER GEAR or SEMI LEVER GEAR SYS status message displayed, open and collar the P110 panel SEMI-LEVER GEAR CONTROL, LEFT circuit breaker and the P210 panel SEMI-LEVER GEAR CONTROL, RIGHT circuit breaker.
2. For the SEMI LEVER GEAR LOCK status message displayed:
 - A. Open and collar the P110 panel SEMI-LEVER GEAR CONTROL, LEFT circuit breaker and the P210 panel SEMI-LEVER GEAR CONTROL, RIGHT circuit breaker.

NOTE: The SEMI LEVER GEAR SYS status message will be displayed during dispatch with the LEFT and RIGHT SEMI-LEVER GEAR CONTROL circuit breakers open.

- B. Clear the airplane sufficiently to accommodate possible downward vertical motion. For a lightly loaded airplane, vertical downward motion may be as much as 2 inches at either main gear and at the tail, and as much as 6 inches at the wing tips. No vertical downward motion is expected for an airplane heavier than 500,000 lbs (226,800 kg).
- C. Pressurize the center hydraulic system (AMM 29-11-00/201).
- D. On Landing Gear Actuation/Indication Maintenance page, confirm both SLG sensors indicate FAR.
- E. Remove power from the center hydraulic system (AMM 29-11-00/201).

OPERATIONS (O)

Dispatch with appropriate performance adjustments applied. Performance is available using the Boeing Takeoff Module (BTM).

NOTE: The SEMI LEVER GEAR SYS status message will remain displayed with the LEFT and RIGHT SEMI-LEVER GEAR CONTROL circuit breakers open.

**32-31-01 Landing Gear Lever Lock Solenoid**

Interval	Installed	Required	Procedure
C	1	0	(O)

May be inoperative provided solenoid is in the locked position.

OPERATIONS (O)

For gear retraction use the following procedure.

1. Push and hold landing gear Lever LOCK OVRD switch.
2. Position Landing Gear Lever to UP.

32-31-02 Landing Gear Selector Valve Electrical Control Circuits

Interval	Installed	Required	Procedure
C	2	1	(M)

One UP/DOWN electrical control circuit may be inoperative provided the remaining UP/DOWN circuit is verified to operate normally each flight day.

MAINTENANCE (M)

Verify that one of the electrical control circuits operates normally each flight day (AMM 32-00-00/901).

NOTE: During this test the GEAR DOOR advisory and GEAR DISAGREE caution messages may be displayed. These messages will not be displayed when this test is completed.

1. Install the landing gear downlock pins (AMM 32-00-30/201).
2. Ensure the landing gear lever is in the DN position.
3. Supply electrical power (AMM 24-22-00/201).
4. Ensure the following circuit breakers are closed.

NOTE: If the circuit breaker can not be closed, disconnect, cap and stow the electrical connector on the affected selector/bypass valve.

- A. P11 panel LDG GEAR EXTD/RETR 1
 - B. P11 panel LDG GEAR EXTD/RETR 2
5. Pressurize the center hydraulic system (AMM 29-11-00/201).
 6. Ensure the downlock pins are installed in all the landing gear before moving the control lever.
 7. Push and hold the landing gear lever LOCK OVRD switch and simultaneously move the landing gear control lever to UP.
 8. Ensure all landing gear doors open.
 9. Move the landing gear lever to DN.
 10. Ensure all landing gear doors close.
 11. Remove power from the center hydraulic system (AMM 29-11-00/201).
 12. Remove the landing gear downlock pins (AMM 32-00-30/201).

OPERATIONS NOTE

False TOO LOW GEAR aural alerts may occur. Confirm landing gear configuration. The aural caution may be inhibited by positioning the Ground Proximity Gear Override switch to OVRD.



32-32-01 Main Gear Door Uplock Spring Assemblies

Interval	Installed	Required	Procedure
B	4	2	(M) (O)

One spring on each door uplock mechanism may be missing and gear retraction allowed provided 270 KIAS/.82 Mach is not exceeded during flight.

MAINTENANCE (M)

Remove any pieces of broken spring (AMM 32-00-00/901).

OPERATIONS (O)

Observe the landing gear EXTEND placard (270K - .82M) speed limit during all phases of flight.

32-32-01 Main Gear Door Uplock Spring Assemblies**32-32-01-01 Spring Retention Cords**

Interval	Installed	Required	Procedure
C	4	0	

**32-32-02 Main Gear Uplock Springs**

Interval	Installed	Required	Procedure
B	4	3	(M) (O)

One spring on one main gear uplock mechanism may be missing and gear retraction allowed provided 270 KIAS/.82 Mach is not exceeded during flight.

MAINTENANCE (M)

Remove any pieces of broken spring (AMM 32-00-00/901).

OPERATIONS (O)

Observe the landing gear EXTEND placard (270K - .82M) speed limit during all phases of flight.

**32-32-03 Main Gear Side Brace Springs (PRR 61350-34 Not Incorporated)**

Interval	Installed	Required	Procedure
B	4	3	(M) (O)

One main gear side brace spring may be missing provided airspeed is limited to 250 knots when extending landing gear using alternate gear extend system.

NOTE: This MMEL item is applicable to airplanes with side brace downlock actuator Boeing part number 293W3152-1 installed. This MMEL item only applies to airplanes without the "toggle lock" style drag braces (line numbers 1 through 40 without SB 777-32-0057 incorporated and later line numbers with SB 777-32-0036 or SB 777-32-0055 incorporated). This MMEL item may be applied with or without incorporation of SB 777-32-0013.

MAINTENANCE (M)

1. For coil type springs, remove the broken spring using one of the following two methods:
 - A. Remove broken spring. Cut the end fittings on each end of the spring in the necked down area, leaving the remaining portion of the end fitting to act as a spacer on the shaft.
 - B. Remove broken spring per AMM 32-32-06/401 and install suitable spacers (BACB28Z10-065 or equivalent) on shafts.
2. For cartridge style springs:
 - A. Remove broken spring per AMM 32-32-06/401 and install suitable spacers (BACB28Z10-065 or equivalent) on shafts.

OPERATIONS (O)

When using the alternate landing gear extension system, observe the speed limit of 250 KIAS.

32-35-01 Landing Gear Alternate Extend System

Interval	Installed	Required	Procedure
B	1	0	(M) (O)

May be inoperative provided:

- Landing gear are secured in the down position.
- Alternate extend system is deactivated.
- Airplane is dispatched in accordance with the appropriate AFM Gear Extended Appendix.

MAINTENANCE (M)

Deactivate alternate extend system and secure the landing gear in the down position (AMM 32-00-00/901).

- Install landing gear downlock pins, including installation procedure steps for airplane flight with landing gear locked in the extended position (AMM 32-00-30/201).
- Open and collar the following P11 panel circuit breakers to deactivate the alternate extend system:
 - LDG GEAR ALTN EXT MOTOR
 - LDG GEAR ALTN EXT CTRL
- Place Alternate Gear Down Dispatch switch (Main Equipment Center above the lower access door) to VMO position.

NOTE: The Warning Electronics System (WES) will set VMO/MMO to 270 KIAS/.73 Mach and VMO GEAR DOWN memo message will be displayed.

- Deactivate the lever lock solenoid for the landing gear control lever by opening and collaring the P210 panel LG LEVER LOCK circuit breaker.

OPERATIONS (O)

- Observe AFM Landing Gear Extended certificate limitations and operational limits.

NOTE: Dispatch requires authorization by the AFM Landing Gear Extended Appendix.

- Base flight plan on landing gear extended.
- The Warning Electronics System (WES) will set VMO/MMO to 270 KIAS/.73 Mach and VMO GEAR DOWN memo message will be displayed.

32-35-01 Landing Gear Alternate Extend System**32-35-01-01 Alternate Extend Hydraulic Pressure Switch**

Interval	Installed	Required	Procedure
B	1	0	(M)

May be inoperative provided:

- a. Inoperative pressure switch is open.
 - b. Landing gear doors are verified to open using the alternate extend system.
-

MAINTENANCE (M)

Verify the landing gear doors open using the alternate extend system (AMM 32-00-00/901).

1. Install landing gear downlock pins (AMM 32-00-30/201).
2. Make sure the following P11 panel circuit breakers are closed:
 - A. LDG GEAR ALTN EXT MOTOR
 - B. LDG GEAR ALTN EXT CTRL
3. On the flight deck, use the Glareshield panel (P55) Display Select Panel (DSP) to select the Landing Gear (GEAR) synoptic on to a Multi-Function Display (MFD).
4. Position Landing Gear Panel (P2-2) Alternate Gear (ALTN GEAR) switch DOWN and hold (maximum 45 seconds) until all landing gear doors are open, indicated by in-transit (three cross-hatched boxes) on the GEAR synoptic.

NOTE: GEAR DOOR advisory message will be displayed.

5. Close the landing gear doors (AMM 32-00-40/201).

NOTE: The Ground Door Release function which closes the landing gear doors must be operative.

6. Remove landing gear downlock pins when maintenance activities are complete (AMM 32-00-30/201).

NOTE: For maintenance ground operations, the landing gear doors must be opened by placing the ALTN GEAR switch DOWN and holding (45 seconds maximum) until all gear doors are open, indicated by in-transit (three cross-hatched boxes) on the GEAR synoptic.

OPERATIONS NOTE

For Non-Normal Checklists that use the alternate gear extend system, the gear should indicate in-transit within approximately 16 seconds.

32-35-02 Ground Door Release Control System

Interval	Installed	Required	Procedure
C	1	0	(M)

May be inoperative provided:

- a. Both door open control switches are verified to be open.
- b. Landing gear doors are verified to open using the alternate extend system.
- c. Landing gear doors are closed before each departure.

MAINTENANCE (M)

Verify both door open control switches are open (not inoperative closed) and the landing gear doors will open using the alternate extend switch on the flight deck (AMM 32-00-00/901).

1. Make sure the following P11 panel circuit breakers are closed:
 - A. LDG GEAR ALTN EXT MOTOR
 - B. LDG GEAR ALTN EXT CTRL
2. Install landing gear downlock pins (AMM 32-00-30/201).
3. Verify both door open control switches are open (not inoperative closed).
 - A. Ensure the Main Wheel Well Electric Service Panel (P56) Arm Doors switch and All Doors Open/Main Landing Gear Doors Close switches are positioned OFF.
 - B. Position Arm Doors switch ARM DOORS and hold. Wait 5 seconds, and then release the switch and ensure it returns to OFF.
 - C. Visually ensure none of the gear doors open.

NOTE: If the gear doors open, a door open control switch is inoperative closed. The airplane may be dispatched using MMEL item 32-35-02-01.
 - D. Position All Doors Open/Main Landing Gear Doors Close switch ALL DOORS OPEN and hold. Wait 5 seconds, and then release the switch and ensure it returns to OFF.
 - E. Visually ensure none of the gear doors open.

NOTE: If the gear doors open, a door open control switch is inoperative closed. The airplane may be dispatched using MMEL item 32-35-02-01.
4. Verify the nose and main landing gear doors open using the alternate extend system.
 - A. Use Glareshield panel (P55) Display Select Panel (DSP) to select the Landing Gear (GEAR) synoptic on to a Multi-Function Display (MFD).



- B. Momentarily position Landing Gear Panel (P2-2) ALTN GEAR switch DOWN. All landing gear doors should open, indicated by in-transit (three cross-hatched boxes) on the GEAR synoptic.

NOTE: If landing gear doors do not open, the airplane may be dispatched using MMEL item 32-35-01.

- C. Close the landing gear doors (AMM 32-00-40/201).

NOTE: The Ground Door Release function which closes the landing gear doors must be operative.

- D. Remove landing gear downlock pins when all maintenance activities are complete (AMM 32-00-30/201).

32-35-02 Ground Door Release Control System

32-35-02-01 Door Open Control Switches

Interval	Installed	Required	Procedure
B	2	0	(M) (O)

May be inoperative provided:

- Inoperative control switch(s) is closed.
- Landing gear are secured in the down position.
- Landing gear alternate extend system is deactivated.
- Airplane is dispatched in accordance with the appropriate AFM Gear Extended Appendix.

MAINTENANCE (M)

Deactivate the gear alternate extend system and secure the landing gear down (AMM 32-00-00/901).

- Install landing gear downlock pins, including installation procedure steps for airplane flight with landing gear locked in the extended position (AMM 32-00-30/201).
- Open and collar the following P11 panel circuit breakers to deactivate the alternate extend system:
 - LDG GEAR ALTN EXT MOTOR
 - LDG GEAR ALTN EXT CTRL
- Place Alternate Gear Down Dispatch switch (Main Equipment Center above the lower access door) to VMO position.

NOTE: The Warning Electronics System (WES) will set VMO/MMO to 270 KIAS/.73 Mach and VMO GEAR DOWN memo message will be displayed.

- Deactivate the lever lock solenoid for the landing gear control lever by opening and collaring the P210 panel LG LEVER LOCK circuit breaker.

**OPERATIONS (O)**

1. Observe AFM Landing Gear Extended certificate limitations and operational limits.

NOTE: Dispatch requires authorization by the AFM Landing Gear Extended Appendix.
2. Base flight plan on landing gear extended.
3. The Warning Electronics System (WES) will set VMO/MMO to 270 KIAS/.73 Mach and VMO GEAR DOWN memo message will be displayed.

**32-41-01 Brake Accumulator Pressure Indicator (Wheel Well)**

Interval	Installed	Required	Procedure
D	1	0	

May be inoperative provided flight deck brake accumulator pressure indicator operates normally.

**32-41-02 Brake Accumulator Pressure Indicator (Flight Deck)**

Interval	Installed	Required	Procedure
C	1	0	(M)

May be inoperative provided brake accumulator charge is verified normal once each flight day.

MAINTENANCE (M)

Verify brake accumulator precharge pressure once each flight day (AMM 12-15-04/301).





- J. On Landing Gear Brakes/Steering Maintenance page, confirm left and right BRAKE METERED PRESS for the NORMAL brake hydraulic system is approximately 3,000 psi.
- K. Release the captain's brake pedals (or parking brake).
- L. Examine each brake's pistons to make sure that all brakes are released.
- M. Remove power and depressurize the right hydraulic system (AMM 29-11-00/201).
- 1) Position Hydraulic Panel (P5) right system hydraulic pumps off.
 - 2) Position Overhead Maintenance Panel (P61) C TAIL FLT CONTROL HYD POWER switch SHUTOFF.
 - 3) Position Overhead Maintenance Panel (P61) R TAIL FLT CONTROL HYD POWER switch NORM.
 - 4) Operate the pitch trim with the alternate pitch trim lever to decrease the pressure in the right hydraulic system to zero.
- N. Ensure the brake accumulator pressure is between 2,800 and 3,000 psi.
- O. Quickly apply the captain's brake pedals full down. There should be no momentary decrease in brake accumulator pressure.
- NOTE: Steady pressure when the captain's brakes are applied indicates the accumulator isolation valve is in the secondary position, which isolates the accumulator.
- P. Release the captain's brake pedals.
- Q. Fully push and hold the captain's brake pedals.
- NOTE: The parking brake can be set to hold the pedals.
- R. Examine each brake's pistons to make sure that all brakes operated.
- S. On Landing Gear Brakes/Steering Maintenance page, confirm left and right BRAKE METERED PRESS for the ALTN brake hydraulic system is approximately 3,000 psi.
- T. Release the captain's brake pedals (or parking brake).
- U. Examine each brake's pistons to make sure that all brakes are released.
- V. Remove power from the center hydraulic system (AMM 29-11-00/201).
- W. Position Overhead Maintenance Panel (P61) C TAIL FLT CONTROL HYD POWER switch NORM.
- X. Remove the nose and main landing gear downlock pins (AMM 32-00-30/201).
- Y. Remove the landing gear door safety pins (AMM 32-00-15/201).

**OPERATIONS (O)**

Establish alternate procedures.

1. During taxi and landing operations, monitor the brake accumulator pressure indicator.
2. If the brake accumulator pressure is less than 2800 psi for longer than three seconds, accomplish the BRAKE SOURCE Non-Normal Checklist.

32-41-03 BRAKE SOURCE Indication System**32-41-03-01 BRAKE SOURCE Light**

Interval	Installed	Required	Procedure
C	1	0	

May be inoperative provided the BRAKE SOURCE alerting message operates normally.

MAINTENANCE NOTE

The BRAKE SOURCE advisory message will not operate for towing operations conducted without AC electrical power applied to the airplane.

32-42-01 Normal Antiskid Valves
32-42-01A Deactivation Tool Used

Interval	Installed	Required	Procedure
C	12	10	(M) (O)

One valve per each six wheel truck may be inoperative provided:

- Associated brake is deactivated using the deactivation assembly.
- AFM performance decrements for brake(s) deactivated are applied.

MAINTENANCE (M)

Deactivate the brake associated with inoperative antiskid valve using DDG item 32-45-01A (M) procedures.

OPERATIONS (O)

NOTE: Operations may be affected at contiguous United States airports impacted by 5G C-Band interference [Ref FAA Airworthiness Directive (AD) 2023-12-05 and associated Boeing Flight Crew Operations Manual Bulletin (OMB)]

Dispatch with takeoff and landing performance based on one or two brakes deactivated.

32-42-01 Normal Antiskid Valves
32-42-01B Brake Line Capped

Interval	Installed	Required	Procedure
C	12	10	(M) (O)

One valve per each six wheel truck may be inoperative provided:

- Associated brake is deactivated by capping the brake line.
- After takeoff, gear remains down for two minutes before retraction.
- AFM performance decrements for brake(s) deactivated are applied.
- Takeoff performance is based on landing gear extended.

MAINTENANCE (M)

Deactivate the brake associated with inoperative antiskid valve using DDG item 32-45-01B (M) procedures.

**OPERATIONS (O)**

NOTE: Operations may be affected at contiguous United States airports impacted by 5G C-Band interference [Ref FAA Airworthiness Directive (AD) 2023-12-05 and associated Boeing Flight Crew Operations Manual Bulletin (OMB)]

1. After takeoff, leave the landing gear extended for a minimum of two minutes to allow the wheels to spin down prior to gear retraction.
2. Base takeoff performance on landing gear extended. Takeoff performance for the landing gear extended is available using AFM-DPI or Boeing Takeoff Module (BTM).
3. Dispatch with takeoff and landing performance also based on one or two brakes deactivated.
4. Increase flight planning fuel by 1,900 lbs (861 kg).

**32-42-02 Antiskid Wheelspeed Transducers**

Interval	Installed	Required	Procedure
C	12	10	(O)

One transducer per each six wheel truck may be inoperative provided:

- AFM performance decrements for brake(s) deactivated are applied.
- No other brakes are deactivated on that truck.

MAINTENANCE NOTE

Brakes associated with inoperative transducers are not required to be deactivated.

OPERATIONS (O)

NOTE: Operations may be affected at contiguous United States airports impacted by 5G C-Band interference [Ref FAA Airworthiness Directive (AD) 2023-12-05 and associated Boeing Flight Crew Operations Manual Bulletin (OMB)]

Dispatch with takeoff and landing performance based on one or two brakes deactivated.

**32-42-03 Alternate Antiskid Valves**

Interval	Installed	Required	Procedure
C	8	0	(M)

May be inoperative provided manual braking capability on the alternate brake system is verified to operate normally.

MAINTENANCE (M)

Verify that pedal braking capability operates normally (AMM 32-00-00/901).

1. Pressurize the center hydraulic system (AMM 29-11-00/201).
2. Remove power on the right hydraulic system (AMM 29-11-00/201).
3. Depress and release the brake pedals. Observe brake piston movement on each brake to check that brakes apply and release.
4. Make sure that all 12 brakes are set.
5. Release the brake pedals.
6. Remove power on the center hydraulic system (AMM 29-11-00/201).

32-42-04 Autobrake System (Including Autobrake Solenoid Valve)
32-42-04A Solenoid Verified Closed

Interval	Installed	Required	Procedure
C	1	0	(M)

May be inoperative provided:

- a. Autobrake solenoid valve is verified closed.
- b. Autobrake selector remains in the OFF position.

MAINTENANCE (M)

Verify that the autobrake solenoid valve is closed and for Brake System Control Unit II (BSCU II) installed, check the program pin configuration (AMM 32-00-00/901).

1. Supply electrical power (AMM 24-22-00/201).
2. Chock the wheels.
3. Release the parking brake.
4. Place the autobrake selector to the OFF position.
5. Pressurize the right hydraulic system (AMM 29-11-00/201).
6. For AUTOBRAKE SOL VALVE status message not displayed, the autobrake solenoid valve is closed.

NOTE: If AUTOBRAKE SOL VALVE Status message is displayed, the autobrake valve module must be deactivated for dispatch. Refer to MMEL item 32-42-04B.

7. For a BSCU II (P/N S294W401-series) installed, use the Antiskid/Autobrake Control System configuration page of the MAT to confirm the pin program configuration of the BSCU primary and secondary channel.

NOTE: The presence of maintenance message 32-30701 or 32-30702, indicates that the BSCU pin programming selection may be faulted.

For -200/-200ER/-300:

- A. Confirm pin program configuration is 0000.

For -200LR/777F/-300ER:

- B. Confirm pin program configuration is 0001.

32-42-04 Autobrake System (Including Autobrake Solenoid Valve)
32-42-04B Control Module Deactivated

Interval	Installed	Required	Procedure
C	1	0	(M)

May be inoperative provided:

- Autobrake hydraulic control module is deactivated.
- Autobrake selector remains in the OFF position.

MAINTENANCE (M)

Deactivate the autobrake hydraulic control module and for Brake System Control Unit II (BSCU II) installed, check the program pin configuration (AMM 32-00-00/901).

- Remove Autobrake Valve Module (AMM 32-42-09/401). The autobrake valve module is installed in the right hand wheel well on the forward end of the center panel.
- Cap and stow the electrical connectors (DM32018).
- Plug and stow the hydraulic tubes. Use two high pressure steel plugs for the hydraulic pressure tubes (MS21913J10, AS5231J10, BACP20AU10J or equivalent) and an aluminum plug for the return tube (MS21913W8, AS5231W08, BACP20AU8D or equivalent).
- Pressurize the right hydraulic system and check for leaks (AMM 29-11-00/201).
- Remove power from the right hydraulic system (AMM 29-11-00/201).
- For a BSCU II (P/N S294W401-series) installed, use the Antiskid/Autobrake Control System configuration page of the MAT to confirm the pin program configuration of the BSCU primary and secondary channel.

NOTE: The presence of maintenance message 32-30701 or 32-30702, indicates that the BSCU pin programming selection may be faulted.

For -200/-200ER/-300:

- Confirm pin program configuration is 0000.

For -200LR/777F/-300ER:

- Confirm pin program configuration is 0001.

**32-44-01 Brake Status Lights (On Nose Gear)****32-44-01A Required By Procedures**

Interval	Installed	Required	Procedure
C	3	0	(O)

May be inoperative provided alternate procedures are established and used.

OPERATIONS (O)

Alternate procedures must be established to ensure ground crew knowledge of brake system status.

32-44-01 Brake Status Lights (On Nose Gear)**32-44-01B Procedures Do Not Require Use**

Interval	Installed	Required	Procedure
D	3	0	

May be inoperative provided procedures do not require their use.

**32-44-02 Gear Retraction Braking System**

Interval	Installed	Required	Procedure
C	1	0	(O)

May be inoperative provided:

- After takeoff, gear remains down for two minutes before retraction.
- Appropriate performance adjustments are applied.

NOTE: In the event of engine failure after V1, retract landing gear after takeoff.

OPERATIONS (O)

NOTE: It is not necessary to apply landing gear extended takeoff performance adjustments.

- Increase flight planning fuel by 1,900 lbs (861 kg).
- After takeoff, leave the landing gear extended for a minimum of two minutes to allow the wheels to spin down prior to gear retraction.
- In case of engine failure after V1, landing gear should be retracted after takeoff.

32-44-03 Parking Brake Set Indication System (Flight Deck)

Interval	Installed	Required	Procedure
C	1	0	(M) (O)

May be inoperative provided:

- a. Alternate procedures are established and used.
- b. Parking brake valve is verified to operate normally once each flight day.
- c. Parking brake and brake antiskid interface indication is verified to operate normally once each flight day.
- d. Parking brake takeoff configuration warning system is verified to operate normally.

MAINTENANCE (M)

NOTE: This MMEL item is for PARKING BRAKE SET memo message not displayed when the parking brake lever is set.

Verify the parking brake takeoff configuration warning operates normally and verify once each flight day the parking brake valve, the parking brake and the brake antiskid interface indication operate normally (AMM 32-00-00/901).

1. Supply electrical power on the airplane (AMM 24-22-00/201).
2. Install chocks around the tires.
3. Install downlock pins in the nose and main landing gear (AMM 32-00-30/201).
4. Pressurize the right hydraulic system (AMM 29-11-00/201).
5. Release the parking brake.
6. For verifying the parking brake valve operates normally once each flight day.
 - A. Gain access to parking brake valve by opening the right inboard fixed trailing edge panel 651DB. The parking brake valve is on the right main landing gear trunnion, outboard and aft of the normal antiskid shuttle valve module.
 - B. Verify that the parking brake valve manual override lever indicates the valve is open.
 - C. Set the parking brake.
 - D. Verify the parking brake valve manual override lever indicates the valve is fully closed.
 - E. Close the right inboard fixed trailing edge panel 651DB.
7. For verifying the parking brake and brake antiskid interface indication operates normally once each flight day.
 - A. Open P11 panel LDG GEAR PARKING BRAKE VALVE circuit breaker.

-
- B. Release the parking brake.
 - C. Verify the ANTISKID advisory message is displayed.
 - D. Close P11 panel LDG GEAR PARKING BRAKE VALVE circuit breaker.
 - E. Verify the ANTISKID advisory message is not displayed.
8. For verifying the parking brake takeoff configuration warning system operates normally.
- A. Set the parking brake.
 - B. Perform the Warning Electronic System (WES) Takeoff Configuration Test (AMM 31-51-00/501).
 - C. During the test, verify the CONFIG PARKING BRAKE warning message is displayed, the master warning light illuminates and the siren sounds.
 - D. Release the parking brake.
 - E. Perform the Warning Electronic System (WES) Takeoff Configuration Test (AMM 31-51-00/501).
 - F. During the test, verify the CONFIG PARKING BRAKE warning message is not displayed.
 - G. Ensure that when the test is completed that PASSED shows adjacent to TEST CONDITION.
9. Remove power from the right hydraulic system (AMM 29-11-00/201).
10. Remove downlock pins in the nose and main landing gear (AMM32-00-30/201).

OPERATIONS (O)

Establish and use alternate procedures to ensure ground crew and flight crew communication regarding parking brake status. Procedures should ensure flight crew manually confirm the position of the parking brake lever.

32-45-01 Wheel Brakes
32-45-01A Deactivation Tool Used

Interval	Installed	Required	Procedure
C	12	10	(M) (O)

One per each six wheel truck may be inoperative provided:

- Associated brake is deactivated with a deactivation assembly.
- AFM performance decrements for brakes deactivated are applied.

MAINTENANCE (M)

Deactivate associated brake using the Deactivation Assembly (AMM 32-00-00/901).

- Supply electrical power (AMM 24-22-00/201).
- Install landing gear downlock pins (AMM 32-00-30/201).
- Remove power on the right and center hydraulic systems (AMM 29-11-00/201).
- Install chocks around the tires.
- Release the parking brake.
- Fully push the brake pedals 12 times to remove the pressure from the brake accumulator.
- Use the MAT to disable the ANTISKID advisory message for the associated brake.

NOTE: The ANTISKID advisory message may reappear if airplane power is cycled or if power to AIMS is cycled. If this happens, repeat the deactivation procedure on the MAT.

- Remove the brake deactivation assembly from its location on the Antiskid Shuttle Valve Module. The assembly is referred to as the Flight Dispatch Disconnect (specification number S294W121-51) vendor part number 80389.
- Remove the cover from the applicable shuttle valve and install the deactivation assembly in place of the valve cover.
- Pressurize the right hydraulic system (AMM 29-11-00/201).
- Fully press the brake pedals. Make sure the applicable brake does not operate by observing brake piston movement.
- Remove power from the right hydraulic system (AMM 29-11-00/201).
- Remove landing gear downlock pins when maintenance and servicing actions are complete (AMM 32-00-30/201).

**OPERATIONS (O)**

NOTE: Operations may be affected at contiguous United States airports impacted by 5G C-Band interference [Ref FAA Airworthiness Directive (AD) 2023-12-05 and associated Boeing Flight Crew Operations Manual Bulletin (OMB)]

Dispatch with takeoff and landing performance based on one or two brakes deactivated.

32-45-01 Wheel Brakes**32-45-01B Brake Line Capped**

Interval	Installed	Required	Procedure
C	12	10	(M) (O)

One per each six wheel truck may be inoperative provided:

- Associated brake is deactivated by capping the brake line.
- After takeoff, gear remains down for two minutes before retraction.
- AFM performance decrements for brakes deactivated are applied.
- Takeoff performance is based on landing gear extended.

MAINTENANCE (M)

Deactivate inoperative brake by capping the associated brake line (AMM 32-00-00/901).

- Supply electrical power (AMM 24-22-00/201).
- Install landing gear downlock pins (AMM 32-00-30/201).
- Remove power on the right and center hydraulic systems (AMM 29-11-00/201).
- Install chocks around the tires.
- Release the parking brake.
- Fully push the brake pedals 12 times to remove the pressure from the brake accumulator.
- Use the MAT to disable the ANTISKID advisory message for the associated brake.

NOTE: The ANTISKID advisory message may reappear if airplane power is cycled or if power to AIMS is cycled. If this happens, repeat the deactivation procedure on the MAT.

- Uncouple the Brake Disconnect - Brake Hose Half and attach the hose to the landing gear truck using P/N BACS38K3D.

NOTE 1: The part of the Brake Disconnect that is attached to the hose is called the Brake Disconnect - Brake Hose Half. The part of the Brake Disconnect that is attached to the brake is called the Brake Disconnect - Coupling.

NOTE 2: The brake lines are self sealing, thus disconnecting the brake line and stowing represents capping.

9. Pressurize the right hydraulic system (AMM 29-11-00/201).
10. Apply brake pressure and check for leaks.
11. Make sure the applicable brake does not operate by observing brake piston movement.
12. Remove power from the right hydraulic system (AMM 29-11-00/201).
13. Remove landing gear downlock pins when maintenance and servicing actions are complete (AMM 32-00-30/201).

OPERATIONS (O)

NOTE: Operations may be affected at contiguous United States airports impacted by 5G C-Band interference [Ref FAA Airworthiness Directive (AD) 2023-12-05 and associated Boeing Flight Crew Operations Manual Bulletin (OMB)]

1. After takeoff, leave the landing gear extended for a minimum of two minutes to allow the wheels to spin down prior to gear retraction.
2. Base takeoff performance on landing gear extended. Takeoff performance for the landing gear extended is available using AFM-DPI or Boeing Takeoff Module (BTM).
3. Dispatch with takeoff and landing performance also based on one or two brakes deactivated.
4. Increase flight planning fuel by 1,900 lb (861 kg).

32-45-01 Wheel Brakes

32-45-01C Center Axle Brake Removed

Interval	Installed	Required	Procedure
C	12	10	(M) (O)

One per each six wheel truck may be removed from the center axle only provided:

- a. Associated brake line is capped.
- b. After takeoff, gear remains down for two minutes before retraction.
- c. AFM performance decrements for brakes deactivated are applied.
- d. Takeoff performance is based on landing gear extended.

MAINTENANCE (M)

Remove inoperative brake and cap the associated brake line (AMM 32-00-00/901).

1. Use the MAT to disable the ANTISKID advisory message for the associated brake.

NOTE: The ANTISKID advisory message may reappear if airplane power is cycled or if power to AIMS is cycled. If this happens, repeat the deactivation procedure on the MAT.

2. Remove the applicable brake (AMM 32-45-07/401).
3. Attach the brake hose to the gear truck. This will ensure that the hose will not interfere with brake rotation when the gear is retracted.
4. Cap and stow the electrical wiring for the brake temperature sensor.

NOTE: Ensure the brake sleeve is installed on the axle before the tire and wheel assembly are installed.

5. Install the tire and wheel assembly (AMM 32-45-01/401).
6. Pressurize the right hydraulic system (AMM 29-11-00/201).
7. Apply brake pressure and check for leaks.
8. Remove power from the right hydraulic system (AMM 29-11-00/201).
9. Remove landing gear downlock pins when maintenance and servicing actions are complete (AMM 32-00-30/201).

OPERATIONS (O)

NOTE: Operations may be affected at contiguous United States airports impacted by 5G C-Band interference [Ref FAA Airworthiness Directive (AD) 2023-12-05 and associated Boeing Flight Crew Operations Manual Bulletin (OMB)]

1. After takeoff, leave the landing gear extended for a minimum of two minutes to allow the wheels to spin down prior to gear retraction.
2. Base takeoff performance on landing gear extended. Takeoff performance for the landing gear extended is available using AFM-DPI or Boeing Takeoff Module (BTM).
3. Dispatch with takeoff and landing performance also based on one or two brakes deactivated.
4. Increase flight planning fuel by 1,900 lb (861 kg).

32-45-02 Wheel Tie Bolts

Interval	Installed	Required	Procedure
A	-	-	(M)

One per wheel may be broken or missing provided:

- Affected wheel is removed, checked for broken parts or damage, and replaced if broken parts or damage is found.
- Associated brake is checked for broken parts or damage, and is replaced or deactivated if broken parts or damage is found.
- After each landing, wheel is inspected for additional broken or missing tie bolts.
- Operations are limited to five departures before repairs are made.

MAINTENANCE (M)

Remove the wheel and tire assembly, and inspect the wheel and brake assemblies for broken parts or damage. Inspect the wheel after each landing and replace the affected wheel after five departures (AMM 32-00-00/901).

- On the main landing gear, deflate affected wheel to 40 psig or less. A defective wheel and tire can burst if the tire is not deflated before removal of the assembly (AMM 32-45-01/401).
- On main landing gear, remove each wheel with a broken, loose or missing tiebolt (AMM 32-45-01/401).
- Inspect the wheel for broken or missing parts or damage. On nose landing gear wheels, use a mirror to fully view the inboard side of the wheel (AMM 32-45-03/601).
- If other bolts are broken, loose or missing, or if other areas of the wheel have been damaged, the affected wheel is to be replaced.
- On main landing gear, inspect the brake for broken or missing parts or damage (AMM 32-45-07/601).
- On main landing gear re-install the affected wheel (AMM 32-45-01/401).
- Inflate tire to correct pressure (AMM 12-15-03/301).

NOTE: Do not move the airplane while the tire is deflated. If the airplane is moved, the deflated tire and associated wheel, and the wheel and tire on the opposite side of the axle must be replaced.

- After each landing (up to 5 maximum), inspect remaining tiebolts on the affected wheel (wheel removal not required) to ensure no other bolts are broken, loose or missing (AMM 32-45-03/601).
- Replace the affected wheel after a maximum of 5 subsequent landings.

**32-45-03****Integral Tire Pressure Indicators**

Interval	Installed	Required	Procedure
D	-	0	

**32-45-04 Nose Gear Spin Brake**

Interval	Installed	Required	Procedure
C	2	0	(O)

May be inoperative or missing provided:

- After takeoff, gear remains down for two minutes before retraction.
- Appropriate performance adjustments are applied.

NOTE: In the event of engine failure after V1, retract landing gear after takeoff.

OPERATIONS (O)

NOTE: It is not necessary to apply landing gear extended takeoff performance adjustments.

- Increase flight planning fuel by 1,900 lb (861 kg).
- After takeoff, leave the landing gear extended for a minimum of two minutes prior to gear retraction.
- In case of engine failure after V1, landing gear should be retracted after takeoff.

**32-46-01 Brake Temperature Indication System**

Interval	Installed	Required	Procedure
C	1	0	(O)

May be inoperative provided AFM Maximum Quick Turnaround Weight limitations are observed.

[For brake cooling fans installed:](#)

NOTE 1: Brake cooling fans will be inoperative when all brake temperature indications are inoperative.

NOTE 2: Any portion of the system that operates normally may be used.

OPERATIONS (O)

Observe AFM Maximum Quick Turnaround Weight limitation.

32-47-01 Brake Cooling Fans

Interval	Installed	Required	Procedure
D	12	0	(M)

May be inoperative provided associated fan shrouds are removed.

MAINTENANCE (M)

Remove the inoperative brake cooling fan shroud (AMM 32-00-00/901).

NOTE: Brake cooling fans (if installed) will be inoperative when all brake temperature indications are inoperative.

1. Install landing gear downlock pins (AMM 32-00-50/201).
2. Install landing gear door safety pins (AMM 32-00-15/201).
3. Install chocks on the main landing gear wheels.
4. Remove center system hydraulic power (AMM 29-11-00/201).
5. Deactivate main landing gear steering system (AMM 32-00-50/201).
6. Open and collar the associated circuit breaker for the inoperative brake cooling fan.
 - A. BRAKE FAN WHEEL 1, 2, 5, 6, 9 and 10 circuit breakers are on the P110 panel.
 - B. BRAKE FAN WHEEL 3, 4, 7, 8, 11 and 12 circuit breakers are on the P210 panel.
7. Remove the finger clamp from the brake cooling fan assembly and from the shroud (Reference AMM 32-47-01 Figure 401).
8. Remove the shroud from the fan assembly and the mounting ring (retain shroud for subsequent fan repair action).
9. Install the finger clamp on the fan assembly.
10. Activate main landing gear steering system (AMM 32-00-50/201).
11. Remove chocks on the main landing gear wheels.
12. Remove landing gear door safety pins (AMM 32-00-15/201).
13. Remove landing gear downlock pins (AMM 32-00-50/201).

**32-49-01 Tire Pressure Indication System**

32-49-01A Required By Procedures

Interval	Installed	Required	Procedure
C	1	0	(M)

May be inoperative provided alternate procedures are established and used.

NOTE: Any portion of the system that operates normally may be used.

MAINTENANCE (M)

1. Use an alternate means to measure tire pressure (AMM 12-15-03/301).
2. If desired, Tire Pressure Indication System may be deactivated by opening and collaring P110 panel LG TIRE PRESSURE INDICATOR circuit breaker.
3. If desired, the tire pressure indication channel for the affected wheel may be deactivated when a wheel with incompatible tire pressure indication hardware is installed. For wheels with compatible hardware no action is suggested.
 - A. For the Messier-Bugatti shipside system without sensor holder installed in the wheel:
 - 1) Remove the tire pressure sensor (AMM 32-49-03-000-801).
 - 2) Reinstall hubcap (AMM 32-45-08-420-801).
 - B. For the Crane shipside system without integral tire fill valve/tire pressure sensor installed in the wheel:
 - 1) Remove hubcap with integral relay transformer (AMM 32-45-08-020-801).
 - 2) For main wheel, install TPIS Deactivation hubcap to restore antiskid wheel speed drive function (AMM 32-45-08-420-801).

[For -200/-200ER/-300:](#)

- a. Use Boeing P/N 294W5140-16, or -18.

[For -200LR/777F/-300ER:](#)

- b. Use Boeing P/N 294W5140-17, or -19.

32-49-01 Tire Pressure Indication System

32-49-01B Procedures Do Not Require Use

Interval	Installed	Required	Procedure
D	1	0	



May be inoperative provided procedures do not require its use.

MAINTENANCE NOTE

1. If desired, Tire Pressure Indication System may be deactivated by opening and collaring P110 panel LG TIRE PRESSURE INDICATOR circuit breaker.
 2. If desired, the tire pressure indication channel for the affected wheel may be deactivated when a wheel with incompatible tire pressure indication hardware is installed. For wheels with compatible hardware no action is suggested.
 - A. For the Messier-Bugatti shipside system without sensor holder installed in the wheel:
 - 1) Remove the tire pressure sensor (AMM 32-49-03-000-801).
 - 2) Reinstall hubcap (AMM 32-45-08-420-801).
 - B. For the Crane shipside system without integral tire fill valve/tire pressure sensor installed in the wheel:
 - 1) Remove hubcap with integral relay transformer (AMM 32-45-08-020-801).
 - 2) For main wheel, install TPIS Deactivation hubcap to restore antiskid wheel speed drive function (AMM 32-45-08-420-801).
- For -200/-200ER/-300:
- a. Use Boeing P/N 294W5140-16, or -18.
- For -200LR/777F/-300ER:
- b. Use Boeing P/N 294W5140-17, or -19.

**32-53-01 Main Gear Steering System (Including Indication)**

Interval	Installed	Required	Procedure
C	1	0	(M)

May be inoperative provided:

- Main gear steering actuators are verified locked in the center position.
 - Main gear steering system is deactivated.
-

NOTE: PRR 62606 is applicable to 777F airplane line number 1768 and-on.

MAINTENANCE (M)

Verify that both main gear steering actuators are locked in the center position and deactivate the main gear steering system (AMM 32-00-00/901).

For 777F with PRR 62606 or equivalent installed:

- Deactivate the Main Gear Steering System.
 - Open and collar P110 panel LG MAIN GEAR STEERING circuit breaker to remove power from the Main Gear Steering Control Unit (MGSCU).
- Verify left and right main gear steering Power Control Actuators (PCA) are hydraulically locked in the centered position.
 - Ensure the Manual Shut-Off Valve handle is in the OFF position.
 - Withdraw the lock pin on the Manual Shut-Off Valve handle.
 - Rotate the Manual Shut-Off Valve handle to the OFF position.
 - Release the lock pin to hold the Manual Shut-Off Valve handle in the OFF position.
 - Ensure the aft axle is in the centered position.
 - Observe the rod end of the PCA. Visually confirm indication fitting at the rod end is abutting/depressing the switch plunger.
 - Observe the clevis end of the PCA. Visually confirm that the switch depression arm on the Aft Torque link assembly is abutting/depressing the switch plunger.

NOTE: If the axis is not centered, proceed to 2.C.

 - Ensure the main gear shutoff PCA Manual Shut-Off Valve handle is in the STEERED position.
 - Pull back the lock pin on the manual shut off handle.
 - Rotate the Manual Shut-Off Valve handle to the STEERED position.
 - Release the lock pin to hold the manual shut off handle in the STEERED position.

NOTE: The main gear steering actuators are biased to the center position with the Shut-Off Valve handle in the STEERED position.

For -200LR/-300ER/777F with PRR 62606 or equivalent not installed:

3. Verify left and right main gear steering Power Control Actuators (PCA) are locked in the centered position.
 - A. Look through the lock indication window on each PCA and if you can not see the metal pin on the PCA, the PCA is locked.
 - B. If you can see the metal pin on either PCA, it is not locked. Use the Push Back method (preferred) or the Check Valve Bleeding method to move the unlocked PCA into the locked position.

1) Push Back Method:

- a. Connect a tow vehicle to the nose landing gear or forward tow fitting on the affected main landing gear (AMM 09-11-00/201).
- b. Move the towing lever, on the forward side of the nose landing gear, to the TOW position.
- c. Release the parking brake.
- d. Move the manual shutoff valve handle on the unlocked PCA to the OFF position to isolate the PCA.
- e. Remove all chocks from around the nose landing gear tires and the main landing gear tires.
- f. Put a chock behind the aft axle tire that is nearest the unlocked PCA (wheel #10 on the left gear, wheel #12 on the right gear).

NOTE: Do not put a chock behind the other aft axle tire (wheel #9 on the left gear, wheel #11 on the right gear).

- g. Use the tow vehicle to slowly push the airplane rearwards onto the chock. Stop airplane movement if the airplane tire moves onto the chock, the PCA locks, or the chock moves.
- h. Look through the lock indication window on the PCA and if you can not see the metal pin on a PCA, the PCA is locked.
- i. If a PCA remains unlocked, place a chock behind the aft axle tire that is farthest from the unlocked PCA, but on the same truck (wheel #9 on the left gear, wheel #11 on the right gear).

NOTE: Do not place a chock behind the other aft axle tire (wheel #10 on the left gear, wheel #12 on the right gear).

- j. Use the tow vehicle to slowly push the airplane rearwards onto the chock. Stop airplane movement if the airplane tire moves onto the chock, the PCA locks, or the chock moves.
 - k. Look through the lock indication window on the PCA and if you can not see the metal pin on the PCA, the PCA is locked.
- 2) Check Valve Bleeding Method:

NOTE: This method loosens the check valve on the unlocked PCA. If you do this, you may void the warranty with the supplier Parker/Abex/NWL. It is recommended that this method be used only after you have tried the Push Back Method. Use caution. The hydraulic fluid trapped in the PCA is under pressure, even when the center hydraulic system is depressurized.

- a. Make sure the parking brake is set.
- b. Lift the aft axle of the main landing gear with the unlocked PCA until the aft tires do not touch the ground (AMM 07-11-03/201).

NOTE: Make sure you use the jack pad that is part of the pivot pin for the aft axle.

- c. Depressurize the center hydraulic system including the reservoir (AMM 29-11-00/201).
- d. Move the manual shutoff valve handle on the unlocked PCA to the OFF position to isolate the PCA.
- e. Use cleaner, TEC 86-2 (or equivalent) to clean the area around the check valve (anti-cavitation check valve) on the unlocked PCA.
- f. Remove the sealant and the lockwire from the check valve on the unlocked PCA.
- g. Put a rag around the unlocked PCA valve to catch the leak of fluid from the valve.
- h. Slowly turn the check valve 1 to 2 turns counter-clockwise until hydraulic fluid starts to leak from under the head of the check valve.

NOTE: Do not remove the check valve from the PCA.

- i. Wait until the hydraulic fluid does not leak from under the check valve.
- j. Push on the right aft tire (wheel #10 on the left gear, wheel #12 on the right gear) in the forward direction to apply a compressive force on the unlocked PCA. You can use a lever between tires #5 and #9 on the left gear and #7 and #11 on the right gear to help apply the force on the PCA.

NOTE: It may be necessary to apply the force for 30 to 60 seconds. Hydraulic fluid may leak from under the head of the check valve when you apply the force.

- k. Look through the lock indication window on the PCA and if you can not see the metal pin on the PCA, the PCA is locked.
- l. If the PCA is not locked, wait approximately 5 minutes and then reapply the force.
- m. Look through the lock indication window on the PCA and if you can not see the metal pin on the PCA, the PCA is locked.
- n. Tighten the check valve to 175-200 pound-inches.
- o. Move the manual shutoff valve handle on the PCA to the STEER position to activate the PCA.
- p. Pressurize the center hydraulic system (AMM 29-11-00/201).
- q. Ensure there is no hydraulic leakage from under the head of the check valve. If the check valve leaks, replace the seals on the check valve.
- r. Lower the aft axle and remove the jack.
- s. Install lockwire on the check valve.
- t. Apply sealant, MIL-A-46146 around the interface between the check valve and the PCA manifold and allow to cure.

NOTE: Application of sealant may be temporarily deferred.

- 4. Deactivate the main gear steering system.
 - A. Shutoff hydraulic power to both PCAs.
 - 1) Pull the ball-lock pin, on the handle for the manual shutoff valve, aft.
 - 2) Move the handle for the manual shutoff valve to the OFF position.
 - 3) Release the ball-lock pin to hold the handle in the OFF position.
 - B. Open and collar P110 panel LG MAIN GEAR STEERING circuit breaker.

NOTE: Removing power from the Main Gear Steering Control Unit (MGSCU) will replace correlated maintenance messages with correlation to maintenance message 32-68840.

OPERATIONS NOTE

- 1. There is negligible change in U-turn widths if the Flight Crew Training Manual Pivot Turn Technique is used.



-
2. Minimum radius turns will result in the following effects:
 - A. Substantially increased tire scuffing.
 - B. Increased thrust required for the turn.
 - C. Somewhat rougher cabin ride, particularly in the area above the main gear.
 - D. For -200/-200ER/-300 airplane line numbers 1 through 596 without Service Bulletin SB 777-32-0058, increased risk of damage to Main Landing Gear Hydraulic Line Forward Support Link Assembly.
 - E. For additional information on SB 777-32-0058, refer to 777-FTD-32-01019.

**32-61-01 Landing Gear Position Indication Systems**

Interval	Installed	Required	Procedure
B	2	0	(M) (O)

May be inoperative provided:

- Landing gear are secured in the down position.
- Airplane is dispatched in accordance with the appropriate AFM Gear Extended Appendix.

MAINTENANCE (M)

Secure the landing gear in the down position (AMM 32-00-00/901).

- Install landing gear downlock pins, including installation procedure steps for airplane flight with landing gear locked in the extended position (AMM 32-00-30/201).
- Place Alternate Gear Down Dispatch switch (Main Equipment Center above the lower access door) to VMO position.

NOTE: The Warning Electronics System (WES) will set VMO/MMO to 270 KIAS/.73 Mach and the VMO GEAR DOWN memo message will be displayed.

- Deactivate the lever lock solenoid for the landing gear control lever by opening and collaring the P210 panel LG LEVER LOCK circuit breaker.

OPERATIONS (O)

- Observe AFM Landing Gear Extended certificate limitations and operational limits.

NOTE: Dispatch requires authorization by the AFM Landing Gear Extended Appendix.

- Base flight plan on landing gear extended.
- The Warning Electronics System (WES) will set VMO/MMO to 270 KIAS/.73 Mach and the VMO GEAR DOWN memo message will be displayed.

32-61-01 Landing Gear Position Indication Systems**32-61-01-01 Truck Tilt Sensors**

Interval	Installed	Required	Procedure
C	4	0	(M) (O)



May be inoperative provided the nose gear lock, nose gear down, main gear side brace and main gear drag brace sensors are verified to operate normally each flight day.

NOTE: The airplane must also be dispatched using MMEL item 27-62-01.

MAINTENANCE (M)

NOTE: An inoperative truck tilt sensor will also result in the AUTO SPEEDBRAKE advisory and status messages being displayed. The AUTO SPEEDBRAKE messages may not display until all three hydraulic systems are pressurized.

Verify the nose gear lock, nose gear down, main gear side brace and main gear drag brace sensors operate normally each flight day (AMM 32-00-00/901).

1. On Landing Gear Actuation Maintenance page:
 - A. Confirm both Nose Gear Lock sensors are NEAR.
 - B. Confirm both Nose Gear, Gear Down sensors are NEAR.
 - C. Confirm all four Main Gear Side Brace sensors are NEAR.
 - D. Confirm all four Main Gear Drag Brace sensors are NEAR.

OPERATIONS (O)

NOTE: An inoperative truck tilt sensor will also result in the AUTO SPEEDBRAKE advisory and status messages being displayed. The AUTO SPEEDBRAKE messages may not display until all three hydraulic systems are pressurized.

After takeoff manually move the autobrake selector from RTO to OFF.

32-61-01 Landing Gear Position Indication Systems

32-61-01-02 Nose Gear Not-Compressed Sensors

32-61-01-02-01 PSEU-1

Interval	Installed	Required	Procedure
C	1	0	(M)

May be inoperative provided:

- a. Nose gear lock, nose gear down, main gear side brace and main gear drag brace sensors are verified to operate normally each flight day.
 - b. PSEU-2 operates normally.
-

MAINTENANCE (M)

Verify the nose gear lock, nose gear down, main gear side brace and main gear drag brace sensors operate normally each flight day (AMM 32-00-00/901).

1. On Landing Gear Actuation Maintenance page:
 - A. Confirm both Nose Gear Lock sensors are NEAR.
 - B. Confirm both Nose Gear, Gear Down sensors are NEAR.
 - C. Confirm all four Main Gear Side Brace sensors are NEAR.
 - D. Confirm all four Main Gear Drag Brace sensors are NEAR.

32-61-01 Landing Gear Position Indication Systems
32-61-01-02 Nose Gear Not-Compressed Sensors
32-61-01-02-02 PSEU-2

Interval	Installed	Required	Procedure
C	1	0	(M)

May be inoperative provided:

- a. Nose gear lock, nose gear down, main gear side brace and main gear drag brace sensors are verified to operate normally each flight day.
- b. Left equipment cooling controller operates normally.
- c. PSEU-1 operates normally.

MAINTENANCE (M)

Verify the nose gear lock, nose gear down, main gear side brace and main gear drag brace sensors operate normally each flight day (AMM 32-00-00/901).

1. On Landing Gear Actuation Maintenance page:
 - A. Confirm both Nose Gear Lock sensors are NEAR.
 - B. Confirm both Nose Gear, Gear Down sensors are NEAR.
 - C. Confirm all four Main Gear Side Brace sensors are NEAR.
 - D. Confirm all four Main Gear Drag Brace sensors are NEAR.

**32-61-02 Landing Gear Door Position Sensors**

Interval	Installed	Required	Procedure
C	6	3	(M)

One per gear (nose, left main, right main) may be inoperative provided sensor is failed in the door not closed position.

MAINTENANCE (M)

Confirm inoperative sensors are not indicating the closed position (AMM 32-00-00/901).

1. Display the Landing Gear Actuation maintenance page and confirm the associated sensor is target FAR or blank.
2. If the associated sensor is target NEAR, disconnect, cap and stow the connector from the associated door sensor.

**32-61-03 Landing Gear Uplock Position Sensors**

Interval	Installed	Required	Procedure
C	6	3	(M)

One per gear may be inoperative provided the sensor is failed in the unlocked position and the associated landing gear door position sensors operate normally.

MAINTENANCE (M)

Confirm inoperative sensors are not indicating the locked position (AMM 32-00-00/901).

1. Display the Landing Gear Actuation maintenance page and confirm the associated uplock sensor is target FAR or blank.
2. If the associated sensor is target NEAR, disconnect, cap and stow the connector from the associated gear uplock sensor.

**32-71-01 Tail Strike Detector Channels****32-71-01A One Inoperative**

Interval	Installed	Required	Procedure
C	2	1	

32-71-01 Tail Strike Detector Channels**32-71-01B Both Inoperative**

Interval	Installed	Required	Procedure
C	2	0	(M) (O)

May be inoperative provided:

- Tail section is visually inspected for damage before each departure.
- Alternate tail strike detection procedures are established and used.
- TAIL STRIKE caution message is disabled.

MAINTENANCE (M)

Before each departure, visually inspect the tail section of the aircraft for damage from a tail strike.

Disable the TAIL STRIKE caution message (AMM DDG 32-00-00/901).

- Remove the Tail Strike Assembly (TSA).
- Disconnect TSA electrical connector.
- Fabricate a 4-6 inch long jumper using 20 gauge wire.
- Install the jumper wire between the ship side sockets 4 and 11.
- Verify the following EICAS messages:
 - TAIL STRIKE caution is not displayed.
 - TAIL STRIKE CHANNEL 1 status is displayed.
 - TAIL STRIKE CHANNEL 2 status is not displayed.
- Secure jumper wire using a bundle strap and bag and stow TSA connector.
- Install the Tail Strike Assembly (TSA).

OPERATIONS (O)

NOTE: The TAIL STRIKE caution message is not available.

During takeoff and flare, flight crew and cabin crew should be alert for a potential tail strike. If a tail strike is perceived by the flight crew or reported from the cabin during takeoff, do the Tail Strike Non-Normal Checklist.

**32-72-01 Tail Skid (-300/-300ER)****32-72-01-01 -300**

Interval	Installed	Required	Procedure
C	1	0	(M) (O)

May be inoperative provided:

- Tail skid is secured in the extended position.
- Water supplies associated with aft drain mast are secured off.
- Appropriate performance adjustments are applied.

MAINTENANCE (M)

Secure tail skid in the fully extended position and secure water supplies associated with the aft drain mast off (AMM 32-00-00/901).

- Open stabilizer compartment service access door 311BL.
- If the tail skid is not fully extended, clear tail skid area and fully extend the tail skid.
 - Manually unlock tail skid lock links and move mechanism over center.

NOTE: The extension speed of the tail skid may be increased by cracking open the bleed valve, if desired.

- When tail skid is fully extended, close the bleed valve.
- Install tail skid downlock pin (AMM 32-00-30).
 - Check tail skid actuator and hydraulic lines for leaks.
 - Close stabilizer compartment service access door 311BL.
 - Close water supply valves to the galley sinks, lavatory sinks and drinking fountains that drain through the aft drain mast.

NOTE: Due to the considerable variability in interior configurations, each operator must determine specific water supply valve shutoff requirements for their fleet.

OPERATIONS (O)

- Reduce performance limited weights by the appropriate adjustments.

Performance Limited Weights	Takeoff & Landing	Enroute Climb
All	1,500 lb (680 kg)	3,500 lb (1,588 kg)

- A. The enroute climb weight penalties listed above are based on single engine operating speeds that approximate maximum lift-to-drag ratio speed. To account for the difference in level off altitude when operating at other speeds, multiply the enroute climb weight penalty listed above by the appropriate factor listed on page 3.00-01-00.2 in the ENROUTE DIVERSION SPEED EFFECTS paragraph.

2. Increase flight planning fuel by 1.0%.

32-72-01 Tail Skid (-300/-300ER)
32-72-01-02 -300ER

Interval	Installed	Required	Procedure
C	1	0	(M) (O)

May be inoperative provided:

- Tail skid is secured in the extended position.
- Water supplies associated with aft drain mast are secured off.
- Appropriate performance adjustments are applied.

NOTE: The tail skid system is not installed on -300ER airplanes delivered from line number 1166 and-on.

MAINTENANCE (M)

Secure tail skid in the fully extended position and secure water supplies associated with the aft drain mast off (AMM 32-00-00/901).

- Open stabilizer compartment service access door 311BL.
- If the tail skid is not fully extended, clear tail skid area and fully extend the tail skid.
 - Manually unlock tail skid lock links and move mechanism over center.

NOTE: The extension speed of the tail skid may be increased by cracking open the bleed valve, if desired.

- When tail skid is fully extended, close the bleed valve.
- Install tail skid downlock pin (AMM 32-00-30).
 - Check tail skid actuator and hydraulic lines for leaks.
 - Close stabilizer compartment service access door 311BL.
 - Close water supply valves to the galley sinks, lavatory sinks and drinking fountains that drain through the aft drain mast.

NOTE: Due to the considerable variability in interior configurations, each operator must determine specific water supply valve shutoff requirements for their fleet.

**OPERATIONS (O)**

1. Reduce performance limited weights by the appropriate adjustments.

Performance Limited Weights	Takeoff & Landing	Enroute Climb
All	1,500 lb (680 kg)	3,500 lb (1,588 kg)

- A. The enroute climb weight penalties listed above are based on single engine operating speeds that approximate maximum lift-to-drag ratio speed. To account for the difference in level off altitude when operating at other speeds, multiply the enroute climb weight penalty listed above by the appropriate factor listed on page 3.00-01-00.2 in the ENROUTE DIVERSION SPEED EFFECTS paragraph.
2. Increase flight planning fuel by 1.0%.

**32-72-02 Tail Skid Position Sensing System (-300/-300ER)****32-72-02-01 -300****32-72-02-01A Tail Skid Operative**

Interval	Installed	Required	Procedure
C	1	0	(M)

May be inoperative provided tail skid extend/retract function is verified to operate normally once each flight day.

MAINTENANCE (M)

Verify the tail skid extend/retract function operates normally each flight day (AMM 32-00-00/901).

1. Supply electrical power on the airplane (AMM 24-22-00/201).
2. Install nose and main landing gear downlock pins (32-00-30/201).
3. Pressurize the center hydraulic system (AMM 29-11-00/201).
4. Position Landing Gear Panel (P2-2) GND PROX GEAR switch OVRD.

NOTE: Selecting OVRD on the GND PROX GEAR switch inhibits the warning siren from sounding when the landing gear handle is positioned to UP. The Master Warning Light will illuminate and the CONFIG GEAR warning message will be displayed when the land gear handle is positioned UP.

5. Move the landing gear control lever to the UP position while simultaneously pressing the landing gear lever LOCK OVRD switch adjacent to the landing gear control lever.
6. Verify visually that the tail skid retracts.
7. Move the landing gear control lever to the DOWN position.
8. Verify visually that the tail skid extends.
9. Position Landing Gear Panel (P2-2) GND PROX GEAR switch off (OVRD not visible).
10. Remove power from the center hydraulic system (AMM 29-11-00/201).
11. Remove landing gear downlock pins (AMM 32-00-30/201).

32-72-02 Tail Skid Position Sensing System (-300/-300ER)**32-72-02-01 -300****32-72-02-01B Tail Skid Inoperative**

Interval	Installed	Required	Procedure
C	1	0	(M) (O)

May be inoperative provided:

- a. Tail skid is secured in the extended position.
- b. Water supplies associated with aft drain mast are secured off.
- c. Appropriate performance adjustments are applied.

MAINTENANCE (M)

Secure the tail skid in the extended position and secure water supplies associated with the aft drain mast off (AMM 32-00-00/901).

1. Use DDG item 32-72-01 (M) procedure to secure the tail skid in the extended position and to secure water supplies associated with the aft drain mast off.

OPERATIONS (O)

1. Reduce performance limited weights by the appropriate adjustments.

Performance Limited Weights	Takeoff & Landing	Enroute Climb
All	1,500 lb (680 kg)	3,500 lb (1,588 kg)

NOTE: The enroute climb weight penalties listed above are based on single engine operating speeds that approximate maximum lift-to-drag ratio speed. To account for the difference in level off altitude when operating at other speeds, multiply the enroute climb weight penalty listed above by the appropriate factor listed on page 3.00-01-00.2 in the ENROUTE DIVERSION SPEED EFFECTS paragraph.

2. Increase flight planning fuel by 1.0%.

32-72-02 Tail Skid Position Sensing System (-300/-300ER)

32-72-02-02 -300ER

32-72-02-02A Tail Skid Operative

Interval	Installed	Required	Procedure
C	1	0	(M)

May be inoperative provided tail skid extend/retract function is verified to operate normally once each flight day.

NOTE: The tail skid system is not installed on -300ER airplanes delivered from line number 1166 and-on.

**MAINTENANCE (M)**

Verify the tail skid extend/retract function operates normally each flight day (AMM 32-00-00/901).

1. Supply electrical power on the airplane (AMM 24-22-00/201).
2. Install nose and main landing gear downlock pins (32-00-30/201).
3. Pressurize the center hydraulic system (AMM 29-11-00/201).
4. Position Landing Gear Panel (P2-2) GND PROX GEAR switch OVRD.

NOTE: Selecting OVRD on the GND PROX GEAR switch inhibits the warning siren from sounding when the landing gear handle is positioned to UP. The Master Warning Light will illuminate and the CONFIG GEAR warning message will be displayed when the land gear handle is positioned UP.

5. Move the landing gear control lever to the UP position while simultaneously pressing the landing gear lever LOCK OVRD switch adjacent to the landing gear control lever.
6. Verify visually that the tail skid retracts.
7. Move the landing gear control lever to the DOWN position.
8. Verify visually that the tail skid extends.
9. Position Landing Gear Panel (P2-2) GND PROX GEAR switch off (OVRD not visible).
10. Remove power from the center hydraulic system (AMM 29-11-00/201).
11. Remove landing gear downlock pins (AMM 32-00-30/201).

32-72-02 Tail Skid Position Sensing System (-300/-300ER)**32-72-02-02 -300ER***********32-72-02-02B Tail Skid Inoperative**

Interval	Installed	Required	Procedure
C	1	0	(M) (O)

May be inoperative provided:

- a. Tail skid is secured in the extended position.
 - b. Water supplies associated with aft drain mast are secured off.
 - c. Appropriate performance adjustments are applied.
-

NOTE: The tail skid system is not installed on -300ER airplanes delivered from line number 1166 and-on.

MAINTENANCE (M)

Secure the tail skid in the extended position and secure water supplies associated with the aft drain mast off (AMM 32-00-00/901).

1. Use DDG item 32-72-01 (M) procedure to secure the tail skid in the extended position and to secure water supplies associated with the aft drain mast off.

OPERATIONS (O)

1. Reduce performance limited weights by the appropriate adjustments.

Performance Limited Weights	Takeoff & Landing	Enroute Climb
All	1,500 lb (680 kg)	3,500 lb (1,588 kg)

NOTE: The enroute climb weight penalties listed above are based on single engine operating speeds that approximate maximum lift-to-drag ratio speed. To account for the difference in level off altitude when operating at other speeds, multiply the enroute climb weight penalty listed above by the appropriate factor listed on page 3.00-01-00.2 in the ENROUTE DIVERSION SPEED EFFECTS paragraph.

2. Increase flight planning fuel by 1.0%.



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33-13-01 Master Brightness Control**33-16-01 Master Dim and Test System****33-21-01 Cabin Interior Illumination**

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**33-11-01 Flight Compartment and Instrument Lighting System**

Interval	Installed	Required	Procedure
C	-	-	

Individual lights or light controls may be inoperative provided:

- Remaining lighting system lights are sufficient to clearly illuminate all required instruments, controls, and other devices for which it is provided.
- Remaining lighting system lights are positioned so that direct rays are shielded from flight crew eyes.
- Lighting configuration and intensity is acceptable to the flight crew.
- Captain's emergency dome light operates normally.

NOTE: Individual button/switch lights and/or annunciations/indications are excluded from this relief.

33-11-01 Flight Compartment and Instrument Lighting System**33-11-01-01 STORM Switch ON Light**

Interval	Installed	Required	Procedure
C	1	0	

**33-13-01 Master Brightness Control**

Interval	Installed	Required	Procedure
C	1	0	

May be inoperative provided the master brightness control switch is selected off.

**33-16-01 Master Dim and Test System**

Interval	Installed	Required	Procedure
B	1	0	

Dim function may be inoperative provided:

- TEST and BRT functions operate normally.
- Light intensity is acceptable to the flight crew.

OPERATIONS NOTE

Only a portion of the flight deck annunciator lights may be affected. The light intensity must be deemed acceptable by each flight crew.

**33-21-01 Cabin Interior Illumination****33-21-01-01 Passenger****33-21-01-01-01 With Powered Floor Proximity Emergency Escape Path Lighting System**

Interval	Installed	Required	Procedure
C	-	-	

Individual lights may be inoperative provided:

- Sufficient lighting remains for crew members to perform their duties.
 - For night operations beyond 60 minutes of landing at a suitable airport, at least 75% of the standby lights operate normally.
-

33-21-01 Cabin Interior Illumination**33-21-01-01 Passenger****33-21-01-01-02 With Photoluminescent Floor Proximity Emergency Escape Path Marking System**

Interval	Installed	Required	Procedure
C	-	-	

Four total ceiling and sidewall lamps may be inoperative in each cabin zone provided:

- No more than two lamps are inoperative in each cabin zone aisle.
- A minimum of two lamps must operate normally between inoperative lamps.
- Inoperative ceiling and sidewall lamps may not be adjacent.

NOTE: A cabin zone is the passenger cabin area between two sets of main entry door pairs.

33-21-01 Cabin Interior Illumination**33-21-01-02 777F**

Interval	Installed	Required	Procedure
C	-	-	

Individual lights may be inoperative provided sufficient lighting remains for supernumeraries/cargo couriers to perform their duties.

33-24-01 Passenger Information Signs (Passenger)
33-24-01A PA System Inoperative

Interval	Installed	Required	Procedure
C	-	0	(M)

May be inoperative provided:

- Associated passenger seat, or lavatory is not occupied from which a Passenger Information sign is not readily legible.
- Associated seat or lavatory must be blocked and placarded - "DO NOT OCCUPY".

NOTE: These provisions are not intended to prohibit lavatory use or inspections by crewmembers.

MAINTENANCE (M)

NOTE: In tests conducted at Boeing for the FAA, a sign was considered readily legible if enough of the sign could be seen to identify it. For example, NO SMOK and FASTEN SEAT B would be considered readily legible.

Block and placard associated seat or close and lock associated lavatory (AMM 33-00-00/901).

- Tapes or ropes of conspicuous contrasting colors must be installed to block access to unusable seats prior to boarding of passengers.
- Conspicuous signs or placards shall be placed in appropriate locations to indicate seats which are not to be occupied by passengers.
- For an associated lavatory, close and lock the lavatory door.

33-24-01 Passenger Information Signs (Passenger)
33-24-01B PA System Operative

Interval	Installed	Required	Procedure
C	-	0	(O)

May be inoperative and the associated passenger seat or lavatory may be occupied provided:

- PA system operates normally and can be clearly heard throughout the cabin during flight.
- PA system is used to alert the cabin crew and to notify passengers when seat belts should be fastened, when smoking is prohibited, and when passengers should return to seats.

OPERATIONS (O)

Each operator shall establish procedures for notifying passengers when seat belts should be fastened, when smoking is prohibited, and when passengers should return to their seats.

33-24-01 Passenger Information Signs (Passenger)
33-24-01C No Passenger Carried

Interval	Installed	Required	Procedure
C	-	0	(O)

May be inoperative provided:

- No passengers are carried.
- A maximum of 19 persons authorized by 14 CFR for non-passenger carrying operations are carried.
- Alternate procedures are established and used.

OPERATIONS (O)

- No passengers may be carried.
- A maximum of 19 persons authorized by 14 CFR 121.583 may be carried.
- Flight crew members and authorized persons must be briefed prior to takeoff and landing regarding alternate procedures for notification when seat belts must be used.

33-24-01 Passenger Information Signs (Passenger)
33-24-01-01 Flight Deck Automatic Function

Interval	Installed	Required	Procedure
C	1	0	(O)

May be inoperative provided:

- Manual control function operates normally.
- Alternate procedures are established and used.

OPERATIONS (O)

Each operator shall establish procedures for when the flight crew shall notify passengers when seat belts should be fastened and smoking prohibited.

NOTE 1: The No Smoking signs automatic function illuminates the NO SMOKING signs when landing gear not up and locked, or cabin altitude above 10,000 feet, or passenger oxygen on.



NOTE 2: The Fasten Seat Belt signs automatic function illuminates the FASTEN SEAT BELT signs when landing gear not up and locked, or flap lever not up, or airplane altitude below an operator defined altitude, or cabin altitude above 10,000 feet, or passenger oxygen on.

33-24-01 Passenger Information Signs (Passenger)**33-24-01-02 Crew/Attendant Rest Areas**

Interval	Installed	Required	Procedure
C	-	0	(O)

May be inoperative provided alternate procedures are established and used.

OPERATIONS (O)

Each operator shall establish procedures for notifying the crew/attendant rest occupants when seat belts should be fastened, when smoking is prohibited, and when to return to their seats.

33-24-01 Passenger Information Signs (Passenger)**33-24-01-03 Aural Tone Function**

Interval	Installed	Required	Procedure
C	1	0	(O)

May be inoperative provided alternate procedures are established and used.

OPERATIONS (O)

Establish alternate procedures.



33-24-02 Supernumerary/Courier Area Lighted Information Signs (777F)

Interval	Installed	Required	Procedure
C	-	0	(O)

May be inoperative provided alternate procedures are established and used.

OPERATIONS (O)

Alternate procedures must be established and used.

33-24-02 Supernumerary/Courier Area Lighted Information Signs (777F)**33-24-02-01 Flight Deck Automatic Function**

Interval	Installed	Required	Procedure
C	1	0	(O)

May be inoperative provided:

- Manual control function operates normally.
 - Alternate procedures are established and used.
-

OPERATIONS (O)

Alternate procedures must be established and used.



33-28-01 Sterile Flight Compartment Light System*********33-28-01A Alternate Procedures Required**

Interval	Installed	Required	Procedure
C	1	0	(O)

May be inoperative provided alternate procedures are established and used.

OPERATIONS (O)

Establish alternate procedures for notifying the cabin crew when a sterile flight compartment condition exists.

33-28-01 Sterile Flight Compartment Light System*********33-28-01B Procedures Do Not Require Use**

Interval	Installed	Required	Procedure
D	1	0	

May be inoperative provided procedures do not require its use.



33-31-01 Main Wheel Well and Nose Wheel Well Service Area Lights**33-31-01A All Operations**

Interval	Installed	Required	Procedure
C	-	0	

NOTE: Relief for the Air Conditioning Compartment, APU Compartment, Stabilizer Compartment and Electrical Equipment Center Service Area Lights was moved to MMEL Item 25-20-01.

33-31-01 Main Wheel Well and Nose Wheel Well Service Area Lights**33-31-01B Day Operations**

Interval	Installed	Required	Procedure
D	-	0	

May be inoperative provided operations are not conducted at night.

NOTE: Relief for the Air Conditioning Compartment, APU Compartment, Stabilizer Compartment and Electrical Equipment Center Service Area Lights was moved to MMEL Item 25-20-01.



33-31-02 Cargo Loading Area Lights

Interval	Installed	Required	Procedure
D	-	0	

**33-37-01 Cargo Compartment Lights**

Interval	Installed	Required	Procedure
C	-	-	

Individual lights may be inoperative provided sufficient lighting remains for ground personnel to perform their duties.

**33-37-02 Main Deck Alert System (777F)****33-37-02A Main Deck Access Allowed in Flight**

Interval	Installed	Required	Procedure
C	1	0	(O)

May be inoperative provided alternate procedures are established and used.

NOTE: Any visual and aural alert system functions that operate normally may be used.

OPERATIONS (O)

Alternate procedures must be established and used. Prior to each flight, the couriers/supernumeraries should be briefed on the alternate procedures.

33-37-02 Main Deck Alert System (777F)**33-37-02B Main Deck Access Prohibited in Flight**

Interval	Installed	Required	Procedure
D	1	0	

May be inoperative provided procedures do not require its use.



33-41-01 Wing Illumination Lights

Interval	Installed	Required	Procedure
C	2	0	

33-41-01 Wing Illumination Lights**33-41-01-01 WING Switch ON Light**

Interval	Installed	Required	Procedure
C	1	0	



Interval	Installed	Required	Procedure
C	2	0	

**33-42-02 Landing Lights****33-42-02A All Operations**

Interval	Installed	Required	Procedure
C	4	2	

Two may be inoperative for night operations.

33-42-02 Landing Lights**33-42-02B Day Operations**

Interval	Installed	Required	Procedure
C	4	0	

May be inoperative provided operations are not conducted at night.

33-42-02 Landing Lights**33-42-02-01 Dim Function**

Interval	Installed	Required	Procedure
C	2	0	



Interval	Installed	Required	Procedure
C	2	0	

**33-43-01 Position Lights****33-43-01A All Operations**

Interval	Installed	Required	Procedure
C	8	4	

May be inoperative provided:

- One stationary red wing tip bulb operates normally.
 - One stationary green wing tip bulb operates normally.
 - One stationary white bulb per wing tip operates normally.
-

33-43-01 Position Lights**33-43-01B Day Operations**

Interval	Installed	Required	Procedure
C	8	0	

May be inoperative between sunrise and sunset.

33-43-01 Position Lights**33-43-01-01 NAV Switch ON Light**

Interval	Installed	Required	Procedure
C	1	0	



33-43-02 Tail Lights

Interval	Installed	Required	Procedure
D	2	0	

**33-44-01 Anti-Collision Lights****33-44-01-01 Red Upper and Lower Fuselage Beacon Lights****33-44-01-01A All Operations**

Interval	Installed	Required	Procedure
C	2	1	

May be inoperative provided white tail and wing tip strobe lights operate normally.

33-44-01 Anti-Collision Lights**33-44-01-01 Red Upper and Lower Fuselage Beacon Lights****33-44-01-01B Day Operations**

Interval	Installed	Required	Procedure
C	2	0	

May be inoperative provided:

- At least one white tail or wing tip strobe light operates normally.
- Operations are not conducted at night.

33-44-01 Anti-Collision Lights**33-44-01-02 White Tail and Wing Tip Strobe Lights****33-44-01-02A All Operations**

Interval	Installed	Required	Procedure
C	3	0	

May be inoperative provided red upper and lower fuselage beacon lights operate normally.

33-44-01 Anti-Collision Lights**33-44-01-02 White Tail and Wing Tip Strobe Lights****33-44-01-02B Day Operations**

Interval	Installed	Required	Procedure
C	3	0	

May be inoperative provided:

- At least one red fuselage beacon light operates normally.



b. Operations are not conducted at night.

33-44-01 Anti-Collision Lights

33-44-01-03 BEACON Switch ON Light

Interval	Installed	Required	Procedure
C	1	0	



33-45-01 LOGO Lights

Interval	Installed	Required	Procedure
D	4	0	

33-45-01 LOGO Lights**33-45-01-01 LOGO Switch ON Light**

Interval	Installed	Required	Procedure
D	1	0	

**33-51-01 Interior Emergency Lights**

Interval	Installed	Required	Procedure
C	-	-	

A random 25% of lights/signs may be inoperative provided:

- a. Inoperative area illumination lights are not adjacent.
- b. Not more than one overhead area light or exit sign at each door is inoperative.
- c. Cross-aisle exit signs operate normally.
- d. Flight deck emergency dome light operates normally.

NOTE: Lights associated with an inoperative door or slide/raft are not required.

OPERATIONS NOTE

1. For Boeing systems, the following are the minimum FAA dispatch requirements for an exit sign to be considered operative:
 - A. Horizontal exit signs - At least 9 of 12 lamps must be operative.
 - B. Vertical exit signs - At least 3 of 5 lamps must be operative.
2. For non-Boeing systems, the operator is responsible for determining from the appropriate document, the dispatch relief allowed.

**33-51-02 Exterior Emergency Slide Lights**

Interval	Installed	Required	Procedure
C	-	0	

May be inoperative provided operations are not conducted at night.

33-51-02 Exterior Emergency Slide Lights**33-51-02-01 -200/-200ER/-200LR**

Interval	Installed	Required	Procedure
A	8	7	

One may be inoperative provided:

- Associated passenger entry door is considered inoperative.
 - Repairs are made within 1 flight day.
-

OPERATIONS NOTE

The airplane must also be dispatched using MMEL item 52-11-01.

33-51-02 Exterior Emergency Slide Lights**33-51-02-02 -300/-300ER****33-51-02-02-01 Doors 1L, 1R, 2L, 2R, 4L, 4R, 5L and 5R**

Interval	Installed	Required	Procedure
A	8	7	

One may be inoperative provided:

- Associated passenger entry door is considered inoperative.
 - Exterior emergency slide lights for doors 3L and 3R operate normally.
 - Repairs are made within 1 flight day.
-

OPERATIONS NOTE

The airplane must also be dispatched using MMEL item 52-11-01.

**33-51-02 Exterior Emergency Slide Lights****33-51-02-02 -300/-300ER****33-51-02-02-02 Doors 3L and 3R**

Interval	Installed	Required	Procedure
A	6	3	

Lights associated with one door may be inoperative provided:

- Associated passenger entry door is considered inoperative.
 - All remaining door exterior emergency slide lights operate normally.
 - Repairs are made within 1 flight day.
-

OPERATIONS NOTE

The airplane must also be dispatched using MMEL item 52-11-01.

33-51-02 Exterior Emergency Slide Lights**33-51-02-03 777F**

Interval	Installed	Required	Procedure
A	2	1	

One may be inoperative provided:

- Associated passenger entry door is considered inoperative.
 - Repairs are made within 1 flight day.
-

OPERATIONS NOTE

The airplane must also be dispatched using MMEL item 52-11-01.

33-51-03 Floor Proximity Emergency Lighting Systems (Passenger)
33-51-03-01 Powered Escape Path Lighting System

Interval	Installed	Required	Procedure
C	-	-	

A random 25% of lights may be inoperative provided inoperative floor proximity lights are not adjacent.

NOTE: Lights associated with an inoperative door slide/raft are not required.

33-51-03 Floor Proximity Emergency Lighting Systems (Passenger)
33-51-03-02 Photoluminescent Escape Path Marking System

Interval	Installed	Required	Procedure
C	-	-	

Four 10-inch segments of photoluminescent strip may be inoperative in each cabin zone provided:

- No more than two 10-inch segments of photoluminescent strip are inoperative in each cabin zone aisle.
- A minimum of 72 inches of photoluminescent strip must operate normally between inoperative segments.
- Inoperative segments of photoluminescent strip must not be laterally adjacent in an aisle.

NOTE 1: Photoluminescent strips associated with an inoperative door slide/raft are not required.

NOTE 2: A cabin zone is the passenger cabin area between two sets of main entry door pairs.

**34-12-01 Mach Indications**34-12-01A *One Inoperative*34-12-01B *Both Inoperative***34-12-02 True Airspeed Indications****34-13-01 Standby Airspeed Indicator System (Non-ISFD System)****34-16-01 Altitude Alerting System**

34-16-01-01 Aural Alert

34-16-01-02 Visual Alert

34-21-01 Dual Total Air Temperature (TAT) Systems**34-21-02 Static Air Temperature (SAT) Indication****34-21-03 Air Data Inertial Reference Unit (ADIRU)**

34-21-03-01 ADIRU Faults

34-21-03-01A *Ground Nav Available*34-21-03-01B *Ground Nav Not Available*

34-21-03-02 ON BAT Light

34-21-03-03 ADIRU Switch Lights

34-21-03-03-01 OFF Light

34-21-03-03-02 ON Light

34-21-04 Pitot Air Data Modules (Primary - ARINC 629)

34-21-04-01 Right Pitot Air Data Module

34-21-04-02 Left Pitot Air Data Module

34-21-04-03 Center Pitot Air Data Module

34-21-05 Static Air Data Modules (Primary - ARINC 629)

34-21-05-01 Right Static Air Data Module

34-21-05-02 Left Static Air Data Module

34-21-05-03 Center Static Air Data Module

34-21-06 Angle of Attack (AOA) Vane Systems**34-21-07 Secondary Attitude Air Data Reference Unit (SAARU)**

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34-46-01-02 Windshear Alert Mode Function (Reactive) (Mode 7)

34-46-01-02A *Predictive Windshear Operative*

34-46-01-02B *Predictive Windshear Inoperative*

34-46-01-03 Terrain Awareness Function

34-46-01-03-01 Terrain Display Functions

34-46-01-03-01A *One Inoperative*

34-46-01-03-01B *Both Inoperative*

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34-53-01A *One Inoperative*

34-53-01B *Both Inoperative*

34-53-01-01 Elementary and Enhanced Downlink Aircraft Reportable
Parameters Not Required By 14 CFR

34-53-01-02 ADS-B Squitter Transmissions

**34-53-02 Automatic Dependent Surveillance-Broadcast (ADS-B)
System (In and Out)**

34-53-02A *Flight Authorized by ATC*

34-53-02B *One Inoperative*

34-53-02C *Enroute Operations Do Not Require Use*

34-53-02-01 ADS-B Out Extended Squitter

34-53-02-01A *Flight Authorized by ATC*

34-53-02-01B *One Inoperative*

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34-61-03-02-02	Line Select Lights, Function Lights, Execute Lights, Alpha/Numeric Lights and Miscellaneous Key Lights



34-12-01 Mach Indications

34-12-01A One Inoperative

Interval	Installed	Required	Procedure
C	2	1	(O)

One may be inoperative provided flight descends to FL 290 or below, if failure of the second indication occurs in flight.

OPERATIONS (O)

If remaining Mach indication fails inflight, descend to FL 290 or below.

34-12-01 Mach Indications

34-12-01B Both Inoperative

Interval	Installed	Required	Procedure
C	2	0	(O)

May be inoperative provided flight remains at FL 290 or below.

OPERATIONS (O)

Flight must remain at or below FL 290.



34-12-02 True Airspeed Indications

Interval	Installed	Required	Procedure
C	2	0	

34-13-01 Standby Airspeed Indicator System (Non-ISFD System)

Interval	Installed	Required	Procedure
B	1	0	(M)

May be inoperative provided:

- a. All ARINC 629 pitot air data modules operate normally.
- b. All ARINC 629 static air data modules operate normally.
- c. All pitot probe heater systems operate normally.
- d. Both AIR DATA/ATT instrument source switches operate normally.
- e. SAARU data is verified available to both PFDs before each departure.

MAINTENANCE (M)

NOTE: The standby airspeed indicator's standby pitot ADM is considered part of the standby airspeed indicator system. An inoperative Standby Pitot ADM (ARINC 429) results in a display flag on the standby airspeed indicator.

Verify SAARU data is available to both PFDs before each departure (AMM 34-00-00/901).

1. With the airplane powered and ADIRU ON and aligned, check SAARU alignment. The Standby Attitude indicator should display sky/ground (blue over brown) shading.
2. Position both AIR DATA/ATT switches ALTN (non-normal position - switch in).
3. Verify the ALTN ATTITUDE advisory message is displayed.
4. Verify the attitude sky/ground (blue over brown) shading and pitch scale is displayed on both Primary Flight Displays (PFD) and no airspeed or altitude failure flags are displayed on the Captain's PFD.
5. Position both AIR DATA/ATT switches off (normal position - switch out).
6. Verify the ALTN ATTITUDE advisory message is not displayed.

OPERATIONS NOTE

Altitude differences may be noted between the standby altimeter display and the primary displays, particularly at higher speeds and altitudes. The standby altimeter defaults to a zero static source error correction when the standby airspeed display is inoperative.

**34-16-01 Altitude Alerting System**

Interval	Installed	Required	Procedure
A	1	0	(O)

May be inoperative provided:

- Autopilot with altitude hold and altitude capture operates normally.
 - Enroute operations do not require its use.
 - Airplane does not depart from a designated airport (as listed in the Operator's MEL) where repair or replacement can be made.
 - Repairs are made within 3 flight days.
-

OPERATIONS (O)

- Enroute operations that require RVSM may not be conducted.
- Use the autopilot and altitude hold for all operations for which it is appropriate.
- Cross-check the MCP ALTITUDE window with altitude indications on captain's and first officer's PFDs upon reaching and departing an assigned altitude. Periodically cross-check altitude indications when maintaining an assigned altitude.
- Pilot monitoring should call out approaching and departing assigned altitudes.
- Flight crew must be aware that the usual alerts for altitude deviations will not occur.

34-16-01 Altitude Alerting System**34-16-01-01 Aural Alert**

Interval	Installed	Required	Procedure
C	1	0	

May be inoperative provided:

- Visual alert operates normally.
 - Autopilot with altitude hold and altitude capture operates normally.
-

34-16-01 Altitude Alerting System**34-16-01-02 Visual Alert**

Interval	Installed	Required	Procedure
C	1	0	



May be inoperative provided:

- a. Aural alert operates normally.
- b. Autopilot with altitude hold and altitude capture operates normally.



34-21-01

Dual Total Air Temperature (TAT) Systems

Interval	Installed	Required	Procedure
C	2	1	



34-21-02 Static Air Temperature (SAT) Indication

Interval	Installed	Required	Procedure
C	1	0	

**34-21-03 Air Data Inertial Reference Unit (ADIRU)****34-21-03-01 ADIRU Faults****34-21-03-01A Ground Nav Available**

Interval	Installed	Required	Procedure
B	1	0	(M)

May be dispatched with faults indicated by ADIRU status message provided:

- Adequate ground navigation facilities are available.
- Non-stabilized magnetic compass operates normally.
- Approach minimums do not require use of triple channel autoland.
- SAARU data is verified available to both PFDs before each departure.

MAINTENANCE (M)

Verify that SAARU data is available to both PFDs before each departure (AMM 34-00-00/901).

- With the airplane powered and ADIRU ON and aligned, check SAARU alignment.
- Position both AIR DATA/ATT switches ALTN (non-normal position - switch in).
- Verify that ALTN ATTITUDE advisory message is displayed.
- Verify that the Attitude Indication on the both Primary Flight Displays (PFD) display the sky/ground (blue over brown) shading and pitch scale. No airspeed or altitude failure flags should be displayed on either PFD.

For ISFD not installed:

NOTE: The SAARU is also the attitude source for the Standby Attitude indicator and it should display sky/ground (blue over brown) shading.

- Position both AIR DATA/ATT switches off (normal position - switch out).
- Verify that ALTN ATTITUDE advisory message is not displayed.

OPERATIONS NOTE

The NO LAND 3 advisory message will be displayed.

34-21-03 Air Data Inertial Reference Unit (ADIRU)**34-21-03-01 ADIRU Faults****34-21-03-01B Ground Nav Not Available**

Interval	Installed	Required	Procedure
B	1	0	(M)

May be dispatched with faults indicated by ADIRU status message provided:

- a. Both GPS receivers operate normally.
- b. Both FMCs operate normally.
- c. Non-stabilized magnetic compass operates normally.
- d. Approach minimums do not require use of triple channel autoland.
- e. SAARU data is verified available to both PFDs before each departure.

MAINTENANCE (M)

Verify that SAARU data is available to both PFDs before each departure (AMM 34-00-00/901).

1. With the airplane powered and ADIRU ON and aligned, check SAARU alignment.
2. Position both AIR DATA/ATT switches ALTN (non-normal position - switch in).
3. Verify that ALTN ATTITUDE advisory message is displayed.
4. Verify that the Attitude Indication on the both Primary Flight Displays (PFD) display the sky/ground (blue over brown) shading and pitch scale. No airspeed or altitude failure flags should be displayed on either PFD.

For ISFD not installed:

NOTE: The SAARU is also the attitude source for the Standby Attitude indicator and it should display sky/ground (blue over brown) shading.

5. Position both AIR DATA/ATT switches off (normal position - switch out).
6. Verify that ALTN ATTITUDE advisory message is not displayed.

OPERATIONS NOTE

The NO LAND 3 advisory message will be displayed.

34-21-03 Air Data Inertial Reference Unit (ADIRU)

34-21-03-02 ON BAT Light

Interval	Installed	Required	Procedure
C	1	0	

**34-21-03 Air Data Inertial Reference Unit (ADIRU)****34-21-03-03 ADIRU Switch Lights****34-21-03-03-01 OFF Light**

Interval	Installed	Required	Procedure
C	1	0	

34-21-03 Air Data Inertial Reference Unit (ADIRU)**34-21-03-03 ADIRU Switch Lights****34-21-03-03-02 ON Light**

Interval	Installed	Required	Procedure
C	1	0	

34-21-04 Pitot Air Data Modules (Primary - ARINC 629)
34-21-04-01 Right Pitot Air Data Module

Interval	Installed	Required	Procedure
B	1	0	(M)

May be inoperative provided:

- a. Left and center pitot probe heater systems operate normally.
- b. Left and center pitot air data modules operate normally.
- c. Left static air data module operates normally.
- d. Standby airspeed indication operates normally.
- e. Right AIR DATA/ATT instrument source switch operates normally.
- f. SAARU data is verified to be available to the right PFD before each departure.
- g. Left and center pitot probes are inspected before each departure.
- h. Approach minimums do not require its use.

MAINTENANCE (M)

NOTE 1: The left, center and right Pitot and Static ADMs are interchangeable.

NOTE 2: An inoperative right Pitot ADM may also result in the AOA VANE R status message. For AOA VANE R status message displayed, the airplane must also be dispatched using MMEL item 34-21-06.

Inspect the left and center pitot probes and verify SAARU is available to the right PFD before each departure (AMM 34-00-00/901).

1. Prior to each departure, visually examine the left and center pitot probe external installations for damage and unwanted material in the pitot probe opening. If damage or unwanted material is noted, perform Pitot Probe - Inspection/Check (AMM 34-11-01/601).
2. Verify SAARU data is available to the right Primary Flight Display (PFD) before each departure.
 - A. With the airplane powered, verify SAARU alignment.

For ISFD not installed:

NOTE: The SAARU is also the attitude source for the Standby Attitude indicator and it should display sky/ground (blue over brown) shading.

- B. Position right AIR DATA/ATT switch ALTN (non-normal position - switch in).
- C. Verify the Attitude Indication on the right PFD displays the sky/ground (blue over brown) shading and pitch scale.
- D. Position right AIR DATA/ATT switch off (normal position - switch out).

OPERATIONS NOTE

1. The NO LAND 3 advisory message will be displayed.
2. If the Captain's AIR DATA/ATT instrument source switch is positioned to ALTN, airspeed may be flagged inoperative, or airspeed and altitude may be in error on the left PFD. This effect is normal for this dispatch condition.
3. An additional failure, such as a plugged or damaged pitot probe, may result in the two remaining pitot sources miscomparing and voted airspeed being declared invalid. Lack of valid voted airspeed will result in display of the NAV AIR DATA SYS advisory or AIR DATA SYS Caution message and single channel airspeed and altitude will be displayed on the PFDs. Additionally, the right PFD airspeed may be flagged inoperative, or airspeed and altitude may be in error.

NOTE: For a right PFD airspeed flagged, selection of the First Officer's AIR DATA/ATT switch to ALTN will display the left PFD airspeed on the right PFD. This single airspeed condition will be indicated by the SGL SOURCE AIR DATA advisory message.

The following display effects with regard to airspeed (A/S) are possible:

Pitot ADM Inop At Dispatch	Pitot Probe Plugged, Damaged, etc.	L PFD A/S Display	Standby A/S Display	R PFD A/S Display
Right	Left	In Error (plugged)	Correct	Flag (ADM failed) or In Error (ADM inop)
Right	Center	Correct	In Error (plugged)	Flag (ADM failed) or In Error (ADM inop)

34-21-04 Pitot Air Data Modules (Primary - ARINC 629)

34-21-04-02 Left Pitot Air Data Module

Interval	Installed	Required	Procedure
B	1	0	(M)

May be inoperative provided:

- a. Right and center pitot probe heater systems operate normally.
- b. Right and center pitot air data modules operate normally.
- c. Right static air data module operates normally.
- d. Standby airspeed indication operates normally.
- e. Left AIR DATA/ATT instrument source switch operates normally.

- f. SAARU data is verified to be available to the left PFD before each departure.
 - g. Right and center pitot probes are inspected before each departure.
 - h. Approach minimums do not require its use.
-

MAINTENANCE (M)

NOTE 1: The left, center and right Pitot and Static ADMs are interchangeable.

NOTE 2: An inoperative left Pitot ADM may also result in the AOA VANE L status message. For AOA VANE L status message displayed, the airplane must also be dispatched using MMEL item 34-21-06.

NOTE 3: The WINDSHEAR REAC status message and WINDSHEAR SYS advisory message may also be displayed. For a WINDSHEAR REAC status message or WINDSHEAR SYS advisory message displayed, the airplane must also be dispatched using MMEL item 34-46-01-02.

NOTE 4: The SATCOM SYSTEM and SATCOM HI GAIN status messages may also be displayed. For a SATCOM SYSTEM or SATCOM HI GAIN status message displayed, the airplane must also be dispatched using MMEL item 23-15-01.

Inspect the right and center pitot probes and verify SAARU is available to the left PFD before each departure (AMM 34-00-00/901).

1. Prior to each departure, visually examine the right and center pitot probe external installations for damage and unwanted material in the pitot probe opening. If damage or unwanted material is noted, perform Pitot Probe - Inspection/Check (AMM 34-11-01/601).
2. Verify SAARU data is available to the left Primary Flight Display (PFD) before each departure.
 - A. With the airplane powered, verify SAARU alignment.

For ISFD not installed:

NOTE: The SAARU is also the attitude source for the Standby Attitude indicator and it should display sky/ground (blue over brown) shading.

- B. Position left AIR DATA/ATT switch ALTN (non-normal position - switch in).
- C. Verify the Attitude Indication on the left PFD displays the sky/ground (blue over brown) shading and pitch scale.
- D. Position left AIR DATA/ATT switch off (normal position - switch out).

OPERATIONS NOTE

1. The NO LAND 3 advisory message will be displayed.
2. If the First Officer's AIR DATA/ATT instrument source switch is positioned to ALTN, airspeed may be flagged inoperative, or airspeed and altitude may be in error on the right PFD. This effect is normal for this dispatch condition.

3. The WINDSHEAR SYS advisory message may also be displayed.
4. An additional failure, such as a plugged or damaged pitot probe, may result in the two remaining pitot sources miscomparing and voted airspeed being declared invalid. Lack of valid voted airspeed will result in display of the NAV AIR DATA SYS advisory or AIR DATA SYS Caution message and single channel airspeed and altitude will be displayed on the PFDs. Additionally, the left PFD airspeed may be flagged inoperative, or airspeed and altitude may be in error.

NOTE: For a left PFD airspeed flagged, selection of the Captain's AIR DATA/ATT switch to ALTN will display the right PFD airspeed on the left PFD. This single airspeed condition will be indicated by the SGL SOURCE AIR DATA advisory message.

The following display effects with regard to airspeed (A/S) are possible:

Pitot ADM Inop At Dispatch	Pitot Probe Plugged, Damaged, etc.	L PFD A/S Display	Standby A/S Display	R PFD A/S Display
Left	Center	Flag (ADM failed) or In Error (ADM inop)	In Error (plugged)	Correct
Left	Right	Flag (ADM failed) or In Error (ADM inop)	Correct	In Error (plugged)

34-21-04 Pitot Air Data Modules (Primary - ARINC 629)

34-21-04-03 Center Pitot Air Data Module

Interval	Installed	Required	Procedure
B	1	0	(M)

May be inoperative provided:

- a. Left and right pitot probe heater systems operate normally.
- b. Left and right pitot air data modules operate normally.
- c. Left and right static air data modules operate normally.
- d. Standby airspeed indication operates normally.
- e. Both AIR DATA/ATT instrument source switches operate normally.
- f. SAARU data is verified to be available to both PFDs before each departure.
- g. Left and right pitot probes are inspected before each departure.

- h. Approach minimums do not require its use.
-

MAINTENANCE (M)

NOTE: The left, center and right Pitot and Static ADMs are interchangeable.

Inspect the left and right pitot probes before each departure and verify SAARU is available to both PFDs (AMM 34-00-00/901).

1. Prior to each departure, visually examine the left and right pitot probe external installations for damage and unwanted material in the pitot probe opening. If damage or unwanted material is noted, perform Pitot Probe - Inspection/Check (AMM 34-11-01/601).
2. Verify SAARU data is available to both Primary Flight Displays (PFD) before each departure.
 - A. With the airplane powered, verify SAARU alignment.

For ISFD not installed:

NOTE: The SAARU is also the attitude source for the Standby Attitude indicator and it should display sky/ground (blue over brown) shading.

- B. Verify SAARU data is available to the left PFD:
 - 1) Position left AIR DATA/ATT switch ALTN (non-normal position - switch in).
 - 2) Verify the Attitude Indication on the left PFD displays the sky/ground (blue over brown) shading and pitch scale.
 - 3) Position left AIR DATA/ATT switch off (normal position - switch out).
- C. Verify SAARU data is available to the right PFD:
 - 1) Position right AIR DATA/ATT Selector ALTN (non-normal position - switch in).
 - 2) Verify the Attitude Indication on the right PFD displays the sky/ground (blue over brown) shading and pitch scale.
 - 3) Position right AIR DATA/ATT switch off (normal position - switch out).

OPERATIONS NOTE

1. The autoland system is not fail-operational and the NO LAND 3 advisory message will be displayed.
2. An additional failure, such as a plugged or damaged pitot probe, may result in the two remaining pitot sources miscomparing and voted airspeed being declared invalid. Lack of valid voted airspeed will result in display of the NAV AIR DATA SYS advisory or AIR DATA SYS Caution message and single channel airspeed and altitude will be displayed on the PFDs.

The following display effects with regard to airspeed (A/S) are possible:



Pitot ADM Inop At Dispatch	Pitot Probe Plugged, Damaged, etc.	L PFD A/S Display	Standby A/S Display	R PFD A/S Display
Center	Left	In Error (plugged)	Correct	Correct
Center	Right	Correct	Correct	In Error (plugged)

34-21-05 Static Air Data Modules (Primary - ARINC 629)
34-21-05-01 Right Static Air Data Module

Interval	Installed	Required	Procedure
C	1	0	(M)

May be inoperative provided:

- a. Left and center static air data modules operate normally.
- b. Left pitot air data module operates normally.
- c. Right AIR DATA/ATT instrument source switch operates normally.
- d. Standby airspeed indication operates normally.
- e. SAARU data is verified to be available to the right PFD before each departure.
- f. Approach minimums do not require its use.

MAINTENANCE (M)

NOTE: The left, center and right Pitot and Static ADMs are interchangeable.

Verify SAARU data is available to the right Primary Flight Display (PFD) (AMM 34-00-00/901).

1. With the airplane powered, verify SAARU alignment.

For ISFD not installed:

NOTE: The SAARU is also the attitude source for the Standby Attitude indicator and it should display sky/ground (blue over brown) shading.

2. Position right AIR DATA/ATT switch ALTN (non-normal position - switch in).
3. Verify the Attitude Indication on the right PFD displays the sky/ground (blue over brown) shading and pitch scale.
4. Position right AIR DATA/ATT switch off (normal position - switch out).

OPERATIONS NOTE

1. The NO LAND 3 advisory message will be displayed.
2. If the Captain's AIR DATA/ATT instrument source switch is positioned to ALTN, airspeed and altitude may be flagged inoperative or may be in error on the left PFD. This effect is normal for this dispatch condition.
3. An additional failure, such as a plugged or disconnected static source, may result in the two remaining static sources miscomparing and voted airspeed and altitude being declared invalid. Lack of valid voted airspeed and altitude will result in display of the NAV AIR DATA SYS advisory or AIR DATA SYS Caution message and single channel airspeed and altitude will be displayed on the PFDs. Additionally, the right PFD airspeed and altitude may be flagged inoperative or may be in error.



NOTE: For a right PFD airspeed and altitude flagged, selection of the First Officer's AIR DATA/ATT switch to ALTN will display the left PFD airspeed and altitude on the right PFD. This single air data condition will be indicated by the SGL SOURCE AIR DATA advisory message.

The following display effects with regard to airspeed (A/S) and altitude (ALT) are possible:

Static ADM Inop At Dispatch	Static Source Plugged, Disconnect, etc.	L PFD A/S & ALT Display	Standby A/S & ALT Display	R PFD A/S & ALT Display
Right	Left	In Error (plugged)	Both Correct	Flags (ADM failed) or In Error (ADM inop)
Right	Center	Both Correct	In Error (plugged)	Flags (ADM failed) or In Error (ADM inop)

34-21-05 Static Air Data Modules (Primary - ARINC 629)

34-21-05-02 Left Static Air Data Module

Interval	Installed	Required	Procedure
C	1	0	(M)

May be inoperative provided:

- Right and center static air data modules operate normally.
- Right pitot air data module operates normally.
- Left AIR DATA/ATT instrument source switch operates normally.
- Standby airspeed indication operates normally.
- SAARU data is verified to be available to the left PFD before each departure.
- Approach minimums do not require its use.

MAINTENANCE (M)

NOTE 1: The left, center and right Pitot and Static ADMs are interchangeable.

NOTE 2: Certain faults of the left static ADM may cause the WINDSHEAR REAC status message and WINDSHEAR SYS advisory message to be displayed. For a WINDSHEAR REAC status message or WINDSHEAR SYS advisory message displayed, the airplane must also be dispatched using MMEL item 34-46-01-02.

NOTE 3: Certain faults of the Left Static ADM may cause the SATCOM SYSTEM and SATCOM HI GAIN status messages to be displayed. For a SATCOM SYSTEM or SATCOM HI GAIN status message displayed, the airplane must also be dispatched using MMEL item 23-15-01.

Verify SAARU data is available to the left Primary Flight Display (PFD) (AMM 34-00-00/901).

1. With the airplane powered, verify SAARU alignment.

For ISFD not installed:

NOTE: The SAARU is also the attitude source for the Standby Attitude indicator and it should display sky/ground (blue over brown) shading.

2. Position left AIR DATA/ATT switch ALTN (non-normal position - switch in).
3. Verify the Attitude Indication on the left PFD displays the sky/ground (blue over brown) shading and pitch scale.
4. Position left AIR DATA/ATT switch off (normal position - switch out).

OPERATIONS NOTE

1. The NO LAND 3 advisory message will be displayed.
2. If the First Officer's AIR DATA/ATT instrument source switch is positioned to ALTN, airspeed and altitude may be flagged inoperative or may be in error on the right PFD. This effect is normal for this dispatch condition.
3. Certain faults of the Left Static ADM may cause the WINDSHEAR SYS advisory message to be displayed.
4. An additional failure, such as a plugged or disconnected static source, may result in the two remaining static sources miscomparing and voted airspeed and altitude being declared invalid. Lack of valid voted airspeed and altitude will result in display of the NAV AIR DATA SYS advisory or AIR DATA SYS Caution message and single channel airspeed and altitude will be displayed on the PFDs. Additionally, the left PFD airspeed and altitude may be flagged inoperative or may be in error.

NOTE: For a left PFD airspeed and altitude flagged, selection of the Captain's AIR DATA/ATT switch to ALTN will display the right PFD airspeed and altitude on the left PFD. This single air data condition will be indicated by the SGL SOURCE AIR DATA advisory message.

The following display effects with regard to airspeed (A/S) and altitude (ALT) are possible:



Static ADM Inop At Dispatch	Static Source Plugged, Disconnectd, etc.	L PFD A/S & ALT Display	Standby A/S & ALT Display	R PFD A/S & ALT Display
Left	Center	Flags (ADM failed) or In Error (ADM inop)	In Error (plugged)	Both Correct
Left	Right	Flags (ADM failed) or In Error (ADM inop)	Both Correct	In Error (plugged)

34-21-05 Static Air Data Modules (Primary - ARINC 629)**34-21-05-03 Center Static Air Data Module**

Interval	Installed	Required	Procedure
C	1	0	(M)

May be inoperative provided:

- Left and right static air data modules operate normally.
- Left and right pitot air data modules operate normally.
- Both AIR DATA/ATT instrument source switches operate normally.
- SAARU data is verified to be available to both PFDs before each departure.
- Standby airspeed indication operates normally.
- Approach minimums do not require its use.

MAINTENANCE (M)

NOTE: The left, center and right Pitot and Static ADMs are interchangeable.

Verify SAARU data is available to both Primary Flight Displays (PFD).

- With the airplane powered, verify SAARU alignment.

For ISFD not installed:

NOTE: The SAARU is also the attitude source for the Standby Attitude indicator and it should display sky/ground (blue over brown) shading.

- Verify SAARU data is available to the left PFD:
 - Position left AIR DATA/ATT switch ALTN (non-normal position - switch in).



- B. Verify the Attitude Indication on the left PFD displays the sky/ground (blue over brown) shading and pitch scale.
- C. Position left AIR DATA/ATT switch off (normal position - switch out).
3. Verify the SAARU data is available to the right PFD:
 - A. Position right AIR DATA/ATT switch to ALTN (non-normal position - switch in).
 - B. Verify the Attitude Indication on the right PFD displays the sky/ground (blue over brown) shading and pitch scale.
 - C. Position right AIR DATA/ATT switch off (normal position - switch out).

OPERATIONS NOTE

1. The NO LAND 3 advisory message will be displayed.
2. An additional failure, such as a plugged or disconnected static source, may result in the two remaining static sources miscomparing and voted airspeed and altitude being declared invalid. Lack of valid voted airspeed and altitude will result in display of the NAV AIR DATA SYS advisory or AIR DATA SYS Caution message and single channel airspeed and altitude will be displayed on the PFDs.

The following display effects with regard to airspeed (A/S) and altitude (ALT) are possible:

Static ADM Inop At Dispatch	Static Source Plugged, Disconnected, etc.	L PFD A/S & ALT Display	Standby A/S & ALT Display	R PFD A/S & ALT Display
Center	Left	In Error (plugged)	Both Correct	Both Correct
Center	Right	Both Correct	Both Correct	In Error (plugged)



34-21-06 Angle of Attack (AOA) Vane Systems

Interval	Installed	Required	Procedure
C	2	1	

OPERATIONS NOTE

If an error greater than four degrees (nose up) occurs on the remaining operational AOA vane, then the stick shaker may activate when stall conditions do not actually exist. In addition, the maximum and minimum speed indications on the PFD may come together. In general, this error will be detected within 20 seconds by the ADIRS and the remaining AOA vane rendered inoperative. The stall warning function will then resume normal operation using other signals within the ADIRS.

34-21-07 Secondary Attitude Air Data Reference Unit (SAARU)
34-21-07-01 Without Integrated Standby Flight Display (ISFD)
Installed

Interval	Installed	Required	Procedure
B	1	0	(O)

May be inoperative provided:

- Operations are conducted in Day VMC only.
- Operations are not conducted into known or forecast over-the-top conditions.
- ADIRU operates normally.
- Left, right and center pitot air data modules operate normally.
- Left, right and center static air data modules operate normally.
- Standby airspeed indicator operates normally.
- One GPS receiver operates normally.
- Non-stabilized magnetic compass operates normally.

For GE:

NOTE: For ENG EEC MODE L or R status or advisory messages displayed, the airplane must also be dispatched using MMEL item 73-21-02-02.

For GE:

MAINTENANCE NOTE

Some SAARU faults may cause the EEC's to revert to ALTN mode after engine start. To determine if this condition exists prior to engine start, set either Overhead Maintenance Panel (P61) EEC MAINT switch to the TEST position for 60 seconds and check EICAS for ENG EEC MODE L or R status or advisory messages displayed. Return the Overhead Maintenance Panel (P61) EEC MAINT switch to the NORM position.

OPERATIONS (O)

NOTE 1: If the Captain's AIR DATA/ATT instrument source switch is positioned to ALTN, attitude will be flagged inoperative and airspeed and altitude may be flagged inoperative on the Captain's PFD. If the First Officer's AIR DATA/ATT instrument source switch is positioned to ALTN, attitude will be flagged inoperative on the First Officer's PFD.

NOTE 2: Selecting ALTN altitude source on the ATC transponder panel with L XPNDR active or selecting NORM altitude source with R XPNDR active will result in no altitude reporting.

- Standby attitude indication will be inoperative.
- The flight must be conducted in day VMC conditions.



3. The flight must not be conducted in known or forecast over-the-top weather conditions.
4. The NO LAND 3 advisory message will be displayed.

34-21-07 Secondary Attitude Air Data Reference Unit (SAARU)
34-21-07-02 With Integrated Standby Flight Display (ISFD)
Installed

Interval	Installed	Required	Procedure
C	1	0	(O)

May be inoperative provided:

- a. Approach minimums do not require the use of the SAARU.
 - b. ADIRU operates normally.
 - c. Left, right and center pitot air data modules operate normally.
 - d. Left, right and center static air data modules operate normally.
 - e. ISFD operates normally.
 - f. One GPS receiver operates normally.
 - g. Non-stabilized magnetic compass operates normally.
-

For GE:

NOTE: For ENG EEC MODE L or R status or advisory messages displayed, the airplane must also be dispatched using MMEL item 73-21-02-02.

For GE:

MAINTENANCE NOTE

Some SAARU faults may cause the EEC's to revert to ALTN mode after engine start. To determine if this condition exists prior to engine start, set either Overhead Maintenance Panel (P61) EEC MAINT switch to the TEST position for 60 seconds and check EICAS for ENG EEC MODE L or R status or advisory messages displayed. Return the Overhead Maintenance Panel (P61) EEC MAINT switch to the NORM position.

OPERATIONS (O)

NOTE 1: If the Captain's AIR DATA/ATT instrument source switch is positioned to ALTN, attitude will be flagged inoperative and airspeed and altitude may be flagged inoperative on the Captain's PFD. If the First Officer's AIR DATA/ATT instrument source switch is positioned to ALTN, attitude will be flagged inoperative on the First Officer's PFD.

NOTE 2: For ATC transponder panel with ALT SOURCE selection, selecting ALTN altitude source will result in no altitude reporting.



NOTE 3: Selecting ALTN altitude source on the ATC transponder panel with L XPNDR active or selecting NORM altitude source with R XPNDR active will result in no altitude reporting.

The NO LAND 3 advisory message will be displayed.

**34-21-08 Integrated Standby Flight Display (ISFD)**

34-21-08-01 Attitude Indication

Interval	Installed	Required	Procedure
B	1	0	

May be inoperative provided:

- Operations are conducted in Day VMC only.
 - Operations are not conducted into known or forecast over-the-top conditions.
-

34-21-08 Integrated Standby Flight Display (ISFD)

34-21-08-02 Airspeed Indication

Interval	Installed	Required	Procedure
B	1	0	(M)

May be inoperative provided:

- All ARINC 629 pitot air data modules operate normally.
 - All ARINC 629 static air data modules operate normally.
 - All pitot probe heater systems operate normally.
 - Both AIR DATA/ATT instrument source switches operate normally.
 - SAARU data is verified available to both PFDs before each departure.
-

MAINTENANCE (M)

Verify that SAARU data is available to both PFDs before each departure (AMM 34-00-00/901).

- With the airplane powered and ADIRU ON and aligned, check SAARU alignment.
 - Position both AIR DATA/ATT switches ALTN (non-normal position - switch in).
 - Verify ALTN ATTITUDE advisory message is displayed.
 - Verify that the Attitude Indication on the both Primary Flight Displays (PFD) display the sky/ground (blue over brown) shading and pitch scale. No airspeed or altitude failure flags should be displayed on either PFD.
 - Position both AIR DATA/ATT switches off (normal position - switch out).
 - Verify ALTN ATTITUDE advisory message is not displayed.

OPERATIONS NOTE

Altitude differences may be noted between the ISFD altimeter indication and the primary displays, particularly at higher speeds and altitudes. The ISFD altimeter defaults to a fixed static source error correction when the ISFD airspeed indication is inoperative.

34-21-08 Integrated Standby Flight Display (ISFD)

34-21-08-03 Approach Mode

Interval	Installed	Required	Procedure
C	1	0	

34-21-08 Integrated Standby Flight Display (ISFD)

34-21-08-04 Heading Display

Interval	Installed	Required	Procedure
C	1	0	

34-21-08 Integrated Standby Flight Display (ISFD)
34-21-08-05 Switch Lights

Interval	Installed	Required	Procedure
C	5	0	

**34-23-01 Non-Stabilized Magnetic Compass (Standby)**

Interval	Installed	Required	Procedure
B	1	0	

May be inoperative provided ADIRU operates normally.

MAINTENANCE NOTE

The ADIRU is operating normally if the ADIRU status message is not displayed.



34-24-01 Standby Attitude Indicator (Non-ISFD System)

Interval	Installed	Required	Procedure
B	1	0	

- May be inoperative provided:
- a. Operations are conducted in Day VMC only.
 - b. Operations are not conducted into known or forecast over-the-top conditions.
-

**34-31-01 Instrument Landing System (ILS)**

Interval	Installed	Required	Procedure
D	3	-	

Any in excess of those required by 14 CFR may be inoperative provided approach minimums do not require their use.

OPERATIONS NOTE

1. See 14 CFR 121.347, 121.349 or other appropriate regulations.
2. For one inoperative ILS, the NO LAND 3 advisory message will be displayed.
3. For more than one inoperative ILS, the NO AUTOLAND advisory message will be displayed.
4. For left ILS inoperative, the Ground Proximity Warning System Mode 5 (Glideslope Deviation) will be inoperative and the airplane must also be dispatched using MMEL 34-46-01-01-02.

For ISFD installed:

5. For center ILS inoperative, the ISFD Approach Mode will be inoperative and the airplane must also be dispatched using MMEL 34-21-08-03.

**34-31-02 Glideslope Antenna Switching**

Interval	Installed	Required	Procedure
D	3	-	

Any in excess of those required by 14 CFR may be inoperative provided approach minimums do not require use of the associated ILS receiver.

OPERATIONS NOTE

1. See 14 CFR 121.347, 121.349 or other appropriate regulations.
2. For one G/S antenna switching inoperative, the NO LAND 3 advisory message will be displayed.
3. For more than one G/S antenna switching inoperative, the NO AUTOLAND advisory message will be displayed.

**34-31-03 Localizer Antenna Switching**

Interval	Installed	Required	Procedure
D	3	-	

Any in excess of those required by 14 CFR may be inoperative provided approach minimums do not require use of the associated ILS receiver.

OPERATIONS NOTE

1. See 14 CFR 121.347, 121.349 or other appropriate regulations.
2. For one LOC antenna switching inoperative, the NO LAND 3 advisory message will be displayed.
3. For more than one LOC antenna switching inoperative, the NO AUTOLAND advisory message will be displayed.

**34-32-01 Marker Beacon System**

Interval	Installed	Required	Procedure
C	1	0	

May be inoperative provided approach minimums do not require its use.

MAINTENANCE NOTE

The Marker Beacon and VOR receivers are contained in the same LRU. If the VOR system fails because of the Marker Beacon, the specific CMC fail message will indicate the Marker Beacon failure.

OPERATIONS NOTE

See 14 CFR 121.349 or other appropriate regulations.

**34-33-01 Radio Altimeter Systems**

Interval	Installed	Required	Procedure
B	3	2	

One may be inoperative provided approach minimums or operating procedures do not require its use.

OPERATIONS NOTE

The NO LAND 3 advisory message will be displayed for a fault indicated by a RADIO ALT X status message and may be displayed for a fault indicated by a RADIO ALT MONITOR X status message.



34-34-01 Microwave Landing System

Interval	Installed	Required	Procedure
D	1	0	

**34-35-01 Para Visual Displays**

Interval	Installed	Required	Procedure
D	-	0	

May be inoperative provided procedures do not require their use.

34-43-01 Weather Radar System
34-43-01A Reactive Windshear Operative

Interval	Installed	Required	Procedure
C	-	0	(O)

May be inoperative provided:

- Weather radar is not required by 14 CFR.
- Reactive windshear alert (GPWS Mode 7) operates normally.
- Alternate procedures are established and used.

MAINTENANCE NOTE

The WINDSHEAR PRED status message is not displayed if the WXR SYS status message is displayed.

OPERATIONS (O)

Alternate procedures must be established and used.

NOTE 1: The WINDSHEAR SYS advisory message will be displayed.

NOTE 2: See 14 CFR 121.357 or other appropriate regulation.

34-43-01 Weather Radar System
34-43-01B Reactive Windshear Inoperative

Interval	Installed	Required	Procedure
B	-	0	(O)

May be inoperative provided:

- Weather radar is not required by 14 CFR.
- Alternate procedures are established and used.

NOTE: Operator's alternate procedures should include reviewing windshear avoidance and windshear recovery procedures.

MAINTENANCE NOTE

The WINDSHEAR PRED status message is not displayed if the WXR SYS status message is displayed.

OPERATIONS (O)

Alternate procedures must be established and used.

NOTE 1: The WINDSHEAR SYS advisory message will be displayed.

NOTE 2: See 14 CFR 121.357 or other appropriate regulation.

**34-43-01 Weather Radar System****34-43-01C One Operative**

Interval	Installed	Required	Procedure
D	2	1	

34-43-01 Weather Radar System**34-43-01-01 Predictive Windshear Function****34-43-01-01A Reactive Windshear Operative**

Interval	Installed	Required	Procedure
C	-	0	(O)

May be inoperative provided:

- Alternate procedures are established and used.
- Reactive windshear alert (GPWS Mode 7) operates normally.

OPERATIONS (O)

Alternate procedures must be established and used.

NOTE: The WINDSHEAR SYS advisory message will be displayed.

34-43-01 Weather Radar System**34-43-01-01 Predictive Windshear Function****34-43-01-01B Reactive Windshear Inoperative**

Interval	Installed	Required	Procedure
B	-	0	(O)

May be inoperative provided alternate procedures are established and used.

NOTE: Operator's alternate procedures should include reviewing windshear avoidance and windshear recovery procedures.

OPERATIONS (O)

Alternate procedures must be established and used.

NOTE: The WINDSHEAR SYS advisory message will be displayed.

**34-43-01 Weather Radar System****34-43-01-02 Auto Tilt Function**

Interval	Installed	Required	Procedure
C	1	0	

May be inoperative provided manual tilt function operates normally.

OPERATIONS NOTE

Manual TILT adjustments may be necessary to optimize weather display.

**34-45-01 Traffic Collision and Avoidance System (TCAS)**

Interval	Installed	Required	Procedure
B	1	0	(M)

May be inoperative provided:

- System is deactivated and secured.
- Enroute or approach procedures do not require its use.

MAINTENANCE (M)

Deactivate and secure the system by opening and collaring the P11 panel TCAS circuit breaker (AMM 34-00-00/901).

34-45-01 Traffic Collision and Avoidance System (TCAS)**34-45-01-01 Resolution Advisory (RA) Display Systems****34-45-01-01A One Inoperative**

Interval	Installed	Required	Procedure
C	2	1	(O)

One may be inoperative on the non-flying pilot side.

OPERATIONS (O)

Observe operative indications and advisories. Respond as appropriate.

34-45-01 Traffic Collision and Avoidance System (TCAS)**34-45-01-01 Resolution Advisory (RA) Display Systems****34-45-01-01B Both Inoperative**

Interval	Installed	Required	Procedure
C	2	0	(O)

May be inoperative provided:

- Traffic Alert (TA) visual display and audio functions operate normally.
- TA only mode is selected by the crew.
- Enroute or approach procedures do not require its use.

OPERATIONS (O)

Select TA mode. Observe operative indications and advisories. Respond as appropriate.

**34-45-01 Traffic Collision and Avoidance System (TCAS)****34-45-01-02 Traffic Alert (TA) Display Systems**

Interval	Installed	Required	Procedure
C	2	0	(O)

May be inoperative provided:

- RA visual display and audio functions operate normally.
- Enroute or approach procedures do not require its use.

OPERATIONS (O)

Observe operative indications and advisories. Respond as appropriate.

34-45-01 Traffic Collision and Avoidance System (TCAS)**34-45-01-03 Audio Functions**

Interval	Installed	Required	Procedure
B	1	0	

May be inoperative provided enroute or approach procedures do not require use of TCAS.

34-45-01 Traffic Collision and Avoidance System (TCAS)**34-45-01-04 Airspace Selection Function**

Interval	Installed	Required	Procedure
C	-	0	

**34-46-01 Ground Proximity Warning System (GPWS)****34-46-01-01 GPWS Function**

Interval	Installed	Required	Procedure
A	1	0	(O)

May be inoperative provided:

- Alternate procedures are established and used.
- Repairs are made within 2 flight days.

For Honeywell weather radar installed:

NOTE: The airplane must also be dispatched using MMEL item 34-43-01-02.

OPERATIONS (O)

Alternate procedures must be established and used.

NOTE 1: These procedures must maintain flight crew awareness of airplane altitude and flight path. They may include flight crew use of all systems available (autopilot, autoland, ILS and FMCS) and approach altitude callouts.

NOTE 2: The GPWS provides the aural signal for the automatic V1 callout and the Engine Fail alert. These aural signals may not be available.

NOTE 3: The GND PROX SYS advisory message will be displayed.

NOTE 4: For PBN operations:

- RNP AR APCH operations are not allowed.
- Refer to DDG Section 4 for additional information.

For Honeywell weather radar installed:

NOTE 5: AUTOTILT FAIL may be annunciated on the ND.

34-46-01 Ground Proximity Warning System (GPWS)**34-46-01-01 GPWS Function****34-46-01-01-01 Modes 1 - 4**

Interval	Installed	Required	Procedure
A	4	0	(O)

May be inoperative provided:

- Alternate procedures are established and used.
- Repairs are made within 2 flight days.

OPERATIONS (O)

Alternate procedures must be established and used.

NOTE: These procedures must maintain flight crew awareness of airplane altitude and flight path. They may include flight crew use of all systems available (autopilot, autoland, ILS and FMCS) and approach altitude callouts.

34-46-01 Ground Proximity Warning System (GPWS)
34-46-01-01 GPWS Function
34-46-01-01-02 Glideslope Deviation (Mode 5)

Interval	Installed	Required	Procedure
B	1	0	

34-46-01 Ground Proximity Warning System (GPWS)
34-46-01-01 GPWS Function
34-46-01-01-03 Advisory Callouts (Mode 6)
34-46-01-01-03A Required By 14 CFR

Interval	Installed	Required	Procedure
B	-	0	(O)

May be inoperative provided alternate procedures are established and used.

OPERATIONS (O)

Alternate procedures must be established and used.

NOTE 1: These procedures must maintain flight crew awareness of airplane altitude and flight path. They may include flight crew use of all systems available (autopilot, autoland, ILS and FMCS) and approach altitude callouts.

NOTE 2: The GPWS provides the aural signal for the automatic V1 callout and the Engine Fail alert. These aural signals may not be available.

NOTE 3: See 14 CFR 121.354 or other appropriate regulation.

**34-46-01 Ground Proximity Warning System (GPWS)****34-46-01-01 GPWS Function****34-46-01-01-03 Advisory Callouts (Mode 6)****34-46-01-01-03B Not Required by 14 CFR**

Interval	Installed	Required	Procedure
C	-	0	(O)

May be inoperative provided:

- Advisory callouts are not required by 14 CFR.
- Alternate procedures are established and used.

OPERATIONS (O)

Alternate procedures must be established and used.

NOTE 1: These procedures must maintain flight crew awareness of airplane altitude and flight path. They may include flight crew use of all systems available (autopilot, autoland, ILS and FMCS) and approach altitude callouts.

NOTE 2: The GPWS provides the aural signal for the automatic V1 callout and the Engine Fail alert. These aural signals may not be available.

NOTE 3: See 14 CFR 121.354 or other appropriate regulation.

34-46-01 Ground Proximity Warning System (GPWS)**34-46-01-02 Windshear Alert Mode Function (Reactive) (Mode 7)****34-46-01-02A Predictive Windshear Operative**

Interval	Installed	Required	Procedure
C	1	0	(O)

May be inoperative provided:

- Alternate procedures are established and used.
- Predictive Windshear operates normally.

OPERATIONS (O)

Alternate procedures must be established and used.

NOTE: The WINDSHEAR SYS advisory message will be displayed.

34-46-01 Ground Proximity Warning System (GPWS)
34-46-01-02 Windshear Alert Mode Function (Reactive) (Mode 7)
34-46-01-02B Predictive Windshear Inoperative

Interval	Installed	Required	Procedure
B	1	0	(O)

May be inoperative provided alternate procedures are established and used.

NOTE: Operator's alternate procedures should include reviewing windshear avoidance and windshear recovery procedures.

OPERATIONS (O)

Establish alternate procedures for reactive windshear inoperative.

NOTE: The WINDSHEAR SYS advisory message will be displayed.

34-46-01 Ground Proximity Warning System (GPWS)
34-46-01-03 Terrain Awareness Function

Interval	Installed	Required	Procedure
B	1	0	(O)

May be inoperative provided alternate procedures are established and used.

[For Honeywell weather radar installed:](#)

NOTE: The airplane must also be dispatched using MMEL item 34-43-01-02.

OPERATIONS (O)

Alternate procedures must be established and used.

NOTE 1: The GND PROX SYS advisory message will be displayed.

NOTE 2: See 14 CFR 121.354 or other appropriate regulation.

NOTE 3: For PBN operations:

RNP AR APCH operations are not allowed.

Refer to DDG Section 4 for additional information.

[For Honeywell weather radar installed:](#)

NOTE 4: AUTOTILT FAIL may be annunciated on the ND.

**34-46-01 Ground Proximity Warning System (GPWS)****34-46-01-03 Terrain Awareness Function****34-46-01-03-01 Terrain Display Functions****34-46-01-03-01A One Inoperative**

Interval	Installed	Required	Procedure
C	2	1	

OPERATIONS NOTE

1. The GND PROX SYS advisory message may be displayed.
2. See 14 CFR 121.354 or other appropriate regulation.

NOTE: For PBN operations:

For GND PROX SYS advisory message displayed,
RNP AR APCH operations are not allowed.
Refer to DDG Section 4 for additional information.

34-46-01 Ground Proximity Warning System (GPWS)**34-46-01-03 Terrain Awareness Function****34-46-01-03-01 Terrain Display Functions****34-46-01-03-01B Both Inoperative**

Interval	Installed	Required	Procedure
B	2	0	

[For Honeywell weather radar installed:](#)

NOTE: The airplane must also be dispatched using MMEL item 34-43-01-02.

OPERATIONS NOTE

1. The GND PROX SYS advisory message may be displayed.
2. For PBN operations:
For GND PROX SYS advisory message displayed,
RNP AR APCH operations are not allowed.
Refer to DDG Section 4 for additional information.

[For Honeywell weather radar installed:](#)

3. AUTOTILT FAIL may be annunciated on the ND.



34-46-01 Ground Proximity Warning System (GPWS)

34-46-01-04 Runway Awareness and Advisory System (RAAS)

Interval	Installed	Required	Procedure
C	1	0	

OPERATIONS NOTE

For RUNWAY SYS advisory message displayed, consider using the RUNWAY OVRD function.

**34-51-01 VOR Navigation Systems**

Interval	Installed	Required	Procedure
D	2	-	

Any in excess of those required by 14 CFR may be inoperative.

OPERATIONS NOTE

1. See 14 CFR 121.347 through 121.351 or other appropriate regulation.
2. For PBN operations:
 - A. RNAV 1, RNAV 2, P-RNAV, B-RNAV and RNAV 5 operations may be allowed provided at least one of the following operates normally:
GPS, VOR or DME.

NOTE: Refer to DDG Section 4 for additional information.

**34-53-01 ATC Transponder/Automatic Altitude Reporting Systems****34-53-01A One Inoperative**

Interval	Installed	Required	Procedure
D	2	1	

Any in excess of those required by 14 CFR may be inoperative.

OPERATIONS NOTE

See 14 CFR 91.215 or other appropriate regulation.

34-53-01 ATC Transponder/Automatic Altitude Reporting Systems**34-53-01B Both Inoperative**

Interval	Installed	Required	Procedure
B	2	0	

May be inoperative provided:

- Enroute operations do not require its use.
- Prior to flight, approval is obtained from ATC facilities having jurisdiction over the planned route of flight.

NOTE: The airplane must also be dispatched using MMEL item 34-53-02.

OPERATIONS NOTE

See 14 CFR 91.215 or other appropriate regulation.

34-53-01 ATC Transponder/Automatic Altitude Reporting Systems**34-53-01-01 Elementary and Enhanced Downlink Aircraft Reportable Parameters Not Required By 14 CFR**

Interval	Installed	Required	Procedure
A	-	0	

May be inoperative provided:

- Enroute operations do not require their use.
- Repairs are made prior to completion of next heavy maintenance visit.

**34-53-01 ATC Transponder/Automatic Altitude Reporting Systems****34-53-01-02 ADS-B Squitter Transmissions**

Interval	Installed	Required	Procedure

Dispatch relief for this equipment moved to item 34-53-02 in revision 23.

**34-53-02 Automatic Dependent Surveillance-Broadcast (ADS-B) System (In and Out)**

34-53-02A Flight Authorized by ATC

Interval	Installed	Required	Procedure
B	2	0	(O)

May be inoperative provided prior to flight, authorization is obtained from ATC facilities having jurisdiction over the planned route of flight using an approved authorization process.

NOTE: Any ADS-B function that operates normally may be used.

OPERATIONS (O)

Prior to flight, obtain authorization from ATC facilities having jurisdiction over the planned route of flight using an approved authorization process.

34-53-02 Automatic Dependent Surveillance-Broadcast (ADS-B) System (In and Out)

34-53-02B One Inoperative

Interval	Installed	Required	Procedure
C	2	1	(O)

May be inoperative provided operative transponder is selected.

OPERATIONS (O)

Select transponder to operative side.

NOTE: See 14 CFR 91.227 or other appropriate regulation.

34-53-02 Automatic Dependent Surveillance-Broadcast (ADS-B) System (In and Out)

34-53-02C Enroute Operations Do Not Require Use

Interval	Installed	Required	Procedure
D	2	0	

May be inoperative provided:

- Enroute operations do not require their use.
- It is not required by 14 CFR.



NOTE: Any ADS-B function that operates normally may be used.

34-53-02 Automatic Dependent Surveillance-Broadcast (ADS-B) System (In and Out)

34-53-02-01 ADS-B Out Extended Squitter

34-53-02-01A Flight Authorized by ATC

Interval	Installed	Required	Procedure
B	2	0	(O)

May be inoperative provided prior to flight, authorization is obtained from ATC facilities having jurisdiction over the planned route of flight using an approved authorization process.

NOTE: Any ADS-B Out function that operates normally may be used.

OPERATIONS (O)

Prior to flight, obtain authorization from ATC facilities having jurisdiction over the planned route of flight using an approved authorization process.

34-53-02 Automatic Dependent Surveillance-Broadcast (ADS-B) System (In and Out)

34-53-02-01 ADS-B Out Extended Squitter

34-53-02-01B One Inoperative

Interval	Installed	Required	Procedure
C	2	1	(O)

One may be inoperative provided operative transponder is selected.

OPERATIONS (O)

Select transponder to operative side.

NOTE: See 14 CFR 91.227 or other appropriate regulation.

34-53-02 Automatic Dependent Surveillance-Broadcast (ADS-B) System (In and Out)
34-53-02-01 ADS-B Out Extended Squitter

34-53-02-01C Enroute Operations Do Not Require Use

Interval	Installed	Required	Procedure
D	2	0	

May be inoperative provided:

- Enroute operations do not require its use.
- It is not required by 14 CFR.

NOTE: Any ADS-B Out function that operates normally may be used.

OPERATIONS NOTE

See 14 CFR 91.227 or other appropriate regulation.

34-53-02 Automatic Dependent Surveillance-Broadcast (ADS-B) System (In and Out)
34-53-02-02 ADS-B In

34-53-02-02A Alternate Procedures Required

Interval	Installed	Required	Procedure
C	2	0	(O)

May be inoperative provided alternate procedures are established and used.

NOTE: Any ADS-B In function that operates normally may be used.

OPERATIONS (O)

Alternate procedures must be established and used.

34-53-02 Automatic Dependent Surveillance-Broadcast (ADS-B) System (In and Out)
34-53-02-02 ADS-B In

34-53-02-02B Enroute Operations Do Not Require Use

Interval	Installed	Required	Procedure
D	2	0	

May be inoperative provided operations do not require its use.



NOTE: Any ADS-B In function that operates normally may be used.

OPERATIONS NOTE

See 14 CFR 91.227 or other appropriate regulation.

**34-55-01 Distance Measuring Equipment (DME)**

Interval	Installed	Required	Procedure
D	2	-	

Any in excess of those required by 14 CFR may be inoperative.

OPERATIONS NOTE

1. See 14 CFR 121.349 or other appropriate regulation.
2. For PBN operations:
 - A. RNAV 1, RNAV 2, P-RNAV, B-RNAV and RNAV 5 operations may be allowed provided at least one of the following operates normally:
GPS, VOR or DME.

NOTE: Refer to DDG Section 4 for additional information.

**34-57-01 ADF Systems**

Interval	Installed	Required	Procedure
D	2	-	

Any in excess of those required by 14 CFR may be inoperative.

OPERATIONS NOTE

See 14 CFR 121.347 through 121.351 or other appropriate regulation.

34-58-01 Global Positioning Systems
34-58-01A Both Inoperative

Interval	Installed	Required	Procedure
C	2	0	

May be inoperative provided:

- Enroute operations do not require their use.
- SAARU operates normally.

NOTE: The airplane must also be dispatched using MMEL item 34-53-02-01.

MAINTENANCE NOTE

- For both GPS inoperative and AIMS power is interrupted, UTC is lost and the AIMS time function will reset to midnight.
- If desired, the left, right, or center MMR may be deactivated by opening and collaring P11 panel MMR-L, MMR-R and/or MMR-C circuit breaker.
- If the GPS L and GPS R status messages are caused by a fault in the center MMR antenna, opening the MMR-C circuit breaker may restore GPS L and R functions.
- The associated ILS function will be inoperative for a deactivated MMR. Relief for ILS functions is available in MMEL item 34-31-01.

OPERATIONS NOTE

- Operators must consult their Operations Specification for route position accuracy requirements such as that required by certain Required Navigation Performance (RNP) routes.
- The NAV UNABLE RNP advisory message may be displayed during lengthy ground operations and flight crew should accomplish the associated Non-Normal Checklist.
- The TERR POS advisory message will be displayed.
- The RWY/POS update on the FMC-CDU TAKEOFF REF page will be active.
- The ADS-B Out data will be inoperative.
- For airplanes equipped with RAAS:
 - RAAS is inoperative.
 - The RUNWAY SYS advisory message will be displayed.
 - The airplane must also be dispatched using MMEL item 34-46-01-04.
- For PBN operations:
 - RNP AR APCH, RNP APCH, RNP 1, RNP 2, RNP 4 and Advanced RNP operations are not allowed.
 - RNAV 1, RNAV 2, P-RNAV, B-RNAV and RNAV 5 may be allowed provided either DME function or VOR function is available.



- C. RNP 10 and RNAV 10 operations may be allowed provided ADIRU operates normally and appropriate time limitations are observed.

NOTE: Operations may be for up to 6.2 hours of flight time starting when system is placed in inertial navigation mode.

NOTE: Refer to DDG Section 4 for additional information.

34-58-01 Global Positioning Systems

34-58-01B One Inoperative

Interval	Installed	Required	Procedure
D	2	1	

One may be inoperative provided enroute operations do not require its use.

NOTE: The airplane must also be dispatched using MMEL item 34-53-02-01.

MAINTENANCE NOTE

1. If desired, one or both MMR not associated with the operative GPS may be deactivated by opening and collaring P11 panel MMR-L, MMR-R and/or MMR-C circuit breaker.
2. The associated ILS function will be inoperative for a deactivated MMR. Relief for ILS functions is available in MMEL item 34-31-01.

For EFB installed:

NOTE: Some functionality of the EFB may be inoperative. For inoperative EFB functions the airplane must also be dispatched using MMEL item 46-11-02-01.

OPERATIONS NOTE

1. The associated ADS-B Out data will be inoperative.
2. For PBN operations:
 - A. RNP AR APCH and RNP 4 operations are not allowed.
 - B. RNP APCH, RNP 1, RNP 2 and Advanced RNP operations may be allowed provided acceptable alternate navigation equipment (e.g. DME/DME, DME/VOR, VOR, ADF) is available to continue non-PBN operations in the event that the remaining GPS becomes inoperative.

NOTE: Refer to DDG Section 4 for additional information

34-61-01 Flight Management Computing Systems (FMCS)
34-61-01-01 With AIMS-1

Interval	Installed	Required	Procedure
C	2	1	(M)

One may be inoperative provided:

- a. Remaining FMC is verified to operate normally.
- b. Enroute operations do not require its use.
- c. Flight remains within 180 minutes of landing at a suitable airport.

MAINTENANCE (M)

Verify remaining FMC is operating normally by positioning the FMC selector to AUTO and verifying the FMC status and advisory messages are not displayed (AMM 34-00-00/901).

OPERATIONS NOTE

1. Flight must remain within 180 minutes of landing at a suitable airport.
2. Operators must consult their Operations Specification for route flight management computing system requirements.
3. For PBN operations:
 - A. RNP AR APCH and RNP 4 operations are not allowed.
 - B. RNP APCH, RNP 1, RNP 2, RNAV 1, RNAV 2, P-RNAV, B-RNAV, RNAV 5 and Advanced RNP operations may be allowed provided alternate navigation equipment (e.g., VOR, ADF) is available to continue non-PBN operations in the event that the remaining FMCS becomes inoperative.
 - C. RNP 10 and RNAV 10 operations may be allowed.

NOTE: Refer to DDG Section 4 for additional information.

34-61-01 Flight Management Computing Systems (FMCS)
34-61-01-02 With AIMS-2

Interval	Installed	Required	Procedure
C	2	1	(M)

One may be inoperative provided:

- a. Remaining FMC is verified to operate normally.
- b. Enroute operations do not require its use.

**MAINTENANCE (M)**

Verify remaining FMC is operating normally by positioning the FMC selector to AUTO and verifying the FMC status and advisory messages are not displayed (AMM 34-00-00/901).

OPERATIONS NOTE

1. Operators must consult their Operations Specification for route flight management computing system requirements.
2. For PBN operations:
 - A. RNP AR APCH and RNP 4 operations are not allowed.
 - B. RNP APCH, RNP 1, RNP 2, RNAV 1, RNAV 2, P-RNAV, B-RNAV, RNAV 5 and Advanced RNP operations may be allowed provided alternate navigation equipment (e.g., VOR, ADF) is available to continue non-PBN operations in the event that the remaining FMCS becomes inoperative.
 - C. RNP 10 and RNAV 10 operations may be allowed.

NOTE: Refer to DDG Section 4 for additional information.

34-61-01 Flight Management Computing Systems (FMCS)**34-61-01-03 Navigation Databases**

Interval	Installed	Required	Procedure
A	-	0	

May be inoperative provided:

- a. Operations do not require its use.
- b. It is not used in a primary navigation system required by 14 CFR.
- c. Alternate procedures are developed and used.
- d. The ICAO Flight Plan is updated (as required) to notify ATC of the navigation equipment status of the aircraft.
- e. Is repaired within 10 flight days.

NOTE: An out-of-currency or out-of-date navigation database is not authorized MMEL relief per 14 CFR.

NOTE: Although this item may not be applied for an out-of-currency Navigation Database, it does not prohibit dispatch in that condition. Dispatch is allowed with an out-of-currency Navigation Database provided alternate procedures are developed and incorporated in another operational manual, in coordination with local regulatory authorities.

OPERATIONS NOTE

For PBN operations:

1. A current navigation database is required for all PBN operations.



NOTE 1: It is assumed that the operator has a preflight procedure to ensure that the navigation database complies with regulatory agency requirements for ensuring the navigation database remains valid for the duration of the flight unless an alternative regulatory agency approved process applies.

NOTE 2: Refer to DDG Section 4 for additional information.



34-61-02 FMC Selector

Interval	Installed	Required	Procedure
C	1	0	

**34-61-03 Control Display Units (CDU)****34-61-03-01 Passenger**

Interval	Installed	Required	Procedure
C	3	2	(O)

Center CDU may be inoperative.

NOTE: Any function that operates normally may be used.

MAINTENANCE NOTE

The left, center and right CDUs are interchangeable.

OPERATIONS (O)

The center CDU is the only CDU that provides access to the Cabin Interphone System. Use the flight deck handset to access the Cabin Interphone System.

34-61-03 Control Display Units (CDU)**34-61-03-01 Passenger****34-61-03-01-01 DSPY, MSG, OFST and EXEC Lights**

Interval	Installed	Required	Procedure
C	12	0	

34-61-03 Control Display Units (CDU)**34-61-03-01 Passenger****34-61-03-01-02 Line Select Lights, Function Lights, Execute Lights, Alpha/Numeric Lights and Miscellaneous Key Lights**

Interval	Installed	Required	Procedure
C	-	0	

34-61-03 Control Display Units (CDU)**34-61-03-02 777F**

Interval	Installed	Required	Procedure
C	3	2	

Center CDU may be inoperative.



NOTE: Any function that operates normally may be used.

MAINTENANCE NOTE

The left, center and right CDUs are interchangeable.

34-61-03 Control Display Units (CDU)

34-61-03-02 777F

34-61-03-02-01 DSPY, MSG, OFST and EXEC Lights

Interval	Installed	Required	Procedure
C	12	0	

34-61-03 Control Display Units (CDU)

34-61-03-02 777F

**34-61-03-02-02 Line Select Lights, Function Lights, Execute Lights,
Alpha/Numeric Lights and Miscellaneous Key Lights**

Interval	Installed	Required	Procedure
C	-	0	

**35-11-01 Crew Oxygen System**

35-11-01-01 Crew Oxygen Pressure Indication System

35-11-01-01-01 Overboard Discharge Indicator Disc

35-11-01-02 Crew Oxygen Solenoid Bleed Valve

35-21-01 Passenger Oxygen System (Chemical or Gaseous) (Passenger)**35-21-02 Passenger Oxygen Service Units (Passenger)**35-21-02A *Seats Are Blocked*35-21-02B *Altitude Restrictions Are Applied***35-21-03 Passenger/Supernumerary Oxygen Automatic Presentation System****35-21-04 Passenger Oxygen ON Light****35-21-05 Crew/Attendant Rest Oxygen Modules (Passenger)**

35-21-05-01 Seat Modules

35-21-05-02 Bunk Modules

35-21-05-03 Vanity Module (Includes Sink With Drink Storage Area)

35-21-06 Supernumerary Seats/Crew Rest/Lavatory Gaseous Oxygen System (777F)

35-21-06-01 Left Circuit

35-21-06-02 Center Circuit

35-21-06-03 Right Circuit

35-21-07 Supernumerary Seats (Only) Gaseous Oxygen System (777F)**35-21-08 Lavatory Chemical Oxygen System (777F)****35-21-09 Crew Rest Chemical Oxygen Systems (777F)****35-22-01 Passenger/Supernumerary Oxygen Pressure Indication System (Gaseous)**

35-22-01-01 Overboard Discharge Indicator Disc

35-31-01 Portable Oxygen Dispensing Units (Or Equivalent) (Bottle and Mask)

35-31-01-01 Tamper Seals or Tags



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35-31-02 Portable Protective Breathing Equipment (PBE)

35-31-02-01 Tamper Seals or Tags

**35-11-01 Crew Oxygen System****35-11-01-01 Crew Oxygen Pressure Indication System**

Interval	Installed	Required	Procedure
B	1	0	(M)

May be inoperative provided:

- Oxygen supply is verified to be above minimum required before each departure.
- Crew oxygen shutoff valve(s) is verified open.

MAINTENANCE (M)

Verify sufficient crew oxygen supply and oxygen shutoff valve(s) open (AMM 35-00-00/901).

- Gain access to the area outboard and above the nose landing gear wheel well, on the left side of the airplane in the Main Equipment Center.
- Verify that the crew oxygen pressure shown on the gauge of the crew oxygen cylinder(s) is above the minimum required for dispatch.
- Verify that the crew oxygen cylinder shutoff valve(s) is open.

35-11-01 Crew Oxygen System**35-11-01-01 Crew Oxygen Pressure Indication System****35-11-01-01-01 Overboard Discharge Indicator Disc**

Interval	Installed	Required	Procedure
B	1	0	

May be damaged or missing.

35-11-01 Crew Oxygen System**35-11-01-02 Crew Oxygen Solenoid Bleed Valve**

Interval	Installed	Required	Procedure
B	1	0	(M)

May be inoperative provided:

- Solenoid bleed valve is deactivated closed.
- Crew oxygen shutoff valve(s) is verified open.

**MAINTENANCE (M)**

Deactivate the solenoid bleed valve closed and verify that the crew oxygen shutoff valve(s) is open (AMM 35-00-00/901).

1. Deactivate the solenoid valve in the closed position.
 - A. Open the OXY PRESS IND circuit breaker on the P210 panel.
 - B. Disconnect, cap and stow the electrical connector DM 35445 from the solenoid bleed valve. The valve is located on the lower left side of the Main Equipment Center, below and forward of the third step of the walkway.
 - C. Close the OXY PRESS IND circuit breaker.
 - D. Verify solenoid bleed valve is in the closed position and crew oxygen cylinder shutoff valve(s) is open.
 - 1) Position the Reset/Test slide lever on a crew oxygen mask to TEST.
 - 2) Push the Emergency/Test selector (red button) on the crew oxygen mask for 10 seconds.
 - 3) Observe that the crew oxygen pressure displayed on the status page does not decrease.

35-21-01 Passenger Oxygen System (Chemical or Gaseous) (Passenger)

Interval	Installed	Required	Procedure
B	1	0	(O)

May be inoperative provided:

- a. Altitude limitations comply with 14 CFR.
- b. Portable oxygen supplies comply with 14 CFR.
- c. Both air conditioning packs operate normally.
- d. Pressurization system operates normally.
- e. Both engine bleed systems operate normally.
- f. Passengers are appropriately briefed.

MAINTENANCE NOTE

1. For a pack to be operating normally, certain items associated with that pack system must be operative. See DDG item 21-51-01 for a listing of those items.
2. For the pressurization system to be operating normally, the following items associated with the pressurization system must be operative:

M MEL Item	Associated Status Message
21-31-01	CABIN ALT AUTO L or R
21-31-02	None
21-31-03	OUTFLOW VALVE FWD or AFT
21-31-04	CPCS REMOTE SENSOR
21-31-05	None
21-31-06	None
21-31-07	None
21-31-08	None
21-32-01	None

3. For an engine bleed system to be operating normally, certain items associated with that system must be operative. See DDG item 36-00-01 for a listing of those items.

OPERATIONS (O)

14 CFR passenger oxygen requirements (14 CFR 121.333(e)(1-3)) vary with the cruise altitude, the cabin pressure altitude to which an airplane may descend in the event of a depressurization during the flight, and the time required to make the emergency descent. If the airplane is operated at FL 250 or below and is able to descend to 14,000 feet cabin pressure altitude or below within four minutes following depressurization, oxygen must be available for 10% of passenger cabin occupants for 30 minutes.

If the airplane is operated at or below FL 250 and is unable to descend to 14,000 or below within four minutes, or if the airplane is operated above FL 250 regardless of the time to descend to 14,000 feet cabin pressure altitude or below following depressurization, oxygen must be available for 10% of the passenger cabin occupants for the duration of the flight with cabin pressure altitudes above 10,000 feet and at or below 14,000 feet, except that there must not be less than a 10-minute supply for the passenger cabin occupants.

If the airplane is able to descend to between 14,000 and 15,000 feet cabin pressure altitude, sufficient oxygen must be available for 30% of passenger cabin occupants. If the airplane cannot descend below 15,000 feet cabin pressure altitude, oxygen must be available for all (100%) of the passenger cabin occupants. In these cases, oxygen must be available for the required number of passengers for the entire duration of the flight flown at the cabin pressure altitudes above 14,000 feet. In all cases, sufficient oxygen must be available for the required number of passengers for at least 10 minutes. Operators should check the appropriate regulations to ensure compliance.

For flights above flight level 250, an undiluted oxygen supply is required for no less than two percent of the passengers for that portion of the flight at cabin pressure altitudes above 8,000 feet following depressurization. In no case will the supply be less than that required for one person.



Airplane Altitude (ft)	Cabin Altitude (ft)	Passenger Oxygen Usage	Passenger Oxygen Supply	14 CFR Paragraph Reference	Remarks
	>10,000 - 14,000	10% after 30 min	After 30 min	121.329(c)(1)	Covers both pressurized and unpressurized aircraft. Requirements based on cabin pressure altitude.
	>14,000 - 15,000	30%	[1]	121.329(c)(2)	
	>15,000	All	[1]	121.329(c)(3)	
<FL250	<14,000	10%	30 min	121.333(e)(1)	Descent to 14,000 ft can be made within 4 min.
<FL250 or >FL250	>10,000 - 14,000	10%	[1] [2]	121.333(e)(2)	Descent to 14,000 ft cannot be made within 4 min. Passenger briefing required.
	>14,000 - 15,000	30%	[1] [2]	121.333(e)(2)	
	>15,000	All	[1] [2]	121.333(e)(2)	
>FL250	>8,000	2%	14 CFR 25.1443(d)	121.333(e)(3) (first aid) [3]	Not less than one person, nor less than two dispensing units.

[1] As required by emergency descent profile following cabin depressurization

[2] A 10 minute minimum supply is required

[3] Medical oxygen requirements are listed in 14 CFR 121.574



35-21-02 Passenger Oxygen Service Units (Passenger)
35-21-02A Seats Are Blocked

Interval	Installed	Required	Procedure
B	-	0	(M)

May be inoperative provided:

- Associated seats are blocked and placarded to prevent occupancy.
- Associated flight attendant seat is considered inoperative.
- Associated lavatory door is locked closed and placarded, INOPERATIVE - DO NOT ENTER.

NOTE: This provision is not intended to prohibit lavatory inspections by crewmembers.

NOTE: For flight attendant seat affected by the inoperative PSU, the airplane must also be dispatched using MMEL item 25-25-01.

MAINTENANCE (M)

Configure the airplane (AMM 35-00-00/901).

- If required, tapes or ropes of conspicuous contrasting colors must be installed to block access to unusable seats prior to boarding of passengers.
- Conspicuous signs or placards shall be placed in appropriate locations to indicate seats which are not to be occupied by passengers or flight attendants.
- For inoperative lavatory oxygen PSU, close, lock and placard associated lavatory to prevent occupancy.

35-21-02 Passenger Oxygen Service Units (Passenger)
35-21-02B Altitude Restrictions Are Applied

Interval	Installed	Required	Procedure
B	-	0	(O)

May be inoperative provided:

- Altitude limitations comply with 14 CFR.
 - Portable oxygen supplies comply with 14 CFR.
 - Both air conditioning packs operate normally.
 - Pressurization system operates normally.
 - Both engine bleed systems operate normally.
 - Passengers are appropriately briefed.
-

MAINTENANCE NOTE

1. For a pack to be operating normally, certain items associated with that pack system must be operative. See DDG item 21-51-01 for a listing of those items.
2. For the pressurization system to be operating normally, the following items associated with the pressurization system must be operative:

M MEL Item	Associated Status Message
21-31-01	CABIN ALT AUTO L or R
21-31-02	None
21-31-03	OUTFLOW VALVE FWD or AFT
21-31-04	CPCS REMOTE SENSOR
21-31-05	None
21-31-06	None
21-31-07	None
21-31-08	None
21-32-01	None

3. For an engine bleed system to be operating normally, certain items associated with that system must be operative. See DDG item 36-00-01 for a listing of those items.

OPERATIONS (O)

Use DDG item 35-21-01 (O) procedure.

**35-21-03 Passenger/Supernumerary Oxygen Automatic Presentation System**

Interval	Installed	Required	Procedure
B	1	0	(O)

May be inoperative provided:

- Manual deployment system operates normally.
- Flight remains at or below FL 300.

OPERATIONS (O)

Flight altitude is limited to FL 300.



35-21-04 Passenger Oxygen ON Light

Interval	Installed	Required	Procedure
C	1	0	

**35-21-05 Crew/Attendant Rest Oxygen Modules (Passenger)**

35-21-05-01 Seat Modules

Interval	Installed	Required	Procedure
B	-	-	(M)

May be inoperative provided associated seat is blocked and placarded to prevent occupancy.

MAINTENANCE (M)

Conspicuous signs or placards shall be placed in appropriate locations to indicate seat(s) which are not to be occupied by cabin crew (AMM 35-00-00/901).

35-21-05 Crew/Attendant Rest Oxygen Modules (Passenger)

35-21-05-02 Bunk Modules

Interval	Installed	Required	Procedure
B	-	-	(M)

May be inoperative provided a conspicuous barrier strap or rope is placed across the associated bunk with a placard attached stating the bunk is not to be used.

MAINTENANCE (M)

Conspicuous signs or placards shall be placed in appropriate locations to indicate bunk(s) which are not to be occupied by cabin crew (AMM 35-00-00/901).

35-21-05 Crew/Attendant Rest Oxygen Modules (Passenger)

35-21-05-03 Vanity Module (Includes Sink With Drink Storage Area)

Interval	Installed	Required	Procedure
B	-	-	(M)

May be inoperative provided a conspicuous barrier strap or rope is placed across the associated vanity entrance with a placard attached stating the vanity is not to be used.

**MAINTENANCE (M)**

Conspicuous signs or placards shall be placed in appropriate locations to indicate associated vanity is not to be used (AMM 35-00-00/901).

**35-21-06 Supernumerary Seats/Crew Rest/Lavatory Gaseous Oxygen System (777F)**

Interval	Installed	Required	Procedure
B	1	0	(M) (O)

May be inoperative provided:

- Supernumerary seats and crew rest bunks are blocked and placarded DO NOT OCCUPY.
- An adequate portable oxygen dispensing unit (bottle and mask) is available in the lavatory.
- Personnel address system operates normally.
- Crew members/supernumeraries are appropriately briefed.

MAINTENANCE (M)

NOTE 1: For SUPRNMRY OXYGEN L status message displayed, the oxygen module over the left seat and the oxygen module in the crew rest are inoperative.

NOTE 2: For SUPRNMRY OXYGEN C status message displayed, the oxygen module over the center seats and the oxygen module in the lavatory are inoperative.

NOTE 3: For SUPRNMRY OXYGEN R status message displayed, the oxygen module over the right seat is inoperative.

NOTE 4: If desired, system may be deactivated (AMM 35-23-00/501).

Block seats and crew rest bunks and placard DO NOT OCCUPY (AMM 35-00-00/901).

OPERATIONS (O)

Crew members/supernumeraries must be briefed prior to takeoff regarding alternate procedures.

35-21-06 Supernumerary Seats/Crew Rest/Lavatory Gaseous Oxygen System (777F)

35-21-06-01 Left Circuit

Interval	Installed	Required	Procedure
B	1	0	(M) (O)

May be inoperative provided:

- Left supernumerary seat is blocked and placarded DO NOT OCCUPY.



- b. Each crew rest bunk without an adequate portable oxygen dispensing unit (bottle and mask) is blocked and placarded DO NOT OCCUPY.
 - c. For flightcrew rest facility used, personnel address system operates normally.
 - d. Crew members/supernumeraries are appropriately briefed.
-

MAINTENANCE (M)

NOTE: For SUPRNMRY OXYGEN L status message displayed, the oxygen module over the left seat and the oxygen module in the crew rest are inoperative.

Block left seat and placard DO NOT OCCUPY. Block crew rest bunks without adequate portable oxygen dispensing units (bottle and mask), and placard DO NOT OCCUPY (AMM 35-00-00/901).

OPERATIONS (O)

Crew members/supernumeraries must be briefed prior to takeoff regarding alternate procedures.

35-21-06 Supernumerary Seats/Crew Rest/Lavatory Gaseous Oxygen System (777F)*********35-21-06-02 Center Circuit**

Interval	Installed	Required	Procedure
B	1	0	(M) (O)

May be inoperative provided:

- a. Each center supernumerary seat is blocked and placarded DO NOT OCCUPY.
 - b. An adequate portable oxygen dispensing unit (bottle and mask) is available in the lavatory.
 - c. Personnel address system operates normally.
 - d. Crew members/supernumeraries are appropriately briefed.
-

MAINTENANCE (M)

NOTE: For SUPRNMRY OXYGEN C status message displayed, the oxygen module over the center seats and the oxygen module in the lavatory are inoperative.

Block each center seat and placard DO NOT OCCUPY (AMM 35-00-00/901).

OPERATIONS (O)

Crew members/supernumeraries must be briefed prior to takeoff regarding alternate procedures.

**35-21-06 Supernumerary Seats/Crew Rest/Lavatory Gaseous Oxygen System (777F)**

35-21-06-03 Right Circuit

Interval	Installed	Required	Procedure
B	1	0	(M) (O)

May be inoperative provided:

- Right supernumerary seat is blocked and placarded DO NOT OCCUPY.
- Crew members/supernumeraries are appropriately briefed.

MAINTENANCE (M)

NOTE: For SUPRNMRY OXYGEN R status message displayed, the oxygen module over the right seat is inoperative.

Block right seat and placard DO NOT OCCUPY (AMM 35-00-00/901).

OPERATIONS (O)

Crew members/supernumeraries must be briefed prior to takeoff regarding alternate procedures.

**35-21-07 Supernumerary Seats (Only) Gaseous Oxygen System (777F)**

Interval	Installed	Required	Procedure
B	-	0	(M) (O)

May be inoperative provided:

- Affected supernumerary seat is blocked and placarded DO NOT OCCUPY.
- Crew members/supernumeraries are appropriately briefed.

MAINTENANCE (M)

NOTE 1: For SUPRNMRY OXYGEN L status message displayed, the oxygen module mask door in the crew rest is inoperative.

NOTE 2: For SUPRNMRY OXYGEN C status message displayed, the oxygen module mask door in the lavatory is inoperative.

Block seats and placard DO NOT OCCUPY (AMM 35-00-00/901).

OPERATIONS (O)

Crew members/supernumeraries must be briefed prior to takeoff regarding alternate procedures.

**35-21-08 Lavatory Chemical Oxygen System (777F)**

Interval	Installed	Required	Procedure
B	1	0	(O)

May be inoperative provided:

- An adequate portable oxygen dispensing unit (bottle and mask) is available in the lavatory.
 - Personnel address system operates normally.
 - Crew members/supernumeraries are appropriately briefed.
-

OPERATIONS (O)

Crew members/supernumeraries must be briefed prior to takeoff regarding alternate procedures.

**35-21-09 Crew Rest Chemical Oxygen Systems (777F)**

Interval	Installed	Required	Procedure
B	-	0	(M) (O)

May be inoperative provided:

- Each crew rest bunk without an adequate portable oxygen dispensing unit (bottle and mask) is blocked and placarded DO NOT OCCUPY.
 - For flightcrew rest facility used, personnel address system operates normally. |
 - Crew members/supernumeraries are appropriately briefed.
-

MAINTENANCE (M)

Block crew rest bunks without adequate portable oxygen dispensing units (bottle and mask), and placard DO NOT OCCUPY (AMM 35-00-00/901).

OPERATIONS (O)

Crew members/supernumeraries must be briefed prior to takeoff regarding alternate procedures.



35-22-01 Passenger/Supernumerary Oxygen Pressure Indication System (Gaseous)

Interval	Installed	Required	Procedure
C	1	0	(M)

May be inoperative provided:

- Oxygen supply is verified to be above minimum required before each departure.
- Associated oxygen shutoff valves are verified open before each departure.

MAINTENANCE (M)

Verify there is sufficient oxygen supply and that the associated oxygen shutoff valves are open before each departure (AMM 35-00-00/901).

- Gain access to the associated oxygen cylinders located in the lower aft cargo compartment in the right sidewall area forward of the lower aft cargo door.
- Verify that the Passenger/Supernumerary Oxygen System average pressure reading based on the direct reading gauge on each oxygen cylinder, meets the minimum required for the flight.
- Verify that the associated oxygen cylinder shutoff valves are open.

35-22-01 Passenger/Supernumerary Oxygen Pressure Indication System (Gaseous)

35-22-01-01 Overboard Discharge Indicator Disc

Interval	Installed	Required	Procedure
B	1	0	

May be damaged or missing.

35-31-01 **Portable Oxygen Dispensing Units (Or Equivalent) (Bottle and Mask)**

Interval	Installed	Required	Procedure
D	-	-	(M)

Any in excess of those required by 14 CFR may be inoperative or removed provided:

- An inoperative or not properly serviced portable oxygen bottle/unit remains in a certified location until removed or serviced at the next suitable maintenance facility.
- Location placarding is removed or obscured.
- Required distribution is maintained.

NOTE 1: Inoperative portable oxygen bottles or units, removed from a certified location or removed from the aircraft, are subject to 49 CFR dangerous goods regulations.

NOTE 2: Medical equipment installed in the aircraft as part of an EMS operation is not considered part of the normal complement of equipment. No MMEL relief applies to that equipment and 14 CFR maintenance and inspection requirements do not apply.

MAINTENANCE (M)

- Ensure required distribution of serviceable units is maintained throughout the airplane (AMM 35-00-00/901).
- Ensure inoperative or not properly serviced portable oxygen bottle/unit remains in a certified location until removed or serviced at the next suitable maintenance facility.

OPERATIONS NOTE

Airlines must determine portable oxygen requirements dependent on their seating configuration and their route system. See 14 CFR 121.329 and 121.333 or other applicable regulations.

35-31-01 **Portable Oxygen Dispensing Units (Or Equivalent) (Bottle and Mask)**

35-31-01-01 **Tamper Seals or Tags**

Interval	Installed	Required	Procedure
C	-	-	(O)



May be inoperative, damaged, or missing provided proper installation and operation is verified at each preflight.

OPERATIONS (O)

Before each departure, verify affected item(s) is properly installed and operate(s) normally.

**35-31-02 Portable Protective Breathing Equipment (PBE)**

Interval	Installed	Required	Procedure
D	-	-	

Any in excess of those required by 14 CFR may be inoperative or removed provided:

- Inoperative PBE remains in a certified location until removed from the aircraft at the next suitable maintenance facility.
- Location placarding is removed or obscured.
- Required distribution is maintained.

NOTE: Inoperative PBEs, removed from a certified location or removed from the aircraft, are subject to 49 CFR dangerous goods regulations.

OPERATIONS NOTE

See 14 CFR 121.337 or other appropriate regulations.

35-31-02 Portable Protective Breathing Equipment (PBE)**35-31-02-01 Tamper Seals or Tags**

Interval	Installed	Required	Procedure
C	-	-	(O)

May be inoperative, damaged, or missing provided proper installation and operation is verified at each preflight.

OPERATIONS (O)

Before each departure, verify affected item(s) is properly installed and operate(s) normally.



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777 Dispatch Deviations Guide

Section 2

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36-11-02-01B *Associated Engine Bleed Not Used*

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36-11-02-02A *Associated Engine Bleed Used*

36-11-02-02B *Associated Engine Bleed Not Used*

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36-20-01 Air Synoptic Display**36-21-01 Manifold Pressure/Flow Sensing Systems****36-21-02 Intermediate Pressure Sensing Systems****36-22-01 Manifold Temperature Sensing Systems**

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36-22-01-02 -200/-200ER (With PRR 61980 or Equivalent Incorporated)

36-22-01-02A *Associated Engine Bleed Used*36-22-01-02B *Associated Engine Bleed Not Used*

36-22-01-03 -300 (Without SB 777-25-0227 and ASCPC SW P/N 3670-GRS-012-00 or later Installed) Or (Without PRR 61980 or Equivalent Incorporated)

36-22-01-03A *Associated Engine Bleed Used*36-22-01-03B *Associated Engine Bleed Not Used*



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36-22-01-04	-300 (With SB 777-25-0227 and ASCPC SW P/N 3670-GRS-012-00 or later Installed) Or (With PRR 61980 or Equivalent Incorporated)
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36-22-01-04B	<i>Associated Engine Bleed Not Used</i>
36-22-01-05	-200LR/-300ER
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**36-00-01 Air Supply and Cabin Pressure Controllers (ASCPC)**
36-00-01-01 Primary (Digital) Controllers

Interval	Installed	Required	Procedure
C	2	1	(M) (O)

Right Primary ASCPC may be inoperative provided:

- a. Right Primary ASCPC is deactivated.
- b. Both ASCPC backup (analog) controllers are verified to operate normally.
- c. Both right CTC channels are verified to operate normally once each flight day.
- d. Left engine bleed system operates normally.
- e. Hydraulic center system demand Air Driven Pump C1 operates normally.
- f. Engine Idle Selection Systems are considered inoperative.
- g. Right isolation system indications are considered inoperative.
- h. APU bleed air shutoff valve indications are considered inoperative.
- i. Auto Cabin Pressure Controls are considered inoperative.
- j. Appropriate performance adjustments are applied.

NOTE 1: The airplane must also be dispatched using MMEL item 73-21-01.

NOTE 2: The airplane must also be dispatched using MMEL item 36-12-01B.

NOTE 3: The airplane must also be dispatched using MMEL item 36-12-03-01.

NOTE 4: The airplane must also be dispatched using MMEL item 21-31-01.

MAINTENANCE (M)

NOTE 1: The left and right ASCPCs are interchangeable.

NOTE 2: For an engine bleed system to be operating normally, the following items associated with that system must be operative:

MMEL Item	Associated Status Message
36-00-01	ASCP PRIMARY CTRL L or R
36-11-01	BLEED PRSOV L or R
36-11-02	BLEED HPSOV L or R
36-11-03	None
36-11-04	None
36-11-05	BLEED FAMV L or R
36-21-01	MAN PRESS SENSOR L or R
36-21-02	BLEED PRESS SENSOR L or R
36-22-01	BLEED TEMP SENSOR L or R

Verify both ASCPC backup controllers operate normally, and verify both right Cabin Temperature Controllers (CTC) channels operate normally once each flight day (AMM 36-00-00/901).

1. Verify (one at a time) that both ASCPC backup (analog) controllers operate normally.
 - A. Verify the L ASCPC backup (analog) controller:
 - 1) Start left engine (AMM 71-00-00/201).
 - 2) Shutdown the APU if running or disconnect ground air cart if connected.
 - 3) Open the P310 panel AIR SPLY L PRI and AIR SPLY L SOL circuit breakers.
 - 4) Verify normal operation of the L ASCPC backup (analog) controller.
 - a. Position the BLEED AIR Panel (P5) L ENG Bleed switch OFF and confirm that the L PACK OFF light illuminates.
 - b. Position the BLEED AIR Panel (P5) L ENG Bleed switch ON and confirm that the L PACK OFF light extinguishes.
 - 5) Shut down left engine (AMM 71-00-00/201).
 - 6) Close the P310 panel AIR SPLY L PRI and AIR SPLY L SOL circuit breakers.
 - B. Verify the R ASCPC backup (analog) controller:
 - 1) Start right engine (AMM 71-00-00/201).
 - 2) Shutdown the APU if running or disconnect ground cart if connected.
 - 3) Open the P210 panel AIR SPLY R PRI and AIR SPLY R SOL circuit breakers.
 - 4) Open the P310 panel AIR SPLY R SEC circuit breaker.



- 5) Verify normal operation of the R ASCPC backup (analog) controller.
 - a. Position the BLEED AIR Panel (P5) R ENG Bleed switch OFF and confirm that the R PACK OFF light illuminates.
 - b. Position the BLEED AIR Panel (P5) R ENG Bleed switch ON and confirm that the R PACK OFF light extinguishes.
- 6) Shut down right engine (AMM 71-00-00/201).
- 7) Close the P210 panel AIR SPLY R PRI and AIR SPLY R SOL circuit breakers.
- 8) Close the P310 panel AIR SPLY R SEC circuit breaker.
2. Deactivate the R ASCPC primary controller.
 - A. Open and collar the P210 panel AIR SPLY R PRI and AIR SPLY R SOL circuit breakers.
 - B. Open and collar the P310 panel AIR SPLY R SEC circuit breaker.

NOTE: The CABIN ALT AUTO R status message will be displayed.
3. Verify that both right CTC channels operate normally once each flight day.

NOTE: If both channels of the right CTC do not operate normally, the right pack may be considered inoperative and the airplane dispatched using MMEL item 21-51-01.

 - A. Supply electrical power on the airplane (AMM 24-22-00/201).
 - B. Remove pneumatic power from the airplane (AMM 36-00-00/201).
 - C. Position right Air Conditioning panel (P5) PACK switch OFF.
 - D. Verify the right CTC channel 1 operates normally.
 - 1) Open the P210 panel R CTC CH 2 (AC) and R CTC CH 2 (DC) circuit breakers.
 - 2) Wait ten (10) seconds.
 - 3) Confirm CABIN TEMP CTRL R status message is not displayed.
 - 4) Close the P210 panel R CTC CH 2 (AC) and R CTC CH 2 (DC) circuit breakers.
 - E. Verify the right CTC channel 2 operates normally.
 - 1) Open the P210 panel R CTC CH 1 (AC) and R CTC CH 1 (DC) circuit breakers.
 - 2) Wait ten (10) seconds.
 - 3) Confirm CABIN TEMP CTRL R status message is not displayed.
 - 4) Close the P210 panel R CTC CH 1 (AC) and R CTC CH 1 (DC) circuit breakers.
 - F. Position right PACK switch AUTO.

**OPERATIONS (O)**

NOTE 1: For APU-to-Pack takeoff function installed, the APU-to-Pack takeoff Supplementary Procedure is not allowed.

NOTE 2: With the right ASCPC inoperative, indications for the APU bleed valve and the right isolation valve will be invalid.

1. Reduce performance limited weights by the appropriate adjustments.

Performance Limited Weights	Takeoff & Landing	Enroute Climb
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For -200 or -200ER/GE90-76B thru -90B:

thru 632,500 lb (286,902 kg)	5,900 lb (2,676 kg)	7,700 lb (3,493 kg)
over 632,500 lb (286,902 kg)	8,500 lb (3,856 kg)	7,900 lb (3,583 kg)

For -200 or -200ER/GE90-94B:

thru 632,500 lb (286,902 kg)	5,900 lb (2,676 kg)	9,400 lb (4,264 kg)
over 632,500 lb (286,902 kg)	9,200 lb (4,173 kg)	9,400 lb (4,264 kg)

For -200 or -200ER/PW4074 thru PW4084:

All	4,400 lb (1,996 kg)	8,600 lb (3,901 kg)
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For -200 or -200ER/PW4090:

thru 632,500 lb (286,902 kg)	6,100 lb (2,767 kg)	7,200 lb (3,266 kg)
over 632,500 lb (286,902 kg)	9,200 lb (4,173 kg)	7,200 lb (3,266 kg)

For -200 or -200ER/RR875 thru RR892:

thru 632,500 lb (286,902 kg)	3,300 lb (1,497 kg)	2,400 lb (1,089 kg)
over 632,500 lb (286,902 kg)	5,100 lb (2,313 kg)	2,400 lb (1,089 kg)

For -200 or -200ER/RR895:

All	3,300 lb (1,497 kg)	2,400 lb (1,089 kg)
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For -200LR or 777F/GE90-100 Series:

All	13,300 lb (6,033 kg)	10,100 lb (4,581 kg)
-----	----------------------	----------------------



For -300/PW4090:

All	5,800 lb (2,631 kg)	7,200 lb (3,266 kg)
-----	---------------------	---------------------

For -300/RR892:

All	3,300 lb (1,497 kg)	2,400 lb (1,089 kg)
-----	---------------------	---------------------

For -300ER/GE90-115B:

All	10,700 lb (4,854 kg)	10,200 lb (4,627 kg)
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NOTE: The enroute climb weight penalties listed above are based on single engine operating speeds that approximate maximum lift-to-drag ratio speed. To account for the difference in level off altitude when operating at other speeds, multiply the enroute climb weight penalty listed above by the appropriate factor listed on page 3.00-01-00.2 in the ENROUTE DIVERSION SPEED EFFECTS paragraph.

2. Increase flight planning fuel by 0.5%.
3. For ETOPS flight planning, increase ETOPS engine inoperative Critical Fuel Reserves.

For PW and RR:

- A. By 0.9%, or by 1.9% when dispatching into known or forecast icing conditions.

For GE:

- B. By 0.6%, or by 1.8% when dispatching into known or forecast icing conditions.
4. The AIR synoptic may display incomplete information or some information may be missing. The ENG IDLE DISAGREE advisory message will be displayed when idle thrust is selected inflight.
5. Operate the wing anti-ice manually.
 - A. Position WING ANTI-ICE selector OFF.
 - B. For flaps up and wing anti-ice required (do not use during takeoff and landing):
 - 1) Position the right ENG Bleed switch OFF.
 - 2) Position WING ANTI-ICE selector ON, then OFF as required.
 - 3) Prior to landing, position the ENG Bleed switch associated with the inoperative ASCP controller ON.
6. If an engine or engine bleed system fails enroute, position the associated ENG Bleed switch OFF.

36-11-01 Pressure Regulating and Shutoff Systems (PRSOV, PRSOVC)
36-11-01-01 -200/-200ER (Without PRR 61980 or Equivalent Incorporated)

Interval	Installed	Required	Procedure
C	2	1	(M) (O)

One valve (PRSOV) and/or the associated controller (PRSOVC) may be inoperative provided:

- a. Associated PRSOV is locked closed.
- b. Opposite engine bleed system operates normally.
- c. Associated engine bleed air switch remains OFF.
- d. Left and right bleed isolation systems operate normally.
- e. Center bleed isolation system operates normally.
- f. Both packs operate normally.
- g. Both outflow valves operate normally.
- h. Airplane remains at FL 350 or below.
- i. APU is used as air source for center system hydraulic demand pumps for takeoff and landing.
- j. Appropriate performance adjustments are applied.

NOTE 1: The airplane must also be dispatched using MMEL item 73-21-01.

NOTE 2: PRR 61980 installs enhancements for dispatch with a single pack, or dispatch with a single engine bleed system, on airplane L/N's 423, 429, 454, 455, 457 thru 461, and 463 and-on. For production equivalent information, see DDG item 21-51-01.

MAINTENANCE (M)

NOTE 1: For a pack to be operating normally, certain items associated with that system must be operative. See DDG item 21-51-01 for a listing of those items.

NOTE 2: For an engine bleed system to be operating normally, certain items associated with that system must be operative. See DDG item 36-00-01 for a listing of those items.

Lock the inoperative Pressure Regulating and Shutoff Valve (PRSOV) in the closed position (AMM 36-00-00/901).

For PW and RR:

1. Deactivate the associated thrust reverser for ground maintenance (AMM 78-31-00/201).
2. Position associated BLEED AIR Panel (P5) ENG Bleed switch OFF.

3. Gain access to the PRSOV by opening the appropriate fan cowl panel (AMM 71-11-04/201) and thrust reverser sleeve (AMM 78-31-00/201).
4. Lock the PRSOV in the closed position by following the MANUAL LOCKING CLOSED instructions placard on the valve.
5. Close the opened thrust reverser sleeve (AMM 78-31-00/201) and fan cowl panel (AMM 71-11-04/201).
6. Activate the thrust reverser (AMM 78-31-00/201).

For GE:

1. Deactivate the leading edge slats in the retracted position (AMM 27-81-00/201).
2. Deactivate the associated thrust reverser for ground maintenance (AMM 78-31-00/201).
3. Position associated BLEED AIR Panel (P5) ENG Bleed switch OFF.
4. Gain access to the PRSOV by opening the appropriate fan cowl panel (AMM 71-11-04/201) and thrust reverser sleeve (AMM 78-31-00/201).
5. Lock the PRSOV in the closed position by following the MANUAL LOCKING CLOSED instructions placard on the valve.
6. Close the opened thrust reverser sleeve (AMM 78-31-00/201) and fan cowl panel (AMM 71-11-04/201).
7. Activate the thrust reverser (AMM 78-31-00/201).
8. Activate the leading edge slats (AMM 27-81-00/201).

OPERATIONS (O)

NOTE: For APU-to-Pack takeoff function installed, the APU-to-Pack Takeoff Supplementary Procedure is not allowed.

1. Reduce the performance limited weights by the appropriate adjustments.

Performance Limited Weights	Takeoff & Landing (Packs On)	Takeoff (Packs Off)
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For -200 or -200ER/GE90-76B thru -90B:

thru 632,500 lb (286,902 kg)	14,300 lb (6,486 kg)	2,000 lb (907 kg)
over 632,500 lb (286,902 kg)	20,700 lb (9,389 kg)	2,000 lb (907 kg)

For -200 or -200ER/GE90-94B:

thru 632,500 lb (286,902 kg)	14,300 lb (6,486 kg)	2,000 lb (907 kg)
over 632,500 lb (286,902 kg)	22,400 lb (10,160 kg)	2,000 lb (907 kg)

For -200 or -200ER/PW4074 thru PW4084:

All	21,800 lb (9,888 kg)	1,500 lb (680 kg)
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**For -200 or -200ER/PW4090:**

thru 632,500 lb (286,902 kg)	21,500 lb (9,752 kg)	2,000 lb (907 kg)
over 632,500 lb (286,902 kg)	26,500 lb (12,020 kg)	2,000 lb (907 kg)

For -200 or -200ER/RR875 thru RR892:

thru 632,500 lb (286,902 kg)	6,400 lb (2,903 kg)	900 lb (408 kg)
over 632,500 lb (286,902 kg)	8,200 lb (3,720 kg)	900 lb (408 kg)

For -200 or -200ER/RR895:

All	6,800 lb (3,084 kg)	900 lb (408 kg)
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2. For ETOPS flight planning, increase the engine inoperative Critical Fuel Reserves by 1.3% when dispatching into known or forecast icing conditions.
3. Flight must remain at or below FL 350.
4. Position associated ENG Bleed switch OFF for the inoperative system.
NOTE: The ENG IDLE DISAGREE advisory message will be displayed when idle thrust is selected inflight.
5. For Takeoff, use the APU as an air source.
 - A. Position APU bleed air switch AUTO.
 - B. Start the APU if it is not running.
 - C. Position L ISLN and R ISLN switches CLOSED.
 - D. Accomplish the BLEED LOSS WING L, R Non-Normal Checklist and do not accomplish the PACK L, R Non-Normal Checklist.
 - E. After takeoff with flaps up:
 - 1) Position L ISLN and R ISLN switches AUTO.
 - 2) Position Wing Anti-Ice switch AUTO.
 - 3) Configure the APU as required.
6. In the event of inflight failure of the opposite engine or the engine bleed air system, use the APU to maintain cabin pressurization at or below FL 220.
 - A. Position APU bleed air switch AUTO.
 - B. Start the APU if it is not running.
 - C. Position both ENG Bleed switches OFF.
 - D. Position WING ANTI-ICE selector OFF (wing anti-ice will not be available for remainder of flight).
 - E. Position L ISLN and R ISLN switches AUTO for remainder of flight (including landing).
7. Prior to Landing, set the APU as a standby air source.
 - A. Position APU bleed air switch AUTO.

- B. Start the APU if it is not running.
- C. Position L ISLN and R ISLN switches AUTO.
- D. In the event of failure of the opposite engine or the engine bleed air system, position both ENG Bleed switches OFF.

36-11-01 Pressure Regulating and Shutoff Systems (PRSOV, PRSOVC)
36-11-01-02 -200/-200ER (With PRR 61980 or Equivalent Incorporated)

Interval	Installed	Required	Procedure
C	2	1	(M) (O)

One valve (PRSOV) and/or the associated controller (PRSOVC) may be inoperative provided:

- a. Associated PRSOV is locked closed.
- b. Opposite engine bleed system operates normally.
- c. Associated engine bleed air switch remains OFF.
- d. Left and right bleed isolation systems operate normally.
- e. Center bleed isolation system operates normally.
- f. Both packs operate normally.
- g. Both outflow valves operate normally.
- h. APU is used as air source for center system hydraulic demand pumps for takeoff and landing.
- i. Appropriate performance adjustments are applied.

NOTE 1: The airplane must also be dispatched using MMEL item 73-21-01.

NOTE 2: PRR 61980 installs enhancements for dispatch with a single pack, or dispatch with a single engine bleed system, on airplane L/N's 423, 429, 454, 455, 457 thru 461, and 463 and-on. For production equivalent information, see DDG item 21-51-01.

MAINTENANCE (M)

NOTE 1: For a pack to be operating normally, certain items associated with that system must be operative. See DDG item 21-51-01 for a listing of those items.

NOTE 2: For an engine bleed system to be operating normally, certain items associated with that system must be operative. See DDG item 36-00-01 for a listing of those items.

Lock the inoperative Pressure Regulating and Shutoff Valve (PRSOV) in the closed position (AMM 36-00-00/901).

**For PW and RR:**

1. Deactivate the associated thrust reverser for ground maintenance (AMM 78-31-00/201).
2. Position associated BLEED AIR Panel (P5) ENG Bleed switch OFF.
3. Gain access to the PRSOV by opening the appropriate fan cowl panel (AMM 71-11-04/201) and thrust reverser sleeve (AMM 78-31-00/201).
4. Lock the PRSOV in the closed position by following the MANUAL LOCKING CLOSED instructions placard on the valve.
5. Close the opened thrust reverser sleeve (AMM 78-31-00/201) and fan cowl panel (AMM 71-11-04/201).
6. Activate the thrust reverser (AMM 78-31-00/201).

For GE:

1. Deactivate the leading edge slats in the retracted position (AMM 27-81-00/201).
2. Deactivate the associated thrust reverser for ground maintenance (AMM 78-31-00/201).
3. Position associated BLEED AIR Panel (P5) ENG Bleed switch OFF.
4. Gain access to the PRSOV by opening the appropriate fan cowl panel (AMM 71-11-04/201) and thrust reverser sleeve (AMM 78-31-00/201).
5. Lock the PRSOV in the closed position by following the MANUAL LOCKING CLOSED instructions placard on the valve.
6. Close the opened thrust reverser sleeve (AMM 78-31-00/201) and fan cowl panel (AMM 71-11-04/201).
7. Activate the thrust reverser (AMM 78-31-00/201).
8. Activate the leading edge slats (AMM 27-81-00/201).

OPERATIONS (O)

NOTE: For the APU-to-Pack takeoff function installed, the APU-to-Pack takeoff Supplementary Procedure is not allowed.

1. Reduce the performance limited weights by the appropriate adjustments.

Performance Limited Weights	Takeoff & Landing (Packs On)	Takeoff (Packs Off)
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For -200 or -200ER/GE90-76B thru -90B:

thru 632,500 lb (286,902 kg)	14,300 lb (6,486 kg)	2,000 lb (907 kg)
over 632,500 lb (286,902 kg)	20,700 lb (9,389 kg)	2,000 lb (907 kg)

For -200 or -200ER/GE90-94B:

thru 632,500 lb (286,902 kg)	14,300 lb (6,486 kg)	2,000 lb (907 kg)
over 632,500 lb (286,902 kg)	22,400 lb (10,160 kg)	2,000 lb (907 kg)

For -200 or -200ER/PW4074 thru PW4084:

All	21,800 lb (9,888 kg)	1,500 lb (680 kg)
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For -200 or -200ER/PW4090:

thru 632,500 lb (286,902 kg)	21,500 lb (9,752 kg)	2,000 lb (907 kg)
over 632,500 lb (286,902 kg)	26,500 lb (12,020 kg)	2,000 lb (907 kg)

For -200 or -200ER/RR875 thru RR892:

thru 632,500 lb (286,902 kg)	6,400 lb (2,903 kg)	900 lb (408 kg)
over 632,500 lb (286,902 kg)	8,200 lb (3,720 kg)	900 lb (408 kg)

For -200 or -200ER/RR895:

All	6,800 lb (3,084 kg)	900 lb (408 kg)
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2. For ETOPS flight planning, increase the engine inoperative Critical Fuel Reserves by 1.3% when dispatching into known or forecast icing conditions.
3. Position associated ENG Bleed switch OFF for the inoperative system.
NOTE: The ENG IDLE DISAGREE advisory message will be displayed when idle thrust is selected inflight.
4. For Takeoff, use the APU as an air source.
 - A. Position APU bleed air switch AUTO.
 - B. Start the APU if it is not running.
 - C. Position L ISLN and R ISLN switches CLOSED.
 - D. Accomplish the BLEED LOSS WING L, R Non-Normal Checklist and do not accomplish the PACK L, R Non-Normal Checklist.
 - E. After takeoff with flaps up:
 - 1) Position L ISLN and R ISLN switches AUTO.
 - 2) Position Wing Anti-Ice switch AUTO.
 - 3) Configure the APU as required.
5. In the event of inflight failure of the opposite engine or the engine bleed air system, use the APU to maintain cabin pressurization at or below FL 220.
 - A. Position APU bleed air switch AUTO.
 - B. Start the APU if it is not running.
 - C. Position both ENG Bleed switches OFF.
 - D. Position WING ANTI-ICE selector OFF (wing anti-ice will not be available for remainder of flight).



- E. Position L ISLN and R ISLN switches AUTO for remainder of flight (including landing).
- 6. Prior to Landing, set the APU as a standby air source.
 - A. Position APU bleed air switch AUTO.
 - B. Start the APU if it is not running.
 - C. Position L ISLN and R ISLN switches AUTO.
 - D. In the event of failure of the opposite engine or the engine bleed air system, position both ENG Bleed switches OFF.

- 36-11-01 Pressure Regulating and Shutoff Systems (PRSOV, PRSOVC)**
- 36-11-01-03 -300 (Without SB 777-25-0227 and ASCPC SW P/N 3670-GRS-012-00 or later Installed) Or (Without PRR 61980 or Equivalent Incorporated)**

Interval	Installed	Required	Procedure
C	2	1	(M) (O)

One valve (PRSOV) and/or the associated controller (PRSOVC) may be inoperative provided:

- a. Associated PRSOV is locked closed.
- b. Opposite engine bleed system operates normally.
- c. Associated engine bleed air switch remains OFF.
- d. Left and right bleed isolation systems operate normally.
- e. Center bleed isolation system operates normally.
- f. Both packs operate normally.
- g. Both outflow valves operate normally.
- h. Procedures are established and used to verify forward cargo compartment remains empty or contains only ballast, empty cargo handling equipment (ballast may be loaded in ULDs), or fly away kits.
- i. APU is used as air source for center system hydraulic demand pumps for takeoff and landing.
- j. Appropriate performance adjustments are applied.

NOTE: Operator MELs must define which items are approved for inclusion in the fly away kits and which materials can be used as ballast.

NOTE 1: The airplane must also be dispatched using MMEL item 73-21-01.

NOTE 2: PRR 61980 installs enhancements for dispatch with a single pack, or dispatch with a single engine bleed system, on airplane L/N's 423, 429, 454, 455, 457 thru 461, and 463 and-on. For production equivalent information, see DDG item 21-51-01.

NOTE 3: All software P/Ns installed in ASCPC-2 are considered later versions than software P/Ns installed in ASCPC-1.

MAINTENANCE (M)

NOTE 1: For a pack to be operating normally, certain items associated with that system must be operative. See DDG item 21-51-01 for a listing of those items.

NOTE 2: For an engine bleed system to be operating normally, certain items associated with that system must be operative. See DDG item 36-00-01 for a listing of those items.

Lock the inoperative Pressure Regulating and Shutoff Valve (PRSOV) in the closed position (AMM 36-00-00/901).

1. Deactivate the associated thrust reverser for ground maintenance (AMM 78-31-00/201).
2. Position associated BLEED AIR Panel (P5) ENG Bleed switch OFF.
3. Gain access to the PRSOV by opening the appropriate fan cowl panel (AMM 71-11-04/201) and thrust reverser sleeve (AMM 78-31-00/201).
4. Lock the PRSOV in the closed position by following the MANUAL LOCKING CLOSED instructions placard on the valve.
5. Close the opened thrust reverser sleeve (AMM 78-31-00/201) and fan cowl panel (AMM 71-11-04/201).
6. Activate the thrust reverser (AMM 78-31-00/201).

OPERATIONS (O)

NOTE: For the APU-to-Pack takeoff function installed, the APU-to-Pack takeoff Supplementary Procedure is not allowed.

1. Reduce the performance limited weights by the appropriate adjustments.

Performance Limited Weights	Takeoff & Landing (Packs On)	Takeoff (Packs Off)
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For -300/PW4090:

All	21,800 lb (9,888 kg)	2,200 lb (998 kg)
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For -300/RR892:

All	7,000 lb (3,175 kg)	900 lb (408 kg)
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2. For ETOPS flight planning, increase the engine inoperative Critical Fuel Reserves by 1.3% when dispatching into known or forecast icing conditions.
3. Forward cargo compartment must remain empty.



NOTE: Does not preclude the carriage of empty cargo containers, pallets, ballast and cargo restraint components.

4. Position associated ENG Bleed switch OFF for the inoperative system.

NOTE: The ENG IDLE DISAGREE advisory message will be displayed when idle thrust is selected inflight.

5. For Takeoff, use the APU as an air source.
- A. Position APU bleed air switch AUTO.
 - B. Start the APU if it is not running.
 - C. Position L ISLN and R ISLN switches CLOSED.
 - D. Accomplish the BLEED LOSS WING L, R Non-Normal Checklist and do not accomplish the PACK L, R Non-Normal Checklist.
 - E. After takeoff with flaps up:
 - 1) Position L ISLN and R ISLN switches AUTO.
 - 2) Position Wing Anti-Ice switch AUTO.
 - 3) Configure the APU as required.
6. In the event of inflight failure of the opposite engine or the engine bleed air system, use the APU to maintain cabin pressurization at or below FL 220.
- A. Position APU bleed air switch AUTO.
 - B. Start the APU if it is not running.
 - C. Position both ENG Bleed switches OFF.
 - D. Position WING ANTI-ICE selector OFF (wing anti-ice will not be available for remainder of flight).
 - E. Position L ISLN and R ISLN switches AUTO for remainder of flight (including landing).
7. Prior to Landing, set the APU as a standby air source.
- A. Position APU bleed air switch AUTO.
 - B. Start the APU if it is not running.
 - C. Position L ISLN and R ISLN switches AUTO.
 - D. In the event of failure of the opposite engine or the engine bleed air system, position both ENG Bleed switches OFF.

36-11-01 Pressure Regulating and Shutoff Systems (PRSOV, PRSOVC)

36-11-01-04 -300 (With SB 777-25-0227 and ASCPC SW P/N 3670-GRS-012-00 or later Installed) Or (With PRR 61980 or Equivalent Incorporated)

Interval	Installed	Required	Procedure
C	2	1	(M) (O)

One valve (PRSOV) and/or the associated controller (PRSOVC) may be inoperative provided:

- a. Associated PRSOV is locked closed.
 - b. Opposite engine bleed system operates normally.
 - c. Associated engine bleed air switch remains OFF.
 - d. Left and right bleed isolation systems operate normally.
 - e. Center bleed isolation system operates normally.
 - f. Both packs operate normally.
 - g. Both outflow valves operate normally.
 - h. APU is used as air source for center system hydraulic demand pumps for takeoff and landing.
 - i. Appropriate performance adjustments are applied.
-

NOTE 1: The airplane must also be dispatched using MMEL item 73-21-01.

NOTE 2: PRR 61980 installs enhancements for dispatch with a single pack, or dispatch with a single engine bleed system, on airplane L/N's 423, 429, 454, 455, 457 thru 461, and 463 and-on. For production equivalent information, see DDG item 21-51-01.

NOTE 3: All software P/Ns installed in ASCPC-2 are considered later versions than software P/Ns installed in ASCPC-1.

MAINTENANCE (M)

NOTE 1: For a pack to be operating normally, certain items associated with that system must be operative. See DDG item 21-51-01 for a listing of those items.

NOTE 2: For an engine bleed system to be operating normally, certain items associated with that system must be operative. See DDG item 36-00-01 for a listing of those items.

Lock the inoperative Pressure Regulating and Shutoff Valve (PRSOV) in the closed position (AMM 36-00-00/901).

1. Deactivate the associated thrust reverser for ground maintenance (AMM 78-31-00/201).
2. Position associated BLEED AIR Panel (P5) ENG Bleed switch OFF.
3. Gain access to the PRSOV by opening the appropriate fan cowl panel (AMM 71-11-04/201) and thrust reverser sleeve (AMM 78-31-00/201).
4. Lock the PRSOV in the closed position by following the MANUAL LOCKING CLOSED instructions placard on the valve.
5. Close the opened thrust reverser sleeve (AMM 78-31-00/201) and fan cowl panel (AMM 71-11-04/201).
6. Activate the thrust reverser (AMM 78-31-00/201).

**OPERATIONS (O)**

NOTE: For the APU-to-Pack takeoff function installed, the APU-to-Pack takeoff Supplementary Procedure is not allowed.

1. Reduce the performance limited weights by the appropriate adjustments.

Performance Limited Weights	Takeoff & Landing (Packs On)	Takeoff (Packs Off)
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For -300/PW4090:

All	21,800 lb (9,888 kg)	2,200 lb (998 kg)
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For -300/RR892:

All	7,000 lb (3,175 kg)	900 lb(408 kg)
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2. For ETOPS flight planning, increase the engine inoperative Critical Fuel Reserves by 1.3% when dispatching into known or forecast icing conditions.
3. Position associated ENG Bleed switch OFF for the inoperative system.
NOTE: The ENG IDLE DISAGREE advisory message will be displayed when idle thrust is selected inflight.
4. For Takeoff, use the APU as an air source.
 - A. Position APU bleed air switch AUTO.
 - B. Start the APU if it is not running.
 - C. Position L ISLN and R ISLN switches CLOSED.
 - D. Accomplish the BLEED LOSS WING L, R Non-Normal Checklist and do not accomplish the PACK L, R Non-Normal Checklist.
 - E. After takeoff with flaps up:
 - 1) Position L ISLN and R ISLN switches AUTO.
 - 2) Position Wing Anti-Ice switch AUTO.
 - 3) Configure the APU as required.
5. In the event of inflight failure of the opposite engine or the engine bleed air system, use the APU to maintain cabin pressurization at or below FL 220.
 - A. Position APU bleed air switch AUTO.
 - B. Start the APU if it is not running.
 - C. Position both ENG Bleed switches OFF.
 - D. Position WING ANTI-ICE selector OFF (wing anti-ice will not be available for remainder of flight).
 - E. Position L ISLN and R ISLN switches AUTO for remainder of flight (including landing).

6. Prior to Landing, set the APU as a standby air source.
 - A. Position APU bleed air switch AUTO.
 - B. Start the APU if it is not running.
 - C. Position L ISLN and R ISLN switches AUTO.
 - D. In the event of failure of the opposite engine or the engine bleed air system, position both ENG Bleed switches OFF.

36-11-01 Pressure Regulating and Shutoff Systems (PRSOV, PRSOVC)
36-11-01-05 -200LR/-300ER

Interval	Installed	Required	Procedure
C	2	1	(M) (O)

One valve (PRSOV) and/or the associated controller (PRSOVC) may be inoperative provided:

- a. Associated PRSOV is locked closed.
- b. Opposite engine bleed system operates normally.
- c. Associated engine bleed air switch remains OFF.
- d. Left and right bleed isolation systems operate normally.
- e. Center bleed isolation system operates normally.
- f. Both packs operate normally.
- g. Both outflow valves operate normally.
- h. APU is used as air source for center system hydraulic demand pumps for takeoff and landing.
- i. Appropriate performance adjustments are applied.

NOTE: The airplane must also be dispatched using MMEL item 73-21-01.

MAINTENANCE (M)

NOTE 1: For a pack to be operating normally, certain items associated with that system must be operative. See DDG item 21-51-01 for a listing of those items.

NOTE 2: For an engine bleed system to be operating normally, certain items associated with that system must be operative. See DDG item 36-00-01 for a listing of those items.

Lock the inoperative Pressure Regulating and Shutoff Valve (PRSOV) in the closed position (AMM 36-00-00/901).

1. Deactivate the leading edge slats in the retracted position (AMM 27-81-00/201).
2. Deactivate the associated thrust reverser for ground maintenance (AMM 78-31-00/201).



3. Position associated BLEED AIR Panel (P5) ENG Bleed switch OFF.
4. Gain access to the PRSOV by opening the appropriate fan cowl panel (AMM 71-11-04/201) and thrust reverser sleeve (AMM 78-31-00/201).
5. Lock the PRSOV in the closed position by following the MANUAL LOCKING CLOSED instructions placard on the valve.
6. Close the opened thrust reverser sleeve (AMM 78-31-00/201) and fan cowl panel (AMM 71-11-04/201).
7. Activate the thrust reverser (AMM 78-31-00/201).
8. Activate the leading edge slats (AMM 27-81-00/201).

OPERATIONS (O)

NOTE: For the APU-to-Pack takeoff function installed, the APU-to-Pack takeoff Supplementary Procedure is not allowed.

1. Reduce the performance limited weights by the appropriate adjustments.

Performance Limited Weights	Takeoff & Landing (Packs On)	Takeoff (Packs Off)
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For -200LR/GE90-100 Series:

All	14,800 lb (6,713 kg)	5,350 lb (2,427 kg)
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For -300ER/GE90-115B:

All	11,600 lb (5,262 kg)	6,200 lb (2,812 kg)
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2. For ETOPS flight planning, increase the engine inoperative Critical Fuel Reserves by 1.3% when dispatching into known or forecast icing conditions.
3. Position associated ENG Bleed switch OFF for the inoperative system.
NOTE: The ENG IDLE DISAGREE advisory message will be displayed when idle thrust is selected inflight.
4. For Takeoff, use the APU as an air source.
 - A. Position APU bleed air switch AUTO.
 - B. Start the APU if it is not running.
 - C. Position L ISLN and R ISLN switches CLOSED.
 - D. Accomplish the BLEED LOSS WING L, R Non-Normal Checklist and do not accomplish the PACK L, R Non-Normal Checklist.
 - E. After takeoff with flaps up:
 - 1) Position L ISLN and R ISLN switches AUTO.
 - 2) Position Wing Anti-Ice switch AUTO.
 - 3) Configure the APU as required.

5. In the event of inflight failure of the opposite engine or the engine bleed air system, use the APU to maintain cabin pressurization at or below FL 220.
 - A. Position APU bleed air switch AUTO.
 - B. Start the APU if it is not running.
 - C. Position both ENG Bleed switches OFF.
 - D. Position WING ANTI-ICE selector OFF (wing anti-ice will not be available for remainder of flight).
 - E. Position L ISLN and R ISLN switches AUTO for remainder of flight (including landing).
6. Prior to Landing, set the APU as a standby air source.
 - A. Position APU bleed air switch AUTO.
 - B. Start the APU if it is not running.
 - C. Position L ISLN and R ISLN switches AUTO.
 - D. In the event of failure of the opposite engine or the engine bleed air system, position both ENG Bleed switches OFF.

36-11-01 Pressure Regulating and Shutoff Systems (PRSOV, PRSOVC)
36-11-01-06 777F

Interval	Installed	Required	Procedure
C	2	1	(M) (O)

One valve (PRSOV) and/or the associated controller (PRSOVC) may be inoperative provided:

- a. Associated PRSOV is locked closed.
- b. Opposite engine bleed system operates normally.
- c. Associated engine bleed air switch remains OFF.
- d. Left and right bleed isolation systems operate normally.
- e. Center bleed isolation system operates normally.
- f. Both packs operate normally.
- g. Both outflow valves operate normally.
- h. Both recirculation fans operate normally.
- i. Both ECSMCs operate normally.
- j. APU is used as air source for center system hydraulic demand pumps for takeoff and landing.
- k. Appropriate performance adjustments are applied.

NOTE: The airplane must also be dispatched using MMEL item 73-21-01.

**MAINTENANCE (M)**

NOTE 1: For a pack to be operating normally, certain items associated with that system must be operative. See DDG item 21-51-01 for a listing of those items.

NOTE 2: For an engine bleed system to be operating normally, certain items associated with that system must be operative. See DDG item 36-00-01 for a listing of those items.

Lock the inoperative Pressure Regulating and Shutoff Valve (PRSOV) in the closed position (AMM 36-00-00/901).

1. Deactivate the leading edge slats in the retracted position (AMM 27-81-00/201).
2. Deactivate the associated thrust reverser for ground maintenance (AMM 78-31-00/201).
3. Position associated BLEED AIR Panel (P5) ENG Bleed switch OFF.
4. Gain access to the PRSOV by opening the appropriate fan cowl panel (AMM 71-11-04/201) and thrust reverser sleeve (AMM 78-31-00/201).
5. Lock the PRSOV in the closed position by following the MANUAL LOCKING CLOSED instructions placard on the valve.
6. Close the opened thrust reverser sleeve (AMM 78-31-00/201) and fan cowl panel (AMM 71-11-04/201).
7. Activate the thrust reverser (AMM 78-31-00/201).
8. Activate the leading edge slats (AMM 27-81-00/201).

OPERATIONS (O)

NOTE 1: For overhead panel (P5) M/D FLOW switch positioned to HIGH, Air conditioning in the lower lobe cargo compartments is degraded. Position M/D FLOW switch to NORM for temperature sensitive cargo carried in the lower lobe cargo compartments.

NOTE 2: For the APU-to-Pack takeoff function installed, the APU-to-Pack takeoff Supplementary Procedure is not allowed.

1. Reduce the performance limited weights by the appropriate adjustments.

Performance Limited Weights	Takeoff & Landing (Packs On)	Takeoff (Packs Off)
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For 777F/GE90-100 Series:

All	14,800 lb (6,713 kg)	5,350 lb (2,427 kg)
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2. For ETOPS flight planning, increase the engine inoperative Critical Fuel Reserves by 1.3% when dispatching into known or forecast icing conditions.
3. Position associated ENG Bleed switch OFF for the inoperative system.



NOTE: The ENG IDLE DISAGREE advisory message will be displayed when idle thrust is selected inflight.

4. For Takeoff, use the APU as an air source.
 - A. Position APU bleed air switch AUTO.
 - B. Start the APU if it is not running.
 - C. Position L ISLN and R ISLN switches CLOSED.
 - D. Accomplish the BLEED LOSS WING L, R Non-Normal Checklist and do not accomplish the PACK L, R Non-Normal Checklist.
 - E. After takeoff with flaps up:
 - 1) Position L ISLN and R ISLN switches AUTO.
 - 2) Position Wing Anti-Ice switch AUTO.
 - 3) Configure the APU as required.
5. In the event of inflight failure of the opposite engine or the engine bleed air system, use the APU to maintain cabin pressurization at or below FL 220.
 - A. Position APU bleed air switch AUTO.
 - B. Start the APU if it is not running.
 - C. Position both ENG Bleed switches OFF.
 - D. Position WING ANTI-ICE selector OFF (wing anti-ice will not be available for remainder of flight).
 - E. Position L ISLN and R ISLN switches AUTO for remainder of flight (including landing).
6. Prior to Landing, set the APU as a standby air source.
 - A. Position APU bleed air switch AUTO.
 - B. Start the APU if it is not running.
 - C. Position L ISLN and R ISLN switches AUTO.
 - D. In the event of failure of the opposite engine or the engine bleed air system, position both ENG Bleed switches OFF.

36-11-01 Pressure Regulating and Shutoff Systems (PRSOV, PRSOVC)

36-11-01-07 ENG Bleed Air Switch Lights

36-11-01-07-01 OFF Lights

Interval	Installed	Required	Procedure
C	2	0	

**36-11-01 Pressure Regulating and Shutoff Systems (PRSOV, PRSOVC)****36-11-01-07 ENG Bleed Air Switch Lights****36-11-01-07-02 ON Lights**

Interval	Installed	Required	Procedure
C	2	0	

36-11-02 High Pressure Shutoff Valves (HPSOV)
36-11-02-01 -200/-200ER (Without PRR 61980 or Equivalent Incorporated)
36-11-02-01A Associated Engine Bleed Used

Interval	Installed	Required	Procedure
C	2	1	(M) (O)

One may be inoperative provided:

- Inoperative HPSOV is locked closed.
- Opposite engine bleed system operates normally.
- Both outflow valves operate normally.
- Airplane remains at FL 350 or below.
- Associated engine bleed air is cycled OFF, then ON at 55% N1 or greater before takeoff.

NOTE: PRR 61980 installs enhancements for dispatch with a single pack, or dispatch with a single engine bleed system, on airplane L/N's 423, 429, 454, 455, 457 thru 461, and 463 and-on. For production equivalent information, see DDG item 21-51-01.

MAINTENANCE (M)

NOTE: For an engine bleed system to be operating normally, certain items associated with that system must be operative. See DDG item 36-00-01 for a listing of those items.

Lock the inoperative High Pressure Shutoff Valve (HPSOV) in the closed position (AMM 36-00-00/901).

For PW:

- Deactivate the associated thrust reverser for ground maintenance (AMM 78-31-00/201).
- Gain access to the HPSOV by opening the appropriate fan cowl panel (AMM 71-11-04/201) and thrust reverser sleeve (AMM 78-31-00/201).
- Lock the HPSOV in the closed position by following the MANUAL LOCKING CLOSED instructions placard on the valve.
- Close the opened thrust reverser sleeve (AMM 78-31-00/201) and fan cowl panel (AMM 71-11-04/201).
- Activate the thrust reverser (AMM 78-31-00/201).
- With both engines shutdown, position associated flight deck Bleed Air panel (P11) ENG bleed switch ON to OFF and back to ON.

For GE:

- Deactivate the leading edge slats in the retracted position (AMM 27-81-00/201).

2. Deactivate the associated thrust reverser for ground maintenance (AMM 78-31-00/201).
3. Gain access to the HPSOV by opening the appropriate fan cowl panel (AMM 71-11-04/201) and thrust reverser sleeve (AMM 78-31-00/201).
4. Lock the HPSOV in the closed position by following the MANUAL LOCKING CLOSED instructions placard on the valve.
5. Close the opened thrust reverser sleeve (AMM 78-31-00/201) and fan cowl panel (AMM 71-11-04/201).
6. Activate the thrust reverser (AMM 78-31-00/201).
7. Activate the leading edge slats (AMM 27-81-00/201).

For RR:

1. Deactivate the associated thrust reverser for ground maintenance (AMM 78-31-00/201).
2. Gain access to the HPSOV by opening the appropriate fan cowl panel (AMM 71-11-04/201) and thrust reverser sleeve (AMM 78-31-00/201).
3. Lock the HPSOV in the closed position by following the MANUAL LOCKING CLOSED instructions placard on the valve.
4. Close the opened thrust reverser sleeve (AMM 78-31-00/201) and fan cowl panel (AMM 71-11-04/201).
5. Activate the thrust reverser (AMM 78-31-00/201).

OPERATIONS (O)

NOTE 1: Whenever power is set below 55% N1, the BLEED OFF ENG L or R advisory message will be displayed for the bleed system with the HPSOV locked closed. Do not accomplish the BLEED OFF ENG L or R Non-Normal Checklist. The engine bleed valve can be reopened by cycling the ENG bleed switch OFF, then ON at 55% N1 or greater.

NOTE 2: The ENG IDLE DISAGREE advisory message will be displayed when idle thrust is selected inflight prior to cycling the ENG bleed switch OFF, then ON at 55% N1 or greater.

NOTE 3: For an Engine Crossbleed Start supplementary procedure conducted, use the engine with the operative HPSOV as the operating engine.

NOTE 4: For the APU-to-Pack takeoff function installed, the APU-to-Pack takeoff Supplementary Procedure is not allowed.

1. Flight must remain at or below FL 350.
2. After engine start, the BLEED OFF ENG L or R advisory message will be displayed for the bleed system with HPSOV locked closed.

NOTE: When this condition occurs, the left and right bleed air manifolds will automatically be pressurized by bleed pressure from the opposite engine. Taxi and descent can be accomplished in this single bleed source configuration to avoid repetitive cycling of the Engine Bleed switch.

3. When setting thrust for takeoff, at 55% N1 or greater, select the associated ENG Bleed switch OFF, then ON.
4. In the event of an engine bleed failure on the opposite engine, maintain a minimum of 76% N1 on the engine with the inoperative HP SOV when the associated Engine Bleed switch is selected ON.
5. Prior to landing:
 - A. If the APU is available:
 - 1) Position APU Bleed switch AUTO.
 - 2) Start the APU if it is not running.
 - 3) Position L ISLN and R ISLN switches AUTO.
 - 4) In the event of failure of the opposite engine or the engine bleed air system, position both ENG Bleed switches OFF.
 - B. If the APU is not available:
 - 1) Prior to selecting flaps 1, if the BLEED OFF ENG L or R advisory message is displayed, open the associated engine bleed valve by cycling the respective ENG bleed switch OFF, then ON.

36-11-02 High Pressure Shutoff Valves (HPSOV)
36-11-02-01 -200/-200ER (Without PRR 61980 or Equivalent Incorporated)
36-11-02-01B Associated Engine Bleed Not Used

Interval	Installed	Required	Procedure
C	2	1	(M) (O)

One may be inoperative provided:

- a. Inoperative HPSOV is locked closed.
- b. Associated PRSOV is locked closed.
- c. Opposite engine bleed system operates normally.
- d. Associated engine bleed air switch remains OFF.
- e. Left and right bleed isolation systems operate normally.
- f. Center bleed isolation system operates normally.
- g. Both packs operate normally.
- h. Both outflow valves operate normally.
- i. Airplane remains at or below FL 350.
- j. APU is used as air source for center system hydraulic demand pumps for takeoff and landing.
- k. Appropriate performance adjustments are applied.

NOTE 1: The airplane must also be dispatched using MMEL item 73-21-01.



NOTE 2: PRR 61980 installs enhancements for dispatch with a single pack, or dispatch with a single engine bleed system, on airplane L/N's 423, 429, 454, 455, 457 thru 461, and 463 and-on. For production equivalent information, see DDG item 21-51-01.

MAINTENANCE (M)

NOTE 1: For a pack to be operating normally, certain items associated with that system must be operative. See DDG item 21-51-01 for a listing of those items.

NOTE 2: For an engine bleed system to be operating normally, certain items associated with that system must be operative. See DDG item 36-00-01 for a listing of those items.

Lock the inoperative High Pressure Shutoff Valve (HPSOV) and associated Pressure Regulating and Shutoff Valve (PRSOV) in the closed position (AMM 36-00-00/901).

For PW and RR:

1. Deactivate the associated thrust reverser for ground maintenance (AMM 78-31-00/201).
2. Position associated BLEED AIR Panel (P5) ENG Bleed switch OFF.
3. Gain access to the HPSOV and PRSOV by opening the appropriate fan cowl panel (AMM 71-11-04/201) and thrust reverser sleeve (AMM 78-31-00/201).
4. Lock the HPSOV and PRSOV in the closed positions by following the MANUAL LOCKING CLOSED instructions placard on the valves.
5. Close the opened thrust reverser sleeve (AMM 78-31-00/201) and fan cowl panel (AMM 71-11-04/201).
6. Activate the thrust reverser (AMM 78-31-00/201).

For GE:

1. Deactivate the leading edge slats in the retracted position (AMM 27-81-00/201).
2. Deactivate the associated thrust reverser for ground maintenance (AMM 78-31-00/201).
3. Position associated BLEED AIR Panel (P5) ENG Bleed switch OFF.
4. Gain access to the HPSOV and PRSOV by opening the appropriate fan cowl panel (AMM 71-11-04/201) and thrust reverser sleeve (AMM 78-31-00/201).
5. Lock the HPSOV and PRSOV in the closed positions by following the MANUAL LOCKING CLOSED instructions placard on the valves.
6. Close the opened thrust reverser sleeve (AMM 78-31-00/201) and fan cowl panel (AMM 71-11-04/201).
7. Activate the thrust reverser (AMM 78-31-00/201).
8. Activate the leading edge slats (AMM 27-81-00/201).

OPERATIONS (O)

NOTE: For the APU-to-Pack takeoff function installed, the APU-to-Pack Takeoff Supplementary Procedure is not allowed.

1. Reduce performance limited weights by the appropriate adjustments.

Performance Limited Weights	Takeoff & Landing (Packs On)	Takeoff (Packs Off)
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For -200 or -200ER/GE90-76B thru -90B:

thru 632,500 lb (286,902 kg)	14,300 lb (6,486 kg)	2,000 lb (907 kg)
over 632,500 lb (286,902 kg)	20,700 lb (9,389 kg)	2,000 lb (907 kg)

For -200 or -200ER/GE90-94B:

thru 632,500 lb (286,902 kg)	14,300 lb (6,486 kg)	2,000 lb (907 kg)
over 632,500 lb (286,902 kg)	22,400 lb (10,160 kg)	2,000 lb (907 kg)

For -200 or -200ER/PW4074 thru PW4084:

All	21,800 lb (9,888 kg)	1,500 lb (680 kg)
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For -200 or -200ER/PW4090:

thru 632,500 lb (286,902 kg)	21,500 lb (9,752 kg)	2,000 lb (907 kg)
over 632,500 lb (286,902 kg)	26,500 lb (12,020 kg)	2,000 lb (907 kg)

For -200 or -200ER/RR875 thru RR892:

thru 632,500 lb (286,902 kg)	6,400 lb (2,903 kg)	900 lb (408 kg)
over 632,500 lb (286,902 kg)	8,200 lb (3,720 kg)	900 lb (408 kg)

For -200 or -200ER/RR895:

All	6,800 lb (3,084 kg)	900 lb (408 kg)
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2. For ETOPS flight planning, increase the engine inoperative Critical Fuel Reserves by 1.3% when dispatching into known or forecast icing conditions.
3. Flight must remain at or below FL 350.
4. Position associated ENG Bleed switch OFF for the inoperative system.

NOTE: The ENG IDLE DISAGREE advisory message will be displayed when idle thrust is selected inflight.



5. For Takeoff, use the APU as an air source.
 - A. Position APU bleed air switch AUTO.
 - B. Start the APU if it is not running.
 - C. Position L ISLN and R ISLN switches CLOSED.
 - D. Accomplish the BLEED LOSS WING L, R Non-Normal Checklist and do not accomplish the PACK L, R Non-Normal Checklist.
 - E. After takeoff with flaps up:
 - 1) Position L ISLN and R ISLN switches AUTO.
 - 2) Position Wing Anti-Ice switch AUTO.
 - 3) Configure the APU as required.
6. In the event of inflight failure of the opposite engine or the engine bleed air system, use the APU to maintain cabin pressurization at or below FL 220.
 - A. Position APU bleed air switch AUTO.
 - B. Start the APU if it is not running.
 - C. Position both ENG Bleed switches OFF.
 - D. Position WING ANTI-ICE selector OFF (wing anti-ice will not be available for remainder of flight).
 - E. Position L ISLN and R ISLN switches AUTO for remainder of flight (including landing).
7. Prior to Landing, use the APU as an air source.
 - A. Position APU bleed air switch AUTO.
 - B. Start the APU if it is not running.
 - C. Position L ISLN and R ISLN switches AUTO.
 - D. In the event of failure of the opposite engine or the engine bleed air system, position both ENG Bleed switches OFF.

36-11-02 High Pressure Shutoff Valves (HPSOV)**36-11-02-02 -200/-200ER (With PRR 61980 or Equivalent Incorporated)****36-11-02-02A Associated Engine Bleed Used**

Interval	Installed	Required	Procedure
C	2	1	(M) (O)

One may be inoperative provided:

- a. Inoperative HPSOV is locked closed.
 - b. Opposite engine bleed system operates normally.
 - c. Both outflow valves operate normally.
 - d. Associated engine bleed air is cycled OFF, then ON at 55% N1 or greater before takeoff.
-

NOTE: PRR 61980 installs enhancements for dispatch with a single pack, or dispatch with a single engine bleed system, on airplane L/N's 423, 429, 454, 455, 457 thru 461, and 463 and-on. For production equivalent information, see DDG item 21-51-01.

MAINTENANCE (M)

NOTE: For an engine bleed system to be operating normally, certain items associated with that system must be operative. See DDG item 36-00-01 for a listing of those items.

Lock the inoperative High Pressure Shutoff Valve (HPSOV) in the closed position (AMM 36-00-00/901).

For PW:

1. Deactivate the associated thrust reverser for ground maintenance (AMM 78-31-00/201).
2. Gain access to the HPSOV by opening the appropriate fan cowl panel (AMM 71-11-04/201) and thrust reverser sleeve (AMM 78-31-00/201).
3. Lock the HPSOV in the closed position by following the MANUAL LOCKING CLOSED instructions placard on the valve.
4. Close the opened thrust reverser sleeve (AMM 78-31-00/201) and fan cowl panel (AMM 71-11-04/201).
5. Activate the thrust reverser (AMM 78-31-00/201).
6. With both engines shutdown, position associated flight deck Bleed Air panel (P11) ENG bleed switch ON to OFF and back to ON.

For GE:

1. Deactivate the leading edge slats in the retracted position (AMM 27-81-00/201).
2. Deactivate the associated thrust reverser for ground maintenance (AMM 78-31-00/201).
3. Gain access to the HPSOV by opening the appropriate fan cowl panel (AMM 71-11-04/201) and thrust reverser sleeve (AMM 78-31-00/201).
4. Lock the HPSOV in the closed position by following the MANUAL LOCKING CLOSED instructions placard on the valve.
5. Close the opened thrust reverser sleeve (AMM 78-31-00/201) and fan cowl panel (AMM 71-11-04/201).
6. Activate the thrust reverser (AMM 78-31-00/201).
7. Activate the leading edge slats (AMM 27-81-00/201).

For RR:

1. Deactivate the associated thrust reverser for ground maintenance (AMM 78-31-00/201).
2. Gain access to the HPSOV by opening the appropriate fan cowl panel (AMM 71-11-04/201) and thrust reverser sleeve (AMM 78-31-00/201).
3. Lock the HPSOV in the closed position by following the MANUAL LOCKING CLOSED instructions placard on the valve.

4. Close the opened thrust reverser sleeve (AMM 78-31-00/201) and fan cowl panel (AMM 71-11-04/201).
5. Activate the thrust reverser (AMM 78-31-00/201).

OPERATIONS (O)

NOTE 1: Whenever power is set below 55% N1, the BLEED OFF ENG L or R advisory message will be displayed for the bleed system with the HPSOV locked closed. Do not accomplish the BLEED OFF ENG L or R Non-Normal Checklist. The engine bleed valve can be reopened by cycling the ENG bleed switch OFF, then ON at 55% N1 or greater.

NOTE 2: The ENG IDLE DISAGREE advisory message will be displayed when idle thrust is selected and the BLEED OFF ENG L or R advisory message is displayed.

NOTE 3: For an Engine Crossbleed Start supplementary procedure conducted, use the engine with the operative HPSOV as the operating engine.

NOTE 4: For the APU-to-Pack takeoff function installed, the APU-to-Pack takeoff Supplementary Procedure is not allowed.

1. After engine start, the BLEED OFF ENG L or R advisory message will be displayed for the bleed system with HPSOV locked closed.

NOTE: When this condition occurs, the left and right bleed air manifolds will automatically be pressurized by bleed pressure from the opposite engine. Taxi and descent can be accomplished in this single bleed source configuration to avoid repetitive cycling of the Engine Bleed switch.

2. When setting thrust for takeoff, at 55% N1 or greater, select the associated ENG Bleed switch OFF, then ON.
3. In the event of an engine bleed failure on the opposite engine, maintain a minimum of 76% N1 on the engine with the inoperative HPSOV when the associated Engine Bleed switch is selected ON.
4. Prior to landing:
 - A. If the APU is available:
 - 1) Position APU Bleed switch AUTO.
 - 2) Start the APU if it is not running.
 - 3) Position L ISLN and R ISLN switches AUTO.
 - 4) In the event of failure of the opposite engine or the engine bleed air system, position both ENG Bleed switches OFF.
 - B. If the APU is not available:
 - 1) Prior to selecting flaps 1, if the BLEED OFF ENG L or R advisory message is displayed, open the associated engine bleed valve by cycling the respective ENG bleed switch OFF, then ON at 55% N1 or greater.

36-11-02 High Pressure Shutoff Valves (HPSOV)
36-11-02-02 -200/-200ER (With PRR 61980 or Equivalent Incorporated)
36-11-02-02B Associated Engine Bleed Not Used

Interval	Installed	Required	Procedure
C	2	1	(M) (O)

One may be inoperative provided:

- a. Inoperative HPSOV is locked closed.
- b. Associated PRSOV is locked closed.
- c. Opposite engine bleed system operates normally.
- d. Associated engine bleed air switch remains OFF.
- e. Left and right bleed isolation systems operate normally.
- f. Center bleed isolation system operates normally.
- g. Both packs operate normally.
- h. Both outflow valves operate normally.
- i. APU is used as air source for center system hydraulic demand pumps for takeoff and landing.
- j. Appropriate performance adjustments are applied.

NOTE 1: The airplane must also be dispatched using MMEL item 73-21-01.

NOTE 2: PRR 61980 installs enhancements for dispatch with a single pack, or dispatch with a single engine bleed system, on airplane L/N's 423, 429, 454, 455, 457 thru 461, and 463 and-on. For production equivalent information, see DDG item 21-51-01.

MAINTENANCE (M)

NOTE 1: For a pack to be operating normally, certain items associated with that system must be operative. See DDG item 21-51-01 for a listing of those items.

NOTE 2: For an engine bleed system to be operating normally, certain items associated with that system must be operative. See DDG item 36-00-01 for a listing of those items.

Lock the inoperative High Pressure Shutoff Valve (HPSOV) and associated Pressure Regulating and Shutoff Valve (PRSOV) in the closed position (AMM 36-00-00/901).

For PW and RR:

1. Deactivate the associated thrust reverser for ground maintenance (AMM 78-31-00/201).
2. Position associated BLEED AIR Panel (P5) ENG Bleed switch OFF.



3. Gain access to the HPSOV and PRSOV by opening the appropriate fan cowl panel (AMM 71-11-04/201) and thrust reverser sleeve (AMM 78-31-00/201).
4. Lock the HPSOV and PRSOV in the closed positions by following the MANUAL LOCKING CLOSED instructions placard on the valves.
5. Close the opened thrust reverser sleeve (AMM 78-31-00/201) and fan cowl panel (AMM 71-11-04/201).
6. Activate the thrust reverser (AMM 78-31-00/201).

For GE:

1. Deactivate the leading edge slats in the retracted position (AMM 27-81-00/201).
2. Deactivate the associated thrust reverser for ground maintenance (AMM 78-31-00/201).
3. Position associated BLEED AIR Panel (P5) ENG Bleed switch OFF.
4. Gain access to the HPSOV and PRSOV by opening the appropriate fan cowl panel (AMM 71-11-04/201) and thrust reverser sleeve (AMM 78-31-00/201).
5. Lock the HPSOV and PRSOV in the closed positions by following the MANUAL LOCKING CLOSED instructions placard on the valves.
6. Close the opened thrust reverser sleeve (AMM 78-31-00/201) and fan cowl panel (AMM 71-11-04/201).
7. Activate the thrust reverser (AMM 78-31-00/201).
8. Activate the leading edge slats (AMM 27-81-00/201).

OPERATIONS (O)

NOTE: For the APU-to-Pack takeoff function installed, the APU-to-Pack Takeoff Supplementary Procedure is not allowed.

1. Reduce the performance limited weights by the appropriate adjustments.

Performance Limited Weights	Takeoff & Landing (Packs On)	Takeoff (Packs Off)
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For -200 or -200ER/GE90-76B thru -90B:

thru 632,500 lb (286,902 kg)	14,300 lb (6,486 kg)	2,000 lb (907 kg)
over 632,500 lb (286,902 kg)	20,700 lb (9,389 kg)	2,000 lb (907 kg)

For -200 or -200ER/GE90-94B:

thru 632,500 lb (286,902 kg)	14,300 lb (6,486 kg)	2,000 lb (907 kg)
over 632,500 lb (286,902 kg)	22,400 lb (10,160 kg)	2,000 lb (907 kg)



For -200 or -200ER/PW4074 thru PW4084:

All	21,800 lb (9,888 kg)	1,500 lb (680 kg)
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For -200 or -200ER/PW4090:

thru 632,500 lb (286,902 kg)	21,500 lb (9,752 kg)	2,000 lb (907 kg)
over 632,500 lb (286,902 kg)	26,500 lb (12,020 kg)	2,000 lb (907 kg)

For -200 or -200ER/RR875 thru RR892:

thru 632,500 lb (286,902 kg)	6,400 lb (2,903 kg)	900 lb (408 kg)
over 632,500 lb (286,902 kg)	8,200 lb (3,720 kg)	900 lb (408 kg)

For -200 or -200ER/RR895:

All	6,800 lb (3,084 kg)	900 lb (408 kg)
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2. For ETOPS flight planning, increase the engine inoperative Critical Fuel Reserves by 1.3% when dispatching into known or forecast icing conditions.
3. Position associated ENG Bleed switch OFF for the inoperative system.
NOTE: The ENG IDLE DISAGREE advisory message will be displayed when idle thrust is selected inflight.
4. For Takeoff, use the APU as an air source.
 - A. Position APU bleed air switch AUTO.
 - B. Start the APU if it is not running.
 - C. Position L ISLN and R ISLN switches CLOSED.
 - D. Accomplish the BLEED LOSS WING L, R Non-Normal Checklist and do not accomplish the PACK L, R Non-Normal Checklist.
 - E. After takeoff with flaps up:
 - 1) Position L ISLN and R ISLN switches AUTO.
 - 2) Position Wing Anti-Ice switch AUTO.
 - 3) Configure the APU as required.
5. In the event of inflight failure of the opposite engine or the engine bleed air system, use the APU to maintain cabin pressurization at or below FL 220.
 - A. Position APU bleed air switch AUTO.
 - B. Start the APU if it is not running.
 - C. Position both ENG Bleed switches OFF.
 - D. Position WING ANTI-ICE selector OFF (wing anti-ice will not be available for remainder of flight).



- E. Position L ISLN and R ISLN switches AUTO for remainder of flight (including landing).
- 6. Prior to Landing, use the APU as an air source.
 - A. Position APU bleed air switch AUTO.
 - B. Start the APU if it is not running.
 - C. Position L ISLN and R ISLN switches AUTO.
 - D. In the event of failure of the opposite engine or the engine bleed air system, position both ENG Bleed switches OFF.

36-11-02 High Pressure Shutoff Valves (HPSOV)

36-11-02-03 -300 (Without SB 777-25-0227 and ASCPC SW P/N 3670-GRS-012-00 or later Installed) Or (Without PRR 61980 or Equivalent Incorporated)

36-11-02-03A Associated Engine Bleed Used

Interval	Installed	Required	Procedure
C	2	1	(M) (O)

One may be inoperative provided:

- a. Inoperative HPSOV is locked closed.
- b. Opposite engine bleed system operates normally.
- c. Both outflow valves operate normally.
- d. Procedures are established and used to verify forward cargo compartment remains empty or contains only ballast, empty cargo handling equipment (ballast may be loaded in ULDs), or fly away kits.
- e. Associated engine bleed air is cycled OFF, then ON at 55% N1 or greater before takeoff.

NOTE: Operator MELs must define which items are approved for inclusion in the fly away kits and which materials can be used as ballast.

NOTE 1: PRR 61980 installs enhancements for dispatch with a single pack, or dispatch with a single engine bleed system, on airplane L/N's 423, 429, 454, 455, 457 thru 461, and 463 and-on. For production equivalent information, see DDG item 21-51-01.

NOTE 2: All software P/Ns installed in ASCPC-2 are considered later versions than software P/Ns installed in ASCPC-1.

MAINTENANCE (M)

NOTE: For an engine bleed system to be operating normally, certain items associated with that system must be operative. See DDG item 36-00-01 for a listing of those items.

Lock the inoperative High Pressure Shutoff Valve (HPSOV) in the closed position (AMM 36-00-00/901).

For PW:

1. Deactivate the associated thrust reverser for ground maintenance (AMM 78-31-00/201).
2. Gain access to the HPSOV by opening the appropriate fan cowl panel (AMM 71-11-04/201) and thrust reverser sleeve (AMM 78-31-00/201).
3. Lock the HPSOV in the closed position by following the MANUAL LOCKING CLOSED instructions placard on the valve.
4. Close the opened thrust reverser sleeve (AMM 78-31-00/201) and fan cowl panel (AMM 71-11-04/201).
5. Activate the thrust reverser (AMM 78-31-00/201).
6. With both engines shutdown, position associated flight deck Bleed Air panel (P11) ENG bleed switch ON to OFF and back to ON.

For RR:

1. Deactivate the associated thrust reverser for ground maintenance (AMM 78-31-00/201).
2. Gain access to the HPSOV by opening the appropriate fan cowl panel (AMM 71-11-04/201) and thrust reverser sleeve (AMM 78-31-00/201).
3. Lock the HPSOV in the closed position by following the MANUAL LOCKING CLOSED instructions placard on the valve.
4. Close the opened thrust reverser sleeve (AMM 78-31-00/201) and fan cowl panel (AMM 71-11-04/201).
5. Activate the thrust reverser (AMM 78-31-00/201).

OPERATIONS (O)

NOTE 1: Whenever power is set below 55% N1, the BLEED OFF ENG L or R advisory message will be displayed for the bleed system with the HPSOV locked closed. Do not accomplish the BLEED OFF ENG L or R Non-Normal Checklist. The engine bleed valve can be reopened by cycling the ENG bleed switch OFF, then ON at 55% N1 or greater.

NOTE 2: The ENG IDLE DISAGREE advisory message will be displayed when idle thrust is selected inflight prior to cycling the ENG bleed switch OFF, then ON at 55% N1 or greater.

NOTE 3: For an Engine Crossbleed Start supplementary procedure conducted, use the engine with the operative HPSOV as the operating engine.

NOTE 4: For the APU-to-Pack takeoff function installed, the APU-to-Pack takeoff Supplementary Procedure is not allowed.

1. Forward cargo compartment must remain empty.

NOTE: Does not preclude the carriage of empty cargo containers, pallets, ballast and cargo restraint components.

2. After engine start, the BLEED OFF ENG L or R advisory message will be displayed for the bleed system with HPSOV locked closed.

NOTE: When this condition occurs, the left and right bleed air manifolds will automatically be pressurized by bleed pressure from the opposite engine. Taxi and descent can be accomplished in this single bleed source configuration to avoid repetitive cycling of the Engine Bleed switch.
3. When setting thrust for takeoff, at 55% N1 or greater, select the associated ENG Bleed switch OFF, then ON.
4. In the event of an engine bleed failure on the opposite engine, maintain a minimum of 76% N1 on the engine with the inoperative HPSOV when the associated Engine Bleed switch is selected ON.
5. Prior to landing:
 - A. If the APU is available:
 - 1) Position APU Bleed switch AUTO.
 - 2) Start the APU if it is not running.
 - 3) Position L ISLN and R ISLN switches AUTO.
 - 4) In the event of failure of the opposite engine or the engine bleed air system, position both ENG Bleed switches OFF.
 - B. If the APU is not available:
 - 1) Prior to selecting flaps 1, if the BLEED OFF ENG L or R advisory message is displayed, open the associated engine bleed valve by cycling the respective ENG bleed switch OFF, then ON.

36-11-02 High Pressure Shutoff Valves (HPSOV)

36-11-02-03 -300 (Without SB 777-25-0227 and ASCPC SW P/N 3670-GRS-012-00 or later Installed) Or (Without PRR 61980 or Equivalent Incorporated)

36-11-02-03B Associated Engine Bleed Not Used

Interval	Installed	Required	Procedure
C	2	1	(M) (O)

One may be inoperative provided:

- a. Inoperative HPSOV is locked closed.
- b. Associated PRSOV is locked closed.
- c. Opposite engine bleed system operates normally.
- d. Associated engine bleed air switch remains OFF.
- e. Left and right bleed isolation systems operate normally.
- f. Center bleed isolation system operates normally.
- g. Both packs operate normally.

- h. Both outflow valves operate normally.
- i. Procedures are established and used to verify forward cargo compartment remains empty or contains only ballast, empty cargo handling equipment (ballast may be loaded in ULDs), or fly away kits.
- j. APU is used as air source for center system hydraulic demand pumps for takeoff and landing.
- k. Appropriate performance adjustments are applied.

NOTE: Operator MELs must define which items are approved for inclusion in the fly away kits and which materials can be used as ballast.

NOTE 1: The airplane must also be dispatched using MMEL item 73-21-01.

NOTE 2: PRR 61980 installs enhancements for dispatch with a single pack, or dispatch with a single engine bleed system, on airplane L/N's 423, 429, 454, 455, 457 thru 461, and 463 and-on. For production equivalent information, see DDG item 21-51-01.

NOTE 3: All software P/Ns installed in ASCPC-2 are considered later versions than software P/Ns installed in ASCPC-1.

MAINTENANCE (M)

NOTE 1: For a pack to be operating normally, certain items associated with that system must be operative. See DDG item 21-51-01 for a listing of those items.

NOTE 2: For an engine bleed system to be operating normally, certain items associated with that system must be operative. See DDG item 36-00-01 for a listing of those items.

Lock the inoperative High Pressure Shutoff Valve (HPSOV) and associated Pressure Regulating and Shutoff Valve (PRSOV) in the closed position (AMM 36-00-00/901).

1. Deactivate the associated thrust reverser for ground maintenance (AMM 78-31-00/201).
2. Position associated BLEED AIR Panel (P5) ENG Bleed switch OFF.
3. Gain access to the HPSOV and PRSOV by opening the appropriate fan cowl panel (AMM 71-11-04/201) and thrust reverser sleeve (AMM 78-31-00/201).
4. Lock the HPSOV and PRSOV in the closed positions by following the MANUAL LOCKING CLOSED instructions placard on the valves.
5. Close the opened thrust reverser sleeve (AMM 78-31-00/201) and fan cowl panel (AMM 71-11-04/201).
6. Activate the thrust reverser (AMM 78-31-00/201).

**OPERATIONS (O)**

NOTE: For the APU-to-Pack takeoff function installed, the APU-to-Pack Takeoff Supplementary Procedure is not allowed.

1. Reduce the performance limited weights by the appropriate adjustments.

Performance Limited Weights	Takeoff & Landing (Packs On)	Takeoff (Packs Off)
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For -300/PW4090:

All	21,800 lb (9,888 kg)	2,200 lb (998 kg)
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For -300/RR892:

All	7,000 lb (3,175 kg)	900 lb (408 kg)
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2. For ETOPS flight planning, increase the engine inoperative Critical Fuel Reserves by 1.3% when dispatching into known or forecast icing conditions.
3. The forward cargo compartment must remain empty, except for ballast, empty cargo containers (ballast may be loaded in ULDs), fly away kits, pallets, and cargo restraint components.
4. Position associated ENG Bleed switch OFF for the inoperative system.
NOTE: The ENG IDLE DISAGREE advisory message will be displayed when idle thrust is selected inflight.
5. For Takeoff, use the APU as an air source.
 - A. Position APU bleed air switch AUTO.
 - B. Start the APU if it is not running.
 - C. Position L ISLN and R ISLN switches CLOSED.
 - D. Accomplish the BLEED LOSS WING L, R Non-Normal Checklist and do not accomplish the PACK L, R Non-Normal Checklist.
 - E. After takeoff with flaps up:
 - 1) Position L ISLN and R ISLN switches AUTO.
 - 2) Position Wing Anti-Ice switch AUTO.
 - 3) Configure the APU as required.
6. In the event of inflight failure of the opposite engine or the engine bleed air system, use the APU to maintain cabin pressurization at or below FL 220.
 - A. Position APU bleed air switch AUTO.
 - B. Start the APU if it is not running.
 - C. Position both ENG Bleed switches OFF.
 - D. Position WING ANTI-ICE selector OFF (wing anti-ice will not be available for remainder of flight).



- E. Position L ISLN and R ISLN switches AUTO for remainder of flight (including landing).
- 7. Prior to Landing, use the APU as an air source.
 - A. Position APU bleed air switch AUTO.
 - B. Start the APU if it is not running.
 - C. Position L ISLN and R ISLN switches AUTO.
 - D. In the event of failure of the opposite engine or the engine bleed air system, position both ENG Bleed switches OFF.

36-11-02 High Pressure Shutoff Valves (HPSOV)

36-11-02-04 -300 (With SB 777-25-0227 and ASCPC SW P/N 3670-GRS-012-00 or later Installed) Or (With PRR 61980 or Equivalent Incorporated)

36-11-02-04A Associated Engine Bleed Used

Interval	Installed	Required	Procedure
C	2	1	(M) (O)

One may be inoperative provided:

- a. Inoperative HPSOV is locked closed.
- b. Opposite engine bleed system operates normally.
- c. Both outflow valves operate normally.
- d. Associated engine bleed air is cycled OFF, then ON at 55% N1 or greater before takeoff.

NOTE 1: PRR 61980 installs enhancements for dispatch with a single pack, or dispatch with a single engine bleed system, on airplane L/N's 423, 429, 454, 455, 457 thru 461, and 463 and-on. For production equivalent information, see DDG item 21-51-01.

NOTE 2: All software P/Ns installed in ASCPC-2 are considered later versions than software P/Ns installed in ASCPC-1.

MAINTENANCE (M)

NOTE: For an engine bleed system to be operating normally, certain items associated with that system must be operative. See DDG item 36-00-01 for a listing of those items.

Lock the inoperative High Pressure Shutoff Valve (HPSOV) in the closed position (AMM 36-00-00/901).

For PW:

- 1. Deactivate the associated thrust reverser for ground maintenance (AMM 78-31-00/201).

2. Gain access to the HPSOV by opening the appropriate fan cowl panel (AMM 71-11-04/201) and thrust reverser sleeve (AMM 78-31-00/201).
3. Lock the HPSOV in the closed position by following the MANUAL LOCKING CLOSED instructions placard on the valve.
4. Close the opened thrust reverser sleeve (AMM 78-31-00/201) and fan cowl panel (AMM 71-11-04/201).
5. Activate the thrust reverser (AMM 78-31-00/201).
6. With both engines shutdown, position associated flight deck Bleed Air panel (P11) ENG bleed switch ON to OFF and back to ON.

For RR:

1. Deactivate the associated thrust reverser for ground maintenance (AMM 78-31-00/201).
2. Gain access to the HPSOV by opening the appropriate fan cowl panel (AMM 71-11-04/201) and thrust reverser sleeve (AMM 78-31-00/201).
3. Lock the HPSOV in the closed position by following the MANUAL LOCKING CLOSED instructions placard on the valve.
4. Close the opened thrust reverser sleeve (AMM 78-31-00/201) and fan cowl panel (AMM 71-11-04/201).
5. Activate the thrust reverser (AMM 78-31-00/201).

OPERATIONS (O)

NOTE 1: Whenever power is set below 55% N1, the BLEED OFF ENG L or R advisory message will be displayed for the bleed system with the HPSOV locked closed. Do not accomplish the BLEED OFF ENG L or R Non-Normal Checklist. The engine bleed valve can be reopened by cycling the ENG bleed switch OFF, then ON at 55% N1 or greater.

NOTE 2: The ENG IDLE DISAGREE advisory message will be displayed when idle thrust is selected inflight prior to cycling the ENG bleed switch OFF, then ON at 55% N1 or greater.

NOTE 3: For an Engine Crossbleed Start supplementary procedure conducted, use the engine with the operative HPSOV as the operating engine.

NOTE 4: For the APU-to-Pack takeoff function installed, the APU-to-Pack takeoff Supplementary Procedure is not allowed.

1. After engine start, the BLEED OFF ENG L or R advisory message will be displayed for the bleed system with HPSOV locked closed.

NOTE: When this condition occurs, the left and right bleed air manifolds will automatically be pressurized by bleed pressure from the opposite engine. Taxi and descent can be accomplished in this single bleed source configuration to avoid repetitive cycling of the Engine Bleed switch.

2. When setting thrust for takeoff, at 55% N1 or greater, select the associated ENG Bleed switch OFF, then ON.

3. In the event of an engine bleed failure on the opposite engine, maintain a minimum of 76% N1 on the engine with the inoperative HPSOV when the associated Engine Bleed switch is selected ON.
4. Prior to landing:
 - A. If the APU is available:
 - 1) Position APU Bleed switch AUTO.
 - 2) Start the APU if it is not running.
 - 3) Position L ISLN and R ISLN switches AUTO.
 - 4) In the event of failure of the opposite engine or the engine bleed air system, position both ENG Bleed switches OFF.
 - B. If the APU is not available:
 - 1) Prior to selecting flaps 1, if the BLEED OFF ENG L or R advisory message is displayed, open the associated engine bleed valve by cycling the respective ENG bleed switch OFF, then ON at 55% N1 or greater.

36-11-02 High Pressure Shutoff Valves (HPSOV)

36-11-02-04 -300 (With SB 777-25-0227 and ASCPC SW P/N 3670-GRS-012-00 or later Installed) Or (With PRR 61980 or Equivalent Incorporated)

36-11-02-04B Associated Engine Bleed Not Used

Interval	Installed	Required	Procedure
C	2	1	(M) (O)

One may be inoperative provided:

- a. Inoperative HPSOV is locked closed.
- b. Associated PRSOV is locked closed.
- c. Opposite engine bleed system operates normally.
- d. Associated engine bleed air switch remains OFF.
- e. Left and right bleed isolation systems operate normally.
- f. Center bleed isolation system operates normally.
- g. Both packs operate normally.
- h. Both outflow valves operate normally.
- i. APU is used as air source for center system hydraulic demand pumps for takeoff and landing.
- j. Appropriate performance adjustments are applied.

NOTE 1: The airplane must also be dispatched using MMEL item 73-21-01.



NOTE 2: PRR 61980 installs enhancements for dispatch with a single pack, or dispatch with a single engine bleed system, on airplane L/N's 423, 429, 454, 455, 457 thru 461, and 463 and-on. For production equivalent information, see DDG item 21-51-01.

NOTE 3: All software P/Ns installed in ASCPC-2 are considered later versions than software P/Ns installed in ASCPC-1.

MAINTENANCE (M)

NOTE 1: For a pack to be operating normally, certain items associated with that system must be operative. See DDG item 21-51-01 for a listing of those items.

NOTE 2: For an engine bleed system to be operating normally, certain items associated with that system must be operative. See DDG item 36-00-01 for a listing of those items.

Lock the inoperative High Pressure Shutoff Valve (HPSOV) and associated Pressure Regulating and Shutoff Valve (PRSOV) in the closed position (AMM 36-00-00/901).

1. Deactivate the associated thrust reverser for ground maintenance (AMM 78-31-00/201).
2. Position associated BLEED AIR Panel (P5) ENG Bleed switch OFF.
3. Gain access to the HPSOV and PRSOV by opening the appropriate fan cowl panel (AMM 71-11-04/201) and thrust reverser sleeve (AMM 78-31-00/201).
4. Lock the HPSOV and PRSOV in the closed positions by following the MANUAL LOCKING CLOSED instructions placard on the valves.
5. Close the opened thrust reverser sleeve (AMM 78-31-00/201) and fan cowl panel (AMM 71-11-04/201).
6. Activate the thrust reverser (AMM 78-31-00/201).

OPERATIONS (O)

NOTE: For the APU-to-Pack takeoff function installed, the APU-to-Pack Takeoff Supplementary Procedure is not allowed.

1. Reduce the performance limited weights by the appropriate adjustments.

Performance Limited Weights	Takeoff & Landing (Packs On)	Takeoff (Packs Off)
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For -300/PW4090:

All	21,800 lb (9,888 kg)	2,200 lb (998 kg)
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For -300/RR892:

All	7,000 lb (3,175 kg)	900 lb (408 kg)
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2. For ETOPS flight planning, increase the engine inoperative Critical Fuel Reserves by 1.3% when dispatching into known or forecast icing conditions.
3. Position associated ENG Bleed switch OFF for the inoperative system.
NOTE: The ENG IDLE DISAGREE advisory message will be displayed when idle thrust is selected inflight.
4. For Takeoff, use the APU as an air source.
 - A. Position APU bleed air switch AUTO.
 - B. Start the APU if it is not running.
 - C. Position L ISLN and R ISLN switches CLOSED.
 - D. Accomplish the BLEED LOSS WING L, R Non-Normal Checklist and do not accomplish the PACK L, R Non-Normal Checklist.
 - E. After takeoff with flaps up:
 - 1) Position L ISLN and R ISLN switches AUTO.
 - 2) Position Wing Anti-Ice switch AUTO.
 - 3) Configure the APU as required.
5. In the event of inflight failure of the opposite engine or the engine bleed air system, use the APU to maintain cabin pressurization at or below FL 220.
 - A. Position APU bleed air switch AUTO.
 - B. Start the APU if it is not running.
 - C. Position both ENG Bleed switches OFF.
 - D. Position WING ANTI-ICE selector OFF (wing anti-ice will not be available for remainder of flight).
 - E. Position L ISLN and R ISLN switches AUTO for remainder of flight (including landing).
6. Prior to Landing, use the APU as an air source.
 - A. Position APU bleed air switch AUTO.
 - B. Start the APU if it is not running.
 - C. Position L ISLN and R ISLN switches AUTO.
 - D. In the event of failure of the opposite engine or the engine bleed air system, position both ENG Bleed switches OFF.

36-11-02 High Pressure Shutoff Valves (HPSOV)**36-11-02-05 -200LR/-300ER****36-11-02-05A Associated Engine Bleed Used**

Interval	Installed	Required	Procedure
C	2	1	(M) (O)

One may be inoperative provided:

- a. Inoperative HPSOV is locked closed.

- b. Opposite engine bleed system operates normally.
 - c. Both outflow valves operate normally.
 - d. Associated engine bleed air is cycled OFF, then ON at 55% N1 or greater before takeoff.
-

MAINTENANCE (M)

NOTE: For an engine bleed system to be operating normally, certain items associated with that system must be operative. See DDG item 36-00-01 for a listing of those items.

Lock the inoperative High Pressure Shutoff Valve (HPSOV) in the closed position (AMM 36-00-00/901).

1. Deactivate the leading edge slats in the retracted position (AMM 27-81-00/201).
2. Deactivate the associated thrust reverser for ground maintenance (AMM 78-31-00/201).
3. Gain access to the HPSOV by opening the appropriate fan cowl panel (AMM 71-11-04/201) and thrust reverser sleeve (AMM 78-31-00/201).
4. Lock the HPSOV in the closed position by following the MANUAL LOCKING CLOSED instructions placard on the valve.
5. Close the opened thrust reverser sleeve (AMM 78-31-00/201) and fan cowl panel (AMM 71-11-04/201).
6. Activate the thrust reverser (AMM 78-31-00/201).
7. Activate the leading edge slats (AMM 27-81-00/201).
8. With both engines shutdown, position associated flight deck Bleed Air panel (P11) ENG bleed switch ON to OFF and back to ON.

OPERATIONS (O)

NOTE 1: Whenever power is set below 55% N1, the BLEED OFF ENG L or R advisory message will be displayed for the bleed system with the HPSOV locked closed. Do not accomplish the BLEED OFF ENG L or R Non-Normal Checklist. The engine bleed valve can be reopened by cycling the ENG bleed switch OFF, then ON at 55% N1 or greater.

NOTE 2: The ENG IDLE DISAGREE advisory message will be displayed when idle thrust is selected inflight prior to cycling the ENG bleed switch OFF, then ON at 55% N1 or greater.

NOTE 3: For an Engine Crossbleed Start supplementary procedure conducted, use the engine with the operative HPSOV as the operating engine.

NOTE 4: For the APU-to-Pack takeoff function installed, the APU-to-Pack takeoff Supplementary Procedure is not allowed.

1. After engine start, the BLEED OFF ENG L or R advisory message will be displayed for the bleed system with HPSOV locked closed.

NOTE: When this condition occurs, the left and right bleed air manifolds will automatically be pressurized by bleed pressure from the opposite engine. Taxi and descent can be accomplished in this single bleed source configuration to avoid repetitive cycling of the Engine Bleed switch.

2. When setting thrust for takeoff, at 55% N1 or greater, select the associated ENG Bleed switch OFF, then ON.
3. In the event of an engine bleed failure on the opposite engine, maintain a minimum of 76% N1 on the engine with the inoperative HPSOV when the associated Engine Bleed switch is selected ON.
4. Prior to landing:
 - A. If the APU is available:
 - 1) Position APU Bleed switch AUTO.
 - 2) Start the APU if it is not running.
 - 3) Position L ISLN and R ISLN switches AUTO.
 - 4) In the event of failure of the opposite engine or the engine bleed air system, position both ENG Bleed switches OFF.
 - B. If the APU is not available:
 - 1) Prior to selecting flaps 1, if the BLEED OFF ENG L or R advisory message is displayed, open the associated engine bleed valve by cycling the respective ENG bleed switch OFF, then ON at 55% N1 or greater.

36-11-02 High Pressure Shutoff Valves (HPSOV)
36-11-02-05 -200LR/-300ER
36-11-02-05B Associated Engine Bleed Not Used

Interval	Installed	Required	Procedure
C	2	1	(M) (O)

One may be inoperative provided:

- a. Inoperative HPSOV is locked closed.
- b. Associated PRSOV is locked closed.
- c. Opposite engine bleed system operates normally.
- d. Associated engine bleed air switch remains OFF.
- e. Left and right bleed isolation systems operate normally.
- f. Center bleed isolation system operates normally.
- g. Both packs operate normally.
- h. Both outflow valves operate normally.
- i. APU is used as air source for center system hydraulic demand pumps for takeoff and landing.



- j. Appropriate performance adjustments are applied.
-

NOTE: The airplane must also be dispatched using MMEL item 73-21-01.

MAINTENANCE (M)

NOTE 1: For a pack to be operating normally, certain items associated with that system must be operative. See DDG item 21-51-01 for a listing of those items.

NOTE 2: For an engine bleed system to be operating normally, certain items associated with that system must be operative. See DDG item 36-00-01 for a listing of those items.

Lock the inoperative High Pressure Shutoff Valve (HPSOV) and associated Pressure Regulating and Shutoff Valve (PRSOV) in the closed position (AMM 36-00-00/901).

1. Deactivate the leading edge slats in the retracted position (AMM 27-81-00/201).
2. Deactivate the associated thrust reverser for ground maintenance (AMM 78-31-00/201).
3. Position associated BLEED AIR Panel (P5) ENG Bleed switch OFF.
4. Gain access to the HPSOV and PRSOV by opening the appropriate fan cowl panel (AMM 71-11-04/201) and thrust reverser sleeve (AMM 78-31-00/201).
5. Lock the HPSOV and PRSOV in the closed positions by following the MANUAL LOCKING CLOSED instructions placard on the valves.
6. Close the opened thrust reverser sleeve (AMM 78-31-00/201) and fan cowl panel (AMM 71-11-04/201).
7. Activate the thrust reverser (AMM 78-31-00/201).
8. Activate the leading edge slats (AMM 27-81-00/201).

OPERATIONS (O)

NOTE: For the APU-to-Pack takeoff function installed, the APU-to-Pack Takeoff Supplementary Procedure is not allowed.

1. Reduce the performance limited weights by the appropriate adjustments.

Performance Limited Weights	Takeoff & Landing (Packs On)	Takeoff (Packs Off)
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For -200LR/GE90-100 Series:

All	14,800 lb (6,713 kg)	5,350 lb (2,427 kg)
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For -300ER/GE90-115B:

All	11,600 lb (5,262 kg)	6,200 lb (2,812 kg)
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2. For ETOPS flight planning, increase the engine inoperative Critical Fuel Reserves by 1.3% when dispatching into known or forecast icing conditions.
3. Position associated ENG Bleed switch OFF for the inoperative system.
NOTE: The ENG IDLE DISAGREE advisory message will be displayed when idle thrust is selected inflight.
4. For Takeoff, use the APU as an air source.
 - A. Position APU bleed air switch AUTO.
 - B. Start the APU if it is not running.
 - C. Position L ISLN and R ISLN switches CLOSED.
 - D. Accomplish the BLEED LOSS WING L, R Non-Normal Checklist and do not accomplish the PACK L, R Non-Normal Checklist.
 - E. After takeoff with flaps up:
 - 1) Position L ISLN and R ISLN switches AUTO.
 - 2) Position Wing Anti-Ice switch AUTO.
 - 3) Configure the APU as required.
5. In the event of inflight failure of the opposite engine or the engine bleed air system, use the APU to maintain cabin pressurization at or below FL 220.
 - A. Position APU bleed air switch AUTO.
 - B. Start the APU if it is not running.
 - C. Position both ENG Bleed switches OFF.
 - D. Position WING ANTI-ICE selector OFF (wing anti-ice will not be available for remainder of flight).
 - E. Position L ISLN and R ISLN switches AUTO for remainder of flight (including landing).
6. Prior to Landing, use the APU as an air source.
 - A. Position APU bleed air switch AUTO.
 - B. Start the APU if it is not running.
 - C. Position L ISLN and R ISLN switches AUTO.
 - D. In the event of failure of the opposite engine or the engine bleed air system, position both ENG Bleed switches OFF.

36-11-02 High Pressure Shutoff Valves (HPSOV)**36-11-02-06 777F****36-11-02-06A Associated Engine Bleed Used**

Interval	Installed	Required	Procedure
C	2	1	(M) (O)

One may be inoperative provided:

- a. Inoperative HPSOV is locked closed.
 - b. Opposite engine bleed system operates normally.
 - c. Both outflow valves operate normally.
 - d. Associated engine bleed air is cycled OFF, then ON at 55% N1 or greater before takeoff.
-

MAINTENANCE (M)

NOTE: For an engine bleed system to be operating normally, certain items associated with that system must be operative. See DDG item 36-00-01 for a listing of those items.

Lock the inoperative High Pressure Shutoff Valve (HPSOV) in the closed position (AMM 36-00-00/901).

1. Deactivate the leading edge slats in the retracted position (AMM 27-81-00/201).
2. Deactivate the associated thrust reverser for ground maintenance (AMM 78-31-00/201).
3. Gain access to the HPSOV by opening the appropriate fan cowl panel (AMM 71-11-04/201) and thrust reverser sleeve (AMM 78-31-00/201).
4. Lock the HPSOV in the closed position by following the MANUAL LOCKING CLOSED instructions placard on the valve.
5. Close the opened thrust reverser sleeve (AMM 78-31-00/201) and fan cowl panel (AMM 71-11-04/201).
6. Activate the thrust reverser (AMM 78-31-00/201).
7. Activate the leading edge slats (AMM 27-81-00/201).
8. With both engines shutdown, position associated flight deck Bleed Air panel (P11) ENG bleed switch ON to OFF and back to ON.

OPERATIONS (O)

NOTE 1: Whenever power is set below 55% N1, the BLEED OFF ENG L or R advisory message will be displayed for the bleed system with the HPSOV locked closed. Do not accomplish the BLEED OFF ENG L or R Non-Normal Checklist. The engine bleed valve can be reopened by cycling the ENG bleed switch OFF, then ON at 55% N1 or greater.

NOTE 2: The ENG IDLE DISAGREE advisory message will be displayed when idle thrust is selected inflight prior to cycling the ENG bleed switch OFF, then ON at 55% N1 or greater.

NOTE 3: For an Engine Crossbleed Start supplementary procedure conducted, use the engine with the operative HPSOV as the operating engine.



NOTE 4: For the APU-to-Pack takeoff function installed, the APU-to-Pack takeoff Supplementary Procedure is not allowed.

1. After engine start, the BLEED OFF ENG L or R advisory message will be displayed for the bleed system with HPSOV locked closed.

NOTE: When this condition occurs, the left and right bleed air manifolds will automatically be pressurized by bleed pressure from the opposite engine. Taxi and descent can be accomplished in this single bleed source configuration to avoid repetitive cycling of the Engine Bleed switch.

2. When setting thrust for takeoff, at 55% N1 or greater, select the associated ENG Bleed switch OFF, then ON.
3. In the event of an engine bleed failure on the opposite engine, maintain a minimum of 76% N1 on the engine with the inoperative HPSOV when the associated Engine Bleed switch is selected ON.
4. Prior to landing:
 - A. If the APU is available:
 - 1) Position APU Bleed switch AUTO.
 - 2) Start the APU if it is not running.
 - 3) Position L ISLN and R ISLN switches AUTO.
 - 4) In the event of failure of the opposite engine or the engine bleed air system, position both ENG Bleed switches OFF.
 - B. If the APU is not available:
 - 1) Prior to selecting flaps 1, if the BLEED OFF ENG L or R advisory message is displayed, open the associated engine bleed valve by cycling the respective ENG bleed switch OFF, then ON.

36-11-02 High Pressure Shutoff Valves (HPSOV)

36-11-02-06 777F

36-11-02-06B Associated Engine Bleed Not Used

Interval	Installed	Required	Procedure
C	2	1	(M) (O)

One may be inoperative provided:

- a. Inoperative HPSOV is locked closed.
- b. Associated PRSOV is locked closed.
- c. Opposite engine bleed system operates normally.
- d. Associated engine bleed air switch remains OFF.
- e. Left and right bleed isolation systems operate normally.
- f. Center bleed isolation system operates normally.
- g. Both packs operate normally.

- h. Both outflow valves operate normally.
 - i. Both recirculation fans operate normally.
 - j. Both ECSMCs operate normally.
 - k. APU is used as air source for center system hydraulic demand pumps for takeoff and landing.
 - l. Appropriate performance adjustments are applied.
-

NOTE: The airplane must also be dispatched using MMEL item 73-21-01.

MAINTENANCE (M)

NOTE 1: For a pack to be operating normally, certain items associated with that system must be operative. See DDG item 21-51-01 for a listing of those items.

NOTE 2: For an engine bleed system to be operating normally, certain items associated with that system must be operative. See DDG item 36-00-01 for a listing of those items.

Lock the inoperative High Pressure Shutoff Valve (HPSOV) and associated Pressure Regulating and Shutoff Valve (PRSOV) in the closed position (AMM 36-00-00/901).

1. Deactivate the leading edge slats in the retracted position (AMM 27-81-00/201).
2. Deactivate the associated thrust reverser for ground maintenance (AMM 78-31-00/201).
3. Position associated BLEED AIR Panel (P5) ENG Bleed switch OFF.
4. Gain access to the HPSOV and PRSOV by opening the appropriate fan cowl panel (AMM 71-11-04/201) and thrust reverser sleeve (AMM 78-31-00/201).
5. Lock the HPSOV and PRSOV in the closed positions by following the MANUAL LOCKING CLOSED instructions placard on the valves.
6. Close the opened thrust reverser sleeve (AMM 78-31-00/201) and fan cowl panel (AMM 71-11-04/201).
7. Activate the thrust reverser (AMM 78-31-00/201).
8. Activate the leading edge slats (AMM 27-81-00/201).

OPERATIONS (O)

NOTE 1: For overhead panel (P5) M/D FLOW switch positioned to HIGH, Air conditioning in the lower lobe cargo compartments is degraded. Position M/D FLOW switch to NORM for temperature sensitive cargo carried in the lower lobe cargo compartments.

NOTE 2: For the APU-to-Pack takeoff function installed, the APU-to-Pack Takeoff Supplementary Procedure is not allowed.

1. Reduce the performance limited weights by the appropriate adjustments.



Performance Limited Weights	Takeoff & Landing (Packs On)	Takeoff (Packs Off)
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For 777F/GE90-100 Series:

All	14,800 lb (6,713 kg)	5,350 lb (2,427 kg)
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2. For ETOPS flight planning, increase the engine inoperative Critical Fuel Reserves by 1.3% when dispatching into known or forecast icing conditions.
3. Position associated ENG Bleed switch OFF for the inoperative system.
NOTE: The ENG IDLE DISAGREE advisory message will be displayed when idle thrust is selected inflight.
4. For Takeoff, use the APU as an air source.
 - A. Position APU bleed air switch AUTO.
 - B. Start the APU if it is not running.
 - C. Position L ISLN and R ISLN switches CLOSED.
 - D. Accomplish the BLEED LOSS WING L, R Non-Normal Checklist and do not accomplish the PACK L, R Non-Normal Checklist.
 - E. After takeoff with flaps up:
 - 1) Position L ISLN and R ISLN switches AUTO.
 - 2) Position Wing Anti-Ice switch AUTO.
 - 3) Configure the APU as required.
5. In the event of inflight failure of the opposite engine or the engine bleed air system, use the APU to maintain cabin pressurization at or below FL 220.
 - A. Position APU bleed air switch AUTO.
 - B. Start the APU if it is not running.
 - C. Position both ENG Bleed switches OFF.
 - D. Position WING ANTI-ICE selector OFF (wing anti-ice will not be available for remainder of flight).
 - E. Position L ISLN and R ISLN switches AUTO for remainder of flight (including landing).
6. Prior to Landing, use the APU as an air source.
 - A. Position APU bleed air switch AUTO.
 - B. Start the APU if it is not running.
 - C. Position L ISLN and R ISLN switches AUTO.
 - D. In the event of failure of the opposite engine or the engine bleed air system, position both ENG Bleed switches OFF.



36-11-03 Intermediate Pressure Check Valves (IPCV)
36-11-03-01 -200/-200ER (Without PRR 61980 or Equivalent Incorporated)

Interval	Installed	Required	Procedure
C	-	-	(M) (O)

May be inoperative provided:

- Inoperative IPCV is open.
- Associated HPSOV is locked closed.
- Associated PRSOV is locked closed.
- Opposite engine bleed system operates normally.
- Associated engine bleed air switch remains OFF.
- Left and right bleed isolation systems operate normally.
- Center bleed isolation system operates normally.
- Both packs operate normally.
- Both outflow valves operate normally.
- Airplane remains at FL 350 or below.
- APU is used as air source for center system hydraulic demand pumps for takeoff and landing.
- Appropriate performance adjustments are applied.

NOTE 1: The airplane must also be dispatched using MMEL item 73-21-01.

NOTE 2: PRR 61980 installs enhancements for dispatch with a single pack, or dispatch with a single engine bleed system, on airplane L/N's 423, 429, 454, 455, 457 thru 461, and 463 and-on. For production equivalent information, see DDG item 21-51-01.

MAINTENANCE (M)

NOTE 1: For a pack to be operating normally, certain items associated with that system must be operative. See DDG item 21-51-01 for a listing of those items.

NOTE 2: For an engine bleed system to be operating normally, certain items associated with that system must be operative. See DDG item 36-00-01 for a listing of those items.

Lock the associated High Pressure Shutoff Valve (HPSOV) and Pressure Regulating and Shutoff Valve (PRSOV) in the closed position (AMM 36-00-00/901).

For PW and RR:

- Deactivate the associated thrust reverser for ground maintenance (AMM 78-31-00/201).



2. Position associated BLEED AIR Panel (P5) ENG Bleed switch OFF.
3. Gain access to the HPSOV and PRSOV by opening the appropriate fan cowl panel (AMM 71-11-04/201) and thrust reverser sleeve (AMM 78-31-00/201).
4. Lock the HPSOV and PRSOV in the closed positions by following the MANUAL LOCKING CLOSED instructions placard on the valves.
5. Close the opened thrust reverser sleeve (AMM 78-31-00/201) and fan cowl panel (AMM 71-11-04/201).
6. Activate the thrust reverser (AMM 78-31-00/201).

For GE:

1. Deactivate the leading edge slats in the retracted position (AMM 27-81-00/201).
2. Deactivate the associated thrust reverser for ground maintenance (AMM 78-31-00/201).
3. Position associated BLEED AIR Panel (P5) ENG Bleed switch OFF.
4. Gain access to the HPSOV and PRSOV by opening the appropriate fan cowl panel (AMM 71-11-04/201) and thrust reverser sleeve (AMM 78-31-00/201).
5. Lock the HPSOV and PRSOV in the closed positions by following the MANUAL LOCKING CLOSED instructions placard on the valves.
6. Close the opened thrust reverser sleeve (AMM 78-31-00/201) and fan cowl panel (AMM 71-11-04/201).
7. Activate the thrust reverser (AMM 78-31-00/201).
8. Activate the leading edge slats (AMM 27-81-00/201).

OPERATIONS (O)

NOTE: For the APU-to-Pack takeoff function installed, the APU-to-Pack Takeoff Supplementary Procedure is not allowed.

1. Reduce performance limited weights by the appropriate adjustments.

Performance Limited Weights	Takeoff & Landing (Packs On)	Takeoff (Packs Off)
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For -200 or -200ER/GE90-76B thru -90B:

thru 632,500 lb (286,902 kg)	14,300 lb (6,486 kg)	2,000 lb (907 kg)
over 632,500 lb (286,902 kg)	20,700 lb (9,389 kg)	2,000 lb (907 kg)

For -200 or -200ER/GE90-94B:

thru 632,500 lb (286,902 kg)	14,300 lb (6,486 kg)	2,000 lb (907 kg)
over 632,500 lb (286,902 kg)	22,400 lb (10,160 kg)	2,000 lb (907 kg)



For -200 or -200ER/PW4074 thru PW4084:

All	21,800 lb (9,888 kg)	1,500 lb (680 kg)
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For -200 or -200ER/PW4090:

thru 632,500 lb (286,902 kg)	21,500 lb (9,752 kg)	2,000 lb (907 kg)
over 632,500 lb (286,902 kg)	26,500 lb (12,020 kg)	2,000 lb (907 kg)

For -200 or -200ER/RR875 thru RR892:

thru 632,500 lb (286,902 kg)	6,400 lb (2,903 kg)	900 lb (408 kg)
over 632,500 lb (286,902 kg)	8,200 lb (3,720 kg)	900 lb (408 kg)

For -200 or -200ER/RR895:

All	6,800 lb (3,084 kg)	900 lb (408 kg)
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2. For ETOPS flight planning, increase the engine inoperative Critical Fuel Reserves by 1.3% when dispatching into known or forecast icing conditions.
3. Flight must remain at or below FL 350.
4. Position associated ENG Bleed switch OFF for the inoperative system.
NOTE: The ENG IDLE DISAGREE advisory message will be displayed when idle thrust is selected inflight.
5. For Takeoff, use the APU as an air source.
 - A. Position APU bleed air switch AUTO.
 - B. Start the APU if it is not running.
 - C. Position L ISLN and R ISLN switches CLOSED.
 - D. Accomplish the BLEED LOSS WING L, R Non-Normal Checklist and do not accomplish the PACK L, R Non-Normal Checklist.
 - E. After takeoff with flaps up:
 - 1) Position L ISLN and R ISLN switches AUTO.
 - 2) Position Wing Anti-Ice switch AUTO.
 - 3) Configure the APU as required.
6. In the event of inflight failure of the opposite engine or the engine bleed air system, use the APU to maintain cabin pressurization at or below FL 220.
 - A. Position APU bleed air switch AUTO.
 - B. Start the APU if it is not running.
 - C. Position both ENG Bleed switches OFF.
 - D. Position WING ANTI-ICE selector OFF (wing anti-ice will not be available for remainder of flight).



- E. Position L ISLN and R ISLN switches AUTO for remainder of flight (including landing).
- 7. Prior to Landing, use the APU as an air source.
 - A. Position APU bleed air switch AUTO.
 - B. Start the APU if it is not running.
 - C. Position L ISLN and R ISLN switches AUTO.
 - D. In the event of failure of the opposite engine or the engine bleed air system, position both ENG Bleed switches OFF.

36-11-03 Intermediate Pressure Check Valves (IPCV)**36-11-03-02 -200/-200ER (With PRR 61980 or Equivalent Incorporated)**

Interval	Installed	Required	Procedure
C	-	-	(M) (O)

May be inoperative provided:

- a. Inoperative IPCV is open.
- b. Associated HPSOV is locked closed.
- c. Associated PRSOV is locked closed.
- d. Opposite engine bleed system operates normally.
- e. Associated engine bleed air switch remains OFF.
- f. Left and right bleed isolation systems operate normally.
- g. Center bleed isolation system operates normally.
- h. Both packs operate normally.
- i. Both outflow valves operate normally.
- j. APU is used as air source for center system hydraulic demand pumps for takeoff and landing.
- k. Appropriate performance adjustments are applied.

NOTE 1: The airplane must also be dispatched using MMEL item 73-21-01.

NOTE 2: PRR 61980 installs enhancements for dispatch with a single pack, or dispatch with a single engine bleed system, on airplane L/N's 423, 429, 454, 455, 457 thru 461, and 463 and-on. For production equivalent information, see DDG item 21-51-01.

MAINTENANCE (M)

NOTE 1: For a pack to be operating normally, certain items associated with that system must be operative. See DDG item 21-51-01 for a listing of those items.



NOTE 2: For an engine bleed system to be operating normally, certain items associated with that system must be operative. See DDG item 36-00-01 for a listing of those items.

Lock the associated High Pressure Shutoff Valve (HPSOV) and Pressure Regulating and Shutoff Valve (PRSOV) in the closed position (AMM 36-00-00/901).

For PW and RR:

1. Deactivate the associated thrust reverser for ground maintenance (AMM 78-31-00/201).
2. Position associated BLEED AIR Panel (P5) ENG Bleed switch OFF.
3. Gain access to the HPSOV and PRSOV by opening the appropriate fan cowl panel (AMM 71-11-04/201) and thrust reverser sleeve (AMM 78-31-00/201).
4. Lock the HPSOV and PRSOV in the closed positions by following the MANUAL LOCKING CLOSED instructions placard on the valves.
5. Close the opened thrust reverser sleeve (AMM 78-31-00/201) and fan cowl panel (AMM 71-11-04/201).
6. Activate the thrust reverser (AMM 78-31-00/201).

For GE:

1. Deactivate the leading edge slats in the retracted position (AMM 27-81-00/201).
2. Deactivate the associated thrust reverser for ground maintenance (AMM 78-31-00/201).
3. Position associated BLEED AIR Panel (P5) ENG Bleed switch OFF.
4. Gain access to the HPSOV and PRSOV by opening the appropriate fan cowl panel (AMM 71-11-04/201) and thrust reverser sleeve (AMM 78-31-00/201).
5. Lock the HPSOV and PRSOV in the closed positions by following the MANUAL LOCKING CLOSED instructions placard on the valves.
6. Close the opened thrust reverser sleeve (AMM 78-31-00/201) and fan cowl panel (AMM 71-11-04/201).
7. Activate the thrust reverser (AMM 78-31-00/201).
8. Activate the leading edge slats (AMM 27-81-00/201).

OPERATIONS (O)

NOTE: For the APU-to-Pack takeoff function installed, the APU-to-Pack Takeoff Supplementary Procedure is not allowed.

1. Reduce the performance limited weights by the appropriate adjustments.

Performance Limited Weights	Takeoff & Landing (Packs On)	Takeoff (Packs Off)
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**For -200 or -200ER/GE90-76B thru -90B:**

thru 632,500 lb (286,902 kg)	14,300 lb (6,486 kg)	2,000 lb (907 kg)
over 632,500 lb (286,902 kg)	20,700 lb (9,389 kg)	2,000 lb (907 kg)

For -200 or -200ER/GE90-94B:

thru 632,500 lb (286,902 kg)	14,300 lb (6,486 kg)	2,000 lb (907 kg)
over 632,500 lb (286,902 kg)	22,400 lb (10,160 kg)	2,000 lb (907 kg)

For -200 or -200ER/PW4074 thru PW4084:

All	21,800 lb (9,888 kg)	1,500 lb (680 kg)
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For -200 or -200ER/PW4090:

thru 632,500 lb (286,902 kg)	21,500 lb (9,752 kg)	2,000 lb (907 kg)
over 632,500 lb (286,902 kg)	26,500 lb (12,020 kg)	2,000 lb (907 kg)

For -200 or -200ER/RR875 thru RR892:

thru 632,500 lb (286,902 kg)	6,400 lb (2,903 kg)	900 lb (408 kg)
over 632,500 lb (286,902 kg)	8,200 lb (3,720 kg)	900 lb (408 kg)

For -200 or -200ER/RR895:

All	6,800 lb (3,084 kg)	900 lb (408 kg)
-----	---------------------	-----------------

2. For ETOPS flight planning, increase the engine inoperative Critical Fuel Reserves by 1.3% when dispatching into known or forecast icing conditions.
3. Position associated ENG Bleed switch OFF for the inoperative system.
NOTE: The ENG IDLE DISAGREE advisory message will be displayed when idle thrust is selected inflight.
4. For Takeoff, use the APU as an air source.
 - A. Position APU bleed air switch AUTO.
 - B. Start the APU if it is not running.
 - C. Position L ISLN and R ISLN switches CLOSED.
 - D. Accomplish the BLEED LOSS WING L, R Non-Normal Checklist and do not accomplish the PACK L, R Non-Normal Checklist.
 - E. After takeoff with flaps up:
 - 1) Position L ISLN and R ISLN switches AUTO.

- 2) Position Wing Anti-Ice switch AUTO.
- 3) Configure the APU as required.
5. In the event of inflight failure of the opposite engine or the engine bleed air system, use the APU to maintain cabin pressurization at or below FL 220.
 - A. Position APU bleed air switch AUTO.
 - B. Start the APU if it is not running.
 - C. Position both ENG Bleed switches OFF.
 - D. Position WING ANTI-ICE selector OFF (wing anti-ice will not be available for remainder of flight).
 - E. Position L ISLN and R ISLN switches AUTO for remainder of flight (including landing).
6. Prior to Landing, use the APU as an air source.
 - A. Position APU bleed air switch AUTO.
 - B. Start the APU if it is not running.
 - C. Position L ISLN and R ISLN switches AUTO.
 - D. In the event of failure of the opposite engine or the engine bleed air system, position both ENG Bleed switches OFF.

36-11-03 Intermediate Pressure Check Valves (IPCV)

36-11-03-03 -300 (Without SB 777-25-0227 and ASCPC SW P/N 3670-GRS-012-00 or later Installed) Or (Without PRR 61980 or Equivalent Incorporated)

Interval	Installed	Required	Procedure
C	-	-	(M) (O)

May be inoperative provided:

- a. Inoperative IPCV is open.
- b. Associated HPSOV is locked closed.
- c. Associated PRSOV is locked closed.
- d. Opposite engine bleed system operates normally.
- e. Associated engine bleed air switch remains OFF.
- f. Left and right bleed isolation systems operate normally.
- g. Center bleed isolation system operates normally.
- h. Both packs operate normally.
- i. Both outflow valves operate normally.
- j. Procedures are established and used to verify forward cargo compartment remains empty or contains only ballast, empty cargo handling equipment (ballast may be loaded in ULDs), or fly away kits.
- k. APU is used as air source for center system hydraulic demand pumps for takeoff and landing.
- l. Appropriate performance adjustments are applied.

NOTE: Operator MELs must define which items are approved for inclusion in the fly away kits and which materials can be used as ballast.

NOTE 1: The airplane must also be dispatched using MMEL item 73-21-01.

NOTE 2: PRR 61980 installs enhancements for dispatch with a single pack, or dispatch with a single engine bleed system, on airplane L/N's 423, 429, 454, 455, 457 thru 461, and 463 and-on. For production equivalent information, see DDG item 21-51-01.

NOTE 3: All software P/Ns installed in ASCPC-2 are considered later versions than software P/Ns installed in ASCPC-1.

MAINTENANCE (M)

NOTE 1: For a pack to be operating normally, certain items associated with that system must be operative. See DDG item 21-51-01 for a listing of those items.

NOTE 2: For an engine bleed system to be operating normally, certain items associated with that system must be operative. See DDG item 36-00-01 for a listing of those items.

Lock the associated High Pressure Shutoff Valve (HPSOV) and Pressure Regulating and Shutoff Valve (PRSOV) in the closed position (AMM 36-00-00/901).

1. Deactivate the associated thrust reverser for ground maintenance (AMM 78-31-00/201).
2. Position associated BLEED AIR Panel (P5) ENG Bleed switch OFF.
3. Gain access to the HPSOV and PRSOV by opening the appropriate fan cowl panel (AMM 71-11-04/201) and thrust reverser sleeve (AMM 78-31-00/201).
4. Lock the HPSOV and PRSOV in the closed positions by following the MANUAL LOCKING CLOSED instructions placard on the valves.
5. Close the opened thrust reverser sleeve (AMM 78-31-00/201) and fan cowl panel (AMM 71-11-04/201).
6. Activate the thrust reverser (AMM 78-31-00/201).

OPERATIONS (O)

NOTE: For the APU-to-Pack takeoff function installed, the APU-to-Pack Takeoff Supplementary Procedure is not allowed.

1. Reduce the performance limited weights by the appropriate adjustments.

Performance Limited Weights	Takeoff & Landing (Packs On)	Takeoff (Packs Off)
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For -300/PW4090:

All	21,800 lb (9,888 kg)	2,200 lb (998 kg)
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For -300/RR892:

All	7,000 lb (3,175 kg)	900 lb (408 kg)
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2. For ETOPS flight planning, increase the engine inoperative Critical Fuel Reserves by 1.3% when dispatching into known or forecast icing conditions.
3. The forward cargo compartment must remain empty, except for ballast, empty cargo containers (ballast may be loaded in ULDs), fly away kits, pallets, and cargo restraint components.
4. Position associated ENG Bleed switch OFF for the inoperative system.
NOTE: The ENG IDLE DISAGREE advisory message will be displayed when idle thrust is selected inflight.
5. For Takeoff, use the APU as an air source.
 - A. Position APU bleed air switch AUTO.
 - B. Start the APU if it is not running.
 - C. Position L ISLN and R ISLN switches CLOSED.
 - D. Accomplish the BLEED LOSS WING L, R Non-Normal Checklist and do not accomplish the PACK L, R Non-Normal Checklist.
 - E. After takeoff with flaps up:
 - 1) Position L ISLN and R ISLN switches AUTO.
 - 2) Position Wing Anti-Ice switch AUTO.
 - 3) Configure the APU as required.
6. In the event of inflight failure of the opposite engine or the engine bleed air system, use the APU to maintain cabin pressurization at or below FL 220.
 - A. Position APU bleed air switch AUTO.
 - B. Start the APU if it is not running.
 - C. Position both ENG Bleed switches OFF.
 - D. Position WING ANTI-ICE selector OFF (wing anti-ice will not be available for remainder of flight).
 - E. Position L ISLN and R ISLN switches AUTO for remainder of flight (including landing).
7. Prior to Landing, use the APU as an air source.
 - A. Position APU bleed air switch AUTO.
 - B. Start the APU if it is not running.
 - C. Position L ISLN and R ISLN switches AUTO.
 - D. In the event of failure of the opposite engine or the engine bleed air system, position both ENG Bleed switches OFF.

- 36-11-03 Intermediate Pressure Check Valves (IPCV)**
- 36-11-03-04 -300 (With SB 777-25-0227 and ASCPC SW P/N 3670-GRS-012-00 or later Installed) Or (With PRR 61980 or Equivalent Incorporated)**

Interval	Installed	Required	Procedure
C	-	-	(M) (O)

May be inoperative provided:

- Inoperative IPCV is open.
- Associated HPSOV is locked closed.
- Associated PRSOV is locked closed.
- Opposite engine bleed system operates normally.
- Associated engine bleed air switch remains OFF.
- Left and right bleed isolation systems operate normally.
- Center bleed isolation system operates normally.
- Both packs operate normally.
- Both outflow valves operate normally.
- APU is used as air source for center system hydraulic demand pumps for takeoff and landing.
- Appropriate performance adjustments are applied.

NOTE 1: The airplane must also be dispatched using MMEL item 73-21-01.

NOTE 2: PRR 61980 installs enhancements for dispatch with a single pack, or dispatch with a single engine bleed system, on airplane L/N's 423, 429, 454, 455, 457 thru 461, and 463 and-on. For production equivalent information, see DDG item 21-51-01.

NOTE 3: All software P/Ns installed in ASCPC-2 are considered later versions than software P/Ns installed in ASCPC-1.

MAINTENANCE (M)

NOTE 1: For a pack to be operating normally, certain items associated with that system must be operative. See DDG item 21-51-01 for a listing of those items.

NOTE 2: For an engine bleed system to be operating normally, certain items associated with that system must be operative. See DDG item 36-00-01 for a listing of those items.



Lock the associated High Pressure Shutoff Valve (HPSOV) and Pressure Regulating and Shutoff Valve (PRSOV) in the closed position (AMM 36-00-00/901).

1. Deactivate the associated thrust reverser for ground maintenance (AMM 78-31-00/201).
2. Position associated BLEED AIR Panel (P5) ENG Bleed switch OFF.
3. Gain access to the HPSOV and PRSOV by opening the appropriate fan cowl panel (AMM 71-11-04/201) and thrust reverser sleeve (AMM 78-31-00/201).
4. Lock the HPSOV and PRSOV in the closed positions by following the MANUAL LOCKING CLOSED instructions placard on the valves.
5. Close the opened thrust reverser sleeve (AMM 78-31-00/201) and fan cowl panel (AMM 71-11-04/201).
6. Activate the thrust reverser (AMM 78-31-00/201).

OPERATIONS (O)

NOTE: For the APU-to-Pack takeoff function installed, the APU-to-Pack Takeoff Supplementary Procedure is not allowed.

1. Reduce the performance limited weights by the appropriate adjustments.

Performance Limited Weights	Takeoff & Landing (Packs On)	Takeoff (Packs Off)
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For -300/PW4090:

All	21,800 lb (9,888 kg)	2,200 lb (998 kg)
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For -300/RR892:

All	7,000 lb (3,175 kg)	900 lb (408 kg)
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2. For ETOPS flight planning, increase the engine inoperative Critical Fuel Reserves by 1.3% when dispatching into known or forecast icing conditions.
3. Position associated ENG Bleed switch OFF for the inoperative system.

NOTE: The ENG IDLE DISAGREE advisory message will be displayed when idle thrust is selected inflight.

4. For Takeoff, use the APU as an air source.
 - A. Position APU bleed air switch AUTO.
 - B. Start the APU if it is not running.
 - C. Position L ISLN and R ISLN switches CLOSED.
 - D. Accomplish the BLEED LOSS WING L, R Non-Normal Checklist and do not accomplish the PACK L, R Non-Normal Checklist.

- E. After takeoff with flaps up:
 - 1) Position L ISLN and R ISLN switches AUTO.
 - 2) Position Wing Anti-Ice switch AUTO.
 - 3) Configure the APU as required.
5. In the event of inflight failure of the opposite engine or the engine bleed air system, use the APU to maintain cabin pressurization at or below FL 220.
 - A. Position APU bleed air switch AUTO.
 - B. Start the APU if it is not running.
 - C. Position both ENG Bleed switches OFF.
 - D. Position WING ANTI-ICE selector OFF (wing anti-ice will not be available for remainder of flight).
 - E. Position L ISLN and R ISLN switches AUTO for remainder of flight (including landing).
6. Prior to Landing, use the APU as an air source.
 - A. Position APU bleed air switch AUTO.
 - B. Start the APU if it is not running.
 - C. Position L ISLN and R ISLN switches AUTO.
 - D. In the event of failure of the opposite engine or the engine bleed air system, position both ENG Bleed switches OFF.

36-11-03 Intermediate Pressure Check Valves (IPCV)
36-11-03-05 -200LR/-300ER

Interval	Installed	Required	Procedure
C	2	1	(M) (O)

May be inoperative provided:

- a. Inoperative IPCV is open.
- b. Associated HPSOV is locked closed.
- c. Associated PRSOV is locked closed.
- d. Opposite engine bleed system operates normally.
- e. Associated engine bleed air switch remains OFF.
- f. Left and right bleed isolation systems operate normally.
- g. Center bleed isolation system operates normally.
- h. Both packs operate normally.
- i. Both outflow valves operate normally.
- j. APU is used as air source for center system hydraulic demand pumps for takeoff and landing.
- k. Appropriate performance adjustments are applied.

NOTE: The airplane must also be dispatched using MMEL item 73-21-01.

**MAINTENANCE (M)**

NOTE 1: For a pack to be operating normally, certain items associated with that system must be operative. See DDG item 21-51-01 for a listing of those items.

NOTE 2: For an engine bleed system to be operating normally, certain items associated with that system must be operative. See DDG item 36-00-01 for a listing of those items.

Lock the associated High Pressure Shutoff Valve (HPSOV) and Pressure Regulating and Shutoff Valve (PRSOV) in the closed position (AMM 36-00-00/901).

1. Deactivate the leading edge slats in the retracted position (AMM 27-81-00/201).
2. Deactivate the associated thrust reverser for ground maintenance (AMM 78-31-00/201).
3. Position associated BLEED AIR Panel (P5) ENG Bleed switch OFF.
4. Gain access to the HPSOV and PRSOV by opening the appropriate fan cowl panel (AMM 71-11-04/201) and thrust reverser sleeve (AMM 78-31-00/201).
5. Lock the HPSOV and PRSOV in the closed positions by following the MANUAL LOCKING CLOSED instructions placard on the valves.
6. Close the opened thrust reverser sleeve (AMM 78-31-00/201) and fan cowl panel (AMM 71-11-04/201).
7. Activate the thrust reverser (AMM 78-31-00/201).
8. Activate the leading edge slats (AMM 27-81-00/201).

OPERATIONS (O)

NOTE: For the APU-to-Pack takeoff function installed, the APU-to-Pack Takeoff Supplementary Procedure is not allowed.

1. Reduce the performance limited weights by the appropriate adjustments.

Performance Limited Weights	Takeoff & Landing (Packs On)	Takeoff (Packs Off)
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For -200LR/GE90-100 Series:

All	14,800 lb (6,713 kg)	5,350 lb (2,427 kg)
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For -300ER/GE90-115B:

All	11,600 lb (5,262 kg)	6,200 lb (2,812 kg)
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2. For ETOPS flight planning, increase the engine inoperative Critical Fuel Reserves by 1.3% when dispatching into known or forecast icing conditions.

3. Position associated ENG Bleed switch OFF for the inoperative system.
NOTE: The ENG IDLE DISAGREE advisory message will be displayed when idle thrust is selected inflight.
4. For Takeoff, use the APU as an air source.
 - A. Position APU bleed air switch AUTO.
 - B. Start the APU if it is not running.
 - C. Position L ISLN and R ISLN switches CLOSED.
 - D. Accomplish the BLEED LOSS WING L, R Non-Normal Checklist and do not accomplish the PACK L, R Non-Normal Checklist.
 - E. After takeoff with flaps up:
 - 1) Position L ISLN and R ISLN switches AUTO.
 - 2) Position Wing Anti-Ice switch AUTO.
 - 3) Configure the APU as required.
5. In the event of inflight failure of the opposite engine or the engine bleed air system, use the APU to maintain cabin pressurization at or below FL 220.
 - A. Position APU bleed air switch AUTO.
 - B. Start the APU if it is not running.
 - C. Position both ENG Bleed switches OFF.
 - D. Position WING ANTI-ICE selector OFF (wing anti-ice will not be available for remainder of flight).
 - E. Position L ISLN and R ISLN switches AUTO for remainder of flight (including landing).
6. Prior to Landing, use the APU as an air source.
 - A. Position APU bleed air switch AUTO.
 - B. Start the APU if it is not running.
 - C. Position L ISLN and R ISLN switches AUTO.
 - D. In the event of failure of the opposite engine or the engine bleed air system, position both ENG Bleed switches OFF.

36-11-03 Intermediate Pressure Check Valves (IPCV)
36-11-03-06 777F

Interval	Installed	Required	Procedure
C	2	1	(M) (O)

May be inoperative provided:

- a. Inoperative IPCV is open.
- b. Associated HPSOV is locked closed.
- c. Associated PRSOV is locked closed.
- d. Opposite engine bleed system operates normally.
- e. Associated engine bleed air switch remains OFF.

- f. Left and right bleed isolation systems operate normally.
 - g. Center bleed isolation system operates normally.
 - h. Both packs operate normally.
 - i. Both outflow valves operate normally.
 - j. Both recirculation fans operate normally.
 - k. Both ECSMCs operate normally.
 - l. APU is used as air source for center system hydraulic demand pumps for takeoff and landing.
 - m. Appropriate performance adjustments are applied.
-

NOTE: The airplane must also be dispatched using MMEL item 73-21-01.

MAINTENANCE (M)

NOTE 1: For a pack to be operating normally, certain items associated with that system must be operative. See DDG item 21-51-01 for a listing of those items.

NOTE 2: For an engine bleed system to be operating normally, certain items associated with that system must be operative. See DDG item 36-00-01 for a listing of those items.

Lock the associated High Pressure Shutoff Valve (HPSOV) and Pressure Regulating and Shutoff Valve (PRSOV) in the closed position (AMM 36-00-00/901).

1. Deactivate the leading edge slats in the retracted position (AMM 27-81-00/201).
2. Deactivate the associated thrust reverser for ground maintenance (AMM 78-31-00/201).
3. Position associated BLEED AIR Panel (P5) ENG Bleed switch OFF.
4. Gain access to the HPSOV and PRSOV by opening the appropriate fan cowl panel (AMM 71-11-04/201) and thrust reverser sleeve (AMM 78-31-00/201).
5. Lock the HPSOV and PRSOV in the closed positions by following the MANUAL LOCKING CLOSED instructions placard on the valves.
6. Close the opened thrust reverser sleeve (AMM 78-31-00/201) and fan cowl panel (AMM 71-11-04/201).
7. Activate the thrust reverser (AMM 78-31-00/201).
8. Activate the leading edge slats (AMM 27-81-00/201).

OPERATIONS (O)

NOTE 1: For overhead panel (P5) M/D FLOW switch positioned to HIGH, Air conditioning in the lower lobe cargo compartments is degraded. Position M/D FLOW switch to NORM for temperature sensitive cargo carried in the lower lobe cargo compartments.



NOTE 2: For the APU-to-Pack takeoff function installed, the APU-to-Pack Takeoff Supplementary Procedure is not allowed.

1. Reduce the performance limited weights by the appropriate adjustments.

Performance Limited Weights	Takeoff & Landing (Packs On)	Takeoff (Packs Off)
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For 777F/GE90-100 Series:

All	14,800 lb (6,713 kg)	5,350 lb (2,427 kg)
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2. For ETOPS flight planning, increase the engine inoperative Critical Fuel Reserves by 1.3% when dispatching into known or forecast icing conditions.
3. Position associated ENG Bleed switch OFF for the inoperative system.
NOTE: The ENG IDLE DISAGREE advisory message will be displayed when idle thrust is selected inflight.
4. For Takeoff, use the APU as an air source.
 - A. Position APU bleed air switch AUTO.
 - B. Start the APU if it is not running.
 - C. Position L ISLN and R ISLN switches CLOSED.
 - D. Accomplish the BLEED LOSS WING L, R Non-Normal Checklist and do not accomplish the PACK L, R Non-Normal Checklist.
 - E. After takeoff with flaps up:
 - 1) Position L ISLN and R ISLN switches AUTO.
 - 2) Position Wing Anti-Ice switch AUTO.
 - 3) Configure the APU as required.
5. In the event of inflight failure of the opposite engine or the engine bleed air system, use the APU to maintain cabin pressurization at or below FL 220.
 - A. Position APU bleed air switch AUTO.
 - B. Start the APU if it is not running.
 - C. Position both ENG Bleed switches OFF.
 - D. Position WING ANTI-ICE selector OFF (wing anti-ice will not be available for remainder of flight).
 - E. Position L ISLN and R ISLN switches AUTO for remainder of flight (including landing).
6. Prior to Landing, use the APU as an air source.
 - A. Position APU bleed air switch AUTO.
 - B. Start the APU if it is not running.
 - C. Position L ISLN and R ISLN switches AUTO.
 - D. In the event of failure of the opposite engine or the engine bleed air system, position both ENG Bleed switches OFF.

36-11-04 Precoolers

Interval	Installed	Required	Procedure
C	2	1	(M) (O)

Airplane may be operated with damage to one precooler provided:

- a. It is verified that precooler airflow is not completely blocked.
- b. Precooler bleed air leakage is verified within normal limits.
- c. Opposite engine bleed system operates normally.

MAINTENANCE (M)

NOTE 1: This MMEL item allows dispatch with damage to one precooler that reduces cooling airflow. If cooling airflow is completely blocked, the associated engine bleed system is considered inoperative and the airplane must be dispatched using MMEL item 36-11-01.

NOTE 2: If precooler bleed leakage is not within normal limits the airplane must be repaired.

NOTE 3: For an engine bleed system to be operating normally, certain items associated with that system must be operative. See DDG item 36-00-01 for a listing of those items.

The damaged precooler must be inspected to verify that:

1. Precooler airflow is not completely blocked (AMM 36-00-00/901).
2. Precooler leakage remains within normal limits (AMM 36-00-00/901).

OPERATIONS (O)

1. For ETOPS flight planning, increase the engine inoperative Critical Fuel Reserves by 1.3% when dispatching into known or forecast icing conditions.
2. In the event of inflight failure of the opposite engine or engine bleed air system, the air demand may exceed the capability of the engine with damaged precooler, which may result in bleed system shutdown. The APU can be used to maintain cabin pressurization at or below FL 220.
 - A. Position APU bleed air switch AUTO.
 - B. Start the APU if it is not running.
 - C. Position both ENG bleed switches OFF.
 - D. Position WING ANTI-ICE selector OFF (wing anti-ice is not available for remainder of flight).

36-11-05 Fan Air Modulation Valves (FAMV)

Interval	Installed	Required	Procedure
C	2	1	(M) (O)

One may be inoperative provided:

- Valve is locked full open.
- Opposite engine bleed system operates normally.
- Appropriate performance adjustments are applied.

MAINTENANCE (M)

NOTE: For an engine bleed system to be operating normally, certain items associated with that system must be operative. See DDG item 36-00-01 for a listing of those items.

Lock the inoperative Fan Air Modulation Valve (FAMV) in the open position (AMM 36-00-00/901).

For PW and RR:

- Deactivate the associated thrust reverser for ground maintenance (AMM 78-31-00/201).
- Gain access to the FAMV by opening the appropriate fan cowl panel (AMM 71-11-04/201) and thrust reverser sleeve (AMM 78-31-00/201).
- Lock the FAMV in the open position by following the MANUAL LOCKING instructions placard on the valve.
- Close the opened thrust reverser sleeve (AMM 78-31-00/201) and fan cowl panel (AMM 71-11-04/201).
- Activate the associated thrust reverser (AMM 78-31-00/201).

For GE:

- Deactivate the leading edge slats in the retracted position (AMM 27-81-00/201).
- Deactivate the associated thrust reverser for ground maintenance (AMM 78-31-00/201).
- Gain access to the FAMV by opening the appropriate fan cowl panel (AMM 71-11-04/201) and thrust reverser sleeve (AMM 78-31-00/201).
- Lock the FAMV in the open position by following the MANUAL LOCKING instructions placard on the valve.
- Close the opened thrust reverser sleeve (AMM 78-31-00/201) and fan cowl panel (AMM 71-11-04/201).
- Activate the associated thrust reverser (AMM 78-31-00/201).
- Activate the leading edge slats (AMM 27-81-00/201).

**OPERATIONS (O)****For Passenger:**

NOTE 1: For left FAMV inoperative, flight deck and forward cabin temperature control will be degraded.

NOTE 2: For right FAMV inoperative, aft cabin temperature control will be degraded.

For 777F:

NOTE 3: For left FAMV inoperative, flight deck, supernumerary and forward main deck temperature control will be degraded.

NOTE 4: For left FAMV inoperative and Aft Cargo Air Conditioning (ACAC) installed, ACAC temperature control will be degraded. Temperature sensitive cargo should not be carried in the aft cargo compartment.

NOTE 5: For right FAMV inoperative, aft main deck temperature control will be degraded.

NOTE 6: For right FAMV inoperative and Forward Cargo Air Conditioning (FCAC) installed, FCAC temperature control will be degraded. Temperature sensitive cargo should not be carried in the forward cargo compartment.

1. Reduce performance limited weights by the appropriate adjustments.

Performance Limited Weights	Takeoff & Landing	Enroute Climb
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For -200 or -200ER/GE90-76B thru -90B:

thru 632,500 lb (286,902 kg)	5,900 lb (2,676 kg)	7,700 lb (3,493 kg)
over 632,500 lb (286,902 kg)	8,500 lb (3,856 kg)	7,900 lb (3,583 kg)

For -200 or -200ER/GE90-94B:

thru 632,500 lb (286,902 kg)	5,900 lb (2,676 kg)	9,400 lb (4,264 kg)
over 632,500 lb (286,902 kg)	9,200 lb (4,173 kg)	9,400 lb (4,264 kg)

For -200 or -200ER/PW4074 thru PW4084:

All	4,400 lb (1,996 kg)	8,600 lb (3,901 kg)
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For -200 or -200ER/PW4090:

thru 632,500 lb (286,902 kg)	6,100 lb (2,767 kg)	7,200 lb (3,266 kg)
over 632,500 lb (286,902 kg)	9,200 lb (4,173 kg)	7,200 lb (3,266 kg)

**For -200 or -200ER/RR875 thru RR892:**

thru 632,500 lb (286,902 kg)	3,300 lb (1,497 kg)	2,400 lb (1,089 kg)
over 632,500 lb (286,902 kg)	5,100 lb (2,313 kg)	2,400 lb (1,089 kg)

For -200 or -200ER/RR895 or -300/RR892:

All	3,300 lb (1,497 kg)	2,400 lb (1,089 kg)
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For -200LR or 777F/GE90-100 Series:

All	13,300 lb (6,033 kg)	10,100 lb (4,581 kg)
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For -300/PW4090:

All	5,800 lb (2,631 kg)	7,200 lb (3,266 kg)
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For -300ER/GE90-115B:

All	10,700 lb (4,854 kg)	10,200 lb (4,627 kg)
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NOTE: The enroute climb weight penalties listed above are based on single engine operating speeds that approximate maximum lift-to-drag ratio speed. To account for the difference in level off altitude when operating at other speeds, multiply the enroute climb weight penalty listed above by the appropriate factor listed on page 3.00-01-00.2 in the ENROUTE DIVERSION SPEED EFFECTS paragraph.

2. Increase flight planning fuel by 0.5%.
3. For ETOPS flight planning, increase ETOPS engine inoperative Critical Fuel Reserves.

For PW and RR:

- A. By 0.9%, or by 1.9% when dispatching into known or forecast icing conditions.

For GE:

- B. By 0.6%, or by 1.8% when dispatching into known or forecast icing conditions.
4. Operate the wing anti-ice manually.
 - A. Position WING ANTI-ICE selector OFF.
 - B. For flaps up and wing anti-ice required (do not use during takeoff and landing):
 - 1) Position the ENG Bleed switch associated with the inoperative FAMV OFF.



NOTE: The ENG IDLE DISAGREE advisory message will be displayed when idle thrust is selected inflight.

- 2) Position WING ANTI-ICE selector ON, then OFF as required.
- 3) Prior to landing, position the ENG Bleed switch associated with the inoperative FAMV ON.

For PW4090-3:

5. Packs Off Takeoffs are not allowed.

36-12-01 Left and Right Isolation Systems (Valve and/or Indication)
36-12-01A Valve Locked Closed

Interval	Installed	Required	Procedure
C	2	1	(M) (O)

One may be inoperative provided:

- Valve is locked closed after engine start.
- Associated valve switch remains in the CLOSED position.
- Center bleed isolation system operates normally.
- Both engine bleed systems operate normally.
- Isolation valve on opposite side is selected CLOSED for takeoff and landing, and selected AUTO with flaps up.
- APU is used as air source for center system hydraulic demand pumps for takeoff and landing.

MAINTENANCE (M)

Lock the inoperative isolation valve in the closed position after engine start (AMM 36-00-00/901).

- Gain access to the inoperative isolation valve by opening the appropriate Blowout Door (195QL for left valve or 196QR for right valve).
- Lock the isolation valve in the closed position by following the MANUAL LOCK instructions placard on the valve.

NOTE: APU or crossbleed air is not available for starting the associated engine when the L or R isolation valve is locked closed. The associated engine must be started and left running before locking the valve closed (AMM 71-00-00/201).

- Position associated Bleed Air Panel (P5) Bleed ISLN switch CLOSED.
- Close the opened Blowout Door.

OPERATIONS (O)

NOTE 1: For the APU-to-Pack takeoff function installed, the APU-to-Pack Takeoff Supplementary Procedure is not allowed.

NOTE 2: APU or crossbleed air is not available for starting the associated engine when the L or R isolation valve is locked closed.

NOTE 3: The flight deck Air Synoptic may display an incorrect or missing valve position.

- For ETOPS flight planning, increase the engine inoperative Critical Fuel Reserves by 1.3% when dispatching into known or forecast icing conditions.



2. For takeoff and landing use the APU as an air source.
 - A. Position APU bleed air switch AUTO.
 - B. Start the APU if it is not running.
 - C. Position L ISLN and R ISLN switches CLOSED.
 - D. After takeoff is completed, position the operative ISLN switch AUTO (inoperative ISLN switch remains CLOSED) and configure the APU as required.

36-12-01 Left and Right Isolation Systems (Valve and/or Indication)

36-12-01B Valve Locked Open

Interval	Installed	Required	Procedure
C	2	1	(M) (O)

One may be inoperative provided:

- a. Associated valve is locked open.
- b. Both center system hydraulic demand pumps operate normally.
- c. Appropriate performance adjustments are applied.

MAINTENANCE (M)

Lock the inoperative isolation valve in the open position (AMM 36-00-00/901).

1. Gain access to the inoperative isolation valve by opening the appropriate Blowout Door (195QL for left valve or 196QR for right valve).
2. Lock the isolation valve in the open position by following the MANUAL LOCK instructions placard on the valve.
3. Position associated Bleed Air Panel (P5) Bleed ISLN switch AUTO.
4. Close the opened Blowout Door.

OPERATIONS (O)

NOTE 1: The flight deck Air Synoptic may display an incorrect or missing valve position.

NOTE 2: For the APU-to-Pack takeoff function installed, the APU-to-Pack Takeoff Supplementary Procedure is not allowed.

1. Reduce performance limited weights by the appropriate adjustments.

Performance Limited Weights	Takeoff Climb	Takeoff Runway/Obstacle
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For -200 or -200ER:

All	1,400 lb (636 kg)	9,600 lb (4,355 kg)
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For -200LR or 777F/GE90-100 Series or -300ER/GE90-115B:

All	1,700 lb (771 kg)	9,600 lb (4,355 kg)
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For -300/PW4090 or RR:

All	400 lb (181 kg)	3,700 lb (1,678 kg)
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2. Position C ISLN switch CLOSED for takeoff and landing.

36-12-01 Left and Right Isolation Systems (Valve and/or Indication)

36-12-01-01 ISLN Switch Lights

36-12-01-01-01 CLOSED Lights

Interval	Installed	Required	Procedure
C	2	0	

36-12-01 Left and Right Isolation Systems (Valve and/or Indication)

36-12-01-01 ISLN Switch Lights

36-12-01-01-02 AUTO Lights

Interval	Installed	Required	Procedure
C	2	0	

**36-12-02 Center Isolation System (Valve and/or Indication)****36-12-02A Valve Locked Closed**

Interval	Installed	Required	Procedure
C	1	0	(M) (O)

May be inoperative provided:

- Valve is locked closed after engine start.
 - Both center system hydraulic demand pumps operate normally.
 - Left and right isolation valves operate normally.
 - Both engine bleed systems operate normally.
 - Appropriate performance adjustments are applied.
-

MAINTENANCE (M)

Lock the center isolation valve in the closed position after engine start (AMM 36-00-00/901).

- Gain access to center isolation valve by opening Bleed Air Valve Door 196NR.
- Lock the isolation valve in the closed position by following the MANUAL LOCK instructions placard on the valve.

NOTE: APU or crossbleed air is not available for starting the right engine when center isolation valve is locked closed. The right engine must be started and left running before locking the valve closed (AMM 71-00-00/201).

- Position Bleed Air Panel (P5) Bleed C ISLN switch CLOSED.
- Close Bleed Air Valve Door 196NR.

OPERATIONS (O)

NOTE 1: For the APU-to-Pack takeoff function installed, the APU-to-Pack Takeoff Supplementary Procedure is not allowed.

NOTE 2: APU or crossbleed air is not available for starting the right engine when the center isolation valve is locked closed.

NOTE 3: The flight deck Air Synoptic may display an incorrect or missing valve position.

- Reduce performance limited weights by the appropriate adjustments.

Performance Limited Weights	Takeoff Climb	Takeoff Runway/Obstacle
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For -200 or -200ER:

All	1,400 lb (636 kg)	9,600 lb (4,355 kg)
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For -200LR or 777F/GE90-100 Series or -300ER/GE90-115B:

All	1,700 lb (771 kg)	9,600 lb (4,355 kg)
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For -300/PW4090 or RR:

All	400 lb (181 kg)	3,700 lb (1,678 kg)
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- For ETOPS flight planning, increase the engine inoperative Critical Fuel Reserves by an additional 1.3% when dispatching into known or forecast icing conditions.

36-12-02 Center Isolation System (Valve and/or Indication)

36-12-02B Valve Locked Open

Interval	Installed	Required	Procedure
C	1	0	(M) (O)

May be inoperative provided:

- Valve is locked open.
- Left and right isolation valves are selected CLOSED for takeoff and landing.
- APU is used as air source for center system hydraulic demand pumps for takeoff and landing.
- Left or right isolation valve is selected AUTO with flaps up.

MAINTENANCE (M)

Lock the center isolation valve in the open position (AMM 36-00-00/901).

- Gain access to center isolation valve by opening Bleed Air Valve Door 196NR.
- Lock the isolation valve in the open position by following the MANUAL LOCK instructions placard on the valve.
- Position Bleed Air Panel (P5) Bleed C ISLN switch AUTO.
- Close Bleed Air Valve Door 196NR.

OPERATIONS (O)

NOTE 1: For the APU-to-Pack takeoff function installed, the APU-to-Pack Takeoff Supplementary Procedure is not allowed.



NOTE 2: The flight deck Air Synoptic may display an incorrect or missing valve position.

1. For takeoff and landing use the APU as an air source.
 - A. Position APU bleed air switch AUTO.
 - B. Start the APU if it is not running.
 - C. Position L ISLN and R ISLN switches CLOSED.
 - D. After takeoff is completed, position L ISLN or R ISLN (one) switch AUTO and configure the APU as required.

36-12-02 Center Isolation System (Valve and/or Indication)**36-12-02-01 ISLN Switch Light****36-12-02-01-01 CLOSED Light**

Interval	Installed	Required	Procedure
C	1	0	

36-12-02 Center Isolation System (Valve and/or Indication)**36-12-02-01 ISLN Switch Light****36-12-02-01-02 AUTO Light**

Interval	Installed	Required	Procedure
C	1	0	

36-12-03 APU Bleed Air Shutoff System (Valve and/or Indications)

Interval	Installed	Required	Procedure
C	1	0	(M) (O)

May be inoperative provided:

- Other procedures do not require APU as a pneumatic source.
- Valve is locked closed.
- APU bleed switch remains OFF.
- Flight remains within 180 minutes of landing at a suitable airport.

NOTE: APU may be used for electrical power.

MAINTENANCE (M)

Lock the APU bleed air shutoff valve in the closed position (AMM 36-00-00/901).

- Gain access to the APU bleed air shutoff valve by opening the Controls Bay Access Door 313AL.
- Lock the APU shutoff valve in the closed position by following the MANUAL LOCK instructions placard on the valve.
- Position the Bleed Air Panel (P5) APU Bleed switch OFF.
- Close the Controls Bay Access Door 313AL.

OPERATIONS (O)

NOTE 1: For the APU-to-Pack takeoff function installed, the APU-to-Pack Takeoff Supplementary Procedure is not allowed.

NOTE 2: APU may be used for electrical power, both inflight and on the ground. Pneumatic starting of the APU is not possible.

- Dispatch is not allowed if the APU is required as a bleed air source by other procedures. MMEL dispatch procedures for the following inoperative equipment may require APU bleed air availability.

MMEL Item	Associated Status Message
29-11-03	HYD PUMP DEM C1 or C2
36-11-01	BLEED PRSOV L or R
36-11-02	BLEED HPSOV L or R
36-11-03	None
36-12-01	BLEED ISLN VALVE L or R
36-12-02	BLEED ISLN VALVE C
36-22-01	BLEED TEMP SENSOR L or R

2. Position APU Bleed switch OFF.

NOTE: The BLEED OFF APU advisory message will be displayed.

3. The flight must remain within 180 minutes of landing at a suitable airport.

36-12-03 APU Bleed Air Shutoff System (Valve and/or Indications)

36-12-03-01 Valve Indications

36-12-03-01A APU Not Required

Interval	Installed	Required	Procedure
C	1	0	(M) (O)

May be inoperative provided:

- a. Other procedures do not require use of the APU.
- b. Backup AC power system is verified to operate normally before each departure.
- c. APU bleed air valve is verified to operate normally.
- d. Flight remains within 180 minutes of landing at a suitable airport.

MAINTENANCE (M)

Verify that the APU bleed air shutoff valve operates normally (AMM 36-00-00/901).

1. Start the APU (AMM 49-11-00/201).
2. Remove all other pneumatic sources from the airplane.
 - A. Remove ground air supply if connected.
 - B. If engine(s) is running, position Bleed Air panel (P5) ENG bleed switch OFF.
3. Position the Bleed Air Panel (P5) APU Bleed switch AUTO.
4. Gain access to the APU bleed air shutoff valve by opening the Controls Bay Access Door 313AL.
5. Observe the APU bleed air shutoff valve knob pointer and ensure the pointer indicates the valves is OPEN.
6. Position the Bleed Air Panel (P5) APU Bleed switch OFF.
7. Observe the APU bleed air shutoff valve knob pointer and ensure the pointer indicates the valve is CLOSED.
8. Close the Controls Bay Access Door 313AL.
9. Position the Bleed Air Panel (P5) APU Bleed switch AUTO.

OPERATIONS (O)

NOTE 1: For the APU-to-Pack takeoff function installed, the APU-to-Pack Takeoff Supplementary Procedure is not allowed.

NOTE 2: Flight deck APU bleed air valve indications on the overhead panel (P5), Air Synoptic and Air Supply Maintenance page may not agree with the actual status of the valve. For the APU bleed air valve commanded open during ground operations and the valve indications incorrectly indicating the valve is closed, the BLEED OFF APU advisory message may be displayed.

1. Dispatch is not allowed if the APU is required by other procedures. MMEL dispatch procedures for the following inoperative equipment may require APU availability.

MMEL Item	Associated Status Message
24-11-01	ELEC GEN SYS L or R IDG VALVE CLOSED L or R
24-25-01	ELEC BACKUP GEN L or R ELEC BACKUP SYS
29-11-03	HYD PUMP DEM C1 or C2
36-11-01	BLEED PRSOV L or R
36-11-02	BLEED HPSOV L or R
36-11-03	None
36-12-01	BLEED ISLN VALVE L or R
36-12-02	BLEED ISLN VALVE C
36-22-01	BLEED TEMP SENSOR L or R

2. The following restrictions imposed for an inoperative APU generator must be observed:
 - A. The flight must remain within 180 minutes of landing at a suitable airport.
 - B. Verify the Backup AC electrical power system operates normally. Confirm the ELEC BACKUP SYS or ELEC BACKUP GEN L or R status and advisory messages are not displayed 30 seconds after both engines have been started.

36-12-03 APU Bleed Air Shutoff System (Valve and/or Indications)

36-12-03-01 Valve Indications

36-12-03-01B APU Bleed Not Required

Interval	Installed	Required	Procedure
C	1	0	(M) (O)

May be inoperative provided:

- a. Other procedures do not require APU as a pneumatic source.

- b. APU bleed air valve is verified to operate normally before each departure.
- c. APU bleed switch remains OFF for flight.
- d. Flight remains within 180 minutes of landing at a suitable airport.

NOTE: APU may be used for electrical power and for pneumatic power for ground operations only.

MAINTENANCE (M)

Verify that the APU bleed air shutoff valve operates normally before each departure (AMM 36-00-00/901).

1. Start the APU (AMM 49-11-00/201).
2. Remove all other pneumatic sources from the airplane.
 - A. Remove ground air supply if connected.
 - B. If engine(s) is running, position Bleed Air panel (P5) ENG bleed switch OFF.
3. Position the Bleed Air Panel (P5) APU Bleed switch AUTO.
4. Gain access to the APU bleed air shutoff valve by opening the Controls Bay Access Door 313AL.
5. Observe the APU bleed air shutoff valve knob pointer and ensure the pointer indicates the valve is OPEN.
6. Position the Bleed Air Panel (P5) APU Bleed switch OFF.
7. Observe the APU bleed air shutoff valve knob pointer and ensure the pointer indicates the valve is CLOSED.
8. Close the Controls Bay Access Door 313AL.
9. Position the Bleed Air Panel (P5) APU Bleed switch AUTO.

OPERATIONS (O)

NOTE 1: For the APU-to-Pack takeoff function installed, the APU-to-Pack Takeoff Supplementary Procedure is not allowed.

NOTE 2: Flight deck APU bleed air valve indications on the overhead panel (P5), Air Synoptic and Air Supply Maintenance page may not agree with the actual status of the valve. For the APU bleed air valve commanded open during ground operations and the valve indications incorrectly indicating the valve is closed, the BLEED OFF APU advisory message may be displayed.

NOTE 3: APU bleed air valve must be verified by maintenance to operate normally before each departure.

1. Dispatch is not allowed if the APU is required as a bleed air source by other procedures and the valve indications are inoperative. MMEL dispatch procedures for the following inoperative equipment may require APU bleed air availability.

MMEL Item	Associated Status Message
29-11-03	HYD PUMP DEM C1 or C2
36-11-01	BLEED PRSOV L or R
36-11-02	BLEED HPSOV L or R
36-11-03	None
36-12-01	BLEED ISLN VALVE L or R
36-12-02	BLEED ISLN VALVE C
36-22-01	BLEED TEMP SENSOR L or R

- Position APU Bleed switch OFF before takeoff. APU Bleed switch remains OFF for the flight.

NOTE: APU may be used for electrical power both inflight and on the ground, and as a pneumatic power source for ground operations only. Inflight pneumatic starting of the APU is not available with the APU Bleed switch OFF.

- The flight must remain within 180 minutes of landing at a suitable airport.

36-12-03 APU Bleed Air Shutoff System (Valve and/or Indications)

36-12-03-02 APU Switch Light

36-12-03-02-01 OFF Light

Interval	Installed	Required	Procedure
C	1	0	

36-12-03 APU Bleed Air Shutoff System (Valve and/or Indications)

36-12-03-02 APU Switch Light

36-12-03-02-02 AUTO Light

Interval	Installed	Required	Procedure
C	1	0	

**36-20-01 Air Synoptic Display**

Interval	Installed	Required	Procedure
C	1	0	

OPERATIONS NOTE

If the Synoptic Display is missing data, selecting an alternate location for the display (L INBD, LWR CTR, or R INBD) may restore the missing data. Synoptic Displays containing missing data may continue to be used to the extent remaining data is useful. Airplane system faults will be annunciated by alerting and status messages.

**36-21-01 Manifold Pressure/Flow Sensing Systems**

Interval	Installed	Required	Procedure
C	2	1	

One may be inoperative provided the associated pack flow control valve inlet pressure sensor operates normally.

MAINTENANCE NOTE

1. For airplanes prior to AIMS Block Point 2003, the MAN PRESS SENSOR L or R status message is displayed for an inoperative manifold pressure sensor.
2. For airplanes incorporating AIMS Block Point 2003 up to AIMS V14, the MAN PRESS SENSOR L or R status message is displayed for an inoperative manifold pressure sensor or for an inoperative manifold flow sensor.
3. For AIMS V14 and on, the MANIFOLD SENSOR L or R status message is displayed for an inoperative manifold pressure sensor or for an inoperative manifold flow sensor.

**36-21-02 Intermediate Pressure Sensing Systems**

Interval	Installed	Required	Procedure
C	2	1	

36-22-01 Manifold Temperature Sensing Systems
36-22-01-01 -200/-200ER (Without PRR 61980 or Equivalent Incorporated)
36-22-01-01A Associated Engine Bleed Used

Interval	Installed	Required	Procedure
C	2	1	(M) (O)

One may be inoperative provided:

- Associated HPSOV is locked closed.
- Opposite engine bleed system operates normally.
- Both outflow valves operate normally.
- Airplane remains at FL 350 or below.
- Associated engine bleed air switch is cycled OFF, then ON at 55% N1 or greater before takeoff.
- Appropriate performance adjustments are applied.

NOTE: PRR 61980 installs enhancements for dispatch with a single pack, or dispatch with a single engine bleed system, on airplane L/N's 423, 429, 454, 455, 457 thru 461, and 463 and-on. For production equivalent information, see DDG item 21-51-01.

MAINTENANCE (M)

NOTE: For an engine bleed system to be operating normally, certain items associated with that system must be operative. See DDG item 36-00-01 for a listing of those items.

Lock the associated High Pressure Shutoff Valve (HPSOV) in the closed position (AMM 36-00-00/901).

For PW and RR:

- Deactivate the associated thrust reverser for ground maintenance (AMM 78-31-00/201).
- Gain access to the HPSOV by opening the appropriate fan cowl panel (AMM 71-11-04/201) and thrust reverser sleeve (AMM 78-31-00/201).
- Lock the HPSOV in the closed position by following the MANUAL LOCKING CLOSED instructions placard on the valve.
- Close the opened thrust reverser sleeve (AMM 78-31-00/201) and fan cowl panel (AMM 71-11-04/201).
- Activate the thrust reverser (AMM 78-31-00/201).

For GE:

- Deactivate the leading edge slats in the retracted position (AMM 27-81-00/201).



2. Deactivate the associated thrust reverser for ground maintenance (AMM 78-31-00/201).
3. Gain access to the HPSOV by opening the appropriate fan cowl panel (AMM 71-11-04/201) and thrust reverser sleeve (AMM 78-31-00/201).
4. Lock the HPSOV in the closed position by following the MANUAL LOCKING CLOSED instructions placard on the valve.
5. Close the opened thrust reverser sleeve (AMM 78-31-00/201) and fan cowl panel (AMM 71-11-04/201).
6. Activate the thrust reverser (AMM 78-31-00/201).
7. Activate the leading edge slats (AMM 27-81-00/201).

OPERATIONS (O)

NOTE 1: Whenever power is set below 55% N1, the BLEED OFF ENG L or R advisory message will be displayed for the bleed system with the HPSOV locked closed. Do not accomplish the BLEED OFF ENG L or R Non-Normal Checklist. The engine bleed valve can be reopened by cycling the ENG bleed switch OFF, then ON at 55% N1 or greater.

NOTE 2: The ENG IDLE DISAGREE advisory message will be displayed when idle thrust is selected inflight prior to cycling the ENG bleed switch OFF, then ON at 55% N1 or greater.

NOTE 3: For APU-to-Pack takeoff function installed, the APU-to-Pack Takeoff Supplementary Procedure is not allowed.

1. Reduce performance limited weights by the appropriate adjustments.

Performance Limited Weights	Takeoff & Landing	Enroute Climb
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For -200 or -200ER/GE90-76B thru -90B:

thru 632,500 lb (286,902 kg)	5,900 lb (2,676 kg)	7,700 lb (3,493 kg)
over 632,500 lb (286,902 kg)	8,500 lb (3,856 kg)	7,900 lb (3,583 kg)

For -200 or -200ER/GE90-94B:

thru 632,500 lb (286,902 kg)	5,900 lb (2,676 kg)	9,400 lb (4,264 kg)
over 632,500 lb (286,902 kg)	9,200 lb (4,173 kg)	9,400 lb (4,264 kg)

For -200 or -200ER/PW4074 thru PW4084:

All	4,400 lb (1,996 kg)	8,600 lb (3,901 kg)
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**For -200 or -200ER/PW4090:**

thru 632,500 lb (286,902 kg)	6,100 lb (2,767 kg)	7,200 lb (3,266 kg)
over 632,500 lb (286,902 kg)	9,200 lb (4,173 kg)	7,200 lb (3,266 kg)

For -200 or -200ER/RR875 thru RR892:

thru 632,500 lb (286,902 kg)	3,300 lb (1,497 kg)	2,400 lb (1,089 kg)
over 632,500 lb (286,902 kg)	5,100 lb (2,313 kg)	2,400 lb (1,089 kg)

For -200 or -200ER/RR895:

All	3,300 lb (1,497 kg)	2,400 lb (1,089 kg)
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NOTE: The enroute climb weight penalties listed above are based on single engine operating speeds that approximate maximum lift-to-drag ratio speed. To account for the difference in level off altitude when operating at other speeds, multiply the enroute climb weight penalty listed above by the appropriate factor listed on page 3.00-01-00.2 in the ENROUTE DIVERSION SPEED EFFECTS paragraph.

2. Increase flight planning fuel by 0.5%.
3. For ETOPS flight planning, increase ETOPS engine inoperative Critical Fuel Reserves.

For PW and RR:

- A. By 0.9%, or by 1.9% when dispatching into known or forecast icing conditions.

For GE:

- B. By 0.6%, or by 1.8% when dispatching into known or forecast icing conditions.
4. Flight must remain at or below FL 350.
 5. After engine start, the BLEED OFF ENG L or R advisory message will be displayed for the bleed system with HPSOV locked closed.

NOTE: When this condition occurs, the left and right bleed air manifolds will automatically be pressurized by bleed pressure from the opposite engine. Taxi and descent can be accomplished in this single bleed source configuration to avoid repetitive cycling of the Engine Bleed switch.

6. When setting thrust for takeoff, at 55% N1 or greater, select the associated ENG bleed switch OFF, then ON.
7. Operate the wing anti-ice manually.
 - A. Position WING ANTI-ICE selector OFF.

- B. For flaps up and wing anti-ice required (do not use during takeoff and landing):
- 1) Position the ENG Bleed switch associated with the inoperative Bleed Temp Sensor OFF.
- NOTE: The ENG IDLE DISAGREE advisory message will be displayed when idle thrust is selected inflight.
- 2) Position WING ANTI-ICE selector ON, then OFF as required.
 - 3) Prior to landing, position the ENG Bleed switch associated with the inoperative Bleed Temp Sensor ON.
8. In the event of an engine bleed failure on the engine opposite to the inoperative Bleed Temp Sensor, maintain a minimum of 76% N1 on the engine with the inoperative Bleed Temp Sensor when the associated Engine Bleed switch is selected ON.
9. Prior to landing:
- A. If the APU is available:
- 1) Position APU Bleed switch AUTO.
 - 2) Start the APU if it is not running.
 - 3) Position L ISLN and R ISLN switches AUTO.
 - 4) In the event of failure of the opposite engine or the engine bleed air system, position both ENG Bleed switches OFF.
- B. If the APU is not available:
- 1) Prior to selecting flaps 1, if the BLEED OFF ENG L or R advisory message is displayed, open the associated engine bleed valve by cycling the respective ENG bleed switch OFF, then ON.

36-22-01 Manifold Temperature Sensing Systems
36-22-01-01 -200/-200ER (Without PRR 61980 or Equivalent Incorporated)
36-22-01-01B Associated Engine Bleed Not Used

Interval	Installed	Required	Procedure
C	2	1	(M) (O)

One may be inoperative provided:

- a. Associated HPSOV is locked closed.
- b. Associated PRSOV is locked closed.
- c. Opposite engine bleed system operates normally.
- d. Associated engine bleed air switch remains OFF.
- e. Left and right bleed isolation systems operate normally.
- f. Center bleed isolation system operates normally.
- g. Both packs operate normally.



- h. Both outflow valves operate normally.
 - i. Airplane remains at FL 350 or below.
 - j. APU is used as air source for center system hydraulic demand pumps for takeoff and landing.
 - k. Appropriate performance adjustments are applied.
-

NOTE 1: The airplane must also be dispatched using MMEL item 73-21-01.

NOTE 2: PRR 61980 installs enhancements for dispatch with a single pack, or dispatch with a single engine bleed system, on airplane L/N's 423, 429, 454, 455, 457 thru 461, and 463 and-on. For production equivalent information, see DDG item 21-51-01.

MAINTENANCE (M)

NOTE 1: For a pack to be operating normally, certain items associated with that system must be operative. See DDG item 21-51-01 for a listing of those items.

NOTE 2: For an engine bleed system to be operating normally, certain items associated with that system must be operative. See DDG item 36-00-01 for a listing of those items.

Lock the associated High Pressure Shutoff Valve (HPSOV) and Pressure Regulating and Shutoff Valve (PRSOV) in the closed position (AMM 36-00-00/901).

For PW and RR:

1. Deactivate the associated thrust reverser for ground maintenance (AMM 78-31-00/201).
2. Position associated BLEED AIR Panel (P5) ENG Bleed switch OFF.
3. Gain access to the HPSOV and PRSOV by opening the appropriate fan cowl panel (AMM 71-11-04/201) and thrust reverser sleeve (AMM 78-31-00/201).
4. Lock the HPSOV and PRSOV in the closed positions by following the MANUAL LOCKING CLOSED instructions placard on the valves.
5. Close the opened thrust reverser sleeve (AMM 78-31-00/201) and fan cowl panel (AMM 71-11-04/201).
6. Activate the thrust reverser (AMM 78-31-00/201).

For GE:

1. Deactivate the leading edge slats in the retracted position (AMM 27-81-00/201).
2. Deactivate the associated thrust reverser for ground maintenance (AMM 78-31-00/201).
3. Position associated BLEED AIR Panel (P5) ENG Bleed switch OFF.

4. Gain access to the HPSOV and PRSOV by opening the appropriate fan cowl panel (AMM 71-11-04/201) and thrust reverser sleeve (AMM 78-31-00/201).
5. Lock the HPSOV and PRSOV in the closed positions by following the MANUAL LOCKING CLOSED instructions placard on the valves.
6. Close the opened thrust reverser sleeve (AMM 78-31-00/201) and fan cowl panel (AMM 71-11-04/201).
7. Activate the thrust reverser (AMM 78-31-00/201).
8. Activate the leading edge slats (AMM 27-81-00/201).

OPERATIONS (O)

NOTE: For the APU-to-Pack takeoff function installed, the APU-to-Pack Takeoff Supplementary Procedure is not allowed.

1. Reduce performance limited weights by the appropriate adjustments.

Performance Limited Weights	Takeoff & Landing (Packs On)	Takeoff (Packs Off)
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For -200 or -200ER/GE90-76B thru -90B:

thru 632,500 lb (286,902 kg)	14,300 lb (6,486 kg)	2,000 lb (907 kg)
over 632,500 lb (286,902 kg)	20,700 lb (9,389 kg)	2,000 lb (907 kg)

For -200 or -200ER/GE90-94B:

thru 632,500 lb (286,902 kg)	14,300 lb (6,486 kg)	2,000 lb (907 kg)
over 632,500 lb (286,902 kg)	22,400 lb (10,160 kg)	2,000 lb (907 kg)

For -200 or -200ER/PW4074 thru PW4084:

All	21,800 lb (9,888 kg)	1,500 lb (680 kg)
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For -200 or -200ER/PW4090:

thru 632,500 lb (286,902 kg)	21,500 lb (9,752 kg)	2,000 lb (907 kg)
over 632,500 lb (286,902 kg)	26,500 lb (12,020 kg)	2,000 lb (907 kg)

For -200 or -200ER/RR875 thru RR892:

thru 632,500 lb (286,902 kg)	6,400 lb (2,903 kg)	900 lb (408 kg)
over 632,500 lb (286,902 kg)	8,200 lb (3,720 kg)	900 lb (408 kg)



For -200 or -200ER/RR895:

All	6,800 lb (3,084 kg)	900 lb (408 kg)
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2. For ETOPS flight planning, increase the engine inoperative Critical Fuel Reserves by 1.3% when dispatching into known or forecast icing conditions.
3. Flight must remain at or below FL 350.
4. Position associated ENG Bleed switch OFF for the inoperative system.
NOTE: The ENG IDLE DISAGREE advisory message will be displayed when idle thrust is selected inflight.
5. For Takeoff, use the APU as an air source.
 - A. Position APU bleed air switch AUTO.
 - B. Start the APU if it is not running.
 - C. Position L ISLN and R ISLN switches CLOSED.
 - D. Accomplish the BLEED LOSS WING L, R Non-Normal Checklist and do not accomplish the PACK L, R Non-Normal Checklist.
 - E. After takeoff with flaps up:
 - 1) Position L ISLN and R ISLN switches AUTO.
 - 2) Position Wing Anti-Ice switch AUTO.
 - 3) Configure the APU as required.
6. In the event of inflight failure of the opposite engine or the engine bleed air system, use the APU to maintain cabin pressurization at or below FL 220.
 - A. Position APU bleed air switch AUTO.
 - B. Start the APU if it is not running.
 - C. Position both ENG Bleed switches OFF.
 - D. Position WING ANTI-ICE selector OFF (wing anti-ice will not be available for remainder of flight).
 - E. Position L ISLN and R ISLN switches AUTO for remainder of flight (including landing).
7. Prior to Landing, use the APU as an air source.
 - A. Position APU bleed air switch AUTO.
 - B. Start the APU if it is not running.
 - C. Position L ISLN and R ISLN switches AUTO.
 - D. In the event of failure of the opposite engine or the engine bleed air system, position both ENG Bleed switches OFF.

**36-22-01 Manifold Temperature Sensing Systems****36-22-01-02 -200/-200ER (With PRR 61980 or Equivalent Incorporated)****36-22-01-02A Associated Engine Bleed Used**

Interval	Installed	Required	Procedure
C	2	1	(M) (O)

One may be inoperative provided:

- Associated HPSOV is locked closed.
 - Opposite engine bleed system operates normally.
 - Both outflow valves operate normally.
 - Associated engine bleed air switch is cycled OFF, then ON at 55% N1 or greater before takeoff.
 - Appropriate performance adjustments are applied.
-

NOTE: PRR 61980 installs enhancements for dispatch with a single pack, or dispatch with a single engine bleed system, on airplane L/N's 423, 429, 454, 455, 457 thru 461, and 463 and-on. For production equivalent information, see DDG item 21-51-01.

MAINTENANCE (M)

NOTE: For an engine bleed system to be operating normally, certain items associated with that system must be operative. See DDG item 36-00-01 for a listing of those items.

Lock the associated High Pressure Shutoff Valve (HPSOV) in the closed position (AMM 36-00-00/901).

For PW and RR:

- Deactivate the associated thrust reverser for ground maintenance (AMM 78-31-00/201).
- Gain access to the HPSOV by opening the appropriate fan cowl panel (AMM 71-11-04/201) and thrust reverser sleeve (AMM 78-31-00/201).
- Lock the HPSOV in the closed position by following the MANUAL LOCKING CLOSED instructions placard on the valve.
- Close the opened thrust reverser sleeve (AMM 78-31-00/201) and fan cowl panel (AMM 71-11-04/201).
- Activate the thrust reverser (AMM 78-31-00/201).

For GE:

- Deactivate the leading edge slats in the retracted position (AMM 27-81-00/201).
- Deactivate the associated thrust reverser for ground maintenance (AMM 78-31-00/201).

3. Gain access to the HPSOV by opening the appropriate fan cowl panel (AMM 71-11-04/201) and thrust reverser sleeve (AMM 78-31-00/201).
4. Lock the HPSOV in the closed position by following the MANUAL LOCKING CLOSED instructions placard on the valve.
5. Close the opened thrust reverser sleeve (AMM 78-31-00/201) and fan cowl panel (AMM 71-11-04/201).
6. Activate the thrust reverser (AMM 78-31-00/201).
7. Activate the leading edge slats (AMM 27-81-00/201).

OPERATIONS (O)

NOTE 1: Whenever power is set below 55% N1, the BLEED OFF ENG L or R advisory message will be displayed for the bleed system with the HPSOV locked closed. Do not accomplish the BLEED OFF ENG L or R Non-Normal Checklist. The engine bleed valve can be reopened by cycling the ENG bleed switch OFF, then ON at 55% N1 or greater.

NOTE 2: The ENG IDLE DISAGREE advisory message will be displayed when idle thrust is selected inflight prior to cycling the ENG bleed switch OFF, then ON at 55% N1 or greater.

NOTE 3: For APU-to-Pack takeoff function installed, the APU-to-Pack Takeoff Supplementary Procedure is not allowed.

1. Reduce performance limited weights by the appropriate adjustments.

Performance Limited Weights	Takeoff & Landing	Enroute Climb
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For -200 or -200ER/GE90-76B thru -90B:

thru 632,500 lb (286,902 kg)	5,900 lb (2,676 kg)	7,700 lb (3,493 kg)
over 632,500 lb (286,902 kg)	8,500 lb (3,856 kg)	7,900 lb (3,583 kg)

For -200 or -200ER/GE90-94B:

thru 632,500 lb (286,902 kg)	5,900 lb (2,676 kg)	9,400 lb (4,264 kg)
over 632,500 lb (286,902 kg)	9,200 lb (4,173 kg)	9,400 lb (4,264 kg)

For -200 or -200ER/PW4074 thru PW4084:

All	4,400 lb (1,996 kg)	8,600 lb (3,901 kg)
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For -200 or -200ER/PW4090:

thru 632,500 lb (286,902 kg)	6,100 lb (2,767 kg)	7,200 lb (3,266 kg)
over 632,500 lb (286,902 kg)	9,200 lb (4,173 kg)	7,200 lb (3,266 kg)

**For -200 or -200ER/RR875 thru RR892:**

thru 632,500 lb (286,902 kg)	3,300 lb (1,497 kg)	2,400 lb (1,089 kg)
over 632,500 lb (286,902 kg)	5,100 lb (2,313 kg)	2,400 lb (1,089 kg)

For -200 or -200ER/RR895:

All	3,300 lb (1,497 kg)	2,400 lb (1,089 kg)
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NOTE: The enroute climb weight penalties listed above are based on single engine operating speeds that approximate maximum lift-to-drag ratio speed. To account for the difference in level off altitude when operating at other speeds, multiply the enroute climb weight penalty listed above by the appropriate factor listed on page 3.00-01-00.2 in the ENROUTE DIVERSION SPEED EFFECTS paragraph.

2. Increase flight planning fuel by 0.5%.
3. For ETOPS flight planning, increase ETOPS engine inoperative Critical Fuel Reserves.

For PW and RR:

- A. By 0.9%, or by 1.9% when dispatching into known or forecast icing conditions.

For GE:

- B. By 0.6%, or by 1.8% when dispatching into known or forecast icing conditions.
4. After engine start, the BLEED OFF ENG L or R advisory message will be displayed for the bleed system with HPSOV locked closed.

NOTE: When this condition occurs, the left and right bleed air manifolds will automatically be pressurized by bleed pressure from the opposite engine. Taxi and descent can be accomplished in this single bleed source configuration to avoid repetitive cycling of the Engine Bleed switch.

5. When setting thrust for takeoff, at 55% N1 or greater, select the associated ENG bleed switch OFF, then ON.
6. Operate the wing anti-ice manually.
 - A. Position WING ANTI-ICE selector OFF.
 - B. For flaps up and wing anti-ice required (do not use during takeoff and landing):
 - 1) Position the ENG Bleed switch associated with the inoperative Bleed Temp Sensor OFF.



NOTE: The ENG IDLE DISAGREE advisory message will be displayed when idle thrust is selected inflight.

- 2) Position WING ANTI-ICE selector ON, then OFF as required.
- 3) Prior to landing, position the ENG Bleed switch associated with the inoperative Bleed Temp Sensor ON.
7. In the event of an engine bleed failure on the engine opposite to the inoperative Bleed Temp Sensor, maintain a minimum of 76% N1 on the engine with the inoperative Bleed Temp Sensor when the associated Engine Bleed switch is selected ON.
8. Prior to landing:
 - A. If the APU is available:
 - 1) Position APU Bleed switch AUTO.
 - 2) Start the APU if it is not running.
 - 3) Position L ISLN and R ISLN switches AUTO.
 - 4) In the event of failure of the opposite engine or the engine bleed air system, position both ENG Bleed switches OFF.
 - B. If the APU is not available:
 - 1) Prior to selecting flaps 1, if the BLEED OFF ENG L or R advisory message is displayed, open the associated engine bleed valve by cycling the respective ENG bleed switch OFF, then ON.

36-22-01 Manifold Temperature Sensing Systems**36-22-01-02 -200/-200ER (With PRR 61980 or Equivalent Incorporated)****36-22-01-02B Associated Engine Bleed Not Used**

Interval	Installed	Required	Procedure
C	2	1	(M) (O)

One may be inoperative provided:

- a. Associated HPSOV is locked closed.
- b. Associated PRSOV is locked closed.
- c. Opposite engine bleed system operates normally.
- d. Associated engine bleed air switch remains OFF.
- e. Left and right bleed isolation systems operate normally.
- f. Center bleed isolation system operates normally.
- g. Both packs operate normally.
- h. Both outflow valves operate normally.
- i. APU is used as air source for center system hydraulic demand pumps for takeoff and landing.

- j. Appropriate performance adjustments are applied.
-

NOTE 1: The airplane must also be dispatched using MMEL item 73-21-01.

NOTE 2: PRR 61980 installs enhancements for dispatch with a single pack, or dispatch with a single engine bleed system, on airplane L/N's 423, 429, 454, 455, 457 thru 461, and 463 and-on. For production equivalent information, see DDG item 21-51-01.

MAINTENANCE (M)

NOTE 1: For a pack to be operating normally, certain items associated with that system must be operative. See DDG item 21-51-01 for a listing of those items.

NOTE 2: For an engine bleed system to be operating normally, certain items associated with that system must be operative. See DDG item 36-00-01 for a listing of those items.

Lock the associated High Pressure Shutoff Valve (HPSOV) and Pressure Regulating and Shutoff Valve (PRSOV) in the closed position (AMM 36-00-00/901).

For PW and RR:

1. Deactivate the associated thrust reverser for ground maintenance (AMM 78-31-00/201).
2. Position associated BLEED AIR Panel (P5) ENG Bleed switch OFF.
3. Gain access to the HPSOV and PRSOV by opening the appropriate fan cowl panel (AMM 71-11-04/201) and thrust reverser sleeve (AMM 78-31-00/201).
4. Lock the HPSOV and PRSOV in the closed positions by following the MANUAL LOCKING CLOSED instructions placard on the valves.
5. Close the opened thrust reverser sleeve (AMM 78-31-00/201) and fan cowl panel (AMM 71-11-04/201).
6. Activate the thrust reverser (AMM 78-31-00/201).

For GE:

1. Deactivate the leading edge slats in the retracted position (AMM 27-81-00/201).
2. Deactivate the associated thrust reverser for ground maintenance (AMM 78-31-00/201).
3. Position associated BLEED AIR Panel (P5) ENG Bleed switch OFF.
4. Gain access to the HPSOV and PRSOV by opening the appropriate fan cowl panel (AMM 71-11-04/201) and thrust reverser sleeve (AMM 78-31-00/201).
5. Lock the HPSOV and PRSOV in the closed positions by following the MANUAL LOCKING CLOSED instructions placard on the valves.
6. Close the opened thrust reverser sleeve (AMM 78-31-00/201) and fan cowl panel (AMM 71-11-04/201).

7. Activate the thrust reverser (AMM 78-31-00/201).
8. Activate the leading edge slats (AMM 27-81-00/201).

OPERATIONS (O)

NOTE: For the APU-to-Pack takeoff function installed, the APU-to-Pack Takeoff Supplementary Procedure is not allowed.

1. Reduce the performance limited weights by the appropriate adjustments.

Performance Limited Weights	Takeoff & Landing (Packs On)	Takeoff (Packs Off)
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For -200 or -200ER/GE90-76B thru -90B:

thru 632,500 lb (286,902 kg)	14,300 lb (6,486 kg)	2,000 lb (907 kg)
over 632,500 lb (286,902 kg)	20,700 lb (9,389 kg)	2,000 lb (907 kg)

For -200 or -200ER/GE90-94B:

thru 632,500 lb (286,902 kg)	14,300 lb (6,486 kg)	2,000 lb (907 kg)
over 632,500 lb (286,902 kg)	22,400 lb (10,160 kg)	2,000 lb (907 kg)

For -200 or -200ER/PW4074 thru PW4084:

All	21,800 lb (9,888 kg)	1,500 lb (680 kg)
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For -200 or -200ER/PW4090:

thru 632,500 lb (286,902 kg)	21,500 lb (9,752 kg)	2,000 lb (907 kg)
over 632,500 lb (286,902 kg)	26,500 lb (12,020 kg)	2,000 lb (907 kg)

For -200 or -200ER/RR875 thru RR892:

thru 632,500 lb (286,902 kg)	6,400 lb (2,903 kg)	900 lb (408 kg)
over 632,500 lb (286,902 kg)	8,200 lb (3,720 kg)	900 lb (408 kg)

For -200 or -200ER/RR895:

All	6,800 lb (3,084 kg)	900 lb (408 kg)
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2. For ETOPS flight planning, increase the engine inoperative Critical Fuel Reserves by 1.3% when dispatching into known or forecast icing conditions.
3. Position associated ENG Bleed switch OFF for the inoperative system.



NOTE: The ENG IDLE DISAGREE advisory message will be displayed when idle thrust is selected inflight.

4. For Takeoff, use the APU as an air source.
 - A. Position APU bleed air switch AUTO.
 - B. Start the APU if it is not running.
 - C. Position L ISLN and R ISLN switches CLOSED.
 - D. Accomplish the BLEED LOSS WING L, R Non-Normal Checklist and do not accomplish the PACK L, R Non-Normal Checklist.
 - E. After takeoff with flaps up:
 - 1) Position L ISLN and R ISLN switches AUTO.
 - 2) Position Wing Anti-Ice switch AUTO.
 - 3) Configure the APU as required.
5. In the event of inflight failure of the opposite engine or the engine bleed air system, use the APU to maintain cabin pressurization at or below FL 220.
 - A. Position APU bleed air switch AUTO.
 - B. Start the APU if it is not running.
 - C. Position both ENG Bleed switches OFF.
 - D. Position WING ANTI-ICE selector OFF (wing anti-ice will not be available for remainder of flight).
 - E. Position L ISLN and R ISLN switches AUTO for remainder of flight (including landing).
6. Prior to Landing, use the APU as an air source.
 - A. Position APU bleed air switch AUTO.
 - B. Start the APU if it is not running.
 - C. Position L ISLN and R ISLN switches AUTO.
 - D. In the event of failure of the opposite engine or the engine bleed air system, position both ENG Bleed switches OFF.

36-22-01 Manifold Temperature Sensing Systems

36-22-01-03 -300 (Without SB 777-25-0227 and ASCPC SW P/N 3670-GRS-012-00 or later Installed) Or (Without PRR 61980 or Equivalent Incorporated)

36-22-01-03A Associated Engine Bleed Used

Interval	Installed	Required	Procedure
C	2	1	(M) (O)

One may be inoperative provided:

- a. Associated HPSOV is locked closed.
- b. Opposite engine bleed system operates normally.
- c. Both outflow valves operate normally.

- d. Procedures are established and used to verify forward cargo compartment remains empty or contains only ballast, empty cargo handling equipment (ballast may be loaded in ULDs), or fly away kits.
- e. Associated engine bleed air switch is cycled OFF, then ON at 55% N1 or greater before takeoff.
- f. Appropriate performance adjustments are applied.

NOTE: Operator MELs must define which items are approved for inclusion in the fly away kits and which materials can be used as ballast.

NOTE 1: PRR 61980 installs enhancements for dispatch with a single pack, or dispatch with a single engine bleed system, on airplane L/N's 423, 429, 454, 455, 457 thru 461, and 463 and-on. For production equivalent information, see DDG item 21-51-01.

NOTE 2: All software P/Ns installed in ASCPC-2 are considered later versions than software P/Ns installed in ASCPC-1.

MAINTENANCE (M)

NOTE: For an engine bleed system to be operating normally, certain items associated with that system must be operative. See DDG item 36-00-01 for a listing of those items.

Lock the associated High Pressure Shutoff Valve (HPSOV) in the closed position (AMM 36-00-00/901).

1. Deactivate the associated thrust reverser for ground maintenance (AMM 78-31-00/201).
2. Gain access to the HPSOV by opening the appropriate fan cowl panel (AMM 71-11-04/201) and thrust reverser sleeve (AMM 78-31-00/201).
3. Lock the HPSOV in the closed position by following the MANUAL LOCKING CLOSED instructions placard on the valve.
4. Close the opened thrust reverser sleeve (AMM 78-31-00/201) and fan cowl panel (AMM 71-11-04/201).
5. Activate the thrust reverser (AMM 78-31-00/201).

OPERATIONS (O)

NOTE 1: Whenever power is set below 55% N1, the BLEED OFF ENG L or R advisory message will be displayed for the bleed system with the HPSOV locked closed. Do not accomplish the BLEED OFF ENG L or R Non-Normal Checklist. The engine bleed valve can be reopened by cycling the ENG bleed switch OFF, then ON at 55% N1 or greater.

NOTE 2: The ENG IDLE DISAGREE advisory message will be displayed when idle thrust is selected inflight prior to cycling the ENG bleed switch OFF, then ON at 55% N1 or greater.



NOTE 3: For the APU-to-Pack takeoff function installed, the APU-to-Pack Takeoff Supplementary Procedure is not allowed.

1. Reduce performance limited weights by the appropriate adjustments.

Performance Limited Weights	Takeoff & Landing	Enroute Climb
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For -300/PW4090:

All	5,800 lb (2,631 kg)	7,200 lb (3,266 kg)
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For -300/RR892:

All	3,300 lb (1,497 kg)	2,400 lb (1,089 kg)
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NOTE: The enroute climb weight penalties listed above are based on single engine operating speeds that approximate maximum lift-to-drag ratio speed. To account for the difference in level off altitude when operating at other speeds, multiply the enroute climb weight penalty listed above by the appropriate factor listed on page 3.00-01-00.2 in the ENROUTE DIVERSION SPEED EFFECTS paragraph.

2. Increase flight planning fuel by 0.5%.
3. For ETOPS flight planning, increase ETOPS engine inoperative Critical Fuel Reserves by 0.9%, or by 1.9% when dispatching into known or forecast icing conditions.
4. The forward cargo compartment must remain empty, except for ballast, empty cargo containers (ballast may be loaded in ULDs), fly away kits, pallets, and cargo restraint components.
5. After engine start, the BLEED OFF ENG L or R advisory message will be displayed for the bleed system with HPSOV locked closed.

NOTE: When this condition occurs, the left and right bleed air manifolds will automatically be pressurized by bleed pressure from the opposite engine. Taxi and descent can be accomplished in this single bleed source configuration to avoid repetitive cycling of the Engine Bleed switch.

6. When setting thrust for takeoff, at 55% N1 or greater, select the associated ENG bleed switch OFF, then ON.
7. Operate the wing anti-ice manually.
 - A. Position WING ANTI-ICE selector OFF.
 - B. For flaps up and wing anti-ice required (do not use during takeoff and landing):
 - 1) Position the ENG Bleed switch associated with the inoperative Bleed Temp Sensor OFF.

NOTE: The ENG IDLE DISAGREE advisory message may be displayed when idle thrust is selected inflight.

- 2) Position WING ANTI-ICE selector ON, then OFF as required.
- 3) Prior to landing, position the ENG Bleed switch associated with the inoperative Bleed Temp Sensor ON.
8. In the event of an engine bleed failure on the engine opposite to the inoperative Bleed Temp Sensor, maintain a minimum of 76% N1 on the engine with the inoperative Bleed Temp Sensor when the associated Engine Bleed switch is selected ON.
9. Prior to landing:
 - A. If the APU is available:
 - 1) Position APU Bleed switch AUTO.
 - 2) Start the APU if it is not running.
 - 3) Position L ISLN and R ISLN switches AUTO.
 - 4) In the event of failure of the opposite engine or the engine bleed air system, position both ENG Bleed switches OFF.
 - B. If the APU is not available:
 - 1) Prior to selecting flaps 1, if the BLEED OFF ENG L or R advisory message is displayed, open the associated engine bleed valve by cycling the respective ENG bleed switch OFF, then ON.

36-22-01 Manifold Temperature Sensing Systems

36-22-01-03 -300 (Without SB 777-25-0227 and ASCPC SW P/N 3670-GRS-012-00 or later Installed) Or (Without PRR 61980 or Equivalent Incorporated)

36-22-01-03B Associated Engine Bleed Not Used

Interval	Installed	Required	Procedure
C	2	1	(M) (O)

One may be inoperative provided:

- a. Associated HPSOV is locked closed.
- b. Associated PRSOV is locked closed.
- c. Opposite engine bleed system operates normally.
- d. Associated engine bleed air switch remains OFF.
- e. Left and right bleed isolation systems operate normally.
- f. Center bleed isolation system operates normally.
- g. Both packs operate normally.
- h. Both outflow valves operate normally.



- i. Procedures are established and used to verify forward cargo compartment remains empty or contains only ballast, empty cargo handling equipment (ballast may be loaded in ULDs), or fly away kits.
- j. APU is used as air source for center system hydraulic demand pumps for takeoff and landing.
- k. Appropriate performance adjustments are applied.

NOTE: Operator MELs must define which items are approved for inclusion in the fly away kits and which materials can be used as ballast.

NOTE 1: The airplane must also be dispatched using MMEL item 73-21-01.

NOTE 2: PRR 61980 installs enhancements for dispatch with a single pack, or dispatch with a single engine bleed system, on airplane L/N's 423, 429, 454, 455, 457 thru 461, and 463 and-on. For production equivalent information, see DDG item 21-51-01.

NOTE 3: All software P/Ns installed in ASCPC-2 are considered later versions than software P/Ns installed in ASCPC-1.

MAINTENANCE (M)

NOTE 1: For a pack to be operating normally, certain items associated with that system must be operative. See DDG item 21-51-01 for a listing of those items.

NOTE 2: For an engine bleed system to be operating normally, certain items associated with that system must be operative. See DDG item 36-00-01 for a listing of those items.

Lock the associated High Pressure Shutoff Valve (HPSOV) and Pressure Regulating and Shutoff Valve (PRSOV) in the closed position (AMM 36-00-00/901).

1. Deactivate the associated thrust reverser for ground maintenance (AMM 78-31-00/201).
2. Position associated BLEED AIR Panel (P5) ENG Bleed switch OFF.
3. Gain access to the HPSOV and PRSOV by opening the appropriate fan cowl panel (AMM 71-11-04/201) and thrust reverser sleeve (AMM 78-31-00/201).
4. Lock the HPSOV and PRSOV in the closed positions by following the MANUAL LOCKING CLOSED instructions placard on the valves.
5. Close the opened thrust reverser sleeve (AMM 78-31-00/201) and fan cowl panel (AMM 71-11-04/201).
6. Activate the thrust reverser (AMM 78-31-00/201).

OPERATIONS (O)

NOTE: For the APU-to-Pack takeoff function installed, the APU-to-Pack Takeoff Supplementary Procedure is not allowed.

1. Reduce the performance limited weights by the appropriate adjustments.

Performance Limited Weights	Takeoff & Landing (Packs On)	Takeoff (Packs Off)
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For -300/PW4090:

All	21,800 lb (9,888 kg)	2,200 lb (998 kg)
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For -300/RR892:

All	7,000 lb (3,175 kg)	900 lb (408 kg)
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2. For ETOPS flight planning, increase the engine inoperative Critical Fuel Reserves by 1.3% when dispatching into known or forecast icing conditions.
3. The forward cargo compartment must remain empty, except for ballast, empty cargo containers (ballast may be loaded in ULDs), fly away kits, pallets, and cargo restraint components.
4. Position associated ENG Bleed switch OFF for the inoperative system.
NOTE: The ENG IDLE DISAGREE advisory message will be displayed when idle thrust is selected inflight.
5. For Takeoff, use the APU as an air source.
 - A. Position APU bleed air switch AUTO.
 - B. Start the APU if it is not running.
 - C. Position L ISLN and R ISLN switches CLOSED.
 - D. Accomplish the BLEED LOSS WING L, R Non-Normal Checklist and do not accomplish the PACK L, R Non-Normal Checklist.
 - E. After takeoff with flaps up:
 - 1) Position L ISLN and R ISLN switches AUTO.
 - 2) Position Wing Anti-Ice switch AUTO.
 - 3) Configure the APU as required.
6. In the event of inflight failure of the opposite engine or the engine bleed air system, use the APU to maintain cabin pressurization at or below FL 220.
 - A. Position APU bleed air switch AUTO.
 - B. Start the APU if it is not running.
 - C. Position both ENG Bleed switches OFF.
 - D. Position WING ANTI-ICE selector OFF (wing anti-ice will not be available for remainder of flight).



- E. Position L ISLN and R ISLN switches AUTO for remainder of flight (including landing).
- 7. Prior to Landing, use the APU as an air source.
 - A. Position APU bleed air switch AUTO.
 - B. Start the APU if it is not running.
 - C. Position L ISLN and R ISLN switches AUTO.
 - D. In the event of failure of the opposite engine or the engine bleed air system, position both ENG Bleed switches OFF.

36-22-01 Manifold Temperature Sensing Systems

36-22-01-04 -300 (With SB 777-25-0227 and ASCPC SW P/N 3670-GRS-012-00 or later Installed) Or (With PRR 61980 or Equivalent Incorporated)

36-22-01-04A Associated Engine Bleed Used

Interval	Installed	Required	Procedure
C	2	1	(M) (O)

One may be inoperative provided:

- a. Associated HPSOV is locked closed.
- b. Opposite engine bleed system operates normally.
- c. Both outflow valves operate normally.
- d. Associated engine bleed air switch is cycled OFF, then ON at 55% N1 or greater before takeoff.
- e. Appropriate performance adjustments are applied.

NOTE 1: PRR 61980 installs enhancements for dispatch with a single pack, or dispatch with a single engine bleed system, on airplane L/N's 423, 429, 454, 455, 457 thru 461, and 463 and-on. For production equivalent information, see DDG item 21-51-01.

NOTE 2: All software P/Ns installed in ASCPC-2 are considered later versions than software P/Ns installed in ASCPC-1.

MAINTENANCE (M)

NOTE: For an engine bleed system to be operating normally, certain items associated with that system must be operative. See DDG item 36-00-01 for a listing of those items.

Lock the associated High Pressure Shutoff Valve (HPSOV) in the closed position (AMM 36-00-00/901).

- 1. Deactivate the associated thrust reverser for ground maintenance (AMM 78-31-00/201).

2. Gain access to the HPSOV by opening the appropriate fan cowl panel (AMM 71-11-04/201) and thrust reverser sleeve (AMM 78-31-00/201).
3. Lock the HPSOV in the closed position by following the MANUAL LOCKING CLOSED instructions placard on the valve.
4. Close the opened thrust reverser sleeve (AMM 78-31-00/201) and fan cowl panel (AMM 71-11-04/201).
5. Activate the thrust reverser (AMM 78-31-00/201).

OPERATIONS (O)

NOTE 1: Whenever power is set below 55% N1, the BLEED OFF ENG L or R advisory message will be displayed for the bleed system with the HPSOV locked closed. Do not accomplish the BLEED OFF ENG L or R Non-Normal Checklist. The engine bleed valve can be reopened by cycling the ENG bleed switch OFF, then ON at 55% N1 or greater.

NOTE 2: The ENG IDLE DISAGREE advisory message will be displayed when idle thrust is selected inflight prior to cycling the ENG bleed switch OFF, then ON at 55% N1 or greater.

NOTE 3: For APU-to-Pack takeoff function installed, the APU-to-Pack takeoff Supplementary Procedure is not allowed.

1. Reduce performance limited weights by the appropriate adjustments.

Performance Limited Weights	Takeoff & Landing	Enroute Climb
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For -300/PW4090:

All	5,800 lb (2,631 kg)	7,200 lb (3,266 kg)
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For -300/RR892:

All	3,300 lb (1,497 kg)	2,400 lb (1,089 kg)
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NOTE: The enroute climb weight penalties listed above are based on single engine operating speeds that approximate maximum lift-to-drag ratio speed. To account for the difference in level off altitude when operating at other speeds, multiply the enroute climb weight penalty listed above by the appropriate factor listed on page 3.00-01-00.2 in the ENROUTE DIVERSION SPEED EFFECTS paragraph.

2. Increase flight planning fuel by 0.5%.
3. For ETOPS flight planning, increase ETOPS engine inoperative Critical Fuel Reserves by 0.9%, or by 1.9% when dispatching into known or forecast icing conditions.

4. After engine start, the BLEED OFF ENG L or R advisory message will be displayed for the bleed system with HPSOV locked closed.

NOTE: When this condition occurs, the left and right bleed air manifolds will automatically be pressurized by bleed pressure from the opposite engine. Taxi and descent can be accomplished in this single bleed source configuration to avoid repetitive cycling of the Engine Bleed switch.

5. When setting thrust for takeoff, at 55% N1 or greater, select the associated ENG bleed switch OFF, then ON.

6. Operate the wing anti-ice manually.

A. Position WING ANTI-ICE selector OFF.

B. For flaps up and wing anti-ice required (do not use during takeoff and landing):

- 1) Position the ENG Bleed switch associated with the inoperative Bleed Temp Sensor OFF.

NOTE: The ENG IDLE DISAGREE advisory message may be displayed when idle thrust is selected inflight.

- 2) Position WING ANTI-ICE selector ON, then OFF as required.

- 3) Prior to landing, position the ENG Bleed switch associated with the inoperative Bleed Temp Sensor ON.

7. In the event of an engine bleed failure on the engine opposite to the inoperative Bleed Temp Sensor, maintain a minimum of 76% N1 on the engine with the inoperative Bleed Temp Sensor when the associated Engine Bleed switch is selected ON.

8. Prior to landing:

A. If the APU is available:

- 1) Position APU Bleed switch AUTO.
- 2) Start the APU if it is not running.
- 3) Position L ISLN and R ISLN switches AUTO.
- 4) In the event of failure of the opposite engine or the engine bleed air system, position both ENG Bleed switches OFF.

B. If the APU is not available:

- 1) Prior to selecting flaps 1, if the BLEED OFF ENG L or R advisory message is displayed, open the associated engine bleed valve by cycling the respective ENG bleed switch OFF, then ON.

36-22-01 Manifold Temperature Sensing Systems

36-22-01-04 -300 (With SB 777-25-0227 and ASCPC SW P/N 3670-GRS-012-00 or later Installed) Or (With PRR 61980 or Equivalent Incorporated)

36-22-01-04B Associated Engine Bleed Not Used

Interval	Installed	Required	Procedure
C	2	1	(M) (O)

One may be inoperative provided:

- a. Associated HPSOV is locked closed.
- b. Associated PRSOV is locked closed.
- c. Opposite engine bleed system operates normally.
- d. Associated engine bleed air switch remains OFF.
- e. Left and right bleed isolation systems operate normally.
- f. Center bleed isolation system operates normally.
- g. Both packs operate normally.
- h. Both outflow valves operate normally.
- i. APU is used as air source for center system hydraulic demand pumps for takeoff and landing.
- j. Appropriate performance adjustments are applied.

NOTE 1: The airplane must also be dispatched using MMEL item 73-21-01.

NOTE 2: PRR 61980 installs enhancements for dispatch with a single pack, or dispatch with a single engine bleed system, on airplane L/N's 423, 429, 454, 455, 457 thru 461, and 463 and-on. For production equivalent information, see DDG item 21-51-01.

NOTE 3: All software P/Ns installed in ASCPC-2 are considered later versions than software P/Ns installed in ASCPC-1.

MAINTENANCE (M)

NOTE 1: For a pack to be operating normally, certain items associated with that system must be operative. See DDG item 21-51-01 for a listing of those items.

NOTE 2: For an engine bleed system to be operating normally, certain items associated with that system must be operative. See DDG item 36-00-01 for a listing of those items.



Lock the associated High Pressure Shutoff Valve (HPSOV) and Pressure Regulating and Shutoff Valve (PRSOV) in the closed position (AMM 36-00-00/901).

1. Deactivate the associated thrust reverser for ground maintenance (AMM 78-31-00/201).
2. Position associated BLEED AIR Panel (P5) ENG Bleed switch OFF.
3. Gain access to the HPSOV and PRSOV by opening the appropriate fan cowl panel (AMM 71-11-04/201) and thrust reverser sleeve (AMM 78-31-00/201).
4. Lock the HPSOV and PRSOV in the closed positions by following the MANUAL LOCKING CLOSED instructions placard on the valves.
5. Close the opened thrust reverser sleeve (AMM 78-31-00/201) and fan cowl panel (AMM 71-11-04/201).
6. Activate the thrust reverser (AMM 78-31-00/201).

OPERATIONS (O)

NOTE: For the APU-to-Pack takeoff function installed, the APU-to-Pack Takeoff Supplementary Procedure is not allowed.

1. Reduce the performance limited weights by the appropriate adjustments.

Performance Limited Weights	Takeoff & Landing (Packs On)	Takeoff (Packs Off)
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For -300/PW4090:

All	21,800 lb (9,888 kg)	2,200 lb (998 kg)
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For -300/RR892:

All	7,000 lb (3,175 kg)	900 lb (408 kg)
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2. For ETOPS flight planning, increase the engine inoperative Critical Fuel Reserves by 1.3% when dispatching into known or forecast icing conditions.
3. Position associated ENG Bleed switch OFF for the inoperative system.

NOTE: The ENG IDLE DISAGREE advisory message will be displayed when idle thrust is selected inflight.

4. For Takeoff, use the APU as an air source.
 - A. Position APU bleed air switch AUTO.
 - B. Start the APU if it is not running.
 - C. Position L ISLN and R ISLN switches CLOSED.
 - D. Accomplish the BLEED LOSS WING L, R Non-Normal Checklist and do not accomplish the PACK L, R Non-Normal Checklist.

- E. After takeoff with flaps up:
 - 1) Position L ISLN and R ISLN switches AUTO.
 - 2) Position Wing Anti-Ice switch AUTO.
 - 3) Configure the APU as required.
5. In the event of inflight failure of the opposite engine or the engine bleed air system, use the APU to maintain cabin pressurization at or below FL 220.
 - A. Position APU bleed air switch AUTO.
 - B. Start the APU if it is not running.
 - C. Position both ENG Bleed switches OFF.
 - D. Position WING ANTI-ICE selector OFF (wing anti-ice will not be available for remainder of flight).
 - E. Position L ISLN and R ISLN switches AUTO for remainder of flight (including landing).
6. Prior to Landing, use the APU as an air source.
 - A. Position APU bleed air switch AUTO.
 - B. Start the APU if it is not running.
 - C. Position L ISLN and R ISLN switches AUTO.
 - D. In the event of failure of the opposite engine or the engine bleed air system, position both ENG Bleed switches OFF.

36-22-01 Manifold Temperature Sensing Systems
36-22-01-05 -200LR/-300ER
36-22-01-05A Associated Engine Bleed Used

Interval	Installed	Required	Procedure
C	2	1	(M) (O)

One may be inoperative provided:

- a. Associated HPSOV is locked closed.
- b. Opposite engine bleed system operates normally.
- c. Both outflow valves operate normally.
- d. Associated engine bleed air switch is cycled OFF, then ON at 55% N1 or greater before takeoff.
- e. Appropriate performance adjustments are applied.

MAINTENANCE (M)

NOTE: For an engine bleed system to be operating normally, certain items associated with that system must be operative. See DDG item 36-00-01 for a listing of those items.



Lock the associated High Pressure Shutoff Valve (HPSOV) in the closed position (AMM 36-00-00/901).

1. Deactivate the leading edge slats in the retracted position (AMM 27-81-00/201).
2. Deactivate the associated thrust reverser for ground maintenance (AMM 78-31-00/201).
3. Gain access to the HPSOV by opening the appropriate fan cowl panel (AMM 71-11-04/201) and thrust reverser sleeve (AMM 78-31-00/201).
4. Lock the HPSOV in the closed position by following the MANUAL LOCKING CLOSED instructions placard on the valve.
5. Close the opened thrust reverser sleeve (AMM 78-31-00/201) and fan cowl panel (AMM 71-11-04/201).
6. Activate the thrust reverser (AMM 78-31-00/201).
7. Activate the leading edge slats (AMM 27-81-00/201).

OPERATIONS (O)

NOTE 1: Whenever power is set below 55% N1, the BLEED OFF ENG L or R advisory message will be displayed for the bleed system with the HPSOV locked closed. Do not accomplish the BLEED OFF ENG L or R Non-Normal Checklist. The engine bleed valve can be reopened by cycling the ENG bleed switch OFF, then ON at 55% N1 or greater.

NOTE 2: The ENG IDLE DISAGREE advisory message will be displayed when idle thrust is selected inflight prior to cycling the ENG bleed switch OFF, then ON at 55% N1 or greater.

NOTE 3: For APU-to-Pack takeoff function installed, the APU-to-Pack takeoff Supplementary Procedure is not allowed.

1. Reduce performance limited weights by the appropriate adjustments.

Performance Limited Weights	Takeoff & Landing	Enroute Climb
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For -200LR/GE90-100 Series:

All	13,300 lb (6,033 kg)	10,100 lb (4,581 kg)
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For -300ER/GE90-115B:

All	10,700 lb (4,854 kg)	10,200 lb (4,627 kg)
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NOTE: The enroute climb weight penalties listed above are based on single engine operating speeds that approximate maximum lift-to-drag ratio speed. To account for the difference in level off altitude when operating at other speeds, multiply the enroute climb weight penalty listed above by the appropriate factor listed on page 3.00-01-00.2 in the ENROUTE DIVERSION SPEED EFFECTS paragraph.

2. Increase flight planning fuel by 0.5%.
3. For ETOPS flight planning, increase ETOPS engine inoperative Critical Fuel Reserves by 0.6%, or by 1.8% when dispatching into known or forecast icing conditions.
4. After engine start, the BLEED OFF ENG L or R advisory message will be displayed for the bleed system with HPSOV locked closed.

NOTE: When this condition occurs, the left and right bleed air manifolds will automatically be pressurized by bleed pressure from the opposite engine. Taxi and descent can be accomplished in this single bleed source configuration to avoid repetitive cycling of the Engine Bleed switch.

5. When setting thrust for takeoff, at 55% N1 or greater, select the associated ENG bleed switch OFF, then ON.
6. Operate the wing anti-ice manually.
 - A. Position WING ANTI-ICE selector OFF.
 - B. For flaps up and wing anti-ice required (do not use during takeoff and landing):

- 1) Position the ENG Bleed switch associated with the inoperative Bleed Temp Sensor OFF.

NOTE: The ENG IDLE DISAGREE advisory message may be displayed when idle thrust is selected inflight.

- 2) Position WING ANTI-ICE selector ON, then OFF as required.
 - 3) Prior to landing, position the ENG Bleed switch associated with the inoperative Bleed Temp Sensor ON.
7. In the event of an engine bleed failure on the engine opposite to the inoperative Bleed Temp Sensor, maintain a minimum of 76% N1 on the engine with the inoperative Bleed Temp Sensor when the associated Engine Bleed switch is selected ON.
8. Prior to landing:
 - A. If the APU is available:
 - 1) Position APU Bleed switch AUTO.
 - 2) Start the APU if it is not running.
 - 3) Position L ISLN and R ISLN switches AUTO.
 - 4) In the event of failure of the opposite engine or the engine bleed air system, position both ENG Bleed switches OFF.



B. If the APU is not available:

- 1) Prior to selecting flaps 1, if the BLEED OFF ENG L or R advisory message is displayed, open the associated engine bleed valve by cycling the respective ENG bleed switch OFF, then ON.

36-22-01 Manifold Temperature Sensing Systems

36-22-01-05 -200LR/-300ER

36-22-01-05B Associated Engine Bleed Not Used

Interval	Installed	Required	Procedure
C	2	1	(M) (O)

One may be inoperative provided:

- a. Associated HPSOV is locked closed.
 - b. Associated PRSOV is locked closed.
 - c. Opposite engine bleed system operates normally.
 - d. Associated engine bleed air switch remains OFF.
 - e. Left and right bleed isolation systems operate normally.
 - f. Center bleed isolation system operates normally.
 - g. Both packs operate normally.
 - h. Both outflow valves operate normally.
 - i. APU is used as air source for center system hydraulic demand pumps for takeoff and landing.
 - j. Appropriate performance adjustments are applied.
-

NOTE: The airplane must also be dispatched using MMEL item 73-21-01.

MAINTENANCE (M)

NOTE 1: For a pack to be operating normally, certain items associated with that system must be operative. See DDG item 21-51-01 for a listing of those items.

NOTE 2: For an engine bleed system to be operating normally, certain items associated with that system must be operative. See DDG item 36-00-01 for a listing of those items.

Lock the associated High Pressure Shutoff Valve (HPSOV) and Pressure Regulating and Shutoff Valve (PRSOV) in the closed position (AMM 36-00-00/901).

1. Deactivate the leading edge slats in the retracted position (AMM 27-81-00/201).
2. Deactivate the associated thrust reverser for ground maintenance (AMM 78-31-00/201).



3. Position associated BLEED AIR Panel (P5) ENG Bleed switch OFF.
4. Gain access to the HPSOV and PRSOV by opening the appropriate fan cowl panel (AMM 71-11-04/201) and thrust reverser sleeve (AMM 78-31-00/201).
5. Lock the HPSOV and PRSOV in the closed positions by following the MANUAL LOCKING CLOSED instructions placard on the valves.
6. Close the opened thrust reverser sleeve (AMM 78-31-00/201) and fan cowl panel (AMM 71-11-04/201).
7. Activate the thrust reverser (AMM 78-31-00/201).
8. Activate the leading edge slats (AMM 27-81-00/201).

OPERATIONS (O)

NOTE: For the APU-to-Pack takeoff function installed, the APU-to-Pack Takeoff Supplementary Procedure is not allowed.

1. Reduce the performance limited weights by the appropriate adjustments.

Performance Limited Weights	Takeoff & Landing (Packs On)	Takeoff (Packs Off)
-----------------------------	------------------------------	---------------------

For -200LR/GE90-100 Series:

All	14,800 lb (6,713 kg)	5,350 lb (2,427 kg)
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For -300ER/GE90-115B:

All	11,600 lb (5,262 kg)	6,200 lb (2,812 kg)
-----	----------------------	---------------------

2. For ETOPS flight planning, increase the engine inoperative Critical Fuel Reserves by 1.3% when dispatching into known or forecast icing conditions.
3. Position associated ENG Bleed switch OFF for the inoperative system.
NOTE: The ENG IDLE DISAGREE advisory message will be displayed when idle thrust is selected inflight.
4. For Takeoff, use the APU as an air source.
 - A. Position APU bleed air switch AUTO.
 - B. Start the APU if it is not running.
 - C. Position L ISLN and R ISLN switches CLOSED.
 - D. Accomplish the BLEED LOSS WING L, R Non-Normal Checklist and do not accomplish the PACK L, R Non-Normal Checklist.
 - E. After takeoff with flaps up:
 - 1) Position L ISLN and R ISLN switches AUTO.
 - 2) Position Wing Anti-Ice switch AUTO.
 - 3) Configure the APU as required.



5. In the event of inflight failure of the opposite engine or the engine bleed air system, use the APU to maintain cabin pressurization at or below FL 220.
 - A. Position APU bleed air switch AUTO.
 - B. Start the APU if it is not running.
 - C. Position both ENG Bleed switches OFF.
 - D. Position WING ANTI-ICE selector OFF (wing anti-ice will not be available for remainder of flight).
 - E. Position L ISLN and R ISLN switches AUTO for remainder of flight (including landing).
6. Prior to Landing, use the APU as an air source.
 - A. Position APU bleed air switch AUTO.
 - B. Start the APU if it is not running.
 - C. Position L ISLN and R ISLN switches AUTO.
 - D. In the event of failure of the opposite engine or the engine bleed air system, position both ENG Bleed switches OFF.

36-22-01 Manifold Temperature Sensing Systems**36-22-01-06 777F****36-22-01-06A Associated Engine Bleed Used**

Interval	Installed	Required	Procedure
C	2	1	(M) (O)

One may be inoperative provided:

- a. Associated HPSOV is locked closed.
- b. Opposite engine bleed system operates normally.
- c. Both outflow valves operate normally.
- d. Associated engine bleed air switch is cycled OFF, then ON at 55% N1 or greater before takeoff.
- e. Appropriate performance adjustments are applied.

MAINTENANCE (M)

NOTE: For an engine bleed system to be operating normally, certain items associated with that system must be operative. See DDG item 36-00-01 for a listing of those items.

Lock the associated High Pressure Shutoff Valve (HPSOV) in the closed position (AMM 36-00-00/901).

1. Deactivate the leading edge slats in the retracted position (AMM 27-81-00/201).
2. Deactivate the associated thrust reverser for ground maintenance (AMM 78-31-00/201).



3. Gain access to the HPSOV by opening the appropriate fan cowl panel (AMM 71-11-04/201) and thrust reverser sleeve (AMM 78-31-00/201).
4. Lock the HPSOV in the closed position by following the MANUAL LOCKING CLOSED instructions placard on the valve.
5. Close the opened thrust reverser sleeve (AMM 78-31-00/201) and fan cowl panel (AMM 71-11-04/201).
6. Activate the thrust reverser (AMM 78-31-00/201).
7. Activate the leading edge slats (AMM 27-81-00/201).

OPERATIONS (O)

NOTE 1: Whenever power is set below 55% N1, the BLEED OFF ENG L or R advisory message will be displayed for the bleed system with the HPSOV locked closed. Do not accomplish the BLEED OFF ENG L or R Non-Normal Checklist. The engine bleed valve can be reopened by cycling the ENG bleed switch OFF, then ON at 55% N1 or greater.

NOTE 2: The ENG IDLE DISAGREE advisory message will be displayed when idle thrust is selected inflight prior to cycling the ENG bleed switch OFF, then ON at 55% N1 or greater.

NOTE 3: For APU-to-Pack takeoff function installed, the APU-to-Pack takeoff Supplementary Procedure is not allowed.

1. Reduce performance limited weights by the appropriate adjustments.

Performance Limited Weights	Takeoff & Landing	Enroute Climb
--------------------------------	----------------------	------------------

For 777F/GE90-100 Series:

All	13,300 lb (6,033 kg)	10,100 lb (4,581 kg)
-----	----------------------	----------------------

NOTE: The enroute climb weight penalties listed above are based on single engine operating speeds that approximate maximum lift-to-drag ratio speed. To account for the difference in level off altitude when operating at other speeds, multiply the enroute climb weight penalty listed above by the appropriate factor listed on page 3.00-01-00.2 in the ENROUTE DIVERSION SPEED EFFECTS paragraph.

2. Increase flight planning fuel by 0.5%.
3. For ETOPS flight planning, increase ETOPS engine inoperative Critical Fuel Reserves by 0.6%, or by 1.8% when dispatching into known or forecast icing conditions.
4. After engine start, the BLEED OFF ENG L or R advisory message will be displayed for the bleed system with HPSOV locked closed.

NOTE: When this condition occurs, the left and right bleed air manifolds will automatically be pressurized by bleed pressure from the opposite engine. Taxi and descent can be accomplished in this single bleed source configuration to avoid repetitive cycling of the Engine Bleed switch.

5. When setting thrust for takeoff, at 55% N1 or greater, select the associated ENG bleed switch OFF, then ON.
6. Operate the wing anti-ice manually.
 - A. Position WING ANTI-ICE selector OFF.
 - B. For flaps up and wing anti-ice required (do not use during takeoff and landing):
 - 1) Position the ENG Bleed switch associated with the inoperative Bleed Temp Sensor OFF.

NOTE: The ENG IDLE DISAGREE advisory message may be displayed when idle thrust is selected in flight.

 - 2) Position WING ANTI-ICE selector ON, then OFF as required.
 - 3) Prior to landing, position the ENG Bleed switch associated with the inoperative Bleed Temp Sensor ON.
7. In the event of an engine bleed failure on the engine opposite to the inoperative Bleed Temp Sensor, maintain a minimum of 76% N1 on the engine with the inoperative Bleed Temp Sensor when the associated Engine Bleed switch is selected ON.
8. Prior to landing:
 - A. If the APU is available:
 - 1) Position APU Bleed switch AUTO.
 - 2) Start the APU if it is not running.
 - 3) Position L ISLN and R ISLN switches AUTO.
 - 4) In the event of failure of the opposite engine or the engine bleed air system, position both ENG Bleed switches OFF.
 - B. If the APU is not available:
 - 1) Prior to selecting flaps 1, if the BLEED OFF ENG L or R advisory message is displayed, open the associated engine bleed valve by cycling the respective ENG bleed switch OFF, then ON.

36-22-01 Manifold Temperature Sensing Systems
36-22-01-06 777F
36-22-01-06B Associated Engine Bleed Not Used

Interval	Installed	Required	Procedure
C	2	1	(M) (O)

One may be inoperative provided:

- a. Associated HPSOV is locked closed.
 - b. Associated PRSOV is locked closed.
 - c. Opposite engine bleed system operates normally.
 - d. Associated engine bleed air switch remains OFF.
 - e. Left and right bleed isolation systems operate normally.
 - f. Center bleed isolation system operates normally.
 - g. Both packs operate normally.
 - h. Both outflow valves operate normally.
 - i. Both recirculation fans operate normally.
 - j. Both ECSMCs operate normally.
 - k. APU is used as air source for center system hydraulic demand pumps for takeoff and landing.
 - l. Appropriate performance adjustments are applied.
-

NOTE: The airplane must also be dispatched using MMEL item 73-21-01.

MAINTENANCE (M)

NOTE 1: For a pack to be operating normally, certain items associated with that system must be operative. See DDG item 21-51-01 for a listing of those items.

NOTE 2: For an engine bleed system to be operating normally, certain items associated with that system must be operative. See DDG item 36-00-01 for a listing of those items.

Lock the associated High Pressure Shutoff Valve (HPSOV) and Pressure Regulating and Shutoff Valve (PRSOV) in the closed position (AMM 36-00-00/901).

1. Deactivate the leading edge slats in the retracted position (AMM 27-81-00/201).
2. Deactivate the associated thrust reverser for ground maintenance (AMM 78-31-00/201).
3. Position associated BLEED AIR Panel (P5) ENG Bleed switch OFF.
4. Gain access to the HPSOV and PRSOV by opening the appropriate fan cowl panel (AMM 71-11-04/201) and thrust reverser sleeve (AMM 78-31-00/201).
5. Lock the HPSOV and PRSOV in the closed positions by following the MANUAL LOCKING CLOSED instructions placard on the valves.
6. Close the opened thrust reverser sleeve (AMM 78-31-00/201) and fan cowl panel (AMM 71-11-04/201).
7. Activate the thrust reverser (AMM 78-31-00/201).
8. Activate the leading edge slats (AMM 27-81-00/201).

**OPERATIONS (O)**

NOTE 1: For overhead panel (P5) M/D FLOW switch positioned to HIGH, Air conditioning in the lower lobe cargo compartments is degraded. Position M/D FLOW switch to NORM for temperature sensitive cargo carried in the lower lobe cargo compartments.

NOTE 2: For the APU-to-Pack takeoff function installed, the APU-to-Pack Takeoff Supplementary Procedure is not allowed.

1. Reduce the performance limited weights by the appropriate adjustments.

Performance Limited Weights	Takeoff & Landing (Packs On)	Takeoff (Packs Off)
--	---	--------------------------------

For 777F/GE90-100 Series:

All	14,800 lb (6,713 kg)	5,350 lb (2,427 kg)
-----	----------------------	---------------------

2. For ETOPS flight planning, increase the engine inoperative Critical Fuel Reserves by 1.3% when dispatching into known or forecast icing conditions.
3. Position associated ENG Bleed switch OFF for the inoperative system.
NOTE: The ENG IDLE DISAGREE advisory message will be displayed when idle thrust is selected inflight.
4. For Takeoff, use the APU as an air source.
 - A. Position APU bleed air switch AUTO.
 - B. Start the APU if it is not running.
 - C. Position L ISLN and R ISLN switches CLOSED.
 - D. Accomplish the BLEED LOSS WING L, R Non-Normal Checklist and do not accomplish the PACK L, R Non-Normal Checklist.
 - E. After takeoff with flaps up:
 - 1) Position L ISLN and R ISLN switches AUTO.
 - 2) Position Wing Anti-Ice switch AUTO.
 - 3) Configure the APU as required.
5. In the event of inflight failure of the opposite engine or the engine bleed air system, use the APU to maintain cabin pressurization at or below FL 220.
 - A. Position APU bleed air switch AUTO.
 - B. Start the APU if it is not running.
 - C. Position both ENG Bleed switches OFF.
 - D. Position WING ANTI-ICE selector OFF (wing anti-ice will not be available for remainder of flight).
 - E. Position L ISLN and R ISLN switches AUTO for remainder of flight (including landing).



6. Prior to Landing, use the APU as an air source.
 - A. Position APU bleed air switch AUTO.
 - B. Start the APU if it is not running.
 - C. Position L ISLN and R ISLN switches AUTO.
 - D. In the event of failure of the opposite engine or the engine bleed air system, position both ENG Bleed switches OFF.



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Section 2

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38-10-01 Potable Water Systems**38-10-01A Inoperative Components Deactivated**

Interval	Installed	Required	Procedure
C	-	-	(M)

Individual components may be inoperative provided:

- Associated components are deactivated or isolated.
- Associated system components are verified not to have leaks.

NOTE: Any portion of system which operates normally may be used.

MAINTENANCE (M)

Configure the airplane (AMM 38-00-00/901).

- Deactivate or isolate inoperative components.
- Verify associated system components do not have leaks.

NOTE: Due to the wide variety of galley, lavatory and drinking fountain installations and various component failure possibilities, operators must develop procedures tailored to their specific installations.

38-10-01 Potable Water Systems**38-10-01B System Not Used**

Interval	Installed	Required	Procedure
C	-	-	(M)

May be inoperative provided:

- System is drained.
 - Procedures are established to ensure that system is not serviced.
-

MAINTENANCE (M)

Configure the airplane (AMM 38-00-00/901).

- Drain potable water system.
- Establish procedures to ensure the potable water system is not serviced.

OPERATIONS NOTE

Establish procedures to ensure the potable water system is not serviced and accounted for in the airplane weight and balance.

**38-10-01 Potable Water Systems****38-10-01-01 Potable Water Indication System**

Interval	Installed	Required	Procedure
D	-	-	

38-30-01 Waste Water Systems
38-30-01A Components Deactivated

Interval	Installed	Required	Procedure
C	-	-	(M)

Individual components may be inoperative provided:

- Associated components are deactivated or isolated.
- Associated system components are verified not to have leaks.

NOTE: Any portion of system which operates normally may be used.

MAINTENANCE (M)

Configure the airplane (AMM 38-00-00/901).

- Deactivate or isolate inoperative components.
- Verify associated system components do not have leaks.

NOTE: Due to the wide variety of lavatory installations and various component failure possibilities, operators must develop procedures tailored to their specific installations.

38-30-01 Waste Water Systems
38-30-01B Lavatory Not Used

Interval	Installed	Required	Procedure
C	-	-	(M)

Associated lavatory system may be inoperative provided:

- Associated components are deactivated or isolated to prevent leaks.
- Associated lavatory door is locked closed and placarded, INOPERATIVE - DO NOT ENTER.

NOTE: These provisions are not intended to prohibit inspections by crewmembers.

MAINTENANCE (M)

Configure the airplane (AMM 38-00-00/901).

- Deactivate or isolate inoperative components.
- If required, drain the associated waste tank. Do not add the precharge chemical to the associated waste tank.
- Close, lock and placard the associated lavatory door.

**38-30-01 Waste Water Systems****38-30-01-01 Vacuum Blowers**

Interval	Installed	Required	Procedure
C	-	-	(M) (O)

May be inoperative provided:

- Associated vacuum blower is deactivated.
- Lavatories affected by inoperative vacuum blower are not used on the ground or at flight altitudes of 16,000 feet or below.

MAINTENANCE (M)

Deactivate the inoperative vacuum blower (AMM DDG 38-00-00/901).

For Passenger:

- For the system 1 (forward tank) vacuum blower inoperative, open and collar the P110 panel VAC BLO LAV SYS 1 circuit breaker.
- For the system 2 and 3 (mid and aft tank) vacuum blower inoperative, open and collar the P210 panel VAC BLO LAV SYS 2 & 3 circuit breaker.

For 777F:

- Open and collar the P210 panel VAC BLO LAV circuit breaker.

OPERATIONS (O)

Establish and use procedures to prevent use of lavatories associated with the inoperative vacuum blower at flight altitudes below 16,000 feet.

38-30-01 Waste Water Systems**38-30-01-02 Waste Tanks**

Interval	Installed	Required	Procedure
C	-	-	(M)

May be inoperative provided:

- Associated waste tank(s) is deactivated.
- Associated components are deactivated or isolated to prevent leaks.
- Associated lavatory door(s) is secured closed and placarded, INOPERATIVE - DO NOT ENTER.

NOTE: These provisions are not intended to prohibit inspections by crewmembers.

MAINTENANCE (M)

Deactivate associated waste tank(s), deactivate or isolate inoperative components, and secure closed and placard lavatory door(s) (AMM DDG 38-00-00/901).

1. Drain the associated waste tank(s). Do not add the precharge chemical to the associated waste tank.
2. Deactivate or isolate inoperative components.
3. Close, lock and placard the associated lavatory door(s).



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45-10-01 Central Maintenance Computing System (CMCS)

Interval	Installed	Required	Procedure
C	1	0	



45-11-01 Maintenance Access Terminal (MAT)

Interval	Installed	Required	Procedure
D	1	0	

**45-13-01 Side Displays**

Interval	Installed	Required	Procedure
D	-	0	(M)

May be inoperative provided inoperative side display(s) is deactivated.

MAINTENANCE (M)

Deactivate the inoperative side display (AMM 45-00-00/901).

1. For the Captain's side display, open and collar P11 panel CAPT SIDE DSPL circuit breaker.
2. For the First Officer's side display, open and collar P11 panel F/O SIDE DSPL circuit breaker.



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46-11-01 Onboard Network System (ONS)

46-11-01A Alternate Procedures Required

Interval	Installed	Required	Procedure
C	1	0	(O)

May be inoperative provided alternate procedures are established and used.

NOTE: Any function which operates normally may be used.

NOTE 1: The Onboard Network System (ONS) includes the Network File Server (NFS), Network Extension Device (NED), Aircraft Interface Device (AID) and Wireless WAN Unit (WWU) components with associated support interfaces.

[For Class 3 EFB installed:](#)

NOTE 2: The airplane must also be dispatched using MMEL item 46-11-02-01B.

MAINTENANCE NOTE

The Maintenance Access Terminal (MAT) functions may not be available including CMCS data collection and loading.

OPERATIONS (O)

Alternate procedures must be established and used.

[For Class 3 EFB installed:](#)

NOTE: The EFB's on-ground connectivity to the AWLU or TWLU and airport networks, the EFB's connectivity to the graphical function of the ARINC 744A airborne printer and the EFB's in-flight connectivity to the SATCOM Swift communication network will be inoperative.

46-11-01 Onboard Network System (ONS)

46-11-01B Procedures Do Not Require Use

Interval	Installed	Required	Procedure
D	1	0	

May be inoperative provided procedures do not require its use.

NOTE 1: The Onboard Network System (ONS) includes the Network File Server (NFS), Network Extension Device (NED), Aircraft Interface Device (AID) and Wireless WAN Unit (WWU) components with associated support interfaces.



For Class 3 EFB installed:

NOTE 2: The airplane must also be dispatched using MMEL item 46-11-02-01B.

MAINTENANCE NOTE

The Maintenance Access Terminal (MAT) functions may not be available including CMCS data collection and loading.

For Class 3 EFB installed:

OPERATIONS NOTE

The EFB's on-ground connectivity to the AWLU or TWLU and airport networks, the EFB's connectivity to the graphical function of the ARINC 744A airborne printer and the EFB's in-flight connectivity to the SATCOM Swift communication network will be inoperative.

**46-11-02 Electronic Flight Bag (EFB) Systems**

46-11-02-01 EFB System (Installed EFB System)

46-11-02-01A Alternate Procedures Required

Interval	Installed	Required	Procedure
C	-	-	(O)

May be inoperative provided alternate procedures are established and used.

NOTE: Any function, program, or document which operates normally may be used.

OPERATIONS (O)

Each operator should develop appropriate procedures to be used when the EFB system is inoperative.

46-11-02 Electronic Flight Bag (EFB) System

46-11-02-01 EFB System (Installed EFB System)

46-11-02-01B Procedures Do Not Require Use

Interval	Installed	Required	Procedure
D	-	0	

May be inoperative provided procedures do not require its use.

46-11-02 Electronic Flight Bag (EFB) System

46-11-02-02 Data Connectivity**46-11-02-02A Alternate Procedures Required**

Interval	Installed	Required	Procedure
C	-	-	(O)

May be inoperative provided alternate procedures are established and used.

**OPERATIONS (O)**

Each operator should develop appropriate procedures to be used when the EFB data connectivity is inoperative.

46-11-02 Electronic Flight Bag (EFB) System*********46-11-02-02 Data Connectivity****46-11-02-02B Procedures Do Not Require Use**

Interval	Installed	Required	Procedure
D	-	0	

May be inoperative provided procedures do not require its use.

46-11-02 Electronic Flight Bag (EFB) System*********46-11-02-03 Power Supply/Power Connection****46-11-02-03A Alternate Procedures Required**

Interval	Installed	Required	Procedure
C	-	-	(O)

May be inoperative provided alternate procedures are established and used.

OPERATIONS (O)

Each operator should develop appropriate procedures to be used when the EFB power connection is inoperative.

46-11-02 Electronic Flight Bag (EFB) System*********46-11-02-03 Power Supply/Power Connection****46-11-02-03B Procedures Do Not Require Use**

Interval	Installed	Required	Procedure
D	-	0	

May be inoperative provided procedures do not require its use.

**46-11-02 Electronic Flight Bag (EFB) System**

46-11-02-04 Mounting Device**46-11-02-04A Alternate Procedures Required**

Interval	Installed	Required	Procedure
C	-	0	(M) (O)

May be inoperative provided:

- Associated EFB and hardware is stowed, secured by an alternate means, or removed from the aircraft.
- Alternate procedures are established and used.

MAINTENANCE (M)

Stow, secure or remove associated EFB and hardware (AMM 46-00-00/901).

OPERATIONS (O)

Each operator should develop appropriate procedures to be used when the EFB mounting hardware is inoperative.

46-11-02 Electronic Flight Bag (EFB) System

46-11-02-04 Mounting Device**46-11-02-04B Procedures Do Not Require Use**

Interval	Installed	Required	Procedure
D	-	0	(M)

May be inoperative provided:

- Associated EFB and hardware is stowed, secured by an alternate means, or removed from the aircraft.
- Procedures do not require its use.

MAINTENANCE (M)

Stow, secure or remove associated EFB and hardware (AMM 46-00-00/901).

**46-11-02 Electronic Flight Bag (EFB) System**

46-11-02-05 Class 3 EFB (STC ST00751DE)

46-11-02-05-01 Display Unit (DU)

46-11-02-05-01A Alternate Procedures Required

Interval	Installed	Required	Procedure
C	4	0	(O)

May be inoperative provided alternate procedures are established and used.

NOTE: Any function, program or document which operates normally may be used.

NOTE: The EFB installation defined by STC ST00751DE is not a Boeing system. Information applicable to this installation is not provided.

46-11-02 Electronic Flight Bag (EFB) System

46-11-02-05 Class 3 EFB (STC ST00751DE)

46-11-02-05-01 Display Unit (DU)

46-11-02-05-01B Procedures Do Not Require Use

Interval	Installed	Required	Procedure
D	4	0	

May be inoperative provided procedures do not require its use.

NOTE: The EFB installation defined by STC ST00751DE is not a Boeing system. Information applicable to this installation is not provided.

**46-11-02 Electronic Flight Bag (EFB) System**

46-11-02-05 Class 3 EFB (STC ST00751DE)

46-11-02-05-02 Remote Processor Unit (RPU)

46-11-02-05-02A Alternate Procedures Required

Interval	Installed	Required	Procedure
C	4	0	(O)

May be inoperative provided alternate procedures are established and used.

NOTE: Any function, program or document which operates normally may be used.

NOTE: The EFB installation defined by STC ST00751DE is not a Boeing system. Information applicable to this installation is not provided.

46-11-02 Electronic Flight Bag (EFB) System

46-11-02-05 Class 3 EFB (STC ST00751DE)

46-11-02-05-02 Remote Processor Unit (RPU)

46-11-02-05-02B Procedures Do Not Require Use

Interval	Installed	Required	Procedure
D	4	0	

May be inoperative provided procedures do not require its use.

NOTE: The EFB installation defined by STC ST00751DE is not a Boeing system. Information applicable to this installation is not provided.

**46-11-02 Electronic Flight Bag (EFB) System**

| 46-11-02-05 Class 3 EFB (STC ST00751DE)

| 46-11-02-05-03 Cabin Wireless LAN Unit (CWLU)

| 46-11-02-05-03A Alternate Procedures Required

Interval	Installed	Required	Procedure
C	2	0	(O)

May be inoperative provided alternate procedures are established and used.

NOTE: Any function, program or document which operates normally may be used.

NOTE: The EFB installation defined by STC ST00751DE is not a Boeing system. Information applicable to this installation is not provided.

46-11-02 Electronic Flight Bag (EFB) System

| 46-11-02-05 Class 3 EFB (STC ST00751DE)

| 46-11-02-05-03 Cabin Wireless LAN Unit (CWLU)

| 46-11-02-05-03B Procedures Do Not Require Use

Interval	Installed	Required	Procedure
D	2	0	

May be inoperative provided procedures do not require its use.

NOTE: The EFB installation defined by STC ST00751DE is not a Boeing system. Information applicable to this installation is not provided.

**46-11-02 Electronic Flight Bag (EFB) System**

46-11-02-05 Class 3 EFB (STC ST00751DE)

46-11-02-05-04 Aircraft Interface Device - Ethernet Switch (AID-ES)

46-11-02-05-04A Alternate Procedures Required

Interval	Installed	Required	Procedure
C	1	0	(O)

May be inoperative provided alternate procedures are established and used.

NOTE: Any function, program or document which operates normally may be used.

NOTE: The EFB installation defined by STC ST00751DE is not a Boeing system. Information applicable to this installation is not provided.

46-11-02 Electronic Flight Bag (EFB) System

46-11-02-05 Class 3 EFB (STC ST00751DE)

46-11-02-05-04 Aircraft Interface Device - Ethernet Switch (AID-ES)

46-11-02-05-04B Procedures Do Not Require Use

Interval	Installed	Required	Procedure
D	1	0	

May be inoperative provided procedures do not require its use.

NOTE: The EFB installation defined by STC ST00751DE is not a Boeing system. Information applicable to this installation is not provided.

**46-11-02 Electronic Flight Bag (EFB) System**

| 46-11-02-05 Class 3 EFB (STC ST00751DE)

| 46-11-02-05-05 Graphical Printer (GPRI)

| 46-11-02-05-05A Alternate Procedures Required

Interval	Installed	Required	Procedure
C	1	0	(O)

May be inoperative provided alternate procedures are established and used.

NOTE: Any function, program or document which operates normally may be used.

NOTE: The EFB installation defined by STC ST00751DE is not a Boeing system. Information applicable to this installation is not provided.

46-11-02 Electronic Flight Bag (EFB) System

| 46-11-02-05 Class 3 EFB (STC ST00751DE)

| 46-11-02-05-05 Graphical Printer (GPRI)

| 46-11-02-05-05B Procedures Do Not Require Use

Interval	Installed	Required	Procedure
D	1	0	

May be inoperative provided procedures do not require its use.

NOTE: The EFB installation defined by STC ST00751DE is not a Boeing system. Information applicable to this installation is not provided.

**46-11-03 Terminal Wireless LAN Unit (TWLU)***********46-11-03A Alternate Procedures Required**

Interval	Installed	Required	Procedure
C	1	0	(O)

May be inoperative provided alternate procedures are established and used.

OPERATIONS (O)

Alternate procedures must be established and used.

46-11-03 Terminal Wireless LAN Unit (TWLU)*********46-11-03B Procedures Do Not Require Use**

Interval	Installed	Required	Procedure
D	1	0	

May be inoperative provided procedures do not require its use.

**46-11-04 Aircraft Wireless LAN Unit (AWLU)**

46-11-04A Alternate Procedures Required

Interval	Installed	Required	Procedure
C	1	0	(O)

May be inoperative provided alternate procedures are established and used.

OPERATIONS (O)

Alternate procedures must be established and used.

46-11-04 Aircraft Wireless LAN Unit (AWLU)

46-11-04B Procedures Do Not Require Use

Interval	Installed	Required	Procedure
D	1	0	

May be inoperative provided procedures do not require its use.



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47-11-01 Nitrogen Generation System (NGS)

47-11-01-01 Nitrogen Generation Performance



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**47-11-01 Nitrogen Generation System (NGS)**

Interval	Installed	Required	Procedure
A	1	0	(M)

May be inoperative provided:

- NGS shutoff valve is deactivated closed.
- Repairs are made within 10 flight days.

MAINTENANCE (M)

Deactivate the NGS shutoff valve in closed position (AMM 47-00-00/901).

- Gain access to the NGS shutoff valve.
- Open and collar the P110 panel NGS CONTROL circuit breaker.
- Position valve's manual lock arm closed.

47-11-01 Nitrogen Generation System (NGS)

47-11-01-01 Nitrogen Generation Performance

Interval	Installed	Required	Procedure
C	1	0	



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**49-11-01 Auxiliary Power Unit (APU)****49-15-01 APU Air Inlet Door Actuation System**49-15-01A *Procedures Do Not Require APU*49-15-01B *APU Required*

49-15-01-01 APU Air Inlet Door Actuator Position Indication Switch

49-42-01 APU Starting System (Includes Electric Starter)49-42-01A *APU Required*49-42-01B *Procedures Do Not Require APU***49-43-01 APU Air Turbine Starter****49-52-01 APU Bleed Air System****49-52-02 APU Bleed Air Check Valve****49-61-01 APU Controller****49-61-02 APU Ground Control Panel****49-70-01 APU FAULT Light****49-71-01 APU EGT Indication****49-73-01 APU RPM Indication****49-94-01 APU OIL QTY Indication**49-94-01A *APU Required*49-94-01B *Procedures Do Not Require APU***49-94-02 APU OIL PRESS Indication**



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49-11-01 Auxiliary Power Unit (APU)

Interval	Installed	Required	Procedure
C	1	0	(O)

APU may be inoperative provided:

- Other procedures do not require its use.
 - Backup AC power system is verified to operate normally before each departure.
 - Flight remains within 180 minutes of landing at a suitable airport.
-

NOTE: APU REMOTE SHUTDOWN Status message indicates that the APU shutdown switch on the nose landing gear Service and APU Shutdown Panel has been activated. The message may be cleared by running the fire/overheat test, which is initiated by pressing the fire/overheat test switch.

MAINTENANCE NOTE

If desired, to prevent automatic starts, the APU may be deactivated (AMM 49-11-00/201).

- Open and collar the P49 panel APUC 2 circuit breaker.
- Open and collar the P310 panel DC FUEL PUMP circuit breaker.
- Open and collar the P310 panel APUC 1 circuit breaker.
- Open and collar the P310 panel APU INLET DR ACTR circuit breaker.
- For APU oil heater installed, open and collar the P320 panel APU OIL HTR ASSY circuit breaker. For APU oil heater not installed, the P320 panel APU OIL HTR ASSY circuit breaker is inoperative and should remain open.

OPERATIONS (O)

- Dispatch is not allowed if the APU is required by other procedures. MMEL dispatch procedures for the following inoperative equipment may require APU availability:



MMEL Item	Associated Status Message
24-11-01	ELEC GEN SYS L or R IDG VALVE CLOSED L or R
24-25-01	ELEC BACKUP GEN L or R ELEC BACKUP SYS
29-11-03	HYD PUMP DEM C1 or C2
36-11-01	BLEED PRSOV L or R
36-11-02	BLEED HPSOV L or R
36-11-03	None
36-12-01	BLEED ISLN VALVE L or R
36-12-02	BLEED ISLN VALVE C
36-22-01	BLEED TEMP SENSOR L or R

2. The flight must remain within 180 minutes of landing at a suitable airport.
3. Verify the Backup AC electrical power system operates normally. Confirm the ELEC BACKUP SYS or ELEC BACKUP GEN L or R status and advisory messages are not displayed 30 seconds after both engines have been started.
4. Position the APU selector OFF.

49-15-01 APU Air Inlet Door Actuation System
49-15-01A Procedures Do Not Require APU

Interval	Installed	Required	Procedure
C	1	0	(O)

May be inoperative with door in any position provided:

- Other procedures do not require the use of the APU.
- APU is not used.
- Backup AC power system is verified to operate normally before each departure.
- Flight remains within 180 minutes of landing at a suitable airport.
- Appropriate performance adjustments are applied.

OPERATIONS (O)

- Dispatch is not allowed if the APU is required by other procedures. MMEL dispatch procedures for the following inoperative equipment may require APU availability (inlet door deactivated full open).

MMEL Item	Associated Status Message
24-11-01	ELEC GEN SYS L or R IDG VALVE CLOSED L or R
24-25-01	ELEC BACKUP GEN L or R ELEC BACKUP SYS
29-11-03	HYD PUMP DEM C1 or C2
36-11-01	BLEED PRSOV L or R
36-11-02	BLEED HPSOV L or R
36-11-03	None
36-12-01	BLEED ISLN VALVE L or R
36-12-02	BLEED ISLN VALVE C
36-22-01	BLEED TEMP SENSOR L or R

- For APU air inlet door not closed, increase flight planning fuel by 2.5%.
- The flight must remain within 180 minutes of landing at a suitable airport.
- Verify the Backup AC electrical power system operates normally. Confirm the ELEC BACKUP SYS or ELEC BACKUP GEN L or R status and advisory messages are not displayed 30 seconds after both engines have been started.
- Position the APU selector OFF.

49-15-01 APU Air Inlet Door Actuation System**49-15-01B APU Required**

Interval	Installed	Required	Procedure
C	1	0	(M) (O)

May be inoperative provided:

- Door is deactivated in the full open position.
- Appropriate performance adjustments are applied.

MAINTENANCE (M)

Deactivate the door in the fully open position (AMM 49-00-00/901).

- Position APU selector on the P5 overhead panel ON.
- Wait 45 seconds then open and collar the P310 Panel APU INLET DR ACTR circuit breaker.
- Position APU selector on the P5 overhead panel OFF.
- On the APU Maintenance page, confirm INLET DOOR POSITION indicates OPEN.
 - If door is confirmed open, the door is deactivated in the fully open position and no additional steps are required.
 - If door is not confirmed open, the following steps must be accomplished to manually position and deactivate the door in the fully open position:
- Open the P49 Panel APUC 2 circuit breaker.
- Open the P310 Panel DC FUEL PUMP and APUC 1 circuit breakers.
- For APU oil heater installed, open the P320 Panel APU OIL HTR ASSY circuit breaker. For APU oil heater not installed, the P320 panel APU OIL HTR ASSY circuit breaker is inoperative and should remain open.
- Open left and right APU Access Doors (315AL and 316AR).
- Remove access cover from the inlet door actuator assembly.
- Turn the transfer cover socket on the inlet door actuator counterclockwise until MAN shows in the window of the transfer socket. The manual drive socket will show. DO NOT apply more than 10 lb-in (1.1 N-m) of torque on the transfer cover socket.
- Turn the manual drive socket counter-clockwise until the torque value suddenly increases. DO NOT apply more than 65 lb-in (7.3 N-m) of torque on the manual drive socket.
- Turn the transfer cover socket clockwise until ELEC shows in the forward window of the transfer cover.
- Re-install access cover on the inlet door actuator assembly.
- Close left and right APU access doors.
- Close the P49 Panel APUC 2 circuit breaker.



16. Close the P310 Panel DC FUEL PUMP and APUC 1 circuit breakers (APU INLET DR ACTR circuit breaker remains open).
17. For APU oil heater installed, close the P320 Panel APU OIL HTR ASSY circuit breaker. For APU oil heater not installed, the P320 panel APU OIL HTR ASSY circuit breaker is inoperative and should remain open.

OPERATIONS (O)

Increase flight planning fuel by 2.5%. If the APU is used, also increase planning fuel by 200 lb (90 kg) per hour of APU usage inflight and 535 lb (240 kg) per hour of APU usage on the ground.

49-15-01 APU Air Inlet Door Actuation System**49-15-01-01 APU Air Inlet Door Actuator Position Indication Switch**

Interval	Installed	Required	Procedure
C	1	0	(M)

May be inoperative provided:

- a. APU inlet door is verified to operate normally before each departure.
- b. Both ELMS P310 panel channels operate normally.

MAINTENANCE (M)

Verify the APU door operates normally before each departure (AMM 49-00-00/901).

1. Supply electrical power on the airplane using external power or starting an engine (AMM 24-22-00/201).
2. For APU off:
 - A. Display the APU Maintenance page and confirm the INLET DOOR POSITION indicates CLOSED.
 - B. Position Electrical Panel (P5) APU selector ON.
 - C. Wait 45 seconds and confirm APU Maintenance page INLET DOOR POSITION indicates OPEN.
 - D. Position Electrical Panel (P5) APU selector OFF.
 - E. Wait 120 seconds and confirm APU Maintenance page INLET DOOR POSITION indicates CLOSED.
3. For APU on:
 - A. Display the APU Maintenance page and confirm the INLET DOOR POSITION indicates OPEN.
 - B. Position Electrical Panel (P5) APU selector OFF.
 - C. Wait 120 seconds and confirm APU Maintenance page INLET DOOR POSITION indicates CLOSED.
 - D. Start the APU (AMM 49-11-00/201).



-
- E. Wait 45 seconds and confirm APU Maintenance page INLET DOOR POSITION indicates OPEN.

49-42-01 APU Starting System (Includes Electric Starter)
49-42-01A APU Required

Interval	Installed	Required	Procedure
C	1	0	(O)

May be dispatched with APU START SYS faults provided APU is started before departure and operated continuously throughout the flight.

MAINTENANCE NOTE

Some electric starter failures may result in damage to the APU if it is used. If the APU fails to start electrically but starts pneumatically, the possibility of APU damage can be reduced by doing the following steps.

1. Perform the APU Magnetic Chip Collector Inspection (AMM Task 49-27-04/201).
2. If the Chip Collector Inspection is satisfactory:
 - A. Position Overhead Maintenance Panel (P61) APU Maintenance switch NORM.
 - B. Position Overhead Panel (P5) APU selector OFF.
 - C. Remove the end cap from the electric starter motor on the APU.
 - D. Use a socket and ratchet on the shaft for the electric starter to turn the APU counterclockwise, facing aft.
 - 1) The APU should turn freely. Verify the APU is turning freely by visual inspection of the power section compressor blades through the bottom access panel of the APU inlet duct (AMM 49-21-00/Figure 601).
 - 2) If the APU does not turn freely, serious damage can occur if the APU is operated (see FIM Chapter 49-40, Task 812, Fault Isolation Procedure - Mechanical Fault).
 - E. Remove the socket and ratchet, and re-install the end cap on the electric starter.

OPERATIONS (O)

1. Increase flight planning fuel by 2.5% to account for the open APU air inlet door. Also increase flight planning fuel by 200 lb (90 kg) per hour of APU usage in-flight and 535 lb (240 kg) per hour of APU usage on the ground.
2. Start the APU before departure and operate the APU continuously throughout the flight.

**49-42-01 APU Starting System (Includes Electric Starter)****49-42-01B Procedures Do Not Require APU**

Interval	Installed	Required	Procedure
C	1	0	(O)

May be dispatched with APU START SYS faults provided:

- Other procedures do not require use of the APU.
- Backup AC power system is verified to operate normally before each departure.
- Flight remains within 180 minutes of landing at a suitable airport.

OPERATIONS (O)

- Dispatch is not allowed if the APU is required by other procedures. MMEL dispatch procedures for the following inoperative equipment may require APU availability.

MMEL Item	Associated Status Message
24-11-01	ELEC GEN SYS L or R IDG VALVE CLOSED L or R
24-25-01	ELEC BACKUP GEN L or R ELEC BACKUP SYS
29-11-03	HYD PUMP DEM C1 or C2
36-11-01	BLEED PRSOV L or R
36-11-02	BLEED HPSOV L or R
36-11-03	None
36-12-01	BLEED ISLN VALVE L or R
36-12-02	BLEED ISLN VALVE C
36-22-01	BLEED TEMP SENSOR L or R

- The flight must remain within 180 minutes of landing at a suitable airport.
- Verify the Backup AC electrical power system operates normally. Confirm the ELEC BACKUP SYS or ELEC BACKUP GEN L or R status and advisory messages are not displayed 30 seconds after both engines have been started.



49-43-01 APU Air Turbine Starter

Interval	Installed	Required	Procedure
C	1	0	

May be inoperative provided electric starter operates normally.

**49-52-01 APU Bleed Air System**

Interval	Installed	Required	Procedure
C	1	0	(O)

May be inoperative provided:

- Other procedures do not require use of the APU as a pneumatic source.
- Flight remains within 180 minutes of landing at a suitable airport.

NOTE: The APU may be used as an electrical power source.

OPERATIONS (O)

NOTE: The APU bleed air system may or may not be completely inoperative when the APU BLEED AIR status message is displayed depending on the actual fault causing the message. The APU bleed air system, if available may be used for ground operation only.

- Dispatch is not allowed if other procedures require APU bleed air. MMEL dispatch procedures for the following inoperative equipment may require APU bleed air availability.

MMEL Item	Associated Status Message
29-11-03	HYD PUMP DEM C1 or C2
36-11-01	BLEED PRSOV L or R
36-11-02	BLEED HPSOV L or R
36-11-03	None
36-12-01	BLEED ISLN VALVE L or R
36-12-02	BLEED ISLN VALVE C
36-22-01	BLEED TEMP SENSOR L or R

- The flight must remain within 180 minutes of landing at a suitable airport.

49-52-02 APU Bleed Air Check Valve

Interval	Installed	Required	Procedure
C	1	0	(O)

May be inoperative provided:

- APU bleed air switch remains OFF except for main engine start.
- Other procedures do not require use of the APU as a pneumatic source.
- Flight remains within 180 minutes of landing at a suitable airport.

NOTE: The APU may be used for electrical power. Air Turbine Starter is not available with APU bleed air shutoff valve closed.

OPERATIONS (O)

- Dispatch is not allowed if other procedures require APU bleed air. MMEL dispatch procedures for the following inoperative equipment may require APU bleed air availability.

MMEL Item	Associated Status Message
29-11-03	HYD PUMP DEM C1 or C2
36-11-01	BLEED PRSOV L or R
36-11-02	BLEED HPSOV L or R
36-11-03	None
36-12-01	BLEED ISLN VALVE L or R
36-12-02	BLEED ISLN VALVE C
36-22-01	BLEED TEMP SENSOR L or R

- APU is available for electrical power only. APU BLEED AIR switch must remain OFF after main engine start.
- Air Turbine Starter for the APU is not available.
- The flight must remain within 180 minutes of landing at a suitable airport.

**49-61-01 APU Controller**

Interval	Installed	Required	Procedure
C	1	0	

May be dispatched with APU CONTROL faults.

MAINTENANCE NOTE

Continued APU operation following display of maintenance messages 49-12990 (APU Lube Filter Clogged) or 49-12950 (APU Generator Filter Clogged) may increase the potential for filter bypass differential pressure to be reached resulting in metal contamination circulating to additional areas of the APU.

**49-61-02 APU Ground Control Panel**

Interval	Installed	Required	Procedure
C	1	0	

MAINTENANCE NOTE

This MMEL item allows dispatch with any or all of the APU ground control panel functions (APU Fire Warning Horn, APU Fire Warning Light, APU Fire Bottle Discharge Switch, APU Shutdown Switch, APU Fire Bottle Armed Light and APU Bottle Discharged Light) inoperative.

**49-70-01 APU FAULT Light**

Interval	Installed	Required	Procedure
C	1	0	

49-71-01 APU EGT Indication

Interval	Installed	Required	Procedure
C	1	0	

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**49-73-01 APU RPM Indication**

Interval	Installed	Required	Procedure
C	1	0	

**49-94-01 APU OIL QTY Indication****49-94-01A APU Required**

Interval	Installed	Required	Procedure
C	1	0	(M)

May be inoperative provided APU oil quantity is verified adequate once each flight day.

MAINTENANCE (M)

Verify the APU oil quantity is adequate once each flight day using the APU oil tank sight gauge (AMM 49-00-00/901).

49-94-01 APU OIL QTY Indication**49-94-01B Procedures Do Not Require APU**

Interval	Installed	Required	Procedure
C	1	0	(O)

May be inoperative provided:

- Other procedures do not require use of APU.
- Backup AC power system is verified to operate normally before each departure.
- Flight remains within 180 minutes of landing at a suitable airport.

OPERATIONS (O)

- Dispatch is not allowed if the APU is required by other procedures. MMEL dispatch procedures for the following inoperative equipment may require APU availability.



MMEL Item	Associated Status Message
24-11-01	ELEC GEN SYS L or R IDG VALVE CLOSED L or R
24-25-01	ELEC BACKUP GEN L or R ELEC BACKUP SYS
29-11-03	HYD PUMP DEM C1 or C2
36-11-01	BLEED PRSOV L or R
36-11-02	BLEED HPSOV L or R
36-11-03	None
36-12-01	BLEED ISLN VALVE L or R
36-12-02	BLEED ISLN VALVE C
36-22-01	BLEED TEMP SENSOR L or R

2. The flight must remain within 180 minutes of landing at a suitable airport.
3. Verify the Backup AC electrical power system operates normally. Confirm the ELEC BACKUP SYS or ELEC BACKUP GEN L or R status and advisory messages are not displayed 30 seconds after both engines have been started.
4. Position the APU selector OFF.

49-94-02 APU OIL PRESS Indication

Interval	Installed	Required	Procedure
C	1	0	

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52-00-01 Doors Synoptic Display

52-09-01 Rigid Cargo Barrier (RCB) Door Seals (777F)

52-11-01 Main Entry Doors/Slides

52-11-01-01 Passenger Configuration

52-11-01-01A *Passengers Carried*

52-11-01-01B *No Passengers Carried*

52-11-01-02 777F

52-11-02 Main Entry Doors Pressure Stop Assemblies

52-11-03 Main Entry Doors Hold-Open Mechanisms

52-11-03-01 -200/-200ER/-200LR

52-11-03-02 -300/-300ER

52-11-03-03 777F

52-11-04 Main Entry Doors Hold-Open Release Handles

52-11-04-01 -200/-200ER/-200LR

52-11-04-02 -300/-300ER

52-11-04-02-01 Doors 1L, 1R, 2L, 2R, 4L, 4R, 5L and 5R

52-11-04-02-02 Doors 3L and 3R

52-11-04-03 777F

52-11-05 Main Entry Door Flight Lock Systems

52-11-05-01 Passenger

52-11-05-02 777F

52-32-01 Main Deck Cargo Door (777F)

52-32-01-01 Cargo Door Lift System (Manual and Electric)

52-32-01-02 Cargo Door Hook/Latch System (Electric)

52-32-01-03 Cargo Door Lock System (Electric)

52-32-02 Main Deck Cargo Door Control Panel Lights (OPENED, CLOSED, LATCHED, UNLOCKED, LOCKED & ARMED) (777F)

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- 52-35-03 Aft Small Cargo DOOR LATCHED Light**
- 52-35-04 Aft Small Cargo Door Arming Relay/Control Switch**
- 52-36-01 Bulk Cargo Door Counterbalance Mechanism**
- 52-37-01 Aft Large Cargo Door Hook/Latch System (Electric Mode)**
- 52-37-02 Aft Large Cargo Door Lift System (Electric Mode)**
- 52-37-03 Aft Large Cargo DOOR LATCHED Light**
- 52-51-01 Flight Deck Door Lock Automatic System (Not 14 CFR 25.795 Compliant)**
 - 52-51-01A *Supplemental Lock Not Installed*
 - 52-51-01B *Supplemental Lock Installed*
- 52-51-02 Boeing Enhanced Flight Deck Security Door Automatic Locking System (14 CFR 25.795 Compliant) (Passenger)**
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 - 52-51-02-01-02 Door Bell Mode
 - 52-51-02-01-03 Switch Guard
 - 52-51-02-02 Flight Deck Door LOCK FAIL Light
 - 52-51-02-03 Flight Deck Door AUTO UNLK Light
 - 52-51-02-04 Flight Deck Door Lock Control Selector
 - 52-51-02-05 Pressure Rate-Of-Change Sensing Module
- 52-51-03 Boeing Enhanced Flight Deck Security Door Dead Bolt (14 CFR 25.795 Compliant) (Passenger)**
- 52-51-04 JAMCO Flight Deck Security Door Automatic Locking System (14 CFR 25.795 Compliant)**
 - 52-51-04-01 Door Automatic Locking Solenoids
 - 52-51-04-02 Door Warning System
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52-51-04-04	FLIGHT DECK DOOR Warning / Caution Light
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52-51-04-05-01	Keypad Indicator Lights
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52-51-06-02B	<i>Electronic Visual Surveillance System Operative</i>
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52-71-02B	<i>Procedures Do Not Require Use</i>



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**52-00-01 Doors Synoptic Display**

Interval	Installed	Required	Procedure
C	1	0	

OPERATIONS NOTE

If the Synoptic Display is missing data, selecting an alternate location for the display (L INBD, LWR CTR, or R INBD) may restore the missing data. Synoptic Displays containing missing data may continue to be used to the extent remaining data is useful. Airplane system faults will be annunciated by alerting and status messages.

**52-09-01 Rigid Cargo Barrier (RCB) Door Seals (777F)**

Interval	Installed	Required	Procedure
C	2	0	(M) (O)

May be inoperative provided:

- a. Procedures are established and used to verify the main deck compartment remains empty or contains only ballast, empty cargo handling equipment (ballast may be loaded in ULDs), or fly away kits.
- b. Associated door is closed and placarded, INOPERATIVE - DO NOT USE.

MAINTENANCE (M)

Empty the main deck cargo, and close and placard the associated RCB door (AMM 52-00-00/901).

1. The main deck cargo compartment must remain empty, except for ballast, empty cargo containers (ballast may be loaded in ULDs), fly away kits, pallets, and cargo restraint components.
2. Close and placard the associated door, INOPERATIVE - DO NOT USE.

OPERATIONS (O)

The main deck cargo compartment must remain empty, except for ballast, empty cargo containers (ballast may be loaded in ULDs), fly away kits, pallets, and cargo restraint components.

52-11-01 Main Entry Doors/Slides
52-11-01-01 Passenger Configuration
52-11-01-01A Passengers Carried

Interval	Installed	Required	Procedure
A	-	-	(M) (O)

One door/slide may be inoperative or slide missing provided:

- All other main entry doors are fully operational.
- Affected door is not used for passenger loading.
- A conspicuous barrier strap or rope and a placard stating that the door is inoperative shall be placed across the inoperative door.
- Emergency exit sign and floor proximity lights associated with the inoperative exit must be covered to obscure the sign and lights.
- Passengers must be briefed not to use affected door.
- All passenger seats halfway to the next exit in each direction from the inoperative door, across the entire width of the airplane, shall be blocked off with conspicuous tapes or ropes that contrast with the airplane interior before loading passengers. Only the seats in these areas shall be blocked; main passenger aisles, cross aisles and exit areas must not be blocked. (For an inoperative forward door/slide, the blocked seating area shall extend from the forward cabin end, rearward to a line halfway between the inoperative forward door and the next set of doors aft of the inoperative one. For an inoperative rear door/slide, the blocked seating area shall extend forward from the aft cabin end to a line halfway between the inoperative door and the next set of doors forward of the inoperative one).
- Conspicuous signs and placards shall be placed in appropriate locations to indicate seats are not to be occupied by passengers.
- Seated capacity must not exceed rated capacity of remaining pairs of exits.
- For extended overwater operations, occupancy shall not exceed the normal rated capacity of the slide/rafts, or the remaining slide/rafts, or the rated overload capacity of the slide/rafts remaining after loss of one additional slide/raft of greatest capacity, whichever is least.
- Blocked seating layouts and evacuation procedures must be developed and approved by the FAA certificate holding office for inclusion in the operator's manual.
- Repairs are made within 1 flight day.

NOTE 1: Weight and Balance Manifest must be revised as necessary to ensure proper loading limits are observed.

NOTE 2: Cabin attendants may be stationed in the vicinity of each door within blocked areas.

NOTE 3: A door with an inoperative EPAS must be considered inoperative.

MAINTENANCE (M)

NOTE: If a passenger entry door's Emergency Power Assist System (EPAS) is inoperative, the associated entry door is considered inoperative.

Cover associated signs and lights, and block associated doors and seats AMM 52-00-00/901).

1. Cover the emergency exit sign of the affected door so that it is not visible.
2. Cover or otherwise obscure the floor proximity escape path marking system lights unique to the inoperative door.
3. Block the inoperative door with a conspicuous barrier strap or rope and placard the door inoperative.
4. Block unusable passenger seats with conspicuous tape or ropes and placard the seats do not occupy.

OPERATIONS (O)

1. Flight crew members and passengers must be briefed prior to takeoff and landing regarding exit that cannot be used for emergency evacuation.
2. Passenger entry door Emergency Power Assist Systems (EPAS) and emergency evacuation slide systems for the affected door should be armed with the mode select handles in the ARMED or AUTOMATIC position prior to taxi, takeoff and landing whenever possible. Girt bar indication flags are visible through windows in the doorway liners when the door and slides are in the ARMED or AUTOMATIC mode.

NOTE: This recommendation includes inoperative doors with escape slides removed.

52-11-01 Main Entry Doors/Slides

52-11-01-01 Passenger

52-11-01-01B No Passengers Carried

Interval	Installed	Required	Procedure
C	-	1	(M) (O)

May be inoperative or slides missing provided:

- a. No passengers are carried.
- b. A maximum of 19 persons authorized by 14 CFR for non-passenger carrying operations are carried.
- c. A conspicuous barrier strap or rope and a placard stating that the door is inoperative shall be placed across the inoperative door.
- d. Emergency exit sign and floor proximity lights associated with the inoperative exit must be covered to obscure the sign and lights.
- e. Inoperative doors are not used for loading.



- f. Each person carried has unobstructed access from their seat to an operative door.
 - g. Safety briefing includes location of inoperative doors and instructions not to use them.
 - h. Alternate procedures are established and used.
-

MAINTENANCE (M)

Block the inoperative doors with conspicuous barrier strap or rope and placard as inoperative, configure the associated emergency exit signs and floor proximity escape path markings as appropriate (AMM 52-00-00/901).

1. Block the inoperative doors with conspicuous barrier strap or rope and placard as inoperative.
2. Cover the emergency exit signs unique to the inoperative doors.
3. Cover the floor proximity escape path marking system lights unique to the inoperative doors.

OPERATIONS (O)

1. No passengers may be carried.
2. A maximum of 19 persons authorized by 14 CFR 121.583 may be carried.
3. Inoperative doors are not to be used for loading.
4. Flight crew members and authorized persons must be briefed prior to takeoff and landing regarding the inoperative doors that cannot be used for emergency evacuation.
5. Passenger entry door Emergency Power Assist Systems (EPAS) and emergency evacuation slide systems for the affected door should be armed with the mode select handles in the ARMED or AUTOMATIC position prior to taxi, takeoff and landing whenever possible. Girt bar indication flags are visible through windows in the doorway liners when the door and slides are in the ARMED or AUTOMATIC mode.

NOTE: This recommendation includes inoperative doors with escape slides removed.

52-11-01 Main Entry Doors/Slides

52-11-01-02 777F

Interval	Installed	Required	Procedure
C	2	1	

One door/slide may be inoperative or slide missing.

**52-11-02 Main Entry Doors Pressure Stop Assemblies**

Interval	Installed	Required	Procedure
C	-	-	(M) (O)

One forward or aft stop per door (total of 8 stops per -200 airplane, or 10 stops per -300 airplane, or 2 stops per 777F) may be missing or inoperative provided:

- There are no visible defects on remaining stops for the affected door(s).
- Both cabin altitude auto controllers operate normally.
- For inoperative or missing pressure stops 1, 2, 7 and 8, limit maximum cabin differential pressure to 2.5 psi.
- For inoperative or missing pressure stops 3, 4, 5 and 6, limit maximum cabin differential pressure to 5.4 psi.

MAINTENANCE (M)

Visually inspect remaining pressure stops on affected door(s) and door frame(s) to ensure there are no other defects (AMM 52-00-00/901).

NOTE: Door stops are numbered 1 to 8 from bottom to top on both forward and aft edges of door.

OPERATIONS (O)

Airplane altitude is limited as follows to assure maximum cabin differential pressure (Delta P) limit is not exceeded by automatic cabin pressure control.

- Before takeoff, set the landing altitude to 8000 feet.
- For a maximum Delta P of 2.5 psi, do not exceed 14,500 feet airplane pressure altitude.
- For a maximum Delta P of 5.4 psi, do not exceed 24,500 feet airplane pressure altitude.
- At top of descent, reset the landing altitude to the actual field elevation for the destination airport.



52-11-03 Main Entry Doors Hold-Open Mechanisms**52-11-03-01 -200/-200ER/-200LR**

Interval	Installed	Required	Procedure
C	8	4	

52-11-03 Main Entry Doors Hold-Open Mechanisms**52-11-03-02 -300/-300ER**

Interval	Installed	Required	Procedure
C	10	6	

May be inoperative provided doors 3L and 3R hold-open mechanisms operate normally.

52-11-03 Main Entry Doors Hold-Open Mechanisms**52-11-03-03 777F**

Interval	Installed	Required	Procedure
C	2	1	



52-11-04 Main Entry Doors Hold-Open Release Handles
52-11-04-01 -200/-200ER/-200LR

Interval	Installed	Required	Procedure
C	8	4	

52-11-04 Main Entry Doors Hold-Open Release Handles
52-11-04-02 -300/-300ER
52-11-04-02-01 Doors 1L, 1R, 2L, 2R, 4L, 4R, 5L and 5R

Interval	Installed	Required	Procedure
C	8	4	

52-11-04 Main Entry Doors Hold-Open Release Handles
52-11-04-02 -300/-300ER
52-11-04-02-02 Doors 3L and 3R

Interval	Installed	Required	Procedure
C	2	1	

One may be inoperative provided supplemental raft is not installed in the associated door bustle.

52-11-04 Main Entry Doors Hold-Open Release Handles
52-11-04-03 777F

Interval	Installed	Required	Procedure
C	2	1	

**52-11-05 Main Entry Door Flight Lock Systems****52-11-05-01 Passenger**

Interval	Installed	Required	Procedure
C	-	0	(M) (O)

May be inoperative provided:

- Each affected door is verified to be capable of being unlatched and opened before each departure.
- A person employed by the operator is designated to monitor the affected door handle(s) when cabin differential pressure is less than 1.5 psi.

MAINTENANCE (M)

Verify the associated passenger entry door(s) is capable of being unlatched and opened. Open and close the associated door(s) using the interior handle. If desired, remove power from the associated flight lock actuator (AMM 52-11-00/201, 52-12-00/201).

- Position the door mode select handle to MANUAL/DISARMED. DO NOT rotate the door handle to the open position with the door mode select lever in the AUTOMATIC/ARMED position. Door will open rapidly and slide will deploy and inflate.
- Rotate the inside door handle to the OPEN position. Open door slightly.
- If desired, gain access to the inoperative flight lock actuator and disconnect, cap and stow the actuator's electrical connector.
- Close door and rotate handle back to CLOSED position.

OPERATIONS (O)

- Each affected door must be monitored by a person employed by the operator when cabin differential pressure is less than 1.5 psi to ensure door handle is not operated by passengers.
- Flight crew should inform cabin crew when monitoring is required.

52-11-05 Main Entry Door Flight Lock Systems**52-11-05-02 777F**

Interval	Installed	Required	Procedure
C	2	0	(M)

May be inoperative provided each affected door is verified to be capable of being unlatched and opened before each departure.

MAINTENANCE (M)

Verify the associated entry door(s) is capable of being unlatched and opened. Open and close the associated door(s) using the interior handle. If desired, remove power from the associated flight lock actuator (AMM 52-11-00/201).

1. Position the door mode select handle to MANUAL/DISARMED. DO NOT rotate the door handle to the open position with the door mode select lever in the AUTOMATIC/ARMED position. Door will open rapidly and slide will deploy and inflate.
2. Rotate the inside door handle to the OPEN position. Open door slightly.
3. If desired, gain access to the inoperative flight lock actuator and disconnect, cap and stow the actuator's electrical connector.
4. Close door and rotate handle back to CLOSED position.

52-32-01 Main Deck Cargo Door (777F)**52-32-01-01 Cargo Door Lift System (Manual and Electric)**

Interval	Installed	Required	Procedure
C	1	0	(M)

May be inoperative provided:

- An alternate procedure is used to close the door.
- Door is verified closed, latched and locked before each departure.

MAINTENANCE (M)

NOTE 1: For an inoperative electrical lift system, the door may be operated manually. Access the lift drive unit manual drive socket and turn counter-clockwise to open or clockwise to close.

NOTE 2: For inoperative electrical and manual lift systems, ground support equipment may be used with a sling attached to the lower attach point to open/close the door.

Manually close, latch, and lock the door and verify all latch cams are in caught position before each departure (AMM 52-00-00/901).

- For manual lift system operative, access the lift actuator manual drive socket.
- Turn the actuator drive clockwise until the door stops.

NOTE: Door will be open one to three inches.

- Continue turning the actuator drive clockwise until the forward or aft lift arm touches the stop.
- Latch the door and confirm.
- Lock the door and confirm.

52-32-01 Main Deck Cargo Door (777F)**52-32-01-02 Cargo Door Hook/Latch System (Electric)**

Interval	Installed	Required	Procedure
C	1	0	(M)

May be inoperative provided:

- Manual function operates normally.
- There is no damage to hook/latch mechanism.
- There is no damage to lock mechanism.
- An alternate procedure is used to latch the door.



- e. Door is verified closed, latched and locked before each departure.
-

MAINTENANCE (M)

Close, manually latch and lock the door and verify all latch cams are in caught position before each departure (AMM 52-00-00/901).

1. Lower the door.
2. Access the hook/latch actuator manual drive socket.
3. Turn the actuator drive counterclockwise until the latched overtravel stop is contacted and the door is latched and hooked.
4. Unload the stop by rotating the actuator drive clockwise until the latch arrows point to the targets in all eight witness ports.
5. Confirm the door is faired with fuselage.
6. Lock the door and confirm.

52-32-01 Main Deck Cargo Door (777F)

52-32-01-03 Cargo Door Lock System (Electric)

Interval	Installed	Required	Procedure
C	1	0	(M)

May be inoperative provided:

- a. Manual function operates normally.
 - b. There is no damage to hook/latch mechanism.
 - c. There is no damage to lock mechanism.
 - d. An alternate procedure is used to lock the door.
 - e. Door is verified closed, latched and locked before each departure.
-

MAINTENANCE (M)

Close, latch and manually lock the door and verify all latch cams are in caught position before each departure (AMM 52-00-00/901).

1. Lower the door.
2. Latch the door and confirm.
3. Access the lock actuator manual drive socket.
4. Turn the actuator drive clockwise until the stop is contacted and back off one turn.
5. Confirm the vent door has closed.
6. Using the witness ports, confirm the lock sectors are located over latch cams but do not touch the cams.



**52-32-02 Main Deck Cargo Door Control Panel Lights
(OPENED, CLOSED, LATCHED, UNLOCKED,
LOCKED & ARMED) (777F)**

Interval	Installed	Required	Procedure
C	12	0	(M)

May be inoperative provided the main deck cargo door is verified to be closed, latched and locked before each departure.

MAINTENANCE (M)

Before each departure, visually verify the main deck cargo door is closed, latched, and locked (AMM 52-00-00/901).

1. Close and latch the door using normal procedures (AMM 52-32-00/201).
2. Confirm the latch arrows point to the targets in all eight witness ports.
3. Confirm the door is aligned and faired with fuselage.
4. Confirm the vent door has closed.
5. Using the witness ports, confirm the lock sectors are located over latch cams but do not touch the cams.

**52-34-01 Forward Cargo Door Hook/Latch System (Electric Mode)**

Interval	Installed	Required	Procedure
C	1	0	(M)

May be inoperative provided:

- Manual mode operates normally.
- There is no damage to the hook/latch mechanism.
- Door is manually latched and locked using the maintenance manual procedure.

MAINTENANCE (M)

Latch and lock the door manually (AMM 52-00-00/901 and AMM 52-34-00/201).

**52-34-02 Forward Cargo Door Lift System (Electric Mode)**

Interval	Installed	Required	Procedure
C	1	0	(M)

May be inoperative provided:

- Manual mode operates normally.
- There is no damage to the door lift system.
- Door is manually closed, latched and locked using the maintenance manual procedure.

MAINTENANCE (M)

Operate the door manually (AMM 52-34-00/201).

**52-34-03 Forward Cargo DOOR LATCHED Light**

Interval	Installed	Required	Procedure
C	1	0	(M)

May be inoperative provided the forward cargo door is verified to be closed, latched and locked before each departure.

MAINTENANCE (M)

Before each departure, verify the forward cargo door is closed, latched, and locked.

1. Close and latch the door using normal procedures (AMM 52-34-00/201).
2. Check that the outside skin of the door is aligned and faired with surrounding body skin.
3. Check that the lock handle is in the unlocked position.
4. Check that the vent door is open.
5. Rotate the lock handle to the CLOSED position.
6. Check that the lock handle is fully closed and faired with the outside skin of the door.
7. Check that the lock handle release lever is faired with the lock handle and outside skin of the door.
8. Check that the vent door is fully closed.
9. Check that the latches are in the fully latched position and that the locks are in the locked position by visual inspection through the witness ports at the lower edge of the door.

**52-35-01 Aft Small Cargo Door Lift/Latch System**

Interval	Installed	Required	Procedure
C	1	0	(M)

May be inoperative provided the associated door is verified to be closed, latched and locked before each departure.

MAINTENANCE (M)

1. Verify the door is closed, latched, and locked (AMM 52-35-00/201).
2. Verify the pressure vent door is closed.

52-35-01 Aft Small Cargo Door Lift/Latch System

52-35-01-01 Electric Mode

Interval	Installed	Required	Procedure
C	1	0	

May be inoperative provided manual mode operates normally.

**52-35-02 Aft Small Cargo Door Hinge System (Electric Mode)**

Interval	Installed	Required	Procedure
C	1	0	(M)

May be inoperative provided:

- Manual mode operates normally.
- There is no damage to the hinge system.
- Door is manually operated using the maintenance manual procedure.

MAINTENANCE (M)

Operate the door manually (AMM 52-35-00/201).

52-35-03 Aft Small Cargo DOOR LATCHED Light

Interval	Installed	Required	Procedure
C	1	0	(M)

May be inoperative provided:

- a. Aft small cargo door indication system operates normally.
- b. Aft small cargo door is verified to be closed, latched and locked before each departure.

MAINTENANCE (M)

Before each departure, verify the aft small cargo door is closed, latched and locked.

1. Closed and latch the aft small cargo door (AMM 52-35-00/201).
2. Verify the pressure vent door is closed.
3. Verify the aft small cargo door alerting message is not displayed by one of the following methods:
 - A. After the small aft cargo door is closed latched and locked:
 - 1) Supply electrical power on the airplane (AMM 24-22-00/201).
 - 2) Close all doors, so that DOORS advisory message is no longer displayed.
 - 3) Verify the DOOR AFT CARGO advisory message is not displayed. On the Display Select Panel push CANC/RCL if necessary to recall messages and view additional pages.
 - B. After all airplane doors are closed, verify with the flight crew the DOOR AFT CARGO advisory message is not displayed.

**52-35-04****Aft Small Cargo Door Arming Relay/Control Switch**

Interval	Installed	Required	Procedure
C	1	0	(M)

May be inoperative provided:

- a. Associated arming control circuit is deactivated.
- b. Aft small cargo door is verified to be closed, latched and locked before each departure.

MAINTENANCE (M)

Deactivate the associated arming control circuit and before each departure, verify the aft small cargo door is closed, latched and locked (AMM 52-00-00/901).

1. Deactivate the associated arming control circuit.
 - A. Open and collar the P039 panel AFT CARGO DOOR circuit breaker.
2. Before each departure, verify the aft small cargo door is closed, latched and locked.
 - A. Close and latch the aft small cargo door (AMM 52-35-00/201).
 - B. Check that the outside skin of the door is aligned and faired with the surrounding body skin.
 - C. Check that the vent door is fully closed.
 - D. Verify the aft small cargo door alerting message is not displayed by one of the following methods:
 - 1) After the small aft cargo door is closed latched and locked:
 - a. Supply electrical power on the airplane (AMM 24-22-00/201).
 - b. Close all doors, so that DOORS advisory message is no longer displayed.
 - c. Verify the DOOR AFT CARGO advisory message is not displayed. On the Display Select Panel push CANC/RCL if necessary to recall messages and view additional pages.
 - 2) After all airplane doors are closed, verify with the flight crew the DOOR AFT CARGO advisory message is not displayed.

**52-36-01 Bulk Cargo Door Counterbalance Mechanism**

Interval	Installed	Required	Procedure
C	1	0	(M)

May be inoperative provided a safety hold open device is used when door is in the open position.

MAINTENANCE (M)

Secure the door in the open position with the safety strap installed in the bulk compartment (AMM 52-36-00/201).

**52-37-01 Aft Large Cargo Door Hook/Latch System (Electric Mode)**

Interval	Installed	Required	Procedure
C	1	0	(M)

May be inoperative provided:

- Manual mode operates normally.
- There is no damage to the hook/latch mechanism.
- Door is manually latched and locked using the maintenance manual procedure.

MAINTENANCE (M)

Latch and lock the door manually (AMM 52-37-00/201).



52-37-02 Aft Large Cargo Door Lift System (Electric Mode)

Interval	Installed	Required	Procedure
C	1	0	(M)

May be inoperative provided:

- a. Manual mode operates normally.
- b. There is no damage to the door lift system.
- c. Door is manually closed, latched and locked using the maintenance manual procedure.

MAINTENANCE (M)

Operate the door manually (AMM 52-37-00/201).

52-37-03 Aft Large Cargo DOOR LATCHED Light

Interval	Installed	Required	Procedure
C	1	0	(M)

May be inoperative provided the aft large cargo door is verified to be closed, latched and locked before each departure.

MAINTENANCE (M)

Before each departure verify that the aft large cargo door is closed, latched, and locked.

1. Close and latch the door per normal procedures (AMM 52-37-00/201).
2. Check that the outside skin of the door is aligned and faired with surrounding body skin.
3. Check that the lock handle is in the unlocked position.
4. Check that the vent door is open.
5. Rotate the lock handle to the CLOSED position.
6. Check that the lock handle is fully closed and faired with the outside skin of the door.
7. Check that the lock handle release lever is faired with the lock handle and outside skin of the door.
8. Check that the vent door is fully closed (the upper edge of the vent door may not be fully faired).
9. Check that the latches are in the fully latched position and that the locks are in the locked position by visual inspection through the witness ports at the lower edge of the door.

**52-51-01 Flight Deck Door Lock Automatic System (Not 14 CFR 25.795 Compliant)**

52-51-01A Supplemental Lock Not Installed

Interval	Installed	Required	Procedure
C	1	0	(M)

May be inoperative provided:

- Door automatic locking electric solenoid is deactivated in the extended position.
 - Door is verified to lock and unlock manually.
-

MAINTENANCE (M)

Deactivate the flight deck door automatic lock electric solenoid in the extended position and verify the door locks and unlocks manually (AMM 52-00-00/901).

- Open and collar P110 panel circuit breaker F/D DR LOCK.
- Open the flight deck door and gain access to the door post.
- Remove the seal assembly to get access to the screws that attach the cover assembly to the door post (AMM 52-51-02/401).
- Remove the cover assembly and strike assembly from the door post.
- Use a strap or lockwire to hold the solenoid pin in the fully extended position so that it holds the door catch in the locked position.
- Install the cover and strike assemblies on the door post.
- Install the seal assembly.
- Close the flight deck door and verify that it is locked. Ensure a flight deck door key is available or someone is in the flight deck before closing the door.
- Verify that the flight door unlocks from the cabin side with a key and from the flight deck side with the door handle.

52-51-01 Flight Deck Door Lock Automatic System (Not 14 CFR 25.795 Compliant)

52-51-01B Supplemental Lock Installed

Interval	Installed	Required	Procedure
C	1	0	

May be inoperative provided supplemental flight deck door security device is installed and operates normally.



**52-51-02 Boeing Enhanced Flight Deck Security Door
Automatic Locking System (14 CFR 25.795
Compliant) (Passenger)**

Interval	Installed	Required	Procedure
A	1	0	(M) (O)

May be inoperative provided:

- Automatic locking system is deactivated.
 - Door dead bolt operates normally and is used to lock the door.
 - Alternate procedures are established and used for locking and unlocking the door using the dead bolt.
 - Repairs are made within 2 flight days.
-

MAINTENANCE (M)

Deactivate the automatic locking system by positioning Flight Deck Access System switch OFF (guard extended) (AMM 52-00-00/901).

NOTE: LOCK FAIL light may remain illuminated when the Flight Deck Access System switch is in the OFF position (guard extended).

OPERATIONS (O)

Each operator must establish and use alternate procedures based on their security requirements for allowing and denying access to the flight deck.

**52-51-02 Boeing Enhanced Flight Deck Security Door
Automatic Locking System (14 CFR 25.795
Compliant) (Passenger)****52-51-02-01 Flight Deck Access System (Keypad, Door Chime)**

Interval	Installed	Required	Procedure
B	1	0	(M) (O)

May be inoperative provided:

- Keypad is deactivated.
 - Alternate procedures are established and used.
-

MAINTENANCE (M)

Deactivate the keypad (AMM 52-00-00/901).

- Open P110 panel F/D DR LOCK circuit breaker.
- Access the chime module.
- Disconnect, cap and stow the electrical connector J1 for the keypad at the chime module.



4. Close P110 panel F/D DR LOCK circuit breaker.

NOTE: Ensure the Flight Deck Access System switch is in the OFF position (guard extended) when flight deck door is to be closed and not occupied.

OPERATIONS (O)

Each operator must establish and use alternate procedures based on their security requirements for allowing and denying access to the flight deck.

NOTE: Ensure the Flight Deck Access System switch is in the OFF position (guard extended) when flight deck door is to be closed and not occupied.

52-51-02 Boeing Enhanced Flight Deck Security Door Automatic Locking System (14 CFR 25.795 Compliant) (Passenger)

52-51-02-01 Flight Deck Access System (Keypad, Door Chime)

52-51-02-01-01 LEDs

Interval	Installed	Required	Procedure
C	3	0	(O)

May be inoperative provided alternate procedures are established and used.

OPERATIONS (O)

Each operator must establish and use alternate procedures based on their security requirements for allowing and denying access to the flight deck.

52-51-02 Boeing Enhanced Flight Deck Security Door Automatic Locking System (14 CFR 25.795 Compliant) (Passenger)

52-51-02-01 Flight Deck Access System (Keypad, Door Chime)

52-51-02-01-02 Door Bell Mode

Interval	Installed	Required	Procedure
C	1	0	(O)

May be inoperative provided alternate procedures are established and used.

OPERATIONS (O)

Each operator must establish and use alternate procedures based on their security requirements for allowing and denying access to the flight deck.



- 52-51-02 Boeing Enhanced Flight Deck Security Door Automatic Locking System (14 CFR 25.795 Compliant) (Passenger)**
- 52-51-02-01 Flight Deck Access System (Keypad, Door Chime)**
- 52-51-02-01-03 Switch Guard**

Interval	Installed	Required	Procedure
C	1	0	

May be inoperative or missing provided the flight deck door LOCK FAIL light operates normally.

- 52-51-02 Boeing Enhanced Flight Deck Security Door Automatic Locking System (14 CFR 25.795 Compliant) (Passenger)**
- 52-51-02-02 Flight Deck Door LOCK FAIL Light**

Interval	Installed	Required	Procedure
B	1	0	(M)

May be inoperative provided automatic lock controls are verified to operate normally.

MAINTENANCE (M)

Verify the automatic lock controls operate normally (AMM 52-00-00/901).

1. With the flight deck door open, supply electrical power on the airplane (AMM 24-22-00/201).
2. Position Flight Deck Access System switch NORM (guard closed).
3. Position the Flight Deck Door Lock selector AUTO.
4. Verify the electric strike is in the locked position (solenoid pin in the electric strike will be extended up such that you can not rotate the strike).
5. Enter keypad access code and verify the door chime sounds.
6. Position the Flight Deck Door Lock selector DENY.
7. Before the DENY time delay has expired, enter the keypad access code and verify the door chime does not sound.
8. Position and hold Flight Deck Door Lock selector UNLKD.
9. Verify the electric strike is in the unlocked position (solenoid pin in the electric strike will retract down such that you can rotate the strike).
10. Position the Flight Deck Door Lock selector AUTO.
11. Verify the electric strike is in the locked position (solenoid pin in the electric strike will be extended up such that you can not rotate the strike).

52-51-02 Boeing Enhanced Flight Deck Security Door Automatic Locking System (14 CFR 25.795 Compliant) (Passenger)
52-51-02-03 Flight Deck Door AUTO UNLK Light

Interval	Installed	Required	Procedure
B	1	0	(M)

May be inoperative provided:

- Automatic lock controls are verified to operate normally.
- Door chime operates normally.

MAINTENANCE (M)

Verify the automatic lock controls operate normally (AMM 52-00-00/901).

- With the flight deck door open, supply electrical power on the airplane (AMM 24-22-00/201).
- Position Flight Deck Access System switch NORM (guard closed).
- Position the Flight Deck Door Lock selector AUTO.
- Verify the electric strike is in the locked position (solenoid pin in the electric strike will be extended up such that you can not rotate the strike).
- Enter keypad access code and verify the door chime sounds.
- Position the Flight Deck Door Lock selector DENY.
- Before the DENY time delay has expired, enter the keypad access code and verify the door chime does not sound.
- Position and hold Flight Deck Door Lock selector UNLKD.
- Verify the electric strike is in the unlocked position (solenoid pin in the electric strike will retract down such that you can rotate the strike).
- Position the Flight Deck Door Lock selector AUTO.
- Verify the electric strike is in the locked position (solenoid pin in the electric strike will be extended up such that you can not rotate the strike).

52-51-02 Boeing Enhanced Flight Deck Security Door Automatic Locking System (14 CFR 25.795 Compliant) (Passenger)
52-51-02-04 Flight Deck Door Lock Control Selector

Interval	Installed	Required	Procedure
B	1	0	(M) (O)

May be inoperative provided:

- Keypad is deactivated.
- Automatic lock is verified to operate normally.



- c. Alternate procedures are established and used.
-

MAINTENANCE (M)

Deactivate the keypad and verify the automatic lock operates normally (AMM 52-00-00/901).

1. Deactivate the keypad.
 - A. Open P110 panel F/D DR LOCK circuit breaker.
 - B. Access the chime module.
 - C. Disconnect, cap and stow the electrical connector J1 for the keypad at the chime module.
 - D. Close P110 panel F/D DR LOCK circuit breaker.
2. Verify the automatic lock operates normally.
 - A. With the flight deck door open, supply electrical power on the airplane (AMM 24-22-00/201).
 - B. Position Flight Deck Access System switch NORM (guard closed).
 - C. Verify the electric strike is in the locked position (solenoid pin in the electric strike will be extended up such that you can not rotate the strike).
 - D. Position Flight Deck Access System switch OFF (guard extended).
 - E. Verify the electric strike is in the unlocked position (solenoid pin in the electric strike will retract down such that you can rotate the strike).
 - F. Position Flight Deck Access System switch NORM (guard closed).
 - G. Verify the electric strike is in the locked position (solenoid pin in the electric strike will be extended up such that you can not rotate the strike).

OPERATIONS (O)

Each operator must establish and use alternate procedures based on their security requirements for allowing and denying access to the flight deck.

NOTE: Ensure the Flight Deck Access System switch is in the OFF position (guard extended) when flight deck door is to be closed and not occupied.

**52-51-02 Boeing Enhanced Flight Deck Security Door
Automatic Locking System (14 CFR 25.795
Compliant) (Passenger)**

52-51-02-05 Pressure Rate-Of-Change Sensing Module

Interval	Installed	Required	Procedure
A	1	0	(M)

May be inoperative provided:

- a. Pressure sensing module is deactivated.

- b. Repairs are made within 2 flight days.
-

MAINTENANCE (M)

Deactivate the pressure sensing module (pressure sensor) (AMM 52-00-00/901).

1. Open P110 panel F/D DR LOCK circuit breaker.
2. Remove the pressure sensor.
3. Disconnect the electrical connector from the pressure sensor.
4. A closed circuit from the pressure sensor is required to keep the automatic locking system operating.
 - A. Fabricate a shorting plug using a connector part number equivalent to Government Designation M83513/04-A01N (detail specification MIL-DTL-83513).
 - B. Trim the excess wire to a 2 to 3 inch length.
 - C. Splice wires 4 and 6 together.
 - D. Insulate splice and wire pigtails.
 - E. Connect the shorting plug to the electrical connector and stow.
5. Install the pressure sensor.
6. Close P110 panel F/D DR LOCK circuit breaker.



52-51-03 Boeing Enhanced Flight Deck Security Door Dead Bolt (14 CFR 25.795 Compliant) (Passenger)

Interval	Installed	Required	Procedure
C	1	0	

May be inoperative provided automatic lock controls operate normally.

52-51-04 JAMCO Flight Deck Security Door Automatic Locking System (14 CFR 25.795 Compliant)

Interval	Installed	Required	Procedure
C	1	0	(M) (O)

May be inoperative provided:

- Automatic locking system is deactivated.
- Mechanical catch (latch) pin lock operates normally and is used to lock the door.
- Alternate procedures are established and used for locking and unlocking the flight deck door using the mechanical catch (latch) pin lock.

NOTE: The Flight Deck Security Door installation manufactured by JAMCO is not a Boeing system. Information applicable to the JAMCO installation is not provided.

52-51-04 JAMCO Flight Deck Security Door Automatic Locking System (14 CFR 25.795 Compliant)

52-51-04-01 Door Automatic Locking Solenoids

Interval	Installed	Required	Procedure
C	2	1	

One may be inoperative provided the remaining locking solenoid operates normally.

NOTE: The Flight Deck Security Door installation manufactured by JAMCO is not a Boeing system. Information applicable to the JAMCO installation is not provided.

52-51-04 JAMCO Flight Deck Security Door Automatic Locking System (14 CFR 25.795 Compliant)

52-51-04-02 Door Warning System
52-51-04-02-01 Speakers

Interval	Installed	Required	Procedure
C	2	1	(M) (O)



One may be inoperative provided remaining speaker is verified to operate normally once each flight day.

NOTE: The Flight Deck Security Door installation manufactured by JAMCO is not a Boeing system. Information applicable to the JAMCO installation is not provided.

52-51-04 JAMCO Flight Deck Security Door Automatic Locking System (14 CFR 25.795 Compliant)

52-51-04-02 Door Warning System

52-51-04-02-02 LED (Green Indicator Lights)

Interval	Installed	Required	Procedure
C	2	0	

NOTE: The Flight Deck Security Door installation manufactured by JAMCO is not a Boeing system. Information applicable to the JAMCO installation is not provided.

52-51-04 JAMCO Flight Deck Security Door Automatic Locking System (14 CFR 25.795 Compliant)

52-51-04-02 Door Warning System

52-51-04-02-03 Aural Warning System

Interval	Installed	Required	Procedure
C	1	0	(M) (O)

May be inoperative provided:

- Door AUTO UNLK light is verified to operate normally.
 - Alternate procedures are established and used.
-

NOTE: The Flight Deck Security Door installation manufactured by JAMCO is not a Boeing system. Information applicable to the JAMCO installation is not provided.

52-51-04 JAMCO Flight Deck Security Door Automatic Locking System (14 CFR 25.795 Compliant)

52-51-04-03 Door Control Panel
52-51-04-03-01 Door LOCK FAIL Light

Interval	Installed	Required	Procedure
C	1	0	(M)

May be inoperative OFF provided automatic lock controls are verified to operate normally.

NOTE: The Flight Deck Security Door installation manufactured by JAMCO is not a Boeing system. Information applicable to the JAMCO installation is not provided.

52-51-04 JAMCO Flight Deck Security Door Automatic Locking System (14 CFR 25.795 Compliant)

52-51-04-03 Door Control Panel
52-51-04-03-02 Door AUTO UNLK Light

Interval	Installed	Required	Procedure
C	1	0	(M) (O)

May be inoperative OFF provided:

- Automatic lock controls are verified to operate normally.
- Aural Warning System operates normally.
- Alternate procedures are established and used.

NOTE: The Flight Deck Security Door installation manufactured by JAMCO is not a Boeing system. Information applicable to the JAMCO installation is not provided.

**52-51-04 JAMCO Flight Deck Security Door Automatic Locking System (14 CFR 25.795 Compliant)**

52-51-04-03 Door Control Panel**52-51-04-03-03 Door HARD LOCK Light**

Interval	Installed	Required	Procedure
C	1	0	(M) (O)

May be inoperative provided:

- Automatic lock controls are verified to operate normally.
 - Alternate procedures are established and used.
-

NOTE: The Flight Deck Security Door installation manufactured by JAMCO is not a Boeing system. Information applicable to the JAMCO installation is not provided.

52-51-04 JAMCO Flight Deck Security Door Automatic Locking System (14 CFR 25.795 Compliant)

52-51-04-03 Door Control Panel**52-51-04-03-04 Door UNLKD Switch / UNLK Switch Position**

Interval	Installed	Required	Procedure
C	1	0	(M) (O)

May be inoperative provided:

- Door can be opened manually from the flight deck.
 - Remaining automatic lock controls are verified to operate normally.
 - Alternate procedures are established and used.
-

NOTE: The Flight Deck Security Door installation manufactured by JAMCO is not a Boeing system. Information applicable to the JAMCO installation is not provided.

52-51-04 JAMCO Flight Deck Security Door Automatic Locking System (14 CFR 25.795 Compliant)

52-51-04-03 Door Control Panel
52-51-04-03-05 Door UNLKD Light

Interval	Installed	Required	Procedure
C	1	0	(M) (O)

May be inoperative provided:

- Automatic lock controls are verified to operate normally.
- Aural warning system operates normally.

NOTE: The Flight Deck Security Door installation manufactured by JAMCO is not a Boeing system. Information applicable to the JAMCO installation is not provided.

52-51-04 JAMCO Flight Deck Security Door Automatic Locking System (14 CFR 25.795 Compliant)

52-51-04-03 Door Control Panel
52-51-04-03-06 Door EMRG ENTRY ACTIVE Light

Interval	Installed	Required	Procedure
C	1	0	(M)

May be inoperative provided the door aural warning system is verified to operate normally.

NOTE: The Flight Deck Security Door installation manufactured by JAMCO is not a Boeing system. Information applicable to the JAMCO installation is not provided.

**52-51-04 JAMCO Flight Deck Security Door Automatic Locking System (14 CFR 25.795 Compliant)**

52-51-04-03 Door Control Panel**52-51-04-03-07 Door OPEN Light**

Interval	Installed	Required	Procedure
C	1	0	(M) (O)

May be inoperative provided automatic lock controls are verified to operate normally.

NOTE: The Flight Deck Security Door installation manufactured by JAMCO is not a Boeing system. Information applicable to the JAMCO installation is not provided.

52-51-04 JAMCO Flight Deck Security Door Automatic Locking System (14 CFR 25.795 Compliant)

52-51-04-04 FLIGHT DECK DOOR Warning / Caution Light

Interval	Installed	Required	Procedure
C	1	0	

NOTE: The Flight Deck Security Door installation manufactured by JAMCO is not a Boeing system. Information applicable to the JAMCO installation is not provided.

52-51-04 JAMCO Flight Deck Security Door Automatic Locking System (14 CFR 25.795 Compliant)

52-51-04-05 Cabin Pushbutton Entry Pad / Keypad

Interval	Installed	Required	Procedure
C	1	0	(O)

May be inoperative provided alternate procedures are established and used.

NOTE: The Flight Deck Security Door installation manufactured by JAMCO is not a Boeing system. Information applicable to the JAMCO installation is not provided.

**52-51-04 JAMCO Flight Deck Security Door Automatic Locking System (14 CFR 25.795 Compliant)**

52-51-04-05 Cabin Pushbutton Entry Pad / Keypad**52-51-04-05-01 Keypad Indicator Lights**

Interval	Installed	Required	Procedure
C	3	0	(M) (O)

May be inoperative provided:

- Keypad is verified to operate normally.
- Alternate procedures are established and used.

NOTE: The Flight Deck Security Door installation manufactured by JAMCO is not a Boeing system. Information applicable to the JAMCO installation is not provided.



**52-51-05 JAMCO Flight Deck Security Door Mechanical Catch
(Latch) Pin Lock (14 CFR 25.795 Compliant)**

Interval	Installed	Required	Procedure
C	1	0	(M)

May be inoperative provided automatic locking system operates normally.

NOTE: The Flight Deck Security Door installation manufactured by JAMCO is not a Boeing system. Information applicable to the JAMCO installation is not provided.



52-51-06 Flight Deck Door Viewing Port (Passenger)
52-51-06-01 Without Electronic Visual Surveillance System

Interval	Installed	Required	Procedure
A	1	0	(O)

May be inoperative provided:

- Alternate procedures are established and used.
 - Repairs are made within 3 flight days.
-

OPERATIONS (O)

Alternate procedures must be established and used.

52-51-06 Flight Deck Door Viewing Port (Passenger)
52-51-06-02 With Electronic Visual Surveillance System

52-51-06-02A Electronic Visual Surveillance System Inoperative

Interval	Installed	Required	Procedure
A	1	0	(O)

May be inoperative provided:

- Alternate procedures are established and used.
 - Repairs are made within 3 flight days.
-

OPERATIONS (O)

Alternate procedures must be established and used.

52-51-06 Flight Deck Door Viewing Port (Passenger)
52-51-06-02 With Electronic Visual Surveillance System

52-51-06-02B Electronic Visual Surveillance System Operative

Interval	Installed	Required	Procedure
C	1	0	(O)

May be inoperative provided:

- A flight deck door visual surveillance system is installed and operates normally.
 - Alternate procedures are established and used.
-

**OPERATIONS (O)**

Alternate procedures must be established and used.

52-71-01 Door Indication Systems

Interval	Installed	Required	Procedure
C	-	0	(M)

May be inoperative provided associated door is verified closed, latched and locked before each departure.

MAINTENANCE (M)

NOTE: The optional procedures to deactivate inoperative passenger entry door sensors and forward, aft and main deck cargo (777F only) door sensors are provided to prevent unexpected display of the CONFIG DOORS warning and potential RTO due to intermittent door sensor faults.

Deactivate the inoperative sensors (if desired), and before each flight, verify the associated door is closed, latched, and locked (AMM 52-00-00/901).

1. For inoperative sensor(s) on Main Entry Doors:
 - A. If desired, deactivate the associated entry door's inoperative sensor(s).
 - 1) Remove, cap and stow electrical connectors from following sensors.
 - a. For entry door 1L, remove electrical connector DS52109 from sensor S52109 (CLOSED & LATCHED DR 1L) and remove electrical connector DS52110 from sensor S52110 (LOCKED DR 1L).
 - b. For entry door 1R, remove electrical connector DS52209 from sensor S52209 (CLOSED & LATCHED DR 1R) and remove electrical connector DS52110 from sensor S52110 (LOCKED DR 1R).
 - c. For entry door 2L, remove electrical connector DS52111 from sensor S52111 (CLOSED & LATCHED DR 2L) and remove electrical connector DS52110 from sensor S52110 (LOCKED DR 2L).
 - d. For entry door 2R, remove electrical connector DS52211 from sensor S52211 (CLOSED & LATCHED DR 2R) and remove electrical connector DS52110 from sensor S52110 (LOCKED DR 2R).
- For -200/-200ER/-200LR:
- e. For entry door 3L, remove electrical connector DS52113 from sensor S52113 (CLOSED & LATCHED DR 3L) and remove electrical connector DS52110 from sensor S52110 (LOCKED DR 3L).

For -300/-300ER:

- f. For entry door 3L, remove electrical connector DS52113 from sensor S52113 (CLOSED & LATCHED DR 3L) and remove electrical connector DS52026 from sensor S52026 (LOCKED DR 3L).

For -200/-200ER/-200LR:

- g. For entry door 3R, remove electrical connector DS52213 from sensor S52213 (CLOSED & LATCHED DR 3R) and remove electrical connector DS52110 from sensor S52110 (LOCKED DR 3R).

For -300/-300ER:

- h. For entry door 3R, remove electrical connector DS52213 from sensor S52213 (CLOSED & LATCHED DR 3R) and remove electrical connector DS52026 from sensor S52026 (LOCKED DR 3R).
- i. For entry door 4L, remove electrical connector DS52115 from sensor S52115 (CLOSED & LATCHED DR 4L) and remove electrical connector DS52110 from sensor S52110 (LOCKED DR 4L).
- j. For entry door 4R, remove electrical connector DS52215 from sensor S52215 (CLOSED & LATCHED DR 4R) and remove electrical connector DS52110 from sensor S52110 (LOCKED DR 4R).

For -300/-300ER:

- k. For entry door 5L, remove electrical connector DS52117 from sensor S52117 (CLOSED & LATCHED DR 5L) and remove electrical connector DS52110 from sensor S52110 (LOCKED DR 5L).

For -300/-300ER:

- l. For entry door 5R, remove electrical connector DS52217 from sensor S52217 (CLOSED & LATCHED DR 5R) and remove electrical connector DS52110 from sensor S52110 (LOCKED DR 5R).

- B. Prior to each flight, verify the associated door is closed, latched and locked using normal procedures.
 - 1) Open the affected door in MANUAL/DISARMED mode, then close and latch.
 - 2) Confirm the door interior liner is aligned and faired with the doorway lining.
 - 3) Confirm the OPEN arrow on the hinge arm linings is aligned and continuous at crossover from forward section of lining to aft section of lining.

- 4) Position the door mode select handle to AUTOMATIC/ARMED, and confirm the yellow flags are visible in the viewing windows immediately forward and aft of door at floor level.
 - 5) Confirm the outside skin of the door is aligned and faired with the fuselage skin.
 - 6) Confirm the vent door is fully closed.
2. For inoperative sensor(s) on Forward Cargo Door:
- A. If desired, deactivate the inoperative door sensor(s).
- 1) Remove, cap and stow electrical connector DS52021 from sensor S52021 (CLOSED).
 - 2) Remove, cap and stow electrical connector DS52042 from sensor S52042 (LATCHED).
 - 3) Remove, cap and stow electrical connector DS52022 from sensor S52022 (LOCKED).
- B. Prior to each flight, verify the door is closed, latched and locked using normal procedures.
- 1) Close the door.
 - 2) Confirm the outside skin of the door is aligned and faired with the fuselage skin.
 - 3) Confirm the lock handle is in the unlocked position.
 - 4) Confirm the vent door is open.
 - 5) Rotate the lock handle back to the CLOSED position.
 - 6) Confirm the lock handle is fully closed and faired with the outside skin of the door.
 - 7) Confirm the lock handle release lever is faired with the lock handle and outside skin of the door.
 - 8) Confirm the vent door is fully closed (the upper edge of the vent door may not be fully faired).
 - 9) Confirm the latches are in the fully latched position and that the locks are in the locked position by visual inspection through the witness ports at the lower edge of the door.
3. For inoperative sensor(s) on Aft Large Cargo Door:
- A. If desired, deactivate the inoperative door sensor(s).
- 1) Remove, cap and stow electrical connector DS52045 from sensor S52045 (CLOSED).
 - 2) Remove, cap and stow electrical connector DS52043 from sensor S52043 (LATCHED).
 - 3) Remove, cap and stow electrical connector DS52044 from sensor S52044 (LOCKED).
- B. Prior to each flight, verify the door is closed, latched and locked using normal procedures.
- 1) Close the door.

- 2) Confirm the outside skin of the door is aligned and faired with the fuselage skin.
 - 3) Confirm the lock handle is in the unlocked position.
 - 4) Confirm the vent door is open.
 - 5) Rotate the lock handle back to the CLOSED position.
 - 6) Confirm the lock handle is fully closed and faired with the outside skin of the door.
 - 7) Confirm the lock handle release lever is faired with the lock handle and outside skin of the door.
 - 8) Confirm the vent door is fully closed (the upper edge of the vent door may not be fully faired).
 - 9) Confirm the latches are in the fully latched position and that the locks are in the locked position by visual inspection through the witness ports at the lower edge of the door.
4. For inoperative sensor(s) on Aft Small Cargo Door:
- A. If desired, deactivate the inoperative door sensor(s).
 - 1) Remove, cap and stow electrical connector DS52023 from sensor S52023 (CLOSED).
 - 2) Remove, cap and stow electrical connector DS52024 from sensor S52024 (LATCHED/LOCKED).
 - B. Prior to each flight, verify the door is closed and latched using normal procedures.
 - 1) Close the door.
 - 2) Confirm the outside skin of the door is aligned and faired with the fuselage skin.
 - 3) Confirm the vent door is fully closed.
 - 4) Open the access cover for the door control panel.
 - 5) Operate the CARGO DOOR CONTROL ARMING switch downward to the ARM position, and confirm the green DOOR LATCHED light illuminates. Release the switch and confirm the light extinguishes.
 - 6) Close the access cover.
5. For inoperative sensor(s) on Bulk Cargo Door:
- A. If desired, deactivate the inoperative door switch.
 - 1) Locate, remove and secure switch S52020 (CLOSED/LATCHED/LOCKED) without cutting its wires.
 - B. Prior to each flight, verify the door is closed and latched using normal procedures.
 - 1) Close the door.
 - 2) Confirm the outside skin of the door is aligned and faired with the fuselage skin.

- 3) Confirm the latch handle is fully closed and faired with the door skin.
- 4) Push on the door to verify it is latched.
6. For inoperative sensor(s) on Forward Access Door:
 - A. If desired, deactivate the inoperative door sensor(s).
 - 1) Remove, cap and stow electrical connector DS52018 from sensor S52018 (CLOSED/LATCHED/LOCKED).
 - B. Prior to each flight, verify the door is closed and latched using normal procedures.
 - 1) Close the door.
 - 2) Confirm the outside skin of the door is aligned and faired with the fuselage skin.
 - 3) Confirm the latch handle is fully closed and faired with the door skin.
 - 4) Push on the door to verify it is latched.
7. For inoperative sensor(s) on Main Equipment Center (E/E Bay) Access Door:
 - A. If desired, deactivate the inoperative door sensor(s).
 - 1) Remove, cap and stow electrical connector DS52019 from sensor S52019 (CLOSED/LATCHED/LOCKED).
 - B. Prior to each flight, verify the door is closed and latched using normal procedures.
 - 1) Close and latch the door.
 - 2) Confirm the outside skin of the door is aligned and faired with the fuselage skin.
 - 3) Confirm the latch handle is fully closed and faired with the door skin.
 - 4) Push on the door to verify it is latched.

For -300/-300ER:

8. For inoperative sensor(s) on Overwing Escape Slide Compartment Door:
 - A. Prior to each flight, verify the door is closed and latched using normal procedures.
 - 1) Open and close the door.
 - 2) Confirm the outside skin of the door is aligned and faired with the fuselage skin.
 - 3) Push on the door to verify it is latched.

For 777F:

9. For inoperative sensor(s) on Main Deck Cargo Door:
 - A. If desired, deactivate the inoperative door sensor(s).
 - 1) Remove, cap and stow electrical connector DS52053 from sensor S52053 (CLOSED).

- 2) Remove, cap and stow electrical connector DS52054 from sensor S52054 (LATCHED).
- 3) Remove, cap and stow electrical connector DS52055 from sensor S52055 (LOCKED).
- B. Prior to each flight, verify the door is closed, latched and locked using normal procedures.
 - 1) Close the door.
 - 2) Confirm the latch arrows point to the targets in all eight witness ports.
 - 3) Confirm the outside skin of the door is aligned and faired with the fuselage skin.
 - 4) Confirm the vent door has closed.
 - 5) Using the witness ports, confirm the lock sectors are located over latch cams but do not touch the cams.

OPERATIONS NOTE

1. The optional procedures to deactivate inoperative passenger entry door sensors and forward, aft and main deck cargo (777F only) door sensors are provided to prevent unexpected display of the CONFIG DOORS warning and potential RTO due to intermittent door sensor faults.
2. The DOOR WING SLIDE L and DOOR WING SLIDE R advisory messages (-300/-300ER) and the DOOR BULK CARGO advisory message may be displayed both on the ground and in flight.
3. For doors with inoperative sensors deactivated:
 - A. The respective door indication status message (if applicable) may be displayed both on the ground and in flight.
 - B. Inoperative door advisory messages may be displayed on the ground only, except the DOOR BULK CARGO advisory message, which may be displayed both on the ground and in flight.
 - C. The DOOR MAIN DECK CARGO, DOOR FWD CARGO and DOOR AFT CARGO (if applicable) caution messages may be displayed on the ground.

The following messages may be displayed after landing if the associated door sensor was deactivated:

CAUTION - DOOR FWD CARGO

CAUTION - DOOR AFT CARGO

ADVISORY - DOOR ENTRY 1L

ADVISORY - DOOR ENTRY 1R

ADVISORY - DOOR AFT CARGO

ADVISORY - DOOR FWD CARGO

ADVISORY - DOOR E/E ACCESS



52-71-02 Main Entry Door Status Annunciation and Alerting Systems (Flight Attendant)

52-71-02A Alternate Procedures Used

Interval	Installed	Required	Procedure
C	-	0	(O)

May be inoperative provided alternate procedures are established and used.

OPERATIONS (O)

NOTE: This item provides relief for the optional ARM/DISARM or AUTO/MANUAL door status annunciations on the attendant switch panel at Door 1 or Door 2, and the door armed alerting module.

Establish alternate procedures for determining the status of the affected entry doors. For the door status annunciation inoperative, door status indications on the door must be used. For an inoperative entry door alerting module, appropriate procedures should be established to ensure an armed door is not inadvertently opened.

52-71-02 Main Entry Door Status Annunciation and Alerting Systems (Flight Attendant)

52-71-02B Procedures Do Not Require Use

Interval	Installed	Required	Procedure
D	-	0	

May be inoperative provided procedures do not require their use.



Intentionally
Blank



777 Dispatch Deviations Guide

Section 2

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56-11-01 Flight Deck Window Indication Systems (Side - No. 2)



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56-11-01 Flight Deck Window Indication Systems (Side - No. 2)

Interval	Installed	Required	Procedure
C	2	0	



Intentionally
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73-21-01 Engine Idle Selection Systems

73-21-02 Electronic Engine Controls (EEC) Normal Mode

73-21-02-01 PW

73-21-02-02 GE

73-21-02-02-01 All (Except GE90-100 Series)

73-21-02-02-02 GE90-100 Series

73-21-02-03 RR

73-21-03 Electronic Engine Controls (EEC) Mode Switch Lights

73-21-03-01 ALTN Lights

73-21-03-02 NORM Lights

73-21-04 EEC C1 Faults

73-21-04-01 GE90-100 Series Without P/N 730A-GEC-A08-06 or
Later EEC Software Installed

73-21-04-02 PW, RR, and GE90-100 Series with P/N
730A-GEC-A08-06 or Later EEC Software Installed

73-21-05 Turbine Overspeed Systems (RR)

73-21-06 Engine Fuel Shutoff Valve Indication Systems

**73-21-07 Engine Thrust Control Malfunction Accommodation
(TCMA) Functions (GE90-100 Series)**

**73-21-08 Engine Fuel High Pressure Shut Off Valve (HPSOV) Run
Solenoid (GE)**

73-21-08-01 Passenger

73-21-08-02 777F

73-31-01 Fuel Flow Indications

73-34-01 Engine Fuel Filter Bypass Warning Systems

73-34-01-01 PW & RR

73-34-01-02 GE



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**73-21-01 Engine Idle Selection Systems**

Interval	Installed	Required	Procedure
C	2	0	(O)

May be inoperative provided appropriate performance adjustments are applied.

OPERATIONS (O)

For -200 or -200ER/PW or -300ER/GE:

Increase descent fuel burn by 275 lb (125 kg) per engine.

For -300/PW:

Increase descent fuel burn by 200 lb (91 kg) per engine.

For -200 or -200ER/GE:

Increase descent fuel burn by 125 lb (57 kg) per engine.

For -200LR or 777F/GE:

Increase descent fuel burn by 375 lb (170 kg) per engine.

For -200 or -200ER/RR:

Increase descent fuel burn by 1,900 lb (861 kg) per engine.

For -300/RR:

Increase descent fuel burn by 1,100 lb (500 kg) per engine.

73-21-02 Electronic Engine Controls (EEC) Normal Mode
73-21-02-01 PW

Interval	Installed	Required	Procedure
C	2	0	(O)

Normal (NORM) mode may be inoperative provided:

- Autothrottle system operates normally.
- Both engines are operated in the alternate (ALTN) mode.
- Appropriate performance adjustments are applied.

MAINTENANCE NOTE

For engines without PW SB 4G-112-73-29 or production equivalent installed (EEC software SCN8C or later), if EEC C1 faults are present on the same engine verify EEC channel B transmissions operate normally.

NOTE: EEC C1 faults are present when the associated ENG EEC C1 L or R status message is displayed.

- Supply EEC maintenance power by positioning associated Overhead Maintenance Panel (P61) EEC MAINT L or R ENG POWER switch TEST.
- Display the Electronic Propulsion Control System (EPCS) maintenance pages.
- Review the EPCS maintenance pages and verify EEC channel B parameters are displayed (not blank).
- Position associated Overhead Maintenance Panel (P61) EEC MAINT L or R ENG POWER switch NORM.

OPERATIONS (O)

NOTE 1: Derated and/or reduced thrust takeoffs, and takeoff thrust bumps are not allowed.

NOTE 2: Thrust rating protection is not provided in alternate mode (for manual thrust settings) and maximum rated thrust is reached at a thrust lever position less than full forward.

NOTE 3: Use of autothrusters when dispatching with EECs in alternate mode makes it easier to maintain thrust at the reference limit.

- Base performance on EECs operating in alternate mode.
- Increase flight planning fuel by 2.3%.
- Position both EEC Mode switches to ALTN.
- For THRUST ASYM COMP advisory and status messages displayed, position the THRUST ASYM COMP switch to OFF then to AUTO to restore Thrust Asymmetry Compensation functionality.

5. For takeoff, advance thrust levers to approximately 50% N1, engage autothrottles and arm VNAV.

73-21-02 Electronic Engine Controls (EEC) Normal Mode**73-21-02-02 GE****73-21-02-02-01 All (Except GE90-100 Series)**

Interval	Installed	Required	Procedure
B	2	0	(O)

Normal (NORM) mode may be inoperative provided:

- a. Autothrottle system operates normally.
- b. LPT Turbine Case Cooling Air Flow Systems on both engines operate normally.
- c. Both engines are operated in the alternate (ALTN) mode.
- d. Engine anti-ice valve may not be inoperative open.

OPERATIONS (O)

NOTE 1: Derated and/or reduced thrust takeoffs, and takeoff thrust bumps are not allowed.

NOTE 2: Thrust rating protection is not provided in alternate mode (for manual thrust settings) and maximum rated thrust is reached at a thrust lever position less than full forward.

1. Increase flight planning fuel by 1.4%.
2. Position both EEC Mode switches to ALTN.
3. Selection of TOB (if available) is not allowed. Select TO for takeoff.
4. For takeoff, engage autothrottles and arm VNAV.

73-21-02 Electronic Engine Controls (EEC) Normal Mode**73-21-02-02 GE****73-21-02-02-02 GE90-100 Series**

Interval	Installed	Required	Procedure
B	2	0	(O)

Normal (NORM) mode may be inoperative provided:

- a. Autothrottle system operates normally.
- b. LPT Turbine Case Cooling Air Flow Systems on both engines operate normally.
- c. Both engines are operated in the alternate (ALTN) mode.

- d. Appropriate performance adjustments are applied.

OPERATIONS (O)

NOTE 1: Derated and/or reduced thrust takeoffs, and takeoff thrust bumps are not allowed.

NOTE 2: Thrust rating protection is not provided in alternate mode (for manual thrust settings) and maximum rated thrust is reached at a thrust lever position less than full forward.

1. For ETOPS flight planning, increase the engine inoperative Critical Fuel Reserves by 1.2%.
2. Base performance on EECs operating in alternate mode.
3. Position both EEC Mode switches to ALTN.
4. Selection of TOB (if available) is not allowed. Select TO for takeoff.
5. For takeoff, engage autothrottles and arm VNAV.

73-21-02 Electronic Engine Controls (EEC) Normal Mode

73-21-02-03 RR

Interval	Installed	Required	Procedure
C	2	0	(O)

Normal (NORM) mode may be inoperative provided:

- a. Autothrottle system operates normally.
- b. N2 indication on affected engine operates normally.
- c. Both engines are operated in the alternate (ALTN) mode.
- d. Appropriate performance adjustments are applied.

OPERATIONS (O)

NOTE 1: Derated and/or reduced thrust takeoffs, and takeoff thrust bumps are not allowed.

NOTE 2: Thrust rating protection is not provided in alternate mode (for manual thrust settings) and maximum rated thrust is reached at a thrust lever position less than full forward.

1. Base performance on EECs operating in alternate mode.
2. Position both EEC Mode switches to ALTN.
3. For THRUST ASYM COMP advisory and status messages displayed, position the THRUST ASYM COMP switch to OFF then to AUTO to restore Thrust Asymmetry Compensation functionality.
4. For takeoff, advance thrust levers to approximately 50% N1, engage autothrottles and arm VNAV.



73-21-03 Electronic Engine Controls (EEC) Mode Switch Lights
73-21-03-01 ALTN Lights

Interval	Installed	Required	Procedure
C	2	0	

73-21-03 Electronic Engine Controls (EEC) Mode Switch Lights
73-21-03-02 NORM Lights

Interval	Installed	Required	Procedure
C	2	0	

**73-21-04 EEC C1 Faults****73-21-04-01 GE90-100 Series Without P/N 730A-GEC-A08-06 or Later EEC Software Installed**

Interval	Installed	Required	Procedure
A	2	0	(M)

May be dispatched with C1 faults provided:

- Prior to each flight EEC faults are reviewed to determine if EEC must be replaced.
- Repairs are made in accordance with times established by engine manufacturer.

MAINTENANCE (M)

NOTE: Certain ENG EEC C1 faults require replacement of the EEC.

Prior to each flight perform the following ENG EEC C1 fault check:

(Required by AD 2020-20-17)

- Refer to FIM Chapter 73-98, Task 805 to determine if the EEC must be replaced.

NOTE: GE Alternative Method of Compliance (AMOC) to AD 2020-20-17 allows FIM Chapter 73-98, Task 805 as an AMOC to the GE GE90-100 SB 73-0117 R01 requirement.

OPERATIONS NOTE

The engine manufacturer's established maximum operating time interval for C1 faults is 150 flight hours or 10 calendar days, whichever occurs first (TCDS E00049EN Note 17).

73-21-04 EEC C1 Faults**73-21-04-02 PW, RR, and GE90-100 Series with P/N 730A-GEC-A08-06 or Later EEC Software Installed**

Interval	Installed	Required	Procedure
A	2	0	

May be dispatched with C1 faults provided repairs are made in accordance with times established by engine manufacturer.

NOTE: This item is also applicable to GE90-70/80/90 Series engines.

OPERATIONS NOTE

For PW:

The engine manufacturer's established maximum operating time interval for C1 faults is 20 days (TCDS E46NE Note 17).

For GE:

The engine manufacturer's established maximum operating time interval for C1 faults is 150 flight hours or 10 calendar days, whichever occurs first (TCDS E00049EN Note 17).

For RR:

The engine manufacturer's established maximum operating time interval for C1 faults is 300 hours or 20 days, whichever occurs first (TCDS E00050EN Note 17).

**73-21-05 Turbine Overspeed Systems (RR)**

Interval	Installed	Required	Procedure
C	2	1	

73-21-06 Engine Fuel Shutoff Valve Indication Systems

Interval	Installed	Required	Procedure
C	2	1	(M)

One may be inoperative provided the associated valve is verified to operate normally each flight day.

MAINTENANCE (M)

Verify that the associated engine fuel shutoff valve is operating normally once each flight day (AMM 73-00-00/901).

For PW:

- If MAT is available, confirm that both maintenance messages are displayed for the associated status message.
 - For status message ENG FUEL VALVE L, 73-40231 and 73-50231.
 - For status message ENG FUEL VALVE R, 73-40232 and 73-50232.
- Start the associated engine and allow engine to stabilize at ground idle (AMM 71-00-00/201).
- After minimum engine operating time, open the P11 panel FUEL SPAR VALVE circuit breaker for the associated engine.
- Position the associated engine FUEL CONTROL switch to CUTOFF and confirm engine shutdown.
- For engine not shutdown, the fuel shutoff valve is not operating normally.
 - Monitor EGT and be prepared to motor the engine.
 - Close the P11 panel FUEL SPAR VALVE circuit breaker.
 - The engine may continue to operate for one minute.
- Confirm the associated FUEL SPAR VALVE L or R status message is displayed.
- Erase the latched FUEL SPAR VALVE L or R status message (AMM 31-61-00/201).
- After engine has stopped rotating, close the P11 panel FUEL SPAR VALVE circuit breaker.

For GE and RR:

- Start the associated engine and allow engine to stabilize at ground idle (AMM 71-00-00/201).
- After minimum engine operating time, open the P11 panel FUEL SPAR VALVE circuit breaker for the associated engine.
- Position the associated engine FUEL CONTROL switch to CUTOFF and confirm engine shutdown.
- For engine not shutdown, the fuel shutoff valve is not operating normally.
 - Monitor EGT and be prepared to motor the engine.



-
- B. Close the P11 panel FUEL SPAR VALVE circuit breaker.
 - C. The engine may continue to operate for one minute.
5. Confirm the associated FUEL SPAR VALVE L or R status message is displayed.
 6. Erase the latched FUEL SPAR VALVE L or R status message (AMM 31-61-00/201).
 7. After engine has stopped rotating, close the P11 panel FUEL SPAR VALVE circuit breaker.

OPERATIONS NOTE

The ENG FUEL VALVE L or R advisory message will be displayed.



73-21-07 Engine Thrust Control Malfunction Accommodation
(TCMA) Functions (GE90-100 Series)

Interval	Installed	Required	Procedure
C	2	1	

**73-21-08 Engine Fuel High Pressure Shut Off Valve (HPSOV)
Run Solenoid (GE)**
73-21-08-01 Passenger

Interval	Installed	Required	Procedure
B	2	1	(M) (O)

May be inoperative provided:

- Associated HPSOV solenoid is deactivated in the open position.
- Associated fuel spar valve is verified to operate normally before each departure.
- Associated Hydro-Mechanical Unit Fuel Metering Valve Shutdown function is verified to operate normally before each departure.
- Autostart system operates normally.
- EEC C1/C2 faults are not present on the associated engine.
- Airplane is not operated in areas of known or forecast volcanic ash contamination.
- A flight attendant is seated at each exit aft of the wing during taxi, takeoff, and landing.
- Flight attendants are appropriately briefed before each departure.

MAINTENANCE (M)

- Verify ENG EEC C1 and ENG EEC C2 EICAS messages are not present.
- Deactivate the associated HPSOV solenoid in the open position (AMM 73-00-00/901).
- Inspect the associated fuel spar valve actuator located in the rear spar and confirm visually that the valve is closed
- Perform an autostart and allow the engine to run at idle for 2 minutes (AMM 71-00-00/201).
- Position the associated engine FUEL CONTROL switch to CUTOFF and confirm engine begins to spool down approximately 12 seconds after the fuel control switch is set to CUTOFF, and the engine is fully shut down within 20 seconds.

NOTE: If the associated HMU FMV shutdown function is not operating normally engine shutdown will take approximately 30-60 seconds; dispatch is not allowed if engine shutdown takes more than 20 seconds.

OPERATIONS (O)

NOTE 1: DO NOT close the associated ENG FUEL VALVE circuit breaker in flight. If the ENG FUEL VALVE circuit breaker is closed and the Fuel Switch is positioned to CUTOFF, the engine cannot be restarted.

NOTE 2: The ENG FUEL VALVE L or R advisory may momentarily display at engine shutdown. The ENG FUEL VALVE L or R status messages will be displayed and latched.

NOTE 3: Shutdown of the affected engine will be delayed by approximately 12 seconds.

NOTE 4: The TCMA function will be inoperative for the associated engine.

NOTE 5: Flight attendant seating requirements in 14 CFR apply.

Flight attendants must be briefed prior to takeoff regarding the engine shutdown delay to ensure that the engine is shut down prior to opening the exits in the event of an evacuation.

73-21-08 Engine Fuel High Pressure Shut Off Valve (HPSOV) Run Solenoid (GE)

73-21-08-02 777F

Interval	Installed	Required	Procedure
B	2	1	(M)

May be inoperative provided:

- Associated HPSOV solenoid is deactivated in the open position.
- Associated fuel spar valve is verified to operate normally before each departure.
- Associated Hydro-Mechanical Unit Fuel Metering Valve Shutdown function is verified to operate normally before each departure.
- Autostart system operates normally.
- EEC C1/C2 faults are not present on the associated engine.
- Airplane is not operated in areas of known or forecast volcanic ash contamination.

MAINTENANCE (M)

- Verify ENG EEC C1 and ENG EEC C2 EICAS messages are not present.
- Deactivate the associated HPSOV solenoid in the open position (AMM 73-00-00/901).
- Inspect the associated fuel spar valve actuator located in the rear spar and confirm visually that the valve is closed
- Perform an autostart and allow the engine to run at idle for 2 minutes (AMM 71-00-00/201).
- Position the associated engine FUEL CONTROL switch to CUTOFF and confirm engine begins to spool down approximately 12 seconds after the fuel control switch is set to CUTOFF, and the engine is fully shut down within 20 seconds.



NOTE: If the associated HMU FMV shutdown function is not operating normally engine shutdown will take approximately 30-60 seconds; dispatch is not allowed if engine shutdown takes more than 20 seconds.

OPERATIONS NOTE

1. DO NOT close the associated ENG FUEL VALVE circuit breaker in flight. If the ENG FUEL VALVE circuit breaker is closed and the Fuel Switch is positioned to CUTOFF, the engine cannot be restarted.
2. The ENG FUEL VALVE L or R advisory may momentarily display at engine shutdown. The ENG FUEL VALVE L or R status messages will be displayed and latched.
3. Shutdown of the affected engine will be delayed by approximately 12 seconds.
4. The TCMA function will be inoperative for the associated engine.

**73-31-01 Fuel Flow Indications**

Interval	Installed	Required	Procedure
C	2	1	

One may be inoperative provided:

- Flight deck fuel tank quantity indicating systems operate normally.
- Flight remains within 180 minutes of landing at a suitable airport.

OPERATIONS NOTE

FMC Calculated Fuel, associated engine Fuel Used and Fuel Used Total indications will be inoperative.

73-34-01 Engine Fuel Filter Bypass Warning Systems
73-34-01-01 PW & RR

Interval	Installed	Required	Procedure
C	2	1	(M)

One may be inoperative provided:

- a. It is verified that the malfunction is in the alerting system.
- b. Fuel is drained from filter plug and checked for contaminants before each departure.

MAINTENANCE (M)

Verify the malfunction is in the alerting system and before each departure check fuel for contamination (AMM 73-00-00/901).

For PW:

1. Verify that the malfunction is in the alerting system.
 - A. If MAT is available, confirm one of the following maintenance message is displayed.
 - 1) For left engine, 73-40921, 73-50921, 73-60911 or 73-60921.
 - 2) For right engine, 73-40922, 73-50922, 73-60912 or 73-60922.
 - B. Do a continuity check of the fuel filter differential pressure switch.
 - 1) Deactivate the associated thrust reverser for ground maintenance (AMM 78-31-00/201).
 - 2) Open the associated left fan cowl panel (AMM 71-11-04/201).
 - 3) Open the associated left thrust reverser (AMM 78-31-00/201).
 - 4) Gain access to the associated engine EEC.
 - 5) Disconnect electrical connectors DM73003C and DM 73003K.
 - 6) Measure the resistance between pin 8 and pin 36 on electrical connector DM73003C.
 - 7) If resistance is less than 100K ohms, the malfunction is in the alerting system.
 - 8) Connect electrical connectors DM73003C and DM73003K to the EEC (AMM 70-11-09/201).
 - 9) Close the associated left thrust reverser (AMM 78-31-00/201).
 - 10) Close the associated left fan cowl panel (AMM 71-11-04/201).
2. Before each departure, check fuel for contaminants at the associated engine fuel pump filter, which is installed in the fuel pump.
 - A. Deactivate the associated thrust reverser for ground maintenance (AMM 78-31-00/201).
 - B. Open the associated left and right fan cowl panels (AMM 71-11-04/201).

- C. Open the associated right thrust reverser (AMM 78-31-00/201).
- D. Drain a small amount of fuel from the fuel filter and examine for contaminants.

NOTE: WATER IS NOT A CONTAMINANT THAT CAN CAUSE A FUEL FILTER BYPASS.

- 1) Remove fuel filter plug and drain a small amount of fuel into a container.
 - 2) Examine the fuel for contaminants (visible particles of solid material).
 - 3) Install fuel filter drain plug with new packing (AMM 73-11-02/401).
- E. Close the associated right thrust reverser (AMM 78-31-00/201).
- F. Close the associated left and right fan cowl panels (AMM 71-11-04/201).
- G. Activate the associated thrust reverser after ground maintenance (AMM 78-31-00/201).

For RR:

- 1. Verify that the malfunction is in the alerting system.
 - A. If MAT is available, check for maintenance messages:
 - 1) For left engine, confirm maintenance message 73-70131 is displayed and 73-90121 is not displayed.
 - 2) For right engine, confirm maintenance message 73-70132 is displayed and 73-90122 is not displayed.
 - B. If MAT is not available, do a continuity check of the fuel filter differential pressure switch:
 - 1) Open the associated right fan cowl panel (AMM 71-11-04/201)
 - 2) Gain access to the associated engine's Electronic Engine Control (EEC).
 - 3) Disconnect electrical connector DM73003E.
 - 4) Measure the resistance between pin 39 and pin 40 on electrical connector DM73003E.
 - 5) If resistance is less than 100 ohms, the malfunction is in the alerting system.
 - 6) Measure the resistance between pin 39 and the connector backshell on the electrical connector DM73003E.
 - 7) If resistance is less than 100K ohms, the malfunction is in the alerting system.
 - 8) Connect electrical connector DM73003E to the EEC (AMM 70-50-02/201).
 - 9) Close the associated right fan cowl panel (AMM 71-11-04/201).

2. Before each departure, check fuel for contaminants at the associated engine low pressure fuel pump filter, which is installed in the fuel oil heat exchanger (FOHE).
 - A. Open the associated right fan cowl panel (AMM 71-11-04/201).
 - B. Drain a small amount of fuel from the fuel filter and examine for contaminants.
 - 1) Remove fuel filter plug and drain a small amount of fuel into a container.

NOTE: WATER IS NOT A CONTAMINANT THAT CAN CAUSE A FUEL FILTER BYPASS.

 - 2) Examine the fuel for contaminants (visible particles of solid material).
 - 3) Install fuel filter drain plug with a new seal (AMM 73-11-13/401).
 - C. Close the associated right fan cowl panel (AMM 71-11-04/201).

73-34-01 Engine Fuel Filter Bypass Warning Systems
73-34-01-02 GE

Interval	Installed	Required	Procedure
C	2	1	(M)

One may be inoperative provided fuel is drained from filter plug and checked for contaminants before each departure.

MAINTENANCE (M)

Check fuel for contamination before each departure (AMM 73-00-00/901).

NOTE: Dispatch is not allowed with ENG FUEL FILTER L or R status message displayed.

1. Deactivate the leading edge slats in the retracted position (AMM 27-81-00/201).
2. Deactivate the associated thrust reverser for ground maintenance (AMM 78-31-00/201).
3. Open the associated left and right fan cowl panels (AMM 71-11-04/201).
4. Open the associated right thrust reverser (AMM 78-31-00/201).
5. Drain a small amount of fuel from the engine fuel filter bowl and examine for contaminants.
 - A. Remove fuel filter bleeder drain plug and drain a small amount of fuel into a container.

NOTE: WATER IS NOT A CONTAMINANT THAT CAN CAUSE A FUEL FILTER BYPASS.
 - B. Examine the fuel for contaminants (visible particles of solid material).



-
- C. Install fuel filter bleeder drain plug with new preformed packing (AMM 73-11-02/401).
 - 6. Close the associated right thrust reverser (AMM 78-31-00/201).
 - 7. Close the associated left and right fan cowl panels (AMM 71-11-04/201).
 - 8. Activate the associated thrust reverser after ground maintenance (AMM 78-31-00/201).
 - 9. Activate the leading edge slats (AMM 27-81-00/201).



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Section 2

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ATA 74

74-00-01 Ignition Systems

74-00-01A *EAI System Operative*

74-00-01B *EAI System Inoperative*



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**74-00-01 Ignition Systems****74-00-01A EAI System Operative**

Interval	Installed	Required	Procedure
B	4	3	

One may be inoperative provided associated engine anti-ice system operates normally.

74-00-01 Ignition Systems**74-00-01B EAI System Inoperative**

Interval	Installed	Required	Procedure
B	4	3	

One may be inoperative provided associated engine anti-ice system is inoperative with valve deactivated open.



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Section 2

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ATA 75

75-23-01 Nacelle Zone Ventilation Valves (PW)

75-23-02 Core Compartment Cooling Valves (GE)

75-24-01 Turbine Case Cooling Air Flow Systems (PW and GE)

75-24-01-01 PW

75-24-01-01-01 HPT Systems

75-24-01-01-02 LPT Systems

75-24-01-02 GE

75-24-01-02-01 LPT Systems

75-24-02 Turbine Cooling Air Systems (PW)

75-24-02A *One Inoperative*

75-24-02B *Both Inoperative*



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**75-23-01 Nacelle Zone Ventilation Valves (PW)**

Interval	Installed	Required	Procedure
C	2	0	(M)

May be inoperative provided the inoperative valve(s) is locked open.

MAINTENANCE (M)

Lock the inoperative nacelle zone ventilation valve in the open position (AMM 75-00-00/901).

1. For left engine nacelle vent valve, open and collar P110 panel L NAC ZONE VENT circuit breaker.
2. For right engine nacelle vent valve, open and collar P210 panel R NAC ZONE VENT circuit breaker.
3. Deactivate the associated engine thrust reverser for ground maintenance (AMM 78-31-00/201).
4. Open associated left fan cowl panel (AMM 71-11-04/201).
5. Open associated left thrust reverser (AMM 78-31-00/201).
6. Gain access to the nacelle zone ventilation valve (engine left thrust reverser half inner wall at 11:00).
7. Lock the valve open by following the MANUALLY LOCKING OPEN instructions placard on the valve.
8. Close associated left thrust reverser (AMM 78-31-00/201).
9. Close associated left fan cowl panel (AMM 71-11-04/201).
10. Activate the associated engine thrust reverser after ground maintenance (AMM 78-31-00/201).

75-23-02 Core Compartment Cooling Valves (GE)

Interval	Installed	Required	Procedure
C	2	0	(M)

May be inoperative provided the inoperative valve(s) is locked open.

MAINTENANCE (M)

NOTE 1: For GE90-100 Series, GE Service Bulletin 75-0022 removes the Core Compartment Cooling (CCC) valve, and GE Service Bulletin 75-0039 installs the improved CCC valve (P/N 2618M97P01).

NOTE 2: For All GE (Except GE90-100 Series), GE Service Bulletin 75-0058 modifies the Cooling (CCC) valve to a locked-open configuration.

Lock the inoperative Core Compartment Cooling (CCC) valve in the open position (AMM 75-00-00/901).

1. Deactivate the leading edge slats in the retracted position (AMM 27-81-00/201).
2. Deactivate the associated engine thrust reverser for ground maintenance (AMM 78-31-00/201).
3. Open the associated left and right fan cowl panels (AMM 71-11-04/201).
4. Open the associated right thrust reverser (AMM 78-31-00/201).
5. Lock the associated CCC valve (located on the forward engine core at 1:00) open by following the TO LOCK OPEN instructions placard on the valve.
6. Close the associated right thrust reverser (AMM 78-31-00/201).
7. Close the associated left and right fan cowl panels (AMM 71-11-04/201).
8. Activate the associated engine thrust reverser after ground maintenance (AMM 78-31-00/201).
9. Activate the leading edge slats (AMM 27-81-00/201).

75-24-01 Turbine Case Cooling Air Flow Systems (PW and GE)**75-24-01-01 PW****75-24-01-01-01 HPT Systems**

Interval	Installed	Required	Procedure
C	2	0	(M) (O)

May be inoperative provided associated turbine case cooling air valve is deactivated in the closed position.

MAINTENANCE (M)

Deactivate the inoperative turbine case cooling air flow valve in the closed position (AMM 75-00-00/901).

1. Deactivate the associated engine thrust reverser for ground maintenance (AMM 78-31-00/201).
2. Open associated left and right fan cowl panels (AMM 71-11-04/201).
3. Open associated right thrust reverser (AMM 78-31-00/201).
4. The turbine case cooling air flow valves are located on the engine LPT case at 2:00.
5. Disconnect the end of the HPT control cable from the HPT valve lever, and then remove the end of the HPT control cable. Install four washers (P/N MS9320-09) on lever bolt and then re-install nut.
6. Valve position (closed) can be checked by looking down the air inlet cooling duct (AC48).
7. Deactivate the HPT air valve actuator torque motor return signals for channels A and B.
 - A. Position associated Overhead Maintenance Panel (P61) EEC MAINT switch to NORM.
 - B. Disconnect electrical connector DM75004 from the TCC valve actuator (M75004).
 - C. Disassemble the connector backshell.
 - D. Disengage the wire from pin location #2 and install in unused socket location #26.
 - E. Disengage the wire from pin location #8 and install in unused socket location #25.
 - F. Reassemble the connector backshell.
 - G. Connect electrical connector DM75004 to the TCC valve actuator (M75004).
 - H. Insert temporary identification marker stating the wiring harness was reworked.
 - I. Erase the associated TURB CASE COOL HPT L or R status message (AMM 31-61-00/201).



- J. Confirm the associated status message TURB CASE COOL HPT L or R is not displayed.
 - K. Position associated Overhead Maintenance Panel (P61) EEC MAINT switch TEST.
 - L. Wait 30 seconds, then confirm the associated TURB CASE COOL HPT L or R status message is displayed.

NOTE: Display of the associated status message ensures the correct wires have been moved in the HPT air valve actuator connector.
 - M. Position associated Overhead Maintenance Panel (P61) EEC MAINT switch NORM.
- 8. Close associated right thrust reverser (AMM 78-31-00/201).
 - 9. Close associated left and right fan cowl panels (AMM 71-11-04/201).
 - 10. Activate the associated engine thrust reverser after ground maintenance (AMM 78-31-00/201).

OPERATIONS (O)

- 1. Do not exceed FL 400.
- 2. Increase flight planning fuel by 0.8% for each inoperative system.
- 3. For ETOPS flight planning, increase engine inoperative Critical Fuel Reserves by 1.0%.

75-24-01 Turbine Case Cooling Air Flow Systems (PW and GE)**75-24-01-01 PW****75-24-01-01-02 LPT Systems**

Interval	Installed	Required	Procedure
C	2	0	(M) (O)

May be inoperative provided:

- a. Associated turbine case cooling air valve is deactivated in the closed position.
 - b. Appropriate performance adjustments are applied.
-

MAINTENANCE (M)

Deactivate the inoperative turbine case cooling air flow valve in the closed position (AMM 75-00-00/901).

- 1. Deactivate the associated engine thrust reverser for ground maintenance (AMM 78-31-00/201).
- 2. Open associated left and right fan cowl panels (AMM 71-11-04/201).
- 3. Open associated right thrust reverser (AMM 78-31-00/201).
- 4. The turbine case cooling air flow valves are located on the engine LPT case at 2:00.

5. For engines with an idler arm on the HPT air valve:
 - A. Disconnect the LPT valve link from the idler arm and stow.
 - B. Install four washers (P/N MS9320-09) on lever bolt and then reinstall nut.
6. For engines without an idler arm on the HPT air valve:
 - A. Disconnect the end of the LPT control cable from the LPT valve lever.
 - B. Remove the end of the LPT control cable.
 - C. Install four washers (P/N MS9320-09) on lever bolt and then re-install nut.
7. Valve position (closed) can be checked by looking down the air inlet cooling duct (AC48).
8. Close associated right thrust reverser (AMM 78-31-00/201).
9. Close associated left and right fan cowl panels (AMM 71-11-04/201).
10. Activate the associated engine thrust reverser after ground maintenance (AMM 78-31-00/201).

OPERATIONS (O)

NOTE: For APU-to-Pack takeoff function installed, the APU-to-Pack takeoff Supplementary Procedure is not allowed.

1. Reduce performance limited weights by the appropriate adjustments.

Performance Limited Weights	Takeoff & Landing	Enroute Climb
--	----------------------------------	--------------------------

For -200 or -200ER/PW4074 thru PW4084:

All	2,200 lb (998 kg)	2,600 lb (1,179 kg)
-----	-------------------	---------------------

For -200 or -200ER/PW4090:

thru 632,500 lb (286,902 kg)	2,900 lb (1,315 kg)	2,300 lb (1,043 kg)
over 632,500 lb (286,902 kg)	3,100 lb (1,406 kg)	2,300 lb (1,043 kg)

For -300/PW4090:

All	3,000 lb (1,361 kg)	2,200 lb (998 kg)
-----	---------------------	-------------------

NOTE: The enroute climb weight penalties listed above are based on single engine operating speeds that approximate maximum lift-to-drag ratio speed. To account for the difference in level off altitude when operating at other speeds, multiply the enroute climb weight penalty listed above by the appropriate factor listed on page 3.00-01-00.2 in the ENROUTE DIVERSION SPEED EFFECTS paragraph.



2. For both valves deactivated closed, increase flight planning fuel by 0.8%.
3. For ETOPS flight planning, increase engine inoperative Critical Fuel Reserves by 0.9%.

75-24-01 Turbine Case Cooling Air Flow Systems (PW and GE)**75-24-01-02 GE****75-24-01-02-01 LPT Systems**

Interval	Installed	Required	Procedure
C	2	0	(M) (O)

May be inoperative provided:

- a. Associated LPT ACC valve is locked in the closed position.
- b. Both EECs operate in the normal mode.

MAINTENANCE (M)

Lock the inoperative LPT Active Clearance Control (ACC) valve in the closed position (AMM 75-00-00/901).

1. Deactivate the leading edge slats in the retracted position (AMM 27-81-00/201).
2. Deactivate the associated engine thrust reverser for ground maintenance (AMM 78-31-00/201).
3. Lock the associated LPT ACC valve closed:
 - A. Open associated left and right fan cowl panels (AMM 71-11-04/201).
 - B. Open associated right thrust reverser (AMM 78-31-00/201).
 - C. Lock the inoperative LPT ACC valve (located on the engine core at 3:00) in the closed position by following the placard instructions on the valve.
 - D. Close associated right thrust reverser (AMM 78-31-00/201).
 - E. Close associated left and right fan cowl panels (AMM 71-11-04/201).
4. Activate the associated engine thrust reverser after ground maintenance (AMM 78-31-00/201).
5. Activate the leading edge slats (AMM 27-81-00/201).

OPERATIONS (O)

1. For both valves deactivated closed, increase flight planning fuel by 0.8%.
2. For ETOPS flight planning, increase engine inoperative Critical Fuel Reserves by 1.8%.

75-24-02 Turbine Cooling Air Systems (PW)
75-24-02A One Inoperative

Interval	Installed	Required	Procedure
C	2	1	(M)

One may be inoperative provided associated Turbine Vane and Blade Cooling (TVBC) air shutoff valves are deactivated in the open position.

MAINTENANCE (M)

Deactivate the associated Turbine Vane and Blade Cooling (TVBC) air shutoff valves in the open position (AMM 75-00-00/901).

1. Deactivate the associated engine thrust reverser for ground maintenance (AMM 78-31-00/201).
2. Open associated left and right fan cowl panels (AMM71-11-04/201).
3. Open associated left and right thrust reversers (AMM 78-31-00/201).
4. Check that the three TVBC air shutoff valves are open (popup indicators are in the out position). The three TVBC air shutoff valves are located as follows: right side of HPC case at 2:30 and left side of HPC case at 9:00 and 11:00.
5. Remove all loop clamps and loosen both end fittings on tubes AA19, AA43 and AA44, and tube AA95 if installed (AMM 75-25-00/Figure 601 and AMM 75-25-01/Figures 401, 402 and 403).

NOTE: Tube AA95 was installed on early PW4074, PW4077 and PW4084, and all PW4090 model engines. Engine serial numbers P777001 thru P777085, and all engine serial numbers P222XXX were delivered with tube AA95 installed. However, for engine serial numbers P777001 thru P777085, incorporation of SB PW4G-112-72-209 removes tube AA95. Engine serial numbers P777086 and higher, and engine serial numbers P208XXX were delivered without tube AA95.

6. Remove tubes AA19, AA43 and AA44, and tube AA95 if installed.
7. Install cap (P/N MS9314-06) on both filter fittings and tighten to 200-225 in-lbs.
8. Install cap (P/N MS9314-06) on all valve fittings and tighten to 200-225 in-lbs.

NOTE: Install cap on valve fitting P/N ST2010-06 located on valve P/N 796641 which is the attach point for tube AA95, if installed.

9. Close associated left and right thrust reversers (AMM 78-31-00/201).
10. Close associated left and right fan cowl panels (AMM71-11-04/201).



11. Activate the associated engine thrust reverser after ground maintenance (AMM 78-31-00/201).

75-24-02 Turbine Cooling Air Systems (PW)**75-24-02B Both Inoperative**

Interval	Installed	Required	Procedure
C	2	0	(M) (O)

May be inoperative provided Turbine Vane and Blade Cooling (TVBC) air shutoff valves are deactivated in the open position.

MAINTENANCE (M)

Use DDG item 75-24-02A (M) procedure.

OPERATIONS (O)

Increase flight planning fuel by 0.6%.



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77-11-01 Engine Pressure Ratio Indicating Systems (PW and RR)

77-11-01-01 PW

77-11-01-02 RR

77-12-01 N2 Tachometer Systems (RR)

77-21-01 Pyrometer Exhaust Gas Temperature (EGT) Indication System (GE Except GE90-100 Series)

77-22-01 Engine Turbine Overheat Sensors (RR)

77-31-01 Engine Vibration Monitor Systems



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**77-11-01 Engine Pressure Ratio Indicating Systems (PW and RR)****77-11-01-01 PW**

Interval	Installed	Required	Procedure
C	2	0	(O)

May be inoperative provided:

- Both engines must be operated in the alternate (ALTN) mode.
 - Appropriate performance adjustments are applied.
-

MAINTENANCE NOTE

For engines without PW SB 4G-112-73-29 or production equivalent installed (EEC software SCN8C or later) and EEC C1 faults present, use DDG item 73-21-02-01 Maintenance Note procedure.

OPERATIONS (O)

Use DDG item 73-21-02-01 (O) procedure.

77-11-01 Engine Pressure Ratio Indicating Systems (PW and RR)**77-11-01-02 RR**

Interval	Installed	Required	Procedure
C	2	0	(O)

May be inoperative provided:

- N2 indication on affected engine operates normally.
 - Both engines must be operated in the alternate (ALTN) mode.
 - Appropriate performance adjustments are applied.
-

OPERATIONS (O)

Use DDG item 73-21-02-03 (O) procedure.

**77-12-01 N2 Tachometer Systems (RR)**

Interval	Installed	Required	Procedure
B	2	1	

One may be inoperative provided EPR and Fuel Flow operate normally.

**77-21-01 Pyrometer Exhaust Gas Temperature (EGT)
Indication System (GE Except GE90-100 Series)**

Interval	Installed	Required	Procedure
C	2	0	(M)

May be inoperative provided:

- a. Inoperative EGT pyrometer indication system is deactivated.
- b. Associated engine EGT indications are verified to be normal.

MAINTENANCE (M)

NOTE: This MMEL item provides relief for failures of the pyrometer EGT indication system that result in fluctuating or erratic flight deck EGT indications.

Deactivate the inoperative pyrometer EGT indication system and verify the associated engine EGT indications are normal (77-00-00/901).

1. Deactivate the inoperative pyrometer EGT indication system by disconnecting the associated pyrometer electronic module.
 - A. Establish electrical power on the airplane (AMM 24-22-00/201).
 - B. Ensure associated EEC MAINT L/R switch on the Aft Overhead Maintenance Panel (P61) is positioned to NORM.
 - C. Retract the leading edge slats if extended (AMM 27-81-00/201).
 - D. Deactivate the leading edge slats (AMM 27-81-00/201).
 - E. Deactivate the associated engine thrust reverser for ground maintenance (AMM-78-31-00/201).
 - F. Open the associated engine left and right fan cowl panels (AMM 71-11-04/201).
 - G. Open the associated engine right thrust reverser (AMM 78-31-00/201).
 - H. Disconnect the electrical connectors from the inoperative pyrometer electronic module that go to the associated EEC.

NOTE: The pyrometer EGT indication system is installed on the right side of the engine at approximately the 3:30 o'clock position (AMM 77-21-03/401).

- 1) Disconnect the channel A and B electrical connectors (DP72103 and DP72203) at the pyrometer electronic module (M77036).

- 2) Install metal screw-on connector covers/caps on to the electrical harness connectors and the pyrometer electronic module receptacles. A total of four caps are required.
 - a. For the electrical harness connector, the metal covers/caps must mate to SNECMA P/N XB1170-61212FN equivalent to EN2997SE61212FN.
 - b. For the pyrometer electronic module receptacle, the metal covers/caps must mate to Pyle National P/N BNY-1712P4D-Y186.

NOTE: PLASTIC CAPS ARE NOT ACCEPTABLE. P/Ns 83723-22R12 and 83723-21R12 or equivalent parts are acceptable provided the caps are metal and screw-on type.

- 3) Stow the disconnected electrical connectors.
- I. Close the associated engine right thrust reverser (AMM 78-31-00/201).
- J. Close the associated engine left and right fan cowl panels (AMM 71-11-04/201).
- K. Activate the associated engine thrust reverser after ground maintenance (AMM-78-31-00/201).
- L. Activate the leading edge slats (AMM 27-81-00/201).
2. Verify associated engine EGT indications are normal.
 - A. Position associated EEC MAINT L/R switch on the Aft Overhead Maintenance Panel (P61) to TEST.
 - B. Verify that the associated engine EGT readout on the flight deck EICAS display is displayed and is not fluctuating, erratic or intermittent.
 - C. Position associated EEC MAINT L/R switch on the Aft Overhead Maintenance Panel (P61) to NORM.



77-22-01 Engine Turbine Overheat Sensors (RR)

Interval	Installed	Required	Procedure
C	4	2	

One per engine may be inoperative.

**77-31-01 Engine Vibration Monitor Systems**

Interval	Installed	Required	Procedure
C	2	1	

MAINTENANCE NOTE

If desired, the inoperative engine vibration monitor system may be deactivated (AMM 77-00-00/901).

1. For left engine vibration monitor inoperative, open and collar P110 panel L ENG VIB MON circuit breaker.
2. For right engine vibration monitor inoperative, open and collar P210 panel R ENG VIB MON circuit breaker.



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78-31-01-02 -200LR/-300ER/777F

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78-34-01B *Interlock Retracted*

78-36-01 Reverser Proximity Sensors



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78-31-01 Thrust Reversers
78-31-01-01 -200/-200ER/-300

Interval	Installed	Required	Procedure
C	2	1	(M)

One may be inoperative provided:

- Both sync locks are verified in the locked position.
- One locking actuator on each sleeve is verified in the locked position.
- Inoperative reverser is secured in the forward thrust position.

MAINTENANCE (M)

Deactivate and secure the inoperative reverser (AMM 78-00-00/901).

- For left engine reverser, open and collar P110 panel L ENG T/R CTRL circuit breaker.
- For right engine reverser, open and collar P210 panel R ENG T/R CTRL circuit breaker.

For GE:

- Deactivate the leading edge slats in the retracted position (AMM 27-81-00/201).
- Gain access to the associated reverser isolation valve (left forward access door - strut aft fairing).
- Deactivate the isolation valve for flight by following the DEACTIVATED MODE instructions placard.
- Close the associated left forward access door - strut aft fairing.
- Open the associated left and right fan cowl panels (AMM 71-11-04/201).
- Remove deactivation pins and install the pins in the left and right reverser sleeves.
- Pressurize the associated hydraulic system (AMM 29-11-00/201).
- Verify that the sync lock and at least one of the two locking actuators on each reverser sleeve has the manual unlock handle in the locked position.
- Depressurize the hydraulic system (AMM 29-11-00/201).
- Verify that associated engine thrust reverser left and right sleeve positions are within limits using the Electronic Propulsion Control System (EPCS) maintenance page 1/2.
 - If the associated engine thrust reverser left and right sleeve (T/RL and T/RR) positions for channels A and B are not within the following allowable values, deactivate the associated RVDT.

NOTE: A dual channel RVDT is installed on the non-locking (upper) hydraulic actuator of each sleeve, which results in four readouts per engine.



For PW:

- 1) 0 to +4.0

For GE:

- 2) -7.0 to +7.0

For RR:

- 3) -5.0 to +5.0

- B. For deactivating the associated RVDT:

NOTE: ONLY ONE RVDT MAY BE DEACTIVATED.

- 1) Disconnect the electrical connector DM78002 for left sleeve or DM78003 for right sleeve from the RVDT.
- 2) Cap the connector plug using conductive cap.
- 3) Stow the connector plug to adjacent structure using wire ties.

13. Close the associated left and right fan cowl panels (71-11-04/201).

For GE:

14. Activate the leading edge slats (AMM 27-81-00/201).

OPERATIONS NOTE

After engine start, the associated ENG REV LIMITED L or R advisory message will be displayed.

78-31-01 Thrust Reversers

78-31-01-02 -200LR/-300ER/777F

Interval	Installed	Required	Procedure
C	2	1	(M) (O)

One may be inoperative provided:

- a. Both sync locks are verified in the locked position.
- b. One locking actuator on each sleeve is verified in the locked position.
- c. Inoperative reverser is secured in the forward thrust position.
- d. Appropriate performance adjustments are applied.

MAINTENANCE (M)

Deactivate and secure the inoperative reverser (AMM 78-00-00/901).

1. For left engine reverser, open and collar P110 panel L ENG T/R CTRL circuit breaker.
2. For right engine reverser, open and collar P210 panel R ENG T/R CTRL circuit breaker.
3. Deactivate the leading edge slats in the retracted position (AMM 27-81-00/201).

4. Gain access to the associated reverser isolation valve (left forward access door - strut aft fairing).
5. Deactivate the isolation valve for flight by following the DEACTIVATED MODE instructions placard.
6. Close the associated left forward access door - strut aft fairing.
7. Open the associated left and right fan cowl panels (AMM 71-11-04/201).
8. Remove deactivation pins and install the pins in the left and right reverser sleeves.
9. Pressurize the associated hydraulic system (AMM 29-11-00/201).
10. Verify that the sync lock and at least one of the two locking actuators on each reverser sleeve has the manual unlock handle in the locked position.
11. Depressurize the hydraulic system (AMM 29-11-00/201).
12. Verify that associated engine thrust reverser left and right sleeve positions are within limits using the Electronic Propulsion Control System (EPCS) maintenance page 1/2.
 - A. If the associated engine thrust reverser left and right sleeve (T/RL and T/RR) positions for channels A and B are not within -7.0 to +7.0, deactivate the associated RVDT.

NOTE: A dual channel RVDT is installed on the non-locking (upper) hydraulic actuator of each sleeve, which results in four readouts per engine.
 - B. For deactivating the associated RVDT:

NOTE: ONLY ONE RVDT MAY BE DEACTIVATED.

 - 1) Disconnect the electrical connector DM78002 for left sleeve or DM78003 for right sleeve from the RVDT.
 - 2) Cap the connector plug using conductive cap.
 - 3) Stow the connector plug to adjacent structure using wire ties.
13. Close the associated left and right fan cowl panels (71-11-04/201).
14. Activate the leading edge slats (AMM 27-81-00/201).

OPERATIONS (O)

NOTE: After engine start, the associated ENG REV LIMITED L or R advisory message will be displayed.

For airplane performance calculations that include the effect of reverse thrust, adjust performance to account for the inoperative thrust reverser.

**78-34-01 Reverse Thrust Lever Interlocks****78-34-01A Interlock Released**

Interval	Installed	Required	Procedure
C	2	1	

One may be inoperative provided the interlock is released.

OPERATIONS NOTE

When using reverse thrust, pull the reverser levers up to the normal interlock position and apply reverse thrust as required when the operative interlock releases.

78-34-01 Reverse Thrust Lever Interlocks**78-34-01B Interlock Retracted**

Interval	Installed	Required	Procedure
C	2	1	(O)

One may be inoperative provided:

- Inoperative interlock is retracted.
- Appropriate performance adjustments are applied.

OPERATIONS (O)

NOTE 1: After engine start, the ENG REV LIMITED L or R advisory message may be displayed for the associated engine.

NOTE 2: When using reverse thrust, pull the reverse levers up to the normal interlock position and maintain idle reverse, unless asymmetric reverse thrust is required.

[For -200/-200ER/-300:](#)

NOTE 3: No performance adjustment is required.

[For -200LR/-300ER/777F:](#)

For airplane performance calculations that include the effect of reverse thrust, adjust performance to account for the inoperative thrust reverser.



78-36-01 Reverser Proximity Sensors

Interval	Installed	Required	Procedure
C	14	12	

One per engine may be inoperative.



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79-21-01 Engine Air/Oil Heat Exchanger Valves (RR)

79-21-02 Engine Fuel/Oil Cooler Bypass Valves (PW)

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79-35-01 Engine Oil Filter Bypass Warning Systems

79-35-01-01 PW

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**79-21-01 Engine Air/Oil Heat Exchanger Valves (RR)**

Interval	Installed	Required	Procedure
C	2	1	(M) (O)

May be inoperative provided:

- Associated valve is locked in the open position.
- Appropriate minimum fuel temperature is maintained during flight.

MAINTENANCE (M)

Lock the inoperative air/oil heat exchanger valve in the open position (AMM 79-00-00/901).

- Gain access to the air/oil heat exchanger valve, which is located behind and below the engine oil tank, by opening the right fan cowl panel on the associated engine (AMM 71-11-04/201).
- Lock the air/oil heat exchanger in the open position.
 - For RR engines not incorporating RR SB 79-C620.
 - Remove the cotter pin from the lockpin on the air/oil heat exchanger valve.
 - Pull the lockpin out and remove the lockpin spacer (retain spacer for restoration).
 - Turn the manual turning device in the clockwise direction until valve is fully open and hold in this position until the lockpin has been installed (Use caution as the valve is heavily spring loaded).
 - Install the lockpin (without spacer) and release the manual turning device.
 - Install the cotter pin through the upper hole in the lockpin.
 - Verify that valve is locked by trying to turn the manual turning device in the counter-clockwise direction.
 - For RR engines incorporating RR SB 79-C620.
 - Loosen the manual lock detent screw.
 - Push the manual lock detent into the air oil heat exchanger body until you feel a click.

NOTE: There should be a red area on the butterfly shaft below the head.

- Tighten the manual lock detent screw to 11-13 in-lbs.
- Close the associated right fan cowl panel (AMM 71-11-04/201).

OPERATIONS (O)

- For ETOPS flight planning, increase engine inoperative Critical Fuel Reserves by 0.6%.



For RR875 thru RR892:

2. During flight, maintain fuel temperature above -40 degrees C.

For RR895:

3. During flight, maintain fuel temperature above -37 degrees C.

79-21-02 Engine Fuel/Oil Cooler Bypass Valves (PW)

Interval	Installed	Required	Procedure
C	2	0	

May be inoperative provided the inoperative valve(s) is in the closed position.

.....

**79-31-01 Engine Oil Quantity Indicating Systems**

Interval	Installed	Required	Procedure
A	2	1	(M)

One may be inoperative provided:

- It is verified before each departure that the oil tank is filled to the recommended capacity.
 - Oil consumption is within limits.
 - Repairs are made within 3 flight days.
-

MAINTENANCE (M)

For GE (Except GE90-70/80/90 Series with SB 72-0324, and GE90-100 Series):

NOTE: Oil quantity indication is operating normally if switching the leads at the sensor restores the oil quantity indication (AMM 79-31-00/601).

Before each departure verify that oil level is filled to the recommended level (AMM 79-00-00/901).

- Before each departure, examine oil level and fill if required (AMM 12-13-01/301).
- Review engine records to verify that oil consumption is within engine manufacturer's limits (AMM 71-00-00/201) or operator's approved limits (if more restrictive).

79-35-01 Engine Oil Filter Bypass Warning Systems
79-35-01-01 PW

Interval	Installed	Required	Procedure
C	2	1	(M)

One may be inoperative provided associated master chip detector is checked for contaminants before each departure.

MAINTENANCE (M)

Perform a Master Magnetic Chip Detector - Inspection/Check before each departure (AMM 79-22-00/601).

79-35-01 Engine Oil Filter Bypass Warning Systems
79-35-01-02 GE

Interval	Installed	Required	Procedure
C	2	1	

79-35-01 Engine Oil Filter Bypass Warning Systems
79-35-01-03 RR

Interval	Installed	Required	Procedure
C	4	2	(M)

One per engine may be inoperative provided:

- a. It is verified that the malfunction is in the alerting system.
- b. Associated master chip detector is checked for contaminants before each departure.

MAINTENANCE (M)

Verify that the malfunction is in the alerting system and inspect the master magnetic chip detector before each departure (AMM 79-00-00/901).

1. Verify that the malfunction is in the alerting system.
 - A. If the MAT is available:
 - 1) Confirm the following maintenance message is displayed:
 - a. For left engine scavenge filter, 79-84911.
 - b. For right engine scavenge filter, 79-84912.
 - c. For left engine HP filter, 79-74411.

-
- d. For right engine HP filter, 79-74412.
 - 2) Confirm the following maintenance message is not displayed:
 - a. For left engine scavenge filter, 79-94861.
 - b. For right engine scavenge filter, 79-94862.
 - c. For left engine HP filter, 79-94361.
 - d. For right engine HP filter, 79-94362.
 - B. If the MAT is not available:
 - 1) Open the left fan cowl panel and gain access to the associated engine Electronic Engine Control (EEC) (AMM 71-11-04/201).
 - 2) For the ENG SCAV OIL FILT L or ENG SCAV OIL FILT R status message displayed, perform a continuity check of the scavenge oil filter differential pressure switch.
 - a. Disconnect electrical terminal DM73003G from the EEC.
 - b. Measure the resistance between pin 23 and pin 24 on electrical connector DM73003G.
 - c. If the resistance is less than 100 ohms, the malfunction is in the alerting system.
 - d. Measure the resistance between pin 24 and ground.
 - e. If the resistance is less than 100 Kohms, the malfunction is in the alerting system.
 - f. Connect the electrical connector DM73003G to the EEC (AMM 70-50-02/201).
 - 3) For the ENG HP OIL FILT L or ENG HP OIL FILT R status message displayed, perform a continuity check of the HP oil filter differential pressure switch.
 - a. Disconnect electrical terminal DM73003A from the EEC.
 - b. Measure the resistance between pin 23 and pin 24 on electrical connector DM73003A.
 - c. If the resistance is less than 100 ohms, the malfunction is in the alerting system.
 - d. Measure the resistance between pin 24 and the connector backshell.
 - e. If the resistance is less than 100 Kohms, the malfunction is in the alerting system.
 - f. Connect the electrical connector DM73003A to the EEC (AMM 70-50-02/201).
 - 4) Close the associated left fan cowl panel (AMM 71-11-04/201).
 2. Perform a Master Magnetic Chip Detector Inspection before each departure (AMM 79-00-00/601).



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80-11-02-01 Autostart Switch Position Indication

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80-11-02-02-02 ON Light

80-11-03 Start Selector Holding/Cutout Systems



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80-11-01 Engine Start Valves

Interval	Installed	Required	Procedure
C	2	1	(M) (O)

One may be inoperative provided:

- Inoperative valve is closed.
- Alternate starting procedures are established and used.

MAINTENANCE (M)

Manually open and close the valve as directed by the flight crew (AMM 80-00-00/901).

For PW:

- Gain access to the guide tube for the manual override receptacle by opening the associated engine center thrust reverser latch access door (415MB for left engine or 425MB right engine).
- Install a starter drive extension (40 inches - one piece) through the guide tube and into the manual override receptacle.
- Establish communications with the flight crew.
- Upon command from the flight crew, use the starter drive extension to turn the starter air valve to the open position and hold. Wear goggles, gloves and protective clothing.
- Upon command from the flight crew, close the starter air valve.
- When the manual override start is complete, remove the starter drive extension from the guide tube and close the center thrust reverser latch access door.

For GE:

- Gain access to the starter air valve manual control handle by opening associated engine center thrust reverser latch access panel (415JB for left engine or 425JB right engine).
- Unstow the starter air valve manual control handle.
- Establish communications with the flight crew.
- Upon command from the flight crew, engage the starter air valve manual control handle, turn the starter air valve to the open position (clockwise - one and one-quarter turns) and hold. Wear goggles, gloves and protective clothing.
- Upon command from the flight crew, close the starter air valve.
- When the manual override start is complete, stow the starter air valve manual control handle and close the center thrust reverser latch access panel.

**For RR**

1. Gain access to the manual override start port located on the left fan cowl panel.
2. Establish communications with the flight crew.
3. Upon command from the flight crew, insert the manual start square drive in the manual override start port, turn the starter air valve to the open position and hold. Wear goggles, gloves and protective clothing.
4. Upon command from the flight crew, close the starter air valve.
5. When the manual override start is complete, remove the manual override square drive from the manual override start port.

OPERATIONS (O)

1. Start the associated engine using normal engine start (Autostart) procedures, with directions to ground crew for manually opening and closing the inoperative engine start valve.
 - A. Establish communications with ground crew.
 - B. Position engine start selector to START and advise ground crew to open start valve.
 - C. Advise ground crew to close start valve as follows:

For PW:

- 1) 42% N2

For GE:

- 2) 62% N2

For RR:

- 3) 50% N3

2. Confirm the ENGINE STARTER CUTOUT L/R caution message is not displayed after engine start.

NOTE: If the ENGINE STARTER CUTOUT L/R caution message is displayed after engine start, the starter air valve may not be closed. The starter air valve must be closed for dispatch.

3. For inflight start of the associated engine, use a windmilling start.



80-11-02 Autostart System

Interval	Installed	Required	Procedure
C	1	0	(O)

May be inoperative provided manual start procedures are established and used.

NOTE: When the AUTOSTART switch is selected OFF, the ENG AUTOSTART OFF advisory message is normally displayed. However, some system failures can result in the ENG AUTOSTART OFF advisory message not being displayed when the AUTOSTART switch is selected OFF. If the autostart system is known to be inoperative, dispatch per the autostart system inoperative provisions of this item. If the autostart system is otherwise operational, consider the autostart switch position indication inoperative and dispatch the airplane using MMEL item 80-11-02-01.

OPERATIONS (O)

1. Use the Manual Engine Start Supplementary Procedure.
2. Leave AUTOSTART switch OFF.

80-11-02 Autostart System**80-11-02-01 Autostart Switch Position Indication**

Interval	Installed	Required	Procedure
C	1	0	

May be inoperative provided engine autostart switch is selected ON.

80-11-02 Autostart System**80-11-02-02 AUTOSTART Switch Lights****80-11-02-02-01 OFF Light**

Interval	Installed	Required	Procedure
C	1	0	

**80-11-02 Autostart System****80-11-02-02 AUTOSTART Switch Lights****80-11-02-02-02 ON Light**

Interval	Installed	Required	Procedure
C	1	0	

80-11-03 Start Selector Holding/Cutout Systems

Interval	Installed	Required	Procedure
C	2	0	(O)

May be inoperative provided alternate start procedures are used.

MAINTENANCE NOTE

For AIMS status displayed due to certain failure modes of the start selector holding/cutout relay, the AIMS status message must be cleared using the following procedure.

1. Remove power from both AIMS cabinets by opening P11 panel AIMS L-1, AIMS L-2, AIMS L-3, AIMS L-4, AIMS R-1, AIMS R-2, AIMS R-3 and AIMS R-4 circuit breakers.
2. Remove the Cargo Fire/Engine Control Module (M80001) from the P5 panel without disconnecting the electrical connectors (AMM 26-22-04/401).
3. Disconnect, cap and stow the associated engine start selector wire:
 - A. For the left engine:
 - 1) Disconnect electrical connector DS80100 from selector S80100.
 - 2) For Engine Control Panel (P5) start selector with a CON position, remove, cap and stow wire W1051-0557-24 from pin 7.
 - 3) For Engine Control Panel (P5) start selector without a CON position, remove, cap and stow wire W1051-0557-24 from pin 2.
 - 4) Re-install electrical connector DS80100 on selector S80100.
 - B. For the right engine:
 - 1) Disconnect electrical connector DS80200 from selector S80200.
 - 2) For Engine Control Panel (P5) start selector with a CON position, remove, cap and stow wire W1052-0557-24 from pin 7.
 - 3) For Engine Control Panel (P5) start selector without a CON position, remove, cap and stow wire W1052-0557-24 from pin 2.
 - 4) Re-install electrical connector DS80200 on selector S80200.
4. Re-install the Cargo Fire/Engine Control Module (M80001) on the P5 panel (AMM 26-22-04/401).

NOTE: It is only necessary to do the installation test for the Engine Start panel. However, if any module electrical connectors were disconnected other than the START/IGNITION or START selector connector, then the associated installation test should be accomplished.

5. Restore power to the AIMS cabinets by closing P11 panel AIMS L-1, AIMS L-2, AIMS L-3, AIMS L-4, AIMS R-1, AIMS R-2, AIMS R-3 and AIMS R-4 circuit breakers.
6. Verify that the AIMS status message has cleared and no additional status messages associated with the Cargo Fire/Engine Control Module are displayed.
7. For engine starting, the start selector must be manually held in the START position.

OPERATIONS (O)

Manually open and close the associated engine start valve.

1. Position start selector to START and hold if required.

For PW:

2. Position start selector to NORM at 42% N2.

For GE:

3. Position start selector to NORM at 62% N2.

For RR:

4. Position start selector to NORM at 50% N3.
5. Confirm the ENGINE STARTER CUTOFF L/R caution message is not displayed after engine start.

NOTE: If the ENGINE STARTER CUTOFF L/R caution message is displayed after engine start, the starter air valve may not be closed. The starter air valve must be closed for dispatch.



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CDL

Introduction

ATA 21 - Air Conditioning

ATA 23 - Communications

ATA 32 - Landing Gear

ATA 33 - Lights

ATA 52 - Doors

ATA 54 - Nacelles and Pylons

ATA 55 - Stabilizer

ATA 56 - Windows

ATA 57 - Wings

ATA 71 - Powerplant

ATA 72 - Engine

ATA 78 - Engine Exhaust



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General Notes

This section contains locations, illustrations and performance information for all of the CDL items from the 777 Airplane Flight Manual (AFM) Appendix Configuration Deviation List (CDL) Revision 5.

Limitations

The associated limitations must be listed on a placard affixed in the cockpit in clear view of the pilot-in-command and other appropriate crew members.

Operation with those missing parts requiring a reduction of VMO/MMO is permitted only when the airplane has the maximum airspeed limit indication and the Mach airspeed warning system programmed for the altitude/speed schedule specified for the applicable missing part.

The pilot in command will be notified of each operation with a missing parts by listing the missing parts in the flight or dispatch release.

The operator will list in the aircraft logbook an appropriate notation covering the missing parts on each flight.

If an additional part is lost in flight the airplane may not depart the airport at which it landed following this event until it again complies with CDL limitations. This, does not preclude the issuance of a ferry permit to allow the airplane to be flown to a point where the necessary repairs or replacements can be made.

Unless otherwise specified, combinations of parts from any system or sub-systems may be missing. Combinations of parts from systems or sub-systems that are not allowed will be specifically listed in the CDL.

Weight Reductions

The performance penalties are cumulative unless specifically designated penalties for combination of missing parts are indicated. Where performance penalties are listed as negligible, no more than three negligible items may be missing without taking further penalty. For each missing item more than three, reduce the takeoff, landing and enroute climb limits by 100 pounds (46 kilograms). Where performance penalties are listed as no penalty, any accumulative number of items listed as no penalty may be missing without further penalty.

Enroute Diversion Speed Effects

The enroute climb weight penalties listed are based on operating speeds that approximate the maximum lift-to-drag ratio speed. To account for the difference in level off altitude when operating at other speeds, multiply the enroute climb weight penalty listed by the following appropriate factor:

Diversion Speed	Factor
LRC	1.5
260 KIAS	1.7
280 KIAS	2.3
300 KIAS	3.1
320 KIAS	3.7

Enroute Fuel Mileage Effects

The drag effects of many 777 CDL items are so small that the changes in flight planning fuel are negligible. For items that have enroute climb weight penalties listed, an increase in flight planning fuel of 0.25% per 1,000 lb (454 kg) of enroute climb weight penalty (non-factored penalty) may be used to account for the drag increase.



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ATA 21

21-32-01 Negative Pressure Relief Vent

21-52-01 ECS Ram Air Exhaust Louver



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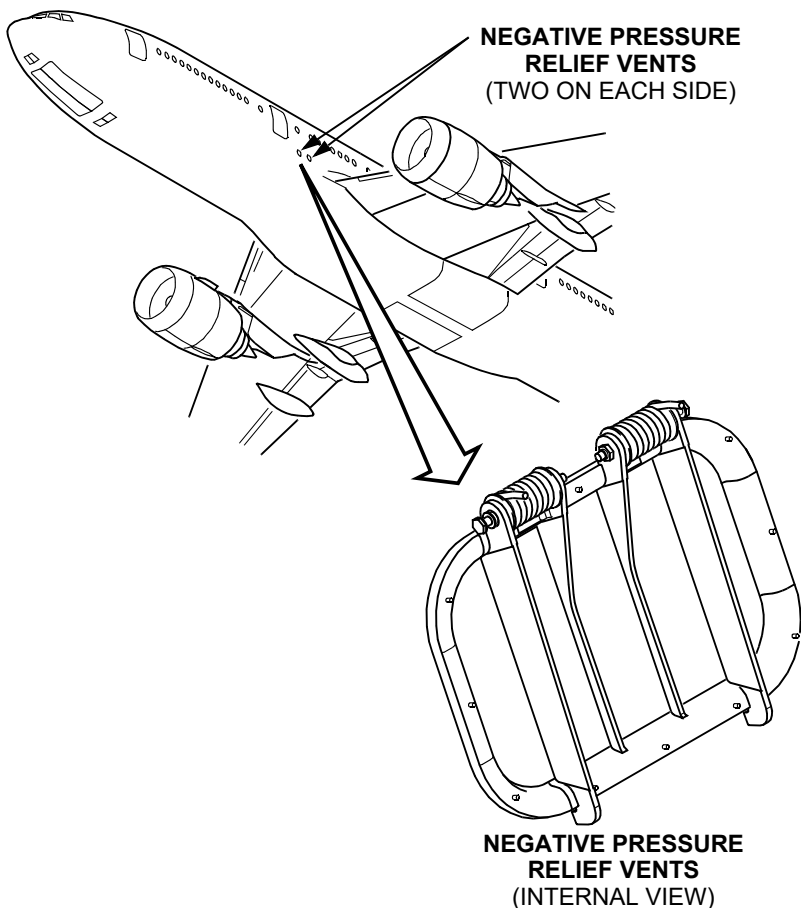
21-32-01 Negative Pressure Relief Vent

Any number may be missing.

Dispatch using MMEL item 21-32-02.

Performance limited weights are reduced by the following:

Number Installed	Takeoff & Landing	Enroute Climb
4	Negligible penalty	Negligible penalty



21-52-01 ECS Ram Air Exhaust Louver

Any number may be missing.

NOTE: PRR 61661 or SB 777-21-0011 removed ram air exhaust louvers on some -200/-300 airplanes.

Performance limited weights are reduced by the following:

Number Installed	Takeoff	Enroute Climb	Approach/Landing Climb
------------------	---------	---------------	------------------------

For -200/-200ER (With louvers installed, or PRR 61661/SB 777-21-0011 Not Incorporated):

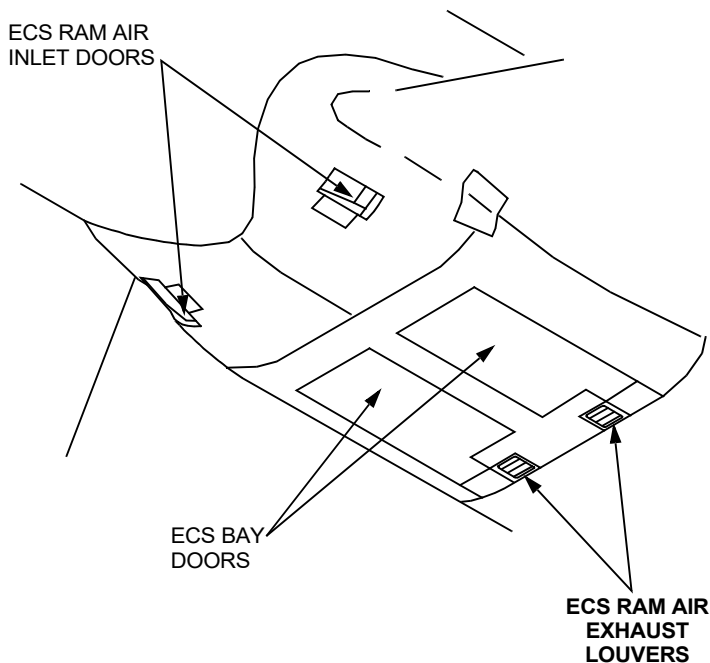
6	Negligible penalty	Negligible penalty	Negligible penalty
---	--------------------	--------------------	--------------------

For -200LR/777F:

6	200 lb (91 kg)	450 lb (205 kg)	150 lb (69 kg)
---	----------------	-----------------	----------------

For -300ER and -300 (with louvers installed):

6	No Penalty	No Penalty	No Penalty
---	------------	------------	------------





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ATA 23

23-61-01 Static Discharger



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23-61-01 Static Discharger

A maximum of five static dischargers may be missing, with the following restrictions:

1. No more than three of the five most outboard dischargers on each wing may be missing.
2. No more than two of the four most outboard dischargers on each horizontal stabilizer may be missing.
3. No more than two of the four most top dischargers on the vertical stabilizer may be missing.

NOTE: For operations with a wing tip fairing removed (CDL item 57-31-02) or for operations with the raked tips removed (CDL item 57-31-03), the static dischargers that are removed with these items are not considered missing. The remaining static dischargers must satisfy the requirements of this item.

Performance limited weights are reduced by the following:

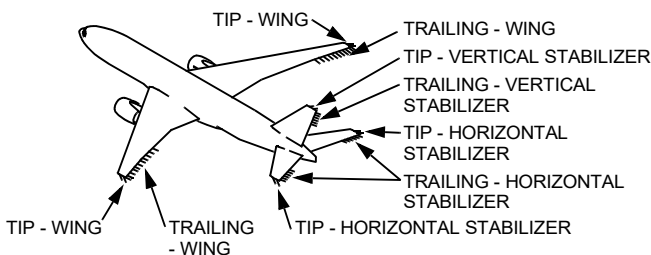
Number Installed	Takeoff & Landing	Enroute Climb
-------------------------	------------------------------	----------------------

For -200/-200ER/-300:

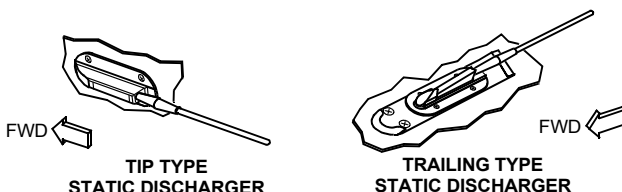
50	No penalty	No penalty
----	------------	------------

For -200LR/777F/-300ER:

54	No penalty	No penalty
----	------------	------------



STATIC DISCHARGER LOCATIONS





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ATA 32

32-12-01 Main Landing Gear Drag Strut Door

32-12-02 Main Landing Gear Trunnion Door

32-45-01 Wheel Hubcaps



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**32-12-01 Main Landing Gear Drag Strut Door**

One may be missing.

NOTE: This item may not be missing in combination with items CDL 32-12-02 or CDL 57-51-01.

The maximum airspeed limits (VMO/MMO) are 270 KIAS/0.73 Mach.

NOTE: Place Alternate Gear Down Dispatch switch (Main Equipment Center, if installed) to VMO position. The Warning Electronics System (WES) will set VMO/MMO to 270 KIAS/.73 Mach and EICAS memo message VMO GEAR DOWN will be displayed.

Performance limited weights are reduced by the following:

Number Installed	Takeoff	Enroute Climb	Approach/Landing Climb
------------------	---------	---------------	------------------------

For -200/-200ER/PW4084:

2	2,400 lb (1,089 kg)	6,000 lb (2,722 kg)	150 lb (69 kg)
---	---------------------	---------------------	----------------

For -200/-200ER/ PW4090:

2	2,700 lb (1,225 kg)	5,400 lb (2,450 kg)	150 lb (69 kg)
---	---------------------	---------------------	----------------

For -200/-200ER/GE90-85B:

2	2,700 lb (1,225 kg)	4,500 lb (2,042 kg)	150 lb (69 kg)
---	---------------------	---------------------	----------------

For -200/-200ER/GE90-90B:

2	3,000 lb (1,361 kg)	5,100 lb (2,314 kg)	200 lb (91 kg)
---	---------------------	---------------------	----------------

For -200/-200ER/GE90-94B:

2	3,000 lb (1,361 kg)	6,000 lb (2,722 kg)	150 lb (69 kg)
---	---------------------	---------------------	----------------

For -200/-200ER/RR884:

2	2,100 lb (953 kg)	4,500 lb (2,042 kg)	150 lb (69 kg)
---	-------------------	---------------------	----------------

For -200/-200ER/RR892:

2	3,300 lb (1,497 kg)	5,100 lb (2,314 kg)	150 lb (69 kg)
---	---------------------	---------------------	----------------



For -200/-200ER/RR895:

2	2,400 lb (1,089 kg)	5,100 lb (2,314 kg)	250 lb (114 kg)
---	---------------------	---------------------	-----------------

For -300/PW4090:

2	2,400 lb (1,089 kg)	5,400 lb (2,450 kg)	150 lb (69 kg)
---	---------------------	---------------------	----------------

For -300/PW4098:

2	3,600 lb (1,633 kg)	5,400 lb (2,450 kg)	150 lb (69 kg)
---	---------------------	---------------------	----------------

For -300/RR892:

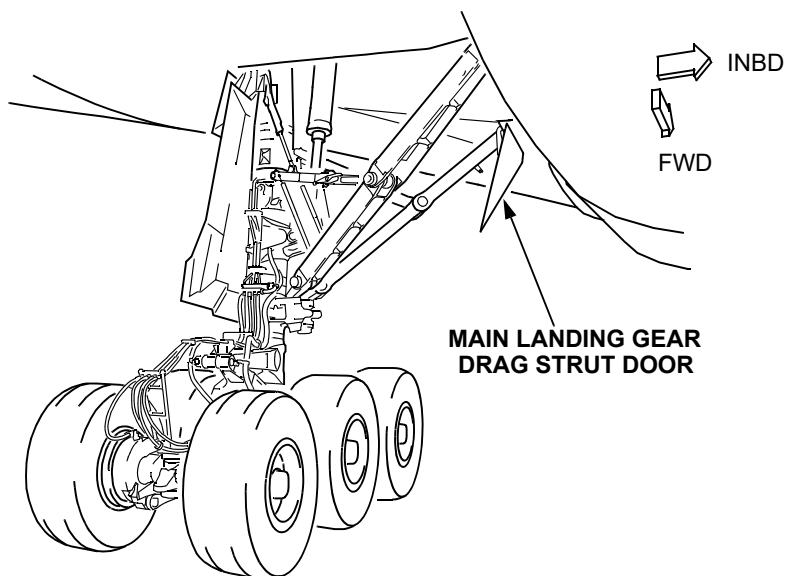
2	2,400 lb (1,089 kg)	4,800 lb (2,178 kg)	150 lb (69 kg)
---	---------------------	---------------------	----------------

For -300ER:

2	2,600 lb (1,180 kg)	5,400 lb (2,450 kg)	150 lb (69 kg)
---	---------------------	---------------------	----------------

For -200LR/777F:

2	2,700 lb (1,225 kg)	5,100 lb (2,314 kg)	200 lb (91 kg)
---	---------------------	---------------------	----------------



LEFT MAIN LANDING GEAR

**32-12-02 Main Landing Gear Trunnion Door**

One may be missing.

NOTE: This item may not be missing in combination with items CDL 32-12-01 or CDL 57-51-01.

The maximum airspeed limits (VMO/MMO) are 270 KIAS/0.73 Mach.

NOTE: Place Alternate Gear Down Dispatch switch (Main Equipment Center, if installed) to VMO position. The Warning Electronics System (WES) will set VMO/MMO to 270 KIAS/.73 Mach and EICAS memo message VMO GEAR DOWN will be displayed.

Performance limited weights are reduced by the following:

Number Installed	Takeoff	Enroute Climb	Approach/Landing Climb
------------------	---------	---------------	------------------------

For -200/-200ER/PW4084:

2	2,450 lb (1,112 kg)	6,050 lb (2,745 kg)	150 lb (69 kg)
---	---------------------	---------------------	----------------

For -200/-200ER/PW4090:

2	2,750 lb (1,248 kg)	5,450 lb (2,473 kg)	150 lb (69 kg)
---	---------------------	---------------------	----------------

For -200/-200ER/GE90-85B:

2	2,750 lb (1,248 kg)	4,550 lb (2,064 kg)	150 lb (69 kg)
---	---------------------	---------------------	----------------

For -200/-200ER/GE90-90B:

2	3,050 lb (1,384 kg)	5,150 lb (2,337 kg)	250 lb (114 kg)
---	---------------------	---------------------	-----------------

For -200/-200ER/GE90-94B:

2	3,050 lb (1,384 kg)	6,050 lb (2,745 kg)	150 lb (69 kg)
---	---------------------	---------------------	----------------

For -200/-200ER/RR884:

2	2,150 lb (976 kg)	4,550 lb (2,064 kg)	150 lb (69 kg)
---	-------------------	---------------------	----------------

For -200/-200ER/RR892:

2	3,350 lb (1,520 kg)	5,150 lb (2,337 kg)	200 lb (91 kg)
---	---------------------	---------------------	----------------

**Section 3****777 Dispatch Deviations Guide****ATA 32****For -200/-200ER/RR895:**

2	2,450 lb (1,112 kg)	5,150 lb (2,337 kg)	250 lb (114 kg)
---	---------------------	---------------------	-----------------

For -300/PW4090:

2	2,450 lb (1,112 kg)	5,450 lb (2,473 kg)	150 lb (69 kg)
---	---------------------	---------------------	----------------

For -300/PW4098:

2	3,650 lb (1,656 kg)	5,450 lb (2,473 kg)	150 lb (69 kg)
---	---------------------	---------------------	----------------

For -300/RR892:

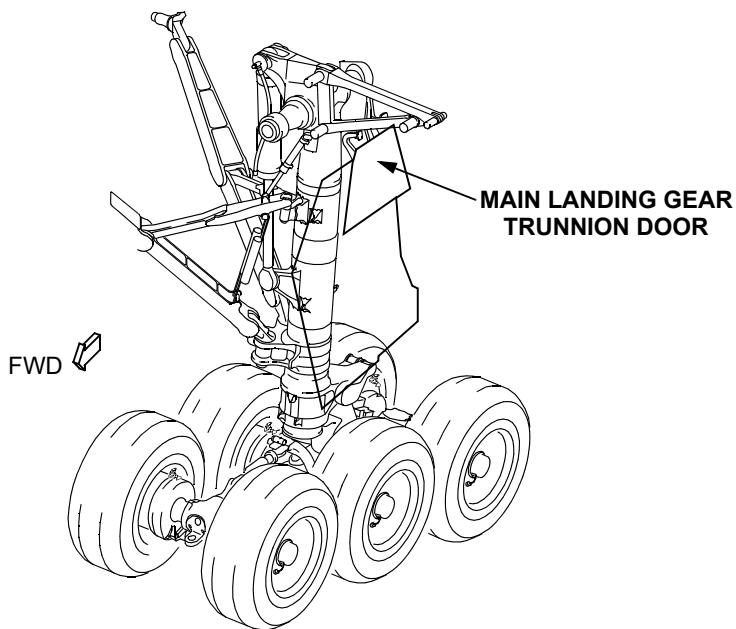
2	2,450 lb (1,112 kg)	4,850 lb (2,200 kg)	150 lb (69 kg)
---	---------------------	---------------------	----------------

For -300ER:

2	2,600 lb (1,180 kg)	5,450 lb (2,473 kg)	150 lb (69 kg)
---	---------------------	---------------------	----------------

For -200LR/777F:

2	2,750 lb (1,248 kg)	5,150 lb (2,337 kg)	200 lb (91 kg)
---	---------------------	---------------------	----------------



LEFT MAIN LANDING GEAR

32-45-01 Wheel Hubcaps

One main wheel hubcap per each six wheel main gear truck may be missing.

NOTE 1: For associated Antiskid Wheelspeed Transducer inoperative, dispatch using MMEL item 32-42-02.

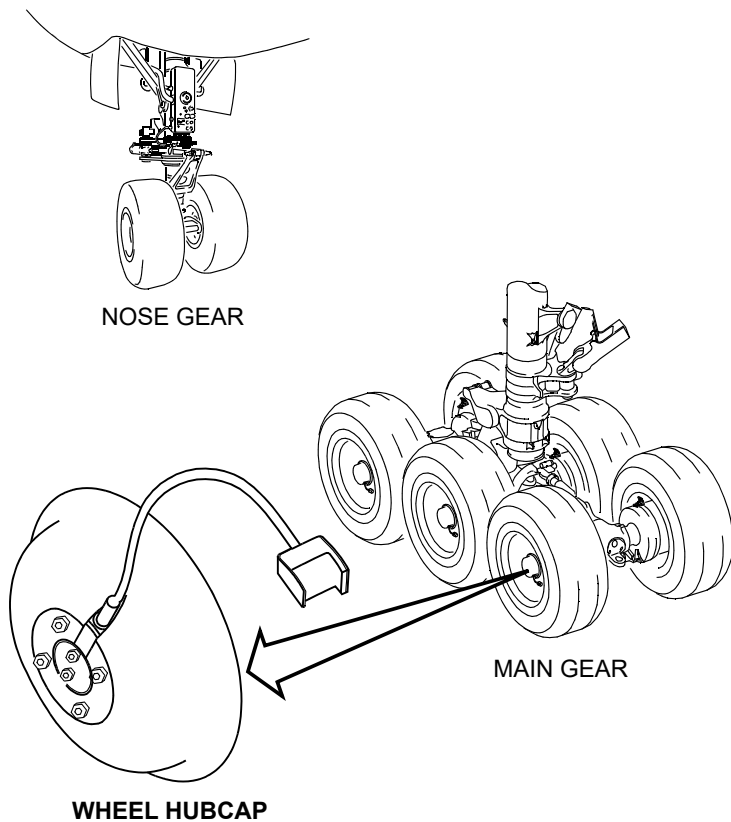
NOTE 2: For associated Tire Pressure Indication System inoperative, dispatch using MMEL item 32-49-01.

One or both nose wheel hubcaps may be missing.

NOTE: For associated Tire Pressure Indication System inoperative, dispatch using MMEL item 32-49-01.

Performance limited weights are reduced by the following:

Number Installed	Takeoff	Enroute Climb	Approach/Landing Climb
14	150 lb (69 kg)	200 lb (91 kg)	150 lb (69 kg)





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ATA 33

33-31-01	External Cargo Loading Light Lens
33-41-01	Wing Illumination Light Lens
33-42-01	Nose Gear Landing & Taxi Light
33-43-01	Wing Tip Rear Position Light Lens
33-43-02	Tail Cone Position Light Lens
33-44-01	Upper Or Lower Fuselage Anti-Collision Light Lens
33-44-02	Tail Cone Anti-Collision Light Lens
33-45-01	Logo Light Lens
33-51-01	External Emergency Light Lens



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33-31-01 External Cargo Loading Light Lens

Any number may be missing.

NOTE: External cargo loading lights operate normally with the lenses missing. If the associated lamp is inoperative, dispatch using MMEL Item 33-31-02.

Performance limited weights are reduced by the following:

Number Installed	Takeoff & Landing	Enroute Climb
-------------------------	------------------------------	----------------------

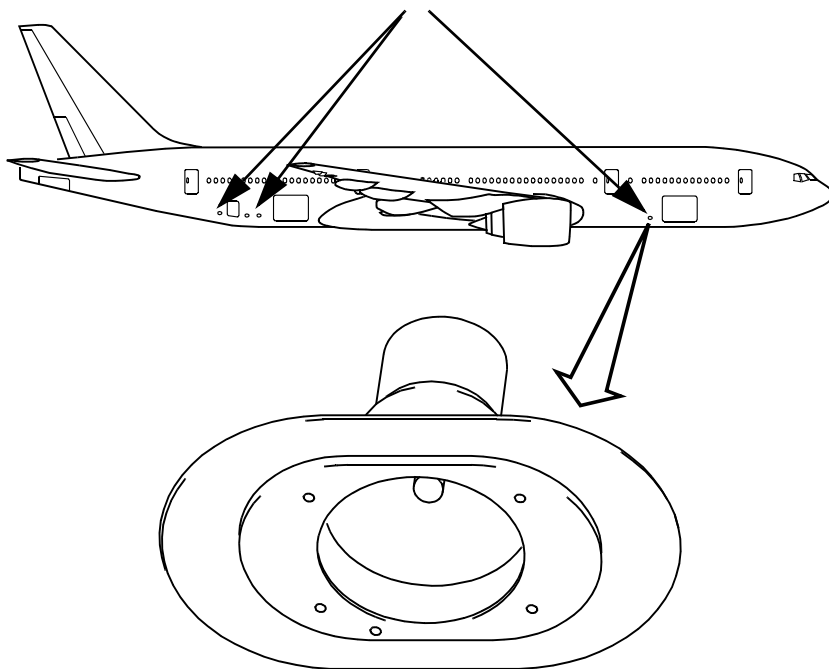
For Passenger:

4	Negligible penalty	Negligible penalty
---	--------------------	--------------------

For 777F:

5	Negligible penalty	Negligible penalty
---	--------------------	--------------------

**EXTERNAL CARGO LOADING LIGHT LENS
(777F MAIN DECK CARGO LIGHT NOT SHOWN)**



EXTERNAL CARGO LOADING LIGHT LENS

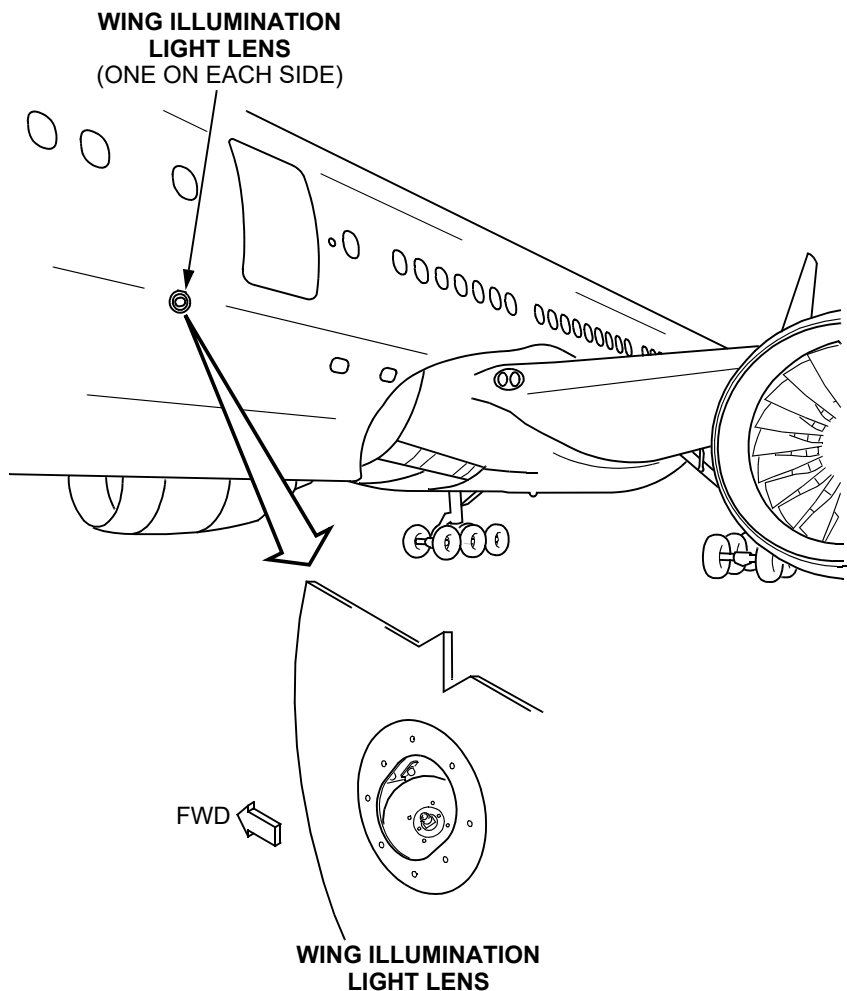
33-41-01 Wing Illumination Light Lens

Any number may be missing.

NOTE: Wing illumination lights operate normally with the lenses missing. If the associated lamp is inoperative, dispatch using MMEL Item 33-41-01.

Performance limited weights are reduced by the following:

Number Installed	Takeoff & Landing	Enroute Climb
2	Negligible penalty	Negligible penalty



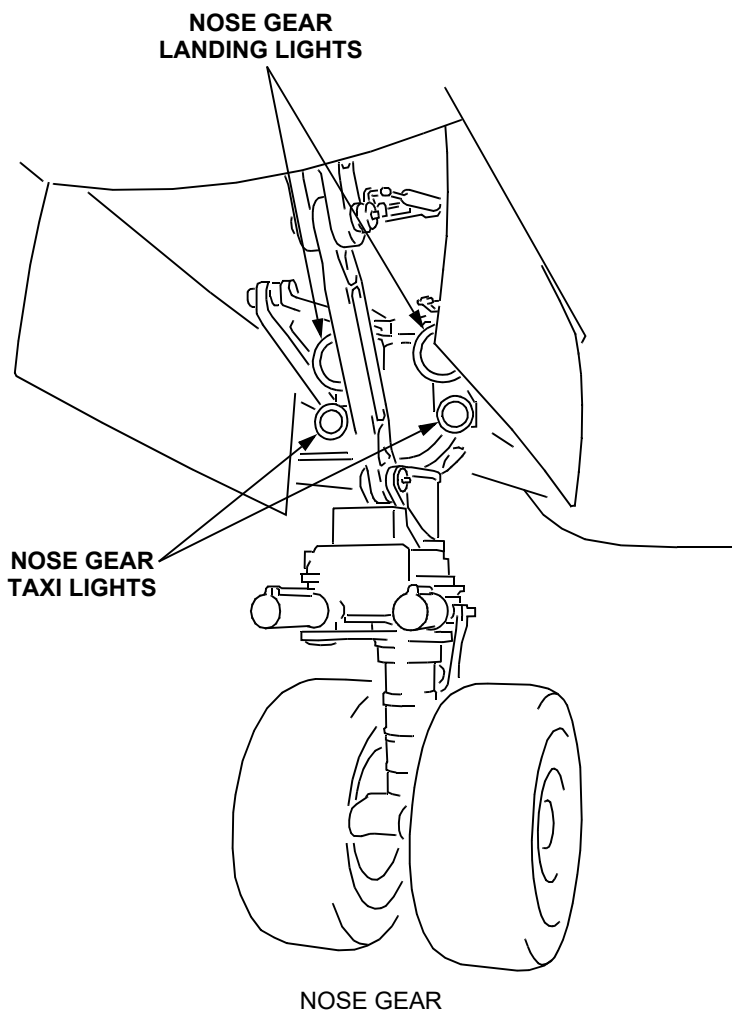
33-42-01 Nose Gear Landing & Taxi Light

Any number may be missing.

NOTE: For nose gear taxi lights inoperative, dispatch using MMEL Item 33-42-01. For nose gear landing lights inoperative, dispatch using MMEL Item 33-42-02.

Performance limited weights are reduced by the following:

Number Installed	Takeoff & Landing	Enroute Climb
4	No penalty	No penalty



33-43-01 Wing Tip Rear Position Light Lens

Any number may be missing.

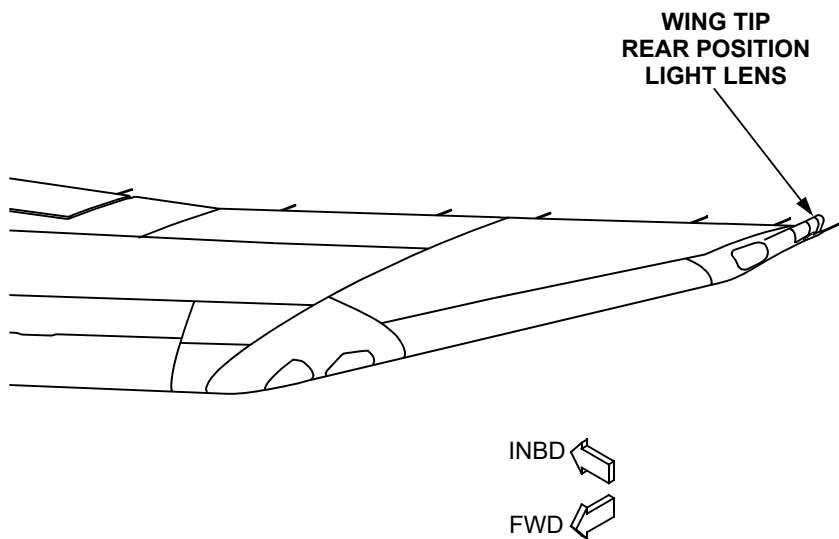
For a missing lens, disconnect electrical power and remove light bulb, reflector, lens cover and lens cover support. Cover the light assembly opening with speed tape.

NOTE 1: For position lights inoperative, dispatch using MMEL Item 33-43-01.

NOTE 2: If the L POS LTS circuit breaker is opened to deactivate the left position lights, the tail lights (if installed) will be inoperative, and MMEL item 33-43-02 may also be applied for dispatch.

Performance limited weights are reduced by the following:

Number Installed	Takeoff & Landing	Enroute Climb
2	No penalty	No penalty



33-43-02 Tail Cone Position Light Lens

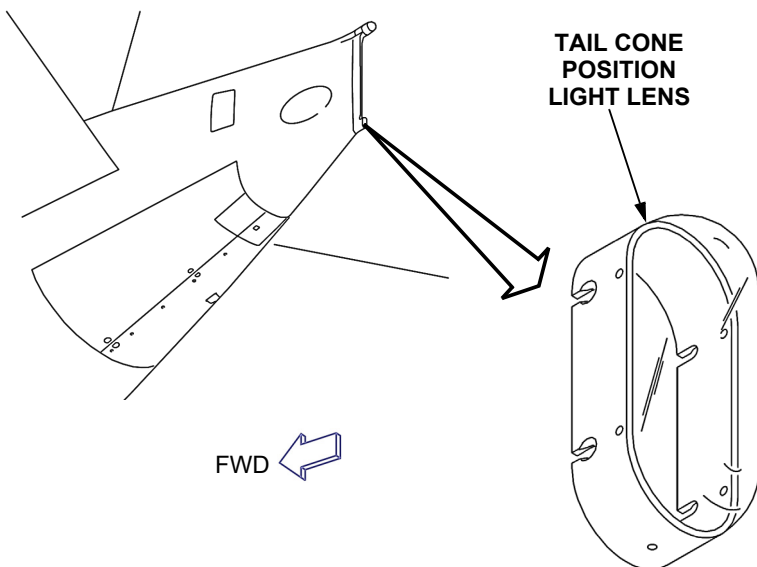
One may be missing.

NOTE 1: With the rest of the light assembly intact, cover the opening with speed tape. Pull and collar the Tail Cone Position Light circuit breaker, C33020.

NOTE 2: For tail cone position light inoperative, dispatch using MMEL Item 33-43-02.

Performance limited weights are reduced by the following:

Number Installed	Takeoff & Landing	Enroute Climb
1	Negligible penalty	Negligible penalty



33-44-01 Upper Or Lower Fuselage Anti-Collision Light Lens

Any number may be missing.

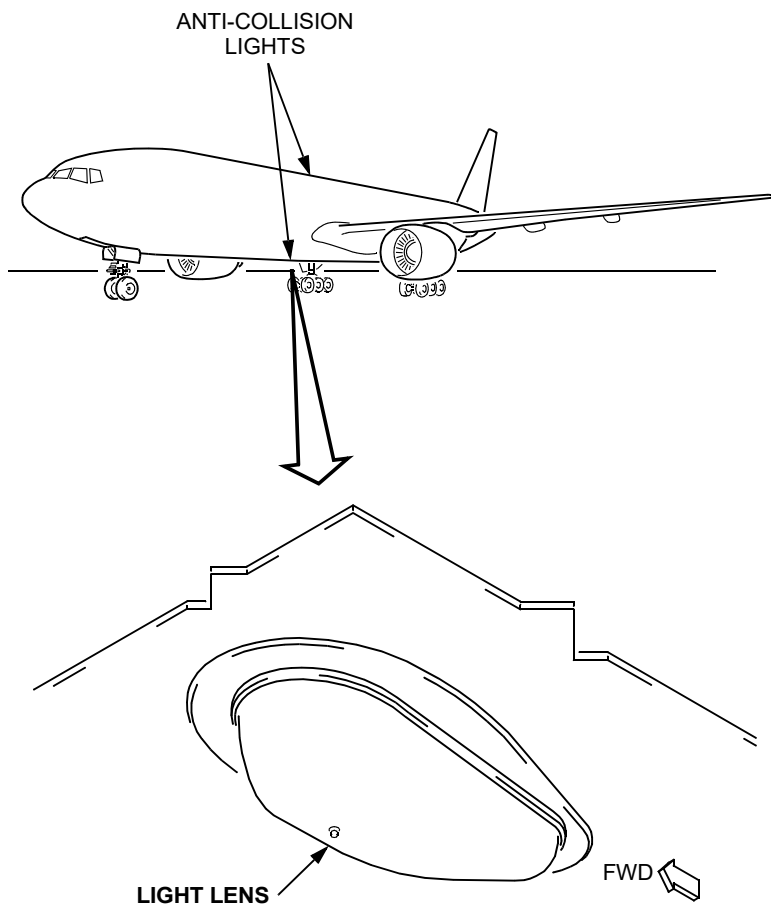
NOTE: This item may not be missing in combination with CDL item 33-44-02.

For a missing lens, disconnect electrical power and remove light bulb, reflector, lens cover and lens cover support. Cover the light assembly opening with speed tape.

NOTE: For anti-collision lights inoperative, dispatch using MMEL Item 33-44-01.

Performance limited weights are reduced by the following:

Number Installed	Takeoff & Landing	Enroute Climb
2	No penalty	No penalty



33-44-02 Tail Cone Anti-Collision Light Lens

One may be missing.

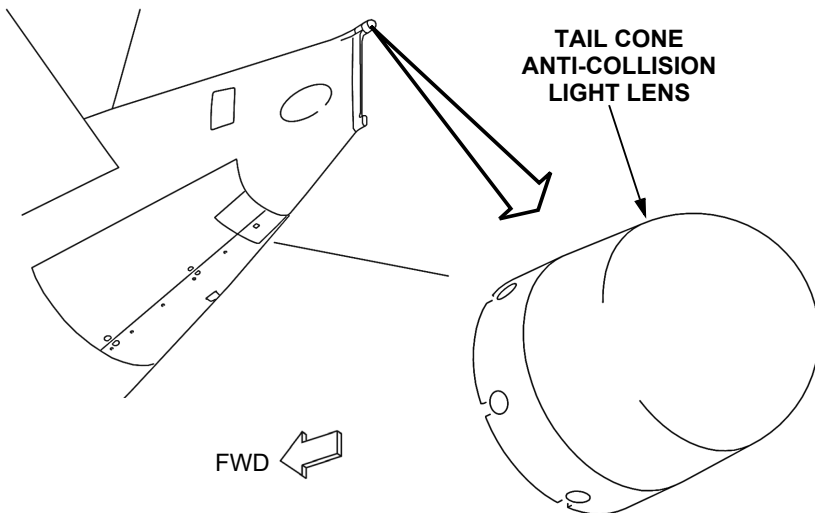
NOTE 1: When the lens is missing, unplug (cap and stow) the connector to the Tail Cone Anti-Collision Light. With the rest of the light assembly intact, cover the opening with speed tape.

NOTE 2: CDL item 33-44-01 may not be missing.

NOTE 3: For tail cone anti-collision light inoperative, dispatch using MMEL Item 33-44-01.

Performance limited weights are reduced by the following:

Number Installed	Takeoff & Landing	Enroute Climb
1	Negligible penalty	Negligible penalty



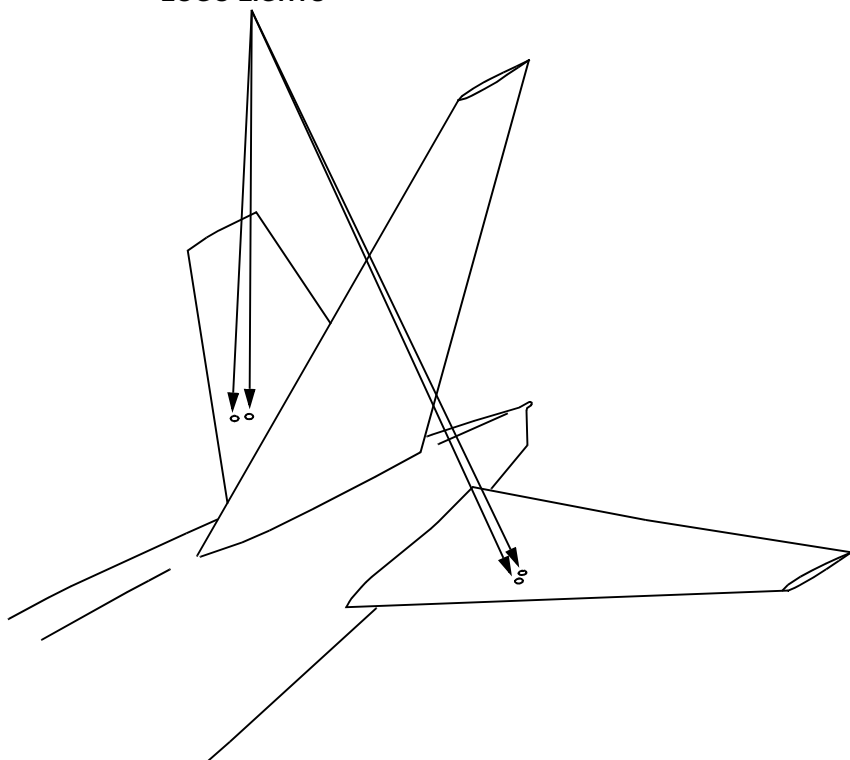
33-45-01 Logo Light Lens

Any number may be missing.

NOTE: Missing lens should be covered by speed tape (Permacel II or 3M 425) and inspected every other flight. Open and collar P210 panel LOGO LT CTRL and LOGO LTS circuit breakers. For logo lights inoperative, dispatch using MMEL Item 33-45-01.

Performance limited weights are reduced by the following:

Number Installed	Takeoff & Landing	Enroute Climb
4	Negligible penalty	Negligible penalty

LOGO LIGHTS


33-51-01 External Emergency Light Lens

One may be missing.

NOTE: Emergency slide lights do not operate normally with the lenses missing. If the associated lens is missing or the lamp is inoperative, dispatch using MMEL Item 33-51-02.

Performance limited weights are reduced by the following:

Number Installed	Takeoff & Landing	Enroute Climb
-------------------------	------------------------------	----------------------

For -200/-200ER/-200LR:

8	Negligible penalty	Negligible penalty
----------	---------------------------	---------------------------

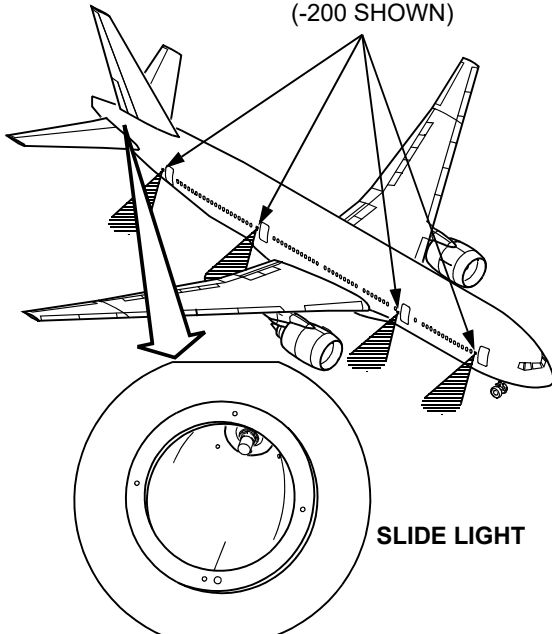
For -300/-300ER:

14	Negligible penalty	Negligible penalty
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For 777F:

2	Negligible penalty	Negligible penalty
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**EMERGENCY LIGHTING - SIDE LIGHT LOCATIONS
(-200 SHOWN)**





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- 52-11-01 Passenger Door Handle Covers**
- 52-35-01 Aft Small Cargo Door Control Door**
- 52-35-02 Aft Cargo Handling Control Door**
- 52-37-01 Aft Large Cargo Door Control Door**
- 52-37-02 Main Cargo Handling Control Door (777F)**
- 52-81-01 Main Landing Gear Door Heat Shield**
- 52-81-02 Main Landing Gear Door Bolt Head Cover Plugs**



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52-11-01 Passenger Door Handle Covers

Any number may be missing.

Performance limited weights are reduced by the following:

Number Installed	Takeoff & Landing	Enroute Climb
-------------------------	------------------------------	----------------------

For -200/-200ER/-200LR:

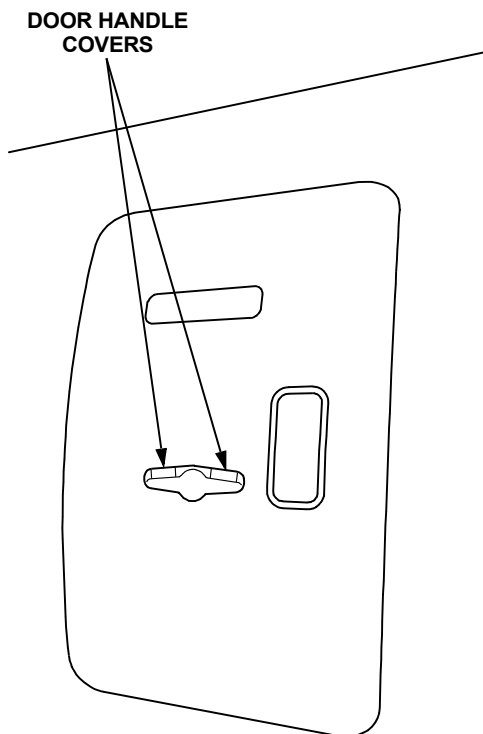
16	Negligible penalty	Negligible penalty
----	--------------------	--------------------

For -300/-300ER:

20	Negligible penalty	Negligible penalty
----	--------------------	--------------------

For 777F:

4	Negligible penalty	Negligible penalty
---	--------------------	--------------------

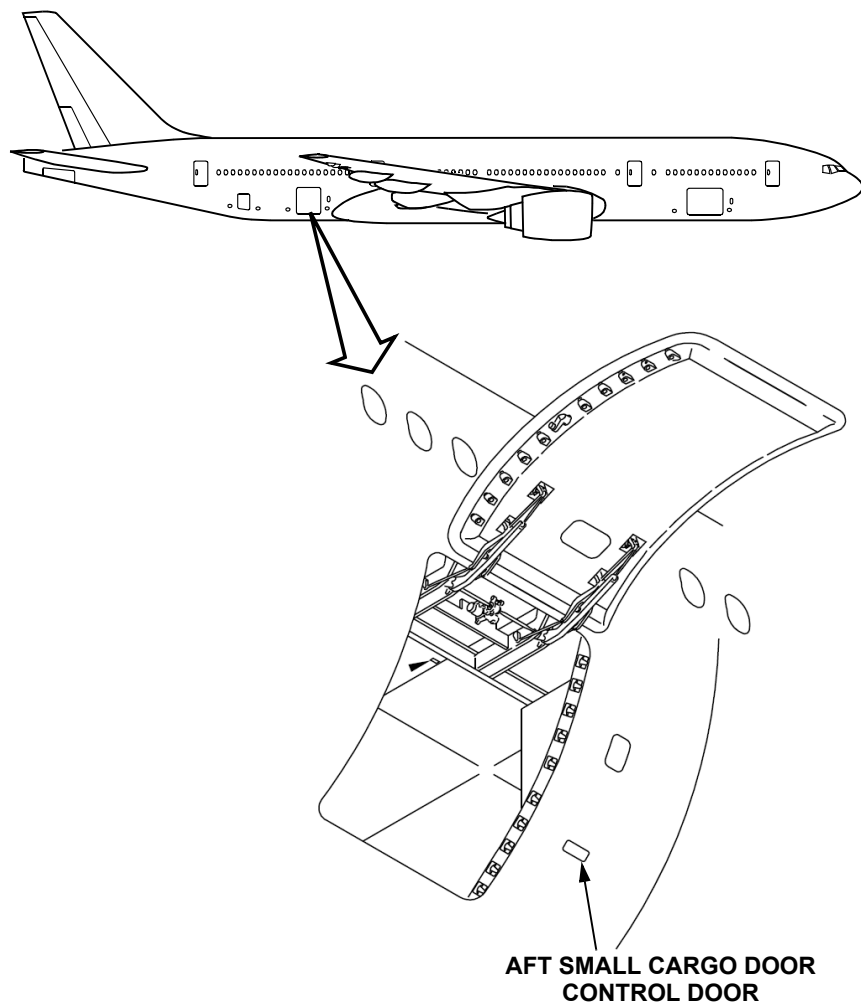


52-35-01 Aft Small Cargo Door Control Door

May be missing.

Performance limited weights are reduced by the following:

Number Installed	Takeoff & Landing	Enroute Climb
1	Negligible penalty	Negligible penalty

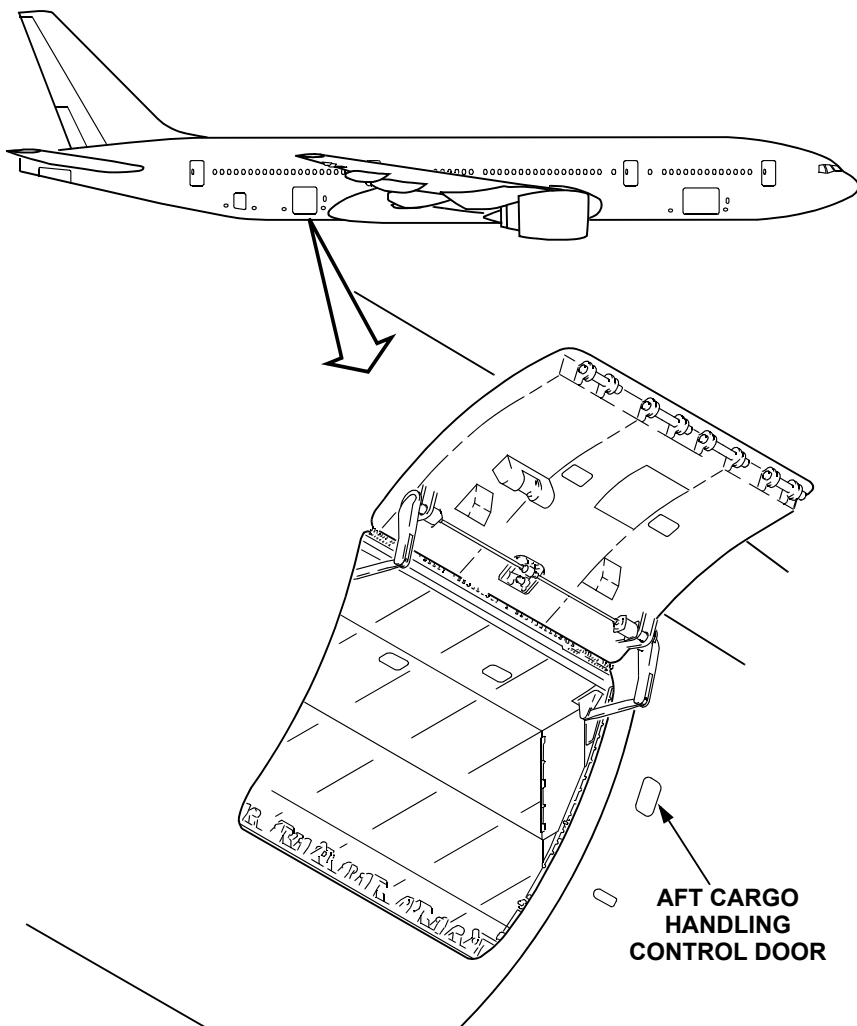


52-35-02 Aft Cargo Handling Control Door

May be missing.

Performance limited weights are reduced by the following:

Number Installed	Takeoff & Landing	Enroute Climb
1	Negligible penalty	Negligible penalty

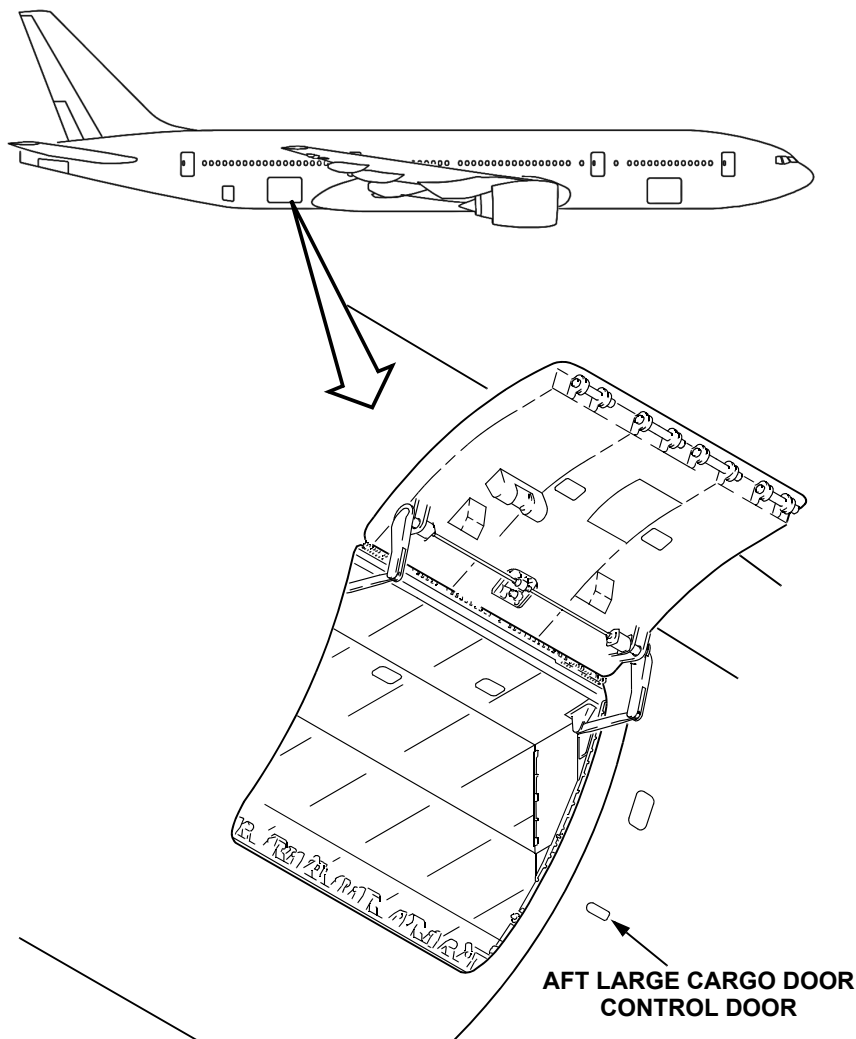


52-37-01 Aft Large Cargo Door Control Door

May be missing.

Performance limited weights are reduced by the following:

Number Installed	Takeoff & Landing	Enroute Climb
1	Negligible penalty	Negligible penalty

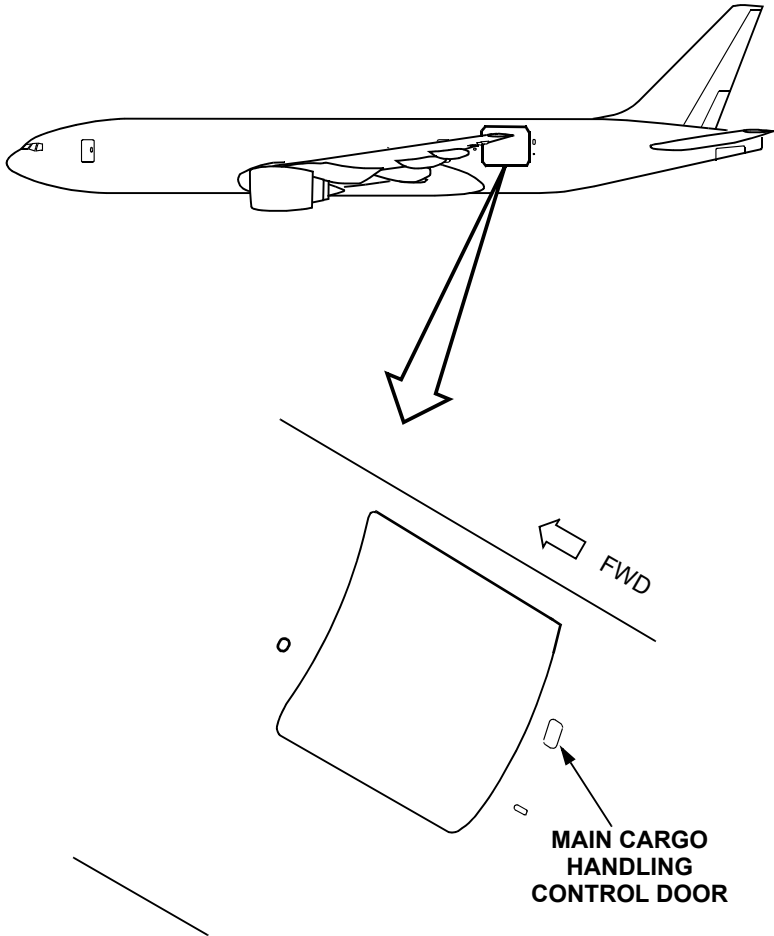


52-37-02 Main Cargo Handling Control Door (777F)

May be missing.

Performance limited weights are reduced by the following:

Number Installed	Takeoff & Landing	Enroute Climb
1	Negligible penalty	Negligible penalty



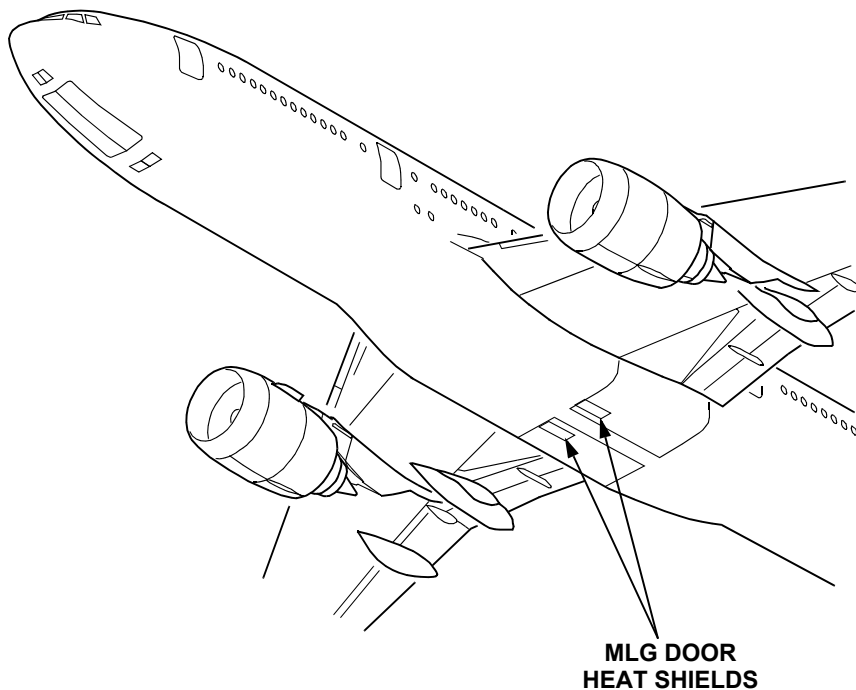
52-81-01 Main Landing Gear Door Heat Shield

One may be missing provided the associated air conditioning pack is considered to be inoperative.

NOTE: Dispatch using MMEL item 21-51-01 for one air conditioning pack inoperative.

Performance limited weights are reduced by the following:

Number Installed	Takeoff & Landing	Enroute Climb
2	Negligible penalty	Negligible penalty

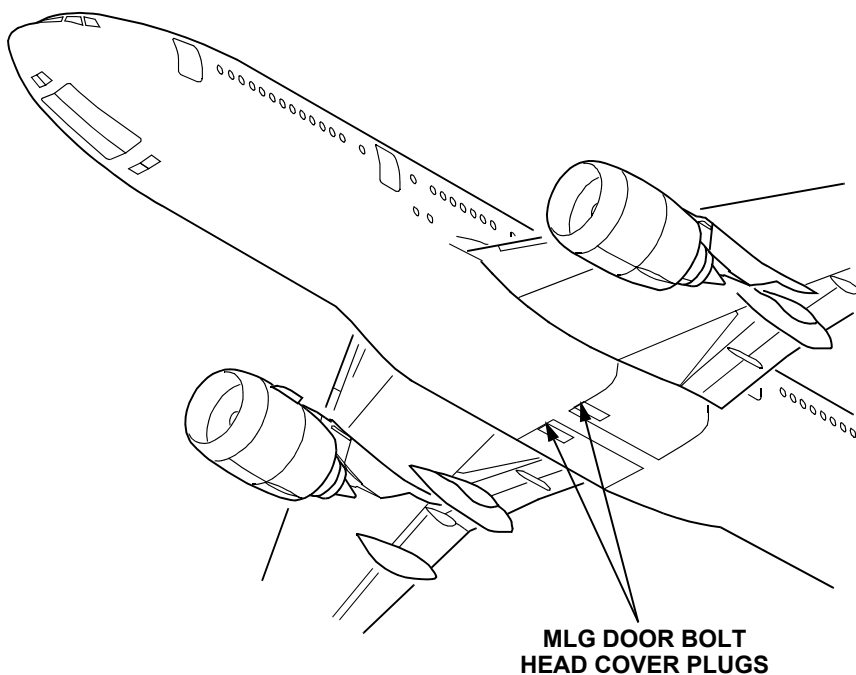


52-81-02 Main Landing Gear Door Bolt Head Cover Plugs

Any number may be missing.

Performance limited weights are reduced by the following:

Number Installed	Takeoff & Landing	Enroute Climb
12	Negligible penalty	Negligible penalty





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54-52-01 Forward Fairing Pressure Relief Door (RR)

54-53-01 Strut Aft Mount Access Panel

54-53-02 Strut Pressure Relief Door



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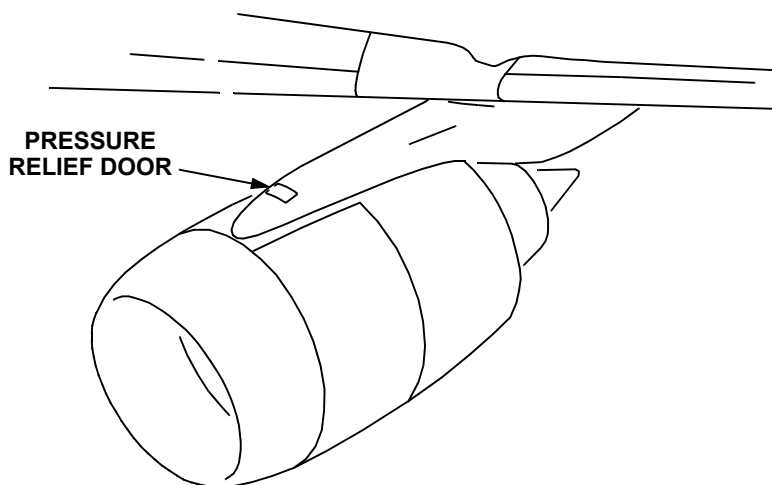
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54-52-01 Forward Fairing Pressure Relief Door (RR)

Any number may be missing.

Performance limited weights are reduced by the following for each missing item:

Number Installed	Takeoff & Landing	Enroute Climb
2	150 lb (69 kg)	200 lb (91 kg)



54-53-01 Strut Aft Mount Access Panel

Any number may be missing.

Performance limited weights are reduced by the following for each missing item:

Number Installed	Takeoff	Enroute Climb	Approach/Landing Climb
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For PW:

4	Negligible penalty	Negligible penalty	Negligible penalty
---	--------------------	--------------------	--------------------

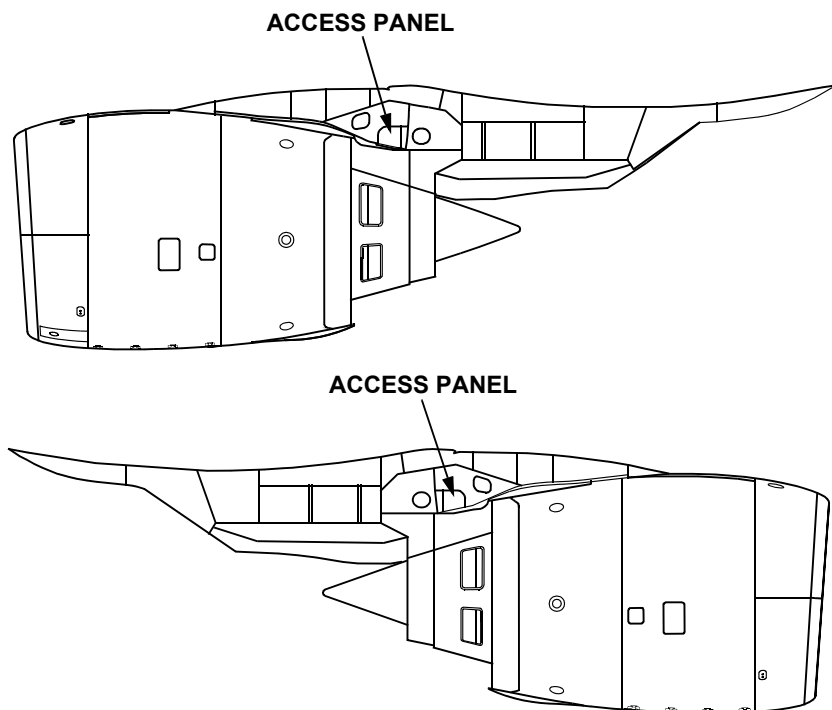
For GE:

6	Negligible penalty	Negligible penalty	Negligible penalty
---	--------------------	--------------------	--------------------

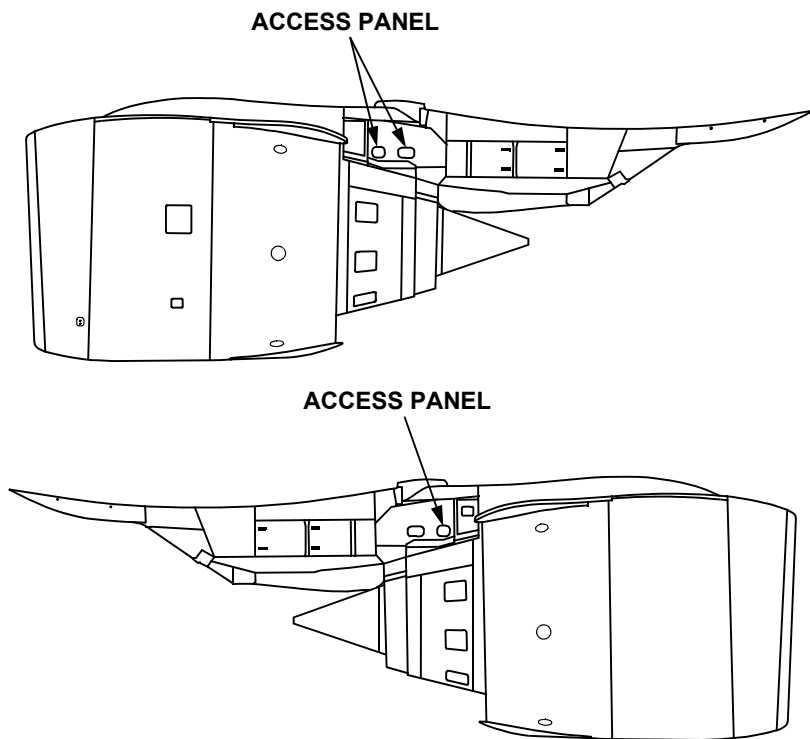
For RR:

2	Negligible penalty	Negligible penalty	Negligible penalty
---	--------------------	--------------------	--------------------

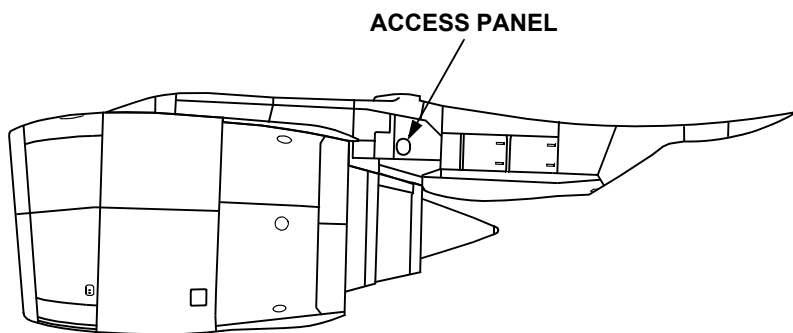
For PW:



For GE:



For RR:



54-53-02 Strut Pressure Relief Door

Any number may be missing.

Performance limited weights are reduced by the following for each missing item:

Number Installed	Takeoff & Landing	Enroute Climb
-------------------------	------------------------------	----------------------

For PW:

8	Negligible penalty	Negligible penalty
---	--------------------	--------------------

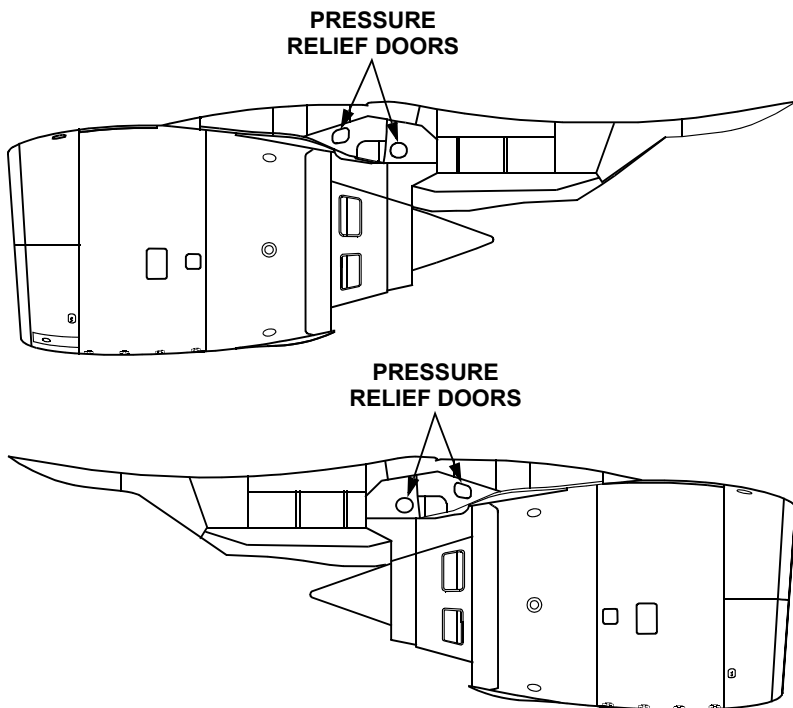
For GE:

4	Negligible penalty	Negligible penalty
---	--------------------	--------------------

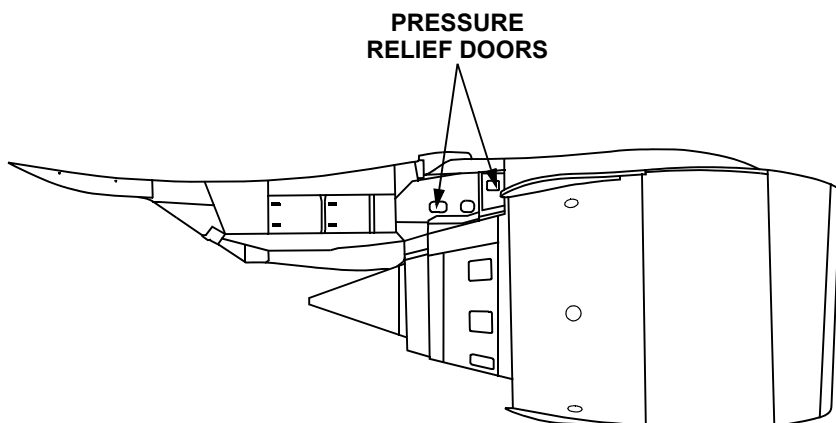
For RR:

2	Negligible penalty	Negligible penalty
---	--------------------	--------------------

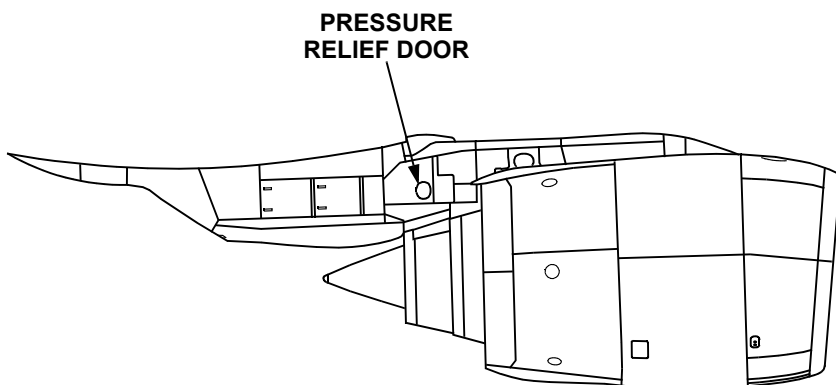
For PW:



For GE:



For RR:





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55-20-01	Elevator Access Cover
55-20-02	Elevator Inboard End Seals
55-20-03	Horizontal Stabilizer-to-Elevator Blade Seals
55-40-01	Vertical Stabilizer Tip Fairing Rudder Seal
55-40-02	Rudder-To-Fuselage Seal
55-40-03	Rudder-To-Tab Streamwise Seal
55-40-04	Rudder Tab-To-Fuselage Seal
55-40-05	Rudder Access Cover
55-40-06	Rudder Tab Access Cover
55-40-07	Rudder Tab Control Rod Fairing
55-40-08	Vertical Stabilizer-To-Fuselage Seal
55-40-09	Vertical Stabilizer-to-Rudder Blade Seals
55-40-10	Rudder-to-Tab Blade Seals



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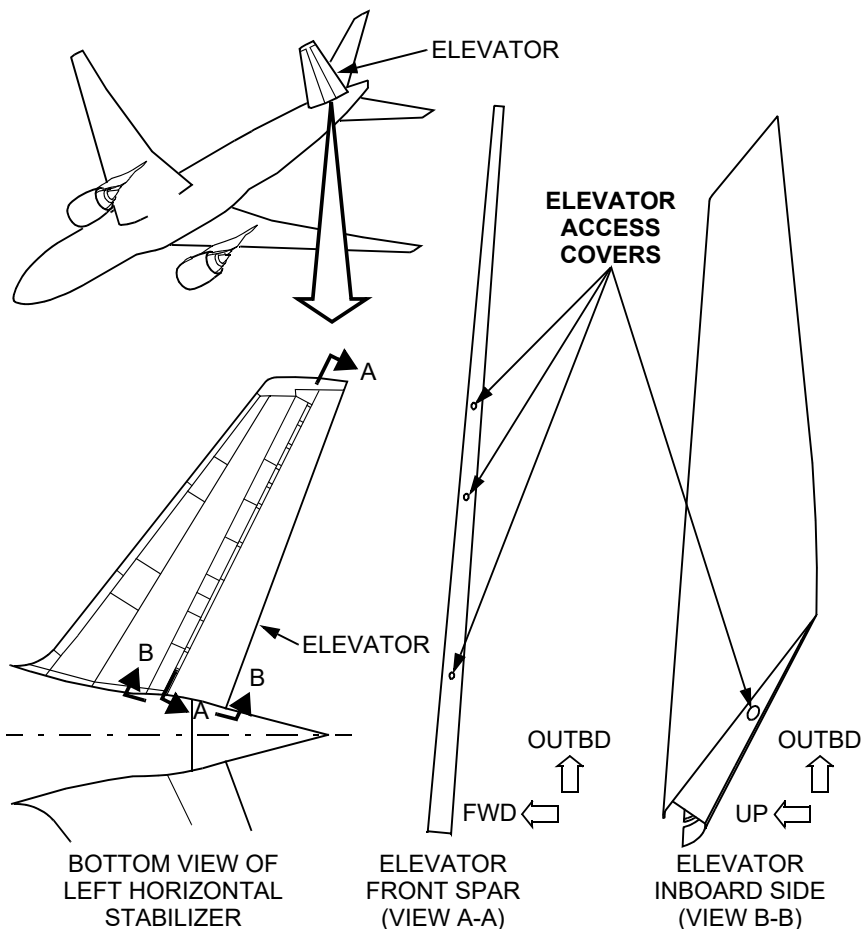
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55-20-01 Elevator Access Cover

Any number may be missing.

Performance limited weights are reduced by the following:

Number Installed	Takeoff & Landing	Enroute Climb
8	Negligible penalty	Negligible penalty



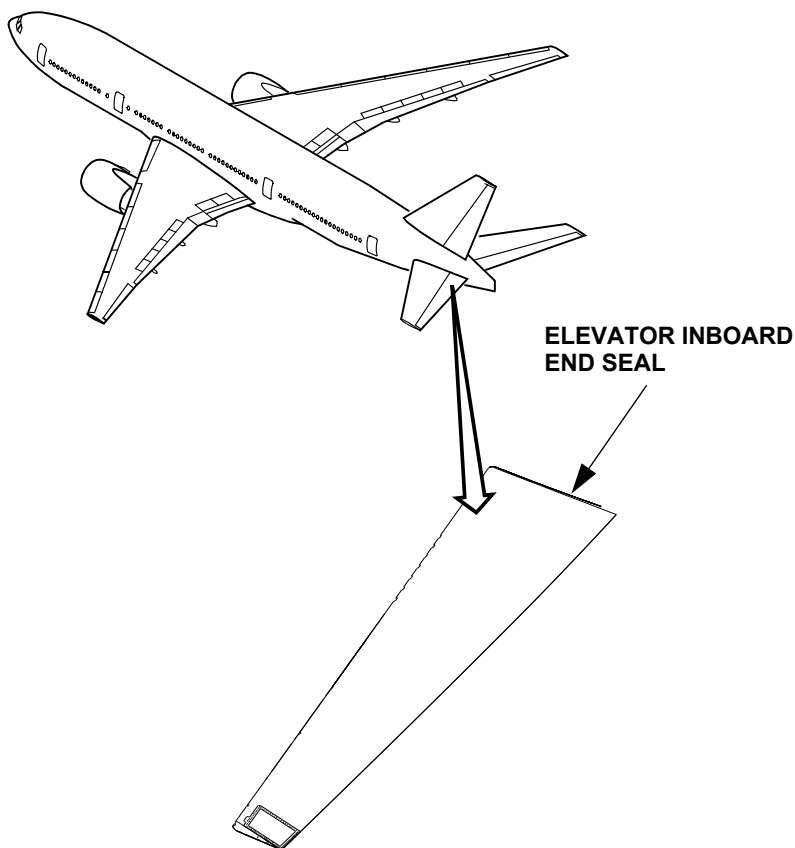
55-20-02 Elevator Inboard End Seals

Any number may be missing.

NOTE: Some line numbers do not have these seals installed.

Performance limited weights are reduced by the following:

Number Installed	Takeoff & Landing	Enroute Climb
2	No penalty	No penalty



**55-20-03 Horizontal Stabilizer-to-Elevator Blade Seals**

Up to 10 ft. (3.0 m) of seal may be missing from the airplane. No more than 5 ft. (1.5 m) of seal may be missing from a semi-span. Seals may not be missing from the upper and lower surfaces of the same semi-span.

Performance limited weights are reduced by the following for each foot (0.3 m) of missing seal:

Number Installed	Takeoff	Enroute Climb	Approach/Landing Climb
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For All with current production configuration:

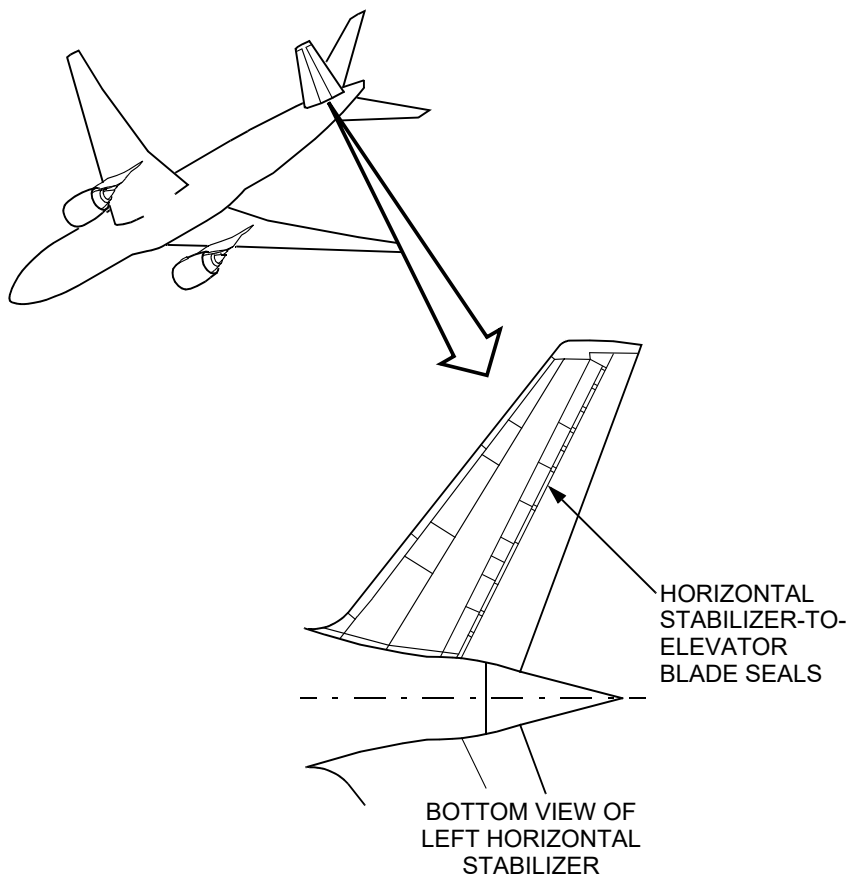
24	50 lb (23 kg)	100 lb (46 kg)	50 lb (23 kg)
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For -200/-200ER/-300 with early production configuration:

52	50 lb (23 kg)	100 lb (46 kg)	50 lb (23 kg)
----	---------------	----------------	---------------

NOTE 1: The 24-seal configuration was used on 777 line number 423 and on for all minor models of 777.

NOTE 2: The 52-seal configuration was used on 777 line number 1 through line number 422. These line numbers were 777-200s or 777-300s

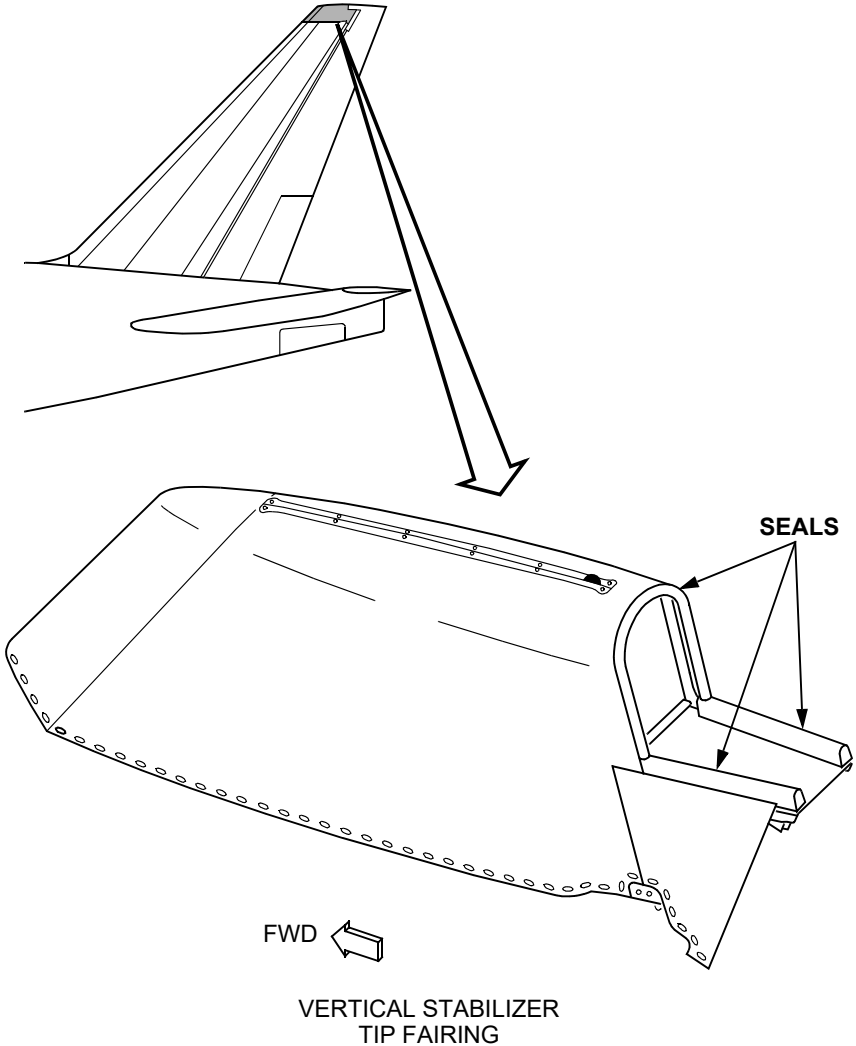


55-40-01 Vertical Stabilizer Tip Fairing Rudder Seal

Any number and length may be missing.

Performance limited weights are reduced by the following:

Number Installed	Takeoff & Landing	Enroute Climb
3	Negligible penalty	Negligible penalty

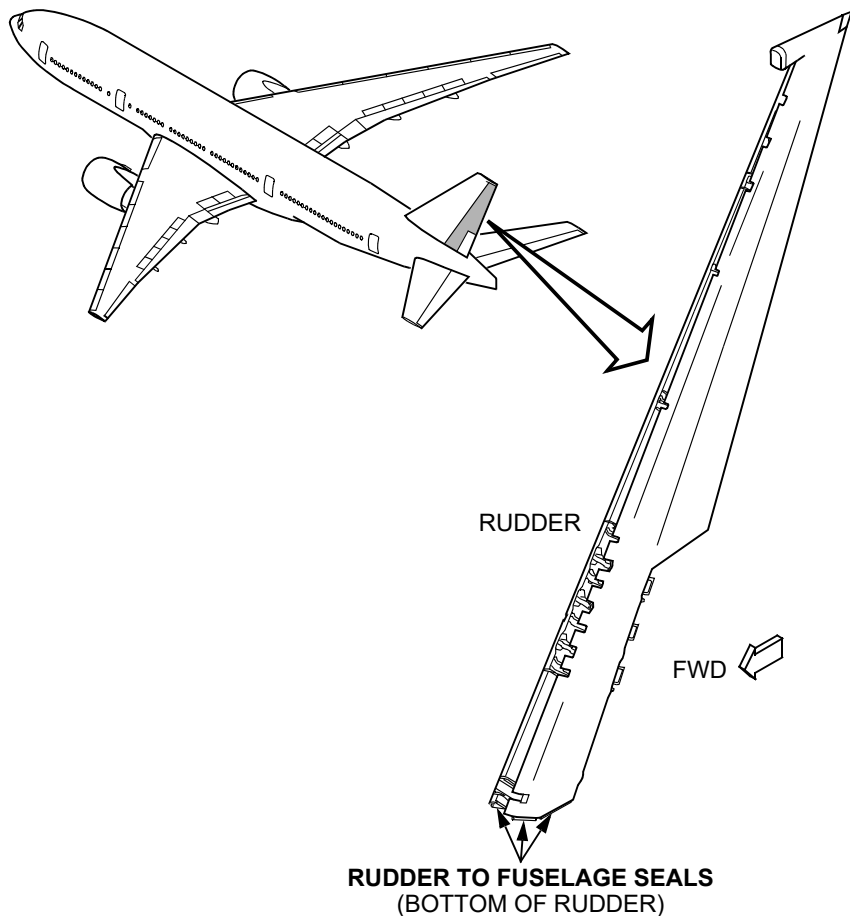


55-40-02 Rudder-To-Fuselage Seal

Any number may be missing.

Performance limited weights are reduced by the following for each missing item:

Number Installed	Takeoff	Enroute Climb	Approach/Landing Climb
4	350 lb (159 kg)	550 lb (249 kg)	300 lb (136 kg)

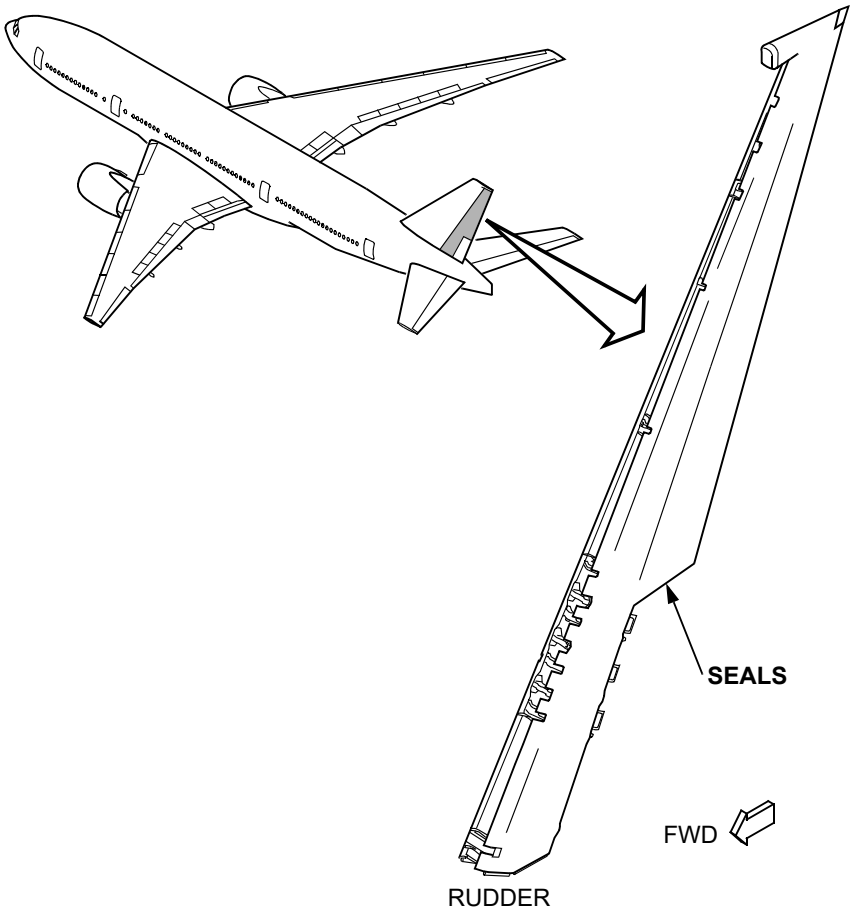


55-40-03 Rudder-To-Tab Streamwise Seal

Any number may be missing.

Performance limited weights are reduced by the following for each missing item:

Number Installed	Takeoff	Enroute Climb	Approach/Landing Climb
2	150 lb (68 kg)	200 lb (91 kg)	Negligible penalty

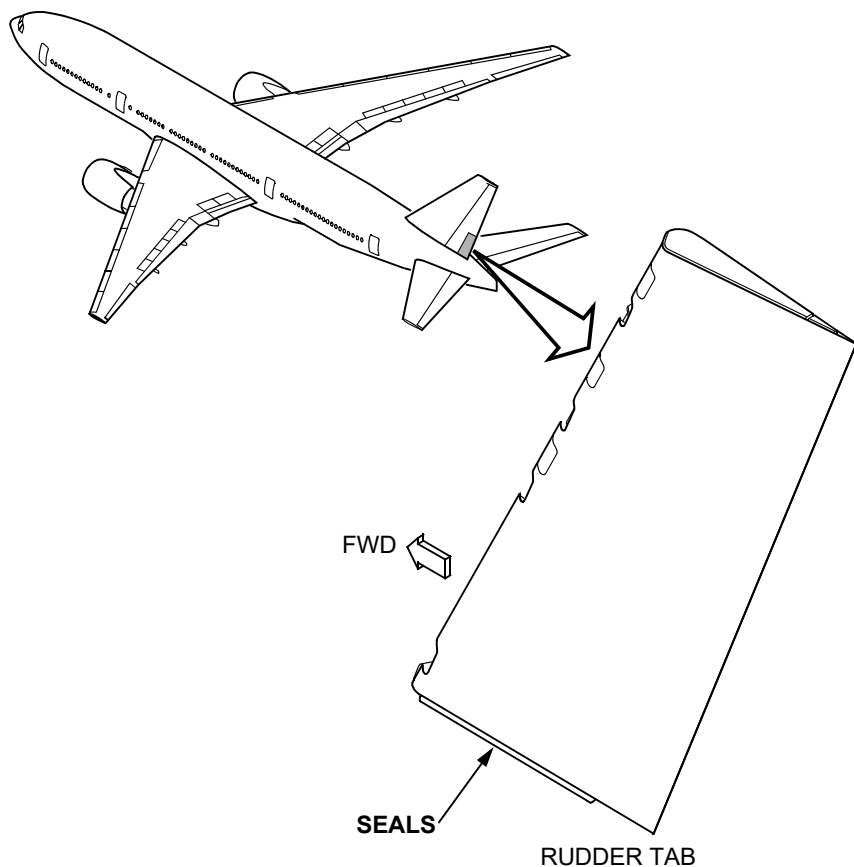


55-40-04 Rudder Tab-To-Fuselage Seal

Any number may be missing.

Performance limited weights are reduced by the following for each missing item:

Number Installed	Takeoff	Enroute Climb	Approach/Landing Climb
2	150 lb (68 kg)	250 lb (113 kg)	Negligible penalty

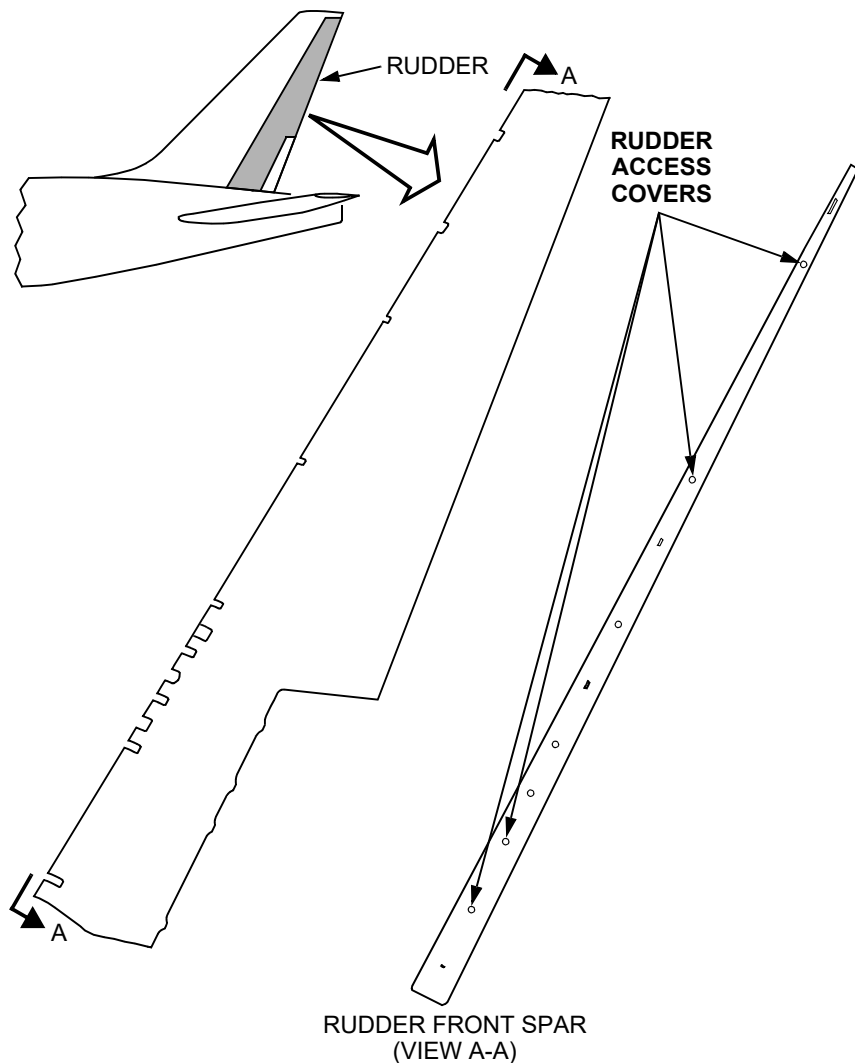


55-40-05 Rudder Access Cover

Any number may be missing.

Performance limited weights are reduced by the following:

Number Installed	Takeoff & Landing	Enroute Climb
4	Negligible penalty	Negligible penalty

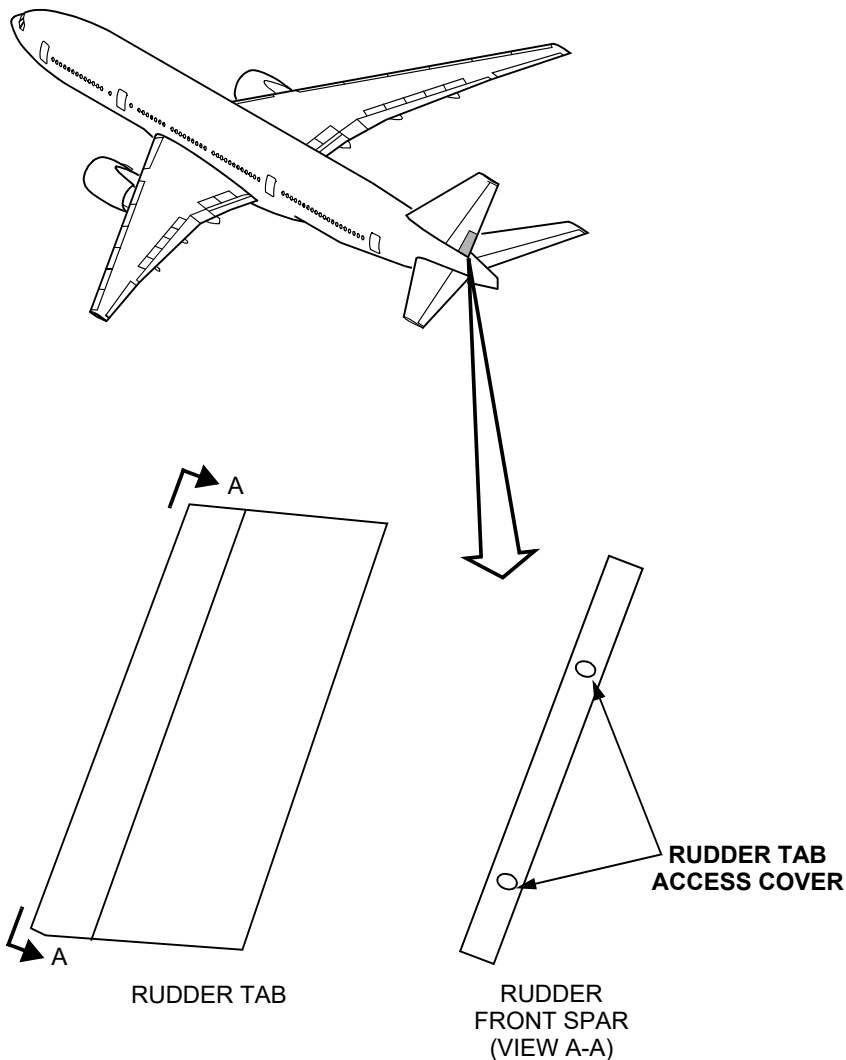


55-40-06 Rudder Tab Access Cover

Any number may be missing.

Performance limited weights are reduced by the following:

Number Installed	Takeoff & Landing	Enroute Climb
2	Negligible penalty	Negligible penalty



55-40-07 Rudder Tab Control Rod Fairing

One may be missing.

Performance limited weights are reduced by the following:

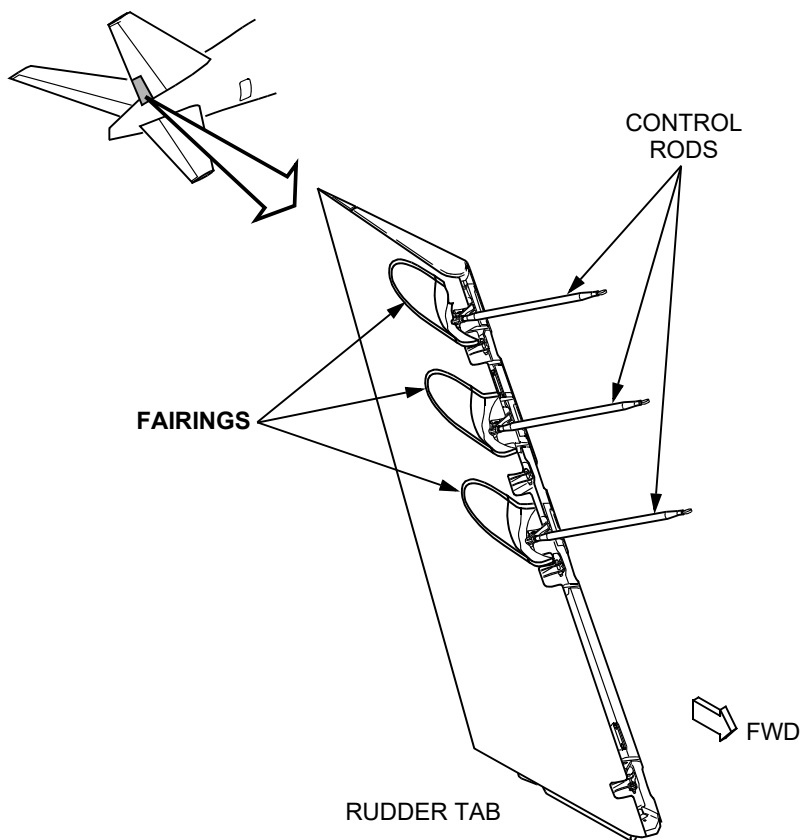
Number Installed	Takeoff	Enroute Climb	Approach/Landing Climb
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For All (Except -200LR/777F):

3	400 lb (182 kg)	650 lb (295 kg)	350 lb (159 kg)
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For -200LR/777F:

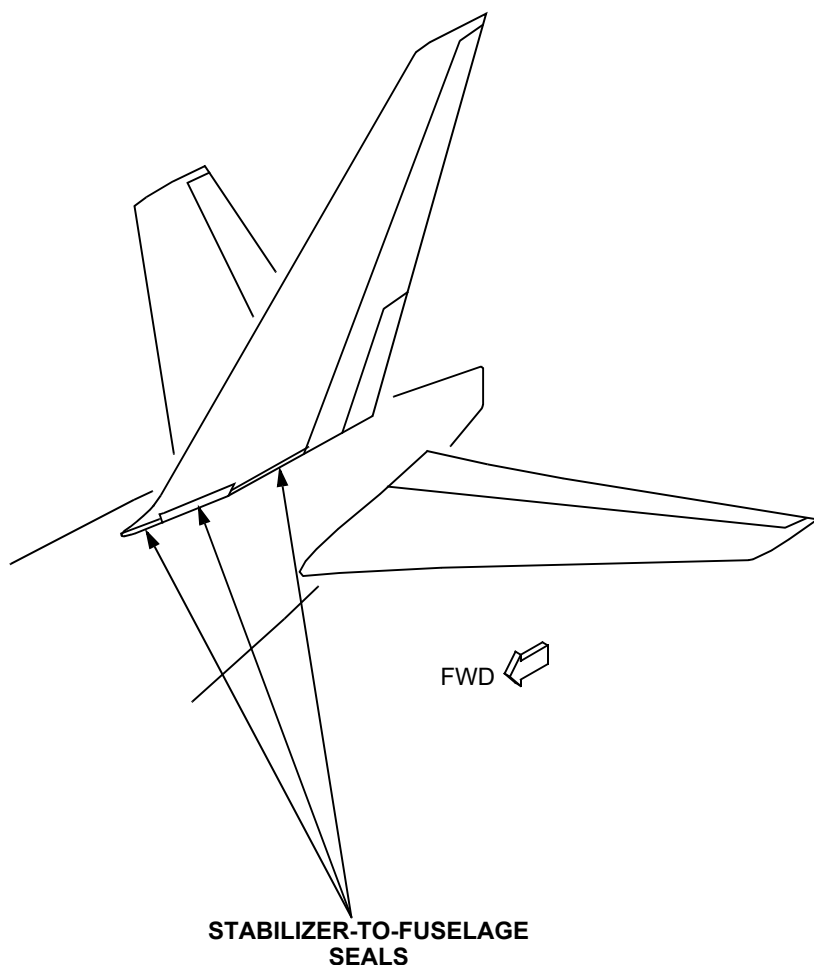
3	300 lb (137 kg)	550 lb (250 kg)	250 lb (114 kg)
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55-40-08 Vertical Stabilizer-To-Fuselage Seal

A maximum of two may be missing. Any length of those two seals may be missing. Performance limited weights are reduced by the following:

Number Installed	Takeoff & Landing	Enroute Climb
9	Negligible penalty	Negligible penalty



**55-40-09 Vertical Stabilizer-to-Rudder Blade Seals**

Up to 7.5 ft. (2.3 m) of seal may be missing from one side of the stabilizer. None of the seal on the opposite side may be missing.

Performance limited weights are reduced by the following for each foot (0.3 m) of missing seal:

Number Installed	Takeoff	Enroute Climb	Approach/Landi ng Climb
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For -200/-200ER/-300 with current production configuration:

13	250 lb (113 kg)	400 lb (181 kg)	250 lb (113 kg)
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For -200/-200ER/-300 with early production configuration:

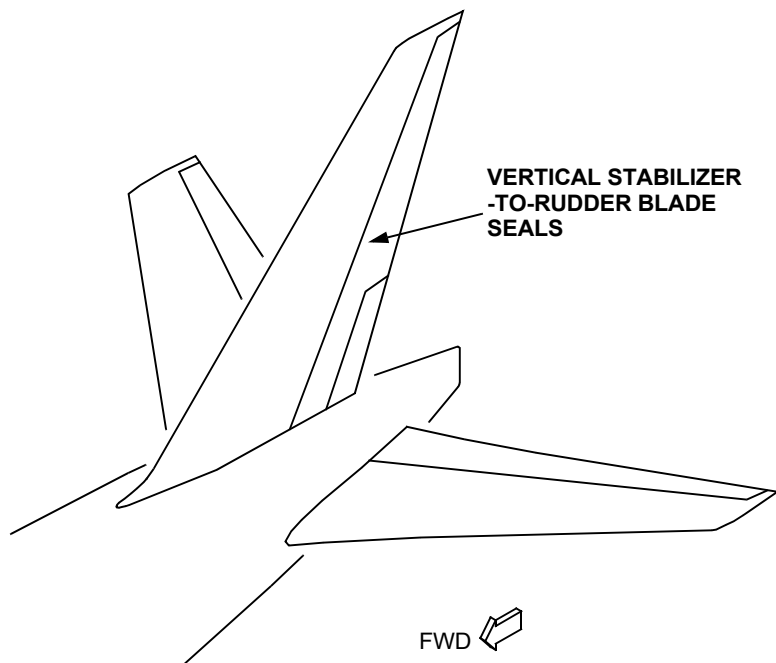
20	250 lb (113 kg)	400 lb (181 kg)	250 lb (113 kg)
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For -200LR/-300ER/F:

13	200 lb (91 kg)	350 lb (158 kg)	200 lb (91 kg)
----	----------------	-----------------	----------------

NOTE 1: The 13-seal configuration was used on 777 line number 423 and on for all minor models of 777.

NOTE 2: The 20-seal configuration was used on 777 line number 1 through line number 422. The line numbers were 777-200s or 777-300s.



**55-40-10 Rudder-to-Tab Blade Seals**

Any amount of the seal may be missing from one side of the stabilizer. None of the seal on the opposite side may be missing.

Performance limited weights are reduced by the following for each foot (0.3 m) of missing seal:

Number Installed	Takeoff	Enroute Climb	Approach/Landi ng Climb
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For -200/-200ER/-300 with current production configuration:

15	150 lb (68 kg)	250 lb (113 kg)	150 lb (68 kg)
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For -200/-200ER/-300 with early production configuration:

16	150 lb (68 kg)	250 lb (113 kg)	150 lb (68 kg)
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For -200LR/-300ER/F with current production configuration:

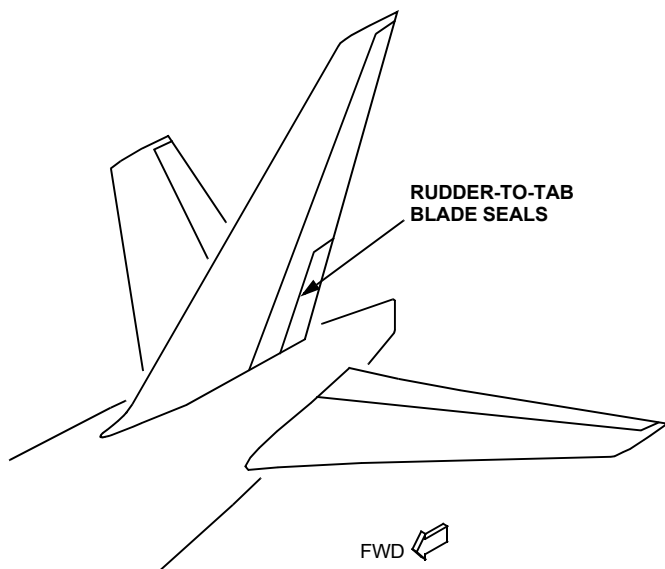
15	100 lb (45 kg)	200 lb (91 kg)	100 lb (45 kg)
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For -200LR/-300ER/F with early production configuration:

16	100 lb (45 kg)	200 lb (91 kg)	100 lb (45 kg)
----	----------------	----------------	----------------

NOTE 1: The 15-seal configuration was used on 777 line number 41 and on for all minor models of 777.

NOTE 2: The 16-seal configuration was used on 777 line number 1 through 40. These line numbers were all 777-200s.





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56-11-01 Number 1 Windshield Aerodynamic Sealant

56-11-02 Number 3 Windshield Aerodynamic Sealant



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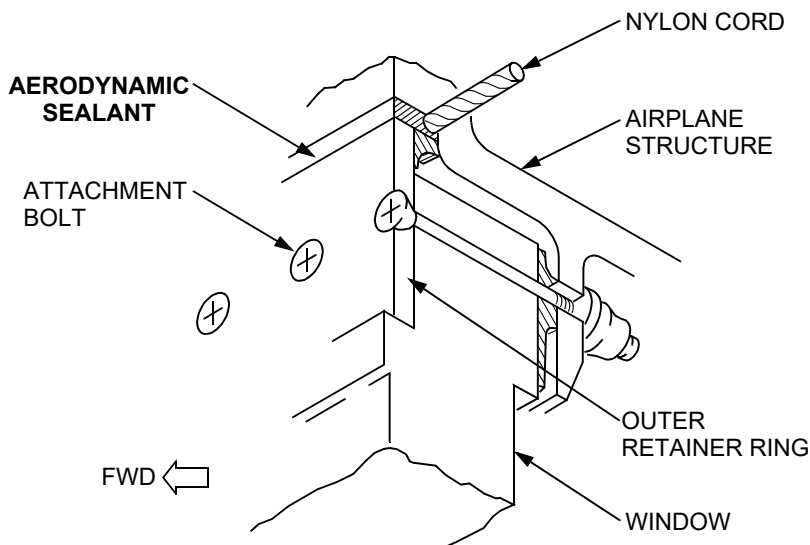
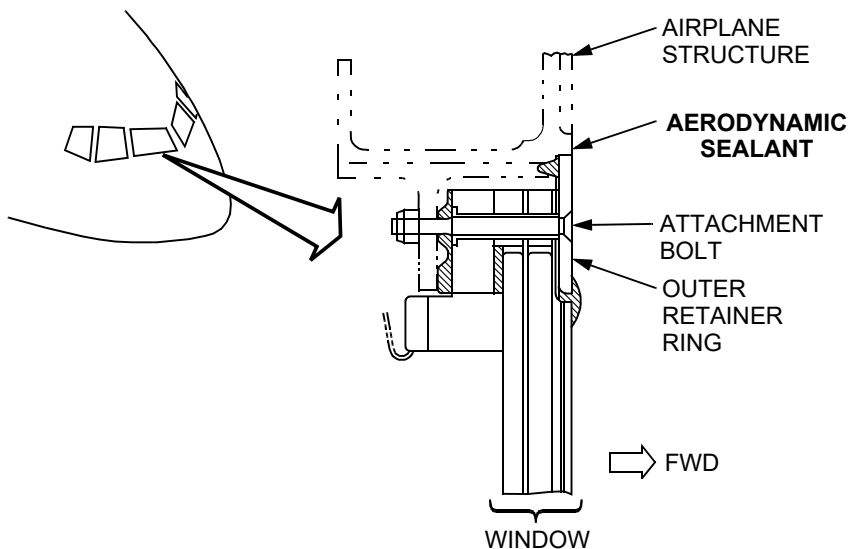
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56-11-01 Number 1 Windshield Aerodynamic Sealant

Any number may be missing.

Performance limited weights are reduced by the following:

Number Installed	Takeoff & Landing	Enroute Climb
2	Negligible penalty	Negligible penalty

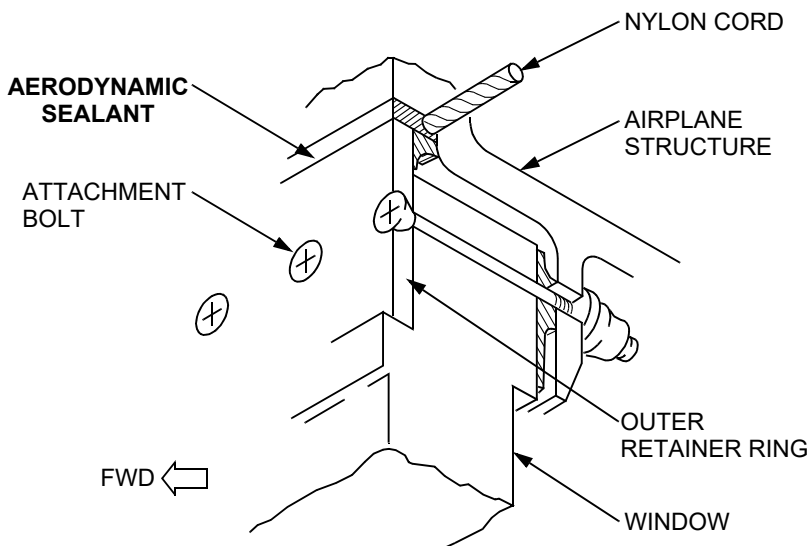
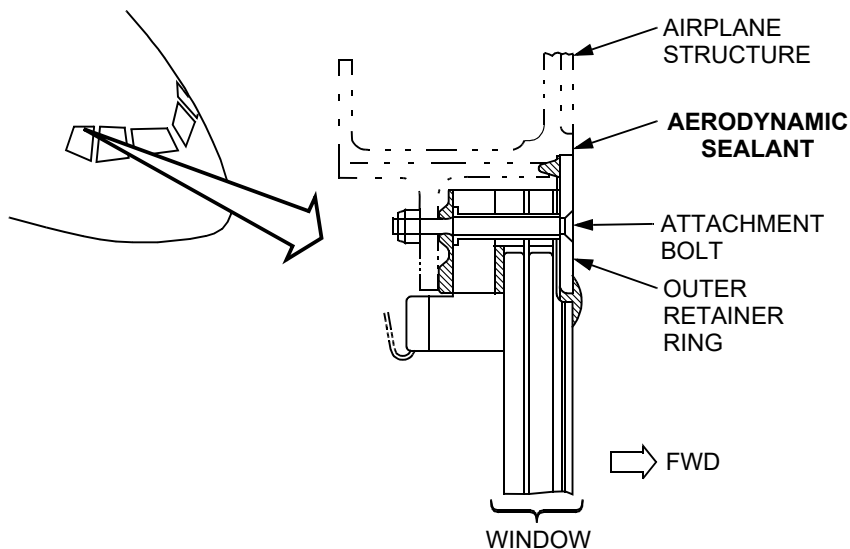


56-11-02 Number 3 Windshield Aerodynamic Sealant

Any number may be missing.

Performance limited weights are reduced by the following:

Number Installed	Takeoff & Landing	Enroute Climb
2	Negligible penalty	Negligible penalty





57-00-01	Wing Vortex Generator
57-00-02	Outboard Flap Vortex Generator
57-26-01	Inboard Flap Center Track Fairing
57-26-02	Inboard Flap Center Track Fairing Seal
57-26-03	Inboard Flap Inboard Support Fairing Seal
57-26-04	Outboard Flap Track Fairing Access Panel
57-26-05	Outboard Flap Track Fairing
57-26-06	Outboard Flap Track Fairing Seal
57-26-07	Outboard Flap Auxiliary Track Fairing
57-26-08	Outboard Flap Auxiliary Track Fairing Seal
57-26-09	Inboard Flap Track Flaperon Fairing Seals
57-26-10	Inboard Flap Inboard Fairing - Aft Segment
57-31-01	Wing Tip Access Panel (-200/-200ER/-300)
57-31-02	Wing Tip Fairing (-200/-200ER/-300)
57-31-03	Raked Tip (-200LR/-300ER/777F)
57-41-01	Outboard Leading Edge Slat Restoration Seal Assembly
57-41-02	Wing Fixed Leading Edge Lower Panel
57-41-03	Slat Linkage Fairing Panel
57-41-04	Slat Spanwise Flow Seal
57-41-05	Outboard Slat Spanwise Bulb Seal
57-41-06	Slat Spanwise Lower Flexible Seal
57-41-07	DELETED (Refuel Station Access Door)
57-41-08	Inboard Slat Inboard End Bulb Seal
57-41-09	Krueger End Seal
57-41-23	Inboard Slat Inboard End Blade Seal
57-51-01	Wing Fixed Trailing Edge Lower Panel
57-51-02	Inboard Flap Inboard Seal

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57-51-03	Overwing Fairing Torque Tube Door
57-51-04	Number 3 & 6 Flap Fairings Seals
57-53-01	Flaperon Inboard Seal
57-53-02	Inboard Flap Leading Edge Seal
57-53-03	Outboard Flap Leading Edge Seal
57-63-01	DELETED (Flaperon Cove Lip Door)
57-63-02	Flaperon Cove Lip Door Streamwise Seal
57-63-03	Flaperon Cove Lip Door Hingewise Seal
57-63-04	Flaperon Outboard Seal
57-63-05	Aileron Streamwise Seal
57-63-06	Aileron Power Control Unit Fairing
57-63-07	Aileron Mass Balance Seal
57-63-08	Aileron Hinge Blade Seals
57-63-09	Aileron Hinge Blade Seal and Seal Retainer
57-71-01	Spoiler Streamwise Seal
57-71-02	Spoiler Leading Edge Seal

57-00-01 Wing Vortex Generator

A maximum of two per wing (four total) may be missing provided the missing vortex generators are not adjacent.

Performance limited weights are reduced by the following:

Number Installed	Takeoff & Landing	Enroute Climb
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For All (Except -200LR/777F/-300ER):

32	No penalty	No penalty
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For -200LR/777F:

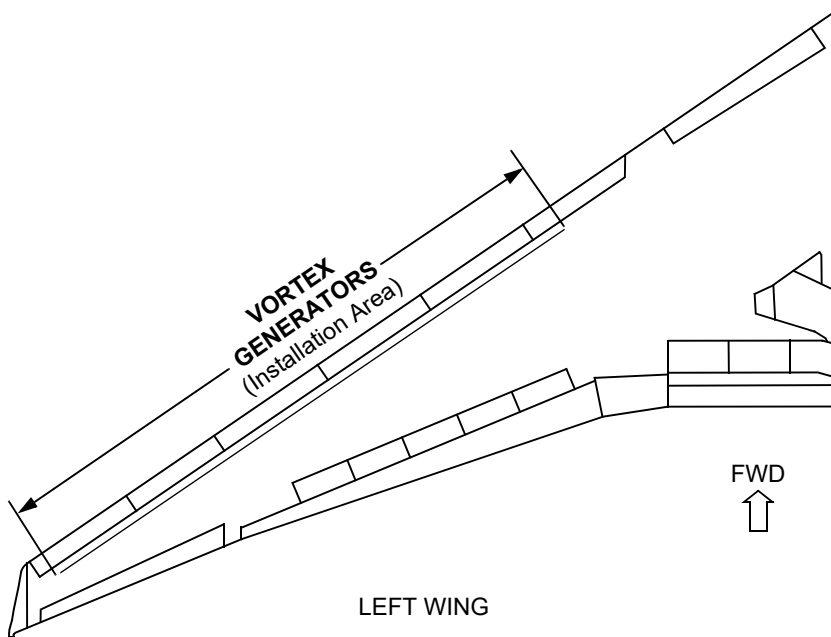
46	No penalty	No penalty
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For -300ER (Without PRR 63202 or SB 777-57-0052):

32	No penalty	No penalty
----	------------	------------

For -300ER (With PRR 63202 or SB 777-57-0052):

46	No penalty	No penalty
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57-00-02 Outboard Flap Vortex Generator

A maximum of four per wing (eight total) may be missing.

VREF Speed is increased by 2 knots.

Performance limited weights are reduced by the following:

Number Installed	Takeoff & Enroute Climb	Approach/Landing Climb	Landing Field Length	Max Quick Turnaround
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For -300:

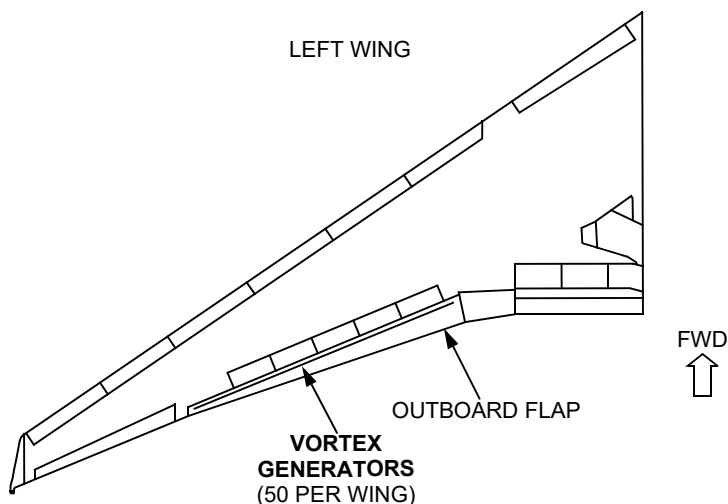
100	No penalty	6,500 lb (2,949 kg)	14,100 lb (6,396 kg)	6,000 lb (2,722 kg)
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For -300ER:

100	No penalty	500 lb (227 kg)	16,400 lb (7,439 kg)	13,400 lb (6,079 kg)
-----	------------	--------------------	-------------------------	-------------------------

For -200LR/777F:

100	No penalty	500 lb (227 kg)	155,000 lb, or increase by 230 feet (70,307 kg, or increase by 71 meters)	13,400 lb (6,079 kg)
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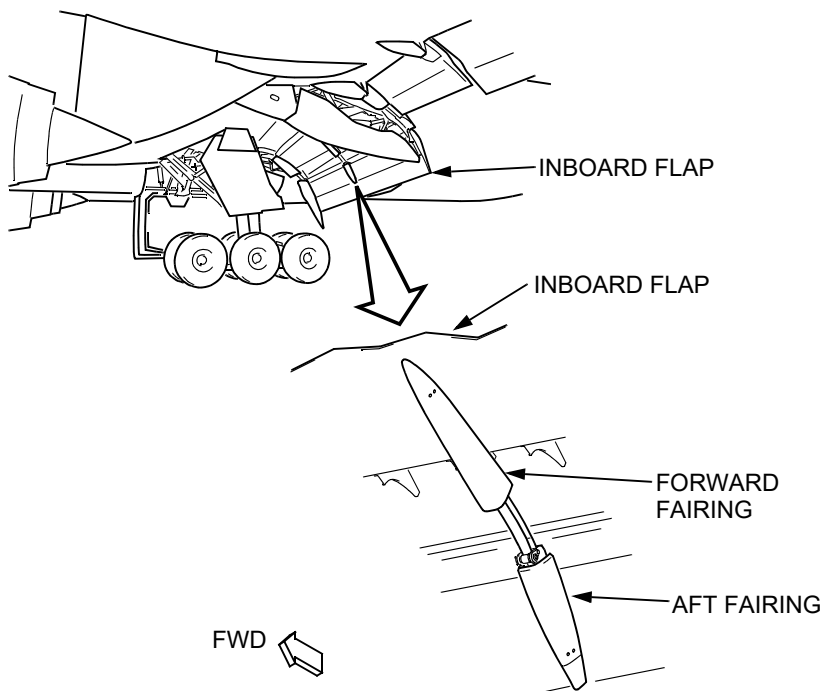
57-26-01 Inboard Flap Center Track Fairing

One may be missing.

NOTE: The aft segment of the track fairing should be removed if the forward segment is missing.

Performance limited weights are reduced by the following:

Number Installed	Takeoff	Enroute Climb	Approach/Landing Climb
2	250 lb (114 kg)	400 lb (182 kg)	250 lb (114 kg)



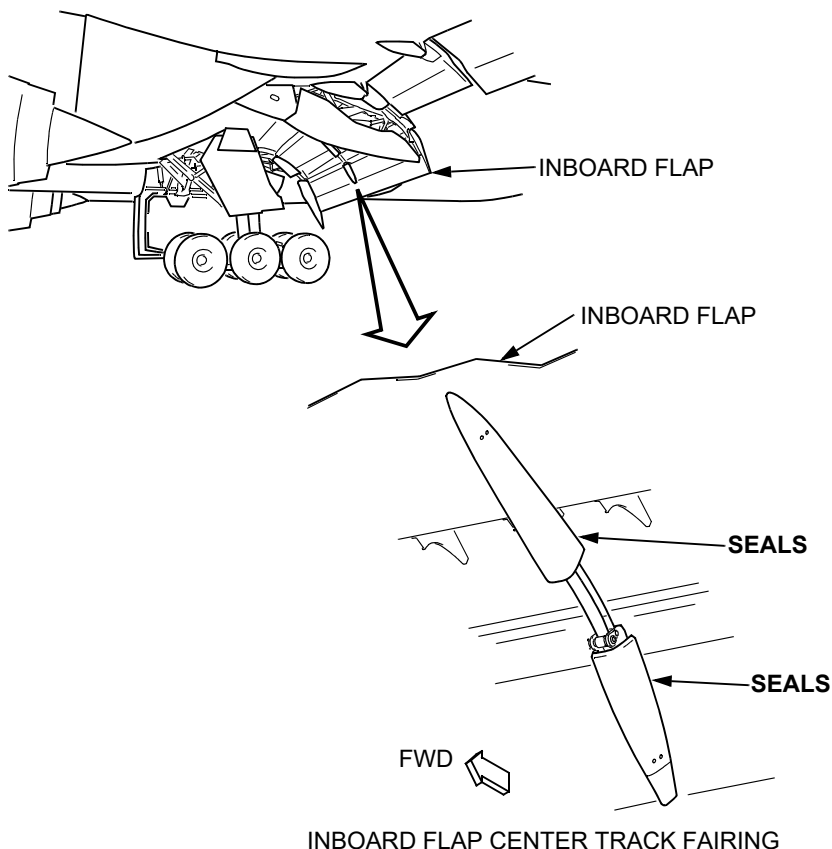
INBOARD FLAP CENTER TRACK FAIRING

57-26-02 Inboard Flap Center Track Fairing Seal

A maximum of eight may be missing.

Performance limited weights are reduced by the following for each foot of missing seal:

Number Installed	Takeoff	Enroute Climb	Approach/Landing Climb
16	100 lb (46 kg)	100 lb (46 kg)	100 lb (46 kg)

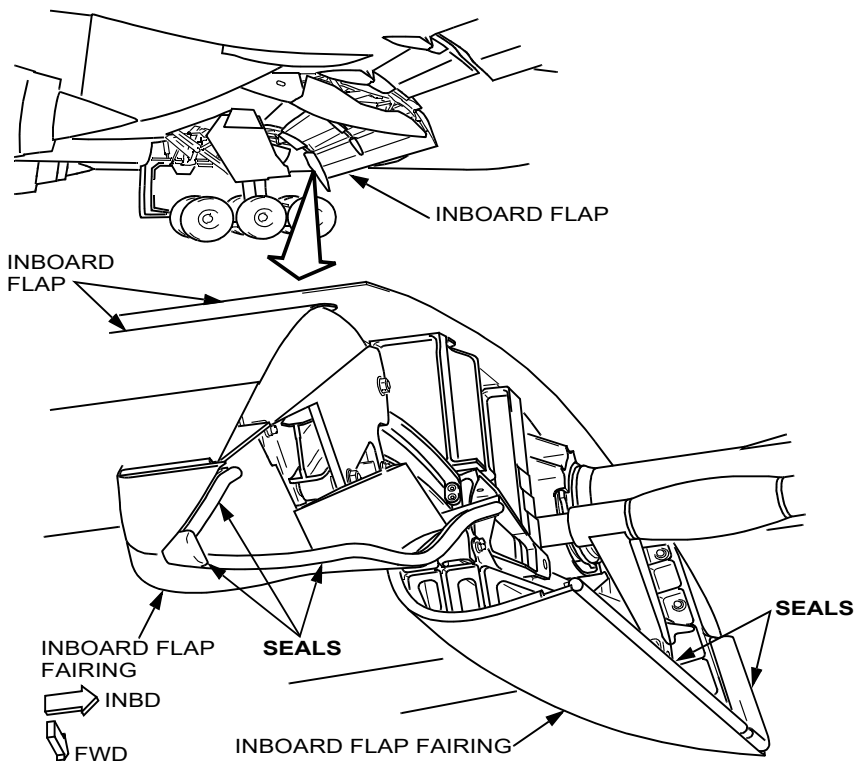


57-26-03 Inboard Flap Inboard Support Fairing Seal

A maximum of eleven may be missing.

Performance limited weights are reduced by the following for each foot of missing seal:

Number Installed	Takeoff	Enroute Climb	Approach/Landing Climb
22	300 lb (137 kg)	450 lb (205 kg)	No Penalty

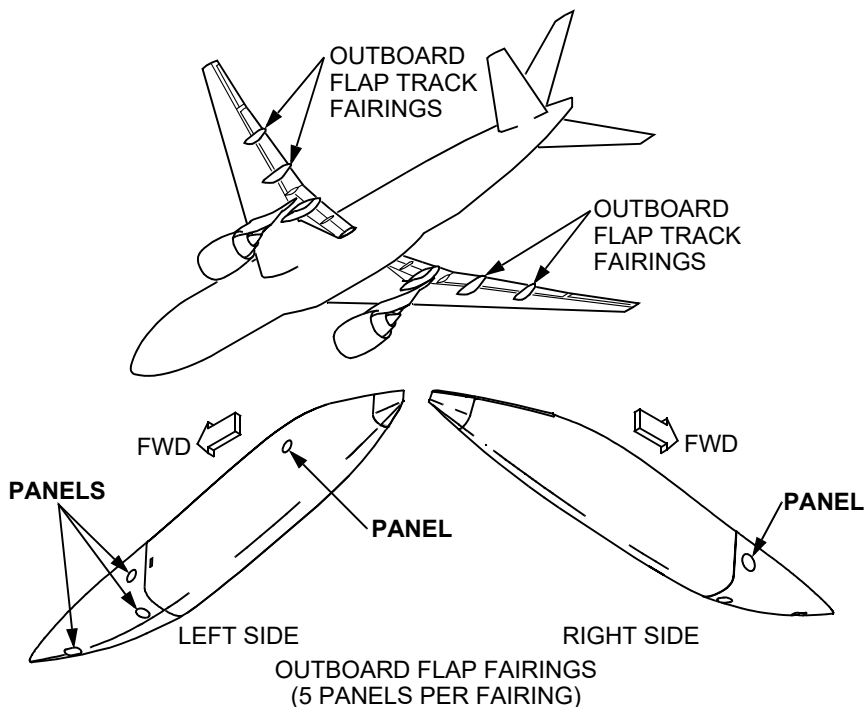


57-26-04 Outboard Flap Track Fairing Access Panel

Any number may be missing.

Performance limited weights are reduced by the following for each missing item:

Number Installed	Takeoff	Enroute Climb	Approach/Landing Climb
20	300 lb (137 kg)	500 lb (227 kg)	300 lb (137 kg)



**57-26-05 Outboard Flap Track Fairing**

One may be missing.

NOTE 1: The aft segment of the track fairing should be removed if the forward segment is missing.

NOTE 2: Support the structure as necessary and remove the bolts to remove the drive link and support link. Reinstall the bolts with sufficient washers to prevent excessive movement during flight.

Performance limited weights are reduced by the following:

Number Installed	Takeoff	Enroute Climb	Approach/Landing Climb
------------------	---------	---------------	------------------------

For -200/-200ER/GE90-76B thru -85B:

4	2,950 lb (1,338 kg)	4,950 lb (2,245 kg)	2,000 lb (907 kg)
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For -200/-200ER/GE90-90B:

4	3,300 lb (1,496 kg)	5,600 lb (2,540 kg)	3,300 lb (1,496 kg)
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For -200/-200ER/GE90-94B:

4	3,300 lb (1,496 kg)	6,550 lb (2,971 kg)	2,000 lb (907 kg)
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For -200/-200ER/PW4074 thru PW4084:

4	2,650 lb (1,202 kg)	6,550 lb (2,971 kg)	2,000 lb (907 kg)
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For -200/-200ER/PW4090:

4	2,950 lb (1,338 kg)	5,900 lb (2,676 kg)	2,000 lb (907 kg)
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For -200/-200ER/RR875 thru RR884:

4	2,300 lb (1,043 kg)	4,950 lb (2,245 kg)	2,000 lb (907 kg)
---	---------------------	---------------------	-------------------



For -200/-200ER/RR892:

4	3,600 lb (1,632 kg)	5,600 lb (2,540 kg)	2,300 lb (1,043 kg)
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For -200/-200ER/RR895:

4	2,650 lb (1,202 kg)	5,600 lb (2,540 kg)	3,600 lb (1,632 kg)
---	---------------------	---------------------	---------------------

For -300/PW4090:

4	2,650 lb (1,202 kg)	5,900 lb (2,676 kg)	2,000 lb (907 kg)
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For -300/RR892:

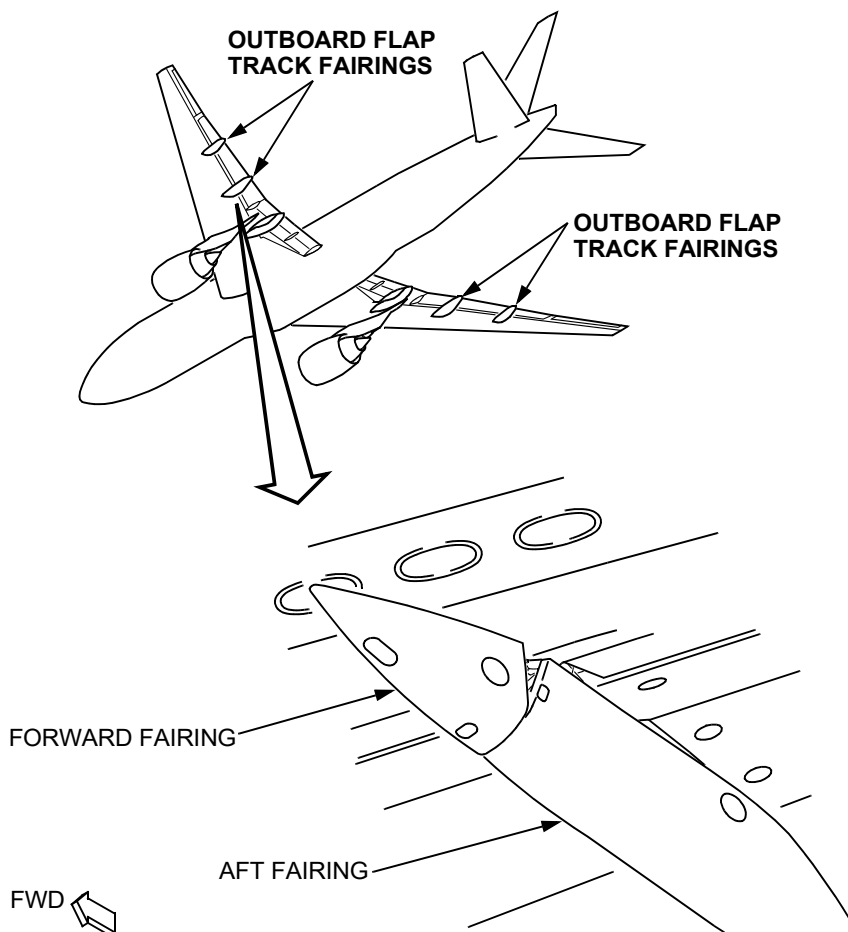
4	2,650 lb (1,202 kg)	5,250 lb (2,381 kg)	2,000 lb (907 kg)
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For -300ER:

4	2,800 lb (1,271 kg)	5,900 lb (2,676 kg)	2,000 lb (907 kg)
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For -200LR/777F:

4	2,950 lb (1,338 kg)	5,600 lb (2,540 kg)	2,650 lb (1,203 kg)
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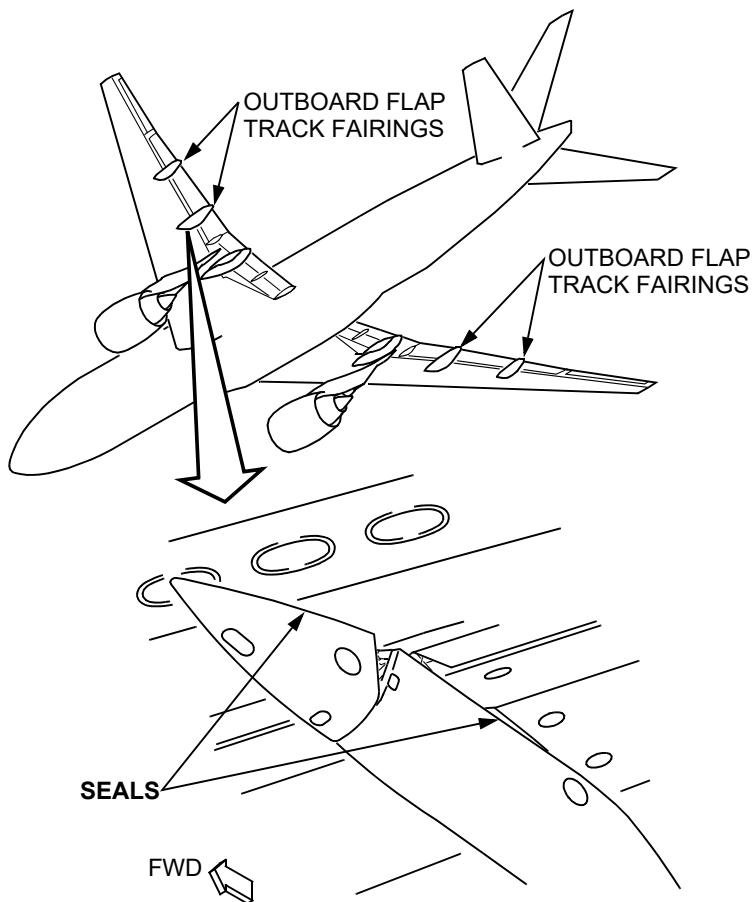


57-26-06 Outboard Flap Track Fairing Seal

A maximum of four per wing (eight total) may be missing.

Performance limited weights are reduced by the following for each foot of missing seal:

Number Installed	Takeoff	Enroute Climb	Approach/Landing Climb
32	100 lb (46 kg)	150 lb (69 kg)	100 lb (46 kg)

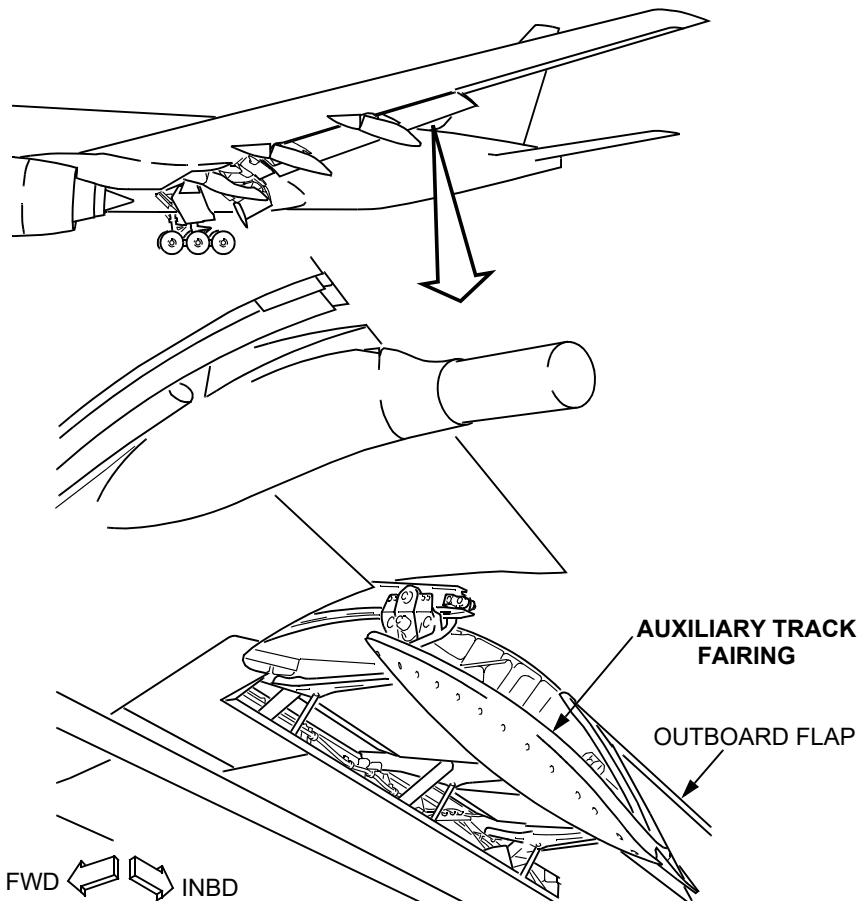


57-26-07 Outboard Flap Auxiliary Track Fairing

One may be missing.

Performance limited weights are reduced by the following:

Number Installed	Takeoff & Landing	Enroute Climb
2	Negligible penalty	150 lb (68 kg)

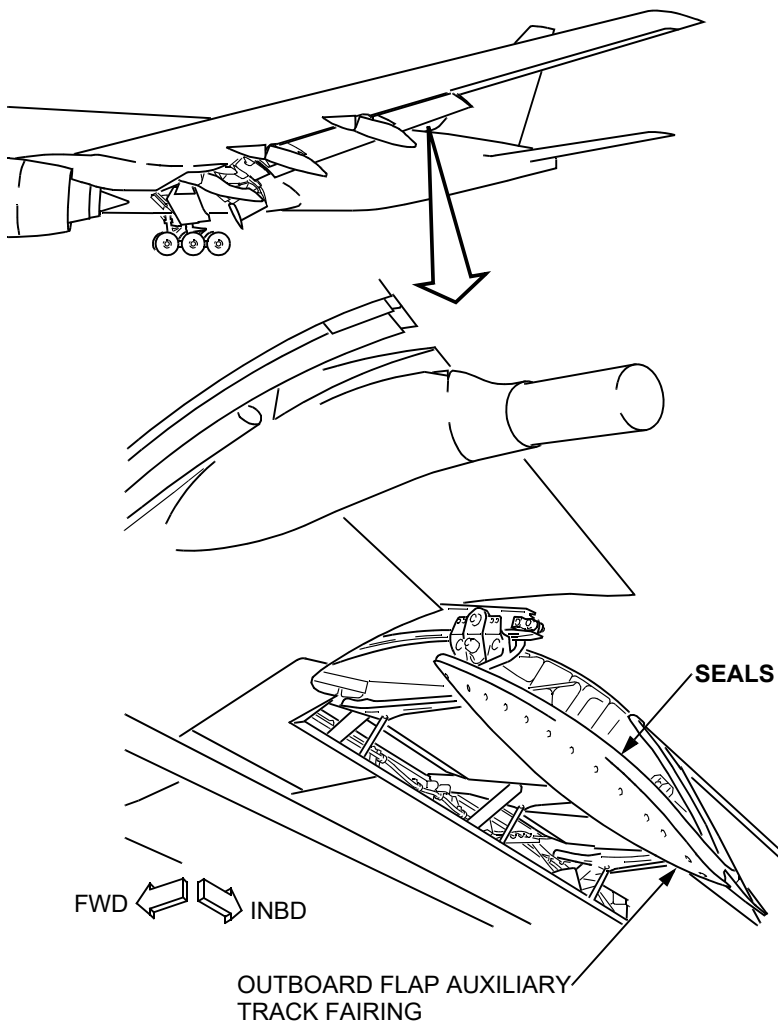


57-26-08 Outboard Flap Auxiliary Track Fairing Seal

A maximum of two may be missing.

Performance limited weights are reduced by the following for each foot of missing seal:

Number Installed	Takeoff	Enroute Climb	Approach/Landing Climb
4	50 lb (23 kg)	50 lb (23 kg)	50 lb (23 kg)



**57-26-09 Inboard Flap Track Flaperon Fairing Seals**

Any number may be missing.

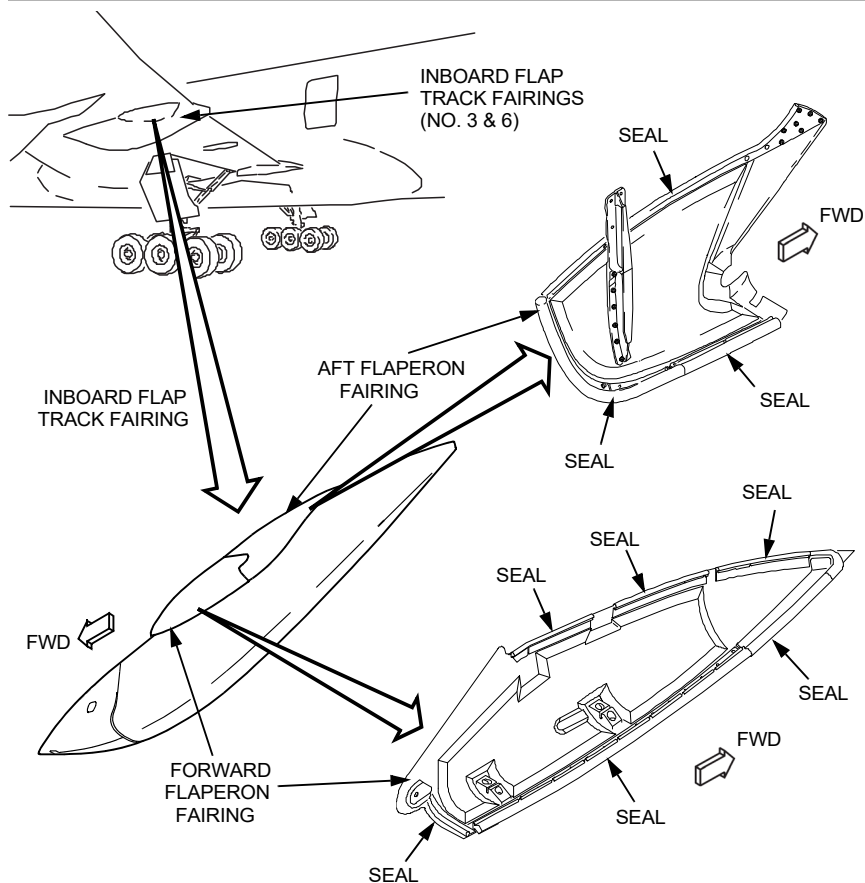
NOTE 1: For missing forward seal(s), the associated aft seals must be removed.

NOTE 2: Each flaperon has 5 forward seals, 1 intermediate seal and 3 aft seals (9 per flaperon).

Performance limited weights are reduced by the following for each flaperon with missing seals:

NOTE: The penalties shown are for one or more seals missing from only one fairing. If one or more seals are missing from both fairings, the penalties are doubled.

Number Installed	Takeoff	Enroute Climb	Approach/Landing Climb
18	400 lb (182 kg)	650 lb (295 kg)	No penalty

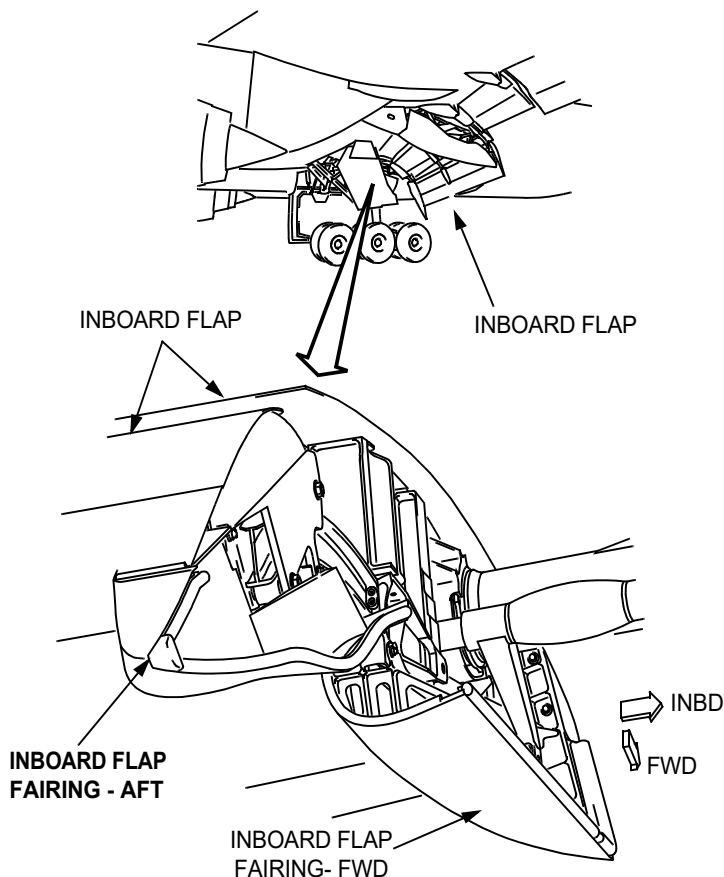


57-26-10 Inboard Flap Inboard Fairing - Aft Segment

Any number may be missing.

Performance limited weights are reduced by the following for each missing item:

Number Installed	Takeoff & Landing	Enroute Climb
2	150 lb (68 kg)	250 lb (113 kg)

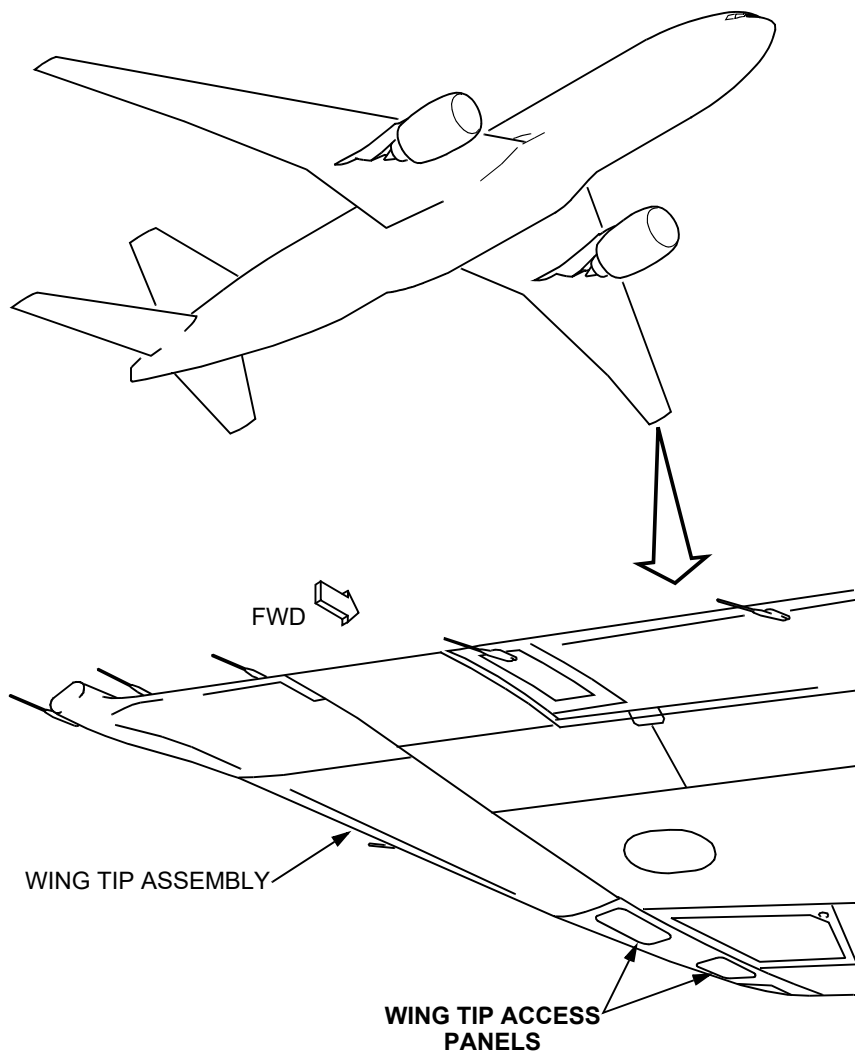


57-31-01 Wing Tip Access Panel (-200/-200ER/-300)

Any number may be missing.

Performance limited weights are reduced by the following:

Number Installed	Takeoff & Landing	Enroute Climb
4	Negligible penalty	Negligible penalty



**57-31-02 Wing Tip Fairing (-200/-200ER/-300)**

One may be missing.

NOTE 1: For inoperative Position Lights, dispatch using MMEL Item 33-43-01.
For inoperative Anti-collision Lights, dispatch using MMEL item 33-44-01.

NOTE 2: Static dischargers that are removed with the wing tip fairing are not considered missing.

Performance limited weights are reduced by the following:

Number Installed	Takeoff	Enroute Climb	Approach/Landing Climb
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For -200/-200ER/GE90-76B thru -85B:

2	17,950 lb (8,141 kg)	5,600 lb (2,540 kg)	11,950 lb (5,420 kg)
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For -200/-200ER/GE90-90B:

2	19,950 lb (9,049 kg)	6,300 lb (2,858 kg)	19,900 lb (9,026 kg)
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For -200/-200ER/GE90-94B:

2	19,950 lb (9,049 kg)	7,400 lb (3,356 kg)	11,950 lb (5,420 kg)
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For -200/-200ER/PW4074 thru PW4084:

2	15,950 lb (7,235 kg)	7,400 lb (3,356 kg)	11,950 lb (5,420 kg)
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For -200/-200ER/PW4090:

2	17,950 lb (8,141 kg)	6,700 lb (3,039 kg)	11,950 lb (5,420 kg)
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For -200/-200ER/RR875 thru RR884:

2	13,950 lb (6,328 kg)	5,600 lb (2,540 kg)	11,950 lb (5,420 kg)
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For -200/-200ER/RR892:

2	21,900 lb (9,933 kg)	6,300 lb (2,858 kg)	13,950 lb (6,328 kg)
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For -200/-200ER/RR895:

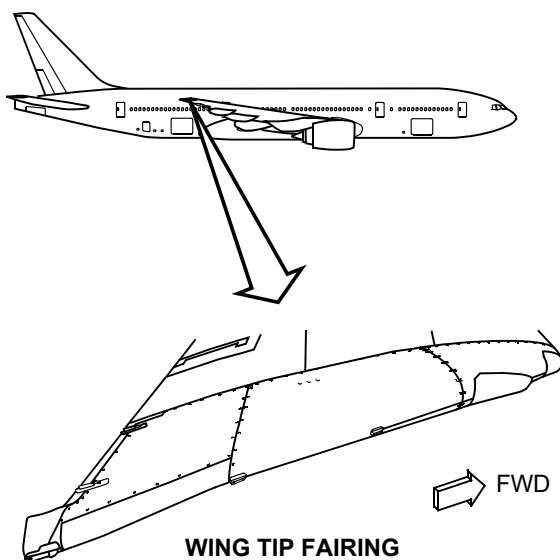
2	15,950 lb (7,235 kg)	6,300 lb (2,858 kg)	21,900 lb (9,933 kg)
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For -300/PW4090:

2	15,950 lb (7,235 kg)	6,700 lb (3,039 kg)	11,950 lb (5,420 kg)
---	----------------------	---------------------	----------------------

For -300/RR892:

2	15,950 lb (7,235 kg)	5,950 lb (2,699 kg)	11,950 lb (5,420 kg)
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57-31-03 Raked Tip (-200LR/-300ER/777F)

Raked wing tips may only be missing as a set.

NOTE 1: For one raked wing tip missing, the other raked wing tip must be removed.

NOTE 2: For inoperative position lights, dispatch using MMEL Item 33-43-01.

NOTE 3: For inoperative anti-collision lights, dispatch using MMEL Item 33-44-01.

NOTE 4: Static dischargers that are removed with the raked wing tips are not considered missing.

Both raked wing tips may be missing provided:

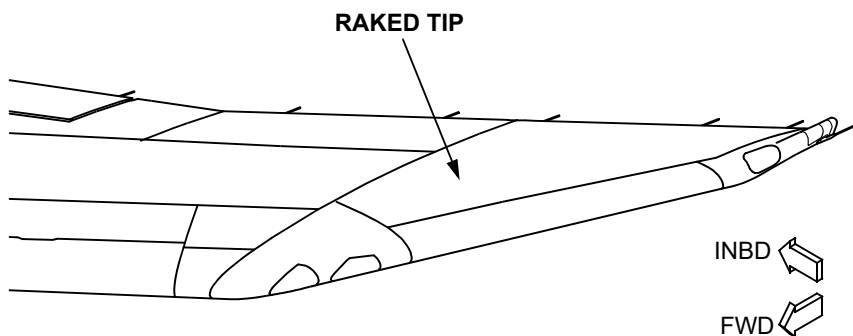
For -200LR/777F:

1. Night time operation with the raked wingtips removed is prohibited.
2. Increase VR, V2 and VREF speeds by 2 knots.
3. Reduce Takeoff performance limited weight by 26,300 lb (11,929 kg).
4. Reduce Enroute Climb performance limited weight by 11,800 lb (5,352 kg).
5. Reduce Approach/Landing Climb performance limited weight by 21,300 lb (9,662 kg).
6. Reduce Landing Field Length limited weight by 151,000 lb (68,492 kg), or increase required Landing Field Length by 230 feet (71 meters).
7. Reduce Maximum Quick Turnaround limited weight by 7,400 lb (3,357 kg).
8. Observe the following Maximum Gross Weight limits:
 - A. Taxi Weight limit is 658,000 lb (298,463 kg).
 - B. Takeoff Weight limit is 656,000 lb (297,556 kg).
 - C. Landing Weight limit is 470,000 lb (213,188 kg).
 - D. Zero Fuel Weight limit is 440,000 lb (199,580 kg).
9. Observe structural Center of Gravity limits for weights above 565,000 lb (256,300 kg) as follows:
 - A. Forward limits:
 - 1) 565,000 lb (256,300 kg) at 16.0 % MAC
 - 2) 582,000 lb (264,000 kg) at 16.3 % MAC
 - 3) 646,300 lb (293,200 kg) at 20.9 % MAC
 - 4) 658,000 lb (298,500 kg) at 21.7 % MAC
 - B. Aft limits:
 - 1) 565,000 lb (256,300 kg) at 44.0 % MAC
 - 2) 642,000 lb (291,200 kg) at 37.4 % MAC
 - 3) 658,000 lb (298,500 kg) at 30.7 % MAC
10. Reduce maximum operating altitude (thrust or buffet limit) by 1,000 feet.

For -300ER:

1. Night time operation with the raked wingtips removed is prohibited.
2. Increase VR, V2 and VREF speeds by 2 knots.

3. Reduce Takeoff performance limited weights by 25,100 lb (11,385 kg).
4. Reduce Landing performance limited weights by 57,500 lb (26,081 kg).
5. Reduce Enroute Climb performance limited weight by 11,700 lb (5,307 kg).
6. Reduce Maximum Quick Turnaround limited weight by 6,900 lb (3,129 kg).
7. Observe the following Maximum Gross Weight limits:
 - A. Taxi Weight limit is 662,000 lb (300,277 kg).
 - B. Takeoff Weight limit is 660,000 lb (299,370 kg).
 - C. Landing Weight limit is 524,000 lb (237,682 kg).
 - D. Zero Fuel Weight limit is 495,000 lb (224,528 kg).
8. Observe Aft Center of Gravity limit of 38.5% MAC for weights above 633,000 lb (287,123 kg).
9. Reduce maximum operating altitude (thrust or buffet limit) by 1,000 feet.



-200LR/-300ER

**57-41-01 Outboard Leading Edge Slat Restoration Seal Assembly**

One symmetrical set of seal assemblies may be missing.

NOTE: For one seal missing, the symmetrical seal assembly on the opposite wing must be removed.

VREF is increased by 2 knots.

Performance limited weights are reduced by the following:

Number Installed	Takeoff	Enroute Climb	Approach/Landing Climb	Landing Field Length	Max Quick Turnaround
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For -200/-200ER/GE90-76B thru -85B:

24	5,400 lb (2,449 kg)	No penalty	3,600 lb (1,632 kg)	14,100 lb, or increase by 150 ft (6,396 kg, or increase by 46 meters)	6,000 lb (2,722 kg)
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For -200/-200ER/GE90-90B:

24	6,000 lb (2,721 kg)	No penalty	6,000 lb (2,721 kg)	14,100 lb, or increase by 150 ft (6,396 kg, or increase by 46 meters)	6,000 lb (2,722 kg)
----	------------------------	------------	------------------------	---	------------------------

For -200/-200ER/GE90-94B:

24	6,000 lb (2,721 kg)	No penalty	3,600 lb (1,632 kg)	14,100 lb, or increase by 150 ft (6,396 kg, or increase by 46 meters)	6,000 lb (2,722 kg)
----	------------------------	------------	------------------------	---	------------------------



For -200/-200ER/PW4074 thru PW4084:

24	4,800 lb (2,177 kg)	No penalty	3,600 lb (1,632 kg)	14,100 lb, or increase by 150 ft (6,396 kg, or increase by 46 meters)	6,000 lb (2,722 kg)
----	------------------------	------------	------------------------	---	------------------------

For -200/-200ER/PW4090:

24	5,400 lb (2,449 kg)	No penalty	3,600 lb (1,632 kg)	14,100 lb, or increase by 150 ft (6,396 kg, or increase by 46 meters)	6,000 lb (2,722 kg)
----	------------------------	------------	------------------------	---	------------------------

For -200/-200ER/RR875 thru RR884:

24	4,200 lb (1,905 kg)	No penalty	3,600 lb (1,632 kg)	14,100 lb, or increase by 150 ft (6,396 kg, or increase by 46 meters)	6,000 lb (2,722 kg)
----	------------------------	------------	------------------------	---	------------------------

For -200/-200ER/RR892:

24	6,600 lb (2,993 kg)	No penalty	4,200 lb (1,905 kg)	14,100 lb, or increase by 150 ft (6,396 kg, or increase by 46 meters)	6,000 lb (2,722 kg)
----	------------------------	------------	------------------------	---	------------------------

**Section 3****777 Dispatch Deviations Guide****ATA 57****For -200/-200ER/RR895:**

24	4,800 lb (2,177 kg)	No penalty	6,600 lb (2,293 kg)	14,100 lb, or increase by 150 ft (6,396 kg, or increase by 46 meters)	6,000 lb (2,722 kg)
----	------------------------	------------	------------------------	---	------------------------

For -300/PW4090:

24	4,800 lb (2,177 kg)	No penalty	3,600 lb (1,632 kg)	14,100 lb, or increase by 150 ft (6,396 kg, or increase by 46 meters)	6,000 lb (2,722 kg)
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For -300/PW4098:

24	7,200 lb (3,265 kg)	No penalty	3,600 lb (1,632 kg)	14,100 lb, or increase by 150 ft (6,396 kg, or increase by 46 meters)	6,000 lb (2,722 kg)
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For -300/RR892:

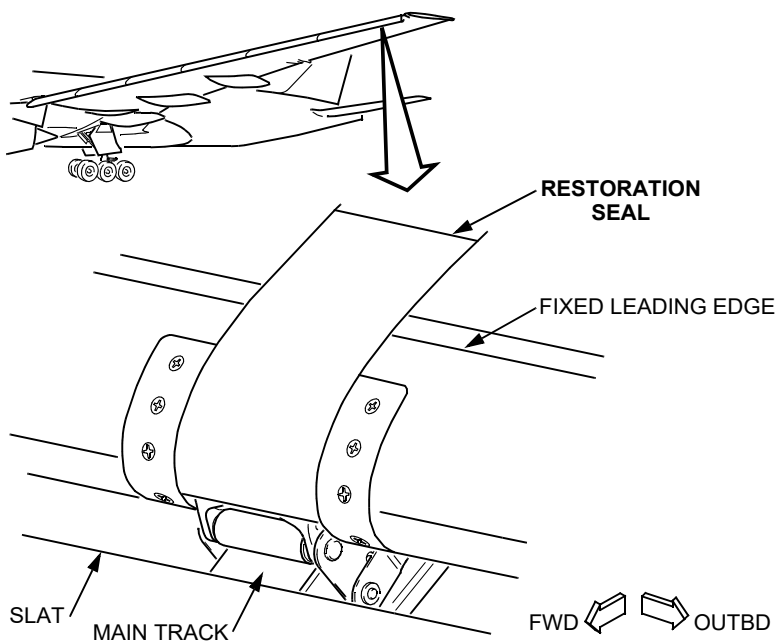
24	4,800 lb (2,177 kg)	No penalty	3,600 lb (1,632 kg)	14,100 lb, or increase by 150 ft (6,396 kg, or increase by 46 meters)	6,000 lb (2,722 kg)
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For -300ER:

26	5,100 lb (2,314 kg)	No penalty	3,600 lb (1,633 kg)	16,400 lb (7,438 kg)	13,400 lb (6,078 kg)
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For -200LR/777F:

26	5,400 lb (2,450 kg)	No penalty	4,800 lb (2,178 kg)	155,000 lb, or increase by 230 feet (70,307 kg, or increase by 71 meters)	13,400 lb (6,078 kg)
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**57-41-02 Wing Fixed Leading Edge Lower Panel**

A maximum of two per wing (four total) may be missing.

Performance limited weights are reduced by the following for each missing item.

Number Installed	Takeoff	Enroute Climb	Approach/Landing Climb
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For -200/-200ER/-300:

Outboard panels

52	250 lb (114 kg)	350 lb (159 kg)	200 lb (91 kg)
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For -200/-200ER/-300:

Inboard panels

18	850 lb (386 kg)	1350 lb (613 kg)	750 lb (341 kg)
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For -300ER:

Outboard panels

56	150 lb (69 kg)	350 lb (159 kg)	150 lb (69 kg)
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For -300ER:

Inboard panels

18	600 lb (273 kg)	1,250 lb (567 kg)	450 lb (205 kg)
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For -200LR/777F:

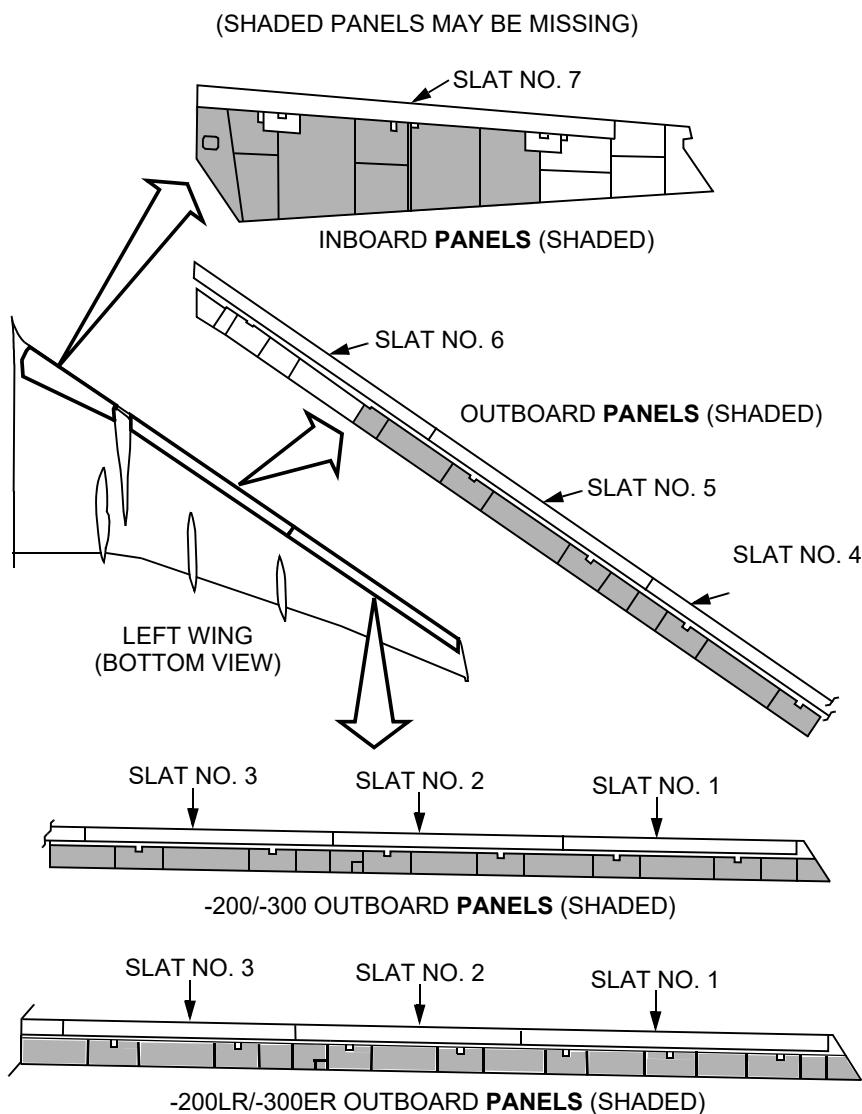
Outboard panels

56	200 lb (91 kg)	300 lb (137 kg)	150 lb (69 kg)
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For -200LR/777F:

Inboard panels

18	650 lb (295 kg)	1,150 lb (522 kg)	550 lb (250 kg)
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57-41-03 Slat Linkage Fairing Panel

A maximum of six per wing (twelve total) may be missing.

Performance limited weights are reduced by the following for each missing item:

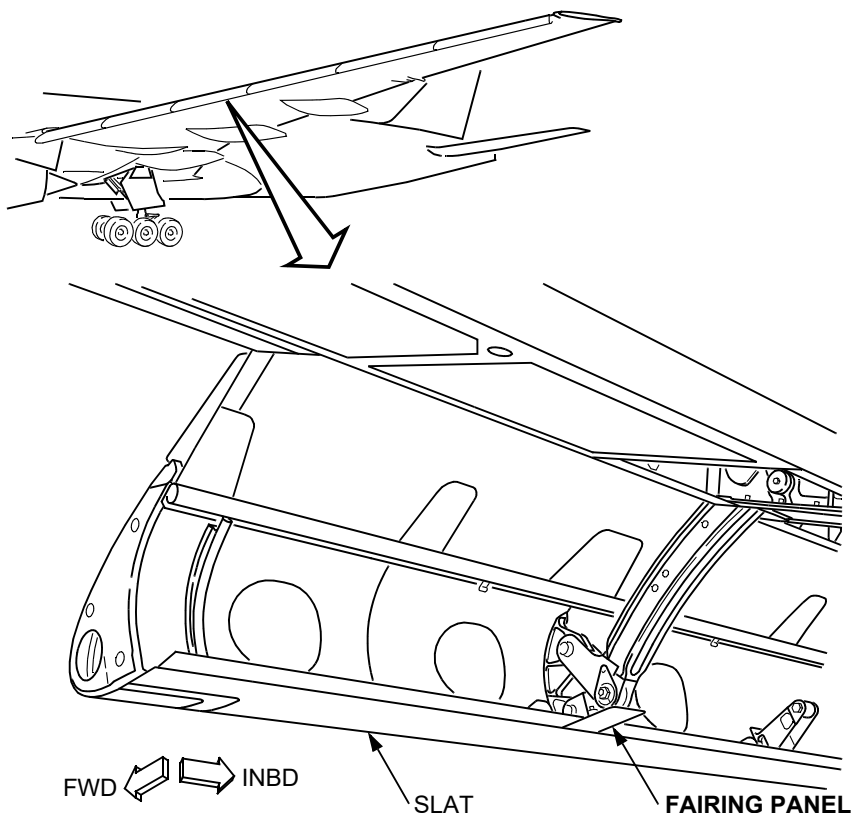
Number Installed	Takeoff	Enroute Climb	Approach/Landing Climb
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For All (Except -200LR/777F/-300ER):

32	Negligible penalty	Negligible penalty	No penalty
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For -200LR/777F/-300ER:

34	Negligible penalty	Negligible penalty	No penalty
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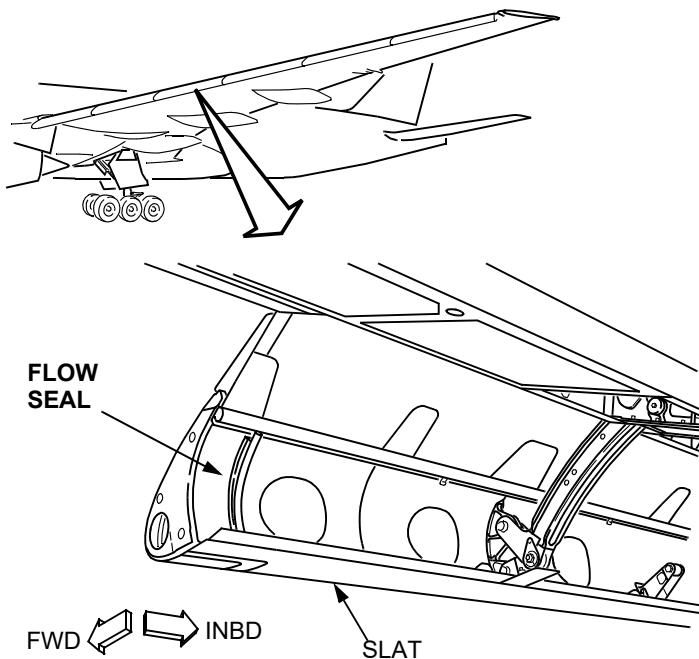


57-41-04 Slat Spanwise Flow Seal

A maximum of two per wing (four total) may be missing.

Performance limited weights are reduced by the following for each missing item:

Number Installed	Takeoff	Enroute Climb	Approach/Landing Climb
28	Negligible penalty	200 lb (91 kg)	No penalty



57-41-05 Outboard Slat Spanwise Bulb Seal

One per wing (two total) may be missing.

NOTE: Airplanes affected by FAA Airworthiness Directive 2002-11-06 must comply with the Alternate Method of Compliance (AMOC) to AD 2002-11-06 (approved by FAA letter: 120S-02-636, dated 09-Aug-02) when operating with missing seals. AD 2002-11-06 is effective for airplane line numbers 1 through 369. Under the AMOC, airplanes affected by AD 2002-11-06 can operate with missing seals for a maximum of 50 flight hours provided the SB 777-57A0034 inspection is performed on the affected slat prior to further flight and at the time of seal replacement.

Performance limited weights are reduced by the following for each foot of missing seal:

Number Installed	Takeoff	Enroute Climb	Approach/Landing Climb
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For -200/-300/PW/GE:

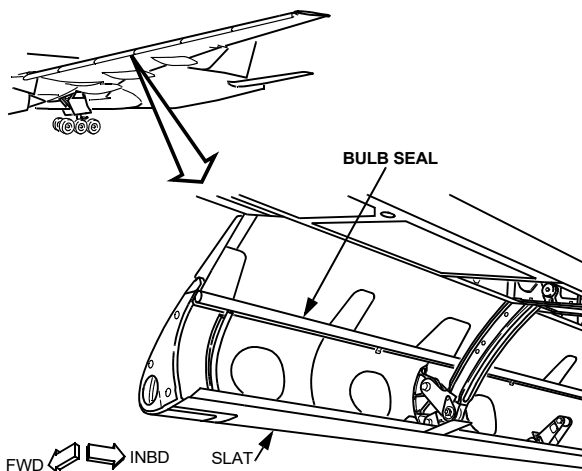
12	350 lb (159 kg)	600 lb (273 kg)	No penalty
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For -200/-300/RR:

12	350 lb (159 kg)	500 lb (227 kg)	No penalty
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For -200LR/-300ER/777F:

12	300 lb (137 kg)	550 lb (250 kg)	No penalty
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57-41-06 Slat Spanwise Lower Flexible Seal

A maximum of four per wing (eight total) may be missing.

Performance limited weights are reduced by the following for each missing item:

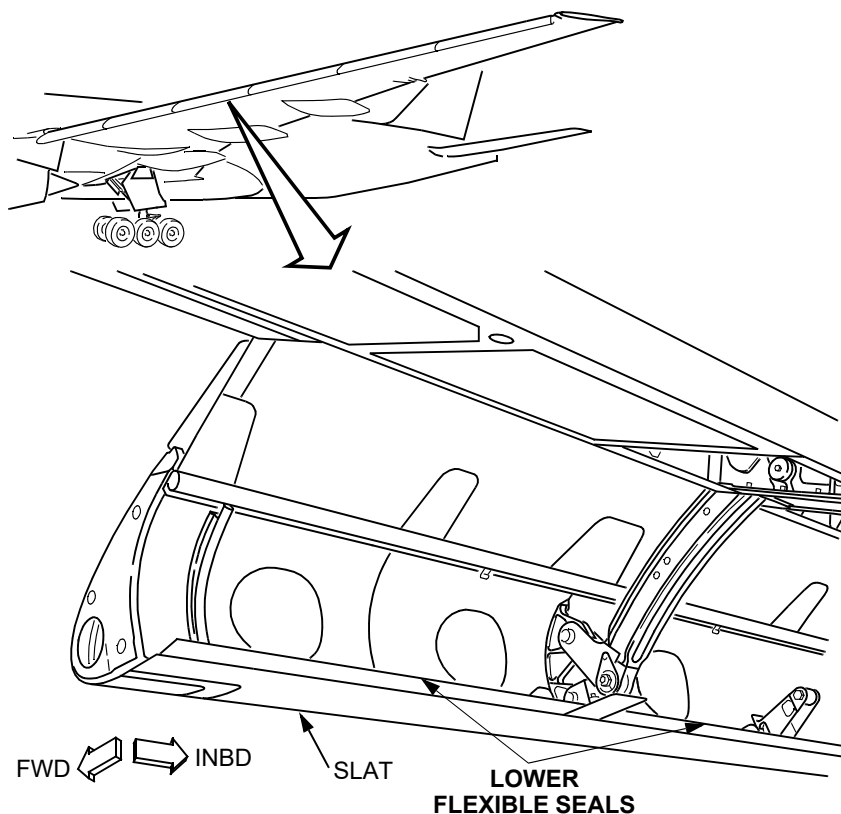
Number Installed	Takeoff	Enroute Climb	Approach/Landing Climb
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For All (Except -200LR/777F/-300ER):

40	Negligible penalty	Negligible penalty	No penalty
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For -200LR/777F/-300ER:

42	Negligible penalty	Negligible penalty	No penalty
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57-41-07 DELETED (Refuel Station Access Door)

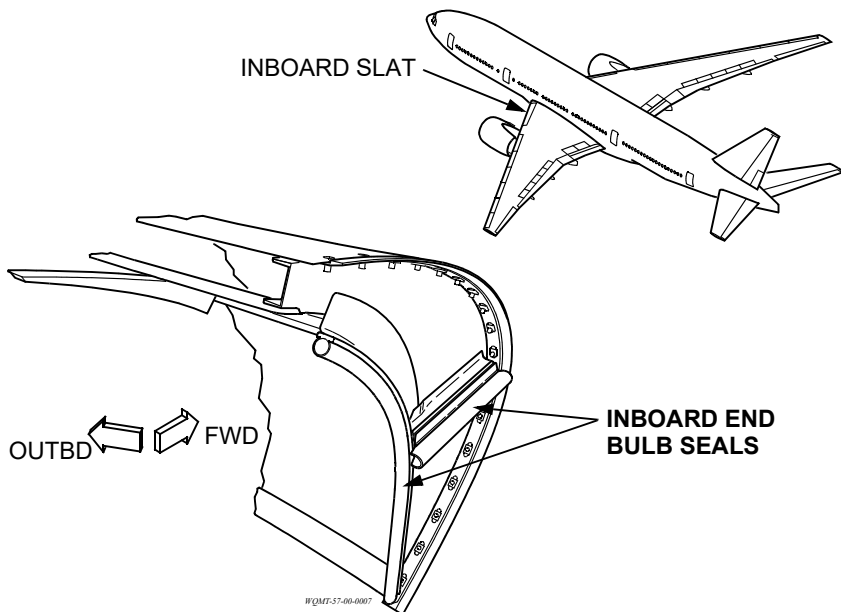
Relief was deleted (AFM Appendix CDL D631W001/D631W002 Revision 8).

57-41-08 Inboard Slat Inboard End Bulb Seal

Any number may be missing.

Performance limited weights are reduced by the following for each missing item:

Number Installed	Takeoff & Landing	Enroute Climb
4	150 lb (69 kg)	250 lb (113 kg)



**57-41-09 Krueger End Seal**

Any number may be missing.

Performance limited weights are reduced by the following for each missing item:

Number Installed	Takeoff	Enroute Climb	Approach/Landing
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For -200/-200ER/RR875 thru RR884:

6	1,050 lb (477 kg)	Negligible penalty	900 lb (409 kg)
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For -200/-200ER/PW4074 thru PW4090/RR895:

6	1350 lb (613 kg)	Negligible penalty	1650 lb (749 kg)
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For -200/-200ER/RR892:

6	1,650 lb (749 kg)	Negligible penalty	1,050 lb (477 kg)
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For -200/-200ER/GE90-76B thru -94B:

6	1,500 lb (681 kg)	Negligible penalty	1,500 lb (681 kg)
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For -300/PW4090/RR892:

6	1,200 lb (545 kg)	Negligible penalty	900 lb (409 kg)
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For -300/PW4098:

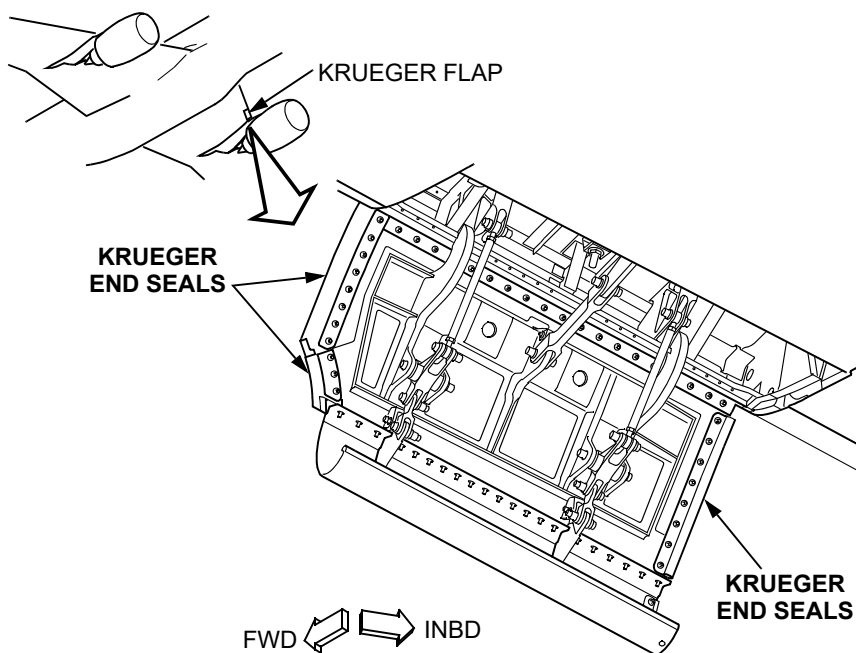
6	1,800 lb (817 kg)	Negligible penalty	900 lb (409 kg)
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For -300ER:

6	1,300 lb (590 kg)	Negligible penalty	900 lb (409 kg)
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For -200LR/777F:

6	1,350 lb (613 kg)	Negligible penalty	1,200 lb (545 kg)
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57-41-23 Inboard Slat Inboard End Blade Seal

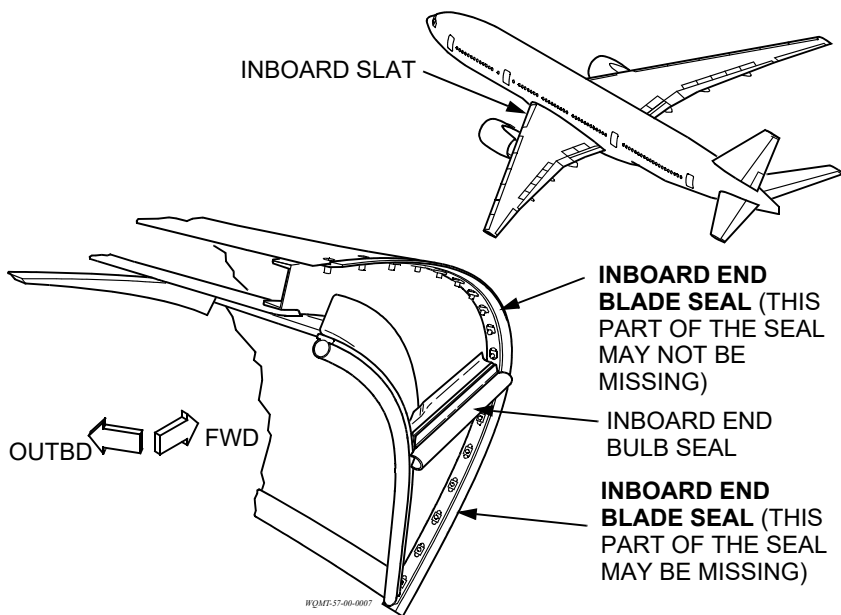
Any number may be missing.

NOTE 1: Seal may be missing only below the crossing Inboard End Bulb Seal (CDL 57-41-08).

NOTE 2: Seal may not be missing in combination with items CDL 57-41-06 or CDL 57-41-08.

Performance limited weights are reduced by the following for each missing item:

Number Installed	Takeoff	Enroute Climb	Approach/Landing Climb
2	Negligible penalty	Negligible penalty	No penalty



**57-51-01 Wing Fixed Trailing Edge Lower Panel**

A maximum of two per wing (four total) may be missing (Inboard or Outboard).

NOTE: This item may not be missing in combination with items CDL 32-12-01 or CDL 32-12-02.

For missing Inboard panels (forward of the inboard flaps), the maximum airspeed limits (VMO/MMO) are 270 KIAS/0.73 Mach.

NOTE: Place Alternate Gear Down Dispatch switch (Main Equipment Center - if installed) to VMO position. The Warning Electronics System (WES) will set VMO/MMO to 270 KIAS/.73 Mach and EICAS memo message VMO GEAR DOWN will be displayed.

Performance limited weights are reduced by the following for each missing item:

Number Installed	Takeoff	Enroute Climb	Approach/Landing Climb
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For -200/-200ER/GE90-76B thru -85B:

30	5,750 lb (2,609 kg)	9,550 lb (4,332 kg)	150 lb (69 kg)
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For -200/-200ER/GE90-90B:

30	6,350 lb (2,881 kg)	10,800 lb (4,899 kg)	250 lb (114 kg)
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For -200/-200ER/GE90-94B:

30	6,350 lb (2,881 kg)	12,700 lb (5,761 kg)	150 lb (69 kg)
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For -200/-200ER/RR875 thru RR884:

30	4,450 lb (2,019 kg)	9,550 lb (4,332 kg)	150 lb (69 kg)
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For -200/RR892:

30	7,000 lb (3,176 kg)	10,800 lb (4,899 kg)	200 lb (91 kg)
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For -200/RR895:

30	5,100 lb (2,314 kg)	10,800 lb (4,899 kg)	300 lb (137 kg)
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For -200/PW4074 thru PW4084:

30	5,100 lb (2,314 kg)	12,700 lb (5,761 kg)	150 lb (69 kg)
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For -200/PW4090:

30	5,750 lb (2,609 kg)	11,450 lb (5,194 kg)	150 lb (69 kg)
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For -300/RR884 or RR892:

30	5,100 lb (2,314 kg)	10,150 lb (4,604 kg)	150 lb (69 kg)
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For -300/PW4090:

30	5,100 lb (2,314 kg)	11,450 lb (5,194 kg)	150 lb (69 kg)
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For -300/PW4098:

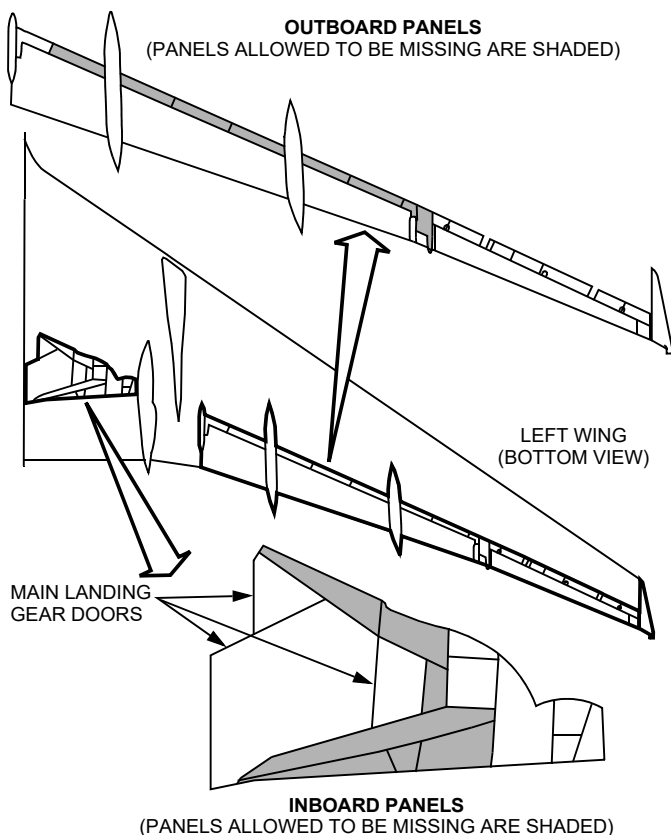
30	7,650 lb (3,470 kg)	11,450 lb (5,194 kg)	150 lb (69 kg)
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For -300ER:

30	5,400 lb (2,450 kg)	11,450 lb (5,194 kg)	150 lb (69 kg)
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For -200LR/777F:

30	5,750 lb (2,609 kg)	10,800 lb (4,899 kg)	200 lb (91 kg)
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57-51-02 Inboard Flap Inboard Seal

Any number may be missing.

Performance limited weights are reduced by the following for each foot of missing seal:

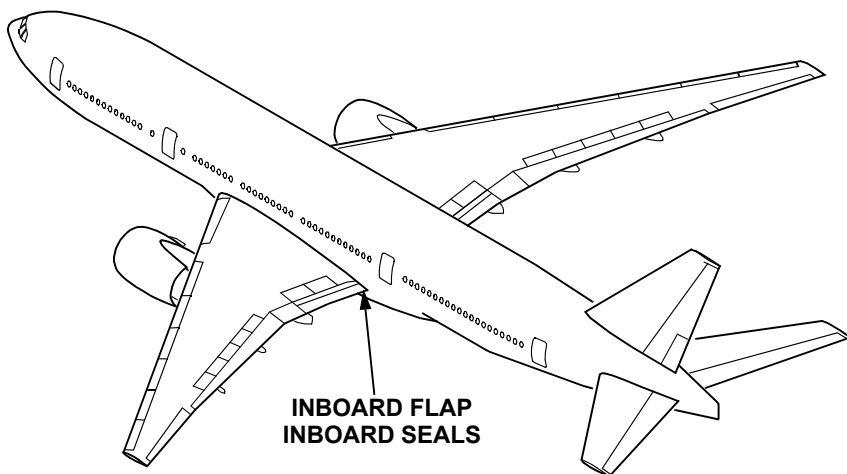
Number Installed	Takeoff	Enroute Climb	Approach/Landing Climb
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For -200/-300:

6	300 lb (137 kg)	450 lb (205 kg)	700 lb (318 kg)
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For -300ER/-200LR/777F:

6	200 lb (91 kg)	400 lb (182 kg)	500 lb (227 kg)
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57-51-03 Overwing Fairing Torque Tube Door

Any number may be missing.

Performance limited weights are reduced by the following for each missing item:

Number Installed	Takeoff	Enroute Climb	Approach/Landing Climb
-------------------------	----------------	----------------------	-------------------------------

For -200/-200ER/GE90-76B thru -85B or GE90-94B or PW4074 thru PW4090 or RR875 thru RR884 or -300/PW4090 or RR892 or -300ER:

2	No penalty	No penalty	7,050 lb (3,198 kg)
---	------------	------------	---------------------

For -200/-200ER/RR892:

2	No penalty	No penalty	8,200 lb (3,719 kg)
---	------------	------------	---------------------

For -200/-200ER/GE90-90B:

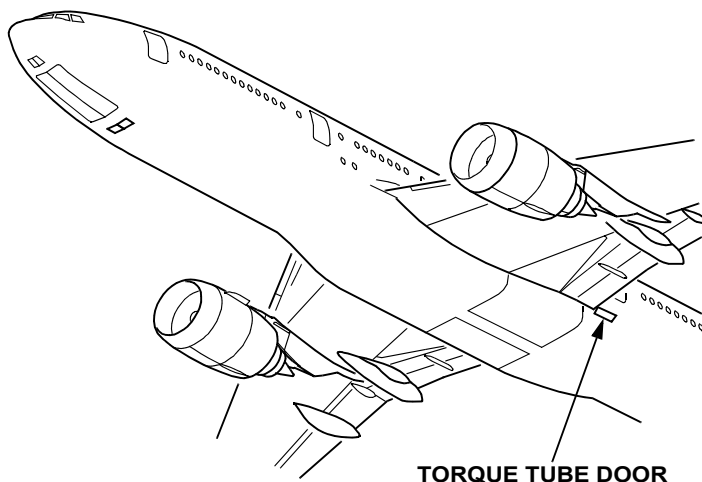
2	No penalty	No penalty	11,750 lb (5,329 kg)
---	------------	------------	----------------------

For -200/-200ER/RR895:

2	No penalty	No penalty	12,900 lb (5,851 kg)
---	------------	------------	----------------------

For -200LR/777F:

2	No penalty	No penalty	9,400 lb (4,264 kg)
---	------------	------------	---------------------



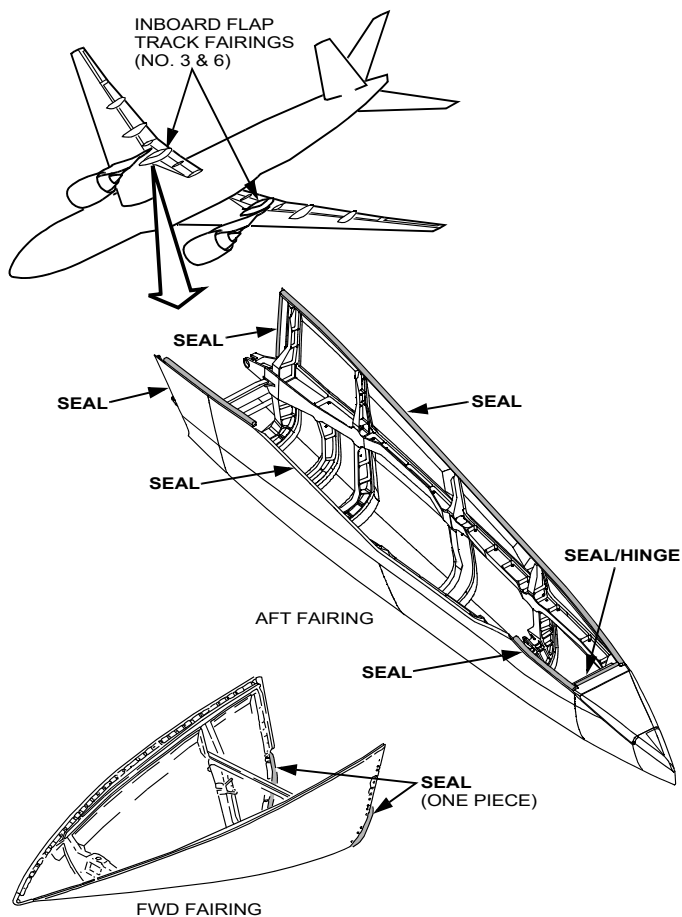
**57-51-04 Number 3 & 6 Flap Fairings Seals**

A maximum of seven seals (including hinge on aft most seal) per fairing (fourteen total) may be missing.

Performance limited weights are reduced by the following:

NOTE: The penalties shown are for one or more seals missing from only one fairing. If one or more seals are missing from both fairings, the penalties are doubled.

Number Installed	Takeoff	Enroute Climb	Approach/Landing Climb
20	150 lb (68 kg)	250 lb (113 kg)	No penalty

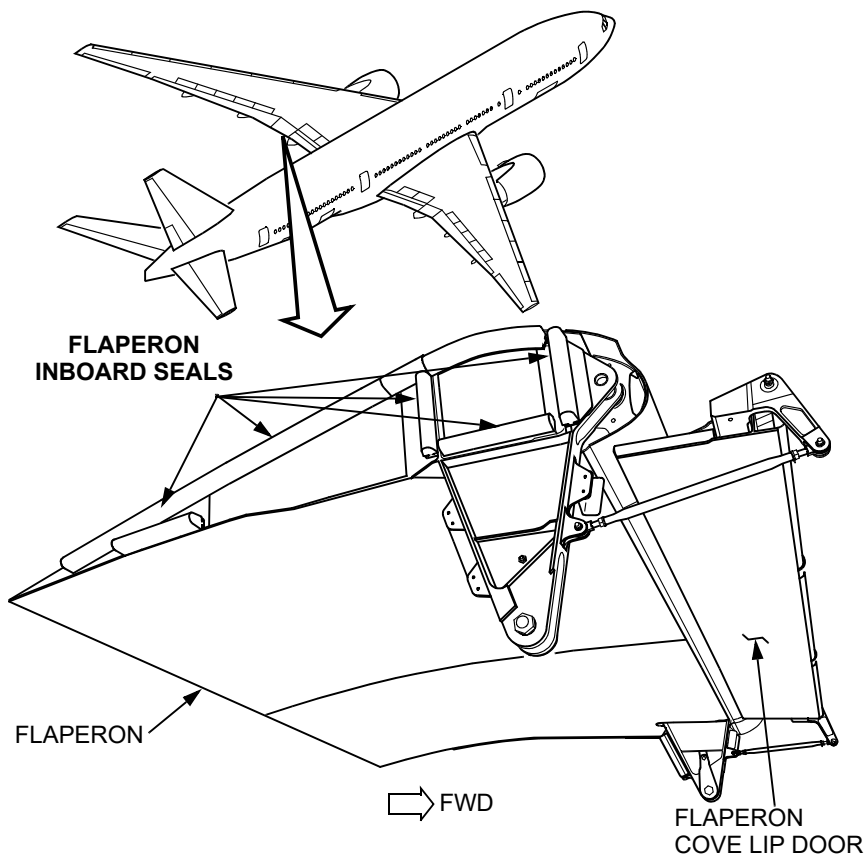


57-53-01 Flaperon Inboard Seal

Any number may be missing.

Performance limited weights are reduced by the following for each foot of missing seal:

Number Installed	Takeoff & Landing	Enroute Climb
12	200 lb (91 kg)	350 lb (159 kg)

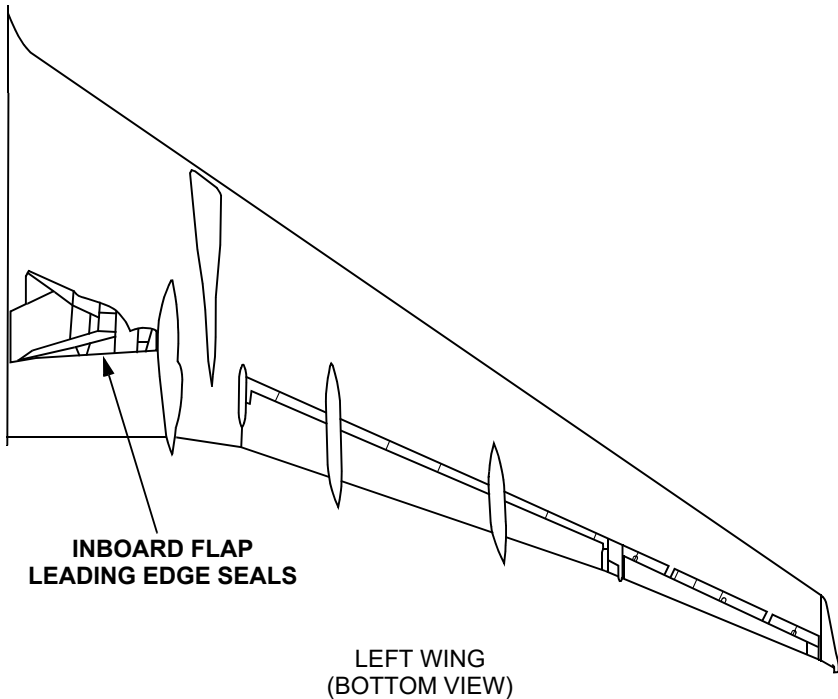


57-53-02 Inboard Flap Leading Edge Seal

A maximum of two per wing (four total) may be missing.

Performance limited weights are reduced by the following for each foot of missing seal:

Number Installed	Takeoff	Enroute Climb	Approach/Landing Climb
20	150 lb (68 kg)	250 lb (113 kg)	No penalty

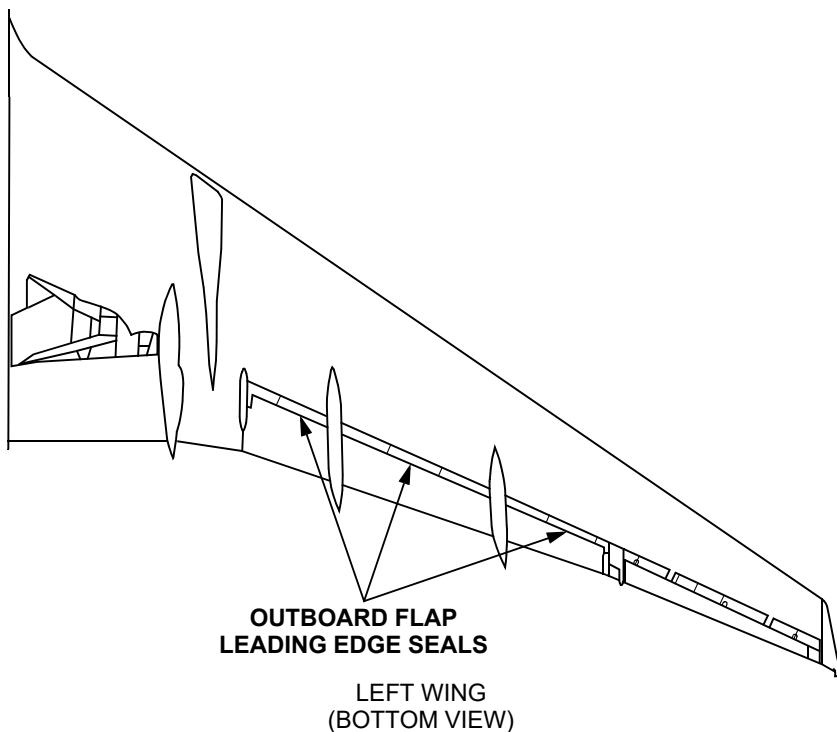


57-53-03 Outboard Flap Leading Edge Seal

One per wing (two total) may be missing.

Performance limited weights are reduced by the following for each foot of missing seal:

Number Installed	Takeoff	Enroute Climb	Approach/Landing Climb
8	150 lb (68 kg)	250 lb (113 kg)	No penalty





57-63-01 DELETED (Flaperon Cove Lip Door)

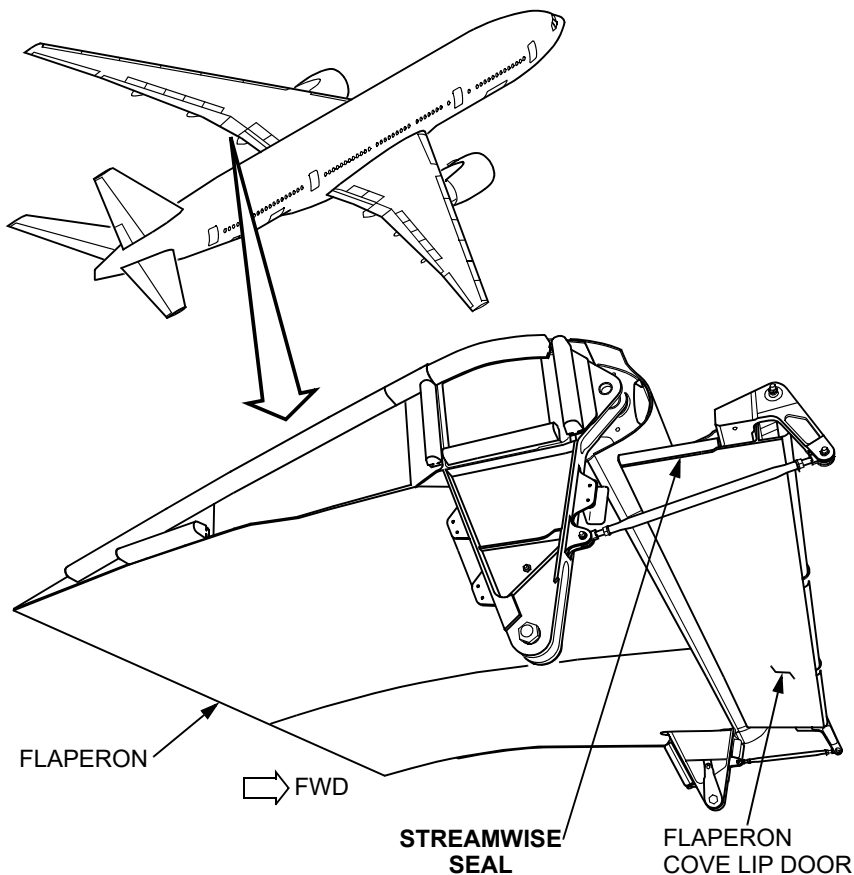
Relief was deleted (AFM Appendix CDL D631W001/D631W002 Revision 8).

57-63-02 Flaperon Cove Lip Door Streamwise Seal

Any number may be missing.

Performance limited weights are reduced by the following for each missing item:

Number Installed	Takeoff	Enroute Climb	Approach/Landing Climb
4	Negligible penalty	150 lb (68 kg)	No penalty



**57-63-03 Flaperon Cove Lip Door Hingewise Seal**

One may be missing.

Performance limited weights are reduced by the following:

Number Installed	Takeoff	Enroute Climb	Approach/Landing Climb
------------------	---------	---------------	------------------------

For -200/-200ER/GE90-76B thru -85B:

2	1,000 lb (454 kg)	1,700 lb (771 kg)	No penalty
---	-------------------	-------------------	------------

For -200/-200ER/GE90-90B:

2	1,100 lb (499 kg)	1,900 lb (862 kg)	No penalty
---	-------------------	-------------------	------------

For -200/-200ER/GE90-94B:

2	1,100 lb (499 kg)	2,200 lb (998 kg)	No penalty
---	-------------------	-------------------	------------

For -200/-200ER/RR875 thru RR884:

2	800 lb (363 kg)	1,700 lb (771 kg)	No penalty
---	-----------------	-------------------	------------

For -200/-200ER/RR892:

2	1,250 lb (567 kg)	1,900 lb (862 kg)	No penalty
---	-------------------	-------------------	------------

For -200/-200ER/RR895:

2	900 lb (409 kg)	1,900 lb (862 kg)	No penalty
---	-----------------	-------------------	------------

For -200/-200ER/PW4074 thru PW4084:

2	900 lb (409 kg)	2,200 lb (998 kg)	No penalty
---	-----------------	-------------------	------------

For -200/-200ER/PW4090:

2	1,000 lb (454 kg)	2,000 lb (908 kg)	No penalty
---	-------------------	-------------------	------------

For -300/RR892:

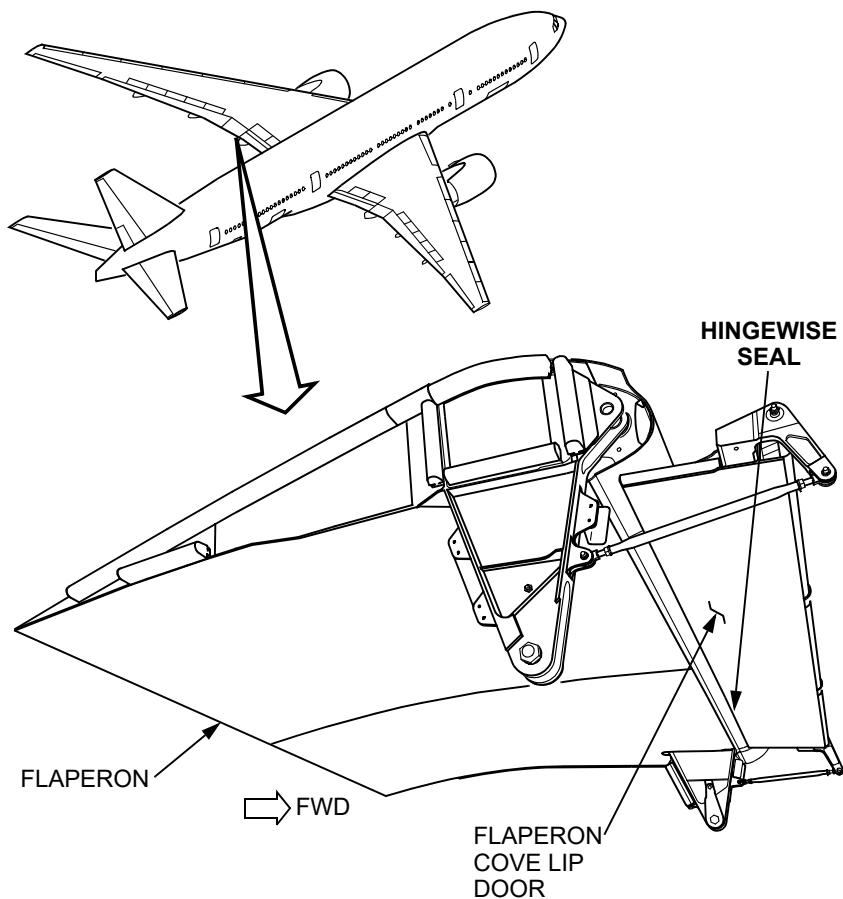
2	900 lb (409 kg)	1,800 lb (817 kg)	No penalty
---	-----------------	-------------------	------------

For -300/PW4090:

2	900 lb (409 kg)	2,000 lb (908 kg)	No penalty
---	-----------------	-------------------	------------

For -200LR/777F or -300ER:

2	1,000 lb (453 kg)	2,000 lb (908 kg)	No penalty
---	-------------------	-------------------	------------

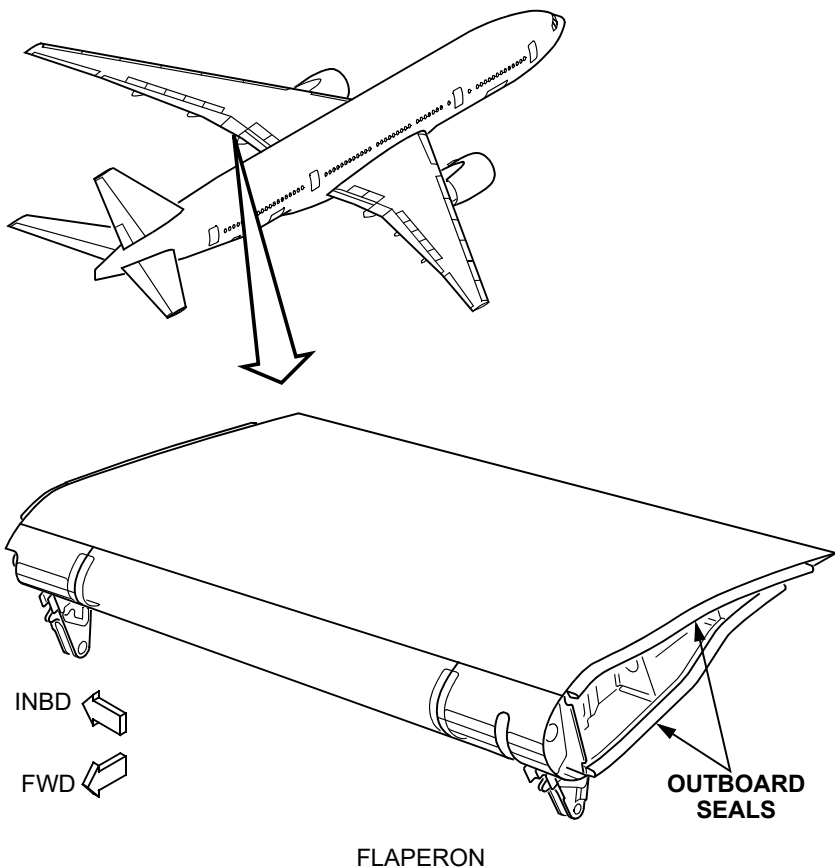


57-63-04 Flaperon Outboard Seal

Any number may be missing.

Performance limited weights are reduced by the following for each foot of missing seal:

Number Installed	Takeoff & Landing	Enroute Climb
4	200 lb (91 kg)	350 lb (159 kg)

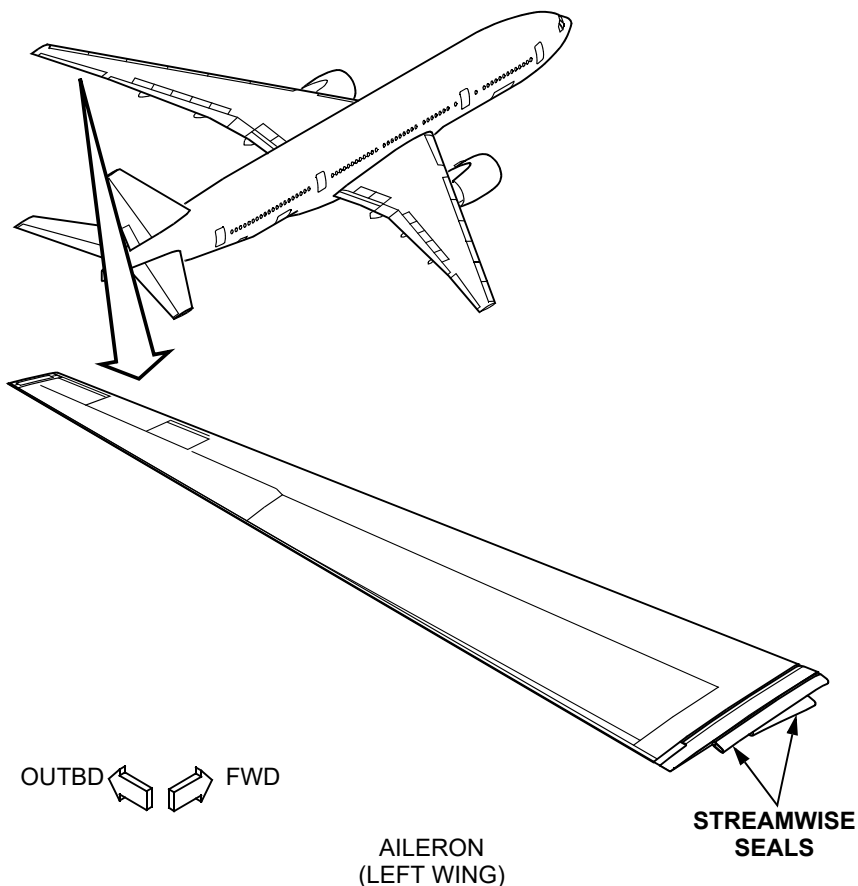


57-63-05 Aileron Streamwise Seal

Any number may be missing.

Performance limited weights are reduced by the following for each foot of missing seal:

Number Installed	Takeoff & Landing	Enroute Climb
8	200 lb (91 kg)	300 lb (136 kg)

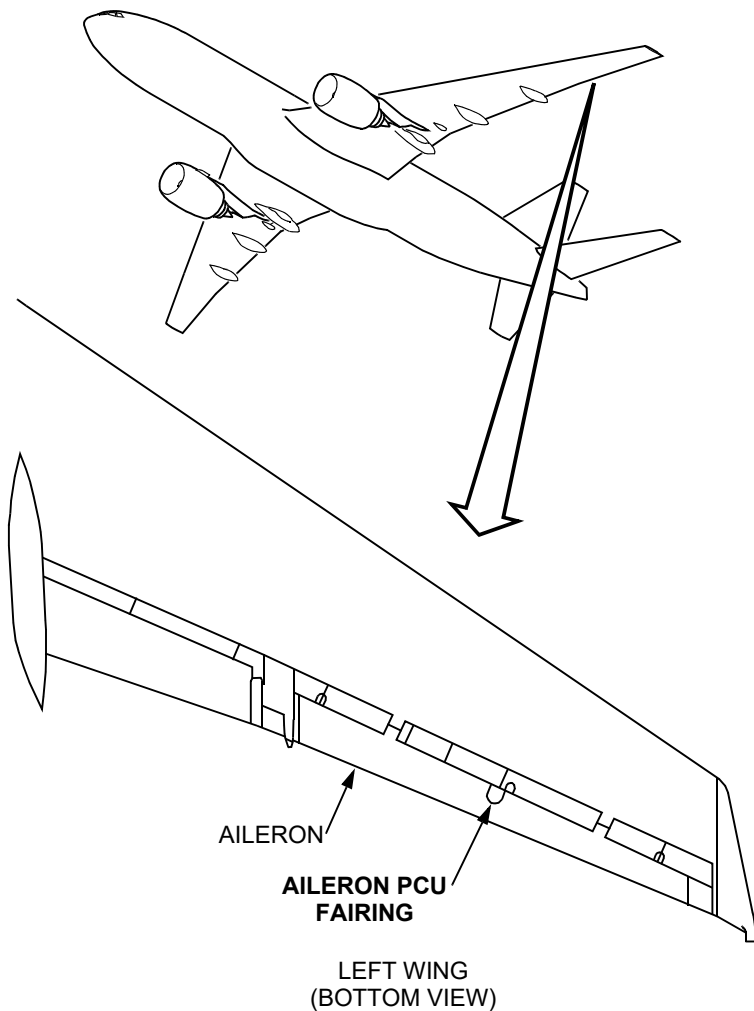


57-63-06 Aileron Power Control Unit Fairing

Any number may be missing.

Performance limited weights are reduced by the following:

Number Installed	Takeoff & Landing	Enroute Climb
2	Negligible penalty	Negligible penalty

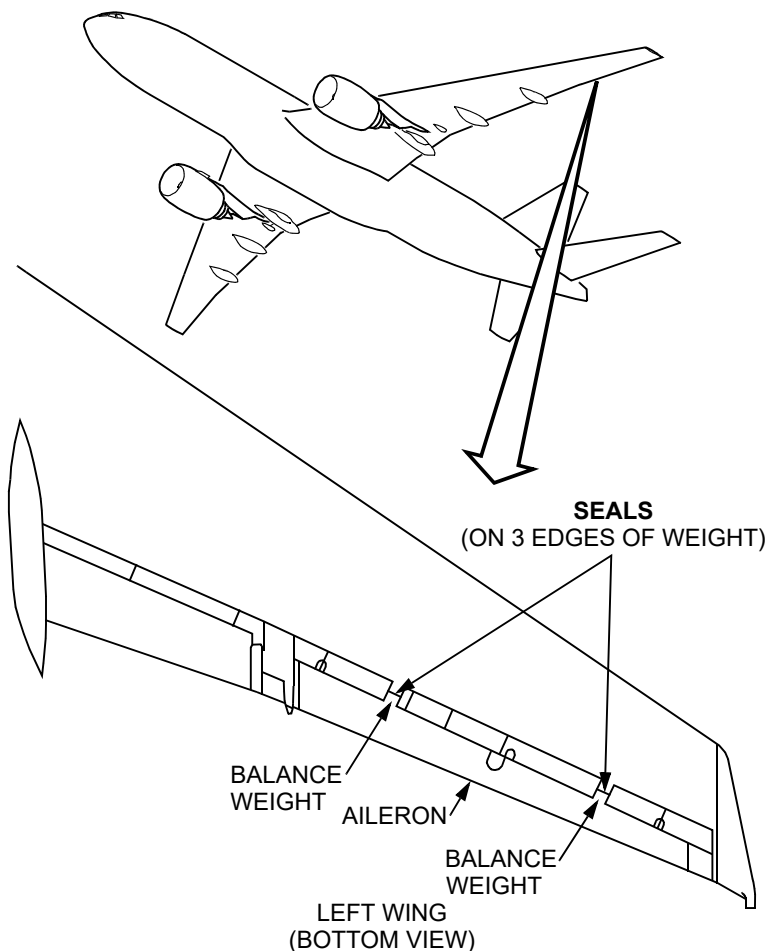


57-63-07 Aileron Mass Balance Seal

Any number may be missing.

Performance limited weights are reduced by the following for each foot of missing seal:

Number Installed	Takeoff	Enroute Climb	Approach/Landing
12	50 lb (23 kg)	100 lb (46 kg)	50 lb (23 kg)



57-63-08 Aileron Hinge Blade Seals

Any number may be missing.

Performance limited weights are reduced by the following for each missing seal:

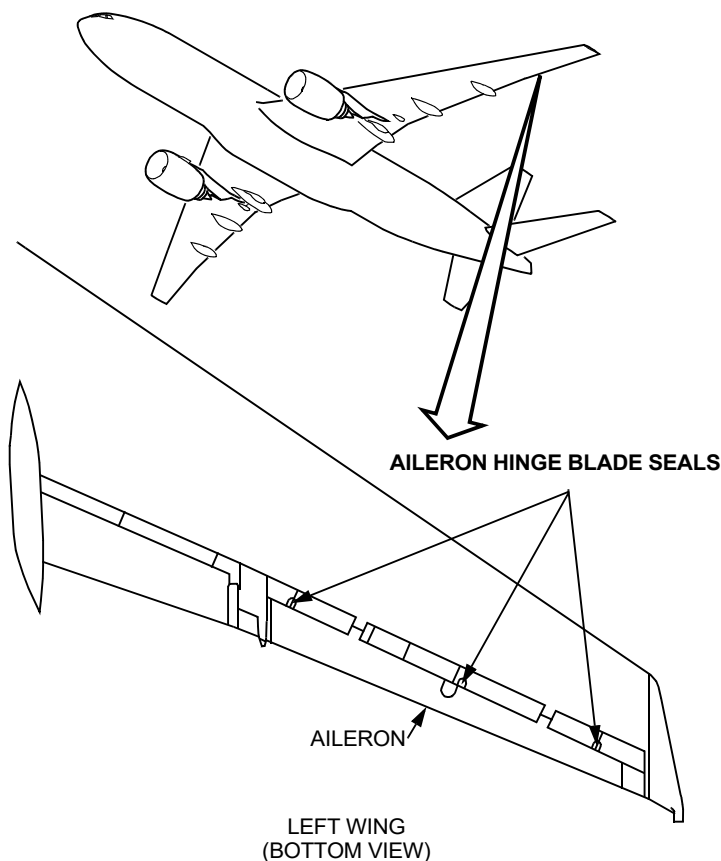
Number Installed	Takeoff	Enroute Climb	Approach/Landing Climb
------------------	---------	---------------	------------------------

For -200/-200ER/-300:

6	200 lb (91 kg)	150 lb (68 kg)	150 lb (68 kg)
---	----------------	----------------	----------------

For -200LR/-300ER/F:

8	150 lb (68 kg)	150 lb (68 kg)	150 lb (68 kg)
---	----------------	----------------	----------------



57-63-09 Aileron Hinge Blade Seal and Seal Retainer

Any number may be missing.

Performance limited weights are reduced by the following for each missing seal and seal retainer:

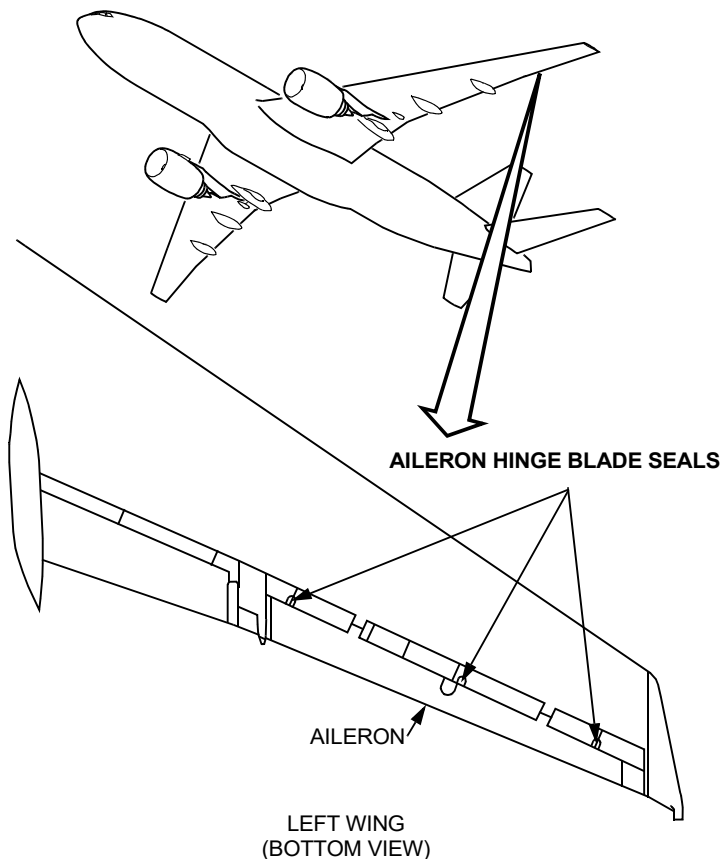
Number Installed	Takeoff	Enroute Climb	Approach/Landing Climb
-------------------------	----------------	----------------------	-------------------------------

For -200/-200ER/-300:

6	500 lb (227 kg)	400 lb (181 kg)	450 lb (204 kg)
---	-----------------	-----------------	-----------------

For -200LR/-300ER/F:

8	350 lb (159 kg)	350 lb (159 kg)	350 lb (159 kg)
---	-----------------	-----------------	-----------------

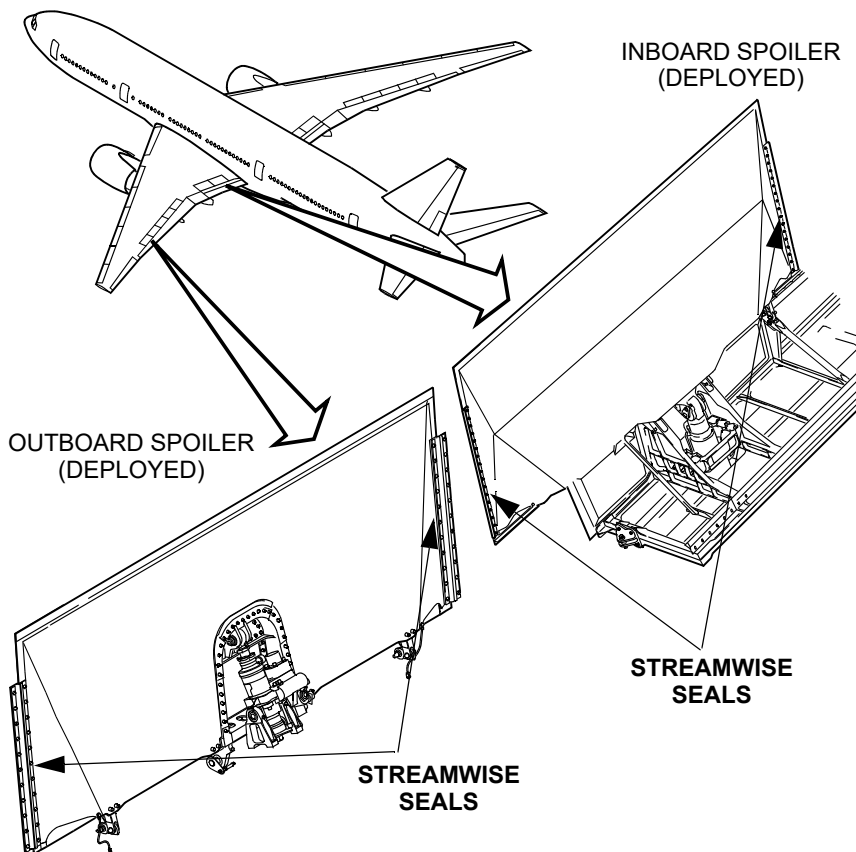


57-71-01 Spoiler Streamwise Seal

A maximum of two per wing (four total) may be missing.

Performance limited weights are reduced by the following for each missing item:

Number Installed	Takeoff & Landing	Enroute Climb
28	600 lb (272 kg)	1,000 lb (454 kg)

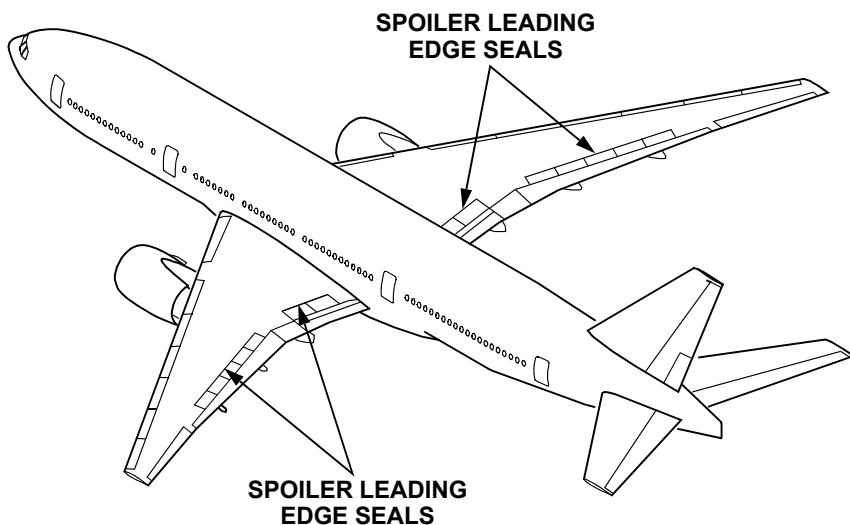


57-71-02 Spoiler Leading Edge Seal

One per wing (two total) may be missing.

Performance limited weights are reduced by the following for each foot of missing seal:

Number Installed	Takeoff & Landing	Enroute Climb
36	150 lb (68 kg)	250 lb (113 kg)





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ATA 71

- 71-11-01 Oil Tank Sight Glass Door/Pressure Relief Door (PW)**
- 71-11-02 Engine Cowl Inlet Streamwise Aerofiller**
- 71-11-03 Fan Cowl Pressure Relief Door (GE, Except GE90-100 Series)**
- 71-11-04 Oil Tank Access Door (GE)**
- 71-11-05 Fan Cowl Hoist Point Plugs (PW and GE)**



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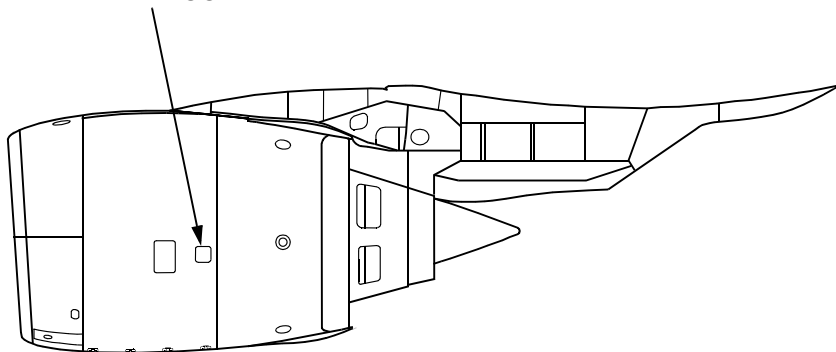
71-11-01 Oil Tank Sight Glass Door/Pressure Relief Door (PW)

Any number may be missing.

Performance limited weights are reduced by the following:

Number Installed	Takeoff & Landing	Enroute Climb
2	Negligible penalty	Negligible penalty

**OIL TANK SIGHT GLASS
DOOR/PRESSURE
RELIEF DOOR**



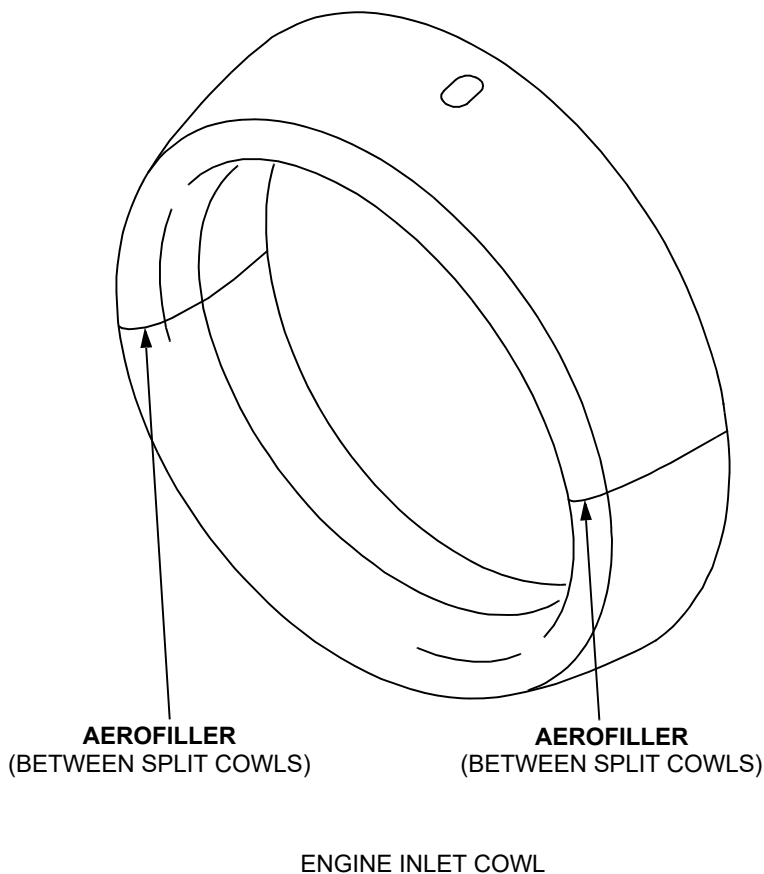
LEFT SIDE OF ENGINE

71-11-02 Engine Cowl Inlet Streamwise Aerofiller

Any number may be missing.

Performance limited weights are reduced by the following:

Number Installed	Takeoff & Landing	Enroute Climb
2	Negligible penalty	Negligible penalty

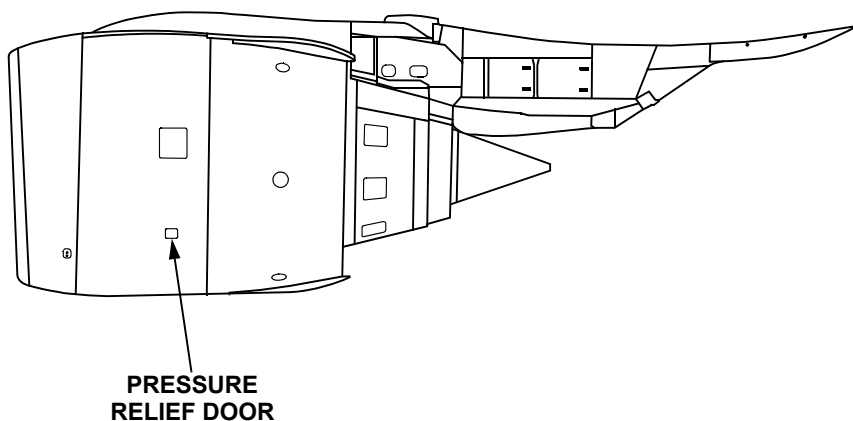


71-11-03 Fan Cowl Pressure Relief Door (GE, Except GE90-100 Series)

Any number may be missing.

Performance limited weights are reduced by the following:

Number Installed	Takeoff & Landing	Enroute Climb
2	Negligible penalty	Negligible penalty



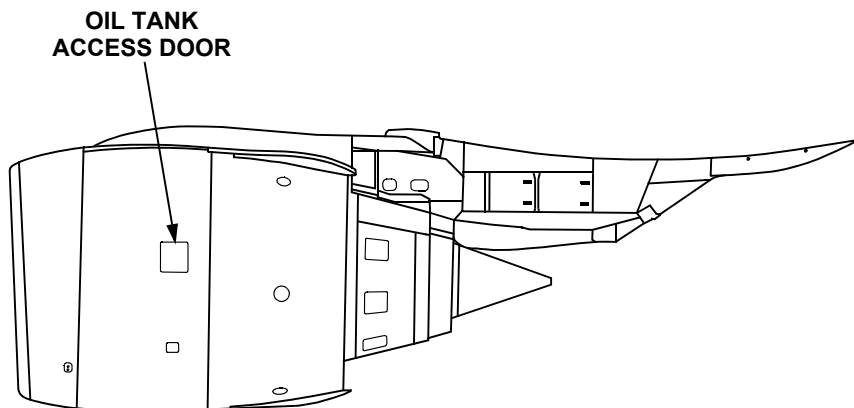
LEFT SIDE OF ENGINE

71-11-04 Oil Tank Access Door (GE)

Any number may be missing.

Performance limited weights are reduced by the following for each missing item:

Number Installed	Takeoff & Landing	Enroute Climb
2	Negligible penalty	200 lb (91 kg)



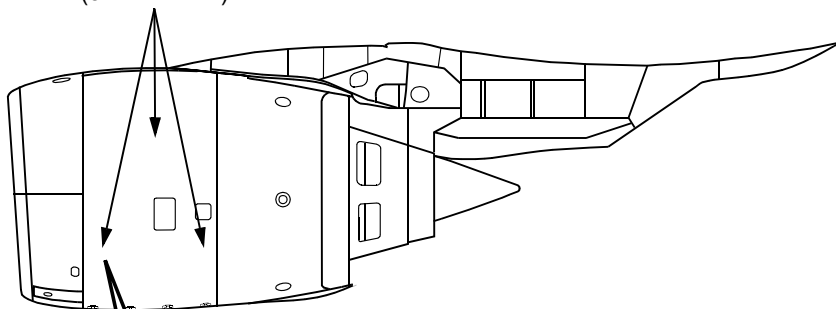
LEFT SIDE OF ENGINE

71-11-05 Fan Cowl Hoist Point Plugs (PW and GE)

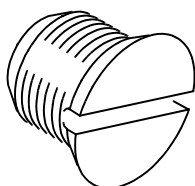
Any number may be missing.

Performance limited weights are reduced by the following for each missing item:

Number Installed	Takeoff & Landing	Enroute Climb
12	Negligible penalty	Negligible penalty

**HOIST POINT PLUGS
(3 PER SIDE)**


LEFT SIDE OF ENGINE



PLUG



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ATA 72

72-00-01 Debris Shield (PW)



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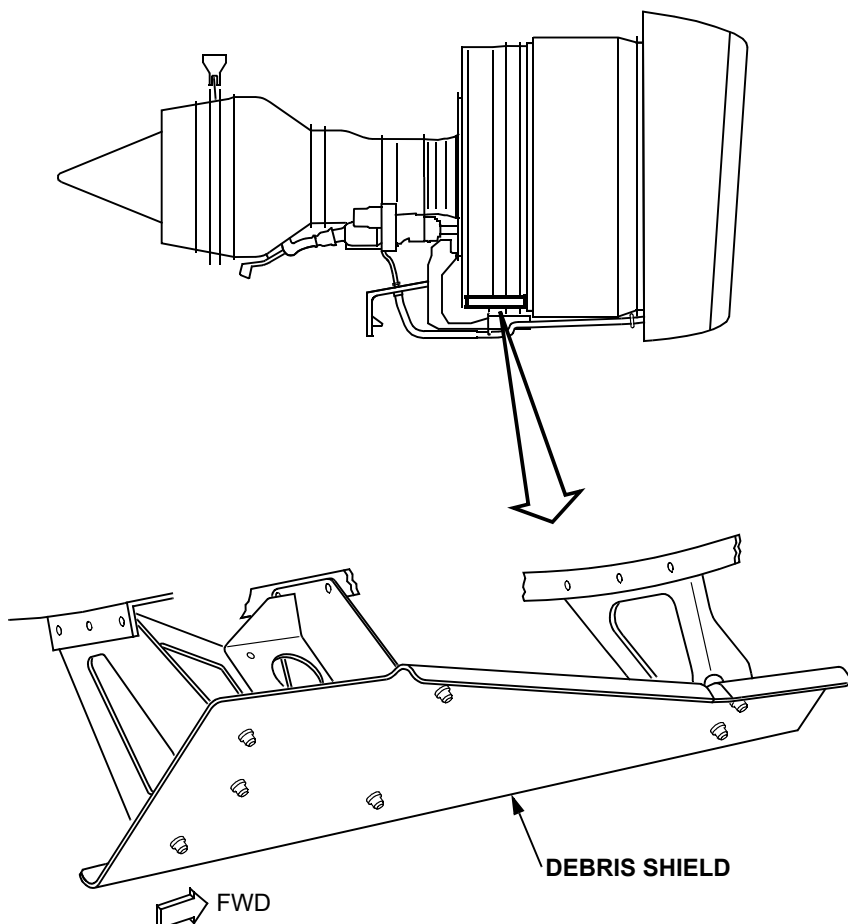
72-00-01 Debris Shield (PW)

The right engine Debris Shield may be missing.

NOTE: The left engine Debris Shield may not be missing.

Performance limited weights are reduced by the following:

Number Installed	Takeoff & Landing	Enroute Climb
2	No penalty	No penalty





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ATA 78

78-31-01	Thrust Reverser Sleeve Actuator Access Panel
78-31-02	Thrust Reverser Blocker Door
78-31-03	Thrust Reverser Cascade Segment
78-31-04	Thrust Reverser Wedge Fairing
78-31-05	Thrust Reverser Blocker Door Seals
78-31-06	Thrust Reverser Hinge Beam Forward Fairing Access Door (GE)



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78-31-01 Thrust Reverser Sleeve Actuator Access Panel

For PW or GE:

The lower two actuator panels on each reverser sleeve (eight total) may be missing.

For PW or GE:

NOTE: The top actuator access panel on each reverser sleeve is not allowed to be missing.

For RR:

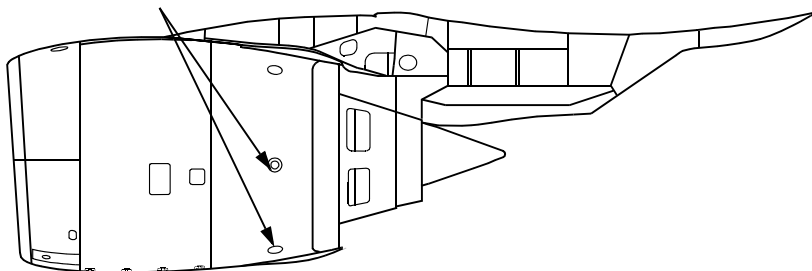
Center actuator panel on each reverser sleeve (four total) may be missing.

For RR:

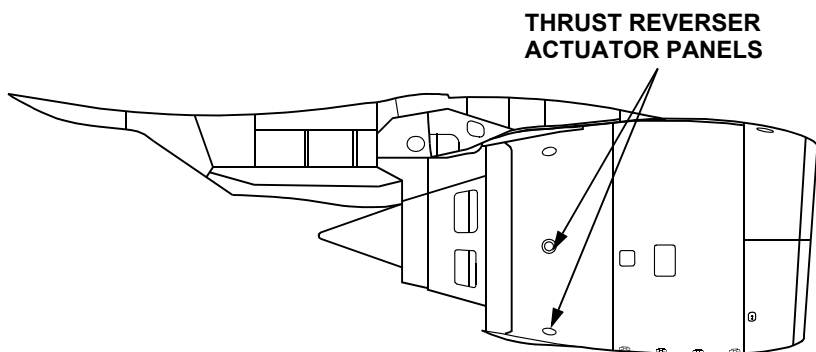
NOTE: The top and bottom actuator access panels on each reverser sleeve are not allowed to be missing.

Performance limited weights are reduced by the following:

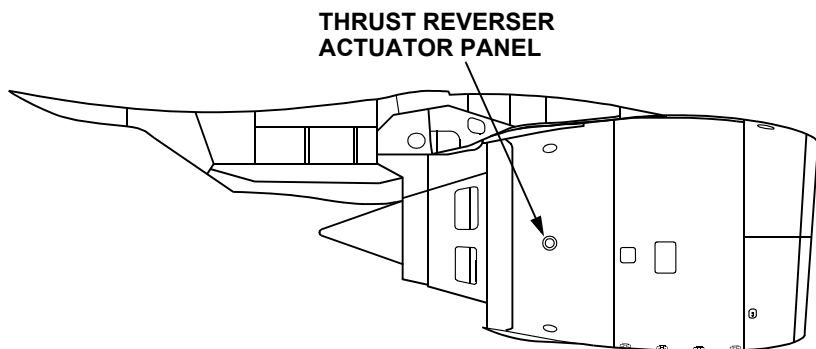
Number Installed	Takeoff & Landing	Enroute Climb
12	Negligible penalty	Negligible penalty

For PW:
**THRUST REVERSER
ACTUATOR PANELS**


For GE:



For RR:



**78-31-02 Thrust Reverser Blocker Door**

One per nacelle half (four total) may be missing.

NOTE: The drag links for missing blocker doors must be removed (AMM 78-31-15/401).

For dry runway takeoff, dry runway landing, wet runway landing, and all enroute climb, performance limited weights are reduced by the following for each missing item:

Number Installed	Takeoff (Dry)	Enroute Climb (Dry, Wet, Contaminated)	Approach and Landing (Dry and Wet)
------------------	---------------	--	------------------------------------

For All/GE (Except -200LR/-300ER):

32	950 lb (430 kg)	750 lb (340 kg)	750 lb (340 kg)
----	-----------------	-----------------	-----------------

For All/PW:

24	1,150 lb (521 kg)	1,250 lb (566 kg)	1,150 lb (521 kg)
----	-------------------	-------------------	-------------------

For All/RR:

24	1,050 lb (476 kg)	600 lb (272 kg)	800 lb (362 kg)
----	-------------------	-----------------	-----------------

For -300ER:

32	1,650 lb (749 kg)	1,400 lb (636 kg)	1,550 lb (704 kg)
----	-------------------	-------------------	-------------------

For -200LR/777F:

32	2,050 lb (930 kg)	1,350 lb (613 kg)	2,000 lb (908 kg)
----	-------------------	-------------------	-------------------

For wet runway takeoff, performance limited weights are reduced by the following for one missing item:

Number Installed	Takeoff (Wet)
------------------	---------------

For -300ER:

32	12,550 lb (5,693 kg)
----	----------------------

For -200LR/777F:

32	12,250 lb (5,557 kg)
----	----------------------



For wet runway takeoff, performance limited weights are reduced by the following for two missing items:

Number Installed	Takeoff (Wet)
------------------	---------------

For -300ER:

32	36,000 lb (16,330 kg)
----	-----------------------

For -200LR/777F:

32	34,700 lb (15,740 kg)
----	-----------------------

For contaminated runway takeoff and landing, performance limited weights are reduced by the following for one missing item:

Number Installed	Takeoff (Contaminated)	Approach and Landing (Contaminated)
------------------	------------------------	-------------------------------------

For -300ER:

32	26,250 lb (11,907 kg)	21,000 lb (9,526 kg)
----	-----------------------	----------------------

For -200LR/777F:

32	25,050 lb (11,363 kg)	26,500 lb (12,021 kg)
----	-----------------------	-----------------------

For contaminated runway takeoff and landing, performance limited weights are reduced by the following for two missing items:

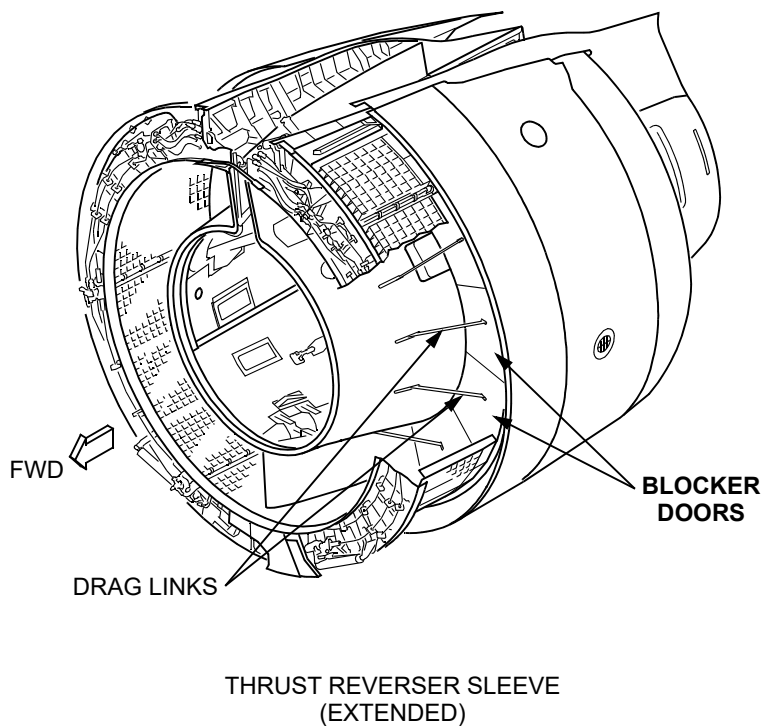
Number Installed	Takeoff (Contaminated)	Approach and Landing (Contaminated)
------------------	------------------------	-------------------------------------

For -300ER:

32	77,100 lb (34,972 kg)	63,000 lb (28,577 kg)
----	-----------------------	-----------------------

For -200LR/777F:

32	73,100 lb (33,158 kg)	79,500 lb (36,061 kg)
----	-----------------------	-----------------------



78-31-03 Thrust Reverser Cascade Segment

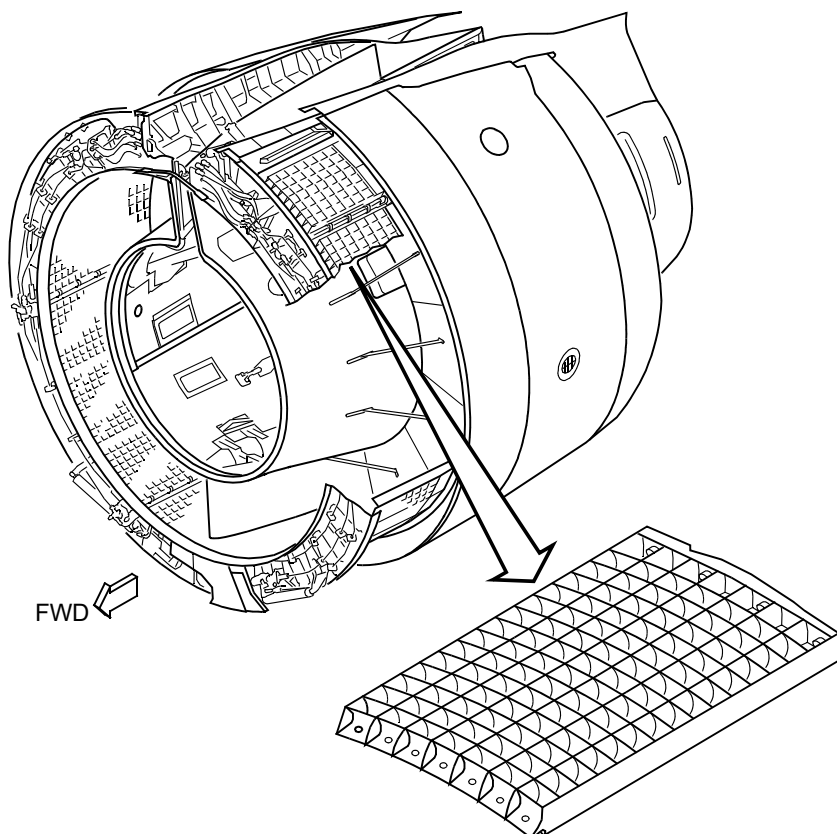
A maximum of six segments on one engine may be missing.

1. Only one engine has segments missing.
2. Missing segments are not adjacent.
3. Top and bottom segments are not missing.
4. Associated thrust reverser is deactivated.

NOTE: Dispatch using MMEL item 78-31-01.

Performance limited weights are reduced by the following:

Number Installed	Takeoff & Landing	Enroute Climb
32	No penalty	No penalty



**THRUST REVERSER
CASCADE SEGMENT**

78-31-04 Thrust Reverser Wedge Fairing

Except for PW4000 reversers P/N 315W3298, a maximum of two per nacelle half (eight total) may be missing.

Performance limited weights are reduced by the following for each missing item:

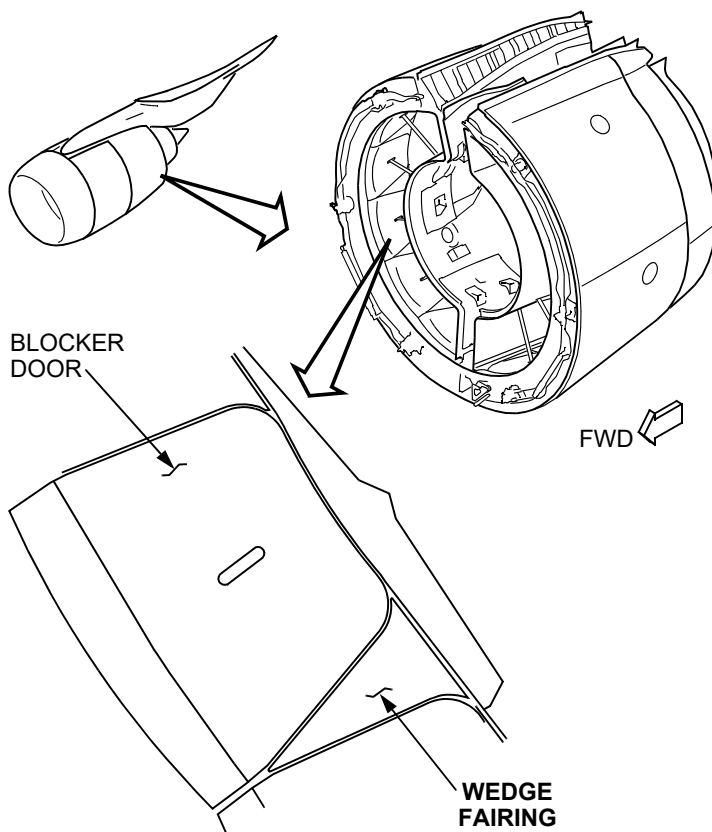
Number Installed	Takeoff	Enroute Climb	Approach/Landing Climb
-------------------------	----------------	----------------------	-------------------------------

For PW (Except PW4000 with P/N 315W3298) / RR:

28	250 lb (114 kg)	250 lb (114 kg)	250 lb (114 kg)
----	-----------------	-----------------	-----------------

For GE (Except GE90-100 Series):

36	200 lb (91 kg)	150 lb (69 kg)	150 lb (69 kg)
----	----------------	----------------	----------------



78-31-05 Thrust Reverser Blocker Door Seals

A maximum of seven may be missing.

Performance limited weights are reduced by the following:

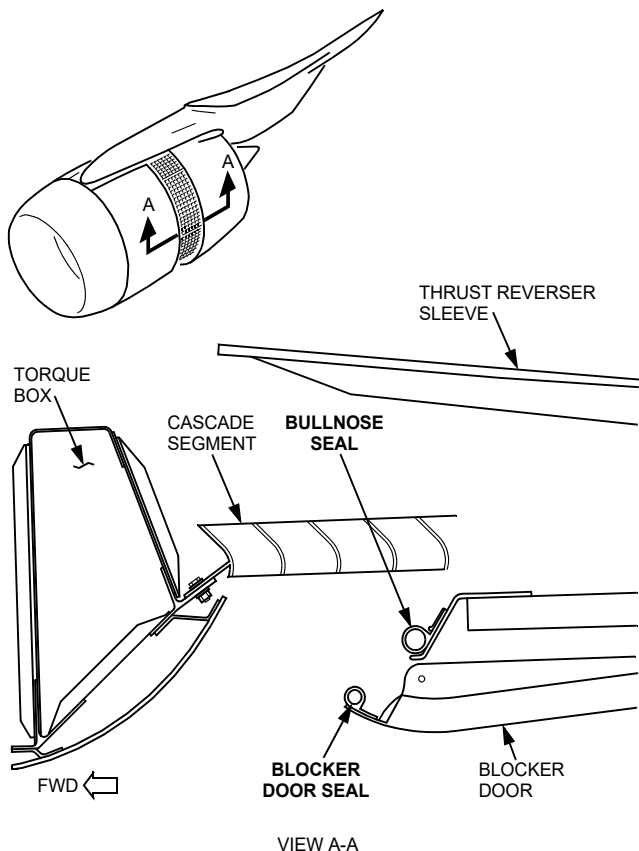
Number Installed	Takeoff & Landing	Enroute Climb
-------------------------	------------------------------	----------------------

For PW/RR:

24	Negligible penalty	Negligible penalty
----	--------------------	--------------------

For GE:

32	Negligible penalty	Negligible penalty
----	--------------------	--------------------



78-31-06 Thrust Reverser Hinge Beam Forward Fairing Access Door (GE)

For -200/-200ER/GE90-90B:

Any number of access doors from the forward fairing may be missing.

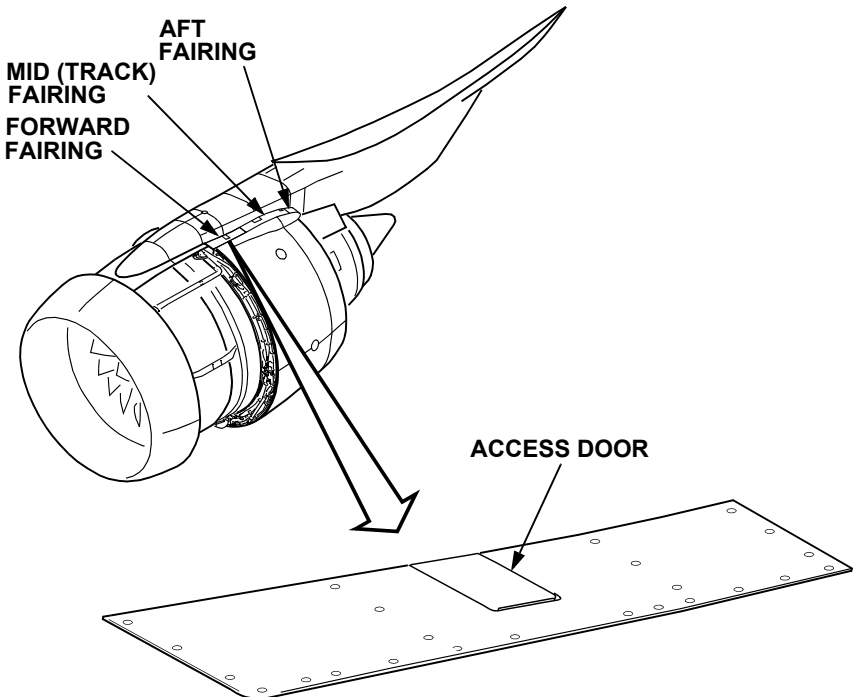
Performance limited weights are reduced by the following for each missing access door:

NOTE: Access doors from the Mid (Track) and Aft Fairings may not be missing.

Number Installed	Takeoff	Enroute Climb	Approach and Landing
------------------	---------	---------------	----------------------

For -200/-200ER/GE90-90B:

4	Negligible	200 lb (91 kg)	Negligible
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**THRUST REVERSER HINGE BEAM FORWARD FAIRING ACCESS DOOR
(GE90-90 SERIES ENGINES)**

**For -200LR/-777F/-300ER/GE90-100 Series:**

Any number of Aft access doors from the forward fairing may be missing.

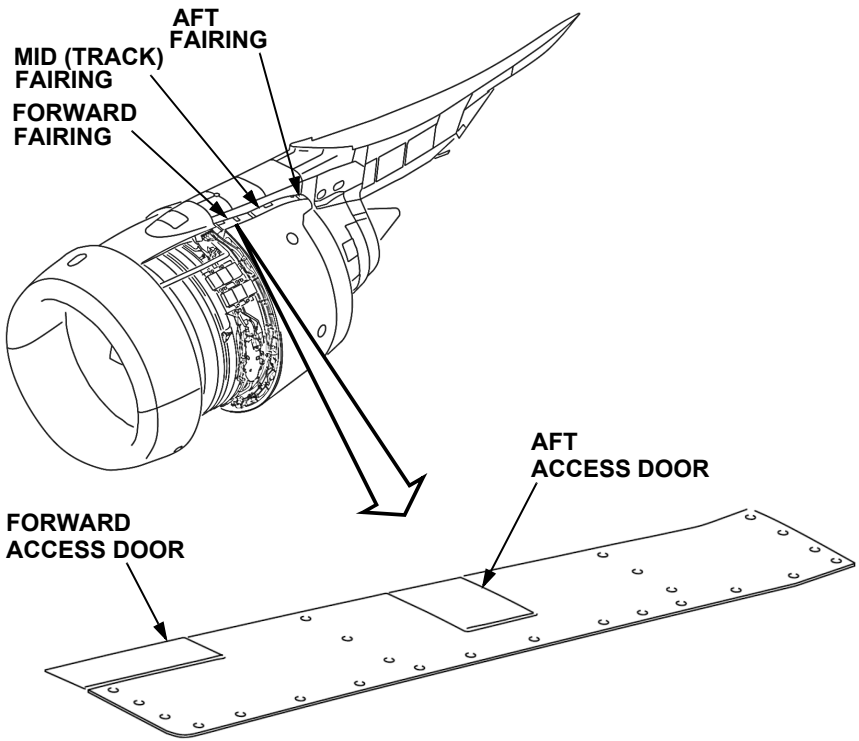
Performance limited weights are reduced by the following for each missing access door:

NOTE: Forward access doors from the forward fairing and access doors from the Mid (Track) and Aft Fairings may not be missing.

Number Installed	Takeoff	Enroute Climb	Approach and Landing
------------------	---------	---------------	----------------------

For -200LR/777F/-300ER:

8	Negligible	200 lb (91 kg)	Negligible
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**THRUST REVERSER HINGE BEAM FORWARD FAIRING ACCESS DOORS
(GE90-100 SERIES ENGINES)**



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PBN Equipment Requirements

Two Primary Flight Displays (PFD) and Navigation Displays (ND) - (31-61-01)

Two Control Display Units (CDU) - (34-61-03)

AIMS Operational Software

Fault Tolerant ADIRU

Autopilot Flight Director Computers (AFDC) - (22-11-01)

Mode Control Panel (MCP) Switches - (22-11-06)

Autopilot Function

Flight Director Systems (22-11-10)

Takeoff/Go-Around (TO/GA) Switches (22-11-09)

VOR Navigation Systems (34-51-01)

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Radio Altimeter Systems (34-33-01)

Enhanced Ground Proximity Warning Systems (EGPWS) (34-46-01)

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Flight Management Computing Systems (FMCS) (34-61-01)

Flight Management Computing Systems (FMCS) - Navigation Database (34-61-01-03)



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Airplane Model Information

Where applicable, airplane model information (annotation) is included above affected text to identify configuration-specific information. The annotation applies only to text aligned immediately below it and any associated indented text. The following example illustrates the model applicability for a Maintenance procedure:

MAINTENANCE (M)

Verify that the associated engine fuel shutoff valve is operating normally once each flight day (AMM 73-00-00/901).

For PW:

1. If MAT is available, confirm that both maintenance messages are displayed for the associated status message.
 - A. For status message ENG FUEL VALVE L, 73-40231 and 73-50231.
 - B. For status message ENG FUEL VALVE R, 73-40232 and 73-50232.
2. Start the associated engine and allow engine to stabilize at ground idle (AMM 71-00-00/201).
3. Open the P11 panel FUEL SPAR VALVE circuit breaker for the associated engine.
4. After minimum engine operating time, position the associated engine FUEL CONTROL switch to CUTOFF and confirm engine shutdown.
5. For engine not shutdown, the fuel shutoff valve is not operating normally.
 - A. Monitor EGT and be prepared to motor the engine.
 - B. Close the P11 panel FUEL SPAR VALVE circuit breaker.

For PW or RR:

- C. The engine may continue to operate for one minute.

For GE:

- D. The engine may continue to operate for 10 seconds.

6. Confirm the associated FUEL SPAR VALVE L or R status message is displayed.
7. Erase the latched FUEL SPAR VALVE L or R status message (AMM 31-61-00/201).
8. After engine has stopped rotating, close the P11 panel FUEL SPAR VALVE circuit breaker.

For the above example procedure, the applicable steps are as follows:

- For PW, steps 1, 1.A, 1.B, 2, 3, 4, 5, 5.A, 5.B, 5.C, 6, 7 and 8 apply.
- For RR, steps 2, 3, 4, 5, 5.A, 5.B, 5.C, 6, 7 and 8 apply.
- For GE, steps 2, 3, 4, 5, 5.A, 5.B, 5.D, 6, 7 and 8 apply.



When airplane model information (annotation) is left justified, aligned with the bolded heading or title and not aligned with an individual paragraph, step, or note, it indicates all the information following that annotation is applicable to the end of the listed steps, or until another airplane model is noted. For this format, all the steps that follow multiple annotations are re-numbered beginning from step 1 for each annotation.



Section 4 Performance Based Navigation

Performance Based Navigation (PBN) Information

Applying dispatch relief for the following items may affect the ability of the airplane to meet the operational capabilities required for PBN operations. The following information lists items that, if inoperative, may affect the ability of the airplane to meet the performance and functional criteria for various PBN operations. This information can aid in identifying potentially affected capabilities associated with a particular dispatch configuration. This section lists equipment requirements that may be found in the document 777 RNP Navigation Capabilities Document D243W018-13, revision "M" (or later). Operators should consult their Operations Specification for requirements applicable to their operations.

PBN Equipment Requirements

Two Primary Flight Displays (PFD) and Navigation Displays (ND) - (31-61-01)

All PBN operations require two PFDs with mini map.

The 777 is equipped with six display units. The PFDs are normally displayed on the outboard display units and the NDs are normally displayed on the inboard display units. Dispatch is only allowed with the lower center display unit inoperative. Therefore, this requirement will always be met for dispatch.

Two Control Display Units (CDU) - (34-61-03)

All PBN operations require two CDUs.

The 777 is equipped with three CDUs. Dispatch is allowed with the center DCU inoperative. Both left and right CDUs must operate normally for dispatch. Therefore, this requirement will always be met for dispatch.

AIMS Operational Software

For RNP AR APCH and RNP APCH:

Airplane must be equipped with AIMS BP05A or later. Navigation Performance Scales and LNAV Go-Around Auto-Engage must be enabled.

Airplanes without Navigation Performance Scales and LNAV Go-Around Auto-Engage (AIMS BP03 and earlier versions) do not support these operations.

Fault Tolerant ADIRU

For AIMS-1 V16 and AIMS-2 V17 or later:

AIR DATA SYS or NAV ADIRU INERTIAL messages must not be displayed.

For all other AIMS-1 or AIMS-2 versions:

NAV AIR DATA SYS or NAV ADIRU INERTIAL messages must not be displayed.

Dispatch is not allowed with any of the above messages displayed.

Therefore, this requirement will always be met for dispatch.

Autopilot Flight Director Computers (AFDC) - (22-11-01)

RNP AR APCH operations require two AFDCs to operate normally. All other PBN operations require a single operational AFDC.

The 777 is equipped with three AFDCs. MMEL dispatch is allowed with one, two, or all three AFDCs inoperative. Therefore, PBN operations may be affected with more than one AFDC inoperative.

PBN operations may be allowed using either the autopilot function or flight director function.

Mode Control Panel (MCP) Switches - (22-11-06)

The following equipment must operate normally in order to use the autopilot function and flight director function:

Approach (APP) Mode Control Panel (MCP) switch must operate normally for RNP AR APCH and RNP APCH.

Lateral navigation (LNAV) Mode Control Panel (MCP) switch must operate normally for all PBN operations.

Vertical navigation (VNAV) Mode Control Panel (MCP) switch must operate normally for RNP AR APCH operations. Other vertical modes may be used for other PBN operations.

Autopilot Function

If the autopilot function is inoperative, PBN operations may be allowed using only the flight director function.

The autopilot function may not be used if any of the following equipment is inoperative:

1. Both autopilot engage switches. (22-11-06)
2. One control wheel autopilot disconnect switch below 1500 feet AGL. (22-11-08)
3. Both control wheel autopilot disconnect switches inoperative. (22-11-08)
4. Warning Electronic System (WES) Autopilot disconnect warning function. (31-51-05)

Flight Director Systems (22-11-10)

If the flight director function is inoperative, PBN operations may be allowed using only the autopilot function.

The flight director function is considered inoperative if any of the following equipment is inoperative:

1. One or more Flight Director Systems (22-11-10).
2. One or more Mode Control Panel (MCP) Flight Director (F/D) switches. (22-11-06)

Both flight director display functions are required for RNP AR APCH. Flight director function for pilot flying (PF) is required for RNP APCH.

Takeoff/Go-Around (TO/GA) Switches (22-11-09)

There are two TO/GA switches installed on the 777. MMEL dispatch is allowed with one or both TO/GA switches inoperative.

For RNP AR APCH, both TO/GA switches must operate normally.

For both TO/GA switches inoperative, PBN operations that require either autopilot or flight director use below 500 feet AGL or MDA, whichever is higher, may not be used. RNP AR APCH, RNP APCH, RNP 1, P-RNAV, RNAV 1 and Advanced RNP operations may not be allowed.

VOR Navigation Systems (34-51-01)

The 777 is equipped with two VORs. For dispatch, either one or both VORs may be inoperative.

- When dispatching with a single GPS system, RNP APCH, RNP 1/RNP 2, and Advanced RNP operations are allowed provided provided at least one of the following operates normally: GPS, VOR or DME.

Global Positioning Systems (GPS) (34-58-01)

The 777 is equipped with two GPSs. For dispatch, either one or both GPSs may be inoperative.

For dispatch with one GPS inoperative:

- RNP AR APCH and RNP 4 operations are not allowed.
- RNP APCH, RNP 1/RNP 2, and; Advanced RNP operations are allowed provided acceptable alternate navigation equipment (e.g. DME/DME, DME/VOR, VOR, ADF) is available to continue non-PBN operations in the event that the remaining GPS becomes inoperative.

For dispatch with both GPSs inoperative:

- RNP AR APCH, RNP APCH, RNP 1, RNP 2, RNP 4 and Advanced RNP operations not allowed.
- RNAV 1, RNAV 2, P-RNAV, B-RNAV, RNAV 5, RNP 10 and RNAV 10 are allowed provided at least one VOR or DME operates normally.

- RNP 10 and RNAV 10 operations may be allowed provided appropriate time limitations are observed.

NOTE: Operations may be for up to 6.2 hours of flight time starting when system is placed in inertial navigation mode

Radio Altimeter Systems (34-33-01)

The 777 is equipped with three radio altimeters. Dispatch is allowed with one radio altimeter inoperative.

- RNP AR APCH requires that two radio altimeters operate normally.
- All other PBN operations are allowed with one radio altimeter inoperative.

Therefore, these requirements will always be met for dispatch.

Enhanced Ground Proximity Warning Systems (EGPWS) (34-46-01)

777 airplanes with Block 98 AIMS (PRR61274) or later are equipped with EGPWS. Dispatch is allowed with GPWS inoperative.

- RNP AR APCH requires EGPWS to operate normally (GND PROX SYS message not displayed, TERR POS message not displayed).
- All other PBN operations do not require EGPWS to be operative.

Distance Measuring Equipment Systems (DME) (34-55-01)

The 777 is equipped with two DMEs. For dispatch, either one or both DMEs may be inoperative.

- When dispatching with a single GPS system, RNP APCH, RNP 1/RNP 2, and Advanced RNP operations are allowed provided provided at least one of the following operates normally: GPS, VOR or DME.

Flight Management Computing Systems (FMCS) (34-61-01)

There are 2 Flight Management Computers (FMC) installed on the 777. All PBN operations require a minimum of 2 FMCs. Dispatch is allowed with one FMC inoperative.

For one FMCS inoperative:

- RNP AR APCH and RNP 4 operations are not allowed.
- RNP APCH, RNP 1, RNP 2, RNAV 1, RNAV 2, P-RNAV, B-RNAV, RNAV 5 and Advanced RNP operations may be allowed provided alternate navigation equipment (e.g., VOR, ADF) is available to continue non-PBN operations in the event that the remaining FMCS becomes inoperative.

- RNP 10 and RNAV 10 operations are allowed.

Flight Management Computing Systems (FMCS) - Navigation Database (34-61-01-03)

- A current navigation database is required for all PBN operations.
- It is assumed that the operator has a preflight procedure to ensure that the navigation database complies with regulatory agency requirements for ensuring the navigation database remains valid for the duration of the flight unless an alternative regulatory agency approved process applies.



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777 FAA DDG Supplement Package

Introduction

The Dispatch Deviations Guide (DDG) is intended to assist airline operations and maintenance organizations in developing the procedures required to operate the airplane in the various nonstandard configurations authorized by the Federal Aviation Administration (FAA) Master Minimum Equipment List (MMEL) and the Airplane Flight Manual (AFM) Appendix Configuration Deviation List (CDL). The DDG is used as a reference by operators to develop their Minimum Equipment List (MEL) and is not a regulatory-approved document.

A DDG is typically revised when the associated model's FAA MMEL is updated by a standard revision. At Boeing's discretion, updated procedural guidance information may be made available via a DDG Supplement prior to a complete DDG revision. A DDG Supplement is only effective for its associated DDG revision.

A DDG Supplement is identified by its associated DDG revision number followed by a dash, then a sequential number. An individual DDG Supplement may be superseded by a newer DDG Supplement. In addition, if Boeing subsequently releases a new DDG revision, then previous DDG Supplements will be incorporated into that DDG revision. The DDG Supplement Record below will be retained for historical reference. DDG Supplement Status may be 'Active', 'Superseded', or 'Incorporated'. Only 'Active' DDG Supplements are contained herein.

DDG Supplement Record

Number	Date	Section	Item	MMEL	Status
34-1	11-MAR-2016	2 (MMEL)	28-21-01	21	Incorporated
34-1	11-MAR-2016	2 (MMEL)	28-31-02	21	Incorporated
34-2	28-APR-2017	2 (MMEL)	24-11-01	21	Incorporated
35-1	11-APR-2018	2 (MMEL)	21-51-01	22	Incorporated
35-2	17-APR-2018	2 (MMEL)	21-52-05	22	Incorporated
35-2	17-APR-2018	2 (MMEL)	21-52-06	22	Incorporated
35-3	13-DEC-2018	2 (MMEL)	21-51-02	22	Incorporated
35-3	13-DEC-2018	2 (MMEL)	80-11-01	22	Incorporated
35-3	13-DEC-2018	2 (MMEL)	80-11-03	22	Incorporated
35-4	21-DEC-2018	3 (CDL)	21-52-01	22	Incorporated
35-4	21-DEC-2018	3 (CDL)	55-20-02	22	Incorporated
35-4	21-DEC-2018	3 (CDL)	55-20-03	22	Incorporated
35-4	21-DEC-2018	3 (CDL)	55-40-01	22	Incorporated
35-4	21-DEC-2018	3 (CDL)	55-40-09	22	Incorporated
35-4	21-DEC-2018	3 (CDL)	55-40-10	22	Incorporated
35-4	21-DEC-2018	3 (CDL)	57-63-08	22	Incorporated
35-4	21-DEC-2018	3 (CDL)	57-63-09	22	Incorporated
35-5	15-MAR-2019	4 (MISC)	N/A	22	Incorporated
35-6	12-JUN-2020	2 (MMEL)	73-21-04	22	Superseded
35-7	17-JUL-2020	2 (MMEL)	73-21-04	22	Superseded
35-8	04-SEP-2020	2 (MMEL)	(multiple)	22a	Incorporated
35-8	04-SEP-2020	2 (MMEL)	23-25-01	22a	Incorporated
35-8	04-SEP-2020	2 (MMEL)	28-22-03	22	Incorporated



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DDG Supplement Record (Cont'd)

Number	Date	Section	Item	MMEL	Status
35-8	04-SEP-2020	2 (MMEL)	73-21-04	22	Incorporated
35-9	30-SEP-2021	2 (MMEL)	46-11-01	22	Incorporated
35-10	12-OCT-2021	1 (EICAS)	N/A	22	Incorporated
35-11	18-MAR-2022	3 (CDL)	57-41-23	22	Incorporated
35-12	14-JUL-2023	2 (MMEL)	32-42-01	22	Incorporated
35-12	14- JUL-2023	2 (MMEL)	32-42-02	22	Incorporated
35-12	14- JUL-2023	2 (MMEL)	32-45-01	22	Incorporated
36-1	21-DEC-2023	2 (MMEL)	21-52-05	23	Superseded
36-1	21-DEC-2023	2 (MMEL)	22-11-08	23	Superseded
36-1	21-DEC-2023	2 (MMEL)	22-11-09	23	Superseded
36-1	21-DEC-2023	2 (MMEL)	22-31-02	23	Active
36-1	21-DEC-2023	2 (MMEL)	23-24-02	23	Superseded
36-1	21-DEC-2023	2 (MMEL)	25-25-06	23	Active
36-1	21-DEC-2023	2 (MMEL)	25-52-01	23	Active
36-1	21-DEC-2023	2 (MMEL)	25-59-01	23	Active
36-1	21-DEC-2023	2 (MMEL)	25-64-01	23	Active
36-1	21-DEC-2023	2 (MMEL)	25-64-02	23	Active
36-1	21-DEC-2023	2 (MMEL)	25-64-03	23	Superseded
36-1	21-DEC-2023	2 (MMEL)	26-26-01	23	Superseded
36-1	21-DEC-2023	2 (MMEL)	27-11-04	23	Superseded
36-1	21-DEC-2023	2 (MMEL)	28-11-01	23	Active
36-1	21-DEC-2023	2 (MMEL)	28-14-03	23	Active
36-1	21-DEC-2023	2 (MMEL)	28-14-07	23	Active
36-1	21-DEC-2023	2 (MMEL)	28-21-02	23	Active
36-1	21-DEC-2023	2 (MMEL)	28-31-04	23	Superseded
36-1	21-DEC-2023	2 (MMEL)	30-21-01	23	Superseded
36-1	21-DEC-2023	2 (MMEL)	30-81-01	23	Active
36-1	21-DEC-2023	2 (MMEL)	31-61-07	23	Superseded
36-1	21-DEC-2023	2 (MMEL)	32-08-01	23	Superseded
36-1	21-DEC-2023	2 (MMEL)	35-31-01	23	Active
36-1	21-DEC-2023	2 (MMEL)	35-31-02	23	Active
36-1	21-DEC-2023	2 (MMEL)	52-71-01	23	Superseded
36-1	21-DEC-2023	4 (MISC)	N/A	23	Superseded
36-2	12-APR-2024	1 (EICAS)	N/A	23a	Active
36-2	12-APR-2024	2 (MMEL)	21-27-09	23a	Active
36-2	12-APR-2024	2 (MMEL)	21-31-01	23a	Active
36-2	12-APR-2024	2 (MMEL)	21-31-03	23a	Active
36-2	12-APR-2024	2 (MMEL)	21-51-01	23a	Active
36-2	12-APR-2024	2 (MMEL)	22-11-01	23a	Active
36-2	12-APR-2024	2 (MMEL)	22-11-06	23a	Active
36-2	12-APR-2024	2 (MMEL)	22-11-08	23a	Active
36-2	12-APR-2024	2 (MMEL)	22-11-09	23a	Active
36-2	12-APR-2024	2 (MMEL)	22-11-10	23a	Active
36-2	12-APR-2024	2 (MMEL)	23-24-02	23a	Superseded
36-2	12-APR-2024	2 (MMEL)	25-64-03	23a	Active
36-2	12-APR-2024	2 (MMEL)	26-26-01	23a	Active
36-2	12-APR-2024	2 (MMEL)	29-11-03	23a	Active



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DDG Supplement Record (Cont'd)

Number	Date	Section	Item	MMEL	Status
36-2	12-APR-2024	2 (MMEL)	30-21-01	23a	Superseded
36-2	12-APR-2024	2 (MMEL)	31-51-05	23a	Active
36-2	12-APR-2024	2 (MMEL)	31-61-04	23a	Active
36-2	12-APR-2024	2 (MMEL)	32-08-01	23a	Active
36-2	12-APR-2024	2 (MMEL)	34-46-01	23a	Superseded
36-2	12-APR-2024	2 (MMEL)	34-51-01	23a	Active
36-2	12-APR-2024	2 (MMEL)	34-55-01	23a	Active
36-2	12-APR-2024	2 (MMEL)	34-58-01	23a	Superseded
36-2	12-APR-2024	2 (MMEL)	34-61-01	23a	Superseded
36-2	12-APR-2024	2 (MMEL)	35-21-06	23a	Superseded
36-2	12-APR-2024	2 (MMEL)	36-00-01	23a	Active
36-2	12-APR-2024	2 (MMEL)	52-71-01	23a	Superseded
36-2	12-APR-2024	2 (MMEL)	73-21-01	23a	Active
36-2	12-APR-2024	2 (MMEL)	73-21-04	23a	Active
36-2	12-APR-2024	4 (MISC)	N/A	23a	Superseded
36-3	29-MAY-2024	2 (MMEL)	23-24-02	23a	Active
36-3	29-MAY-2024	2 (MMEL)	23-70-03	23a	Active
36-3	29-MAY-2024	2 (MMEL)	25-53-02	23a	Superseded
36-3	29-MAY-2024	2 (MMEL)	25-58-01	23a	Active
36-3	29-MAY-2024	2 (MMEL)	27-11-04	23a	Active
36-3	29-MAY-2024	2 (MMEL)	30-21-01	23a	Active
36-3	29-MAY-2024	2 (MMEL)	34-43-01	23a	Superseded
36-3	29-MAY-2024	2 (MMEL)	34-46-01	23a	Active
36-3	29-MAY-2024	2 (MMEL)	34-58-01	23a	Superseded
36-3	29-MAY-2024	2 (MMEL)	36-12-02	23a	Active
36-3	29-MAY-2024	2 (MMEL)	46-11-01	23a	Active
36-3	29-MAY-2024	2 (MMEL)	52-71-01	23a	Active
36-3	29-MAY-2024	3 (CDL)	78-31-04	23a	Active
36-3	29-MAY-2024	4 (MISC)	N/A	23a	Superseded
36-4	19-DEC-2024	2 (MMEL)	28-31-01	23a	Superseded
36-4	19-DEC-2024	2 (MMEL)	28-31-02	23a	Superseded
36-4	19-DEC-2024	2 (MMEL)	28-31-03	23a	Superseded
36-4	19-DEC-2024	2 (MMEL)	28-31-04	23a	Superseded
36-4	19-DEC-2024	2 (MMEL)	31-61-07	23a	Active
36-4	19-DEC-2024	2 (MMEL)	32-45-01	23a	Active
36-4	19-DEC-2024	2 (MMEL)	32-53-01	23a	Active
36-4	19-DEC-2024	2 (MMEL)	34-43-01	23a	Active
36-4	19-DEC-2024	2 (MMEL)	34-58-01	23a	Superseded
36-4	19-DEC-2024	2 (MMEL)	34-61-01	23a	Superseded
36-4	19-DEC-2024	2 (MMEL)	38-30-01	23a	Active
36-4	19-DEC-2024	4 (MISC)	N/A	23a	Superseded
36-5	17-JAN-2025	2 (MMEL)	28-31-01	23a	Active
36-5	17-JAN-2025	2 (MMEL)	28-31-02	23a	Superseded
36-5	17-JAN-2025	2 (MMEL)	28-31-03	23a	Superseded
36-5	17-JAN-2025	2 (MMEL)	28-31-04	23a	Active
36-6	26-SEP-2025	2 (MMEL)	21-52-05	23a	Active
36-6	26-SEP-2025	2 (MMEL)	24-25-01	23a	Active
36-6	26-SEP-2025	2 (MMEL)	25-53-02	23a	Active



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DDG Supplement Record (Cont'd)

Number	Date	Section	Item	MMEL	Status
36-6	26-SEP-2025	2 (MMEL)	27-11-03	23a	Active
36-6	26-SEP-2025	2 (MMEL)	27-61-03	23a	Active
36-6	26-SEP-2025	2 (MMEL)	28-21-03	23a	Active
36-6	26-SEP-2025	2 (MMEL)	28-31-02	23a	Active
36-6	26-SEP-2025	2 (MMEL)	28-31-03	23a	Active
36-6	26-SEP-2025	2 (MMEL)	28-41-01	23a	Active
36-6	26-SEP-2025	2 (MMEL)	28-41-03	23a	Active
36-6	26-SEP-2025	2 (MMEL)	32-49-01	23a	Active
36-6	26-SEP-2025	2 (MMEL)	34-58-01	23a	Active
36-6	26-SEP-2025	2 (MMEL)	34-61-01	23a	Active
36-6	26-SEP-2025	2 (MMEL)	35-21-06	23a	Active
36-6	26-SEP-2025	2 (MMEL)	78-31-01	23a	Active
36-6	26-SEP-2025	4 (MISC)	N/A	23a	Active



777 FAA DDG Supplement Package

Supplement Number: 36-1

Issue Date: 21-DEC-2023

Section Number: 2 (MMEL)

Item Number: 21-52-05

Item Name: Ram Air Inlet Door Systems

DDG Effectivity: 777 FAA DDG (D630W003-TBC), Revision 36

MMEL Effectivity: 777 FAA MMEL, Revision 23

Description: Modified DDG top note for option 21-52-05A.

Reason: The DDG top note is applicable to both passenger and freighter airplanes.

Disposition: For reference in the Operator's MEL.

Attachments: DDG page 2.21-52-05.1-3

21-52-05 Ram Air Inlet Door Systems
21-52-05A Associated Pack Used

Interval	Installed	Required	Procedure
C	2	1	(M) (O)

One may be inoperative provided:

- a. Associated door is secured full open.
- b. Appropriate performance adjustments are applied.

NOTE: To avoid cold flight deck temperatures, the left Trim Air Control System and left Trim Air PRSOV should operate normally when the left Ram Air Inlet Door is secured full open.

MAINTENANCE (M)

Secure the inoperative ram air inlet door full open (AMM 21-00-00/901).

1. For left ram air inlet door inoperative:
 - A. Open P110 panel L CTC CHAN 1 (AC) circuit breaker.
 - B. Open P110 panel L CTC CHAN 1 (DC) circuit breaker.
 - C. Open P110 panel L CTC CHAN 2 (AC) circuit breaker.
 - D. Open P110 panel L CTC CHAN 2 (DC) circuit breaker.
 - E. Open the ram air inlet actuator door 191LL.
2. For right ram air inlet door inoperative:
 - A. Open P210 panel R CTC CHAN 1 (AC) circuit breaker.
 - B. Open P210 panel R CTC CHAN 1 (DC) circuit breaker.
 - C. Open P210 panel R CTC CHAN 2 (AC) circuit breaker.
 - D. Open P210 panel R CTC CHAN 2 (DC) circuit breaker.
 - E. Open the ram air inlet actuator door 192LR
3. Disconnect, cap, and stow the electrical connector from the actuator.
4. Disconnect the actuator rod end from the bellcrank.
5. Loosen the screws on the fixed link, and rotate the fixed link 180 degrees.
6. Position the inlet door to the fully open position.
7. Connect the fixed link to the bellcrank.
8. Tighten the screws on the fixed link to 18-28 pound-inches.
9. Close the ram air inlet actuator access door.
10. Close the associated circuit breakers.

OPERATIONS (O)

1. Reduce performance limited weights by the appropriate adjustments.

Performance Limited Weights	Takeoff & Landing	Enroute Climb
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For -200 or -200ER/GE90-76B thru -90B:

thru 632,500 lb (286,902 kg)	400 lb (181 kg)	2,100 lb (953 kg)
over 632,500 lb (286,902 kg)	500 lb (227 kg)	2,200 lb (998 kg)

For -200 or -200ER/GE90-94B:

thru 632,500 lb (286,902 kg)	400 lb (181 kg)	2,600 lb (1,179 kg)
over 632,500 lb (286,902 kg)	700 lb (317 kg)	2,600 lb (1,179 kg)

For -200 or -200ER/PW4074 thru PW4084:

All	400 lb (181 kg)	2,300 lb (1,043 kg)
-----	-----------------	---------------------

For -200 or -200ER/PW4090:

thru 632,500 lb (286,902 kg)	400 lb (181 kg)	2,300 lb (1,043 kg)
over 632,500 lb (286,902 kg)	500 lb (227 kg)	2,300 lb (1,043 kg)

For -200 or -200ER/RR875 thru RR895:

thru 632,500 lb (286,902 kg)	400 lb (181 kg)	2,200 lb (998 kg)
over 632,500 lb (286,902 kg)	600 lb (272 kg)	2,200 lb (998 kg)

For -200LR or 777F/GE90-100 Series:

All	500 lb (227 kg)	2,300 lb (1,043 kg)
-----	-----------------	---------------------

For -300/PW4090:

All	400 lb (181 kg)	2,300 lb (1,043 kg)
-----	-----------------	---------------------

For -300/RR892:

All	400 lb (181 kg)	2,100 lb (953 kg)
-----	-----------------	-------------------

For -300ER/GE90-115B:

All	400 lb (181 kg)	2,300 lb (1,043 kg)
-----	-----------------	---------------------

NOTE: The enroute climb weight penalties are based on single engine operating speeds that approximate maximum lift-to-drag ratio speed. To account for the difference in level off altitude when operating at other speeds, multiply the enroute climb weight penalty listed above by the appropriate factor listed on page 3.00-01-00.2 in the ENROUTE DIVERSION SPEED EFFECTS paragraph.

2. Increase flight planning fuel by 0.7%

21-52-05 Ram Air Inlet Door Systems

21-52-05B Associated Pack Not Used

Interval	Installed	Required	Procedure
C	2	1	(M) (O)

One may be inoperative provided:

- a. Associated inlet door is verified closed.
- b. Associated pack is not used.

MAINTENANCE (M)

Verify the ram air inlet door is closed for dispatch.

1. On the AIR CONDITIONING maintenance page, verify that the associated RAM AIR INLET indicates 0.00.

NOTE: If the ram air inlet door can not be closed for dispatch, install a ram air inlet door cover plate (AMM 21-00-00/901)

OPERATIONS (O)

Position associated pack switch OFF.

NOTE: The airplane must also be dispatched using MMEL item 21-51-01 Operations procedure.



777 FAA DDG Supplement Package

Supplement Number: 36-1

Issue Date: 21-DEC-2023

Section Number: 2 (MMEL)

Item Number: 22-11-08

Item Name: Control Wheel Autopilot Disconnect Switches

DDG Effectivity: 777 FAA DDG (D630W003-TBC), Revision 36

MMEL Effectivity: 777 FAA MMEL, Revision 23

Description: Modified option 22-11-08A Operations note. Added DDG top note and Operations notes to option 22-11-08B.

Reason: Modified option 22-11-08A to align with proviso format. Added PBN guidance to option 22-11-08B.

Disposition: For reference in the Operator's MEL.

Attachments: DDG page 2.22-11-08.1

**22-11-08 Control Wheel Autopilot Disconnect Switches****22-11-08A One Inoperative**

Interval	Installed	Required	Procedure
C	2	1	

One may be inoperative provided:

- Autopilot is not used at 1,500 feet AGL or below.
- Approach minimums do not require the use of the autopilot.

NOTE: Capability to perform Performance Based Navigation operations may be affected.

OPERATIONS NOTE

- Do not use autopilot at 1,500 feet AGL or below.
- For PBN operations:
PBN operations may be conducted provided flight director is used at 1,500 feet AGL or below.

22-11-08 Control Wheel Autopilot Disconnect Switches**22-11-08B Both Inoperative**

Interval	Installed	Required	Procedure
B	2	0	

May be inoperative provided:

- Autopilot is not used.
- Approach minimums do not require use of the autopilot.
- Number of flight segments and segment duration is acceptable to flight crew.
- Enroute operations do not require use of the autopilot.

NOTE: Capability to perform Performance Based Navigation operations may be affected.

OPERATIONS NOTE

- Enroute operations that require RVSM may not be conducted.
- For PBN operations:
PBN operations may be conducted provided flight director is used.



777 FAA DDG Supplement Package

Supplement Number: 36-1

Issue Date: 21-DEC-2023

Section Number: 2 (MMEL)

Item Number: 22-11-09

Item Name: Takeoff/Go-Around (TO/GA) Switches

DDG Effectivity: 777 FAA DDG (D630W003-TBC), Revision 36

MMEL Effectivity: 777 FAA MMEL, Revision 23

Description: Modified option 22-11-09B Operations note.

Reason: Clarified PBN requirements.

Disposition: For reference in the Operator's MEL.

Attachments: DDG page 2.22-11-09.1

**22-11-09 Takeoff/Go-Around (TO/GA) Switches****22-11-09A One Inoperative**

Interval	Installed	Required	Procedure
C	2	1	

One may be inoperative provided approach minimums do not require its use.

OPERATIONS NOTE

1. The autoland system is not fail-operational. The NO LAND 3 advisory message may be displayed.
2. For PBN operations:
RNP AR APCH operations are not allowed.

NOTE: Refer to DDG Section 4 for additional information.

22-11-09 Takeoff/Go-Around (TO/GA) Switches**22-11-09B Both Inoperative**

Interval	Installed	Required	Procedure
C	2	0	

May be inoperative provided:

- a. Both thrust levers are operated manually for takeoff and go-around.
- b. Autopilot and flight director are not used on approach below 500 feet AGL or MDA, whichever is higher.

NOTE: Flight director go-around and windshear guidance are not available with both TO/GA switches inoperative.

OPERATIONS NOTE

1. The autoland system is not fail-operational. The NO LAND 3 or NO AUTOLAND advisory message may be displayed.
2. Flight director go-around and windshear guidance are not available with both TO/GA switches inoperative.
3. For PBN operations:
 - A. RNP AR APCH operations are not allowed.
 - B. PBN operations are not allowed if autopilot or flight director use is required below 500 feet AGL or applicable approach minimums, which ever is higher.
 - C. RNP APCH, RNP 1, P-RNAV, RNAV 1 and Advanced RNP operations may not be allowed.

NOTE: Refer to DDG Section 4 for additional information.



777 FAA DDG Supplement Package

Supplement Number: 36-1

Issue Date: 21-DEC-2023

Section Number: 2 (MMEL)

Item Number: 22-31-02

Item Name: Autothrottle Servo Motors

DDG Effectivity: 777 FAA DDG (D630W003-TBC), Revision 36

MMEL Effectivity: 777 FAA MMEL, Revision 23

Description: Deleted Operations note 4 in subitem options 22-31-02-01A and 22-31-02-02A. Deleted Operations note 3 in subitem options 22-31-02-01B and 22-31-02-02B.

Reason: The autothrottle disconnect caution will not be annunciated if the affected servo is deactivated.

Disposition: For reference in the Operator's MEL.

Attachments: DDG pages 2.22-31-02.1-4

22-31-02 Autothrottle Servo Motors

22-31-02-01 PW and RR

22-31-02-01A One Inoperative

Interval	Installed	Required	Procedure
C	2	1	(M) (O)

One may be inoperative provided:

- Associated servo motor is deactivated.
- Both thrust levers are manually set for takeoff and go-around thrust.
- Flight remains within 180 minutes of landing at a suitable airport.

MAINTENANCE (M)

Deactivate the inoperative autothrottle servo motor (AMM 22-00-00/901).

- For the left servo motor inoperative:
 - Open and collar the P11 panel A/T SERVO L circuit breaker.
 - Disconnect, cap and stow the electrical connector from the left autothrottle servo motor.
- For the right servo motor inoperative:
 - Open and collar the P11 panel A/T SERVO R circuit breaker.
 - Disconnect, cap and stow the electrical connector from the right autothrottle servo motor.

OPERATIONS (O)

NOTE 1: The AUTOTHROTTLE L or R advisory message will be displayed.

NOTE 2: The autothrottle cannot be engaged on the ground.

NOTE 3: The EPR trim function of the EEC will operate normally, provided both autothrottle arm switches are positioned to A/T ARM.

- The flight must remain within 180 minutes of landing at a suitable airport.
- For takeoff:
 - Set both autothrottle switches to A/T ARM.
 - Set both thrust levers manually.
- After takeoff, if desired set the deactivated autothrottle servo motor A/T ARM switch OFF to restore functionality of the operative autothrottle servo motor.
- For go-around, set both thrust levers manually.

**22-31-02 Autothrottle Servo Motors****22-31-02-01 PW and RR****22-31-02-01B Both Inoperative**

Interval	Installed	Required	Procedure
C	2	0	(M) (O)

May be inoperative provided:

- Both servo motors are deactivated.
- Approach minimums do not require use of the autothrottles.
- Flight remains within 180 minutes of landing at a suitable airport.

MAINTENANCE (M)

Deactivate the inoperative autothrottle servo motors (AMM 22-00-00/901).

- Open and collar the P11 panel A/T SERVO L circuit breaker.
- Open and collar the P11 panel A/T SERVO R circuit breaker.

OPERATIONS (O)

NOTE 1: The AUTOTHROTTLE L and R advisory messages will be displayed.

NOTE 2: The EPR trim function of the EEC will operate normally, provided both autothrottle arm switches are positioned to A/T ARM.

- The flight must remain within 180 minutes of landing at a suitable airport.
- For takeoff, set both autothrottle switches to A/T ARM.

22-31-02 Autothrottle Servo Motors**22-31-02-02 GE****22-31-02-02A One Inoperative**

Interval	Installed	Required	Procedure
C	2	1	(M) (O)

One may be inoperative provided:

- EEC normal mode operates normally.
- Associated servo motor is deactivated.
- Both thrust levers are manually set for takeoff and go-around thrust.
- Flight remains within 180 minutes of landing at a suitable airport.

MAINTENANCE (M)

Deactivate the inoperative autothrottle servo motor (AMM 22-00-00/901).

1. For the left servo motor inoperative:
 - A. Open and collar the P11 panel A/T SERVO L circuit breaker.
 - B. Disconnect, cap and stow the electrical connector from the left autothrottle servo motor.
2. For the right servo motor inoperative:
 - A. Open and collar the P11 panel A/T SERVO R circuit breaker.
 - B. Disconnect, cap and stow the electrical connector from the right autothrottle servo motor.

OPERATIONS (O)

NOTE 1: The AUTOTHROTTLE L or R advisory message will be displayed.

NOTE 2: The autothrottle cannot be engaged on the ground.

NOTE 3: The N1 trim function of the EEC will operate normally, provided both autothrottle arm switches are positioned to A/T ARM.

1. The flight must remain within 180 minutes of landing at a suitable airport.
2. For takeoff:
 - A. Set both autothrottle switches to A/T ARM.
 - B. Set both thrust levers manually.
3. After takeoff, if desired set the deactivated autothrottle servo motor A/T ARM switch OFF to restore functionality of the operative autothrottle servo motor.
4. For go-around, set both thrust levers manually.

22-31-02 Autothrottle Servo Motors

22-31-02-02 GE

22-31-02-02B Both Inoperative

Interval	Installed	Required	Procedure
C	2	0	(M) (O)

May be inoperative provided:

- a. EEC normal mode operates normally.
- b. Both servo motors are deactivated.
- c. Approach minimums do not require use of the autothrottles.
- d. Flight remains within 180 minutes of landing at a suitable airport.

MAINTENANCE (M)

Deactivate the inoperative autothrottle servo motors (AMM 22-00-00/901).

1. Open and collar the P11 panel A/T SERVO L circuit breaker.



2. Open and collar the P11 panel A/T SERVO R circuit breaker.

OPERATIONS (O)

NOTE 1: The AUTOTHROTTLE L and R advisory messages will be displayed.

NOTE 2: The N1 trim function of the EEC will operate normally, provided both autothrottle arm switches are positioned to A/T ARM.

1. The flight must remain within 180 minutes of landing at a suitable airport.
2. For takeoff, set both autothrottle switches to A/T ARM.

Supplement 36-1



777 FAA DDG Supplement Package

Supplement Number: 36-1

Issue Date: 21-DEC-2023

Section Number: 2 (MMEL)

Item Number: 23-24-02

Item Name: Emergency Locator Transmitter (ELT) (Fixed)

DDG Effectivity: 777 FAA DDG (D630W003-TBC), Revision 36

MMEL Effectivity: 777 FAA MMEL, Revision 23

Description: Changed 'ELT-DT' to 'ELT' in Maintenance (M) procedure step 1 of options 23-24-02A and 23-24-02C.

Reason: Item 23-24-02 is applicable to the ELT not ELT-DT.

Disposition: For reference in the Operator's MEL.

Attachments: DDG pages 2.23-24-02.1-3

**23-24-02 Emergency Locator Transmitter (ELT) (Fixed)**

23-24-02A Required By 14 CFR - Inoperative

Interval	Installed	Required	Procedure
A	1	0	(M)

May be inoperative provided:

- System is deactivated.
- Repairs are made within 90 consecutive calendar-days.

NOTE: Dispatch relief for the Portable (survival) ELT's is in MMEL item 25-63-03.

MAINTENANCE (M)

Deactivate the inoperative ELT (AMM 23-00-00/901).

- Gain access to the inoperative ELT in the section 47 crown.
- Disconnect, cap and stow the associated control/indication electrical connector.
- Position the ELT front panel switch to ARM.

OPERATIONS NOTE

See 14 CFR 91.207, 121.353, 121.339 or other appropriate regulations.

23-24-02 Emergency Locator Transmitter (ELT) (Fixed)

23-24-02B Required By 14 CFR - Missing

Interval	Installed	Required	Procedure
A	1	0	(M)

May be missing provided:

- Placard stating "ELT not installed" is placed in view of the pilot.
- Repairs are made within 90 consecutive calendar-days.

NOTE: Dispatch relief for the Portable (survival) ELT's is in MMEL item 25-63-03.

MAINTENANCE (M)

Placard "ELT not installed" and place it in view of the pilot (AMM DDG 23-00-00/901).

OPERATIONS NOTE

See 14 CFR 91.207, 121.353, 121.339 or other appropriate regulations.

**23-24-02 Emergency Locator Transmitter (ELT) (Fixed)**

23-24-02C Not Required By 14 CFR - Inoperative

Interval	Installed	Required	Procedure
D	1	0	(M)

Any in excess of those required by 14 CFR may be inoperative provided system is deactivated.

NOTE: Dispatch relief for the Portable (survival) ELT's is in MMEL item 25-63-03.

MAINTENANCE (M)

Deactivate the inoperative ELT (AMM 23-00-00/901).

1. Gain access to the inoperative ELT in the section 47 crown.
2. Disconnect, cap and stow the associated control/indication electrical connector.
3. Position the ELT front panel switch to ARM.

OPERATIONS NOTE

See 14 CFR 91.207, 121.353, 121.339 or other appropriate regulations.

23-24-02 Emergency Locator Transmitter (ELT) (Fixed)

23-24-02D Not Required By 14 CFR - Missing

Interval	Installed	Required	Procedure
D	1	0	

Any in excess of those required by 14 CFR may be missing.

NOTE: Dispatch relief for the Portable (survival) ELT's is in MMEL item 25-63-03.

OPERATIONS NOTE

See 14 CFR 91.207, 121.353, 121.339 or other appropriate regulations.

23-24-02 Emergency Locator Transmitter (ELT) (Fixed)

23-24-02-01 Remote ELT Switch

Interval	Installed	Required	Procedure
D	1	0	(M)

May be inoperative provided:

- a. Remote ELT Switch is deactivated.
- b. ELT Switch is placed in the ARMED mode.

MAINTENANCE (M)

Deactivate the remote ELT switch and set ELT front panel switch to ARM (AMM 23-00-00/901).

1. Position the remote ELT switch to ARMED.
2. Gain access to the inoperative ELT in the section 47 crown.
3. Position the ELT front panel switch to ARM.

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Superseded



777 FAA DDG Supplement Package

Supplement Number: 36-1

Issue Date: 21-DEC-2023

Section Number: 2 (MMEL)

Item Number: 25-25-06

Item Name: Passenger Mini-Suite

DDG Effectivity: 777 FAA DDG (D630W003-TBC), Revision 36

MMEL Effectivity: 777 FAA MMEL, Revision 23

Description: Added DDG top note to subitem option 25-25-06-01B.

Reason: The note identifies the corresponding MMEL item required to comply with proviso requiring the affected seat to be considered inoperative.

Disposition: For reference in the Operator's MEL.

Attachments: DDG page 2.25-25-06.1

**25-25-06 Passenger Mini-Suite**

25-25-06-01 Entry Door(s)**25-25-06-01A Seat Occupied**

Interval	Installed	Required	Procedure
D	-	0	(M)

One or more may be inoperative and seat occupied provided that the affected door(s) is secured in the fully open position or removed.

MAINTENANCE (M)

Secure the associated suite entry door(s) in the fully open position (AMM DDG 25-00-00 /901).

25-25-06 Passenger Mini-Suite

25-25-06-01 Entry Door(s)**25-25-06-01B Seat Not Occupied**

Interval	Installed	Required	Procedure
D	-	0	

One or more may be inoperative provided:

- Affected seat is not occupied.
- Affected seat is considered inoperative

NOTE: The airplane must also be dispatched using MMEL item 25-25-02-01.



777 FAA DDG Supplement Package

Supplement Number: 36-1

Issue Date: 21-DEC-2023

Section Number: 2 (MMEL)

Item Number: 25-52-01

Item Name: Lower Cargo Compartment Lining (Forward and Aft)

DDG Effectivity: 777 FAA DDG (D630W003-TBC), Revision 36

MMEL Effectivity: 777 FAA MMEL, Revision 23

Description: Modified DDG top note.

Reason: Added item defined as part of lower cargo compartment lining.

Disposition: For reference in the Operator's MEL.

Attachments: DDG page 2.25-52-01.1

**25-52-01 Lower Cargo Compartment Lining (Forward and Aft)**

Interval	Installed	Required	Procedure
C	-	-	(O)

May be damaged or missing provided procedures are established and used to verify the associated cargo compartment remains empty or contains only ballast, empty cargo handling equipment (ballast may be loaded in ULDs), or fly away kits.

NOTE: Operator MELs must define which items are approved for inclusion in the fly away kits and which materials can be used as ballast.

NOTE: Lower Cargo Compartment Linings include: sloping sidewall liners, all endwalls, decompression panels, ceiling liners and mounted equipment, all cargo door liners, vertical sidewall liners including cheek panels and mounted equipment, waste tank enclosure panels, access panels, ceiling light assemblies, the main equipment center bulkhead liner, firestops, bilge curtains, bulkhead liners, endwall/enclosure liners and panels, and all fasteners attaching cargo linings. In addition, some of the cargo floor panels and walkways are included in this item because they are containment barriers for smoke and fire suppression. The forward cargo compartment floor panels and walkways that are containment barriers are located at the aft end of the forward compartment. The aft compartment floor panels and walkways that are containment barriers are located at the aft end of the aft compartment, just forward of the bulk cargo compartment. All of the bulk cargo compartment floor panels are containment barriers.

OPERATIONS (O)

The associated cargo compartment must remain empty, except for ballast, empty cargo containers (ballast may be loaded in ULDs), fly away kits, pallets, and cargo restraint components.

NOTE: Both the aft and bulk cargo compartments must remain empty when there are damaged or missing linings in either the aft or the bulk cargo compartment.



777 FAA DDG Supplement Package

Supplement Number: 36-1

Issue Date: 21-DEC-2023

Section Number: 2 (MMEL)

Item Number: 25-59-01

Item Name: Main Deck Cargo Compartment Lining (777F)

DDG Effectivity: 777 FAA DDG (D630W003-TBC), Revision 36

MMEL Effectivity: 777 FAA MMEL, Revision 23

Description: Modified DDG top note.

Reason: Added item defined as part of main deck cargo compartment lining.

Disposition: For reference in the Operator's MEL.

Attachments: DDG page 2.25-59-01.1

**25-59-01 Main Deck Cargo Compartment Lining (777F)**

Interval	Installed	Required	Procedure
C	-	-	(O)

May be damaged or missing provided procedures are established and used to verify the main deck cargo compartment remains empty or contains only ballast, empty cargo handling equipment (ballast may be loaded in ULDs), or fly away kits.

NOTE: Operator MELs must define which items are approved for inclusion in the fly away kits and which materials can be used as ballast.

NOTE: Main Deck Cargo Compartment Linings include: Ceilings, sidewalls, cover strips, endwalls, dado panels, window reveals, insulation, decompression panel, cargo handling control panels, electrical panels, cargo door edge trim, cargo door lining, smoke detector assemblies, markers, liner support rails and brackets, fabric parts, ECS duct and ceiling light assemblies, and all fasteners attaching cargo linings.

OPERATIONS (O)

The main deck cargo compartment must remain empty, except for ballast, empty cargo containers (ballast may be loaded in ULDs), fly away kits, pallets, and cargo restraint components.



777 FAA DDG Supplement Package

Supplement Number: 36-1

Issue Date: 21-DEC-2023

Section Number: 2 (MMEL)

Item Number: 25-64-01

Item Name: Megaphones (Passenger)

DDG Effectivity: 777 FAA DDG (D630W003-TBC), Revision 36

MMEL Effectivity: 777 FAA MMEL, Revision 23

Description: Modified subitem 25-64-01-01 Operations (O) procedure.

Reason: Clarified equipment that is affected by the inoperative tamper seals or tags.

Disposition: For reference in the Operator's MEL.

Attachments: DDG pages 2.25-64-01.1-2

**25-64-01 Megaphones (Passenger)****25-64-01A Passengers Carried**

Interval	Installed	Required	Procedure
D	-	-	

Any in excess of those required by 14 CFR may be inoperative or removed provided:

- Inoperative megaphone remains in a certified location until removed from the aircraft at the next suitable maintenance facility.
- Location placarding is removed or obscured.
- Required distribution is maintained.

OPERATIONS NOTE

See 14 CFR 121.309 or other appropriate regulations.

25-64-01 Megaphones (Passenger)**25-64-01B No Passengers Carried**

Interval	Installed	Required	Procedure
C	-	0	(O)

May be inoperative or missing provided:

- No passengers are carried.
- A maximum of 19 persons authorized by 14 CFR for non-passenger carrying operations are carried.
- Alternate procedures are established and used.

OPERATIONS (O)

- No passengers may be carried.
- A maximum of 19 persons authorized by 14 CFR 121.583 may be carried.
- Establish alternate procedures for use when megaphones are not available.

25-64-01 Megaphones (Passenger)**25-64-01-01 Tamper Seals or Tags**

Interval	Installed	Required	Procedure
C	-	-	(O)



May be inoperative, damaged, or missing provided proper installation and operation is verified at each preflight.

OPERATIONS (O)

Before each departure, verify affected Megaphone(s) is properly installed and operate(s) normally.

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777 FAA DDG Supplement Package

Supplement Number: 36-1

Issue Date: 21-DEC-2023

Section Number: 2 (MMEL)

Item Number: 25-64-02

Item Name: Portable Flashlights/Flashlight Holders

DDG Effectivity: 777 FAA DDG (D630W003-TBC), Revision 36

MMEL Effectivity: 777 FAA MMEL, Revision 23

Description: Modified subitem 25-64-02-01 Operations (O) procedure.

Reason: Clarified equipment that is affected by the inoperative tamper seals or tags.

Disposition: For reference in the Operator's MEL.

Attachments: DDG pages 2.25-64-02.1-2

**25-64-02 Portable Flashlights/Flashlight Holders****25-64-02A Required by 14 CFR**

Interval	Installed	Required	Procedure
C	-	-	

May be inoperative or removed provided:

- Crewmember assigned to the affected position has an equivalent operative flashlight readily available.
 - Inoperative flashlight remains in a certified location or is removed from the aircraft.
 - Location placarding is removed or obscured.
-

25-64-02 Portable Flashlights/Flashlight Holders**25-64-02B Any In Excess Of Those Required By 14 CFR**

Interval	Installed	Required	Procedure
D	-	-	

Any in excess of those required by 14 CFR may be inoperative or removed provided:

- Inoperative flashlight remains in a certified location until removed from the aircraft at the next suitable maintenance facility.
 - Location placarding is removed or obscured.
-

25-64-02 Portable Flashlights/Flashlight Holders**25-64-02C No Passenger Carried**

Interval	Installed	Required	Procedure
C	-	0	(O)

May be inoperative or missing provided:

- No passengers are carried.
 - A maximum of 19 persons authorized by 14 CFR for non-passenger carrying operations are carried.
 - Alternate procedures are established and used.
-

OPERATIONS (O)

- No passengers may be carried.



2. A maximum of 19 persons authorized by 14 CFR 121.583 may be carried.
3. Establish alternate procedures for use when flashlights are not available.

25-64-02 Portable Flashlights/Flashlight Holders**25-64-02-01 Tamper Seals or Tags**

Interval	Installed	Required	Procedure
C	-	-	(O)

May be inoperative, damaged, or missing provided proper installation and operation is verified at each preflight.

OPERATIONS (O)

Before each departure, verify affected Portable Flashlight(s)/Flashlight Holder(s) is properly installed and operate(s) normally.



777 FAA DDG Supplement Package

Supplement Number: 36-1

Issue Date: 21-DEC-2023

Section Number: 2 (MMEL)

Item Number: 25-64-03

Item Name: Emergency Medical Equipment

DDG Effectivity: 777 FAA DDG (D630W003-TBC), Revision 36

MMEL Effectivity: 777 FAA MMEL, Revision 23

Description: Modified Operations (O) procedure for sub subitems 25-64-03-01-01, 25-64-03-02-01, and 25-64-03-03-01.

Reason: Clarified equipment that is affected by the inoperative tamper seals or tags.

Disposition: For reference in the Operator's MEL.

Attachments: DDG pages 2.25-64-03.1-4

**25-64-03 Emergency Medical Equipment****25-64-03-01 First Aid Kit (FAK) and/or Associated Equipment****25-64-03-01A Required By 14 CFR**

Interval	Installed	Required	Procedure
A	-	-	(O)

If more than one is required by 14 CFR, only one of the required first aid kits may be incomplete, missing or inoperative provided:

- FAK is resealed in a manner that will identify it as a unit that cannot be mistaken for a fully serviceable unit.
- Location placarding is removed or obscured.
- Repairs or replacements are made within one flight.

NOTE: Medical equipment installed in the aircraft as part of an EMS operation is not considered part of the normal complement of equipment. No MMEL relief applies to that equipment and 14 CFR maintenance and inspection requirements do not apply.

OPERATIONS (O)

NOTE: See 14 CFR 121.803 or other appropriate regulations.

Each operator must develop their own procedures based on the emergency medical equipment installed on their airplanes.

25-64-03 Emergency Medical Equipment**25-64-03-01 First Aid Kit (FAK) and/or Associated Equipment****25-64-03-01B Any In Excess Of Those Required By 14 CFR**

Interval	Installed	Required	Procedure
D	-	-	

Any in excess of those required by 14 CFR may be incomplete, inoperative or removed.

OPERATIONS NOTE

See 14 CFR 121.803 or other appropriate regulations.

25-64-03 Emergency Medical Equipment
25-64-03-01 First Aid Kit (FAK) and/or Associated Equipment
25-64-03-01-01 Tamper Seals or Tags

Interval	Installed	Required	Procedure
C	-	-	(O)

May be inoperative, damaged, or missing provided proper FAK servicing is verified at each preflight.

OPERATIONS (O)

Before each departure, verify affected FAK(s) and/or associated equipment has been properly serviced.

25-64-03 Emergency Medical Equipment
25-64-03-02 Emergency Medical Kit (EMK) and/or Associated Equipment

25-64-03-02A Required By 14 CFR

Interval	Installed	Required	Procedure
A	-	0	(O)

May be incomplete, missing or inoperative provided:

- EMK is resealed in a manner that will identify it as a unit that cannot be mistaken for a fully serviceable unit.
- Location placarding is removed or obscured.
- Repairs or replacements are made within one flight.

NOTE: Medical equipment installed in the aircraft as part of an EMS operation is not considered part of the normal complement of equipment. No MMEL relief applies to that equipment and 14 CFR maintenance and inspection requirements do not apply.

OPERATIONS (O)

NOTE: See 14 CFR 121.803 or other appropriate regulations.

Each operator must develop their own procedures based on the emergency medical equipment installed on their airplanes.

**25-64-03 Emergency Medical Equipment****25-64-03-02 Emergency Medical Kit (EMK) and/or Associated Equipment**

25-64-03-02B Any In Excess Of Those Required By 14 CFR

Interval	Installed	Required	Procedure
D	-	-	

Any in excess of those required by 14 CFR may be incomplete, inoperative or removed.

OPERATIONS NOTE

See 14 CFR 121.803 or other appropriate regulations.

25-64-03 Emergency Medical Equipment**25-64-03-02 Emergency Medical Kit (EMK) and/or Associated Equipment****25-64-03-02-01 Tamper Seals or Tags**

Interval	Installed	Required	Procedure
C	-	-	(O)

May be inoperative, damaged, or missing provided proper EMK servicing is verified at each preflight.

OPERATIONS (O)

Before each departure, verify affected EMK(s) and/or associated equipment has been properly serviced.

25-64-03 Emergency Medical Equipment**25-64-03-03 Automated External Defibrillators (AED) and/or Associated Equipment**

25-64-03-03A Required By 14 CFR

Interval	Installed	Required	Procedure
A	-	0	(O)

May be incomplete, missing or inoperative provided:

- AED is resealed in a manner that will identify it as a unit that cannot be mistaken for a fully serviceable unit.



- b. Location placarding is removed or obscured.
- c. Repairs or replacements are made within one flight.

NOTE: Medical equipment installed in the aircraft as part of an Emergency Medical Service (EMS) operation is not considered part of the normal complement of equipment. No MMEL relief applies to that equipment and 14 CFR maintenance and inspection requirements do not apply.

OPERATIONS (O)

NOTE: See 14 CFR 121.803 or other appropriate regulations.

Each operator must develop their own procedures based on the emergency medical equipment installed on their airplanes.

25-64-03 Emergency Medical Equipment

25-64-03-03 Automated External Defibrillators (AED) and/or Associated Equipment

25-64-03-03B Any In Excess Of Those Required By 14 CFR

Interval	Installed	Required	Procedure
D	-	-	

Any in excess of those required by 14 CFR may be incomplete, inoperative or removed.

OPERATIONS NOTE

See 14 CFR 121.803 or other appropriate regulations.

25-64-03 Emergency Medical Equipment

25-64-03-03 Automated External Defibrillators (AED) and/or Associated Equipment

25-64-03-03-01 Tamper Seals or Tags

Interval	Installed	Required	Procedure
C	-	-	(O)

May be inoperative, damaged, or missing provided proper servicing is verified at each preflight.

OPERATIONS (O)

Before each departure, verify affected AED(s) and/or associated equipment has been properly serviced.



777 FAA DDG Supplement Package

Supplement Number: 36-1

Issue Date: 21-DEC-2023

Section Number: 2 (MMEL)

Item Number: 26-26-01

Item Name: Portable Fire Extinguishers

DDG Effectivity: 777 FAA DDG (D630W003-TBC), Revision 36

MMEL Effectivity: 777 FAA MMEL, Revision 23

Description: Modified subitem 26-26-01-01 Operations (O) procedure.

Reason: Corrected procedure and clarified equipment that is affected by the inoperative tamper seals or tags.

Disposition: For reference in the Operator's MEL.

Attachments: DDG page 2.26-26-01.1

26-26-01 Portable Fire Extinguishers

Interval	Installed	Required	Procedure
D	-	-	

Any in excess of those required by 14 CFR may be inoperative or removed provided:

- Inoperative fire extinguisher remains in a certified location until removed from the aircraft at the next suitable maintenance facility.
- Location placarding is removed or obscured.
- Required distribution is maintained.

NOTE: Inoperative fire extinguishers removed from a certified location or removed from the aircraft, are subject to 49 CFR dangerous goods regulations.

MAINTENANCE NOTE

- Stow the out-of-service extinguisher in a place (possibly in a bag) where it will not be used by mistake, until it can be removed from the airplane at the next available maintenance base.
- Ensure required distribution of serviceable extinguishers is maintained throughout the airplane.

OPERATIONS NOTE

See 14 CFR 121.309 or other appropriate regulations.

26-26-01 Portable Fire Extinguishers

26-26-01-01 Tamper Seals or Tags

Interval	Installed	Required	Procedure
C	-	-	(O)

May be inoperative, damaged or missing provided proper installation and servicing is verified at each preflight.

OPERATIONS (O)

Before each departure, verify affected Portable Fire Extinguisher(s) is properly installed and serviced.



777 FAA DDG Supplement Package

Supplement Number: 36-1

Issue Date: 21-DEC-2023

Section Number: 2 (MMEL)

Item Number: 27-11-04

Item Name: Flaperon Power Control Units (PCUs)

DDG Effectivity: 777 FAA DDG (D630W003-TBC), Revision 36

MMEL Effectivity: 777 FAA MMEL, Revision 23

Description: Added DDG top note and Operations note to subitem 27-11-04-01.

Reason: The DDG top note identifies additional MMEL item application requirement resulting from deactivation of the right flaperon inboard PCU. The Operations note is included for crew awareness of the resulting EICAS status message in this dispatch configuration.

Disposition: For reference in the Operator's MEL.

Attachments: DDG pages 2.27-11-04.1-3

27-11-04 Flaperon Power Control Units (PCUs)
27-11-04-01 Passenger (With PRRs 61384-15 and 62200, or Equivalent Installed)

Interval	Installed	Required	Procedure
A	4	3	(M)

One may be inoperative provided:

- Inoperative PCU is deactivated in the bypass mode.
- Flaperons are verified to operate normally.
- Repairs are made within 3 flight days.

NOTE 1: For an inoperative right flaperon inboard PCU, the airplane must also be dispatched using MMEL item 27-02-05.

NOTE 2: PRRs 61384-15 and 62200 Part-B are applicable to airplane production L/Ns 718, 772, 774 and 776 and-on. This item is also applicable to airplanes with the AIMS-1 or AIMS-2 platforms with BP V14 or later AIMS OPS installed together with the following PFC software installed: PFC OPS 2762-PFC-741-00, or PFC OPS 276D-PFC-841-00, or PFC OPS 2768-PFC-910-00 or later installed.

MAINTENANCE (M)

Deactivate the inoperative flaperon PCU in the bypass mode and verify flaperons operate normally (AMM 27-00-00/901).

- For an inoperative left flaperon outboard PCU, open and collar the Power Supply Assembly Left (PSA-L) in E1-6 rack FLPRN PCU circuit breaker.
- For an inoperative left flaperon inboard PCU, open and collar the Power Supply Assembly Right (PSA-R) in E5-1 rack FLPRN PCU circuit breaker.
- For an inoperative right flaperon outboard PCU, open and collar the Power Supply Assembly Center (PSA-C) in E2-6 rack FLPRN PCU circuit breaker.
- For an inoperative right flaperon inboard PCU:
 - Open P110 panel FLPRN PCU RIB(BYP)/SPLR 4 & 11 circuit breaker.
 - Open P110 panel ACE-L2 PWR circuit breaker.
 - Set Maintenance Panel (P61) left, center and right WING FLT CONTROL HYD POWER switches to SHUT OFF.
 - Disconnect, cap and stow the PCU electrical connector D4422P (WD 27-11-18). The connector is located in the Main Equipment Center on the left side of the E1-1 shelf near the forward cargo access door (station 408, connector position 71).



NOTE: The ACTUATOR PRESS SNSR status message will be displayed. The airplane must also be dispatched using MMEL item 27-02-05.

- E. Close the open circuit breakers.
 - F. Set Maintenance Panel (P61) left, center and right WING FLT CONTROL HYD POWER switches to NORM.
5. Check the operation of the flaperons.

NOTE: Do not do an Actuator Confidence Test, or an ACE Monitor Test, or L or R Flaperon Actuator Confidence Tests.

6. Confirm the FLIGHT CONTROL SYS status message is not displayed.

OPERATIONS NOTE

For an inoperative right flaperon inboard PCU, the ACTUATOR PRESS SNSR status message will be displayed.

27-11-04 Flaperon Power Control Units (PCUs)

27-11-04-02 777F

Interval	Installed	Required	Procedure
A	4	3	(M)

One may be inoperative provided:

- a. Inoperative PCU is deactivated in the bypass mode.
- b. Flaperons are verified to operate normally.
- c. Repairs are made within 3 flight days.

MAINTENANCE (M)

Deactivate the inoperative flaperon PCU in the bypass mode and verify flaperons operate normally (AMM 27-00-00/901).

- 1. For an inoperative left flaperon outboard PCU, open and collar the Power Supply Assembly Left (PSA-L) in E1-6 rack FLPRN PCU circuit breaker.
- 2. For an inoperative left flaperon inboard PCU, open and collar the Power Supply Assembly Right (PSA-R) in E5-1 rack FLPRN PCU circuit breaker.
- 3. For an inoperative right flaperon outboard PCU, open and collar the Power Supply Assembly Center (PSA-C) in E2-6 rack FLPRN PCU circuit breaker.
- 4. For an inoperative right flaperon inboard PCU:
 - A. Open P110 panel FLPRN PCU RIB(BYP)/SPLR 4 & 11 circuit breaker.
 - B. Open P110 panel ACE-L2 PWR circuit breaker.
 - C. Set Maintenance Panel (P61) left, center and right WING FLT CONTROL HYD POWER switches to SHUT OFF.



- D. Disconnect, cap and stow the PCU electrical connector D4422P (WD 27-11-18). The connector is located in the Main Equipment Center on the left side of the E1-1 shelf near the forward cargo access door (station 408, connector position 71).

NOTE: The ACTUATOR PRESS SNSR status message will be displayed. The airplane must also be dispatched using MMEL item 27-02-05.

- E. Close the open circuit breakers.
- F. Set Maintenance Panel (P61) left, center and right WING FLT CONTROL HYD POWER switches to NORM.
5. Check the operation of the flaperons.

NOTE: Do not do an Actuator Confidence Test, or an ACE Monitor Test, or L or R Flaperon Actuator Confidence Tests.

6. Confirm the FLIGHT CONTROL SYS status message is not displayed.



777 FAA DDG Supplement Package

Supplement Number: 36-1

Issue Date: 21-DEC-2023

Section Number: 2 (MMEL)

Item Number: 28-11-01

Item Name: Sump Drain Valves

DDG Effectivity: 777 FAA DDG (D630W003-TBC), Revision 36

MMEL Effectivity: 777 FAA MMEL, Revision 23

Description: Modified the AMM references in subitem 28-11-01-02 and subitem option 28-11-01-03A Maintenance (M) procedures.

Reason: These AMM references were incorrectly modified in the previous DDG revision.

Disposition: For reference in the Operator's MEL.

Attachments: DDG pages 2.28-11-01.1-2

28-11-01 Sump Drain Valves
28-11-01-01 Surge Tanks

Interval	Installed	Required	Procedure
C	2	0	(M)

May be inoperative provided there is no evidence of leakage.

MAINTENANCE (M)

Visually inspect exterior surface location of the inoperative sump drain valve to verify that there is no evidence of fuel leakage (AMM 28-00-00/901).

28-11-01 Sump Drain Valves
28-11-01-02 Main and Center Tanks

Interval	Installed	Required	Procedure
C	4	3	(M)

One may be inoperative provided:

- There is no evidence of leakage.
- Alternate procedures are used to prevent water accumulation in associated tank.

MAINTENANCE (M)

Verify that there is no evidence of leakage and use alternate procedures to prevent water accumulation in associated tank (AMM 28-00-00/901).

- Visually inspect exterior surface location of the inoperative sump drain valve to verify that there is no evidence of fuel leakage.
- Use alternate procedures to prevent water accumulation in the associated tank.

28-11-01 Sump Drain Valves
28-11-01-03 Auxiliary Tank(s)

28-11-01-03A Auxiliary Tank(s) Usable

Interval	Installed	Required	Procedure
C	-	0	(M)

May be inoperative provided:

- There is no evidence of leakage.



- b. Alternate procedures are used to prevent water accumulation in associated tank(s).
-

MAINTENANCE (M)

Verify that there is no evidence of leakage and use alternate procedures to prevent water accumulation in associated tank (AMM 28-00-00/901).

1. Visually inspect exterior surface location of the inoperative sump drain valve to verify that there is no evidence of fuel leakage.
2. Use alternate procedures to prevent water accumulation in the auxiliary fuel tank(s).

28-11-01 Sump Drain Valves

28-11-01-03 Auxiliary Tank(s)

28-11-01-03B Auxiliary Tank(s) Empty

Interval	Installed	Required	Procedure
D	-	0	

May be inoperative provided all auxiliary tanks remain empty.

MAINTENANCE NOTE

For auxiliary tank(s) not empty, defuel the auxiliary tank(s) (AMM 28-26-00/201).



777 FAA DDG Supplement Package

Supplement Number: 36-1

Issue Date: 21-DEC-2023

Section Number: 2 (MMEL)

Item Number: 28-14-03

Item Name: Auxiliary Tank Refuel Transfer Valve(s)

DDG Effectivity: 777 FAA DDG (D630W003-TBC), Revision 36

MMEL Effectivity: 777 FAA MMEL, Revision 23

Description: Modified the AMM reference in the option 28-14-03A Maintenance (M) procedure.

Reason: This AMM reference was incorrectly modified in the previous DDG revision.

Disposition: For reference in the Operator's MEL.

Attachments: DDG pages 2.28-14-03.1-2

28-14-03 Auxiliary Tank Refuel Transfer Valve(s)

28-14-03A Auxiliary Tank(s) Usable

Interval	Installed	Required	Procedure
C	-	0	(M) (O)

May be inoperative provided:

- Inoperative valve is deactivated closed.
- The auxiliary tank fuel pump transfer system operates normally.
- Auxiliary tank fuel quantity indication operates normally.
- Maximum zero fuel weight limit is reduced by the weight of the associated auxiliary tank fuel.

MAINTENANCE (M)

NOTE: For the auxiliary tank fuel pump transfer system to be operating normally, certain items associated with that system must be operative. See DDG item 28-14-01 for a listing of those items.

Deactivate the inoperative refuel transfer valve closed (AMM 28-00-00/901).

- For auxiliary tank 1 refuel transfer valve inoperative, open and collar P310 panel AUX RFUEL XFER VLV 1 circuit breaker.
- For auxiliary tank 2 installed with refuel transfer valve inoperative, open and collar P310 panel AUX RFUEL XFER VLV 2 circuit breaker.
- For auxiliary tank 3 installed with refuel transfer valve inoperative, open and collar P310 panel AUX RFUEL XFER VLV 3 circuit breaker.
- Gain access to the inoperative auxiliary tank refuel transfer valve.
- Manually close the inoperative auxiliary tank refuel transfer valve.

NOTE: Valve may be manually opened for refuel.

OPERATIONS (O)

Reduce the maximum fuel weight by the weight of the auxiliary tank fuel.

28-14-03 Auxiliary Tank Refuel Transfer Valve(s)

28-14-03B Auxiliary Tank(s) Empty

Interval	Installed	Required	Procedure
D	-	0	

May be inoperative provided all auxiliary tanks remain empty.

**MAINTENANCE NOTE**

For auxiliary tank(s) not empty, defuel the auxiliary tank(s) (AMM 28-26-00/201).

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777 FAA DDG Supplement Package

Supplement Number: 36-1

Issue Date: 21-DEC-2023

Section Number: 2 (MMEL)

Item Number: 28-14-07

Item Name: Auxiliary Tank Fuel Backup Transfer System

DDG Effectivity: 777 FAA DDG (D630W003-TBC), Revision 36

MMEL Effectivity: 777 FAA MMEL, Revision 23

Description: Modified the AMM reference in the option 28-14-07A Maintenance (M) procedure.

Reason: This AMM reference was incorrectly modified in the previous DDG revision.

Disposition: For reference in the Operator's MEL.

Attachments: DDG page 2.28-14-07.1

28-14-07 Auxiliary Tank Fuel Backup Transfer System

28-14-07A Auxiliary Tank(s) Usable

Interval	Installed	Required	Procedure
C	1	0	(M) (O)

May be inoperative provided:

- Associated auxiliary tank fuel backup transfer PRSOV is deactivated.
- Auxiliary tank fuel vent valves operate normally.
- Auxiliary tank fuel pump transfer system operates normally.
- Maximum zero fuel weight limit is reduced by the weight of auxiliary tank fuel.

MAINTENANCE (M)

NOTE: For the auxiliary tank fuel pump transfer system to be operating normally, certain items associated with that system must be operative. See DDG item 28-14-01 for a listing of those items.

Deactivate the associated auxiliary tank fuel backup transfer PRSOV (AMM 28-00-00/901).

- Open and collar P110 panel AIR SUPPLY-AUX FUEL TANK PRESS CTRL circuit breaker.

OPERATIONS (O)

Reduce the maximum fuel weight by the weight of the auxiliary tank fuel.

28-14-07 Auxiliary Tank Fuel Backup Transfer System

28-14-07B Auxiliary Tank(s) Empty

Interval	Installed	Required	Procedure
D	1	0	

May be inoperative provided:

- Auxiliary tank fuel vent valves operate normally.
- All auxiliary tanks remain empty.

MAINTENANCE NOTE

For auxiliary tank(s) not empty, defuel the auxiliary tank(s) (AMM 28-26-00/201).



777 FAA DDG Supplement Package

Supplement Number: 36-1

Issue Date: 21-DEC-2023

Section Number: 2 (MMEL)

Item Number: 28-21-02

Item Name: Auxiliary Tank Fuel Quantity Indication (Auxiliary Refuel Panel)

DDG Effectivity: 777 FAA DDG (D630W003-TBC), Revision 36

MMEL Effectivity: 777 FAA MMEL, Revision 23

Description: Modified the AMM reference in the option 28-21-02A Maintenance (M) procedure.

Reason: This AMM reference was incorrectly modified in the previous DDG revision.

Disposition: For reference in the Operator's MEL.

Attachments: DDG page 2.28-21-02.1

**28-21-02 Auxiliary Tank Fuel Quantity Indication (Auxiliary Refuel Panel)**

28-21-02A Auxiliary Tank(s) Usable

Interval	Installed	Required	Procedure
C	1	0	(M)

May be inoperative provided alternate procedures are used for refueling.

MAINTENANCE (M)

Verification of auxiliary tank fuel quantity may be accomplished using one of the following methods (AMM 28-00-00/901):

1. Determine the fuel quantity using the associated tank's Electronic Fuel Level Indication (EFLI).
2. Determine the fuel quantity using the flight deck fuel quantity indication system.
3. Defuel the associated tank and service with a known quantity of fuel.

28-21-02 Auxiliary Tank Fuel Quantity Indication (Auxiliary Refuel Panel)

28-21-02B Auxiliary Tank(s) Empty

Interval	Installed	Required	Procedure
D	1	0	

May be inoperative provided all auxiliary tanks remain empty.

MAINTENANCE NOTE

For auxiliary tank(s) not empty, defuel the auxiliary tank(s) (AMM 28-26-00/201).



777 FAA DDG Supplement Package

Supplement Number: 36-1

Issue Date: 21-DEC-2023

Section Number: 2 (MMEL)

Item Number: 28-31-04

Item Name: Fuel Jettison Nozzle Valves

DDG Effectivity: 777 FAA DDG (D630W003-TBC), Revision 36

MMEL Effectivity: 777 FAA MMEL, Revision 23

Description: Modified the AMM references in the subitem option 28-31-04-01B and 28-31-04-02B Maintenance (M) procedures.

Reason: These AMM references were incorrectly modified in the previous DDG revision.

Disposition: For reference in the Operator's MEL.

Attachments: DDG pages 2.28-31-04.1-5

28-31-04 Fuel Jettison Nozzle Valves
28-31-04-01 Without PRR 62777-79S or Equivalent Installed
28-31-04-01A One Inoperative

Interval	Installed	Required	Procedure
C	2	1	(M)

One nozzle valve may be inoperative provided it is secured closed.

NOTE: The PRR 62777-79S is applicable to airplane line numbers 1099 and-on. Service Bulletins 777-28-0078 (ELMS2) and 777-28-0082 (ELMS1) each provide the production equivalent of PRR 62777-79S.

MAINTENANCE (M)

Secure the inoperative fuel jettison nozzle valve in the closed position (AMM 28-00-00/901).

1. For left jettison nozzle valve inoperative:
 - A. Open and collar P110 panel L FUEL JETT NOZ VLV circuit breaker.
 - B. Gain access to the left jettison nozzle valve actuator.
 - C. Disconnect, cap and stow the electrical connector (DV28166).
 - D. Confirm the manual override handle is in the CLOSED position and lockwire.
2. For right jettison nozzle valve inoperative:
 - A. Open and collar P210 panel R FUEL JETT NOZ VLV circuit breaker.
 - B. Gain access to the right jettison nozzle valve actuator.
 - C. Disconnect, cap and stow the electrical connector (DV28266).
 - D. Confirm the manual override handle is in the CLOSED position and lockwire.

28-31-04 Fuel Jettison Nozzle Valves
28-31-04-01 Without PRR 62777-79S or Equivalent Installed
28-31-04-01B Both Inoperative

Interval	Installed	Required	Procedure
C	2	0	(M) (O)

May be inoperative provided:

- a. Both jettison nozzle valves are secured closed.
- b. Jettison system is deactivated.
- c. Appropriate performance penalties are applied.



NOTE: The PRR 62777-79S is applicable to airplane line numbers 1099 and-on. Service Bulletins 777-28-0078 (ELMS2) and 777-28-0082 (ELMS1) each provide the production equivalent of PRR 62777-79S.

MAINTENANCE (M)

Secure the fuel jettison nozzle valves in the closed position and deactivate the jettison system (AMM 28-00-00/901).

1. For left jettison nozzle valve:
 - A. Open and collar P110 panel L FUEL JETT NOZ VLV circuit breaker.
 - B. Gain access to the left jettison nozzle valve actuator.
 - C. Disconnect, cap and stow the electrical connector (DV28166).
 - D. Confirm the manual override handle is in the CLOSED position and lockwire.
2. For right jettison nozzle valve:
 - A. Open and collar P210 panel R FUEL JETT NOZ VLV circuit breaker.
 - B. Gain access to the right jettison nozzle valve actuator.
 - C. Disconnect, cap and stow the electrical connector (DV28266).
 - D. Confirm the manual override handle is in the CLOSED position and lockwire.
3. Deactivate the fuel jettison system.
 - A. Open and collar P110 panel L FUEL JETT PUMP CTRL and L FUEL JETT ISLN VLV circuit breakers.
 - B. Open and collar P210 panel R FUEL JETT PUMP CTRL and R FUEL JETT ISLN VLV circuit breakers.

OPERATIONS (O)

NOTE: The Fuel Synoptic will incorrectly indicate jettison flow from the center tank when the center tank override pumps are on with fuel in the center tank. The FUEL JETT NOZZLE L and R advisory messages will be displayed and overhead panel L and R fuel jettison nozzle VALVE lights will be illuminated.

For -200ER/-300:

Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 4,000 lb (1,800 kg) (for fuel that would be burned during a return-to-land procedure).

For -200LR/777F/-300ER:

Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 5,400 lb (2,450 kg) (for fuel that would be burned during a return-to-land procedure).

28-31-04 Fuel Jettison Nozzle Valves
28-31-04-01 Without PRR 62777-79S or Equivalent Installed
28-31-04-01-01 Switch VALVE Lights

Interval	Installed	Required	Procedure
C	2	0	

28-31-04 Fuel Jettison Nozzle Valves
28-31-04-01 Without PRR 62777-79S or Equivalent Installed
28-31-04-01-02 Switch ON Lights

Interval	Installed	Required	Procedure
C	2	0	

28-31-04 Fuel Jettison Nozzle Valves
28-31-04-02 With PRR 62777-79S or Equivalent Installed
28-31-04-02A One Inoperative

Interval	Installed	Required	Procedure
C	2	1	(M)

One nozzle valve may be inoperative provided it is secured closed.

NOTE: The PRR 62777-79S is applicable to airplane line numbers 1099 and-on. Service Bulletins 777-28-0078 (ELMS2) and 777-28-0082 (ELMS1) each provide the production equivalent of PRR 62777-79S.

MAINTENANCE (M)

Secure the inoperative fuel jettison nozzle valve in the closed position (AMM 28-00-00/901).

1. For left jettison nozzle valve inoperative:
 - A. Open and collar P110 panel L FUEL JETT NOZ VLV circuit breaker.
 - B. Gain access to the left jettison nozzle valve actuator.
 - C. Disconnect, cap and stow the electrical connector (DV28166).
 - D. Confirm the manual override handle is in the CLOSED position and lockwire.
2. For right jettison nozzle valve inoperative:
 - A. Open and collar P210 panel R FUEL JETT NOZ VLV circuit breaker.

- B. Gain access to the right jettison nozzle valve actuator.
- C. Disconnect, cap and stow the electrical connector (DV28266).
- D. Confirm the manual override handle is in the CLOSED position and lockwire.

28-31-04 Fuel Jettison Nozzle Valves

28-31-04-02 With PRR 62777-79S or Equivalent Installed

28-31-04-02B Both Inoperative

Interval	Installed	Required	Procedure
C	2	0	(M) (O)

May be inoperative provided:

- a. Both jettison nozzle valves are secured closed.
- b. Jettison system is deactivated.
- c. Fuel scavenge system is considered inoperative.
- d. Appropriate performance penalties are applied.

NOTE 1: The PRR 62777-79S is applicable to airplane line numbers 1099 and-on. Service Bulletins 777-28-0078 (ELMS2) and 777-28-0082 (ELMS1) each provide the production equivalent of PRR 62777-79S.

NOTE 2: The airplane must also be dispatched using MMEL item 28-22-04.

MAINTENANCE (M)

Secure the fuel jettison nozzle valves in the closed position and deactivate the jettison system (AMM 28-00-00/901).

1. For left jettison nozzle valve:
 - A. Open and collar P110 panel L FUEL JETT NOZ VLV circuit breaker.
 - B. Gain access to the left jettison nozzle valve actuator.
 - C. Disconnect, cap and stow the electrical connector (DV28166).
 - D. Confirm the manual override handle is in the CLOSED position and lockwire.
2. For right jettison nozzle valve:
 - A. Open and collar P210 panel R FUEL JETT NOZ VLV circuit breaker.
 - B. Gain access to the right jettison nozzle valve actuator.
 - C. Disconnect, cap and stow the electrical connector (DV28266).
 - D. Confirm the manual override handle is in the CLOSED position and lockwire.
3. Deactivate the fuel jettison system.
 - A. Open and collar P110 panel L FUEL JETT PUMP CTRL and L FUEL JETT ISLN VLV circuit breakers.



- B. Open and collar P210 panel R FUEL JETT PUMP CTRL and R FUEL JETT ISLN VLV circuit breakers.

OPERATIONS (O)

NOTE: The Fuel Synoptic will incorrectly indicate jettison flow from the center tank when the center tank override pumps are on with fuel in the center tank. The FUEL JETT NOZZLE L and R advisory messages will be displayed and overhead panel L and R fuel jettison nozzle VALVE lights will be illuminated.

For -200ER/-300:

Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 4,000 lb (1,800 kg) (for fuel that would be burned during a return-to-land procedure).

For -200LR/777F/-300ER:

Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 5,400 lb (2,450 kg) (for fuel that would be burned during a return-to-land procedure).

28-31-04 Fuel Jettison Nozzle Valves**28-31-04-02 With PRR 62777-79S or Equivalent Installed****28-31-04-02-01 Switch VALVE Lights**

Interval	Installed	Required	Procedure
C	2	0	

28-31-04 Fuel Jettison Nozzle Valves**28-31-04-02 With PRR 62777-79S or Equivalent Installed****28-31-04-02-02 Switch ON Lights**

Interval	Installed	Required	Procedure
C	2	0	



777 FAA DDG Supplement Package

Supplement Number: 36-1

Issue Date: 21-DEC-2023

Section Number: 2 (MMEL)

Item Number: 30-21-01

Item Name: Engine Anti-Ice Systems

DDG Effectivity: 777 FAA DDG (D630W003-TBC), Revision 36

MMEL Effectivity: 777 FAA MMEL, Revision 23

Description: Added Maintenance (M) procedure step to subitem options 30-21-01-01B and 30-21-01-01D.

Reason: Maintenance (M) procedure step was added to provide safety to maintenance personnel.

Disposition: For reference in the Operator's MEL.

Attachments: DDG pages 2.30-21-01.1-24

30-21-01 Engine Anti-Ice Systems

Interval	Installed	Required	Procedure
C	2	1	(M)

One may be inoperative provided:

- Valve is locked closed.
- Associated engine anti-ice control switch remains OFF.
- Airplane is not operated in known or forecast icing conditions.
- Flight remains within 120 minutes of landing at a suitable airport.

MAINTENANCE (M)

Lock associated EAI valve in the closed position (AMM 30-00-00/901).

- Deactivate the associated EAI valve controller:
 - For the right EAI valve, open P210 panel EAI CTRL R circuit breaker.
 - For the left EAI valve, open P310 panel EAI CTRL L circuit breaker.

For GE:

- Deactivate the leading edge slats in the retracted position (AMM 27-81-00/201).
- Deactivate the associated thrust reverser for ground maintenance (AMM 78-31-00/201).
- Lock the EAI valve in the closed position.
 - Remove the locking screw from the normal position.
 - Set the crank to the locked (closed) position.
 - Install the locking screw through the crank into the locked position.
 - Position associated ANTI-ICE Panel (P5) Engine Anti-Ice selector OFF.
- Close the associated circuit breaker(s).
- Activate the thrust reverser (AMM 78-31-00/201).

For GE:

- Activate the leading edge slats (AMM 27-81-00/201).
- Activate the associated EAI valve controller by closing the associated circuit breaker.
- Position the associated ANTI-ICE Panel (P5) Engine Anti-Ice selector OFF.

**30-21-01 Engine Anti-Ice Systems****30-21-01-01 PW and RR****30-21-01-01A OAT 38 Degrees C or Below, With Deactivation Kit Installed**

Interval	Installed	Required	Procedure
C	2	1	(M) (O)

For operations at ambient temperature 38 degrees C or below, one may be inoperative provided:

- Valve is deactivated open using deactivation kit.
- Engine fan case overheat detection system operates normally.
- Appropriate performance adjustments are applied.

MAINTENANCE (M)

Deactivate the associated EAI valve by installing the EAI valve lock-open kit, P/N K28703 (AMM 30-00-00/901).

- Remove inoperative EAI valve (AMM 30-21-02/401).
- Modify valve with lock-open kit (AMM 30-00-00/901).
- Re-install modified valve (AMM 30-21-02/401).

For PW:

- Visually inspect the area near the Cowl Thermal Anti-Ice duct for signs of leaks and the fan cowl in general for flammable fluid leaks. Dispatch is not allowed if any of the duct clamps or supports are not secure or any air leaks from the EAI duct are present, or if significant flammable fluid leakage is present.

OPERATIONS (O)

- Reduce performance limited weights by the appropriate adjustments.

Performance Limited Weights	Takeoff & Landing	Enroute Climb
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For -200 or -200ER/PW4074 thru PW4084:

All	19,600 lb (8,891 kg)	19,400 lb (8,800 kg)
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For -200 or -200ER/PW4090:

thru 632,500 lb (286,902 kg)	12,200 lb (5,534 kg)	13,800 lb (6,260 kg)
over 632,500 lb (286,902 kg)	17,300 lb (7,847 kg)	13,800 lb (6,260 kg)

For -200 or -200ER/RR875 thru RR892:

thru 632,500 lb (286,902 kg)	20,100 lb (9,117 kg)	21,600 lb (9,798 kg)
over 632,500 lb (286,902 kg)	26,500 lb (12,020 kg)	21,600 lb (9,798 kg)

For -200 or -200ER/RR895:

All	21,700 lb (9,843 kg)	23,000 lb (10,433 kg)
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For -300/PW4090:

All	12,400 lb (5,625 kg)	13,000 lb (5,897 kg)
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For -300/RR892:

All	22,000 lb (9,979 kg)	23,500 lb (10,659 kg)
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NOTE: The enroute climb weight penalties listed above are based on single engine operating speeds that approximate maximum lift-to-drag ratio speed. To account for the difference in level off altitude when operating at other speeds, multiply the enroute climb weight penalty listed above by the appropriate factor listed on page 3.00-01-00.2 in the ENROUTE DIVERSION SPEED EFFECTS paragraph.

For PW:

2. Increase flight planning fuel by 1.0% for an inoperative EAI Valve or EAI Pressure Sensor, or by 1.3% for an inoperative EAI Card.

For RR:

3. Increase flight planning fuel by 0.5% for an inoperative EAI Valve or EAI Pressure Sensor, or by 0.8% for an inoperative EAI Card.
4. For ETOPS flight planning:

For PW:

- A. Increase the ETOPS engine inoperative Critical Fuel Reserves by 1.6%.

For RR:

- B. Increase the ETOPS engine inoperative Critical Fuel Reserves by 0.9%.
5. Position ENGINE ANTI-ICE selector(s) as follows:
 - A. For EAI Valve or EAI Pressure Sensor inoperative, position associated EAI selector ON.

NOTE 1: When an EAI selector is positioned to ON and the temperature is greater than 10 degrees C, the ANTI-ICE ON advisory message will be displayed. Do not accomplish the ANTI-ICE ON Non-Normal Checklist.

NOTE 2: Inflight, the ENG IDLE DISAGREE advisory message may be displayed when engine thrust is reduced to idle.

- B. For EAI Card inoperative, position both EAI selectors ON.

NOTE 1: When an EAI selector is positioned to ON and the temperature is greater than 10 degrees C, the ANTI-ICE ON advisory message will be displayed. Do not accomplish the ANTI-ICE ON Non-Normal Checklist.

NOTE 2: The associated ANTI-ICE ENG L or R advisory message will be displayed, the EAI indication will not be displayed on EICAS, and the associated EAI valve will be displayed as invalid (low intensity white) and the green flow bar will not be displayed on the Air synoptic.

6. For the inoperative EAI valve, if the ANTI-ICE LEAK ENG L or R caution message is subsequently displayed:

For PW:

- A. Do not accomplish the two minute delay associated with the ANTI-ICE LEAK ENG L or R Non-Normal Checklist. Complete the remainder of the checklist as soon as practical.

For RR:

- B. Do not accomplish the initial one minute delay associated with the ANTI-ICE LEAK ENG L or R Non-Normal Checklist. Complete the remainder of the checklist as soon as practical.

30-21-01 Engine Anti-Ice Systems

30-21-01-01 PW and RR

30-21-01-01B OAT 38 Degrees C or Below, Without Deactivation Kit Installed

Interval	Installed	Required	Procedure
C	2	1	(M) (O)

For operations at ambient temperature 38 degrees C or below, one may be inoperative provided:

- Valve is electrically disconnected and verified open.
- Engine fan case overheat detection system operates normally.
- Associated EAI pressure sensor system operates normally.
- Associated EAI card operates normally.

- e. Appropriate performance adjustments are applied.
-

MAINTENANCE (M)

For EAI VALVE L or R status message displayed, deactivate the EAI valve by disconnecting the electrical connector and verify EAI valve is open when the associated engine is running (AMM 30-00-00/901).

1. Deactivate the associated EAI valve controller.
 - A. For the right EAI valve, open P210 panel EAI CTRL R circuit breaker.
 - B. For the left EAI valve, open P310 panel EAI CTRL L circuit breaker.
2. Deactivate the leading edge slats in the retracted position (AMM 27-81-00/201).
3. Deactivate the associated thrust reverser for ground maintenance (AMM 78-31-00/201).
4. Deactivate the EAI valve in the open position.
 - A. On the associated engine, disconnect, cap and stow the torque motor electrical connector DM30211 or DM30211A from the EAI valve controller, M30211.
 - B. Install a protective cover over the electrical connector on the valve controller.

For PW:

- C. Visually inspect the area near the Cowl Thermal Anti-Ice duct for signs of leaks and the fan cowl in general for flammable fluid leaks. Dispatch is not allowed if any of the duct clamps or supports are not secure or any air leaks from the EAI duct are present, or if significant flammable fluid leakage is present.
5. Activate the thrust reverser (AMM 78-31-00/201).
6. Activate the associated EAI valve controller by closing the associated circuit breaker.
7. Verify the EAI valve is open when the associated engine is running.
 - A. Position associated ANTI-ICE Panel (P5) Engine Anti-Ice selector OFF.
 - B. Start the associated engine (71-00-00/201).
 - C. Confirm the EAI indication (green EAI) is displayed above the N1 indication on the EICAS display for the associated engine (indicates the EAI valve is open).

NOTE: If the EAI indication is not displayed, the valve may be stuck closed and a thrust setting slightly above minimum idle may open the valve.

- D. Shutdown the associated engine (71-00-00/201).

OPERATIONS (O)

1. Reduce performance limited weights by the appropriate adjustments.



Performance Limited Weights	Takeoff & Landing	Enroute Climb
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For -200 or -200ER/PW4074 thru PW4084:

All	19,600 lb (8,891 kg)	19,400 lb (8,800 kg)
-----	----------------------	----------------------

For -200 or -200ER/PW4090:

thru 632,500 lb (286,902 kg)	12,200 lb (5,534 kg)	13,800 lb (6,260 kg)
over 632,500 lb (286,902 kg)	17,300 lb (7,847 kg)	13,800 lb (6,260 kg)

For -200 or -200ER/RR875 thru RR892:

thru 632,500 lb (286,902 kg)	20,100 lb (9,117 kg)	21,600 lb (9,798 kg)
over 632,500 lb (286,902 kg)	26,500 lb (12,020 kg)	21,600 lb (9,798 kg)

For -200 or -200ER/RR895:

All	21,700 lb (9,843 kg)	23,000 lb (10,433 kg)
-----	----------------------	-----------------------

For -300/PW4090:

All	12,400 lb (5,625 kg)	13,000 lb (5,897 kg)
-----	----------------------	----------------------

For -300/RR892:

All	22,000 lb (9,979 kg)	23,500 lb (10,659 kg)
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NOTE: The enroute climb weight penalties listed above are based on single engine operating speeds that approximate maximum lift-to-drag ratio speed. To account for the difference in level off altitude when operating at other speeds, multiply the enroute climb weight penalty listed above by the appropriate factor listed on page 3.00-01-00.2 in the ENROUTE DIVERSION SPEED EFFECTS paragraph.

For PW:

2. Increase flight planning fuel by 1.0%.

For RR:

3. Increase flight planning fuel by 0.5%.
4. For ETOPS flight planning:

For PW:

- A. Increase the ETOPS engine inoperative Critical Fuel Reserves by 1.6%.

For RR:

- B. Increase the ETOPS engine inoperative Critical Fuel Reserves by 0.9%.
5. Verify the EAI valve is open when the associated engine is running before each departure.
 - A. Position associated ANTI-ICE Panel (P5) Engine Anti-Ice selector OFF.
 - B. After the associated engine has been started, confirm the associated EAI indication (green EAI) is displayed above the N1 indication.

NOTE 1: If the EAI indication is not displayed, the valve may be stuck closed and a thrust setting slightly above minimum idle may open the valve.

NOTE 2: On the Air synoptic, the EAI valve will indicate failed open because the EAI selector is OFF and the green flow bar will be displayed.

6. Position associated ENGINE ANTI-ICE selector ON.

NOTE 1: When an EAI selector is positioned to ON and the temperature is greater than 10 degrees C, the ANTI-ICE ON advisory message will be displayed. Do not accomplish the ANTI-ICE ON Non-Normal Checklist.

NOTE 2: Inflight, the ENG IDLE DISAGREE advisory message may be displayed when engine thrust is reduced to idle.

NOTE 3: The ANTI-ICE ENG advisory message may be displayed during ground operations prior to takeoff. Flight crew should accomplish the associated non-normal checklist. If ANTI-ICE ENG clears, takeoff is allowed.

7. For the inoperative EAI valve, if the ANTI-ICE LEAK ENG L or R caution message is subsequently displayed:

For PW:

- A. Do not accomplish the two minute delay associated with the ANTI-ICE LEAK ENG L or R Non-Normal Checklist. Complete the remainder of the checklist as soon as practical.

**For RR:**

- B. Do not accomplish the initial one minute delay associated with the ANTI-ICE LEAK ENG L or R Non-Normal Checklist. Complete the remainder of the checklist as soon as practical.

30-21-01 Engine Anti-Ice Systems**30-21-01-01 PW and RR****30-21-01-01C OAT Above 38 Degrees C, With Deactivation Kit Installed**

Interval	Installed	Required	Procedure
A	2	1	(M) (O)

For operations at ambient temperature above 38 degrees C, one may be inoperative provided:

- Valve is deactivated open using deactivation kit.
- Engine fan case overheat detection system operates normally.
- Appropriate performance adjustments are applied.
- Repairs are made within 3 flight days.

MAINTENANCE (M)

Deactivate the associated EAI valve by installing the EAI valve lock-open kit, P/N K28703 (AMM 30-00-00/901).

- Remove inoperative EAI valve (AMM 30-21-02/401).
- Modify valve with lock-open kit (AMM 30-00-00/901).
- Re-install modified valve (AMM 30-21-02/401).

For PW:

- Visually inspect the area near the Cowl Thermal Anti-Ice duct for signs of leaks and the fan cowl in general for flammable fluid leaks. Dispatch is not allowed if any of the duct clamps or supports are not secure or any air leaks from the EAI duct are present, or if significant flammable fluid leakage is present.

OPERATIONS (O)

- Reduce performance limited weights by the appropriate adjustments.

Performance Limited Weights	Takeoff & Landing	Enroute Climb
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For -200 or -200ER/PW4074 thru PW4084:

All	19,600 lb (8,891 kg)	19,400 lb (8,800 kg)
-----	----------------------	----------------------



For -200 or -200ER/PW4090:

thru 632,500 lb (286,902 kg)	12,200 lb (5,534 kg)	13,800 lb (6,260 kg)
over 632,500 lb (286,902 kg)	17,300 lb (7,847 kg)	13,800 lb (6,260 kg)

For -200 or -200ER/RR875 thru RR892:

thru 632,500 lb (286,902 kg)	20,100 lb (9,117 kg)	21,600 lb (9,798 kg)
over 632,500 lb (286,902 kg)	26,500 lb (12,020 kg)	21,600 lb (9,798 kg)

For -200 or -200ER/RR895:

All	21,700 lb (9,843 kg)	23,000 lb (10,433 kg)
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For -300/PW4090:

All	12,400 lb (5,625 kg)	13,000 lb (5,897 kg)
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For -300/RR892:

All	22,000 lb (9,979 kg)	23,500 lb (10,659 kg)
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NOTE: The enroute climb weight penalties listed above are based on single engine operating speeds that approximate maximum lift-to-drag ratio speed. To account for the difference in level off altitude when operating at other speeds, multiply the enroute climb weight penalty listed above by the appropriate factor listed on page 3.00-01-00.2 in the ENROUTE DIVERSION SPEED EFFECTS paragraph.

For PW:

2. Increase flight planning fuel by 1.0% for an inoperative EAI Valve or EAI Pressure Sensor, or by 1.3% for an inoperative EAI Card.

For RR:

3. Increase flight planning fuel by 0.5% for an inoperative EAI Valve or EAI Pressure Sensor, or by 0.8% for an inoperative EAI Card.
4. For ETOPS flight planning:

For PW:

- A. Increase the ETOPS engine inoperative Critical Fuel Reserves by 1.6%.

**For RR:**

- B. Increase the ETOPS engine inoperative Critical Fuel Reserves by 0.9%.
5. Position ENGINE ANTI-ICE selector(s) as follows:
- A. For EAI Valve or EAI Pressure Sensor inoperative, position associated EAI selector ON.
- NOTE 1: When an EAI selector is positioned to ON and the temperature is greater than 10 degrees C, the ANTI-ICE ON advisory message will be displayed. Do not accomplish the ANTI-ICE ON Non-Normal Checklist.
- NOTE 2: Inflight, the ENG IDLE DISAGREE advisory message may be displayed when engine thrust is reduced to idle.
- B. For EAI Card inoperative, position both EAI selectors ON.
- NOTE 1: When an EAI selector is positioned to ON and the temperature is greater than 10 degrees C, the ANTI-ICE ON advisory message will be displayed. Do not accomplish the ANTI-ICE ON Non-Normal Checklist.
- NOTE 2: The associated ANTI-ICE ENG L or R advisory message will be displayed, the EAI indication will not be displayed on EICAS, and the associated EAI valve will be displayed as invalid (low intensity white) and the green flow bar will not be displayed on the Air synoptic.
6. For the inoperative EAI valve, if the ANTI-ICE LEAK ENG L or R caution message is subsequently displayed:

For PW:

- A. Do not accomplish the two minute delay associated with the ANTI-ICE LEAK ENG L or R Non-Normal Checklist. Complete the remainder of the checklist as soon as practical.

For RR:

- B. Do not accomplish the initial one minute delay associated with the ANTI-ICE LEAK ENG L or R Non-Normal Checklist. Complete the remainder of the checklist as soon as practical.

30-21-01 Engine Anti-Ice Systems**30-21-01-01 PW and RR****30-21-01-01D OAT Above 38 Degrees C, Without Deactivation Kit Installed**

Interval	Installed	Required	Procedure
A	2	1	(M) (O)

For operations at ambient temperature above 38 degrees C, one may be inoperative provided:

- a. Valve is electrically disconnected and verified open.
- b. Engine fan case overheat detection system operates normally.
- c. Associated EAI pressure sensor system operates normally.
- d. Associated EAI card operates normally.
- e. Appropriate performance adjustments are applied.
- f. Repairs are made within 3 flight days.

MAINTENANCE (M)

For EAI VALVE L or R status message displayed, deactivate the EAI valve by disconnecting the electrical connector and verify EAI valve is open when the associated engine is running (AMM 30-00-00/901).

1. Deactivate the associated EAI valve controller.
 - A. For the right EAI valve, open P210 panel EAI CTRL R circuit breaker.
 - B. For the left EAI valve, open P310 panel EAI CTRL L circuit breaker.
2. Deactivate the leading edge slats in the retracted position (AMM 27-81-00/201).
3. Deactivate the associated thrust reverser for ground maintenance (AMM 78-31-00/201).
4. Deactivate the EAI valve in the open position.
 - A. On the associated engine, disconnect, cap and stow the torque motor electrical connector DM30211 or DM30211A from the EAI valve controller, M30211.
 - B. Install a protective cover over the electrical connector on the valve controller.

For PW:

- C. Visually inspect the area near the Cowl Thermal Anti-Ice duct for signs of leaks and the fan cowl in general for flammable fluid leaks. Dispatch is not allowed if any of the duct clamps or supports are not secure or any air leaks from the EAI duct are present, or if significant flammable fluid leakage is present.
5. Activate the thrust reverser (AMM 78-31-00/201).
6. Activate the associated EAI valve controller by closing the associated circuit breaker.
7. Verify the EAI valve is open when the associated engine is running.
 - A. Position associated ANTI-ICE Panel (P5) Engine Anti-Ice selector OFF.
 - B. Start the associated engine (71-00-00/201).
 - C. Confirm the EAI indication (green EAI) is displayed above the N1 indication on the EICAS display for the associated engine (indicates the EAI valve is open).



NOTE: If the EAI indication is not displayed, the valve may be stuck closed and a thrust setting slightly above minimum idle may open the valve.

D. Shutdown the associated engine (71-00-00/201).

OPERATIONS (O)

1. Reduce performance limited weights by the appropriate adjustments.

Performance Limited Weights	Takeoff & Landing	Enroute Climb
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For -200 or -200ER/PW4074 thru PW4084:

All	19,600 lb (8,891 kg)	19,400 lb (8,800 kg)
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For -200 or -200ER/PW4090:

thru 632,500 lb (286,902 kg)	12,200 lb (5,534 kg)	13,800 lb (6,260 kg)
over 632,500 lb (286,902 kg)	17,300 lb (7,847 kg)	13,800 lb (6,260 kg)

For -200 or -200ER/RR875 thru RR892:

thru 632,500 lb (286,902 kg)	20,100 lb (9,117 kg)	21,600 lb (9,798 kg)
over 632,500 lb (286,902 kg)	26,500 lb (12,020 kg)	21,600 lb (9,798 kg)

For -200 or -200ER/RR895:

All	21,700 lb (9,843 kg)	23,000 lb (10,433 kg)
-----	----------------------	-----------------------

For -300/PW4090:

All	12,400 lb (5,625 kg)	13,000 lb (5,897 kg)
-----	----------------------	----------------------

For -300/RR892:

All	22,000 lb (9,979 kg)	23,500 lb (10,659 kg)
-----	----------------------	-----------------------

NOTE: The enroute climb weight penalties listed above are based on single engine operating speeds that approximate maximum lift-to-drag ratio speed. To account for the difference in level off altitude when operating at other speeds, multiply the enroute climb weight penalty listed above by the appropriate factor listed on page 3.00-01-00.2 in the ENROUTE DIVERSION SPEED EFFECTS paragraph.

For PW:

2. Increase flight planning fuel by 1.0%.

For RR:

3. Increase flight planning fuel by 0.5%.
4. For ETOPS flight planning:

For PW:

- A. Increase the ETOPS engine inoperative Critical Fuel Reserves by 1.6%.

For RR:

- B. Increase the ETOPS engine inoperative Critical Fuel Reserves by 0.9%.
5. Verify the EAI valve is open when the associated engine is running before each departure.
 - A. Position associated ANTI-ICE Panel (P5) Engine Anti-Ice selector OFF.
 - B. After the associated engine has been started, confirm the associated EAI indication (green EAI) is displayed above the N1 indication.

NOTE 1: If the EAI indication is not displayed, the valve may be stuck closed and a thrust setting slightly above minimum idle may open the valve.

NOTE 2: On the Air synoptic, the EAI valve will indicate failed open because the EAI selector is OFF and the green flow bar will be displayed.

6. Position associated ENGINE ANTI-ICE selector ON.

NOTE 1: When an EAI selector is positioned to ON and the temperature is greater than 10 degrees C, the ANTI-ICE ON advisory message will be displayed. Do not accomplish the ANTI-ICE ON Non-Normal Checklist.

NOTE 2: Inflight, the ENG IDLE DISAGREE advisory message may be displayed when engine thrust is reduced to idle.

NOTE 3: The ANTI-ICE ENG advisory message may be displayed during ground operations prior to takeoff. Flight crew should accomplish the associated non-normal checklist. If ANTI-ICE ENG clears, takeoff is allowed.



7. For the inoperative EAI valve, if the ANTI-ICE LEAK ENG L or R caution message is subsequently displayed:

For PW:

- A. Do not accomplish the two minute delay associated with the ANTI-ICE LEAK ENG L or R Non-Normal Checklist. Complete the remainder of the checklist as soon as practical.

For RR:

- B. Do not accomplish the initial one minute delay associated with the ANTI-ICE LEAK ENG L or R Non-Normal Checklist. Complete the remainder of the checklist as soon as practical.

30-21-01 Engine Anti-Ice Systems

30-21-01-02 GE

30-21-01-02A OAT 38 Degrees C or Below, With Deactivation Kit Installed

Interval	Installed	Required	Procedure
C	2	1	(M) (O)

For operations at ambient temperature 38 degrees C or below, one may be inoperative provided:

- Valve is deactivated open using deactivation kit.
- Engine fan case overheat detection system operates normally.
- Turbine case cooling air flow systems operate normally.
- Appropriate performance adjustments are applied.

Except for GE90-100 Series:

- EECs must be operated in the normal mode.

MAINTENANCE (M)

Deactivate the associated EAI valve by installing the EAI valve lock-open kit, P/N K28703 (AMM 30-00-00/901).

- Remove inoperative EAI valve (AMM 30-21-02/401).
- Modify valve with lock-open kit (AMM 30-00-00/901).
- Re-install modified valve (AMM 30-21-02/401).
- Visually inspect the area near the Cowl Thermal Anti-Ice duct for signs of leaks and the fan cowl in general for flammable fluid leaks. Dispatch is not allowed if any of the duct clamps or supports are not secure or any air leaks from the EAI duct are present, or if significant flammable fluid leakage is present.

OPERATIONS (O)

- Reduce performance limited weights by the appropriate adjustments.



Performance Limited Weights	Takeoff & Landing	Enroute Climb
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For -200 or -200ER/GE90-76B thru -90B:

thru 632,500 lb (286,902 kg)	9,100 lb (4,128 kg)	11,800 lb (5,352 kg)
over 632,500 lb (286,902 kg)	13,200 lb (5,987 kg)	11,800 lb (5,352 kg)

For -200 or -200ER/GE90-94B:

thru 632,500 lb (286,902 kg)	9,100 lb (4,128 kg)	13,900 lb (6,305 kg)
over 632,500 lb (286,902 kg)	14,300 lb (6,486 kg)	13,900 lb (6,305 kg)

For -200LR or 777F/GE90-100 Series:

All	13,300 lb (6,033 kg)	12,100 lb (5,489 kg)
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For -300ER/GE90-115B:

All	10,400 lb (4,717 kg)	11,800 lb (5,353 kg)
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NOTE: The enroute climb weight penalties listed above are based on single engine operating speeds that approximate maximum lift-to-drag ratio speed. To account for the difference in level off altitude when operating at other speeds, multiply the enroute climb weight penalty listed above by the appropriate factor listed on page 3.00-01-00.2 in the ENROUTE DIVERSION SPEED EFFECTS paragraph.

2. Increase flight planning fuel by 1.0% for an inoperative EAI Valve or EAI Pressure Sensor, or by 1.3% for an inoperative EAI Card.
3. For ETOPS flight planning, increase engine inoperative Critical Fuel Reserves by 1.8%.
4. Position ENGINE ANTI-ICE selector(s) as follows:
 - A. For EAI Valve or EAI Pressure Sensor inoperative, position associated EAI selector ON.

NOTE 1: When an EAI selector is positioned to ON and the temperature is greater than 10 degrees C, the ANTI-ICE ON advisory message will be displayed. Do not accomplish the ANTI-ICE ON Non-Normal Checklist.

NOTE 2: Inflight, the ENG IDLE DISAGREE advisory message may be displayed when engine thrust is reduced to idle.

- B. For EAI Card inoperative, position both EAI selectors ON.

NOTE 1: When an EAI selector is positioned to ON and the temperature is greater than 10 degrees C, the ANTI-ICE ON advisory message will be displayed. Do not accomplish the ANTI-ICE ON Non-Normal Checklist.

NOTE 2: The associated ANTI-ICE ENG L or R advisory message will be displayed, the EAI indication will not be displayed on EICAS, and the associated EAI valve will be displayed as invalid (low intensity white) and the green flow bar will not be displayed on the Air synoptic.

5. For the inoperative EAI valve, if the ANTI-ICE LEAK ENG L or R caution message is subsequently displayed, do not accomplish the two minute delay associated with the ANTI-ICE LEAK ENG L or R Non-Normal Checklist. Complete the remainder of the checklist as soon as practical.

30-21-01 Engine Anti-Ice Systems

30-21-01-02 GE

30-21-01-02B OAT 38 Degrees C or Below, Without Deactivation Kit Installed

Interval	Installed	Required	Procedure
C	2	1	(M) (O)

For operations at ambient temperature 38 degrees C or below, one may be inoperative provided:

- Valve is electrically disconnected and verified open.
- Engine fan case overheat detection system operates normally.
- Associated EAI pressure sensor system operates normally.
- Associated EAI card operates normally.
- Turbine case cooling air flow systems operate normally.
- Appropriate performance adjustments are applied.

[Except for GE90-100 Series:](#)

- EECs must be operated in the normal mode.

MAINTENANCE (M)

For EAI VALVE L or R status message displayed, deactivate the EAI valve by disconnecting the electrical connector and verify EAI valve is open when the associated engine is running (AMM 30-00-00/901).

- Deactivate the associated EAI valve controller.
 - For the right EAI valve, open P210 panel EAI CTRL R circuit breaker.

- B. For the left EAI valve, open P310 panel EAI CTRL L circuit breaker.
2. Deactivate the leading edge slats in the retracted position (AMM 27-81-00/201).
3. Deactivate the associated thrust reverser for ground maintenance (AMM 78-31-00/201).
4. Deactivate the EAI valve in the open position.
 - A. On the associated engine, disconnect, cap and stow the torque motor electrical connector DM30211 or DM30211A from the EAI valve controller, M30211.
 - B. Install a protective cover over the electrical connector on the valve controller.
 - C. Visually inspect the area near the Cowl Thermal Anti-Ice duct for signs of leaks and the fan cowl in general for flammable fluid leaks. Dispatch is not allowed if any of the duct clamps or supports are not secure or any air leaks from the EAI duct are present, or if significant flammable fluid leakage is present.
5. Activate the thrust reverser (AMM 78-31-00/201).
6. Activate the leading edge slats (AMM 27-81-00/201).
7. Activate the associated EAI valve controller by closing the associated circuit breaker.
8. Verify the EAI valve is open when the associated engine is running.
 - A. Position associated ANTI-ICE Panel (P5) Engine Anti-Ice selector OFF.
 - B. Start the associated engine (71-00-00/201).
 - C. Confirm the EAI indication (green EAI) is displayed above the N1 indication on the EICAS display for the associated engine (indicates the EAI valve is open).

NOTE: If the EAI indication is not displayed, the valve may be stuck closed and a thrust setting slightly above minimum idle may open the valve.
 - D. Shutdown the associated engine (71-00-00/201).

OPERATIONS (O)

1. Reduce performance limited weights by the appropriate adjustments.

Performance Limited Weights	Takeoff & Landing	Enroute Climb
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For -200 or -200ER/GE90-76B thru -90B:

thru 632,500 lb (286,902 kg)	9,100 lb (4,128 kg)	11,800 lb (5,352 kg)
over 632,500 lb (286,902 kg)	13,200 lb (5,987 kg)	11,800 lb (5,352 kg)

For -200 or -200ER/GE90-94B:

thru 632,500 lb (286,902 kg)	9,100 lb (4,128 kg)	13,900 lb (6,305 kg)
over 632,500 lb (286,902 kg)	14,300 lb (6,486 kg)	13,900 lb (6,305 kg)

For -200LR or 777F/GE90-100 Series:

All	13,300 lb (6,033 kg)	12,100 lb (5,489 kg)
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For -300ER/GE90-115B:

All	10,400 lb (4,717 kg)	11,800 lb (5,353 kg)
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NOTE: The enroute climb weight penalties listed above are based on single engine operating speeds that approximate maximum lift-to-drag ratio speed. To account for the difference in level off altitude when operating at other speeds, multiply the enroute climb weight penalty listed above by the appropriate factor listed on page 3.00-01-00.2 in the ENROUTE DIVERSION SPEED EFFECTS paragraph.

2. Increase flight planning fuel by 1.0%.
3. For ETOPS flight planning, increase engine inoperative Critical Fuel Reserves by 1.8%.
4. Verify the EAI valve is open when the associated engine is running before each departure.
 - A. Position associated ANTI-ICE Panel (P5) Engine Anti-Ice selector OFF.
 - B. After the associated engine has been started, confirm the associated EAI indication (green EAI) is displayed above the N1 indication.

NOTE 1: If the EAI indication is not displayed, the valve may be stuck closed and a thrust setting slightly above minimum idle may open the valve.

NOTE 2: On the Air synoptic, the EAI valve will indicate failed open because the EAI selector is OFF and the green flow bar will be displayed.

5. Position associated ENGINE ANTI-ICE selector ON.

NOTE 1: When an EAI selector is positioned to ON and the temperature is greater than 10 degrees C, the ANTI-ICE ON advisory message will be displayed. Do not accomplish the ANTI-ICE ON Non-Normal Checklist.

NOTE 2: Inflight, the ENG IDLE DISAGREE advisory message may be displayed when engine thrust is reduced to idle.

NOTE 3: The ANTI-ICE ENG advisory message may be displayed during ground operations prior to takeoff. Flight crew should accomplish the associated non-normal checklist. If ANTI-ICE ENG clears, takeoff is allowed.

6. For the inoperative EAI valve, if the ANTI-ICE LEAK ENG L or R caution message is subsequently displayed, do not accomplish the two minute delay associated with the ANTI-ICE LEAK ENG L or R Non-Normal Checklist. Complete the remainder of the checklist as soon as practical.

30-21-01 Engine Anti-Ice Systems

30-21-01-02 GE

30-21-01-02C OAT Above 38 Degrees C, With Deactivation Kit Installed

Interval	Installed	Required	Procedure
A	2	1	(M) (O)

For operations at ambient temperature above 38 degrees C, one may be inoperative provided:

- a. Valve is deactivated open using deactivation kit.
- b. Engine fan case overheat detection system operates normally.
- c. Turbine case cooling air flow systems operate normally.
- d. Appropriate performance adjustments are applied.
- e. Repairs are made within 3 flight days.

Except for GE90-100 Series:

- f. EECs must be operated in the normal mode.

MAINTENANCE (M)

Deactivate the associated EAI valve by installing the EAI valve lock-open kit, P/N K28703 (AMM 30-00-00/901).

1. Remove inoperative EAI valve (AMM 30-21-02/401).
2. Modify valve with lock-open kit (AMM 30-00-00/901).
3. Re-install modified valve (AMM 30-21-02/401).
4. Visually inspect the area near the Cowl Thermal Anti-Ice duct for signs of leaks and the fan cowl in general for flammable fluid leaks. Dispatch is not allowed if any of the duct clamps or supports are not secure or any air leaks from the EAI duct are present, or if significant flammable fluid leakage is present.

OPERATIONS (O)

1. Reduce performance limited weights by the appropriate adjustments.



Performance Limited Weights	Takeoff & Landing	Enroute Climb
--------------------------------	----------------------	------------------

For -200 or -200ER/GE90-76B thru -90B:

thru 632,500 lb (286,902 kg)	9,100 lb (4,128 kg)	11,800 lb (5,352 kg)
over 632,500 lb (286,902 kg)	13,200 lb (5,987 kg)	11,800 lb (5,352 kg)

For -200 or -200ER/GE90-94B:

thru 632,500 lb (286,902 kg)	9,100 lb (4,128 kg)	13,900 lb (6,305 kg)
over 632,500 lb (286,902 kg)	14,300 lb (6,486 kg)	13,900 lb (6,305 kg)

For -200LR or 777F/GE90-100 Series:

All	13,300 lb (6,033 kg)	12,100 lb (5,489 kg)
-----	----------------------	----------------------

For -300ER/GE90-115B:

All	10,400 lb (4,717 kg)	11,800 lb (5,353 kg)
-----	----------------------	----------------------

NOTE: The enroute climb weight penalties listed above are based on single engine operating speeds that approximate maximum lift-to-drag ratio speed. To account for the difference in level off altitude when operating at other speeds, multiply the enroute climb weight penalty listed above by the appropriate factor listed on page 3.00-01-00.2 in the ENROUTE DIVERSION SPEED EFFECTS paragraph.

2. Increase flight planning fuel by 1.0% for an inoperative EAI Valve or EAI Pressure Sensor, or by 1.3% for an inoperative EAI Card.
3. For ETOPS flight planning, increase engine inoperative Critical Fuel Reserves by 1.8%.
4. Position ENGINE ANTI-ICE selector(s) as follows:
 - A. For EAI Valve or EAI Pressure Sensor inoperative, position associated EAI selector ON.

NOTE 1: When an EAI selector is positioned to ON and the temperature is greater than 10 degrees C, the ANTI-ICE ON advisory message will be displayed. Do not accomplish the ANTI-ICE ON Non-Normal Checklist.

NOTE 2: Inflight, the ENG IDLE DISAGREE advisory message may be displayed when engine thrust is reduced to idle.

- B. For EAI Card inoperative, position both EAI selectors ON.

NOTE 1: When an EAI selector is positioned to ON and the temperature is greater than 10 degrees C, the ANTI-ICE ON advisory message will be displayed. Do not accomplish the ANTI-ICE ON Non-Normal Checklist.

NOTE 2: The associated ANTI-ICE ENG L or R advisory message will be displayed, the EAI indication will not be displayed on EICAS, and the associated EAI valve will be displayed as invalid (low intensity white) and the green flow bar will not be displayed on the Air synoptic.

5. For the inoperative EAI valve, if the ANTI-ICE LEAK ENG L or R caution message is subsequently displayed, do not accomplish the two minute delay associated with the ANTI-ICE LEAK ENG L or R Non-Normal Checklist. Complete the remainder of the checklist as soon as practical.

30-21-01 Engine Anti-Ice Systems

30-21-01-02 GE

30-21-01-02D OAT Above 38 Degrees C, Without Deactivation Kit Installed

Interval	Installed	Required	Procedure
A	2	1	(M) (O)

For operations at ambient temperature above 38 degrees C, one may be inoperative provided:

- Valve is electrically disconnected and verified open.
- Engine fan case overheat detection system operates normally.
- Associated EAI pressure sensor system operates normally.
- Associated EAI card operates normally.
- Turbine case cooling air flow systems operate normally.
- Appropriate performance adjustments are applied.
- Repairs are made within 3 flight days.

Except for GE90-100 Series:

- EECs must be operated in the normal mode.

MAINTENANCE (M)

For EAI VALVE L or R status message displayed, deactivate the EAI valve by disconnecting the electrical connector and verify EAI valve is open when the associated engine is running (AMM 30-00-00/901).

1. Deactivate the associated EAI valve controller.
 - A. For the right EAI valve, open P210 panel EAI CTRL R circuit breaker.
 - B. For the left EAI valve, open P310 panel EAI CTRL L circuit breaker.
2. Deactivate the leading edge slats in the retracted position (AMM 27-81-00/201).
3. Deactivate the associated thrust reverser for ground maintenance (AMM 78-31-00/201).
4. Deactivate the EAI valve in the open position.
 - A. On the associated engine, disconnect, cap and stow the torque motor electrical connector DM30211 or DM30211A from the EAI valve controller, M30211.
 - B. Install a protective cover over the electrical connector on the valve controller.
 - C. Visually inspect the area near the Cowl Thermal Anti-Ice duct for signs of leaks and the fan cowl in general for flammable fluid leaks. Dispatch is not allowed if any of the duct clamps or supports are not secure or any air leaks from the EAI duct are present, or if significant flammable fluid leakage is present.
5. Activate the thrust reverser (AMM 78-31-00/201).
6. Activate the leading edge slats (AMM 27-81-00/201).
7. Activate the associated EAI valve controller by closing the associated circuit breaker.
8. Verify the EAI valve is open when the associated engine is running.
 - A. Position associated ANTI-ICE Panel (P5) Engine Anti-Ice selector OFF.
 - B. Start the associated engine (71-00-00/201).
 - C. Confirm the EAI indication (green EAI) is displayed above the N1 indication on the EICAS display for the associated engine (indicates the EAI valve is open).

NOTE: If the EAI indication is not displayed, the valve may be stuck closed and a thrust setting slightly above minimum idle may open the valve.

- D. Shutdown the associated engine (71-00-00/201).

OPERATIONS (O)

1. Reduce performance limited weights by the appropriate adjustments.

Performance Limited Weights	Takeoff & Landing	Enroute Climb
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For -200 or -200ER/GE90-76B thru -90B:

thru 632,500 lb (286,902 kg)	9,100 lb (4,128 kg)	11,800 lb (5,352 kg)
over 632,500 lb (286,902 kg)	13,200 lb (5,987 kg)	11,800 lb (5,352 kg)

For -200 or -200ER/GE90-94B:

thru 632,500 lb (286,902 kg)	9,100 lb (4,128 kg)	13,900 lb (6,305 kg)
over 632,500 lb (286,902 kg)	14,300 lb (6,486 kg)	13,900 lb (6,305 kg)

For -200LR or 777F/GE90-100 Series:

All	13,300 lb (6,033 kg)	12,100 lb (5,489 kg)
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For -300ER/GE90-115B:

All	10,400 lb (4,717 kg)	11,800 lb (5,353 kg)
-----	----------------------	----------------------

NOTE: The enroute climb weight penalties listed above are based on single engine operating speeds that approximate maximum lift-to-drag ratio speed. To account for the difference in level off altitude when operating at other speeds, multiply the enroute climb weight penalty listed above by the appropriate factor listed on page 3.00-01-00.2 in the ENROUTE DIVERSION SPEED EFFECTS paragraph.

2. Increase flight planning fuel by 1.0%.
3. For ETOPS flight planning, increase engine inoperative Critical Fuel Reserves by 1.8%.
4. Verify the EAI valve is open when the associated engine is running before each departure.
 - A. Position associated ANTI-ICE Panel (P5) Engine Anti-Ice selector OFF.
 - B. After the associated engine has been started, confirm the associated EAI indication (green EAI) is displayed above the N1 indication.

NOTE 1: If the EAI indication is not displayed, the valve may be stuck closed and a thrust setting slightly above minimum idle may open the valve.

NOTE 2: On the Air synoptic, the EAI valve will indicate failed open because the EAI selector is OFF and the green flow bar will be displayed.



5. Position associated ENGINE ANTI-ICE selector ON.

NOTE 1: When an EAI selector is positioned to ON and the temperature is greater than 10 degrees C, the ANTI-ICE ON advisory message will be displayed. Do not accomplish the ANTI-ICE ON Non-Normal Checklist.

NOTE 2: Inflight, the ENG IDLE DISAGREE advisory message may be displayed when engine thrust is reduced to idle.

NOTE 3: The ANTI-ICE ENG advisory message may be displayed during ground operations prior to takeoff. Flight crew should accomplish the associated non-normal checklist. If ANTI-ICE ENG clears, takeoff is allowed.

6. For the inoperative EAI valve, if the ANTI-ICE LEAK ENG L or R caution message is subsequently displayed, do not accomplish the two minute delay associated with the ANTI-ICE LEAK ENG L or R Non-Normal Checklist. Complete the remainder of the checklist as soon as practical.



777 FAA DDG Supplement Package

Supplement Number: 36-1

Issue Date: 21-DEC-2023

Section Number: 2 (MMEL)

Item Number: 30-81-01

Item Name: Ice Detectors

DDG Effectivity: 777 FAA DDG (D630W003-TBC), Revision 36

MMEL Effectivity: 777 FAA MMEL, Revision 23

Description: Added Operations (O) procedure steps.

Reason: Operations (O) procedure positions WAI and both EAI selectors OFF since Wing and Engine Anti-Ice are operated manually.

Disposition: For reference in the Operator's MEL.

Attachments: DDG page 2.30-81-01.1

**30-81-01 Ice Detectors**

Interval	Installed	Required	Procedure
C	2	0	(O)

May be inoperative provided engine and wing anti-ice systems are operated manually.

OPERATIONS (O)

If one or both ice detectors are inoperative, operate the Wing and Engine Anti-Ice systems manually.

1. Position WING ANTI-ICE selector and both ENGINE ANTI-ICE selectors OFF before departure.
2. For wing anti-ice required:
 - A. Position WING ANTI-ICE selector ON, then OFF as required.
3. For engine anti-ice required:
 - A. Position ENGINE ANTI-ICE selector(s) ON, then OFF as required.



777 FAA DDG Supplement Package

Supplement Number: 36-1

Issue Date: 21-DEC-2023

Section Number: 2 (MMEL)

Item Number: 31-61-07

Item Name: Electronic Checklist (ECL) System

DDG Effectivity: 777 FAA DDG (D630W003-TBC), Revision 36

MMEL Effectivity: 777 FAA MMEL, Revision 23

Description: Added a Maintenance note.

Reason: Note clarifies no maintenance action is required for display of CHECKLIST DISABLED or CHECKLIST NOT AVAILABLE messages on the MFD.

Disposition: For reference in the Operator's MEL.

Attachments: DDG page 2.31-61-07.1

31-61-07 Electronic Checklist (ECL) System

Interval	Installed	Required	Procedure
C	1	0	(M) (O)

May be inoperative provided electronic checklist is deactivated and established suitable checklist procedures are used.

MAINTENANCE (M)

Deactivate the electronic checklist (AMM 31-00-00/901).

1. Display the Maintenance Task Maintenance page.
2. Toggle the Checklist Function ENABLE/DISABLE button to DISABLE.

NOTE: No maintenance action is required if the CHECKLIST DISABLED or CHECKLIST NOT AVAILABLE message is displayed on an MFD.

OPERATIONS (O)

Use established suitable checklists.

NOTE: The rectangular icon symbol that precedes some alert messages will not be displayed.

31-61-07 Electronic Checklist (ECL) System

31-61-07-01 ECL Closed Loop Switch Indications

Interval	Installed	Required	Procedure
C	-	0	(O)

May be inoperative provided ECL line item override procedures are used when required to complete checklists.

OPERATIONS (O)

Use the Line Item Override (ITEM OVRD) Key as required.



777 FAA DDG Supplement Package

Supplement Number: 36-1

Issue Date: 21-DEC-2023

Section Number: 2 (MMEL)

Item Number: 32-08-01

Item Name: Proximity Sensor Electronic Unit (PSEU) Channels

DDG Effectivity: 777 FAA DDG (D630W003-TBC), Revision 36

MMEL Effectivity: 777 FAA MMEL, Revision 23

Description: Added DDG top note to option 32-08-01A. Modified option titles for option 32-08-01C and 32-08-01D.

Reason: The note identifies the corresponding MMEL item required to comply with the proviso requiring the flight lock system to be considered inoperative. Option titles for 32-08-01C and 32-08-01D were modified to refer to the correct inoperative PSEU channels corresponding to the associated provisos.

Disposition: For reference in the Operator's MEL.

Attachments: DDG pages 2,32-08-01.1-3

32-08-01 Proximity Sensor Electronic Unit (PSEU) Channels
32-08-01A One Inoperative - PSEU 1

Interval	Installed	Required	Procedure
C	4	3	

One may be inoperative in PSEU 1 provided the flight lock system is considered inoperative.

NOTE: The airplane must also be dispatched using MMEL item 52-11-05.

32-08-01 Proximity Sensor Electronic Unit (PSEU) Channels
32-08-01B One Inoperative - PSEU 2

Interval	Installed	Required	Procedure
C	4	3	

One may be inoperative in PSEU 2.

32-08-01 Proximity Sensor Electronic Unit (PSEU) Channels
32-08-01C Two Inoperative - PSEU 1 On Right System Bus, PSEU 2 On Left System Bus

Interval	Installed	Required	Procedure
C	4	2	(M) (O)

One may be inoperative in each PSEU provided:

- PSEU 1 transmission is verified on the left ARINC 629 systems bus.
- PSEU 2 transmission is verified on the right ARINC 629 systems bus.
- Flight lock system is considered inoperative.

NOTE: The airplane must also be dispatched using MMEL item 52-11-05.

MAINTENANCE (M)

Verify PSEU 1 transmission is on the left ARINC 629 systems bus and PSEU 2 transmission is on the right ARINC 629 systems bus (AMM 32-00-00/901).

- Verify PSEU 1 and PSEU 2 data is displayed on the Landing Gear ACTN/INDN maintenance page.
- Open P110 panel PSEU 1 PRI PWR circuit breaker.
- Verify PSEU 1 data is removed from the Landing Gear ACTN/INDN display. PSEU 1 is transmitting on the left ARINC 629 systems bus.



4. Open P310 panel PSEU 2 ALTN PWR circuit breaker.
5. Verify PSEU 2 data is removed from the Landing Gear ACTN/INDN display. PSEU 2 is transmitting on the right ARINC 629 systems bus.
6. Close the PSEU 1 PRI PWR and the PSEU 2 ALTN PWR circuit breakers.

OPERATIONS (O)

The autobrake selector may not automatically move from RTO to OFF after takeoff. If autobrake selector remains in RTO, manually move to OFF.

32-08-01 Proximity Sensor Electronic Unit (PSEU) Channels
32-08-01D Two Inoperative - PSEU 1 On Left System Bus, PSEU 2 On Right System Bus

Interval	Installed	Required	Procedure
C	4	2	(M) (O)

One may be inoperative in each PSEU provided:

- a. PSEU 1 transmission is verified on the right ARINC 629 systems bus.
- b. PSEU 2 transmission is verified on the left ARINC 629 systems bus.
- c. After takeoff, gear remains down for two minutes before retraction.
- d. Takeoff performance is based on landing gear extended.
- e. Flight lock system is considered inoperative

NOTE: The airplane must also be dispatched using MMEL item 52-11-05.

MAINTENANCE (M)

Verify that PSEU 1 transmission is on the right ARINC 629 systems bus and PSEU 2 transmission is on the left ARINC 629 systems bus (AMM 32-00-00/901).

1. Verify PSEU 1 and PSEU 2 data is displayed on the Landing Gear ACTN/INDN maintenance page.
2. Open P310 panel PSEU 1 ALTN PWR circuit breaker.
3. Verify PSEU 1 data is removed from the Landing Gear ACTN/INDN display. PSEU 1 is transmitting on the right ARINC 629 bus.
4. Open P210 panel PSEU 2 PRI PWR circuit breaker.
5. Verify PSEU 2 data is removed from the Landing Gear ACTN/INDN display. PSEU 2 is transmitting on the left ARINC 629 systems bus.
6. Close the PSEU 1 ALTN PWR and PSEU 2 PRI PWR circuit breakers.

OPERATIONS (O)

1. The autobrake selector may not automatically move from RTO to OFF after takeoff. If autobrake selector remains in RTO, manually move to OFF.
2. After takeoff, leave the landing gear extended for a minimum of two minutes to allow the wheels to spin down prior to gear retraction.

3. Base takeoff performance on landing gear extended. Takeoff performance for the landing gear extended is available using AFM-DPI or Boeing Takeoff Module (BTM).
4. Increase flight planning fuel by 1,900 lbs (861 kg).

32-08-01 Proximity Sensor Electronic Unit (PSEU) Channels

32-08-01E Two Inoperative - Gear Down Dispatch

Interval	Installed	Required	Procedure
C	4	2	(M) (O)

One may be inoperative in each PSEU provided:

- a. Landing gear are secured in the down position.
- b. Airplane is dispatched in accordance with the appropriate AFM Gear Extended Appendix.
- c. Flight lock system is considered inoperative

NOTE: The airplane must also be dispatched using MMEL item 52-11-05.

MAINTENANCE (M)

Secure the landing gear in the down position (AMM 32-00-00/901).

1. Install landing gear downlock pins, including installation procedure steps for airplane flight with landing gear locked in the extended position (AMM 32-00-30/201).
2. Place Alternate Gear Down Dispatch switch (Main Equipment Center above lower access door) to VMO position.

NOTE: The Warning Electronics System (WES) will set VMO/MMO to 270 KIAS/.73 Mach and VMO GEAR DOWN memo message will be displayed.

3. Deactivate the lever lock solenoid for the landing gear control lever by opening and collaring the P210 panel LG LEVER LOCK circuit breaker.

OPERATIONS (O)

1. The autobrake selector may not automatically move from RTO to OFF after takeoff. If autobrake selector remains in RTO, manually move to OFF.
2. Observe AFM Landing Gear Extended certificate limitations and operational limits.

NOTE: Dispatch requires authorization by the AFM Landing Gear Extended Appendix.

3. Base flight plan on landing gear extended.
4. The Warning Electronics System (WES) will set VMO/MMO to 270 KIAS/.73 Mach and VMO GEAR DOWN memo message will be displayed.



777 FAA DDG Supplement Package

Supplement Number: 36-1

Issue Date: 21-DEC-2023

Section Number: 2 (MMEL)

Item Number: 35-31-01

Item Name: Portable Oxygen Dispensing Units (Or Equivalent) (Bottle and Mask)

DDG Effectivity: 777 FAA DDG (D630W003-TBC), Revision 36

MMEL Effectivity: 777 FAA MMEL, Revision 23

Description: Modified subitem 35-31-01-01 Operations (O) procedure.

Reason: Clarified equipment that is affected by the inoperative tamper seals or tags.

Disposition: For reference in the Operator's MEL.

Attachments: DDG pages 2.35-31-01.1-2

35-31-01 Portable Oxygen Dispensing Units (Or Equivalent) (Bottle and Mask)

Interval	Installed	Required	Procedure
D	-	-	(M)

Any in excess of those required by 14 CFR may be inoperative or removed provided:

- An inoperative or not properly serviced portable oxygen bottle/unit remains in a certified location until removed or serviced at the next suitable maintenance facility.
- Location placarding is removed or obscured.
- Required distribution is maintained.

NOTE 1: Inoperative portable oxygen bottles or units, removed from a certified location or removed from the aircraft, are subject to 49 CFR dangerous goods regulations.

NOTE 2: Medical equipment installed in the aircraft as part of an EMS operation is not considered part of the normal complement of equipment. No MMEL relief applies to that equipment and 14 CFR maintenance and inspection requirements do not apply.

MAINTENANCE (M)

- Ensure required distribution of serviceable units is maintained throughout the airplane (AMM 35-00-00/901).
- Ensure inoperative or not properly serviced portable oxygen bottle/unit remains in a certified location until removed or serviced at the next suitable maintenance facility.

OPERATIONS NOTE

Airlines must determine portable oxygen requirements dependent on their seating configuration and their route system. See 14 CFR 121.329 and 121.333 or other applicable regulations.

35-31-01 Portable Oxygen Dispensing Units (Or Equivalent) (Bottle and Mask)

35-31-01-01 Tamper Seals or Tags

Interval	Installed	Required	Procedure
C	-	-	(O)



May be inoperative, damaged, or missing provided proper installation and operation is verified at each preflight.

OPERATIONS (O)

Before each departure, verify affected Portable Oxygen Dispensing Unit(s) is properly installed and operate(s) normally.

Supplement 36-1



777 FAA DDG Supplement Package

Supplement Number: 36-1

Issue Date: 21-DEC-2023

Section Number: 2 (MMEL)

Item Number: 35-31-02

Item Name: Portable Protective Breathing Equipment (PBE)

DDG Effectivity: 777 FAA DDG (D630W003-TBC), Revision 36

MMEL Effectivity: 777 FAA MMEL, Revision 23

Description: Modified subitem 35-31-02-01 Operations (O) procedure.

Reason: Clarified equipment that is affected by the inoperative tamper seals or tags.

Disposition: For reference in the Operator's MEL.

Attachments: DDG pages 2.35-31-02.1

35-31-02 Portable Protective Breathing Equipment (PBE)

Interval	Installed	Required	Procedure
D	-	-	

Any in excess of those required by 14 CFR may be inoperative or removed provided:

- Inoperative PBE remains in a certified location until removed from the aircraft at the next suitable maintenance facility.
- Location placarding is removed or obscured.
- Required distribution is maintained.

NOTE: Inoperative PBEs, removed from a certified location or removed from the aircraft, are subject to 49 CFR dangerous goods regulations.

OPERATIONS NOTE

See 14 CFR 121.337 or other appropriate regulations.

35-31-02 Portable Protective Breathing Equipment (PBE)

35-31-02-01 Tamper Seals or Tags

Interval	Installed	Required	Procedure
C	-	-	(O)

May be inoperative, damaged, or missing provided proper installation and operation is verified at each preflight.

OPERATIONS (O)

Before each departure, verify affected PBE(s) is properly installed and operate(s) normally.



777 FAA DDG Supplement Package

Supplement Number: 36-1

Issue Date: 21-DEC-2023

Section Number: 2 (MMEL)

Item Number: 52-71-01

Item Name: Door Indication Systems

DDG Effectivity: 777 FAA DDG (D630W003-TBC), Revision 36

MMEL Effectivity: 777 FAA MMEL, Revision 23

Description: Added Operations note.

Reason: Added Operations note to clarify bulk cargo door switch faults do not display the CONFIG DOORS warning.

Disposition: For reference in the Operator's MEL.

Attachments: DDG pages 2.52-71-01.1-6

52-71-01 Door Indication Systems

Interval	Installed	Required	Procedure
C	-	0	(M)

May be inoperative provided associated door is verified closed, latched and locked before each departure.

MAINTENANCE (M)

NOTE: The optional procedures to deactivate inoperative passenger entry door sensors and forward, aft and main deck cargo (777F only) door sensors are provided to prevent unexpected display of the CONFIG DOORS warning and potential RTO due to intermittent door sensor faults.

Deactivate the inoperative sensors (if desired), and before each flight, verify the associated door is closed, latched, and locked (AMM 52-00-00/901).

1. For inoperative sensor(s) on Main Entry Doors:
 - A. If desired, deactivate the associated entry door's inoperative sensor(s).
 - 1) Remove, cap and stow electrical connectors from following sensors.
 - a. For entry door 1L, remove electrical connector DS52109 from sensor S52109 (CLOSED & LATCHED DR 1L) and remove electrical connector DS52110 from sensor S52110 (LOCKED DR 1L).
 - b. For entry door 1R, remove electrical connector DS52209 from sensor S52209 (CLOSED & LATCHED DR 1R) and remove electrical connector DS52110 from sensor S52110 (LOCKED DR 1R).
 - c. For entry door 2L, remove electrical connector DS52111 from sensor S52111 (CLOSED & LATCHED DR 2L) and remove electrical connector DS52110 from sensor S52110 (LOCKED DR 2L).
 - d. For entry door 2R, remove electrical connector DS52211 from sensor S52211 (CLOSED & LATCHED DR 2R) and remove electrical connector DS52110 from sensor S52110 (LOCKED DR 2R).
- For -200/-200ER/-200LR:
- e. For entry door 3L, remove electrical connector DS52113 from sensor S52113 (CLOSED & LATCHED DR 3L) and remove electrical connector DS52110 from sensor S52110 (LOCKED DR 3L).

For -300/-300ER:

- f. For entry door 3L, remove electrical connector DS52113 from sensor S52113 (CLOSED & LATCHED DR 3L) and remove electrical connector DS52026 from sensor S52026 (LOCKED DR 3L).

For -200/-200ER/-200LR:

- g. For entry door 3R, remove electrical connector DS52213 from sensor S52213 (CLOSED & LATCHED DR 3R) and remove electrical connector DS52110 from sensor S52110 (LOCKED DR 3R).

For -300/-300ER:

- h. For entry door 3R, remove electrical connector DS52213 from sensor S52213 (CLOSED & LATCHED DR 3R) and remove electrical connector DS52026 from sensor S52026 (LOCKED DR 3R).
- i. For entry door 4L, remove electrical connector DS52115 from sensor S52115 (CLOSED & LATCHED DR 4L) and remove electrical connector DS52110 from sensor S52110 (LOCKED DR 4L).
- j. For entry door 4R, remove electrical connector DS52215 from sensor S52215 (CLOSED & LATCHED DR 4R) and remove electrical connector DS52110 from sensor S52110 (LOCKED DR 4R).

For -300/-300ER:

- k. For entry door 5L, remove electrical connector DS52117 from sensor S52117 (CLOSED & LATCHED DR 5L) and remove electrical connector DS52110 from sensor S52110 (LOCKED DR 5L).

For -300/-300ER:

- l. For entry door 5R, remove electrical connector DS52217 from sensor S52217 (CLOSED & LATCHED DR 5R) and remove electrical connector DS52110 from sensor S52110 (LOCKED DR 5R).

- B. Prior to each flight, verify the associated door is closed, latched and locked using normal procedures.
 - 1) Open the affected door in MANUAL/DISARMED mode, then close and latch.
 - 2) Confirm the door interior liner is aligned and faired with the doorway lining.
 - 3) Confirm the OPEN arrow on the hinge arm linings is aligned and continuous at crossover from forward section of lining to aft section of lining.

- 4) Position the door mode select handle to AUTOMATIC/ARMED, and confirm the yellow flags are visible in the viewing windows immediately forward and aft of door at floor level.
 - 5) Confirm the outside skin of the door is aligned and faired with the fuselage skin.
 - 6) Confirm the vent door is fully closed.
2. For inoperative sensor(s) on Forward Cargo Door:
- A. If desired, deactivate the inoperative door sensor(s).
 - 1) Remove, cap and stow electrical connector DS52021 from sensor S52021 (CLOSED).
 - 2) Remove, cap and stow electrical connector DS52042 from sensor S52042 (LATCHED).
 - 3) Remove, cap and stow electrical connector DS52022 from sensor S52022 (LOCKED).
 - B. Prior to each flight, verify the door is closed, latched and locked using normal procedures.
 - 1) Close the door.
 - 2) Confirm the outside skin of the door is aligned and faired with the fuselage skin.
 - 3) Confirm the lock handle is in the unlocked position.
 - 4) Confirm the vent door is open.
 - 5) Rotate the lock handle back to the CLOSED position.
 - 6) Confirm the lock handle is fully closed and faired with the outside skin of the door.
 - 7) Confirm the lock handle release lever is faired with the lock handle and outside skin of the door.
 - 8) Confirm the vent door is fully closed (the upper edge of the vent door may not be fully faired).
 - 9) Confirm the latches are in the fully latched position and that the locks are in the locked position by visual inspection through the witness ports at the lower edge of the door.
3. For inoperative sensor(s) on Aft Large Cargo Door:
- A. If desired, deactivate the inoperative door sensor(s).
 - 1) Remove, cap and stow electrical connector DS52045 from sensor S52045 (CLOSED).
 - 2) Remove, cap and stow electrical connector DS52043 from sensor S52043 (LATCHED).
 - 3) Remove, cap and stow electrical connector DS52044 from sensor S52044 (LOCKED).
 - B. Prior to each flight, verify the door is closed, latched and locked using normal procedures.
 - 1) Close the door.

- 2) Confirm the outside skin of the door is aligned and faired with the fuselage skin.
 - 3) Confirm the lock handle is in the unlocked position.
 - 4) Confirm the vent door is open.
 - 5) Rotate the lock handle back to the CLOSED position.
 - 6) Confirm the lock handle is fully closed and faired with the outside skin of the door.
 - 7) Confirm the lock handle release lever is faired with the lock handle and outside skin of the door.
 - 8) Confirm the vent door is fully closed (the upper edge of the vent door may not be fully faired).
 - 9) Confirm the latches are in the fully latched position and that the locks are in the locked position by visual inspection through the witness ports at the lower edge of the door.
4. For inoperative sensor(s) on Aft Small Cargo Door:
- A. If desired, deactivate the inoperative door sensor(s).
 - 1) Remove, cap and stow electrical connector DS52023 from sensor S52023 (CLOSED).
 - 2) Remove, cap and stow electrical connector DS52024 from sensor S52024 (LATCHED/LOCKED).
 - B. Prior to each flight, verify the door is closed and latched using normal procedures.
 - 1) Close the door.
 - 2) Confirm the outside skin of the door is aligned and faired with the fuselage skin.
 - 3) Confirm the vent door is fully closed.
 - 4) Open the access cover for the door control panel.
 - 5) Operate the CARGO DOOR CONTROL ARMING switch downward to the ARM position, and confirm the green DOOR LATCHED light illuminates. Release the switch and confirm the light extinguishes.
 - 6) Close the access cover.
5. For inoperative sensor(s) on Bulk Cargo Door:
- A. If desired, deactivate the inoperative door switch.
 - 1) Locate, remove and secure switch S52020 (CLOSED/LATCHED/LOCKED) without cutting its wires.
 - B. Prior to each flight, verify the door is closed and latched using normal procedures.
 - 1) Close the door.
 - 2) Confirm the outside skin of the door is aligned and faired with the fuselage skin.

- 3) Confirm the latch handle is fully closed and faired with the door skin.
- 4) Push on the door to verify it is latched.
6. For inoperative sensor(s) on Forward Access Door:
 - A. If desired, deactivate the inoperative door sensor(s).
 - 1) Remove, cap and stow electrical connector DS52018 from sensor S52018 (CLOSED/LATCHED/LOCKED).
 - B. Prior to each flight, verify the door is closed and latched using normal procedures.
 - 1) Close the door.
 - 2) Confirm the outside skin of the door is aligned and faired with the fuselage skin.
 - 3) Confirm the latch handle is fully closed and faired with the door skin.
 - 4) Push on the door to verify it is latched.
7. For inoperative sensor(s) on Main Equipment Center (E/E Bay) Access Door:
 - A. If desired, deactivate the inoperative door sensor(s).
 - 1) Remove, cap and stow electrical connector DS52019 from sensor S52019 (CLOSED/LATCHED/LOCKED).
 - B. Prior to each flight, verify the door is closed and latched using normal procedures.
 - 1) Close and latch the door.
 - 2) Confirm the outside skin of the door is aligned and faired with the fuselage skin.
 - 3) Confirm the latch handle is fully closed and faired with the door skin.
 - 4) Push on the door to verify it is latched.

For -300/-300ER:

8. For inoperative sensor(s) on Overwing Escape Slide Compartment Door:
 - A. Prior to each flight, verify the door is closed and latched using normal procedures.
 - 1) Open and close the door.
 - 2) Confirm the outside skin of the door is aligned and faired with the fuselage skin.
 - 3) Push on the door to verify it is latched.

For 777F:

9. For inoperative sensor(s) on Main Deck Cargo Door:
 - A. If desired, deactivate the inoperative door sensor(s).
 - 1) Remove, cap and stow electrical connector DS52053 from sensor S52053 (CLOSED).



- 2) Remove, cap and stow electrical connector DS52054 from sensor S52054 (LATCHED).
- 3) Remove, cap and stow electrical connector DS52055 from sensor S52055 (LOCKED).
- B. Prior to each flight, verify the door is closed, latched and locked using normal procedures.
 - 1) Close the door.
 - 2) Confirm the latch arrows point to the targets in all eight witness ports.
 - 3) Confirm the outside skin of the door is aligned and faired with the fuselage skin.
 - 4) Confirm the vent door has closed.
 - 5) Using the witness ports, confirm the lock sectors are located over latch cams but do not touch the cams.

OPERATIONS NOTE

1. The optional procedures to deactivate inoperative passenger entry door sensors and forward, aft and main deck cargo (777F only) door sensors are provided to prevent unexpected display of the CONFIG DOORS warning and potential RTO due to intermittent door sensor faults.
2. The DOOR WING SLIDE L and DOOR WING SLIDE R advisory messages (-300/-300ER) and the DOOR BULK CARGO advisory message may be displayed both on the ground and in flight.
3. For doors with inoperative sensors deactivated:
 - A. The respective door indication status message (if applicable) may be displayed both on the ground and in flight.
 - B. Inoperative door advisory messages may be displayed on the ground only, except the DOOR BULK CARGO advisory message, which may be displayed both on the ground and in flight.
 - C. The DOOR MAIN DECK CARGO, DOOR FWD CARGO and DOOR AFT CARGO (if applicable) caution messages may be displayed on the ground.
4. The CONFIG DOORS warning is not displayed for bulk cargo door switch faults.

The following messages may be displayed after landing if the associated door sensor was deactivated:

CAUTION - DOOR FWD CARGO

CAUTION - DOOR AFT CARGO

ADVISORY - DOOR ENTRY 1L

ADVISORY - DOOR ENTRY 1R

ADVISORY - DOOR AFT CARGO

ADVISORY - DOOR FWD CARGO

ADVISORY - DOOR E/E ACCESS



777 FAA DDG Supplement Package

Supplement Number: 36-1

Issue Date: 21-DEC-2023

Section Number: 4 (MISC)

Item Number: N/A

Item Name: Performance Based Navigation

DDG Effectivity: 777 FAA DDG (D630W003-TBC), Revision 36

MMEL Effectivity: 777 FAA MMEL, Revision 23

Description: Modified PBN guidance.

Reason: Modified PBN guidance for the Mode Control Panel (MCP) Switches. Corrected the PRR number reference for Enhanced Ground Proximity Warning System (EGPWS)

Disposition: For reference in the Operator's MEL.

Attachments: DDG pages 4.00-02-00.1-5



Section 4 Performance Based Navigation

Performance Based Navigation (PBN) Information

Applying dispatch relief for the following items may affect the ability of the airplane to meet the operational capabilities required for PBN operations. The following information lists items that, if inoperative, may affect the ability of the airplane to meet the performance and functional criteria for various PBN operations. This information can aid in identifying potentially affected capabilities associated with a particular dispatch configuration.

This section lists equipment requirements that may be found in the document 777 RNP Navigation Capabilities Document D243W018-13, revision "M" (or later). Operators should consult their Operations Specification for requirements applicable to their operations.

PBN Equipment Requirements

Two Primary Flight Displays (PFD) and Navigation Displays (ND) - (31-61-01)

All PBN operations require two PFDs with mini map.

The 777 is equipped with six display units. The PFDs are normally displayed on the outboard display units and the NDs are normally displayed on the inboard display units. Dispatch is only allowed with the lower center display unit inoperative. Therefore, this requirement will always be met for dispatch.

Two Control Display Units (CDU) - (34-61-03)

All PBN operations require two CDUs.

The 777 is equipped with three CDUs. Dispatch is allowed with the center DCU inoperative. Both left and right CDUs must operate normally for dispatch. Therefore, this requirement will always be met for dispatch.

AIMS Operational Software

For RNP AR APCH and RNP APCH:

Airplane must be equipped with AIMS BP05A or later. Navigation Performance Scales and LNAV Go-Around Auto-Engage must be enabled.

Airplanes without Navigation Performance Scales and LNAV Go-Around Auto-Engage (AIMS BP03 and earlier versions) do not support these operations.

Fault Tolerant ADIRU

For AIMS-1 V16 and AIMS-2 V17 or later:

AIR DATA SYS or NAV ADIRU INERTIAL messages must not be displayed.

For all other AIMS-1 or AIMS-2 versions:

NAV AIR DATA SYS or NAV ADIRU INERTIAL messages must not be displayed.

Dispatch is not allowed with any of the above messages displayed.

Therefore, this requirement will always be met for dispatch.

Autopilot Flight Director Computers (AFDC) - (22-11-01)

RNP AR APCH operations require two AFDCs to operate normally. All other PBN operations require a single operational AFDC.

The 777 is equipped with three AFDCs. MMEL dispatch is allowed with one, two, or all three AFDCs inoperative. Therefore, PBN operations may be affected with more than one AFDC inoperative.

PBN operations may be allowed using either the autopilot function or flight director function.

Mode Control Panel (MCP) Switches - (22-11-06)

Lateral navigation (LNAV) Mode Control Panel (MCP) switch must operate normally for all PBN operations.

Approach (APP) Mode Control Panel (MCP) switch must operate normally for RNP AR APCH and RNP APCH.

Vertical navigation (VNAV) Mode Control Panel (MCP) switch must operate normally for RNP AR APCH operations. Other vertical modes may be used for other PBN operations.

Autopilot Function

If the autopilot function is inoperative, PBN operations may be allowed using only the flight director function.

The autopilot function may not be used if any of the following equipment is inoperative:

1. Both autopilot engage switches. (22-11-06)
2. One control wheel autopilot disconnect switch below 1500 feet AGL. (22-11-08)
3. Both control wheel autopilot disconnect switches inoperative. (22-11-08)
4. Warning Electronic System (WES) Autopilot disconnect warning function. (31-51-05)

Flight Director Systems (22-11-10)

If the flight director function is inoperative, PBN operations may be allowed using only the autopilot function.

The flight director function is considered inoperative if any of the following equipment is inoperative:

1. One or more Flight Director Systems (22-11-10).
2. One or more Mode Control Panel (MCP) Flight Director (F/D) switches. (22-11-06)

Both flight director display functions are required for RNP AR APCH. Flight director function for pilot flying (PF) is required for RNP APCH.

Takeoff/Go-Around (TO/GA) Switches (22-11-09)

There are two TO/GA switches installed on the 777. MMEL dispatch is allowed with one or both TO/GA switches inoperative.

For RNP AR APCH, both TO/GA switches must operate normally.

For both TO/GA switches inoperative, PBN operations that require either autopilot or flight director use below 500 feet AGL or MDA, whichever is higher, may not be used. RNP AR APCH, RNP APCH, RNP 1, P-RNAV, RNAV 1 and Advanced RNP operations may not be allowed.

VOR Navigation Systems (34-51-01)

The 777 is equipped with two VORs. For dispatch, either one or both VORs may be inoperative.

- When dispatching with a single GPS system, RNP APCH, RNP 1/RNP 2, and Advanced RNP operations are allowed provided at least one of the following operates normally: GPS, VOR or DME.

Global Positioning Systems (GPS) (34-58-01)

The 777 is equipped with two GPSs. For dispatch, either one or both GPSs may be inoperative.

For dispatch with one GPS inoperative:

- RNP AR APCH and RNP 4 operations are not allowed.
- RNP APCH, RNP 1/RNP 2, and; Advanced RNP operations are allowed provided acceptable alternate navigation equipment (e.g. DME/DME, DME/VOR, VOR, ADF) is available to continue non-PBN operations in the event that the remaining GPS becomes inoperative.

For dispatch with both GPSs inoperative:

- RNP AR APCH, RNP APCH, RNP 1, RNP 2, RNP 4 and Advanced RNP operations not allowed.
- RNAV 1, RNAV 2, P-RNAV, B-RNAV, RNAV 5, RNP 10 and RNAV 10 are allowed provided at least one VOR or DME operates normally.
- RNP 10 and RNAV 10 operations may be allowed provided appropriate time limitations are observed.

NOTE: Operations may be for up to 6.2 hours of flight time starting when system is placed in inertial navigation mode

Radio Altimeter Systems (34-33-01)

The 777 is equipped with three radio altimeters. Dispatch is allowed with one radio altimeter inoperative.

- RNP AR APCH requires that two radio altimeters operate normally.
- All other PBN operations are allowed with one radio altimeter inoperative.

Therefore, these requirements will always be met for dispatch.

Enhanced Ground Proximity Warning Systems (EGPWS) (34-46-01)

777 airplanes with PRR 61724 or later are equipped with EGPWS. Dispatch is allowed with GPWS inoperative.

- RNP AR APCH requires EGPWS to operate normally (GND PROX SYS message not displayed, TERR POS message not displayed).
- All other PBN operations do not require EGPWS to be operative.

Distance Measuring Equipment Systems (DME) (34-55-01)

The 777 is equipped with two DMEs. For dispatch, either one or both DMEs may be inoperative.

- When dispatching with a single GPS system, RNP APCH, RNP 1/RNP 2, and Advanced RNP operations are allowed provided at least one of the following operates normally: GPS, VOR or DME.

Flight Management Computing Systems (FMCS) (34-61-01)

There are 2 Flight Management Computers (FMC) installed on the 777. All PBN operations require a minimum of 2 FMCS. Dispatch is allowed with one FMCS inoperative.

For one FMCS inoperative:

- RNP AR APCH and RNP 4 operations are not allowed.
- RNP APCH, RNP 1, RNP 2, RNAV 1, RNAV 2, P-RNAV, B-RNAV, RNAV 5 and Advanced RNP operations may be allowed provided alternate navigation equipment (e.g., VOR, ADF) is available to continue non-PBN operations in the event that the remaining FMCS becomes inoperative.
- RNP 10 and RNAV 10 operations are allowed.

**Flight Management Computing Systems (FMCS) -
Navigation Database (34-61-01-03)**

- A current navigation database is required for all PBN operations.
- It is assumed that the operator has a preflight procedure to ensure that the navigation database complies with regulatory agency requirements for ensuring the navigation database remains valid for the duration of the flight unless an alternative regulatory agency approved process applies.

Supplement 36A
Superseded



777 FAA DDG Supplement Package

Supplement Number: 36-2

Issue Date: 12-APR-2024

Section Number: 1 (EICAS)

Item Number: N/A

Item Name: EICAS Messages Cross Reference List

DDG Effectivity: 777 FAA DDG (D630W003-TBC), Revision 36

MMEL Effectivity: 777 FAA MMEL, Revision 23a

Description: PACK FCV L and PACK FCV R EICAS status messages were deleted from the EICAS cross reference list since they were deleted in AIMS bp v14.

Added cross reference to 22-11-01B (AFDC Two inoperative) for SINGLE SOURCE F/D Advisory message. The message is set when both PFDs are using the same AFDC source, so two of the three AFDC are inoperative.

Reason: Boeing is providing awareness to operators.

Disposition: For reference in the Operator's MEL.

Attachments: DDG page 1.00-02-00.40-44



Cross Reference List

<u>Message</u>	<u>Level</u>	<u>MMEL Item</u>
OVERHEAT CIRCUIT R2	Status	26-11-02
OVERHEAT ENG L	Caution	None
OVERHEAT ENG R	Caution	None
OVERSPEED	Warning	N/A
OVERSPEED SYS	Status	None
OVERSPEED VFE	Status	None
OVERSPEED VMO/MMO	Status	None
OVRD VALVE MOTOR L	Status	21-27-03
OVRD VALVE MOTOR R	Status	21-27-03

<u>Message</u>	<u>Level</u>	<u>MMEL Item</u>
PACK ACM L	Status	21-52-01
PACK ACM R	Status	21-52-01
PACK FCV LWR L	Status	21-51-02-02
PACK FCV LWR R	Status	21-51-02-02
PACK FCV UPR L	Status	21-51-02-01
PACK FCV UPR R	Status	21-51-02-01
PACK L	Advisory	21-51-01
PACK L	Status	21-51-01
PACK L + R	Caution	None
PACK MODE L	Advisory	21-52-01 21-52-02 21-52-07
PACK MODE R	Advisory	21-52-01 21-52-02 21-52-07
PACK OUT TEMP SNSR L	Status	21-52-04
PACK OUT TEMP SNSR R	Status	21-52-04

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Cross Reference List

<u>Message</u>	<u>Level</u>	<u>MMEL Item</u>
PACK R	Advisory	21-51-01
PACK R	Status	21-51-01
PARKING BRAKE SET	Memo	N/A
PASS OXYGEN C	Status	35-21-01
PASS OXYGEN L	Status	35-21-01
PASS OXYGEN LOW	Advisory	35-21-01
PASS OXYGEN ON	Advisory	N/A
PASS OXYGEN R	Status	35-21-01
PASS SIGNS ON	Memo	N/A
PASSENGER ADDRESS	Status	23-31-01
PDCU L	Status	23-93-03
PDCU R	Status	23-93-03
PFC CHANNEL	Status	27-02-03
PFC CONTROL LANES	Status	27-02-02
PFCS INTERFACE	Status	27-02-04
PILOT RESPONSE	Advisory	N/A
PILOT RESPONSE	Caution	N/A
PILOT RESPONSE	Warning	N/A
PITCH DOWN AUTHORITY	Caution	None
PITCH UP AUTHORITY	Caution	None
PITOT ADM C	Status	34-21-04-03
PITOT ADM L	Status	34-21-04-02
PITOT ADM R	Status	34-21-04-01
PR SNS REST UPR FWD	Status	21-29-07
PRI FLIGHT COMPUTERS	Caution	None
PRINTER	Comm	N/A



Cross Reference List

<u>Message</u>	<u>Level</u>	<u>MMEL Item</u>
PSEU 1	Status	None
PSEU 1 CHANNEL	Status	32-08-01
PSEU 2	Status	None
PSEU 2 CHANNEL	Status	32-08-01
PVD BOTH ON	Memo	N/A
PVD CAPT ON	Memo	N/A
PVD F/O ON	Memo	N/A
PVD SYS CAPT	Advisory	34-35-01
PVD SYS CAPT	Status	34-35-01
PVD SYS F/O	Advisory	34-35-01
PVD SYS F/O	Status	34-35-01

<u>Message</u>	<u>Level</u>	<u>MMEL Item</u>
R GEAR DOOR SENSOR	Status	32-61-02
R GEAR UPLOCK SENSOR	Status	32-61-03
RADIO ALT C	Status	34-33-01
RADIO ALT L	Status	34-33-01
RADIO ALT R	Status	34-33-01
RADIO ALT MONITOR C	Status	34-33-01
RADIO ALT MONITOR L	Status	34-33-01
RADIO ALT MONITOR R	Status	34-33-01
RADIO TRANSMIT	Advisory	N/A
RAM AIR EXIT DOOR L	Status	21-52-06
RAM AIR EXIT DOOR R	Status	21-52-06
RAM AIR INLET DOOR L	Status	21-52-05
RAM AIR INLET DOOR R	Status	21-52-05
RAT DEPLOY INOP	Status	None

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Cross Reference List

<u>Message</u>	<u>Level</u>	<u>MMEL Item</u>
RAT GEN HEAT	Status	29-21-01
RAT UNLOCKED	Advisory	None
RECIRC FAN	Status	21-25-01
RECIRC FANS OFF	Memo	N/A
REMOTE LIGHT SENSOR	Status	31-61-03
RESERVE BRAKES/STRG	Advisory	29-11-06
REVERSER INTERLOCK L	Status	78-34-01
REVERSER INTERLOCK R	Status	78-34-01
RSCF ASG CARD L	Status	31-09-01 31-09-01-02
RSCF ASG CARD R	Status	31-09-01 31-09-01-02
RSCF PWR SUPPLY 1	Status	31-09-02-02B
RSCF PWR SUPPLY 2	Status	31-09-02-02A
RUNWAY OVRD	Advisory	N/A
RUNWAY POS	Advisory	N/A
RUNWAY SYS	Advisory	34-46-01-04
RUNWAY SYS	Status	34-46-01-04

<u>Message</u>	<u>Level</u>	<u>MMEL Item</u>
SAARU	Status	34-21-07
SATCOM	Advisory	23-15-01
SATCOM COOLING FAN L	Status	21-27-09
SATCOM COOLING FAN R	Status	21-27-09
SATCOM DATALINK	Advisory	23-15-01-02
SATCOM DATALINK OFF	Memo	N/A
SATCOM HI GAIN	Status	23-15-01-01



Cross Reference List

<u>Message</u>	<u>Level</u>	<u>MMEL Item</u>
SATCOM MESSAGE	Comm	N/A
SATCOM MESSAGE	Comm	N/A
SATCOM SYSTEM	Status	23-15-01
SATCOM VOICE	Advisory	23-15-01
SATVOICE AVAIL	Comm	N/A
SATVOICE LOST	Advisory	N/A
SEATBELTS ON	Memo	N/A
SELCAL	Comm	N/A
SEMI LEVER GEAR	Status	32-30-02
SEMI LEVER GEAR LOCK	Status	32-30-02
SEMI LEVER GEAR SYS	Status	32-30-02
SGL SOURCE AIR DATA	Advisory	None
SGL SOURCE DISPLAYS	Caution	None
SGL SOURCE RAD ALT	Advisory	None
SINGLE SOURCE F/D	Advisory	22-11-01B 22-11-10
SINGLE SOURCE ILS	Caution	34-31-01
SLATS DRIVE	Caution	None
SLATS PRIMARY FAIL	Caution	None
SLATS SECONDARY FAIL	Status	27-03-03
SMOKE COMPT UPR DR 1	Caution	N/A
SMOKE COMPT UPR DR 2	Caution	N/A
SMOKE COMPT UPR DR 3	Caution	N/A
SMOKE COMPT UPR DR 4	Caution	N/A
SMOKE COMPT UPR DR 5	Caution	N/A
SMOKE CREW REST F/D	Caution	N/A

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777 FAA DDG Supplement Package

Supplement Number: 36-2

Issue Date: 12-APR-2024

Section Number: 2 (MMEL)

Item Number: 21-27-09

Item Name: SATCOM Backup Cooling Fans (Passenger)

DDG Effectivity: 777 FAA DDG (D630W003-TBC), Revision 36

MMEL Effectivity: 777 FAA MMEL, Revision 23a

Description: Modified Maintenance (M) procedures for all item options, modified the provisos, deleted Operational (O) procedure, and added a top note for item option 21-27-09B.

Reason: Maintenance (M) procedures were modified to clarify the affected system and to be consistent with the provisos. Item option 21-27-09B was modified to allow operators with dual SATCOM systems to dispatch with one backup cooling fan inoperative and not require ETOPS limitation.

Disposition: For reference in the Operator's MEL.

Attachments: DDG page 2.21-27-09.1-3

21-27-09 SATCOM Backup Cooling Fans (Passenger)

21-27-09A Both Lavatory/Galley Vent Fans Operational

Interval	Installed	Required	Procedure
C	-	0	(M)

May be inoperative provided:

- Inoperative cooling fan(s) is deactivated.
- Both lavatory/galley ventilation fans operate normally.

MAINTENANCE (M)

Deactivate the inoperative SATCOM backup cooling fan(s) (AMM 21-00-00/901).

- For the left backup cooling fan inoperative:
 - Open and collar the P110 panel L SATCOM BACKUP CLG FAN CTRL circuit breaker.
 - Open and collar the P110 panel L SATCOM BACKUP CLG FAN circuit breaker.
- For right backup cooling fan inoperative:
 - Open and collar the P210 panel R SATCOM BACKUP CLG FAN CTRL circuit breaker.
 - Open and collar the P210 panel R SATCOM BACKUP CLG FAN circuit breaker.

21-27-09 SATCOM Backup Cooling Fans (Passenger)

21-27-09B Lavatory/Galley Fan(s) Inoperative

Interval	Installed	Required	Procedure
C	-	0	(M)

May be inoperative provided:

- Inoperative SATCOM backup cooling fan(s) is deactivated.
- Associated SATCOM system(s) is considered inoperative.

NOTE: The airplane must also be dispatched using MMEL item 23-15-01.

**MAINTENANCE (M)**

Deactivate the inoperative SATCOM backup cooling fan(s) and associated SATCOM system(s) (AMM 21-00-00/901).

1. For the left backup cooling fan inoperative:
 - A. Open and collar the P110 panel L SATCOM BACKUP CLG FAN CTRL circuit breaker.
 - B. Open and collar the P110 panel L SATCOM BACKUP CLG FAN circuit breaker.
 - C. Open and collar the P110 panel SATL HIGH L circuit breaker.
 - D. Open and collar the P110 panel SATL SDU L circuit breaker.
 - E. For a low gain antenna system installed, open and collar P110 panel SATL LOW circuit breaker.
2. For right backup cooling fan inoperative:
 - A. Open and collar the P210 panel R SATCOM BACKUP CLG FAN CTRL circuit breaker.
 - B. Open and collar the P210 panel R SATCOM BACKUP CLG FAN circuit breaker.
 - C. Open and collar the P210 panel SATL HIGH R circuit breaker.
 - D. Open and collar the P210 panel SATL SDU R circuit breaker.

21-27-09 SATCOM Backup Cooling Fans (Passenger)

21-27-09C Procedures Do Not Require Use of SATCOM

Interval	Installed	Required	Procedure
D	-	0	(M)

May be inoperative provided:

- a. Inoperative SATCOM backup cooling fan(s) is deactivated.
- b. Procedures do not require the use of SATCOM.

MAINTENANCE (M)

Deactivate the inoperative SATCOM backup cooling fan(s) and associated SATCOM system(s) (AMM 21-00-00/901).

1. For the left backup cooling fan inoperative:
 - A. Open and collar the P110 panel L SATCOM BACKUP CLG FAN CTRL circuit breaker.
 - B. Open and collar the P110 panel L SATCOM BACKUP CLG FAN circuit breaker.
 - C. Open and collar the P110 panel SATL HIGH L circuit breaker.
 - D. Open and collar the P110 panel SATL SDU L circuit breaker.

- E. For a low gain antenna system installed, open and collar P110 panel SATL LOW circuit breaker.
- 2. For right backup cooling fan inoperative:
 - A. Open and collar the P210 panel R SATCOM BACKUP CLG FAN CTRL circuit breaker.
 - B. Open and collar the P210 panel R SATCOM BACKUP CLG FAN circuit breaker.
 - C. Open and collar the P210 panel SATL HIGH R circuit breaker.
 - D. Open and collar the P210 panel SATL SDU R circuit breaker.



777 FAA DDG Supplement Package

Supplement Number: 36-2

Issue Date: 12-APR-2024

Section Number: 2 (MMEL)

Item Number: 21-31-01

Item Name: Auto Cabin Pressure Controls (L and R)

DDG Effectivity: 777 FAA DDG (D630W003-TBC), Revision 36

MMEL Effectivity: 777 FAA MMEL, Revision 23a

Description: Added a proviso and DDG top notes for subitem 21-31-01-03.

Reason: To correct the DDG assembly error.

Disposition: For reference in the Operator's MEL.

Attachments: DDG page 2.21-31-01.1-5

**21-31-01 Auto Cabin Pressure Controls (L and R)****21-31-01-01 Passenger With P/N 3676-GRS-014-00 or Later ASCPC Software Installed****21-31-01-01A Manual Control Operates on Both Outflow Valves**

Interval	Installed	Required	Procedure
C	2	1	(M)

One may be inoperative provided:

- Manual cabin pressure control is verified to operate normally on both outflow valves before each departure.
- Left or right AFDC operates normally.

MAINTENANCE (M)

NOTE: ASCPC software P/N 3676-GRS-014-00 is installed by Boeing Service Bulletin 777-36-0013. All software P/Ns installed in ASCPC-2 are considered later versions than software P/Ns installed in ASCPC-1.

Before each departure, verify manual cabin pressure control operates normally on both outflow valves (AMM 21-00-00/901).

- Position Pressurization Panel (P5) OUTFLOW VALVE switches to MAN.

NOTE: Outflow valve position indications on EICAS may blank when MAN is selected.

- Hold each Pressurization Panel (P5) OUTFLOW VALVE MANUAL switch CLOSE until the EICAS outflow valve position indicator is CL, or if EICAS indication is not available hold the switch CLOSE for 30 seconds.
- Confirm the outflow valve doors are closed.
- Hold each Pressurization Panel (P5) OUTFLOW VALVE MANUAL switch OPEN until the EICAS outflow valve position indicator is OP, or if EICAS indication is not available hold the switch OPEN for 30 seconds.
- Confirm the outflow valve doors are full open.
- Position Pressurization Panel (P5) OUTFLOW VALVE switches to AUTO.

OPERATIONS NOTE

Leave OUTFLOW VALVE switches in AUTO unless required to accomplish a Non-Normal Checklist.

- 21-31-01 Auto Cabin Pressure Controls (L and R)**
- 21-31-01-01 Passenger With P/N 3676-GRS-014-00 or Later ASCPC Software Installed**
- 21-31-01-01B One Outflow Valve Deactivated / Manual Control Operates on Other Valve**

Interval	Installed	Required	Procedure
C	2	1	(M) (O)

One may be inoperative provided:

- One outflow valve is deactivated 7% open.
- Manual cabin pressure control is verified to operate normally on the remaining outflow valve.
- Left or right AFDC operates normally.
- One pack operates normally, and the other pack operates normally or in standby cooling mode.
- Both CTCs operate normally.
- Both engine bleed systems operate normally.

For OFCR installed:

- For aft outflow valve deactivated 7% open, the OFCR is considered inoperative.

For OFAR installed:

- For aft outflow valve deactivated 7% open, the OFAR is considered inoperative.

For OFCR or OFAR installed:

NOTE: For aft outflow valve deactivated, the airplane must also be dispatched using MMEL item 25-29-04.

MAINTENANCE (M)

Use DDG item 21-31-03-02 (M) procedure.

OPERATIONS (O)

- Use DDG item 21-31-03-02 (O) procedure.
- Leave OUTFLOW VALVE switches in AUTO unless required to accomplish a Non-Normal Checklist.

- 21-31-01 Auto Cabin Pressure Controls (L and R)**
- 21-31-01-02 777F**

Interval	Installed	Required	Procedure
C	2	1	(M)

One may be inoperative provided:

- Manual cabin pressure control is verified to operate normally on both outflow valves before each departure.
- Left or right AFDC operates normally.

MAINTENANCE (M)

Before each departure, verify manual cabin pressure control operates normally on both outflow valves (AMM 21-00-00/901).

- Position Pressurization Panel (P5) OUTFLOW VALVE switches to MAN.
NOTE: Outflow valve position indications on EICAS may blank when MAN is selected.
- Hold each Pressurization Panel (P5) OUTFLOW VALVE MANUAL switch CLOSE until the EICAS outflow valve position indicator is CL, or if EICAS indication is not available hold the switch CLOSE for 30 seconds.
- Confirm the outflow valve doors are closed.
- Hold each Pressurization Panel (P5) OUTFLOW VALVE MANUAL switch OPEN until the EICAS outflow valve position indicator is OP, or if EICAS indication is not available hold the switch OPEN for 30 seconds.
- Confirm the outflow valve doors are full open.
- Position Pressurization Panel (P5) OUTFLOW VALVE switches to AUTO.

OPERATIONS NOTE

Leave OUTFLOW VALVE switches in AUTO unless required to accomplish a Non-Normal Checklist.

21-31-01 Auto Cabin Pressure Controls (L and R)

21-31-01-03 All

Interval	Installed	Required	Procedure
C	2	0	(M) (O)

May be inoperative provided:

- Flight is conducted unpressurized.
- Procedures are established and used to verify cargo compartments remain empty or contain only ballast, empty cargo handling equipment (ballast may be loaded in ULDs), or fly away kits.
- For extended overwater flight, manual cabin pressure control is verified to operate normally on both outflow valves.

[For LLAR, OFCR, OFAR, MDCR or MDFCR installed:](#)

- The rest area(s) is considered inoperative.

[For first class suites installed:](#)

- First class suites are considered inoperative.

NOTE: Operator MELs must define which items are approved for inclusion in the fly away kits and which materials can be used as ballast.

NOTE 1: First class suites refers to the High Wall First Class Passenger Suite installation (option number 2525E407E69).

For LLAR, OFCR, OFAR, MDCR or MDFCR installed:

NOTE 2: The airplane must also be dispatched using MMEL item 25-29-04.

For first class suites installed:

NOTE 3: The airplane must also be dispatched using MMEL item 25-25-05.

MAINTENANCE (M)

For extended overwater flight, verify manual cabin pressure control operates normally on both outflow valves (AMM 21-00-00/901).

1. Position Pressurization Panel (P5) OUTFLOW VALVE switches to MAN.

NOTE: Outflow valve position indications on EICAS may blank when MAN is selected.

2. Hold each Pressurization Panel (P5) OUTFLOW VALVE MANUAL switch CLOSE until the EICAS outflow valve position indicator is CL, or if EICAS indication is not available hold the switch CLOSE for 30 seconds.
3. Confirm the outflow valve doors are closed.
4. Hold each Pressurization Panel (P5) OUTFLOW VALVE MANUAL switch OPEN until the EICAS outflow valve position indicator is OP, or if EICAS indication is not available hold the switch OPEN for 30 seconds.
5. Confirm the outflow valve doors are full open.
6. Leave Pressurization Panel (P5) OUTFLOW VALVE switches in MAN.

OPERATIONS (O)

1. Cargo compartments must remain empty, except for ballast, empty cargo containers (ballast may be loaded in ULDs), fly away kits, pallets, and cargo restraint components.
2. Position both outflow valves full open.
 - A. Position both OUTFLOW VALVE switches to MAN.

NOTE 1: The CABIN ALTITUDE AUTO caution message will be displayed.

NOTE 2: Outflow valve position indications on EICAS may blank when MAN is selected.
 - B. Hold each OUTFLOW VALVE MANUAL switch OPEN for 30 seconds.
3. Maintain the outflow valves full open unless required to accomplish a Non-Normal Checklist.
4. For flight with passengers:
 - A. Limit enroute altitude to 10,000 feet.



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- B. To avoid passenger discomfort, limit climb (after takeoff obstacle clearance has been assured) and descent rates to 500 fpm.

Supplement 36-2



777 FAA DDG Supplement Package

Supplement Number: 36-2

Issue Date: 12-APR-2024

Section Number: 2 (MMEL)

Item Number: 21-31-03

Item Name: Outflow Valves (FWD and AFT)

DDG Effectivity: 777 FAA DDG (D630W003-TBC), Revision 36

MMEL Effectivity: 777 FAA MMEL, Revision 23a

Description: Added notes in the Maintenance (M) procedure steps for subitem 21-31-03-01, 21-31-03-02, 21-31-03-03-01, and 21-31-03-03-02.

Reason: To emphasize the importance of following the correct Maintenance (M) procedure steps to avoid in-flight depressurization.

Disposition: For reference in the Operator's MEL.

Attachments: DDG page 2.21-31-03.1-17

21-31-03 Outflow Valves (FWD and AFT)
21-31-03-01 Passenger Without P/N 3676-GRS-014-00 or Later ASCPC Software Installed

Interval	Installed	Required	Procedure
C	2	1	(M) (O)

One may be inoperative provided:

- Valve is deactivated 7% open.
- Manual cabin pressure control is verified to operate normally on the remaining outflow valve.
- Both auto cabin pressure controls operate normally.
- One pack operates normally, and the other pack operates normally or in standby cooling mode.
- Both CTCs operate normally.
- Both engine bleed systems operate normally.

For OFCR installed:

- For aft outflow valve deactivated 7% open, the OFCR is considered inoperative.

For OFAR installed:

- For aft outflow valve deactivated 7% open, the OFAR is considered inoperative.

For OFCR or OFAR installed:

NOTE: For aft outflow valve deactivated, the airplane must also be dispatched using MMEL item 25-29-04.

MAINTENANCE (M)

NOTE 1: ASCPC software P/N 3676-GRS-014-00 is installed by Boeing Service Bulletin 777-36-0013. All software P/Ns installed in ASCPC-2 are considered later versions than software P/Ns installed in ASCPC-1.

NOTE 2: For a pack to be operating normally, certain items associated with that system must be operative. See DDG item 21-51-01 for a listing of those items.

NOTE 3: A pack will operate in standby cooling mode with an inoperative ACM, or an inoperative ECV, or an inoperative Condenser Inlet Temperature Control System.

NOTE 4: For an engine bleed system to be operating normally, certain items associated with that system must be operative. See DDG item 36-00-01 for a listing of those items.



Deactivate inoperative outflow valve 7% open and verify manual cabin pressure control operates normally on the remaining operative outflow valve (AMM 21-00-00/901).

NOTE: For ground operations with OAT at or above 24 degrees C and forward outflow valve deactivated 7% open, supply conditioned air to the airplane.

1. Supply electrical power on the airplane (AMM 24-22-00/201).
2. Deactivate inoperative outflow valve 7% open.
 - A. For manual control operative and valve position displayed on the Air Supply Maintenance page:
 - 1) Position the associated Pressurization Panel (P5) OUTFLOW VALVE switch to MAN.
 - 2) Using the associated Pressurization Panel (P5) OUTFLOW VALVE MANUAL switch, position the inoperative outflow valve to indicate 0.07 on the Air Supply Maintenance page.
 - 3) For the forward outflow valve inoperative:
 - a. Open and collar the P210 panel R FWD OFL VLV circuit breaker.
 - b. Open and collar the P310 panel L FWD OFL VLV circuit breaker.
 - 4) For the aft outflow valve inoperative:
 - a. Open and collar the P210 panel R AFT OFL VLV circuit breaker.
 - b. Open and collar the P310 panel L AFT OFL VLV circuit breaker.

NOTE: The circuit breakers must be opened prior to positioning the OUTFLOW VALVE switch to AUTO. If the OUTFLOW VALVE switch is positioned to AUTO before opening the circuit breakers the associated outflow valve will move to the fully open position and the airplane will be unable to pressurize in flight.

- 5) Position the associated Pressurization Panel (P5) OUTFLOW VALVE switch to AUTO.
- 6) Confirm the gap between the two outflow doors is 0.375 ± 0.031 inches.

NOTE: Confirming the gap between the outflow doors ensures the outflow valve is deactivated in the correct position for pressurized flight.

- B. For OFV (P/N 2214420-X) installed with manual control inoperative or valve position on the Air Supply Maintenance page not available:
- 1) For the forward outflow valve inoperative:
 - a. Open and collar the P210 panel R FWD OFL VLV circuit breaker.
 - b. Open and collar the P310 panel L FWD OFL VLV circuit breaker.
 - 2) For the aft outflow valve inoperative:
 - a. Open and collar the P210 panel R AFT OFL VLV circuit breaker.
 - b. Open and collar the P310 panel L AFT OFL VLV circuit breaker.
 - 3) Gain access to the associated outflow valve assembly.
 - 4) Disconnect the electrical connector from the back of the valve control unit.
 - 5) Remove one DC motor from the outflow valve assembly.
 - 6) Move the outflow valve doors until the gap between the two doors is 0.375 ± 0.031 inches.
 - 7) Re-install the DC motor.
 - 8) Reconnect the electrical connector on the back of the valve control unit.
 - 9) Position the associated Pressurization Panel (P5) OUTFLOW VALVE switch to AUTO.
3. Verify manual cabin pressure control operates normally on the remaining outflow valve.
- A. Position the associated Pressurization Panel (P5) OUTFLOW VALVE switch to MAN.

NOTE: Outflow valve position indications on EICAS may blank when MAN is selected.
 - B. Hold the associated Pressurization Panel (P5) OUTFLOW VALVE MANUAL switch CLOSE until the EICAS outflow valve position indicator is CL, or if EICAS indication is not available hold the switch CLOSE for 30 seconds.
 - C. Confirm the associated outflow valve door is closed.
 - D. Hold the associated Pressurization Panel (P5) OUTFLOW VALVE MANUAL switch OPEN until the EICAS outflow valve position indicator is OP, or if EICAS indication is not available hold the switch OPEN for 30 seconds.
 - E. Confirm the associated outflow valve door is full open.
 - F. Position the associated Pressurization Panel (P5) OUTFLOW VALVE switch to AUTO.

**OPERATIONS (O)**

NOTE 1: Aft cabin temperature control may be degraded when dispatching with the aft outflow valve deactivated 7% open.

NOTE 2: The associated OUTFLOW VALVE AFT or FWD advisory message will be displayed.

NOTE 3: The outflow valve position indication pointer will be missing for the deactivated outflow valve on the EICAS pressurization indications.

1. During passenger loading, if both packs are operating, select right pack OFF prior to closing the main entry door(s). This avoids uncomfortable changes in cabin pressure.
2. After all main entry doors are closed at the parking stand or departure gate, if both packs are operating, select right pack OFF before reopening a main entry door. After leaving the parking stand or gate, select right pack AUTO.
3. If the operative outflow valve fails during flight and cannot be controlled manually it may be necessary to select packs OFF to depressurize the airplane for landing.
 - A. During descent when below 15,000 ft. MSL, select one pack OFF.
 - B. During descent when below 10,000 ft. MSL, select remaining pack OFF.

For Forward Cargo Air Conditioning (FCAC) installed:

4. For aft outflow valve deactivated and FIRE CARGO AFT warning message, accomplish the associated non-normal checklist and also set FWD CARGO A/C control OFF.
5. Prior to stand or gate arrival, if both packs are operating, select right pack OFF. After main entry doors are open at the stand or gate, select packs as desired.

21-31-03 Outflow Valves (FWD and AFT)**21-31-03-02 Passenger With P/N 3676-GRS-014-00 or Later
ASCPC Software Installed**

Interval	Installed	Required	Procedure
C	2	1	(M) (O)

One may be inoperative provided:

- a. Valve is deactivated 7% open.
- b. Manual cabin pressure control is verified to operate normally on the remaining outflow valve.
- c. Left or right auto cabin pressure control operates normally.
- d. One pack operates normally, and the other pack operates normally or in standby cooling mode.
- e. Both CTCs operate normally.

f. Both engine bleed systems operate normally.

For OFCR installed:

g. For aft outflow valve deactivated 7% open, the OFCR is considered inoperative.

For OFAR installed:

h. For aft outflow valve deactivated 7% open, the OFAR is considered inoperative.

For OFCR or OFAR installed:

NOTE: For aft outflow valve deactivated, the airplane must also be dispatched using MMEL item 25-29-04.

MAINTENANCE (M)

NOTE 1: ASCPC software P/N 3676-GRS-014-00 is installed by Boeing Service Bulletin 777-36-0013. All software P/Ns installed in ASCPC-2 are considered later versions than software P/Ns installed in ASCPC-1.

NOTE 2: For a pack to be operating normally, certain items associated with that system must be operative. See DDG item 21-51-01 for a listing of those items.

NOTE 3: A pack will operate in standby cooling mode with an inoperative ACM, or an inoperative ECV, or an inoperative Condenser Inlet Temperature Control System.

NOTE 4: For an engine bleed system to be operating normally, certain items associated with that system must be operative. See DDG item 36-00-01 for a listing of those items.

Deactivate inoperative outflow valve 7% open and verify manual cabin pressure control operates normally on the remaining outflow valve (AMM 21-00-00/901).

1. Supply electrical power on the airplane (AMM 24-22-00/201).
2. Deactivate inoperative outflow valve 7% open.

NOTE: For ground operations with OAT at or above 24 degrees C and forward outflow valve deactivated 7% open, supply conditioned air to the airplane.

A. For manual control operative and valve position displayed on the Air Supply Maintenance page:

- 1) Position the associated Pressurization Panel (P5) OUTFLOW VALVE switch to MAN.
- 2) Using the associated Pressurization Panel (P5) OUTFLOW VALVE MANUAL switch, position the inoperative outflow valve to indicate 0.07 on the Air Supply Maintenance page.
- 3) For the forward outflow valve inoperative:
 - a. Open and collar the P210 panel R FWD OFL VLV circuit breaker.

- b. Open and collar the P310 panel L FWD OFL VLV circuit breaker.
- 4) For the aft outflow valve inoperative:
 - a. Open and collar the P210 panel R AFT OFL VLV circuit breaker.
 - b. Open and collar the P310 panel L AFT OFL VLV circuit breaker.

NOTE: The circuit breakers must be opened prior to positioning the OUTFLOW VALVE switch to AUTO. If the OUTFLOW VALVE switch is positioned to AUTO before opening the circuit breakers the associated outflow valve will move to the fully open position and the airplane will be unable to pressurize in flight.

- 5) Position the associated Pressurization Panel (P5) OUTFLOW VALVE switch to AUTO.
- 6) Confirm the gap between the two outflow doors is 0.375 ± 0.031 inches.

NOTE: Confirming the gap between the outflow doors ensures the outflow valve is deactivated in the correct position for pressurized flight.

B. For OFV (P/N 2214420-X) installed with manual control inoperative or valve position on the Air Supply Maintenance page not available:

- 1) For the forward outflow valve inoperative:
 - a. Open and collar the P210 panel R FWD OFL VLV circuit breaker.
 - b. Open and collar the P310 panel L FWD OFL VLV circuit breaker.
- 2) For the aft outflow valve inoperative:
 - a. Open and collar the P210 panel R AFT OFL VLV circuit breaker.
 - b. Open and collar the P310 panel L AFT OFL VLV circuit breaker.
- 3) Gain access to the associated outflow valve assembly.
- 4) Disconnect the electrical connector from the back of the valve control unit.
- 5) Remove one DC motor from the outflow valve assembly.
- 6) Move the outflow valve doors until the gap between the two doors is 0.375 ± 0.031 inches.
- 7) Re-install the DC motor.
- 8) Reconnect the electrical connector on the back of the valve control unit.

- 9) Position the associated Pressurization Panel (P5) OUTFLOW VALVE switch to AUTO.
3. Verify manual cabin pressure control operates normally on the remaining outflow valve.
 - A. Position the associated Pressurization Panel (P5) OUTFLOW VALVE switch to MAN.

NOTE: Outflow valve position indications on EICAS may blank when MAN is selected.
 - B. Hold the associated Pressurization Panel (P5) OUTFLOW VALVE MANUAL switch CLOSE until the EICAS outflow valve position indicator is CL, or if EICAS indication is not available hold the switch CLOSE for 30 seconds.
 - C. Confirm the associated outflow valve door is closed.
 - D. Hold the associated Pressurization Panel (P5) OUTFLOW VALVE MANUAL switch OPEN until the EICAS outflow valve position indicator is OP, or if EICAS indication is not available hold the switch OPEN for 30 seconds.
 - E. Confirm the associated outflow valve door is full open.
 - F. Position the associated Pressurization Panel (P5) OUTFLOW VALVE switch to AUTO.

OPERATIONS (O)

NOTE 1: Aft cabin temperature control may be degraded when dispatching with the aft outflow valve deactivated 7% open.

NOTE 2: The associated OUTFLOW VALVE AFT or FWD advisory messages will be displayed.

NOTE 3: The outflow valve position indication pointer will be missing for the deactivated outflow valve on the EICAS pressurization indications.

1. During passenger loading, if both packs are operating, select right pack OFF prior to closing the main entry door(s). This avoids uncomfortable changes in cabin pressure.
2. After all main entry doors are closed at the parking stand or departure gate, if both packs are operating, select right pack OFF before reopening a main entry door. After leaving the parking stand or gate, select right pack AUTO.
3. If the operative outflow valve fails during flight and cannot be controlled manually it may be necessary to select packs OFF to depressurize the airplane for landing.
 - A. During descent when below 15,000 ft. MSL, select one pack OFF.
 - B. During descent when below 10,000 ft. MSL, select remaining pack OFF.

**For Forward Cargo Air Conditioning (FCAC) installed:**

4. For aft outflow valve deactivated and FIRE CARGO AFT warning message, accomplish the associated non-normal checklist and also set FWD CARGO A/C control OFF.
5. Prior to stand or gate arrival, if both packs are operating, select right pack OFF. After main entry doors are open at the stand or gate, select packs as desired.

21-31-03 Outflow Valves (FWD and AFT)**21-31-03-03 777F****21-31-03-03-01 With P/N 3677-GRS-105-00 ASCPC Software Installed**

Interval	Installed	Required	Procedure
C	2	1	(M) (O)

One may be inoperative provided:

- a. Valve is deactivated 7% open.
- b. Manual cabin pressure control is verified to operate normally on the remaining outflow valve.
- c. Left or right auto cabin pressure control operates normally.
- d. One pack operates normally, and the other pack operates normally or in standby cooling mode.
- e. Both CTCs operate normally.
- f. Both engine bleed systems operate normally.
- g. Rigid cargo barrier doors remain closed inflight.

MAINTENANCE (M)

NOTE 1: For a pack to be operating normally, certain items associated with that system must be operative. See DDG item 21-51-01 for a listing of those items.

NOTE 2: A pack will operate in standby cooling mode with an inoperative ACM, or an inoperative ECV, or an inoperative Condenser Inlet Temperature Control System.

NOTE 3: For an engine bleed system to be operating normally, certain items associated with that system must be operative. See DDG item 36-00-01 for a listing of those items.

Deactivate inoperative outflow valve 7% open and verify manual cabin pressure control operates normally on the remaining operative outflow valve (AMM 21-00-00/901).

NOTE: For ground operations with OAT at or above 24 degrees C and forward outflow valve deactivated 7% open, supply conditioned air to the airplane.

1. Supply electrical power on the airplane (AMM 24-22-00/201).
2. Deactivate inoperative outflow valve 7% open.
 - A. For manual control operative and valve position displayed on the Air Supply Maintenance page:
 - 1) Position the associated Pressurization Panel (P5) OUTFLOW VALVE switch to MAN.
 - 2) Using the associated Pressurization Panel (P5) OUTFLOW VALVE MANUAL switch, position the inoperative outflow valve to indicate 0.07 on the Air Supply Maintenance page.
 - 3) For the forward outflow valve inoperative:
 - a. Open and collar the P210 panel R FWD OFL VLV circuit breaker.
 - b. Open and collar the P310 panel L FWD OFL VLV circuit breaker.
 - 4) For the aft outflow valve inoperative:
 - a. Open and collar the P210 panel R AFT OFL VLV circuit breaker.
 - b. Open and collar the P310 panel L AFT OFL VLV circuit breaker.

NOTE: The circuit breakers must be opened prior to positioning the OUTFLOW VALVE switch to AUTO. If the OUTFLOW VALVE switch is positioned to AUTO before opening the circuit breakers the associated outflow valve will move to the fully open position and the airplane will be unable to pressurize in flight.

- 5) Position the associated Pressurization Panel (P5) OUTFLOW VALVE switch to AUTO.
 - 6) Confirm the gap between the two outflow doors is 0.375 ± 0.031 inches.

NOTE: Confirming the gap between the outflow doors ensures the outflow valve is deactivated in the correct position for pressurized flight.

- B. For OFV (P/N 2214420-X) installed with manual control inoperative, or valve position on the Air Supply Maintenance page not available:
 - 1) For the forward outflow valve inoperative:
 - a. Open and collar the P210 panel R FWD OFL VLV circuit breaker.
 - b. Open and collar the P310 panel L FWD OFL VLV circuit breaker.

- 2) For the aft outflow valve inoperative:
 - a. Open and collar the P210 panel R AFT OFL VLV circuit breaker.
 - b. Open and collar the P310 panel L AFT OFL VLV circuit breaker.
 - 3) Gain access to the associated outflow valve assembly.
 - 4) Disconnect the electrical connector from the back of the valve control unit.
 - 5) Remove one DC motor from the outflow valve assembly.
 - 6) Move the outflow valve doors until the gap between the two doors is 0.375 ± 0.031 inches.
 - 7) Re-install the DC motor.
 - 8) Reconnect the electrical connector on the back of the valve control unit.
 - 9) Position the associated Pressurization Panel (P5) OUTFLOW VALVE switch to AUTO.
3. Verify manual cabin pressure control operates normally on the remaining outflow valve.
- A. Position the associated Pressurization Panel (P5) OUTFLOW VALVE switch to MAN.

NOTE: Outflow valve position indications on EICAS may blank when MAN is selected.
 - B. Hold the associated Pressurization Panel (P5) OUTFLOW VALVE MANUAL switch CLOSE until the EICAS outflow valve position indicator is CL, or if EICAS indication is not available hold the switch CLOSE for 30 seconds.
 - C. Confirm the associated outflow valve door is closed.
 - D. Hold the associated Pressurization Panel (P5) OUTFLOW VALVE MANUAL switch OPEN until the EICAS outflow valve position indicator is OP, or if EICAS indication is not available hold the switch OPEN for 30 seconds.
 - E. Confirm the associated outflow valve door is full open.
 - F. Position the associated Pressurization Panel (P5) OUTFLOW VALVE switch to AUTO.

OPERATIONS (O)

NOTE 1: The associated OUTFLOW VALVE AFT or FWD advisory message will be displayed.

NOTE 2: The outflow valve position indication pointer will be missing for the deactivated outflow valve on the EICAS pressurization indications.

1. During passenger loading, if both packs are operating, select right pack OFF prior to closing the main entry door(s). This avoids uncomfortable changes in cabin pressure.

2. After all main entry doors are closed at the parking stand or departure gate, if both packs are operating, select right pack OFF before reopening a main entry door. After leaving the parking stand or gate, select right pack AUTO.
3. Rigid cargo barrier doors remain closed inflight.
4. If the operative outflow valve fails during flight and cannot be controlled manually it may be necessary to select packs OFF to depressurize the airplane for landing.
 - A. During descent when below 15,000 ft. MSL, select one pack OFF.
 - B. During descent when below 10,000 ft. MSL, select remaining pack OFF.
5. For forward outflow valve deactivated:
 - A. For FIRE CARGO MAIN DECK warning message, the FWD OUTFLOW VALVE is already considered closed.

For Aft Cargo Air Conditioning (ACAC) installed:

 - B. For FIRE CARGO FWD warning message, accomplish the associated non-normal checklist and also set LWR CARGO TEMP AFT A/C control OFF.
6. For aft outflow valve deactivated:

For Aft Cargo Air Conditioning (ACAC) installed:

 - A. For FIRE CARGO AFT warning message, accomplish the associated non-normal checklist and also set LWR CARGO TEMP FWD A/C control OFF.

For Aft Cargo Air Conditioning (ACAC) not installed:

 - B. For FIRE CARGO AFT warning message, accomplish the associated non-normal checklist and also set FWD CARGO A/C control OFF.
7. Prior to stand or gate arrival, if both packs are operating, select right pack OFF. After main entry doors are open at the stand or gate, select packs as desired.

21-31-03 Outflow Valves (FWD and AFT)

21-31-03-03 777F

21-31-03-03-02 With P/N 3674-GRS-106-00 or Later ASCPC Software Installed

Interval	Installed	Required	Procedure
C	2	1	(M) (O)

One may be inoperative provided:

- a. Valve is deactivated 7% open.
- b. Manual cabin pressure control is verified to operate normally on the remaining outflow valve.
- c. Left or right auto cabin pressure control operates normally.



- d. One pack operates normally, and the other pack operates normally or in standby cooling mode.
 - e. Both CTCs operate normally.
 - f. Both engine bleed systems operate normally.
-

MAINTENANCE (M)

NOTE 1: PRR 62273, Part C, installs ASCPC-2 software P/N 3674-GRS-106-00 on L/N 998 and-on. Service Bulletin 777-36-0032 provides the production equivalent of PRR 62273 by installing ASCPC-2 software P/N 3674-GRS-106-00.

NOTE 2: For a pack to be operating normally, certain items associated with that system must be operative. See DDG item 21-51-01 for a listing of those items.

NOTE 3: A pack will operate in standby cooling mode with an inoperative ACM, or an inoperative ECV, or an inoperative Condenser Inlet Temperature Control System.

NOTE 4: For an engine bleed system to be operating normally, certain items associated with that system must be operative. See DDG item 36-00-01 for a listing of those items.

Deactivate inoperative outflow valve 7% open and verify manual cabin pressure control operates normally on the remaining operative outflow valve (AMM 21-00-00/901).

NOTE: For ground operations with OAT at or above 24 degrees C and forward outflow valve deactivated 7% open, supply conditioned air to the airplane.

1. Supply electrical power on the airplane (AMM 24-22-00/201).
2. Deactivate inoperative outflow valve 7% open.
 - A. For manual control operative and valve position displayed on the Air Supply Maintenance page:
 - 1) Position the associated Pressurization Panel (P5) OUTFLOW VALVE switch to MAN.
 - 2) Using the associated Pressurization Panel (P5) OUTFLOW VALVE MANUAL switch, position the inoperative outflow valve to indicate 0.07 on the Air Supply Maintenance page.
 - 3) For the forward outflow valve inoperative:
 - a. Open and collar the P210 panel R FWD OFL VLV circuit breaker.
 - b. Open and collar the P310 panel L FWD OFL VLV circuit breaker.

- 4) For the aft outflow valve inoperative:
 - a. Open and collar the P210 panel R AFT OFL VLV circuit breaker.
 - b. Open and collar the P310 panel L AFT OFL VLV circuit breaker.

NOTE: The circuit breakers must be opened prior to positioning the OUTFLOW VALVE switch to AUTO. If the OUTFLOW VALVE switch is positioned to AUTO before opening the circuit breakers the associated outflow valve will move to the fully open position and the airplane will be unable to pressurize in flight.

- 5) Position the associated Pressurization Panel (P5) OUTFLOW VALVE switch to AUTO.
- 6) Confirm the gap between the two outflow doors is 0.375 ± 0.031 inches.

NOTE: Confirming the gap between the outflow doors ensures the outflow valve is deactivated in the correct position for pressurized flight.

- B. For OFV (P/N 2214420-X) installed with manual control inoperative, or valve position on the Air Supply Maintenance page not available:

- 1) For the forward outflow valve inoperative:
 - a. Open and collar the P210 panel R FWD OFL VLV circuit breaker.
 - b. Open and collar the P310 panel L FWD OFL VLV circuit breaker.
- 2) For the aft outflow valve inoperative:
 - a. Open and collar the P210 panel R AFT OFL VLV circuit breaker.
 - b. Open and collar the P310 panel L AFT OFL VLV circuit breaker.
- 3) Gain access to the associated outflow valve assembly.
- 4) Disconnect the electrical connector from the back of the valve control unit.
- 5) Remove one DC motor from the outflow valve assembly.
- 6) Move the outflow valve doors until the gap between the two doors is 0.375 ± 0.031 inches.
- 7) Re-install the DC motor.
- 8) Reconnect the electrical connector on the back of the valve control unit.
- 9) Position the associated Pressurization Panel (P5) OUTFLOW VALVE switch to AUTO.



3. Verify manual cabin pressure control operates normally on the remaining outflow valve.
 - A. Position the associated Pressurization Panel (P5) OUTFLOW VALVE switch to MAN.

NOTE: Outflow valve position indications on EICAS may blank when MAN is selected.
 - B. Hold the associated Pressurization Panel (P5) OUTFLOW VALVE MANUAL switch CLOSE until the EICAS outflow valve position indicator is CL, or if EICAS indication is not available hold the switch CLOSE for 30 seconds.
 - C. Confirm the associated outflow valve door is closed.
 - D. Hold the associated Pressurization Panel (P5) OUTFLOW VALVE MANUAL switch OPEN until the EICAS outflow valve position indicator is OP, or if EICAS indication is not available hold the switch OPEN for 30 seconds.
 - E. Confirm the associated outflow valve door is full open.
 - F. Position the associated Pressurization Panel (P5) OUTFLOW VALVE switch to AUTO.

OPERATIONS (O)

NOTE 1: The associated OUTFLOW VALVE AFT or FWD advisory message will be displayed.

NOTE 2: The outflow valve position indication pointer will be missing for the deactivated outflow valve on the EICAS pressurization indications.

1. During passenger loading, if both packs are operating, select right pack OFF prior to closing the main entry door(s). This avoids uncomfortable changes in cabin pressure.
2. After all main entry doors are closed at the parking stand or departure gate, if both packs are operating, select right pack OFF before reopening a main entry door. After leaving the parking stand or gate, select right pack AUTO.
3. If the operative outflow valve fails during flight and cannot be controlled manually it may be necessary to select packs OFF to depressurize the airplane for landing.
 - A. During descent when below 15,000 ft. MSL, select one pack OFF.
 - B. During descent when below 10,000 ft. MSL, select remaining pack OFF.
4. For forward outflow valve deactivated:

For Aft Cargo Air Conditioning (ACAC) installed:

 - A. For FIRE CARGO FWD warning message, accomplish the associated non-normal checklist and also set LWR CARGO TEMP AFT A/C control OFF.
5. For aft outflow valve deactivated:

For Aft Cargo Air Conditioning (ACAC) installed:

- A. For FIRE CARGO AFT warning message, accomplish the associated non-normal checklist and also set LWR CARGO TEMP FWD A/C control OFF.

For Aft Cargo Air Conditioning (ACAC) not installed:

- B. For FIRE CARGO AFT warning message, accomplish the associated non-normal checklist and also set FWD CARGO A/C control OFF.
6. Prior to stand or gate arrival, if both packs are operating, select right pack OFF. After main entry doors are open at the stand or gate, select packs as desired.

21-31-03 Outflow Valves (FWD and AFT)

21-31-03-04 All

Interval	Installed	Required	Procedure
C	2	0	(M) (O)

May be inoperative provided:

- a. Flight is conducted unpressurized.
- b. Procedures are established and used to verify cargo compartments remain empty or contain only ballast, empty cargo handling equipment (ballast may be loaded in ULDs), or fly away kits.
- c. Extended overwater flight is prohibited.

For LLAR, OFCR, OFAR, MDCR or MDFCR installed:

- d. The rest area(s) is considered inoperative.

For first class suites installed:

- e. First class suites are considered inoperative.

NOTE: Operator MELs must define which items are approved for inclusion in the fly away kits and which materials can be used as ballast.

NOTE 1: First class suites refers to the High Wall First Class Passenger Suite installation (option number 2525E407E69).

For LLAR, OFCR, OFAR, MDCR or MDFCR installed:

NOTE 2: The airplane must also be dispatched using MMEL item 25-29-04.

For first class suites installed:

NOTE 3: The airplane must also be dispatched using MMEL item 25-25-05.

MAINTENANCE (M)

NOTE: For both outflow valve status messages displayed due to faults in a single channel in both valves, it may be possible to achieve operation of one outflow valve on both channels by exchanging components. Consult airplane CMC data, and appropriate fault isolation manual (FIM) and AMM.

Deactivate both outflow valves full open (AMM 21-31-00/201).

1. Determine position of the outflow valves.
2. For outflow valve(s) in the full open position:
 - A. Open and collar the P210 panel R FWD OFL VLV circuit breaker.
 - B. Open and collar the P310 panel L FWD OFL VLV circuit breaker.
 - C. Open and collar the P210 panel R AFT OFL VLV circuit breaker.
 - D. Open and collar the P310 panel L AFT OFL VLV circuit breaker.
 - E. Position the associated Pressurization Panel (P5) OUTFLOW VALVE switch(es) to AUTO.
3. For outflow valve(s) not in the full open position and manual control is available:
 - A. Position the associated Pressurization Panel (P5) OUTFLOW VALVE switch(es) to MAN.
 - B. Hold the associated Pressurization Panel (P5) OUTFLOW VALVE MANUAL switch(es) OPEN until the EICAS outflow valve position indicator is OP, or if EICAS indication is not available hold the switch OPEN for 30 seconds.
 - C. For the EICAS outflow valve position indication(s) not available, confirm the associated outflow door(s) is full open.
 - D. Open and collar the P210 panel R FWD OFL VLV circuit breaker.
 - E. Open and collar the P310 panel L FWD OFL VLV circuit breaker.
 - F. Open and collar the P210 panel R AFT OFL VLV circuit breaker.
 - G. Open and collar the P310 panel L AFT OFL VLV circuit breaker.
 - H. Position the associated Pressurization Panel (P5) OUTFLOW VALVE switch(es) to AUTO.
4. For outflow valve(s) not in the full open position and manual control is not available:
 - A. Open and collar the P210 panel R FWD OFL VLV circuit breaker.
 - B. Open and collar the P310 panel L FWD OFL VLV circuit breaker.
 - C. Open and collar the P210 panel R AFT OFL VLV circuit breaker.
 - D. Open and collar the P310 panel L AFT OFL VLV circuit breaker.
 - E. Gain access to the associated outflow valve assembly.
 - F. Remove the primary and secondary pushrods.
 - G. Position the valve door(s) in the full open position.
 - H. Install the hold-open rod, part number J21006.



- I. Position the associated Pressurization Panel (P5) OUTFLOW VALVE switch(es) to AUTO.

OPERATIONS (O)

NOTE: The CABIN ALTITUDE AUTO caution message will be displayed.

1. The flight is conducted unpressurized.
2. Extended overwater flight is prohibited.
3. Cargo compartments must remain empty, except for ballast, empty cargo containers (ballast may be loaded in ULDs), fly away kits, pallets, and cargo restraint components.
4. For flight with passengers:
 - A. Limit enroute altitude to 10,000 feet.
 - B. To avoid passenger discomfort, limit climb (after takeoff obstacle clearance has been assured) and descent rates to 500 fpm.

21-31-03 Outflow Valves (FWD and AFT)**21-31-03-05 OUTFLOW VALVE Switch Lights****21-31-03-05-01 MAN Lights**

Interval	Installed	Required	Procedure
C	2	0	

21-31-03 Outflow Valves (FWD and AFT)**21-31-03-05 OUTFLOW VALVE Switch Lights****21-31-03-05-02 AUTO Lights**

Interval	Installed	Required	Procedure
C	2	0	



777 FAA DDG Supplement Package

Supplement Number: 36-2

Issue Date: 12-APR-2024

Section Number: 2 (MMEL)

Item Number: 21-51-01

Item Name: Air Conditioning Packs

DDG Effectivity: 777 FAA DDG (D630W003-TBC), Revision 36

MMEL Effectivity: 777 FAA MMEL, Revision 23a

Description: Modified the pack operative table in the Maintenance (M) procedures/note for subitem 21-51-01-01 through 21-51-01-05B.

Reason: To include the updated the EICAS messages (post AIMS v14) and subitem numbers associated with the Pack Flow Control/Shutoff Valves (FCVs) item.

Disposition: For reference in the Operator's MEL.

Attachments: DDG page 2.21-51-01.1-18

**21-51-01 Air Conditioning Packs****21-51-01-01 -200/-200ER (Without PRR 61980 or Production Equivalent Incorporated)**

Interval	Installed	Required	Procedure
C	2	1	(M) (O)

One may be inoperative provided:

- Remaining pack operates normally.
- Both engine bleed systems operate normally.
- Both outflow valves operate normally.
- Airplane remains at FL 350 or below.
- Flight remains within 60 minutes of landing at a suitable airport.
- Appropriate performance adjustments are applied.
- For FSI OHAR installed, the FSI OHAR is deactivated closed.
- JAMCO flight deck security door part number XXX 9721-1 is not installed.

NOTE: These provisions are not intended to prohibit attendant rest facility inspections by crew members.

NOTE 1: The Overhead Attendant Rest (OHAR) installation manufactured by Flight Structures, Inc. (FSI) is not a Boeing system. Information applicable to the FSI installation is not provided.

NOTE 2: PRR 61980 installs enhancements for dispatch with a single pack, or dispatch with a single engine bleed system, on airplane L/N's 423, 429, 454, 455, 457 thru 461, and 463 and-on. Installation of the following Service Bulletins provides the production equivalent of PRR 61980.

Boeing Service Bulletin	Item
777-21-0075	ECSMC Software P/N 210E-BCE-01A-09 or later
777-21-0086	Return Air Grill Baffles
777-25-0227 or 777-25-0327	Fwd Cargo Floor Panels and Roller Trays
777-29-0027	HYDIM Software P/N 297A-BCD-000-27 or later
777-31-0086	WEU Software P/N 3105-BCE-01C-10 or later
777-36-0016	ASCPC-1 Software P/N 3677-GRS-015-00 or later

MAINTENANCE (M)

NOTE 1: For a pack to be operating normally, the following associated items must be operative (AMM 21-00-00/901):

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MMEL Item	Associated Status Message
21-51-01	PACK L or R
21-51-02-01	PACK FCV UPR L or R
21-51-02-02	PACK FCV LWR L or R
21-52-01	PACK ACM L or R
21-52-02	ECON COOL VALVE L or R
21-52-03	CMPRSR SENSOR L or R
21-52-04	PACK OUT TEMP SNSR L or R
21-52-05	RAM AIR INLET DOOR L or R
21-52-06	RAM AIR EXIT DOOR L or R
21-52-07	COND TEMP SENSOR L or R

NOTE 2: For an engine bleed system to be operating normally, certain items associated with that system must be operative. See DDG item 36-00-01 for a listing of those items.

1. For APU pneumatic duct pressure seal P/N 93C10353-9 or earlier installed, inspect the APU pneumatic duct pressure seal (boot) per 777-SL-36-005.

NOTE: Airplanes L/N 212 and on were delivered with improved seal P/N 93C10353-13 installed and do not require this inspection.

OPERATIONS (O)

NOTE 1: Aft and bulk cargo heating may be affected. Depending on cargo type and ambient conditions, compartment temperatures may be as cold as 55 degrees F (13 degrees C) with HIGH selected.

NOTE 2: If the air cycle machine fails on the remaining pack, the pack will operate continuously in standby cooling mode, regardless of altitude or outside air temperature, to maintain cabin pressurization. A single pack in standby cooling mode is capable of maintaining cabin pressurization at cruise altitude.

For Lower Lobe Attendant Rest (LLAR), Overhead Flight Crew Rest (OFCR), Overhead Flight Attendant Rest (OFAR), Main Deck Crew Rest (MDCR) or Main Deck Flight Crew Rest (MDFCR) installed:

NOTE 3: For left pack inoperative or off, temperature control in the crew/attendant rest(s) may be degraded.

For Forward Cargo Air Conditioning (FCAC) installed:

NOTE 4: FCAC temperature control is degraded. Temperature sensitive cargo should not be carried in the forward cargo compartment.



NOTE 5: For APU-to-Pack takeoff function installed, if left pack is inoperative the APU-to-Pack Takeoff Supplementary Procedure is not allowed.

1. Position associated pack switch OFF.
2. Reduce performance limited weights by the appropriate adjustments.

NOTE: The takeoff performance limited weight reductions may be avoided by performing a Packs Off Takeoff or for a right pack inoperative, performing an APU-to-Pack Takeoff (if installed) Supplementary Procedure.

Performance Limited Weights	Takeoff & Landing	Enroute Climb
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For -200 or -200ER/GE90-76B thru -90B:

thru 632,500 lb (286,902 kg)	14,300 lb (6,486 kg)	7,700 lb (3,493 kg)
over 632,500 lb (286,902 kg)	20,700 lb (9,389 kg)	7,900 lb (3,583 kg)

For -200 or -200ER/GE90-94B:

thru 632,500 lb (286,902 kg)	14,300 lb (6,486 kg)	9,400 lb (4,264 kg)
over 632,500 lb (286,902 kg)	22,400 lb (10,160 kg)	9,400 lb (4,264 kg)

For -200 or -200ER/PW4074 thru PW4084:

All	21,800 lb (9,888 kg)	8,600 lb (3,901 kg)
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For -200 or -200ER/PW4090:

thru 632,500 lb (286,902 kg)	21,500 lb (9,752 kg)	7,200 lb (3,266 kg)
over 632,500 lb (286,902 kg)	26,500 lb (12,020 kg)	7,200 lb (3,266 kg)

For -200 or -200ER/RR875 thru RR892:

thru 632,500 lb (286,902 kg)	6,400 lb (2,903 kg)	No penalty
over 632,500 lb (286,902 kg)	8,200 lb (3,720 kg)	No penalty

For -200 or -200ER/RR895:

All	6,800 lb (3,084 kg)	No penalty
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NOTE: The enroute climb weight penalties are based on single engine operating speeds that approximate maximum lift-to-drag ratio speed. To account for the difference in level off altitude when operating at other speeds, multiply the enroute climb weight penalty listed above by the appropriate factor listed on page 3.00-01-00.2 in the ENROUTE DIVERSION SPEED EFFECTS paragraph.

3. Flight must remain within 60 minutes of landing at a suitable airport.
4. Flight must remain at or below FL 350.
5. Operate the wing anti-ice manually.
 - A. Position WING ANTI-ICE selector OFF.
 - B. For flaps up and wing anti-ice required (do not use during takeoff and landing):
 - 1) Position the ENG Bleed switch associated with the inoperative pack OFF.

NOTE: The ENG IDLE DISAGREE advisory message will be displayed when idle thrust is selected inflight.
 - 2) Position WING ANTI-ICE selector ON, then OFF as required.
 - 3) Prior to landing, position the ENG Bleed switch associated with the inoperative pack ON.
6. For a FIRE CARGO AFT or FIRE CARGO FWD warning message, accomplish the associated Non-Normal Checklist and the following:
 - A. If above FL 330, descend to FL 330.
 - B. Position both ENGINE ANTI-ICE selectors ON at top-of-descent.

For Overhead Flight Attendant Rest (OFAR) installed:

7. For smoke detected in OFAR:
 - A. Position AFT and BULK CARGO TEMP selectors OFF.

For Forward Cargo Air Conditioning (FCAC) installed:

- B. Position FWD CARGO A/C control OFF.

21-51-01 Air Conditioning Packs

21-51-01-02 -300 (Without PRR 61980 or Production Equivalent Incorporated)

Interval	Installed	Required	Procedure
C	2	1	(O)

One may be inoperative provided:

- a. Remaining pack operates normally.
- b. Both engine bleed systems operate normally.
- c. Both outflow valves operate normally.



- d. Procedures are established and used to verify forward cargo compartment remains empty or contains only ballast, empty cargo handling equipment, (ballast may be loaded in ULDs), or fly away kits.
- e. Flight remains within 60 minutes of landing at a suitable airport.
- f. Appropriate performance adjustments are applied.

NOTE: Operator MELs must define which items are approved for inclusion in the fly away kits and which materials can be used as ballast.

NOTE: PRR 61980 installs enhancements for dispatch with a single pack, or dispatch with a single engine bleed system, on airplane L/N's 423, 429, 454, 455, 457 thru 461, and 463 and-on. Installation of the following Service Bulletins provides the production equivalent of PRR 61980.

Boeing Service Bulletin	Item
777-21-0075	ECSMC Software P/N 210E-BCE-01A-09 or later
777-21-0086	Return Air Grill Baffles
777-25-0227 or 777-25-0327	Fwd Cargo Floor Panels and Roller Trays
777-29-0027	HYDIM Software P/N 297A-BCD-000-27 or later
777-31-0086	WEU Software P/N 3105-BCE-01C-10 or later
777-36-0016	ASCPC-1 Software P/N 3677-GRS-015-00 or later

MAINTENANCE NOTE

1. For a pack to be operating normally, the following associated items must be operative:

MMEL Item	Associated Status Message
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21-51-01	PACK L or R
21-51-02-01	PACK FCV UPR L or R
21-51-02-02	PACK FCV LWR L or R
21-52-01	PACK ACM L or R
21-52-02	ECON COOL VALVE L or R
21-52-03	CMPRSR SENSOR L or R
21-52-04	PACK OUT TEMP SNSR L or R
21-52-05	RAM AIR INLET DOOR L or R
21-52-06	RAM AIR EXIT DOOR L or R
21-52-07	COND TEMP SENSOR L or R

- For an engine bleed system to be operating normally, certain items associated with that system must be operative. See DDG item 36-00-01 for a listing of those items.

OPERATIONS (O)

NOTE 1: Aft and bulk cargo heating may be affected. Depending on cargo type and ambient conditions, compartment temperatures may be as cold as 55 degrees F (13 degrees C) with HIGH selected.

NOTE 2: If the air cycle machine fails on the remaining pack, the pack will operate continuously in standby cooling mode, regardless of altitude or outside air temperature, to maintain cabin pressurization. A single pack in standby cooling mode is capable of maintaining cabin pressurization at cruise altitude.

For Lower Lobe Attendant Rest (LLAR), Overhead Flight Crew Rest (OFCR), Overhead Flight Attendant Rest (OFAR), Main Deck Crew Rest (MDCR) or Main Deck Flight Crew Rest (MDFCR) installed:

NOTE 3: For left pack inoperative or off, temperature control in the associated crew rest may be degraded.

NOTE 4: For APU-to-Pack takeoff function installed, if left pack is inoperative the APU-to-Pack Takeoff Supplementary Procedure is not allowed.

- Position associated pack switch OFF.
- Reduce takeoff performance limited weights by the appropriate adjustments.

NOTE: The takeoff performance limited weight reductions may be avoided by performing a Packs Off Takeoff or for a right pack inoperative, performing an APU-to-Pack Takeoff (if installed) Supplementary Procedure.



Performance Limited Weights	Takeoff & Landing	Enroute Climb
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For -300/PW4090:

All	21,800 lb (9,888 kg)	7,200 lb (3,266 kg)
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For -300/RR892:

All	7,000 lb (3,175 kg)	No penalty
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NOTE: The enroute climb weight penalties are based on single engine operating speeds that approximate maximum lift-to-drag ratio speed. To account for the difference in level off altitude when operating at other speeds, multiply the enroute climb weight penalty listed above by the appropriate factor listed on page 3.00-01-00.2 in the ENROUTE DIVERSION SPEED EFFECTS paragraph.

3. Flight must remain within 60 minutes of landing at a suitable airport.
4. The forward cargo compartment must remain empty, except for ballast, empty cargo containers (ballast may be loaded in ULDs), fly away kits, pallets, and cargo restraint components.
5. Operate the wing anti-ice manually.
 - A. Position WING ANTI-ICE selector OFF.
 - B. For flaps up and wing anti-ice required (do not use during takeoff and landing):
 - 1) Position the ENG Bleed switch associated with the inoperative pack OFF.

NOTE: The ENG IDLE DISAGREE advisory message will be displayed when idle thrust is selected inflight.
 - 2) Position WING ANTI-ICE selector ON, then OFF as required.
 - 3) Prior to landing, position the ENG Bleed switch associated with the inoperative pack ON.
6. For a FIRE CARGO AFT or FIRE CARGO FWD warning message, accomplish the associated Non-Normal Checklist and the following:
 - A. If above FL 330, descend to FL 330.
 - B. Position both ENGINE ANTI-ICE selectors ON at top-of-descent.

For Overhead Flight Attendant Rest (OFAR) installed:

7. For smoke detected in OFAR:
 - A. Position AFT and BULK CARGO TEMP selectors OFF.

For Forward Cargo Air Conditioning (FCAC) installed:

- B. Position FWD CARGO A/C control OFF.

21-51-01 Air Conditioning Packs
21-51-01-03 -200/-200ER/-300 (With PRR 61980 or Production Equivalent Incorporated)

Interval	Installed	Required	Procedure
C	2	1	(M) (O)

One may be inoperative provided:

- Remaining pack operates normally.
- Both engine bleed systems operate normally.
- Both outflow valves operate normally.
- Flight remains within 60 minutes of landing at a suitable airport.
- Appropriate performance adjustments are applied.
- For FSI OHAR installed on -200, the FSI OHAR is deactivated closed.

NOTE: These provisions are not intended to prohibit attendant rest facility inspections by crewmembers.

NOTE 1: The Overhead Attendant Rest (OHAR) installation manufactured by Flight Structures, Inc. (FSI) is not a Boeing system. Information applicable to the FSI installation is not provided.

NOTE 2: PRR 61980 installs enhancements for dispatch with a single pack, or dispatch with a single engine bleed system, on airplane L/N's 423, 429, 454, 455, 457 thru 461, and 463 and-on. Installation of the following Service Bulletins provides the production equivalent of PRR 61980.

Boeing Service Bulletin	Item
777-21-0075	ECSMC Software P/N 210E-BCE-01A-09 or later
777-21-0086	Return Air Grill Baffles
777-25-0227 or 777-25-0327	Fwd Cargo Floor Panels and Roller Trays
777-29-0027	HYDIM Software P/N 297A-BCD-000-27 or later
777-31-0086	WEU Software P/N 3105-BCE-01C-10 or later
777-36-0016	ASCPC-1 Software P/N 3677-GRS-015-00 or later

MAINTENANCE (M)

NOTE 1: For a pack to be operating normally, the following associated items must be operative (AMM 21-00-00/901):

MMEL Item	Associated Status Message
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21-51-01	PACK L or R
21-51-02-01	PACK FCV UPR L or R
21-51-02-02	PACK FCV LWR L or R
21-52-01	PACK ACM L or R
21-52-02	ECON COOL VALVE L or R
21-52-03	CMPRSR SENSOR L or R
21-52-04	PACK OUT TEMP SNSR L or R
21-52-05	RAM AIR INLET DOOR L or R
21-52-06	RAM AIR EXIT DOOR L or R
21-52-07	COND TEMP SENSOR L or R

NOTE 2: For an engine bleed system to be operating normally, certain items associated with that system must be operative. See DDG item 36-00-01 for a listing of those items.

For -200/-200ER:

1. For APU pneumatic duct pressure seal P/N 93C10353-9 or earlier installed, inspect the APU pneumatic duct pressure seal (boot) per 777-SL-36-005.

NOTE: Airplanes L/N 212 and on were delivered with improved seal P/N 93C10353-13 installed and do not require this inspection.

OPERATIONS (O)

NOTE 1: Aft and bulk cargo heating may be affected. Depending on cargo type and ambient conditions, compartment temperatures may be as cold as 55 degrees F (13 degrees C) with HIGH selected.

NOTE 2: If the air cycle machine fails on the remaining pack, the pack will operate continuously in standby cooling mode, regardless of altitude or outside air temperature, to maintain cabin pressurization. A single pack in standby cooling mode is capable of maintaining cabin pressurization at cruise altitude.

For Lower Lobe Attendant Rest (LLAR), Overhead Flight Crew Rest (OFCR), Overhead Flight Attendant Rest (OFAR), Main Deck Crew Rest (MDCR) or Main Deck Flight Crew Rest (MDFCR) installed:

NOTE 3: For left pack inoperative or off, temperature control in the associated crew rest may be degraded.

For Forward Cargo Air Conditioning (FCAC) installed:

NOTE 4: FCAC temperature control is degraded. Temperature sensitive cargo should not be carried in the forward cargo compartment.



NOTE 5: For APU-to-Pack takeoff function installed, if left pack is inoperative the APU-to-Pack Takeoff Supplementary Procedure is not allowed.

1. Position associated pack switch OFF.
2. Reduce performance limited weights by the appropriate adjustments.

NOTE: The takeoff performance limited weight reductions may be avoided by performing a Packs Off Takeoff or for a right pack inoperative, performing an APU-to-Pack Takeoff (if installed) Supplementary Procedure.

Performance Limited Weights	Takeoff & Landing	Enroute Climb
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For -200 or -200ER/GE90-76B thru -90B:

thru 632,500 lb (286,902 kg)	14,300 lb (6,486 kg)	7,700 lb (3,493 kg)
over 632,500 lb (286,902 kg)	20,700 lb (9,389 kg)	7,900 lb (3,583 kg)

For -200 or -200ER/GE90-94B:

thru 632,500 lb (286,902 kg)	14,300 lb (6,486 kg)	9,400 lb (4,264 kg)
over 632,500 lb (286,902 kg)	22,400 lb (10,160 kg)	9,400 lb (4,264 kg)

For -200 or -200ER/PW4074 thru PW4084:

All	21,800 lb (9,888 kg)	8,600 lb (3,901 kg)
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For -200 or -200ER/PW4090:

thru 632,500 lb (286,902 kg)	21,500 lb (9,752 kg)	7,200 lb (3,266 kg)
over 632,500 lb (286,902 kg)	26,500 lb (12,020 kg)	7,200 lb (3,266 kg)

For -200 or -200ER/RR875 thru RR892:

thru 632,500 lb (286,902 kg)	6,400 lb (2,903 kg)	No penalty
over 632,500 lb (286,902 kg)	8,200 lb (3,720 kg)	No penalty

For -200 or -200ER/RR895:

All	6,800 lb (3,084 kg)	No penalty
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For -300/PW4090:

All	21,800 lb (9,888 kg)	7,200 lb (3,266 kg)
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For -300/RR892:

All	7,000 lb (3,175 kg)	No penalty
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NOTE: The enroute climb weight penalties are based on single engine operating speeds that approximate maximum lift-to-drag ratio speed. To account for the difference in level off altitude when operating at other speeds, multiply the enroute climb weight penalty listed above by the appropriate factor listed on page 3.00-01-00.2 in the ENROUTE DIVERSION SPEED EFFECTS paragraph.

3. Flight must remain within 60 minutes of landing at a suitable airport.
4. Operate the wing anti-ice manually.
 - A. Position WING ANTI-ICE selector OFF.
 - B. For flaps up and wing anti-ice required (do not use during takeoff and landing):
 - 1) Position the ENG Bleed switch associated with the inoperative pack OFF.

NOTE: The ENG IDLE DISAGREE advisory message will be displayed when idle thrust is selected inflight.
 - 2) Position WING ANTI-ICE selector ON, then OFF as required.
 - 3) Prior to landing, position the ENG Bleed switch associated with the inoperative pack ON.
5. For a FIRE CARGO AFT or FIRE CARGO FWD warning message, accomplish the associated Non-Normal Checklist and the following:
 - A. If above FL 330, descend to FL 330.
 - B. Position both ENGINE ANTI-ICE selectors ON at top-of-descent.

For Overhead Flight Attendant Rest (OFAR) installed:

6. For smoke detected in OFAR:
 - A. Position AFT and BULK CARGO TEMP selectors OFF.

For Forward Cargo Air Conditioning (FCAC) installed:

- B. Position FWD CARGO A/C control OFF.

21-51-01 Air Conditioning Packs

21-51-01-04 -200LR/-300ER

Interval	Installed	Required	Procedure
C	2	1	(O)

One may be inoperative provided:

- a. Remaining pack operates normally.

- b. Both engine bleed systems operate normally.
- c. Both outflow valves operate normally.
- d. Flight remains within 60 minutes of landing at a suitable airport.
- e. Appropriate performance adjustments are applied.

MAINTENANCE NOTE

1. For a pack to be operating normally, the following associated items must be operative:

MMEL Item	Associated Status Message
21-51-01	PACK L or R
21-51-02-01	PACK FCV UPR L or R
21-51-02-02	PACK FCV LWR L or R
21-52-01	PACK ACM L or R
21-52-02	ECON COOL VALVE L or R
21-52-03	CMPRSR SENSOR L or R
21-52-04	PACK OUT TEMP SNSR L or R
21-52-05	RAM AIR INLET DOOR L or R
21-52-06	RAM AIR EXIT DOOR L or R
21-52-07	COND TEMP SENSOR L or R

2. For an engine bleed system to be operating normally, certain items associated with that system must be operative. See DDG item 36-00-01 for a listing of those items.

OPERATIONS (O)

NOTE 1: Aft and bulk cargo heating may be affected. Depending on cargo type and ambient conditions, compartment temperatures may be as cold as 55 degrees F (13 degrees C) with HIGH selected.

NOTE 2: If the air cycle machine fails on the remaining pack, the pack will operate continuously in standby cooling mode, regardless of altitude or outside air temperature, to maintain cabin pressurization. A single pack in standby cooling mode is capable of maintaining cabin pressurization at cruise altitude.

For Lower Lobe Attendant Rest (LLAR), Overhead Flight Crew Rest (OFCR), Overhead Flight Attendant Rest (OFAR), Main Deck Crew Rest (MDCR) or Main Deck Flight Crew Rest (MDFCR) installed:

NOTE 3: For left pack inoperative or off, temperature control in the associated crew rest may be degraded.

**For Forward Cargo Air Conditioning (FCAC) installed:**

NOTE 4: FCAC temperature control is degraded. Temperature sensitive cargo should not be carried in the forward cargo compartment.

NOTE 5: For APU-to-Pack takeoff function installed, if left pack is inoperative the APU-to-Pack Takeoff Supplementary Procedure is not allowed.

1. Position associated pack switch OFF.
2. Reduce performance limited weights by the appropriate adjustments.

NOTE: The takeoff performance limited weight reductions may be avoided by performing a Packs Off Takeoff or for a right pack inoperative, performing an APU-to-Pack Takeoff (if installed) Supplementary Procedure.

Performance Limited Weights	Takeoff & Landing	Enroute Climb
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For -200LR/GE90-100 Series:

All	14,800 lb (6,713 kg)	No penalty
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For -300ER/GE90-115B:

All	11,600 lb (5,262 kg)	No penalty
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3. Flight must remain within 60 minutes of landing at a suitable airport.
4. Operate the wing anti-ice manually.
 - A. Position WING ANTI-ICE selector OFF.
 - B. For flaps up and wing anti-ice required (do not use during takeoff and landing):
 - 1) Position the ENG Bleed switch associated with the inoperative pack OFF.

NOTE: The ENG IDLE DISAGREE advisory message will be displayed when idle thrust is selected inflight.
 - 2) Position WING ANTI-ICE selector ON, then OFF as required.
 - 3) Prior to landing, position the ENG Bleed switch associated with the inoperative pack ON.
5. For a FIRE CARGO AFT or FIRE CARGO FWD warning message, accomplish the associated Non-Normal Checklist and the following:
 - A. If above FL 330, descend to FL 330.
 - B. Position both ENGINE ANTI-ICE selectors ON at top-of-descent.

For Overhead Flight Attendant Rest (OFAR) installed:

6. For smoke detected in OFAR:
 - A. Position AFT and BULK CARGO TEMP selectors OFF.

For Forward Cargo Air Conditioning (FCAC) installed:

B. Position FWD CARGO A/C control OFF.

21-51-01 Air Conditioning Packs

21-51-01-05 777F

21-51-01-05A AVS Operative

Interval	Installed	Required	Procedure
C	2	1	(O)

One may be inoperative provided:

- Remaining pack operates normally.
- Both engine bleed systems operate normally.
- Both outflow valves operate normally.
- Alternate ventilation system operates normally.
- For left pack inoperative, right CTC operates normally.
- Appropriate performance adjustments are applied.

MAINTENANCE NOTE

NOTE 1: For a pack to be operating normally, the following associated items must be operative:

MMEL Item	Associated Status Message
21-51-01	PACK L or R
21-51-02-01	PACK FCV UPR L or R
21-51-02-02	PACK FCV LWR L or R
21-52-01	PACK ACM L or R
21-52-02	ECON COOL VALVE L or R
21-52-03	CMPRSR SENSOR L or R
21-52-04	PACK OUT TEMP SNSR L or R
21-52-05	RAM AIR INLET DOOR L or R
21-52-06	RAM AIR EXIT DOOR L or R
21-52-07	COND TEMP SENSOR L or R

NOTE 2: For an engine bleed system to be operating normally, certain items associated with that system must be operative. See DDG item 36-00-01 for a listing of those items.

**OPERATIONS (O)**

NOTE 1: If the air cycle machine fails on the remaining pack, the pack will operate continuously in standby cooling mode, regardless of altitude or outside air temperature, to maintain cabin pressurization. A single pack in standby cooling mode is capable of maintaining cabin pressurization at cruise altitude.

NOTE 2: For APU-to-Pack takeoff function installed, if left pack is inoperative the APU-to-Pack Takeoff Supplementary Procedure is not allowed.

For Forward Cargo Air Conditioning (FCAC) installed:

NOTE 3: FCAC temperature control is degraded. Temperature sensitive cargo should not be carried in the forward cargo compartment.

For Aft Cargo Air Conditioning (ACAC) installed:

NOTE 4: ACAC temperature control is degraded. Temperature sensitive cargo should not be carried in the aft cargo compartment.

1. Position associated pack switch OFF.
2. Reduce performance limited weights by the appropriate adjustments.

NOTE: The takeoff performance limited weight reductions may be avoided by performing a Packs Off Takeoff or for a right pack inoperative, performing an APU-to-Pack Takeoff (if installed) Supplementary Procedure.

Performance Limited Weights	Takeoff & Landing	Enroute Climb
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For 777F/GE90-100 Series:

All	14,800 lb (6,713 kg)	No penalty
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3. Operate the wing anti-ice manually.
 - A. Position WING ANTI-ICE selector OFF.
 - B. For flaps up and wing anti-ice required (do not use during takeoff and landing):
 - 1) Position the ENG Bleed switch associated with the inoperative pack OFF.

NOTE: The ENG IDLE DISAGREE advisory message will be displayed when idle thrust is selected inflight.
 - 2) Position WING ANTI-ICE selector ON, then OFF as required.
 - 3) Prior to landing, position the ENG Bleed switch associated with the inoperative pack ON.
4. For a FIRE CARGO AFT or FIRE CARGO FWD warning message, accomplish the associated Non-Normal Checklist and the following:
 - A. If above FL 330, descend to FL 330.

B. Position both ENGINE ANTI-ICE selectors ON at top-of-descent.

21-51-01 Air Conditioning Packs

21-51-01-05 777F

21-51-01-05B AVS Inoperative

Interval	Installed	Required	Procedure
C	2	1	(O)

One may be inoperative provided:

- Remaining pack operates normally.
- Both engine bleed systems operate normally.
- Both outflow valves operate normally.
- Flight remains within 60 minutes of landing at a suitable airport.
- For left pack inoperative, right CTC operates normally.
- Appropriate performance adjustments are applied.

MAINTENANCE NOTE

NOTE 1: For a pack to be operating normally, the following associated items must be operative:

MMEL Item	Associated Status Message
21-51-01	PACK L or R
21-51-02-01	PACK FCV UPR L or R
21-51-02-02	PACK FCV LWR L or R
21-52-01	PACK ACM L or R
21-52-02	ECON COOL VALVE L or R
21-52-03	CMPRSR SENSOR L or R
21-52-04	PACK OUT TEMP SNSR L or R
21-52-05	RAM AIR INLET DOOR L or R
21-52-06	RAM AIR EXIT DOOR L or R
21-52-07	COND TEMP SENSOR L or R

NOTE 2: For an engine bleed system to be operating normally, certain items associated with that system must be operative. See DDG item 36-00-01 for a listing of those items.

**OPERATIONS (O)**

NOTE 1: If the air cycle machine fails on the remaining pack, the pack will operate continuously in standby cooling mode, regardless of altitude or outside air temperature, to maintain cabin pressurization. A single pack in standby cooling mode is capable of maintaining cabin pressurization at cruise altitude.

NOTE 2: For APU-to-Pack takeoff function installed, if left pack is inoperative the APU-to-Pack Takeoff Supplementary Procedure is not allowed.

For Forward Cargo Air Conditioning (FCAC) installed:

NOTE 3: FCAC temperature control is degraded. Temperature sensitive cargo should not be carried in the forward cargo compartment.

For Aft Cargo Air Conditioning (ACAC) installed:

NOTE 4: ACAC temperature control is degraded. Temperature sensitive cargo should not be carried in the aft cargo compartment.

1. Position associated pack switch OFF.
2. Reduce performance limited weights by the appropriate adjustments.

NOTE: The takeoff performance limited weight reductions may be avoided by performing a Packs Off Takeoff or for a right pack inoperative, performing an APU-to-Pack Takeoff (if installed) Supplementary Procedure.

Performance Limited Weights	Takeoff & Landing	Enroute Climb
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For 777F/GE90-100 Series:

All	14,800 lb (6,713 kg)	No penalty
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3. Flight must remain within 60 minutes of landing at a suitable airport.
4. Operate the wing anti-ice manually.
 - A. Position WING ANTI-ICE selector OFF.
 - B. For flaps up and wing anti-ice required (do not use during takeoff and landing):
 - 1) Position the ENG Bleed switch associated with the inoperative pack OFF.

NOTE: The ENG IDLE DISAGREE advisory message will be displayed when idle thrust is selected inflight.
 - 2) Position WING ANTI-ICE selector ON, then OFF as required.
 - 3) Prior to landing, position the ENG Bleed switch associated with the inoperative pack ON.



5. For a FIRE CARGO AFT or FIRE CARGO FWD warning message, accomplish the associated Non-Normal Checklist and the following:
- A. If above FL 330, descend to FL 330.
 - B. Position both ENGINE ANTI-ICE selectors ON at top-of-descent.

21-51-01 Air Conditioning Packs**21-51-01-06 DELETED (All)**

Interval	Installed	Required	Procedure

Deleted relief in revision 23

21-51-01 Air Conditioning Packs**21-51-01-07 PACK Switch Lights****21-51-01-07-01 OFF Lights**

Interval	Installed	Required	Procedure
C	2	0	

21-51-01 Air Conditioning Packs**21-51-01-07 PACK Switch Lights****21-51-01-07-02 AUTO Lights**

Interval	Installed	Required	Procedure
C	2	0	



777 FAA DDG Supplement Package

Supplement Number: 36-2

Issue Date: 12-APR-2024

Section Number: 2 (MMEL)

Item Number: 22-11-01

Item Name: Autopilot Flight Director Computers (AFDC)

DDG Effectivity: 777 FAA DDG (D630W003-TBC), Revision 36

MMEL Effectivity: 777 FAA MMEL, Revision 23a

Description: Modified PBN guidance.

Reason: To correct and add more clarification for PBN operation capabilities.

Disposition: For reference in the Operator's MEL.

Attachments: DDG page 2.22-11-01.1-3

22-11-01 Autopilot Flight Director Computers (AFDC)
22-11-01A One Inoperative

Interval	Installed	Required	Procedure
C	3	2	(M) (O)

One may be inoperative provided:

- Associated AFDC is deactivated.
- Associated AFDC backdrive actuator is deactivated.
- Approach minimums do not require use of the associated autopilot.

MAINTENANCE (M)

NOTE: The AFDCs are interchangeable. The failed AFDC may be moved to the center position to maintain left and right backdrive functions operational.

Deactivate the inoperative AFDC and associated backdrive actuator (AMM 22-00-00/901).

- For the center AFDC inoperative, open and collar the P11 panel AFDC C circuit breaker.
- For the left AFDC inoperative:
 - Open and collar the P11 panel AFDC L circuit breaker.
 - Open and collar the P11 panel AFDC BACK DRIVE L circuit breaker.
- For the right AFDC inoperative:
 - Open and collar the P11 panel AFDC R circuit breaker.
 - Open and collar the P11 panel AFDC BACK DRIVE R circuit breaker.

OPERATIONS (O)

- Align the ADIRU.
- For the left or right AFDC inoperative, confirm the autopilot system is operational by engaging and disengaging the autopilot on the ground with flaps UP before each departure.
- The autoland system is not fail-operational and the NO LAND 3 advisory message will be displayed.

22-11-01 Autopilot Flight Director Computers (AFDC)
22-11-01B Two Inoperative

Interval	Installed	Required	Procedure
C	3	1	(M) (O)

Center and one other AFDC may be inoperative provided:

- Associated AFDC is deactivated.



- b. For left or right AFDC inoperative, the associated AFDC backdrive actuator is deactivated.
- c. Radio altimeter associated with the operative AFDC operates normally.
- d. Both flap/slat control lanes operate normally.
- e. All warning electronics system channels operate normally.
- f. Approach minimums do not require use of the autopilots.

MAINTENANCE (M)

NOTE: The AFDCs are interchangeable.

Deactivate the inoperative AFDCs and associated backdrive actuator (AMM 22-00-00/901).

1. Open and collar the P11 panel AFDC C circuit breaker.
2. For the left AFDC inoperative:
 - A. Open and collar the P11 panel AFDC L circuit breaker.
 - B. Open and collar the P11 panel AFDC BACK DRIVE L circuit breaker.
3. For the right AFDC inoperative:
 - A. Open and collar the P11 panel AFDC R circuit breaker.
 - B. Open and collar the P11 panel AFDC BACK DRIVE R circuit breaker.

OPERATIONS (O)

1. Align the ADIRU.
2. Confirm the autopilot system is operational by engaging and disengaging the autopilot on the ground with flaps UP before each departure.
3. The autoland system is inoperative and the NO AUTOLAND advisory message will be displayed. Autopilot may be engaged, however only one flight director is available.
4. For PBN operations:
 - A. RNP AR APCH operations are not allowed.
 - B. Other PBN operations may be conducted.

NOTE: Refer to DDG Section 4 for additional information.

22-11-01 Autopilot Flight Director Computers (AFDC)

22-11-01C All Inoperative

Interval	Installed	Required	Procedure
B	3	0	(M) (O)

May be inoperative provided:

- a. All AFDCs are deactivated.
- b. Both AFDC backdrive actuators are deactivated.
- c. Approach minimums do not require use of the autopilots.

- d. Number of flight segments and segment duration is acceptable to flight crew.
 - e. Enroute operations do not require use of the autopilots.
-

MAINTENANCE (M)

Deactivate the inoperative AFDCs and associated backdrive actuators (AMM 22-00-00/901).

1. Open and collar the P11 panel AFDC C circuit breaker.
2. Open and collar the P11 panel AFDC L circuit breaker.
3. Open and collar the P11 panel AFDC BACK DRIVE L circuit breaker.
4. Open and collar the P11 panel AFDC R circuit breaker.
5. Open and collar the P11 panel AFDC BACK DRIVE R circuit breaker.
6. Simultaneously open and close the P11 panel MCP L and MCP R circuit breakers to set the MCP IAS-MACH, HDG-TRK and V/S-FPA windows to their default states (IAS, HDG and V/S).

OPERATIONS (O)

1. Autopilot cannot be engaged, flight director is not available, and the autoland system is inoperative. No advisory messages will be displayed.
2. The selected speed and the speed bug will not be available on the PFD.
3. The autothrottle is available. However, the autothrottle engaged light will not illuminate when the autothrottle is engaged.
4. For PBN operations:
 - PBN operations are not allowed.

NOTE: Refer to DDG Section 4 for additional information.



777 FAA DDG Supplement Package

Supplement Number: 36-2

Issue Date: 12-APR-2024

Section Number: 2 (MMEL)

Item Number: 22-11-06

Item Name: Mode Control Panel Switches

DDG Effectivity: 777 FAA DDG (D630W003-TBC), Revision 36

MMEL Effectivity: 777 FAA MMEL, Revision 23a

Description: Modified PBN guidance.

Reason: To correct and add more clarification for PBN operation capabilities.

Disposition: For reference in the Operator's MEL.

Attachments: DDG page 2.22-11-06.1-7

**22-11-06 Mode Control Panel Switches****22-11-06-01 A/P Engage Switches****22-11-06-01A One Inoperative**

Interval	Installed	Required	Procedure
C	2	1	

OPERATIONS NOTE

For PBN operations:

1. RNP AR APCH operations using autopilot are not allowed.
2. RNP AR APCH operations may be conducted using flight director provided its use is appropriate for the intended operation.
3. Other PBN operations using autopilot may be conducted.

NOTE: Refer to DDG Section 4 for additional information.

22-11-06 Mode Control Panel Switches**22-11-06-01 A/P Engage Switches****22-11-06-01B Both Inoperative**

Interval	Installed	Required	Procedure
B	2	0	(O)

May be inoperative provided:

- a. Approach minimums do not require use of the autopilots.
- b. Number of flight segments and segment duration is acceptable to flight crew.
- c. Enroute operations do not require use of the autopilots.

OPERATIONS (O)

1. Autopilot cannot be engaged. Flight director may be used if available.
2. For PBN operations:
 - A. PBN operations using autopilot are not allowed.
 - B. PBN operations may be conducted using flight director provided its use is appropriate for the intended operation.

NOTE: Refer to DDG Section 4 for additional information.

22-11-06 Mode Control Panel Switches

22-11-06-02 Autothrottle Arm Switches (A/T ARM L, R)

22-11-06-02-01 PW and RR

22-11-06-02-01A One Inoperative

Interval	Installed	Required	Procedure
C	2	1	(M) (O)

One may be inoperative provided:

- Associated servo motor is deactivated.
- Both thrust levers are manually set for takeoff and go-around thrust.
- Flight remains within 180 minutes of landing at a suitable airport.

MAINTENANCE (M)

Deactivate the associated autothrottle servo motor (AMM 22-00-00/901).

- For the left A/T ARM switch inoperative, open and collar P11 panel A/T SERVO L circuit breaker.
- For the right A/T ARM switch inoperative, open and collar P11 panel A/T SERVO R circuit breaker.

OPERATIONS (O)

- The AUTOTHROTTLE L or R advisory message will be displayed.
- The autothrottle cannot be engaged on the ground.
- Set both thrust levers manually for takeoff and go-around.
- The flight must remain within 180 minutes of landing at a suitable airport.
- For an inoperative switch failed to the armed state, the EPR trim function will operate normally, provided the operative autothrottle arm switch is positioned to A/T ARM.

NOTE: The autothrottle disconnect caution will be annunciated if the flight crew selects FLCH, VNAV, or TO/GA. Pressing the autothrottle disconnect switch will clear the caution.

- For an inoperative switch failed to the off state, autothrottle functionality is available in flight for the operative autothrottle.

**22-11-06 Mode Control Panel Switches****22-11-06-02 Autothrottle Arm Switches (A/T ARM L, R)****22-11-06-02-01 PW and RR****22-11-06-02-01B Both Inoperative**

Interval	Installed	Required	Procedure
C	2	0	

May be inoperative provided:

- Approach minimums do not require autothrottle use.
- Flight remains within 180 minutes of landing at a suitable airport.

OPERATIONS NOTE

- Autothrottle is inoperative.
- The flight must remain within 180 minutes of landing at a suitable airport.

22-11-06 Mode Control Panel Switches**22-11-06-02 Autothrottle Arm Switches (A/T ARM L, R)****22-11-06-02-02 GE****22-11-06-02-02A One Inoperative**

Interval	Installed	Required	Procedure
C	2	1	(M) (O)

One may be inoperative provided:

- Associated servo motor is deactivated.
- EEC normal mode operates normally.
- Both thrust levers are manually set for takeoff and go-around thrust.
- Flight remains within 180 minutes of landing at a suitable airport.

MAINTENANCE (M)

Deactivate the associated autothrottle servo motor (AMM 22-00-00/901).

- For the left A/T ARM switch inoperative, open and collar P11 panel A/T SERVO L circuit breaker.
- For the right A/T ARM switch inoperative, open and collar P11 panel A/T SERVO R circuit breaker.

OPERATIONS (O)

- The AUTOTHROTTLE L or R advisory message will be displayed.
- The autothrottle cannot be engaged on the ground.
- Set both thrust levers manually for takeoff and go-around.

4. The flight must remain within 180 minutes of landing at a suitable airport.
5. For an inoperative switch failed to the armed state, the N1 trim function will operate normally, provided the operative autothrottle arm switch is positioned to A/T ARM.

NOTE: The autothrottle disconnect caution will be annunciated if the flight crew selects FLCH, VNAV, or TO/GA. Pressing the autothrottle disconnect switch will clear the caution.

6. For an inoperative switch failed to the off state, autothrottle functionality is available in flight for the operative autothrottle.

22-11-06 Mode Control Panel Switches
22-11-06-02 Autothrottle Arm Switches (A/T ARM L, R)
22-11-06-02-02 GE
22-11-06-02-02B Both Inoperative

Interval	Installed	Required	Procedure
C	2	0	

May be inoperative provided:

- a. EEC normal mode operates normally.
- b. Approach minimums do not require autothrottle use.
- c. Flight remains within 180 minutes of landing at a suitable airport.

OPERATIONS NOTE

1. Autothrottle is inoperative.
2. The flight must remain within 180 minutes of landing at a suitable airport.

22-11-06 Mode Control Panel Switches
22-11-06-03 A/T Engage Switch
22-11-06-03-01 PW and RR

Interval	Installed	Required	Procedure
C	1	0	

May be inoperative provided:

- a. Approach minimums do not require autothrottle use.
- b. Flight remains within 180 minutes of landing at a suitable airport.

OPERATIONS NOTE

1. Any autothrottle mode which operates normally may be used.
2. The flight must remain within 180 minutes of landing at a suitable airport.

**22-11-06 Mode Control Panel Switches****22-11-06-03 A/T Engage Switch****22-11-06-03-02 GE**

Interval	Installed	Required	Procedure
C	1	0	

May be inoperative provided:

- EEC normal mode operates normally.
- Approach minimums do not require autothrottle use.
- Flight remains within 180 minutes of landing at a suitable airport.

OPERATIONS NOTE

- Any autothrottle mode which operates normally may be used.
- The flight must remain within 180 minutes of landing at a suitable airport.

22-11-06 Mode Control Panel Switches**22-11-06-04 F/D Switches**

Interval	Installed	Required	Procedure
C	2	0	

May be inoperative provided approach minimums do not require flight director use.

OPERATIONS NOTE

For PBN operations:

- One F/D switch inoperative:
 - RNP AR APCH operations using flight director are not allowed.
 - RNP AR APCH operations may be conducted using autopilot provided its use is appropriate for the intended operation.
 - Other PBN operations using flight director may be conducted provided flight director operates normally on the side of the pilot flying (PF).
- Both F/D switches inoperative:
 - PBN operations using flight director are not allowed.
 - PBN operations may be conducted using autopilot provided its use is appropriate for the intended operation.

NOTE: Refer to DDG Section 4 for additional information.

**22-11-06 Mode Control Panel Switches****22-11-06-05 IAS - MACH Reference Switch**

Interval	Installed	Required	Procedure
C	1	0	

May be inoperative provided IAS is displayed in associated window.

MAINTENANCE NOTE

For MACH displayed in the associated MCP window, simultaneously open and close P11 panel MCP L and MCP R circuit breakers to cause IAS to be displayed.

22-11-06 Mode Control Panel Switches**22-11-06-06 HDG - TRK Reference Switch**

Interval	Installed	Required	Procedure
C	1	0	

May be inoperative provided HDG is displayed in associated window.

MAINTENANCE NOTE

For TRK displayed in the associated MCP window, simultaneously open and close P11 panel MCP L and MCP R circuit breakers to cause HDG to be displayed.

22-11-06 Mode Control Panel Switches**22-11-06-07 V/S - FPA Reference Switch**

Interval	Installed	Required	Procedure
C	1	0	

MAINTENANCE NOTE

For FPA displayed in the associated MCP window, simultaneously open and close P11 panel MCP L and MCP R circuit breakers to cause V/S to be displayed.

22-11-06 Mode Control Panel Switches**22-11-06-08 APP Switch**

Interval	Installed	Required	Procedure
C	1	0	



May be inoperative provided approach minimums do not require use of the autopilot flight director system approach mode.

22-11-06 Mode Control Panel Switches**22-11-06-09 CLB CON, LNAV, VNAV, FLCH, VS/FPA, Heading/Track HOLD, Altitude HOLD, and LOC Switches**

Interval	Installed	Required	Procedure
C	8	0	

May be inoperative provided enroute operations do not require their use.

OPERATIONS NOTE

1. For enroute operations such as RVSM, any one of the following autopilot pitch modes will provide acceptable altitude hold capability: Altitude Hold (ALT), Vertical Speed (V/S) or Flight Path Angle (FPA) with 000 selected, Vertical Navigation (VNAV) with FMC flight plan, or VNAV with FMC flight plan combined with altitude intervention.
2. For PBN operations:
 - A. For lateral navigation (LNAV) switch inoperative:
PBN operations are not allowed.
 - B. For vertical navigation switch (VNAV) inoperative:
 - 1) RNP AR APCH operations are not allowed.
 - 2) RNP APCH operations may be conducted using other vertical modes provided appropriate approach minimums are used.
 - 3) Other PBN operations may be conducted.

NOTE: Refer to DDG Section 4 for additional information.

22-11-06 Mode Control Panel Switches**22-11-06-10 Arm/Engage Lights**

Interval	Installed	Required	Procedure
C	11	0	

May be inoperative provided the associated mode indications on both PFDs operate normally.



777 FAA DDG Supplement Package

Supplement Number: 36-2

Issue Date: 12-APR-2024

Section Number: 2 (MMEL)

Item Number: 22-11-08

Item Name: Control Wheel Autopilot Disconnect Switches

DDG Effectivity: 777 FAA DDG (D630W003-TBC), Revision 36

MMEL Effectivity: 777 FAA MMEL, Revision 23a

Description: Modified PBN guidance.

Reason: To correct and add more clarification for PBN operation capabilities.

Disposition: For reference in the Operator's MEL.

Attachments: DDG page 2.22-11-08.1

22-11-08 Control Wheel Autopilot Disconnect Switches
22-11-08A One Inoperative

Interval	Installed	Required	Procedure
C	2	1	

One may be inoperative provided:

- Autopilot is not used at 1,500 feet AGL or below.
- Approach minimums do not require the use of the autopilot.

OPERATIONS NOTE

- Do not use autopilot at 1,500 feet AGL or below.
- For PBN operations:
 - PBN operations using autopilot at 1,500 feet AGL or below are not allowed.
 - PBN operations may be conducted using flight director provided its use is appropriate for the intended operation.

NOTE: Refer to DDG Section 4 for additional information.

22-11-08 Control Wheel Autopilot Disconnect Switches
22-11-08B Both Inoperative

Interval	Installed	Required	Procedure
B	2	0	

May be inoperative provided:

- Autopilot is not used.
- Approach minimums do not require use of the autopilot.
- Number of flight segments and segment duration is acceptable to flight crew.
- Enroute operations do not require use of the autopilot.

OPERATIONS NOTE

- Enroute operations that require RVSM may not be conducted.
- For PBN operations:
 - PBN operations using autopilot are not allowed.
 - PBN operations may be conducted using flight director provided its use is appropriate for the intended operation.

NOTE: Refer to DDG Section 4 for additional information.



777 FAA DDG Supplement Package

Supplement Number: 36-2

Issue Date: 12-APR-2024

Section Number: 2 (MMEL)

Item Number: 22-11-09

Item Name: Takeoff/Go-Around (TO/GA) Switches

DDG Effectivity: 777 FAA DDG (D630W003-TBC), Revision 36

MMEL Effectivity: 777 FAA MMEL, Revision 23a

Description: Modified PBN guidance.

Reason: To correct and add more clarification for PBN operation capabilities.

Disposition: For reference in the Operator's MEL.

Attachments: DDG page 2.22-11-09.1

**22-11-09 Takeoff/Go-Around (TO/GA) Switches****22-11-09A One Inoperative**

Interval	Installed	Required	Procedure
C	2	1	

One may be inoperative provided approach minimums do not require its use.

OPERATIONS NOTE

1. The autoland system is not fail-operational. The NO LAND 3 advisory message may be displayed.
2. For PBN operations:
 - A. RNP AR APCH operations are not allowed.
 - B. Other PBN operations may be conducted.

NOTE: Refer to DDG Section 4 for additional information.

22-11-09 Takeoff/Go-Around (TO/GA) Switches**22-11-09B Both Inoperative**

Interval	Installed	Required	Procedure
C	2	0	

May be inoperative provided:

- a. Both thrust levers are operated manually for takeoff and go-around.
- b. Autopilot and flight director are not used on approach below 500 feet AGL or MDA, whichever is higher.

NOTE: Flight director go-around and windshear guidance are not available with both TO/GA switches inoperative.

OPERATIONS NOTE

1. The autoland system is not fail-operational. The NO LAND 3 or NO AUTOLAND advisory message may be displayed.
2. Flight director go-around and windshear guidance are not available with both TO/GA switches inoperative.
3. For PBN operations:

PBN operations using autopilot or flight director below 500 feet AGL or MDA are not allowed.

NOTE: Refer to DDG Section 4 for additional information.



777 FAA DDG Supplement Package

Supplement Number: 36-2

Issue Date: 12-APR-2024

Section Number: 2 (MMEL)

Item Number: 22-11-10

Item Name: Flight Director Systems

DDG Effectivity: 777 FAA DDG (D630W003-TBC), Revision 36

MMEL Effectivity: 777 FAA MMEL, Revision 23a

Description: Modified PBN guidance.

Reason: To correct and add more clarification for PBN operation capabilities.

Disposition: For reference in the Operator's MEL.

Attachments: DDG page 2.22-11-10.1

**22-11-10 Flight Director Systems**

Interval	Installed	Required	Procedure
C	2	0	

May be inoperative provided approach minimums do not require their use.

OPERATIONS NOTE

For PBN operations:

1. One Flight Director system inoperative:
 - A. RNP AR APCH operations using flight director are not allowed.
 - B. RNP AR APCH operations may be conducted using autopilot provided its use is appropriate for the intended operation.
 - C. Other PBN operations using flight director may be conducted provided flight director operates normally on the side of the pilot flying (PF).
2. Both Flight Director systems inoperative:
 - A. PBN operations using flight director are not allowed.
 - B. PBN operations may be conducted using autopilot provided its use is appropriate for the intended operation.

NOTE: Refer to DDG Section 4 for additional information.



777 FAA DDG Supplement Package

Supplement Number: 36-2

Issue Date: 12-APR-2024

Section Number: 2 (MMEL)

Item Number: 23-24-02

Item Name: Emergency Locator Transmitter (ELT) (Fixed)

DDG Effectivity: 777 FAA DDG (D630W003-TBC), Revision 36

MMEL Effectivity: 777 FAA MMEL, Revision 23a

Description: Added note and modified ELT and Remote ELT Switch Maintenance procedures.

Reason: To included maintenance procedures for various ELT models.

Disposition: For reference in the Operator's MEL.

Attachments: DDG page 2.23-24-02.1-3

**23-24-02 Emergency Locator Transmitter (ELT) (Fixed)**

23-24-02A Required By 14 CFR - Inoperative

Interval	Installed	Required	Procedure
A	1	0	(M)

May be inoperative provided:

- System is deactivated.
- Repairs are made within 90 consecutive calendar-days.

NOTE: Dispatch relief for the Portable (survival) ELT's is in MMEL item 25-63-03.

MAINTENANCE (M)

Deactivate the inoperative ELT (AMM 23-00-00/901).

- Gain access to the inoperative ELT in the section 47 crown.
- For ACR/Artex model ELTs, remove the ELT, disconnect the wire harnesses connected to the ELT, re-install disconnected battery and ELT. Install a placard stating "ELT not installed" in view of pilot.
- For ELTA and Honeywell model ELTs, set the ELT (LRU) front panel switch to OFF.

OPERATIONS NOTE

See 14 CFR 91.207, 121.353, 121.339 or other appropriate regulations.

23-24-02 Emergency Locator Transmitter (ELT) (Fixed)

23-24-02B Required By 14 CFR - Missing

Interval	Installed	Required	Procedure
A	1	0	(M)

May be missing provided:

- Placard stating "ELT not installed" is placed in view of the pilot.
- Repairs are made within 90 consecutive calendar-days.

NOTE: Dispatch relief for the Portable (survival) ELT's is in MMEL item 25-63-03.

MAINTENANCE (M)

Placard "ELT not installed" and place it in view of the pilot (AMM DDG 23-00-00/901).

**OPERATIONS NOTE**

See 14 CFR 91.207, 121.353, 121.339 or other appropriate regulations.

23-24-02 Emergency Locator Transmitter (ELT) (Fixed)*********23-24-02C Not Required By 14 CFR - Inoperative**

Interval	Installed	Required	Procedure
D	1	0	(M)

Any in excess of those required by 14 CFR may be inoperative provided system is deactivated.

NOTE: Dispatch relief for the Portable (survival) ELT's is in MMEL item 25-63-03.

MAINTENANCE (M)

Deactivate the inoperative ELT (AMM 23-00-00/901).

1. Gain access to the inoperative ELT in the section 47 crown.
2. For ACR/Artex model ELTs, remove the ELT, disconnect the wire harnesses connected to the ELT, re-install disconnected battery and ELT. Install a placard stating "ELT not installed" in view of pilot.
3. For ELTA and Honeywell model ELTs, set the ELT (LRU) front panel switch to OFF.

OPERATIONS NOTE

See 14 CFR 91.207, 121.353, 121.339 or other appropriate regulations.

23-24-02 Emergency Locator Transmitter (ELT) (Fixed)*********23-24-02D Not Required By 14 CFR - Missing**

Interval	Installed	Required	Procedure
D	1	0	

Any in excess of those required by 14 CFR may be missing.

NOTE: Dispatch relief for the Portable (survival) ELT's is in MMEL item 25-63-03.

OPERATIONS NOTE

See 14 CFR 91.207, 121.353, 121.339 or other appropriate regulations.

23-24-02 Emergency Locator Transmitter (ELT) (Fixed)

23-24-02-01 Remote ELT Switch

Interval	Installed	Required	Procedure
D	1	0	(M)

May be inoperative provided:

- Remote ELT Switch is deactivated.
- ELT Switch is placed in the ARMED mode.

NOTE: Depending on ELT model, remote ELT switch nomenclature may be ARM or ARMED.

MAINTENANCE (M)

NOTE: If remote ELT switch is failed so that the ELT system is activated, deactivate the ELT system and dispatch per MMEL 23-24-02A or 23-24-02C.

If ELT does not activate with the remote ELT switch set to ON, deactivate the remote ELT switch (AMM 23-00-00/901).

- Placard the inoperative remote ELT switch.
- Set the remote ELT switch to the armed position.
 - For ACR/Artex and Honeywell model ELTs, set the remote ELT switch to ARM.
 - For ELTA model ELTs, set the remote ELT switch to ARMED.
- Gain access to the inoperative ELT (LRU) in the section 47 crown.
- Set the ELT front panel switch to the armed position.
 - For ACR/Artex model ELTs, set the ELT front panel switch to OFF.
 - For ELTA model ELTs, set the ELT front panel switch to ARMED.
 - For Honeywell model ELTs, set the ELT front panel switch to ARM.



777 FAA DDG Supplement Package

Supplement Number: 36-2

Issue Date: 12-APR-2024

Section Number: 2 (MMEL)

Item Number: 25-64-03

Item Name: Emergency Medical Equipment

DDG Effectivity: 777 FAA DDG (D630W003-TBC), Revision 36

MMEL Effectivity: 777 FAA MMEL, Revision 23a

Description: Modified the provisos.

Reason: To align with the FAA policy letter PL-132.

Disposition: For reference in the Operator's MEL.

Attachments: DDG page 2. 25-64-03.1-4

25-64-03 Emergency Medical Equipment
25-64-03-01 First Aid Kit (FAK) and/or Associated Equipment
25-64-03-01A Required By 14 CFR

Interval	Installed	Required	Procedure
A	-	-	(O)

If more than one is required by 14 CFR, only one of the required FAKs may be incomplete or removed provided:

- The FAK is labeled or placarded in a manner that will identify it as a unit that cannot be mistaken for a fully serviceable unit.
- Location placarding is removed or obscured.
- Repairs or replacements are made within one flight.

NOTE: Medical equipment installed in the aircraft as part of an EMS operation is not considered part of the normal complement of equipment. No MMEL relief applies to that equipment and 14 CFR maintenance and inspection requirements do not apply.

OPERATIONS (O)

NOTE: See 14 CFR 121.803 or other appropriate regulations.

Each operator must develop their own procedures based on the emergency medical equipment installed on their airplanes.

25-64-03 Emergency Medical Equipment
25-64-03-01 First Aid Kit (FAK) and/or Associated Equipment
25-64-03-01B Any In Excess Of Those Required By 14 CFR

Interval	Installed	Required	Procedure
D	-	-	

Any in excess of those required by 14 CFR may be incomplete or removed.

OPERATIONS NOTE

See 14 CFR 121.803 or other appropriate regulations.

**25-64-03 Emergency Medical Equipment****25-64-03-01 First Aid Kit (FAK) and/or Associated Equipment****25-64-03-01-01 Tamper Seals or Tags**

Interval	Installed	Required	Procedure
C	-	-	(O)

May be inoperative, damaged, or missing provided proper FAK servicing is verified at each preflight.

OPERATIONS (O)

Before each departure, verify affected FAK(s) and/or associated equipment has been properly serviced.

25-64-03 Emergency Medical Equipment**25-64-03-02 Emergency Medical Kit (EMK) and/or Associated Equipment**

25-64-03-02A Required By 14 CFR

Interval	Installed	Required	Procedure
A	-	0	(O)

May be incomplete or removed provided:

- EMK is labeled or placarded in a manner that will identify it as a unit that cannot be mistaken for a fully serviceable unit.
- Location placarding is removed or obscured.
- Repairs or replacements are made within one flight.

NOTE: Medical equipment installed in the aircraft as part of an EMS operation is not considered part of the normal complement of equipment. No MMEL relief applies to that equipment and 14 CFR maintenance and inspection requirements do not apply.

OPERATIONS (O)

NOTE: See 14 CFR 121.803 or other appropriate regulations.

Each operator must develop their own procedures based on the emergency medical equipment installed on their airplanes.

**25-64-03 Emergency Medical Equipment****25-64-03-02 Emergency Medical Kit (EMK) and/or Associated Equipment**

25-64-03-02B Any In Excess Of Those Required By 14 CFR

Interval	Installed	Required	Procedure
D	-	-	

Any in excess of those required by 14 CFR may be incomplete or removed.

OPERATIONS NOTE

See 14 CFR 121.803 or other appropriate regulations.

25-64-03 Emergency Medical Equipment**25-64-03-02 Emergency Medical Kit (EMK) and/or Associated Equipment****25-64-03-02-01 Tamper Seals or Tags**

Interval	Installed	Required	Procedure
C	-	-	(O)

May be inoperative, damaged, or missing provided proper EMK servicing is verified at each preflight.

OPERATIONS (O)

Before each departure, verify affected EMK(s) and/or associated equipment has been properly serviced.

25-64-03 Emergency Medical Equipment**25-64-03-03 Automated External Defibrillators (AED) and/or Associated Equipment**

25-64-03-03A Required By 14 CFR

Interval	Installed	Required	Procedure
A	-	0	(O)

May be incomplete, inoperative, or removed provided:

- AED is labeled or placarded in a manner that will identify it as a unit that cannot be mistaken for a fully serviceable unit.
- Location placarding is removed or obscured.



- c. Repairs or replacements are made within one flight.

NOTE: Medical equipment installed in the aircraft as part of an Emergency Medical Service (EMS) operation is not considered part of the normal complement of equipment. No MMEL relief applies to that equipment and 14 CFR maintenance and inspection requirements do not apply.

OPERATIONS (O)

NOTE: See 14 CFR 121.803 or other appropriate regulations.

Each operator must develop their own procedures based on the emergency medical equipment installed on their airplanes.

25-64-03 Emergency Medical Equipment

25-64-03-03 Automated External Defibrillators (AED) and/or Associated Equipment

25-64-03-03B Any In Excess Of Those Required By 14 CFR

Interval	Installed	Required	Procedure
D	-	-	

Any in excess of those required by 14 CFR may be incomplete, inoperative or removed.

OPERATIONS NOTE

See 14 CFR 121.803 or other appropriate regulations.

25-64-03 Emergency Medical Equipment

25-64-03-03 Automated External Defibrillators (AED) and/or Associated Equipment

25-64-03-03-01 Tamper Seals or Tags

Interval	Installed	Required	Procedure
C	-	-	(O)

May be inoperative, damaged, or missing provided proper servicing is verified at each preflight.

OPERATIONS (O)

Before each departure, verify affected AED(s) and/or associated equipment has been properly serviced.



777 FAA DDG Supplement Package

Supplement Number: 36-2

Issue Date: 12-APR-2024

Section Number: 2 (MMEL)

Item Number: 26-26-01

Item Name: Portable Fire Extinguishers

DDG Effectivity: 777 FAA DDG (D630W003-TBC), Revision 36

MMEL Effectivity: 777 FAA MMEL, Revision 23a

Description: Modified Maintenance Note.

Reason: To align with the MMEL provisos in ensuring that the fire extinguisher remains in the certified location and is not used.

Disposition: For reference in the Operator's MEL.

Attachments: DDG page 2. 26-26-01.1

26-26-01 Portable Fire Extinguishers

Interval	Installed	Required	Procedure
D	-	-	

Any in excess of those required by 14 CFR may be inoperative or removed provided:

- Inoperative fire extinguisher remains in a certified location until removed from the aircraft at the next suitable maintenance facility.
- Location placarding is removed or obscured.
- Required distribution is maintained.

NOTE: Inoperative fire extinguishers removed from a certified location or removed from the aircraft, are subject to 49 CFR dangerous goods regulations.

MAINTENANCE NOTE

- Ensure inoperative fire extinguisher remains in the installed location until removed from the aircraft at the next suitable maintenance facility.
- Obscure or remove location placarding from the inoperative fire extinguisher.
- Ensure required distribution of serviceable extinguishers is maintained throughout the airplane.

OPERATIONS NOTE

See 14 CFR 121.309 or other appropriate regulations.

26-26-01 Portable Fire Extinguishers

26-26-01-01 Tamper Seals or Tags

Interval	Installed	Required	Procedure
C	-	-	(O)

May be inoperative, damaged or missing provided proper installation and servicing is verified at each preflight.

OPERATIONS (O)

Before each departure, verify affected Portable Fire Extinguisher(s) is properly installed and serviced.



777 FAA DDG Supplement Package

Supplement Number: 36-2

Issue Date: 12-APR-2024

Section Number: 2 (MMEL)

Item Number: 29-11-03

Item Name: Center System Demand Air Driven Pumps (ADP)

DDG Effectivity: 777 FAA DDG (D630W003-TBC), Revision 36

MMEL Effectivity: 777 FAA MMEL, Revision 23a

Description: Modified a note in the Operations (O) procedures for subitem option 29-11-03-01B.

Reason: To include the updated referenced MMEL item option number.

Disposition: For reference in the Operator's MEL.

Attachments: DDG page 2. 29-11-03.1-12

29-11-03 Center System Demand Air Driven Pumps (ADP)
29-11-03A ADP Verified Not Running

Interval	Installed	Required	Procedure
C	2	1	(M) (O)

One may be inoperative provided:

- a. Inoperative ADP is selected off.
- b. Both center system ACMPs operate normally.
- c. Remaining ADP heater operates normally.
- d. Remaining ADP AUTO function operates normally.
- e. Associated pump is verified not running when selected OFF.
- f. APU is used as ADP air source during takeoff.
- g. Appropriate performance adjustments are applied.

MAINTENANCE (M)

Verify the inoperative center system demand ADP is not running when selected OFF and the remaining center system demand ADP heater operates normally.

1. Verify the inoperative center system demand ADP is not running when selected OFF.
 - A. Position Hydraulic Panel (P5) DEMAND AIR C1 and C2 selectors OFF.
 - B. Position Hydraulic Panel (P5) PRIMARY ELEC C1 and C2 switches off.
 - C. On the Air Synoptic page, confirm the left and right DUCT PRESS are greater than 21.
 - D. On the Hydraulic Synoptic page, confirm the center system hydraulic system pressure is less than 200, or on the Hydraulic Maintenance page, confirm the associated Demand Pump PRESS is less than 200 and RUN is NO.

NOTE: For ADP running when selected OFF, the pump must be deactivated. Airplane may be dispatched using MMEL item 29-11-03B.

 - E. Position the operative Hydraulic Panel (P5) DEMAND AIR C1 or C2 selector AUTO.
 - F. Position Hydraulic Panel (P5) PRIMARY C1 ELEC and PRIMARY C2 ELEC switches ON.
2. Verify the remaining center system demand ADP heater is operating normally.



For airplanes with SB 777-29-0004 (PRR 61570) incorporated:

- A. Confirm the associated HYD PUMP DEM C1 or C2 status message is not displayed.

For airplanes with SB 777-29-0004 (PRR 61570) not incorporated:

- A. Press and release E6-2 shelf RAT GEN HEATER TEST switch.
- B. Within 60 seconds of pressing RAT GEN HEATER TEST switch, confirm the operative ADP heater current is 1.5 to 2.2 amps.

NOTE 1: C1 ADP wire number is W5601-1004-16 and ground block is GB46015.

NOTE 2: C2 ADP wire number is W5135-1004-16 and ground block is GB46012.

OPERATIONS (O)

NOTE: For APU-to-Pack takeoff function installed, the APU-to-Pack Takeoff Supplementary Procedure is not allowed.

1. Reduce the performance limited weights by the appropriate adjustments.

Performance Limited Weights	Takeoff Climb	Takeoff Runway/Obstacle
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For -200 or -200ER/GE or PW or RR:

All	1,400 lb (636 kg)	9,600 lb (4,355 kg)
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For -200LR or 777F/GE90-100 Series or -300ER/GE90-115B:

All	1,700 lb (771 kg)	9,600 lb (4,355 kg)
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For -300/PW4090 or RR:

All	400 lb (181 kg)	3,700 lb (1,678 kg)
-----	-----------------	---------------------

2. Position the inoperative center system DEMAND pump selector OFF.
3. For takeoff, configure the pneumatic system with the APU supplying air for the remaining operative center system demand pump.
 - A. Position APU Bleed air switch AUTO.
 - B. Start the APU if it is not operating.
 - C. Position Bleed Isolation L ISLN and R ISLN switches CLOSED.

NOTE: The BLEED ISLN CLOSED L and BLEED ISLN CLOSED R advisory messages will be displayed.
 - D. After takeoff is completed, position Bleed Isolation L ISLN and R ISLN switches AUTO and configure the APU as required.

29-11-03 Center System Demand Air Driven Pumps (ADP)
29-11-03B ADP Deactivated

Interval	Installed	Required	Procedure
C	2	1	(M) (O)

One may be inoperative provided:

- Inoperative ADP is deactivated.
- Both center system ACMPs operate normally.
- Remaining ADP heater operates normally.
- Remaining ADP AUTO function operates normally.
- APU is used as ADP air source during takeoff.
- Appropriate performance adjustments are applied.

MAINTENANCE (M)

Deactivate the inoperative ADP and verify the remaining center system demand ADP heater operates normally (AMM 29-00-00/901).

- Deactivate the inoperative ADP.
 - Depressurize the center hydraulic system (AMM 29-11-00/201).
NOTE: It is not necessary to depressurize the reservoir.
 - Pressurize the pneumatic system (AMM 36-00-00/201).
 - Disconnect the Logic Speed Control Unit (LSCU).
 - For C1 ADP inoperative:
 - Open the P310 panel C HYD AIR PUMP 1 circuit breaker.
 - Gain access to the right systems card file (P84) in the main equipment center and position the power switch for the HYDIM CR to OFF.
 - Gain access to the C1 ADP by opening access door 197JL.
 - Disconnect, cap and stow electrical connector DM29306A from the C1 ADP LSCU. Cap the LSCU receptacle.
 - Close access door 197JL.
 - Position the power switch for the HYDIM CR to ON.
 - Close the P310 panel C HYD AIR PUMP 1 circuit breaker.
 - For C2 ADP inoperative:
 - Open the P310 panel C HYD AIR PUMP 2 circuit breaker.
 - Gain access to the left systems card file (P85) and position the power switch for the HYDIM CL to OFF.
 - Gain access to the C2 ADP by opening access door 197JL.



- d. Disconnect, cap and stow electrical connector DM29321A from the C2 ADP LSCU. Cap the LSCU receptacle.
 - e. Close access door 197JL.
 - f. Position the power switch for the HYDIM CL to ON.
 - g. Close the P310 panel C HYD AIR PUMP 2 circuit breaker.
 - 3) On the Hydraulic Maintenance page, confirm the associated Demand Pump PRESS is less than 200 and RUN is NO, or on the Hydraulic Synoptic page, confirm the center system hydraulic system pressure is less than 200.
- D. Depressurize the pneumatic pressure (AMM 36-00-00/201).
2. Verify the remaining center system demand ADP heater is operating normally.

For airplanes with SB 777-29-0004 (PRR 61570) incorporated:

- A. Confirm the associated HYD PUMP DEM C1 or C2 status message is not displayed.

For airplanes with SB 777-29-0004 (PRR 61570) not incorporated:

- A. Press and release E6-2 shelf RAT GEN HEATER TEST switch.
- B. Within 60 seconds of pressing RAT GEN HEATER TEST switch, confirm the operative ADP heater current is 1.5 to 2.2 amps.

NOTE 1: C1 ADP wire number is W5601-1004-16 and ground block is GB46015.

NOTE 2: C2 ADP wire number is W5135-1004-16 and ground block is GB46012.

OPERATIONS (O)

NOTE: For APU-to-Pack takeoff function installed, the APU-to-Pack Takeoff Supplementary Procedure is not allowed.

1. Reduce the performance limited weights by the appropriate adjustments.

Performance Limited Weights	Takeoff Climb	Takeoff Runway/Obstacle
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For -200 or -200ER/GE or PW or RR:

All	1,400 lb (636 kg)	9,600 lb (4,355 kg)
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For -200LR or 777F/GE90-100 Series or -300ER/GE90-115B:

All	1,700 lb (771 kg)	9,600 lb (4,355 kg)
-----	-------------------	---------------------

For -300/PW4090 or RR:

All	400 lb (181 kg)	3,700 lb (1,678 kg)
-----	-----------------	---------------------

2. Position the inoperative center system DEMAND pump selector OFF.
3. For takeoff, configure the pneumatic system with the APU supplying air for the remaining operative center system demand pump.
 - A. Position APU Bleed air switch AUTO.
 - B. Start the APU if it is not operating.
 - C. Position Bleed Isolation L ISLN and R ISLN switches CLOSED.

NOTE: The BLEED ISLN CLOSED L and BLEED ISLN CLOSED R advisory messages will be displayed.

 - D. After takeoff is completed, position Bleed Isolation L ISLN and R ISLN switches AUTO and configure the APU as required.

29-11-03 Center System Demand Air Driven Pumps (ADP)
29-11-03C Other ADP Auto/Heater Function Inoperative

Interval	Installed	Required	Procedure
C	2	1	(M) (O)

One may be inoperative provided:

- a. Inoperative ADP is selected off.
- b. Both center system ACMPs operate normally.
- c. Associated pump is verified not running when selected OFF.
- d. Remaining ADP is selected ON for the entire flight.
- e. APU is used as ADP air source during takeoff.
- f. Appropriate performance adjustments are applied.

MAINTENANCE (M)

Verify the inoperative center system demand ADP is not running when selected OFF. If required, disconnect the ADP so it does not run (AMM 29-00-00/901).

1. Depressurize the center hydraulic system (AMM 29-11-00/201).

NOTE: It is not necessary to depressurize the reservoir.
2. Pressurize the pneumatic system (AMM 36-00-00/201).
3. On the Hydraulic Maintenance page, confirm the associated Demand Pump PRESS is less than 200 and RUN is NO, or on the Hydraulic Synoptic page, confirm the center system hydraulic system pressure is less than 200
4. For the ADP running when selected OFF, disconnect the Logic Speed Control Unit (LSCU).
 - A. For C1 ADP inoperative:
 - 1) Open the P310 panel C HYD AIR PUMP 1 circuit breaker.



- 2) Gain access to the right systems card file (P84) in the main equipment center and position the power switch for the HYDIM CR to OFF.
 - 3) Gain access to the C1 ADP by opening access door 197JL.
 - 4) Disconnect, cap and stow electrical connector DM29306A from the C1 ADP LSCU. Cap the LSCU receptacle.
 - 5) Close access door 197JL.
 - 6) Position the power switch for the HYDIM CR to ON.
 - 7) Close the P310 panel C HYD AIR PUMP 1 circuit breaker.
- B. For C2 ADP inoperative:
- 1) Open the P310 panel C HYD AIR PUMP 2 circuit breaker.
 - 2) Gain access to the left systems card file (P85) and position the power switch for the HYDIM CL to OFF.
 - 3) Gain access to the C2 ADP by opening access door 197JL.
 - 4) Disconnect, cap and stow electrical connector DM29321A from the C2 ADP LSCU. Cap the LSCU receptacle.
 - 5) Close access door 197JL.
 - 6) Position the power switch for the HYDIM CL to ON.
 - 7) Close the P310 panel C HYD AIR PUMP 2 circuit breaker.
- C. On the Hydraulic Maintenance page, confirm the associated Demand Pump PRESS is less than 200 and RUN is NO, or on the Hydraulic Synoptic page, confirm the center system hydraulic system pressure is less than 200.
5. Depressurize the pneumatic pressure (AMM 36-00-00/201).

OPERATIONS (O)

NOTE: For APU-to-Pack takeoff function installed, the APU-to-Pack Takeoff Supplementary Procedure is not allowed.

1. Reduce the performance limited weights by the appropriate adjustments.

Performance Limited Weights	Takeoff Climb	Takeoff Runway/Obstacle
-----------------------------	---------------	-------------------------

For -200 or -200ER/GE or PW or RR:

All	1,400 lb (636 kg)	9,600 lb (4,355 kg)
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For -200LR or 777F/GE90-100 Series or -300ER/GE90-115B:

All	1,700 lb (771 kg)	9,600 lb (4,355 kg)
-----	-------------------	---------------------

For -300/PW4090 or RR:

All	400 lb (181 kg)	3,700 lb (1,678 kg)
-----	-----------------	---------------------

2. Position the inoperative center system DEMAND pump selector OFF.
3. Position remaining operative center system DEMAND pump selector ON prior to takeoff and leave ON for the entire flight.

NOTE: Do not position center system DEMAND pump selector ON before engine start.

4. For takeoff, configure the pneumatic system with the APU supplying air for the remaining operative center system demand pump.
 - A. Position APU Bleed air switch AUTO.
 - B. Start the APU if it is not operating.
 - C. Position Bleed Isolation L ISLN and R ISLN switches CLOSED.

NOTE: The BLEED ISLN CLOSED L and BLEED ISLN CLOSED R advisory messages will be displayed.

- D. After takeoff is completed, position Bleed Isolation L ISLN and R ISLN switches AUTO and configure the APU as required.

29-11-03 Center System Demand Air Driven Pumps (ADP)

29-11-03-01 ADP Auto/Heater Function (-200/-200ER With PRR 61570 or SB 777-29-0004 Incorporated, and -200LR/-300/-300ER/777F)

29-11-03-01A Other ADP Operates Normally

Interval	Installed	Required	Procedure
C	2	1	(M) (O)

One may be inoperative provided:

- a. Remaining ADP operates normally.
- b. Associated ADP ON and OFF selector positions are verified to operate normally.
- c. Associated ADP is selected ON for the entire flight.

MAINTENANCE (M)

Verify the associated center system demand ADP selector ON and OFF positions operate normally (AMM 29-00-00/901).

1. Supply electrical power on the airplane (AMM 24-22-00/201).
2. Pressurize the pneumatic system (AMM 36-00-00/201).
3. For the associated HYD PUMP DEM C1 or C2 status message displayed:
 - A. Position the associated Hydraulic Panel (P5) DEMAND AIR C1 or C2 selector ON.
 - B. Confirm the associated HYD PUMP DEM C1 or C2 status message clears.



- C. Position the associated Hydraulic Panel (P5) DEMAND AIR C1 or C2 selector OFF.
 - D. Confirm the associated HYD PUMP DEM C1 or C2 status message is displayed.
 - E. Depressurize the pneumatic system (AMM 36-00-00/201).
4. For the associated HYD PUMP DEM C1 or C2 status message not displayed:
- A. Position the associated Hydraulic Panel (P5) DEMAND AIR C1 or C2 selector ON.
 - B. On the Hydraulic Maintenance page, confirm the associated Demand Pump PRESS is greater than 2800, or on the Hydraulic Synoptic page, confirm the center system hydraulic system pressure is greater than 2800.
 - C. Position the associated Hydraulic Panel (P5) DEMAND AIR C1 or C2 selector OFF.
 - D. On the Hydraulic Maintenance page, confirm the associated Demand Pump PRESS is less than 200, or on the Hydraulic Synoptic page, confirm the center system hydraulic system pressure is less than 200.
 - E. Depressurize the pneumatic system (AMM 36-00-00/201).

OPERATIONS (O)

1. Position associated center system DEMAND pump selector ON prior to takeoff and leave ON for the entire flight.

NOTE: Do not position center system DEMAND pump selector ON before engine start.

2. Position remaining center system DEMAND pump selector AUTO.

29-11-03 Center System Demand Air Driven Pumps (ADP)**29-11-03-01 ADP Auto/Heater Function (-200/-200ER With PRR 61570 or SB 777-29-0004 Incorporated, and -200LR/-300/-300ER/777F)****29-11-03-01B Other ADP Inoperative**

Interval	Installed	Required	Procedure
C	2	1	(M) (O)

One may be inoperative provided:

- a. Both center system ACMPs operate normally.
- b. Associated ADP ON and OFF selector positions are verified to operate normally.
- c. Associated ADP is selected ON for the entire flight.
- d. APU is used as ADP air source during takeoff.

- e. Appropriate performance adjustments are applied.

MAINTENANCE (M)

Verify the associated center system demand ADP selector ON and OFF positions operate normally (AMM 29-00-00/901).

1. Supply electrical power on the airplane (AMM 24-22-00/201).
2. Pressurize the pneumatic system (AMM 36-00-00/201).
3. For the associated HYD PUMP DEM C1 or C2 status message displayed:
 - A. Position the associated Hydraulic Panel (P5) DEMAND AIR C1 or C2 selector ON.
 - B. Confirm the associated HYD PUMP DEM C1 or C2 status message clears.
 - C. Position the associated Hydraulic Panel (P5) DEMAND AIR C1 or C2 selector OFF.
 - D. Confirm the associated HYD PUMP DEM C1 or C2 status message is displayed.
 - E. Depressurize the pneumatic system (AMM 36-00-00/201).
4. For the associated HYD PUMP DEM C1 or C2 status message not displayed:
 - A. Position the associated Hydraulic Panel (P5) DEMAND AIR C1 or C2 selector ON.
 - B. On the Hydraulic Maintenance page, confirm the associated Demand Pump PRESS is greater than 2800, or on the Hydraulic Synoptic page, confirm the center system hydraulic system pressure is greater than 2800.
 - C. Position the associated Hydraulic Panel (P5) DEMAND AIR C1 or C2 selector OFF.
 - D. On the Hydraulic Maintenance page, confirm the associated Demand Pump PRESS is less than 200, or on the Hydraulic Synoptic page, confirm the center system hydraulic system pressure is less than 200.
 - E. Depressurize the pneumatic system (AMM 36-00-00/201).

OPERATIONS (O)

NOTE: For APU-to-Pack takeoff function installed, the APU-to-Pack Takeoff Supplementary Procedure is not allowed.

1. Reduce the performance limited weights by the appropriate adjustments.

NOTE: For airplane also dispatched using MMEL item 29-11-03C, it is not required to add the following penalties.

Performance Limited Weights	Takeoff Climb	Takeoff Runway/Obstacle
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For -200 or -200ER/GE or PW or RR:

All	1,400 lb (636 kg)	9,600 lb (4,355 kg)
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For -200LR or 777F/GE90-100 Series or -300ER/GE90-115B:

All	1,700 lb (771 kg)	9,600 lb (4,355 kg)
-----	-------------------	---------------------

For -300/PW4090 or RR:

All	400 lb (181 kg)	3,700 lb (1,678 kg)
-----	-----------------	---------------------

2. Position inoperative center system DEMAND pump selector OFF.
3. Position associated center system DEMAND pump selector ON prior to takeoff and leave ON for the entire flight.

NOTE: Do not position center system DEMAND pump selector ON before engine start.

4. For takeoff, configure the pneumatic system with the APU supplying air for the operating center system demand pump.
 - A. Position APU Bleed air switch AUTO.
 - B. Start the APU if it is not operating.
 - C. Position Bleed Isolation L ISLN and R ISLN switches CLOSED.

NOTE: The BLEED ISLN CLOSED L and BLEED ISLN CLOSED R advisory messages will be displayed.

- D. After takeoff is completed, position Bleed Isolation L ISLN and R ISLN switches AUTO and configure the APU as required.

29-11-03 Center System Demand Air Driven Pumps (ADP)

29-11-03-01 ADP Auto/Heater Function (-200/-200ER With PRR 61570 or SB 777-29-0004 Incorporated, and -200LR/-300/-300ER/777F)

29-11-03-01C Both Auto/Heater Functions Inoperative

Interval	Installed	Required	Procedure
C	2	0	(M) (O)

May be inoperative provided:

- a. Both center system ACMPs operate normally.
- b. ADP ON and OFF selector positions are verified to operate normally for both pumps.
- c. One ADP is selected ON for the entire flight.
- d. APU is used as ADP air source during takeoff.

- e. Appropriate performance adjustments are applied.

MAINTENANCE (M)

Verify both center system demand ADP selector's ON and OFF positions operate normally (AMM 29-00-00/901).

1. Supply electrical power on the airplane (AMM 24-22-00/201).
2. Pressurize the pneumatic system (AMM 36-00-00/201).
3. For the associated HYD PUMP DEM C1 and/or C2 status message displayed:
 - A. Position the associated Hydraulic Panel (P5) DEMAND AIR C1 or C2 selector ON.
 - B. Confirm the associated HYD PUMP DEM C1 or C2 status message clears.
 - C. Position the associated Hydraulic Panel (P5) DEMAND AIR C1 or C2 selector OFF.
 - D. Confirm the associated HYD PUMP DEM C1 or C2 status message is displayed.
 - E. Depressurize the pneumatic system (AMM 36-00-00/201).
4. For the associated HYD PUMP DEM C1 and/or C2 status message not displayed:
 - A. Position the associated Hydraulic Panel (P5) DEMAND AIR C1 or C2 selector ON.
 - B. On the Hydraulic Maintenance page, confirm the associated Demand Pump PRESS is greater than 2800, or on the Hydraulic Synoptic page, confirm the center system hydraulic system pressure is greater than 2800.
 - C. Position the associated Hydraulic Panel (P5) DEMAND AIR C1 or C2 selector OFF.
 - D. On the Hydraulic Maintenance page, confirm the associated Demand Pump PRESS is less than 200, or on the Hydraulic Synoptic page, confirm the center system hydraulic system pressure is less than 200.
 - E. Depressurize the pneumatic system (AMM 36-00-00/201).

OPERATIONS (O)

NOTE: For APU-to-Pack takeoff function installed, the APU-to-Pack Takeoff Supplementary Procedure is not allowed.

1. Reduce the performance limited weights by the appropriate adjustments.

Performance Limited Weights	Takeoff Climb	Takeoff Runway/Obstacle
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For -200 or -200ER/GE or PW or RR:

All	1,400 lb (636 kg)	9,600 lb (4,355 kg)
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For -200LR or 777F/GE90-100 Series or -300ER/GE90-115B:

All	1,700 lb (771 kg)	9,600 lb (4,355 kg)
-----	-------------------	---------------------

For -300/PW4090 or RR:

All	400 lb (181 kg)	3,700 lb (1,678 kg)
-----	-----------------	---------------------

2. Position one center system DEMAND pump selector ON prior to takeoff and leave ON for the entire flight.

NOTE: Do not position center system DEMAND pump selector ON before engine start.
3. Position remaining center system DEMAND pump selector AUTO.
4. For takeoff, configure the pneumatic system with the APU supplying air for the operating center system demand pump.
 - A. Position APU Bleed air switch AUTO.
 - B. Start the APU if it is not operating.
 - C. Position Bleed Isolation L ISLN and R ISLN switches CLOSED.

NOTE: The BLEED ISLN CLOSED L and BLEED ISLN CLOSED R advisory messages will be displayed.

- D. After takeoff is completed, position Bleed Isolation L ISLN and R ISLN switches AUTO and configure the APU as required.



777 FAA DDG Supplement Package

Supplement Number: 36-2

Issue Date: 12-APR-2024

Section Number: 2 (MMEL)

Item Number: 30-21-01

Item Name: Engine Anti-Ice Systems

DDG Effectivity: 777 FAA DDG (D630W003-TBC), Revision 36

MMEL Effectivity: 777 FAA MMEL, Revision 23a

Description: Modified Maintenance (M) procedures for the item and subitem options 30-21-01-01B and 30-21-01-01D

Reason: To ensure that the Maintenance (M) procedures steps are applied to all engine options (GE, PW, and RR) to provide safety to maintenance personnel.

Disposition: For reference in the Operator's MEL.

Attachments: DDG page 2.30-21-01.1-24

30-21-01 Engine Anti-Ice Systems

Interval	Installed	Required	Procedure
C	2	1	(M)

One may be inoperative provided:

- Valve is locked closed.
- Associated engine anti-ice control switch remains OFF.
- Airplane is not operated in known or forecast icing conditions.
- Flight remains within 120 minutes of landing at a suitable airport.

MAINTENANCE (M)

Lock associated EAI valve in the closed position (AMM 30-00-00/901).

- Deactivate the associated EAI valve controller:
 - For the right EAI valve, open P210 panel EAI CTRL R circuit breaker.
 - For the left EAI valve, open P310 panel EAI CTRL L circuit breaker.
- Deactivate the leading edge slats in the retracted position (AMM 27-81-00/201).
- Deactivate the associated thrust reverser for ground maintenance (AMM 78-31-00/201).
- Lock the EAI valve in the closed position.
 - Remove the locking screw from the normal position.
 - Set the crank to the locked (closed) position.
 - Install the locking screw through the crank into the locked position.
 - Position associated ANTI-ICE Panel (P5) Engine Anti-Ice selector OFF.
- Close the associated circuit breaker(s).
- Activate the thrust reverser (AMM 78-31-00/201).
- Activate the leading edge slats (AMM 27-81-00/201).
- Activate the associated EAI valve controller by closing the associated circuit breaker.
- Position the associated ANTI-ICE Panel (P5) Engine Anti-Ice selector OFF.

30-21-01 Engine Anti-Ice Systems
30-21-01-01 PW and RR
30-21-01-01A OAT 38 Degrees C or Below, With Deactivation Kit Installed

Interval	Installed	Required	Procedure
C	2	1	(M) (O)



For operations at ambient temperature 38 degrees C or below, one may be inoperative provided:

- Valve is deactivated open using deactivation kit.
- Engine fan case overheat detection system operates normally.
- Appropriate performance adjustments are applied.

MAINTENANCE (M)

Deactivate the associated EAI valve by installing the EAI valve lock-open kit, P/N K28703 (AMM 30-00-00/901).

- Remove inoperative EAI valve (AMM 30-21-02/401).
- Modify valve with lock-open kit (AMM 30-00-00/901).
- Re-install modified valve (AMM 30-21-02/401).

For PW:

- Visually inspect the area near the Cowl Thermal Anti-Ice duct for signs of leaks and the fan cowl in general for flammable fluid leaks. Dispatch is not allowed if any of the duct clamps or supports are not secure or any air leaks from the EAI duct are present, or if significant flammable fluid leakage is present.

OPERATIONS (O)

- Reduce performance limited weights by the appropriate adjustments.

Performance Limited Weights	Takeoff & Landing	Enroute Climb
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For -200 or -200ER/PW4074 thru PW4084:

All	19,600 lb (8,891 kg)	19,400 lb (8,800 kg)
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For -200 or -200ER/PW4090:

thru 632,500 lb (286,902 kg)	12,200 lb (5,534 kg)	13,800 lb (6,260 kg)
over 632,500 lb (286,902 kg)	17,300 lb (7,847 kg)	13,800 lb (6,260 kg)

For -200 or -200ER/RR875 thru RR892:

thru 632,500 lb (286,902 kg)	20,100 lb (9,117 kg)	21,600 lb (9,798 kg)
over 632,500 lb (286,902 kg)	26,500 lb (12,020 kg)	21,600 lb (9,798 kg)



For -200 or -200ER/RR895:

All	21,700 lb (9,843 kg)	23,000 lb (10,433 kg)
-----	----------------------	-----------------------

For -300/PW4090:

All	12,400 lb (5,625 kg)	13,000 lb (5,897 kg)
-----	----------------------	----------------------

For -300/RR892:

All	22,000 lb (9,979 kg)	23,500 lb (10,659 kg)
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NOTE: The enroute climb weight penalties listed above are based on single engine operating speeds that approximate maximum lift-to-drag ratio speed. To account for the difference in level off altitude when operating at other speeds, multiply the enroute climb weight penalty listed above by the appropriate factor listed on page 3.00-01-00.2 in the ENROUTE DIVERSION SPEED EFFECTS paragraph.

For PW:

2. Increase flight planning fuel by 1.0% for an inoperative EAI Valve or EAI Pressure Sensor, or by 1.3% for an inoperative EAI Card.

For RR:

3. Increase flight planning fuel by 0.5% for an inoperative EAI Valve or EAI Pressure Sensor, or by 0.8% for an inoperative EAI Card.
4. For ETOPS flight planning:

For PW:

- A. Increase the ETOPS engine inoperative Critical Fuel Reserves by 1.6%.

For RR:

- B. Increase the ETOPS engine inoperative Critical Fuel Reserves by 0.9%.
5. Position ENGINE ANTI-ICE selector(s) as follows:
 - A. For EAI Valve or EAI Pressure Sensor inoperative, position associated EAI selector ON.

NOTE 1: When an EAI selector is positioned to ON and the temperature is greater than 10 degrees C, the ANTI-ICE ON advisory message will be displayed. Do not accomplish the ANTI-ICE ON Non-Normal Checklist.

NOTE 2: Inflight, the ENG IDLE DISAGREE advisory message may be displayed when engine thrust is reduced to idle.

- B. For EAI Card inoperative, position both EAI selectors ON.

NOTE 1: When an EAI selector is positioned to ON and the temperature is greater than 10 degrees C, the ANTI-ICE ON advisory message will be displayed. Do not accomplish the ANTI-ICE ON Non-Normal Checklist.

NOTE 2: The associated ANTI-ICE ENG L or R advisory message will be displayed, the EAI indication will not be displayed on EICAS, and the associated EAI valve will be displayed as invalid (low intensity white) and the green flow bar will not be displayed on the Air synoptic.

6. For the inoperative EAI valve, if the ANTI-ICE LEAK ENG L or R caution message is subsequently displayed:

For PW:

- A. Do not accomplish the two minute delay associated with the ANTI-ICE LEAK ENG L or R Non-Normal Checklist. Complete the remainder of the checklist as soon as practical.

For RR:

- B. Do not accomplish the initial one minute delay associated with the ANTI-ICE LEAK ENG L or R Non-Normal Checklist. Complete the remainder of the checklist as soon as practical.

30-21-01 Engine Anti-Ice Systems

30-21-01-01 PW and RR

30-21-01-01B OAT 38 Degrees C or Below, Without Deactivation Kit Installed

Interval	Installed	Required	Procedure
C	2	1	(M) (O)

For operations at ambient temperature 38 degrees C or below, one may be inoperative provided:

- a. Valve is electrically disconnected and verified open.
- b. Engine fan case overheat detection system operates normally.
- c. Associated EAI pressure sensor system operates normally.
- d. Associated EAI card operates normally.
- e. Appropriate performance adjustments are applied.

MAINTENANCE (M)

For EAI VALVE L or R status message displayed, deactivate the EAI valve by disconnecting the electrical connector and verify EAI valve is open when the associated engine is running (AMM 30-00-00/901).

1. Deactivate the associated EAI valve controller.
 - A. For the right EAI valve, open P210 panel EAI CTRL R circuit breaker.

- B. For the left EAI valve, open P310 panel EAI CTRL L circuit breaker.
2. Deactivate the leading edge slats in the retracted position (AMM 27-81-00/201).
3. Deactivate the associated thrust reverser for ground maintenance (AMM 78-31-00/201).
4. Deactivate the EAI valve in the open position.
 - A. On the associated engine, disconnect, cap and stow the torque motor electrical connector DM30211 or DM30211A from the EAI valve controller, M30211.
 - B. Install a protective cover over the electrical connector on the valve controller.

For PW:

- C. Visually inspect the area near the Cowl Thermal Anti-Ice duct for signs of leaks and the fan cowl in general for flammable fluid leaks. Dispatch is not allowed if any of the duct clamps or supports are not secure or any air leaks from the EAI duct are present, or if significant flammable fluid leakage is present.
5. Activate the thrust reverser (AMM 78-31-00/201).
6. Activate the leading edge slats (AMM 27-81-00/201).
7. Activate the associated EAI valve controller by closing the associated circuit breaker.
8. Verify the EAI valve is open when the associated engine is running.
 - A. Position associated ANTI-ICE Panel (P5) Engine Anti-Ice selector OFF.
 - B. Start the associated engine (71-00-00/201).
 - C. Confirm the EAI indication (green EAI) is displayed above the N1 indication on the EICAS display for the associated engine (indicates the EAI valve is open).

NOTE: If the EAI indication is not displayed, the valve may be stuck closed and a thrust setting slightly above minimum idle may open the valve.

- D. Shutdown the associated engine (71-00-00/201).

OPERATIONS (O)

1. Reduce performance limited weights by the appropriate adjustments.

Performance Limited Weights	Takeoff & Landing	Enroute Climb
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For -200 or -200ER/PW4074 thru PW4084:

All	19,600 lb (8,891 kg)	19,400 lb (8,800 kg)
-----	----------------------	----------------------



For -200 or -200ER/PW4090:

thru 632,500 lb (286,902 kg)	12,200 lb (5,534 kg)	13,800 lb (6,260 kg)
over 632,500 lb (286,902 kg)	17,300 lb (7,847 kg)	13,800 lb (6,260 kg)

For -200 or -200ER/RR875 thru RR892:

thru 632,500 lb (286,902 kg)	20,100 lb (9,117 kg)	21,600 lb (9,798 kg)
over 632,500 lb (286,902 kg)	26,500 lb (12,020 kg)	21,600 lb (9,798 kg)

For -200 or -200ER/RR895:

All	21,700 lb (9,843 kg)	23,000 lb (10,433 kg)
-----	----------------------	-----------------------

For -300/PW4090:

All	12,400 lb (5,625 kg)	13,000 lb (5,897 kg)
-----	----------------------	----------------------

For -300/RR892:

All	22,000 lb (9,979 kg)	23,500 lb (10,659 kg)
-----	----------------------	-----------------------

NOTE: The enroute climb weight penalties listed above are based on single engine operating speeds that approximate maximum lift-to-drag ratio speed. To account for the difference in level off altitude when operating at other speeds, multiply the enroute climb weight penalty listed above by the appropriate factor listed on page 3.00-01-00.2 in the ENROUTE DIVERSION SPEED EFFECTS paragraph.

For PW:

2. Increase flight planning fuel by 1.0%.

For RR:

3. Increase flight planning fuel by 0.5%.
4. For ETOPS flight planning:

For PW:

- A. Increase the ETOPS engine inoperative Critical Fuel Reserves by 1.6%.

For RR:

- B. Increase the ETOPS engine inoperative Critical Fuel Reserves by 0.9%.

5. Verify the EAI valve is open when the associated engine is running before each departure.
 - A. Position associated ANTI-ICE Panel (P5) Engine Anti-Ice selector OFF.
 - B. After the associated engine has been started, confirm the associated EAI indication (green EAI) is displayed above the N1 indication.

NOTE 1: If the EAI indication is not displayed, the valve may be stuck closed and a thrust setting slightly above minimum idle may open the valve.

NOTE 2: On the Air synoptic, the EAI valve will indicate failed open because the EAI selector is OFF and the green flow bar will be displayed.

6. Position associated ENGINE ANTI-ICE selector ON.

NOTE 1: When an EAI selector is positioned to ON and the temperature is greater than 10 degrees C, the ANTI-ICE ON advisory message will be displayed. Do not accomplish the ANTI-ICE ON Non-Normal Checklist.

NOTE 2: Inflight, the ENG IDLE DISAGREE advisory message may be displayed when engine thrust is reduced to idle.

NOTE 3: The ANTI-ICE ENG advisory message may be displayed during ground operations prior to takeoff. Flight crew should accomplish the associated non-normal checklist. If ANTI-ICE ENG clears, takeoff is allowed.

7. For the inoperative EAI valve, if the ANTI-ICE LEAK ENG L or R caution message is subsequently displayed:

For PW:

- A. Do not accomplish the two minute delay associated with the ANTI-ICE LEAK ENG L or R Non-Normal Checklist. Complete the remainder of the checklist as soon as practical.

For RR:

- B. Do not accomplish the initial one minute delay associated with the ANTI-ICE LEAK ENG L or R Non-Normal Checklist. Complete the remainder of the checklist as soon as practical.

30-21-01 Engine Anti-Ice Systems

30-21-01-01 PW and RR

30-21-01-01C OAT Above 38 Degrees C, With Deactivation Kit Installed

Interval	Installed	Required	Procedure
A	2	1	(M) (O)



For operations at ambient temperature above 38 degrees C, one may be inoperative provided:

- Valve is deactivated open using deactivation kit.
- Engine fan case overheat detection system operates normally.
- Appropriate performance adjustments are applied.
- Repairs are made within 3 flight days.

MAINTENANCE (M)

Deactivate the associated EAI valve by installing the EAI valve lock-open kit, P/N K28703 (AMM 30-00-00/901).

- Remove inoperative EAI valve (AMM 30-21-02/401).
- Modify valve with lock-open kit (AMM 30-00-00/901).
- Re-install modified valve (AMM 30-21-02/401).

For PW:

- Visually inspect the area near the Cowl Thermal Anti-Ice duct for signs of leaks and the fan cowl in general for flammable fluid leaks. Dispatch is not allowed if any of the duct clamps or supports are not secure or any air leaks from the EAI duct are present, or if significant flammable fluid leakage is present.

OPERATIONS (O)

- Reduce performance limited weights by the appropriate adjustments.

Performance Limited Weights	Takeoff & Landing	Enroute Climb
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For -200 or -200ER/PW4074 thru PW4084:

All	19,600 lb (8,891 kg)	19,400 lb (8,800 kg)
-----	----------------------	----------------------

For -200 or -200ER/PW4090:

thru 632,500 lb (286,902 kg)	12,200 lb (5,534 kg)	13,800 lb (6,260 kg)
over 632,500 lb (286,902 kg)	17,300 lb (7,847 kg)	13,800 lb (6,260 kg)

For -200 or -200ER/RR875 thru RR892:

thru 632,500 lb (286,902 kg)	20,100 lb (9,117 kg)	21,600 lb (9,798 kg)
over 632,500 lb (286,902 kg)	26,500 lb (12,020 kg)	21,600 lb (9,798 kg)



For -200 or -200ER/RR895:

All	21,700 lb (9,843 kg)	23,000 lb (10,433 kg)
-----	----------------------	-----------------------

For -300/PW4090:

All	12,400 lb (5,625 kg)	13,000 lb (5,897 kg)
-----	----------------------	----------------------

For -300/RR892:

All	22,000 lb (9,979 kg)	23,500 lb (10,659 kg)
-----	----------------------	-----------------------

NOTE: The enroute climb weight penalties listed above are based on single engine operating speeds that approximate maximum lift-to-drag ratio speed. To account for the difference in level off altitude when operating at other speeds, multiply the enroute climb weight penalty listed above by the appropriate factor listed on page 3.00-01-00.2 in the ENROUTE DIVERSION SPEED EFFECTS paragraph.

For PW:

2. Increase flight planning fuel by 1.0% for an inoperative EAI Valve or EAI Pressure Sensor, or by 1.3% for an inoperative EAI Card.

For RR:

3. Increase flight planning fuel by 0.5% for an inoperative EAI Valve or EAI Pressure Sensor, or by 0.8% for an inoperative EAI Card.
4. For ETOPS flight planning:

For PW:

- A. Increase the ETOPS engine inoperative Critical Fuel Reserves by 1.6%.

For RR:

- B. Increase the ETOPS engine inoperative Critical Fuel Reserves by 0.9%.
5. Position ENGINE ANTI-ICE selector(s) as follows:
 - A. For EAI Valve or EAI Pressure Sensor inoperative, position associated EAI selector ON.

NOTE 1: When an EAI selector is positioned to ON and the temperature is greater than 10 degrees C, the ANTI-ICE ON advisory message will be displayed. Do not accomplish the ANTI-ICE ON Non-Normal Checklist.

NOTE 2: Inflight, the ENG IDLE DISAGREE advisory message may be displayed when engine thrust is reduced to idle.

- B. For EAI Card inoperative, position both EAI selectors ON.



NOTE 1: When an EAI selector is positioned to ON and the temperature is greater than 10 degrees C, the ANTI-ICE ON advisory message will be displayed. Do not accomplish the ANTI-ICE ON Non-Normal Checklist.

NOTE 2: The associated ANTI-ICE ENG L or R advisory message will be displayed, the EAI indication will not be displayed on EICAS, and the associated EAI valve will be displayed as invalid (low intensity white) and the green flow bar will not be displayed on the Air synoptic.

6. For the inoperative EAI valve, if the ANTI-ICE LEAK ENG L or R caution message is subsequently displayed:

For PW:

- A. Do not accomplish the two minute delay associated with the ANTI-ICE LEAK ENG L or R Non-Normal Checklist. Complete the remainder of the checklist as soon as practical.

For RR:

- B. Do not accomplish the initial one minute delay associated with the ANTI-ICE LEAK ENG L or R Non-Normal Checklist. Complete the remainder of the checklist as soon as practical.

30-21-01 Engine Anti-Ice Systems

30-21-01-01 PW and RR

30-21-01-01D OAT Above 38 Degrees C, Without Deactivation Kit Installed

Interval	Installed	Required	Procedure
A	2	1	(M) (O)

For operations at ambient temperature above 38 degrees C, one may be inoperative provided:

- Valve is electrically disconnected and verified open.
 - Engine fan case overheat detection system operates normally.
 - Associated EAI pressure sensor system operates normally.
 - Associated EAI card operates normally.
 - Appropriate performance adjustments are applied.
 - Repairs are made within 3 flight days.
-

MAINTENANCE (M)

For EAI VALVE L or R status message displayed, deactivate the EAI valve by disconnecting the electrical connector and verify EAI valve is open when the associated engine is running (AMM 30-00-00/901).

1. Deactivate the associated EAI valve controller.
 - A. For the right EAI valve, open P210 panel EAI CTRL R circuit breaker.
 - B. For the left EAI valve, open P310 panel EAI CTRL L circuit breaker.
2. Deactivate the leading edge slats in the retracted position (AMM 27-81-00/201).
3. Deactivate the associated thrust reverser for ground maintenance (AMM 78-31-00/201).
4. Deactivate the EAI valve in the open position.
 - A. On the associated engine, disconnect, cap and stow the torque motor electrical connector DM30211 or DM30211A from the EAI valve controller, M30211.
 - B. Install a protective cover over the electrical connector on the valve controller.

For PW:

- C. Visually inspect the area near the Cowl Thermal Anti-Ice duct for signs of leaks and the fan cowl in general for flammable fluid leaks. Dispatch is not allowed if any of the duct clamps or supports are not secure or any air leaks from the EAI duct are present, or if significant flammable fluid leakage is present.
5. Activate the thrust reverser (AMM 78-31-00/201).
6. Activate the leading edge slats (AMM 27-81-00/201).
7. Activate the associated EAI valve controller by closing the associated circuit breaker.
8. Verify the EAI valve is open when the associated engine is running.
 - A. Position associated ANTI-ICE Panel (P5) Engine Anti-Ice selector OFF.
 - B. Start the associated engine (71-00-00/201).
 - C. Confirm the EAI indication (green EAI) is displayed above the N1 indication on the EICAS display for the associated engine (indicates the EAI valve is open).

NOTE: If the EAI indication is not displayed, the valve may be stuck closed and a thrust setting slightly above minimum idle may open the valve.

- D. Shutdown the associated engine (71-00-00/201).

OPERATIONS (O)

1. Reduce performance limited weights by the appropriate adjustments.



Performance Limited Weights	Takeoff & Landing	Enroute Climb
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For -200 or -200ER/PW4074 thru PW4084:

All	19,600 lb (8,891 kg)	19,400 lb (8,800 kg)
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For -200 or -200ER/PW4090:

thru 632,500 lb (286,902 kg)	12,200 lb (5,534 kg)	13,800 lb (6,260 kg)
over 632,500 lb (286,902 kg)	17,300 lb (7,847 kg)	13,800 lb (6,260 kg)

For -200 or -200ER/RR875 thru RR892:

thru 632,500 lb (286,902 kg)	20,100 lb (9,117 kg)	21,600 lb (9,798 kg)
over 632,500 lb (286,902 kg)	26,500 lb (12,020 kg)	21,600 lb (9,798 kg)

For -200 or -200ER/RR895:

All	21,700 lb (9,843 kg)	23,000 lb (10,433 kg)
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For -300/PW4090:

All	12,400 lb (5,625 kg)	13,000 lb (5,897 kg)
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For -300/RR892:

All	22,000 lb (9,979 kg)	23,500 lb (10,659 kg)
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NOTE: The enroute climb weight penalties listed above are based on single engine operating speeds that approximate maximum lift-to-drag ratio speed. To account for the difference in level off altitude when operating at other speeds, multiply the enroute climb weight penalty listed above by the appropriate factor listed on page 3.00-01-00.2 in the ENROUTE DIVERSION SPEED EFFECTS paragraph.

For PW:

2. Increase flight planning fuel by 1.0%.

For RR:

3. Increase flight planning fuel by 0.5%.
4. For ETOPS flight planning:

For PW:

- A. Increase the ETOPS engine inoperative Critical Fuel Reserves by 1.6%.

For RR:

- B. Increase the ETOPS engine inoperative Critical Fuel Reserves by 0.9%.
5. Verify the EAI valve is open when the associated engine is running before each departure.
 - A. Position associated ANTI-ICE Panel (P5) Engine Anti-Ice selector OFF.
 - B. After the associated engine has been started, confirm the associated EAI indication (green EAI) is displayed above the N1 indication.

NOTE 1: If the EAI indication is not displayed, the valve may be stuck closed and a thrust setting slightly above minimum idle may open the valve.

NOTE 2: On the Air synoptic, the EAI valve will indicate failed open because the EAI selector is OFF and the green flow bar will be displayed.

6. Position associated ENGINE ANTI-ICE selector ON.

NOTE 1: When an EAI selector is positioned to ON and the temperature is greater than 10 degrees C, the ANTI-ICE ON advisory message will be displayed. Do not accomplish the ANTI-ICE ON Non-Normal Checklist.

NOTE 2: Inflight, the ENG IDLE DISAGREE advisory message may be displayed when engine thrust is reduced to idle.

NOTE 3: The ANTI-ICE ENG advisory message may be displayed during ground operations prior to takeoff. Flight crew should accomplish the associated non-normal checklist. If ANTI-ICE ENG clears, takeoff is allowed.

7. For the inoperative EAI valve, if the ANTI-ICE LEAK ENG L or R caution message is subsequently displayed:

For PW:

- A. Do not accomplish the two minute delay associated with the ANTI-ICE LEAK ENG L or R Non-Normal Checklist. Complete the remainder of the checklist as soon as practical.

**For RR:**

- B. Do not accomplish the initial one minute delay associated with the ANTI-ICE LEAK ENG L or R Non-Normal Checklist. Complete the remainder of the checklist as soon as practical.

30-21-01 Engine Anti-Ice Systems**30-21-01-02 GE****30-21-01-02A OAT 38 Degrees C or Below, With Deactivation Kit Installed**

Interval	Installed	Required	Procedure
C	2	1	(M) (O)

For operations at ambient temperature 38 degrees C or below, one may be inoperative provided:

- Valve is deactivated open using deactivation kit.
- Engine fan case overheat detection system operates normally.
- Turbine case cooling air flow systems operate normally.
- Appropriate performance adjustments are applied.

Except for GE90-100 Series:

- EECs must be operated in the normal mode.

MAINTENANCE (M)

Deactivate the associated EAI valve by installing the EAI valve lock-open kit, P/N K28703 (AMM 30-00-00/901).

- Remove inoperative EAI valve (AMM 30-21-02/401).
- Modify valve with lock-open kit (AMM 30-00-00/901).
- Re-install modified valve (AMM 30-21-02/401).
- Visually inspect the area near the Cowl Thermal Anti-Ice duct for signs of leaks and the fan cowl in general for flammable fluid leaks. Dispatch is not allowed if any of the duct clamps or supports are not secure or any air leaks from the EAI duct are present, or if significant flammable fluid leakage is present.

OPERATIONS (O)

- Reduce performance limited weights by the appropriate adjustments.

Performance Limited Weights	Takeoff & Landing	Enroute Climb
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**For -200 or -200ER/GE90-76B thru -90B:**

thru 632,500 lb (286,902 kg)	9,100 lb (4,128 kg)	11,800 lb (5,352 kg)
over 632,500 lb (286,902 kg)	13,200 lb (5,987 kg)	11,800 lb (5,352 kg)

For -200 or -200ER/GE90-94B:

thru 632,500 lb (286,902 kg)	9,100 lb (4,128 kg)	13,900 lb (6,305 kg)
over 632,500 lb (286,902 kg)	14,300 lb (6,486 kg)	13,900 lb (6,305 kg)

For -200LR or 777F/GE90-100 Series:

All	13,300 lb (6,033 kg)	12,100 lb (5,489 kg)
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For -300ER/GE90-115B:

All	10,400 lb (4,717 kg)	11,800 lb (5,353 kg)
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NOTE: The enroute climb weight penalties listed above are based on single engine operating speeds that approximate maximum lift-to-drag ratio speed. To account for the difference in level off altitude when operating at other speeds, multiply the enroute climb weight penalty listed above by the appropriate factor listed on page 3.00-01-00.2 in the ENROUTE DIVERSION SPEED EFFECTS paragraph.

2. Increase flight planning fuel by 1.0% for an inoperative EAI Valve or EAI Pressure Sensor, or by 1.3% for an inoperative EAI Card.
3. For ETOPS flight planning, increase engine inoperative Critical Fuel Reserves by 1.8%.
4. Position ENGINE ANTI-ICE selector(s) as follows:
 - A. For EAI Valve or EAI Pressure Sensor inoperative, position associated EAI selector ON.

NOTE 1: When an EAI selector is positioned to ON and the temperature is greater than 10 degrees C, the ANTI-ICE ON advisory message will be displayed. Do not accomplish the ANTI-ICE ON Non-Normal Checklist.

NOTE 2: Inflight, the ENG IDLE DISAGREE advisory message may be displayed when engine thrust is reduced to idle.

- B. For EAI Card inoperative, position both EAI selectors ON.

NOTE 1: When an EAI selector is positioned to ON and the temperature is greater than 10 degrees C, the ANTI-ICE ON advisory message will be displayed. Do not accomplish the ANTI-ICE ON Non-Normal Checklist.

NOTE 2: The associated ANTI-ICE ENG L or R advisory message will be displayed, the EAI indication will not be displayed on EICAS, and the associated EAI valve will be displayed as invalid (low intensity white) and the green flow bar will not be displayed on the Air synoptic.

5. For the inoperative EAI valve, if the ANTI-ICE LEAK ENG L or R caution message is subsequently displayed, do not accomplish the two minute delay associated with the ANTI-ICE LEAK ENG L or R Non-Normal Checklist. Complete the remainder of the checklist as soon as practical.

30-21-01 Engine Anti-Ice Systems

30-21-01-02 GE

30-21-01-02B OAT 38 Degrees C or Below, Without Deactivation Kit Installed

Interval	Installed	Required	Procedure
C	2	1	(M) (O)

For operations at ambient temperature 38 degrees C or below, one may be inoperative provided:

- a. Valve is electrically disconnected and verified open.
- b. Engine fan case overheat detection system operates normally.
- c. Associated EAI pressure sensor system operates normally.
- d. Associated EAI card operates normally.
- e. Turbine case cooling air flow systems operate normally.
- f. Appropriate performance adjustments are applied.

Except for GE90-100 Series:

- g. EECs must be operated in the normal mode.

MAINTENANCE (M)

For EAI VALVE L or R status message displayed, deactivate the EAI valve by disconnecting the electrical connector and verify EAI valve is open when the associated engine is running (AMM 30-00-00/901).

1. Deactivate the associated EAI valve controller.
 - A. For the right EAI valve, open P210 panel EAI CTRL R circuit breaker.
 - B. For the left EAI valve, open P310 panel EAI CTRL L circuit breaker.
2. Deactivate the leading edge slats in the retracted position (AMM 27-81-00/201).

3. Deactivate the associated thrust reverser for ground maintenance (AMM 78-31-00/201).
4. Deactivate the EAI valve in the open position.
 - A. On the associated engine, disconnect, cap and stow the torque motor electrical connector DM30211 or DM30211A from the EAI valve controller, M30211.
 - B. Install a protective cover over the electrical connector on the valve controller.
 - C. Visually inspect the area near the Cowl Thermal Anti-Ice duct for signs of leaks and the fan cowl in general for flammable fluid leaks. Dispatch is not allowed if any of the duct clamps or supports are not secure or any air leaks from the EAI duct are present, or if significant flammable fluid leakage is present.
5. Activate the thrust reverser (AMM 78-31-00/201).
6. Activate the leading edge slats (AMM 27-81-00/201).
7. Activate the associated EAI valve controller by closing the associated circuit breaker.
8. Verify the EAI valve is open when the associated engine is running.
 - A. Position associated ANTI-ICE Panel (P5) Engine Anti-Ice selector OFF.
 - B. Start the associated engine (71-00-00/201).
 - C. Confirm the EAI indication (green EAI) is displayed above the N1 indication on the EICAS display for the associated engine (indicates the EAI valve is open).

NOTE: If the EAI indication is not displayed, the valve may be stuck closed and a thrust setting slightly above minimum idle may open the valve.
 - D. Shutdown the associated engine (71-00-00/201).

OPERATIONS (O)

1. Reduce performance limited weights by the appropriate adjustments.

Performance Limited Weights	Takeoff & Landing	Enroute Climb
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For -200 or -200ER/GE90-76B thru -90B:

thru 632,500 lb (286,902 kg)	9,100 lb (4,128 kg)	11,800 lb (5,352 kg)
over 632,500 lb (286,902 kg)	13,200 lb (5,987 kg)	11,800 lb (5,352 kg)



For -200 or -200ER/GE90-94B:

thru 632,500 lb (286,902 kg)	9,100 lb (4,128 kg)	13,900 lb (6,305 kg)
over 632,500 lb (286,902 kg)	14,300 lb (6,486 kg)	13,900 lb (6,305 kg)

For -200LR or 777F/GE90-100 Series:

All	13,300 lb (6,033 kg)	12,100 lb (5,489 kg)
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For -300ER/GE90-115B:

All	10,400 lb (4,717 kg)	11,800 lb (5,353 kg)
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NOTE: The enroute climb weight penalties listed above are based on single engine operating speeds that approximate maximum lift-to-drag ratio speed. To account for the difference in level off altitude when operating at other speeds, multiply the enroute climb weight penalty listed above by the appropriate factor listed on page 3.00-01-00.2 in the ENROUTE DIVERSION SPEED EFFECTS paragraph.

2. Increase flight planning fuel by 1.0%.
3. For ETOPS flight planning, increase engine inoperative Critical Fuel Reserves by 1.8%.
4. Verify the EAI valve is open when the associated engine is running before each departure.
 - A. Position associated ANTI-ICE Panel (P5) Engine Anti-Ice selector OFF.
 - B. After the associated engine has been started, confirm the associated EAI indication (green EAI) is displayed above the N1 indication.

NOTE 1: If the EAI indication is not displayed, the valve may be stuck closed and a thrust setting slightly above minimum idle may open the valve.

NOTE 2: On the Air synoptic, the EAI valve will indicate failed open because the EAI selector is OFF and the green flow bar will be displayed.

5. Position associated ENGINE ANTI-ICE selector ON.

NOTE 1: When an EAI selector is positioned to ON and the temperature is greater than 10 degrees C, the ANTI-ICE ON advisory message will be displayed. Do not accomplish the ANTI-ICE ON Non-Normal Checklist.

NOTE 2: Inflight, the ENG IDLE DISAGREE advisory message may be displayed when engine thrust is reduced to idle.

NOTE 3: The ANTI-ICE ENG advisory message may be displayed during ground operations prior to takeoff. Flight crew should accomplish the associated non-normal checklist. If ANTI-ICE ENG clears, takeoff is allowed.

6. For the inoperative EAI valve, if the ANTI-ICE LEAK ENG L or R caution message is subsequently displayed, do not accomplish the two minute delay associated with the ANTI-ICE LEAK ENG L or R Non-Normal Checklist. Complete the remainder of the checklist as soon as practical.

30-21-01 Engine Anti-Ice Systems

30-21-01-02 GE

30-21-01-02C OAT Above 38 Degrees C, With Deactivation Kit Installed

Interval	Installed	Required	Procedure
A	2	1	(M) (O)

For operations at ambient temperature above 38 degrees C, one may be inoperative provided:

- a. Valve is deactivated open using deactivation kit.
- b. Engine fan case overheat detection system operates normally.
- c. Turbine case cooling air flow systems operate normally.
- d. Appropriate performance adjustments are applied.
- e. Repairs are made within 3 flight days.

Except for GE90-100 Series:

- f. EECs must be operated in the normal mode.

MAINTENANCE (M)

Deactivate the associated EAI valve by installing the EAI valve lock-open kit, P/N K28703 (AMM 30-00-00/901).

1. Remove inoperative EAI valve (AMM 30-21-02/401).
2. Modify valve with lock-open kit (AMM 30-00-00/901).
3. Re-install modified valve (AMM 30-21-02/401).
4. Visually inspect the area near the Cowl Thermal Anti-Ice duct for signs of leaks and the fan cowl in general for flammable fluid leaks. Dispatch is not allowed if any of the duct clamps or supports are not secure or any air leaks from the EAI duct are present, or if significant flammable fluid leakage is present.

OPERATIONS (O)

1. Reduce performance limited weights by the appropriate adjustments.



Performance Limited Weights	Takeoff & Landing	Enroute Climb
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For -200 or -200ER/GE90-76B thru -90B:

thru 632,500 lb (286,902 kg)	9,100 lb (4,128 kg)	11,800 lb (5,352 kg)
over 632,500 lb (286,902 kg)	13,200 lb (5,987 kg)	11,800 lb (5,352 kg)

For -200 or -200ER/GE90-94B:

thru 632,500 lb (286,902 kg)	9,100 lb (4,128 kg)	13,900 lb (6,305 kg)
over 632,500 lb (286,902 kg)	14,300 lb (6,486 kg)	13,900 lb (6,305 kg)

For -200LR or 777F/GE90-100 Series:

All	13,300 lb (6,033 kg)	12,100 lb (5,489 kg)
-----	----------------------	----------------------

For -300ER/GE90-115B:

All	10,400 lb (4,717 kg)	11,800 lb (5,353 kg)
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NOTE: The enroute climb weight penalties listed above are based on single engine operating speeds that approximate maximum lift-to-drag ratio speed. To account for the difference in level off altitude when operating at other speeds, multiply the enroute climb weight penalty listed above by the appropriate factor listed on page 3.00-01-00.2 in the ENROUTE DIVERSION SPEED EFFECTS paragraph.

2. Increase flight planning fuel by 1.0% for an inoperative EAI Valve or EAI Pressure Sensor, or by 1.3% for an inoperative EAI Card.
3. For ETOPS flight planning, increase engine inoperative Critical Fuel Reserves by 1.8%.
4. Position ENGINE ANTI-ICE selector(s) as follows:
 - A. For EAI Valve or EAI Pressure Sensor inoperative, position associated EAI selector ON.

NOTE 1: When an EAI selector is positioned to ON and the temperature is greater than 10 degrees C, the ANTI-ICE ON advisory message will be displayed. Do not accomplish the ANTI-ICE ON Non-Normal Checklist.

NOTE 2: Inflight, the ENG IDLE DISAGREE advisory message may be displayed when engine thrust is reduced to idle.

- B. For EAI Card inoperative, position both EAI selectors ON.

NOTE 1: When an EAI selector is positioned to ON and the temperature is greater than 10 degrees C, the ANTI-ICE ON advisory message will be displayed. Do not accomplish the ANTI-ICE ON Non-Normal Checklist.

NOTE 2: The associated ANTI-ICE ENG L or R advisory message will be displayed, the EAI indication will not be displayed on EICAS, and the associated EAI valve will be displayed as invalid (low intensity white) and the green flow bar will not be displayed on the Air synoptic.

5. For the inoperative EAI valve, if the ANTI-ICE LEAK ENG L or R caution message is subsequently displayed, do not accomplish the two minute delay associated with the ANTI-ICE LEAK ENG L or R Non-Normal Checklist. Complete the remainder of the checklist as soon as practical.

30-21-01 Engine Anti-Ice Systems

30-21-01-02 GE

30-21-01-02D OAT Above 38 Degrees C, Without Deactivation Kit Installed

Interval	Installed	Required	Procedure
A	2	1	(M) (O)

For operations at ambient temperature above 38 degrees C, one may be inoperative provided:

- Valve is electrically disconnected and verified open.
- Engine fan case overheat detection system operates normally.
- Associated EAI pressure sensor system operates normally.
- Associated EAI card operates normally.
- Turbine case cooling air flow systems operate normally.
- Appropriate performance adjustments are applied.
- Repairs are made within 3 flight days.

Except for GE90-100 Series:

- EECs must be operated in the normal mode.

MAINTENANCE (M)

For EAI VALVE L or R status message displayed, deactivate the EAI valve by disconnecting the electrical connector and verify EAI valve is open when the associated engine is running (AMM 30-00-00/901).

1. Deactivate the associated EAI valve controller.
 - A. For the right EAI valve, open P210 panel EAI CTRL R circuit breaker.
 - B. For the left EAI valve, open P310 panel EAI CTRL L circuit breaker.
2. Deactivate the leading edge slats in the retracted position (AMM 27-81-00/201).
3. Deactivate the associated thrust reverser for ground maintenance (AMM 78-31-00/201).
4. Deactivate the EAI valve in the open position.
 - A. On the associated engine, disconnect, cap and stow the torque motor electrical connector DM30211 or DM30211A from the EAI valve controller, M30211.
 - B. Install a protective cover over the electrical connector on the valve controller.
 - C. Visually inspect the area near the Cowl Thermal Anti-Ice duct for signs of leaks and the fan cowl in general for flammable fluid leaks. Dispatch is not allowed if any of the duct clamps or supports are not secure or any air leaks from the EAI duct are present, or if significant flammable fluid leakage is present.
5. Activate the thrust reverser (AMM 78-31-00/201).
6. Activate the leading edge slats (AMM 27-81-00/201).
7. Activate the associated EAI valve controller by closing the associated circuit breaker.
8. Verify the EAI valve is open when the associated engine is running.
 - A. Position associated ANTI-ICE Panel (P5) Engine Anti-Ice selector OFF.
 - B. Start the associated engine (71-00-00/201).
 - C. Confirm the EAI indication (green EAI) is displayed above the N1 indication on the EICAS display for the associated engine (indicates the EAI valve is open).

NOTE: If the EAI indication is not displayed, the valve may be stuck closed and a thrust setting slightly above minimum idle may open the valve.

- D. Shutdown the associated engine (71-00-00/201).

OPERATIONS (O)

1. Reduce performance limited weights by the appropriate adjustments.

Performance Limited Weights	Takeoff & Landing	Enroute Climb
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For -200 or -200ER/GE90-76B thru -90B:

thru 632,500 lb (286,902 kg)	9,100 lb (4,128 kg)	11,800 lb (5,352 kg)
over 632,500 lb (286,902 kg)	13,200 lb (5,987 kg)	11,800 lb (5,352 kg)

For -200 or -200ER/GE90-94B:

thru 632,500 lb (286,902 kg)	9,100 lb (4,128 kg)	13,900 lb (6,305 kg)
over 632,500 lb (286,902 kg)	14,300 lb (6,486 kg)	13,900 lb (6,305 kg)

For -200LR or 777F/GE90-100 Series:

All	13,300 lb (6,033 kg)	12,100 lb (5,489 kg)
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For -300ER/GE90-115B:

All	10,400 lb (4,717 kg)	11,800 lb (5,353 kg)
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NOTE: The enroute climb weight penalties listed above are based on single engine operating speeds that approximate maximum lift-to-drag ratio speed. To account for the difference in level off altitude when operating at other speeds, multiply the enroute climb weight penalty listed above by the appropriate factor listed on page 3.00-01-00.2 in the ENROUTE DIVERSION SPEED EFFECTS paragraph.

2. Increase flight planning fuel by 1.0%.
3. For ETOPS flight planning, increase engine inoperative Critical Fuel Reserves by 1.8%.
4. Verify the EAI valve is open when the associated engine is running before each departure.
 - A. Position associated ANTI-ICE Panel (P5) Engine Anti-Ice selector OFF.
 - B. After the associated engine has been started, confirm the associated EAI indication (green EAI) is displayed above the N1 indication.

NOTE 1: If the EAI indication is not displayed, the valve may be stuck closed and a thrust setting slightly above minimum idle may open the valve.

NOTE 2: On the Air synoptic, the EAI valve will indicate failed open because the EAI selector is OFF and the green flow bar will be displayed.



5. Position associated ENGINE ANTI-ICE selector ON.

NOTE 1: When an EAI selector is positioned to ON and the temperature is greater than 10 degrees C, the ANTI-ICE ON advisory message will be displayed. Do not accomplish the ANTI-ICE ON Non-Normal Checklist.

NOTE 2: Inflight, the ENG IDLE DISAGREE advisory message may be displayed when engine thrust is reduced to idle.

NOTE 3: The ANTI-ICE ENG advisory message may be displayed during ground operations prior to takeoff. Flight crew should accomplish the associated non-normal checklist. If ANTI-ICE ENG clears, takeoff is allowed.

6. For the inoperative EAI valve, if the ANTI-ICE LEAK ENG L or R caution message is subsequently displayed, do not accomplish the two minute delay associated with the ANTI-ICE LEAK ENG L or R Non-Normal Checklist. Complete the remainder of the checklist as soon as practical.



777 FAA DDG Supplement Package

Supplement Number: 36-2

Issue Date: 12-APR-2024

Section Number: 2 (MMEL)

Item Number: 31-51-05

Item Name: Warning Electronic System (WES) Autopilot Disconnect Warning Function

DDG Effectivity: 777 FAA DDG (D630W003-TBC), Revision 36

MMEL Effectivity: 777 FAA MMEL, Revision 23a

Description: Modified PBN guidance.

Reason: To correct and add more clarification for PBN operation capabilities.

Disposition: For reference in the Operator's MEL.

Attachments: DDG page 2.31-51-05.1

31-51-05 Warning Electronic System (WES) Autopilot Disconnect Warning Function

Interval	Installed	Required	Procedure
B	1	0	(O)

May be inoperative provided:

- Autopilots are not used.
- Approach minimums do not require use of the autopilots.
- Number of flight segments and segment duration is acceptable to flight crew.
- Enroute operations do not require use of the autopilots.
- Flight remains within 180 minutes of landing at a suitable airport.

OPERATIONS (O)

Do not use the autopilots and observe the following restrictions:

- Approach minimums must not require autopilot use.
- Enroute operations must not require autopilot use.
- The flight must remain within 180 minutes of landing at a suitable airport.
- For PBN operations:
 - PBN operations using autopilot are not allowed.
 - PBN operations may be conducted using flight director provided its use is appropriate for the intended operation.

NOTE: Refer to DDG Section 4 for additional information.

31-51-05 Warning Electronic System (WES) Autopilot Disconnect Warning Function

31-51-05-01 WES Autopilot Interfaces

Interval	Installed	Required	Procedure
C	3	2	(O)

One may be inoperative provided the WES autopilot disconnect warning function is verified to operate normally.

OPERATIONS (O)

Verify the Warning Electronic System (WES) autopilot disconnect warning function operates normally.

- Engage, then manually disengage the autopilots.
- Confirm illumination of the master warning lights, activation of an aural indication and display of the AUTOPILOT DISC warning message.



777 FAA DDG Supplement Package

Supplement Number: 36-2

Issue Date: 12-APR-2024

Section Number: 2 (MMEL)

Item Number: 31-61-04

Item Name: Display Select Panel (DSP)

DDG Effectivity: 777 FAA DDG (D630W003-TBC), Revision 36

MMEL Effectivity: 777 FAA MMEL, Revision 23a

Description: Reformatted into subitems for -200 and -300 series airplanes and added notes.

Reason: To account for effects on the Flight Deck Visual Surveillance System and Ground Maneuver Camera System.

Disposition: For reference in the Operator's MEL.

Attachments: DDG page 2.31-61-04.1-2

**31-61-04 Display Select Panel (DSP)****31-61-04-01 -200/-200ER/-200LR/777F**

Interval	Installed	Required	Procedure
C	1	0	(M) (O)

May be inoperative provided the left and right CDU DSP backup features are verified to operate normally.

For Flight Deck Door Visual Surveillance System (FDDVSS) installed:

NOTE: For airplanes with installed EFB, if FDDVSS cannot be selected on the installed EFB using the VIDEO application, FDDVSS is inoperative and the airplane must also be dispatched using MMEL item 23-70-01.

NOTE: For airplanes without installed EFB, FDDVSS is inoperative and the airplane must also be dispatched using MMEL item 23-70-01.

MAINTENANCE (M)

Verify the left and right CDU DSP backup features operate normally (AMM 31-00-00/901).

1. Press DSP CTL (Display Select Panel Control select key) on the main menu of the CDU to ON.
2. Confirm that the DSP prompt is displayed on the main menu of the CDU beside the Display Select Panel page select key. This indicates display select control is available through the CDU.

OPERATIONS (O)

Use the CDU Display Select Panel (DSP) backup feature.

31-61-04 Display Select Panel (DSP)**31-61-04-02 -300/-300ER**

Interval	Installed	Required	Procedure
C	1	0	(M) (O)

May be inoperative provided:

- a. The left and right CDU DSP backup features are verified to operate normally.
- b. The ground maneuver camera system is considered inoperative.

NOTE 1: The airplane must also be dispatched using MMEL item 31-61-10.

For Flight Deck Door Visual Surveillance System (FDDVSS) installed:

NOTE 2: For airplanes with installed EFB, if FDDVSS cannot be selected on the installed EFB using the VIDEO application, FDDVSS is inoperative and the airplane must also be dispatched using MMEL item 23-70-01.



NOTE 3: For airplanes without installed EFB, FDDVSS is inoperative and the airplane must also be dispatched using MMEL item 23-70-01.

MAINTENANCE (M)

Verify the left and right CDU DSP backup features operate normally (AMM 31-00-00/901).

1. Press DSP CTL (Display Select Panel Control select key) on the main menu of the CDU to ON.
2. Confirm that the DSP prompt is displayed on the main menu of the CDU beside the Display Select Panel page select key. This indicates display select control is available through the CDU.

OPERATIONS (O)

Use the CDU Display Select Panel (DSP) backup feature.

31-61-04 Display Select Panel (DSP)

31-61-04-03 Multifunction Display Lights

Interval	Installed	Required	Procedure
C	3	0	

31-61-04 Display Select Panel (DSP)

31-61-04-04 Illuminated Switch Lights

31-61-04-04-01 -200/-200ER/-200LR/777F

Interval	Installed	Required	Procedure
C	15	0	

31-61-04 Display Select Panel (DSP)

31-61-04-04 Illuminated Switch Lights

31-61-04-04-02 -300/-300ER

Interval	Installed	Required	Procedure
C	16	0	



777 FAA DDG Supplement Package

Supplement Number: 36-2

Issue Date: 12-APR-2024

Section Number: 2 (MMEL)

Item Number: 32-08-01

Item Name: Proximity Sensor Electronic Unit (PSEU) Channels

DDG Effectivity: 777 FAA DDG (D630W003-TBC), Revision 36

MMEL Effectivity: 777 FAA MMEL, Revision 23a

Description: Modified Maintenance (M) procedures steps for item option 32-08-01C and 32-08-01D.

Reason: To include the correct circuit breaker references for PSEU 1 and PSEU 2 verification purpose.

Disposition: For reference in the Operator's MEL.

Attachments: DDG page 2. 32-08-01.1-3

32-08-01 Proximity Sensor Electronic Unit (PSEU) Channels
32-08-01A One Inoperative - PSEU 1

Interval	Installed	Required	Procedure
C	4	3	

One may be inoperative in PSEU 1 provided the flight lock system is considered inoperative.

NOTE: The airplane must also be dispatched using MMEL item 52-11-05.

32-08-01 Proximity Sensor Electronic Unit (PSEU) Channels
32-08-01B One Inoperative - PSEU 2

Interval	Installed	Required	Procedure
C	4	3	

One may be inoperative in PSEU 2.

32-08-01 Proximity Sensor Electronic Unit (PSEU) Channels
32-08-01C Two Inoperative - PSEU 1 On Right System Bus, PSEU 2 On Left System Bus

Interval	Installed	Required	Procedure
C	4	2	(M) (O)

One may be inoperative in each PSEU provided:

- PSEU 1 transmission is verified on the left ARINC 629 systems bus.
- PSEU 2 transmission is verified on the right ARINC 629 systems bus.
- Flight lock system is considered inoperative.

NOTE: The airplane must also be dispatched using MMEL item 52-11-05.

MAINTENANCE (M)

Verify PSEU 1 transmission is on the left ARINC 629 systems bus and PSEU 2 transmission is on the right ARINC 629 systems bus (AMM 32-00-00/901).

- Verify PSEU 1 and PSEU 2 data is displayed on the Landing Gear ACTN/INDN maintenance page.
- Open P110 panel PSEU 1 PRI PWR circuit breaker.
- Verify PSEU 1 data is removed from the Landing Gear ACTN/INDN display. PSEU 1 is transmitting on the left ARINC 629 systems bus.



4. Open P210 panel PSEU 2 PRI PWR circuit breaker.
5. Verify PSEU 2 data is removed from the Landing Gear ACTN/INDN display. PSEU 2 is transmitting on the right ARINC 629 systems bus.
6. Close the PSEU 1 PRI PWR and the PSEU 2 PRI PWR circuit breakers.

OPERATIONS (O)

The autobrake selector may not automatically move from RTO to OFF after takeoff. If autobrake selector remains in RTO, manually move to OFF.

32-08-01 Proximity Sensor Electronic Unit (PSEU) Channels
32-08-01D Two Inoperative - PSEU 1 On Left System Bus, PSEU 2 On Right System Bus

Interval	Installed	Required	Procedure
C	4	2	(M) (O)

One may be inoperative in each PSEU provided:

- a. PSEU 1 transmission is verified on the right ARINC 629 systems bus.
- b. PSEU 2 transmission is verified on the left ARINC 629 systems bus.
- c. After takeoff, gear remains down for two minutes before retraction.
- d. Takeoff performance is based on landing gear extended.
- e. Flight lock system is considered inoperative

NOTE: The airplane must also be dispatched using MMEL item 52-11-05.

MAINTENANCE (M)

Verify that PSEU 1 transmission is on the right ARINC 629 systems bus and PSEU 2 transmission is on the left ARINC 629 systems bus (AMM 32-00-00/901).

1. Verify PSEU 1 and PSEU 2 data is displayed on the Landing Gear ACTN/INDN maintenance page.
2. Open P310 panel PSEU 1 ALTN PWR circuit breaker.
3. Verify PSEU 1 data is removed from the Landing Gear ACTN/INDN display. PSEU 1 is transmitting on the right ARINC 629 bus.
4. Open P310 panel PSEU 2 ALTN PWR circuit breaker.
5. Verify PSEU 2 data is removed from the Landing Gear ACTN/INDN display. PSEU 2 is transmitting on the left ARINC 629 systems bus.
6. Close the PSEU 1 ALTN PWR and PSEU 2 ALTN PWR circuit breakers.

OPERATIONS (O)

1. The autobrake selector may not automatically move from RTO to OFF after takeoff. If autobrake selector remains in RTO, manually move to OFF.
2. After takeoff, leave the landing gear extended for a minimum of two minutes to allow the wheels to spin down prior to gear retraction.



3. Base takeoff performance on landing gear extended. Takeoff performance for the landing gear extended is available using AFM-DPI or Boeing Takeoff Module (BTM).
4. Increase flight planning fuel by 1,900 lbs (861 kg).

32-08-01 Proximity Sensor Electronic Unit (PSEU) Channels**32-08-01E Two Inoperative - Gear Down Dispatch**

Interval	Installed	Required	Procedure
C	4	2	(M) (O)

One may be inoperative in each PSEU provided:

- a. Landing gear are secured in the down position.
- b. Airplane is dispatched in accordance with the appropriate AFM Gear Extended Appendix.
- c. Flight lock system is considered inoperative

NOTE: The airplane must also be dispatched using MMEL item 52-11-05.

MAINTENANCE (M)

Secure the landing gear in the down position (AMM 32-00-00/901).

1. Install landing gear downlock pins, including installation procedure steps for airplane flight with landing gear locked in the extended position (AMM 32-00-30/201).
2. Place Alternate Gear Down Dispatch switch (Main Equipment Center above lower access door) to VMO position.

NOTE: The Warning Electronics System (WES) will set VMO/MMO to 270 KIAS/.73 Mach and VMO GEAR DOWN memo message will be displayed.

3. Deactivate the lever lock solenoid for the landing gear control lever by opening and collaring the P210 panel LG LEVER LOCK circuit breaker.

OPERATIONS (O)

1. The autobrake selector may not automatically move from RTO to OFF after takeoff. If autobrake selector remains in RTO, manually move to OFF.
2. Observe AFM Landing Gear Extended certificate limitations and operational limits.

NOTE: Dispatch requires authorization by the AFM Landing Gear Extended Appendix.

3. Base flight plan on landing gear extended.
4. The Warning Electronics System (WES) will set VMO/MMO to 270 KIAS/.73 Mach and VMO GEAR DOWN memo message will be displayed.



777 FAA DDG Supplement Package

Supplement Number: 36-2

Issue Date: 12-APR-2024

Section Number: 2 (MMEL)

Item Number: 34-46-01

Item Name: Ground Proximity Warning System (GPWS)

DDG Effectivity: 777 FAA DDG (D630W003-TBC), Revision 36

MMEL Effectivity: 777 FAA MMEL, Revision 23a

Description: Modified PBN guidance.

Reason: To correct and add more clarification for PBN operation capabilities.

Disposition: For reference in the Operator's MEL.

Attachments: DDG page 2.34-46-01.1-6

**34-46-01 Ground Proximity Warning System (GPWS)****34-46-01-01 GPWS Function**

Interval	Installed	Required	Procedure
A	1	0	(O)

May be inoperative provided:

- a. Alternate procedures are established and used.
 - b. Repairs are made within 2 flight days.
-

[For Honeywell weather radar installed:](#)

NOTE: The airplane must also be dispatched using MMEL item 34-43-01-02.

OPERATIONS (O)

NOTE 1: The GPWS provides the aural signal for the automatic V1 callout and the Engine Fail alert. These aural signals may not be available.

NOTE 2: The GND PROX SYS advisory message will be displayed.

[For Honeywell weather radar installed:](#)

NOTE 3: AUTOTILT FAIL may be annunciated on the ND.

1. Alternate procedures must be established and used.

NOTE: These procedures must maintain flight crew awareness of airplane altitude and flight path. They may include flight crew use of all systems available (autopilot, autoland, ILS and FMCS) and approach altitude callouts.

2. For PBN operations:
 - A. RNP AR APCH operations are not allowed.
 - B. Other PBN operations may be conducted.

NOTE: Refer to DDG Section 4 for additional information.

34-46-01 Ground Proximity Warning System (GPWS)**34-46-01-01 GPWS Function****34-46-01-01-01 Modes 1 - 4**

Interval	Installed	Required	Procedure
A	4	0	(O)

May be inoperative provided:

- a. Alternate procedures are established and used.
 - b. Repairs are made within 2 flight days.
-

**OPERATIONS (O)**

1. Alternate procedures must be established and used.

NOTE: These procedures must maintain flight crew awareness of airplane altitude and flight path. They may include flight crew use of all systems available (autopilot, autoland, ILS and FMCS) and approach altitude callouts.

2. For PBN operations:

For GND PROX SYS advisory message displayed,

- A. RNP AR APCH operations are not allowed.
- B. Other PBN operations may be conducted.

NOTE: Refer to DDG Section 4 for additional information.

34-46-01 Ground Proximity Warning System (GPWS)**34-46-01-01 GPWS Function****34-46-01-01-02 Glideslope Deviation (Mode 5)**

Interval	Installed	Required	Procedure
B	1	0	

OPERATIONS NOTE

For PBN operations:

For GND PROX SYS advisory message displayed,

1. RNP AR APCH operations are not allowed.
2. Other PBN operations may be conducted.

NOTE: Refer to DDG Section 4 for additional information.

34-46-01 Ground Proximity Warning System (GPWS)**34-46-01-01 GPWS Function****34-46-01-01-03 Advisory Callouts (Mode 6)****34-46-01-01-03A Required By 14 CFR**

Interval	Installed	Required	Procedure
B	-	0	(O)

May be inoperative provided alternate procedures are established and used.

OPERATIONS (O)

NOTE 1: The GPWS provides the aural signal for the automatic V1 callout and the Engine Fail alert. These aural signals may not be available.

NOTE 2: See 14 CFR 121.354 or other appropriate regulation.

1. Alternate procedures must be established and used.

NOTE: These procedures must maintain flight crew awareness of airplane altitude and flight path. They may include flight crew use of all systems available (autopilot, autoland, ILS and FMCS) and approach altitude callouts.

2. For PBN operations:

For GND PROX SYS advisory message displayed,

- A. RNP AR APCH operations are not allowed.
- B. Other PBN operations may be conducted.

NOTE: Refer to DDG Section 4 for additional information.

34-46-01 Ground Proximity Warning System (GPWS)

34-46-01-01 GPWS Function

34-46-01-01-03 Advisory Callouts (Mode 6)

34-46-01-01-03B Not Required by 14 CFR

Interval	Installed	Required	Procedure
C	-	0	(O)

May be inoperative provided:

- a. Advisory callouts are not required by 14 CFR.
- b. Alternate procedures are established and used.

OPERATIONS (O)

NOTE 1: The GPWS provides the aural signal for the automatic V1 callout and the Engine Fail alert. These aural signals may not be available.

NOTE 2: See 14 CFR 121.354 or other appropriate regulation.

1. Alternate procedures must be established and used.

NOTE: These procedures must maintain flight crew awareness of airplane altitude and flight path. They may include flight crew use of all systems available (autopilot, autoland, ILS and FMCS) and approach altitude callouts.

2. For PBN operations:

For GND PROX SYS advisory message displayed,

- A. RNP AR APCH operations are not allowed.
- B. Other PBN operations may be conducted.

NOTE: Refer to DDG Section 4 for additional information.

**34-46-01 Ground Proximity Warning System (GPWS)****34-46-01-02 Windshear Alert Mode Function (Reactive) (Mode 7)****34-46-01-02A Predictive Windshear Operative**

Interval	Installed	Required	Procedure
C	1	0	(O)

May be inoperative provided:

- Alternate procedures are established and used.
- Predictive Windshear operates normally.

OPERATIONS (O)

Alternate procedures must be established and used.

NOTE: The WINDSHEAR SYS advisory message will be displayed.

34-46-01 Ground Proximity Warning System (GPWS)**34-46-01-02 Windshear Alert Mode Function (Reactive) (Mode 7)****34-46-01-02B Predictive Windshear Inoperative**

Interval	Installed	Required	Procedure
B	1	0	(O)

May be inoperative provided alternate procedures are established and used.

NOTE: Operator's alternate procedures should include reviewing windshear avoidance and windshear recovery procedures.

OPERATIONS (O)

Establish alternate procedures for reactive windshear inoperative.

NOTE: The WINDSHEAR SYS advisory message will be displayed.

34-46-01 Ground Proximity Warning System (GPWS)**34-46-01-03 Terrain Awareness Function**

Interval	Installed	Required	Procedure
B	1	0	(O)

May be inoperative provided alternate procedures are established and used.

[For Honeywell weather radar installed:](#)

NOTE: The airplane must also be dispatched using MMEL item 34-43-01-02.

OPERATIONS (O)

NOTE 1: The GND PROX SYS advisory message will be displayed.

NOTE 2: See 14 CFR 121.354 or other appropriate regulation.

[For Honeywell weather radar installed:](#)

NOTE 3: AUTOTILT FAIL may be annunciated on the ND.

1. Alternate procedures must be established and used.
2. For PBN operations:
 - A. RNP AR APCH operations are not allowed.
 - B. Other PBN operations may be conducted.

NOTE: Refer to DDG Section 4 for additional information.

34-46-01 Ground Proximity Warning System (GPWS)

34-46-01-03 Terrain Awareness Function

34-46-01-03-01 Terrain Display Functions

34-46-01-03-01A One Inoperative

Interval	Installed	Required	Procedure
C	2	1	

OPERATIONS NOTE

1. The GND PROX SYS advisory message may be displayed.
2. See 14 CFR 121.354 or other appropriate regulation.
3. For PBN operations:

For GND PROX SYS advisory message displayed,

 - A. RNP AR APCH operations are not allowed.
 - B. Other PBN operations may be conducted.

NOTE: Refer to DDG Section 4 for additional information.

34-46-01 Ground Proximity Warning System (GPWS)

34-46-01-03 Terrain Awareness Function

34-46-01-03-01 Terrain Display Functions

34-46-01-03-01B Both Inoperative

Interval	Installed	Required	Procedure
B	2	0	



[For Honeywell weather radar installed:](#)

NOTE: The airplane must also be dispatched using MMEL item 34-43-01-02.

OPERATIONS NOTE

1. The GND PROX SYS advisory message may be displayed.
2. For PBN operations:
For GND PROX SYS advisory message displayed,
 - A. RNP AR APCH operations are not allowed.
 - B. Other PBN operations may be conducted.

NOTE: Refer to DDG Section 4 for additional information

[For Honeywell weather radar installed:](#)

3. AUTOTILT FAIL may be annunciated on the ND.

34-46-01 Ground Proximity Warning System (GPWS)

34-46-01-04 Runway Awareness and Advisory System (RAAS)

Interval	Installed	Required	Procedure
C	1	0	

OPERATIONS NOTE

For RUNWAY SYS advisory message displayed, consider using the RUNWAY OVRD function.



777 FAA DDG Supplement Package

Supplement Number: 36-2

Issue Date: 12-APR-2024

Section Number: 2 (MMEL)

Item Number: 34-51-01

Item Name: VOR Navigation Systems

DDG Effectivity: 777 FAA DDG (D630W003-TBC), Revision 36

MMEL Effectivity: 777 FAA MMEL, Revision 23a

Description: Modified PBN guidance.

Reason: To correct and add more clarification for PBN operation capabilities.

Disposition: For reference in the Operator's MEL.

Attachments: DDG page 2.34-51-01.1

**34-51-01 VOR Navigation Systems**

Interval	Installed	Required	Procedure
D	2	-	

Any in excess of those required by 14 CFR may be inoperative.

OPERATIONS NOTE

1. See 14 CFR 121.347 through 121.351 or other appropriate regulation.
2. For PBN operations:
 - A. RNAV 1, RNAV 2, P-RNAV, B-RNAV and RNAV 5 operations may be conducted provided one GPS, one DME/DME or one VOR/DME operates normally.
 - B. Other PBN operations may be conducted.

NOTE: Refer to DDG Section 4 for additional information.



777 FAA DDG Supplement Package

Supplement Number: 36-2

Issue Date: 12-APR-2024

Section Number: 2 (MMEL)

Item Number: 34-55-01

Item Name: Distance Measuring Equipment (DME)

DDG Effectivity: 777 FAA DDG (D630W003-TBC), Revision 36

MMEL Effectivity: 777 FAA MMEL, Revision 23a

Description: Modified PBN guidance.

Reason: To correct and add more clarification for PBN operation capabilities.

Disposition: For reference in the Operator's MEL.

Attachments: DDG page 2.34-55-01.1

**34-55-01 Distance Measuring Equipment (DME)**

Interval	Installed	Required	Procedure
D	2	-	

Any in excess of those required by 14 CFR may be inoperative.

OPERATIONS NOTE

1. See 14 CFR 121.349 or other appropriate regulation.
2. For PBN operations:
 - A. RNAV 1, RNAV 2, P-RNAV, B-RNAV and RNAV 5 operations may be conducted provided one GPS, one DME/DME or one VOR/DME operates normally.
 - B. Other PBN operations may be conducted.

NOTE: Refer to DDG Section 4 for additional information.



777 FAA DDG Supplement Package

Supplement Number: 36-2

Issue Date: 12-APR-2024

Section Number: 2 (MMEL)

Item Number: 34-58-01

Item Name: Global Positioning Systems

DDG Effectivity: 777 FAA DDG (D630W003-TBC), Revision 36

MMEL Effectivity: 777 FAA MMEL, Revision 23a

Description: Modified PBN guidance.

Reason: To correct and add more clarification for PBN operation capabilities.

Disposition: For reference in the Operator's MEL.

Attachments: DDG page 2.34-58-01.1-3

34-58-01 Global Positioning Systems

34-58-01A Both Inoperative

Interval	Installed	Required	Procedure
C	2	0	

May be inoperative provided:

- Enroute operations do not require their use.
- SAARU operates normally.

NOTE: The airplane must also be dispatched using MMEL item 34-53-02-01.

MAINTENANCE NOTE

- For both GPS inoperative and AIMS power is interrupted, UTC is lost and the AIMS time function will reset to midnight.
- If desired, the left, right, or center MMR may be deactivated by opening and collaring P11 panel MMR-L, MMR-R and/or MMR-C circuit breaker.
- If the GPS L and GPS R status messages are caused by a fault in the center MMR antenna, opening the MMR-C circuit breaker may restore GPS L and R functions.
- The associated ILS function will be inoperative for a deactivated MMR. Relief for ILS functions is available in MMEL item 34-31-01.

OPERATIONS NOTE

- Operators must consult their Operations Specification for route position accuracy requirements such as that required by certain Required Navigation Performance (RNP) routes.
- The NAV UNABLE RNP advisory message may be displayed during lengthy ground operations and flight crew should accomplish the associated Non-Normal Checklist.
- The TERR POS advisory message will be displayed.
- The RWY/POS update on the FMC-CDU TAKEOFF REF page will be active.
- The ADS-B Out data will be inoperative.
- For airplanes equipped with RAAS:
 - RAAS is inoperative.
 - The RUNWAY SYS advisory message will be displayed.
 - The airplane must also be dispatched using MMEL item 34-46-01-04.
- For PBN operations:
 - RNP AR APCH, RNP APCH, RNP 1, RNP 2, RNP 4 and Advanced RNP operations are not allowed.



- B. RNAV 1, RNAV 2, P-RNAV, B-RNAV and RNAV 5 operations may be conducted provided one DME/DME or one VOR/DME operates normally.
- C. RNP 10 and RNAV 10 operations may be conducted provided appropriate time limitations are observed. Refer to the RNP Navigation Capabilities Document for the time limitations.

NOTE: Refer to DDG Section 4 for additional information.

34-58-01 Global Positioning Systems

34-58-01B One Inoperative

Interval	Installed	Required	Procedure
D	2	1	

One may be inoperative provided enroute operations do not require its use.

NOTE: The airplane must also be dispatched using MMEL item 34-53-02-01.

MAINTENANCE NOTE

1. If desired, one or both MMR not associated with the operative GPS may be deactivated by opening and collaring P11 panel MMR-L, MMR-R and/or MMR-C circuit breaker.
2. The associated ILS function will be inoperative for a deactivated MMR. Relief for ILS functions is available in MMEL item 34-31-01.

For EFB installed:

NOTE: Some functionality of the EFB may be inoperative. For inoperative EFB functions the airplane must also be dispatched using MMEL item 46-11-02-01.

OPERATIONS NOTE

1. The associated ADS-B Out data will be inoperative.
2. For PBN operations:
 - A. RNP AR APCH and RNP 4 operations are not allowed.
 - B. RNP 2 operations over oceanic and remote continental airspace are not allowed.
 - C. RNP APCH, RNP 1, other RNP 2, and Advanced RNP operations may be allowed provided alternate navigation equipment (e.g. DME/DME, VOR/DME, VOR, ADF) is available to continue to a suitable airport if the remaining GPS becomes inoperative.
 - D. RNAV 1, RNAV 2, P-RNAV, B-RNAV and RNAV 5 operations may be conducted.
 - E. RNP 10 and RNAV 10 operations may be conducted provided appropriate time limitations are observed. Refer to the RNP Navigation Capabilities Document for the time limitations.



NOTE: Refer to DDG Section 4 for additional information.

Supplement 36-2
Superseded



777 FAA DDG Supplement Package

Supplement Number: 36-2

Issue Date: 12-APR-2024

Section Number: 2 (MMEL)

Item Number: 34-61-01

Item Name: Flight Management Computing Systems (FMCS)

DDG Effectivity: 777 FAA DDG (D630W003-TBC), Revision 36

MMEL Effectivity: 777 FAA MMEL, Revision 23a

Description: Modified PBN guidance.

Reason: To correct and add more clarification for PBN operation capabilities.

Disposition: For reference in the Operator's MEL.

Attachments: DDG page 2.34-61-01.1-3

**34-61-01 Flight Management Computing Systems (FMCS)****34-61-01-01 With AIMS-1**

Interval	Installed	Required	Procedure
C	2	1	(M)

One may be inoperative provided:

- Remaining FMC is verified to operate normally.
- Enroute operations do not require its use.
- Flight remains within 180 minutes of landing at a suitable airport.

MAINTENANCE (M)

Verify remaining FMC is operating normally by positioning the FMC selector to AUTO and verifying the FMC status and advisory messages are not displayed (AMM 34-00-00/901).

OPERATIONS NOTE

- Flight must remain within 180 minutes of landing at a suitable airport.
- Operators must consult their Operations Specification for route flight management computing system requirements.
- For PBN operations:
 - RNP AR APCH and RNP 4 operations are not allowed.
 - RNP 2 operations over oceanic and remote continental airspace are not allowed.
 - RNP APCH, RNP 1, other RNP 2, RNAV 1, RNAV 2, P-RNAV, B-RNAV, RNAV 5 and Advanced RNP operations may be conducted provided alternate navigation equipment (e.g., VOR, ADF) is available to continue to a suitable airport if the remaining FMCS becomes inoperative.
 - RNP 10 and RNAV 10 operations may be conducted using the alternate navigation capability in the multifunction control display units (MCDU).

NOTE: Refer to DDG Section 4 for additional information.

34-61-01 Flight Management Computing Systems (FMCS)**34-61-01-02 With AIMS-2**

Interval	Installed	Required	Procedure
C	2	1	(M)

One may be inoperative provided:

- Remaining FMC is verified to operate normally.

- b. Enroute operations do not require its use.

MAINTENANCE (M)

Verify remaining FMC is operating normally by positioning the FMC selector to AUTO and verifying the FMC status and advisory messages are not displayed (AMM 34-00-00/901).

OPERATIONS NOTE

1. Operators must consult their Operations Specification for route flight management computing system requirements.
2. For PBN operations:
 - A. RNP AR APCH and RNP 4 operations are not allowed.
 - B. RNP 2 operations over oceanic and remote continental airspace are not allowed.
 - C. RNP APCH, RNP 1, other RNP 2, RNAV 1, RNAV 2, P-RNAV, B-RNAV, RNAV 5 and Advanced RNP operations may be conducted provided alternate navigation equipment (e.g., VOR, ADF) is available to continue to a suitable airport if the remaining FMCS becomes inoperative.
 - D. RNP 10 and RNAV 10 operations may be conducted using the alternate navigation capability in the multifunction control display units (MCDU).

NOTE: Refer to DDG Section 4 for additional information.

34-61-01 Flight Management Computing Systems (FMCS)

34-61-01-03 Navigation Databases

Interval	Installed	Required	Procedure
A	-	0	

May be inoperative provided:

- a. Operations do not require its use.
- b. It is not used in a primary navigation system required by 14 CFR.
- c. Alternate procedures are developed and used.
- d. The ICAO Flight Plan is updated (as required) to notify ATC of the navigation equipment status of the aircraft.
- e. Is repaired within 10 flight days.

NOTE: An out-of-currency or out-of-date navigation database is not authorized MMEL relief per 14 CFR.



NOTE: Although this item may not be applied for an out-of-currency Navigation Database, it does not prohibit dispatch in that condition. Dispatch is allowed with an out-of-currency Navigation Database provided alternate procedures are developed and incorporated in another operational manual, in coordination with local regulatory authorities.

OPERATIONS NOTE

For PBN operations:

■ The FMCS navigation database must be current for all PBN operations.

NOTE 1: It is assumed that the operator has a preflight procedure to ensure that the navigation database complies with regulatory agency requirements for ensuring the navigation database remains valid for the duration of the flight unless an alternative regulatory agency approved process applies.

NOTE 2: Refer to DDG Section 4 for additional information.



777 FAA DDG Supplement Package

Supplement Number: 36-2

Issue Date: 12-APR-2024

Section Number: 2 (MMEL)

Item Number: 35-21-06

Item Name: Supernumerary Seats/Crew Rest/Lavatory Gaseous Oxygen System (777F)

DDG Effectivity: 777 FAA DDG (D630W003-TBC), Revision 36

MMEL Effectivity: 777 FAA MMEL, Revision 23a

Description: Modified Maintenance (M) procedures note 4.

Reason: To include the correct AMM reference for deactivation of the supernumerary oxygen system.

Disposition: For reference in the Operator's MEL.

Attachments: DDG page 2.35-21-06.1-3

35-21-06 Supernumerary Seats/Crew Rest/Lavatory Gaseous Oxygen System (777F)

Interval	Installed	Required	Procedure
B	1	0	(M) (O)

May be inoperative provided:

- Supernumerary seats and crew rest bunks are blocked and placarded DO NOT OCCUPY.
- An adequate portable oxygen dispensing unit (bottle and mask) is available in the lavatory.
- Personnel address system operates normally.
- Crew members/supernumeraries are appropriately briefed.

MAINTENANCE (M)

NOTE 1: For SUPRNMRY OXYGEN L status message displayed, the oxygen module over the left seat and the oxygen module in the crew rest are inoperative.

NOTE 2: For SUPRNMRY OXYGEN C status message displayed, the oxygen module over the center seats and the oxygen module in the lavatory are inoperative.

NOTE 3: For SUPRNMRY OXYGEN R status message displayed, the oxygen module over the right seat is inoperative.

NOTE 4: If desired, system may be deactivated (AMM 35-23-00)/201).

Block seats and crew rest bunks and placard DO NOT OCCUPY (AMM 35-00-00/901).

OPERATIONS (O)

Crew members/supernumeraries must be briefed prior to takeoff regarding alternate procedures.

35-21-06 Supernumerary Seats/Crew Rest/Lavatory Gaseous Oxygen System (777F)

35-21-06-01 Left Circuit

Interval	Installed	Required	Procedure
B	1	0	(M) (O)

May be inoperative provided:

- Left supernumerary seat is blocked and placarded DO NOT OCCUPY.



- b. Each crew rest bunk without an adequate portable oxygen dispensing unit (bottle and mask) is blocked and placarded DO NOT OCCUPY.
 - c. For flightcrew rest facility used, personnel address system operates normally.
 - d. Crew members/supernumeraries are appropriately briefed.
-

MAINTENANCE (M)

NOTE: For SUPRNMRY OXYGEN L status message displayed, the oxygen module over the left seat and the oxygen module in the crew rest are inoperative.

Block left seat and placard DO NOT OCCUPY. Block crew rest bunks without adequate portable oxygen dispensing units (bottle and mask), and placard DO NOT OCCUPY (AMM 35-00-00/901).

OPERATIONS (O)

Crew members/supernumeraries must be briefed prior to takeoff regarding alternate procedures.

35-21-06 Supernumerary Seats/Crew Rest/Lavatory Gaseous Oxygen System (777F)

35-21-06-02 Center Circuit

Interval	Installed	Required	Procedure
B	1	0	(M) (O)

May be inoperative provided:

- a. Each center supernumerary seat is blocked and placarded DO NOT OCCUPY.
 - b. An adequate portable oxygen dispensing unit (bottle and mask) is available in the lavatory.
 - c. Personnel address system operates normally.
 - d. Crew members/supernumeraries are appropriately briefed.
-

MAINTENANCE (M)

NOTE: For SUPRNMRY OXYGEN C status message displayed, the oxygen module over the center seats and the oxygen module in the lavatory are inoperative.

Block each center seat and placard DO NOT OCCUPY (AMM 35-00-00/901).

OPERATIONS (O)

Crew members/supernumeraries must be briefed prior to takeoff regarding alternate procedures.

**35-21-06 Supernumerary Seats/Crew Rest/Lavatory Gaseous Oxygen System (777F)**

35-21-06-03 Right Circuit

Interval	Installed	Required	Procedure
B	1	0	(M) (O)

May be inoperative provided:

- Right supernumerary seat is blocked and placarded DO NOT OCCUPY.
- Crew members/supernumeraries are appropriately briefed.

MAINTENANCE (M)

NOTE: For SUPRNMRY OXYGEN R status message displayed, the oxygen module over the right seat is inoperative.

Block right seat and placard DO NOT OCCUPY (AMM 35-00-00/901).

OPERATIONS (O)

Crew members/supernumeraries must be briefed prior to takeoff regarding alternate procedures.



777 FAA DDG Supplement Package

Supplement Number: 36-2

Issue Date: 12-APR-2024

Section Number: 2 (MMEL)

Item Number: 36-00-01

Item Name: Air Supply and Cabin Pressure Controllers (ASCPC)

DDG Effectivity: 777 FAA DDG (D630W003-TBC), Revision 36

MMEL Effectivity: 777 FAA MMEL, Revision 23a

Description: Modified proviso.

Reason: Only the (associated) Right Engine Idle Selection System and Right Auto Cabin Pressure Control need to be considered inoperative.

Disposition: For reference in the Operator's MEL.

Attachments: DDG page 2.36-00-01.1-5

**36-00-01 Air Supply and Cabin Pressure Controllers (ASCPC)**
36-00-01-01 Primary (Digital) Controllers

Interval	Installed	Required	Procedure
C	2	1	(M) (O)

Right Primary ASCPC may be inoperative provided:

- Right Primary ASCPC is deactivated.
- Both ASCPC backup (analog) controllers are verified to operate normally.
- Both right CTC channels are verified to operate normally once each flight day.
- Left engine bleed system operates normally.
- Hydraulic center system demand Air Driven Pump C1 operates normally.
- Right engine Idle Selection System is considered inoperative.
- Right isolation system indications are considered inoperative
- APU bleed air shutoff valve indications are considered inoperative.
- Right Auto Cabin Pressure Control is considered inoperative.
- Appropriate performance adjustments are applied.

NOTE 1: The airplane must also be dispatched using MMEL item 73-21-01.

NOTE 2: The airplane must also be dispatched using MMEL item 36-12-01B.

NOTE 3: The airplane must also be dispatched using MMEL item 36-12-03-01.

NOTE 4: The airplane must also be dispatched using MMEL item 21-31-01.

MAINTENANCE (M)

NOTE 1: The left and right ASCPCs are interchangeable.

NOTE 2: For an engine bleed system to be operating normally, the following items associated with that system must be operative:

MMEL Item	Associated Status Message
36-00-01	ASCP PRIMARY CTRL L or R
36-11-01	BLEED PRSOV L or R
36-11-02	BLEED HPSOV L or R
36-11-03	None
36-11-04	None
36-11-05	BLEED FAMV L or R
36-21-01	MAN PRESS SENSOR L or R
36-21-02	BLEED PRESS SENSOR L or R
36-22-01	BLEED TEMP SENSOR L or R

Verify both ASCPC backup controllers operate normally, and verify both right Cabin Temperature Controllers (CTC) channels operate normally once each flight day (AMM 36-00-00/901).

1. Verify (one at a time) that both ASCPC backup (analog) controllers operate normally.
 - A. Verify the L ASCPC backup (analog) controller:
 - 1) Start left engine (AMM 71-00-00/201).
 - 2) Shutdown the APU if running or disconnect ground air cart if connected.
 - 3) Open the P310 panel AIR SPLY L PRI and AIR SPLY L SOL circuit breakers.
 - 4) Verify normal operation of the L ASCPC backup (analog) controller.
 - a. Position the BLEED AIR Panel (P5) L ENG Bleed switch OFF and confirm that the L PACK OFF light illuminates.
 - b. Position the BLEED AIR Panel (P5) L ENG Bleed switch ON and confirm that the L PACK OFF light extinguishes.
 - 5) Shut down left engine (AMM 71-00-00/201).
 - 6) Close the P310 panel AIR SPLY L PRI and AIR SPLY L SOL circuit breakers.
 - B. Verify the R ASCPC backup (analog) controller:
 - 1) Start right engine (AMM 71-00-00/201).
 - 2) Shutdown the APU if running or disconnect ground cart if connected.
 - 3) Open the P210 panel AIR SPLY R PRI and AIR SPLY R SOL circuit breakers.
 - 4) Open the P310 panel AIR SPLY R SEC circuit breaker.

- 5) Verify normal operation of the R ASCPC backup (analog) controller.
 - a. Position the BLEED AIR Panel (P5) R ENG Bleed switch OFF and confirm that the R PACK OFF light illuminates.
 - b. Position the BLEED AIR Panel (P5) R ENG Bleed switch ON and confirm that the R PACK OFF light extinguishes.
- 6) Shut down right engine (AMM 71-00-00/201).
- 7) Close the P210 panel AIR SPLY R PRI and AIR SPLY R SOL circuit breakers.
- 8) Close the P310 panel AIR SPLY R SEC circuit breaker.
2. Deactivate the R ASCPC primary controller.
 - A. Open and collar the P210 panel AIR SPLY R PRI and AIR SPLY R SOL circuit breakers.
 - B. Open and collar the P310 panel AIR SPLY R SEC circuit breaker.

NOTE: The CABIN ALT AUTO R status message will be displayed.
3. Verify that both right CTC channels operate normally once each flight day.

NOTE: If both channels of the right CTC do not operate normally, the right pack may be considered inoperative and the airplane dispatched using MMEL item 21-51-01.

 - A. Supply electrical power on the airplane (AMM 24-22-00/201).
 - B. Remove pneumatic power from the airplane (AMM 36-00-00/201).
 - C. Position right Air Conditioning panel (P5) PACK switch OFF.
 - D. Verify the right CTC channel 1 operates normally.
 - 1) Open the P210 panel R CTC CH 2 (AC) and R CTC CH 2 (DC) circuit breakers.
 - 2) Wait ten (10) seconds.
 - 3) Confirm CABIN TEMP CTRL R status message is not displayed.
 - 4) Close the P210 panel R CTC CH 2 (AC) and R CTC CH 2 (DC) circuit breakers.
 - E. Verify the right CTC channel 2 operates normally.
 - 1) Open the P210 panel R CTC CH 1 (AC) and R CTC CH 1 (DC) circuit breakers.
 - 2) Wait ten (10) seconds.
 - 3) Confirm CABIN TEMP CTRL R status message is not displayed.
 - 4) Close the P210 panel R CTC CH 1 (AC) and R CTC CH 1 (DC) circuit breakers.
 - F. Position right PACK switch AUTO.

**OPERATIONS (O)**

NOTE 1: For APU-to-Pack takeoff function installed, the APU-to-Pack takeoff Supplementary Procedure is not allowed.

NOTE 2: With the right ASCPC inoperative, indications for the APU bleed valve and the right isolation valve will be invalid.

1. Reduce performance limited weights by the appropriate adjustments.

Performance Limited Weights	Takeoff & Landing	Enroute Climb
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For -200 or -200ER/GE90-76B thru -90B:

thru 632,500 lb (286,902 kg)	5,900 lb (2,676 kg)	7,700 lb (3,493 kg)
over 632,500 lb (286,902 kg)	8,500 lb (3,856 kg)	7,900 lb (3,583 kg)

For -200 or -200ER/GE90-94B:

thru 632,500 lb (286,902 kg)	5,900 lb (2,676 kg)	9,400 lb (4,264 kg)
over 632,500 lb (286,902 kg)	9,200 lb (4,173 kg)	9,400 lb (4,264 kg)

For -200 or -200ER/PW4074 thru PW4084:

All	4,400 lb (1,996 kg)	8,600 lb (3,901 kg)
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For -200 or -200ER/PW4090:

thru 632,500 lb (286,902 kg)	6,100 lb (2,767 kg)	7,200 lb (3,266 kg)
over 632,500 lb (286,902 kg)	9,200 lb (4,173 kg)	7,200 lb (3,266 kg)

For -200 or -200ER/RR875 thru RR892:

thru 632,500 lb (286,902 kg)	3,300 lb (1,497 kg)	2,400 lb (1,089 kg)
over 632,500 lb (286,902 kg)	5,100 lb (2,313 kg)	2,400 lb (1,089 kg)

For -200 or -200ER/RR895:

All	3,300 lb (1,497 kg)	2,400 lb (1,089 kg)
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For -200LR or 777F/GE90-100 Series:

All	13,300 lb (6,033 kg)	10,100 lb (4,581 kg)
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For -300/PW4090:

All	5,800 lb (2,631 kg)	7,200 lb (3,266 kg)
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For -300/RR892:

All	3,300 lb (1,497 kg)	2,400 lb (1,089 kg)
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For -300ER/GE90-115B:

All	10,700 lb (4,854 kg)	10,200 lb (4,627 kg)
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NOTE: The enroute climb weight penalties listed above are based on single engine operating speeds that approximate maximum lift-to-drag ratio speed. To account for the difference in level off altitude when operating at other speeds, multiply the enroute climb weight penalty listed above by the appropriate factor listed on page 3.00-01-00.2 in the ENROUTE DIVERSION SPEED EFFECTS paragraph.

2. Increase flight planning fuel by 0.5%.
3. For ETOPS flight planning, increase ETOPS engine inoperative Critical Fuel Reserves.

For PW and RR:

- A. By 0.9%, or by 1.9% when dispatching into known or forecast icing conditions.

For GE:

- B. By 0.6%, or by 1.8% when dispatching into known or forecast icing conditions.
4. The AIR synoptic may display incomplete information or some information may be missing. The ENG IDLE DISAGREE advisory message will be displayed when idle thrust is selected inflight.
5. Operate the wing anti-ice manually.
 - A. Position WING ANTI-ICE selector OFF.
 - B. For flaps up and wing anti-ice required (do not use during takeoff and landing):
 - 1) Position the right ENG Bleed switch OFF.
 - 2) Position WING ANTI-ICE selector ON, then OFF as required.
 - 3) Prior to landing, position the ENG Bleed switch associated with the inoperative ASCP controller ON.
6. If an engine or engine bleed system fails enroute, position the associated ENG Bleed switch OFF.



777 FAA DDG Supplement Package

Supplement Number: 36-2

Issue Date: 12-APR-2024

Section Number: 2 (MMEL)

Item Number: 52-71-01

Item Name: Door Indication Systems

DDG Effectivity: 777 FAA DDG (D630W003-TBC), Revision 36

MMEL Effectivity: 777 FAA MMEL, Revision 23a

Description: Modified Operations (O) Note.

Reason: To remove redundancy.

Disposition: For reference in the Operator's MEL.

Attachments: DDG page 2.52-71-01.1-6

52-71-01 Door Indication Systems

Interval	Installed	Required	Procedure
C	-	0	(M)

May be inoperative provided associated door is verified closed, latched and locked before each departure.

MAINTENANCE (M)

NOTE: The optional procedures to deactivate inoperative passenger entry door sensors and forward, aft and main deck cargo (777F only) door sensors are provided to prevent unexpected display of the CONFIG DOORS warning and potential RTO due to intermittent door sensor faults.

Deactivate the inoperative sensors (if desired), and before each flight, verify the associated door is closed, latched, and locked (AMM 52-00-00/901).

1. For inoperative sensor(s) on Main Entry Doors:
 - A. If desired, deactivate the associated entry door's inoperative sensor(s).
 - 1) Remove, cap and stow electrical connectors from following sensors.
 - a. For entry door 1L, remove electrical connector DS52109 from sensor S52109 (CLOSED & LATCHED DR 1L) and remove electrical connector DS52110 from sensor S52110 (LOCKED DR 1L).
 - b. For entry door 1R, remove electrical connector DS52209 from sensor S52209 (CLOSED & LATCHED DR 1R) and remove electrical connector DS52110 from sensor S52110 (LOCKED DR 1R).
 - c. For entry door 2L, remove electrical connector DS52111 from sensor S52111 (CLOSED & LATCHED DR 2L) and remove electrical connector DS52110 from sensor S52110 (LOCKED DR 2L).
 - d. For entry door 2R, remove electrical connector DS52211 from sensor S52211 (CLOSED & LATCHED DR 2R) and remove electrical connector DS52110 from sensor S52110 (LOCKED DR 2R).
 - For -200/-200ER/-200LR:
 - e. For entry door 3L, remove electrical connector DS52113 from sensor S52113 (CLOSED & LATCHED DR 3L) and remove electrical connector DS52110 from sensor S52110 (LOCKED DR 3L).

For -300/-300ER:

- f. For entry door 3L, remove electrical connector DS52113 from sensor S52113 (CLOSED & LATCHED DR 3L) and remove electrical connector DS52026 from sensor S52026 (LOCKED DR 3L).

For -200/-200ER/-200LR:

- g. For entry door 3R, remove electrical connector DS52213 from sensor S52213 (CLOSED & LATCHED DR 3R) and remove electrical connector DS52110 from sensor S52110 (LOCKED DR 3R).

For -300/-300ER:

- h. For entry door 3R, remove electrical connector DS52213 from sensor S52213 (CLOSED & LATCHED DR 3R) and remove electrical connector DS52026 from sensor S52026 (LOCKED DR 3R).
- i. For entry door 4L, remove electrical connector DS52115 from sensor S52115 (CLOSED & LATCHED DR 4L) and remove electrical connector DS52110 from sensor S52110 (LOCKED DR 4L).
- j. For entry door 4R, remove electrical connector DS52215 from sensor S52215 (CLOSED & LATCHED DR 4R) and remove electrical connector DS52110 from sensor S52110 (LOCKED DR 4R).

For -300/-300ER:

- k. For entry door 5L, remove electrical connector DS52117 from sensor S52117 (CLOSED & LATCHED DR 5L) and remove electrical connector DS52110 from sensor S52110 (LOCKED DR 5L).

For -300/-300ER:

- l. For entry door 5R, remove electrical connector DS52217 from sensor S52217 (CLOSED & LATCHED DR 5R) and remove electrical connector DS52110 from sensor S52110 (LOCKED DR 5R).

- B. Prior to each flight, verify the associated door is closed, latched and locked using normal procedures.
 - 1) Open the affected door in MANUAL/DISARMED mode, then close and latch.
 - 2) Confirm the door interior liner is aligned and faired with the doorway lining.
 - 3) Confirm the OPEN arrow on the hinge arm linings is aligned and continuous at crossover from forward section of lining to aft section of lining.

- 4) Position the door mode select handle to AUTOMATIC/ARMED, and confirm the yellow flags are visible in the viewing windows immediately forward and aft of door at floor level.
 - 5) Confirm the outside skin of the door is aligned and faired with the fuselage skin.
 - 6) Confirm the vent door is fully closed.
2. For inoperative sensor(s) on Forward Cargo Door:
- A. If desired, deactivate the inoperative door sensor(s).
 - 1) Remove, cap and stow electrical connector DS52021 from sensor S52021 (CLOSED).
 - 2) Remove, cap and stow electrical connector DS52042 from sensor S52042 (LATCHED).
 - 3) Remove, cap and stow electrical connector DS52022 from sensor S52022 (LOCKED).
 - B. Prior to each flight, verify the door is closed, latched and locked using normal procedures.
 - 1) Close the door.
 - 2) Confirm the outside skin of the door is aligned and faired with the fuselage skin.
 - 3) Confirm the lock handle is in the unlocked position.
 - 4) Confirm the vent door is open.
 - 5) Rotate the lock handle back to the CLOSED position.
 - 6) Confirm the lock handle is fully closed and faired with the outside skin of the door.
 - 7) Confirm the lock handle release lever is faired with the lock handle and outside skin of the door.
 - 8) Confirm the vent door is fully closed (the upper edge of the vent door may not be fully faired).
 - 9) Confirm the latches are in the fully latched position and that the locks are in the locked position by visual inspection through the witness ports at the lower edge of the door.
3. For inoperative sensor(s) on Aft Large Cargo Door:
- A. If desired, deactivate the inoperative door sensor(s).
 - 1) Remove, cap and stow electrical connector DS52045 from sensor S52045 (CLOSED).
 - 2) Remove, cap and stow electrical connector DS52043 from sensor S52043 (LATCHED).
 - 3) Remove, cap and stow electrical connector DS52044 from sensor S52044 (LOCKED).
 - B. Prior to each flight, verify the door is closed, latched and locked using normal procedures.
 - 1) Close the door.

- 2) Confirm the outside skin of the door is aligned and faired with the fuselage skin.
 - 3) Confirm the lock handle is in the unlocked position.
 - 4) Confirm the vent door is open.
 - 5) Rotate the lock handle back to the CLOSED position.
 - 6) Confirm the lock handle is fully closed and faired with the outside skin of the door.
 - 7) Confirm the lock handle release lever is faired with the lock handle and outside skin of the door.
 - 8) Confirm the vent door is fully closed (the upper edge of the vent door may not be fully faired).
 - 9) Confirm the latches are in the fully latched position and that the locks are in the locked position by visual inspection through the witness ports at the lower edge of the door.
4. For inoperative sensor(s) on Aft Small Cargo Door:
- A. If desired, deactivate the inoperative door sensor(s).
 - 1) Remove, cap and stow electrical connector DS52023 from sensor S52023 (CLOSED).
 - 2) Remove, cap and stow electrical connector DS52024 from sensor S52024 (LATCHED/LOCKED).
 - B. Prior to each flight, verify the door is closed and latched using normal procedures.
 - 1) Close the door.
 - 2) Confirm the outside skin of the door is aligned and faired with the fuselage skin.
 - 3) Confirm the vent door is fully closed.
 - 4) Open the access cover for the door control panel.
 - 5) Operate the CARGO DOOR CONTROL ARMING switch downward to the ARM position, and confirm the green DOOR LATCHED light illuminates. Release the switch and confirm the light extinguishes.
 - 6) Close the access cover.
5. For inoperative sensor(s) on Bulk Cargo Door:
- A. If desired, deactivate the inoperative door switch.
 - 1) Locate, remove and secure switch S52020 (CLOSED/LATCHED/LOCKED) without cutting its wires.
 - B. Prior to each flight, verify the door is closed and latched using normal procedures.
 - 1) Close the door.
 - 2) Confirm the outside skin of the door is aligned and faired with the fuselage skin.

- 3) Confirm the latch handle is fully closed and faired with the door skin.
- 4) Push on the door to verify it is latched.
6. For inoperative sensor(s) on Forward Access Door:
 - A. If desired, deactivate the inoperative door sensor(s).
 - 1) Remove, cap and stow electrical connector DS52018 from sensor S52018 (CLOSED/LATCHED/LOCKED).
 - B. Prior to each flight, verify the door is closed and latched using normal procedures.
 - 1) Close the door.
 - 2) Confirm the outside skin of the door is aligned and faired with the fuselage skin.
 - 3) Confirm the latch handle is fully closed and faired with the door skin.
 - 4) Push on the door to verify it is latched.
7. For inoperative sensor(s) on Main Equipment Center (E/E Bay) Access Door:
 - A. If desired, deactivate the inoperative door sensor(s).
 - 1) Remove, cap and stow electrical connector DS52019 from sensor S52019 (CLOSED/LATCHED/LOCKED).
 - B. Prior to each flight, verify the door is closed and latched using normal procedures.
 - 1) Close and latch the door.
 - 2) Confirm the outside skin of the door is aligned and faired with the fuselage skin.
 - 3) Confirm the latch handle is fully closed and faired with the door skin.
 - 4) Push on the door to verify it is latched.

For -300/-300ER:

8. For inoperative sensor(s) on Overwing Escape Slide Compartment Door:
 - A. Prior to each flight, verify the door is closed and latched using normal procedures.
 - 1) Open and close the door.
 - 2) Confirm the outside skin of the door is aligned and faired with the fuselage skin.
 - 3) Push on the door to verify it is latched.

For 777F:

9. For inoperative sensor(s) on Main Deck Cargo Door:
 - A. If desired, deactivate the inoperative door sensor(s).
 - 1) Remove, cap and stow electrical connector DS52053 from sensor S52053 (CLOSED).



- 2) Remove, cap and stow electrical connector DS52054 from sensor S52054 (LATCHED).
- 3) Remove, cap and stow electrical connector DS52055 from sensor S52055 (LOCKED).
- B. Prior to each flight, verify the door is closed, latched and locked using normal procedures.
 - 1) Close the door.
 - 2) Confirm the latch arrows point to the targets in all eight witness ports.
 - 3) Confirm the outside skin of the door is aligned and faired with the fuselage skin.
 - 4) Confirm the vent door has closed.
 - 5) Using the witness ports, confirm the lock sectors are located over latch cams but do not touch the cams.

OPERATIONS NOTE

1. The optional procedures to deactivate inoperative passenger entry door sensors and forward, aft and main deck cargo (777F only) door sensors are provided to prevent unexpected display of the CONFIG DOORS warning and potential RTO due to intermittent door sensor faults.
2. The DOOR WING SLIDE L and DOOR WING SLIDE R advisory messages (-300/-300ER) and the DOOR BULK CARGO advisory message may be displayed both on the ground and in flight.
3. For doors with inoperative sensors deactivated:
 - A. The respective door indication status message (if applicable) may be displayed both on the ground and in flight.
 - B. Inoperative door advisory messages may be displayed on the ground only, except the DOOR BULK CARGO advisory message, which may be displayed both on the ground and in flight.
 - C. The DOOR MAIN DECK CARGO, DOOR FWD CARGO and DOOR AFT CARGO (if applicable) caution messages may be displayed on the ground.
4. The CONFIG DOORS warning is not displayed for bulk cargo door switch faults.



777 FAA DDG Supplement Package

Supplement Number: 36-2

Issue Date: 12-APR-2024

Section Number: 2 (MMEL)

Item Number: 73-21-01

Item Name: Engine Idle Selection Systems

DDG Effectivity: 777 FAA DDG (D630W003-TBC), Revision 36

MMEL Effectivity: 777 FAA MMEL, Revision 23a

Description: Modified Operations (O) procedures.

Reason: To clarify that the performance penalty is to be applied on the affected engine(s) only.

Disposition: For reference in the Operator's MEL.

Attachments: DDG page 2.73-21-01.1

**73-21-01 Engine Idle Selection Systems**

Interval	Installed	Required	Procedure
C	2	0	(O)

May be inoperative provided appropriate performance adjustments are applied.

OPERATIONS (O)

For -200 or -200ER/PW or -300ER/GE:

Increase descent fuel burn by 275 lb (125 kg) per affected engine.

For -300/PW:

Increase descent fuel burn by 200 lb (91 kg) per affected engine.

For -200 or -200ER/GE:

Increase descent fuel burn by 125 lb (57 kg) per affected engine.

For -200LR or 777F/GE:

Increase descent fuel burn by 375 lb (170 kg) per affected engine.

For -200 or -200ER/RR:

Increase descent fuel burn by 1,900 lb (861 kg) per affected engine.

For -300/RR:

Increase descent fuel burn by 1,100 lb (500 kg) per affected engine.



777 FAA DDG Supplement Package

Supplement Number: 36-2

Issue Date: 12-APR-2024

Section Number: 2 (MMEL)

Item Number: 73-21-04

Item Name: EEC C1 Faults

DDG Effectivity: 777 FAA DDG (D630W003-TBC), Revision 36

MMEL Effectivity: 777 FAA MMEL, Revision 23a

Description: Modified title and removed DDG note for subitem 73-21-04-02.

Reason: To include GE90-70/80/90 series in the subitem title instead of the DDG note.

Disposition: For reference in the Operator's MEL.

Attachments: DDG page 2.73-21-04.1-2

**73-21-04 EEC C1 Faults****73-21-04-01 GE90-100 Series Without P/N 730A-GEC-A08-06 or Later EEC Software Installed**

Interval	Installed	Required	Procedure
A	2	0	(M)

May be dispatched with C1 faults provided:

- Prior to each flight EEC faults are reviewed to determine if EEC must be replaced.
- Repairs are made in accordance with times established by engine manufacturer.

MAINTENANCE (M)

NOTE: Certain ENG EEC C1 faults require replacement of the EEC.

Prior to each flight perform the following ENG EEC C1 fault check:
(Required by AD 2020-20-17)

- Refer to FIM Chapter 73-98, Task 805 to determine if the EEC must be replaced.

NOTE: GE Alternative Method of Compliance (AMOC) to AD 2020-20-17 allows FIM Chapter 73-98, Task 805 as an AMOC to the GE GE90-100 SB 73-0117 R01 requirement.

OPERATIONS NOTE

The engine manufacturer's established maximum operating time interval for C1 faults is 150 flight hours or 10 calendar days, whichever occurs first (TCDS E00049EN Note 17).

73-21-04 EEC C1 Faults**73-21-04-02 PW, RR, GE90-70/80/90 Series, and GE90-100 Series with P/N 730A-GEC-A08-06 or Later EEC Software Installed**

Interval	Installed	Required	Procedure
A	2	0	

May be dispatched with C1 faults provided repairs are made in accordance with times established by engine manufacturer.

OPERATIONS NOTE**For PW:**

The engine manufacturer's established maximum operating time interval for C1 faults is 20 days (TCDS E46NE Note 17).

For GE:

The engine manufacturer's established maximum operating time interval for C1 faults is 150 flight hours or 10 calendar days, whichever occurs first (TCDS E00049EN Note 17).

For RR:

The engine manufacturer's established maximum operating time interval for C1 faults is 300 hours or 20 days, whichever occurs first (TCDS E00050EN Note 17).



777 FAA DDG Supplement Package

Supplement Number: 36-2

Issue Date: 12-APR-2024

Section Number: 4 (MISC)

Item Number: N/A

Item Name: Performance Based Navigation

DDG Effectivity: 777 FAA DDG (D630W003-TBC), Revision 36

MMEL Effectivity: 777 FAA MMEL, Revision 23a

Description: Modified PBN guidance.

Reason: To correct and add more clarification for PBN operation capabilities.

Disposition: For reference in the Operator's MEL.

Attachments: DDG pages 4.00-02-00.1-7



777 Dispatch Deviations Guide

Section 4 Performance Based Navigation

Performance Based Navigation (PBN) Information

Applying dispatch relief for the following items may affect the ability of the airplane to meet the operational capabilities required for PBN operations. The following information lists items that, if inoperative, may affect the ability of the airplane to meet the performance and functional criteria for various PBN operations. This information can aid in identifying potentially affected capabilities associated with a particular dispatch configuration.

This section lists equipment requirements that may be found in the document 777 RNP Navigation Capabilities Document D243W018-13, revision "M" (or later). Operators should consult their Operations Specification for requirements applicable to their operations.

PBN Equipment Requirements

Display Units (DUs) - (31-61-01)

PBN operations require two PFDs with mini map and EICAS display. The 777 is equipped with six display units. The PFDs are normally displayed on the outboard display units and the NDs are normally displayed on the inboard display units. EICAS is normally displayed on the upper center display unit. MMEL dispatch is only allowed with the lower center display unit inoperative. Therefore, these requirements will always be met for dispatch.

Multi-Purpose Control and Display Units - Control Display Units (CDU) - (34-61-03)

The 777 is equipped with three CDUs. MMEL dispatch is only allowed with the center CDU inoperative. Both left and right CDUs must operate normally for dispatch. PBN operations are allowed with two CDUs. Therefore, this requirement will always be met for dispatch.

AIMS Operational Software

For RNP AR APCH and RNP APCH:

Airplane must be equipped with AIMS BP05A or later. Navigation Performance Scales (NPS) and LNAV Go-Around Auto-Engage must be enabled.

Airplanes without Navigation Performance Scales and LNAV Go-Around Auto-Engage (AIMS BP03 and earlier versions) do not support these operations.

Fault Tolerant ADIRU

For AIMS-1 V16 and AIMS-2 V17 or later:

AIR DATA SYS or NAV ADIRU INERTIAL messages must not be displayed.

For all other AIMS-1 or AIMS-2 versions:

NAV AIR DATA SYS or NAV ADIRU INERTIAL messages must not be displayed.

MMEL dispatch is not allowed with any of the above messages displayed. Therefore, this requirement will always be met for dispatch.

Autopilot Flight Director Computers (AFDC) - (22-11-01)

The 777 autopilot flight director system (AFDS) is equipped with three autopilot flight director computers (AFDCs). Each of the three AFDCs can provide both flight director guidance, which can be displayed on the primary flight displays (PFDs), and send autopilot commands to the flight control actuation system when the autopilot is engaged.

MMEL dispatch is allowed with one, two, or all three AFDCs inoperative. Therefore, PBN operations may be affected with more than one AFDC inoperative.

For dispatch with two Autopilot Flight Director Computers inoperative (22-11-01B):

- RNP AR APCH operations are not allowed.
- Other PBN operations may be conducted.

For dispatch with all (3) Autopilot Flight Director Computers inoperative (22-11-01C):

- PBN operations are not allowed.

Autopilot and Flight Director

PBN operations may be conducted using either the autopilot or flight director. The minimum authorized RNP value may be different depending on whether autopilot or flight director is used.

Autopilot

When the autopilot is engaged, autopilot commands from the AFDCs are sent to the Primary Flight Computers to generate flight control surface commands.

If the autopilot function is inoperative, PBN operations may be conducted using the flight director.

The autopilot will be impacted if any of the following equipment is inoperative:

1. Mode Control Panel Switches - A/P Engage Switches
 - A. For dispatch with one autopilot engage switch inoperative (22-11-06-01A):
 - RNP AR APCH operations using autopilot are not allowed.
 - RNP AR APCH operations may be conducted using flight director provided its use is appropriate for the intended operation.
 - Other PBN operations using autopilot may be conducted.
 - B. For dispatch with both autopilot engage switches inoperative (22-11-06-01B):
 - PBN operations using autopilot are not allowed.
 - PBN operations may be conducted using flight director provided its use is appropriate for the intended operation.
2. Control Wheel Autopilot Disconnect Switches
 - A. For dispatch with one control wheel autopilot disconnect switch inoperative (22-11-08A):
 - PBN operations using autopilot at 1,500 feet AGL or below are not allowed.
 - PBN operations may be conducted using flight director provided its use is appropriate for the intended operation.
 - B. For dispatch with both control wheel autopilot disconnect switches inoperative (22-11-08B):
 - PBN operations using autopilot are not allowed.
 - PBN operations may be conducted using flight director provided its use is appropriate for the intended operation.
3. Warning Electronic System (WES) Autopilot Disconnect Warning Function
 - A. For dispatch with WES autopilot disconnect warning function inoperative (31-51-05):
 - PBN operations using autopilot are not allowed.
 - PBN operations may be conducted using flight director provided its use is appropriate for the intended operation.

Flight Director

When the flight director is selected on, flight director guidance from the AFDCs is sent to and displayed on the Primary Flight Displays (PFDs). The flight crew uses the flight director guidance to manually control the airplane in accordance with the command guidance from the AFDCs.

Flight director guidance can be displayed on one or both PFDs.

Flight director guidance is selected on and off by using the flight director switches on the mode control panel (MCP).

The flight director guidance will be impacted if any of the following equipment is inoperative:

1. Mode Control Panel Switches - F/D Switches
 - A. For dispatch with one Flight Director switch inoperative (22-11-06-04):
 - RNP AR APCH operations using flight director are not allowed.
 - RNP AR APCH operations may be conducted using autopilot provided its use is appropriate for the intended operation.
 - Other PBN operations using flight director may be conducted provided flight director operates normally on the side of the pilot flying (PF).
 - B. For dispatch with both Flight Director switches inoperative (22-11-06-04):
 - PBN operations using flight director are not allowed.
 - PBN operations may be conducted using autopilot provided its use is appropriate for the intended operation.
2. Flight Director Systems
 - A. For dispatch with one Flight Director system inoperative (22-11-10):
 - RNP AR APCH operations using flight director are not allowed.
 - RNP AR APCH operations may be conducted using autopilot provided its use is appropriate for the intended operation.
 - Other PBN operations using flight director may be conducted provided flight director operates normally on the side of the pilot flying (PF).
 - B. For dispatch with both Flight Director systems inoperative (22-11-10):
 - PBN operations using flight director are not allowed.
 - PBN operations may be conducted using autopilot provided its use is appropriate for the intended operation.

LNAV/VNAV Mode Control Panel (MCP) Switches - (22-11-06-09)

For dispatch with lateral navigation (LNAV) switch inoperative (22-11-06-09):

- PBN operations are not allowed.

For dispatch with vertical navigation (VNAV) switch inoperative. (22-11-06-09):

- RNP AR APCH operations are not allowed.
- RNP APCH operations may be conducted using other vertical modes provided appropriate approach minimums are used.
- Other PBN operations may be conducted.

Takeoff/Go-Around (TO/GA) Switches - (22-11-09)

There are two TO/GA switches installed on the 777. MMEL dispatch is allowed with one or both TO/GA switches inoperative.

For dispatch with one TO/GA switch inoperative (22-11-09A):

- RNP AR APCH operations are not allowed.
- Other PBN operations may be conducted.

For dispatch with both TO/GA switches inoperative (22-11-09B):

- PBN operations using autopilot or flight director below 500 feet AGL or MDA are not allowed.

Global Positioning Systems (GPS) - (34-58-01)

The 777 is equipped with two GPSs. MMEL dispatch is allowed with one or both GPSs inoperative.

For dispatch with one GPS inoperative (34-58-01B):

- RNP AR APCH and RNP 4 operations are not allowed.
- RNP 2 operations over oceanic and remote continental airspace are not allowed.
- RNP APCH, RNP 1, other RNP 2, and Advanced RNP operations may be conducted provided alternate navigation equipment (e.g. DME/DME, VOR/DME, VOR, ADF) is available to continue to a suitable airport if the remaining GPS becomes inoperative.
- RNAV 1, RNAV 2, P-RNAV, B-RNAV and RNAV 5 operations may be conducted.
- RNP 10 and RNAV 10 operations may be conducted provided appropriate time limitations are observed. Refer to the RNP Navigation Capabilities Document for the time limitations.

For dispatch with both GPSs inoperative (34-58-01A):

- RNP AR APCH, RNP APCH, RNP 1, RNP 2, RNP 4 and Advanced RNP operations are not allowed.
- RNAV 1, RNAV 2, P-RNAV, B-RNAV and RNAV 5 operations may be conducted provided one DME/DME or one VOR/DME operates normally.
- RNP 10 and RNAV 10 operations may be conducted provided appropriate time limitations are observed. Refer to the RNP Navigation Capabilities Document for the time limitations.

VOR Navigation Systems - (34-51-01)

The 777 is equipped with two VORs. MMEL dispatch is allowed with one or both VORs inoperative.

- RNAV 1, RNAV 2, P-RNAV, B-RNAV and RNAV 5 operations may be conducted provided one GPS, one DME/DME or one VOR/DME operates normally.
- Other PBN operations may be conducted.

Distance Measuring Equipment Systems (DME) - (34-55-01)

The 777 is equipped with two DMEs. MMEL dispatch is allowed with one or both DMEs inoperative.

- RNAV 1, RNAV 2, P-RNAV, B-RNAV and RNAV 5 operations may be conducted provided one GPS, one DME/DME or one VOR/DME operates normally.
- Other PBN operations may be conducted.

Radio Altimeter Systems - (34-33-01)

The 777 is equipped with three radio altimeters. MMEL dispatch is allowed with one radio altimeter inoperative.

PBN operations are allowed with two radio altimeters. Therefore, this requirement will always be met for dispatch.

Ground Proximity Warning Systems (GPWS) - (34-46-01), Terrain Awareness Function (34-46-01-03)

777 airplanes are equipped with a single ground proximity warning system (GPWS). Airplanes with PRR 61724 or equivalent are equipped with enhanced GPWS (EGPWS) that includes a terrain awareness function.

MMEL dispatch is allowed with the GPWS (EGPWS) inoperative.

For dispatch with either GND PROX SYS or TERR POS Advisory message displayed:

- RNP AR APCH operations are not allowed.

- Other PBN operations may be conducted.

Flight Management Computing Systems (FMCS) - (34-61-01)

There are two flight management computing systems (FMCSs) installed on the 777. MMEL dispatch is allowed with one FMCS inoperative.

For dispatch with one FMCS inoperative:

- RNP AR APCH and RNP 4 operations are not allowed.
- RNP 2 operations over oceanic and remote continental airspace are not allowed.
- RNP APCH, RNP 1, other RNP 2, RNAV 1, RNAV 2, P-RNAV, B-RNAV, RNAV 5 and Advanced RNP operations may be conducted provided alternate navigation equipment (e.g., VOR, ADF) is available to continue to a suitable airport if the remaining FMCS becomes inoperative.
- RNP 10 and RNAV 10 operations may be conducted using the alternate navigation capability in the multifunction control display units (MCDU).

Flight Management Computing Systems (FMCS) - Navigation Database (34-61-01-03)

The FMCS navigation database must be current for PBN operations.

It is assumed that the operator has a preflight procedure to ensure that the navigation database complies with regulatory agency requirements for ensuring the navigation database remains valid for the duration of the flight unless an alternative regulatory agency approved process applies.



777 FAA DDG Supplement Package

Supplement Number: 36-3

Issue Date: 29-MAY-2024

Section Number: 2 (MMEL)

Item Number: 23-24-02

Item Name: Emergency Locator Transmitter (ELT) (Fixed)

DDG Effectivity: 777 FAA DDG (D630W003-TBC), Revision 36

MMEL Effectivity: 777 FAA MMEL, Revision 23a

Description: Modified option A and C Maintenance (M) procedures for deactivation of the ACR/Artex model ELT. Deleted DDG top note and modified Maintenance step 2 for subitem -01.

Reason: Provided more detailed ACR/Artex model ELT deactivation procedure steps for clarity. Remote ELT switch position label for the armed position is not only dependent on ELT model, but on ELT control panel model installed.

Disposition: For reference in the Operator's MEL.

Attachments: DDG page 2.23-24-02.1-3

**23-24-02 Emergency Locator Transmitter (ELT) (Fixed)**

23-24-02A Required By 14 CFR - Inoperative

Interval	Installed	Required	Procedure
A	1	0	(M)

May be inoperative provided:

- System is deactivated.
- Repairs are made within 90 consecutive calendar-days.

NOTE: Dispatch relief for the Portable (survival) ELT's is in MMEL item 25-63-03.

MAINTENANCE (M)

Deactivate the inoperative ELT (AMM 23-00-00/901).

- Gain access to the inoperative ELT in the section 47 crown.
- For ACR/Artex model ELTs,
 - Remove the ELT.
 - Separate the battery pack from the ELT.
 - Disconnect, cap and stow the electrical connectors between the battery pack and ELT.
 - Attach the battery pack to the ELT and re-install ELT.
 - Install a placard stating "ELT not installed" in view of pilot.
- For ELTA and Honeywell model ELTs, set the ELT (LRU) front panel switch to OFF.

OPERATIONS NOTE

See 14 CFR 91.207, 121.353, 121.339 or other appropriate regulations.

23-24-02 Emergency Locator Transmitter (ELT) (Fixed)

23-24-02B Required By 14 CFR - Missing

Interval	Installed	Required	Procedure
A	1	0	(M)

May be missing provided:

- Placard stating "ELT not installed" is placed in view of the pilot.
- Repairs are made within 90 consecutive calendar-days.

NOTE: Dispatch relief for the Portable (survival) ELT's is in MMEL item 25-63-03.

**MAINTENANCE (M)**

Placard "ELT not installed" and place it in view of the pilot (AMM DDG 23-00-00/901).

OPERATIONS NOTE

See 14 CFR 91.207, 121.353, 121.339 or other appropriate regulations.

23-24-02 Emergency Locator Transmitter (ELT) (Fixed)*********23-24-02C Not Required By 14 CFR - Inoperative**

Interval	Installed	Required	Procedure
D	1	0	(M)

Any in excess of those required by 14 CFR may be inoperative provided system is deactivated.

NOTE: Dispatch relief for the Portable (survival) ELT's is in MMEL item 25-63-03.

MAINTENANCE (M)

Deactivate the inoperative ELT (AMM 23-00-00/901).

1. Gain access to the inoperative ELT in the section 47 crown.
2. For ACR/Artex model ELTs,
 - A. Remove the ELT.
 - B. Separate the battery pack from the ELT.
 - C. Disconnect, cap and stow the electrical connectors between the battery pack and ELT.
 - D. Attach the battery pack to the ELT and re-install ELT.
 - E. Install a placard stating "ELT not installed" in view of pilot.
3. For ELTA and Honeywell model ELTs, set the ELT (LRU) front panel switch to OFF.

OPERATIONS NOTE

See 14 CFR 91.207, 121.353, 121.339 or other appropriate regulations.

23-24-02 Emergency Locator Transmitter (ELT) (Fixed)*********23-24-02D Not Required By 14 CFR - Missing**

Interval	Installed	Required	Procedure
D	1	0	

Any in excess of those required by 14 CFR may be missing.

NOTE: Dispatch relief for the Portable (survival) ELT's is in MMEL item 25-63-03.

OPERATIONS NOTE

See 14 CFR 91.207, 121.353, 121.339 or other appropriate regulations.

23-24-02 Emergency Locator Transmitter (ELT) (Fixed) ***

23-24-02-01 Remote ELT Switch

Interval	Installed	Required	Procedure
D	1	0	(M)

May be inoperative provided:

- Remote ELT Switch is deactivated.
- ELT Switch is placed in the ARMED mode.

MAINTENANCE (M)

NOTE: If remote ELT switch is failed so that the ELT system is activated, deactivate the ELT system and dispatch per MMEL 23-24-02A or 23-24-02C.

If ELT does not activate with the remote ELT switch set to ON, deactivate the remote ELT switch (AMM 23-00-00/901).

- Placard the inoperative remote ELT switch.
- Set the remote ELT switch to the armed position.

NOTE: The armed position may be labeled ARM or ARMED.

- Gain access to the inoperative ELT (LRU) in the section 47 crown.
- Set the ELT front panel switch to the armed position.
 - For ACR/Artex model ELTs, set the ELT front panel switch to OFF.
 - For ELTA model ELTs, set the ELT front panel switch to ARMED.
 - For Honeywell model ELTs, set the ELT front panel switch to ARM.



777 FAA DDG Supplement Package

Supplement Number: 36-3

Issue Date: 29-MAY-2024

Section Number: 2 (MMEL)

Item Number: 23-70-03

Item Name: Cabin Video Monitoring System (CVMS)

DDG Effectivity: 777 FAA DDG (D630W003-TBC), Revision 36

MMEL Effectivity: 777 FAA MMEL, Revision 23a

Description: Added DDG top note.

Reason: Added note to clarify the Cargo Visual Surveillance System is considered part of the CVMS.

Disposition: For reference in the Operator's MEL.

Attachments: DDG page 2. 23-70-03.1

**23-70-03 Cabin Video Monitoring System (CVMS)**

Interval	Installed	Required	Procedure
D	1	0	

NOTE: Any portion of the system that operates normally may be used.

NOTE 1: The Cabin Video Monitoring System (CVMS) includes the Cargo Visual Surveillance System (CVSS).

NOTE 2: Dispatch relief for the Flight Deck Door Visual Surveillance Systems is available in 23-70-01.

NOTE 3: Dispatch relief for the Direct View Camera System is available in 23-70-02.

NOTE 4: Dispatch relief for the Ground Maneuver Camera System is available in 31-61-10.



777 FAA DDG Supplement Package

Supplement Number: 36-3

Issue Date: 29-MAY-2024

Section Number: 2 (MMEL)

Item Number: 25-53-02

Item Name: Lower Cargo Restraint Systems

DDG Effectivity: 777 FAA DDG (D630W003-TBC), Revision 36

MMEL Effectivity: 777 FAA MMEL, Revision 23a

Description: Added AMM procedure reference to the option A Maintenance (M) procedure. Modified option A Maintenance procedure step 2.

Reason: AMM procedure reference was unintentionally removed in revision 36. Clarified the Maintenance procedure step refers to the restraint components for the affected cargo compartment.

Disposition: For reference in the Operator's MEL.

Attachments: DDG page 2. 25-53-02.1-2

25-53-02 Lower Cargo Restraint Systems
25-53-02A Cargo Compartment Used

Interval	Installed	Required	Procedure
A	-	-	(M)

May be inoperative or missing provided:

- Approved cargo loading limits are observed. The only source documents are the Type Certificate (TC), Supplemental Type Certificate (STC), Airplane Flight Manual (AFM), Airplane Flight Manual Supplement (AFMS) or the TC/STC Weight and Balance Manual (WBM).
- Repairs are made within 120 consecutive calendar-days.

MAINTENANCE (M)

Use acceptable cargo loading limits from an approved source (AMM 25-00-00/901).

- Refer to the Weight and Balance Manual for cargo loading limits.
- If the number of inoperative or missing restraint components for the affected cargo compartment exceeds the allowance in the Weight and Balance Manual, the affected cargo compartment cannot be used. Refer to MMEL Item 25-53-02B.
- Deactivation procedures may or may not be required. Develop deactivation procedures depending on system configuration.

25-53-02 Lower Cargo Restraint Systems
25-53-02B Cargo Compartment(s) Remains Empty

Interval	Installed	Required	Procedure
A	-	-	

May be inoperative or missing provided:

- Cargo compartment remains empty.
- Repairs are made within 120 consecutive calendar-days.

25-53-02 Lower Cargo Restraint Systems
25-53-02C Individual Cargo Areas

Interval	Installed	Required	Procedure
A	-	-	



Individual cargo areas may be inoperative provided:

- a. Aircraft is operated in accordance with Original Equipment Manufacturer (OEM) W&B source document.
- b. Repairs are made within 120 consecutive calendar-days.

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Superseded



777 FAA DDG Supplement Package

Supplement Number: 36-3

Issue Date: 29-MAY-2024

Section Number: 2 (MMEL)

Item Number: 25-58-01

Item Name: Main Deck Cargo Restraint Systems (777F)

DDG Effectivity: 777 FAA DDG (D630W003-TBC), Revision 36

MMEL Effectivity: 777 FAA MMEL, Revision 23a

Description: Modified option A Maintenance procedure step 2.

Reason: Clarified the Maintenance procedure step refers to the restraint components for the affected cargo compartment.

Disposition: For reference in the Operator's MEL.

Attachments: DDG page 2. 25-58-01.1-2

25-58-01 Main Deck Cargo Restraint Systems (777F)
25-58-01A Cargo Compartment Used

Interval	Installed	Required	Procedure
A	-	-	(M)

May be inoperative or missing provided:

- Approved cargo loading limits are observed. The only source documents are the Type Certificate (TC), Supplemental Type Certificate (STC), Airplane Flight Manual (AFM), Airplane Flight Manual Supplement (AFMS) or the TC/STC Weight and Balance Manual (WBM).
- Repairs are made within 120 consecutive calendar days.

MAINTENANCE (M)

Use acceptable cargo loading limits from an approved source (AMM 25-00-00/901).

- Refer to the Weight and Balance Manual for cargo loading limits.
- If the number of inoperative or missing restraint components for the affected cargo compartment exceeds the allowance in the Weight and Balance Manual, the affected cargo compartment cannot be used. Refer to MMEL Item 25-58-01B.
- Deactivation procedures may or may not be required. Develop deactivation procedures depending on system configuration.

25-58-01 Main Deck Cargo Restraint Systems (777F)
25-58-01B Cargo Compartment(s) Remains Empty

Interval	Installed	Required	Procedure
A	-	-	

May be inoperative or missing provided:

- Cargo compartment remains empty.
- Repairs are made within 120 consecutive calendar days.

25-58-01 Main Deck Cargo Restraint Systems (777F)
25-58-01C Individual Cargo Areas

Interval	Installed	Required	Procedure
A	-	-	



Individual cargo areas may be inoperative provided:

- a. Aircraft is operated in accordance with Original Equipment Manufacturer (OEM) W&B source document.
- b. Repairs are made within 120 consecutive calendar-days.

Supplement 36-3



777 FAA DDG Supplement Package

Supplement Number: 36-3

Issue Date: 29-MAY-2024

Section Number: 2 (MMEL)

Item Number: 27-11-04

Item Name: Flaperon Power Control Units (PCUs)

DDG Effectivity: 777 FAA DDG (D630W003-TBC), Revision 36

MMEL Effectivity: 777 FAA MMEL, Revision 23a

Description: Added DDG top note and Operations note to subitem 27-11-04-02.

Reason: The DDG top note identifies additional MMEL item application requirement resulting from deactivation of the right flaperon inboard PCU. The Operations note is included for crew awareness of the resulting EICAS status message in this dispatch configuration.

Disposition: For reference in the Operator's MEL.

Attachments: DDG page 2. 27-11-04.1-3

27-11-04 Flaperon Power Control Units (PCUs)
27-11-04-01 Passenger (With PRRs 61384-15 and 62200, or Equivalent Installed)

Interval	Installed	Required	Procedure
A	4	3	(M)

One may be inoperative provided:

- Inoperative PCU is deactivated in the bypass mode.
- Flaperons are verified to operate normally.
- Repairs are made within 3 flight days.

NOTE 1: For an inoperative right flaperon inboard PCU, the airplane must also be dispatched using MMEL item 27-02-05.

NOTE 2: PRRs 61384-15 and 62200 Part-B are applicable to airplane production L/Ns 718, 772, 774 and 776 and-on. This item is also applicable to airplanes with the AIMS-1 or AIMS-2 platforms with BP V14 or later AIMS OPS installed together with the following PFC software installed: PFC OPS 2762-PFC-741-00, or PFC OPS 276D-PFC-841-00, or PFC OPS 2768-PFC-910-00 or later installed.

MAINTENANCE (M)

Deactivate the inoperative flaperon PCU in the bypass mode and verify flaperons operate normally (AMM 27-00-00/901).

- For an inoperative left flaperon outboard PCU, open and collar the Power Supply Assembly Left (PSA-L) in E1-6 rack FLPRN PCU circuit breaker.
- For an inoperative left flaperon inboard PCU, open and collar the Power Supply Assembly Right (PSA-R) in E5-1 rack FLPRN PCU circuit breaker.
- For an inoperative right flaperon outboard PCU, open and collar the Power Supply Assembly Center (PSA-C) in E2-6 rack FLPRN PCU circuit breaker.
- For an inoperative right flaperon inboard PCU:
 - Open P110 panel FLPRN PCU RIB(BYP)/SPLR 4 & 11 circuit breaker.
 - Open P110 panel ACE-L2 PWR circuit breaker.
 - Set Maintenance Panel (P61) left, center and right WING FLT CONTROL HYD POWER switches to SHUT OFF.
 - Disconnect, cap and stow the PCU electrical connector D4422P (WD 27-11-18). The connector is located in the Main Equipment Center on the left side of the E1-1 shelf near the forward cargo access door (station 408, connector position 71).



NOTE: The ACTUATOR PRESS SNSR status message will be displayed. The airplane must also be dispatched using MMEL item 27-02-05.

- E. Close the open circuit breakers.
 - F. Set Maintenance Panel (P61) left, center and right WING FLT CONTROL HYD POWER switches to NORM.
5. Check the operation of the flaperons.

NOTE: Do not do an Actuator Confidence Test, or an ACE Monitor Test, or L or R Flaperon Actuator Confidence Tests.

6. Confirm the FLIGHT CONTROL SYS status message is not displayed.

OPERATIONS NOTE

For an inoperative right flaperon inboard PCU, the ACTUATOR PRESS SNSR status message will be displayed.

27-11-04 Flaperon Power Control Units (PCUs)

27-11-04-02 777F

Interval	Installed	Required	Procedure
A	4	3	(M)

One may be inoperative provided:

- a. Inoperative PCU is deactivated in the bypass mode.
- b. Flaperons are verified to operate normally.
- c. Repairs are made within 3 flight days.

NOTE: For an inoperative right flaperon inboard PCU, the airplane must also be dispatched using MMEL item 27-02-05.

MAINTENANCE (M)

Deactivate the inoperative flaperon PCU in the bypass mode and verify flaperons operate normally (AMM 27-00-00/901).

- 1. For an inoperative left flaperon outboard PCU, open and collar the Power Supply Assembly Left (PSA-L) in E1-6 rack FLPRN PCU circuit breaker.
- 2. For an inoperative left flaperon inboard PCU, open and collar the Power Supply Assembly Right (PSA-R) in E5-1 rack FLPRN PCU circuit breaker.
- 3. For an inoperative right flaperon outboard PCU, open and collar the Power Supply Assembly Center (PSA-C) in E2-6 rack FLPRN PCU circuit breaker.
- 4. For an inoperative right flaperon inboard PCU:
 - A. Open P110 panel FLPRN PCU RIB(BYP)/SPLR 4 & 11 circuit breaker.
 - B. Open P110 panel ACE-L2 PWR circuit breaker.

- C. Set Maintenance Panel (P61) left, center and right WING FLT CONTROL HYD POWER switches to SHUT OFF.
- D. Disconnect, cap and stow the PCU electrical connector D4422P (WD 27-11-18). The connector is located in the Main Equipment Center on the left side of the E1-1 shelf near the forward cargo access door (station 408, connector position 71).

NOTE: The ACTUATOR PRESS SNSR status message will be displayed. The airplane must also be dispatched using MMEL item 27-02-05.

- E. Close the open circuit breakers.
 - F. Set Maintenance Panel (P61) left, center and right WING FLT CONTROL HYD POWER switches to NORM.
5. Check the operation of the flaperons.

NOTE: Do not do an Actuator Confidence Test, or an ACE Monitor Test, or L or R Flaperon Actuator Confidence Tests.

6. Confirm the FLIGHT CONTROL SYS status message is not displayed.

OPERATIONS NOTE

For an inoperative right flaperon inboard PCU, the ACTUATOR PRESS SNSR status message will be displayed.



777 FAA DDG Supplement Package

Supplement Number: 36-3

Issue Date: 29-MAY-2024

Section Number: 2 (MMEL)

Item Number: 30-21-01

Item Name: Engine Anti-Ice Systems

DDG Effectivity: 777 FAA DDG (D630W003-TBC), Revision 36

MMEL Effectivity: 777 FAA MMEL, Revision 23a

Description: Added Maintenance (M) and Operations (O) notes to all subitem 30-21-01-01 and 30-21-01-02 options.

Reason: Added notes to support the proviso, "Engine fan case overheat detection system operates normally."

Disposition: For reference in the Operator's MEL.

Attachments: DDG page 2. 30-21-01.1-25

30-21-01 Engine Anti-Ice Systems

Interval	Installed	Required	Procedure
C	2	1	(M)

One may be inoperative provided:

- Valve is locked closed.
- Associated engine anti-ice control switch remains OFF.
- Airplane is not operated in known or forecast icing conditions.
- Flight remains within 120 minutes of landing at a suitable airport.

MAINTENANCE (M)

Lock associated EAI valve in the closed position (AMM 30-00-00/901).

- Deactivate the associated EAI valve controller:
 - For the right EAI valve, open P210 panel EAI CTRL R circuit breaker.
 - For the left EAI valve, open P310 panel EAI CTRL L circuit breaker.
- Deactivate the leading edge slats in the retracted position (AMM 27-81-00/201).
- Deactivate the associated thrust reverser for ground maintenance (AMM 78-31-00/201).
- Lock the EAI valve in the closed position.
 - Remove the locking screw from the normal position.
 - Set the crank to the locked (closed) position.
 - Install the locking screw through the crank into the locked position.
 - Position associated ANTI-ICE Panel (P5) Engine Anti-Ice selector OFF.
- Close the associated circuit breaker(s).
- Activate the thrust reverser (AMM 78-31-00/201).
- Activate the leading edge slats (AMM 27-81-00/201).
- Activate the associated EAI valve controller by closing the associated circuit breaker.
- Position the associated ANTI-ICE Panel (P5) Engine Anti-Ice selector OFF.

30-21-01 Engine Anti-Ice Systems

30-21-01-01 PW and RR

30-21-01-01A OAT 38 Degrees C or Below, With Deactivation Kit Installed

Interval	Installed	Required	Procedure
C	2	1	(M) (O)



For operations at ambient temperature 38 degrees C or below, one may be inoperative provided:

- Valve is deactivated open using deactivation kit.
- Engine fan case overheat detection system operates normally.
- Appropriate performance adjustments are applied.

MAINTENANCE (M)

NOTE: Dispatch is not allowed with EAI LOOPS FANCASE L or R status message displayed.

Deactivate the associated EAI valve by installing the EAI valve lock-open kit, P/N K28703 (AMM 30-00-00/901).

- Remove inoperative EAI valve (AMM 30-21-02/401).
- Modify valve with lock-open kit (AMM 30-00-00/901).
- Re-install modified valve (AMM 30-21-02/401).

For PW:

- Visually inspect the area near the Cowl Thermal Anti-Ice duct for signs of leaks and the fan cowl in general for flammable fluid leaks. Dispatch is not allowed if any of the duct clamps or supports are not secure or any air leaks from the EAI duct are present, or if significant flammable fluid leakage is present.

OPERATIONS (O)

NOTE: Dispatch is not allowed with EAI LOOPS FANCASE L or R status message displayed.

- Reduce performance limited weights by the appropriate adjustments.

Performance Limited Weights	Takeoff & Landing	Enroute Climb
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For -200 or -200ER/PW4074 thru PW4084:

All	19,600 lb (8,891 kg)	19,400 lb (8,800 kg)
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For -200 or -200ER/PW4090:

thru 632,500 lb (286,902 kg)	12,200 lb (5,534 kg)	13,800 lb (6,260 kg)
over 632,500 lb (286,902 kg)	17,300 lb (7,847 kg)	13,800 lb (6,260 kg)



For -200 or -200ER/RR875 thru RR892:

thru 632,500 lb (286,902 kg)	20,100 lb (9,117 kg)	21,600 lb (9,798 kg)
over 632,500 lb (286,902 kg)	26,500 lb (12,020 kg)	21,600 lb (9,798 kg)

For -200 or -200ER/RR895:

All	21,700 lb (9,843 kg)	23,000 lb (10,433 kg)
-----	----------------------	-----------------------

For -300/PW4090:

All	12,400 lb (5,625 kg)	13,000 lb (5,897 kg)
-----	----------------------	----------------------

For -300/RR892:

All	22,000 lb (9,979 kg)	23,500 lb (10,659 kg)
-----	----------------------	-----------------------

NOTE: The enroute climb weight penalties listed above are based on single engine operating speeds that approximate maximum lift-to-drag ratio speed. To account for the difference in level off altitude when operating at other speeds, multiply the enroute climb weight penalty listed above by the appropriate factor listed on page 3.00-01-00.2 in the ENROUTE DIVERSION SPEED EFFECTS paragraph.

For PW:

2. Increase flight planning fuel by 1.0% for an inoperative EAI Valve or EAI Pressure Sensor, or by 1.3% for an inoperative EAI Card.

For RR:

3. Increase flight planning fuel by 0.5% for an inoperative EAI Valve or EAI Pressure Sensor, or by 0.8% for an inoperative EAI Card.
4. For ETOPS flight planning:

For PW:

- A. Increase the ETOPS engine inoperative Critical Fuel Reserves by 1.6%.

For RR:

- B. Increase the ETOPS engine inoperative Critical Fuel Reserves by 0.9%.
5. Position ENGINE ANTI-ICE selector(s) as follows:
 - A. For EAI Valve or EAI Pressure Sensor inoperative, position associated EAI selector ON.



NOTE 1: When an EAI selector is positioned to ON and the temperature is greater than 10 degrees C, the ANTI-ICE ON advisory message will be displayed. Do not accomplish the ANTI-ICE ON Non-Normal Checklist.

NOTE 2: Inflight, the ENG IDLE DISAGREE advisory message may be displayed when engine thrust is reduced to idle.

- B. For EAI Card inoperative, position both EAI selectors ON.

NOTE 1: When an EAI selector is positioned to ON and the temperature is greater than 10 degrees C, the ANTI-ICE ON advisory message will be displayed. Do not accomplish the ANTI-ICE ON Non-Normal Checklist.

NOTE 2: The associated ANTI-ICE ENG L or R advisory message will be displayed, the EAI indication will not be displayed on EICAS, and the associated EAI valve will be displayed as invalid (low intensity white) and the green flow bar will not be displayed on the Air synoptic.

6. For the inoperative EAI valve, if the ANTI-ICE LEAK ENG L or R caution message is subsequently displayed:

For PW:

- A. Do not accomplish the two minute delay associated with the ANTI-ICE LEAK ENG L or R Non-Normal Checklist. Complete the remainder of the checklist as soon as practical.

For RR:

- B. Do not accomplish the initial one minute delay associated with the ANTI-ICE LEAK ENG L or R Non-Normal Checklist. Complete the remainder of the checklist as soon as practical.

30-21-01 Engine Anti-Ice Systems

30-21-01-01 PW and RR

30-21-01-01B OAT 38 Degrees C or Below, Without Deactivation Kit Installed

Interval	Installed	Required	Procedure
C	2	1	(M) (O)

For operations at ambient temperature 38 degrees C or below, one may be inoperative provided:

- Valve is electrically disconnected and verified open.
- Engine fan case overheat detection system operates normally.
- Associated EAI pressure sensor system operates normally.
- Associated EAI card operates normally.

- e. Appropriate performance adjustments are applied.
-

MAINTENANCE (M)

NOTE: Dispatch is not allowed with EAI LOOPS FANCASE L or R status message displayed.

For EAI VALVE L or R status message displayed, deactivate the EAI valve by disconnecting the electrical connector and verify EAI valve is open when the associated engine is running (AMM 30-00-00/901).

1. Deactivate the associated EAI valve controller.
 - A. For the right EAI valve, open P210 panel EAI CTRL R circuit breaker.
 - B. For the left EAI valve, open P310 panel EAI CTRL L circuit breaker.
2. Deactivate the leading edge slats in the retracted position (AMM 27-81-00/201).
3. Deactivate the associated thrust reverser for ground maintenance (AMM 78-31-00/201).
4. Deactivate the EAI valve in the open position.
 - A. On the associated engine, disconnect, cap and stow the torque motor electrical connector DM30211 or DM30211A from the EAI valve controller, M30211.
 - B. Install a protective cover over the electrical connector on the valve controller.

For PW:

- C. Visually inspect the area near the Cowl Thermal Anti-Ice duct for signs of leaks and the fan cowl in general for flammable fluid leaks. Dispatch is not allowed if any of the duct clamps or supports are not secure or any air leaks from the EAI duct are present, or if significant flammable fluid leakage is present.
5. Activate the thrust reverser (AMM 78-31-00/201).
6. Activate the leading edge slats (AMM 27-81-00/201).
7. Activate the associated EAI valve controller by closing the associated circuit breaker.
8. Verify the EAI valve is open when the associated engine is running.
 - A. Position associated ANTI-ICE Panel (P5) Engine Anti-Ice selector OFF.
 - B. Start the associated engine (71-00-00/201).
 - C. Confirm the EAI indication (green EAI) is displayed above the N1 indication on the EICAS display for the associated engine (indicates the EAI valve is open).

NOTE: If the EAI indication is not displayed, the valve may be stuck closed and a thrust setting slightly above minimum idle may open the valve.



D. Shutdown the associated engine (71-00-00/201).

OPERATIONS (O)

NOTE: Dispatch is not allowed with EAI LOOPS FANCASE L or R status message displayed.

1. Reduce performance limited weights by the appropriate adjustments.

Performance Limited Weights	Takeoff & Landing	Enroute Climb
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For -200 or -200ER/PW4074 thru PW4084:

All	19,600 lb (8,891 kg)	19,400 lb (8,800 kg)
-----	----------------------	----------------------

For -200 or -200ER/PW4090:

thru 632,500 lb (286,902 kg)	12,200 lb (5,534 kg)	13,800 lb (6,260 kg)
over 632,500 lb (286,902 kg)	17,300 lb (7,847 kg)	13,800 lb (6,260 kg)

For -200 or -200ER/RR875 thru RR892:

thru 632,500 lb (286,902 kg)	20,100 lb (9,117 kg)	21,600 lb (9,798 kg)
over 632,500 lb (286,902 kg)	26,500 lb (12,020 kg)	21,600 lb (9,798 kg)

For -200 or -200ER/RR895:

All	21,700 lb (9,843 kg)	23,000 lb (10,433 kg)
-----	----------------------	-----------------------

For -300/PW4090:

All	12,400 lb (5,625 kg)	13,000 lb (5,897 kg)
-----	----------------------	----------------------

For -300/RR892:

All	22,000 lb (9,979 kg)	23,500 lb (10,659 kg)
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NOTE: The enroute climb weight penalties listed above are based on single engine operating speeds that approximate maximum lift-to-drag ratio speed. To account for the difference in level off altitude when operating at other speeds, multiply the enroute climb weight penalty listed above by the appropriate factor listed on page 3.00-01-00.2 in the ENROUTE DIVERSION SPEED EFFECTS paragraph.

For PW:

2. Increase flight planning fuel by 1.0%.

For RR:

3. Increase flight planning fuel by 0.5%.
4. For ETOPS flight planning:

For PW:

- A. Increase the ETOPS engine inoperative Critical Fuel Reserves by 1.6%.

For RR:

- B. Increase the ETOPS engine inoperative Critical Fuel Reserves by 0.9%.
5. Verify the EAI valve is open when the associated engine is running before each departure.
 - A. Position associated ANTI-ICE Panel (P5) Engine Anti-Ice selector OFF.
 - B. After the associated engine has been started, confirm the associated EAI indication (green EAI) is displayed above the N1 indication.

NOTE 1: If the EAI indication is not displayed, the valve may be stuck closed and a thrust setting slightly above minimum idle may open the valve.

NOTE 2: On the Air synoptic, the EAI valve will indicate failed open because the EAI selector is OFF and the green flow bar will be displayed.

6. Position associated ENGINE ANTI-ICE selector ON.

NOTE 1: When an EAI selector is positioned to ON and the temperature is greater than 10 degrees C, the ANTI-ICE ON advisory message will be displayed. Do not accomplish the ANTI-ICE ON Non-Normal Checklist.

NOTE 2: Inflight, the ENG IDLE DISAGREE advisory message may be displayed when engine thrust is reduced to idle.

NOTE 3: The ANTI-ICE ENG advisory message may be displayed during ground operations prior to takeoff. Flight crew should accomplish the associated non-normal checklist. If ANTI-ICE ENG clears, takeoff is allowed.



7. For the inoperative EAI valve, if the ANTI-ICE LEAK ENG L or R caution message is subsequently displayed:

For PW:

- A. Do not accomplish the two minute delay associated with the ANTI-ICE LEAK ENG L or R Non-Normal Checklist. Complete the remainder of the checklist as soon as practical.

For RR:

- B. Do not accomplish the initial one minute delay associated with the ANTI-ICE LEAK ENG L or R Non-Normal Checklist. Complete the remainder of the checklist as soon as practical.

30-21-01 Engine Anti-Ice Systems

30-21-01-01 PW and RR

30-21-01-01C OAT Above 38 Degrees C, With Deactivation Kit Installed

Interval	Installed	Required	Procedure
A	2	1	(M) (O)

For operations at ambient temperature above 38 degrees C, one may be inoperative provided:

- Valve is deactivated open using deactivation kit.
- Engine fan case overheat detection system operates normally.
- Appropriate performance adjustments are applied.
- Repairs are made within 3 flight days.

MAINTENANCE (M)

NOTE: Dispatch is not allowed with EAI LOOPS FANCASE L or R status message displayed.

Deactivate the associated EAI valve by installing the EAI valve lock-open kit, P/N K28703 (AMM 30-00-00/901).

- Remove inoperative EAI valve (AMM 30-21-02/401).
- Modify valve with lock-open kit (AMM 30-00-00/901).
- Re-install modified valve (AMM 30-21-02/401).

For PW:

- Visually inspect the area near the Cowl Thermal Anti-Ice duct for signs of leaks and the fan cowl in general for flammable fluid leaks. Dispatch is not allowed if any of the duct clamps or supports are not secure or any air leaks from the EAI duct are present, or if significant flammable fluid leakage is present.

**OPERATIONS (O)**

NOTE: Dispatch is not allowed with EAI LOOPS FANCASE L or R status message displayed.

1. Reduce performance limited weights by the appropriate adjustments.

Performance Limited Weights	Takeoff & Landing	Enroute Climb
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For -200 or -200ER/PW4074 thru PW4084:

All	19,600 lb (8,891 kg)	19,400 lb (8,800 kg)
-----	----------------------	----------------------

For -200 or -200ER/PW4090:

thru 632,500 lb (286,902 kg)	12,200 lb (5,534 kg)	13,800 lb (6,260 kg)
over 632,500 lb (286,902 kg)	17,300 lb (7,847 kg)	13,800 lb (6,260 kg)

For -200 or -200ER/RR875 thru RR892:

thru 632,500 lb (286,902 kg)	20,100 lb (9,117 kg)	21,600 lb (9,798 kg)
over 632,500 lb (286,902 kg)	26,500 lb (12,020 kg)	21,600 lb (9,798 kg)

For -200 or -200ER/RR895:

All	21,700 lb (9,843 kg)	23,000 lb (10,433 kg)
-----	----------------------	-----------------------

For -300/PW4090:

All	12,400 lb (5,625 kg)	13,000 lb (5,897 kg)
-----	----------------------	----------------------

For -300/RR892:

All	22,000 lb (9,979 kg)	23,500 lb (10,659 kg)
-----	----------------------	-----------------------



NOTE: The enroute climb weight penalties listed above are based on single engine operating speeds that approximate maximum lift-to-drag ratio speed. To account for the difference in level off altitude when operating at other speeds, multiply the enroute climb weight penalty listed above by the appropriate factor listed on page 3.00-01-00.2 in the ENROUTE DIVERSION SPEED EFFECTS paragraph.

For PW:

2. Increase flight planning fuel by 1.0% for an inoperative EAI Valve or EAI Pressure Sensor, or by 1.3% for an inoperative EAI Card.

For RR:

3. Increase flight planning fuel by 0.5% for an inoperative EAI Valve or EAI Pressure Sensor, or by 0.8% for an inoperative EAI Card.
4. For ETOPS flight planning:

For PW:

- A. Increase the ETOPS engine inoperative Critical Fuel Reserves by 1.6%.

For RR:

- B. Increase the ETOPS engine inoperative Critical Fuel Reserves by 0.9%.
5. Position ENGINE ANTI-ICE selector(s) as follows:
 - A. For EAI Valve or EAI Pressure Sensor inoperative, position associated EAI selector ON.

NOTE 1: When an EAI selector is positioned to ON and the temperature is greater than 10 degrees C, the ANTI-ICE ON advisory message will be displayed. Do not accomplish the ANTI-ICE ON Non-Normal Checklist.

NOTE 2: Inflight, the ENG IDLE DISAGREE advisory message may be displayed when engine thrust is reduced to idle.

- B. For EAI Card inoperative, position both EAI selectors ON.

NOTE 1: When an EAI selector is positioned to ON and the temperature is greater than 10 degrees C, the ANTI-ICE ON advisory message will be displayed. Do not accomplish the ANTI-ICE ON Non-Normal Checklist.

NOTE 2: The associated ANTI-ICE ENG L or R advisory message will be displayed, the EAI indication will not be displayed on EICAS, and the associated EAI valve will be displayed as invalid (low intensity white) and the green flow bar will not be displayed on the Air synoptic.

6. For the inoperative EAI valve, if the ANTI-ICE LEAK ENG L or R caution message is subsequently displayed:

For PW:

- A. Do not accomplish the two minute delay associated with the ANTI-ICE LEAK ENG L or R Non-Normal Checklist. Complete the remainder of the checklist as soon as practical.

For RR:

- B. Do not accomplish the initial one minute delay associated with the ANTI-ICE LEAK ENG L or R Non-Normal Checklist. Complete the remainder of the checklist as soon as practical.

30-21-01 Engine Anti-Ice Systems

30-21-01-01 PW and RR

30-21-01-01D OAT Above 38 Degrees C, Without Deactivation Kit Installed

Interval	Installed	Required	Procedure
A	2	1	(M) (O)

For operations at ambient temperature above 38 degrees C, one may be inoperative provided:

- a. Valve is electrically disconnected and verified open.
- b. Engine fan case overheat detection system operates normally.
- c. Associated EAI pressure sensor system operates normally.
- d. Associated EAI card operates normally.
- e. Appropriate performance adjustments are applied.
- f. Repairs are made within 3 flight days.

MAINTENANCE (M)

NOTE: Dispatch is not allowed with EAI LOOPS FANCASE L or R status message displayed.

For EAI VALVE L or R status message displayed, deactivate the EAI valve by disconnecting the electrical connector and verify EAI valve is open when the associated engine is running (AMM 30-00-00/901).

1. Deactivate the associated EAI valve controller.
 - A. For the right EAI valve, open P210 panel EAI CTRL R circuit breaker.
 - B. For the left EAI valve, open P310 panel EAI CTRL L circuit breaker.
2. Deactivate the leading edge slats in the retracted position (AMM 27-81-00/201).
3. Deactivate the associated thrust reverser for ground maintenance (AMM 78-31-00/201).



4. Deactivate the EAI valve in the open position.
 - A. On the associated engine, disconnect, cap and stow the torque motor electrical connector DM30211 or DM30211A from the EAI valve controller, M30211.
 - B. Install a protective cover over the electrical connector on the valve controller.

For PW:

- C. Visually inspect the area near the Cowl Thermal Anti-Ice duct for signs of leaks and the fan cowl in general for flammable fluid leaks. Dispatch is not allowed if any of the duct clamps or supports are not secure or any air leaks from the EAI duct are present, or if significant flammable fluid leakage is present.
5. Activate the thrust reverser (AMM 78-31-00/201).
6. Activate the leading edge slats (AMM 27-81-00/201).
7. Activate the associated EAI valve controller by closing the associated circuit breaker.
8. Verify the EAI valve is open when the associated engine is running.
 - A. Position associated ANTI-ICE Panel (P5) Engine Anti-Ice selector OFF.
 - B. Start the associated engine (71-00-00/201).
 - C. Confirm the EAI indication (green EAI) is displayed above the N1 indication on the EICAS display for the associated engine (indicates the EAI valve is open).

NOTE: If the EAI indication is not displayed, the valve may be stuck closed and a thrust setting slightly above minimum idle may open the valve.

- D. Shutdown the associated engine (71-00-00/201).

OPERATIONS (O)

NOTE: Dispatch is not allowed with EAI LOOPS FANCASE L or R status message displayed.

1. Reduce performance limited weights by the appropriate adjustments.

Performance Limited Weights	Takeoff & Landing	Enroute Climb
--------------------------------	----------------------	------------------

For -200 or -200ER/PW4074 thru PW4084:

All	19,600 lb (8,891 kg)	19,400 lb (8,800 kg)
-----	----------------------	----------------------



For -200 or -200ER/PW4090:

thru 632,500 lb (286,902 kg)	12,200 lb (5,534 kg)	13,800 lb (6,260 kg)
over 632,500 lb (286,902 kg)	17,300 lb (7,847 kg)	13,800 lb (6,260 kg)

For -200 or -200ER/RR875 thru RR892:

thru 632,500 lb (286,902 kg)	20,100 lb (9,117 kg)	21,600 lb (9,798 kg)
over 632,500 lb (286,902 kg)	26,500 lb (12,020 kg)	21,600 lb (9,798 kg)

For -200 or -200ER/RR895:

All	21,700 lb (9,843 kg)	23,000 lb (10,433 kg)
-----	----------------------	-----------------------

For -300/PW4090:

All	12,400 lb (5,625 kg)	13,000 lb (5,897 kg)
-----	----------------------	----------------------

For -300/RR892:

All	22,000 lb (9,979 kg)	23,500 lb (10,659 kg)
-----	----------------------	-----------------------

NOTE: The enroute climb weight penalties listed above are based on single engine operating speeds that approximate maximum lift-to-drag ratio speed. To account for the difference in level off altitude when operating at other speeds, multiply the enroute climb weight penalty listed above by the appropriate factor listed on page 3.00-01-00.2 in the ENROUTE DIVERSION SPEED EFFECTS paragraph.

For PW:

2. Increase flight planning fuel by 1.0%.

For RR:

3. Increase flight planning fuel by 0.5%.
4. For ETOPS flight planning:

For PW:

- A. Increase the ETOPS engine inoperative Critical Fuel Reserves by 1.6%.

For RR:

- B. Increase the ETOPS engine inoperative Critical Fuel Reserves by 0.9%.



5. Verify the EAI valve is open when the associated engine is running before each departure.
 - A. Position associated ANTI-ICE Panel (P5) Engine Anti-Ice selector OFF.
 - B. After the associated engine has been started, confirm the associated EAI indication (green EAI) is displayed above the N1 indication.

NOTE 1: If the EAI indication is not displayed, the valve may be stuck closed and a thrust setting slightly above minimum idle may open the valve.

NOTE 2: On the Air synoptic, the EAI valve will indicate failed open because the EAI selector is OFF and the green flow bar will be displayed.

6. Position associated ENGINE ANTI-ICE selector ON.

NOTE 1: When an EAI selector is positioned to ON and the temperature is greater than 10 degrees C, the ANTI-ICE ON advisory message will be displayed. Do not accomplish the ANTI-ICE ON Non-Normal Checklist.

NOTE 2: Inflight, the ENG IDLE DISAGREE advisory message may be displayed when engine thrust is reduced to idle.

NOTE 3: The ANTI-ICE ENG advisory message may be displayed during ground operations prior to takeoff. Flight crew should accomplish the associated non-normal checklist. If ANTI-ICE ENG clears, takeoff is allowed.

7. For the inoperative EAI valve, if the ANTI-ICE LEAK ENG L or R caution message is subsequently displayed:

For PW:

- A. Do not accomplish the two minute delay associated with the ANTI-ICE LEAK ENG L or R Non-Normal Checklist. Complete the remainder of the checklist as soon as practical.

For RR:

- B. Do not accomplish the initial one minute delay associated with the ANTI-ICE LEAK ENG L or R Non-Normal Checklist. Complete the remainder of the checklist as soon as practical.

30-21-01 Engine Anti-Ice Systems

30-21-01-02 GE

30-21-01-02A OAT 38 Degrees C or Below, With Deactivation Kit Installed

Interval	Installed	Required	Procedure
C	2	1	(M) (O)

For operations at ambient temperature 38 degrees C or below, one may be inoperative provided:

- Valve is deactivated open using deactivation kit.
- Engine fan case overheat detection system operates normally.
- Turbine case cooling air flow systems operate normally.
- Appropriate performance adjustments are applied.

Except for GE90-100 Series:

- EECs must be operated in the normal mode.

MAINTENANCE (M)

NOTE: Dispatch is not allowed with EAI LOOPS FANCASE L or R status message displayed.

Deactivate the associated EAI valve by installing the EAI valve lock-open kit, P/N K28703 (AMM 30-00-00/901).

- Remove inoperative EAI valve (AMM 30-21-02/401).
- Modify valve with lock-open kit (AMM 30-00-00/901).
- Re-install modified valve (AMM 30-21-02/401).
- Visually inspect the area near the Cowl Thermal Anti-Ice duct for signs of leaks and the fan cowl in general for flammable fluid leaks. Dispatch is not allowed if any of the duct clamps or supports are not secure or any air leaks from the EAI duct are present, or if significant flammable fluid leakage is present.

OPERATIONS (O)

NOTE: Dispatch is not allowed with EAI LOOPS FANCASE L or R status message displayed.

- Reduce performance limited weights by the appropriate adjustments.

Performance Limited Weights	Takeoff & Landing	Enroute Climb
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For -200 or -200ER/GE90-76B thru -90B:

thru 632,500 lb (286,902 kg)	9,100 lb (4,128 kg)	11,800 lb (5,352 kg)
over 632,500 lb (286,902 kg)	13,200 lb (5,987 kg)	11,800 lb (5,352 kg)



For -200 or -200ER/GE90-94B:

thru 632,500 lb (286,902 kg)	9,100 lb (4,128 kg)	13,900 lb (6,305 kg)
over 632,500 lb (286,902 kg)	14,300 lb (6,486 kg)	13,900 lb (6,305 kg)

For -200LR or 777F/GE90-100 Series:

All	13,300 lb (6,033 kg)	12,100 lb (5,489 kg)
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For -300ER/GE90-115B:

All	10,400 lb (4,717 kg)	11,800 lb (5,353 kg)
-----	----------------------	----------------------

NOTE: The enroute climb weight penalties listed above are based on single engine operating speeds that approximate maximum lift-to-drag ratio speed. To account for the difference in level off altitude when operating at other speeds, multiply the enroute climb weight penalty listed above by the appropriate factor listed on page 3.00-01-00.2 in the ENROUTE DIVERSION SPEED EFFECTS paragraph.

2. Increase flight planning fuel by 1.0% for an inoperative EAI Valve or EAI Pressure Sensor, or by 1.3% for an inoperative EAI Card.
3. For ETOPS flight planning, increase engine inoperative Critical Fuel Reserves by 1.8%.
4. Position ENGINE ANTI-ICE selector(s) as follows:
 - A. For EAI Valve or EAI Pressure Sensor inoperative, position associated EAI selector ON.

NOTE 1: When an EAI selector is positioned to ON and the temperature is greater than 10 degrees C, the ANTI-ICE ON advisory message will be displayed. Do not accomplish the ANTI-ICE ON Non-Normal Checklist.

NOTE 2: Inflight, the ENG IDLE DISAGREE advisory message may be displayed when engine thrust is reduced to idle.

- B. For EAI Card inoperative, position both EAI selectors ON.

NOTE 1: When an EAI selector is positioned to ON and the temperature is greater than 10 degrees C, the ANTI-ICE ON advisory message will be displayed. Do not accomplish the ANTI-ICE ON Non-Normal Checklist.

NOTE 2: The associated ANTI-ICE ENG L or R advisory message will be displayed, the EAI indication will not be displayed on EICAS, and the associated EAI valve will be displayed as invalid (low intensity white) and the green flow bar will not be displayed on the Air synoptic.

5. For the inoperative EAI valve, if the ANTI-ICE LEAK ENG L or R caution message is subsequently displayed, do not accomplish the two minute delay associated with the ANTI-ICE LEAK ENG L or R Non-Normal Checklist. Complete the remainder of the checklist as soon as practical.

30-21-01 Engine Anti-Ice Systems

30-21-01-02 GE

30-21-01-02B OAT 38 Degrees C or Below, Without Deactivation Kit Installed

Interval	Installed	Required	Procedure
C	2	1	(M) (O)

For operations at ambient temperature 38 degrees C or below, one may be inoperative provided:

- a. Valve is electrically disconnected and verified open.
- b. Engine fan case overheat detection system operates normally.
- c. Associated EAI pressure sensor system operates normally.
- d. Associated EAI card operates normally.
- e. Turbine case cooling air flow systems operate normally.
- f. Appropriate performance adjustments are applied.

Except for GE90-100 Series:

- g. EECs must be operated in the normal mode.

MAINTENANCE (M)

NOTE: Dispatch is not allowed with EAI LOOPS FANCASE L or R status message displayed.

For EAI VALVE L or R status message displayed, deactivate the EAI valve by disconnecting the electrical connector and verify EAI valve is open when the associated engine is running (AMM 30-00-00/901).

1. Deactivate the associated EAI valve controller.
 - A. For the right EAI valve, open P210 panel EAI CTRL R circuit breaker.
 - B. For the left EAI valve, open P310 panel EAI CTRL L circuit breaker.
2. Deactivate the leading edge slats in the retracted position (AMM 27-81-00/201).
3. Deactivate the associated thrust reverser for ground maintenance (AMM 78-31-00/201).



4. Deactivate the EAI valve in the open position.
 - A. On the associated engine, disconnect, cap and stow the torque motor electrical connector DM30211 or DM30211A from the EAI valve controller, M30211.
 - B. Install a protective cover over the electrical connector on the valve controller.
 - C. Visually inspect the area near the Cowl Thermal Anti-Ice duct for signs of leaks and the fan cowl in general for flammable fluid leaks. Dispatch is not allowed if any of the duct clamps or supports are not secure or any air leaks from the EAI duct are present, or if significant flammable fluid leakage is present.
5. Activate the thrust reverser (AMM 78-31-00/201).
6. Activate the leading edge slats (AMM 27-81-00/201).
7. Activate the associated EAI valve controller by closing the associated circuit breaker.
8. Verify the EAI valve is open when the associated engine is running.
 - A. Position associated ANTI-ICE Panel (P5) Engine Anti-Ice selector OFF.
 - B. Start the associated engine (71-00-00/201).
 - C. Confirm the EAI indication (green EAI) is displayed above the N1 indication on the EICAS display for the associated engine (indicates the EAI valve is open).

NOTE: If the EAI indication is not displayed, the valve may be stuck closed and a thrust setting slightly above minimum idle may open the valve.
 - D. Shutdown the associated engine (71-00-00/201).

OPERATIONS (O)

NOTE: Dispatch is not allowed with EAI LOOPS FAN CASE L or R status message displayed.

1. Reduce performance limited weights by the appropriate adjustments.

Performance Limited Weights	Takeoff & Landing	Enroute Climb
--------------------------------	----------------------	------------------

For -200 or -200ER/GE90-76B thru -90B:

thru 632,500 lb (286,902 kg)	9,100 lb (4,128 kg)	11,800 lb (5,352 kg)
over 632,500 lb (286,902 kg)	13,200 lb (5,987 kg)	11,800 lb (5,352 kg)

For -200 or -200ER/GE90-94B:

thru 632,500 lb (286,902 kg)	9,100 lb (4,128 kg)	13,900 lb (6,305 kg)
over 632,500 lb (286,902 kg)	14,300 lb (6,486 kg)	13,900 lb (6,305 kg)

For -200LR or 777F/GE90-100 Series:

All	13,300 lb (6,033 kg)	12,100 lb (5,489 kg)
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For -300ER/GE90-115B:

All	10,400 lb (4,717 kg)	11,800 lb (5,353 kg)
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NOTE: The enroute climb weight penalties listed above are based on single engine operating speeds that approximate maximum lift-to-drag ratio speed. To account for the difference in level off altitude when operating at other speeds, multiply the enroute climb weight penalty listed above by the appropriate factor listed on page 3.00-01-00.2 in the ENROUTE DIVERSION SPEED EFFECTS paragraph.

2. Increase flight planning fuel by 1.0%.
3. For ETOPS flight planning, increase engine inoperative Critical Fuel Reserves by 1.8%.
4. Verify the EAI valve is open when the associated engine is running before each departure.
 - A. Position associated ANTI-ICE Panel (P5) Engine Anti-Ice selector OFF.
 - B. After the associated engine has been started, confirm the associated EAI indication (green EAI) is displayed above the N1 indication.

NOTE 1: If the EAI indication is not displayed, the valve may be stuck closed and a thrust setting slightly above minimum idle may open the valve.

NOTE 2: On the Air synoptic, the EAI valve will indicate failed open because the EAI selector is OFF and the green flow bar will be displayed.

5. Position associated ENGINE ANTI-ICE selector ON.

NOTE 1: When an EAI selector is positioned to ON and the temperature is greater than 10 degrees C, the ANTI-ICE ON advisory message will be displayed. Do not accomplish the ANTI-ICE ON Non-Normal Checklist.



NOTE 2: Inflight, the ENG IDLE DISAGREE advisory message may be displayed when engine thrust is reduced to idle.

NOTE 3: The ANTI-ICE ENG advisory message may be displayed during ground operations prior to takeoff. Flight crew should accomplish the associated non-normal checklist. If ANTI-ICE ENG clears, takeoff is allowed.

6. For the inoperative EAI valve, if the ANTI-ICE LEAK ENG L or R caution message is subsequently displayed, do not accomplish the two minute delay associated with the ANTI-ICE LEAK ENG L or R Non-Normal Checklist. Complete the remainder of the checklist as soon as practical.

30-21-01 Engine Anti-Ice Systems

30-21-01-02 GE

30-21-01-02C OAT Above 38 Degrees C, With Deactivation Kit Installed

Interval	Installed	Required	Procedure
A	2	1	(M) (O)

For operations at ambient temperature above 38 degrees C, one may be inoperative provided:

- a. Valve is deactivated open using deactivation kit.
- b. Engine fan case overheat detection system operates normally.
- c. Turbine case cooling air flow systems operate normally.
- d. Appropriate performance adjustments are applied.
- e. Repairs are made within 3 flight days.

Except for GE90-100 Series:

- f. EECs must be operated in the normal mode.

MAINTENANCE (M)

NOTE: Dispatch is not allowed with EAI LOOPS FANCASE L or R status message displayed.

Deactivate the associated EAI valve by installing the EAI valve lock-open kit, P/N K28703 (AMM 30-00-00/901).

1. Remove inoperative EAI valve (AMM 30-21-02/401).
2. Modify valve with lock-open kit (AMM 30-00-00/901).
3. Re-install modified valve (AMM 30-21-02/401).
4. Visually inspect the area near the Cowl Thermal Anti-Ice duct for signs of leaks and the fan cowl in general for flammable fluid leaks. Dispatch is not allowed if any of the duct clamps or supports are not secure or any air leaks from the EAI duct are present, or if significant flammable fluid leakage is present.

OPERATIONS (O)

NOTE: Dispatch is not allowed with EAI LOOPS FANCASE L or R status message displayed.

1. Reduce performance limited weights by the appropriate adjustments.

Performance Limited Weights	Takeoff & Landing	Enroute Climb
-----------------------------	-------------------	---------------

For -200 or -200ER/GE90-76B thru -90B:

thru 632,500 lb (286,902 kg)	9,100 lb (4,128 kg)	11,800 lb (5,352 kg)
over 632,500 lb (286,902 kg)	13,200 lb (5,987 kg)	11,800 lb (5,352 kg)

For -200 or -200ER/GE90-94B:

thru 632,500 lb (286,902 kg)	9,100 lb (4,128 kg)	13,900 lb (6,305 kg)
over 632,500 lb (286,902 kg)	14,300 lb (6,486 kg)	13,900 lb (6,305 kg)

For -200LR or 777F/GE90-100 Series:

All	13,300 lb (6,033 kg)	12,100 lb (5,489 kg)
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For -300ER/GE90-115B:

All	10,400 lb (4,717 kg)	11,800 lb (5,353 kg)
-----	----------------------	----------------------

NOTE: The enroute climb weight penalties listed above are based on single engine operating speeds that approximate maximum lift-to-drag ratio speed. To account for the difference in level off altitude when operating at other speeds, multiply the enroute climb weight penalty listed above by the appropriate factor listed on page 3.00-01-00.2 in the ENROUTE DIVERSION SPEED EFFECTS paragraph.

2. Increase flight planning fuel by 1.0% for an inoperative EAI Valve or EAI Pressure Sensor, or by 1.3% for an inoperative EAI Card.
3. For ETOPS flight planning, increase engine inoperative Critical Fuel Reserves by 1.8%.
4. Position ENGINE ANTI-ICE selector(s) as follows:
 - A. For EAI Valve or EAI Pressure Sensor inoperative, position associated EAI selector ON.



NOTE 1: When an EAI selector is positioned to ON and the temperature is greater than 10 degrees C, the ANTI-ICE ON advisory message will be displayed. Do not accomplish the ANTI-ICE ON Non-Normal Checklist.

NOTE 2: Inflight, the ENG IDLE DISAGREE advisory message may be displayed when engine thrust is reduced to idle.

- B. For EAI Card inoperative, position both EAI selectors ON.

NOTE 1: When an EAI selector is positioned to ON and the temperature is greater than 10 degrees C, the ANTI-ICE ON advisory message will be displayed. Do not accomplish the ANTI-ICE ON Non-Normal Checklist.

NOTE 2: The associated ANTI-ICE ENG L or R advisory message will be displayed, the EAI indication will not be displayed on EICAS, and the associated EAI valve will be displayed as invalid (low intensity white) and the green flow bar will not be displayed on the Air synoptic.

5. For the inoperative EAI valve, if the ANTI-ICE LEAK ENG L or R caution message is subsequently displayed, do not accomplish the two minute delay associated with the ANTI-ICE LEAK ENG L or R Non-Normal Checklist. Complete the remainder of the checklist as soon as practical.

30-21-01 Engine Anti-Ice Systems

30-21-01-02 GE

30-21-01-02D OAT Above 38 Degrees C, Without Deactivation Kit Installed

Interval	Installed	Required	Procedure
A	2	1	(M) (O)

For operations at ambient temperature above 38 degrees C, one may be inoperative provided:

- Valve is electrically disconnected and verified open.
- Engine fan case overheat detection system operates normally.
- Associated EAI pressure sensor system operates normally.
- Associated EAI card operates normally.
- Turbine case cooling air flow systems operate normally.
- Appropriate performance adjustments are applied.
- Repairs are made within 3 flight days.

Except for GE90-100 Series:

- EECs must be operated in the normal mode.

MAINTENANCE (M)

NOTE: Dispatch is not allowed with EAI LOOPS FANCASE L or R status message displayed.

For EAI VALVE L or R status message displayed, deactivate the EAI valve by disconnecting the electrical connector and verify EAI valve is open when the associated engine is running (AMM 30-00-00/901).

1. Deactivate the associated EAI valve controller.
 - A. For the right EAI valve, open P210 panel EAI CTRL R circuit breaker.
 - B. For the left EAI valve, open P310 panel EAI CTRL L circuit breaker.
2. Deactivate the leading edge slats in the retracted position (AMM 27-81-00/201).
3. Deactivate the associated thrust reverser for ground maintenance (AMM 78-31-00/201).
4. Deactivate the EAI valve in the open position.
 - A. On the associated engine, disconnect, cap and stow the torque motor electrical connector DM30211 or DM30211A from the EAI valve controller, M30211.
 - B. Install a protective cover over the electrical connector on the valve controller.
 - C. Visually inspect the area near the Cowl Thermal Anti-Ice duct for signs of leaks and the fan cowl in general for flammable fluid leaks. Dispatch is not allowed if any of the duct clamps or supports are not secure or any air leaks from the EAI duct are present, or if significant flammable fluid leakage is present.
5. Activate the thrust reverser (AMM 78-31-00/201).
6. Activate the leading edge slats (AMM 27-81-00/201).
7. Activate the associated EAI valve controller by closing the associated circuit breaker.
8. Verify the EAI valve is open when the associated engine is running.
 - A. Position associated ANTI-ICE Panel (P5) Engine Anti-Ice selector OFF.
 - B. Start the associated engine (71-00-00/201).
 - C. Confirm the EAI indication (green EAI) is displayed above the N1 indication on the EICAS display for the associated engine (indicates the EAI valve is open).

NOTE: If the EAI indication is not displayed, the valve may be stuck closed and a thrust setting slightly above minimum idle may open the valve.

- D. Shutdown the associated engine (71-00-00/201).

**OPERATIONS (O)**

NOTE: Dispatch is not allowed with EAI LOOPS FANCASE L or R status message displayed.

1. Reduce performance limited weights by the appropriate adjustments.

Performance Limited Weights	Takeoff & Landing	Enroute Climb
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For -200 or -200ER/GE90-76B thru -90B:

thru 632,500 lb (286,902 kg)	9,100 lb (4,128 kg)	11,800 lb (5,352 kg)
over 632,500 lb (286,902 kg)	13,200 lb (5,987 kg)	11,800 lb (5,352 kg)

For -200 or -200ER/GE90-94B:

thru 632,500 lb (286,902 kg)	9,100 lb (4,128 kg)	13,900 lb (6,305 kg)
over 632,500 lb (286,902 kg)	14,300 lb (6,486 kg)	13,900 lb (6,305 kg)

For -200LR or 777F/GE90-100 Series:

All	13,300 lb (6,033 kg)	12,100 lb (5,489 kg)
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For -300ER/GE90-115B:

All	10,400 lb (4,717 kg)	11,800 lb (5,353 kg)
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NOTE: The enroute climb weight penalties listed above are based on single engine operating speeds that approximate maximum lift-to-drag ratio speed. To account for the difference in level off altitude when operating at other speeds, multiply the enroute climb weight penalty listed above by the appropriate factor listed on page 3.00-01-00.2 in the ENROUTE DIVERSION SPEED EFFECTS paragraph.

2. Increase flight planning fuel by 1.0%.
3. For ETOPS flight planning, increase engine inoperative Critical Fuel Reserves by 1.8%.
4. Verify the EAI valve is open when the associated engine is running before each departure.
 - A. Position associated ANTI-ICE Panel (P5) Engine Anti-Ice selector OFF.

- B. After the associated engine has been started, confirm the associated EAI indication (green EAI) is displayed above the N1 indication.

NOTE 1: If the EAI indication is not displayed, the valve may be stuck closed and a thrust setting slightly above minimum idle may open the valve.

NOTE 2: On the Air synoptic, the EAI valve will indicate failed open because the EAI selector is OFF and the green flow bar will be displayed.

5. Position associated ENGINE ANTI-ICE selector ON.

NOTE 1: When an EAI selector is positioned to ON and the temperature is greater than 10 degrees C, the ANTI-ICE ON advisory message will be displayed. Do not accomplish the ANTI-ICE ON Non-Normal Checklist.

NOTE 2: Inflight, the ENG IDLE DISAGREE advisory message may be displayed when engine thrust is reduced to idle.

NOTE 3: The ANTI-ICE ENG advisory message may be displayed during ground operations prior to takeoff. Flight crew should accomplish the associated non-normal checklist. If ANTI-ICE ENG clears, takeoff is allowed.

6. For the inoperative EAI valve, if the ANTI-ICE LEAK ENG L or R caution message is subsequently displayed, do not accomplish the two minute delay associated with the ANTI-ICE LEAK ENG L or R Non-Normal Checklist. Complete the remainder of the checklist as soon as practical.



777 FAA DDG Supplement Package

Supplement Number: 36-3

Issue Date: 29-MAY-2024

Section Number: 2 (MMEL)

Item Number: 34-43-01

Item Name: Weather Radar System

DDG Effectivity: 777 FAA DDG (D630W003-TBC), Revision 36

MMEL Effectivity: 777 FAA MMEL, Revision 23a

Description: Added DDG top note.

Reason: Clarified which weather radar system models are applicable to this item.

Disposition: For reference in the Operator's MEL.

Attachments: DDG page 2.34-43-01.1-3

34-43-01 Weather Radar System
34-43-01A Reactive Windshear Operative

Interval	Installed	Required	Procedure
C	-	0	(O)

May be inoperative provided:

- Weather radar is not required by 14 CFR.
- Reactive windshear alert (GPWS Mode 7) operates normally.
- Alternate procedures are established and used.

MAINTENANCE NOTE

The WINDSHEAR PRED status message is not displayed if the WXR SYS status message is displayed.

OPERATIONS (O)

Alternate procedures must be established and used.

NOTE 1: The WINDSHEAR SYS advisory message will be displayed.

NOTE 2: See 14 CFR 121.357 or other appropriate regulation.

34-43-01 Weather Radar System
34-43-01B Reactive Windshear Inoperative

Interval	Installed	Required	Procedure
B	-	0	(O)

May be inoperative provided:

- Weather radar is not required by 14 CFR.
- Alternate procedures are established and used.

NOTE: Operator's alternate procedures should include reviewing windshear avoidance and windshear recovery procedures.

MAINTENANCE NOTE

The WINDSHEAR PRED status message is not displayed if the WXR SYS status message is displayed.

OPERATIONS (O)

Alternate procedures must be established and used.

NOTE 1: The WINDSHEAR SYS advisory message will be displayed.

NOTE 2: See 14 CFR 121.357 or other appropriate regulation.

**34-43-01 Weather Radar System****34-43-01C One Operative**

Interval	Installed	Required	Procedure
D	2	1	

34-43-01 Weather Radar System**34-43-01-01 Predictive Windshear Function****34-43-01-01A Reactive Windshear Operative**

Interval	Installed	Required	Procedure
C	-	0	(O)

May be inoperative provided:

- Alternate procedures are established and used.
- Reactive windshear alert (GPWS Mode 7) operates normally.

OPERATIONS (O)

Alternate procedures must be established and used.

NOTE: The WINDSHEAR SYS advisory message will be displayed.

34-43-01 Weather Radar System**34-43-01-01 Predictive Windshear Function****34-43-01-01B Reactive Windshear Inoperative**

Interval	Installed	Required	Procedure
B	-	0	(O)

May be inoperative provided alternate procedures are established and used.

NOTE: Operator's alternate procedures should include reviewing windshear avoidance and windshear recovery procedures.

OPERATIONS (O)

Alternate procedures must be established and used.

NOTE: The WINDSHEAR SYS advisory message will be displayed.

**34-43-01 Weather Radar System****34-43-01-02 Auto Tilt Function**

Interval	Installed	Required	Procedure
C	1	0	

May be inoperative provided manual tilt function operates normally.

NOTE: This item is applicable to the Collins WRT-2100 and Honeywell RDR-4B weather radar systems with the auto tilt function.

OPERATIONS NOTE

Manual TILT adjustments may be necessary to optimize weather display.



777 FAA DDG Supplement Package

Supplement Number: 36-3

Issue Date: 29-MAY-2024

Section Number: 2 (MMEL)

Item Number: 34-46-01

Item Name: Ground Proximity Warning System (GPWS)

DDG Effectivity: 777 FAA DDG (D630W003-TBC), Revision 36

MMEL Effectivity: 777 FAA MMEL, Revision 23a

Description: Modified DDG top notes and Operations notes.

Reason: Clarified the AUTOTILT FAIL message may display for the RDR-4B weather radar if the auto tilt function is installed and that dispatch using item 34-43-01-02 is only required for display of the AUTOTILT FAIL message. For subitems -01-01, -01-02, -01-03A and -01-03B clarified the GND PROX SYS advisory message 'will' be displayed and modified associated PBN (O) notes, accordingly.

Disposition: For reference in the Operator's MEL.

Attachments: DDG page 2.34-46-01.1-6

**34-46-01 Ground Proximity Warning System (GPWS)****34-46-01-01 GPWS Function**

Interval	Installed	Required	Procedure
A	1	0	(O)

May be inoperative provided:

- a. Alternate procedures are established and used.
- b. Repairs are made within 2 flight days.

NOTE: For AUTOTILT FAIL annunciated on the ND, the airplane must also be dispatched using MMEL item 34-43-01-02.

OPERATIONS (O)

NOTE 1: The GPWS provides the aural signal for the automatic V1 callout and the Engine Fail alert. These aural signals may not be available.

NOTE 2: The GND PROX SYS advisory message will be displayed.

[For Honeywell RDR-4B weather radar with auto tilt function installed:](#)

NOTE 3: AUTOTILT FAIL may be annunciated on the ND.

1. Alternate procedures must be established and used.

NOTE: These procedures must maintain flight crew awareness of airplane altitude and flight path. They may include flight crew use of all systems available (autopilot, autoland, ILS and FMCS) and approach altitude callouts.

2. For PBN operations:

- A. RNP AR APCH operations are not allowed.
- B. Other PBN operations may be conducted.

NOTE: Refer to DDG Section 4 for additional information.

34-46-01 Ground Proximity Warning System (GPWS)**34-46-01-01 GPWS Function****34-46-01-01-01 Modes 1 - 4**

Interval	Installed	Required	Procedure
A	4	0	(O)

May be inoperative provided:

- a. Alternate procedures are established and used.
- b. Repairs are made within 2 flight days.

OPERATIONS (O)

NOTE: The GND PROX SYS advisory message will be displayed.

1. Alternate procedures must be established and used.

NOTE: These procedures must maintain flight crew awareness of airplane altitude and flight path. They may include flight crew use of all systems available (autopilot, autoland, ILS and FMCS) and approach altitude callouts.

2. For PBN operations:
 - A. RNP AR APCH operations are not allowed.
 - B. Other PBN operations may be conducted.

NOTE: Refer to DDG Section 4 for additional information.

34-46-01 Ground Proximity Warning System (GPWS)

34-46-01-01 GPWS Function

34-46-01-01-02 Glideslope Deviation (Mode 5)

Interval	Installed	Required	Procedure
B	1	0	

OPERATIONS NOTE

NOTE: The GND PROX SYS advisory message will be displayed.

For PBN operations:

1. RNP AR APCH operations are not allowed.
2. Other PBN operations may be conducted.

NOTE: Refer to DDG Section 4 for additional information.

34-46-01 Ground Proximity Warning System (GPWS)

34-46-01-01 GPWS Function

34-46-01-01-03 Advisory Callouts (Mode 6)

34-46-01-01-03A Required By 14 CFR

Interval	Installed	Required	Procedure
B	-	0	(O)

May be inoperative provided alternate procedures are established and used.

**OPERATIONS (O)**

NOTE 1: The GPWS provides the aural signal for the automatic V1 callout and the Engine Fail alert. These aural signals may not be available.

NOTE 2: See 14 CFR 121.354 or other appropriate regulation.

NOTE 3: The GND PROX SYS advisory message will be displayed.

1. Alternate procedures must be established and used.

NOTE: These procedures must maintain flight crew awareness of airplane altitude and flight path. They may include flight crew use of all systems available (autopilot, autoland, ILS and FMCS) and approach altitude callouts.

2. For PBN operations:

- A. RNP AR APCH operations are not allowed.
- B. Other PBN operations may be conducted.

NOTE: Refer to DDG Section 4 for additional information.

34-46-01 Ground Proximity Warning System (GPWS)**34-46-01-01 GPWS Function****34-46-01-01-03 Advisory Callouts (Mode 6)****34-46-01-01-03B Not Required by 14 CFR**

Interval	Installed	Required	Procedure
C	-	0	(O)

May be inoperative provided:

- a. Advisory callouts are not required by 14 CFR.
- b. Alternate procedures are established and used.

OPERATIONS (O)

NOTE 1: The GPWS provides the aural signal for the automatic V1 callout and the Engine Fail alert. These aural signals may not be available.

NOTE 2: See 14 CFR 121.354 or other appropriate regulation.

NOTE 3: The GND PROX SYS advisory message will be displayed.

1. Alternate procedures must be established and used.

NOTE: These procedures must maintain flight crew awareness of airplane altitude and flight path. They may include flight crew use of all systems available (autopilot, autoland, ILS and FMCS) and approach altitude callouts.

2. For PBN operations:

- A. RNP AR APCH operations are not allowed.



B. Other PBN operations may be conducted.

NOTE: Refer to DDG Section 4 for additional information.

34-46-01 Ground Proximity Warning System (GPWS)**34-46-01-02 Windshear Alert Mode Function (Reactive) (Mode 7)****34-46-01-02A Predictive Windshear Operative**

Interval	Installed	Required	Procedure
C	1	0	(O)

May be inoperative provided:

- Alternate procedures are established and used.
- Predictive Windshear operates normally.

OPERATIONS (O)

Alternate procedures must be established and used.

NOTE: The WINDSHEAR SYS advisory message will be displayed.

34-46-01 Ground Proximity Warning System (GPWS)**34-46-01-02 Windshear Alert Mode Function (Reactive) (Mode 7)****34-46-01-02B Predictive Windshear Inoperative**

Interval	Installed	Required	Procedure
B	1	0	(O)

May be inoperative provided alternate procedures are established and used.

NOTE: Operator's alternate procedures should include reviewing windshear avoidance and windshear recovery procedures.

OPERATIONS (O)

Establish alternate procedures for reactive windshear inoperative.

NOTE: The WINDSHEAR SYS advisory message will be displayed.

34-46-01 Ground Proximity Warning System (GPWS)**34-46-01-03 Terrain Awareness Function**

Interval	Installed	Required	Procedure
B	1	0	(O)

May be inoperative provided alternate procedures are established and used.



NOTE: For AUTOTILT FAIL annunciated on the ND, the airplane must also be dispatched using MMEL item 34-43-01-02.

OPERATIONS (O)

NOTE 1: The GND PROX SYS advisory message will be displayed.

NOTE 2: See 14 CFR 121.354 or other appropriate regulation.

[For Honeywell RDR-4B weather radar with auto tilt function installed:](#)

NOTE 3: AUTOTILT FAIL may be annunciated on the ND.

1. Alternate procedures must be established and used.
2. For PBN operations:
 - A. RNP AR APCH operations are not allowed.
 - B. Other PBN operations may be conducted.

NOTE: Refer to DDG Section 4 for additional information.

34-46-01 Ground Proximity Warning System (GPWS)

34-46-01-03 Terrain Awareness Function

34-46-01-03-01 Terrain Display Functions

34-46-01-03-01A One Inoperative

Interval	Installed	Required	Procedure
C	2	1	

OPERATIONS NOTE

1. The GND PROX SYS advisory message may be displayed.
2. See 14 CFR 121.354 or other appropriate regulation.
3. For PBN operations:

For GND PROX SYS advisory message displayed,

 - A. RNP AR APCH operations are not allowed.
 - B. Other PBN operations may be conducted.

NOTE: Refer to DDG Section 4 for additional information.

34-46-01 Ground Proximity Warning System (GPWS)

34-46-01-03 Terrain Awareness Function

34-46-01-03-01 Terrain Display Functions

34-46-01-03-01B Both Inoperative

Interval	Installed	Required	Procedure
B	2	0	

NOTE: For AUTOTILT FAIL annunciated on the ND, the airplane must also be dispatched using MMEL item 34-43-01-02.

OPERATIONS NOTE

1. The GND PROX SYS advisory message may be displayed.
2. For PBN operations:
For GND PROX SYS advisory message displayed,
 - A. RNP AR APCH operations are not allowed.
 - B. Other PBN operations may be conducted.

NOTE: Refer to DDG Section 4 for additional information

[For Honeywell RDR-4B weather radar with auto tilt function installed:](#)

3. AUTOTILT FAIL may be annunciated on the ND.

34-46-01 Ground Proximity Warning System (GPWS)

34-46-01-04 Runway Awareness and Advisory System (RAAS)

Interval	Installed	Required	Procedure
C	1	0	

OPERATIONS NOTE

For RUNWAY SYS advisory message displayed, consider using the RUNWAY OVRD function.



777 FAA DDG Supplement Package

Supplement Number: 36-3

Issue Date: 29-MAY-2024

Section Number: 2 (MMEL)

Item Number: 34-58-01

Item Name: Global Positioning Systems

DDG Effectivity: 777 FAA DDG (D630W003-TBC), Revision 36

MMEL Effectivity: 777 FAA MMEL, Revision 23a

Description: Modified PBN guidance.

Reason: To incorporate updated PBN operation capabilities.

Disposition: For reference in the Operator's MEL.

Attachments: DDG page 2.34-58-01.1-2

**34-58-01 Global Positioning Systems****34-58-01A Both Inoperative**

Interval	Installed	Required	Procedure
C	2	0	

May be inoperative provided:

- Enroute operations do not require their use.
- SAARU operates normally.

NOTE: The airplane must also be dispatched using MMEL item 34-53-02-01.

MAINTENANCE NOTE

- For both GPS inoperative and AIMS power is interrupted, UTC is lost and the AIMS time function will reset to midnight.
- If desired, the left, right, or center MMR may be deactivated by opening and collaring P11 panel MMR-L, MMR-R and/or MMR-C circuit breaker.
- If the GPS L and GPS R status messages are caused by a fault in the center MMR antenna, opening the MMR-C circuit breaker may restore GPS L and R functions.
- The associated ILS function will be inoperative for a deactivated MMR. Relief for ILS functions is available in MMEL item 34-31-01.

OPERATIONS NOTE

- Operators must consult their Operations Specification for route position accuracy requirements such as that required by certain Required Navigation Performance (RNP) routes.
- The NAV UNABLE RNP advisory message may be displayed during lengthy ground operations and flight crew should accomplish the associated Non-Normal Checklist.
- The TERR POS advisory message will be displayed.
- The RWY/POS update on the FMC-CDU TAKEOFF REF page will be active.
- The ADS-B Out data will be inoperative.
- For airplanes equipped with RAAS:
 - RAAS is inoperative.
 - The RUNWAY SYS advisory message will be displayed.
 - The airplane must also be dispatched using MMEL item 34-46-01-04.
- For PBN operations:
 - RNP AR APCH, RNP APCH, RNP 1, RNP 2, RNP 4 and Advanced RNP operations are not allowed.

- B. RNAV 1, RNAV 2, P-RNAV, B-RNAV and RNAV 5 operations may be conducted provided one DME/DME or one VOR/DME operates normally.
- C. RNP 10 and RNAV 10 operations may be conducted provided appropriate time limitations are observed. Refer to the RNP Navigation Capabilities Document for the time limitations.

NOTE: Refer to DDG Section 4 for additional information.

34-58-01 Global Positioning Systems

34-58-01B One Inoperative

Interval	Installed	Required	Procedure
D	2	1	

One may be inoperative provided enroute operations do not require its use.

NOTE: The airplane must also be dispatched using MMEL item 34-53-02-01.

MAINTENANCE NOTE

1. If desired, one or both MMR not associated with the operative GPS may be deactivated by opening and collaring P11 panel MMR-L, MMR-R and/or MMR-C circuit breaker.
2. The associated ILS function will be inoperative for a deactivated MMR. Relief for ILS functions is available in MMEL item 34-31-01.

For EFB installed:

NOTE: Some functionality of the EFB may be inoperative. For inoperative EFB functions the airplane must also be dispatched using MMEL item 46-11-02-01.

OPERATIONS NOTE

1. The associated ADS-B Out data will be inoperative.
2. For PBN operations:
 - A. RNP AR APCH and RNP 4 operations are not allowed.
 - B. RNP 2 operations over oceanic and remote continental airspace are not allowed.
 - C. RNP APCH, RNP 1, other RNP 2, Advanced RNP, RNP 10, and RNAV 10 operations may be allowed provided alternate navigation equipment (e.g. DME/DME, VOR/DME, VOR, ADF) is available to continue to a suitable airport if the remaining GPS becomes inoperative.
 - D. RNAV 1, RNAV 2, P-RNAV, B-RNAV and RNAV 5 operations may be conducted.

NOTE: Refer to DDG Section 4 for additional information.



777 FAA DDG Supplement Package

Supplement Number: 36-3

Issue Date: 29-MAY-2024

Section Number: 2 (MMEL)

Item Number: 36-12-02

Item Name: Center Isolation System (Valve and/or Indication)

DDG Effectivity: 777 FAA DDG (D630W003-TBC), Revision 36

MMEL Effectivity: 777 FAA MMEL, Revision 23a

Description: Added Operations procedure step to option B.

Reason: Provided Operations procedure step to enable cross-bleed engine start in the event of an inflight engine failure.

Disposition: For reference in the Operator's MEL.

Attachments: DDG page 2.36-12-02.1-3

36-12-02 Center Isolation System (Valve and/or Indication)

36-12-02A Valve Locked Closed

Interval	Installed	Required	Procedure
C	1	0	(M) (O)

May be inoperative provided:

- Valve is locked closed after engine start.
- Both center system hydraulic demand pumps operate normally.
- Left and right isolation valves operate normally.
- Both engine bleed systems operate normally.
- Appropriate performance adjustments are applied.

MAINTENANCE (M)

Lock the center isolation valve in the closed position after engine start (AMM 36-00-00/901).

- Gain access to center isolation valve by opening Bleed Air Valve Door 196NR.
- Lock the isolation valve in the closed position by following the MANUAL LOCK instructions placard on the valve.

NOTE: APU or crossbleed air is not available for starting the right engine when center isolation valve is locked closed. The right engine must be started and left running before locking the valve closed (AMM 71-00-00/201).

- Position Bleed Air Panel (P5) Bleed C ISLN switch CLOSED.
- Close Bleed Air Valve Door 196NR.

OPERATIONS (O)

NOTE 1: For the APU-to-Pack takeoff function installed, the APU-to-Pack Takeoff Supplementary Procedure is not allowed.

NOTE 2: APU or crossbleed air is not available for starting the right engine when the center isolation valve is locked closed.

NOTE 3: The flight deck Air Synoptic may display an incorrect or missing valve position.

- Reduce performance limited weights by the appropriate adjustments.

Performance Limited Weights	Takeoff Climb	Takeoff Runway/Obstacle
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For -200 or -200ER:

All	1,400 lb (636 kg)	9,600 lb (4,355 kg)
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For -200LR or 777F/GE90-100 Series or -300ER/GE90-115B:

All	1,700 lb (771 kg)	9,600 lb (4,355 kg)
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For -300/PW4090 or RR:

All	400 lb (181 kg)	3,700 lb (1,678 kg)
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2. For ETOPS flight planning, increase the engine inoperative Critical Fuel Reserves by an additional 1.3% when dispatching into known or forecast icing conditions.

36-12-02 Center Isolation System (Valve and/or Indication)

36-12-02B Valve Locked Open

Interval	Installed	Required	Procedure
C	1	0	(M) (O)

May be inoperative provided:

- a. Valve is locked open.
- b. Left and right isolation valves are selected CLOSED for takeoff and landing.
- c. APU is used as air source for center system hydraulic demand pumps for takeoff and landing.
- d. Left or right isolation valve is selected AUTO with flaps up.

MAINTENANCE (M)

Lock the center isolation valve in the open position (AMM 36-00-00/901).

1. Gain access to center isolation valve by opening Bleed Air Valve Door 196NR.
2. Lock the isolation valve in the open position by following the MANUAL LOCK instructions placard on the valve.
3. Position Bleed Air Panel (P5) Bleed C ISLN switch AUTO.
4. Close Bleed Air Valve Door 196NR.

OPERATIONS (O)

NOTE 1: For the APU-to-Pack takeoff function installed, the APU-to-Pack Takeoff Supplementary Procedure is not allowed.



NOTE 2: The flight deck Air Synoptic may display an incorrect or missing valve position.

1. For takeoff and landing use the APU as an air source.
 - A. Position APU bleed air switch AUTO.
 - B. Start the APU if it is not running.
 - C. Position L ISLN and R ISLN switches CLOSED.
 - D. After takeoff is completed, position L ISLN or R ISLN (one) switch AUTO and configure the APU as required.
2. In the event of inflight engine failure, accomplish the associated Non-Normal Checklist.

NOTE 1: To enable the crossbleed function, position L ISLN and R ISLN (both) switches AUTO.

NOTE 2: After crossbleed engine start, position L ISLN or R ISLN (one) switch CLOSED.

36-12-02 Center Isolation System (Valve and/or Indication)**36-12-02-01 ISLN Switch Light****36-12-02-01-01 CLOSED Light**

Interval	Installed	Required	Procedure
C	1	0	

36-12-02 Center Isolation System (Valve and/or Indication)**36-12-02-01 ISLN Switch Light****36-12-02-01-02 AUTO Light**

Interval	Installed	Required	Procedure
C	1	0	



777 FAA DDG Supplement Package

Supplement Number: 36-3

Issue Date: 29-MAY-2024

Section Number: 2 (MMEL)

Item Number: 46-11-01

Item Name: Onboard Network System (ONS)

DDG Effectivity: 777 FAA DDG (D630W003-TBC), Revision 36

MMEL Effectivity: 777 FAA MMEL, Revision 23a

Description: Modified DDG top note.

Reason: Modified note to update MMEL item number reference.

Disposition: For reference in the Operator's MEL.

Attachments: DDG page 2.46-11-01.1-2

46-11-01 Onboard Network System (ONS)

46-11-01A Alternate Procedures Required

Interval	Installed	Required	Procedure
C	1	0	(O)

May be inoperative provided alternate procedures are established and used.

NOTE: Any function which operates normally may be used.

NOTE 1: The Onboard Network System (ONS) includes the Network File Server (NFS), Network Extension Device (NED), Aircraft Interface Device (AID) and Wireless WAN Unit (WWU) components with associated support interfaces.

For Class 3 EFB installed:

NOTE 2: The airplane must also be dispatched using MMEL item 46-11-02-01.

MAINTENANCE NOTE

The Maintenance Access Terminal (MAT) functions may not be available including CMCS data collection and loading.

OPERATIONS (O)

Alternate procedures must be established and used.

For Class 3 EFB installed:

NOTE: The EFB's on-ground connectivity to the AWLU or TWLU and airport networks, the EFB's connectivity to the graphical function of the ARINC 744A airborne printer and the EFB's in-flight connectivity to the SATCOM Swift communication network will be inoperative.

46-11-01 Onboard Network System (ONS)

46-11-01B Procedures Do Not Require Use

Interval	Installed	Required	Procedure
D	1	0	

May be inoperative provided procedures do not require its use.

NOTE 1: The Onboard Network System (ONS) includes the Network File Server (NFS), Network Extension Device (NED), Aircraft Interface Device (AID) and Wireless WAN Unit (WWU) components with associated support interfaces.

[For Class 3 EFB installed:](#)

NOTE 2: The airplane must also be dispatched using MMEL item 46-11-02-01.

MAINTENANCE NOTE

The Maintenance Access Terminal (MAT) functions may not be available including CMCS data collection and loading.

[For Class 3 EFB installed:](#)

OPERATIONS NOTE

The EFB's on-ground connectivity to the AWLU or TWLU and airport networks, the EFB's connectivity to the graphical function of the ARINC 744A airborne printer and the EFB's in-flight connectivity to the SATCOM Swift communication network will be inoperative.



777 FAA DDG Supplement Package

Supplement Number: 36-3

Issue Date: 29-MAY-2024

Section Number: 2 (MMEL)

Item Number: 52-71-01

Item Name: Door Indication Systems

DDG Effectivity: 777 FAA DDG (D630W003-TBC), Revision 36

MMEL Effectivity: 777 FAA MMEL, Revision 23a

Description: Modified Maintenance step 1 annotations.

Reason: Corrected annotations for entry doors 2L, 2R, 4L and 4R.

Disposition: For reference in the Operator's MEL.

Attachments: DDG page 2.52-71-01.1-7

52-71-01 Door Indication Systems

Interval	Installed	Required	Procedure
C	-	0	(M)

May be inoperative provided associated door is verified closed, latched and locked before each departure.

MAINTENANCE (M)

NOTE: The optional procedures to deactivate inoperative passenger entry door sensors and forward, aft and main deck cargo (777F only) door sensors are provided to prevent unexpected display of the CONFIG DOORS warning and potential RTO due to intermittent door sensor faults.

Deactivate the inoperative sensors (if desired), and before each flight, verify the associated door is closed, latched, and locked (AMM 52-00-00/901).

1. For inoperative sensor(s) on Main Entry Doors:
 - A. If desired, deactivate the associated entry door's inoperative sensor(s).
 - 1) Remove, cap and stow electrical connectors from following sensors.
 - a. For entry door 1L, remove electrical connector DS52109 from sensor S52109 (CLOSED & LATCHED DR 1L) and remove electrical connector DS52110 from sensor S52110 (LOCKED DR 1L).
 - b. For entry door 1R, remove electrical connector DS52209 from sensor S52209 (CLOSED & LATCHED DR 1R) and remove electrical connector DS52110 from sensor S52110 (LOCKED DR 1R).
 - For -200/-200ER/-200LR:
 - c. For entry door 2L, remove electrical connector DS52111 from sensor S52111 (CLOSED & LATCHED DR 2L) and remove electrical connector DS52110 from sensor S52110 (LOCKED DR 2L).
 - For -300/-300ER:
 - d. For entry door 2L, remove electrical connector DS52111 from sensor S52111 (CLOSED & LATCHED DR 2L) and remove electrical connector DS52110 from sensor S52110 (LOCKED DR 2L).

For -200/-200ER/-200LR:

- e. For entry door 2R, remove electrical connector DS52211 from sensor S52211 (CLOSED & LATCHED DR 2R) and remove electrical connector DS52110 from sensor S52110 (LOCKED DR 2R).

For -300/-300ER:

- f. For entry door 2R, remove electrical connector DS52211 from sensor S52211 (CLOSED & LATCHED DR 2R) and remove electrical connector DS52110 from sensor S52110 (LOCKED DR 2R).

For -200/-200ER/-200LR:

- g. For entry door 3L, remove electrical connector DS52113 from sensor S52113 (CLOSED & LATCHED DR 3L) and remove electrical connector DS52110 from sensor S52110 (LOCKED DR 3L).

For -300/-300ER:

- h. For entry door 3L, remove electrical connector DS52113 from sensor S52113 (CLOSED & LATCHED DR 3L) and remove electrical connector DS52026 from sensor S52026 (LOCKED DR 3L).

For -200/-200ER/-200LR:

- i. For entry door 3R, remove electrical connector DS52213 from sensor S52213 (CLOSED & LATCHED DR 3R) and remove electrical connector DS52110 from sensor S52110 (LOCKED DR 3R).

For -300/-300ER:

- j. For entry door 3R, remove electrical connector DS52213 from sensor S52213 (CLOSED & LATCHED DR 3R) and remove electrical connector DS52026 from sensor S52026 (LOCKED DR 3R).

For -200/-200ER/-200LR:

- k. For entry door 4L, remove electrical connector DS52115 from sensor S52115 (CLOSED & LATCHED DR 4L) and remove electrical connector DS52110 from sensor S52110 (LOCKED DR 4L).

For -300/-300ER:

- l. For entry door 4L, remove electrical connector DS52115 from sensor S52115 (CLOSED & LATCHED DR 4L) and remove electrical connector DS52110 from sensor S52110 (LOCKED DR 4L).

For -200/-200ER/-200LR:

- m. For entry door 4R, remove electrical connector DS52215 from sensor S52215 (CLOSED & LATCHED DR 4R) and remove electrical connector DS52110 from sensor S52110 (LOCKED DR 4R).

For -300/-300ER:

- n. For entry door 4R, remove electrical connector DS52215 from sensor S52215 (CLOSED & LATCHED DR 4R) and remove electrical connector DS52110 from sensor S52110 (LOCKED DR 4R).

For -300/-300ER:

- o. For entry door 5L, remove electrical connector DS52117 from sensor S52117 (CLOSED & LATCHED DR 5L) and remove electrical connector DS52110 from sensor S52110 (LOCKED DR 5L).

For -300/-300ER:

- p. For entry door 5R, remove electrical connector DS52217 from sensor S52217 (CLOSED & LATCHED DR 5R) and remove electrical connector DS52110 from sensor S52110 (LOCKED DR 5R).

- B. Prior to each flight, verify the associated door is closed, latched and locked using normal procedures.
 - 1) Open the affected door in MANUAL/DISARMED mode, then close and latch.
 - 2) Confirm the door interior liner is aligned and faired with the doorway lining.
 - 3) Confirm the OPEN arrow on the hinge arm linings is aligned and continuous at crossover from forward section of lining to aft section of lining.
 - 4) Position the door mode select handle to AUTOMATIC/ARMED, and confirm the yellow flags are visible in the viewing windows immediately forward and aft of door at floor level.
 - 5) Confirm the outside skin of the door is aligned and faired with the fuselage skin.
 - 6) Confirm the vent door is fully closed.

2. For inoperative sensor(s) on Forward Cargo Door:

- A. If desired, deactivate the inoperative door sensor(s).
 - 1) Remove, cap and stow electrical connector DS52021 from sensor S52021 (CLOSED).
 - 2) Remove, cap and stow electrical connector DS52042 from sensor S52042 (LATCHED).
 - 3) Remove, cap and stow electrical connector DS52022 from sensor S52022 (LOCKED).

- B. Prior to each flight, verify the door is closed, latched and locked using normal procedures.
 - 1) Close the door.
 - 2) Confirm the outside skin of the door is aligned and faired with the fuselage skin.
 - 3) Confirm the lock handle is in the unlocked position.
 - 4) Confirm the vent door is open.
 - 5) Rotate the lock handle back to the CLOSED position.
 - 6) Confirm the lock handle is fully closed and faired with the outside skin of the door.
 - 7) Confirm the lock handle release lever is faired with the lock handle and outside skin of the door.
 - 8) Confirm the vent door is fully closed (the upper edge of the vent door may not be fully faired).
 - 9) Confirm the latches are in the fully latched position and that the locks are in the locked position by visual inspection through the witness ports at the lower edge of the door.
- 3. For inoperative sensor(s) on Aft Large Cargo Door:
 - A. If desired, deactivate the inoperative door sensor(s).
 - 1) Remove, cap and stow electrical connector DS52045 from sensor S52045 (CLOSED).
 - 2) Remove, cap and stow electrical connector DS52043 from sensor S52043 (LATCHED).
 - 3) Remove, cap and stow electrical connector DS52044 from sensor S52044 (LOCKED).
 - B. Prior to each flight, verify the door is closed, latched and locked using normal procedures.
 - 1) Close the door.
 - 2) Confirm the outside skin of the door is aligned and faired with the fuselage skin.
 - 3) Confirm the lock handle is in the unlocked position.
 - 4) Confirm the vent door is open.
 - 5) Rotate the lock handle back to the CLOSED position.
 - 6) Confirm the lock handle is fully closed and faired with the outside skin of the door.
 - 7) Confirm the lock handle release lever is faired with the lock handle and outside skin of the door.
 - 8) Confirm the vent door is fully closed (the upper edge of the vent door may not be fully faired).
 - 9) Confirm the latches are in the fully latched position and that the locks are in the locked position by visual inspection through the witness ports at the lower edge of the door.

4. For inoperative sensor(s) on Aft Small Cargo Door:
 - A. If desired, deactivate the inoperative door sensor(s).
 - 1) Remove, cap and stow electrical connector DS52023 from sensor S52023 (CLOSED).
 - 2) Remove, cap and stow electrical connector DS52024 from sensor S52024 (LATCHED/LOCKED).
 - B. Prior to each flight, verify the door is closed and latched using normal procedures.
 - 1) Close the door.
 - 2) Confirm the outside skin of the door is aligned and faired with the fuselage skin.
 - 3) Confirm the vent door is fully closed.
 - 4) Open the access cover for the door control panel.
 - 5) Operate the CARGO DOOR CONTROL ARMING switch downward to the ARM position, and confirm the green DOOR LATCHED light illuminates. Release the switch and confirm the light extinguishes.
 - 6) Close the access cover.
5. For inoperative sensor(s) on Bulk Cargo Door:
 - A. If desired, deactivate the inoperative door switch.
 - 1) Locate, remove and secure switch S52020 (CLOSED/LATCHED/LOCKED) without cutting its wires.
 - B. Prior to each flight, verify the door is closed and latched using normal procedures.
 - 1) Close the door.
 - 2) Confirm the outside skin of the door is aligned and faired with the fuselage skin.
 - 3) Confirm the latch handle is fully closed and faired with the door skin.
 - 4) Push on the door to verify it is latched.
6. For inoperative sensor(s) on Forward Access Door:
 - A. If desired, deactivate the inoperative door sensor(s).
 - 1) Remove, cap and stow electrical connector DS52018 from sensor S52018 (CLOSED/LATCHED/LOCKED).
 - B. Prior to each flight, verify the door is closed and latched using normal procedures.
 - 1) Close the door.
 - 2) Confirm the outside skin of the door is aligned and faired with the fuselage skin.
 - 3) Confirm the latch handle is fully closed and faired with the door skin.
 - 4) Push on the door to verify it is latched.

7. For inoperative sensor(s) on Main Equipment Center (E/E Bay) Access Door:
- A. If desired, deactivate the inoperative door sensor(s).
 - 1) Remove, cap and stow electrical connector DS52019 from sensor S52019 (CLOSED/LATCHED/LOCKED).
 - B. Prior to each flight, verify the door is closed and latched using normal procedures.
 - 1) Close and latch the door.
 - 2) Confirm the outside skin of the door is aligned and faired with the fuselage skin.
 - 3) Confirm the latch handle is fully closed and faired with the door skin.
 - 4) Push on the door to verify it is latched.

For -300/-300ER:

8. For inoperative sensor(s) on Overwing Escape Slide Compartment Door:
- A. Prior to each flight, verify the door is closed and latched using normal procedures.
 - 1) Open and close the door.
 - 2) Confirm the outside skin of the door is aligned and faired with the fuselage skin.
 - 3) Push on the door to verify it is latched.

For 777F:

9. For inoperative sensor(s) on Main Deck Cargo Door:
- A. If desired, deactivate the inoperative door sensor(s).
 - 1) Remove, cap and stow electrical connector DS52053 from sensor S52053 (CLOSED).
 - 2) Remove, cap and stow electrical connector DS52054 from sensor S52054 (LATCHED).
 - 3) Remove, cap and stow electrical connector DS52055 from sensor S52055 (LOCKED).
 - B. Prior to each flight, verify the door is closed, latched and locked using normal procedures.
 - 1) Close the door.
 - 2) Confirm the latch arrows point to the targets in all eight witness ports.
 - 3) Confirm the outside skin of the door is aligned and faired with the fuselage skin.
 - 4) Confirm the vent door has closed.
 - 5) Using the witness ports, confirm the lock sectors are located over latch cams but do not touch the cams.

OPERATIONS NOTE

1. The optional procedures to deactivate inoperative passenger entry door sensors and forward, aft and main deck cargo (777F only) door sensors are provided to prevent unexpected display of the CONFIG DOORS warning and potential RTO due to intermittent door sensor faults.
2. The DOOR WING SLIDE L and DOOR WING SLIDE R advisory messages (-300/-300ER) and the DOOR BULK CARGO advisory message may be displayed both on the ground and in flight.
3. For doors with inoperative sensors deactivated:
 - A. The respective door indication status message (if applicable) may be displayed both on the ground and in flight.
 - B. Inoperative door advisory messages may be displayed on the ground only, except the DOOR BULK CARGO advisory message, which may be displayed both on the ground and in flight.
 - C. The DOOR MAIN DECK CARGO, DOOR FWD CARGO and DOOR AFT CARGO (if applicable) caution messages may be displayed on the ground.
4. The CONFIG DOORS warning is not displayed for bulk cargo door switch faults.



777 FAA DDG Supplement Package

Supplement Number: 36-3

Issue Date: 29-MAY-2024

Section Number: 3 (CDL)

Item Number: 78-31-04

Item Name: Thrust Reverser Wedge Fairing

DDG Effectivity: 777 FAA DDG (D630W003-TBC), Revision 36

MMEL Effectivity: 777 FAA MMEL, Revision 23a

Description: Modified item statement.

Reason: Statement modified to exclude GE90-100 series engines.

Disposition: For reference in the Operator's MEL.

Attachments: DDG page 3.78-31-04.1

78-31-04 Thrust Reverser Wedge Fairing

Except for PW4000 reversers P/N 315W3298 and GE90-100 Series, a maximum of two per nacelle half (eight total) may be missing.

Performance limited weights are reduced by the following for each missing item:

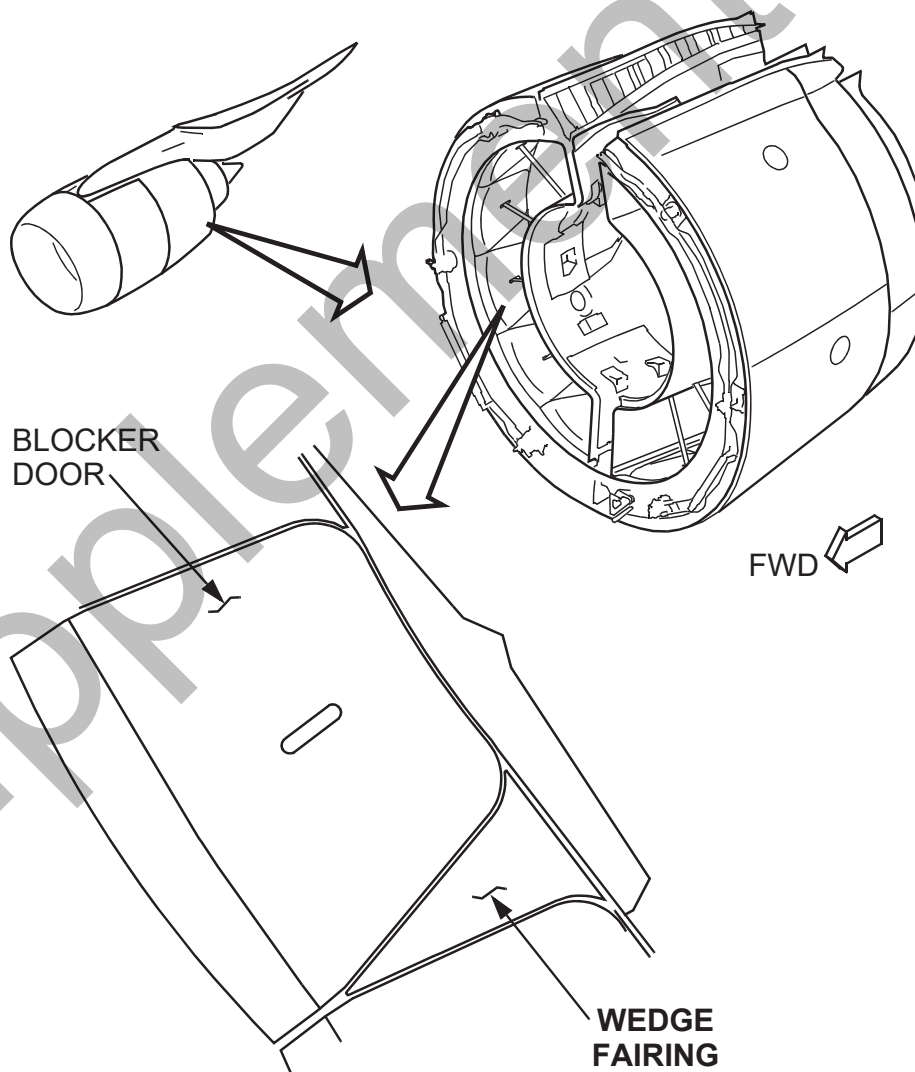
Number Installed	Takeoff	Enroute Climb	Approach/Landing Climb
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For PW (Except PW4000 with P/N 315W3298) / RR:

28	250 lb (114 kg)	250 lb (114 kg)	250 lb (114 kg)
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For GE (Except GE90-100 Series):

36	200 lb (91 kg)	150 lb (69 kg)	150 lb (69 kg)
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777 FAA DDG Supplement Package

Supplement Number: 36-3

Issue Date: 24-MAY-2024

Section Number: 4 (MISC)

Item Number: N/A

Item Name: Performance Based Navigation

DDG Effectivity: 777 FAA DDG (D630W003-TBC), Revision 36

MMEL Effectivity: 777 FAA MMEL, Revision 23a

Description: Modified PBN guidance.

Reason: To incorporate updated PBN operation capabilities.

Disposition: For reference in the Operator's MEL.

Attachments: DDG pages 4.00-02-00.1-7



Section 4 Performance Based Navigation

Performance Based Navigation (PBN) Information

Applying dispatch relief for the following items may affect the ability of the airplane to meet the operational capabilities required for PBN operations. The following information lists items that, if inoperative, may affect the ability of the airplane to meet the performance and functional criteria for various PBN operations. This information can aid in identifying potentially affected capabilities associated with a particular dispatch configuration.

This section lists equipment requirements that may be found in the document 777 RNP Navigation Capabilities Document D243W018-13, revision "O" (or later). Operators should consult their Operations Specification for requirements applicable to their operations.

PBN Equipment Requirements

Display Units (DUs) - (31-61-01)

PBN operations require two PFDs with mini map and EICAS display. The 777 is equipped with six display units. The PFDs are normally displayed on the outboard display units and the NDs are normally displayed on the inboard display units. EICAS is normally displayed on the upper center display unit. MMEL dispatch is only allowed with the lower center display unit inoperative. Therefore, these requirements will always be met for dispatch.

Multi-Purpose Control and Display Units - Control Display Units (CDU) - (34-61-03)

The 777 is equipped with three CDUs. MMEL dispatch is only allowed with the center CDU inoperative. Both left and right CDUs must operate normally for dispatch. PBN operations are allowed with two CDUs. Therefore, this requirement will always be met for dispatch.

AIMS Operational Software

For RNP AR APCH and RNP APCH:

Airplane must be equipped with AIMS BP05A or later. Navigation Performance Scales (NPS) and LNAV Go-Around Auto-Engage must be enabled.

Airplanes without Navigation Performance Scales and LNAV Go-Around Auto-Engage (AIMS BP03 and earlier versions) do not support these operations.

Fault Tolerant ADIRU

For AIMS-1 V16 and AIMS-2 V17 or later:

AIR DATA SYS or NAV ADIRU INERTIAL messages must not be displayed.

For all other AIMS-1 or AIMS-2 versions:

NAV AIR DATA SYS or NAV ADIRU INERTIAL messages must not be displayed.

MMEL dispatch is not allowed with any of the above messages displayed. Therefore, this requirement will always be met for dispatch.

Autopilot Flight Director Computers (AFDC) - (22-11-01)

The 777 autopilot flight director system (AFDS) is equipped with three autopilot flight director computers (AFDCs). Each of the three AFDCs can provide both flight director guidance, which can be displayed on the primary flight displays (PFDs), and send autopilot commands to the flight control actuation system when the autopilot is engaged.

MMEL dispatch is allowed with one, two, or all three AFDCs inoperative. Therefore, PBN operations may be affected with more than one AFDC inoperative.

For dispatch with two Autopilot Flight Director Computers inoperative (22-11-01B):

- RNP AR APCH operations are not allowed.
- Other PBN operations may be conducted.

For dispatch with all (3) Autopilot Flight Director Computers inoperative (22-11-01C):

- PBN operations are not allowed.

Autopilot and Flight Director

PBN operations may be conducted using either the autopilot or flight director. The minimum authorized RNP value may be different depending on whether autopilot or flight director is used.

Autopilot

When the autopilot is engaged, autopilot commands from the AFDCs are sent to the Primary Flight Computers to generate flight control surface commands.

If the autopilot function is inoperative, PBN operations may be conducted using the flight director.

The autopilot will be impacted if any of the following equipment is inoperative:

1. Mode Control Panel Switches - A/P Engage Switches
 - A. For dispatch with one autopilot engage switch inoperative (22-11-06-01A):
 - RNP AR APCH operations using autopilot are not allowed.
 - RNP AR APCH operations may be conducted using flight director provided its use is appropriate for the intended operation.
 - Other PBN operations using autopilot may be conducted.
 - B. For dispatch with both autopilot engage switches inoperative (22-11-06-01B):
 - PBN operations using autopilot are not allowed.
 - PBN operations may be conducted using flight director provided its use is appropriate for the intended operation.
2. Control Wheel Autopilot Disconnect Switches
 - A. For dispatch with one control wheel autopilot disconnect switch inoperative (22-11-08A):
 - PBN operations using autopilot at 1,500 feet AGL or below are not allowed.
 - PBN operations may be conducted using flight director provided its use is appropriate for the intended operation.
 - B. For dispatch with both control wheel autopilot disconnect switches inoperative (22-11-08B):
 - PBN operations using autopilot are not allowed.
 - PBN operations may be conducted using flight director provided its use is appropriate for the intended operation.
3. Warning Electronic System (WES) Autopilot Disconnect Warning Function
 - A. For dispatch with WES autopilot disconnect warning function inoperative (31-51-05):
 - PBN operations using autopilot are not allowed.
 - PBN operations may be conducted using flight director provided its use is appropriate for the intended operation.

Flight Director

When the flight director is selected on, flight director guidance from the AFDCs is sent to and displayed on the Primary Flight Displays (PFDs). The flight crew uses the flight director guidance to manually control the airplane in accordance with the command guidance from the AFDCs.

Flight director guidance can be displayed on one or both PFDs.

Flight director guidance is selected on and off by using the flight director switches on the mode control panel (MCP).

The flight director guidance will be impacted if any of the following equipment is inoperative:

1. Mode Control Panel Switches - F/D Switches
 - A. For dispatch with one Flight Director switch inoperative (22-11-06-04):
 - RNP AR APCH operations using flight director are not allowed.
 - RNP AR APCH operations may be conducted using autopilot provided its use is appropriate for the intended operation.
 - Other PBN operations using flight director may be conducted provided flight director operates normally on the side of the pilot flying (PF).
 - B. For dispatch with both Flight Director switches inoperative (22-11-06-04):
 - PBN operations using flight director are not allowed.
 - PBN operations may be conducted using autopilot provided its use is appropriate for the intended operation.
2. Flight Director Systems
 - A. For dispatch with one Flight Director system inoperative (22-11-10):
 - RNP AR APCH operations using flight director are not allowed.
 - RNP AR APCH operations may be conducted using autopilot provided its use is appropriate for the intended operation.
 - Other PBN operations using flight director may be conducted provided flight director operates normally on the side of the pilot flying (PF).
 - B. For dispatch with both Flight Director systems inoperative (22-11-10):
 - PBN operations using flight director are not allowed.
 - PBN operations may be conducted using autopilot provided its use is appropriate for the intended operation.

LNAV/VNAV Mode Control Panel (MCP) Switches - (22-11-06-09)

For dispatch with lateral navigation (LNAV) switch inoperative (22-11-06-09):

- PBN operations are not allowed.

For dispatch with vertical navigation (VNAV) switch inoperative. (22-11-06-09):

- RNP AR APCH operations are not allowed.
- RNP APCH operations may be conducted using other vertical modes provided appropriate approach minimums are used.
- Other PBN operations may be conducted.

Takeoff/Go-Around (TO/GA) Switches - (22-11-09)

There are two TO/GA switches installed on the 777. MMEL dispatch is allowed with one or both TO/GA switches inoperative.

For dispatch with one TO/GA switch inoperative (22-11-09A):

- RNP AR APCH operations are not allowed.
- Other PBN operations may be conducted.

For dispatch with both TO/GA switches inoperative (22-11-09B):

- PBN operations using autopilot or flight director below 500 feet AGL or MDA are not allowed.

Global Positioning Systems (GPS) - (34-58-01)

The 777 is equipped with two GPSs. MMEL dispatch is allowed with one or both GPSs inoperative.

For dispatch with one GPS inoperative (34-58-01B):

- RNP AR APCH and RNP 4 operations are not allowed.
- RNP 2 operations over oceanic and remote continental airspace are not allowed.
- RNP APCH, RNP 1, other RNP 2, Advanced RNP, RNP 10, and RNAV 10 operations may be conducted provided alternate navigation equipment (e.g. DME/DME, VOR/DME, VOR, ADF) is available to continue to a suitable airport if the remaining GPS becomes inoperative.
- RNAV 1, RNAV 2, P-RNAV, B-RNAV and RNAV 5 operations may be conducted.

For dispatch with both GPSs inoperative (34-58-01A):

- RNP AR APCH, RNP APCH, RNP 1, RNP 2, RNP 4 and Advanced RNP operations are not allowed.

- RNAV 1, RNAV 2, P-RNAV, B-RNAV and RNAV 5 operations may be conducted provided one DME/DME or one VOR/DME operates normally.
- RNP 10 and RNAV 10 operations may be conducted provided appropriate time limitations are observed. Refer to the RNP Navigation Capabilities Document for the time limitations.

VOR Navigation Systems - (34-51-01)

The 777 is equipped with two VORs. MMEL dispatch is allowed with one or both VORs inoperative.

- RNAV 1, RNAV 2, P-RNAV, B-RNAV and RNAV 5 operations may be conducted provided one GPS, one DME/DME or one VOR/DME operates normally.
- Other PBN operations may be conducted.

Distance Measuring Equipment Systems (DME) - (34-55-01)

The 777 is equipped with two DMEs. MMEL dispatch is allowed with one or both DMEs inoperative.

- RNAV 1, RNAV 2, P-RNAV, B-RNAV and RNAV 5 operations may be conducted provided one GPS, one DME/DME or one VOR/DME operates normally.
- Other PBN operations may be conducted.

Radio Altimeter Systems - (34-33-01)

The 777 is equipped with three radio altimeters. MMEL dispatch is allowed with one radio altimeter inoperative.

PBN operations are allowed with two radio altimeters. Therefore, this requirement will always be met for dispatch.

Ground Proximity Warning Systems (GPWS) - (34-46-01), Terrain Awareness Function (34-46-01-03)

777 airplanes are equipped with a single ground proximity warning system (GPWS). Airplanes with PRR 61724 or equivalent are equipped with enhanced GPWS (EGPWS) that includes a terrain awareness function.

MMEL dispatch is allowed with the GPWS (EGPWS) inoperative.

For dispatch with either GND PROX SYS or TERR POS Advisory message displayed:

- RNP AR APCH operations are not allowed.
- Other PBN operations may be conducted.

Flight Management Computing Systems (FMCS) - (34-61-01)

There are two flight management computing systems (FMCSs) installed on the 777. MMEL dispatch is allowed with one FMCS inoperative.

For dispatch with one FMCS inoperative:

- RNP AR APCH and RNP 4 operations are not allowed.
- RNP 2 operations over oceanic and remote continental airspace are not allowed.
- RNP APCH, RNP 1, other RNP 2, RNAV 1, RNAV 2, P-RNAV, B-RNAV, RNAV 5 and Advanced RNP operations may be conducted provided alternate navigation equipment (e.g., VOR, ADF) is available to continue to a suitable airport if the remaining FMCS becomes inoperative.
- RNP 10 and RNAV 10 operations may be conducted using the alternate navigation capability in the multifunction control display units (MCDU).

Flight Management Computing Systems (FMCS) - Navigation Database (34-61-01-03)

The FMCS navigation database must be current for PBN operations.

It is assumed that the operator has a preflight procedure to ensure that the navigation database complies with regulatory agency requirements for ensuring the navigation database remains valid for the duration of the flight unless an alternative regulatory agency approved process applies.



777 FAA DDG Supplement Package

Supplement Number: 36-4

Issue Date: 19-DEC-2024

Section Number: 2 (MMEL)

Item Number: 28-31-01

Item Name: Fuel Jettison System

DDG Effectivity: 777 FAA DDG (D630W003-TBC), Revision 36

MMEL Effectivity: 777 FAA MMEL, Revision 23a

Description: Modified Operations (O) procedures for subitems 28-31-01-01 and 28-31-01-02.

Reason: Operations (O) procedures were modified to reflect updated performance adjustments.

Disposition: For reference in the Operator's MEL.

Attachments: DDG pages 2.28-31-01.1-5

28-31-01 Fuel Jettison System
28-31-01-01 Without PRR 62777-79S or Equivalent Installed

Interval	Installed	Required	Procedure
C	1	0	(M) (O)

May be inoperative provided:

- Jettison system is deactivated.
- Jettison nozzle valves are secured closed.
- Appropriate performance adjustments are applied.

NOTE: The PRR 62777-79S is applicable to airplane line numbers 1099 and-on. Service Bulletins 777-28-0078 (ELMS2) and 777-28-0082 (ELMS1) each provide the production equivalent of PRR 62777-79S.

MAINTENANCE (M)

Deactivate the jettison system and secure the jettison nozzle valves in the closed position (AMM 28-00-00/901).

- Open and collar P110 panel L FUEL JETT PUMP CTRL, L FUEL JETT ISLN VLV and L FUEL JETT NOZ VLV circuit breakers.
- Open and collar P210 panel R FUEL JETT PUMP CTRL, R FUEL JETT ISLN VLV and R FUEL JETT NOZ VLV circuit breakers.
- For left jettison nozzle valve:
 - Gain access to the left jettison nozzle valve actuator through associated access door.
 - Disconnect, cap and stow the electrical connector (DV28166).
 - Confirm the manual override handle is in the CLOSED position and lockwire.
 - Close access door.
- For right jettison nozzle valve:
 - Gain access to the right jettison nozzle valve actuator through associated access door.
 - Disconnect, cap and stow the electrical connector (DV28266).
 - Confirm the manual override handle is in the CLOSED position and lockwire.
 - Close access door.

OPERATIONS (O)

NOTE: The Fuel Synoptic will incorrectly indicate jettison flow from the center tank when the center tank override pumps are on with fuel in the center tank. The FUEL JETT NOZZLE L and R advisory messages will be displayed and overhead panel L and R fuel jettison nozzle VALVE lights will be illuminated.

For -200 or -200ER/PW4074 thru PW4084:

5. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 3,400 lb (1,542 kg) (for fuel that would be burned during a return-to-land procedure).

For -200 or -200ER/PW4090:

6. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 3,800 lb (1,724 kg) (for fuel that would be burned during a return-to-land procedure).

For -200 or -200ER/GE90-76B thru -90B:

7. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 3,300 lb (1,497 kg) (for fuel that would be burned during a return-to-land procedure).

For -200 or -200ER/GE90-94B or RR875 thru RR892:

8. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 3,700 lb (1,678 kg) (for fuel that would be burned during a return-to-land procedure).

For -200 or -200ER/RR895:

9. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 4,000 lb (1,814 kg) (for fuel that would be burned during a return-to-land procedure).

For -200LR or 777F/GE90-100 Series:

10. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 4,000 lb (1,814 kg) (for fuel that would be burned during a return-to-land procedure).

For -300/RR884 thru RR892:

11. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 3,900 lb (1,769 kg) (for fuel that would be burned during a return-to-land procedure).

For -300/PW4090:

12. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 4,100 lb (1,860 kg) (for fuel that would be burned during a return-to-land procedure).

For -300ER/GE90-115B:

13. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 4,200 lb (1,905 kg) (for fuel that would be burned during a return-to-land procedure).

28-31-01 Fuel Jettison System
28-31-01-01 Without PRR 62777-79S or Equivalent Installed
28-31-01-01-01 ARM Switch FAULT Light

Interval	Installed	Required	Procedure
C	1	0	

28-31-01 Fuel Jettison System
28-31-01-01 Without PRR 62777-79S or Equivalent Installed
28-31-01-01-02 ARM Switch ARMED Light

Interval	Installed	Required	Procedure
C	1	0	

28-31-01 Fuel Jettison System
28-31-01-02 With PRR 62777-79S or Equivalent Installed

Interval	Installed	Required	Procedure
C	1	0	(M) (O)

May be inoperative provided:

- Jettison system is deactivated.
- Jettison nozzle valves are secured closed.
- Fuel scavenge system is considered inoperative.
- Appropriate performance adjustments are applied.

NOTE 1: The PRR 62777-79S is applicable to airplane line numbers 1099 and-on. Service Bulletins 777-28-0078 (ELMS2) and 777-28-0082 (ELMS1) each provide the production equivalent of PRR 62777-79S.

NOTE 2: The airplane must also be dispatched using MMEL item 28-22-04.

MAINTENANCE (M)

Deactivate the jettison system and secure the jettison nozzle valves in the closed position (AMM 28-00-00/901).

- Open and collar P110 panel L FUEL JETT PUMP CTRL, L FUEL JETT ISLN VLV and L FUEL JETT NOZ VLV circuit breakers.
- Open and collar P210 panel R FUEL JETT PUMP CTRL, R FUEL JETT ISLN VLV and R FUEL JETT NOZ VLV circuit breakers.

3. For left jettison nozzle valve:
 - A. Gain access to the left jettison nozzle valve actuator through associated access door.
 - B. Disconnect, cap and stow the electrical connector (DV28166).
 - C. Confirm the manual override handle is in the CLOSED position and lockwire.
 - D. Close access door.
4. For right jettison nozzle valve:
 - A. Gain access to the right jettison nozzle valve actuator through associated access door.
 - B. Disconnect, cap and stow the electrical connector (DV28266).
 - C. Confirm the manual override handle is in the CLOSED position and lockwire.
 - D. Close access door.

OPERATIONS (O)

NOTE: The Fuel Synoptic will incorrectly indicate jettison flow from the center tank when the center tank override pumps are on with fuel in the center tank. The FUEL JETT NOZZLE L and R advisory messages will be displayed and overhead panel L and R fuel jettison nozzle VALVE lights will be illuminated.

For -200 or -200ER/PW4074 thru PW4084:

5. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 3,400 lb (1,542 kg) (for fuel that would be burned during a return-to-land procedure).

For -200 or -200ER/PW4090:

6. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 3,800 lb (1,724 kg) (for fuel that would be burned during a return-to-land procedure).

For -200 or -200ER/GE90-76B thru -90B:

7. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 3,300 lb (1,497 kg) (for fuel that would be burned during a return-to-land procedure).

For -200 or -200ER/GE90-94B or RR875 thru RR892:

8. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 3,700 lb (1,678 kg) (for fuel that would be burned during a return-to-land procedure).

For -200 or -200ER/RR895:

9. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 4,000 lb (1,814 kg) (for fuel that would be burned during a return-to-land procedure).

For -200LR or 777F/GE90-100 Series:

10. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 4,000 lb (1,814 kg) (for fuel that would be burned during a return-to-land procedure).

For -300/RR884 thru RR892:

11. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 3,900 lb (1,769 kg) (for fuel that would be burned during a return-to-land procedure).

For -300/PW4090:

12. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 4,100 lb (1,860 kg) (for fuel that would be burned during a return-to-land procedure).

For -300ER/GE90-115B:

13. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 4,200 lb (1,905 kg) (for fuel that would be burned during a return-to-land procedure).

28-31-01 Fuel Jettison System
28-31-01-02 With PRR 62777-79S or Equivalent Installed
28-31-01-02-01 ARM Switch FAULT Light

Interval	Installed	Required	Procedure
C	1	0	

28-31-01 Fuel Jettison System
28-31-01-02 With PRR 62777-79S or Equivalent Installed
28-31-01-02-02 ARM Switch ARMED Light

Interval	Installed	Required	Procedure
C	1	0	



777 FAA DDG Supplement Package

Supplement Number: 36-4

Issue Date: 19-DEC-2024

Section Number: 2 (MMEL)

Item Number: 28-31-02

Item Name: Center Tank Jettison Isolation Valves

DDG Effectivity: 777 FAA DDG (D630W003-TBC), Revision 36

MMEL Effectivity: 777 FAA MMEL, Revision 23a

Description: Modified Operations (O) procedures for subitem options 28-31-02-02B and 28-31-02-03B.

Reason: Operations (O) procedures were modified to reflect updated performance adjustments.

Disposition: For reference in the Operator's MEL.

Attachments: DDG pages 2.28-31-02.1-7

28-31-02 Center Tank Jettison Isolation Valves
28-31-02-01 -200

Interval	Installed	Required	Procedure
C	2	0	(M)

May be inoperative provided inoperative valve(s) is secured closed.

MAINTENANCE (M)

Deactivate and secure the inoperative jettison isolation valve in the closed position (AMM 28-00-00/901).

1. Open the landing gear doors and install safety pins (AMM 32-00-15/201).
2. For left jettison isolation valve inoperative:
 - A. Open and collar P110 panel L FUEL JETT ISLN VLV circuit breaker.
 - B. Gain access to the jettison isolation valve actuator located in the left main wheel well (rear spar of the left center tank).
 - C. Disconnect, cap and stow the electrical connector (DV28165).
 - D. Move the manual override handle to the CLOSED position and lockwire.
3. For right jettison isolation valve inoperative:
 - A. Open and collar P210 panel R FUEL JETT ISLN VLV circuit breaker.
 - B. Gain access to the jettison isolation valve actuator located in the right main wheel well (rear spar of the right center tank).
 - C. Disconnect, cap and stow the electrical connector (DV28265).
 - D. Move the manual override handle to the CLOSED position and lockwire.
4. Remove the landing gear door safety pins and close the landing gear doors (AMM 32-00-15/201).

28-31-02 Center Tank Jettison Isolation Valves
28-31-02-02 -200ER/-200LR/-300/-300ER/777F (Without PRR 62777-79S or Equivalent Installed)
28-31-02-02A Fuel Jettison System Deactivated

Interval	Installed	Required	Procedure
C	2	0	(M)

May be inoperative provided:

- a. Inoperative valve(s) is secured closed.
- b. Fuel jettison system is considered inoperative.

NOTE 1: The PRR 62777-79S is applicable to airplane line numbers 1099 and-on. Service Bulletins 777-28-0078 (ELMS2) and 777-28-0082 (ELMS1) each provide the production equivalent of PRR 62777-79S.

NOTE 2: The airplane must also be dispatched using MMEL item 28-31-01.

MAINTENANCE (M)

Secure the inoperative jettison isolation valve in the closed position (AMM 28-00-00/901).

1. Open the landing gear doors and install safety pins (AMM 32-00-15/201).
2. For left jettison isolation valve:
 - A. Open and collar P110 panel L FUEL JETT ISLN VLV circuit breaker.
 - B. Gain access to the jettison isolation valve actuator located in the left main wheel well (rear spar of the left center tank).
 - C. Disconnect, cap and stow the electrical connector (DV28165).
 - D. Move the manual override handle to the CLOSED position and lockwire.
3. For right jettison isolation valve:
 - A. Open and collar P210 panel R FUEL JETT ISLN VLV circuit breaker.
 - B. Gain access to the jettison isolation valve actuator located in the right main wheel well (rear spar of the right center tank).
 - C. Disconnect, cap and stow the electrical connector (DV28265).
 - D. Move the manual override handle to the CLOSED position and lockwire.
4. Remove the landing gear door safety pins and close the landing gear doors (AMM 32-00-15/201).

28-31-02 Center Tank Jettison Isolation Valves

28-31-02-02 -200ER/-200LR/-300/-300ER/777F (Without PRR 62777-79S or Equivalent Installed)

28-31-02-02B Main Tank Jettison Operative

Interval	Installed	Required	Procedure
C	2	0	(M) (O)

May be inoperative provided:

- a. Inoperative valve(s) is secured closed.
- b. Appropriate gross weight and center-of-gravity adjustments are applied.

NOTE: The PRR 62777-79S is applicable to airplane line numbers 1099 and-on. Service Bulletins 777-28-0078 (ELMS2) and 777-28-0082 (ELMS1) each provide the production equivalent of PRR 62777-79S.

MAINTENANCE (M)

Deactivate and secure the inoperative jettison isolation valve in the closed position (AMM 28-00-00/901).

1. Open the landing gear doors and install safety pins (AMM 32-00-15/201).
2. For left jettison isolation valve inoperative:
 - A. Open and collar P110 panel L FUEL JETT ISLN VLV circuit breaker.
 - B. Gain access to the jettison isolation valve actuator located in the left main wheel well (rear spar of the left center tank).
 - C. Disconnect, cap and stow the electrical connector (DV28165).
 - D. Move the manual override handle to the CLOSED position and lockwire.
3. For right jettison isolation valve inoperative:
 - A. Open and collar P210 panel R FUEL JETT ISLN VLV circuit breaker.
 - B. Gain access to the jettison isolation valve actuator located in the right main wheel well (rear spar of the right center tank).
 - C. Disconnect, cap and stow the electrical connector (DV28265).
 - D. Move the manual override handle to the CLOSED position and lockwire.
4. Remove the landing gear door safety pins and close the landing gear doors (AMM 32-00-15/201).

OPERATIONS (O)

For -200 or -200ER/PW4074 thru PW4084:

5. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 3,400 lb (1,542 kg) (for fuel that would be burned during a return-to-land procedure).

For -200 or -200ER/PW4090:

6. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 3,800 lb (1,724 kg) (for fuel that would be burned during a return-to-land procedure).

For -200 or -200ER/GE90-76B thru -90B:

7. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 3,300 lb (1,497 kg) (for fuel that would be burned during a return-to-land procedure).

For -200 or -200ER/GE90-94B or RR875 thru RR892:

8. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 3,700 lb (1,678 kg) (for fuel that would be burned during a return-to-land procedure).

For -200 or -200ER/RR895:

9. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 4,000 lb (1,814 kg) (for fuel that would be burned during a return-to-land procedure).

For -200LR or 777F/GE90-100 Series:

10. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 4,000 lb (1,814 kg) (for fuel that would be burned during a return-to-land procedure).

For -300/RR884 thru RR892:

11. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 3,900 lb (1,769 kg) (for fuel that would be burned during a return-to-land procedure).

For -300/PW4090:

12. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 4,100 lb (1,860 kg) (for fuel that would be burned during a return-to-land procedure).

For -300ER/GE90-115B:

13. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 4,200 lb (1,905 kg) (for fuel that would be burned during a return-to-land procedure).

28-31-02 Center Tank Jettison Isolation Valves
28-31-02-03 -200ER/-200LR/-300/-300ER/777F (With PRR 62777-79S or Equivalent Installed)
28-31-02-03A Fuel Jettison System Deactivated

Interval	Installed	Required	Procedure
C	2	0	(M)

May be inoperative provided:

- a. Inoperative valve(s) is secured closed.
- b. Fuel jettison system is considered inoperative.
- c. Fuel scavenge system is considered inoperative.

NOTE 1: The PRR 62777-79S is applicable to airplane line numbers 1099 and-on. Service Bulletins 777-28-0078 (ELMS2) and 777-28-0082 (ELMS1) each provide the production equivalent of PRR 62777-79S.

NOTE 2: The airplane must also be dispatched using MMEL item 28-31-01.

MAINTENANCE (M)

Secure the inoperative jettison isolation valve in the closed position (AMM 28-00-00/901).

1. Open the landing gear doors and install safety pins (AMM 32-00-15/201).
2. For left jettison isolation valve:
 - A. Open and collar P110 panel L FUEL JETT ISLN VLV circuit breaker.

- B. Gain access to the jettison isolation valve actuator located in the left main wheel well (rear spar of the left center tank).
 - C. Disconnect, cap and stow the electrical connector (DV28165).
 - D. Move the manual override handle to the CLOSED position and lockwire.
3. For right jettison isolation valve:
 - A. Open and collar P210 panel R FUEL JETT ISLN VLV circuit breaker.
 - B. Gain access to the jettison isolation valve actuator located in the right main wheel well (rear spar of the right center tank).
 - C. Disconnect, cap and stow the electrical connector (DV28265).
 - D. Move the manual override handle to the CLOSED position and lockwire.
 4. Remove the landing gear door safety pins and close the landing gear doors (AMM 32-00-15/201).

28-31-02 Center Tank Jettison Isolation Valves
28-31-02-03 -200ER/-200LR/-300/-300ER/777F (With PRR 62777-79S or Equivalent Installed)
28-31-02-03B Main Tank Jettison Operative

Interval	Installed	Required	Procedure
C	2	0	(M) (O)

May be inoperative provided:

- a. Inoperative valve(s) is secured closed.
- b. Appropriate gross weight and center-of-gravity adjustments are applied.

NOTE: The PRR 62777-79S is applicable to airplane line numbers 1099 and-on. Service Bulletins 777-28-0078 (ELMS2) and 777-28-0082 (ELMS1) each provide the production equivalent of PRR 62777-79S.

MAINTENANCE (M)

Deactivate and secure the inoperative jettison isolation valve in the closed position (AMM 28-00-00/901).

1. Open the landing gear doors and install safety pins (AMM 32-00-15/201).
2. For left jettison isolation valve inoperative:
 - A. Open and collar P110 panel L FUEL JETT ISLN VLV circuit breaker.
 - B. Gain access to the jettison isolation valve actuator located in the left main wheel well (rear spar of the left center tank).
 - C. Disconnect, cap and stow the electrical connector (DV28165).
 - D. Move the manual override handle to the CLOSED position and lockwire.

3. For right jettison isolation valve inoperative:
 - A. Open and collar P210 panel R FUEL JETT ISLN VLV circuit breaker.
 - B. Gain access to the jettison isolation valve actuator located in the right main wheel well (rear spar of the right center tank).
 - C. Disconnect, cap and stow the electrical connector (DV28265).
 - D. Move the manual override handle to the CLOSED position and lockwire.
4. Remove the landing gear door safety pins and close the landing gear doors (AMM 32-00-15/201).

OPERATIONS (O)

For -200 or -200ER/PW4074 thru PW4084:

5. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 3,400 lb (1,542 kg) (for fuel that would be burned during a return-to-land procedure).

For -200 or -200ER/PW4090:

6. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 3,800 lb (1,724 kg) (for fuel that would be burned during a return-to-land procedure).

For -200 or -200ER/GE90-76B thru -90B:

7. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 3,300 lb (1,497 kg) (for fuel that would be burned during a return-to-land procedure).

For -200 or -200ER/GE90-94B or RR875 thru RR892:

8. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 3,700 lb (1,678 kg) (for fuel that would be burned during a return-to-land procedure).

For -200 or -200ER/RR895:

9. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 4,000 lb (1,814 kg) (for fuel that would be burned during a return-to-land procedure).

For -200LR or 777F/GE90-100 Series:

10. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 4,000 lb (1,814 kg) (for fuel that would be burned during a return-to-land procedure).

For -300/RR884 thru RR892:

11. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 3,900 lb (1,769 kg) (for fuel that would be burned during a return-to-land procedure).

For -300/PW4090:

12. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 4,100 lb (1,860 kg) (for fuel that would be burned during a return-to-land procedure).

For -300ER/GE90-115B:

13. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 4,200 lb (1,905 kg) (for fuel that would be burned during a return-to-land procedure).

Supplement 36-4
Superseded



777 FAA DDG Supplement Package

Supplement Number: 36-4

Issue Date: 19-DEC-2024

Section Number: 2 (MMEL)

Item Number: 28-31-03

Item Name: Main Tanks Jettison Pumps

DDG Effectivity: 777 FAA DDG (D630W003-TBC), Revision 36

MMEL Effectivity: 777 FAA MMEL, Revision 23a

Description: Modified Operations (O) procedures for subitems 28-31-03-01 and 28-31-03-02.

Reason: Operations (O) procedures were modified to reflect updated performance adjustments.

Disposition: For reference in the Operator's MEL.

Attachments: DDG pages 2.28-31-03.1-4

28-31-03 Main Tanks Jettison Pumps

28-31-03-01 Without PRR 62777-79S or Equivalent Installed

Interval	Installed	Required	Procedure
C	2	0	(M) (O)

May be inoperative provided:

- Inoperative pump is deactivated.
- Appropriate performance adjustments are applied.

NOTE: The PRR 62777-79S is applicable to airplane line numbers 1099 and-on. Service Bulletins 777-28-0078 (ELMS2) and 777-28-0082 (ELMS1) each provide the production equivalent of PRR 62777-79S.

MAINTENANCE (M)

Deactivate the inoperative pump(s) (AMM 28-00-00/901).

- For left main tank jettison pump inoperative, open and collar P110 panel L FUEL JETT PUMP CTRL circuit breaker.
- For right main tank jettison pump inoperative, open and collar P210 panel R FUEL JETT PUMP CTRL circuit breaker.

OPERATIONS (O)

NOTE: For one main tank jettison pump inoperative, fuel jettison of the opposite jettison pump is inhibited to prevent fuel imbalance.

[For -200 or -200ER/PW4074 thru PW4084:](#)

- Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 3,400 lb (1,542 kg) (for fuel that would be burned during a return-to-land procedure).

[For -200 or -200ER/PW4090:](#)

- Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 3,800 lb (1,724 kg) (for fuel that would be burned during a return-to-land procedure).

[For -200 or -200ER/GE90-76B thru -90B:](#)

- Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 3,300 lb (1,497 kg) (for fuel that would be burned during a return-to-land procedure).

[For -200 or -200ER/GE90-94B or RR875 thru RR892:](#)

- Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 3,700 lb (1,678 kg) (for fuel that would be burned during a return-to-land procedure).

For -200 or -200ER/RR895:

7. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 4,000 lb (1,814 kg) (for fuel that would be burned during a return-to-land procedure).

For -200LR or 777F/GE90-100 Series:

8. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 4,000 lb (1,814 kg) (for fuel that would be burned during a return-to-land procedure).

For -300/RR884 thru RR892:

9. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 3,900 lb (1,769 kg) (for fuel that would be burned during a return-to-land procedure).

For -300/PW4090:

10. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 4,100 lb (1,860 kg) (for fuel that would be burned during a return-to-land procedure).

For -300ER/GE90-115B:

11. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 4,200 lb (1,905 kg) (for fuel that would be burned during a return-to-land procedure).

28-31-03 Main Tanks Jettison Pumps
28-31-03-02 With PRR 62777-79S or Equivalent Installed

Interval	Installed	Required	Procedure
C	2	0	(M) (O)

May be inoperative provided:

- a. Inoperative pump is deactivated.
- b. Fuel scavenge system is considered inoperative.
- c. Appropriate performance adjustments are applied.

NOTE 1: The PRR 62777-79S is applicable to airplane line numbers 1099 and-on. Service Bulletins 777-28-0078 (ELMS2) and 777-28-0082 (ELMS1) each provide the production equivalent of PRR 62777-79S.

NOTE 2: The airplane must also be dispatched using MMEL item 28-22-04.

MAINTENANCE (M)

Deactivate the inoperative pump(s) (AMM 28-00-00/901).

1. For left main tank jettison pump inoperative, open and collar P110 panel L FUEL JETT PUMP CTRL circuit breaker.
2. For right main tank jettison pump inoperative, open and collar P210 panel R FUEL JETT PUMP CTRL circuit breaker.

OPERATIONS (O)

NOTE: For one main tank jettison pump inoperative:

1. Fuel jettison of the opposite jettison pump is inhibited to prevent fuel imbalance.
2. Fuel scavenge of the opposite jettison pump operates normally.
3. Fuel scavenge rate is approximately half the normal fuel scavenge rate.

For -200 or -200ER/PW4074 thru PW4084:

4. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 3,400 lb (1,542 kg) (for fuel that would be burned during a return-to-land procedure).

For -200 or -200ER/PW4090:

5. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 3,800 lb (1,724 kg) (for fuel that would be burned during a return-to-land procedure).

For -200 or -200ER/GE90-76B thru -90B:

6. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 3,300 lb (1,497 kg) (for fuel that would be burned during a return-to-land procedure).

For -200 or -200ER/GE90-94B or RR875 thru RR892:

7. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 3,700 lb (1,678 kg) (for fuel that would be burned during a return-to-land procedure).

For -200 or -200ER/RR895:

8. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 4,000 lb (1,814 kg) (for fuel that would be burned during a return-to-land procedure).

For -200LR or 777F/GE90-100 Series:

9. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 4,000 lb (1,814 kg) (for fuel that would be burned during a return-to-land procedure).

For -300/RR884 thru RR892:

10. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 3,900 lb (1,769 kg) (for fuel that would be burned during a return-to-land procedure).

For -300/PW4090:

11. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 4,100 lb (1,860 kg) (for fuel that would be burned during a return-to-land procedure).

For -300ER/GE90-115B:

12. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 4,200 lb (1,905 kg) (for fuel that would be burned during a return-to-land procedure).

Supplement 36-4
Superseded



777 FAA DDG Supplement Package

Supplement Number: 36-4

Issue Date: 19-DEC-2024

Section Number: 2 (MMEL)

Item Number: 28-31-04

Item Name: Fuel Jettison Nozzle Valves

DDG Effectivity: 777 FAA DDG (D630W003-TBC), Revision 36

MMEL Effectivity: 777 FAA MMEL, Revision 23a

Description: **Note: This item supersedes content in a previous supplement.**

- 1) Modified the AMM references in the subitem options 28-31-04-01B and 28-31-04-02B Maintenance (M) procedures.
- 2) Modified Operations (O) procedures for subitem options 28-31-04-01B and 28-31-04-02B.

Reason:

- 1) The AMM references were incorrectly modified in the previous DDG revision.
- 2) Operations (O) procedures were modified to reflect updated performance adjustments.

Disposition: For reference in the Operator's MEL.

Attachments: DDG pages 2.28-31-04.1-7

28-31-04 Fuel Jettison Nozzle Valves
28-31-04-01 Without PRR 62777-79S or Equivalent Installed
28-31-04-01A One Inoperative

Interval	Installed	Required	Procedure
C	2	1	(M)

One nozzle valve may be inoperative provided it is secured closed.

NOTE: The PRR 62777-79S is applicable to airplane line numbers 1099 and-on. Service Bulletins 777-28-0078 (ELMS2) and 777-28-0082 (ELMS1) each provide the production equivalent of PRR 62777-79S.

MAINTENANCE (M)

Secure the inoperative fuel jettison nozzle valve in the closed position (AMM 28-00-00/901).

1. For left jettison nozzle valve inoperative:
 - A. Open and collar P110 panel L FUEL JETT NOZ VLV circuit breaker.
 - B. Gain access to the left jettison nozzle valve actuator.
 - C. Disconnect, cap and stow the electrical connector (DV28166).
 - D. Confirm the manual override handle is in the CLOSED position and lockwire.
2. For right jettison nozzle valve inoperative:
 - A. Open and collar P210 panel R FUEL JETT NOZ VLV circuit breaker.
 - B. Gain access to the right jettison nozzle valve actuator.
 - C. Disconnect, cap and stow the electrical connector (DV28266).
 - D. Confirm the manual override handle is in the CLOSED position and lockwire.

28-31-04 Fuel Jettison Nozzle Valves
28-31-04-01 Without PRR 62777-79S or Equivalent Installed
28-31-04-01B Both Inoperative

Interval	Installed	Required	Procedure
C	2	0	(M) (O)

May be inoperative provided:

- a. Both jettison nozzle valves are secured closed.
- b. Jettison system is deactivated.
- c. Appropriate performance penalties are applied.

NOTE: The PRR 62777-79S is applicable to airplane line numbers 1099 and-on. Service Bulletins 777-28-0078 (ELMS2) and 777-28-0082 (ELMS1) each provide the production equivalent of PRR 62777-79S.

MAINTENANCE (M)

Secure the fuel jettison nozzle valves in the closed position and deactivate the jettison system (AMM 28-00-00/901).

1. For left jettison nozzle valve:
 - A. Open and collar P110 panel L FUEL JETT NOZ VLV circuit breaker.
 - B. Gain access to the left jettison nozzle valve actuator.
 - C. Disconnect, cap and stow the electrical connector (DV28166).
 - D. Confirm the manual override handle is in the CLOSED position and lockwire.
2. For right jettison nozzle valve:
 - A. Open and collar P210 panel R FUEL JETT NOZ VLV circuit breaker.
 - B. Gain access to the right jettison nozzle valve actuator.
 - C. Disconnect, cap and stow the electrical connector (DV28266).
 - D. Confirm the manual override handle is in the CLOSED position and lockwire.
3. Deactivate the fuel jettison system.
 - A. Open and collar P110 panel L FUEL JETT PUMP CTRL and L FUEL JETT ISLN VLV circuit breakers.
 - B. Open and collar P210 panel R FUEL JETT PUMP CTRL and R FUEL JETT ISLN VLV circuit breakers.

OPERATIONS (O)

NOTE: The Fuel Synoptic will incorrectly indicate jettison flow from the center tank when the center tank override pumps are on with fuel in the center tank. The FUEL JETT NOZZLE L and R advisory messages will be displayed and overhead panel L and R fuel jettison nozzle VALVE lights will be illuminated.

For -200 or -200ER/PW4074 thru PW4084:

4. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 3,400 lb (1,542 kg) (for fuel that would be burned during a return-to-land procedure).

For -200 or -200ER/PW4090:

5. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 3,800 lb (1,724 kg) (for fuel that would be burned during a return-to-land procedure).

For -200 or -200ER/GE90-76B thru -90B:

6. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 3,300 lb (1,497 kg) (for fuel that would be burned during a return-to-land procedure).

For -200 or -200ER/GE90-94B or RR875 thru RR892:

7. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 3,700 lb (1,678 kg) (for fuel that would be burned during a return-to-land procedure).

For -200 or -200ER/RR895:

8. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 4,000 lb (1,814 kg) (for fuel that would be burned during a return-to-land procedure).

For -200LR or 777F/GE90-100 Series:

9. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 4,000 lb (1,814 kg) (for fuel that would be burned during a return-to-land procedure).

For -300/RR884 thru RR892:

10. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 3,900 lb (1,769 kg) (for fuel that would be burned during a return-to-land procedure).

For -300/PW4090:

11. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 4,100 lb (1,860 kg) (for fuel that would be burned during a return-to-land procedure).

For -300ER/GE90-115B:

12. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 4,200 lb (1,905 kg) (for fuel that would be burned during a return-to-land procedure).

28-31-04 Fuel Jettison Nozzle Valves
28-31-04-01 Without PRR 62777-79S or Equivalent Installed
28-31-04-01-01 Switch VALVE Lights

Interval	Installed	Required	Procedure
C	2	0	

28-31-04 Fuel Jettison Nozzle Valves
28-31-04-01 Without PRR 62777-79S or Equivalent Installed
28-31-04-01-02 Switch ON Lights

Interval	Installed	Required	Procedure
C	2	0	

28-31-04 Fuel Jettison Nozzle Valves**28-31-04-02 With PRR 62777-79S or Equivalent Installed****28-31-04-02A One Inoperative**

Interval	Installed	Required	Procedure
C	2	1	(M)

One nozzle valve may be inoperative provided it is secured closed.

NOTE: The PRR 62777-79S is applicable to airplane line numbers 1099 and-on. Service Bulletins 777-28-0078 (ELMS2) and 777-28-0082 (ELMS1) each provide the production equivalent of PRR 62777-79S.

MAINTENANCE (M)

Secure the inoperative fuel jettison nozzle valve in the closed position (AMM 28-00-00/901).

1. For left jettison nozzle valve inoperative:
 - A. Open and collar P110 panel L FUEL JETT NOZ VLV circuit breaker.
 - B. Gain access to the left jettison nozzle valve actuator.
 - C. Disconnect, cap and stow the electrical connector (DV28166).
 - D. Confirm the manual override handle is in the CLOSED position and lockwire.
2. For right jettison nozzle valve inoperative:
 - A. Open and collar P210 panel R FUEL JETT NOZ VLV circuit breaker.
 - B. Gain access to the right jettison nozzle valve actuator.
 - C. Disconnect, cap and stow the electrical connector (DV28266).
 - D. Confirm the manual override handle is in the CLOSED position and lockwire.

28-31-04 Fuel Jettison Nozzle Valves**28-31-04-02 With PRR 62777-79S or Equivalent Installed****28-31-04-02B Both Inoperative**

Interval	Installed	Required	Procedure
C	2	0	(M) (O)

May be inoperative provided:

- a. Both jettison nozzle valves are secured closed.
- b. Jettison system is deactivated.
- c. Fuel scavenge system is considered inoperative.

- d. Appropriate performance penalties are applied.
-

NOTE 1: The PRR 62777-79S is applicable to airplane line numbers 1099 and-on. Service Bulletins 777-28-0078 (ELMS2) and 777-28-0082 (ELMS1) each provide the production equivalent of PRR 62777-79S.

NOTE 2: The airplane must also be dispatched using MMEL item 28-22-04.

MAINTENANCE (M)

Secure the fuel jettison nozzle valves in the closed position and deactivate the jettison system (AMM 28-00-00/901).

1. For left jettison nozzle valve:
 - A. Open and collar P110 panel L FUEL JETT NOZ VLV circuit breaker.
 - B. Gain access to the left jettison nozzle valve actuator.
 - C. Disconnect, cap and stow the electrical connector (DV28166).
 - D. Confirm the manual override handle is in the CLOSED position and lockwire.
2. For right jettison nozzle valve:
 - A. Open and collar P210 panel R FUEL JETT NOZ VLV circuit breaker.
 - B. Gain access to the right jettison nozzle valve actuator.
 - C. Disconnect, cap and stow the electrical connector (DV28266).
 - D. Confirm the manual override handle is in the CLOSED position and lockwire.
3. Deactivate the fuel jettison system.
 - A. Open and collar P110 panel L FUEL JETT PUMP CTRL and L FUEL JETT ISLN VLV circuit breakers.
 - B. Open and collar P210 panel R FUEL JETT PUMP CTRL and R FUEL JETT ISLN VLV circuit breakers.

OPERATIONS (O)

NOTE: The Fuel Synoptic will incorrectly indicate jettison flow from the center tank when the center tank override pumps are on with fuel in the center tank. The FUEL JETT NOZZLE L and R advisory messages will be displayed and overhead panel L and R fuel jettison nozzle VALVE lights will be illuminated.

For -200 or -200ER/PW4074 thru PW4084:

4. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 3,400 lb (1,542 kg) (for fuel that would be burned during a return-to-land procedure).

For -200 or -200ER/PW4090:

5. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 3,800 lb (1,724 kg) (for fuel that would be burned during a return-to-land procedure).

For -200 or -200ER/GE90-76B thru -90B:

6. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 3,300 lb (1,497 kg) (for fuel that would be burned during a return-to-land procedure).

For -200 or -200ER/GE90-94B or RR875 thru RR892:

7. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 3,700 lb (1,678 kg) (for fuel that would be burned during a return-to-land procedure).

For -200 or -200ER/RR895:

8. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 4,000 lb (1,814 kg) (for fuel that would be burned during a return-to-land procedure).

For -200LR or 777F/GE90-100 Series:

9. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 4,000 lb (1,814 kg) (for fuel that would be burned during a return-to-land procedure).

For -300/RR884 thru RR892:

10. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 3,900 lb (1,769 kg) (for fuel that would be burned during a return-to-land procedure).

For -300/PW4090:

11. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 4,100 lb (1,860 kg) (for fuel that would be burned during a return-to-land procedure).

For -300ER/GE90-115B:

12. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 4,200 lb (1,905 kg) (for fuel that would be burned during a return-to-land procedure).

28-31-04 Fuel Jettison Nozzle Valves
28-31-04-02 With PRR 62777-79S or Equivalent Installed
28-31-04-02-01 Switch VALVE Lights

Interval	Installed	Required	Procedure
C	2	0	



28-31-04 Fuel Jettison Nozzle Valves
28-31-04-02 With PRR 62777-79S or Equivalent Installed
28-31-04-02-02 Switch ON Lights

Interval	Installed	Required	Procedure
C	2	0	

Supplement 36-4
Superseded



777 FAA DDG Supplement Package

Supplement Number: 36-4

Issue Date: 19-DEC-2024

Section Number: 2 (MMEL)

Item Number: 31-61-07

Item Name: Electronic Checklist (ECL) System

DDG Effectivity: 777 FAA DDG (D630W003-TBC), Revision 36

MMEL Effectivity: 777 FAA MMEL, Revision 23a

Description: **Note: This item supersedes content in a previous supplement.**

- 1) Added a Maintenance (M) procedure note.
- 2) Added an Operations (O) procedure note.

Reason:

- 1) Maintenance (M) procedure note was added to clarify no maintenance action is required for display of CHECKLIST DISABLED or CHECKLIST NOT AVAILABLE messages on the MFD.
- 2) Operations (O) procedure note was added to clarify the CHKL NON-NORMAL and CHKL INCOMPLETE NORM messages will not annunciate due to ECL system deactivated.

Disposition: For reference in the Operator's MEL.

Attachments: DDG page 2.31-61-07.1

31-61-07 Electronic Checklist (ECL) System

Interval	Installed	Required	Procedure
C	1	0	(M) (O)

May be inoperative provided electronic checklist is deactivated and established suitable checklist procedures are used.

MAINTENANCE (M)

Deactivate the electronic checklist (AMM 31-00-00/901).

1. Display the Maintenance Task Maintenance page.
2. Toggle the Checklist Function ENABLE/DISABLE button to DISABLE.

NOTE: No maintenance action is required if the CHECKLIST DISABLED or CHECKLIST NOT AVAILABLE message is displayed on an MFD.

OPERATIONS (O)

Use established suitable checklists.

NOTE 1: The rectangular icon symbol that precedes some alert messages will not be displayed.

NOTE 2: When the ECL is disabled, the CHKL NON-NORMAL and CHKL INCOMPLETE NORM messages will not annunciate.

31-61-07 Electronic Checklist (ECL) System
31-61-07-01 ECL Closed Loop Switch Indications

Interval	Installed	Required	Procedure
C	-	0	(O)

May be inoperative provided ECL line item override procedures are used when required to complete checklists.

OPERATIONS (O)

Use the Line Item Override (ITEM OVRD) Key as required.



777 FAA DDG Supplement Package

Supplement Number: 36-4

Issue Date: 19-DEC-2024

Section Number: 2 (MMEL)

Item Number: 32-45-01

Item Name: Wheel Brakes

DDG Effectivity: 777 FAA DDG (D630W003-TBC), Revision 36

MMEL Effectivity: 777 FAA MMEL, Revision 23a

Description: Modified Maintenance (M) procedure steps for item option 32-45-01C.

Reason: Maintenance (M) procedure steps were modified to be consistent with the AMM.

Disposition: For reference in the Operator's MEL.

Attachments: DDG pages 2.32-45-01.1-4

32-45-01 Wheel Brakes
32-45-01A Deactivation Tool Used

Interval	Installed	Required	Procedure
C	12	10	(M) (O)

One per each six wheel truck may be inoperative provided:

- a. Associated brake is deactivated with a deactivation assembly.
- b. AFM performance decrements for brakes deactivated are applied.

MAINTENANCE (M)

Deactivate associated brake using the Deactivation Assembly (AMM 32-00-00/901).

1. Supply electrical power (AMM 24-22-00/201).
2. Install landing gear downlock pins (AMM 32-00-30/201).
3. Remove power on the right and center hydraulic systems (AMM 29-11-00/201).
4. Install chocks around the tires.
5. Release the parking brake.
6. Fully push the brake pedals 12 times to remove the pressure from the brake accumulator.
7. Use the MAT to disable the ANTISKID advisory message for the associated brake.

NOTE: The ANTISKID advisory message may reappear if airplane power is cycled or if power to AIMS is cycled. If this happens, repeat the deactivation procedure on the MAT.

8. Remove the brake deactivation assembly from its location on the Antiskid Shuttle Valve Module. The assembly is referred to as the Flight Dispatch Disconnect (specification number S294W121-51) vendor part number 80389.
9. Remove the cover from the applicable shuttle valve and install the deactivation assembly in place of the valve cover.
10. Pressurize the right hydraulic system (AMM 29-11-00/201).
11. Fully press the brake pedals. Make sure the applicable brake does not operate by observing brake piston movement.
12. Remove power from the right hydraulic system (AMM 29-11-00/201).
13. Remove landing gear downlock pins when maintenance and servicing actions are complete (AMM 32-00-30/201).

OPERATIONS (O)

NOTE: Operations may be affected at contiguous United States airports impacted by 5G C-Band interference [Ref FAA Airworthiness Directive (AD) 2023-12-05 and associated Boeing Flight Crew Operations Manual Bulletin (OMB)]

Dispatch with takeoff and landing performance based on one or two brakes deactivated.

32-45-01 Wheel Brakes

32-45-01B Brake Line Capped

Interval	Installed	Required	Procedure
C	12	10	(M) (O)

One per each six wheel truck may be inoperative provided:

- Associated brake is deactivated by capping the brake line.
- After takeoff, gear remains down for two minutes before retraction.
- AFM performance decrements for brakes deactivated are applied.
- Takeoff performance is based on landing gear extended.

MAINTENANCE (M)

Deactivate inoperative brake by capping the associated brake line (AMM 32-00-00/901).

- Supply electrical power (AMM 24-22-00/201).
- Install landing gear downlock pins (AMM 32-00-30/201).
- Remove power on the right and center hydraulic systems (AMM 29-11-00/201).
- Install chocks around the tires.
- Release the parking brake.
- Fully push the brake pedals 12 times to remove the pressure from the brake accumulator.
- Use the MAT to disable the ANTISKID advisory message for the associated brake.

NOTE: The ANTISKID advisory message may reappear if airplane power is cycled or if power to AIMS is cycled. If this happens, repeat the deactivation procedure on the MAT.

- Uncouple the Brake Disconnect - Brake Hose Half and attach the hose to the landing gear truck using P/N BACS38K3D.

NOTE 1: The part of the Brake Disconnect that is attached to the hose is called the Brake Disconnect - Brake Hose Half. The part of the Brake Disconnect that is attached to the brake is called the Brake Disconnect - Coupling.

NOTE 2: The brake lines are self sealing, thus disconnecting the brake line and stowing represents capping.

9. Pressurize the right hydraulic system (AMM 29-11-00/201).
10. Apply brake pressure and check for leaks.
11. Make sure the applicable brake does not operate by observing brake piston movement.
12. Remove power from the right hydraulic system (AMM 29-11-00/201).
13. Remove landing gear downlock pins when maintenance and servicing actions are complete (AMM 32-00-30/201).

OPERATIONS (O)

NOTE: Operations may be affected at contiguous United States airports impacted by 5G C-Band interference [Ref FAA Airworthiness Directive (AD) 2023-12-05 and associated Boeing Flight Crew Operations Manual Bulletin (OMB)]

1. After takeoff, leave the landing gear extended for a minimum of two minutes to allow the wheels to spin down prior to gear retraction.
2. Base takeoff performance on landing gear extended. Takeoff performance for the landing gear extended is available using AFM-DPI or Boeing Takeoff Module (BTM).
3. Dispatch with takeoff and landing performance also based on one or two brakes deactivated.
4. Increase flight planning fuel by 1,900 lb (861 kg).

32-45-01

Wheel Brakes

32-45-01C

Center Axle Brake Removed

Interval	Installed	Required	Procedure
C	12	10	(M) (O)

One per each six wheel truck may be removed from the center axle only provided:

- a. Associated brake line is capped.
- b. After takeoff, gear remains down for two minutes before retraction.
- c. AFM performance decrements for brakes deactivated are applied.
- d. Takeoff performance is based on landing gear extended.

MAINTENANCE (M)

Remove inoperative brake and cap the associated brake line (AMM 32-00-00/901).

1. Supply electrical power (AMM 24-22-00/201).
2. Install landing gear downlock pins (AMM 32-00-30/201).
3. Use the MAT to disable the ANTISKID advisory message for the associated brake.

NOTE: The ANTISKID advisory message may reappear if airplane power is cycled or if power to AIMS is cycled. If this happens, repeat the deactivation procedure on the MAT.

4. Remove the applicable brake (AMM 32-45-07/401).
5. Attach the brake hose to the gear truck. This will ensure that the hose will not interfere with brake rotation when the gear is retracted.
6. Cap and stow the electrical wiring for the brake temperature sensor.

NOTE: Ensure the brake sleeve is installed on the axle before the tire and wheel assembly are installed.

7. Install the tire and wheel assembly (AMM 32-45-01/401).
8. Pressurize the right hydraulic system (AMM 29-11-00/201).
9. Apply brake pressure and check for leaks.
10. Remove power from the right hydraulic system (AMM 29-11-00/201).
11. Remove landing gear downlock pins when maintenance and servicing actions are complete (AMM 32-00-30/201).

OPERATIONS (O)

NOTE: Operations may be affected at contiguous United States airports impacted by 5G C-Band interference [Ref FAA Airworthiness Directive (AD) 2023-12-05 and associated Boeing Flight Crew Operations Manual Bulletin (OMB)]

1. After takeoff, leave the landing gear extended for a minimum of two minutes to allow the wheels to spin down prior to gear retraction.
2. Base takeoff performance on landing gear extended. Takeoff performance for the landing gear extended is available using AFM-DPI or Boeing Takeoff Module (BTM).
3. Dispatch with takeoff and landing performance also based on one or two brakes deactivated.
4. Increase flight planning fuel by 1,900 lb (861 kg).



777 FAA DDG Supplement Package

Supplement Number: 36-4

Issue Date: 19-DEC-2024

Section Number: 2 (MMEL)

Item Number: 32-53-01

Item Name: Main Gear Steering System (Including Indication)

DDG Effectivity: 777 FAA DDG (D630W003-TBC), Revision 36

MMEL Effectivity: 777 FAA MMEL, Revision 23a

Description: Modified DDG top note and Maintenance (M) procedure.

Reason:

- 1) DDG top note was modified to clarify PRR 62606 applicability and installation.
- 2) Maintenance (M) procedure annotation and steps were modified to correct discrepancies for airplanes with PRR 62606 or equivalent installed.

Disposition: For reference in the Operator's MEL.

Attachments: DDG pages 2.32-53-01.1-5

32-53-01 Main Gear Steering System (Including Indication)

Interval	Installed	Required	Procedure
C	1	0	(M)

May be inoperative provided:

- a. Main gear steering actuators are verified locked in the center position.
- b. Main gear steering system is deactivated.

NOTE 1: PRR 62606 is applicable to 777F airplane line number 1768 and-on.

NOTE 2: PRR 62606 installation can be identified by the presence of a scissor/torsion link on the clevis end of the Power Control Actuator (PCA).

MAINTENANCE (M)

Verify that both main gear steering actuators are locked in the center position and deactivate the main gear steering system (AMM 32-00-00/901).

For -200LR/-300ER/777F with PRR 62606 or equivalent installed:

1. Deactivate the Main Gear Steering System.
 - A. Open and collar P110 panel LG MAIN GEAR STEERING circuit breaker to remove power from the Main Gear Steering Control Unit (MGSCU).
 - B. Depressurize the center hydraulic system.
2. Verify left and right main gear steering Power Control Actuators (PCA) are hydraulically locked in the centered position.
 - A. For each main landing gear ensure the PCA Manual Shut-Off Valve handle is in the STEER position.
 - 1) If the Manual Shut-Off Valve handle is not locked in the STEER position:
 - a. Withdraw the lock pin on the Manual Shut-Off Valve handle.
 - b. Rotate the Manual Shut-Off Valve handle to the STEER position.
 - c. Release the lock pin to hold the Manual Shut-Off Valve handle in the STEER position.
 - 2) Pressurize the center hydraulic system.

NOTE: The main gear steering actuators are biased to the center position with the Shut-Off Valve handle in the STEER position.

- B. For each main landing gear verify the aft axle is in the centered position.
 - 1) Observe the rod end of the PCA. Visually confirm indication fitting at the rod end is abutting/depressing the switch plunger.
 - 2) Observe the clevis end of the PCA. Visually confirm that the indicator tab on the Aft Torque link assembly is abutting/depressing the switch plunger.

For airplanes with PRR 62606 or equivalent not installed:

- 3. Verify left and right main gear steering Power Control Actuators (PCA) are locked in the centered position.
 - A. Look through the lock indication window on each PCA and if you can not see the metal pin on the PCA, the PCA is locked.
 - B. If you can see the metal pin on either PCA, it is not locked. Use the Push Back method (preferred) or the Check Valve Bleeding method to move the unlocked PCA into the locked position.
 - 1) Push Back Method:
 - a. Connect a tow vehicle to the nose landing gear or forward tow fitting on the affected main landing gear (AMM 09-11-00/201).
 - b. Move the towing lever, on the forward side of the nose landing gear, to the TOW position.
 - c. Release the parking brake.
 - d. Move the manual shutoff valve handle on the unlocked PCA to the OFF position to isolate the PCA.
 - e. Remove all chocks from around the nose landing gear tires and the main landing gear tires.
 - f. Put a chock behind the aft axle tire that is nearest the unlocked PCA (wheel #10 on the left gear, wheel #12 on the right gear).
 - NOTE: Do not put a chock behind the other aft axle tire (wheel #9 on the left gear, wheel #11 on the right gear).
 - g. Use the tow vehicle to slowly push the airplane rearwards onto the chock. Stop airplane movement if the airplane tire moves onto the chock, the PCA locks, or the chock moves.
 - h. Look through the lock indication window on the PCA and if you can not see the metal pin on a PCA, the PCA is locked.
 - i. If a PCA remains unlocked, place a chock behind the aft axle tire that is farthest from the unlocked PCA, but on the same truck (wheel #9 on the left gear, wheel #11 on the right gear).

NOTE: Do not place a chock behind the other aft axle tire (wheel #10 on the left gear, wheel #12 on the right gear).

- j. Use the tow vehicle to slowly push the airplane rearwards onto the chock. Stop airplane movement if the airplane tire moves onto the chock, the PCA locks, or the chock moves.
- k. Look through the lock indication window on the PCA and if you can not see the metal pin on the PCA, the PCA is locked.

2) Check Valve Bleeding Method:

NOTE: This method loosens the check valve on the unlocked PCA. If you do this, you may void the warranty with the supplier Parker/Abex/NWL. It is recommended that this method be used only after you have tried the Push Back Method. Use caution. The hydraulic fluid trapped in the PCA is under pressure, even when the center hydraulic system is depressurized.

- a. Make sure the parking brake is set.
- b. Lift the aft axle of the main landing gear with the unlocked PCA until the aft tires do not touch the ground (AMM 07-11-03/201).

NOTE: Make sure you use the jack pad that is part of the pivot pin for the aft axle.

- c. Depressurize the center hydraulic system including the reservoir (AMM 29-11-00/201).
- d. Move the manual shutoff valve handle on the unlocked PCA to the OFF position to isolate the PCA.
- e. Use cleaner, TEC 86-2 (or equivalent) to clean the area around the check valve (anti-cavitation check valve) on the unlocked PCA.
- f. Remove the sealant and the lockwire from the check valve on the unlocked PCA.
- g. Put a rag around the unlocked PCA valve to catch the leak of fluid from the valve.
- h. Slowly turn the check valve 1 to 2 turns counter-clockwise until hydraulic fluid starts to leak from under the head of the check valve.

NOTE: Do not remove the check valve from the PCA.

- i. Wait until the hydraulic fluid does not leak from under the check valve.

- j. Push on the right aft tire (wheel #10 on the left gear, wheel #12 on the right gear) in the forward direction to apply a compressive force on the unlocked PCA. You can use a lever between tires #5 and #9 on the left gear and #7 and #11 on the right gear to help apply the force on the PCA.

NOTE: It may be necessary to apply the force for 30 to 60 seconds. Hydraulic fluid may leak from under the head of the check valve when you apply the force.

- k. Look through the lock indication window on the PCA and if you can not see the metal pin on the PCA, the PCA is locked.
- l. If the PCA is not locked, wait approximately 5 minutes and then reapply the force.
- m. Look through the lock indication window on the PCA and if you can not see the metal pin on the PCA, the PCA is locked.
- n. Tighten the check valve to 175-200 pound-inches.
- o. Move the manual shutoff valve handle on the PCA to the STEER position to activate the PCA.
- p. Pressurize the center hydraulic system (AMM 29-11-00/201).
- q. Ensure there is no hydraulic leakage from under the head of the check valve. If the check valve leaks, replace the seals on the check valve.
- r. Lower the aft axle and remove the jack.
- s. Install lockwire on the check valve.
- t. Apply sealant, MIL-A-46146 around the interface between the check valve and the PCA manifold and allow to cure.

NOTE: Application of sealant may be temporarily deferred.

4. Deactivate the main gear steering system.

A. Shutoff hydraulic power to both PCAs.

- 1) Pull the ball-lock pin, on the handle for the manual shutoff valve, aft.
- 2) Move the handle for the manual shutoff valve to the OFF position.
- 3) Release the ball-lock pin to hold the handle in the OFF position.

B. Open and collar P110 panel LG MAIN GEAR STEERING circuit breaker.

NOTE: Removing power from the Main Gear Steering Control Unit (MGSCU) will replace correlated maintenance messages with correlation to maintenance message 32-68840.

OPERATIONS NOTE

1. There is negligible change in U-turn widths if the Flight Crew Training Manual Pivot Turn Technique is used.
2. Minimum radius turns will result in the following effects:
 - A. Substantially increased tire scuffing.
 - B. Increased thrust required for the turn.
 - C. Somewhat rougher cabin ride, particularly in the area above the main gear.
 - D. For -200/-200ER/-300 airplane line numbers 1 through 596 without Service Bulletin SB 777-32-0058, increased risk of damage to Main Landing Gear Hydraulic Line Forward Support Link Assembly.
 - E. For additional information on SB 777-32-0058, refer to 777-FTD-32-01019.



777 FAA DDG Supplement Package

Supplement Number: 36-4

Issue Date: 19-DEC-2024

Section Number: 2 (MMEL)

Item Number: 34-43-01

Item Name: Weather Radar System

DDG Effectivity: 777 FAA DDG (D630W003-TBC), Revision 36

MMEL Effectivity: 777 FAA MMEL, Revision 23a

Description: **Note: This item supersedes content in a previous supplement.**

Modified DDG top note from previous supplement for subitem 34-43-01-02.

Reason: DDG top note was modified to correct the nomenclature for the Collins weather radar system model.

Disposition: For reference in the Operator's MEL.

Attachments: DDG pages 2.34-43-01.1-3

34-43-01 Weather Radar System
34-43-01A Reactive Windshear Operative

Interval	Installed	Required	Procedure
C	-	0	(O)

May be inoperative provided:

- Weather radar is not required by 14 CFR.
- Reactive windshear alert (GPWS Mode 7) operates normally.
- Alternate procedures are established and used.

MAINTENANCE NOTE

The WINDSHEAR PRED status message is not displayed if the WXR SYS status message is displayed.

OPERATIONS (O)

Alternate procedures must be established and used.

NOTE 1: The WINDSHEAR SYS advisory message will be displayed.

NOTE 2: See 14 CFR 121.357 or other appropriate regulation.

34-43-01 Weather Radar System
34-43-01B Reactive Windshear Inoperative

Interval	Installed	Required	Procedure
B	-	0	(O)

May be inoperative provided:

- Weather radar is not required by 14 CFR.
- Alternate procedures are established and used.

NOTE: Operator's alternate procedures should include reviewing windshear avoidance and windshear recovery procedures.

MAINTENANCE NOTE

The WINDSHEAR PRED status message is not displayed if the WXR SYS status message is displayed.

OPERATIONS (O)

Alternate procedures must be established and used.

NOTE 1: The WINDSHEAR SYS advisory message will be displayed.

NOTE 2: See 14 CFR 121.357 or other appropriate regulation.

34-43-01 Weather Radar System**34-43-01C One Operative**

Interval	Installed	Required	Procedure
D	2	1	

34-43-01 Weather Radar System**34-43-01-01 Predictive Windshear Function****34-43-01-01A Reactive Windshear Operative**

Interval	Installed	Required	Procedure
C	-	0	(O)

May be inoperative provided:

- Alternate procedures are established and used.
- Reactive windshear alert (GPWS Mode 7) operates normally.

OPERATIONS (O)

Alternate procedures must be established and used.

NOTE: The WINDSHEAR SYS advisory message will be displayed.

34-43-01 Weather Radar System**34-43-01-01 Predictive Windshear Function****34-43-01-01B Reactive Windshear Inoperative**

Interval	Installed	Required	Procedure
B	-	0	(O)

May be inoperative provided alternate procedures are established and used.

NOTE: Operator's alternate procedures should include reviewing windshear avoidance and windshear recovery procedures.

OPERATIONS (O)

Alternate procedures must be established and used.

NOTE: The WINDSHEAR SYS advisory message will be displayed.

34-43-01 Weather Radar System
34-43-01-02 Auto Tilt Function

Interval	Installed	Required	Procedure
C	1	0	

May be inoperative provided manual tilt function operates normally.

NOTE: This item is applicable to the Collins WXR-2100 and Honeywell RDR-4B weather radar systems with the auto tilt function.

OPERATIONS NOTE

Manual TILT adjustments may be necessary to optimize weather display.



777 FAA DDG Supplement Package

Supplement Number: 36-4

Issue Date: 19-DEC-2024

Section Number: 2 (MMEL)

Item Number: 34-58-01

Item Name: Global Positioning Systems

DDG Effectivity: 777 FAA DDG (D630W003-TBC), Revision 36

MMEL Effectivity: 777 FAA MMEL, Revision 23a

Description: **Note: This item supersedes content in a previous supplement.**

- 1) Modified PBN guidance for both item options.
- 2) Modified PBN guidance for item option 34-58-01B.
- 3) Modified PBN guidance from previous supplement for item option 34-58-01B.

Reason:

- 1) PBN guidance was modified to correct and add more clarification for PBN operation capabilities.
- 2) PBN guidance was modified to incorporate updated PBN operation capabilities.
- 3) PBN guidance was modified to clarify specific RNP 2 operations.

Disposition: For reference in the Operator's MEL.

Attachments: DDG pages 2.34-58-01.1-2

34-58-01 Global Positioning Systems
34-58-01A Both Inoperative

Interval	Installed	Required	Procedure
C	2	0	

May be inoperative provided:

- Enroute operations do not require their use.
- SAARU operates normally.

NOTE: The airplane must also be dispatched using MMEL item 34-53-02-01.

MAINTENANCE NOTE

- For both GPS inoperative and AIMS power is interrupted, UTC is lost and the AIMS time function will reset to midnight.
- If desired, the left, right, or center MMR may be deactivated by opening and collaring P11 panel MMR-L, MMR-R and/or MMR-C circuit breaker.
- If the GPS L and GPS R status messages are caused by a fault in the center MMR antenna, opening the MMR-C circuit breaker may restore GPS L and R functions.
- The associated ILS function will be inoperative for a deactivated MMR. Relief for ILS functions is available in MMEL item 34-31-01.

OPERATIONS NOTE

- Operators must consult their Operations Specification for route position accuracy requirements such as that required by certain Required Navigation Performance (RNP) routes.
- The NAV UNABLE RNP advisory message may be displayed during lengthy ground operations and flight crew should accomplish the associated Non-Normal Checklist.
- The TERR POS advisory message will be displayed.
- The RWY/POS update on the FMC-CDU TAKEOFF REF page will be active.
- The ADS-B Out data will be inoperative.
- For airplanes equipped with RAAS:
 - RAAS is inoperative.
 - The RUNWAY SYS advisory message will be displayed.
 - The airplane must also be dispatched using MMEL item 34-46-01-04.
- For PBN operations:
 - RNP AR APCH, RNP APCH, RNP 1, RNP 2, RNP 4 and Advanced RNP operations are not allowed.

- B. RNAV 1, RNAV 2, P-RNAV, B-RNAV and RNAV 5 operations may be conducted provided one DME/DME or one VOR/DME operates normally.
- C. RNP 10 and RNAV 10 operations may be conducted provided appropriate time limitations are observed. Refer to the RNP Navigation Capabilities Document for the time limitations.

NOTE: Refer to DDG Section 4 for additional information.

34-58-01 Global Positioning Systems
34-58-01B One Inoperative

Interval	Installed	Required	Procedure
D	2	1	

One may be inoperative provided enroute operations do not require its use.

NOTE: The airplane must also be dispatched using MMEL item 34-53-02-01.

MAINTENANCE NOTE

1. If desired, one or both MMR not associated with the operative GPS may be deactivated by opening and collaring P11 panel MMR-L, MMR-R and/or MMR-C circuit breaker.
2. The associated ILS function will be inoperative for a deactivated MMR. Relief for ILS functions is available in MMEL item 34-31-01.

For EFB installed:

NOTE: Some functionality of the EFB may be inoperative. For inoperative EFB functions the airplane must also be dispatched using MMEL item 46-11-02-01.

OPERATIONS NOTE

1. The associated ADS-B Out data will be inoperative.
2. For PBN operations:
 - A. RNP AR APCH and RNP 4 operations are not allowed.
 - B. RNP 2 operations over oceanic and remote continental airspace are not allowed.
 - C. RNP APCH, RNP 1, RNP 2 operations other than those over oceanic or remote continental airspace, Advanced RNP, RNP 10, and RNAV 10 operations may be allowed provided alternate navigation equipment (e.g. DME/DME, VOR/DME, VOR, ADF) is available to continue to a suitable airport if the remaining GPS becomes inoperative.
 - D. RNAV 1, RNAV 2, P-RNAV, B-RNAV and RNAV 5 operations may be conducted.

NOTE: Refer to DDG Section 4 for additional information.



777 FAA DDG Supplement Package

Supplement Number: 36-4

Issue Date: 19-DEC-2024

Section Number: 2 (MMEL)

Item Number: 34-61-01

Item Name: Flight Management Computing Systems (FMCS)

DDG Effectivity: 777 FAA DDG (D630W003-TBC), Revision 36

MMEL Effectivity: 777 FAA MMEL, Revision 23a

Description: **Note: This item supersedes content in a previous supplement.**

- 1) Modified PBN guidance for all subitems.
- 2) Modified PBN guidance from previous supplement for subitems 34-61-01-01 and 34-61-01-02.

Reason:

- 1) PBN guidance was modified to correct and add more clarification for PBN operation capabilities.
- 2) PBN guidance was modified to clarify specific RNP 2 operations.

Disposition: For reference in the Operator's MEL.

Attachments: DDG pages 2,34-61-01.1-3

34-61-01 Flight Management Computing Systems (FMCS)
34-61-01-01 With AIMS-1

Interval	Installed	Required	Procedure
C	2	1	(M)

One may be inoperative provided:

- a. Remaining FMC is verified to operate normally.
- b. Enroute operations do not require its use.
- c. Flight remains within 180 minutes of landing at a suitable airport.

MAINTENANCE (M)

Verify remaining FMC is operating normally by positioning the FMC selector to AUTO and verifying the FMC status and advisory messages are not displayed (AMM 34-00-00/901).

OPERATIONS NOTE

1. Flight must remain within 180 minutes of landing at a suitable airport.
2. Operators must consult their Operations Specification for route flight management computing system requirements.
3. For PBN operations:
 - A. RNP AR APCH and RNP 4 operations are not allowed.
 - B. RNP 2 operations over oceanic and remote continental airspace are not allowed.
 - C. RNP APCH, RNP 1, RNP 2 operations other than those over oceanic or remote continental airspace, RNAV 1, RNAV 2, P-RNAV, B-RNAV, RNAV 5 and Advanced RNP operations may be conducted provided alternate navigation equipment (e.g., VOR, ADF) is available to continue to a suitable airport if the remaining FMCS becomes inoperative.
 - D. RNP 10 and RNAV 10 operations may be conducted using the alternate navigation capability in the multifunction control display units (MCDU).

NOTE: Refer to DDG Section 4 for additional information.

34-61-01 Flight Management Computing Systems (FMCS)
34-61-01-02 With AIMS-2

Interval	Installed	Required	Procedure
C	2	1	(M)

One may be inoperative provided:

- a. Remaining FMC is verified to operate normally.

- b. Enroute operations do not require its use.

MAINTENANCE (M)

Verify remaining FMC is operating normally by positioning the FMC selector to AUTO and verifying the FMC status and advisory messages are not displayed (AMM 34-00-00/901).

OPERATIONS NOTE

1. Operators must consult their Operations Specification for route flight management computing system requirements.
2. For PBN operations:
 - A. RNP AR APCH and RNP 4 operations are not allowed.
 - B. RNP 2 operations over oceanic and remote continental airspace are not allowed.
 - C. RNP APCH, RNP 1, RNP 2 operations other than those over oceanic or remote continental airspace, RNAV 1, RNAV 2, P-RNAV, B-RNAV, RNAV 5 and Advanced RNP operations may be conducted provided alternate navigation equipment (e.g., VOR, ADF) is available to continue to a suitable airport if the remaining FMCS becomes inoperative.
 - D. RNP 10 and RNAV 10 operations may be conducted using the alternate navigation capability in the multifunction control display units (MCDU).

NOTE: Refer to DDG Section 4 for additional information.

34-61-01 Flight Management Computing Systems (FMCS)

34-61-01-03 Navigation Databases

Interval	Installed	Required	Procedure
A	-	0	

May be inoperative provided:

- a. Operations do not require its use.
- b. It is not used in a primary navigation system required by 14 CFR.
- c. Alternate procedures are developed and used.
- d. The ICAO Flight Plan is updated (as required) to notify ATC of the navigation equipment status of the aircraft.
- e. Is repaired within 10 flight days.

NOTE: An out-of-currency or out-of-date navigation database is not authorized MMEL relief per 14 CFR.

NOTE: Although this item may not be applied for an out-of-currency Navigation Database, it does not prohibit dispatch in that condition. Dispatch is allowed with an out-of-currency Navigation Database provided alternate procedures are developed and incorporated in another operational manual, in coordination with local regulatory authorities.

OPERATIONS NOTE

For PBN operations:

- The FMCS navigation database must be current for all PBN operations.

NOTE 1: It is assumed that the operator has a preflight procedure to ensure that the navigation database complies with regulatory agency requirements for ensuring the navigation database remains valid for the duration of the flight unless an alternative regulatory agency approved process applies.

NOTE 2: Refer to DDG Section 4 for additional information.



777 FAA DDG Supplement Package

Supplement Number: 36-4

Issue Date: 19-DEC-2024

Section Number: 2 (MMEL)

Item Number: 38-30-01

Item Name: Waste Water Systems

DDG Effectivity: 777 FAA DDG (D630W003-TBC), Revision 36

MMEL Effectivity: 777 FAA MMEL, Revision 23a

Description: Modified Operations (O) procedure for sub item 38-30-01-01.

Reason: Operations (O) procedure was modified to be consistent with the proviso.

Disposition: For reference in the Operator's MEL.

Attachments: DDG pages 2.38-30-01.1-3

38-30-01 Waste Water Systems
38-30-01A Components Deactivated

Interval	Installed	Required	Procedure
C	-	-	(M)

Individual components may be inoperative provided:

- Associated components are deactivated or isolated.
- Associated system components are verified not to have leaks.

NOTE: Any portion of system which operates normally may be used.

MAINTENANCE (M)

Configure the airplane (AMM 38-00-00/901).

- Deactivate or isolate inoperative components.
- Verify associated system components do not have leaks.

NOTE: Due to the wide variety of lavatory installations and various component failure possibilities, operators must develop procedures tailored to their specific installations.

38-30-01 Waste Water Systems
38-30-01B Lavatory Not Used

Interval	Installed	Required	Procedure
C	-	-	(M)

Associated lavatory system may be inoperative provided:

- Associated components are deactivated or isolated to prevent leaks.
- Associated lavatory door is locked closed and placarded, INOPERATIVE - DO NOT ENTER.

NOTE: These provisions are not intended to prohibit inspections by crewmembers.

MAINTENANCE (M)

Configure the airplane (AMM 38-00-00/901).

- Deactivate or isolate inoperative components.
- If required, drain the associated waste tank. Do not add the precharge chemical to the associated waste tank.
- Close, lock and placard the associated lavatory door.

38-30-01 Waste Water Systems
38-30-01-01 Vacuum Blowers

Interval	Installed	Required	Procedure
C	-	-	(M) (O)

May be inoperative provided:

- Associated vacuum blower is deactivated.
- Lavatories affected by inoperative vacuum blower are not used on the ground or at flight altitudes of 16,000 feet or below.

MAINTENANCE (M)

Deactivate the inoperative vacuum blower (AMM DDG 38-00-00/901).

For Passenger:

- For the system 1 (forward tank) vacuum blower inoperative, open and collar the P110 panel VAC BLO LAV SYS 1 circuit breaker.
- For the system 2 and 3 (mid and aft tank) vacuum blower inoperative, open and collar the P210 panel VAC BLO LAV SYS 2 & 3 circuit breaker.

For 777F:

- Open and collar the P210 panel VAC BLO LAV circuit breaker.

OPERATIONS (O)

Establish and use procedures to prevent use of lavatories associated with the inoperative vacuum blower at flight altitude of 16,000 feet or below.

38-30-01 Waste Water Systems
38-30-01-02 Waste Tanks

Interval	Installed	Required	Procedure
C	-	-	(M)

May be inoperative provided:

- Associated waste tank(s) is deactivated.
- Associated components are deactivated or isolated to prevent leaks.
- Associated lavatory door(s) is secured closed and placarded, INOPERATIVE - DO NOT ENTER.

NOTE: These provisions are not intended to prohibit inspections by crewmembers.

MAINTENANCE (M)

Deactivate associated waste tank(s), deactivate or isolate inoperative components, and secure closed and placard lavatory door(s) (AMM DDG 38-00-00/901).

1. Drain the associated waste tank(s). Do not add the precharge chemical to the associated waste tank.
2. Deactivate or isolate inoperative components.
3. Close, lock and placard the associated lavatory door(s).

Supplement 36-4



777 FAA DDG Supplement Package

Supplement Number: 36-4

Issue Date: 19-DEC-2024

Section Number: 4 (MISC)

Item Number: N/A

Item Name: Performance Based Navigation

DDG Effectivity: 777 FAA DDG (D630W003-TBC), Revision 36

MMEL Effectivity: 777 FAA MMEL, Revision 23a

Description: **Note: This item supersedes content in a previous supplement.**

Modified PBN guidance.

Reason:

- 1) PBN guidance was modified for the Mode Control Panel (MCP) Switches. Corrected the PRR number reference for Enhanced Ground Proximity Warning System (EGPWS).
- 2) PBN guidance was modified to correct and add more clarification for PBN operation capabilities.
- 3) PBN guidance was modified to incorporate updated PBN operation capabilities.
- 4) PBN guidance was modified to clarify verbiage used and to be consistent with RNP capabilities document.

Disposition: For reference in the Operator's MEL.

Attachments: DDG pages 4.00-02-00.1-7



Section 4 Performance Based Navigation

Performance Based Navigation (PBN) Information

Applying dispatch relief for the following items may affect the ability of the airplane to meet the operational capabilities required for PBN operations. The following information lists items that, if inoperative, may affect the ability of the airplane to meet the performance and functional criteria for various PBN operations. This information can aid in identifying potentially affected capabilities associated with a particular dispatch configuration. This section lists equipment requirements that may be found in the document 777 RNP Navigation Capabilities Document D243W018-13, revision "O" (or later). Operators should consult their Operations Specification for requirements applicable to their operations.

PBN Equipment Requirements

Display Units (DUs) - (31-61-01)

PBN operations require two PFDs with mini map and EICAS display. The 777 is equipped with six display units. The PFDs are normally displayed on the outboard display units and the NDs are normally displayed on the inboard display units. EICAS is normally displayed on the upper center display unit. MMEL dispatch is only allowed with the lower center display unit inoperative. Therefore, these requirements will always be met for dispatch.

Multi-Purpose Control and Display Units - Control Display Units (CDU) - (34-61-03)

The 777 is equipped with three CDUs. MMEL dispatch is only allowed with the center CDU inoperative. Both left and right CDUs must operate normally for dispatch. PBN operations are allowed with two CDUs. Therefore, this requirement will always be met for dispatch.

AIMS Operational Software

For RNP AR APCH:

Airplane must be equipped with AIMS BP05A or later. Navigation Performance Scales (NPS) and LNAV Go-Around Auto-Engage must be enabled.

Airplanes without Navigation Performance Scales and LNAV Go-Around Auto-Engage (AIMS BP03 and earlier versions) do not support these operations.

Fault Tolerant ADIRU

For AIMS-1 V16 and AIMS-2 V17 or later:

AIR DATA SYS or NAV ADIRU INERTIAL messages must not be displayed.

For all other AIMS-1 or AIMS-2 versions:

NAV AIR DATA SYS or NAV ADIRU INERTIAL messages must not be displayed.

MMEL dispatch is not allowed with any of the above messages displayed. Therefore, this requirement will always be met for dispatch.

Autopilot Flight Director Computers (AFDC) - (22-11-01)

The 777 autopilot flight director system (AFDS) is equipped with three autopilot flight director computers (AFDCs). Each of the three AFDCs can provide both flight director guidance, which can be displayed on the primary flight displays (PFDs), and send autopilot commands to the flight control actuation system when the autopilot is engaged.

MMEL dispatch is allowed with one, two, or all three AFDCs inoperative. Therefore, PBN operations may be affected with more than one AFDC inoperative.

For dispatch with two Autopilot Flight Director Computers) inoperative (22-11-01B):

- RNP AR APCH operations are not allowed.
- Other PBN operations may be conducted.

For dispatch with all (3) Autopilot Flight Director Computers) inoperative (22-11-01C):

- PBN operations are not allowed.

Autopilot and Flight Director

PBN operations may be conducted using either the autopilot or flight director. The minimum authorized RNP value may be different depending on whether autopilot or flight director is used.

Autopilot

When the autopilot is engaged, autopilot commands from the AFDCs are sent to the Primary Flight Computers to generate flight control surface commands.

If the autopilot function is inoperative, PBN operations may be conducted using the flight director.

The autopilot will be impacted if any of the following equipment is inoperative:

1. Mode Control Panel Switches - A/P Engage Switches
 - A. For dispatch with one autopilot engage switch inoperative (22-11-06-01A):
 - RNP AR APCH operations using autopilot are not allowed.
 - RNP AR APCH operations may be conducted using flight director provided its use is appropriate for the intended operation.
 - Other PBN operations using autopilot may be conducted.
 - B. For dispatch with both autopilot engage switches inoperative (22-11-06-01B):
 - PBN operations using autopilot are not allowed.
 - PBN operations may be conducted using flight director provided its use is appropriate for the intended operation.
2. Control Wheel Autopilot Disconnect Switches
 - A. For dispatch with one control wheel autopilot disconnect switch inoperative (22-11-08A):
 - PBN operations using autopilot at 1,500 feet AGL or below are not allowed.
 - PBN operations may be conducted using flight director provided its use is appropriate for the intended operation.
 - B. For dispatch with both control wheel autopilot disconnect switches inoperative (22-11-08B):
 - PBN operations using autopilot are not allowed.
 - PBN operations may be conducted using flight director provided its use is appropriate for the intended operation.
3. Warning Electronic System (WES) Autopilot Disconnect Warning Function
 - A. For dispatch with WES autopilot disconnect warning function inoperative (31-51-05):
 - PBN operations using autopilot are not allowed.
 - PBN operations may be conducted using flight director provided its use is appropriate for the intended operation.

Flight Director

When the flight director is selected on, flight director guidance from the AFDCs is sent to and displayed on the Primary Flight Displays (PFDs). The flight crew uses the flight director guidance to manually control the airplane in accordance with the command guidance from the AFDCs.

Flight director guidance can be displayed on one or both PFDs.

Flight director guidance is selected on and off by using the flight director switches on the mode control panel (MCP).

The flight director guidance will be impacted if any of the following equipment is inoperative:

1. Mode Control Panel Switches - F/D Switches

- A. For dispatch with one Flight Director switch inoperative (22-11-06-04):
- RNP AR APCH operations using flight director are not allowed.
 - RNP AR APCH operations may be conducted using autopilot provided its use is appropriate for the intended operation.
 - Other PBN operations using flight director may be conducted provided flight director operates normally on the side of the pilot flying (PF).
- B. For dispatch with both Flight Director switches inoperative (22-11-06-04):
- PBN operations using flight director are not allowed.
 - PBN operations may be conducted using autopilot provided its use is appropriate for the intended operation.

2. Flight Director Systems

- A. For dispatch with one Flight Director system inoperative (22-11-10):
- RNP AR APCH operations using flight director are not allowed.
 - RNP AR APCH operations may be conducted using autopilot provided its use is appropriate for the intended operation.
 - Other PBN operations using flight director may be conducted provided flight director operates normally on the side of the pilot flying (PF).
- B. For dispatch with both Flight Director systems inoperative (22-11-10):
- PBN operations using flight director are not allowed.
 - PBN operations may be conducted using autopilot provided its use is appropriate for the intended operation.

LNAV/VNAV Mode Control Panel (MCP) Switches - (22-11-06-09)

For dispatch with lateral navigation (LNAV) switch inoperative (22-11-06-09):

- PBN operations are not allowed.

For dispatch with vertical navigation (VNAV) switch inoperative. (22-11-06-09):

- RNP AR APCH operations are not allowed.
- RNP APCH operations may be conducted using other vertical modes provided appropriate approach minimums are used.
- Other PBN operations may be conducted.

Takeoff/Go-Around (TO/GA) Switches - (22-11-09)

There are two TO/GA switches installed on the 777. MMEL dispatch is allowed with one or both TO/GA switches inoperative.

For dispatch with one TO/GA switch inoperative (22-11-09A):

- RNP AR APCH operations are not allowed.
- Other PBN operations may be conducted.

For dispatch with both TO/GA switches inoperative (22-11-09B):

- PBN operations using autopilot or flight director below 500 feet AGL or MDA are not allowed.

Global Positioning Systems (GPS) - (34-58-01)

The 777 is equipped with two GPSs. MMEL dispatch is allowed with one or both GPSs inoperative.

For dispatch with one GPS inoperative (34-58-01B):

- RNP AR APCH and RNP 4 operations are not allowed.
- RNP 2 operations over oceanic and remote continental airspace are not allowed.
- RNP APCH, RNP 1, RNP 2 operations other than those over oceanic or remote continental airspace, Advanced RNP, RNP 10, and RNAV 10 operations may be conducted provided alternate navigation equipment (e.g. DME/DME, VOR/DME, VOR, ADF) is available to continue to a suitable airport if the remaining GPS becomes inoperative.
- RNAV 1, RNAV 2, P-RNAV, B-RNAV and RNAV 5 operations may be conducted.

For dispatch with both GPSs inoperative (34-58-01A):

- RNP AR APCH, RNP APCH, RNP 1, RNP 2, RNP 4 and Advanced RNP operations are not allowed.

- RNAV 1, RNAV 2, P-RNAV, B-RNAV and RNAV 5 operations may be conducted provided one DME/DME or one VOR/DME operates normally.
- RNP 10 and RNAV 10 operations may be conducted provided appropriate time limitations are observed. Refer to the RNP Navigation Capabilities Document for the time limitations.

VOR Navigation Systems - (34-51-01)

The 777 is equipped with two VORs. MMEL dispatch is allowed with one or both VORs inoperative.

- RNAV 1, RNAV 2, P-RNAV, B-RNAV and RNAV 5 operations may be conducted provided one GPS, one DME/DME or one VOR/DME operates normally.
- Other PBN operations may be conducted.

Distance Measuring Equipment Systems (DME) - (34-55-01)

The 777 is equipped with two DMEs. MMEL dispatch is allowed with one or both DMEs inoperative.

- RNAV 1, RNAV 2, P-RNAV, B-RNAV and RNAV 5 operations may be conducted provided one GPS, one DME/DME or one VOR/DME operates normally.
- Other PBN operations may be conducted.

Radio Altimeter Systems - (34-33-01)

The 777 is equipped with three radio altimeters. MMEL dispatch is allowed with one radio altimeter inoperative.

PBN operations are allowed with two radio altimeters. Therefore, this requirement will always be met for dispatch.

Ground Proximity Warning Systems (GPWS) - (34-46-01), Terrain Awareness Function (34-46-01-03)

777 airplanes are equipped with a single ground proximity warning system (GPWS). Airplanes with PRR 61724 or equivalent are equipped with enhanced GPWS (EGPWS) that includes a terrain awareness function.

MMEL dispatch is allowed with the GPWS (EGPWS) inoperative.

For dispatch with either GND PROX SYS or TERR POS Advisory message displayed:

- RNP AR APCH operations are not allowed.
- Other PBN operations may be conducted.

Flight Management Computing Systems (FMCS) - (34-61-01)

There are two flight management computing systems (FMCSs) installed on the 777. MMEL dispatch is allowed with one FMCS inoperative.

For dispatch with one FMCS inoperative:

- RNP AR APCH and RNP 4 operations are not allowed.
- RNP 2 operations over oceanic and remote continental airspace are not allowed.
- RNP APCH, RNP 1, RNP 2 operations other than those over oceanic or remote continental airspace, RNAV 1, RNAV 2, P-RNAV, B-RNAV, RNAV 5 and Advanced RNP operations may be conducted provided alternate navigation equipment (e.g., VOR, ADF) is available to continue to a suitable airport if the remaining FMCS becomes inoperative.
- RNP 10 and RNAV 10 operations may be conducted using the alternate navigation capability in the multifunction control display units (MCDU).

Flight Management Computing Systems (FMCS) - Navigation Database (34-61-01-03)

The FMCS navigation database must be current for PBN operations.

It is assumed that the operator has a preflight procedure to ensure that the navigation database complies with regulatory agency requirements for ensuring the navigation database remains valid for the duration of the flight unless an alternative regulatory agency approved process applies.



777 FAA DDG Supplement Package

Supplement Number: 36-5

Issue Date: 17-JAN-2025

Section Number: 2 (MMEL)

Item Number: 28-31-01

Item Name: Fuel Jettison System

DDG Effectivity: 777 FAA DDG (D630W003-TBC), Revision 36

MMEL Effectivity: 777 FAA MMEL, Revision 23a

Description: **Note: This item supersedes content in a previous supplement.**

Modified Operations (O) procedures for subitems 28-31-01-01 and 28-31-01-02.

Reason: Operations (O) procedures were modified to reflect updated performance adjustments. Operations (O) procedure numbering error introduced in previous supplement was corrected.

Disposition: For reference in the Operator's MEL.

Attachments: DDG pages 2.28-31-01.1-5

28-31-01 Fuel Jettison System
28-31-01-01 Without PRR 62777-79S or Equivalent Installed

Interval	Installed	Required	Procedure
C	1	0	(M) (O)

May be inoperative provided:

- Jettison system is deactivated.
- Jettison nozzle valves are secured closed.
- Appropriate performance adjustments are applied.

NOTE: The PRR 62777-79S is applicable to airplane line numbers 1099 and-on. Service Bulletins 777-28-0078 (ELMS2) and 777-28-0082 (ELMS1) each provide the production equivalent of PRR 62777-79S.

MAINTENANCE (M)

Deactivate the jettison system and secure the jettison nozzle valves in the closed position (AMM 28-00-00/901).

- Open and collar P110 panel L FUEL JETT PUMP CTRL, L FUEL JETT ISLN VLV and L FUEL JETT NOZ VLV circuit breakers.
- Open and collar P210 panel R FUEL JETT PUMP CTRL, R FUEL JETT ISLN VLV and R FUEL JETT NOZ VLV circuit breakers.
- For left jettison nozzle valve:
 - Gain access to the left jettison nozzle valve actuator through associated access door.
 - Disconnect, cap and stow the electrical connector (DV28166).
 - Confirm the manual override handle is in the CLOSED position and lockwire.
 - Close access door.
- For right jettison nozzle valve:
 - Gain access to the right jettison nozzle valve actuator through associated access door.
 - Disconnect, cap and stow the electrical connector (DV28266).
 - Confirm the manual override handle is in the CLOSED position and lockwire.
 - Close access door.

OPERATIONS (O)

NOTE: The Fuel Synoptic will incorrectly indicate jettison flow from the center tank when the center tank override pumps are on with fuel in the center tank. The FUEL JETT NOZZLE L and R advisory messages will be displayed and overhead panel L and R fuel jettison nozzle VALVE lights will be illuminated.

For -200 or -200ER/PW4074 thru PW4084:

1. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 3,400 lb (1,542 kg) (for fuel that would be burned during a return-to-land procedure).

For -200 or -200ER/PW4090:

1. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 3,800 lb (1,724 kg) (for fuel that would be burned during a return-to-land procedure).

For -200 or -200ER/GE90-76B thru -90B:

1. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 3,300 lb (1,497 kg) (for fuel that would be burned during a return-to-land procedure).

For -200 or -200ER/GE90-94B or RR875 thru RR892:

1. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 3,700 lb (1,678 kg) (for fuel that would be burned during a return-to-land procedure).

For -200 or -200ER/RR895:

1. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 4,000 lb (1,814 kg) (for fuel that would be burned during a return-to-land procedure).

For -200LR or 777F/GE90-100 Series:

1. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 4,000 lb (1,814 kg) (for fuel that would be burned during a return-to-land procedure).

For -300/RR884 thru RR892:

1. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 3,900 lb (1,769 kg) (for fuel that would be burned during a return-to-land procedure).

For -300/PW4090:

1. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 4,100 lb (1,860 kg) (for fuel that would be burned during a return-to-land procedure).

For -300ER/GE90-115B:

1. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 4,200 lb (1,905 kg) (for fuel that would be burned during a return-to-land procedure).

28-31-01 Fuel Jettison System
28-31-01-01 Without PRR 62777-79S or Equivalent Installed
28-31-01-01-01 ARM Switch FAULT Light

Interval	Installed	Required	Procedure
C	1	0	

28-31-01 Fuel Jettison System
28-31-01-01 Without PRR 62777-79S or Equivalent Installed
28-31-01-01-02 ARM Switch ARMED Light

Interval	Installed	Required	Procedure
C	1	0	

28-31-01 Fuel Jettison System
28-31-01-02 With PRR 62777-79S or Equivalent Installed

Interval	Installed	Required	Procedure
C	1	0	(M) (O)

May be inoperative provided:

- Jettison system is deactivated.
- Jettison nozzle valves are secured closed.
- Fuel scavenge system is considered inoperative.
- Appropriate performance adjustments are applied.

NOTE 1: The PRR 62777-79S is applicable to airplane line numbers 1099 and-on. Service Bulletins 777-28-0078 (ELMS2) and 777-28-0082 (ELMS1) each provide the production equivalent of PRR 62777-79S.

NOTE 2: The airplane must also be dispatched using MMEL item 28-22-04.

MAINTENANCE (M)

Deactivate the jettison system and secure the jettison nozzle valves in the closed position (AMM 28-00-00/901).

- Open and collar P110 panel L FUEL JETT PUMP CTRL, L FUEL JETT ISLN VLV and L FUEL JETT NOZ VLV circuit breakers.
- Open and collar P210 panel R FUEL JETT PUMP CTRL, R FUEL JETT ISLN VLV and R FUEL JETT NOZ VLV circuit breakers.

3. For left jettison nozzle valve:
 - A. Gain access to the left jettison nozzle valve actuator through associated access door.
 - B. Disconnect, cap and stow the electrical connector (DV28166).
 - C. Confirm the manual override handle is in the CLOSED position and lockwire.
 - D. Close access door.
4. For right jettison nozzle valve:
 - A. Gain access to the right jettison nozzle valve actuator through associated access door.
 - B. Disconnect, cap and stow the electrical connector (DV28266).
 - C. Confirm the manual override handle is in the CLOSED position and lockwire.
 - D. Close access door.

OPERATIONS (O)

NOTE: The Fuel Synoptic will incorrectly indicate jettison flow from the center tank when the center tank override pumps are on with fuel in the center tank. The FUEL JETT NOZZLE L and R advisory messages will be displayed and overhead panel L and R fuel jettison nozzle VALVE lights will be illuminated.

For -200 or -200ER/PW4074 thru PW4084:

1. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 3,400 lb (1,542 kg) (for fuel that would be burned during a return-to-land procedure).

For -200 or -200ER/PW4090:

1. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 3,800 lb (1,724 kg) (for fuel that would be burned during a return-to-land procedure).

For -200 or -200ER/GE90-76B thru -90B:

1. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 3,300 lb (1,497 kg) (for fuel that would be burned during a return-to-land procedure).

For -200 or -200ER/GE90-94B or RR875 thru RR892:

1. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 3,700 lb (1,678 kg) (for fuel that would be burned during a return-to-land procedure).

For -200 or -200ER/RR895:

1. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 4,000 lb (1,814 kg) (for fuel that would be burned during a return-to-land procedure).

For -200LR or 777F/GE90-100 Series:

1. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 4,000 lb (1,814 kg) (for fuel that would be burned during a return-to-land procedure).

For -300/RR884 thru RR892:

1. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 3,900 lb (1,769 kg) (for fuel that would be burned during a return-to-land procedure).

For -300/PW4090:

1. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 4,100 lb (1,860 kg) (for fuel that would be burned during a return-to-land procedure).

For -300ER/GE90-115B:

1. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 4,200 lb (1,905 kg) (for fuel that would be burned during a return-to-land procedure).

28-31-01 Fuel Jettison System
28-31-01-02 With PRR 62777-79S or Equivalent Installed
28-31-01-02-01 ARM Switch FAULT Light

Interval	Installed	Required	Procedure
C	1	0	

28-31-01 Fuel Jettison System
28-31-01-02 With PRR 62777-79S or Equivalent Installed
28-31-01-02-02 ARM Switch ARMED Light

Interval	Installed	Required	Procedure
C	1	0	



777 FAA DDG Supplement Package

Supplement Number: 36-5

Issue Date: 17-JAN-2025

Section Number: 2 (MMEL)

Item Number: 28-31-02

Item Name: Center Tank Jettison Isolation Valves

DDG Effectivity: 777 FAA DDG (D630W003-TBC), Revision 36

MMEL Effectivity: 777 FAA MMEL, Revision 23a

Description: **Note: This item supersedes content in a previous supplement.**

Modified Operations (O) procedures for subitem options 28-31-02-02B and 28-31-02-03B.

Reason: Operations (O) procedures were modified to reflect updated performance adjustments and CG adjustment information was reincluded to correct the inadvertent removal in previous supplement.

Disposition: For reference in the Operator's MEL.

Attachments: DDG pages 2,28-31-02.1-9

**28-31-02 Center Tank Jettison Isolation Valves****28-31-02-01 -200**

Interval	Installed	Required	Procedure
C	2	0	(M)

May be inoperative provided inoperative valve(s) is secured closed.

MAINTENANCE (M)

Deactivate and secure the inoperative jettison isolation valve in the closed position (AMM 28-00-00/901).

1. Open the landing gear doors and install safety pins (AMM 32-00-15/201).
2. For left jettison isolation valve inoperative:
 - A. Open and collar P110 panel L FUEL JETT ISLN VLV circuit breaker.
 - B. Gain access to the jettison isolation valve actuator located in the left main wheel well (rear spar of the left center tank).
 - C. Disconnect, cap and stow the electrical connector (DV28165).
 - D. Move the manual override handle to the CLOSED position and lockwire.
3. For right jettison isolation valve inoperative:
 - A. Open and collar P210 panel R FUEL JETT ISLN VLV circuit breaker.
 - B. Gain access to the jettison isolation valve actuator located in the right main wheel well (rear spar of the right center tank).
 - C. Disconnect, cap and stow the electrical connector (DV28265).
 - D. Move the manual override handle to the CLOSED position and lockwire.
4. Remove the landing gear door safety pins and close the landing gear doors (AMM 32-00-15/201).

28-31-02 Center Tank Jettison Isolation Valves**28-31-02-02 -200ER/-200LR/-300/-300ER/777F (Without PRR 62777-79S or Equivalent Installed)****28-31-02-02A Fuel Jettison System Deactivated**

Interval	Installed	Required	Procedure
C	2	0	(M)

May be inoperative provided:

- a. Inoperative valve(s) is secured closed.
- b. Fuel jettison system is considered inoperative.



NOTE 1: The PRR 62777-79S is applicable to airplane line numbers 1099 and-on. Service Bulletins 777-28-0078 (ELMS2) and 777-28-0082 (ELMS1) each provide the production equivalent of PRR 62777-79S.

NOTE 2: The airplane must also be dispatched using MMEL item 28-31-01.

MAINTENANCE (M)

Secure the inoperative jettison isolation valve in the closed position (AMM 28-00-00/901).

1. Open the landing gear doors and install safety pins (AMM 32-00-15/201).
2. For left jettison isolation valve:
 - A. Open and collar P110 panel L FUEL JETT ISLN VLV circuit breaker.
 - B. Gain access to the jettison isolation valve actuator located in the left main wheel well (rear spar of the left center tank).
 - C. Disconnect, cap and stow the electrical connector (DV28165).
 - D. Move the manual override handle to the CLOSED position and lockwire.
3. For right jettison isolation valve:
 - A. Open and collar P210 panel R FUEL JETT ISLN VLV circuit breaker.
 - B. Gain access to the jettison isolation valve actuator located in the right main wheel well (rear spar of the right center tank).
 - C. Disconnect, cap and stow the electrical connector (DV28265).
 - D. Move the manual override handle to the CLOSED position and lockwire.
4. Remove the landing gear door safety pins and close the landing gear doors (AMM 32-00-15/201).

28-31-02 Center Tank Jettison Isolation Valves

28-31-02-02 -200ER/-200LR/-300/-300ER/777F (Without PRR 62777-79S or Equivalent Installed)

28-31-02-02B Main Tank Jettison Operative

Interval	Installed	Required	Procedure
C	2	0	(M) (O)

May be inoperative provided:

- a. Inoperative valve(s) is secured closed.
 - b. Appropriate gross weight and center-of-gravity adjustments are applied.
-

NOTE: The PRR 62777-79S is applicable to airplane line numbers 1099 and-on. Service Bulletins 777-28-0078 (ELMS2) and 777-28-0082 (ELMS1) each provide the production equivalent of PRR 62777-79S.

MAINTENANCE (M)

Deactivate and secure the inoperative jettison isolation valve in the closed position (AMM 28-00-00/901).

1. Open the landing gear doors and install safety pins (AMM 32-00-15/201).
2. For left jettison isolation valve inoperative:
 - A. Open and collar P110 panel L FUEL JETT ISLN VLV circuit breaker.
 - B. Gain access to the jettison isolation valve actuator located in the left main wheel well (rear spar of the left center tank).
 - C. Disconnect, cap and stow the electrical connector (DV28165).
 - D. Move the manual override handle to the CLOSED position and lockwire.
3. For right jettison isolation valve inoperative:
 - A. Open and collar P210 panel R FUEL JETT ISLN VLV circuit breaker.
 - B. Gain access to the jettison isolation valve actuator located in the right main wheel well (rear spar of the right center tank).
 - C. Disconnect, cap and stow the electrical connector (DV28265).
 - D. Move the manual override handle to the CLOSED position and lockwire.
4. Remove the landing gear door safety pins and close the landing gear doors (AMM 32-00-15/201).

OPERATIONS (O)

For -200/PW4074 thru PW4084:

1. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 3,400 lb (1,542 kg) (for fuel that would be burned during a return-to-land procedure).

For -200/PW4090:

1. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 3,800 lb (1,724 kg) (for fuel that would be burned during a return-to-land procedure).

For -200/GE90-76B thru -90B:

1. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 3,300 lb (1,497 kg) (for fuel that would be burned during a return-to-land procedure).

For -200/GE90-94B or RR875 thru RR892:

1. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 3,700 lb (1,678 kg) (for fuel that would be burned during a return-to-land procedure).

For -200/RR895:

1. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 4,000 lb (1,814 kg) (for fuel that would be burned during a return-to-land procedure).

For -200ER/PW4074 thru PW4084:

1. For dispatch with center tank fuel greater than 22,000 lb (9,980 kg) and Gross Weight exceeding 460,000 lb (208,650 kg), Center-of-Gravity (CG) must be aft of 22% MAC for takeoff
2. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 3,400 lb (1,542 kg) (for fuel that would be burned during a return-to-land procedure).

For -200ER/PW4090:

1. For dispatch with center tank fuel greater than 22,000 lb (9,980 kg) and Gross Weight exceeding 460,000 lb (208,650 kg), Center-of-Gravity (CG) must be aft of 22% MAC for takeoff.
2. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 3,800 lb (1,724 kg) (for fuel that would be burned during a return-to-land procedure).

For -200ER/GE90-76B thru -90B:

1. For dispatch with center tank fuel greater than 22,000 lb (9,980 kg) and Gross Weight exceeding 460,000 lb (208,650 kg), Center-of-Gravity (CG) must be aft of 22% MAC for takeoff.
2. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 3,300 lb (1,497 kg) (for fuel that would be burned during a return-to-land procedure).

For -200ER/GE90-94B or RR875 thru RR892:

1. For dispatch with center tank fuel greater than 22,000 lb (9,980 kg) and Gross Weight exceeding 460,000 lb (208,650 kg), Center-of-Gravity (CG) must be aft of 22% MAC for takeoff.
2. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 3,700 lb (1,678 kg) (for fuel that would be burned during a return-to-land procedure).

For -200ER/RR895:

1. For dispatch with center tank fuel greater than 22,000 lb (9,980 kg) and Gross Weight exceeding 460,000 lb (208,650 kg), Center-of-Gravity (CG) must be aft of 22% MAC for takeoff.
2. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 4,000 lb (1,814 kg) (for fuel that would be burned during a return-to-land procedure).

For -200LR or 777F/GE90-100 Series:

1. For dispatch with center tank fuel and Gross Weight exceeding 523,000 lb (237,230 kg), Center-of-Gravity (CG) must be aft of 25% MAC for takeoff.
2. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 4,000 lb (1,814 kg) (for fuel that would be burned during a return-to-land procedure).

For -300/RR884 thru RR892:

1. For dispatch with center tank fuel and Gross Weight exceeding 524,000 lb (238,000 kg), Center-of-Gravity (CG) must be aft of 17% MAC for takeoff.
2. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 3,900 lb (1,769 kg) (for fuel that would be burned during a return-to-land procedure).

For -300/PW4090:

1. For dispatch with center tank fuel and Gross Weight exceeding 524,000 lb (238,000 kg), Center-of-Gravity (CG) must be aft of 17% MAC for takeoff.
2. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 4,100 lb (1,860 kg) (for fuel that would be burned during a return-to-land procedure).

For -300ER/GE90-115B:

1. For dispatch with center tank fuel and Gross Weight exceeding 556,000 lb (252,200 kg), Center-of-Gravity (CG) must be aft of 21% MAC for takeoff.
2. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 4,200 lb (1,905 kg) (for fuel that would be burned during a return-to-land procedure).

28-31-02 Center Tank Jettison Isolation Valves
28-31-02-03 -200ER/-200LR/-300/-300ER/777F (With PRR 62777-79S or Equivalent Installed)
28-31-02-03A Fuel Jettison System Deactivated

Interval	Installed	Required	Procedure
C	2	0	(M)

May be inoperative provided:

- a. Inoperative valve(s) is secured closed.
- b. Fuel jettison system is considered inoperative.
- c. Fuel scavenge system is considered inoperative.

NOTE 1: The PRR 62777-79S is applicable to airplane line numbers 1099 and-on. Service Bulletins 777-28-0078 (ELMS2) and 777-28-0082 (ELMS1) each provide the production equivalent of PRR 62777-79S.

NOTE 2: The airplane must also be dispatched using MMEL item 28-31-01.

MAINTENANCE (M)

Secure the inoperative jettison isolation valve in the closed position (AMM 28-00-00/901).

1. Open the landing gear doors and install safety pins (AMM 32-00-15/201).

2. For left jettison isolation valve:
 - A. Open and collar P110 panel L FUEL JETT ISLN VLV circuit breaker.
 - B. Gain access to the jettison isolation valve actuator located in the left main wheel well (rear spar of the left center tank).
 - C. Disconnect, cap and stow the electrical connector (DV28165).
 - D. Move the manual override handle to the CLOSED position and lockwire.
3. For right jettison isolation valve:
 - A. Open and collar P210 panel R FUEL JETT ISLN VLV circuit breaker.
 - B. Gain access to the jettison isolation valve actuator located in the right main wheel well (rear spar of the right center tank).
 - C. Disconnect, cap and stow the electrical connector (DV28265).
 - D. Move the manual override handle to the CLOSED position and lockwire.
4. Remove the landing gear door safety pins and close the landing gear doors (AMM 32-00-15/201).

28-31-02 Center Tank Jettison Isolation Valves
28-31-02-03 -200ER/-200LR/-300/-300ER/777F (With PRR 62777-79S or Equivalent Installed)
28-31-02-03B Main Tank Jettison Operative

Interval	Installed	Required	Procedure
C	2	0	(M) (O)

May be inoperative provided:

- a. Inoperative valve(s) is secured closed.
- b. Appropriate gross weight and center-of-gravity adjustments are applied.

NOTE: The PRR 62777-79S is applicable to airplane line numbers 1099 and-on. Service Bulletins 777-28-0078 (ELMS2) and 777-28-0082 (ELMS1) each provide the production equivalent of PRR 62777-79S.

MAINTENANCE (M)

Deactivate and secure the inoperative jettison isolation valve in the closed position (AMM 28-00-00/901).

1. Open the landing gear doors and install safety pins (AMM 32-00-15/201).
2. For left jettison isolation valve inoperative:
 - A. Open and collar P110 panel L FUEL JETT ISLN VLV circuit breaker.
 - B. Gain access to the jettison isolation valve actuator located in the left main wheel well (rear spar of the left center tank).
 - C. Disconnect, cap and stow the electrical connector (DV28165).

- D. Move the manual override handle to the CLOSED position and lockwire.
3. For right jettison isolation valve inoperative:
 - A. Open and collar P210 panel R FUEL JETT ISLN VLV circuit breaker.
 - B. Gain access to the jettison isolation valve actuator located in the right main wheel well (rear spar of the right center tank).
 - C. Disconnect, cap and stow the electrical connector (DV28265).
 - D. Move the manual override handle to the CLOSED position and lockwire.
4. Remove the landing gear door safety pins and close the landing gear doors (AMM 32-00-15/201).

OPERATIONS (O)**For -200/PW4074 thru PW4084:**

1. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 3,400 lb (1,542 kg) (for fuel that would be burned during a return-to-land procedure).

For -200/PW4090:

1. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 3,800 lb (1,724 kg) (for fuel that would be burned during a return-to-land procedure).

For -200/GE90-76B thru -90B:

1. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 3,300 lb (1,497 kg) (for fuel that would be burned during a return-to-land procedure).

For -200/GE90-94B or RR875 thru RR892:

1. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 3,700 lb (1,678 kg) (for fuel that would be burned during a return-to-land procedure).

For -200/RR895:

1. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 4,000 lb (1,814 kg) (for fuel that would be burned during a return-to-land procedure).

For -200ER/PW4074 thru PW4084:

1. For dispatch with center tank fuel greater than 22,000 lb (9,980 kg) and Gross Weight exceeding 460,000 lb (208,650 kg), Center-of-Gravity (CG) must be aft of 22% MAC for takeoff.
2. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 3,400 lb (1,542 kg) (for fuel that would be burned during a return-to-land procedure).

For -200ER/PW4090:

1. For dispatch with center tank fuel greater than 22,000 lb (9,980 kg) and Gross Weight exceeding 460,000 lb (208,650 kg), Center-of-Gravity (CG) must be aft of 22% MAC for takeoff.
2. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 3,800 lb (1,724 kg) (for fuel that would be burned during a return-to-land procedure).

For -200ER/GE90-76B thru -90B:

1. For dispatch with center tank fuel greater than 22,000 lb (9,980 kg) and Gross Weight exceeding 460,000 lb (208,650 kg), Center-of-Gravity (CG) must be aft of 22% MAC for takeoff.
2. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 3,300 lb (1,497 kg) (for fuel that would be burned during a return-to-land procedure).

For -200ER/GE90-94B or RR875 thru RR892:

1. For dispatch with center tank fuel greater than 22,000 lb (9,980 kg) and Gross Weight exceeding 460,000 lb (208,650 kg), Center-of-Gravity (CG) must be aft of 22% MAC for takeoff.
2. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 3,700 lb (1,678 kg) (for fuel that would be burned during a return-to-land procedure).

For -200ER/RR895:

1. For dispatch with center tank fuel greater than 22,000 lb (9,980 kg) and Gross Weight exceeding 460,000 lb (208,650 kg), Center-of-Gravity (CG) must be aft of 22% MAC for takeoff.
2. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 4,000 lb (1,814 kg) (for fuel that would be burned during a return-to-land procedure).

For -200LR or 777F/GE90-100 Series:

1. For dispatch with center tank fuel and Gross Weight exceeding 523,000 lb (237,230 kg), Center-of-Gravity (CG) must be aft of 25% MAC for takeoff.
2. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 4,000 lb (1,814 kg) (for fuel that would be burned during a return-to-land procedure).

For -300/RR884 thru RR892:

1. For dispatch with center tank fuel and Gross Weight exceeding 524,000 lb (238,000 kg), Center-of-Gravity (CG) must be aft of 17% MAC for takeoff.
2. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 3,900 lb (1,769 kg) (for fuel that would be burned during a return-to-land procedure).

For -300/PW4090:

1. For dispatch with center tank fuel and Gross Weight exceeding 524,000 lb (238,000 kg), Center-of-Gravity (CG) must be aft of 17% MAC for takeoff.

2. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 4,100 lb (1,860 kg) (for fuel that would be burned during a return-to-land procedure).

For -300ER/GE90-115B:

1. For dispatch with center tank fuel and Gross Weight exceeding 556,000 lb (252,200 kg), Center-of-Gravity (CG) must be aft of 21% MAC for takeoff.
2. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 4,200 lb (1,905 kg) (for fuel that would be burned during a return-to-land procedure).

Supplement 36-15
Superseded



777 FAA DDG Supplement Package

Supplement Number: 36-5

Issue Date: 17-JAN-2025

Section Number: 2 (MMEL)

Item Number: 28-31-03

Item Name: Main Tanks Jettison Pumps

DDG Effectivity: 777 FAA DDG (D630W003-TBC), Revision 36

MMEL Effectivity: 777 FAA MMEL, Revision 23a

Description: **Note: This item supersedes content in a previous supplement.**

Modified Operations (O) procedures for subitems 28-31-03-01 and 28-31-03-02.

Reason: Operations (O) procedures were modified to reflect updated performance adjustments. Operations (O) procedure numbering error introduced in previous supplement was corrected.

Disposition: For reference in the Operator's MEL.

Attachments: DDG pages 2,28-31-03.1-4

28-31-03 Main Tanks Jettison Pumps
28-31-03-01 Without PRR 62777-79S or Equivalent Installed

Interval	Installed	Required	Procedure
C	2	0	(M) (O)

May be inoperative provided:

- Inoperative pump is deactivated.
- Appropriate performance adjustments are applied.

NOTE: The PRR 62777-79S is applicable to airplane line numbers 1099 and-on. Service Bulletins 777-28-0078 (ELMS2) and 777-28-0082 (ELMS1) each provide the production equivalent of PRR 62777-79S.

MAINTENANCE (M)

Deactivate the inoperative pump(s) (AMM 28-00-00/901).

- For left main tank jettison pump inoperative, open and collar P110 panel L FUEL JETT PUMP CTRL circuit breaker.
- For right main tank jettison pump inoperative, open and collar P210 panel R FUEL JETT PUMP CTRL circuit breaker.

OPERATIONS (O)

NOTE: For one main tank jettison pump inoperative, fuel jettison of the opposite jettison pump is inhibited to prevent fuel imbalance.

For -200 or -200ER/PW4074 thru PW4084:

- Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 3,400 lb (1,542 kg) (for fuel that would be burned during a return-to-land procedure).

For -200 or -200ER/PW4090:

- Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 3,800 lb (1,724 kg) (for fuel that would be burned during a return-to-land procedure).

For -200 or -200ER/GE90-76B thru -90B:

- Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 3,300 lb (1,497 kg) (for fuel that would be burned during a return-to-land procedure).

For -200 or -200ER/GE90-94B or RR875 thru RR892:

- Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 3,700 lb (1,678 kg) (for fuel that would be burned during a return-to-land procedure).

For -200 or -200ER/RR895:

1. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 4,000 lb (1,814 kg) (for fuel that would be burned during a return-to-land procedure).

For -200LR or 777F/GE90-100 Series:

1. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 4,000 lb (1,814 kg) (for fuel that would be burned during a return-to-land procedure).

For -300/RR884 thru RR892:

1. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 3,900 lb (1,769 kg) (for fuel that would be burned during a return-to-land procedure).

For -300/PW4090:

1. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 4,100 lb (1,860 kg) (for fuel that would be burned during a return-to-land procedure).

For -300ER/GE90-115B:

1. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 4,200 lb (1,905 kg) (for fuel that would be burned during a return-to-land procedure).

28-31-03 Main Tanks Jettison Pumps
28-31-03-02 With PRR 62777-79S or Equivalent Installed

Interval	Installed	Required	Procedure
C	2	0	(M) (O)

May be inoperative provided:

- a. Inoperative pump is deactivated.
- b. Fuel scavenge system is considered inoperative.
- c. Appropriate performance adjustments are applied.

NOTE 1: The PRR 62777-79S is applicable to airplane line numbers 1099 and-on. Service Bulletins 777-28-0078 (ELMS2) and 777-28-0082 (ELMS1) each provide the production equivalent of PRR 62777-79S.

NOTE 2: The airplane must also be dispatched using MMEL item 28-22-04.

MAINTENANCE (M)

Deactivate the inoperative pump(s) (AMM 28-00-00/901).

1. For left main tank jettison pump inoperative, open and collar P110 panel L FUEL JETT PUMP CTRL circuit breaker.
2. For right main tank jettison pump inoperative, open and collar P210 panel R FUEL JETT PUMP CTRL circuit breaker.

OPERATIONS (O)

NOTE: For one main tank jettison pump inoperative:

1. Fuel jettison of the opposite jettison pump is inhibited to prevent fuel imbalance.
2. Fuel scavenge of the opposite jettison pump operates normally.
3. Fuel scavenge rate is approximately half the normal fuel scavenge rate.

For -200 or -200ER/PW4074 thru PW4084:

1. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 3,400 lb (1,542 kg) (for fuel that would be burned during a return-to-land procedure).

For -200 or -200ER/PW4090:

1. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 3,800 lb (1,724 kg) (for fuel that would be burned during a return-to-land procedure).

For -200 or -200ER/GE90-76B thru -90B:

1. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 3,300 lb (1,497 kg) (for fuel that would be burned during a return-to-land procedure).

For -200 or -200ER/GE90-94B or RR875 thru RR892:

1. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 3,700 lb (1,678 kg) (for fuel that would be burned during a return-to-land procedure).

For -200 or -200ER/RR895:

1. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 4,000 lb (1,814 kg) (for fuel that would be burned during a return-to-land procedure).

For -200LR or 777F/GE90-100 Series:

1. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 4,000 lb (1,814 kg) (for fuel that would be burned during a return-to-land procedure).

For -300/RR884 thru RR892:

1. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 3,900 lb (1,769 kg) (for fuel that would be burned during a return-to-land procedure).

For -300/PW4090:

1. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 4,100 lb (1,860 kg) (for fuel that would be burned during a return-to-land procedure).

For -300ER/GE90-115B:

1. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 4,200 lb (1,905 kg) (for fuel that would be burned during a return-to-land procedure).

Supplement 36-5
Superseded



777 FAA DDG Supplement Package

Supplement Number: 36-5

Issue Date: 17-JAN-2025

Section Number: 2 (MMEL)

Item Number: 28-31-04

Item Name: Fuel Jettison Nozzle Valves

DDG Effectivity: 777 FAA DDG (D630W003-TBC), Revision 36

MMEL Effectivity: 777 FAA MMEL, Revision 23a

Description: **Note: This item supersedes content in a previous supplement.**

- 1) Modified the AMM references in the subitem options 28-31-04-01B and 28-31-04-02B Maintenance (M) procedures.
- 2) Modified Operations (O) procedures for subitem options 28-31-04-01B and 28-31-04-02B.

Reason:

- 1) The AMM references were incorrectly modified in the previous DDG revision.
- 2) Operations (O) procedures were modified to reflect updated performance adjustments. Operations (O) procedure numbering error introduced in previous supplement was corrected.

Disposition: For reference in the Operator's MEL.

Attachments: DDG pages 2.28-31-04.1-7

28-31-04 Fuel Jettison Nozzle Valves
28-31-04-01 Without PRR 62777-79S or Equivalent Installed
28-31-04-01A One Inoperative

Interval	Installed	Required	Procedure
C	2	1	(M)

One nozzle valve may be inoperative provided it is secured closed.

NOTE: The PRR 62777-79S is applicable to airplane line numbers 1099 and-on. Service Bulletins 777-28-0078 (ELMS2) and 777-28-0082 (ELMS1) each provide the production equivalent of PRR 62777-79S.

MAINTENANCE (M)

Secure the inoperative fuel jettison nozzle valve in the closed position (AMM 28-00-00/901).

1. For left jettison nozzle valve inoperative:
 - A. Open and collar P110 panel L FUEL JETT NOZ VLV circuit breaker.
 - B. Gain access to the left jettison nozzle valve actuator.
 - C. Disconnect, cap and stow the electrical connector (DV28166).
 - D. Confirm the manual override handle is in the CLOSED position and lockwire.
2. For right jettison nozzle valve inoperative:
 - A. Open and collar P210 panel R FUEL JETT NOZ VLV circuit breaker.
 - B. Gain access to the right jettison nozzle valve actuator.
 - C. Disconnect, cap and stow the electrical connector (DV28266).
 - D. Confirm the manual override handle is in the CLOSED position and lockwire.

28-31-04 Fuel Jettison Nozzle Valves
28-31-04-01 Without PRR 62777-79S or Equivalent Installed
28-31-04-01B Both Inoperative

Interval	Installed	Required	Procedure
C	2	0	(M) (O)

May be inoperative provided:

- a. Both jettison nozzle valves are secured closed.
- b. Jettison system is deactivated.
- c. Appropriate performance penalties are applied.

NOTE: The PRR 62777-79S is applicable to airplane line numbers 1099 and-on. Service Bulletins 777-28-0078 (ELMS2) and 777-28-0082 (ELMS1) each provide the production equivalent of PRR 62777-79S.

MAINTENANCE (M)

Secure the fuel jettison nozzle valves in the closed position and deactivate the jettison system (AMM 28-00-00/901).

1. For left jettison nozzle valve:
 - A. Open and collar P110 panel L FUEL JETT NOZ VLV circuit breaker.
 - B. Gain access to the left jettison nozzle valve actuator.
 - C. Disconnect, cap and stow the electrical connector (DV28166).
 - D. Confirm the manual override handle is in the CLOSED position and lockwire.
2. For right jettison nozzle valve:
 - A. Open and collar P210 panel R FUEL JETT NOZ VLV circuit breaker.
 - B. Gain access to the right jettison nozzle valve actuator.
 - C. Disconnect, cap and stow the electrical connector (DV28266).
 - D. Confirm the manual override handle is in the CLOSED position and lockwire.
3. Deactivate the fuel jettison system.
 - A. Open and collar P110 panel L FUEL JETT PUMP CTRL and L FUEL JETT ISLN VLV circuit breakers.
 - B. Open and collar P210 panel R FUEL JETT PUMP CTRL and R FUEL JETT ISLN VLV circuit breakers.

OPERATIONS (O)

NOTE: The Fuel Synoptic will incorrectly indicate jettison flow from the center tank when the center tank override pumps are on with fuel in the center tank. The FUEL JETT NOZZLE L and R advisory messages will be displayed and overhead panel L and R fuel jettison nozzle VALVE lights will be illuminated.

For -200 or -200ER/PW4074 thru PW4084:

1. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 3,400 lb (1,542 kg) (for fuel that would be burned during a return-to-land procedure).

For -200 or -200ER/PW4090:

1. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 3,800 lb (1,724 kg) (for fuel that would be burned during a return-to-land procedure).

For -200 or -200ER/GE90-76B thru -90B:

1. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 3,300 lb (1,497 kg) (for fuel that would be burned during a return-to-land procedure).

For -200 or -200ER/GE90-94B or RR875 thru RR892:

1. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 3,700 lb (1,678 kg) (for fuel that would be burned during a return-to-land procedure).

For -200 or -200ER/RR895:

1. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 4,000 lb (1,814 kg) (for fuel that would be burned during a return-to-land procedure).

For -200LR or 777F/GE90-100 Series:

1. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 4,000 lb (1,814 kg) (for fuel that would be burned during a return-to-land procedure).

For -300/RR884 thru RR892:

1. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 3,900 lb (1,769 kg) (for fuel that would be burned during a return-to-land procedure).

For -300/PW4090:

1. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 4,100 lb (1,860 kg) (for fuel that would be burned during a return-to-land procedure).

For -300ER/GE90-115B:

1. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 4,200 lb (1,905 kg) (for fuel that would be burned during a return-to-land procedure).

28-31-04 Fuel Jettison Nozzle Valves
28-31-04-01 Without PRR 62777-79S or Equivalent Installed
28-31-04-01-01 Switch VALVE Lights

Interval	Installed	Required	Procedure
C	2	0	

28-31-04 Fuel Jettison Nozzle Valves
28-31-04-01 Without PRR 62777-79S or Equivalent Installed
28-31-04-01-02 Switch ON Lights

Interval	Installed	Required	Procedure
C	2	0	

28-31-04 Fuel Jettison Nozzle Valves
28-31-04-02 With PRR 62777-79S or Equivalent Installed
28-31-04-02A One Inoperative

Interval	Installed	Required	Procedure
C	2	1	(M)

One nozzle valve may be inoperative provided it is secured closed.

NOTE: The PRR 62777-79S is applicable to airplane line numbers 1099 and-on. Service Bulletins 777-28-0078 (ELMS2) and 777-28-0082 (ELMS1) each provide the production equivalent of PRR 62777-79S.

MAINTENANCE (M)

Secure the inoperative fuel jettison nozzle valve in the closed position (AMM 28-00-00/901).

1. For left jettison nozzle valve inoperative:
 - A. Open and collar P110 panel L FUEL JETT NOZ VLV circuit breaker.
 - B. Gain access to the left jettison nozzle valve actuator.
 - C. Disconnect, cap and stow the electrical connector (DV28166).
 - D. Confirm the manual override handle is in the CLOSED position and lockwire.
2. For right jettison nozzle valve inoperative:
 - A. Open and collar P210 panel R FUEL JETT NOZ VLV circuit breaker.
 - B. Gain access to the right jettison nozzle valve actuator.
 - C. Disconnect, cap and stow the electrical connector (DV28266).
 - D. Confirm the manual override handle is in the CLOSED position and lockwire.

28-31-04 Fuel Jettison Nozzle Valves
28-31-04-02 With PRR 62777-79S or Equivalent Installed
28-31-04-02B Both Inoperative

Interval	Installed	Required	Procedure
C	2	0	(M) (O)

May be inoperative provided:

- a. Both jettison nozzle valves are secured closed.
- b. Jettison system is deactivated.
- c. Fuel scavenge system is considered inoperative.

- d. Appropriate performance penalties are applied.
-

NOTE 1: The PRR 62777-79S is applicable to airplane line numbers 1099 and-on. Service Bulletins 777-28-0078 (ELMS2) and 777-28-0082 (ELMS1) each provide the production equivalent of PRR 62777-79S.

NOTE 2: The airplane must also be dispatched using MMEL item 28-22-04.

MAINTENANCE (M)

Secure the fuel jettison nozzle valves in the closed position and deactivate the jettison system (AMM 28-00-00/901).

1. For left jettison nozzle valve:
 - A. Open and collar P110 panel L FUEL JETT NOZ VLV circuit breaker.
 - B. Gain access to the left jettison nozzle valve actuator.
 - C. Disconnect, cap and stow the electrical connector (DV28166).
 - D. Confirm the manual override handle is in the CLOSED position and lockwire.
2. For right jettison nozzle valve:
 - A. Open and collar P210 panel R FUEL JETT NOZ VLV circuit breaker.
 - B. Gain access to the right jettison nozzle valve actuator.
 - C. Disconnect, cap and stow the electrical connector (DV28266).
 - D. Confirm the manual override handle is in the CLOSED position and lockwire.
3. Deactivate the fuel jettison system.
 - A. Open and collar P110 panel L FUEL JETT PUMP CTRL and L FUEL JETT ISLN VLV circuit breakers.
 - B. Open and collar P210 panel R FUEL JETT PUMP CTRL and R FUEL JETT ISLN VLV circuit breakers.

OPERATIONS (O)

NOTE: The Fuel Synoptic will incorrectly indicate jettison flow from the center tank when the center tank override pumps are on with fuel in the center tank. The FUEL JETT NOZZLE L and R advisory messages will be displayed and overhead panel L and R fuel jettison nozzle VALVE lights will be illuminated.

For -200 or -200ER/PW4074 thru PW4084:

1. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 3,400 lb (1,542 kg) (for fuel that would be burned during a return-to-land procedure).

For -200 or -200ER/PW4090:

1. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 3,800 lb (1,724 kg) (for fuel that would be burned during a return-to-land procedure).

For -200 or -200ER/GE90-76B thru -90B:

1. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 3,300 lb (1,497 kg) (for fuel that would be burned during a return-to-land procedure).

For -200 or -200ER/GE90-94B or RR875 thru RR892:

1. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 3,700 lb (1,678 kg) (for fuel that would be burned during a return-to-land procedure).

For -200 or -200ER/RR895:

1. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 4,000 lb (1,814 kg) (for fuel that would be burned during a return-to-land procedure).

For -200LR or 777F/GE90-100 Series:

1. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 4,000 lb (1,814 kg) (for fuel that would be burned during a return-to-land procedure).

For -300/RR884 thru RR892:

1. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 3,900 lb (1,769 kg) (for fuel that would be burned during a return-to-land procedure).

For -300/PW4090:

1. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 4,100 lb (1,860 kg) (for fuel that would be burned during a return-to-land procedure).

For -300ER/GE90-115B:

1. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 4,200 lb (1,905 kg) (for fuel that would be burned during a return-to-land procedure).

28-31-04 Fuel Jettison Nozzle Valves
28-31-04-02 With PRR 62777-79S or Equivalent Installed
28-31-04-02-01 Switch VALVE Lights

Interval	Installed	Required	Procedure
C	2	0	



28-31-04 Fuel Jettison Nozzle Valves
28-31-04-02 With PRR 62777-79S or Equivalent Installed
28-31-04-02-02 Switch ON Lights

Interval	Installed	Required	Procedure
C	2	0	

Supplement 36-5



777 FAA DDG Supplement Package

Supplement Number: 36-6

Issue Date: 26-SEP-2025

Section Number: 2 (MMEL)

Item Number: 21-52-05

Item Name: Ram Air Inlet Door Systems

DDG Effectivity: 777 FAA DDG (D630W003-TBC), Revision 36

MMEL Effectivity: 777 FAA MMEL, Revision 23a

Description: **Note: This item supersedes content in a previous supplement.**

Modified DDG top note for option 21-52-05A and modified Maintenance (M) procedures for option 21-52-05B.

Reason:

- 1) To clarify that the DDG top note is applicable to both passenger and freighter airplanes.
- 2) The Maintenance (M) procedure was modified to reflect the correct RAM AIR INLET indication and to clarify the requirement for the cover plate.

Disposition: For reference in the Operator's MEL.

Attachments: DDG pages 2.21-52-05.1-3

**21-52-05 Ram Air Inlet Door Systems****21-52-05A Associated Pack Used**

Interval	Installed	Required	Procedure
C	2	1	(M) (O)

One may be inoperative provided:

- Associated door is secured full open.
- Appropriate performance adjustments are applied.

NOTE: To avoid cold flight deck temperatures, the left Trim Air Control System and left Trim Air PRSOV should operate normally when the left Ram Air Inlet Door is secured full open.

MAINTENANCE (M)

Secure the inoperative ram air inlet door full open (AMM 21-00-00/901).

- For left ram air inlet door inoperative:
 - Open P110 panel L CTC CHAN 1 (AC) circuit breaker.
 - Open P110 panel L CTC CHAN 1 (DC) circuit breaker.
 - Open P110 panel L CTC CHAN 2 (AC) circuit breaker.
 - Open P110 panel L CTC CHAN 2 (DC) circuit breaker.
 - Open the ram air inlet actuator door 191LL.
- For right ram air inlet door inoperative:
 - Open P210 panel R CTC CHAN 1 (AC) circuit breaker.
 - Open P210 panel R CTC CHAN 1 (DC) circuit breaker.
 - Open P210 panel R CTC CHAN 2 (AC) circuit breaker.
 - Open P210 panel R CTC CHAN 2 (DC) circuit breaker.
 - Open the ram air inlet actuator door 192LR.
- Disconnect, cap, and stow the electrical connector from the actuator.
- Disconnect the actuator rod end from the bellcrank.
- Loosen the screws on the fixed link, and rotate the fixed link 180 degrees.
- Position the inlet door to the fully open position.
- Connect the fixed link to the bellcrank.
- Tighten the screws on the fixed link to 18-28 pound-inches.
- Close the ram air inlet actuator access door.
- Close the associated circuit breakers.

OPERATIONS (O)

- Reduce performance limited weights by the appropriate adjustments.

Performance Limited Weights	Takeoff & Landing	Enroute Climb
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For -200 or -200ER/GE90-76B thru -90B:

thru 632,500 lb (286,902 kg)	400 lb (181 kg)	2,100 lb (953 kg)
over 632,500 lb (286,902 kg)	500 lb (227 kg)	2,200 lb (998 kg)

For -200 or -200ER/GE90-94B:

thru 632,500 lb (286,902 kg)	400 lb (181 kg)	2,600 lb (1,179 kg)
over 632,500 lb (286,902 kg)	700 lb (317 kg)	2,600 lb (1,179 kg)

For -200 or -200ER/PW4074 thru PW4084:

All	400 lb (181 kg)	2,300 lb (1,043 kg)
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For -200 or -200ER/PW4090:

thru 632,500 lb (286,902 kg)	400 lb (181 kg)	2,300 lb (1,043 kg)
over 632,500 lb (286,902 kg)	500 lb (227 kg)	2,300 lb (1,043 kg)

For -200 or -200ER/RR875 thru RR895:

thru 632,500 lb (286,902 kg)	400 lb (181 kg)	2,200 lb (998 kg)
over 632,500 lb (286,902 kg)	600 lb (272 kg)	2,200 lb (998 kg)

For -200LR or 777F/GE90-100 Series:

All	500 lb (227 kg)	2,300 lb (1,043 kg)
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For -300/PW4090:

All	400 lb (181 kg)	2,300 lb (1,043 kg)
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For -300/RR892:

All	400 lb (181 kg)	2,100 lb (953 kg)
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For -300ER/GE90-115B:

All	400 lb (181 kg)	2,300 lb (1,043 kg)
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NOTE: The enroute climb weight penalties are based on single engine operating speeds that approximate maximum lift-to-drag ratio speed. To account for the difference in level off altitude when operating at other speeds, multiply the enroute climb weight penalty listed above by the appropriate factor listed on page 3.00-01-00.2 in the ENROUTE DIVERSION SPEED EFFECTS paragraph.

2. Increase flight planning fuel by 0.7%

21-52-05 Ram Air Inlet Door Systems

21-52-05B Associated Pack Not Used

Interval	Installed	Required	Procedure
C	2	1	(M) (O)

One may be inoperative provided:

- a. Associated inlet door is verified closed.
- b. Associated pack is not used.

MAINTENANCE (M)

Verify the ram air inlet door is closed for dispatch.

1. On the AIR CONDITIONING maintenance page, verify that the associated RAM AIR INLET indicates 0.97 or greater.

NOTE: If the ram air inlet door cannot be verified closed for dispatch, install a ram air inlet door cover plate (AMM 21-00-00/901)

OPERATIONS (O)

Position associated pack switch OFF.

NOTE: The airplane must also be dispatched using MMEL item 21-51-01 Operations procedure.



777 FAA DDG Supplement Package

Supplement Number: 36-6

Issue Date: 26-SEP-2025

Section Number: 2 (MMEL)

Item Number: 24-25-01

Item Name: Backup Electrical Power System (Backup Generators, Converter and CCBs)

DDG Effectivity: 777 FAA DDG (D630W003-TBC), Revision 36

MMEL Effectivity: 777 FAA MMEL, Revision 23a

Description: Added DDG top note and modified Maintenance Note and Operations Note for option 24-25-01A.

Reason: To clarify the notes related to dispatch instead of maintenance actions as DDG top notes and added AMM procedure reference to the Maintenance Note. The Operations Note was modified to provide clarification on the expected EICAS messages.

Disposition: For reference in the Operator's MEL.

Attachments: DDG pages 2.24-25-01.1-3

24-25-01 Backup Electrical Power System (Backup Generators, Converter and CCBs)

24-25-01A Generator Installed

Interval	Installed	Required	Procedure
C	1	0	

May be inoperative provided:

- Both engine driven generators operate normally.
- APU generator operates normally.

NOTE 1: A sheared generator shaft is indicated by associated maintenance message 24-46000 or 24-46010, and the PFCS INTERFACE status message.

NOTE 2: For the PFCS INTERFACE status message displayed, the airplane must also be dispatched using MMEL item 27-02-04.

MAINTENANCE NOTE

- To prevent damage to the inoperative backup generator, confirm the generator's oil system status is normal when its associated shaft is not sheared (AMM 24-00-00/901).
 - Confirm oil level status and oil filter status indicate NORMAL using Electrical Maintenance page 1/2 or using the sightglass.
 - Confirm associated 24-46080 or 24-46090 maintenance message for low oil pressure is not displayed or check the differential pressure indicator at the backup generator.

NOTE: Backup generator P/N 757066G (S281W401-12) and P/N 762620D (S281W401-15) and earlier require compliance with Airworthiness Directive (AD) 2000-13-04. Information for various backup generator configurations, service bulletins, PRRs and AD compliance is contained in Boeing Service Letter 777-SL-24-030-A.

OPERATIONS NOTE

- The Electrical Synoptic Display may have incomplete or missing information.
- If an engine with an inoperative Backup Generator is started first, the ELEC BACKUP SYS advisory message will be displayed.
- For ELEC BACKUP SYS advisory message displayed after both engines are started, confirm FLIGHT CONTROL SYS status message is not displayed.

**24-25-01 Backup Electrical Power System (Backup Generators, Converter and CCBs)****24-25-01B Generator Removed**

Interval	Installed	Required	Procedure
C	1	0	(M)

May be inoperative provided:

- One backup generator is removed and an appropriate dispatch kit is installed.
- Both engine driven generators operate normally.
- APU generator operates normally.

NOTE 1: The airplane must also be dispatched using MMEL item 27-02-04.

NOTE 2: The airplane must have PRR 61703 installed, or the service equivalent SB 777-27-0023 installed, to remove a backup generator.

MAINTENANCE (M)

Remove inoperative backup generator and install associated dispatch kit (AMM 24-00-00/901).

1. Remove the backup generator (AMM 24-25-01/401).

NOTE: The PFCS INTERFACE status message will be displayed.

2. Remove the union (if installed) and check valve from the oil-in port on the removed backup generator.
3. Install dispatch kit (AMM 24-25-16/401).

For PW:

- A. Kit J24012.

For GE:

- B. Kit J24013.

For RR:

- C. Kit J24011.

OPERATIONS NOTE

1. The following weight and balance information is provided:
 - A. Backup generator:
 - 1) Weight: 43.5 lb (19.8 kg)
 - 2) Balance Arm: 979.8 in.
 - B. Dispatch kit:

For PW and RR:

- 1) Weight: 9.1 lb (4.2 kg)
- 2) Balance Arm: 983.6 in.

For GE:

- 1) Weight: 14 lb (6.3 kg)
- 2) Balance Arm: 983.6 in.
2. The Electrical Synoptic Display may have incomplete or missing information.
3. For ELEC BACKUP SYS advisory message displayed after engine start, confirm FLIGHT CONTROL SYS status message is not displayed.

24-25-01 Backup Electrical Power System (Backup Generators, Converter and CCBs)

24-25-01-01 BACKUP GEN Switch Lights

24-25-01-01-01 OFF Lights

Interval	Installed	Required	Procedure
C	2	0	

24-25-01 Backup Electrical Power System (Backup Generators, Converter and CCBs)

24-25-01-01 BACKUP GEN Switch Lights

24-25-01-01-02 ON Lights

Interval	Installed	Required	Procedure
C	2	0	



777 FAA DDG Supplement Package

Supplement Number: 36-6

Issue Date: 26-SEP-2025

Section Number: 2 (MMEL)

Item Number: 25-53-02

Item Name: Lower Cargo Restraint Systems

DDG Effectivity: 777 FAA DDG (D630W003-TBC), Revision 36

MMEL Effectivity: 777 FAA MMEL, Revision 23a

Description: **Note: This item supersedes content in a previous supplement.**

- 1) Added AMM procedure reference to option A.
- 2) Added Maintenance notes and moved existing Maintenance steps to Operations notes.

Reason:

- 1) The AMM procedure reference was unintentionally removed in revision 36.
- 2) To provide guidance for maintaining structural integrity and stabilizing cargo floor beams and frames with missing lower cargo restraint system equipment and to clarify the number of inoperative or missing restraint components refers to the affected cargo compartment.

These structural requirements are planned to be added at the next MMEL revision.

Disposition: For reference in the Operator's MEL.

Attachments: DDG pages 2.25-53-02.1-3

25-53-02 Lower Cargo Restraint Systems

25-53-02A Cargo Compartment Used

Interval	Installed	Required	Procedure
A	-	-	(M)

May be inoperative or missing provided:

- Approved cargo loading limits are observed. The only source documents are the Type Certificate (TC), Supplemental Type Certificate (STC), Airplane Flight Manual (AFM), Airplane Flight Manual Supplement (AFMS) or the TC/STC Weight and Balance Manual (WBM).
- Repairs are made within 120 consecutive calendar-days.

MAINTENANCE (M)

NOTE 1: The cargo handling system doorsill hardware must be installed for flight. If hardware is missing, fasteners common to the doorsill and airplane structure (sill chord/web, or sill chord/web/stub-frame) must be reinstalled prior to flight. Use additional washers with existing grip length bolt if required, or replace bolt with shorter grip length. (AMM 25-00-00/901)

NOTE 2: The cargo handling system left and right BL 6.5, 19.5, 41.5 and 58.5 roller trays must remain installed to stabilize cargo floor beams and frames.

NOTE 3: The barrier net separating the aft and bulk cargo compartments is considered a cargo restraint component. Refer to the Weight and Balance Manual for approved cargo-loading limits if the barrier net is inoperative or missing.

NOTE 4: The door protector net in the bulk cargo compartment must be installed to keep shifted cargo from interfering with the operation of the cargo door.

NOTE 5: Deactivation procedures may or may not be required. Develop deactivation procedures depending on system configuration.

OPERATIONS NOTE

NOTE: Use acceptable cargo loading limits from an approved source.

- Refer to the Weight and Balance Manual for cargo loading limits.
- If the number of inoperative or missing restraint components for the affected cargo compartment exceeds the allowance in the Weight and Balance Manual, the affected cargo compartment cannot be used. Refer to MMEL Item 25-53-02B.

**25-53-02 Lower Cargo Restraint Systems****25-53-02B Cargo Compartment(s) Remains Empty**

Interval	Installed	Required	Procedure
A	-	-	

May be inoperative or missing provided:

- Cargo compartment remains empty.
- Repairs are made within 120 consecutive calendar-days.

MAINTENANCE NOTE

NOTE 1: The cargo handling system doorsill hardware must be installed for flight. If hardware is missing, fasteners common to the doorsill and airplane structure (sill chord/web, or sill chord/web/stub-frame) must be reinstalled prior to flight. Use additional washers with existing grip length bolt if required, or replace bolt with shorter grip length. (AMM 25-00-00/901)

NOTE 2: The cargo handling system left and right BL 6.5, 19.5, 41.5 and 58.5 roller trays must remain installed to stabilize cargo floor beams and frames.

25-53-02 Lower Cargo Restraint Systems**25-53-02C Individual Cargo Areas**

Interval	Installed	Required	Procedure
A	-	-	

Individual cargo areas may be inoperative provided:

- Aircraft is operated in accordance with Original Equipment Manufacturer (OEM) W&B source document.
- Repairs are made within 120 consecutive calendar-days.

MAINTENANCE NOTE

NOTE 1: The cargo handling system doorsill hardware must be installed for flight. If hardware is missing, fasteners common to the doorsill and airplane structure (sill chord/web, or sill chord/web/stub-frame) must be reinstalled prior to flight. Use additional washers with existing grip length bolt if required, or replace bolt with shorter grip length. (AMM 25-00-00/901)

NOTE 2: The cargo handling system left and right BL 6.5, 19.5, 41.5 and 58.5 roller trays must remain installed to stabilize cargo floor beams and frames.

NOTE 3: Deactivation procedures may or may not be required. Develop deactivation procedures depending on system configuration.

OPERATIONS NOTE

NOTE: Use acceptable cargo loading limits from an approved source.

1. Refer to the Weight and Balance Manual for cargo loading limits.
2. If the number of inoperative or missing restraint components for the affected cargo compartment exceeds the allowance in the Weight and Balance Manual, the affected cargo compartment cannot be used. Refer to MMEL Item 25-53-02B.



777 FAA DDG Supplement Package

Supplement Number: 36-6

Issue Date: 26-SEP-2025

Section Number: 2 (MMEL)

Item Number: 27-11-03

Item Name: Aileron Power Control Units (PCUs)

DDG Effectivity: 777 FAA DDG (D630W003-TBC), Revision 36

MMEL Effectivity: 777 FAA MMEL, Revision 23a

Description: Modified Operations (O) procedures for subitem 27-11-03-02.

Reason: To reflect updated Center of Gravity (CG) limits observation step for different engine thrust ratings.

Disposition: For reference in the Operator's MEL.

Attachments: DDG pages 2.27-11-03.1-4

27-11-03 Aileron Power Control Units (PCUs)
27-11-03-01 Passenger (With PRRs 61384-15 and 62200 or Equivalent Installed)

Interval	Installed	Required	Procedure
A	4	3	(M)

One may be inoperative provided:

- Inoperative PCU is deactivated in the bypass mode.
- Inoperative PCU blocking mode is verified to operate normally.
- Ailerons are verified to operate normally.
- Repairs are made within 3 flight days.

NOTE: PRRs 61384-15 and 62200 Part-B are applicable to airplane production L/Ns 718, 772, 774 and 776 and-on. This item is also applicable to airplanes with the AIMS-1 or AIMS-2 platforms with BP V14 or later AIMS OPS installed together with the following PFC software installed: PFC OPS 2762-PFC-741-00, or PFC OPS 276D-PFC-841-00, or PFC OPS 2768-PFC-910-00 or later installed.

MAINTENANCE (M)

Deactivate the inoperative aileron PCU in the bypass mode, verify blocking mode operates normally and verify ailerons operate normally (AMM 27-00-00/901).

- For an inoperative left aileron PCU:
 - Open the Power Supply Assembly Center (PSA-C) in E2-6 rack AIL PCU circuit breaker.
 - Open P110 panel AIL PCU LIB(BLK)/LOB(BYP) circuit breaker.
 - Gain access to the left aileron PCUs.
 - Verify blocking mode operates normally on the inoperative PCU.
 - Deactivate the inoperative PCU in bypass mode.
 - For an inoperative left inboard aileron PCU, collar the Power Supply Assembly Center (PSA-C) in E2-6 rack AIL PCU circuit breaker.
 - For an inoperative left outboard aileron PCU, collar the P110 panel AIL PCU LIB(BLK)/LOB(BYP) circuit breaker.
 - Close the operative aileron PCU circuit breaker.
- For an inoperative right aileron PCU:
 - Open the Power Supply Assembly Right (PSA-R) in E5-1 rack AIL PCU circuit breaker.
 - Open the Power Supply Assembly Left (PSA-L) in E1-6 rack AIL PCU circuit breaker.
 - Gain access to the right aileron PCUs.



- D. Verify blocking mode operates normally on the inoperative PCU.
- E. Deactivate the inoperative PCU in bypass mode.
 - 1) For an inoperative right inboard aileron PCU, collar the Power Supply Assembly Right (PSA-R) in E5-1 rack AIL PCU circuit breaker.
 - 2) For an inoperative right outboard aileron PCU, collar the Power Supply Assembly Left (PSA-L) in E1-6 rack AIL PCU circuit breaker.
 - 3) Close the operative aileron PCU circuit breaker.
3. Check the operation of the ailerons.

NOTE: Do not do an Actuator Confidence Test, or an ACE Monitor Test, or L or R Aileron Actuator Confidence Tests.
4. Confirm the FLIGHT CONTROL SYS status message is not displayed.

OPERATIONS NOTE

When airspeed exceeds approximately 250 KIAS, air loads may cause the associated aileron to move slightly to a trailing-edge-up position while the opposite aileron will normally be commanded to a slight trailing-edge-down position. Some trim input may be necessary to correct for the resulting roll.

27-11-03 Aileron Power Control Units (PCUs)**27-11-03-02 777F**

Interval	Installed	Required	Procedure
A	4	3	(M) (O)

One may be inoperative provided:

- a. Inoperative PCU is deactivated in the bypass mode.
- b. Inoperative PCU blocking mode is verified to operate normally.
- c. Ailerons are verified to operate normally.
- d. Appropriate performance adjustments are applied.
- e. Repairs are made within 3 flight days.

MAINTENANCE (M)

Deactivate the inoperative aileron PCU in the bypass mode, verify blocking mode operates normally and verify ailerons operate normally (AMM 27-00-00/901).

1. For an inoperative left aileron PCU:
 - A. Open the Power Supply Assembly Center (PSA-C) in E2-6 rack AIL PCU circuit breaker.
 - B. Open P110 panel AIL PCU LIB(BLK)/LOB(BYP) circuit breaker.
 - C. Gain access to the left aileron PCUs.
 - D. Verify blocking mode operates normally on the inoperative PCU.

- E. Deactivate the inoperative PCU in bypass mode.
 - 1) For an inoperative left inboard aileron PCU, collar the Power Supply Assembly Center (PSA-C) in E2-6 rack AIL PCU circuit breaker.
 - 2) For an inoperative left outboard aileron PCU, collar the P110 panel AIL PCU LIB(BLK)/LOB(BYP) circuit breaker.
 - 3) Close the operative aileron PCU circuit breaker.
2. For an inoperative right aileron PCU:
 - A. Open the Power Supply Assembly Right (PSA-R) in E5-1 rack AIL PCU circuit breaker.
 - B. Open the Power Supply Assembly Left (PSA-L) in E1-6 rack AIL PCU circuit breaker.
 - C. Gain access to the right aileron PCUs.
 - D. Verify blocking mode operates normally on the inoperative PCU.
 - E. Deactivate the inoperative PCU in bypass mode.
 - 1) For an inoperative right inboard aileron PCU, collar the Power Supply Assembly Right (PSA-R) in E5-1 rack AIL PCU circuit breaker.
 - 2) For an inoperative right outboard aileron PCU, collar the Power Supply Assembly Left (PSA-L) in E1-6 rack AIL PCU circuit breaker.
 - 3) Close the operative aileron PCU circuit breaker.
3. Check the operation of the ailerons.

NOTE: Do not do an Actuator Confidence Test, or an ACE Monitor Test, or L or R Aileron Actuator Confidence Tests.
4. Confirm the FLIGHT CONTROL SYS status message is not displayed.

OPERATIONS (O)

NOTE: When airspeed exceeds approximately 250 KIAS, air loads may cause the associated aileron to move slightly to a trailing-edge-up position while the opposite aileron will normally be commanded to a slight trailing-edge-down position. Some trim input may be necessary to correct for the resulting roll.

1. Observe Maximum Gross Weight limits.
 - A. Taxi Weight limit is 722,300 lb (327,629 kg).
 - B. Takeoff Weight limit is 720,300 lb (326,722 kg).
 - C. Landing Weight limit is 575,000 lb (260,815 kg).
 - D. Zero Fuel Weight limit is 547,000 lb (248,115 kg).

For 777F/GE90-110B:

2. Observe Center of Gravity (CG) limits.
 - A. Forward Takeoff and Landing CG limits:
 - 1) 280,000 lb (127,006 kg) at 20.0% MAC.

- 2) 722,300 lb (327,629 kg) at 20.0% MAC.
- B. Forward Zero Fuel CG limits:
 - 1) 280,000 lb (127,006 kg) at 20.0% MAC.
 - 2) 460,000 lb (208,652 kg) at 20.0% MAC.
 - 3) 547,000 lb (248,115 kg) at 28.0% MAC.
- C. Aft Takeoff, Landing, and Zero Fuel CG limits:
 - 1) 280,000 lb (127,006 kg) at 24.8% MAC.
 - 2) 321,900 lb (146,011 kg) at 33.3% MAC.
 - 3) 450,000 lb (204,117 kg) at 39.5% MAC.
 - 4) 547,000 lb (248,115 kg) at 42.6% MAC.
 - 5) 575,000 lb (260,815 kg) at 43.3% MAC.
 - 6) 603,600 lb (273,788 kg) at 44.0% MAC.
 - 7) 609,000 lb (276,237 kg) at 44.0% MAC.
 - 8) 705,300 lb (319,918 kg) at 41.2% MAC.
 - 9) 722,300 lb (327,629 kg) at 39.9% MAC.

For 777F/GE90-115B:

3. Observe Center of Gravity (CG) limits.
 - A. Forward Takeoff and Landing CG limits:
 - 1) 280,000 lb (127,006 kg) at 20.0% MAC.
 - 2) 722,300 lb (327,629 kg) at 20.0% MAC.
 - B. Forward Zero Fuel CG limits:
 - 1) 280,000 lb (127,006 kg) at 20.0% MAC.
 - 2) 460,000 lb (208,652 kg) at 20.0% MAC.
 - 3) 547,000 lb (248,115 kg) at 28.0% MAC.
 - C. Aft Takeoff, Landing, and Zero Fuel CG limits:
 - 1) 280,000 lb (127,006 kg) at 20.7% MAC.
 - 2) 325,000 lb (147,418 kg) at 30.8% MAC.
 - 3) 340,550 lb (154,471 kg) at 33.5% MAC.
 - 4) 450,000 lb (204,117 kg) at 38.9% MAC.
 - 5) 547,000 lb (248,115 kg) at 42.0% MAC.
 - 6) 575,000 lb (260,815 kg) at 42.7% MAC.
 - 7) 615,600 lb (279,231 kg) at 43.7% MAC.
 - 8) 705,300 lb (319,918 kg) at 41.2% MAC.
 - 9) 722,300 lb (327,629 kg) at 39.9% MAC.



777 FAA DDG Supplement Package

Supplement Number: 36-6

Issue Date: 26-SEP-2025

Section Number: 2 (MMEL)

Item Number: 27-61-03

Item Name: Spoiler Power Control Units (PCUs)

DDG Effectivity: 777 FAA DDG (D630W003-TBC), Revision 36

MMEL Effectivity: 777 FAA MMEL, Revision 23a

Description: Modified Operations (O) procedures for subitem 27-61-03-02.

Reason: To reflect updated Center of Gravity (CG) limits observation step for different thrust ratings.

Disposition: For reference in the Operator's MEL.

Attachments: DDG pages 2.27-61-03.1-6

27-61-03 Spoiler Power Control Units (PCUs)
27-61-03-01 Passenger (With PRRs 61384-15 and 62200, or Equivalent Installed)

Interval	Installed	Required	Procedure
A	14	12	(M) (O)

One symmetrical pair (except 4/11) may be inoperative provided:

- Symmetrical pair is deactivated in the retracted position.
- Remaining spoilers are verified to operate normally.
- Appropriate performance adjustments are applied.
- Repairs are made within 3 flight days.

NOTE 1: Dispatch relief does not apply for a fault condition that is reported to cause asymmetric roll input, or for a spoiler panel detached from the actuator, or for a spoiler not in the retracted position.

NOTE 2: PRRs 61384-15 and 62200 Part-B are applicable to airplane production L/Ns 718, 772, 774 and 776 and-on. This item is also applicable to airplanes with the AIMS-1 or AIMS-2 platforms with BP V14 or later AIMS OPS installed together with the following PFC software installed: PFC OPS 2762-PFC-741-00, or PFC OPS 276D-PFC-841-00, or PFC OPS 2768-PFC-910-00 or later installed.

MAINTENANCE (M)

Deactivate the symmetrical spoiler pair in the retracted position and verify remaining spoilers operate normally (AMM 27-00-00/901).

- Confirm speedbrake lever is in DOWN position.
- For spoilers 1/14 or 7/8, open the Power Supply Assembly Center (PSA-C) in E2-6 rack ACE PWR circuit breaker.
- For spoilers 2/13, open the Power Supply Assembly Left (PSA-L) in E1-6 rack ACE PWR circuit breaker.
- For spoilers 3/12 or 6/9, open the Power Supply Assembly Right (PSA-R) in E5-1 rack ACE PWR circuit breaker.
- For spoilers 5/10, open P110 panel ACE-L2 PWR circuit breaker.
- Depressurize all three hydraulic systems.
- Set Maintenance Panel (P61) left, center and right WING FLT CONTROL HYD POWER switches to SHUT OFF.
- Gain access to the spoiler pair PCUs.
- Disconnect, cap and stow the inoperative PCU electrical connectors.
- Close any opened access panels.
- Close the open circuit breakers.



12. Set Maintenance Panel (P61) left, center and right WING FLT CONTROL HYD POWER switches to NORM.

13. Check the operation of the spoilers.

NOTE: Do not do the ACE Monitor Test.

14. Confirm the FLIGHT CONTROL SYS status message is not displayed.

OPERATIONS (O)

NOTE: The SPOILERS advisory message will be displayed.

Apply the appropriate performance adjustments.

Takeoff Runway/Obstacle	Minimum Takeoff Field Length	Landing Field Length	Minimum Landing Field Length
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For -200 or -200ER/GE90-76B thru -90B:

11,500 lb (5,216 kg)	6,900 ft (2,103 m)	25,000 lb (11,340 kg)	6,600 ft (2,012 m)
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For -200 or -200ER/GE90-94B:

11,000 lb (4,990 kg)	7,200 ft (2,195 m)	25,000 lb (11,340 kg)	7,100 ft (2,164 m)
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For -200 or -200ER/PW4074 thru PW4084:

11,000 lb (4,990 kg)	6,100 ft (1,859 m)	25,000 lb (11,340 kg)	No penalty (No penalty)
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For -200 or -200ER/PW4090:

11,000 lb (4,990 kg)	6,300 ft (1,920 m)	25,000 lb (11,340 kg)	6,700 ft (2,042 m)
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For -200 or -200ER/RR875 thru RR892:

11,000 lb (4,990 kg)	6,200 ft (1,890 m)	24,000 lb (10,886 kg)	6,500 ft (1,981 m)
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For -200 or -200ER/RR895:

11,000 lb (4,990 kg)	6,600 ft (2,012 m)	25,000 lb (11,340 kg)	6,800 ft (2,073 m)
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**Section 2****777 Dispatch Deviations Guide****ATA 27****For -200LR/GE90-110B:**

11,000 lb (4,990 kg)	7,600 ft (2,316 m)	31,000 lb (14,061 kg)	6,900 ft (2,103 m)
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For -200LR/GE90-115B:

12,000 lb (5,443 kg)	7,800 ft (2,377 m)	27,000 lb (12,247 kg)	7,100 ft (2,164 m)
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For -300/PW4090:

12,000 lb (5,443 kg)	5,000 ft (1,524 m)	26,000 lb (11,794 kg)	No penalty (No penalty)
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For -300/RR892:

11,000 lb (4,990 kg)	4,800 ft (1,463 m)	25,000 lb (11,340 kg)	No penalty (No penalty)
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For -300ER/GE90-115B:

12,000 lb (5,443 kg)	6,100 ft (1,859 m)	31,000 lb (14,062 kg)	No penalty (No penalty)
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27-61-03 Spoiler Power Control Units (PCUs)**27-61-03-02 777F**

Interval	Installed	Required	Procedure
A	14	12	(M) (O)

One symmetrical pair (except 4/11) may be inoperative provided:

- Symmetrical pair is deactivated in the retracted position.
- Remaining spoilers are verified to operate normally.
- Appropriate performance adjustments are applied.
- Repairs are made within 3 flight days.

NOTE: Dispatch relief does not apply for a fault condition that is reported to cause asymmetric roll input, or for a spoiler panel detached from the actuator, or for a spoiler not in the retracted position.

MAINTENANCE (M)

Deactivate the symmetrical spoiler pair in the retracted position and verify remaining spoilers operate normally (AMM 27-00-00/901).

1. Confirm speedbrake lever is in DOWN position.
2. For spoilers 1/14 or 7/8, open the Power Supply Assembly Center (PSA-C) in E2-6 rack ACE PWR circuit breaker.
3. For spoilers 2/13, open the Power Supply Assembly Left (PSA-L) in E1-6 rack ACE PWR circuit breaker.
4. For spoilers 3/12 or 6/9, open the Power Supply Assembly Right (PSA-R) in E5-1 rack ACE PWR circuit breaker.
5. For spoilers 5/10, open P110 panel ACE-L2 PWR circuit breaker.
6. Depressurize all three hydraulic systems.
7. Set Maintenance Panel (P61) left, center and right WING FLT CONTROL HYD POWER switches to SHUT OFF.
8. Gain access to the spoiler pair PCUs.
9. Disconnect, cap and stow the inoperative PCU electrical connectors.
10. Close any opened access panels.
11. Close the open circuit breakers.
12. Set Maintenance Panel (P61) left, center and right WING FLT CONTROL HYD POWER switches to NORM.
13. Check the operation of the spoilers.

NOTE: Do not do the ACE Monitor Test.

14. Confirm the FLIGHT CONTROL SYS status message is not displayed.

OPERATIONS (O)

NOTE: The SPOILERS advisory message will be displayed.

1. Observe Maximum Gross Weight limits.
 - A. Taxi Weight limit is 722,300 lb (327,629 kg).
 - B. Takeoff Weight limit is 720,300 lb (326,722 kg).
 - C. Landing Weight limit is 575,000 lb (260,815 kg).
 - D. Zero Fuel Weight limit is 547,000 lb (248,115 kg).

For 777F/GE90-110B:

2. Observe Center of Gravity (CG) limits.
 - A. Forward Takeoff and Landing CG limits:
 - 1) 280,000 lb (127,006 kg) at 20.0% MAC.
 - 2) 722,300 lb (327,629 kg) at 20.0% MAC.
 - B. Forward Zero Fuel CG limits:
 - 1) 280,000 lb (127,006 kg) at 20.0% MAC.
 - 2) 460,000 lb (208,652 kg) at 20.0% MAC.
 - 3) 547,000 lb (248,115 kg) at 28.0% MAC.



C. Aft Takeoff, Landing, and Zero Fuel CG limits:

- 1) 280,000 lb (127,006 kg) at 24.8% MAC.
- 2) 321,900 lb (146,011 kg) at 33.3% MAC.
- 3) 450,000 lb (204,117 kg) at 39.5% MAC.
- 4) 547,000 lb (248,115 kg) at 42.6% MAC.
- 5) 575,000 lb (260,815 kg) at 43.3% MAC.
- 6) 603,600 lb (273,788 kg) at 44.0% MAC.
- 7) 609,000 lb (276,237 kg) at 44.0% MAC.
- 8) 705,300 lb (319,918 kg) at 41.2% MAC.
- 9) 722,300 lb (327,629 kg) at 39.9% MAC.

For 777F/GE90-115B:

3. Observe Center of Gravity (CG) limits.

A. Forward Takeoff and Landing CG limits:

- 1) 280,000 lb (127,006 kg) at 20.0% MAC.
- 2) 722,300 lb (327,629 kg) at 20.0% MAC.

B. Forward Zero Fuel CG limits:

- 1) 280,000 lb (127,006 kg) at 20.0% MAC.
- 2) 460,000 lb (208,652 kg) at 20.0% MAC.
- 3) 547,000 lb (248,115 kg) at 28.0% MAC.

C. Aft Takeoff, Landing, and Zero Fuel CG limits:

- 1) 280,000 lb (127,006 kg) at 20.7% MAC.
- 2) 325,000 lb (147,418 kg) at 30.8% MAC.
- 3) 340,550 lb (154,471 kg) at 33.5% MAC.
- 4) 450,000 lb (204,117 kg) at 38.9% MAC.
- 5) 547,000 lb (248,115 kg) at 42.0% MAC.
- 6) 575,000 lb (260,815 kg) at 42.7% MAC.
- 7) 615,600 lb (279,231 kg) at 43.7% MAC.
- 8) 705,300 lb (319,918 kg) at 41.2% MAC.
- 9) 722,300 lb (327,629 kg) at 39.9% MAC.

4. Apply appropriate performance adjustments.

Takeoff Runway/Obstacle	Minimum Takeoff Field Length	Landing Field Length	Minimum Landing Field Length
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For 777F/GE90-110B:

11,000 lb (4,990 kg)	7,600 ft (2,316 m)	31,000 lb (14,061 kg)	6,900 ft (2,103 m)
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For 777F/GE90-115B:

12,000 lb (5,443 kg)	7,800 ft (2,377 m)	27,000 lb (12,247 kg)	7,100 ft (2,164 m)
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Supplement 36-6



777 FAA DDG Supplement Package

Supplement Number: 36-6

Issue Date: 26-SEP-2025

Section Number: 2 (MMEL)

Item Number: 28-21-03

Item Name: Refuel Adapters

DDG Effectivity: 777 FAA DDG (D630W003-TBC), Revision 36

MMEL Effectivity: 777 FAA MMEL, Revision 23a

Description: Modified Maintenance (M) procedure note.

Reason: To reflect the correct failure modes that are applicable for the item.

Disposition: For reference in the Operator's MEL.

Attachments: DDG pages 2.28-21-03.1

28-21-03 Refuel Adapters

Interval	Installed	Required	Procedure
C	-	1	(M)

May be inoperative provided there are no fuel leaks.

MAINTENANCE (M)

Immediately after refuel operations, inspect the area around the associated refuel adapter and verify fuel leaks are not present.

NOTE 1: Dispatch is not allowed for an inoperative poppet valve. A poppet valve is not operating normally if it leaks immediately after refueling.

NOTE 2: This item is applicable to failures of the refuel/defuel handle, break-away flange, or a break-away flange that is beyond wear limits.

NOTE 3: An inoperative adapter should not be used in refuel or defuel operations.



777 FAA DDG Supplement Package

Supplement Number: 36-6

Issue Date: 26-SEP-2025

Section Number: 2 (MMEL)

Item Number: 28-31-02

Item Name: Center Tank Jettison Isolation Valves

DDG Effectivity: 777 FAA DDG (D630W003-TBC), Revision 36

MMEL Effectivity: 777 FAA MMEL, Revision 23a

Description: **Note: This item supersedes content in a previous supplement.**

Modified Operations (O) procedures for subitem options 28-31-02-02B and 28-31-02-03B.

Reason: To reflect updated performance adjustments, reincluded CG adjustment information that was inadvertently removed, and removed inapplicable model variant information that was inadvertently included.

Disposition: For reference in the Operator's MEL.

Attachments: DDG pages 2.28-31-02.1-8

**28-31-02 Center Tank Jettison Isolation Valves****28-31-02-01 -200**

Interval	Installed	Required	Procedure
C	2	0	(M)

May be inoperative provided inoperative valve(s) is secured closed.

MAINTENANCE (M)

Deactivate and secure the inoperative jettison isolation valve in the closed position (AMM 28-00-00/901).

1. Open the landing gear doors and install safety pins (AMM 32-00-15/201).
2. For left jettison isolation valve inoperative:
 - A. Open and collar P110 panel L FUEL JETT ISLN VLV circuit breaker.
 - B. Gain access to the jettison isolation valve actuator located in the left main wheel well (rear spar of the left center tank).
 - C. Disconnect, cap and stow the electrical connector (DV28165).
 - D. Move the manual override handle to the CLOSED position and lockwire.
3. For right jettison isolation valve inoperative:
 - A. Open and collar P210 panel R FUEL JETT ISLN VLV circuit breaker.
 - B. Gain access to the jettison isolation valve actuator located in the right main wheel well (rear spar of the right center tank).
 - C. Disconnect, cap and stow the electrical connector (DV28265).
 - D. Move the manual override handle to the CLOSED position and lockwire.
4. Remove the landing gear door safety pins and close the landing gear doors (AMM 32-00-15/201).

28-31-02 Center Tank Jettison Isolation Valves**28-31-02-02 -200ER/-200LR/-300/-300ER/777F (Without PRR 62777-79S or Equivalent Installed)****28-31-02-02A Fuel Jettison System Deactivated**

Interval	Installed	Required	Procedure
C	2	0	(M)

May be inoperative provided:

- a. Inoperative valve(s) is secured closed.
- b. Fuel jettison system is considered inoperative.

NOTE 1: The PRR 62777-79S is applicable to airplane line numbers 1099 and-on. Service Bulletins 777-28-0078 (ELMS2) and 777-28-0082 (ELMS1) each provide the production equivalent of PRR 62777-79S.

NOTE 2: The airplane must also be dispatched using MMEL item 28-31-01.

MAINTENANCE (M)

Secure the inoperative jettison isolation valve in the closed position (AMM 28-00-00/901).

1. Open the landing gear doors and install safety pins (AMM 32-00-15/201).
2. For left jettison isolation valve:
 - A. Open and collar P110 panel L FUEL JETT ISLN VLV circuit breaker.
 - B. Gain access to the jettison isolation valve actuator located in the left main wheel well (rear spar of the left center tank).
 - C. Disconnect, cap and stow the electrical connector (DV28165).
 - D. Move the manual override handle to the CLOSED position and lockwire.
3. For right jettison isolation valve:
 - A. Open and collar P210 panel R FUEL JETT ISLN VLV circuit breaker.
 - B. Gain access to the jettison isolation valve actuator located in the right main wheel well (rear spar of the right center tank).
 - C. Disconnect, cap and stow the electrical connector (DV28265).
 - D. Move the manual override handle to the CLOSED position and lockwire.
4. Remove the landing gear door safety pins and close the landing gear doors (AMM 32-00-15/201).

28-31-02 Center Tank Jettison Isolation Valves

28-31-02-02 -200ER/-200LR/-300/-300ER/777F (Without PRR 62777-79S or Equivalent Installed)

28-31-02-02B Main Tank Jettison Operative

Interval	Installed	Required	Procedure
C	2	0	(M) (O)

May be inoperative provided:

- a. Inoperative valve(s) is secured closed.
- b. Appropriate gross weight and center-of-gravity adjustments are applied.

NOTE: The PRR 62777-79S is applicable to airplane line numbers 1099 and-on. Service Bulletins 777-28-0078 (ELMS2) and 777-28-0082 (ELMS1) each provide the production equivalent of PRR 62777-79S.

MAINTENANCE (M)

Deactivate and secure the inoperative jettison isolation valve in the closed position (AMM 28-00-00/901).

1. Open the landing gear doors and install safety pins (AMM 32-00-15/201).
2. For left jettison isolation valve inoperative:
 - A. Open and collar P110 panel L FUEL JETT ISLN VLV circuit breaker.
 - B. Gain access to the jettison isolation valve actuator located in the left main wheel well (rear spar of the left center tank).
 - C. Disconnect, cap and stow the electrical connector (DV28165).
 - D. Move the manual override handle to the CLOSED position and lockwire.
3. For right jettison isolation valve inoperative:
 - A. Open and collar P210 panel R FUEL JETT ISLN VLV circuit breaker.
 - B. Gain access to the jettison isolation valve actuator located in the right main wheel well (rear spar of the right center tank).
 - C. Disconnect, cap and stow the electrical connector (DV28265).
 - D. Move the manual override handle to the CLOSED position and lockwire.
4. Remove the landing gear door safety pins and close the landing gear doors (AMM 32-00-15/201).

OPERATIONS (O)

For -200ER/PW4074 thru PW4084:

1. For dispatch with center tank fuel greater than 22,000 lb (9,980 kg) and Gross Weight exceeding 460,000 lb (208,650 kg), Center-of-Gravity (CG) must be aft of 22% MAC for takeoff.
2. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 3,400 lb (1,542 kg) (for fuel that would be burned during a return-to-land procedure).

For -200ER/PW4090:

1. For dispatch with center tank fuel greater than 22,000 lb (9,980 kg) and Gross Weight exceeding 460,000 lb (208,650 kg), Center-of-Gravity (CG) must be aft of 22% MAC for takeoff.
2. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 3,800 lb (1,724 kg) (for fuel that would be burned during a return-to-land procedure).

For -200ER/GE90-76B thru -90B:

1. For dispatch with center tank fuel greater than 22,000 lb (9,980 kg) and Gross Weight exceeding 460,000 lb (208,650 kg), Center-of-Gravity (CG) must be aft of 22% MAC for takeoff.
2. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 3,300 lb (1,497 kg) (for fuel that would be burned during a return-to-land procedure).

For -200ER/GE90-94B or RR875 thru RR892:

1. For dispatch with center tank fuel greater than 22,000 lb (9,980 kg) and Gross Weight exceeding 460,000 lb (208,650 kg), Center-of-Gravity (CG) must be aft of 22% MAC for takeoff.
2. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 3,700 lb (1,678 kg) (for fuel that would be burned during a return-to-land procedure).

For -200ER/RR895:

1. For dispatch with center tank fuel greater than 22,000 lb (9,980 kg) and Gross Weight exceeding 460,000 lb (208,650 kg), Center-of-Gravity (CG) must be aft of 22% MAC for takeoff.
2. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 4,000 lb (1,814 kg) (for fuel that would be burned during a return-to-land procedure).

For -200LR or 777F/GE90-100 Series:

1. For dispatch with center tank fuel and Gross Weight exceeding 523,000 lb (237,230 kg), Center-of-Gravity (CG) must be aft of 25% MAC for takeoff.
2. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 4,000 lb (1,814 kg) (for fuel that would be burned during a return-to-land procedure).

For -300/RR884 thru RR892:

1. For dispatch with center tank fuel and Gross Weight exceeding 524,000 lb (238,000 kg), Center-of-Gravity (CG) must be aft of 17% MAC for takeoff.
2. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 3,900 lb (1,769 kg) (for fuel that would be burned during a return-to-land procedure).

For -300/PW4090:

1. For dispatch with center tank fuel and Gross Weight exceeding 524,000 lb (238,000 kg), Center-of-Gravity (CG) must be aft of 17% MAC for takeoff.
2. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 4,100 lb (1,860 kg) (for fuel that would be burned during a return-to-land procedure).

For -300ER/GE90-115B:

1. For dispatch with center tank fuel and Gross Weight exceeding 556,000 lb (252,200 kg), Center-of-Gravity (CG) must be aft of 21% MAC for takeoff.
2. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 4,200 lb (1,905 kg) (for fuel that would be burned during a return-to-land procedure).

**28-31-02 Center Tank Jettison Isolation Valves****28-31-02-03 -200ER/-200LR/-300/-300ER/777F (With PRR 62777-79S or Equivalent Installed)****28-31-02-03A Fuel Jettison System Deactivated**

Interval	Installed	Required	Procedure
C	2	0	(M)

May be inoperative provided:

- Inoperative valve(s) is secured closed.
- Fuel jettison system is considered inoperative.
- Fuel scavenge system is considered inoperative.

NOTE 1: The PRR 62777-79S is applicable to airplane line numbers 1099 and-on. Service Bulletins 777-28-0078 (ELMS2) and 777-28-0082 (ELMS1) each provide the production equivalent of PRR 62777-79S.

NOTE 2: The airplane must also be dispatched using MMEL item 28-31-01.

MAINTENANCE (M)

Secure the inoperative jettison isolation valve in the closed position (AMM 28-00-00/901).

- Open the landing gear doors and install safety pins (AMM 32-00-15/201).
- For left jettison isolation valve:
 - Open and collar P110 panel L FUEL JETT ISLN VLV circuit breaker.
 - Gain access to the jettison isolation valve actuator located in the left main wheel well (rear spar of the left center tank).
 - Disconnect, cap and stow the electrical connector (DV28165).
 - Move the manual override handle to the CLOSED position and lockwire.
- For right jettison isolation valve:
 - Open and collar P210 panel R FUEL JETT ISLN VLV circuit breaker.
 - Gain access to the jettison isolation valve actuator located in the right main wheel well (rear spar of the right center tank).
 - Disconnect, cap and stow the electrical connector (DV28265).
 - Move the manual override handle to the CLOSED position and lockwire.
- Remove the landing gear door safety pins and close the landing gear doors (AMM 32-00-15/201).

28-31-02 Center Tank Jettison Isolation Valves
28-31-02-03 -200ER/-200LR/-300/-300ER/777F (With PRR 62777-79S or Equivalent Installed)
28-31-02-03B Main Tank Jettison Operative

Interval	Installed	Required	Procedure
C	2	0	(M) (O)

May be inoperative provided:

- Inoperative valve(s) is secured closed.
- Appropriate gross weight and center-of-gravity adjustments are applied.

NOTE: The PRR 62777-79S is applicable to airplane line numbers 1099 and-on. Service Bulletins 777-28-0078 (ELMS2) and 777-28-0082 (ELMS1) each provide the production equivalent of PRR 62777-79S.

MAINTENANCE (M)

Deactivate and secure the inoperative jettison isolation valve in the closed position (AMM 28-00-00/901).

- Open the landing gear doors and install safety pins (AMM 32-00-15/201).
- For left jettison isolation valve inoperative:
 - Open and collar P110 panel L FUEL JETT ISLN VLV circuit breaker.
 - Gain access to the jettison isolation valve actuator located in the left main wheel well (rear spar of the left center tank).
 - Disconnect, cap and stow the electrical connector (DV28165).
 - Move the manual override handle to the CLOSED position and lockwire.
- For right jettison isolation valve inoperative:
 - Open and collar P210 panel R FUEL JETT ISLN VLV circuit breaker.
 - Gain access to the jettison isolation valve actuator located in the right main wheel well (rear spar of the right center tank).
 - Disconnect, cap and stow the electrical connector (DV28265).
 - Move the manual override handle to the CLOSED position and lockwire.
- Remove the landing gear door safety pins and close the landing gear doors (AMM 32-00-15/201).

OPERATIONS (O)

For -200ER/PW4074 thru PW4084:

- For dispatch with center tank fuel greater than 22,000 lb (9,980 kg) and Gross Weight exceeding 460,000 lb (208,650 kg), Center-of-Gravity (CG) must be aft of 22% MAC for takeoff.

2. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 3,400 lb (1,542 kg) (for fuel that would be burned during a return-to-land procedure).

For -200ER/PW4090:

1. For dispatch with center tank fuel greater than 22,000 lb (9,980 kg) and Gross Weight exceeding 460,000 lb (208,650 kg), Center-of-Gravity (CG) must be aft of 22% MAC for takeoff.
2. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 3,800 lb (1,724 kg) (for fuel that would be burned during a return-to-land procedure).

For -200ER/GE90-76B thru -90B:

1. For dispatch with center tank fuel greater than 22,000 lb (9,980 kg) and Gross Weight exceeding 460,000 lb (208,650 kg), Center-of-Gravity (CG) must be aft of 22% MAC for takeoff.
2. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 3,300 lb (1,497 kg) (for fuel that would be burned during a return-to-land procedure).

For -200ER/GE90-94B or RR875 thru RR892:

1. For dispatch with center tank fuel greater than 22,000 lb (9,980 kg) and Gross Weight exceeding 460,000 lb (208,650 kg), Center-of-Gravity (CG) must be aft of 22% MAC for takeoff.
2. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 3,700 lb (1,678 kg) (for fuel that would be burned during a return-to-land procedure).

For -200ER/RR895:

1. For dispatch with center tank fuel greater than 22,000 lb (9,980 kg) and Gross Weight exceeding 460,000 lb (208,650 kg), Center-of-Gravity (CG) must be aft of 22% MAC for takeoff.
2. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 4,000 lb (1,814 kg) (for fuel that would be burned during a return-to-land procedure).

For -200LR or 777F/GE90-100 Series:

1. For dispatch with center tank fuel and Gross Weight exceeding 523,000 lb (237,230 kg), Center-of-Gravity (CG) must be aft of 25% MAC for takeoff.
2. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 4,000 lb (1,814 kg) (for fuel that would be burned during a return-to-land procedure).

For -300/RR884 thru RR892:

1. For dispatch with center tank fuel and Gross Weight exceeding 524,000 lb (238,000 kg), Center-of-Gravity (CG) must be aft of 17% MAC for takeoff.
2. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 3,900 lb (1,769 kg) (for fuel that would be burned during a return-to-land procedure).

For -300/PW4090:

1. For dispatch with center tank fuel and Gross Weight exceeding 524,000 lb (238,000 kg), Center-of-Gravity (CG) must be aft of 17% MAC for takeoff.
2. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 4,100 lb (1,860 kg) (for fuel that would be burned during a return-to-land procedure).

For -300ER/GE90-115B:

1. For dispatch with center tank fuel and Gross Weight exceeding 556,000 lb (252,200 kg), Center-of-Gravity (CG) must be aft of 21% MAC for takeoff.
2. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus 4,200 lb (1,905 kg) (for fuel that would be burned during a return-to-land procedure).



777 FAA DDG Supplement Package

Supplement Number: 36-6

Issue Date: 26-SEP-2025

Section Number: 2 (MMEL)

Item Number: 28-31-03

Item Name: Main Tanks Jettison Pumps

DDG Effectivity: 777 FAA DDG (D630W003-TBC), Revision 36

MMEL Effectivity: 777 FAA MMEL, Revision 23a

Description: **Note: This item supersedes content in a previous supplement.**

Modified Operations (O) procedures for subitems 28-31-03-01 and 28-31-03-02.

Reason: To reflect updated performance adjustments, corrected Operations (O) procedure numbering error, and reincluded center tank fuel performance adjustment information that was inadvertently removed.

Disposition: For reference in the Operator's MEL.

Attachments: DDG pages 2.28-31-03.1-4

**28-31-03 Main Tanks Jettison Pumps****28-31-03-01 Without PRR 62777-79S or Equivalent Installed**

Interval	Installed	Required	Procedure
C	2	0	(M) (O)

May be inoperative provided:

- Inoperative pump is deactivated.
- Appropriate performance adjustments are applied.

NOTE: The PRR 62777-79S is applicable to airplane line numbers 1099 and-on.
Service Bulletins 777-28-0078 (ELMS2) and 777-28-0082 (ELMS1) each provide the production equivalent of PRR 62777-79S.

MAINTENANCE (M)

Deactivate the inoperative pump(s) (AMM 28-00-00/901).

- For left main tank jettison pump inoperative, open and collar P110 panel L FUEL JETT PUMP CTRL circuit breaker.
- For right main tank jettison pump inoperative, open and collar P210 panel R FUEL JETT PUMP CTRL circuit breaker.

OPERATIONS (O)

NOTE: For one main tank jettison pump inoperative, fuel jettison of the opposite jettison pump is inhibited to prevent fuel imbalance.

For -200 or -200ER/PW4074 thru PW4084:

- Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus center tank fuel (which would be jettisoned during a return-to-land procedure), plus 3,400 lb (1,542 kg) (for fuel that would be burned during a return-to-land procedure).

For -200 or -200ER/PW4090:

- Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus center tank fuel (which would be jettisoned during a return-to-land procedure), plus 3,800 lb (1,724 kg) (for fuel that would be burned during a return-to-land procedure).

For -200 or -200ER/GE90-76B thru -90B:

- Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus center tank fuel (which would be jettisoned during a return-to-land procedure), plus 3,300 lb (1,497 kg) (for fuel that would be burned during a return-to-land procedure).

For -200 or -200ER/GE90-94B or RR875 thru RR892:

1. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus center tank fuel (which would be jettisoned during a return-to-land procedure), plus 3,700 lb (1,678 kg) (for fuel that would be burned during a return-to-land procedure).

For -200 or -200ER/RR895:

1. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus center tank fuel (which would be jettisoned during a return-to-land procedure), plus 4,000 lb (1,814 kg) (for fuel that would be burned during a return-to-land procedure).

For -200LR or 777F/GE90-100 Series:

1. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus center tank fuel (which would be jettisoned during a return-to-land procedure), plus 4,000 lb (1,814 kg) (for fuel that would be burned during a return-to-land procedure).

For -300/RR884 thru RR892:

1. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus center tank fuel (which would be jettisoned during a return-to-land procedure), plus 3,900 lb (1,769 kg) (for fuel that would be burned during a return-to-land procedure).

For -300/PW4090:

1. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus center tank fuel (which would be jettisoned during a return-to-land procedure), plus 4,100 lb (1,860 kg) (for fuel that would be burned during a return-to-land procedure).

For -300ER/GE90-115B:

1. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus center tank fuel (which would be jettisoned during a return-to-land procedure), plus 4,200 lb (1,905 kg) (for fuel that would be burned during a return-to-land procedure).

28-31-03 Main Tanks Jettison Pumps

28-31-03-02 With PRR 62777-79S or Equivalent Installed

Interval	Installed	Required	Procedure
C	2	0	(M) (O)

May be inoperative provided:

- a. Inoperative pump is deactivated.
- b. Fuel scavenge system is considered inoperative.
- c. Appropriate performance adjustments are applied.



NOTE 1: The PRR 62777-79S is applicable to airplane line numbers 1099 and-on. Service Bulletins 777-28-0078 (ELMS2) and 777-28-0082 (ELMS1) each provide the production equivalent of PRR 62777-79S.

NOTE 2: The airplane must also be dispatched using MMEL item 28-22-04.

MAINTENANCE (M)

Deactivate the inoperative pump(s) (AMM 28-00-00/901).

1. For left main tank jettison pump inoperative, open and collar P110 panel L FUEL JETT PUMP CTRL circuit breaker.
2. For right main tank jettison pump inoperative, open and collar P210 panel R FUEL JETT PUMP CTRL circuit breaker.

OPERATIONS (O)

NOTE: For one main tank jettison pump inoperative:

1. Fuel jettison of the opposite jettison pump is inhibited to prevent fuel imbalance.
2. Fuel scavenge of the opposite jettison pump operates normally.
3. Fuel scavenge rate is approximately half the normal fuel scavenge rate.

For -200 or -200ER/PW4074 thru PW4084:

1. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus center tank fuel (which would be jettisoned during a return-to-land procedure), plus 3,400 lb (1,542 kg) (for fuel that would be burned during a return-to-land procedure).

For -200 or -200ER/PW4090:

1. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus center tank fuel (which would be jettisoned during a return-to-land procedure), plus 3,800 lb (1,724 kg) (for fuel that would be burned during a return-to-land procedure).

For -200 or -200ER/GE90-76B thru -90B:

1. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus center tank fuel (which would be jettisoned during a return-to-land procedure), plus 3,300 lb (1,497 kg) (for fuel that would be burned during a return-to-land procedure).

For -200 or -200ER/GE90-94B or RR875 thru RR892:

1. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus center tank fuel (which would be jettisoned during a return-to-land procedure), plus 3,700 lb (1,678 kg) (for fuel that would be burned during a return-to-land procedure).

For -200 or -200ER/RR895:

1. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus center tank fuel (which would be jettisoned during a return-to-land procedure), plus 4,000 lb (1,814 kg) (for fuel that would be burned during a return-to-land procedure).

For -200LR or 777F/GE90-100 Series:

1. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus center tank fuel (which would be jettisoned during a return-to-land procedure), plus 4,000 lb (1,814 kg) (for fuel that would be burned during a return-to-land procedure).

For -300/RR884 thru RR892:

1. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus center tank fuel (which would be jettisoned during a return-to-land procedure), plus 3,900 lb (1,769 kg) (for fuel that would be burned during a return-to-land procedure).

For -300/PW4090:

1. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus center tank fuel (which would be jettisoned during a return-to-land procedure), plus 4,100 lb (1,860 kg) (for fuel that would be burned during a return-to-land procedure).

For -300ER/GE90-115B:

1. Limit the takeoff weight to the maximum climb limited landing weight at the departure airport, plus center tank fuel (which would be jettisoned during a return-to-land procedure), plus 4,200 lb (1,905 kg) (for fuel that would be burned during a return-to-land procedure).



777 FAA DDG Supplement Package

Supplement Number: 36-6

Issue Date: 26-SEP-2025

Section Number: 2 (MMEL)

Item Number: 28-41-01

Item Name: Fuel Quantity Indications (Flight Deck)

DDG Effectivity: 777 FAA DDG (D630W003-TBC), Revision 36

MMEL Effectivity: 777 FAA MMEL, Revision 23a

Description: Modified Operations (O) procedure notes for all subitems.

Reason: To clarify that the fuel quantity will not always be blank prior to manual initialization.

Disposition: For reference in the Operator's MEL.

Attachments: DDG pages 2.28-41-01.1-9

**28-41-01 Fuel Quantity Indications (Flight Deck)****28-41-01-01 Center Tank**

Interval	Installed	Required	Procedure
B	1	0	(M) (O)

May be inoperative provided:

- Main tank fuel quantity indications operate normally.
- Fuel quantity in center tank is verified by an alternate procedure.
- All pumps for the center tank operate normally.
- Fuel flow Indications operate normally.
- FMC FUEL is initialized with the known total fuel quantity.
- Flight remains within 180 minutes of landing at a suitable airport.

For 200ER, -200LR, 777F, -300 and -300ER:

- Appropriate gross weight and center-of-gravity adjustments are applied.

NOTE: The airplane must also be dispatched using MMEL item 28-41-03.

MAINTENANCE (M)

Verify the fuel quantity in the center tank using one of the following methods (AMM 28-00-00/901):

- Use the center tank's Integrated Refuel Panel (IRP) (P28) quantity indication (if operating).
- Refuel using the FMC Calculated Fuel on CDU Progress page 2 from previous flight and operating fuel quantity indications to determine the center tank quantity. Then refuel the center tank with a known quantity of fuel. A fuel tanker's fuel indication may be used if verified by the center tank's fuel measuring sticks.
- Defuel the center tank (AMM 28-26-00/201) and service with a known quantity of fuel (AMM 12-11-01/301).
- Determine the fuel quantity using the center tank's measuring sticks (Fuel Measuring Stick Manual).

NOTE: The fuel quantity measuring sticks may not give accurate results until 5 to 15 minutes after refueling operations are complete.

OPERATIONS (O)

NOTE: FUEL IN CENTER and FUEL LOW CENTER advisory messages will not be displayed when appropriate.

- Total fuel quantity must be calculated and manually initialized on the FMC PERF INIT page.

NOTE 1: The total fuel quantity indication may be inoperative or inaccurate.

NOTE 2: The FMC CALCULATED fuel quantity may be blank prior to manual initialization.

- A. Manually enter the total fuel quantity on the PERF INIT page. MANUAL will be displayed next to the entered fuel quantity.

NOTE: The manually entered total fuel quantity and the takeoff speeds may be blank after engine start.

- B. Before and after engine start, verify the fuel quantity entered is displayed under the CALCULATED fuel quantity display on the PROGRESS page 2.
2. Flight crew should monitor FMC Calculated fuel and the operative fuel tank quantity indications to ensure usages are normal and to derive the amount of fuel in the tank with the inoperative quantity indication when necessary.
3. Apply appropriate gross weight and center-of-gravity restrictions.

For -200ER:

- A. For gross weight exceeding 495,000 lb (225,000 kg), Center-of-Gravity must be aft of 20% MAC for takeoff.

For -200LR/777F:

- B. For gross weight exceeding 563,000 lb (255,000 kg), Center-of-Gravity must be aft of 22.2% MAC for takeoff.

For -300:

- C. For gross weight exceeding 535,000 lb (243,000 kg), Center-of-Gravity must be aft of 16% MAC for takeoff.

For -300ER:

- D. For gross weight exceeding 586,000 lb (266,000 kg), Center-of-Gravity must be aft of 19.4% MAC for takeoff.

4. The flight must remain with 180 minutes of landing at a suitable airport.
5. The FUEL LOW CENTER advisory message will not operate. When FUEL PUMP CENTER L or R advisory messages are displayed, accomplish the FUEL PUMP CENTER L or R Non-Normal Checklist.
6. If it is necessary to jettison fuel enroute:

NOTE: Selection of fuel jettison with a fuel quantity indication inoperative causes all FMC performance and VNAV data to become invalid. If VNAV is the current active pitch mode of the Autopilot Flight Director System (AFDS), the AFDS pitch mode will fail with a resulting Master Caution light, AUTOPILOT caution message and amber bar through the pitch annunciation on the Primary Flight Display (PFD).

- A. Fuel jettison system maximum landing weight and fuel to remain automatic shutoff functions will be inoperative.
- B. FUEL AUTO JETTISON caution message will be displayed. Accomplish the Manual FUEL AUTO JETTISON Non-Normal Checklist.

28-41-01 Fuel Quantity Indications (Flight Deck)
28-41-01-02 Main Tanks
28-41-01-02-01 Without PRR 62777-79S or Equivalent Installed

Interval	Installed	Required	Procedure
B	2	1	(M) (O)

One may be inoperative provided:

- Center tank fuel quantity indication operates normally.
- Fuel quantity in associated tank is verified by an alternate procedure.
- All pumps for the associated tank operate normally.
- Fuel flow Indications operate normally.
- FMC FUEL is initialized with the known total fuel quantity.
- Flight remains within 180 minutes of landing at a suitable airport.

For 200ER, -200LR, 777F, -300 and -300ER:

- Appropriate gross weight and center-of-gravity adjustments are applied.

NOTE 1: The airplane must also be dispatched using MMEL item 28-41-03.

NOTE 2: The PRR 62777-79S is applicable to airplane line numbers 1099 and-on.
Service Bulletins 777-28-0078 (ELMS2) and 777-28-0082 (ELMS1) each provide the production equivalent of PRR 62777-79S.

MAINTENANCE (M)

Verify the fuel quantity in the tank associated with the inoperative quantity indication using one of the following methods (AMM 28-00-00/901):

- Use the associated tank's Integrated Refuel Panel (IRP) (P28) quantity indication (if operating).
- Refuel using the FMC Calculated Fuel on CDU Progress page 2 from previous flight and operating fuel quantity indications to determine the associated tank quantity. Then refuel the associated tank with a known quantity of fuel. A fuel tanker's fuel indication may be used if verified by the associated tank's fuel measuring sticks.
- Defuel the associated tank (AMM 28-26-00/201) and service with a known quantity of fuel (AMM 12-11-01/301).
- Determine the fuel quantity using the associated tank's measuring sticks (Fuel Measuring Stick Manual).

NOTE: The fuel quantity measuring sticks may not give accurate results until 5 to 15 minutes after refueling operations are complete.

OPERATIONS (O)

NOTE 1: FUEL IMBALANCE advisory message may be displayed erroneously or may not display with an actual fuel imbalance.

NOTE 2: If the FUEL IMBALANCE advisory message displays, to prevent unnecessary fuel balancing, the crew should not accomplish the FUEL IMBALANCE checklist unless an actual fuel imbalance can be confirmed.

1. Total fuel quantity must be calculated and manually initialized on the FMC PERF INIT page.

NOTE 1: The total fuel quantity indication may be inoperative or inaccurate.

NOTE 2: The FMC CALCULATED fuel quantity may be blank prior to manual initialization.

- A. Manually enter the total fuel quantity on the PERF INIT page. MANUAL will be displayed next to the entered fuel quantity.

NOTE: The manually entered total fuel quantity and the takeoff speeds may blank after engine start.

- B. Before and after engine start, verify the fuel quantity entered is displayed under the CALCULATED fuel quantity display on the PROGRESS page 2.
2. Flight crew should monitor FMC Calculated fuel and the operative fuel tank quantity indications to ensure usages are normal and to derive the amount of fuel in the tank with the inoperative quantity indication when necessary.
 3. Apply appropriate gross weight and center-of-gravity restrictions.

For -200ER:

- A. For gross weight exceeding 495,000 lb (225,000 kg), Center-of-Gravity must be aft of 20% MAC for takeoff.

For -200LR/777F:

- B. For gross weight exceeding 563,000 lb (255,000 kg), Center-of-Gravity must be aft of 22.2% MAC for takeoff.

For -300:

- C. For gross weight exceeding 535,000 lb (243,000 kg), Center-of-Gravity must be aft of 16% MAC for takeoff.

For -300ER:

- D. For gross weight exceeding 586,000 lb (266,000 kg), Center-of-Gravity must be aft of 19.4% MAC for takeoff.

4. The flight must remain with 180 minutes of landing at a suitable airport.
5. If it is necessary to jettison fuel enroute:



NOTE: Selection of fuel jettison with a fuel quantity indication inoperative causes all FMC performance and VNAV data to become invalid. If VNAV is the current active pitch mode of the Autopilot Flight Director System (AFDS), the AFDS pitch mode will fail with a resulting Master Caution light, AUTOPILOT caution message and amber bar through the pitch annunciation on the Primary Flight Display (PFD).

- A. Fuel jettison system maximum landing weight and fuel to remain automatic shutoff functions will be inoperative.
- B. FUEL AUTO JETTISON caution message will be displayed. Accomplish the Manual FUEL AUTO JETTISON Non-Normal Checklist.

28-41-01 Fuel Quantity Indications (Flight Deck)**28-41-01-02 Main Tanks****28-41-01-02-02 With PRR 62777-79S or Equivalent Installed**

Interval	Installed	Required	Procedure
B	2	1	(M) (O)

One may be inoperative provided:

- a. Center tank fuel quantity indication operates normally.
- b. Fuel quantity in associated tank is verified by an alternate procedure.
- c. All pumps for the associated tank operate normally.
- d. Fuel flow Indications operate normally.
- e. FMC FUEL is initialized with the known total fuel quantity.
- f. Flight remains within 180 minutes of landing at a suitable airport.
- g. Fuel scavenge system is considered inoperative.

For 200ER, -200LR, 777F, -300 and -300ER:

- h. Appropriate gross weight and center-of-gravity adjustments are applied.

NOTE 1: The airplane must also be dispatched using MMEL item 28-41-03.

NOTE 2: The airplane must also be dispatched using MMEL item 28-22-04.

NOTE 3: The PRR 62777-79S is applicable to airplane line numbers 1099 and-on. Service Bulletins 777-28-0078 (ELMS2) and 777-28-0082 (ELMS1) each provide the production equivalent of PRR 62777-79S.

MAINTENANCE (M)

Verify the fuel quantity in the tank associated with the inoperative quantity indication using one of the following methods (AMM 28-00-00/901):

1. Use the associated tank's Integrated Refuel Panel (IRP) (P28) quantity indication (if operating).

2. Refuel using the FMC Calculated Fuel on CDU Progress page 2 from previous flight and operating fuel quantity indications to determine the associated tank quantity. Then refuel the associated tank with a known quantity of fuel. A fuel tanker's fuel indication may be used if verified by the associated tank's fuel measuring sticks.
3. Defuel the associated tank (AMM 28-26-00/201) and service with a known quantity of fuel (AMM 12-11-01/301).
4. Determine the fuel quantity using the associated tank's measuring sticks (Fuel Measuring Stick Manual).

NOTE: The fuel quantity measuring sticks may not give accurate results until 5 to 15 minutes after refueling operations are complete.

OPERATIONS (O)

NOTE 1: FUEL IMBALANCE advisory message may be displayed erroneously or may not display with an actual fuel imbalance.

NOTE 2: If the FUEL IMBALANCE advisory message displays, to prevent unnecessary fuel balancing, the crew should not accomplish the FUEL IMBALANCE checklist unless an actual fuel imbalance can be confirmed.

1. Total fuel quantity must be calculated and manually initialized on the FMC PERF INIT page.

NOTE 1: The total fuel quantity indication may be inoperative or inaccurate.

NOTE 2: The FMC CALCULATED fuel quantity may be blank prior to manual initialization.

- A. Manually enter the total fuel quantity on the PERF INIT page. MANUAL will be displayed next to the entered fuel quantity.

NOTE: The manually entered total fuel quantity and the takeoff speeds may be blank after engine start.

- B. Before and after engine start, verify the fuel quantity entered is displayed under the CALCULATED fuel quantity display on the PROGRESS page 2.

2. Flight crew should monitor FMC Calculated fuel and the operative fuel tank quantity indications to ensure usages are normal and to derive the amount of fuel in the tank with the inoperative quantity indication when necessary.
3. Apply appropriate gross weight and center-of-gravity restrictions.

For -200ER:

- A. For gross weight exceeding 495,000 lb (225,000 kg), Center-of-Gravity must be aft of 20% MAC for takeoff.

For -200LR/777F:

- B. For gross weight exceeding 563,000 lb (255,000 kg), Center-of-Gravity must be aft of 22.2% MAC for takeoff.

**For -300:**

- C. For gross weight exceeding 535,000 lb (243,000 kg), Center-of-Gravity must be aft of 16% MAC for takeoff.

For -300ER:

- D. For gross weight exceeding 586,000 lb (266,000 kg), Center-of-Gravity must be aft of 19.4% MAC for takeoff.
4. The flight must remain with 180 minutes of landing at a suitable airport.
5. If it is necessary to jettison fuel enroute:

NOTE: Selection of fuel jettison with a fuel quantity indication inoperative causes all FMC performance and VNAV data to become invalid. If VNAV is the current active pitch mode of the Autopilot Flight Director System (AFDS), the AFDS pitch mode will fail with a resulting Master Caution light, AUTOPILOT caution message and amber bar through the pitch annunciation on the Primary Flight Display (PFD).

- A. Fuel jettison system maximum landing weight and fuel to remain automatic shutoff functions will be inoperative.
- B. FUEL AUTO JETTISON caution message will be displayed. Accomplish the Manual FUEL AUTO JETTISON Non-Normal Checklist.

28-41-01 Fuel Quantity Indications (Flight Deck)**28-41-01-03 Auxiliary Tank(s)**

Interval	Installed	Required	Procedure
B	1	0	(M) (O)

May be inoperative provided:

- Center and main tank fuel quantity indications operate normally.
- Fuel quantity in auxiliary tank(s) is verified by an alternate procedure.
- Auxiliary tank fuel pump transfer system operates normally.
- Auxiliary tank fuel backup transfer system operates normally.
- Fuel flow indications operate normally.
- FMC FUEL is initialized with the known total fuel quantity.
- Flight remains within 180 minutes of landing at a suitable airport.
- Appropriate gross weight and center-of-gravity adjustments are applied.

NOTE: The airplane must also be dispatched using MMEL item 28-41-03.

MAINTENANCE (M)

NOTE 1: For the auxiliary tank fuel pump transfer system to be operating normally, certain items associated with that system must be operative. See DDG item 28-14-01 for a listing of those items.

NOTE 2: For the auxiliary tank fuel backup transfer system to be operating normally, certain items associated with that system must be operative. See DDG item 28-14-04 for a listing of those items.

Verify the fuel quantity in auxiliary tank(s) and prior to each flight confirm auxiliary fuel cell configuration fields are present on the FQIS Maintenance Page (AMM 28-00-00/901).

1. Verify the fuel quantity in auxiliary tank(s) using one of the following methods:
 - A. Use the associated tank's auxiliary refuel panel quantity indication (if operating).
 - B. Determine the fuel quantity using the associated tank's Electronic Fuel Level Indication (EFLI).
 - C. Defuel the associated tank(s) and service with a known quantity of fuel.
2. Prior to each flight, confirm the appropriate auxiliary fuel cell configuration fields are present on the Fuel Quantity Maintenance Page 2/3.

OPERATIONS (O)

NOTE: FUEL IN AUX and FUEL LOW AUX advisory messages and FUEL AUX XFR Caution message will not be displayed when appropriate.

1. Total fuel quantity must be calculated and manually initialized on the FMC PERF INIT page.

NOTE 1: The total fuel quantity indication may be inoperative or inaccurate.

NOTE 2: The FMC CALCULATED fuel quantity may be blank prior to manual initialization.

- A. Manually enter the total fuel quantity on the PERF INIT page. MANUAL will be displayed next to the entered fuel quantity.

NOTE: The manually entered total fuel quantity and the takeoff speeds may be blank after engine start.

- B. Before and after engine start, verify the fuel quantity entered is displayed under the CALCULATED fuel quantity display on the PROGRESS page 2.
2. Flight Crew must confirm that auxiliary tank fuel transfer has actually occurred by monitoring the center tank fuel quantity.

NOTE 1: Fuel transfer from the auxiliary tank(s) to the center tank begins 55 minutes after takeoff.



NOTE 2: The fuel system synoptic may also be used to monitoring fuel transfer to center tank.

3. Apply appropriate gross weight and center-of-gravity restrictions.

For -200LR:

- A. For gross weight exceeding 563,000 lb (255,000 kg), Center-of-Gravity must be aft of 22.2% MAC for takeoff.

4. If it is necessary to jettison fuel enroute:

NOTE: Selection of fuel jettison with a fuel quantity indication inoperative causes all FMC performance and VNAV data to become invalid. If VNAV is the current active pitch mode of the Autopilot Flight Director System (AFDS), the AFDS pitch mode will fail with a resulting Master Caution light, AUTOPILOT caution message and amber bar through the pitch annunciation on the Primary Flight Display (PFD).

- A. Fuel jettison system maximum landing weight and fuel to remain automatic shutoff functions will be inoperative.
- B. FUEL AUTO JETTISON caution message will be displayed. Accomplish the Manual FUEL AUTO JETTISON Non-Normal Checklist.



777 FAA DDG Supplement Package

Supplement Number: 36-6

Issue Date: 26-SEP-2025

Section Number: 2 (MMEL)

Item Number: 28-41-03

Item Name: Total Fuel Quantity Indication

DDG Effectivity: 777 FAA DDG (D630W003-TBC), Revision 36

MMEL Effectivity: 777 FAA MMEL, Revision 23a

Description: Modified Operations (O) procedure notes.

Reason: To clarify that the fuel quantity will not always be blank prior to manual initialization.

Disposition: For reference in the Operator's MEL.

Attachments: DDG pages 2.28-41-03.1

28-41-03 Total Fuel Quantity Indication

Interval	Installed	Required	Procedure
C	1	0	(O)

May be inoperative provided:

- Fuel flow indications operate normally.
- FMC FUEL is initialized with the known total fuel quantity.
- Flight remains within 180 minutes of landing at a suitable airport.

OPERATIONS (O)

- Total fuel quantity must be calculated and manually initialized on the FMC PERF INIT page.

NOTE 1: The total fuel quantity indication will be inoperative.

NOTE 2: The FMC CALCULATED fuel quantity may be blank prior to manual initialization.

- Manually enter the total fuel quantity on the PERF INIT page. MANUAL will be displayed next to the entered fuel quantity.

NOTE: The manually entered total fuel quantity and the takeoff speeds may blank after engine start.

- Before and after engine start, verify the fuel quantity entered is displayed under the CALCULATED fuel quantity display on the PROGRESS page 2.

- Flight crew should monitor FMC Calculated fuel and the operative fuel tank quantity indications to ensure usages are normal.
- The flight must remain within 180 minutes of landing at a suitable airport.
- If it is necessary to jettison fuel enroute:

NOTE: Selection of fuel jettison with Total Fuel Quantity Indication inoperative causes all FMC Performance and VNAV data to become invalid. If VNAV is the current active pitch mode of the Autopilot Flight Director System (AFDS), the AFDS pitch mode will fail with a resulting Master Caution light, AUTOPILOT caution message and amber bar through the pitch annunciation on the Primary Flight Display (PFD).

- Fuel jettison system maximum landing weight and fuel to remain automatic shutoff functions will be inoperative.
- FUEL AUTO JETTISON caution message will be displayed. Accomplish the FUEL AUTO JETTISON Non-Normal Checklist.



777 FAA DDG Supplement Package

Supplement Number: 36-6

Issue Date: 26-SEP-2025

Section Number: 2 (MMEL)

Item Number: 32-49-01

Item Name: Tire Pressure Indication System

DDG Effectivity: 777 FAA DDG (D630W003-TBC), Revision 36

MMEL Effectivity: 777 FAA MMEL, Revision 23a

Description: Modified option A Maintenance (M) procedure and option B Maintenance Note, added Operations Note to both options.

Reason: To specify an alternate procedure for measuring tire pressure and clarify the reason deactivation of the TPIS may be desired.

Disposition: For reference in the Operator's MEL.

Attachments: DDG pages 2.32-49-01.1-2

32-49-01 Tire Pressure Indication System

32-49-01A Required By Procedures

Interval	Installed	Required	Procedure
C	1	0	(M)

May be inoperative provided alternate procedures are established and used.

NOTE: Any portion of the system that operates normally may be used.

MAINTENANCE (M)

1. For tire pressure check Scheduled Maintenance Task, use a tire pressure gauge to measure tire pressure (AMM 32-00-00/901)

NOTE 1: To prevent display of the TIRE PRESS Advisory message, the Tire Pressure Indication System may be deactivated by opening and collaring P110 panel LG TIRE PRESSURE INDICATOR circuit breaker.

NOTE 2: If desired, the tire pressure indication channel for the affected wheel may be deactivated when a wheel with incompatible tire pressure indication hardware is installed. For wheels with compatible hardware no action is suggested.

- A. For the Messier-Bugatti shipside system without sensor holder installed in the wheel:
 - 1) Remove the tire pressure sensor (AMM 32-49-03-000-801).
 - 2) Reinstall hubcap (AMM 32-45-08-420-801).
- B. For the Crane shipside system without integral tire fill valve/tire pressure sensor installed in the wheel:
 - 1) Remove hubcap with integral relay transformer (AMM 32-45-08-020-801).
 - 2) For main wheel, install TPIS Deactivation hubcap to restore antiskid wheel speed drive function (AMM 32-45-08-420-801).

For -200/-200ER/-300:

- a. Use Boeing P/N 294W5140-16, or -18.

For -200LR/777F/-300ER:

- b. Use Boeing P/N 294W5140-17, or -19.

OPERATIONS NOTE

NOTE: TIRE PRESS Advisory message may be displayed.

**32-49-01 Tire Pressure Indication System**

32-49-01B Procedures Do Not Require Use

Interval	Installed	Required	Procedure
D	1	0	

May be inoperative provided procedures do not require its use.

MAINTENANCE NOTE

1. To prevent display of the TIRE PRESS Advisory message, the Tire Pressure Indication System may be deactivated by opening and collaring P110 panel LG TIRE PRESSURE INDICATOR circuit breaker.
2. If desired, the tire pressure indication channel for the affected wheel may be deactivated when a wheel with incompatible tire pressure indication hardware is installed. For wheels with compatible hardware no action is suggested.
 - A. For the Messier-Bugatti shipside system without sensor holder installed in the wheel:
 - 1) Remove the tire pressure sensor (AMM 32-49-03-000-801).
 - 2) Reinstall hubcap (AMM 32-45-08-420-801).
 - B. For the Crane shipside system without integral tire fill valve/tire pressure sensor installed in the wheel:
 - 1) Remove hubcap with integral relay transformer (AMM 32-45-08-020-801).
 - 2) For main wheel, install TPIS Deactivation hubcap to restore antiskid wheel speed drive function (AMM 32-45-08-420-801).

For -200/-200ER/-300:

- a. Use Boeing P/N 294W5140-16, or -18.

For -200LR/777F/-300ER:

- b. Use Boeing P/N 294W5140-17, or -19.

OPERATIONS NOTE

NOTE: TIRE PRESS Advisory message may be displayed.



777 FAA DDG Supplement Package

Supplement Number: 36-6

Issue Date: 26-SEP-2025

Section Number: 2 (MMEL)

Item Number: 34-58-01

Item Name: Global Positioning Systems

DDG Effectivity: 777 FAA DDG (D630W003-TBC), Revision 36

MMEL Effectivity: 777 FAA MMEL, Revision 23a

Description: **Note: This item supersedes content in a previous supplement.**
Modified PBN guidance.

Reason: To correct and add more clarification for PBN operation capabilities, to clarify specific RNP 2 operations, and to incorporate updated PBN operation capabilities for RNP 4 operations.

Disposition: For reference in the Operator's MEL.

Attachments: DDG pages 2.34-58-01.1-2

34-58-01 Global Positioning Systems

34-58-01A Both Inoperative

Interval	Installed	Required	Procedure
C	2	0	

May be inoperative provided:

- Enroute operations do not require their use.
- SAARU operates normally.

NOTE: The airplane must also be dispatched using MMEL item 34-53-02-01.

MAINTENANCE NOTE

- For both GPS inoperative and AIMS power is interrupted, UTC is lost and the AIMS time function will reset to midnight.
- If desired, the left, right, or center MMR may be deactivated by opening and collaring P11 panel MMR-L, MMR-R and/or MMR-C circuit breaker.
- If the GPS L and GPS R status messages are caused by a fault in the center MMR antenna, opening the MMR-C circuit breaker may restore GPS L and R functions.
- The associated ILS function will be inoperative for a deactivated MMR. Relief for ILS functions is available in MMEL item 34-31-01.

OPERATIONS NOTE

- Operators must consult their Operations Specification for route position accuracy requirements such as that required by certain Required Navigation Performance (RNP) routes.
- The NAV UNABLE RNP advisory message may be displayed during lengthy ground operations and flight crew should accomplish the associated Non-Normal Checklist.
- The TERR POS advisory message will be displayed.
- The RWY/POS update on the FMC-CDU TAKEOFF REF page will be active.
- The ADS-B Out data will be inoperative.
- For airplanes equipped with RAAS:
 - RAAS is inoperative.
 - The RUNWAY SYS advisory message will be displayed.
 - The airplane must also be dispatched using MMEL item 34-46-01-04.
- For PBN operations:
 - RNP AR APCH, RNP APCH, RNP 1, RNP 2, RNP 4 and Advanced RNP operations are not allowed.



- B. RNAV 1, RNAV 2, P-RNAV, B-RNAV and RNAV 5 operations may be conducted provided one DME/DME or one VOR/DME operates normally.
- C. RNP 10 and RNAV 10 operations may be conducted provided appropriate time limitations are observed. Refer to the RNP Navigation Capabilities Document for the time limitations.

NOTE: Refer to DDG Section 4 for additional information.

34-58-01 Global Positioning Systems

34-58-01B One Inoperative

Interval	Installed	Required	Procedure
D	2	1	

One may be inoperative provided enroute operations do not require its use.

NOTE: The airplane must also be dispatched using MMEL item 34-53-02-01.

MAINTENANCE NOTE

1. If desired, one or both MMR not associated with the operative GPS may be deactivated by opening and collaring P11 panel MMR-L, MMR-R and/or MMR-C circuit breaker.
2. The associated ILS function will be inoperative for a deactivated MMR. Relief for ILS functions is available in MMEL item 34-31-01.

For EFB installed:

NOTE: Some functionality of the EFB may be inoperative. For inoperative EFB functions the airplane must also be dispatched using MMEL item 46-11-02-01.

OPERATIONS NOTE

1. The associated ADS-B Out data will be inoperative.
2. For PBN operations:
 - A. RNP AR APCH operations are not allowed.
 - B. RNP 2 and RNP 4 operations over oceanic and remote continental airspace are not allowed.
 - C. RNP APCH, RNP 1, RNP 2 and RNP 4 operations other than those over oceanic or remote continental airspace, Advanced RNP, RNP 10, and RNAV 10 operations may be allowed provided alternate navigation equipment (e.g. DME/DME, VOR/DME, VOR, ADF) is available to continue to a suitable airport if the remaining GPS becomes inoperative.
 - D. RNAV 1, RNAV 2, P-RNAV, B-RNAV and RNAV 5 operations may be conducted.

NOTE: Refer to DDG Section 4 for additional information.



777 FAA DDG Supplement Package

Supplement Number: 36-6

Issue Date: 26-SEP-2025

Section Number: 2 (MMEL)

Item Number: 34-61-01

Item Name: Flight Management Computing Systems (FMCS)

DDG Effectivity: 777 FAA DDG (D630W003-TBC), Revision 36

MMEL Effectivity: 777 FAA MMEL, Revision 23a

Description: **Note: This item supersedes content in a previous supplement.**
Modified PBN guidance.

Reason: To correct and add more clarification for PBN operation capabilities, and to clarify specific RNP 2 operations.

Disposition: For reference in the Operator's MEL.

Attachments: DDG pages 2.34-61-01.1-3

34-61-01 Flight Management Computing Systems (FMCS)
34-61-01-01 With AIMS-1

Interval	Installed	Required	Procedure
C	2	1	(M)

One may be inoperative provided:

- a. Remaining FMC is verified to operate normally.
- b. Enroute operations do not require its use.
- c. Flight remains within 180 minutes of landing at a suitable airport.

MAINTENANCE (M)

Verify remaining FMC is operating normally by positioning the FMC selector to AUTO and verifying the FMC status and advisory messages are not displayed (AMM 34-00-00/901).

OPERATIONS NOTE

1. Flight must remain within 180 minutes of landing at a suitable airport.
2. Operators must consult their Operations Specification for route flight management computing system requirements.
3. For PBN operations:
 - A. RNP AR APCH and RNP 4 operations are not allowed.
 - B. RNP 2 operations over oceanic and remote continental airspace are not allowed.
 - C. RNP APCH, RNP 1, RNP 2 operations other than those over oceanic or remote continental airspace, RNAV 1, RNAV 2, P-RNAV, B-RNAV, RNAV 5 and Advanced RNP operations may be conducted provided alternate navigation equipment (e.g., VOR, ADF) is available to continue to a suitable airport if the remaining FMCS becomes inoperative.
 - D. RNP 10 and RNAV 10 operations may be conducted using the alternate navigation capability in the multifunction control display units (MCDU) if the remaining FMCS becomes inoperative.

NOTE: Refer to DDG Section 4 for additional information.

34-61-01 Flight Management Computing Systems (FMCS)
34-61-01-02 With AIMS-2

Interval	Installed	Required	Procedure
C	2	1	(M)

One may be inoperative provided:

- a. Remaining FMC is verified to operate normally.



- b. Enroute operations do not require its use.

MAINTENANCE (M)

Verify remaining FMC is operating normally by positioning the FMC selector to AUTO and verifying the FMC status and advisory messages are not displayed (AMM 34-00-00/901).

OPERATIONS NOTE

1. Operators must consult their Operations Specification for route flight management computing system requirements.
2. For PBN operations:
 - A. RNP AR APCH and RNP 4 operations are not allowed.
 - B. RNP 2 operations over oceanic and remote continental airspace are not allowed.
 - C. RNP APCH, RNP 1, RNP 2 operations other than those over oceanic or remote continental airspace, RNAV 1, RNAV 2, P-RNAV, B-RNAV, RNAV 5 and Advanced RNP operations may be conducted provided alternate navigation equipment (e.g., VOR, ADF) is available to continue to a suitable airport if the remaining FMCS becomes inoperative.
 - D. RNP 10 and RNAV 10 operations may be conducted using the alternate navigation capability in the multifunction control display units (MCDU) if the remaining FMCS becomes inoperative.

NOTE: Refer to DDG Section 4 for additional information.

34-61-01 Flight Management Computing Systems (FMCS)

34-61-01-03 Navigation Databases

Interval	Installed	Required	Procedure
A	-	0	

May be inoperative provided:

- a. Operations do not require its use.
- b. It is not used in a primary navigation system required by 14 CFR.
- c. Alternate procedures are developed and used.
- d. The ICAO Flight Plan is updated (as required) to notify ATC of the navigation equipment status of the aircraft.
- e. Is repaired within 10 flight days.

NOTE: An out-of-currency or out-of-date navigation database is not authorized MMEL relief per 14 CFR.

NOTE: Although this item may not be applied for an out-of-currency Navigation Database, it does not prohibit dispatch in that condition. Dispatch is allowed with an out-of-currency Navigation Database provided alternate procedures are developed and incorporated in another operational manual, in coordination with local regulatory authorities.

OPERATIONS NOTE

For PBN operations:

The FMCS navigation database must be current for all PBN operations.

NOTE 1: It is assumed that the operator has a preflight procedure to ensure that the navigation database complies with regulatory agency requirements for ensuring the navigation database remains valid for the duration of the flight unless an alternative regulatory agency approved process applies.

NOTE 2: Refer to DDG Section 4 for additional information.



777 FAA DDG Supplement Package

Supplement Number: 36-6

Issue Date: 26-SEP-2025

Section Number: 2 (MMEL)

Item Number: 35-21-06

Item Name: Supernumerary Seats/Crew Rest/Lavatory Gaseous Oxygen System (777F)

DDG Effectivity: 777 FAA DDG (D630W003-TBC), Revision 36

MMEL Effectivity: 777 FAA MMEL, Revision 23a

Description: **Note: This item supersedes content in a previous supplement.**

Modified Maintenance (M) procedure note 4.

Reason: To include the correct AMM reference for deactivation of the supernumerary oxygen system and to clarify the condition that requires system deactivation.

Disposition: For reference in the Operator's MEL.

Attachments: DDG pages 2.35-21-06.1-3

35-21-06 Supernumerary Seats/Crew Rest/Lavatory Gaseous Oxygen System (777F)

Interval	Installed	Required	Procedure
B	1	0	(M) (O)

May be inoperative provided:

- Supernumerary seats and crew rest bunks are blocked and placarded DO NOT OCCUPY.
- An adequate portable oxygen dispensing unit (bottle and mask) is available in the lavatory.
- Personnel address system operates normally.
- Crew members/supernumeraries are appropriately briefed.

MAINTENANCE (M)

NOTE 1: For SUPRNMRY OXYGEN L status message displayed, the oxygen module over the left seat and the oxygen module in the crew rest are inoperative.

NOTE 2: For SUPRNMRY OXYGEN C status message displayed, the oxygen module over the center seats and the oxygen module in the lavatory are inoperative.

NOTE 3: For SUPRNMRY OXYGEN R status message displayed, the oxygen module over the right seat is inoperative.

NOTE 4: SUPRNMRY OXYGEN LOW message could be due to consumption or leakage. If leak is suspected the system must be deactivated (AMM 35-23-00/201).

Block seats and crew rest bunks and placard DO NOT OCCUPY (AMM 35-00-00/901).

OPERATIONS (O)

Crew members/supernumeraries must be briefed prior to takeoff regarding alternate procedures.

35-21-06 Supernumerary Seats/Crew Rest/Lavatory Gaseous Oxygen System (777F)

35-21-06-01 Left Circuit

Interval	Installed	Required	Procedure
B	1	0	(M) (O)



May be inoperative provided:

- Left supernumerary seat is blocked and placarded DO NOT OCCUPY.
- Each crew rest bunk without an adequate portable oxygen dispensing unit (bottle and mask) is blocked and placarded DO NOT OCCUPY.
- For flightcrew rest facility used, personnel address system operates normally.
- Crew members/supernumeraries are appropriately briefed.

MAINTENANCE (M)

NOTE: For SUPRNMRY OXYGEN L status message displayed, the oxygen module over the left seat and the oxygen module in the crew rest are inoperative.

Block left seat and placard DO NOT OCCUPY. Block crew rest bunks without adequate portable oxygen dispensing units (bottle and mask), and placard DO NOT OCCUPY (AMM 35-00-00/901).

OPERATIONS (O)

Crew members/supernumeraries must be briefed prior to takeoff regarding alternate procedures.

35-21-06 Supernumerary Seats/Crew Rest/Lavatory Gaseous Oxygen System (777F)

35-21-06-02 Center Circuit

Interval	Installed	Required	Procedure
B	1	0	(M) (O)

May be inoperative provided:

- Each center supernumerary seat is blocked and placarded DO NOT OCCUPY.
- An adequate portable oxygen dispensing unit (bottle and mask) is available in the lavatory.
- Personnel address system operates normally.
- Crew members/supernumeraries are appropriately briefed.

MAINTENANCE (M)

NOTE: For SUPRNMRY OXYGEN C status message displayed, the oxygen module over the center seats and the oxygen module in the lavatory are inoperative.

Block each center seat and placard DO NOT OCCUPY (AMM 35-00-00/901).

OPERATIONS (O)

Crew members/supernumeraries must be briefed prior to takeoff regarding alternate procedures.

35-21-06 Supernumerary Seats/Crew Rest/Lavatory Gaseous Oxygen System (777F)

35-21-06-03 Right Circuit

Interval	Installed	Required	Procedure
B	1	0	(M) (O)

May be inoperative provided:

- Right supernumerary seat is blocked and placarded DO NOT OCCUPY.
- Crew members/supernumeraries are appropriately briefed.

MAINTENANCE (M)

NOTE: For SUPRNMRY OXYGEN R status message displayed, the oxygen module over the right seat is inoperative.

Block right seat and placard DO NOT OCCUPY (AMM 35-00-00/901).

OPERATIONS (O)

Crew members/supernumeraries must be briefed prior to takeoff regarding alternate procedures.



777 FAA DDG Supplement Package

Supplement Number: 36-6

Issue Date: 26-SEP-2025

Section Number: 2 (MMEL)

Item Number: 78-31-01

Item Name: Thrust Reversers

DDG Effectivity: 777 FAA DDG (D630W003-TBC), Revision 36

MMEL Effectivity: 777 FAA MMEL, Revision 23a

Description: Added Maintenance (M) warning.

Reason: To warn operators of the risks of installing pressure caps or plugs on thrust reverser system hydraulic lines when deactivating the thrust reverser.

Disposition: For reference in the Operator's MEL.

Attachments: DDG pages 2.78-31-01.1-4

78-31-01 Thrust Reversers

78-31-01-01 -200/-200ER/-300

Interval	Installed	Required	Procedure
C	2	1	(M)

One may be inoperative provided:

- Both sync locks are verified in the locked position.
- One locking actuator on each sleeve is verified in the locked position.
- Inoperative reverser is secured in the forward thrust position.

MAINTENANCE (M)

WARNING: Do not install pressure caps or plugs on thrust reverser system hydraulic lines when deactivating the thrust reverser for flight dispatch. Capping or plugging the thrust reverser system hydraulic lines can reduce the safety of the deactivated system and may cause an uncommanded deployment.

Deactivate and secure the inoperative reverser (AMM 78-00-00/901).

- For left engine reverser, open and collar P110 panel L ENG T/R CTRL circuit breaker.
- For right engine reverser, open and collar P210 panel R ENG T/R CTRL circuit breaker.

For GE:

- Deactivate the leading edge slats in the retracted position (AMM 27-81-00/201).
- Gain access to the associated reverser isolation valve (left forward access door - strut aft fairing).
- Deactivate the isolation valve for flight by following the DEACTIVATED MODE instructions placard.
- Close the associated left forward access door - strut aft fairing.
- Open the associated left and right fan cowl panels (AMM 71-11-04/201).
- Remove deactivation pins and install the pins in the left and right reverser sleeves.
- Pressurize the associated hydraulic system (AMM 29-11-00/201).
- Verify that the sync lock and at least one of the two locking actuators on each reverser sleeve has the manual unlock handle in the locked position.
- Depressurize the hydraulic system (AMM 29-11-00/201).



12. Verify that associated engine thrust reverser left and right sleeve positions are within limits using the Electronic Propulsion Control System (EPCS) maintenance page 1/2.

- A. If the associated engine thrust reverser left and right sleeve (T/RL and T/RR) positions for channels A and B are not within the following allowable values, deactivate the associated RVDT.

NOTE: A dual channel RVDT is installed on the non-locking (upper) hydraulic actuator of each sleeve, which results in four readouts per engine.

For PW:

- 1) 0 to +4.0

For GE:

- 2) -7.0 to +7.0

For RR:

- 3) -5.0 to +5.0

- B. For deactivating the associated RVDT:

NOTE: ONLY ONE RVDT MAY BE DEACTIVATED.

- 1) Disconnect the electrical connector DM78002 for left sleeve or DM78003 for right sleeve from the RVDT.
- 2) Cap the connector plug using conductive cap.
- 3) Stow the connector plug to adjacent structure using wire ties.

13. Close the associated left and right fan cowl panels (71-11-04/201).

For GE:

14. Activate the leading edge slats (AMM 27-81-00/201).

OPERATIONS NOTE

After engine start, the associated ENG REV LIMITED L or R advisory message will be displayed.

78-31-01 Thrust Reversers

78-31-01-02 -200LR/-300ER/777F

Interval	Installed	Required	Procedure
C	2	1	(M) (O)

One may be inoperative provided:

- a. Both sync locks are verified in the locked position.
 - b. One locking actuator on each sleeve is verified in the locked position.
 - c. Inoperative reverser is secured in the forward thrust position.
 - d. Appropriate performance adjustments are applied.
-

MAINTENANCE (M)

WARNING: Do not install pressure caps or plugs on thrust reverser system hydraulic lines when deactivating the thrust reverser for flight dispatch. Capping or plugging the thrust reverser system hydraulic lines can reduce the safety of the deactivated system and may cause an uncommanded deployment.

Deactivate and secure the inoperative reverser (AMM 78-00-00/901).

1. For left engine reverser, open and collar P110 panel L ENG T/R CTRL circuit breaker.
2. For right engine reverser, open and collar P210 panel R ENG T/R CTRL circuit breaker.
3. Deactivate the leading edge slats in the retracted position (AMM 27-81-00/201).
4. Gain access to the associated reverser isolation valve (left forward access door - strut aft fairing).
5. Deactivate the isolation valve for flight by following the DEACTIVATED MODE instructions placard.
6. Close the associated left forward access door - strut aft fairing.
7. Open the associated left and right fan cowl panels (AMM 71-11-04/201).
8. Remove deactivation pins and install the pins in the left and right reverser sleeves.
9. Pressurize the associated hydraulic system (AMM 29-11-00/201).
10. Verify that the sync lock and at least one of the two locking actuators on each reverser sleeve has the manual unlock handle in the locked position.
11. Depressurize the hydraulic system (AMM 29-11-00/201).
12. Verify that associated engine thrust reverser left and right sleeve positions are within limits using the Electronic Propulsion Control System (EPCS) maintenance page 1/2.
 - A. If the associated engine thrust reverser left and right sleeve (T/RL and T/RR) positions for channels A and B are not within -7.0 to +7.0, deactivate the associated RVDT.

NOTE: A dual channel RVDT is installed on the non-locking (upper) hydraulic actuator of each sleeve, which results in four readouts per engine.

- B. For deactivating the associated RVDT:

NOTE: ONLY ONE RVDT MAY BE DEACTIVATED.

- 1) Disconnect the electrical connector DM78002 for left sleeve or DM78003 for right sleeve from the RVDT.
 - 2) Cap the connector plug using conductive cap.
 - 3) Stow the connector plug to adjacent structure using wire ties.



13. Close the associated left and right fan cowl panels (71-11-04/201).
14. Activate the leading edge slats (AMM 27-81-00/201).

OPERATIONS (O)

NOTE: After engine start, the associated ENG REV LIMITED L or R advisory message will be displayed.

For airplane performance calculations that include the effect of reverse thrust, adjust performance to account for the inoperative thrust reverser.

Supplement 36-6



777 FAA DDG Supplement Package

Supplement Number: 36-6

Issue Date: 26-SEP-2025

Section Number: 4 (MMEL)

Item Number: N/A

Item Name: Performance Based Navigation

DDG Effectivity: 777 FAA DDG (D630W003-TBC), Revision 36

MMEL Effectivity: 777 FAA MMEL, Revision 23a

Description: **Note: This item supersedes content in a previous supplement.**
Modified PBN guidance.

Reason: To correct the PRR number reference for Enhanced Ground Proximity Warning System (EGPWS), updated wording for the Mode Control Panel (MCP) Switches, to incorporate updated PBN operation capabilities, to correct and add more clarification for PBN operations and to be consistent with the RNP capabilities document.

Disposition: For reference in the Operator's MEL.

Attachments: DDG pages 4.00-02-00.1-7



Section 4 Performance Based Navigation

Performance Based Navigation (PBN) Information

Applying dispatch relief for the following items may affect the ability of the airplane to meet the operational capabilities required for PBN operations. The following information lists items that, if inoperative, may affect the ability of the airplane to meet the performance and functional criteria for various PBN operations. This information can aid in identifying potentially affected capabilities associated with a particular dispatch configuration.

This section lists equipment requirements that may be found in the document 777 RNP Navigation Capabilities Document D243W018-13, revision "O" (or later). Operators should consult their Operations Specification for requirements applicable to their operations.

PBN Equipment Requirements

Display Units (DUs) - (31-61-01)

PBN operations require two PFDs with mini map and EICAS display. The 777 is equipped with six display units. The PFDs are normally displayed on the outboard display units and the NDs are normally displayed on the inboard display units. EICAS is normally displayed on the upper center display unit. MMEL dispatch is only allowed with the lower center display unit inoperative. Therefore, these requirements will always be met for dispatch.

Multi-Purpose Control and Display Units - Control Display Units (CDU) - (34-61-03)

The 777 is equipped with three CDUs. MMEL dispatch is only allowed with the center CDU inoperative. Both left and right CDUs must operate normally for dispatch. PBN operations are allowed with two CDUs. Therefore, this requirement will always be met for dispatch.

AIMS Operational Software

For RNP AR APCH:

Airplane must be equipped with AIMS BP05A or later. Navigation Performance Scales (NPS) and LNAV Go-Around Auto-Engage must be enabled.

Airplanes without Navigation Performance Scales and LNAV Go-Around Auto-Engage (AIMS BP03 and earlier versions) do not support these operations.

Fault Tolerant ADIRU

For AIMS-1 V16 and AIMS-2 V17 or later:

AIR DATA SYS or NAV ADIRU INERTIAL messages must not be displayed.

For all other AIMS-1 or AIMS-2 versions:

NAV AIR DATA SYS or NAV ADIRU INERTIAL messages must not be displayed.

MMEL dispatch is not allowed with any of the above messages displayed. Therefore, this requirement will always be met for dispatch.

Autopilot Flight Director Computers (AFDC) - (22-11-01)

The 777 autopilot flight director system (AFDS) is equipped with three autopilot flight director computers (AFDCs). Each of the three AFDCs can provide both flight director guidance, which can be displayed on the primary flight displays (PFDs), and send autopilot commands to the flight control actuation system when the autopilot is engaged.

MMEL dispatch is allowed with one, two, or all three AFDCs inoperative. Therefore, PBN operations may be affected with more than one AFDC inoperative.

For dispatch with two Autopilot Flight Director Computers) inoperative (22-11-01B):

- RNP AR APCH operations are not allowed.
- Other PBN operations may be conducted.

For dispatch with all (3) Autopilot Flight Director Computers) inoperative (22-11-01C):

- PBN operations are not allowed.

Autopilot and Flight Director

PBN operations may be conducted using either the autopilot or flight director. The minimum authorized RNP value may be different depending on whether autopilot or flight director is used.

Autopilot

When the autopilot is engaged, autopilot commands from the AFDCs are sent to the Primary Flight Computers to generate flight control surface commands.

If the autopilot function is inoperative, PBN operations may be conducted using the flight director.

The autopilot will be impacted if any of the following equipment is inoperative:

1. Mode Control Panel Switches - A/P Engage Switches
 - A. For dispatch with one autopilot engage switch inoperative (22-11-06-01A):
 - RNP AR APCH operations using autopilot are not allowed.
 - RNP AR APCH operations may be conducted using flight director provided its use is appropriate for the intended operation.
 - Other PBN operations using autopilot may be conducted.
 - B. For dispatch with both autopilot engage switches inoperative (22-11-06-01B):
 - PBN operations using autopilot are not allowed.
 - PBN operations may be conducted using flight director provided its use is appropriate for the intended operation.
2. Control Wheel Autopilot Disconnect Switches
 - A. For dispatch with one control wheel autopilot disconnect switch inoperative (22-11-08A):
 - PBN operations using autopilot at 1,500 feet AGL or below are not allowed.
 - PBN operations may be conducted using flight director provided its use is appropriate for the intended operation.
 - B. For dispatch with both control wheel autopilot disconnect switches inoperative (22-11-08B):
 - PBN operations using autopilot are not allowed.
 - PBN operations may be conducted using flight director provided its use is appropriate for the intended operation.
3. Warning Electronic System (WES) Autopilot Disconnect Warning Function
 - A. For dispatch with WES autopilot disconnect warning function inoperative (31-51-05):
 - PBN operations using autopilot are not allowed.
 - PBN operations may be conducted using flight director provided its use is appropriate for the intended operation.

Flight Director

When the flight director is selected on, flight director guidance from the AFDCs is sent to and displayed on the Primary Flight Displays (PFDs). The flight crew uses the flight director guidance to manually control the airplane in accordance with the command guidance from the AFDCs.

Flight director guidance can be displayed on one or both PFDs.

Flight director guidance is selected on and off by using the flight director switches on the mode control panel (MCP).

The flight director guidance will be impacted if any of the following equipment is inoperative:

1. Mode Control Panel Switches - F/D Switches
 - A. For dispatch with one Flight Director switch inoperative (22-11-06-04):
 - RNP AR APCH operations using flight director are not allowed.
 - RNP AR APCH operations may be conducted using autopilot provided its use is appropriate for the intended operation.
 - Other PBN operations using flight director may be conducted provided flight director operates normally on the side of the pilot flying (PF).
 - B. For dispatch with both Flight Director switches inoperative (22-11-06-04):
 - PBN operations using flight director are not allowed.
 - PBN operations may be conducted using autopilot provided its use is appropriate for the intended operation.
2. Flight Director Systems
 - A. For dispatch with one Flight Director system inoperative (22-11-10):
 - RNP AR APCH operations using flight director are not allowed.
 - RNP AR APCH operations may be conducted using autopilot provided its use is appropriate for the intended operation.
 - Other PBN operations using flight director may be conducted provided flight director operates normally on the side of the pilot flying (PF).
 - B. For dispatch with both Flight Director systems inoperative (22-11-10):
 - PBN operations using flight director are not allowed.
 - PBN operations may be conducted using autopilot provided its use is appropriate for the intended operation.

LNAV/VNAV Mode Control Panel (MCP) Switches - (22-11-06-09)

For dispatch with lateral navigation (LNAV) switch inoperative (22-11-06-09):

- PBN operations are not allowed.

For dispatch with vertical navigation (VNAV) switch inoperative. (22-11-06-09):

- RNP AR APCH operations are not allowed.
- RNP APCH operations may be conducted using other vertical modes provided appropriate approach minimums are used.
- Other PBN operations may be conducted.

Takeoff/Go-Around (TO/GA) Switches - (22-11-09)

There are two TO/GA switches installed on the 777. MMEL dispatch is allowed with one or both TO/GA switches inoperative.

For dispatch with one TO/GA switch inoperative (22-11-09A):

- RNP AR APCH operations are not allowed.
- Other PBN operations may be conducted.

For dispatch with both TO/GA switches inoperative (22-11-09B):

- PBN operations using autopilot or flight director below 500 feet AGL or MDA are not allowed.

Global Positioning Systems (GPS) - (34-58-01)

The 777 is equipped with two GPSs. MMEL dispatch is allowed with one or both GPSs inoperative.

For dispatch with one GPS inoperative (34-58-01B):

- RNP AR APCH operations are not allowed.
- RNP 2 and RNP 4 operations over oceanic and remote continental airspace are not allowed.
- RNP APCH, RNP 1, RNP 2 and RNP 4 operations other than those over oceanic or remote continental airspace, Advanced RNP, RNP 10, and RNAV 10 operations may be conducted provided alternate navigation equipment (e.g. DME/DME, VOR/DME, VOR, ADF) is available to continue to a suitable airport if the remaining GPS becomes inoperative.
- RNAV 1, RNAV 2, P-RNAV, B-RNAV and RNAV 5 operations may be conducted.

For dispatch with both GPSs inoperative (34-58-01A):

- RNP AR APCH, RNP APCH, RNP 1, RNP 2, RNP 4 and Advanced RNP operations are not allowed.

- RNAV 1, RNAV 2, P-RNAV, B-RNAV and RNAV 5 operations may be conducted provided one DME/DME or one VOR/DME operates normally.
- RNP 10 and RNAV 10 operations may be conducted provided appropriate time limitations are observed. Refer to the RNP Navigation Capabilities Document for the time limitations.

VOR Navigation Systems - (34-51-01)

The 777 is equipped with two VORs. MMEL dispatch is allowed with one or both VORs inoperative.

- RNAV 1, RNAV 2, P-RNAV, B-RNAV and RNAV 5 operations may be conducted provided one GPS, one DME/DME or one VOR/DME operates normally.
- Other PBN operations may be conducted.

Distance Measuring Equipment Systems (DME) - (34-55-01)

The 777 is equipped with two DMEs. MMEL dispatch is allowed with one or both DMEs inoperative.

- RNAV 1, RNAV 2, P-RNAV, B-RNAV and RNAV 5 operations may be conducted provided one GPS, one DME/DME or one VOR/DME operates normally.
- Other PBN operations may be conducted.

Radio Altimeter Systems - (34-33-01)

The 777 is equipped with three radio altimeters. MMEL dispatch is allowed with one radio altimeter inoperative.

PBN operations are allowed with two radio altimeters. Therefore, this requirement will always be met for dispatch.

Ground Proximity Warning Systems (GPWS) - (34-46-01), Terrain Awareness Function (34-46-01-03)

777 airplanes are equipped with a single ground proximity warning system (GPWS). Airplanes with PRR 61724 or equivalent are equipped with enhanced GPWS (EGPWS) that includes a terrain awareness function.

MMEL dispatch is allowed with the GPWS (EGPWS) inoperative.

For dispatch with either GND PROX SYS or TERR POS Advisory message displayed:

- RNP AR APCH operations are not allowed.
- Other PBN operations may be conducted.

Flight Management Computing Systems (FMCS) - (34-61-01)

There are two flight management computing systems (FMCSs) installed on the 777. MMEL dispatch is allowed with one FMCS inoperative.

For dispatch with one FMCS inoperative:

- RNP AR APCH and RNP 4 operations are not allowed.
- RNP 2 operations over oceanic and remote continental airspace are not allowed.
- RNP APCH, RNP 1, RNP 2 operations other than those over oceanic or remote continental airspace, RNAV 1, RNAV 2, P-RNAV, B-RNAV, RNAV 5 and Advanced RNP operations may be conducted provided alternate navigation equipment (e.g., VOR, ADF) is available to continue to a suitable airport if the remaining FMCS becomes inoperative.
- RNP 10 and RNAV 10 operations may be conducted using the alternate navigation capability in the multifunction control display units (MCDU) if the remaining FMCS becomes inoperative.

Flight Management Computing Systems (FMCS) - Navigation Database (34-61-01-03)

The FMCS navigation database must be current for PBN operations.

It is assumed that the operator has a preflight procedure to ensure that the navigation database complies with regulatory agency requirements for ensuring the navigation database remains valid for the duration of the flight unless an alternative regulatory agency approved process applies.